



```
In [ ]: from google.colab import drive
drive.mount('/content/drive')
```

Thesis Project by Group CLS

THESIS 1-3

Thesis Project

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Section 1. Introduction

1.1 Problem Overview

Large-scale global optimization (LSGO) presents a critical challenge in computational intelligence, where the "curse of dimensionality" leads to an explosive increase in local optima and a high risk of premature convergence. While foundational algorithms like Differential Evolution (DE) are robust, their dependence on crossover and random mutation may restrict guided exploration — reducing particle diversity over time and leading to premature convergence.

This notebook implements and evaluates the proposed **Diversity-Preserving Level-Based Learning Differential Evolution (DP-LLDE)** algorithm, tested against the **CEC 2017 Large-Scale Global Optimization Benchmark** across 29 benchmark functions (F1-F30, excluding deprecated F2) at **10 dimensions**.

1.2 Research Objectives

1. Design and implement DP-LLDE, integrating sequential relative particle mutation and dynamic level-based learning into the Differential Evolution framework.
2. Evaluate DP-LLDE on the CEC 2017 benchmark suite (F1-F30) at 10

dimensions.

3. Examine convergence behavior, solution quality, and the effect of diversity-preserving strategies across unimodal, multimodal, hybrid, and composition functions.

1.3 Algorithm Overview - DP-LLDE

DP-LLDE extends LBLDE (Qiao et al., 2022) with three core contributions:

1. **Sequential Level Assignment** — each individual is assigned a learning source level drawn randomly from $\{2, \dots, NL\}$ on the first generation, decaying toward the elite level (level 1) in subsequent generations.
2. **Diversity-Aware Exemplar Selection** — if two exemplars e_1, e_2 fall below a diversity threshold, one is replaced with the most distant candidate in that level, preventing exemplar collapse.
3. **Normalized Mutation Pressure (Eq. 10):**

$$V_i = e_1 + F_i \times (e_2 - X_i) \times \frac{NL - k}{NL}$$

The factor $(NL-k)/NL$ decays to zero as $k \rightarrow NL$, naturally reducing mutation pressure as learning quality improves.

Parameter	Value
Population Size (NP)	100
Number of Levels (NL)	4
Initial μ_{CR}	0.35
Max FES	$10,000 \times D$
Dimensions (D)	10
Seed	42

Section 2. Benchmark Function Description

2.1 Overview

The **CEC 2017 Large-Scale Global Optimization Benchmark** (Awad et al., 2017) is a standardized suite of 30 test functions designed to evaluate optimization algorithms on complex, high-dimensional search spaces. The benchmark tests an algorithm's ability to avoid false local optima, convergence speed, and solution accuracy.

All functions are defined over the search space $[-100, 100]^D$ with their respective global optima offset by a function-specific bias value.

Library used: `cec2017` by Duncan Tilley — provides the official shift vectors, rotation matrices, and shuffle permutations from the CEC 2017 specification.

2.2 Unimodal Function Overview

Unimodal functions contain a **single global optimum and no local optima**. They are used to test the convergence speed and accuracy of an algorithm in a straightforward landscape.

No.	Function Name	Global Optimum
F1	Shifted and Rotated Bent Cigar	100
F2	Shifted and Rotated Sum of Different Power (<i>Deprecated</i>)	200
F3	Shifted and Rotated Zakharov	300

Note: F2 is officially deprecated from the CEC 2017 benchmark but is included here for completeness as it remains implemented in the `cec2017` library.

2.3 Multimodal Function Overview

Multimodal functions contain **multiple local optima**, making them significantly harder to solve. They are used to evaluate an algorithm's capacity to avoid premature convergence and escape false local optima.

No.	Function Name	Global Optimum
F4	Shifted and Rotated Rosenbrock's	400
F5	Shifted and Rotated Rastrigin's	500

No.	Function Name	Global Optimum
F6	Shifted and Rotated Expanded Schaffer's F6	600
F7	Shifted and Rotated Lunacek Bi-Rastrigin	700
F8	Shifted and Rotated Non-Continuous Rastrigin's	800
F9	Shifted and Rotated Levy	900
F10	Shifted and Rotated Schwefel's	1000

2.4 Hybrid Function Overview

Hybrid functions divide the search space into several regions, each governed by a **different basic function**. They assess an algorithm's adaptability to heterogeneous landscapes and mixed search behaviors.

No.	Component Functions	Global Optimum
F11	Zakharov; Rosenbrock; Rastrigin	1100
F12	Elliptic; Schwefel; Bent Cigar	1200
F13	Bent Cigar; Rosenbrock; Bi-Rastrigin	1300
F14	Elliptic; Ackley; Schaffer F7; Rastrigin	1400
F15	Bent Cigar; HGBat; Rastrigin; Rosenbrock	1500
F16	Schaffer F6; HGBat; Rosenbrock; Schwefel	1600
F17	Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin	1700
F18	Elliptic; Ackley; Rastrigin; HGBat; Discus	1800
F19	Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer F6	1900
F20	HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer F7	2000

2.5 Composite Function Overview

Composition functions **combine multiple shifted and rotated basic functions** into a single search space, creating highly complex and deceptive optimization landscapes. They are the most challenging category in the CEC 2017 suite and significantly test an algorithm's overall robustness and global search capability.

No.	Component Functions	Global Optimum
F21	Rosenbrock; Elliptic; Rastrigin	2100
F22	Rastrigin; Griewank; Schwefel	2200
F23	Rosenbrock; Ackley; Schwefel; Rastrigin	2300
F24	Ackley; Elliptic; Griewank; Rastrigin	2400
F25	Rastrigin; HappyCat; Ackley; Discus; Rosenbrock	2500
F26	Schaffer F6; Schwefel; Griewank; Rosenbrock; Rastrigin	2600
F27	HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer F6	2700
F28	Ackley; Griewank; Discus; Rosenbrock; HappyCat; Schaffer F6	2800
F29	F15; F16; F17	2900
F30	F15; F18; F19	3000

Section 3. List of Required Libraries

3.1 List of Libraries and Versions

The following libraries and modules were used for optimization, benchmarking, and visualization in this project:

Library / Module	Purpose	Version
<code>python</code>	Core programming language	3.10+
<code>numpy</code>	Numerical computing and array operations	2.0.2
<code>matplotlib</code>	Plotting and saving convergence graphs	3.10.0
<code>cec2017</code>	CEC 2017 benchmark functions (simple, hybrid, composition)	Latest (pip)
<code>cec2017.simple</code>	Simple benchmark functions F1-F10	—
<code>cec2017.hybrid</code>	Hybrid benchmark functions F11-F20	—
<code>cec2017.composition</code>	Composition benchmark functions F21-F30	—
<code>dplldc</code>	DP-LLDE optimization algorithm (uploaded locally)	Custom
<code>google.colab.files</code>	File upload utility for loading <code>dplldc.py</code> into the Colab environment	Built-in
<code>sys</code>	Clears cached module before re-importing <code>dplldc.py</code>	Built-in

Section 4. Environment Setup and Validation

4.1 Installation

Install `cec2017`

```
In [ ]: !pip install cec2017 --quiet
```

ERROR: Could not find a version that satisfies the requirement cec2017 (from versions: none)
ERROR: No matching distribution found for cec2017

4.2 File Validation

Import `simple.py`, `hybrid.py`, and `composition.py` python files for benchmark functions

```
In [ ]: from cec2017 import simple, hybrid, composition
print('cec2017 loaded!')
```

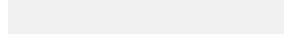
cec2017 loaded!

Import `sys` control over warning messages for verification that `dpllde.py` exists

```
In [ ]: from google.colab import files
import sys

uploaded = files.upload()

if 'dpllde.py' in uploaded:
    if 'dpllde' in sys.modules:
        del sys.modules['dpllde']
    import dpllde
    from dpllde import DPLLDE
    print('dpllde.py uploaded and imported successfully!')
else:
    print('ERROR: Please upload a file named dpllde.py')
```

 Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

ERROR: Please upload a file named dpllde.py

```
In [ ]: import dpllde
        from dpllde import DPLLDE
```

Section 5. Configuration and Helper Functions

5.1 Library Imports

Import libraries `numpy`, `matplotlib`, and `cec2017` for numerical computation, visualization, and CEC 2017 benchmark functions.

```
In [ ]: import numpy as np
import matplotlib.pyplot as plt
from cec2017 import simple, hybrid, composition
```

5.2 Global Parameters

Define global parameters

```
In [ ]: D = 30 # Change this when testing different dimensions
```

```
In [ ]: bounds = np.array([[-100.0, 100.0]] * D)
MAX_FES = 10000 * D
NP, NL, MU_CR = 100, 4, 0.35
SEED = 42
```

Store all results for summary table under `all_results` dict

```
In [ ]: all_results = {}
```

5.3 Function Definitions

Define `make_wrapper` to adapt CEC 2017 functions to accept a 1D input array from DP-LLDE, returning a large penalty value if the output is invalid (nan/inf).

```
In [ ]: def make_wrapper(fn):
    """Wrap a cec2017 function (expects (1,D) input) to accept (D,) input."""
    def wrapper(x):
        val = fn(x.reshape(1, -1))
        if hasattr(val, '__len__'):
            val = float(val[0])
        else:
            val = float(val)
        return val if np.isfinite(val) else 1e18
    return wrapper
```

Define `run_and_plot` to execute a single DP-LLDE optimization run on a given benchmark function, print the results, store them for the summary table, and save the convergence graph.

```
In [ ]: def run_and_plot(fn_name, fn_eval, f_global, fn_label):
    """Run DP-LLDE on a function and plot convergence + diversity graphs."""
    print('=' * 80)
    print(f'DP-LLDE on CEC 2017 {fn_name} – Single Run')
    print('=' * 80)
    print(f'Function      : {fn_label} ({D}D)')
    print(f'Bounds        : [-100, 100]')
    print(f'Global optimum : {f_global}')

    opt = DPLLDE(
        objective_func=fn_eval,
        bounds=bounds,
        NP=NP, NL=NL, mu_CR_ini=MU_CR,
        max_fes=MAX_FES,
        seed=SEED
    )
    # ← Unpack 4 values now
    best_sol, best_fit, history, diversity_history = opt.optimize(verbose=True)

    error = abs(best_fit - f_global)
    print(f'\n{"="*80}')
    print('Results:')
    print(f'  Best Fitness      : {best_fit:.6e}')
    print(f'  Global Optimum    : {f_global:.6e}')
    print(f'  Error (|best-optimum|) : {error:.6e}')
    print(f'  Best Solution (5 dim) : {best_sol[:5]}')
    print(f'  D^(0) (init. diversity): {opt.D_0:.4e}')
    print(f'{"="*80}')

    # Store for summary
    all_results[fn_name] = {
        'label': fn_label,
        'best_fit': best_fit,
        'f_global': f_global,
        'error': error
    }

    fes_axis = np.linspace(0, MAX_FES, len(history))
    gen_axis = np.arange(len(diversity_history)) # Generations (0, 1, 2, ...)

    # — Two-panel figure —
    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 10))

    # Convergence curve (top) – FEs on x-axis
    ax1.plot(fes_axis, history, color='steelblue', linewidth=1.5)
    ax1.set_ylabel('Best Fitness', fontsize=12)
    ax1.set_xlabel('Function Evaluations (FEs)', fontsize=12)
    ax1.set_title(
        f'DP-LLDE – CEC 2017 {fn_name}: {fn_label} ({D}D)',
```



```

        fontsize=12
    )
    ax1.grid(True, which='both', linestyle='--', alpha=0.6)

    # Diversity curve (bottom) – Generations on x-axis
    ax2.plot(gen_axis, diversity_history, color='darkorange', linewidth=1.5)
    ax2.set_xlabel('Generation', fontsize=12)
    ax2.set_ylabel('Population Diversity\n(Mean Std Dev)', fontsize=12)
    ax2.set_title(f'Diversity Across Generations – {fn_name}', fontsize=12)
    ax2.grid(True, which='both', linestyle='--', alpha=0.6)

    plt.tight_layout()
    fname = f'convergence_{fn_name}.png'
    plt.savefig(fname, dpi=150)
    plt.show()
    print(f'Graph saved as {fname}\n')

```

Section 6. CEC2017 Benchmark Functions

6.1 Unimodal Functions (F1-F3)

```

In [ ]: # =====
# F1 – Shifted and Rotated Bent Cigar
# =====
run_and_plot('F1', make_wrapper(simple.f1), 100.0, 'Shifted and Rotated Bent C

```

=====

=

DP-LLDE on CEC 2017 F1 – Single Run

=====

=

Function : Shifted and Rotated Bent Cigar (30D)
Bounds : [-100, 100]
Global optimum : 100.0
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 1.189140e+12

Generation 10:	Best = 2.850124e+11	FES = 1100	$\mu_F = 0.559$	$\mu_{CR} = 0.357$
Diversity = 2.3657e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 8.494643e+10	FES = 2100	$\mu_F = 0.601$	$\mu_{CR} = 0.358$
Diversity = 1.1027e+01	$k_t = [4, 2, 4, 4]$			
Generation 30:	Best = 3.025903e+10	FES = 3100	$\mu_F = 0.633$	$\mu_{CR} = 0.353$
Diversity = 6.4056e+00	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 1.486022e+10	FES = 4100	$\mu_F = 0.673$	$\mu_{CR} = 0.354$
Diversity = 3.8525e+00	$k_t = [4, 3, 4, 2]$			
Generation 50:	Best = 7.984105e+09	FES = 5100	$\mu_F = 0.707$	$\mu_{CR} = 0.353$
Diversity = 2.5072e+00	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 3.467744e+09	FES = 6100	$\mu_F = 0.725$	$\mu_{CR} = 0.357$
Diversity = 1.6868e+00	$k_t = [3, 3, 4, 2]$			
Generation 70:	Best = 1.734313e+09	FES = 7100	$\mu_F = 0.733$	$\mu_{CR} = 0.358$
Diversity = 1.2202e+00	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 1.105761e+09	FES = 8100	$\mu_F = 0.748$	$\mu_{CR} = 0.354$
Diversity = 9.3497e-01	$k_t = [2, 4, 4, 3]$			
Generation 90:	Best = 7.119041e+08	FES = 9100	$\mu_F = 0.760$	$\mu_{CR} = 0.356$
Diversity = 8.8911e-01	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 2.897901e+08	FES = 10100	$\mu_F = 0.769$	$\mu_{CR} = 0.355$
Diversity = 8.4351e-01	$k_t = [3, 4, 4, 4]$			
Generation 110:	Best = 1.520550e+08	FES = 11100	$\mu_F = 0.764$	$\mu_{CR} = 0.351$
Diversity = 7.6951e-01	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 8.636296e+07	FES = 12100	$\mu_F = 0.755$	$\mu_{CR} = 0.351$
Diversity = 7.0949e-01	$k_t = [3, 2, 2, 4]$			
Generation 130:	Best = 5.192206e+07	FES = 13100	$\mu_F = 0.753$	$\mu_{CR} = 0.351$
Diversity = 7.8792e-01	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 2.648906e+07	FES = 14100	$\mu_F = 0.750$	$\mu_{CR} = 0.345$
Diversity = 9.2673e-01	$k_t = [3, 3, 4, 4]$			
Generation 150:	Best = 1.575593e+07	FES = 15100	$\mu_F = 0.721$	$\mu_{CR} = 0.341$
Diversity = 7.4216e-01	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 9.776117e+06	FES = 16100	$\mu_F = 0.722$	$\mu_{CR} = 0.322$
Diversity = 7.3504e-01	$k_t = [4, 2, 3, 2]$			
Generation 170:	Best = 7.272944e+06	FES = 17100	$\mu_F = 0.716$	$\mu_{CR} = 0.316$
Diversity = 7.1096e-01	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 5.723017e+06	FES = 18100	$\mu_F = 0.725$	$\mu_{CR} = 0.282$
Diversity = 6.1318e-01	$k_t = [3, 2, 4, 3]$			
Generation 190:	Best = 4.398792e+06	FES = 19100	$\mu_F = 0.749$	$\mu_{CR} = 0.255$
Diversity = 5.5735e-01	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 3.153908e+06	FES = 20100	$\mu_F = 0.747$	$\mu_{CR} = 0.218$
Diversity = 6.5085e-01	$k_t = [3, 2, 2, 2]$			
Generation 210:	Best = 2.275547e+06	FES = 21100	$\mu_F = 0.756$	$\mu_{CR} = 0.194$
Diversity = 6.9384e-01	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 1.367460e+06	FES = 22100	$\mu_F = 0.743$	$\mu_{CR} = 0.173$
Diversity = 6.6654e-01	$k_t = [3, 4, 4, 3]$			

Generation	230: Best = 9.436236e+05	FES = 23100	$\mu_F = 0.740$	$\mu_{CR} = 0.158$
Diversity =	6.3688e-01	$k_t = [1, 1, 1, 1]$		
Generation	240: Best = 5.543813e+05	FES = 24100	$\mu_F = 0.742$	$\mu_{CR} = 0.147$
Diversity =	6.7110e-01	$k_t = [4, 4, 4, 3]$		
Generation	250: Best = 4.210762e+05	FES = 25100	$\mu_F = 0.754$	$\mu_{CR} = 0.146$
Diversity =	7.9415e-01	$k_t = [1, 1, 1, 1]$		
Generation	260: Best = 2.202167e+05	FES = 26100	$\mu_F = 0.757$	$\mu_{CR} = 0.139$
Diversity =	7.6373e-01	$k_t = [4, 2, 3, 2]$		
Generation	270: Best = 1.519180e+05	FES = 27100	$\mu_F = 0.750$	$\mu_{CR} = 0.135$
Diversity =	6.7256e-01	$k_t = [1, 1, 1, 1]$		
Generation	280: Best = 1.239644e+05	FES = 28100	$\mu_F = 0.754$	$\mu_{CR} = 0.121$
Diversity =	6.1889e-01	$k_t = [3, 4, 2, 2]$		
Generation	290: Best = 8.982482e+04	FES = 29100	$\mu_F = 0.753$	$\mu_{CR} = 0.110$
Diversity =	6.9788e-01	$k_t = [1, 1, 1, 1]$		
Generation	300: Best = 5.668576e+04	FES = 30100	$\mu_F = 0.752$	$\mu_{CR} = 0.102$
Diversity =	6.7793e-01	$k_t = [3, 4, 2, 3]$		
Generation	310: Best = 4.370161e+04	FES = 31100	$\mu_F = 0.758$	$\mu_{CR} = 0.102$
Diversity =	6.8306e-01	$k_t = [1, 1, 1, 1]$		
Generation	320: Best = 3.206880e+04	FES = 32100	$\mu_F = 0.761$	$\mu_{CR} = 0.101$
Diversity =	8.4880e-01	$k_t = [4, 2, 2, 2]$		
Generation	330: Best = 2.537306e+04	FES = 33100	$\mu_F = 0.762$	$\mu_{CR} = 0.105$
Diversity =	1.0015e+00	$k_t = [1, 1, 1, 1]$		
Generation	340: Best = 1.810456e+04	FES = 34100	$\mu_F = 0.753$	$\mu_{CR} = 0.110$
Diversity =	8.5276e-01	$k_t = [3, 4, 3, 3]$		
Generation	350: Best = 1.253533e+04	FES = 35100	$\mu_F = 0.752$	$\mu_{CR} = 0.109$
Diversity =	7.3176e-01	$k_t = [1, 1, 1, 1]$		
Generation	360: Best = 7.971989e+03	FES = 36100	$\mu_F = 0.767$	$\mu_{CR} = 0.111$
Diversity =	7.5465e-01	$k_t = [4, 4, 3, 4]$		
Generation	370: Best = 5.651986e+03	FES = 37100	$\mu_F = 0.781$	$\mu_{CR} = 0.108$
Diversity =	6.2716e-01	$k_t = [1, 1, 1, 1]$		
Generation	380: Best = 4.206443e+03	FES = 38100	$\mu_F = 0.773$	$\mu_{CR} = 0.105$
Diversity =	5.1239e-01	$k_t = [4, 2, 2, 3]$		
Generation	390: Best = 2.922289e+03	FES = 39100	$\mu_F = 0.774$	$\mu_{CR} = 0.109$
Diversity =	4.2849e-01	$k_t = [1, 1, 1, 1]$		
Generation	400: Best = 2.222821e+03	FES = 40100	$\mu_F = 0.783$	$\mu_{CR} = 0.105$
Diversity =	4.2604e-01	$k_t = [3, 3, 4, 4]$		
Generation	410: Best = 1.658840e+03	FES = 41100	$\mu_F = 0.779$	$\mu_{CR} = 0.111$
Diversity =	3.5549e-01	$k_t = [1, 1, 1, 1]$		
Generation	420: Best = 1.086789e+03	FES = 42100	$\mu_F = 0.785$	$\mu_{CR} = 0.108$
Diversity =	3.7304e-01	$k_t = [3, 3, 3, 4]$		
Generation	430: Best = 9.318922e+02	FES = 43100	$\mu_F = 0.784$	$\mu_{CR} = 0.106$
Diversity =	3.0788e-01	$k_t = [1, 1, 1, 1]$		
Generation	440: Best = 6.853954e+02	FES = 44100	$\mu_F = 0.774$	$\mu_{CR} = 0.101$
Diversity =	2.6220e-01	$k_t = [2, 3, 4, 3]$		
Generation	450: Best = 5.783864e+02	FES = 45100	$\mu_F = 0.793$	$\mu_{CR} = 0.100$
Diversity =	2.2485e-01	$k_t = [1, 1, 1, 1]$		
Generation	460: Best = 4.364513e+02	FES = 46100	$\mu_F = 0.798$	$\mu_{CR} = 0.102$
Diversity =	2.3944e-01	$k_t = [3, 3, 2, 2]$		
Generation	470: Best = 3.569304e+02	FES = 47100	$\mu_F = 0.806$	$\mu_{CR} = 0.102$
Diversity =	2.2823e-01	$k_t = [1, 1, 1, 1]$		
Generation	480: Best = 2.906322e+02	FES = 48100	$\mu_F = 0.801$	$\mu_{CR} = 0.103$
Diversity =	1.8339e-01	$k_t = [4, 2, 3, 3]$		
Generation	490: Best = 2.381750e+02	FES = 49100	$\mu_F = 0.797$	$\mu_{CR} = 0.105$
Diversity =	1.6868e-01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 2.166106e+02	FES = 50100	$\mu_F = 0.794$	$\mu_{CR} = 0.108$
Diversity = 1.5922e-01	$k_t = [2, 3, 2, 4]$		
Generation 510: Best = 1.996092e+02	FES = 51100	$\mu_F = 0.802$	$\mu_{CR} = 0.099$
Diversity = 1.2212e-01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.775399e+02	FES = 52100	$\mu_F = 0.801$	$\mu_{CR} = 0.110$
Diversity = 1.0610e-01	$k_t = [4, 4, 2, 3]$		
Generation 530: Best = 1.694352e+02	FES = 53100	$\mu_F = 0.803$	$\mu_{CR} = 0.106$
Diversity = 1.1497e-01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.544345e+02	FES = 54100	$\mu_F = 0.803$	$\mu_{CR} = 0.098$
Diversity = 9.5219e-02	$k_t = [4, 4, 4, 3]$		
Generation 550: Best = 1.462071e+02	FES = 55100	$\mu_F = 0.791$	$\mu_{CR} = 0.095$
Diversity = 9.5443e-02	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.434434e+02	FES = 56100	$\mu_F = 0.788$	$\mu_{CR} = 0.093$
Diversity = 8.8352e-02	$k_t = [2, 3, 3, 3]$		
Generation 570: Best = 1.338327e+02	FES = 57100	$\mu_F = 0.779$	$\mu_{CR} = 0.095$
Diversity = 8.1937e-02	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.325777e+02	FES = 58100	$\mu_F = 0.764$	$\mu_{CR} = 0.095$
Diversity = 7.4559e-02	$k_t = [2, 3, 4, 2]$		
Generation 590: Best = 1.323069e+02	FES = 59100	$\mu_F = 0.792$	$\mu_{CR} = 0.102$
Diversity = 7.8206e-02	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.295288e+02	FES = 60100	$\mu_F = 0.805$	$\mu_{CR} = 0.104$
Diversity = 7.2940e-02	$k_t = [3, 2, 4, 2]$		
Generation 610: Best = 1.240949e+02	FES = 61100	$\mu_F = 0.807$	$\mu_{CR} = 0.104$
Diversity = 7.2084e-02	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.226374e+02	FES = 62100	$\mu_F = 0.785$	$\mu_{CR} = 0.100$
Diversity = 7.0239e-02	$k_t = [2, 2, 4, 2]$		
Generation 630: Best = 1.202426e+02	FES = 63100	$\mu_F = 0.795$	$\mu_{CR} = 0.103$
Diversity = 6.7652e-02	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.155480e+02	FES = 64100	$\mu_F = 0.777$	$\mu_{CR} = 0.101$
Diversity = 5.9867e-02	$k_t = [4, 2, 3, 2]$		
Generation 650: Best = 1.149902e+02	FES = 65100	$\mu_F = 0.784$	$\mu_{CR} = 0.100$
Diversity = 5.4338e-02	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.134308e+02	FES = 66100	$\mu_F = 0.786$	$\mu_{CR} = 0.099$
Diversity = 4.4736e-02	$k_t = [3, 2, 2, 2]$		
Generation 670: Best = 1.119505e+02	FES = 67100	$\mu_F = 0.774$	$\mu_{CR} = 0.107$
Diversity = 4.0116e-02	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.092137e+02	FES = 68100	$\mu_F = 0.769$	$\mu_{CR} = 0.102$
Diversity = 3.6331e-02	$k_t = [3, 4, 3, 3]$		
Generation 690: Best = 1.090541e+02	FES = 69100	$\mu_F = 0.764$	$\mu_{CR} = 0.097$
Diversity = 3.6031e-02	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.077713e+02	FES = 70100	$\mu_F = 0.768$	$\mu_{CR} = 0.104$
Diversity = 3.1821e-02	$k_t = [3, 4, 2, 4]$		
Generation 710: Best = 1.072138e+02	FES = 71100	$\mu_F = 0.792$	$\mu_{CR} = 0.109$
Diversity = 2.8496e-02	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 1.063416e+02	FES = 72100	$\mu_F = 0.806$	$\mu_{CR} = 0.111$
Diversity = 2.8182e-02	$k_t = [4, 4, 3, 4]$		
Generation 730: Best = 1.054195e+02	FES = 73100	$\mu_F = 0.811$	$\mu_{CR} = 0.117$
Diversity = 2.3152e-02	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 1.050466e+02	FES = 74100	$\mu_F = 0.820$	$\mu_{CR} = 0.110$
Diversity = 2.1007e-02	$k_t = [2, 4, 4, 4]$		
Generation 750: Best = 1.043974e+02	FES = 75100	$\mu_F = 0.821$	$\mu_{CR} = 0.114$
Diversity = 2.0517e-02	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 1.039071e+02	FES = 76100	$\mu_F = 0.816$	$\mu_{CR} = 0.124$
Diversity = 1.8079e-02	$k_t = [2, 3, 4, 4]$		

Generation 770: Best = 1.031368e+02	FES = 77100	$\mu_F = 0.829$	$\mu_{CR} = 0.116$
Diversity = 1.6125e-02	k_t = [1, 1, 1, 1]		
Generation 780: Best = 1.030729e+02	FES = 78100	$\mu_F = 0.828$	$\mu_{CR} = 0.111$
Diversity = 1.7808e-02	k_t = [2, 3, 2, 2]		
Generation 790: Best = 1.030286e+02	FES = 79100	$\mu_F = 0.824$	$\mu_{CR} = 0.108$
Diversity = 1.7697e-02	k_t = [1, 1, 1, 1]		
Generation 800: Best = 1.026889e+02	FES = 80100	$\mu_F = 0.818$	$\mu_{CR} = 0.110$
Diversity = 1.6357e-02	k_t = [2, 2, 2, 2]		
Generation 810: Best = 1.019333e+02	FES = 81100	$\mu_F = 0.825$	$\mu_{CR} = 0.102$
Diversity = 1.6192e-02	k_t = [1, 1, 1, 1]		
Generation 820: Best = 1.018088e+02	FES = 82100	$\mu_F = 0.815$	$\mu_{CR} = 0.098$
Diversity = 1.4890e-02	k_t = [2, 3, 4, 2]		
Generation 830: Best = 1.017040e+02	FES = 83100	$\mu_F = 0.804$	$\mu_{CR} = 0.098$
Diversity = 1.3859e-02	k_t = [1, 1, 1, 1]		
Generation 840: Best = 1.015379e+02	FES = 84100	$\mu_F = 0.812$	$\mu_{CR} = 0.098$
Diversity = 1.2447e-02	k_t = [2, 4, 3, 4]		
Generation 850: Best = 1.014972e+02	FES = 85100	$\mu_F = 0.814$	$\mu_{CR} = 0.095$
Diversity = 1.2751e-02	k_t = [1, 1, 1, 1]		
Generation 860: Best = 1.014019e+02	FES = 86100	$\mu_F = 0.809$	$\mu_{CR} = 0.088$
Diversity = 1.1083e-02	k_t = [4, 3, 3, 4]		
Generation 870: Best = 1.011466e+02	FES = 87100	$\mu_F = 0.825$	$\mu_{CR} = 0.089$
Diversity = 9.8259e-03	k_t = [1, 1, 1, 1]		
Generation 880: Best = 1.011280e+02	FES = 88100	$\mu_F = 0.831$	$\mu_{CR} = 0.091$
Diversity = 9.3732e-03	k_t = [4, 2, 2, 4]		
Generation 890: Best = 1.009984e+02	FES = 89100	$\mu_F = 0.835$	$\mu_{CR} = 0.093$
Diversity = 1.0200e-02	k_t = [1, 1, 1, 1]		
Generation 900: Best = 1.009005e+02	FES = 90100	$\mu_F = 0.821$	$\mu_{CR} = 0.096$
Diversity = 9.5444e-03	k_t = [4, 4, 2, 3]		
Generation 910: Best = 1.008062e+02	FES = 91100	$\mu_F = 0.802$	$\mu_{CR} = 0.099$
Diversity = 8.4140e-03	k_t = [1, 1, 1, 1]		
Generation 920: Best = 1.007572e+02	FES = 92100	$\mu_F = 0.804$	$\mu_{CR} = 0.104$
Diversity = 8.2038e-03	k_t = [3, 3, 3, 2]		
Generation 930: Best = 1.006461e+02	FES = 93100	$\mu_F = 0.800$	$\mu_{CR} = 0.097$
Diversity = 7.6944e-03	k_t = [1, 1, 1, 1]		
Generation 940: Best = 1.005031e+02	FES = 94100	$\mu_F = 0.790$	$\mu_{CR} = 0.094$
Diversity = 7.5090e-03	k_t = [4, 2, 2, 3]		
Generation 950: Best = 1.004541e+02	FES = 95100	$\mu_F = 0.786$	$\mu_{CR} = 0.094$
Diversity = 7.4773e-03	k_t = [1, 1, 1, 1]		
Generation 960: Best = 1.004313e+02	FES = 96100	$\mu_F = 0.805$	$\mu_{CR} = 0.097$
Diversity = 6.6353e-03	k_t = [3, 4, 4, 4]		
Generation 970: Best = 1.003488e+02	FES = 97100	$\mu_F = 0.796$	$\mu_{CR} = 0.102$
Diversity = 6.1942e-03	k_t = [1, 1, 1, 1]		
Generation 980: Best = 1.002897e+02	FES = 98100	$\mu_F = 0.780$	$\mu_{CR} = 0.112$
Diversity = 7.7731e-03	k_t = [2, 4, 2, 2]		
Generation 990: Best = 1.002500e+02	FES = 99100	$\mu_F = 0.790$	$\mu_{CR} = 0.110$
Diversity = 7.6808e-03	k_t = [1, 1, 1, 1]		
Generation 1000: Best = 1.002065e+02	FES = 100100	$\mu_F = 0.797$	$\mu_{CR} = 0.110$
Diversity = 6.8497e-03	k_t = [2, 4, 3, 3]		
Generation 1010: Best = 1.001877e+02	FES = 101100	$\mu_F = 0.786$	$\mu_{CR} = 0.114$
Diversity = 5.8468e-03	k_t = [1, 1, 1, 1]		
Generation 1020: Best = 1.001215e+02	FES = 102100	$\mu_F = 0.785$	$\mu_{CR} = 0.112$
Diversity = 4.9613e-03	k_t = [3, 2, 4, 4]		
Generation 1030: Best = 1.000916e+02	FES = 103100	$\mu_F = 0.786$	$\mu_{CR} = 0.104$
Diversity = 4.8678e-03	k_t = [1, 1, 1, 1]		

Generation 1040: Best = 1.000829e+02 FES = 104100 μ_F = 0.793 μ_{CR} = 0.094
 Diversity = 4.7051e-03 k_t = [4, 4, 2, 4]
 Generation 1050: Best = 1.000797e+02 FES = 105100 μ_F = 0.807 μ_{CR} = 0.096
 Diversity = 4.6900e-03 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 1.000728e+02 FES = 106100 μ_F = 0.813 μ_{CR} = 0.096
 Diversity = 4.5223e-03 k_t = [4, 3, 4, 2]
 Generation 1070: Best = 1.000613e+02 FES = 107100 μ_F = 0.828 μ_{CR} = 0.099
 Diversity = 4.1152e-03 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 1.000535e+02 FES = 108100 μ_F = 0.817 μ_{CR} = 0.103
 Diversity = 3.5698e-03 k_t = [3, 4, 4, 2]
 Generation 1090: Best = 1.000477e+02 FES = 109100 μ_F = 0.825 μ_{CR} = 0.096
 Diversity = 3.5026e-03 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.000442e+02 FES = 110100 μ_F = 0.820 μ_{CR} = 0.099
 Diversity = 3.1336e-03 k_t = [3, 4, 4, 4]
 Generation 1110: Best = 1.000413e+02 FES = 111100 μ_F = 0.825 μ_{CR} = 0.096
 Diversity = 2.7925e-03 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.000262e+02 FES = 112100 μ_F = 0.824 μ_{CR} = 0.102
 Diversity = 2.5128e-03 k_t = [2, 4, 2, 4]
 Generation 1130: Best = 1.000252e+02 FES = 113100 μ_F = 0.823 μ_{CR} = 0.104
 Diversity = 2.2863e-03 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.000209e+02 FES = 114100 μ_F = 0.830 μ_{CR} = 0.102
 Diversity = 2.1145e-03 k_t = [3, 3, 3, 2]
 Generation 1150: Best = 1.000165e+02 FES = 115100 μ_F = 0.824 μ_{CR} = 0.111
 Diversity = 2.1025e-03 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.000119e+02 FES = 116100 μ_F = 0.805 μ_{CR} = 0.115
 Diversity = 1.8687e-03 k_t = [2, 4, 3, 3]
 Generation 1170: Best = 1.000104e+02 FES = 117100 μ_F = 0.803 μ_{CR} = 0.107
 Diversity = 1.6423e-03 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.000075e+02 FES = 118100 μ_F = 0.793 μ_{CR} = 0.110
 Diversity = 1.4146e-03 k_t = [4, 4, 4, 4]
 Generation 1190: Best = 1.000069e+02 FES = 119100 μ_F = 0.793 μ_{CR} = 0.101
 Diversity = 1.0947e-03 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.000063e+02 FES = 120100 μ_F = 0.791 μ_{CR} = 0.098
 Diversity = 1.0870e-03 k_t = [4, 3, 4, 4]
 Generation 1210: Best = 1.000050e+02 FES = 121100 μ_F = 0.798 μ_{CR} = 0.101
 Diversity = 9.1568e-04 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.000045e+02 FES = 122100 μ_F = 0.805 μ_{CR} = 0.105
 Diversity = 6.9762e-04 k_t = [4, 2, 4, 4]
 Generation 1230: Best = 1.000041e+02 FES = 123100 μ_F = 0.801 μ_{CR} = 0.101
 Diversity = 6.1595e-04 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.000030e+02 FES = 124100 μ_F = 0.801 μ_{CR} = 0.100
 Diversity = 6.0223e-04 k_t = [4, 4, 3, 3]
 Generation 1250: Best = 1.000027e+02 FES = 125100 μ_F = 0.806 μ_{CR} = 0.097
 Diversity = 6.3220e-04 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.000026e+02 FES = 126100 μ_F = 0.818 μ_{CR} = 0.108
 Diversity = 4.9428e-04 k_t = [3, 3, 4, 4]
 Generation 1270: Best = 1.000025e+02 FES = 127100 μ_F = 0.812 μ_{CR} = 0.103
 Diversity = 4.0441e-04 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.000021e+02 FES = 128100 μ_F = 0.808 μ_{CR} = 0.106
 Diversity = 4.0146e-04 k_t = [3, 3, 3, 4]
 Generation 1290: Best = 1.000018e+02 FES = 129100 μ_F = 0.792 μ_{CR} = 0.106
 Diversity = 3.4760e-04 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.000016e+02 FES = 130100 μ_F = 0.795 μ_{CR} = 0.102
 Diversity = 3.0974e-04 k_t = [2, 4, 2, 3]

Generation 1310: Best = 1.000014e+02 FES = 131100 μ_F = 0.790 μ_{CR} = 0.109
 Diversity = 3.1927e-04 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.000012e+02 FES = 132100 μ_F = 0.786 μ_{CR} = 0.120
 Diversity = 2.9131e-04 k_t = [3, 2, 3, 2]
 Generation 1330: Best = 1.000012e+02 FES = 133100 μ_F = 0.787 μ_{CR} = 0.120
 Diversity = 2.8242e-04 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.000010e+02 FES = 134100 μ_F = 0.811 μ_{CR} = 0.109
 Diversity = 2.4940e-04 k_t = [3, 4, 4, 3]
 Generation 1350: Best = 1.000007e+02 FES = 135100 μ_F = 0.810 μ_{CR} = 0.110
 Diversity = 2.8929e-04 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.000006e+02 FES = 136100 μ_F = 0.801 μ_{CR} = 0.100
 Diversity = 3.6557e-04 k_t = [2, 4, 4, 3]
 Generation 1370: Best = 1.000005e+02 FES = 137100 μ_F = 0.793 μ_{CR} = 0.099
 Diversity = 2.9514e-04 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.000004e+02 FES = 138100 μ_F = 0.788 μ_{CR} = 0.104
 Diversity = 2.4788e-04 k_t = [3, 4, 4, 2]
 Generation 1390: Best = 1.000004e+02 FES = 139100 μ_F = 0.797 μ_{CR} = 0.104
 Diversity = 2.0501e-04 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.000003e+02 FES = 140100 μ_F = 0.790 μ_{CR} = 0.107
 Diversity = 2.0018e-04 k_t = [3, 4, 2, 2]
 Generation 1410: Best = 1.000003e+02 FES = 141100 μ_F = 0.783 μ_{CR} = 0.111
 Diversity = 2.0042e-04 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.000002e+02 FES = 142100 μ_F = 0.785 μ_{CR} = 0.106
 Diversity = 1.5140e-04 k_t = [2, 4, 4, 4]
 Generation 1430: Best = 1.000002e+02 FES = 143100 μ_F = 0.778 μ_{CR} = 0.102
 Diversity = 1.3904e-04 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.000002e+02 FES = 144100 μ_F = 0.782 μ_{CR} = 0.107
 Diversity = 1.2094e-04 k_t = [2, 2, 3, 2]
 Generation 1450: Best = 1.000002e+02 FES = 145100 μ_F = 0.775 μ_{CR} = 0.103
 Diversity = 1.0056e-04 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.000001e+02 FES = 146100 μ_F = 0.773 μ_{CR} = 0.104
 Diversity = 9.1219e-05 k_t = [2, 3, 2, 2]
 Generation 1470: Best = 1.000001e+02 FES = 147100 μ_F = 0.781 μ_{CR} = 0.109
 Diversity = 1.0545e-04 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.000001e+02 FES = 148100 μ_F = 0.785 μ_{CR} = 0.107
 Diversity = 8.8679e-05 k_t = [4, 4, 4, 4]
 Generation 1490: Best = 1.000001e+02 FES = 149100 μ_F = 0.784 μ_{CR} = 0.101
 Diversity = 8.9303e-05 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.000001e+02 FES = 150100 μ_F = 0.792 μ_{CR} = 0.100
 Diversity = 9.7172e-05 k_t = [3, 3, 3, 4]
 Generation 1510: Best = 1.000001e+02 FES = 151100 μ_F = 0.802 μ_{CR} = 0.103
 Diversity = 9.1799e-05 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.000001e+02 FES = 152100 μ_F = 0.807 μ_{CR} = 0.099
 Diversity = 7.6726e-05 k_t = [4, 3, 3, 3]
 Generation 1530: Best = 1.000001e+02 FES = 153100 μ_F = 0.802 μ_{CR} = 0.103
 Diversity = 6.0154e-05 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.000001e+02 FES = 154100 μ_F = 0.798 μ_{CR} = 0.101
 Diversity = 6.4942e-05 k_t = [4, 2, 3, 3]
 Generation 1550: Best = 1.000000e+02 FES = 155100 μ_F = 0.809 μ_{CR} = 0.107
 Diversity = 6.1735e-05 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.000000e+02 FES = 156100 μ_F = 0.807 μ_{CR} = 0.108
 Diversity = 6.5133e-05 k_t = [3, 2, 4, 4]
 Generation 1570: Best = 1.000000e+02 FES = 157100 μ_F = 0.798 μ_{CR} = 0.098
 Diversity = 9.3264e-05 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.000000e+02 FES = 158100 μ_F = 0.803 μ_{CR} = 0.095
 Diversity = 8.0741e-05 k_t = [3, 2, 4, 3]
 Generation 1590: Best = 1.000000e+02 FES = 159100 μ_F = 0.813 μ_{CR} = 0.103
 Diversity = 7.5654e-05 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.000000e+02 FES = 160100 μ_F = 0.806 μ_{CR} = 0.105
 Diversity = 5.1633e-05 k_t = [4, 4, 2, 4]
 Generation 1610: Best = 1.000000e+02 FES = 161100 μ_F = 0.801 μ_{CR} = 0.104
 Diversity = 4.6422e-05 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.000000e+02 FES = 162100 μ_F = 0.797 μ_{CR} = 0.118
 Diversity = 3.8338e-05 k_t = [4, 3, 2, 3]
 Generation 1630: Best = 1.000000e+02 FES = 163100 μ_F = 0.810 μ_{CR} = 0.112
 Diversity = 3.3952e-05 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.000000e+02 FES = 164100 μ_F = 0.815 μ_{CR} = 0.103
 Diversity = 3.3814e-05 k_t = [2, 3, 3, 3]
 Generation 1650: Best = 1.000000e+02 FES = 165100 μ_F = 0.805 μ_{CR} = 0.106
 Diversity = 2.9377e-05 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.000000e+02 FES = 166100 μ_F = 0.790 μ_{CR} = 0.103
 Diversity = 2.7663e-05 k_t = [2, 3, 2, 4]
 Generation 1670: Best = 1.000000e+02 FES = 167100 μ_F = 0.810 μ_{CR} = 0.099
 Diversity = 2.3247e-05 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.000000e+02 FES = 168100 μ_F = 0.807 μ_{CR} = 0.101
 Diversity = 2.0604e-05 k_t = [2, 3, 4, 4]
 Generation 1690: Best = 1.000000e+02 FES = 169100 μ_F = 0.798 μ_{CR} = 0.100
 Diversity = 1.6065e-05 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.000000e+02 FES = 170100 μ_F = 0.810 μ_{CR} = 0.104
 Diversity = 1.9453e-05 k_t = [2, 4, 4, 2]
 Generation 1710: Best = 1.000000e+02 FES = 171100 μ_F = 0.804 μ_{CR} = 0.104
 Diversity = 1.8465e-05 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.000000e+02 FES = 172100 μ_F = 0.799 μ_{CR} = 0.114
 Diversity = 1.4704e-05 k_t = [4, 3, 2, 3]
 Generation 1730: Best = 1.000000e+02 FES = 173100 μ_F = 0.788 μ_{CR} = 0.105
 Diversity = 1.3666e-05 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.000000e+02 FES = 174100 μ_F = 0.772 μ_{CR} = 0.104
 Diversity = 1.3645e-05 k_t = [4, 3, 4, 4]
 Generation 1750: Best = 1.000000e+02 FES = 175100 μ_F = 0.777 μ_{CR} = 0.104
 Diversity = 1.4812e-05 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.000000e+02 FES = 176100 μ_F = 0.782 μ_{CR} = 0.104
 Diversity = 1.2078e-05 k_t = [3, 3, 2, 3]
 Generation 1770: Best = 1.000000e+02 FES = 177100 μ_F = 0.801 μ_{CR} = 0.097
 Diversity = 1.0794e-05 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.000000e+02 FES = 178100 μ_F = 0.797 μ_{CR} = 0.108
 Diversity = 1.0868e-05 k_t = [2, 4, 3, 3]
 Generation 1790: Best = 1.000000e+02 FES = 179100 μ_F = 0.798 μ_{CR} = 0.115
 Diversity = 1.3914e-05 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 1.000000e+02 FES = 180100 μ_F = 0.797 μ_{CR} = 0.112
 Diversity = 1.1763e-05 k_t = [2, 4, 3, 2]
 Generation 1810: Best = 1.000000e+02 FES = 181100 μ_F = 0.786 μ_{CR} = 0.110
 Diversity = 1.0356e-05 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 1.000000e+02 FES = 182100 μ_F = 0.803 μ_{CR} = 0.119
 Diversity = 1.0289e-05 k_t = [2, 3, 4, 3]
 Generation 1830: Best = 1.000000e+02 FES = 183100 μ_F = 0.806 μ_{CR} = 0.121
 Diversity = 1.0706e-05 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 1.000000e+02 FES = 184100 μ_F = 0.804 μ_{CR} = 0.117
 Diversity = 9.9045e-06 k_t = [2, 2, 2, 4]

Generation 1850: Best = 1.000000e+02	FES = 185100	$\mu_F = 0.801$	$\mu_{CR} = 0.117$
Diversity = 8.3653e-06	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 1.000000e+02	FES = 186100	$\mu_F = 0.787$	$\mu_{CR} = 0.111$
Diversity = 7.3815e-06	$k_t = [4, 2, 3, 2]$		
Generation 1870: Best = 1.000000e+02	FES = 187100	$\mu_F = 0.785$	$\mu_{CR} = 0.110$
Diversity = 6.8102e-06	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 1.000000e+02	FES = 188100	$\mu_F = 0.790$	$\mu_{CR} = 0.111$
Diversity = 6.0062e-06	$k_t = [3, 4, 4, 4]$		
Generation 1890: Best = 1.000000e+02	FES = 189100	$\mu_F = 0.798$	$\mu_{CR} = 0.106$
Diversity = 5.8602e-06	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 1.000000e+02	FES = 190100	$\mu_F = 0.808$	$\mu_{CR} = 0.106$
Diversity = 5.3663e-06	$k_t = [3, 2, 4, 2]$		
Generation 1910: Best = 1.000000e+02	FES = 191100	$\mu_F = 0.802$	$\mu_{CR} = 0.108$
Diversity = 5.6566e-06	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 1.000000e+02	FES = 192100	$\mu_F = 0.780$	$\mu_{CR} = 0.109$
Diversity = 4.5995e-06	$k_t = [2, 2, 4, 2]$		
Generation 1930: Best = 1.000000e+02	FES = 193100	$\mu_F = 0.785$	$\mu_{CR} = 0.106$
Diversity = 4.4507e-06	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 1.000000e+02	FES = 194100	$\mu_F = 0.792$	$\mu_{CR} = 0.106$
Diversity = 3.8470e-06	$k_t = [2, 4, 4, 4]$		
Generation 1950: Best = 1.000000e+02	FES = 195100	$\mu_F = 0.801$	$\mu_{CR} = 0.106$
Diversity = 4.0086e-06	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 1.000000e+02	FES = 196100	$\mu_F = 0.795$	$\mu_{CR} = 0.099$
Diversity = 3.1618e-06	$k_t = [3, 4, 3, 2]$		
Generation 1970: Best = 1.000000e+02	FES = 197100	$\mu_F = 0.782$	$\mu_{CR} = 0.101$
Diversity = 2.7331e-06	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 1.000000e+02	FES = 198100	$\mu_F = 0.785$	$\mu_{CR} = 0.102$
Diversity = 2.6200e-06	$k_t = [3, 3, 4, 3]$		
Generation 1990: Best = 1.000000e+02	FES = 199100	$\mu_F = 0.790$	$\mu_{CR} = 0.099$
Diversity = 2.4731e-06	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 1.000000e+02	FES = 200100	$\mu_F = 0.791$	$\mu_{CR} = 0.100$
Diversity = 2.2474e-06	$k_t = [4, 4, 4, 4]$		
Generation 2010: Best = 1.000000e+02	FES = 201100	$\mu_F = 0.796$	$\mu_{CR} = 0.099$
Diversity = 2.2631e-06	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 1.000000e+02	FES = 202100	$\mu_F = 0.800$	$\mu_{CR} = 0.101$
Diversity = 2.9032e-06	$k_t = [4, 3, 2, 2]$		
Generation 2030: Best = 1.000000e+02	FES = 203100	$\mu_F = 0.790$	$\mu_{CR} = 0.100$
Diversity = 3.1864e-06	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 1.000000e+02	FES = 204100	$\mu_F = 0.800$	$\mu_{CR} = 0.108$
Diversity = 3.0254e-06	$k_t = [4, 2, 4, 2]$		
Generation 2050: Best = 1.000000e+02	FES = 205100	$\mu_F = 0.786$	$\mu_{CR} = 0.101$
Diversity = 2.4954e-06	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 1.000000e+02	FES = 206100	$\mu_F = 0.789$	$\mu_{CR} = 0.108$
Diversity = 1.7296e-06	$k_t = [4, 3, 2, 4]$		
Generation 2070: Best = 1.000000e+02	FES = 207100	$\mu_F = 0.784$	$\mu_{CR} = 0.109$
Diversity = 1.7300e-06	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 1.000000e+02	FES = 208100	$\mu_F = 0.800$	$\mu_{CR} = 0.115$
Diversity = 1.6925e-06	$k_t = [4, 4, 3, 3]$		
Generation 2090: Best = 1.000000e+02	FES = 209100	$\mu_F = 0.802$	$\mu_{CR} = 0.116$
Diversity = 1.7264e-06	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 1.000000e+02	FES = 210100	$\mu_F = 0.782$	$\mu_{CR} = 0.114$
Diversity = 1.6098e-06	$k_t = [4, 2, 2, 2]$		
Generation 2110: Best = 1.000000e+02	FES = 211100	$\mu_F = 0.778$	$\mu_{CR} = 0.106$
Diversity = 1.5280e-06	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 1.000000e+02 FES = 212100 μ_F = 0.771 μ_{CR} = 0.103
 Diversity = 1.4058e-06 k_t = [3, 4, 3, 4]
 Generation 2130: Best = 1.000000e+02 FES = 213100 μ_F = 0.767 μ_{CR} = 0.107
 Diversity = 1.1132e-06 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.000000e+02 FES = 214100 μ_F = 0.787 μ_{CR} = 0.111
 Diversity = 1.0582e-06 k_t = [3, 4, 3, 3]
 Generation 2150: Best = 1.000000e+02 FES = 215100 μ_F = 0.790 μ_{CR} = 0.108
 Diversity = 1.0974e-06 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.000000e+02 FES = 216100 μ_F = 0.789 μ_{CR} = 0.111
 Diversity = 1.0083e-06 k_t = [4, 2, 3, 2]
 Generation 2170: Best = 1.000000e+02 FES = 217100 μ_F = 0.798 μ_{CR} = 0.114
 Diversity = 1.0284e-06 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.000000e+02 FES = 218100 μ_F = 0.800 μ_{CR} = 0.112
 Diversity = 7.4071e-07 k_t = [3, 3, 3, 3]
 Generation 2190: Best = 1.000000e+02 FES = 219100 μ_F = 0.820 μ_{CR} = 0.107
 Diversity = 7.3567e-07 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 1.000000e+02 FES = 220100 μ_F = 0.826 μ_{CR} = 0.101
 Diversity = 7.3681e-07 k_t = [4, 3, 4, 3]
 Generation 2210: Best = 1.000000e+02 FES = 221100 μ_F = 0.812 μ_{CR} = 0.105
 Diversity = 7.2391e-07 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 1.000000e+02 FES = 222100 μ_F = 0.802 μ_{CR} = 0.103
 Diversity = 6.2022e-07 k_t = [4, 3, 4, 3]
 Generation 2230: Best = 1.000000e+02 FES = 223100 μ_F = 0.789 μ_{CR} = 0.100
 Diversity = 5.5885e-07 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.000000e+02 FES = 224100 μ_F = 0.784 μ_{CR} = 0.104
 Diversity = 4.9497e-07 k_t = [4, 3, 2, 3]
 Generation 2250: Best = 1.000000e+02 FES = 225100 μ_F = 0.780 μ_{CR} = 0.108
 Diversity = 4.4984e-07 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.000000e+02 FES = 226100 μ_F = 0.779 μ_{CR} = 0.104
 Diversity = 4.4014e-07 k_t = [2, 2, 3, 3]
 Generation 2270: Best = 1.000000e+02 FES = 227100 μ_F = 0.776 μ_{CR} = 0.101
 Diversity = 4.1538e-07 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.000000e+02 FES = 228100 μ_F = 0.779 μ_{CR} = 0.102
 Diversity = 3.1547e-07 k_t = [2, 3, 4, 2]
 Generation 2290: Best = 1.000000e+02 FES = 229100 μ_F = 0.791 μ_{CR} = 0.103
 Diversity = 3.6359e-07 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.000000e+02 FES = 230100 μ_F = 0.798 μ_{CR} = 0.103
 Diversity = 3.6016e-07 k_t = [4, 2, 2, 4]
 Generation 2310: Best = 1.000000e+02 FES = 231100 μ_F = 0.794 μ_{CR} = 0.103
 Diversity = 3.0782e-07 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.000000e+02 FES = 232100 μ_F = 0.788 μ_{CR} = 0.107
 Diversity = 2.8185e-07 k_t = [4, 2, 2, 4]
 Generation 2330: Best = 1.000000e+02 FES = 233100 μ_F = 0.785 μ_{CR} = 0.102
 Diversity = 3.0581e-07 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.000000e+02 FES = 234100 μ_F = 0.790 μ_{CR} = 0.100
 Diversity = 3.1775e-07 k_t = [2, 3, 4, 3]
 Generation 2350: Best = 1.000000e+02 FES = 235100 μ_F = 0.782 μ_{CR} = 0.101
 Diversity = 2.9476e-07 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.000000e+02 FES = 236100 μ_F = 0.803 μ_{CR} = 0.100
 Diversity = 2.8768e-07 k_t = [3, 2, 2, 2]
 Generation 2370: Best = 1.000000e+02 FES = 237100 μ_F = 0.787 μ_{CR} = 0.094
 Diversity = 2.6883e-07 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.000000e+02 FES = 238100 μ_F = 0.798 μ_{CR} = 0.102
 Diversity = 2.5499e-07 k_t = [3, 3, 3, 2]

Generation 2390: Best = 1.000000e+02 FES = 239100 μ_F = 0.796 μ_{CR} = 0.108
 Diversity = 2.4733e-07 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.000000e+02 FES = 240100 μ_F = 0.809 μ_{CR} = 0.117
 Diversity = 2.8400e-07 k_t = [3, 2, 3, 3]
 Generation 2410: Best = 1.000000e+02 FES = 241100 μ_F = 0.815 μ_{CR} = 0.109
 Diversity = 2.6452e-07 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.000000e+02 FES = 242100 μ_F = 0.806 μ_{CR} = 0.102
 Diversity = 2.3251e-07 k_t = [3, 2, 3, 2]
 Generation 2430: Best = 1.000000e+02 FES = 243100 μ_F = 0.797 μ_{CR} = 0.095
 Diversity = 2.0287e-07 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.000000e+02 FES = 244100 μ_F = 0.807 μ_{CR} = 0.102
 Diversity = 1.7301e-07 k_t = [3, 4, 3, 4]
 Generation 2450: Best = 1.000000e+02 FES = 245100 μ_F = 0.799 μ_{CR} = 0.103
 Diversity = 1.6442e-07 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.000000e+02 FES = 246100 μ_F = 0.810 μ_{CR} = 0.109
 Diversity = 1.5436e-07 k_t = [3, 4, 3, 4]
 Generation 2470: Best = 1.000000e+02 FES = 247100 μ_F = 0.811 μ_{CR} = 0.108
 Diversity = 1.3172e-07 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.000000e+02 FES = 248100 μ_F = 0.814 μ_{CR} = 0.104
 Diversity = 1.2825e-07 k_t = [4, 3, 2, 2]
 Generation 2490: Best = 1.000000e+02 FES = 249100 μ_F = 0.801 μ_{CR} = 0.104
 Diversity = 1.3156e-07 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.000000e+02 FES = 250100 μ_F = 0.799 μ_{CR} = 0.102
 Diversity = 1.0730e-07 k_t = [4, 2, 2, 3]
 Generation 2510: Best = 1.000000e+02 FES = 251100 μ_F = 0.801 μ_{CR} = 0.098
 Diversity = 9.9081e-08 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.000000e+02 FES = 252100 μ_F = 0.812 μ_{CR} = 0.108
 Diversity = 9.6758e-08 k_t = [2, 3, 4, 3]
 Generation 2530: Best = 1.000000e+02 FES = 253100 μ_F = 0.808 μ_{CR} = 0.108
 Diversity = 1.0753e-07 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.000000e+02 FES = 254100 μ_F = 0.812 μ_{CR} = 0.103
 Diversity = 9.0890e-08 k_t = [2, 3, 2, 2]
 Generation 2550: Best = 1.000000e+02 FES = 255100 μ_F = 0.791 μ_{CR} = 0.096
 Diversity = 7.1794e-08 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.000000e+02 FES = 256100 μ_F = 0.789 μ_{CR} = 0.097
 Diversity = 7.9417e-08 k_t = [2, 4, 3, 3]
 Generation 2570: Best = 1.000000e+02 FES = 257100 μ_F = 0.784 μ_{CR} = 0.099
 Diversity = 8.6007e-08 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.000000e+02 FES = 258100 μ_F = 0.799 μ_{CR} = 0.102
 Diversity = 8.9892e-08 k_t = [4, 3, 3, 4]
 Generation 2590: Best = 1.000000e+02 FES = 259100 μ_F = 0.810 μ_{CR} = 0.112
 Diversity = 1.1728e-07 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.000000e+02 FES = 260100 μ_F = 0.811 μ_{CR} = 0.113
 Diversity = 1.0257e-07 k_t = [4, 3, 3, 2]
 Generation 2610: Best = 1.000000e+02 FES = 261100 μ_F = 0.814 μ_{CR} = 0.111
 Diversity = 9.6685e-08 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.000000e+02 FES = 262100 μ_F = 0.810 μ_{CR} = 0.101
 Diversity = 8.6208e-08 k_t = [2, 4, 3, 2]
 Generation 2630: Best = 1.000000e+02 FES = 263100 μ_F = 0.797 μ_{CR} = 0.099
 Diversity = 8.4240e-08 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.000000e+02 FES = 264100 μ_F = 0.809 μ_{CR} = 0.109
 Diversity = 7.5042e-08 k_t = [4, 4, 4, 4]
 Generation 2650: Best = 1.000000e+02 FES = 265100 μ_F = 0.790 μ_{CR} = 0.108
 Diversity = 5.6748e-08 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.000000e+02 FES = 266100 μ_F = 0.786 μ_{CR} = 0.109
 Diversity = 5.1827e-08 k_t = [4, 2, 2, 2]
 Generation 2670: Best = 1.000000e+02 FES = 267100 μ_F = 0.794 μ_{CR} = 0.103
 Diversity = 4.9012e-08 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.000000e+02 FES = 268100 μ_F = 0.800 μ_{CR} = 0.110
 Diversity = 4.3869e-08 k_t = [4, 2, 3, 4]
 Generation 2690: Best = 1.000000e+02 FES = 269100 μ_F = 0.801 μ_{CR} = 0.112
 Diversity = 3.5218e-08 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.000000e+02 FES = 270100 μ_F = 0.804 μ_{CR} = 0.107
 Diversity = 4.2672e-08 k_t = [2, 2, 3, 2]
 Generation 2710: Best = 1.000000e+02 FES = 271100 μ_F = 0.810 μ_{CR} = 0.099
 Diversity = 4.2546e-08 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.000000e+02 FES = 272100 μ_F = 0.786 μ_{CR} = 0.093
 Diversity = 3.5322e-08 k_t = [4, 4, 4, 2]
 Generation 2730: Best = 1.000000e+02 FES = 273100 μ_F = 0.775 μ_{CR} = 0.096
 Diversity = 3.4802e-08 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.000000e+02 FES = 274100 μ_F = 0.783 μ_{CR} = 0.100
 Diversity = 3.2694e-08 k_t = [2, 3, 4, 2]
 Generation 2750: Best = 1.000000e+02 FES = 275100 μ_F = 0.791 μ_{CR} = 0.097
 Diversity = 3.0041e-08 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.000000e+02 FES = 276100 μ_F = 0.785 μ_{CR} = 0.103
 Diversity = 3.5866e-08 k_t = [2, 2, 3, 2]
 Generation 2770: Best = 1.000000e+02 FES = 277100 μ_F = 0.797 μ_{CR} = 0.102
 Diversity = 3.2471e-08 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.000000e+02 FES = 278100 μ_F = 0.798 μ_{CR} = 0.101
 Diversity = 3.2342e-08 k_t = [4, 4, 3, 4]
 Generation 2790: Best = 1.000000e+02 FES = 279100 μ_F = 0.803 μ_{CR} = 0.094
 Diversity = 2.6354e-08 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.000000e+02 FES = 280100 μ_F = 0.798 μ_{CR} = 0.095
 Diversity = 2.8080e-08 k_t = [4, 3, 3, 3]
 Generation 2810: Best = 1.000000e+02 FES = 281100 μ_F = 0.802 μ_{CR} = 0.103
 Diversity = 2.7648e-08 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.000000e+02 FES = 282100 μ_F = 0.808 μ_{CR} = 0.098
 Diversity = 2.8494e-08 k_t = [2, 2, 3, 2]
 Generation 2830: Best = 1.000000e+02 FES = 283100 μ_F = 0.814 μ_{CR} = 0.092
 Diversity = 3.0797e-08 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.000000e+02 FES = 284100 μ_F = 0.807 μ_{CR} = 0.100
 Diversity = 3.2539e-08 k_t = [2, 3, 3, 2]
 Generation 2850: Best = 1.000000e+02 FES = 285100 μ_F = 0.806 μ_{CR} = 0.099
 Diversity = 3.7076e-08 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.000000e+02 FES = 286100 μ_F = 0.799 μ_{CR} = 0.098
 Diversity = 3.2843e-08 k_t = [4, 4, 4, 3]
 Generation 2870: Best = 1.000000e+02 FES = 287100 μ_F = 0.803 μ_{CR} = 0.100
 Diversity = 3.1264e-08 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.000000e+02 FES = 288100 μ_F = 0.811 μ_{CR} = 0.095
 Diversity = 3.1772e-08 k_t = [3, 3, 4, 4]
 Generation 2890: Best = 1.000000e+02 FES = 289100 μ_F = 0.808 μ_{CR} = 0.095
 Diversity = 2.8187e-08 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.000000e+02 FES = 290100 μ_F = 0.816 μ_{CR} = 0.099
 Diversity = 2.1516e-08 k_t = [2, 4, 2, 3]
 Generation 2910: Best = 1.000000e+02 FES = 291100 μ_F = 0.819 μ_{CR} = 0.100
 Diversity = 1.9131e-08 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.000000e+02 FES = 292100 μ_F = 0.807 μ_{CR} = 0.102
 Diversity = 1.6804e-08 k_t = [2, 4, 2, 3]

```

Generation 2930: Best = 1.000000e+02  FES = 293100   $\mu_F$  = 0.799   $\mu_{CR}$  = 0.095
Diversity = 1.2415e-08  k_t = [1, 1, 1, 1]
Generation 2940: Best = 1.000000e+02  FES = 294100   $\mu_F$  = 0.792   $\mu_{CR}$  = 0.094
Diversity = 1.0436e-08  k_t = [2, 3, 3, 4]
Generation 2950: Best = 1.000000e+02  FES = 295100   $\mu_F$  = 0.802   $\mu_{CR}$  = 0.089
Diversity = 8.8809e-09  k_t = [1, 1, 1, 1]
Generation 2960: Best = 1.000000e+02  FES = 296100   $\mu_F$  = 0.796   $\mu_{CR}$  = 0.090
Diversity = 8.6988e-09  k_t = [3, 2, 2, 2]
Generation 2970: Best = 1.000000e+02  FES = 297100   $\mu_F$  = 0.796   $\mu_{CR}$  = 0.093
Diversity = 8.8918e-09  k_t = [1, 1, 1, 1]
Generation 2980: Best = 1.000000e+02  FES = 298100   $\mu_F$  = 0.798   $\mu_{CR}$  = 0.091
Diversity = 8.5789e-09  k_t = [3, 2, 2, 4]
Generation 2990: Best = 1.000000e+02  FES = 299100   $\mu_F$  = 0.805   $\mu_{CR}$  = 0.089
Diversity = 1.0423e-08  k_t = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 1.000000e+02

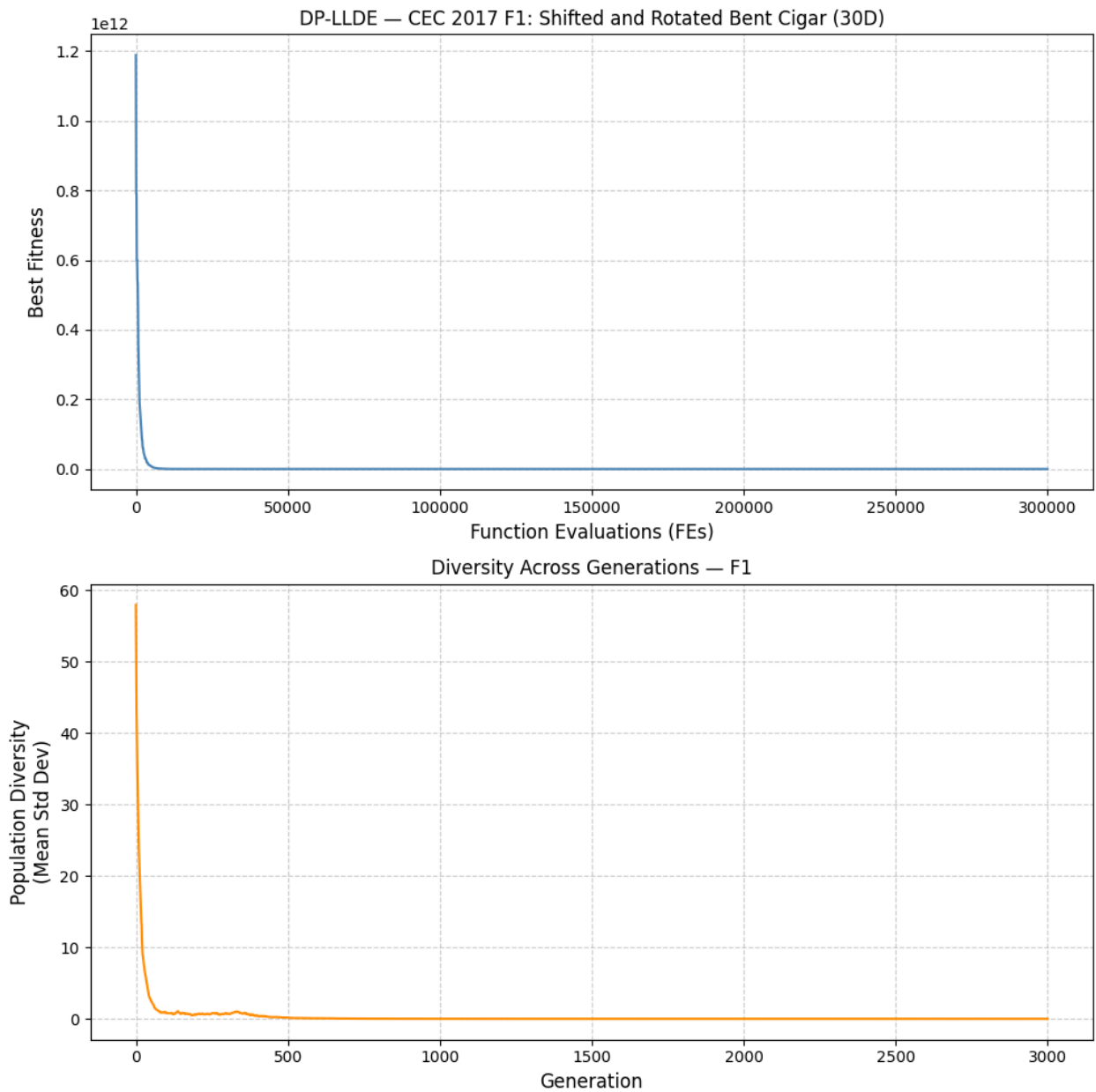
Total Function Evals = 300000

Total Successful Updates = 86061

```

=====
=
Results:
  Best Fitness      : 1.000000e+02
  Global Optimum    : 1.000000e+02
  Error (|best-optimum|) : 2.304574e-09
  Best Solution (5 dim) : [-55.27639845 -70.42955972 -29.61018187 -58.32676328
22.08960188]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F1.png

```
In [ ]: # =====
# F2 – Shifted and Rotated Sum of Different Power
# =====
run_and_plot('F2', make_wrapper(simple.f2), 200.0, 'Shifted and Rotated Sum of
```

=====

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DP-LLDE on CEC 2017 F2 – Single Run

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=

Function : Shifted and Rotated Sum of Different Power (Deprecated) (30D)
Bounds : [-100, 100]
Global optimum : 200.0
WARNING: f2 has been deprecated from the CEC 2017 benchmark suite
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 3.044577e+48
Generation 10: Best = 2.325879e+36 FES = 1100 $\mu_F = 0.574$ $\mu_{CR} = 0.357$
Diversity = 3.5682e+01 $k_t = [1, 1, 1, 1]$
Generation 20: Best = 1.574412e+30 FES = 2100 $\mu_F = 0.605$ $\mu_{CR} = 0.350$
Diversity = 2.4878e+01 $k_t = [4, 2, 4, 4]$
Generation 30: Best = 5.608552e+29 FES = 3100 $\mu_F = 0.601$ $\mu_{CR} = 0.341$
Diversity = 1.6459e+01 $k_t = [1, 1, 1, 1]$
Generation 40: Best = 4.510329e+27 FES = 4100 $\mu_F = 0.611$ $\mu_{CR} = 0.342$
Diversity = 1.3251e+01 $k_t = [4, 3, 4, 2]$
Generation 50: Best = 3.114273e+25 FES = 5100 $\mu_F = 0.606$ $\mu_{CR} = 0.353$
Diversity = 9.0862e+00 $k_t = [1, 1, 1, 1]$
Generation 60: Best = 3.143354e+24 FES = 6100 $\mu_F = 0.619$ $\mu_{CR} = 0.359$
Diversity = 6.0939e+00 $k_t = [3, 3, 4, 3]$
Generation 70: Best = 2.025410e+22 FES = 7100 $\mu_F = 0.647$ $\mu_{CR} = 0.362$
Diversity = 4.9731e+00 $k_t = [1, 1, 1, 1]$
Generation 80: Best = 3.113567e+20 FES = 8100 $\mu_F = 0.674$ $\mu_{CR} = 0.364$
Diversity = 4.0023e+00 $k_t = [2, 2, 4, 3]$
Generation 90: Best = 4.196655e+18 FES = 9100 $\mu_F = 0.685$ $\mu_{CR} = 0.381$
Diversity = 3.4072e+00 $k_t = [1, 1, 1, 1]$
Generation 100: Best = 1.061416e+18 FES = 10100 $\mu_F = 0.688$ $\mu_{CR} = 0.389$
Diversity = 2.6147e+00 $k_t = [2, 2, 3, 3]$
Generation 110: Best = 4.882588e+17 FES = 11100 $\mu_F = 0.704$ $\mu_{CR} = 0.396$
Diversity = 1.9933e+00 $k_t = [1, 1, 1, 1]$
Generation 120: Best = 2.120042e+17 FES = 12100 $\mu_F = 0.726$ $\mu_{CR} = 0.392$
Diversity = 1.8109e+00 $k_t = [4, 3, 3, 2]$
Generation 130: Best = 1.130196e+17 FES = 13100 $\mu_F = 0.737$ $\mu_{CR} = 0.389$
Diversity = 1.7050e+00 $k_t = [1, 1, 1, 1]$
Generation 140: Best = 7.024139e+16 FES = 14100 $\mu_F = 0.740$ $\mu_{CR} = 0.396$
Diversity = 1.5027e+00 $k_t = [4, 2, 2, 3]$
Generation 150: Best = 4.338149e+16 FES = 15100 $\mu_F = 0.747$ $\mu_{CR} = 0.389$
Diversity = 1.5315e+00 $k_t = [1, 1, 1, 1]$
Generation 160: Best = 3.301796e+16 FES = 16100 $\mu_F = 0.742$ $\mu_{CR} = 0.396$
Diversity = 1.3989e+00 $k_t = [3, 3, 3, 3]$
Generation 170: Best = 2.008699e+16 FES = 17100 $\mu_F = 0.757$ $\mu_{CR} = 0.395$
Diversity = 1.4915e+00 $k_t = [1, 1, 1, 1]$
Generation 180: Best = 1.290205e+16 FES = 18100 $\mu_F = 0.765$ $\mu_{CR} = 0.400$
Diversity = 1.6380e+00 $k_t = [3, 4, 2, 3]$
Generation 190: Best = 8.590127e+15 FES = 19100 $\mu_F = 0.760$ $\mu_{CR} = 0.402$
Diversity = 1.6252e+00 $k_t = [1, 1, 1, 1]$
Generation 200: Best = 5.718944e+15 FES = 20100 $\mu_F = 0.757$ $\mu_{CR} = 0.405$
Diversity = 1.5731e+00 $k_t = [4, 3, 3, 2]$
Generation 210: Best = 3.196316e+15 FES = 21100 $\mu_F = 0.767$ $\mu_{CR} = 0.411$
Diversity = 1.5277e+00 $k_t = [1, 1, 1, 1]$
Generation 220: Best = 2.172316e+15 FES = 22100 $\mu_F = 0.758$ $\mu_{CR} = 0.408$

Diversity = 1.4801e+00	k_t = [4, 4, 4, 2]				
Generation 230: Best = 1.516624e+15	FES = 23100	$\mu_F = 0.754$	$\mu_{CR} = 0.406$		
Diversity = 1.4257e+00	k_t = [1, 1, 1, 1]				
Generation 240: Best = 1.157555e+15	FES = 24100	$\mu_F = 0.746$	$\mu_{CR} = 0.404$		
Diversity = 1.3099e+00	k_t = [4, 2, 4, 2]				
Generation 250: Best = 9.180143e+14	FES = 25100	$\mu_F = 0.758$	$\mu_{CR} = 0.394$		
Diversity = 1.2460e+00	k_t = [1, 1, 1, 1]				
Generation 260: Best = 7.224736e+14	FES = 26100	$\mu_F = 0.755$	$\mu_{CR} = 0.396$		
Diversity = 1.2350e+00	k_t = [2, 2, 4, 4]				
Generation 270: Best = 5.899374e+14	FES = 27100	$\mu_F = 0.762$	$\mu_{CR} = 0.400$		
Diversity = 1.2496e+00	k_t = [1, 1, 1, 1]				
Generation 280: Best = 5.101985e+14	FES = 28100	$\mu_F = 0.782$	$\mu_{CR} = 0.395$		
Diversity = 1.1523e+00	k_t = [2, 4, 3, 4]				
Generation 290: Best = 4.358459e+14	FES = 29100	$\mu_F = 0.779$	$\mu_{CR} = 0.399$		
Diversity = 1.2352e+00	k_t = [1, 1, 1, 1]				
Generation 300: Best = 3.825283e+14	FES = 30100	$\mu_F = 0.771$	$\mu_{CR} = 0.399$		
Diversity = 1.2076e+00	k_t = [4, 3, 2, 3]				
Generation 310: Best = 3.366398e+14	FES = 31100	$\mu_F = 0.766$	$\mu_{CR} = 0.393$		
Diversity = 1.1861e+00	k_t = [1, 1, 1, 1]				
Generation 320: Best = 3.024225e+14	FES = 32100	$\mu_F = 0.767$	$\mu_{CR} = 0.402$		
Diversity = 1.1338e+00	k_t = [3, 3, 2, 2]				
Generation 330: Best = 2.706662e+14	FES = 33100	$\mu_F = 0.751$	$\mu_{CR} = 0.398$		
Diversity = 1.1871e+00	k_t = [1, 1, 1, 1]				
Generation 340: Best = 2.532799e+14	FES = 34100	$\mu_F = 0.756$	$\mu_{CR} = 0.402$		
Diversity = 1.1005e+00	k_t = [2, 2, 3, 4]				
Generation 350: Best = 2.283277e+14	FES = 35100	$\mu_F = 0.757$	$\mu_{CR} = 0.391$		
Diversity = 1.0589e+00	k_t = [1, 1, 1, 1]				
Generation 360: Best = 1.969094e+14	FES = 36100	$\mu_F = 0.776$	$\mu_{CR} = 0.394$		
Diversity = 1.0700e+00	k_t = [4, 4, 3, 3]				
Generation 370: Best = 1.683778e+14	FES = 37100	$\mu_F = 0.779$	$\mu_{CR} = 0.399$		
Diversity = 1.1181e+00	k_t = [1, 1, 1, 1]				
Generation 380: Best = 1.347020e+14	FES = 38100	$\mu_F = 0.784$	$\mu_{CR} = 0.397$		
Diversity = 1.2295e+00	k_t = [2, 3, 4, 3]				
Generation 390: Best = 9.474870e+13	FES = 39100	$\mu_F = 0.778$	$\mu_{CR} = 0.397$		
Diversity = 1.2962e+00	k_t = [1, 1, 1, 1]				
Generation 400: Best = 7.421049e+13	FES = 40100	$\mu_F = 0.778$	$\mu_{CR} = 0.389$		
Diversity = 1.1983e+00	k_t = [3, 4, 3, 3]				
Generation 410: Best = 5.672196e+13	FES = 41100	$\mu_F = 0.780$	$\mu_{CR} = 0.377$		
Diversity = 1.1905e+00	k_t = [1, 1, 1, 1]				
Generation 420: Best = 4.251767e+13	FES = 42100	$\mu_F = 0.778$	$\mu_{CR} = 0.374$		
Diversity = 1.2487e+00	k_t = [2, 4, 4, 2]				
Generation 430: Best = 3.046520e+13	FES = 43100	$\mu_F = 0.760$	$\mu_{CR} = 0.374$		
Diversity = 1.1531e+00	k_t = [1, 1, 1, 1]				
Generation 440: Best = 2.269837e+13	FES = 44100	$\mu_F = 0.763$	$\mu_{CR} = 0.376$		
Diversity = 1.2302e+00	k_t = [2, 3, 2, 3]				
Generation 450: Best = 1.693127e+13	FES = 45100	$\mu_F = 0.748$	$\mu_{CR} = 0.376$		
Diversity = 1.2517e+00	k_t = [1, 1, 1, 1]				
Generation 460: Best = 1.372742e+13	FES = 46100	$\mu_F = 0.751$	$\mu_{CR} = 0.376$		
Diversity = 1.1844e+00	k_t = [4, 2, 4, 2]				
Generation 470: Best = 1.131663e+13	FES = 47100	$\mu_F = 0.755$	$\mu_{CR} = 0.368$		
Diversity = 1.1710e+00	k_t = [1, 1, 1, 1]				
Generation 480: Best = 9.890303e+12	FES = 48100	$\mu_F = 0.751$	$\mu_{CR} = 0.354$		
Diversity = 1.1242e+00	k_t = [4, 2, 2, 4]				
Generation 490: Best = 8.450700e+12	FES = 49100	$\mu_F = 0.760$	$\mu_{CR} = 0.349$		

Diversity = 1.0636e+00	k_t = [1, 1, 1, 1]					
Generation 500: Best = 7.685552e+12	FES = 50100	$\mu_F = 0.751$	$\mu_{CR} = 0.346$			
Diversity = 9.5958e-01	k_t = [2, 3, 4, 4]					
Generation 510: Best = 6.844212e+12	FES = 51100	$\mu_F = 0.752$	$\mu_{CR} = 0.345$			
Diversity = 1.0218e+00	k_t = [1, 1, 1, 1]					
Generation 520: Best = 6.349806e+12	FES = 52100	$\mu_F = 0.746$	$\mu_{CR} = 0.352$			
Diversity = 1.0441e+00	k_t = [3, 3, 4, 3]					
Generation 530: Best = 5.805539e+12	FES = 53100	$\mu_F = 0.740$	$\mu_{CR} = 0.356$			
Diversity = 9.7981e-01	k_t = [1, 1, 1, 1]					
Generation 540: Best = 5.438049e+12	FES = 54100	$\mu_F = 0.750$	$\mu_{CR} = 0.355$			
Diversity = 9.8815e-01	k_t = [4, 2, 2, 4]					
Generation 550: Best = 5.244387e+12	FES = 55100	$\mu_F = 0.738$	$\mu_{CR} = 0.355$			
Diversity = 1.0014e+00	k_t = [1, 1, 1, 1]					
Generation 560: Best = 5.093333e+12	FES = 56100	$\mu_F = 0.740$	$\mu_{CR} = 0.366$			
Diversity = 9.1835e-01	k_t = [3, 4, 3, 2]					
Generation 570: Best = 4.972637e+12	FES = 57100	$\mu_F = 0.740$	$\mu_{CR} = 0.364$			
Diversity = 9.6660e-01	k_t = [1, 1, 1, 1]					
Generation 580: Best = 4.824839e+12	FES = 58100	$\mu_F = 0.749$	$\mu_{CR} = 0.362$			
Diversity = 9.1346e-01	k_t = [4, 2, 2, 3]					
Generation 590: Best = 4.747283e+12	FES = 59100	$\mu_F = 0.758$	$\mu_{CR} = 0.356$			
Diversity = 9.2940e-01	k_t = [1, 1, 1, 1]					
Generation 600: Best = 4.589080e+12	FES = 60100	$\mu_F = 0.758$	$\mu_{CR} = 0.344$			
Diversity = 8.8056e-01	k_t = [3, 3, 4, 2]					
Generation 610: Best = 4.505388e+12	FES = 61100	$\mu_F = 0.759$	$\mu_{CR} = 0.349$			
Diversity = 9.3262e-01	k_t = [1, 1, 1, 1]					
Generation 620: Best = 4.351213e+12	FES = 62100	$\mu_F = 0.757$	$\mu_{CR} = 0.346$			
Diversity = 1.0041e+00	k_t = [2, 3, 4, 3]					
Generation 630: Best = 4.205743e+12	FES = 63100	$\mu_F = 0.759$	$\mu_{CR} = 0.340$			
Diversity = 1.0334e+00	k_t = [1, 1, 1, 1]					
Generation 640: Best = 4.139377e+12	FES = 64100	$\mu_F = 0.748$	$\mu_{CR} = 0.354$			
Diversity = 9.7378e-01	k_t = [4, 2, 4, 3]					
Generation 650: Best = 4.004199e+12	FES = 65100	$\mu_F = 0.747$	$\mu_{CR} = 0.353$			
Diversity = 9.8362e-01	k_t = [1, 1, 1, 1]					
Generation 660: Best = 3.908794e+12	FES = 66100	$\mu_F = 0.749$	$\mu_{CR} = 0.351$			
Diversity = 8.9094e-01	k_t = [4, 4, 2, 3]					
Generation 670: Best = 3.817454e+12	FES = 67100	$\mu_F = 0.733$	$\mu_{CR} = 0.349$			
Diversity = 9.6105e-01	k_t = [1, 1, 1, 1]					
Generation 680: Best = 3.746416e+12	FES = 68100	$\mu_F = 0.729$	$\mu_{CR} = 0.357$			
Diversity = 9.2369e-01	k_t = [3, 4, 2, 2]					
Generation 690: Best = 3.699059e+12	FES = 69100	$\mu_F = 0.732$	$\mu_{CR} = 0.359$			
Diversity = 8.9155e-01	k_t = [1, 1, 1, 1]					
Generation 700: Best = 3.654133e+12	FES = 70100	$\mu_F = 0.739$	$\mu_{CR} = 0.361$			
Diversity = 8.4492e-01	k_t = [4, 3, 3, 4]					
Generation 710: Best = 3.616230e+12	FES = 71100	$\mu_F = 0.747$	$\mu_{CR} = 0.363$			
Diversity = 8.6074e-01	k_t = [1, 1, 1, 1]					
Generation 720: Best = 3.595184e+12	FES = 72100	$\mu_F = 0.748$	$\mu_{CR} = 0.364$			
Diversity = 8.1915e-01	k_t = [3, 4, 2, 2]					
Generation 730: Best = 3.559359e+12	FES = 73100	$\mu_F = 0.755$	$\mu_{CR} = 0.367$			
Diversity = 9.2039e-01	k_t = [1, 1, 1, 1]					
Generation 740: Best = 3.533613e+12	FES = 74100	$\mu_F = 0.756$	$\mu_{CR} = 0.370$			
Diversity = 1.0237e+00	k_t = [4, 3, 4, 2]					
Generation 750: Best = 3.513380e+12	FES = 75100	$\mu_F = 0.760$	$\mu_{CR} = 0.369$			
Diversity = 1.0161e+00	k_t = [1, 1, 1, 1]					
Generation 760: Best = 3.503911e+12	FES = 76100	$\mu_F = 0.738$	$\mu_{CR} = 0.379$			

Diversity = 9.7483e-01	k_t = [4, 3, 4, 3]				
Generation 770: Best = 3.486640e+12	FES = 77100	$\mu_F = 0.739$	$\mu_{CR} = 0.382$		
Diversity = 9.1803e-01	k_t = [1, 1, 1, 1]				
Generation 780: Best = 3.465044e+12	FES = 78100	$\mu_F = 0.763$	$\mu_{CR} = 0.386$		
Diversity = 1.0471e+00	k_t = [4, 3, 4, 3]				
Generation 790: Best = 3.445072e+12	FES = 79100	$\mu_F = 0.761$	$\mu_{CR} = 0.380$		
Diversity = 1.3574e+00	k_t = [1, 1, 1, 1]				
Generation 800: Best = 3.418528e+12	FES = 80100	$\mu_F = 0.761$	$\mu_{CR} = 0.373$		
Diversity = 1.5721e+00	k_t = [3, 4, 4, 2]				
Generation 810: Best = 3.395129e+12	FES = 81100	$\mu_F = 0.755$	$\mu_{CR} = 0.381$		
Diversity = 1.6633e+00	k_t = [1, 1, 1, 1]				
Generation 820: Best = 3.387904e+12	FES = 82100	$\mu_F = 0.738$	$\mu_{CR} = 0.381$		
Diversity = 1.6120e+00	k_t = [4, 3, 4, 2]				
Generation 830: Best = 3.370821e+12	FES = 83100	$\mu_F = 0.727$	$\mu_{CR} = 0.366$		
Diversity = 1.5637e+00	k_t = [1, 1, 1, 1]				
Generation 840: Best = 3.349515e+12	FES = 84100	$\mu_F = 0.723$	$\mu_{CR} = 0.365$		
Diversity = 1.3837e+00	k_t = [3, 3, 4, 4]				
Generation 850: Best = 3.321088e+12	FES = 85100	$\mu_F = 0.726$	$\mu_{CR} = 0.368$		
Diversity = 1.5116e+00	k_t = [1, 1, 1, 1]				
Generation 860: Best = 3.305592e+12	FES = 86100	$\mu_F = 0.721$	$\mu_{CR} = 0.370$		
Diversity = 1.4750e+00	k_t = [3, 2, 3, 2]				
Generation 870: Best = 3.286737e+12	FES = 87100	$\mu_F = 0.730$	$\mu_{CR} = 0.373$		
Diversity = 1.2967e+00	k_t = [1, 1, 1, 1]				
Generation 880: Best = 3.272479e+12	FES = 88100	$\mu_F = 0.720$	$\mu_{CR} = 0.378$		
Diversity = 1.1008e+00	k_t = [2, 2, 3, 4]				
Generation 890: Best = 3.264350e+12	FES = 89100	$\mu_F = 0.727$	$\mu_{CR} = 0.378$		
Diversity = 1.1767e+00	k_t = [1, 1, 1, 1]				
Generation 900: Best = 3.253189e+12	FES = 90100	$\mu_F = 0.732$	$\mu_{CR} = 0.378$		
Diversity = 1.1334e+00	k_t = [3, 4, 3, 3]				
Generation 910: Best = 3.244472e+12	FES = 91100	$\mu_F = 0.744$	$\mu_{CR} = 0.376$		
Diversity = 1.0956e+00	k_t = [1, 1, 1, 1]				
Generation 920: Best = 3.231197e+12	FES = 92100	$\mu_F = 0.750$	$\mu_{CR} = 0.371$		
Diversity = 1.0098e+00	k_t = [4, 3, 2, 2]				
Generation 930: Best = 3.224238e+12	FES = 93100	$\mu_F = 0.743$	$\mu_{CR} = 0.363$		
Diversity = 1.0304e+00	k_t = [1, 1, 1, 1]				
Generation 940: Best = 3.212555e+12	FES = 94100	$\mu_F = 0.753$	$\mu_{CR} = 0.361$		
Diversity = 1.0268e+00	k_t = [3, 4, 4, 4]				
Generation 950: Best = 3.203642e+12	FES = 95100	$\mu_F = 0.759$	$\mu_{CR} = 0.362$		
Diversity = 1.1690e+00	k_t = [1, 1, 1, 1]				
Generation 960: Best = 3.194830e+12	FES = 96100	$\mu_F = 0.771$	$\mu_{CR} = 0.365$		
Diversity = 1.1864e+00	k_t = [4, 4, 2, 4]				
Generation 970: Best = 3.187472e+12	FES = 97100	$\mu_F = 0.760$	$\mu_{CR} = 0.378$		
Diversity = 1.2135e+00	k_t = [1, 1, 1, 1]				
Generation 980: Best = 3.177131e+12	FES = 98100	$\mu_F = 0.747$	$\mu_{CR} = 0.377$		
Diversity = 1.0926e+00	k_t = [4, 2, 2, 4]				
Generation 990: Best = 3.169854e+12	FES = 99100	$\mu_F = 0.755$	$\mu_{CR} = 0.374$		
Diversity = 1.1517e+00	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 3.160256e+12	FES = 100100	$\mu_F = 0.728$	$\mu_{CR} = 0.377$		
Diversity = 9.5693e-01	k_t = [4, 2, 3, 3]				
Generation 1010: Best = 3.152967e+12	FES = 101100	$\mu_F = 0.744$	$\mu_{CR} = 0.373$		
Diversity = 1.0972e+00	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 3.145843e+12	FES = 102100	$\mu_F = 0.747$	$\mu_{CR} = 0.363$		
Diversity = 1.0733e+00	k_t = [2, 2, 2, 2]				
Generation 1030: Best = 3.141265e+12	FES = 103100	$\mu_F = 0.736$	$\mu_{CR} = 0.364$		

Diversity = 1.1225e+00	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 3.136577e+12	FES = 104100	$\mu_F = 0.734$	$\mu_{CR} = 0.367$	
Diversity = 1.0069e+00	k_t = [4, 4, 4, 4]			
Generation 1050: Best = 3.134063e+12	FES = 105100	$\mu_F = 0.741$	$\mu_{CR} = 0.366$	
Diversity = 9.7498e-01	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 3.130318e+12	FES = 106100	$\mu_F = 0.742$	$\mu_{CR} = 0.367$	
Diversity = 9.0313e-01	k_t = [4, 3, 2, 2]			
Generation 1070: Best = 3.127144e+12	FES = 107100	$\mu_F = 0.744$	$\mu_{CR} = 0.368$	
Diversity = 8.7588e-01	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 3.125927e+12	FES = 108100	$\mu_F = 0.740$	$\mu_{CR} = 0.375$	
Diversity = 8.0118e-01	k_t = [3, 4, 3, 4]			
Generation 1090: Best = 3.124530e+12	FES = 109100	$\mu_F = 0.743$	$\mu_{CR} = 0.374$	
Diversity = 8.3659e-01	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 3.123212e+12	FES = 110100	$\mu_F = 0.733$	$\mu_{CR} = 0.378$	
Diversity = 7.9492e-01	k_t = [2, 3, 4, 4]			
Generation 1110: Best = 3.121481e+12	FES = 111100	$\mu_F = 0.732$	$\mu_{CR} = 0.383$	
Diversity = 8.6508e-01	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 3.119940e+12	FES = 112100	$\mu_F = 0.738$	$\mu_{CR} = 0.385$	
Diversity = 9.4446e-01	k_t = [4, 2, 4, 2]			
Generation 1130: Best = 3.118870e+12	FES = 113100	$\mu_F = 0.740$	$\mu_{CR} = 0.381$	
Diversity = 9.1832e-01	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 3.117360e+12	FES = 114100	$\mu_F = 0.746$	$\mu_{CR} = 0.380$	
Diversity = 9.8521e-01	k_t = [2, 2, 3, 3]			
Generation 1150: Best = 3.113688e+12	FES = 115100	$\mu_F = 0.754$	$\mu_{CR} = 0.379$	
Diversity = 1.0935e+00	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 3.110830e+12	FES = 116100	$\mu_F = 0.759$	$\mu_{CR} = 0.367$	
Diversity = 9.9995e-01	k_t = [2, 2, 3, 4]			
Generation 1170: Best = 3.107046e+12	FES = 117100	$\mu_F = 0.762$	$\mu_{CR} = 0.366$	
Diversity = 1.1282e+00	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 3.105283e+12	FES = 118100	$\mu_F = 0.762$	$\mu_{CR} = 0.369$	
Diversity = 1.0509e+00	k_t = [2, 4, 2, 3]			
Generation 1190: Best = 3.102978e+12	FES = 119100	$\mu_F = 0.755$	$\mu_{CR} = 0.363$	
Diversity = 1.0498e+00	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 3.101423e+12	FES = 120100	$\mu_F = 0.753$	$\mu_{CR} = 0.359$	
Diversity = 1.0712e+00	k_t = [4, 3, 2, 4]			
Generation 1210: Best = 3.100540e+12	FES = 121100	$\mu_F = 0.749$	$\mu_{CR} = 0.349$	
Diversity = 1.1044e+00	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 3.099756e+12	FES = 122100	$\mu_F = 0.736$	$\mu_{CR} = 0.348$	
Diversity = 9.8395e-01	k_t = [4, 4, 4, 4]			
Generation 1230: Best = 3.099204e+12	FES = 123100	$\mu_F = 0.743$	$\mu_{CR} = 0.337$	
Diversity = 1.0873e+00	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 3.098685e+12	FES = 124100	$\mu_F = 0.733$	$\mu_{CR} = 0.332$	
Diversity = 1.1883e+00	k_t = [4, 2, 2, 3]			
Generation 1250: Best = 3.098334e+12	FES = 125100	$\mu_F = 0.730$	$\mu_{CR} = 0.323$	
Diversity = 1.2308e+00	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 3.098047e+12	FES = 126100	$\mu_F = 0.731$	$\mu_{CR} = 0.323$	
Diversity = 1.3256e+00	k_t = [2, 3, 4, 2]			
Generation 1270: Best = 3.097385e+12	FES = 127100	$\mu_F = 0.729$	$\mu_{CR} = 0.319$	
Diversity = 1.4432e+00	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 3.096951e+12	FES = 128100	$\mu_F = 0.734$	$\mu_{CR} = 0.299$	
Diversity = 1.3756e+00	k_t = [2, 3, 2, 3]			
Generation 1290: Best = 3.096599e+12	FES = 129100	$\mu_F = 0.742$	$\mu_{CR} = 0.295$	
Diversity = 1.4494e+00	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 3.096265e+12	FES = 130100	$\mu_F = 0.734$	$\mu_{CR} = 0.260$	

Diversity = 1.5578e+00	k_t = [2, 3, 3, 4]			
Generation 1310: Best = 3.095798e+12	FES = 131100	$\mu_F = 0.736$	$\mu_{CR} = 0.256$	
Diversity = 1.6812e+00	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 3.095489e+12	FES = 132100	$\mu_F = 0.750$	$\mu_{CR} = 0.236$	
Diversity = 1.5159e+00	k_t = [2, 2, 4, 4]			
Generation 1330: Best = 3.095254e+12	FES = 133100	$\mu_F = 0.772$	$\mu_{CR} = 0.226$	
Diversity = 1.5918e+00	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 3.094911e+12	FES = 134100	$\mu_F = 0.781$	$\mu_{CR} = 0.227$	
Diversity = 1.6831e+00	k_t = [3, 2, 3, 4]			
Generation 1350: Best = 3.094665e+12	FES = 135100	$\mu_F = 0.773$	$\mu_{CR} = 0.221$	
Diversity = 1.5811e+00	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 3.094380e+12	FES = 136100	$\mu_F = 0.765$	$\mu_{CR} = 0.224$	
Diversity = 1.4515e+00	k_t = [2, 4, 2, 4]			
Generation 1370: Best = 3.094056e+12	FES = 137100	$\mu_F = 0.780$	$\mu_{CR} = 0.226$	
Diversity = 1.4495e+00	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 3.093792e+12	FES = 138100	$\mu_F = 0.769$	$\mu_{CR} = 0.207$	
Diversity = 1.4318e+00	k_t = [3, 2, 4, 2]			
Generation 1390: Best = 3.093468e+12	FES = 139100	$\mu_F = 0.770$	$\mu_{CR} = 0.199$	
Diversity = 1.4169e+00	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 3.093078e+12	FES = 140100	$\mu_F = 0.768$	$\mu_{CR} = 0.190$	
Diversity = 1.2694e+00	k_t = [4, 3, 4, 4]			
Generation 1410: Best = 3.092604e+12	FES = 141100	$\mu_F = 0.768$	$\mu_{CR} = 0.180$	
Diversity = 1.2252e+00	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 3.092101e+12	FES = 142100	$\mu_F = 0.776$	$\mu_{CR} = 0.179$	
Diversity = 1.3249e+00	k_t = [3, 4, 3, 2]			
Generation 1430: Best = 3.091519e+12	FES = 143100	$\mu_F = 0.774$	$\mu_{CR} = 0.172$	
Diversity = 1.3793e+00	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 3.090753e+12	FES = 144100	$\mu_F = 0.764$	$\mu_{CR} = 0.170$	
Diversity = 1.3832e+00	k_t = [2, 3, 2, 2]			
Generation 1450: Best = 3.089943e+12	FES = 145100	$\mu_F = 0.771$	$\mu_{CR} = 0.165$	
Diversity = 1.4579e+00	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 3.089153e+12	FES = 146100	$\mu_F = 0.766$	$\mu_{CR} = 0.160$	
Diversity = 1.3176e+00	k_t = [2, 4, 2, 4]			
Generation 1470: Best = 3.087385e+12	FES = 147100	$\mu_F = 0.769$	$\mu_{CR} = 0.168$	
Diversity = 1.1958e+00	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 3.086324e+12	FES = 148100	$\mu_F = 0.762$	$\mu_{CR} = 0.161$	
Diversity = 1.1980e+00	k_t = [2, 2, 4, 2]			
Generation 1490: Best = 3.085033e+12	FES = 149100	$\mu_F = 0.774$	$\mu_{CR} = 0.156$	
Diversity = 1.4238e+00	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 3.083200e+12	FES = 150100	$\mu_F = 0.777$	$\mu_{CR} = 0.157$	
Diversity = 1.5134e+00	k_t = [3, 4, 3, 2]			
Generation 1510: Best = 3.082193e+12	FES = 151100	$\mu_F = 0.772$	$\mu_{CR} = 0.154$	
Diversity = 1.5589e+00	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 3.081569e+12	FES = 152100	$\mu_F = 0.773$	$\mu_{CR} = 0.150$	
Diversity = 1.5406e+00	k_t = [3, 4, 3, 4]			
Generation 1530: Best = 3.081014e+12	FES = 153100	$\mu_F = 0.774$	$\mu_{CR} = 0.144$	
Diversity = 1.4980e+00	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 3.080181e+12	FES = 154100	$\mu_F = 0.773$	$\mu_{CR} = 0.144$	
Diversity = 1.5786e+00	k_t = [3, 3, 2, 3]			
Generation 1550: Best = 3.079414e+12	FES = 155100	$\mu_F = 0.783$	$\mu_{CR} = 0.149$	
Diversity = 1.7332e+00	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 3.078727e+12	FES = 156100	$\mu_F = 0.784$	$\mu_{CR} = 0.147$	
Diversity = 1.7605e+00	k_t = [2, 4, 3, 4]			
Generation 1570: Best = 3.078216e+12	FES = 157100	$\mu_F = 0.781$	$\mu_{CR} = 0.146$	

Diversity = 1.6417e+00	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 3.077869e+12	FES = 158100	$\mu_F = 0.770$	$\mu_{CR} = 0.134$	
Diversity = 1.6754e+00	k_t = [4, 2, 3, 4]			
Generation 1590: Best = 3.077117e+12	FES = 159100	$\mu_F = 0.758$	$\mu_{CR} = 0.139$	
Diversity = 1.6107e+00	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 3.076621e+12	FES = 160100	$\mu_F = 0.760$	$\mu_{CR} = 0.143$	
Diversity = 1.7031e+00	k_t = [3, 2, 2, 4]			
Generation 1610: Best = 3.076174e+12	FES = 161100	$\mu_F = 0.766$	$\mu_{CR} = 0.146$	
Diversity = 1.8255e+00	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 3.075490e+12	FES = 162100	$\mu_F = 0.777$	$\mu_{CR} = 0.138$	
Diversity = 1.7932e+00	k_t = [3, 3, 3, 3]			
Generation 1630: Best = 3.074795e+12	FES = 163100	$\mu_F = 0.783$	$\mu_{CR} = 0.135$	
Diversity = 1.6881e+00	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 3.074372e+12	FES = 164100	$\mu_F = 0.779$	$\mu_{CR} = 0.136$	
Diversity = 1.6103e+00	k_t = [2, 4, 2, 2]			
Generation 1650: Best = 3.073984e+12	FES = 165100	$\mu_F = 0.768$	$\mu_{CR} = 0.139$	
Diversity = 1.6928e+00	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 3.073601e+12	FES = 166100	$\mu_F = 0.751$	$\mu_{CR} = 0.142$	
Diversity = 1.8265e+00	k_t = [4, 2, 3, 2]			
Generation 1670: Best = 3.072945e+12	FES = 167100	$\mu_F = 0.762$	$\mu_{CR} = 0.146$	
Diversity = 1.7162e+00	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 3.072157e+12	FES = 168100	$\mu_F = 0.762$	$\mu_{CR} = 0.149$	
Diversity = 1.6275e+00	k_t = [2, 4, 3, 4]			
Generation 1690: Best = 3.071185e+12	FES = 169100	$\mu_F = 0.757$	$\mu_{CR} = 0.140$	
Diversity = 1.7114e+00	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 3.070657e+12	FES = 170100	$\mu_F = 0.765$	$\mu_{CR} = 0.152$	
Diversity = 1.5982e+00	k_t = [3, 4, 3, 4]			
Generation 1710: Best = 3.069444e+12	FES = 171100	$\mu_F = 0.776$	$\mu_{CR} = 0.159$	
Diversity = 1.6504e+00	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 3.068495e+12	FES = 172100	$\mu_F = 0.778$	$\mu_{CR} = 0.155$	
Diversity = 1.7699e+00	k_t = [2, 4, 2, 2]			
Generation 1730: Best = 3.067428e+12	FES = 173100	$\mu_F = 0.770$	$\mu_{CR} = 0.153$	
Diversity = 1.7350e+00	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 3.067091e+12	FES = 174100	$\mu_F = 0.763$	$\mu_{CR} = 0.153$	
Diversity = 1.6815e+00	k_t = [4, 3, 4, 2]			
Generation 1750: Best = 3.066242e+12	FES = 175100	$\mu_F = 0.766$	$\mu_{CR} = 0.157$	
Diversity = 1.7249e+00	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 3.065339e+12	FES = 176100	$\mu_F = 0.758$	$\mu_{CR} = 0.141$	
Diversity = 1.6717e+00	k_t = [4, 3, 4, 3]			
Generation 1770: Best = 3.064540e+12	FES = 177100	$\mu_F = 0.758$	$\mu_{CR} = 0.150$	
Diversity = 1.6301e+00	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 3.064124e+12	FES = 178100	$\mu_F = 0.748$	$\mu_{CR} = 0.145$	
Diversity = 1.5241e+00	k_t = [2, 2, 4, 3]			
Generation 1790: Best = 3.063537e+12	FES = 179100	$\mu_F = 0.761$	$\mu_{CR} = 0.141$	
Diversity = 1.5185e+00	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 3.063310e+12	FES = 180100	$\mu_F = 0.764$	$\mu_{CR} = 0.141$	
Diversity = 1.5273e+00	k_t = [2, 2, 4, 3]			
Generation 1810: Best = 3.062873e+12	FES = 181100	$\mu_F = 0.750$	$\mu_{CR} = 0.151$	
Diversity = 1.4986e+00	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 3.062565e+12	FES = 182100	$\mu_F = 0.753$	$\mu_{CR} = 0.154$	
Diversity = 1.5203e+00	k_t = [2, 4, 3, 2]			
Generation 1830: Best = 3.061833e+12	FES = 183100	$\mu_F = 0.756$	$\mu_{CR} = 0.150$	
Diversity = 1.6648e+00	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 3.061448e+12	FES = 184100	$\mu_F = 0.761$	$\mu_{CR} = 0.149$	

Diversity = 1.4046e+00	k_t = [2, 2, 3, 4]			
Generation 1850: Best = 3.060920e+12	FES = 185100	$\mu_F = 0.755$	$\mu_{CR} = 0.147$	
Diversity = 1.3607e+00	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 3.060245e+12	FES = 186100	$\mu_F = 0.746$	$\mu_{CR} = 0.143$	
Diversity = 1.2127e+00	k_t = [2, 2, 3, 4]			
Generation 1870: Best = 3.059825e+12	FES = 187100	$\mu_F = 0.751$	$\mu_{CR} = 0.146$	
Diversity = 1.2249e+00	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 3.059147e+12	FES = 188100	$\mu_F = 0.755$	$\mu_{CR} = 0.155$	
Diversity = 1.2215e+00	k_t = [4, 3, 4, 2]			
Generation 1890: Best = 3.058681e+12	FES = 189100	$\mu_F = 0.746$	$\mu_{CR} = 0.160$	
Diversity = 1.2710e+00	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 3.058253e+12	FES = 190100	$\mu_F = 0.747$	$\mu_{CR} = 0.170$	
Diversity = 1.1338e+00	k_t = [2, 2, 4, 2]			
Generation 1910: Best = 3.057501e+12	FES = 191100	$\mu_F = 0.759$	$\mu_{CR} = 0.177$	
Diversity = 1.2113e+00	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 3.056729e+12	FES = 192100	$\mu_F = 0.764$	$\mu_{CR} = 0.173$	
Diversity = 1.1890e+00	k_t = [4, 2, 2, 2]			
Generation 1930: Best = 3.055010e+12	FES = 193100	$\mu_F = 0.753$	$\mu_{CR} = 0.173$	
Diversity = 1.1675e+00	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 3.054062e+12	FES = 194100	$\mu_F = 0.763$	$\mu_{CR} = 0.173$	
Diversity = 1.2388e+00	k_t = [4, 2, 3, 2]			
Generation 1950: Best = 3.053461e+12	FES = 195100	$\mu_F = 0.765$	$\mu_{CR} = 0.166$	
Diversity = 1.3609e+00	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 3.052956e+12	FES = 196100	$\mu_F = 0.764$	$\mu_{CR} = 0.164$	
Diversity = 1.2990e+00	k_t = [4, 2, 2, 2]			
Generation 1970: Best = 3.052447e+12	FES = 197100	$\mu_F = 0.759$	$\mu_{CR} = 0.160$	
Diversity = 1.3016e+00	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 3.051472e+12	FES = 198100	$\mu_F = 0.755$	$\mu_{CR} = 0.168$	
Diversity = 1.2588e+00	k_t = [3, 3, 3, 3]			
Generation 1990: Best = 3.050990e+12	FES = 199100	$\mu_F = 0.756$	$\mu_{CR} = 0.162$	
Diversity = 1.2707e+00	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 3.050215e+12	FES = 200100	$\mu_F = 0.756$	$\mu_{CR} = 0.162$	
Diversity = 1.1735e+00	k_t = [2, 3, 3, 2]			
Generation 2010: Best = 3.049721e+12	FES = 201100	$\mu_F = 0.759$	$\mu_{CR} = 0.168$	
Diversity = 1.4115e+00	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 3.049271e+12	FES = 202100	$\mu_F = 0.747$	$\mu_{CR} = 0.167$	
Diversity = 1.3380e+00	k_t = [4, 4, 3, 4]			
Generation 2030: Best = 3.048941e+12	FES = 203100	$\mu_F = 0.748$	$\mu_{CR} = 0.164$	
Diversity = 1.3389e+00	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 3.048582e+12	FES = 204100	$\mu_F = 0.739$	$\mu_{CR} = 0.170$	
Diversity = 1.2844e+00	k_t = [4, 2, 3, 3]			
Generation 2050: Best = 3.048186e+12	FES = 205100	$\mu_F = 0.747$	$\mu_{CR} = 0.173$	
Diversity = 1.2352e+00	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 3.047625e+12	FES = 206100	$\mu_F = 0.748$	$\mu_{CR} = 0.182$	
Diversity = 1.1965e+00	k_t = [2, 4, 2, 2]			
Generation 2070: Best = 3.047454e+12	FES = 207100	$\mu_F = 0.748$	$\mu_{CR} = 0.190$	
Diversity = 1.2561e+00	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 3.047092e+12	FES = 208100	$\mu_F = 0.752$	$\mu_{CR} = 0.192$	
Diversity = 1.0895e+00	k_t = [4, 3, 4, 4]			
Generation 2090: Best = 3.046954e+12	FES = 209100	$\mu_F = 0.765$	$\mu_{CR} = 0.185$	
Diversity = 1.0650e+00	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 3.046671e+12	FES = 210100	$\mu_F = 0.769$	$\mu_{CR} = 0.180$	
Diversity = 1.1767e+00	k_t = [4, 2, 4, 4]			
Generation 2110: Best = 3.046435e+12	FES = 211100	$\mu_F = 0.762$	$\mu_{CR} = 0.185$	

Diversity = 1.2341e+00	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 3.046261e+12	FES = 212100	$\mu_F = 0.751$	$\mu_{CR} = 0.179$	
Diversity = 1.3402e+00	k_t = [4, 2, 4, 2]			
Generation 2130: Best = 3.046138e+12	FES = 213100	$\mu_F = 0.755$	$\mu_{CR} = 0.184$	
Diversity = 1.5113e+00	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 3.045967e+12	FES = 214100	$\mu_F = 0.753$	$\mu_{CR} = 0.172$	
Diversity = 1.4956e+00	k_t = [2, 3, 2, 2]			
Generation 2150: Best = 3.045764e+12	FES = 215100	$\mu_F = 0.755$	$\mu_{CR} = 0.169$	
Diversity = 1.5500e+00	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 3.045605e+12	FES = 216100	$\mu_F = 0.770$	$\mu_{CR} = 0.158$	
Diversity = 1.3995e+00	k_t = [2, 3, 4, 4]			
Generation 2170: Best = 3.045400e+12	FES = 217100	$\mu_F = 0.760$	$\mu_{CR} = 0.154$	
Diversity = 1.3506e+00	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 3.045129e+12	FES = 218100	$\mu_F = 0.756$	$\mu_{CR} = 0.156$	
Diversity = 1.2669e+00	k_t = [2, 3, 2, 4]			
Generation 2190: Best = 3.044883e+12	FES = 219100	$\mu_F = 0.760$	$\mu_{CR} = 0.157$	
Diversity = 1.3722e+00	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 3.044555e+12	FES = 220100	$\mu_F = 0.758$	$\mu_{CR} = 0.148$	
Diversity = 1.4547e+00	k_t = [2, 3, 2, 2]			
Generation 2210: Best = 3.044132e+12	FES = 221100	$\mu_F = 0.757$	$\mu_{CR} = 0.154$	
Diversity = 1.5247e+00	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 3.043737e+12	FES = 222100	$\mu_F = 0.762$	$\mu_{CR} = 0.151$	
Diversity = 1.3982e+00	k_t = [4, 3, 2, 3]			
Generation 2230: Best = 3.043421e+12	FES = 223100	$\mu_F = 0.767$	$\mu_{CR} = 0.153$	
Diversity = 1.3581e+00	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 3.043129e+12	FES = 224100	$\mu_F = 0.773$	$\mu_{CR} = 0.146$	
Diversity = 1.3828e+00	k_t = [4, 3, 2, 3]			
Generation 2250: Best = 3.042802e+12	FES = 225100	$\mu_F = 0.756$	$\mu_{CR} = 0.146$	
Diversity = 1.4472e+00	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 3.042639e+12	FES = 226100	$\mu_F = 0.739$	$\mu_{CR} = 0.152$	
Diversity = 1.3925e+00	k_t = [4, 2, 2, 4]			
Generation 2270: Best = 3.042337e+12	FES = 227100	$\mu_F = 0.751$	$\mu_{CR} = 0.155$	
Diversity = 1.4125e+00	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 3.041941e+12	FES = 228100	$\mu_F = 0.748$	$\mu_{CR} = 0.167$	
Diversity = 1.2845e+00	k_t = [3, 2, 3, 4]			
Generation 2290: Best = 3.041418e+12	FES = 229100	$\mu_F = 0.752$	$\mu_{CR} = 0.167$	
Diversity = 1.3109e+00	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 3.041228e+12	FES = 230100	$\mu_F = 0.751$	$\mu_{CR} = 0.164$	
Diversity = 1.2722e+00	k_t = [4, 3, 3, 3]			
Generation 2310: Best = 3.040875e+12	FES = 231100	$\mu_F = 0.755$	$\mu_{CR} = 0.158$	
Diversity = 1.1981e+00	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 3.040632e+12	FES = 232100	$\mu_F = 0.757$	$\mu_{CR} = 0.164$	
Diversity = 1.1755e+00	k_t = [2, 2, 4, 3]			
Generation 2330: Best = 3.040169e+12	FES = 233100	$\mu_F = 0.771$	$\mu_{CR} = 0.172$	
Diversity = 1.1714e+00	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 3.039645e+12	FES = 234100	$\mu_F = 0.772$	$\mu_{CR} = 0.177$	
Diversity = 1.1481e+00	k_t = [4, 4, 3, 2]			
Generation 2350: Best = 3.039220e+12	FES = 235100	$\mu_F = 0.771$	$\mu_{CR} = 0.175$	
Diversity = 1.2493e+00	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 3.038598e+12	FES = 236100	$\mu_F = 0.782$	$\mu_{CR} = 0.170$	
Diversity = 1.2069e+00	k_t = [2, 2, 2, 3]			
Generation 2370: Best = 3.037971e+12	FES = 237100	$\mu_F = 0.769$	$\mu_{CR} = 0.170$	
Diversity = 1.2910e+00	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 3.037572e+12	FES = 238100	$\mu_F = 0.774$	$\mu_{CR} = 0.174$	

Diversity = 1.2824e+00 k_t = [3, 4, 3, 2]
 Generation 2390: Best = 3.037378e+12 FES = 239100 μ_F = 0.768 μ_{CR} = 0.167
 Diversity = 1.2433e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 3.037094e+12 FES = 240100 μ_F = 0.760 μ_{CR} = 0.162
 Diversity = 1.1494e+00 k_t = [3, 2, 4, 4]
 Generation 2410: Best = 3.037022e+12 FES = 241100 μ_F = 0.754 μ_{CR} = 0.166
 Diversity = 1.1531e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 3.036974e+12 FES = 242100 μ_F = 0.764 μ_{CR} = 0.169
 Diversity = 1.0838e+00 k_t = [4, 4, 2, 4]
 Generation 2430: Best = 3.036942e+12 FES = 243100 μ_F = 0.771 μ_{CR} = 0.167
 Diversity = 1.1312e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 3.036898e+12 FES = 244100 μ_F = 0.776 μ_{CR} = 0.163
 Diversity = 1.2362e+00 k_t = [3, 4, 4, 3]
 Generation 2450: Best = 3.036860e+12 FES = 245100 μ_F = 0.775 μ_{CR} = 0.154
 Diversity = 1.2692e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 3.036827e+12 FES = 246100 μ_F = 0.768 μ_{CR} = 0.150
 Diversity = 1.1809e+00 k_t = [4, 3, 3, 2]
 Generation 2470: Best = 3.036807e+12 FES = 247100 μ_F = 0.759 μ_{CR} = 0.151
 Diversity = 1.1673e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 3.036793e+12 FES = 248100 μ_F = 0.750 μ_{CR} = 0.145
 Diversity = 1.0098e+00 k_t = [4, 3, 3, 4]
 Generation 2490: Best = 3.036783e+12 FES = 249100 μ_F = 0.757 μ_{CR} = 0.144
 Diversity = 1.0699e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 3.036776e+12 FES = 250100 μ_F = 0.753 μ_{CR} = 0.146
 Diversity = 1.0888e+00 k_t = [3, 2, 2, 3]
 Generation 2510: Best = 3.036773e+12 FES = 251100 μ_F = 0.757 μ_{CR} = 0.149
 Diversity = 1.0420e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 3.036769e+12 FES = 252100 μ_F = 0.762 μ_{CR} = 0.152
 Diversity = 1.1784e+00 k_t = [3, 3, 3, 2]
 Generation 2530: Best = 3.036766e+12 FES = 253100 μ_F = 0.758 μ_{CR} = 0.152
 Diversity = 1.3009e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 3.036764e+12 FES = 254100 μ_F = 0.766 μ_{CR} = 0.154
 Diversity = 1.3651e+00 k_t = [3, 3, 4, 4]
 Generation 2550: Best = 3.036763e+12 FES = 255100 μ_F = 0.755 μ_{CR} = 0.143
 Diversity = 1.2251e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 3.036762e+12 FES = 256100 μ_F = 0.748 μ_{CR} = 0.146
 Diversity = 1.2713e+00 k_t = [3, 2, 4, 2]
 Generation 2570: Best = 3.036761e+12 FES = 257100 μ_F = 0.752 μ_{CR} = 0.147
 Diversity = 1.2070e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 3.036761e+12 FES = 258100 μ_F = 0.753 μ_{CR} = 0.144
 Diversity = 1.2797e+00 k_t = [2, 2, 2, 3]
 Generation 2590: Best = 3.036761e+12 FES = 259100 μ_F = 0.754 μ_{CR} = 0.131
 Diversity = 1.4848e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 3.036760e+12 FES = 260100 μ_F = 0.763 μ_{CR} = 0.134
 Diversity = 1.3399e+00 k_t = [4, 2, 2, 4]
 Generation 2610: Best = 3.036760e+12 FES = 261100 μ_F = 0.769 μ_{CR} = 0.127
 Diversity = 1.3903e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 3.036760e+12 FES = 262100 μ_F = 0.762 μ_{CR} = 0.124
 Diversity = 1.3668e+00 k_t = [2, 2, 3, 2]
 Generation 2630: Best = 3.036759e+12 FES = 263100 μ_F = 0.765 μ_{CR} = 0.125
 Diversity = 1.4507e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 3.036759e+12 FES = 264100 μ_F = 0.765 μ_{CR} = 0.129
 Diversity = 1.4397e+00 k_t = [2, 4, 3, 4]
 Generation 2650: Best = 3.036759e+12 FES = 265100 μ_F = 0.773 μ_{CR} = 0.123

Diversity = 1.4378e+00	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 3.036759e+12	FES = 266100	$\mu_F = 0.775$	$\mu_{CR} = 0.115$	
Diversity = 1.3966e+00	k_t = [3, 3, 4, 2]			
Generation 2670: Best = 3.036759e+12	FES = 267100	$\mu_F = 0.761$	$\mu_{CR} = 0.116$	
Diversity = 1.5656e+00	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 3.036758e+12	FES = 268100	$\mu_F = 0.750$	$\mu_{CR} = 0.125$	
Diversity = 1.3951e+00	k_t = [2, 2, 4, 3]			
Generation 2690: Best = 3.036758e+12	FES = 269100	$\mu_F = 0.742$	$\mu_{CR} = 0.126$	
Diversity = 1.3185e+00	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 3.036758e+12	FES = 270100	$\mu_F = 0.755$	$\mu_{CR} = 0.134$	
Diversity = 1.1557e+00	k_t = [4, 4, 2, 4]			
Generation 2710: Best = 3.036758e+12	FES = 271100	$\mu_F = 0.755$	$\mu_{CR} = 0.130$	
Diversity = 1.3563e+00	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 3.036758e+12	FES = 272100	$\mu_F = 0.757$	$\mu_{CR} = 0.122$	
Diversity = 1.4763e+00	k_t = [4, 3, 3, 2]			
Generation 2730: Best = 3.036758e+12	FES = 273100	$\mu_F = 0.747$	$\mu_{CR} = 0.124$	
Diversity = 1.5776e+00	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 3.036758e+12	FES = 274100	$\mu_F = 0.768$	$\mu_{CR} = 0.128$	
Diversity = 1.5693e+00	k_t = [2, 3, 2, 3]			
Generation 2750: Best = 3.036758e+12	FES = 275100	$\mu_F = 0.772$	$\mu_{CR} = 0.121$	
Diversity = 1.5672e+00	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 3.036758e+12	FES = 276100	$\mu_F = 0.770$	$\mu_{CR} = 0.121$	
Diversity = 1.5094e+00	k_t = [3, 3, 4, 3]			
Generation 2770: Best = 3.036758e+12	FES = 277100	$\mu_F = 0.766$	$\mu_{CR} = 0.122$	
Diversity = 1.5885e+00	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 3.036758e+12	FES = 278100	$\mu_F = 0.774$	$\mu_{CR} = 0.126$	
Diversity = 1.6734e+00	k_t = [2, 3, 3, 3]			
Generation 2790: Best = 3.036758e+12	FES = 279100	$\mu_F = 0.785$	$\mu_{CR} = 0.124$	
Diversity = 1.5084e+00	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 3.036758e+12	FES = 280100	$\mu_F = 0.772$	$\mu_{CR} = 0.128$	
Diversity = 1.4330e+00	k_t = [4, 2, 2, 2]			
Generation 2810: Best = 3.036758e+12	FES = 281100	$\mu_F = 0.776$	$\mu_{CR} = 0.126$	
Diversity = 1.5180e+00	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 3.036758e+12	FES = 282100	$\mu_F = 0.767$	$\mu_{CR} = 0.125$	
Diversity = 1.3108e+00	k_t = [2, 4, 3, 4]			
Generation 2830: Best = 3.036758e+12	FES = 283100	$\mu_F = 0.766$	$\mu_{CR} = 0.121$	
Diversity = 1.3372e+00	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 3.036758e+12	FES = 284100	$\mu_F = 0.777$	$\mu_{CR} = 0.124$	
Diversity = 1.3501e+00	k_t = [2, 4, 4, 4]			
Generation 2850: Best = 3.036758e+12	FES = 285100	$\mu_F = 0.781$	$\mu_{CR} = 0.130$	
Diversity = 1.3040e+00	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 3.036758e+12	FES = 286100	$\mu_F = 0.781$	$\mu_{CR} = 0.130$	
Diversity = 1.3311e+00	k_t = [2, 2, 3, 4]			
Generation 2870: Best = 3.036758e+12	FES = 287100	$\mu_F = 0.777$	$\mu_{CR} = 0.131$	
Diversity = 1.3194e+00	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 3.036758e+12	FES = 288100	$\mu_F = 0.784$	$\mu_{CR} = 0.129$	
Diversity = 1.3731e+00	k_t = [4, 2, 4, 2]			
Generation 2890: Best = 3.036758e+12	FES = 289100	$\mu_F = 0.778$	$\mu_{CR} = 0.125$	
Diversity = 1.4370e+00	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 3.036758e+12	FES = 290100	$\mu_F = 0.789$	$\mu_{CR} = 0.117$	
Diversity = 1.3621e+00	k_t = [4, 2, 4, 3]			
Generation 2910: Best = 3.036758e+12	FES = 291100	$\mu_F = 0.780$	$\mu_{CR} = 0.112$	
Diversity = 1.3543e+00	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 3.036758e+12	FES = 292100	$\mu_F = 0.784$	$\mu_{CR} = 0.118$	

```

Diversity = 1.1779e+00 k_t = [2, 3, 3, 3]
Generation 2930: Best = 3.036758e+12 FES = 293100  $\mu_F$  = 0.779  $\mu_{CR}$  = 0.121
Diversity = 1.3013e+00 k_t = [1, 1, 1, 1]
Generation 2940: Best = 3.036758e+12 FES = 294100  $\mu_F$  = 0.787  $\mu_{CR}$  = 0.126
Diversity = 1.0493e+00 k_t = [3, 4, 3, 4]
Generation 2950: Best = 3.036758e+12 FES = 295100  $\mu_F$  = 0.800  $\mu_{CR}$  = 0.127
Diversity = 1.1492e+00 k_t = [1, 1, 1, 1]
Generation 2960: Best = 3.036758e+12 FES = 296100  $\mu_F$  = 0.805  $\mu_{CR}$  = 0.128
Diversity = 1.1136e+00 k_t = [3, 4, 3, 2]
Generation 2970: Best = 3.036758e+12 FES = 297100  $\mu_F$  = 0.803  $\mu_{CR}$  = 0.133
Diversity = 1.1962e+00 k_t = [1, 1, 1, 1]
Generation 2980: Best = 3.036758e+12 FES = 298100  $\mu_F$  = 0.805  $\mu_{CR}$  = 0.137
Diversity = 1.0684e+00 k_t = [2, 2, 3, 4]
Generation 2990: Best = 3.036758e+12 FES = 299100  $\mu_F$  = 0.800  $\mu_{CR}$  = 0.133
Diversity = 1.0782e+00 k_t = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 3.036758e+12

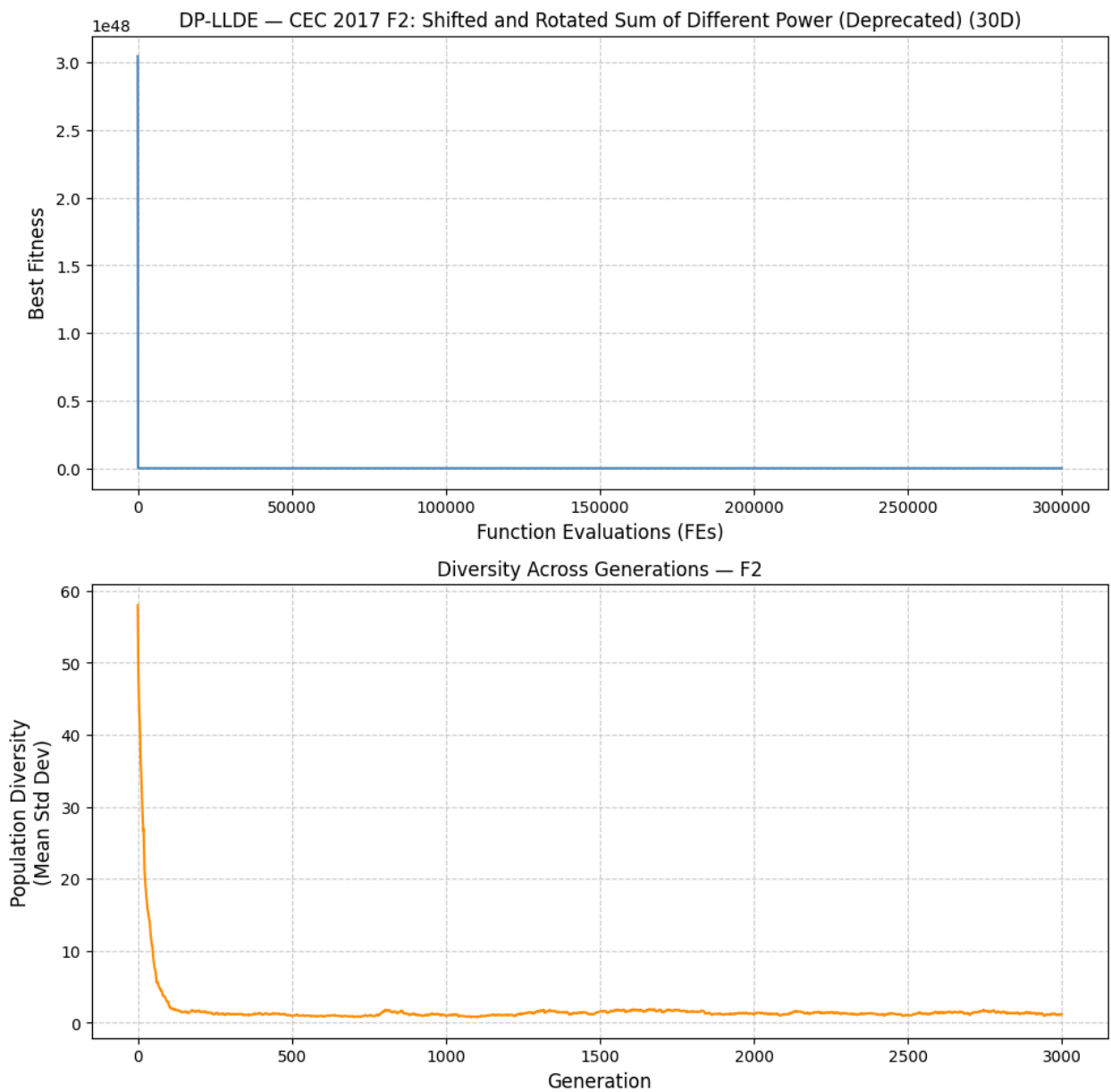
Total Function Evals = 300000

Total Successful Updates = 119452

```

=====
=
Results:
  Best Fitness      : 3.036758e+12
  Global Optimum    : 2.000000e+02
  Error (|best-optimum|) : 3.036758e+12
  Best Solution (5 dim) : [-17.44408274 -4.51973722 -76.33393735 -2.63440069
3.78889821]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F2.png

```
In [ ]: # =====
# F3 – Shifted and Rotated Zakharov
# =====
run_and_plot('F3', make_wrapper(simple.f3), 300.0, 'Shifted and Rotated Zakharov')
```

=====

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DP-LLDE on CEC 2017 F3 – Single Run

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Function : Shifted and Rotated Zakharov (30D)

Bounds : [-100, 100]

Global optimum : 300.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 4.672334e+07

Generation 10: Best = 1.279874e+05 FES = 1100 $\mu_F = 0.565$ $\mu_{CR} = 0.342$

Diversity = 4.5633e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 8.096259e+04 FES = 2100 $\mu_F = 0.595$ $\mu_{CR} = 0.308$

Diversity = 3.6764e+01 $k_t = [4, 4, 2, 4]$

Generation 30: Best = 5.715107e+04 FES = 3100 $\mu_F = 0.627$ $\mu_{CR} = 0.307$

Diversity = 2.3433e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 4.864267e+04 FES = 4100 $\mu_F = 0.650$ $\mu_{CR} = 0.309$

Diversity = 1.6520e+01 $k_t = [2, 4, 4, 2]$

Generation 50: Best = 3.551525e+04 FES = 5100 $\mu_F = 0.655$ $\mu_{CR} = 0.297$

Diversity = 1.2576e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 3.241629e+04 FES = 6100 $\mu_F = 0.665$ $\mu_{CR} = 0.282$

Diversity = 9.2625e+00 $k_t = [3, 3, 4, 4]$

Generation 70: Best = 2.810881e+04 FES = 7100 $\mu_F = 0.694$ $\mu_{CR} = 0.294$

Diversity = 6.9836e+00 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 2.714451e+04 FES = 8100 $\mu_F = 0.716$ $\mu_{CR} = 0.255$

Diversity = 5.6533e+00 $k_t = [2, 3, 4, 4]$

Generation 90: Best = 2.603355e+04 FES = 9100 $\mu_F = 0.736$ $\mu_{CR} = 0.188$

Diversity = 4.7725e+00 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 2.589798e+04 FES = 10100 $\mu_F = 0.741$ $\mu_{CR} = 0.151$

Diversity = 4.0696e+00 $k_t = [4, 4, 3, 2]$

Generation 110: Best = 2.552479e+04 FES = 11100 $\mu_F = 0.750$ $\mu_{CR} = 0.168$

Diversity = 3.4733e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 2.486184e+04 FES = 12100 $\mu_F = 0.762$ $\mu_{CR} = 0.156$

Diversity = 2.7927e+00 $k_t = [3, 2, 4, 3]$

Generation 130: Best = 2.411188e+04 FES = 13100 $\mu_F = 0.786$ $\mu_{CR} = 0.117$

Diversity = 2.6117e+00 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 2.335790e+04 FES = 14100 $\mu_F = 0.791$ $\mu_{CR} = 0.088$

Diversity = 2.5072e+00 $k_t = [3, 2, 4, 2]$

Generation 150: Best = 2.227362e+04 FES = 15100 $\mu_F = 0.798$ $\mu_{CR} = 0.081$

Diversity = 2.5801e+00 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 2.120139e+04 FES = 16100 $\mu_F = 0.812$ $\mu_{CR} = 0.087$

Diversity = 2.7360e+00 $k_t = [3, 4, 2, 4]$

Generation 170: Best = 2.068451e+04 FES = 17100 $\mu_F = 0.814$ $\mu_{CR} = 0.093$

Diversity = 2.8800e+00 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 1.992780e+04 FES = 18100 $\mu_F = 0.826$ $\mu_{CR} = 0.091$

Diversity = 2.8637e+00 $k_t = [3, 3, 3, 4]$

Generation 190: Best = 1.930326e+04 FES = 19100 $\mu_F = 0.808$ $\mu_{CR} = 0.096$

Diversity = 2.9166e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 1.824466e+04 FES = 20100 $\mu_F = 0.801$ $\mu_{CR} = 0.108$

Diversity = 2.7318e+00 $k_t = [2, 2, 4, 2]$

Generation 210: Best = 1.795331e+04 FES = 21100 $\mu_F = 0.788$ $\mu_{CR} = 0.107$

Diversity = 2.8103e+00 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 1.741784e+04 FES = 22100 $\mu_F = 0.777$ $\mu_{CR} = 0.091$

Diversity = 2.4202e+00 $k_t = [2, 3, 2, 2]$

Generation 230: Best = 1.627748e+04	FES = 23100	$\mu_F = 0.768$	$\mu_{CR} = 0.096$
Diversity = 2.4036e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 1.583210e+04	FES = 24100	$\mu_F = 0.775$	$\mu_{CR} = 0.097$
Diversity = 2.0665e+00	$k_t = [4, 4, 4, 3]$		
Generation 250: Best = 1.548760e+04	FES = 25100	$\mu_F = 0.782$	$\mu_{CR} = 0.106$
Diversity = 2.1232e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.487695e+04	FES = 26100	$\mu_F = 0.795$	$\mu_{CR} = 0.153$
Diversity = 2.1138e+00	$k_t = [4, 4, 3, 4]$		
Generation 270: Best = 1.465187e+04	FES = 27100	$\mu_F = 0.804$	$\mu_{CR} = 0.130$
Diversity = 2.1764e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.454108e+04	FES = 28100	$\mu_F = 0.799$	$\mu_{CR} = 0.114$
Diversity = 1.9974e+00	$k_t = [4, 2, 4, 2]$		
Generation 290: Best = 1.367677e+04	FES = 29100	$\mu_F = 0.819$	$\mu_{CR} = 0.126$
Diversity = 2.1284e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.300678e+04	FES = 30100	$\mu_F = 0.820$	$\mu_{CR} = 0.116$
Diversity = 2.2021e+00	$k_t = [4, 3, 4, 3]$		
Generation 310: Best = 1.167897e+04	FES = 31100	$\mu_F = 0.812$	$\mu_{CR} = 0.108$
Diversity = 2.4479e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.149013e+04	FES = 32100	$\mu_F = 0.796$	$\mu_{CR} = 0.113$
Diversity = 2.6125e+00	$k_t = [4, 2, 4, 2]$		
Generation 330: Best = 1.090496e+04	FES = 33100	$\mu_F = 0.789$	$\mu_{CR} = 0.110$
Diversity = 2.4698e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.048328e+04	FES = 34100	$\mu_F = 0.779$	$\mu_{CR} = 0.106$
Diversity = 2.3066e+00	$k_t = [3, 2, 3, 4]$		
Generation 350: Best = 9.842152e+03	FES = 35100	$\mu_F = 0.789$	$\mu_{CR} = 0.103$
Diversity = 2.0328e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 9.620441e+03	FES = 36100	$\mu_F = 0.783$	$\mu_{CR} = 0.103$
Diversity = 1.9100e+00	$k_t = [3, 3, 2, 2]$		
Generation 370: Best = 9.238378e+03	FES = 37100	$\mu_F = 0.787$	$\mu_{CR} = 0.116$
Diversity = 1.9145e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 8.905691e+03	FES = 38100	$\mu_F = 0.786$	$\mu_{CR} = 0.125$
Diversity = 1.8590e+00	$k_t = [4, 4, 4, 4]$		
Generation 390: Best = 8.576247e+03	FES = 39100	$\mu_F = 0.795$	$\mu_{CR} = 0.107$
Diversity = 1.8258e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 8.351290e+03	FES = 40100	$\mu_F = 0.807$	$\mu_{CR} = 0.106$
Diversity = 1.8310e+00	$k_t = [4, 2, 4, 4]$		
Generation 410: Best = 7.902389e+03	FES = 41100	$\mu_F = 0.809$	$\mu_{CR} = 0.097$
Diversity = 1.5808e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 7.645403e+03	FES = 42100	$\mu_F = 0.806$	$\mu_{CR} = 0.119$
Diversity = 1.6563e+00	$k_t = [4, 3, 2, 4]$		
Generation 430: Best = 7.060835e+03	FES = 43100	$\mu_F = 0.800$	$\mu_{CR} = 0.122$
Diversity = 1.5092e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 6.931741e+03	FES = 44100	$\mu_F = 0.821$	$\mu_{CR} = 0.131$
Diversity = 1.6655e+00	$k_t = [3, 3, 4, 3]$		
Generation 450: Best = 6.601166e+03	FES = 45100	$\mu_F = 0.812$	$\mu_{CR} = 0.127$
Diversity = 1.7117e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 6.046345e+03	FES = 46100	$\mu_F = 0.823$	$\mu_{CR} = 0.139$
Diversity = 1.7522e+00	$k_t = [2, 3, 2, 4]$		
Generation 470: Best = 5.830474e+03	FES = 47100	$\mu_F = 0.825$	$\mu_{CR} = 0.129$
Diversity = 1.7929e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 5.613665e+03	FES = 48100	$\mu_F = 0.831$	$\mu_{CR} = 0.138$
Diversity = 1.9285e+00	$k_t = [2, 2, 3, 3]$		
Generation 490: Best = 5.449976e+03	FES = 49100	$\mu_F = 0.820$	$\mu_{CR} = 0.125$
Diversity = 1.9026e+00	$k_t = [1, 1, 1, 1]$		

Generation 500:	Best = 5.050500e+03	FES = 50100	$\mu_F = 0.811$	$\mu_{CR} = 0.121$
Diversity = 1.7600e+00	$k_t = [2, 3, 3, 2]$			
Generation 510:	Best = 4.900766e+03	FES = 51100	$\mu_F = 0.795$	$\mu_{CR} = 0.122$
Diversity = 1.7851e+00	$k_t = [1, 1, 1, 1]$			
Generation 520:	Best = 4.742341e+03	FES = 52100	$\mu_F = 0.800$	$\mu_{CR} = 0.112$
Diversity = 1.7373e+00	$k_t = [3, 3, 2, 2]$			
Generation 530:	Best = 4.351779e+03	FES = 53100	$\mu_F = 0.788$	$\mu_{CR} = 0.109$
Diversity = 1.5986e+00	$k_t = [1, 1, 1, 1]$			
Generation 540:	Best = 4.351779e+03	FES = 54100	$\mu_F = 0.792$	$\mu_{CR} = 0.106$
Diversity = 1.4454e+00	$k_t = [3, 4, 4, 4]$			
Generation 550:	Best = 4.102629e+03	FES = 55100	$\mu_F = 0.782$	$\mu_{CR} = 0.121$
Diversity = 1.2843e+00	$k_t = [1, 1, 1, 1]$			
Generation 560:	Best = 4.012022e+03	FES = 56100	$\mu_F = 0.786$	$\mu_{CR} = 0.149$
Diversity = 1.2900e+00	$k_t = [2, 4, 2, 2]$			
Generation 570:	Best = 3.844381e+03	FES = 57100	$\mu_F = 0.778$	$\mu_{CR} = 0.161$
Diversity = 1.2636e+00	$k_t = [1, 1, 1, 1]$			
Generation 580:	Best = 3.694156e+03	FES = 58100	$\mu_F = 0.773$	$\mu_{CR} = 0.148$
Diversity = 1.3038e+00	$k_t = [4, 3, 2, 2]$			
Generation 590:	Best = 3.616622e+03	FES = 59100	$\mu_F = 0.784$	$\mu_{CR} = 0.150$
Diversity = 1.2226e+00	$k_t = [1, 1, 1, 1]$			
Generation 600:	Best = 3.517490e+03	FES = 60100	$\mu_F = 0.790$	$\mu_{CR} = 0.117$
Diversity = 1.1550e+00	$k_t = [4, 3, 4, 3]$			
Generation 610:	Best = 3.315899e+03	FES = 61100	$\mu_F = 0.785$	$\mu_{CR} = 0.118$
Diversity = 1.1575e+00	$k_t = [1, 1, 1, 1]$			
Generation 620:	Best = 3.296554e+03	FES = 62100	$\mu_F = 0.780$	$\mu_{CR} = 0.113$
Diversity = 1.0973e+00	$k_t = [3, 3, 3, 4]$			
Generation 630:	Best = 3.195650e+03	FES = 63100	$\mu_F = 0.764$	$\mu_{CR} = 0.112$
Diversity = 9.4186e-01	$k_t = [1, 1, 1, 1]$			
Generation 640:	Best = 3.113533e+03	FES = 64100	$\mu_F = 0.760$	$\mu_{CR} = 0.112$
Diversity = 8.8951e-01	$k_t = [3, 3, 4, 3]$			
Generation 650:	Best = 3.053777e+03	FES = 65100	$\mu_F = 0.768$	$\mu_{CR} = 0.107$
Diversity = 8.8965e-01	$k_t = [1, 1, 1, 1]$			
Generation 660:	Best = 2.923059e+03	FES = 66100	$\mu_F = 0.766$	$\mu_{CR} = 0.119$
Diversity = 9.9428e-01	$k_t = [3, 3, 4, 2]$			
Generation 670:	Best = 2.797427e+03	FES = 67100	$\mu_F = 0.786$	$\mu_{CR} = 0.113$
Diversity = 1.0816e+00	$k_t = [1, 1, 1, 1]$			
Generation 680:	Best = 2.685537e+03	FES = 68100	$\mu_F = 0.776$	$\mu_{CR} = 0.110$
Diversity = 1.0514e+00	$k_t = [4, 4, 3, 4]$			
Generation 690:	Best = 2.621862e+03	FES = 69100	$\mu_F = 0.784$	$\mu_{CR} = 0.112$
Diversity = 1.1201e+00	$k_t = [1, 1, 1, 1]$			
Generation 700:	Best = 2.535189e+03	FES = 70100	$\mu_F = 0.781$	$\mu_{CR} = 0.103$
Diversity = 1.0621e+00	$k_t = [4, 4, 2, 4]$			
Generation 710:	Best = 2.461515e+03	FES = 71100	$\mu_F = 0.776$	$\mu_{CR} = 0.094$
Diversity = 9.6661e-01	$k_t = [1, 1, 1, 1]$			
Generation 720:	Best = 2.430838e+03	FES = 72100	$\mu_F = 0.777$	$\mu_{CR} = 0.105$
Diversity = 8.2040e-01	$k_t = [2, 2, 3, 3]$			
Generation 730:	Best = 2.355400e+03	FES = 73100	$\mu_F = 0.767$	$\mu_{CR} = 0.112$
Diversity = 8.3127e-01	$k_t = [1, 1, 1, 1]$			
Generation 740:	Best = 2.309903e+03	FES = 74100	$\mu_F = 0.764$	$\mu_{CR} = 0.112$
Diversity = 8.3510e-01	$k_t = [2, 4, 4, 2]$			
Generation 750:	Best = 2.270013e+03	FES = 75100	$\mu_F = 0.777$	$\mu_{CR} = 0.136$
Diversity = 8.8153e-01	$k_t = [1, 1, 1, 1]$			
Generation 760:	Best = 2.195057e+03	FES = 76100	$\mu_F = 0.793$	$\mu_{CR} = 0.106$
Diversity = 8.7364e-01	$k_t = [4, 3, 2, 2]$			

Generation 770: Best = 2.132826e+03	FES = 77100	$\mu_F = 0.781$	$\mu_{CR} = 0.123$
Diversity = 8.7998e-01	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 2.078351e+03	FES = 78100	$\mu_F = 0.782$	$\mu_{CR} = 0.113$
Diversity = 9.6271e-01	$k_t = [4, 2, 2, 3]$		
Generation 790: Best = 2.048849e+03	FES = 79100	$\mu_F = 0.782$	$\mu_{CR} = 0.124$
Diversity = 9.6904e-01	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 1.949766e+03	FES = 80100	$\mu_F = 0.780$	$\mu_{CR} = 0.118$
Diversity = 9.1088e-01	$k_t = [3, 4, 4, 4]$		
Generation 810: Best = 1.888711e+03	FES = 81100	$\mu_F = 0.772$	$\mu_{CR} = 0.095$
Diversity = 9.3744e-01	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 1.818007e+03	FES = 82100	$\mu_F = 0.777$	$\mu_{CR} = 0.097$
Diversity = 1.0423e+00	$k_t = [4, 3, 3, 4]$		
Generation 830: Best = 1.784938e+03	FES = 83100	$\mu_F = 0.780$	$\mu_{CR} = 0.082$
Diversity = 8.8817e-01	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 1.768959e+03	FES = 84100	$\mu_F = 0.770$	$\mu_{CR} = 0.102$
Diversity = 7.8830e-01	$k_t = [2, 3, 2, 4]$		
Generation 850: Best = 1.751363e+03	FES = 85100	$\mu_F = 0.760$	$\mu_{CR} = 0.110$
Diversity = 7.4720e-01	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 1.724813e+03	FES = 86100	$\mu_F = 0.736$	$\mu_{CR} = 0.092$
Diversity = 7.1449e-01	$k_t = [3, 3, 2, 2]$		
Generation 870: Best = 1.724813e+03	FES = 87100	$\mu_F = 0.708$	$\mu_{CR} = 0.082$
Diversity = 6.8182e-01	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 1.705040e+03	FES = 88100	$\mu_F = 0.717$	$\mu_{CR} = 0.081$
Diversity = 6.5739e-01	$k_t = [2, 4, 4, 3]$		
Generation 890: Best = 1.671847e+03	FES = 89100	$\mu_F = 0.708$	$\mu_{CR} = 0.094$
Diversity = 6.4446e-01	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 1.653276e+03	FES = 90100	$\mu_F = 0.723$	$\mu_{CR} = 0.102$
Diversity = 6.6616e-01	$k_t = [3, 4, 2, 4]$		
Generation 910: Best = 1.619067e+03	FES = 91100	$\mu_F = 0.714$	$\mu_{CR} = 0.086$
Diversity = 6.1762e-01	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 1.604543e+03	FES = 92100	$\mu_F = 0.712$	$\mu_{CR} = 0.090$
Diversity = 6.2371e-01	$k_t = [3, 3, 3, 2]$		
Generation 930: Best = 1.597733e+03	FES = 93100	$\mu_F = 0.692$	$\mu_{CR} = 0.102$
Diversity = 6.0318e-01	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 1.571289e+03	FES = 94100	$\mu_F = 0.695$	$\mu_{CR} = 0.120$
Diversity = 5.6701e-01	$k_t = [2, 2, 2, 4]$		
Generation 950: Best = 1.568698e+03	FES = 95100	$\mu_F = 0.694$	$\mu_{CR} = 0.113$
Diversity = 5.5024e-01	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 1.536242e+03	FES = 96100	$\mu_F = 0.693$	$\mu_{CR} = 0.108$
Diversity = 5.3849e-01	$k_t = [4, 3, 3, 4]$		
Generation 970: Best = 1.529771e+03	FES = 97100	$\mu_F = 0.678$	$\mu_{CR} = 0.109$
Diversity = 4.7249e-01	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 1.512830e+03	FES = 98100	$\mu_F = 0.706$	$\mu_{CR} = 0.124$
Diversity = 4.4842e-01	$k_t = [3, 3, 4, 4]$		
Generation 990: Best = 1.511030e+03	FES = 99100	$\mu_F = 0.710$	$\mu_{CR} = 0.124$
Diversity = 4.4112e-01	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 1.505877e+03	FES = 100100	$\mu_F = 0.680$	$\mu_{CR} = 0.115$
Diversity = 4.3563e-01	$k_t = [3, 4, 2, 3]$		
Generation 1010: Best = 1.495551e+03	FES = 101100	$\mu_F = 0.669$	$\mu_{CR} = 0.103$
Diversity = 4.4445e-01	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 1.461129e+03	FES = 102100	$\mu_F = 0.684$	$\mu_{CR} = 0.105$
Diversity = 4.4082e-01	$k_t = [3, 4, 2, 2]$		
Generation 1030: Best = 1.451053e+03	FES = 103100	$\mu_F = 0.686$	$\mu_{CR} = 0.122$
Diversity = 4.3672e-01	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 1.451053e+03 FES = 104100 μ_F = 0.693 μ_{CR} = 0.105
 Diversity = 4.3244e-01 k_t = [3, 4, 4, 3]
 Generation 1050: Best = 1.451053e+03 FES = 105100 μ_F = 0.729 μ_{CR} = 0.114
 Diversity = 4.2872e-01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 1.441004e+03 FES = 106100 μ_F = 0.711 μ_{CR} = 0.123
 Diversity = 4.5489e-01 k_t = [3, 3, 4, 2]
 Generation 1070: Best = 1.427066e+03 FES = 107100 μ_F = 0.716 μ_{CR} = 0.117
 Diversity = 4.7735e-01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 1.390375e+03 FES = 108100 μ_F = 0.707 μ_{CR} = 0.124
 Diversity = 5.1510e-01 k_t = [4, 3, 3, 2]
 Generation 1090: Best = 1.371815e+03 FES = 109100 μ_F = 0.703 μ_{CR} = 0.112
 Diversity = 5.5413e-01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.371815e+03 FES = 110100 μ_F = 0.708 μ_{CR} = 0.097
 Diversity = 5.1637e-01 k_t = [2, 4, 4, 4]
 Generation 1110: Best = 1.336629e+03 FES = 111100 μ_F = 0.688 μ_{CR} = 0.106
 Diversity = 4.7406e-01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.336345e+03 FES = 112100 μ_F = 0.678 μ_{CR} = 0.103
 Diversity = 4.3501e-01 k_t = [2, 2, 2, 4]
 Generation 1130: Best = 1.331406e+03 FES = 113100 μ_F = 0.653 μ_{CR} = 0.100
 Diversity = 4.2173e-01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.308354e+03 FES = 114100 μ_F = 0.656 μ_{CR} = 0.111
 Diversity = 3.9398e-01 k_t = [4, 3, 4, 3]
 Generation 1150: Best = 1.308354e+03 FES = 115100 μ_F = 0.641 μ_{CR} = 0.109
 Diversity = 3.8633e-01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.307025e+03 FES = 116100 μ_F = 0.630 μ_{CR} = 0.110
 Diversity = 3.9247e-01 k_t = [4, 4, 2, 2]
 Generation 1170: Best = 1.287138e+03 FES = 117100 μ_F = 0.646 μ_{CR} = 0.099
 Diversity = 3.6745e-01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.285258e+03 FES = 118100 μ_F = 0.665 μ_{CR} = 0.102
 Diversity = 3.5012e-01 k_t = [2, 4, 4, 2]
 Generation 1190: Best = 1.285138e+03 FES = 119100 μ_F = 0.673 μ_{CR} = 0.119
 Diversity = 3.4077e-01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.275026e+03 FES = 120100 μ_F = 0.677 μ_{CR} = 0.135
 Diversity = 3.2579e-01 k_t = [3, 3, 3, 4]
 Generation 1210: Best = 1.249419e+03 FES = 121100 μ_F = 0.706 μ_{CR} = 0.121
 Diversity = 3.2935e-01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.249419e+03 FES = 122100 μ_F = 0.727 μ_{CR} = 0.115
 Diversity = 3.2831e-01 k_t = [4, 4, 3, 4]
 Generation 1230: Best = 1.249014e+03 FES = 123100 μ_F = 0.739 μ_{CR} = 0.105
 Diversity = 3.0222e-01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.231662e+03 FES = 124100 μ_F = 0.719 μ_{CR} = 0.103
 Diversity = 2.9539e-01 k_t = [3, 4, 2, 2]
 Generation 1250: Best = 1.227658e+03 FES = 125100 μ_F = 0.732 μ_{CR} = 0.104
 Diversity = 3.0290e-01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.216918e+03 FES = 126100 μ_F = 0.728 μ_{CR} = 0.105
 Diversity = 3.3786e-01 k_t = [4, 2, 4, 2]
 Generation 1270: Best = 1.212814e+03 FES = 127100 μ_F = 0.726 μ_{CR} = 0.091
 Diversity = 3.3591e-01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.199526e+03 FES = 128100 μ_F = 0.726 μ_{CR} = 0.086
 Diversity = 3.5178e-01 k_t = [4, 4, 3, 2]
 Generation 1290: Best = 1.194737e+03 FES = 129100 μ_F = 0.725 μ_{CR} = 0.100
 Diversity = 3.7352e-01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.187909e+03 FES = 130100 μ_F = 0.721 μ_{CR} = 0.100
 Diversity = 3.4528e-01 k_t = [4, 2, 2, 4]

Generation 1310: Best = 1.181989e+03 FES = 131100 μ_F = 0.701 μ_{CR} = 0.100
 Diversity = 3.4336e-01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.179186e+03 FES = 132100 μ_F = 0.708 μ_{CR} = 0.103
 Diversity = 3.0709e-01 k_t = [2, 3, 3, 4]
 Generation 1330: Best = 1.172104e+03 FES = 133100 μ_F = 0.720 μ_{CR} = 0.105
 Diversity = 2.9731e-01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.172104e+03 FES = 134100 μ_F = 0.696 μ_{CR} = 0.099
 Diversity = 2.8601e-01 k_t = [4, 3, 2, 4]
 Generation 1350: Best = 1.165884e+03 FES = 135100 μ_F = 0.698 μ_{CR} = 0.104
 Diversity = 2.7847e-01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.160161e+03 FES = 136100 μ_F = 0.669 μ_{CR} = 0.095
 Diversity = 2.6130e-01 k_t = [3, 4, 4, 3]
 Generation 1370: Best = 1.157111e+03 FES = 137100 μ_F = 0.689 μ_{CR} = 0.095
 Diversity = 2.6479e-01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.147774e+03 FES = 138100 μ_F = 0.695 μ_{CR} = 0.095
 Diversity = 2.6394e-01 k_t = [2, 3, 2, 4]
 Generation 1390: Best = 1.147051e+03 FES = 139100 μ_F = 0.694 μ_{CR} = 0.095
 Diversity = 2.5372e-01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.145946e+03 FES = 140100 μ_F = 0.695 μ_{CR} = 0.091
 Diversity = 2.4174e-01 k_t = [3, 3, 3, 4]
 Generation 1410: Best = 1.142847e+03 FES = 141100 μ_F = 0.714 μ_{CR} = 0.089
 Diversity = 2.3211e-01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.136392e+03 FES = 142100 μ_F = 0.718 μ_{CR} = 0.093
 Diversity = 2.3653e-01 k_t = [2, 3, 3, 2]
 Generation 1430: Best = 1.134541e+03 FES = 143100 μ_F = 0.709 μ_{CR} = 0.099
 Diversity = 2.4583e-01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.132179e+03 FES = 144100 μ_F = 0.673 μ_{CR} = 0.107
 Diversity = 2.3525e-01 k_t = [3, 2, 3, 2]
 Generation 1450: Best = 1.126167e+03 FES = 145100 μ_F = 0.676 μ_{CR} = 0.106
 Diversity = 2.3424e-01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.123825e+03 FES = 146100 μ_F = 0.682 μ_{CR} = 0.099
 Diversity = 2.2134e-01 k_t = [2, 3, 2, 4]
 Generation 1470: Best = 1.121198e+03 FES = 147100 μ_F = 0.700 μ_{CR} = 0.113
 Diversity = 2.0604e-01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.118768e+03 FES = 148100 μ_F = 0.705 μ_{CR} = 0.107
 Diversity = 2.0884e-01 k_t = [2, 3, 2, 2]
 Generation 1490: Best = 1.118220e+03 FES = 149100 μ_F = 0.694 μ_{CR} = 0.114
 Diversity = 2.0023e-01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.116991e+03 FES = 150100 μ_F = 0.681 μ_{CR} = 0.115
 Diversity = 1.7277e-01 k_t = [4, 3, 4, 4]
 Generation 1510: Best = 1.113075e+03 FES = 151100 μ_F = 0.681 μ_{CR} = 0.108
 Diversity = 1.7143e-01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.110433e+03 FES = 152100 μ_F = 0.686 μ_{CR} = 0.102
 Diversity = 1.6097e-01 k_t = [4, 3, 2, 4]
 Generation 1530: Best = 1.109120e+03 FES = 153100 μ_F = 0.684 μ_{CR} = 0.095
 Diversity = 1.4205e-01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.107578e+03 FES = 154100 μ_F = 0.700 μ_{CR} = 0.100
 Diversity = 1.3918e-01 k_t = [3, 2, 4, 4]
 Generation 1550: Best = 1.105042e+03 FES = 155100 μ_F = 0.704 μ_{CR} = 0.097
 Diversity = 1.4971e-01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.102720e+03 FES = 156100 μ_F = 0.727 μ_{CR} = 0.102
 Diversity = 1.5334e-01 k_t = [3, 4, 3, 3]
 Generation 1570: Best = 1.099351e+03 FES = 157100 μ_F = 0.731 μ_{CR} = 0.106
 Diversity = 1.6144e-01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.098005e+03 FES = 158100 μ_F = 0.729 μ_{CR} = 0.104
 Diversity = 1.4877e-01 k_t = [2, 2, 4, 3]
 Generation 1590: Best = 1.097015e+03 FES = 159100 μ_F = 0.731 μ_{CR} = 0.103
 Diversity = 1.5213e-01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.091401e+03 FES = 160100 μ_F = 0.714 μ_{CR} = 0.100
 Diversity = 1.4950e-01 k_t = [3, 2, 2, 4]
 Generation 1610: Best = 1.091288e+03 FES = 161100 μ_F = 0.712 μ_{CR} = 0.101
 Diversity = 1.5124e-01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.088683e+03 FES = 162100 μ_F = 0.733 μ_{CR} = 0.104
 Diversity = 1.5029e-01 k_t = [4, 4, 4, 3]
 Generation 1630: Best = 1.088606e+03 FES = 163100 μ_F = 0.726 μ_{CR} = 0.107
 Diversity = 1.6017e-01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.084391e+03 FES = 164100 μ_F = 0.748 μ_{CR} = 0.104
 Diversity = 1.5736e-01 k_t = [2, 3, 3, 4]
 Generation 1650: Best = 1.084391e+03 FES = 165100 μ_F = 0.749 μ_{CR} = 0.099
 Diversity = 1.5512e-01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.081007e+03 FES = 166100 μ_F = 0.773 μ_{CR} = 0.098
 Diversity = 1.5398e-01 k_t = [4, 3, 2, 4]
 Generation 1670: Best = 1.080268e+03 FES = 167100 μ_F = 0.784 μ_{CR} = 0.102
 Diversity = 1.5387e-01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.080012e+03 FES = 168100 μ_F = 0.752 μ_{CR} = 0.105
 Diversity = 1.6003e-01 k_t = [2, 2, 2, 3]
 Generation 1690: Best = 1.078055e+03 FES = 169100 μ_F = 0.748 μ_{CR} = 0.098
 Diversity = 1.6072e-01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.076402e+03 FES = 170100 μ_F = 0.733 μ_{CR} = 0.105
 Diversity = 1.4705e-01 k_t = [3, 4, 2, 2]
 Generation 1710: Best = 1.075665e+03 FES = 171100 μ_F = 0.744 μ_{CR} = 0.097
 Diversity = 1.4444e-01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.075447e+03 FES = 172100 μ_F = 0.753 μ_{CR} = 0.091
 Diversity = 1.3130e-01 k_t = [4, 2, 3, 4]
 Generation 1730: Best = 1.073494e+03 FES = 173100 μ_F = 0.738 μ_{CR} = 0.102
 Diversity = 1.1623e-01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.071754e+03 FES = 174100 μ_F = 0.749 μ_{CR} = 0.102
 Diversity = 1.2367e-01 k_t = [2, 2, 2, 2]
 Generation 1750: Best = 1.070132e+03 FES = 175100 μ_F = 0.751 μ_{CR} = 0.111
 Diversity = 1.2912e-01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.069312e+03 FES = 176100 μ_F = 0.772 μ_{CR} = 0.119
 Diversity = 1.2568e-01 k_t = [4, 3, 3, 4]
 Generation 1770: Best = 1.068606e+03 FES = 177100 μ_F = 0.745 μ_{CR} = 0.105
 Diversity = 1.1811e-01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.067072e+03 FES = 178100 μ_F = 0.746 μ_{CR} = 0.114
 Diversity = 1.2011e-01 k_t = [2, 2, 4, 4]
 Generation 1790: Best = 1.066501e+03 FES = 179100 μ_F = 0.764 μ_{CR} = 0.110
 Diversity = 1.1416e-01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 1.065780e+03 FES = 180100 μ_F = 0.753 μ_{CR} = 0.098
 Diversity = 1.1298e-01 k_t = [4, 3, 4, 2]
 Generation 1810: Best = 1.065304e+03 FES = 181100 μ_F = 0.767 μ_{CR} = 0.099
 Diversity = 1.1802e-01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 1.064748e+03 FES = 182100 μ_F = 0.771 μ_{CR} = 0.097
 Diversity = 1.0922e-01 k_t = [4, 3, 2, 3]
 Generation 1830: Best = 1.063719e+03 FES = 183100 μ_F = 0.763 μ_{CR} = 0.109
 Diversity = 1.0724e-01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 1.062480e+03 FES = 184100 μ_F = 0.773 μ_{CR} = 0.098
 Diversity = 1.0834e-01 k_t = [2, 4, 3, 2]

Generation 1850: Best = 1.060912e+03	FES = 185100	$\mu_F = 0.767$	$\mu_{CR} = 0.107$
Diversity = 1.1440e-01	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 1.060644e+03	FES = 186100	$\mu_F = 0.755$	$\mu_{CR} = 0.112$
Diversity = 1.1511e-01	k_t = [4, 4, 4, 3]		
Generation 1870: Best = 1.060056e+03	FES = 187100	$\mu_F = 0.744$	$\mu_{CR} = 0.108$
Diversity = 1.2266e-01	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 1.059308e+03	FES = 188100	$\mu_F = 0.764$	$\mu_{CR} = 0.101$
Diversity = 1.1702e-01	k_t = [2, 2, 3, 3]		
Generation 1890: Best = 1.057998e+03	FES = 189100	$\mu_F = 0.754$	$\mu_{CR} = 0.105$
Diversity = 1.1971e-01	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 1.056130e+03	FES = 190100	$\mu_F = 0.745$	$\mu_{CR} = 0.111$
Diversity = 1.1682e-01	k_t = [2, 3, 4, 2]		
Generation 1910: Best = 1.053635e+03	FES = 191100	$\mu_F = 0.754$	$\mu_{CR} = 0.112$
Diversity = 1.2648e-01	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 1.052763e+03	FES = 192100	$\mu_F = 0.753$	$\mu_{CR} = 0.102$
Diversity = 1.2772e-01	k_t = [4, 3, 2, 3]		
Generation 1930: Best = 1.052501e+03	FES = 193100	$\mu_F = 0.755$	$\mu_{CR} = 0.110$
Diversity = 1.2757e-01	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 1.050878e+03	FES = 194100	$\mu_F = 0.759$	$\mu_{CR} = 0.116$
Diversity = 1.2601e-01	k_t = [2, 3, 3, 3]		
Generation 1950: Best = 1.050878e+03	FES = 195100	$\mu_F = 0.753$	$\mu_{CR} = 0.124$
Diversity = 1.2303e-01	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 1.050637e+03	FES = 196100	$\mu_F = 0.746$	$\mu_{CR} = 0.108$
Diversity = 1.2005e-01	k_t = [4, 3, 3, 2]		
Generation 1970: Best = 1.049631e+03	FES = 197100	$\mu_F = 0.742$	$\mu_{CR} = 0.108$
Diversity = 1.1559e-01	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 1.048703e+03	FES = 198100	$\mu_F = 0.746$	$\mu_{CR} = 0.111$
Diversity = 1.2133e-01	k_t = [2, 2, 4, 2]		
Generation 1990: Best = 1.046652e+03	FES = 199100	$\mu_F = 0.749$	$\mu_{CR} = 0.114$
Diversity = 1.3128e-01	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 1.045822e+03	FES = 200100	$\mu_F = 0.761$	$\mu_{CR} = 0.125$
Diversity = 1.3752e-01	k_t = [2, 4, 3, 2]		
Generation 2010: Best = 1.045292e+03	FES = 201100	$\mu_F = 0.756$	$\mu_{CR} = 0.126$
Diversity = 1.3810e-01	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 1.044069e+03	FES = 202100	$\mu_F = 0.731$	$\mu_{CR} = 0.123$
Diversity = 1.3656e-01	k_t = [4, 2, 4, 2]		
Generation 2030: Best = 1.043185e+03	FES = 203100	$\mu_F = 0.701$	$\mu_{CR} = 0.125$
Diversity = 1.3664e-01	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 1.042623e+03	FES = 204100	$\mu_F = 0.707$	$\mu_{CR} = 0.115$
Diversity = 1.3353e-01	k_t = [3, 4, 3, 2]		
Generation 2050: Best = 1.041531e+03	FES = 205100	$\mu_F = 0.702$	$\mu_{CR} = 0.104$
Diversity = 1.2447e-01	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 1.039644e+03	FES = 206100	$\mu_F = 0.704$	$\mu_{CR} = 0.115$
Diversity = 1.1763e-01	k_t = [4, 2, 3, 3]		
Generation 2070: Best = 1.039391e+03	FES = 207100	$\mu_F = 0.711$	$\mu_{CR} = 0.115$
Diversity = 1.2499e-01	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 1.037367e+03	FES = 208100	$\mu_F = 0.719$	$\mu_{CR} = 0.112$
Diversity = 1.2177e-01	k_t = [3, 4, 4, 3]		
Generation 2090: Best = 1.036224e+03	FES = 209100	$\mu_F = 0.702$	$\mu_{CR} = 0.110$
Diversity = 1.2215e-01	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 1.035063e+03	FES = 210100	$\mu_F = 0.715$	$\mu_{CR} = 0.113$
Diversity = 1.2546e-01	k_t = [2, 2, 2, 4]		
Generation 2110: Best = 1.034863e+03	FES = 211100	$\mu_F = 0.716$	$\mu_{CR} = 0.111$
Diversity = 1.1461e-01	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 1.034331e+03 FES = 212100 μ_F = 0.727 μ_{CR} = 0.115
 Diversity = 1.1073e-01 k_t = [4, 3, 4, 4]
 Generation 2130: Best = 1.033027e+03 FES = 213100 μ_F = 0.723 μ_{CR} = 0.122
 Diversity = 9.7534e-02 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.032498e+03 FES = 214100 μ_F = 0.723 μ_{CR} = 0.117
 Diversity = 9.9259e-02 k_t = [4, 2, 4, 2]
 Generation 2150: Best = 1.031498e+03 FES = 215100 μ_F = 0.734 μ_{CR} = 0.106
 Diversity = 1.0478e-01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.029431e+03 FES = 216100 μ_F = 0.735 μ_{CR} = 0.107
 Diversity = 1.1721e-01 k_t = [3, 3, 4, 3]
 Generation 2170: Best = 1.028196e+03 FES = 217100 μ_F = 0.742 μ_{CR} = 0.111
 Diversity = 1.3270e-01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.027983e+03 FES = 218100 μ_F = 0.757 μ_{CR} = 0.117
 Diversity = 1.3131e-01 k_t = [3, 2, 2, 3]
 Generation 2190: Best = 1.027654e+03 FES = 219100 μ_F = 0.749 μ_{CR} = 0.122
 Diversity = 1.3539e-01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 1.026655e+03 FES = 220100 μ_F = 0.728 μ_{CR} = 0.120
 Diversity = 1.4253e-01 k_t = [3, 3, 2, 2]
 Generation 2210: Best = 1.026138e+03 FES = 221100 μ_F = 0.699 μ_{CR} = 0.123
 Diversity = 1.3687e-01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 1.024342e+03 FES = 222100 μ_F = 0.691 μ_{CR} = 0.127
 Diversity = 1.3145e-01 k_t = [3, 3, 4, 2]
 Generation 2230: Best = 1.024026e+03 FES = 223100 μ_F = 0.691 μ_{CR} = 0.147
 Diversity = 1.2851e-01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.022626e+03 FES = 224100 μ_F = 0.701 μ_{CR} = 0.141
 Diversity = 1.1845e-01 k_t = [2, 3, 3, 3]
 Generation 2250: Best = 1.021712e+03 FES = 225100 μ_F = 0.685 μ_{CR} = 0.130
 Diversity = 1.1928e-01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.021108e+03 FES = 226100 μ_F = 0.691 μ_{CR} = 0.133
 Diversity = 1.1775e-01 k_t = [4, 4, 4, 3]
 Generation 2270: Best = 1.020351e+03 FES = 227100 μ_F = 0.686 μ_{CR} = 0.136
 Diversity = 1.1412e-01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.019201e+03 FES = 228100 μ_F = 0.680 μ_{CR} = 0.146
 Diversity = 1.1755e-01 k_t = [4, 2, 2, 3]
 Generation 2290: Best = 1.018731e+03 FES = 229100 μ_F = 0.699 μ_{CR} = 0.141
 Diversity = 1.2118e-01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.017755e+03 FES = 230100 μ_F = 0.691 μ_{CR} = 0.146
 Diversity = 1.1504e-01 k_t = [2, 2, 4, 4]
 Generation 2310: Best = 1.016895e+03 FES = 231100 μ_F = 0.691 μ_{CR} = 0.143
 Diversity = 1.1340e-01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.015843e+03 FES = 232100 μ_F = 0.710 μ_{CR} = 0.140
 Diversity = 1.1684e-01 k_t = [2, 4, 3, 2]
 Generation 2330: Best = 1.015140e+03 FES = 233100 μ_F = 0.709 μ_{CR} = 0.127
 Diversity = 1.1907e-01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.014059e+03 FES = 234100 μ_F = 0.711 μ_{CR} = 0.109
 Diversity = 1.2179e-01 k_t = [4, 4, 2, 2]
 Generation 2350: Best = 1.013208e+03 FES = 235100 μ_F = 0.708 μ_{CR} = 0.115
 Diversity = 1.1774e-01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.013169e+03 FES = 236100 μ_F = 0.706 μ_{CR} = 0.108
 Diversity = 1.0406e-01 k_t = [4, 2, 2, 4]
 Generation 2370: Best = 1.012087e+03 FES = 237100 μ_F = 0.707 μ_{CR} = 0.123
 Diversity = 1.0132e-01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.010975e+03 FES = 238100 μ_F = 0.708 μ_{CR} = 0.130
 Diversity = 1.0238e-01 k_t = [3, 4, 4, 3]

Generation 2390: Best = 1.009390e+03 FES = 239100 μ_F = 0.722 μ_{CR} = 0.120
 Diversity = 1.0866e-01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.008561e+03 FES = 240100 μ_F = 0.723 μ_{CR} = 0.113
 Diversity = 1.0829e-01 k_t = [4, 2, 3, 2]
 Generation 2410: Best = 1.006607e+03 FES = 241100 μ_F = 0.709 μ_{CR} = 0.106
 Diversity = 1.0771e-01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.006530e+03 FES = 242100 μ_F = 0.726 μ_{CR} = 0.112
 Diversity = 1.0245e-01 k_t = [2, 3, 3, 4]
 Generation 2430: Best = 1.005407e+03 FES = 243100 μ_F = 0.735 μ_{CR} = 0.117
 Diversity = 1.0529e-01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.004847e+03 FES = 244100 μ_F = 0.731 μ_{CR} = 0.120
 Diversity = 1.0878e-01 k_t = [3, 4, 2, 2]
 Generation 2450: Best = 1.004058e+03 FES = 245100 μ_F = 0.713 μ_{CR} = 0.110
 Diversity = 1.1298e-01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.002854e+03 FES = 246100 μ_F = 0.709 μ_{CR} = 0.113
 Diversity = 1.0912e-01 k_t = [4, 3, 4, 4]
 Generation 2470: Best = 1.001031e+03 FES = 247100 μ_F = 0.730 μ_{CR} = 0.110
 Diversity = 1.1207e-01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.000545e+03 FES = 248100 μ_F = 0.747 μ_{CR} = 0.119
 Diversity = 1.1914e-01 k_t = [4, 4, 4, 2]
 Generation 2490: Best = 1.000369e+03 FES = 249100 μ_F = 0.739 μ_{CR} = 0.112
 Diversity = 1.2097e-01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 9.998933e+02 FES = 250100 μ_F = 0.709 μ_{CR} = 0.110
 Diversity = 1.1173e-01 k_t = [3, 3, 4, 2]
 Generation 2510: Best = 9.991234e+02 FES = 251100 μ_F = 0.713 μ_{CR} = 0.114
 Diversity = 1.1513e-01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 9.983642e+02 FES = 252100 μ_F = 0.715 μ_{CR} = 0.115
 Diversity = 1.0988e-01 k_t = [4, 4, 3, 3]
 Generation 2530: Best = 9.978288e+02 FES = 253100 μ_F = 0.725 μ_{CR} = 0.131
 Diversity = 1.1323e-01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 9.968689e+02 FES = 254100 μ_F = 0.746 μ_{CR} = 0.118
 Diversity = 1.0835e-01 k_t = [3, 3, 2, 4]
 Generation 2550: Best = 9.945914e+02 FES = 255100 μ_F = 0.735 μ_{CR} = 0.119
 Diversity = 1.0254e-01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 9.939505e+02 FES = 256100 μ_F = 0.731 μ_{CR} = 0.120
 Diversity = 1.0526e-01 k_t = [2, 4, 2, 4]
 Generation 2570: Best = 9.924908e+02 FES = 257100 μ_F = 0.727 μ_{CR} = 0.111
 Diversity = 1.1119e-01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 9.923437e+02 FES = 258100 μ_F = 0.733 μ_{CR} = 0.103
 Diversity = 1.0962e-01 k_t = [2, 4, 2, 4]
 Generation 2590: Best = 9.911978e+02 FES = 259100 μ_F = 0.725 μ_{CR} = 0.105
 Diversity = 1.0463e-01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 9.909612e+02 FES = 260100 μ_F = 0.722 μ_{CR} = 0.104
 Diversity = 9.8023e-02 k_t = [4, 4, 4, 4]
 Generation 2610: Best = 9.905918e+02 FES = 261100 μ_F = 0.710 μ_{CR} = 0.111
 Diversity = 9.7017e-02 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 9.900708e+02 FES = 262100 μ_F = 0.700 μ_{CR} = 0.120
 Diversity = 9.1588e-02 k_t = [2, 4, 2, 2]
 Generation 2630: Best = 9.898357e+02 FES = 263100 μ_F = 0.735 μ_{CR} = 0.121
 Diversity = 8.6918e-02 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 9.890240e+02 FES = 264100 μ_F = 0.732 μ_{CR} = 0.119
 Diversity = 8.6771e-02 k_t = [3, 3, 4, 2]
 Generation 2650: Best = 9.865213e+02 FES = 265100 μ_F = 0.716 μ_{CR} = 0.125
 Diversity = 9.1192e-02 k_t = [1, 1, 1, 1]

Generation 2660: Best = 9.849609e+02 FES = 266100 μ_F = 0.711 μ_{CR} = 0.118
 Diversity = 9.5216e-02 k_t = [4, 2, 2, 2]
 Generation 2670: Best = 9.847905e+02 FES = 267100 μ_F = 0.722 μ_{CR} = 0.121
 Diversity = 1.0748e-01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 9.844677e+02 FES = 268100 μ_F = 0.727 μ_{CR} = 0.110
 Diversity = 1.0301e-01 k_t = [2, 4, 4, 4]
 Generation 2690: Best = 9.837155e+02 FES = 269100 μ_F = 0.723 μ_{CR} = 0.111
 Diversity = 9.9188e-02 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 9.833751e+02 FES = 270100 μ_F = 0.713 μ_{CR} = 0.106
 Diversity = 1.0424e-01 k_t = [3, 3, 2, 2]
 Generation 2710: Best = 9.825862e+02 FES = 271100 μ_F = 0.715 μ_{CR} = 0.099
 Diversity = 1.0620e-01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 9.819129e+02 FES = 272100 μ_F = 0.726 μ_{CR} = 0.099
 Diversity = 1.1027e-01 k_t = [2, 4, 3, 2]
 Generation 2730: Best = 9.807199e+02 FES = 273100 μ_F = 0.714 μ_{CR} = 0.108
 Diversity = 1.0682e-01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 9.795019e+02 FES = 274100 μ_F = 0.720 μ_{CR} = 0.122
 Diversity = 1.0351e-01 k_t = [4, 2, 2, 3]
 Generation 2750: Best = 9.783592e+02 FES = 275100 μ_F = 0.726 μ_{CR} = 0.115
 Diversity = 1.1307e-01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 9.766321e+02 FES = 276100 μ_F = 0.725 μ_{CR} = 0.120
 Diversity = 1.1606e-01 k_t = [4, 3, 4, 2]
 Generation 2770: Best = 9.757508e+02 FES = 277100 μ_F = 0.727 μ_{CR} = 0.111
 Diversity = 1.2142e-01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 9.747146e+02 FES = 278100 μ_F = 0.730 μ_{CR} = 0.099
 Diversity = 1.1155e-01 k_t = [4, 3, 4, 4]
 Generation 2790: Best = 9.741463e+02 FES = 279100 μ_F = 0.709 μ_{CR} = 0.100
 Diversity = 1.0418e-01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 9.741463e+02 FES = 280100 μ_F = 0.708 μ_{CR} = 0.103
 Diversity = 1.0046e-01 k_t = [2, 4, 2, 4]
 Generation 2810: Best = 9.729413e+02 FES = 281100 μ_F = 0.714 μ_{CR} = 0.111
 Diversity = 1.0218e-01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 9.721271e+02 FES = 282100 μ_F = 0.711 μ_{CR} = 0.114
 Diversity = 1.0806e-01 k_t = [4, 2, 3, 3]
 Generation 2830: Best = 9.712888e+02 FES = 283100 μ_F = 0.714 μ_{CR} = 0.116
 Diversity = 1.1144e-01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 9.692121e+02 FES = 284100 μ_F = 0.700 μ_{CR} = 0.111
 Diversity = 1.0697e-01 k_t = [4, 3, 3, 3]
 Generation 2850: Best = 9.682465e+02 FES = 285100 μ_F = 0.726 μ_{CR} = 0.109
 Diversity = 1.1201e-01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 9.674638e+02 FES = 286100 μ_F = 0.708 μ_{CR} = 0.111
 Diversity = 1.1203e-01 k_t = [2, 3, 2, 2]
 Generation 2870: Best = 9.655579e+02 FES = 287100 μ_F = 0.698 μ_{CR} = 0.104
 Diversity = 1.1716e-01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 9.634539e+02 FES = 288100 μ_F = 0.691 μ_{CR} = 0.108
 Diversity = 1.2257e-01 k_t = [3, 4, 4, 2]
 Generation 2890: Best = 9.634539e+02 FES = 289100 μ_F = 0.695 μ_{CR} = 0.115
 Diversity = 1.3138e-01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 9.632550e+02 FES = 290100 μ_F = 0.722 μ_{CR} = 0.110
 Diversity = 1.2681e-01 k_t = [3, 4, 3, 2]
 Generation 2910: Best = 9.627316e+02 FES = 291100 μ_F = 0.727 μ_{CR} = 0.105
 Diversity = 1.1528e-01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 9.615713e+02 FES = 292100 μ_F = 0.704 μ_{CR} = 0.107
 Diversity = 1.1077e-01 k_t = [4, 4, 3, 3]

Generation 2930: Best = 9.609191e+02 FES = 293100 μ_F = 0.685 μ_{CR} = 0.115
 Diversity = 1.1219e-01 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 9.585524e+02 FES = 294100 μ_F = 0.677 μ_{CR} = 0.122
 Diversity = 1.0567e-01 k_t = [4, 4, 2, 2]
 Generation 2950: Best = 9.582518e+02 FES = 295100 μ_F = 0.683 μ_{CR} = 0.128
 Diversity = 1.0965e-01 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 9.570251e+02 FES = 296100 μ_F = 0.700 μ_{CR} = 0.136
 Diversity = 1.0438e-01 k_t = [2, 4, 4, 4]
 Generation 2970: Best = 9.556302e+02 FES = 297100 μ_F = 0.704 μ_{CR} = 0.131
 Diversity = 1.0714e-01 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 9.555232e+02 FES = 298100 μ_F = 0.706 μ_{CR} = 0.120
 Diversity = 1.0411e-01 k_t = [4, 3, 2, 4]
 Generation 2990: Best = 9.549270e+02 FES = 299100 μ_F = 0.697 μ_{CR} = 0.117
 Diversity = 1.0072e-01 k_t = [1, 1, 1, 1]

Optimization complete.

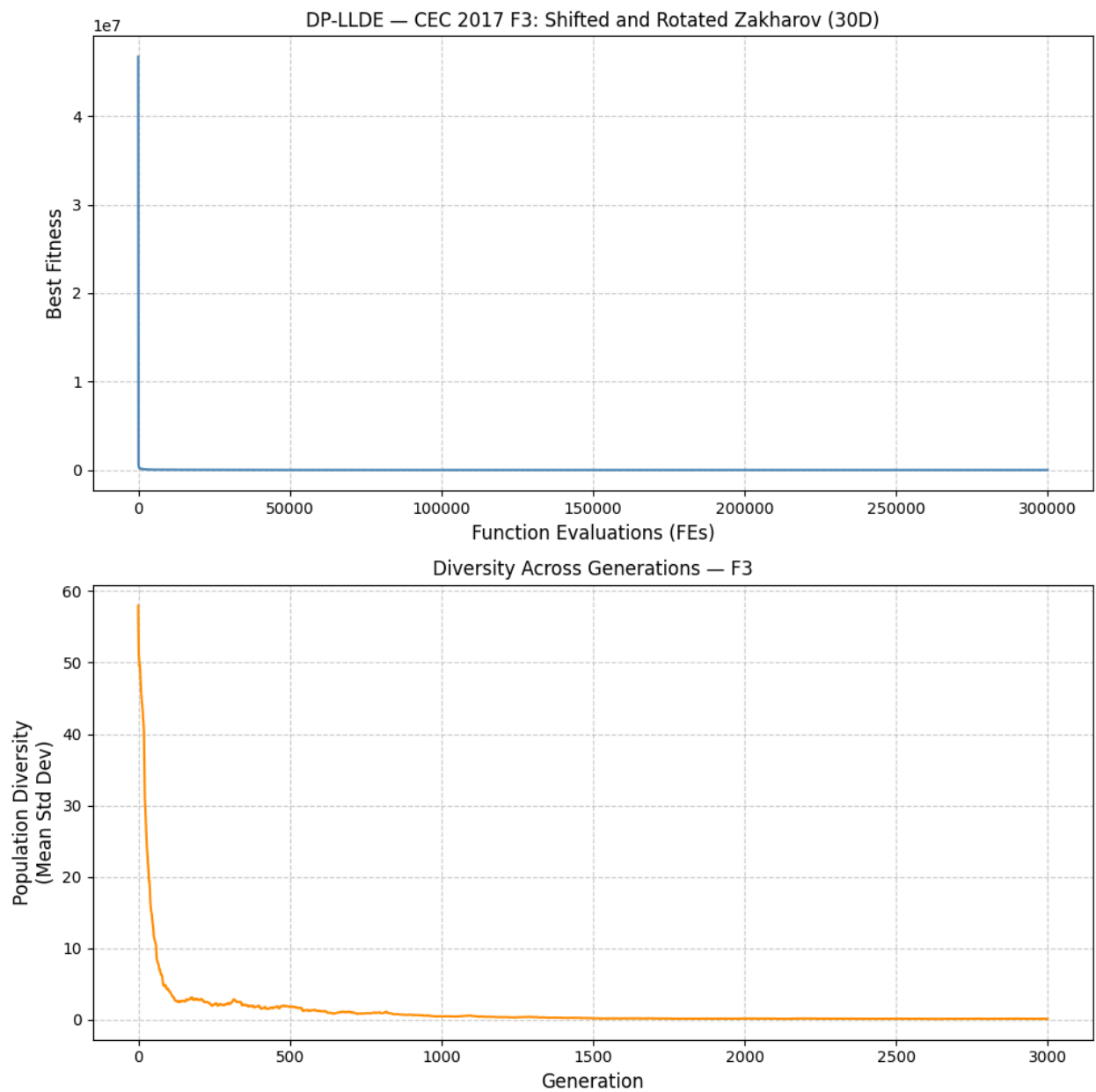
Best Fitness = 9.542100e+02

Total Function Evals = 300000

Total Successful Updates = 55020

```

=====
=
Results:
  Best Fitness      : 9.542100e+02
  Global Optimum    : 3.000000e+02
  Error (|best-optimum|) : 6.542100e+02
  Best Solution (5 dim) : [-54.94644766  4.80050863  38.8660576  9.13187886
-39.92002625]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F3.png

6.2 Multimodal Functions (F4-F10)

```
In [ ]: # =====
# F4 — Shifted and Rotated Rosenbrock
# =====
run_and_plot('F4', make_wrapper(simple.f4), 400.0, "Shifted and Rotated Rosenb
```


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DP-LLDE on CEC 2017 F4 – Single Run

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=

Function : Shifted and Rotated Rosenbrock's (30D)
Bounds : [-100, 100]
Global optimum : 400.0
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 3.421845e+04

Generation 10:	Best = 3.593564e+03	FES = 1100	$\mu_F = 0.563$	$\mu_{CR} = 0.357$
Diversity = 2.7337e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 1.216871e+03	FES = 2100	$\mu_F = 0.596$	$\mu_{CR} = 0.360$
Diversity = 1.3468e+01	$k_t = [4, 2, 4, 4]$			
Generation 30:	Best = 7.438781e+02	FES = 3100	$\mu_F = 0.620$	$\mu_{CR} = 0.356$
Diversity = 7.5457e+00	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 6.568019e+02	FES = 4100	$\mu_F = 0.637$	$\mu_{CR} = 0.359$
Diversity = 4.4676e+00	$k_t = [4, 3, 4, 2]$			
Generation 50:	Best = 6.120693e+02	FES = 5100	$\mu_F = 0.658$	$\mu_{CR} = 0.366$
Diversity = 3.0264e+00	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 5.797112e+02	FES = 6100	$\mu_F = 0.670$	$\mu_{CR} = 0.366$
Diversity = 1.8628e+00	$k_t = [3, 4, 3, 3]$			
Generation 70:	Best = 5.600629e+02	FES = 7100	$\mu_F = 0.687$	$\mu_{CR} = 0.369$
Diversity = 1.3624e+00	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 5.510680e+02	FES = 8100	$\mu_F = 0.723$	$\mu_{CR} = 0.373$
Diversity = 1.1598e+00	$k_t = [2, 4, 4, 4]$			
Generation 90:	Best = 5.425865e+02	FES = 9100	$\mu_F = 0.733$	$\mu_{CR} = 0.375$
Diversity = 9.0732e-01	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 5.406473e+02	FES = 10100	$\mu_F = 0.731$	$\mu_{CR} = 0.372$
Diversity = 7.3620e-01	$k_t = [2, 2, 4, 4]$			
Generation 110:	Best = 5.389885e+02	FES = 11100	$\mu_F = 0.720$	$\mu_{CR} = 0.380$
Diversity = 7.3846e-01	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 5.370231e+02	FES = 12100	$\mu_F = 0.719$	$\mu_{CR} = 0.378$
Diversity = 6.7493e-01	$k_t = [3, 2, 3, 4]$			
Generation 130:	Best = 5.359607e+02	FES = 13100	$\mu_F = 0.716$	$\mu_{CR} = 0.368$
Diversity = 7.0628e-01	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 5.351403e+02	FES = 14100	$\mu_F = 0.716$	$\mu_{CR} = 0.372$
Diversity = 7.6633e-01	$k_t = [4, 2, 2, 2]$			
Generation 150:	Best = 5.338949e+02	FES = 15100	$\mu_F = 0.702$	$\mu_{CR} = 0.356$
Diversity = 7.9694e-01	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 5.333233e+02	FES = 16100	$\mu_F = 0.688$	$\mu_{CR} = 0.345$
Diversity = 6.5062e-01	$k_t = [3, 3, 3, 3]$			
Generation 170:	Best = 5.324886e+02	FES = 17100	$\mu_F = 0.688$	$\mu_{CR} = 0.341$
Diversity = 5.6224e-01	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 5.315413e+02	FES = 18100	$\mu_F = 0.688$	$\mu_{CR} = 0.338$
Diversity = 4.8110e-01	$k_t = [4, 4, 4, 4]$			
Generation 190:	Best = 5.308519e+02	FES = 19100	$\mu_F = 0.697$	$\mu_{CR} = 0.330$
Diversity = 5.4152e-01	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 5.302150e+02	FES = 20100	$\mu_F = 0.702$	$\mu_{CR} = 0.344$
Diversity = 4.9310e-01	$k_t = [3, 4, 3, 2]$			
Generation 210:	Best = 5.295720e+02	FES = 21100	$\mu_F = 0.691$	$\mu_{CR} = 0.334$
Diversity = 4.4681e-01	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 5.288812e+02	FES = 22100	$\mu_F = 0.690$	$\mu_{CR} = 0.341$
Diversity = 4.6426e-01	$k_t = [4, 4, 4, 2]$			

Generation 230: Best = 5.281057e+02	FES = 23100	$\mu_F = 0.674$	$\mu_{CR} = 0.322$
Diversity = 4.1056e-01	k_t = [1, 1, 1, 1]		
Generation 240: Best = 5.272880e+02	FES = 24100	$\mu_F = 0.693$	$\mu_{CR} = 0.307$
Diversity = 4.6021e-01	k_t = [4, 2, 4, 2]		
Generation 250: Best = 5.258727e+02	FES = 25100	$\mu_F = 0.699$	$\mu_{CR} = 0.308$
Diversity = 4.6369e-01	k_t = [1, 1, 1, 1]		
Generation 260: Best = 5.248271e+02	FES = 26100	$\mu_F = 0.709$	$\mu_{CR} = 0.294$
Diversity = 4.6601e-01	k_t = [2, 4, 2, 3]		
Generation 270: Best = 5.240045e+02	FES = 27100	$\mu_F = 0.712$	$\mu_{CR} = 0.293$
Diversity = 4.4645e-01	k_t = [1, 1, 1, 1]		
Generation 280: Best = 5.232648e+02	FES = 28100	$\mu_F = 0.691$	$\mu_{CR} = 0.299$
Diversity = 3.5971e-01	k_t = [4, 4, 2, 4]		
Generation 290: Best = 5.224085e+02	FES = 29100	$\mu_F = 0.711$	$\mu_{CR} = 0.290$
Diversity = 3.7500e-01	k_t = [1, 1, 1, 1]		
Generation 300: Best = 5.213185e+02	FES = 30100	$\mu_F = 0.722$	$\mu_{CR} = 0.278$
Diversity = 3.9162e-01	k_t = [3, 3, 2, 2]		
Generation 310: Best = 5.207022e+02	FES = 31100	$\mu_F = 0.731$	$\mu_{CR} = 0.272$
Diversity = 4.2298e-01	k_t = [1, 1, 1, 1]		
Generation 320: Best = 5.197731e+02	FES = 32100	$\mu_F = 0.733$	$\mu_{CR} = 0.258$
Diversity = 4.0805e-01	k_t = [3, 2, 2, 4]		
Generation 330: Best = 5.185593e+02	FES = 33100	$\mu_F = 0.737$	$\mu_{CR} = 0.245$
Diversity = 4.9276e-01	k_t = [1, 1, 1, 1]		
Generation 340: Best = 5.174888e+02	FES = 34100	$\mu_F = 0.747$	$\mu_{CR} = 0.237$
Diversity = 6.2560e-01	k_t = [2, 3, 4, 2]		
Generation 350: Best = 5.164312e+02	FES = 35100	$\mu_F = 0.744$	$\mu_{CR} = 0.219$
Diversity = 6.2865e-01	k_t = [1, 1, 1, 1]		
Generation 360: Best = 5.155035e+02	FES = 36100	$\mu_F = 0.749$	$\mu_{CR} = 0.205$
Diversity = 6.6125e-01	k_t = [4, 4, 3, 4]		
Generation 370: Best = 5.139920e+02	FES = 37100	$\mu_F = 0.752$	$\mu_{CR} = 0.202$
Diversity = 6.7763e-01	k_t = [1, 1, 1, 1]		
Generation 380: Best = 5.128048e+02	FES = 38100	$\mu_F = 0.757$	$\mu_{CR} = 0.185$
Diversity = 6.8379e-01	k_t = [3, 2, 2, 4]		
Generation 390: Best = 5.115639e+02	FES = 39100	$\mu_F = 0.754$	$\mu_{CR} = 0.186$
Diversity = 8.7921e-01	k_t = [1, 1, 1, 1]		
Generation 400: Best = 5.105527e+02	FES = 40100	$\mu_F = 0.755$	$\mu_{CR} = 0.169$
Diversity = 8.4416e-01	k_t = [2, 2, 4, 4]		
Generation 410: Best = 5.095210e+02	FES = 41100	$\mu_F = 0.759$	$\mu_{CR} = 0.155$
Diversity = 8.5595e-01	k_t = [1, 1, 1, 1]		
Generation 420: Best = 5.078755e+02	FES = 42100	$\mu_F = 0.760$	$\mu_{CR} = 0.149$
Diversity = 1.0476e+00	k_t = [4, 4, 3, 2]		
Generation 430: Best = 5.061094e+02	FES = 43100	$\mu_F = 0.762$	$\mu_{CR} = 0.144$
Diversity = 1.3162e+00	k_t = [1, 1, 1, 1]		
Generation 440: Best = 5.037201e+02	FES = 44100	$\mu_F = 0.758$	$\mu_{CR} = 0.146$
Diversity = 1.3862e+00	k_t = [2, 2, 3, 3]		
Generation 450: Best = 4.994409e+02	FES = 45100	$\mu_F = 0.764$	$\mu_{CR} = 0.148$
Diversity = 1.0371e+00	k_t = [1, 1, 1, 1]		
Generation 460: Best = 4.897106e+02	FES = 46100	$\mu_F = 0.774$	$\mu_{CR} = 0.154$
Diversity = 9.3118e-01	k_t = [2, 3, 3, 3]		
Generation 470: Best = 4.825638e+02	FES = 47100	$\mu_F = 0.773$	$\mu_{CR} = 0.165$
Diversity = 8.6667e-01	k_t = [1, 1, 1, 1]		
Generation 480: Best = 4.762115e+02	FES = 48100	$\mu_F = 0.784$	$\mu_{CR} = 0.167$
Diversity = 8.1527e-01	k_t = [4, 3, 2, 4]		
Generation 490: Best = 4.688819e+02	FES = 49100	$\mu_F = 0.786$	$\mu_{CR} = 0.173$
Diversity = 8.0689e-01	k_t = [1, 1, 1, 1]		

Generation 500:	Best = 4.608029e+02	FES = 50100	$\mu_F = 0.779$	$\mu_{CR} = 0.185$
Diversity = 7.8666e-01	$k_t = [3, 2, 2, 4]$			
Generation 510:	Best = 4.544337e+02	FES = 51100	$\mu_F = 0.779$	$\mu_{CR} = 0.193$
Diversity = 8.2760e-01	$k_t = [1, 1, 1, 1]$			
Generation 520:	Best = 4.487222e+02	FES = 52100	$\mu_F = 0.791$	$\mu_{CR} = 0.204$
Diversity = 7.8989e-01	$k_t = [2, 4, 2, 2]$			
Generation 530:	Best = 4.429765e+02	FES = 53100	$\mu_F = 0.802$	$\mu_{CR} = 0.203$
Diversity = 8.0670e-01	$k_t = [1, 1, 1, 1]$			
Generation 540:	Best = 4.368091e+02	FES = 54100	$\mu_F = 0.791$	$\mu_{CR} = 0.198$
Diversity = 8.1784e-01	$k_t = [3, 3, 2, 2]$			
Generation 550:	Best = 4.346989e+02	FES = 55100	$\mu_F = 0.783$	$\mu_{CR} = 0.199$
Diversity = 8.1175e-01	$k_t = [1, 1, 1, 1]$			
Generation 560:	Best = 4.298054e+02	FES = 56100	$\mu_F = 0.780$	$\mu_{CR} = 0.192$
Diversity = 7.7241e-01	$k_t = [3, 3, 2, 2]$			
Generation 570:	Best = 4.214831e+02	FES = 57100	$\mu_F = 0.792$	$\mu_{CR} = 0.198$
Diversity = 8.4810e-01	$k_t = [1, 1, 1, 1]$			
Generation 580:	Best = 4.195404e+02	FES = 58100	$\mu_F = 0.782$	$\mu_{CR} = 0.203$
Diversity = 7.9354e-01	$k_t = [4, 3, 4, 2]$			
Generation 590:	Best = 4.160237e+02	FES = 59100	$\mu_F = 0.765$	$\mu_{CR} = 0.206$
Diversity = 7.4346e-01	$k_t = [1, 1, 1, 1]$			
Generation 600:	Best = 4.121373e+02	FES = 60100	$\mu_F = 0.761$	$\mu_{CR} = 0.195$
Diversity = 7.1489e-01	$k_t = [4, 4, 3, 3]$			
Generation 610:	Best = 4.110409e+02	FES = 61100	$\mu_F = 0.727$	$\mu_{CR} = 0.192$
Diversity = 7.0738e-01	$k_t = [1, 1, 1, 1]$			
Generation 620:	Best = 4.094663e+02	FES = 62100	$\mu_F = 0.716$	$\mu_{CR} = 0.177$
Diversity = 6.2512e-01	$k_t = [3, 2, 3, 3]$			
Generation 630:	Best = 4.087704e+02	FES = 63100	$\mu_F = 0.704$	$\mu_{CR} = 0.169$
Diversity = 6.2022e-01	$k_t = [1, 1, 1, 1]$			
Generation 640:	Best = 4.077934e+02	FES = 64100	$\mu_F = 0.706$	$\mu_{CR} = 0.159$
Diversity = 6.1458e-01	$k_t = [4, 3, 2, 2]$			
Generation 650:	Best = 4.071450e+02	FES = 65100	$\mu_F = 0.698$	$\mu_{CR} = 0.145$
Diversity = 6.1527e-01	$k_t = [1, 1, 1, 1]$			
Generation 660:	Best = 4.066640e+02	FES = 66100	$\mu_F = 0.689$	$\mu_{CR} = 0.139$
Diversity = 5.5898e-01	$k_t = [4, 4, 4, 4]$			
Generation 670:	Best = 4.060607e+02	FES = 67100	$\mu_F = 0.698$	$\mu_{CR} = 0.140$
Diversity = 5.2583e-01	$k_t = [1, 1, 1, 1]$			
Generation 680:	Best = 4.055306e+02	FES = 68100	$\mu_F = 0.709$	$\mu_{CR} = 0.125$
Diversity = 4.9078e-01	$k_t = [4, 4, 4, 4]$			
Generation 690:	Best = 4.052018e+02	FES = 69100	$\mu_F = 0.709$	$\mu_{CR} = 0.114$
Diversity = 5.4621e-01	$k_t = [1, 1, 1, 1]$			
Generation 700:	Best = 4.049984e+02	FES = 70100	$\mu_F = 0.712$	$\mu_{CR} = 0.119$
Diversity = 4.5744e-01	$k_t = [2, 4, 4, 3]$			
Generation 710:	Best = 4.047037e+02	FES = 71100	$\mu_F = 0.696$	$\mu_{CR} = 0.115$
Diversity = 5.0557e-01	$k_t = [1, 1, 1, 1]$			
Generation 720:	Best = 4.044856e+02	FES = 72100	$\mu_F = 0.697$	$\mu_{CR} = 0.122$
Diversity = 4.8204e-01	$k_t = [2, 4, 2, 2]$			
Generation 730:	Best = 4.041018e+02	FES = 73100	$\mu_F = 0.698$	$\mu_{CR} = 0.115$
Diversity = 6.1982e-01	$k_t = [1, 1, 1, 1]$			
Generation 740:	Best = 4.039820e+02	FES = 74100	$\mu_F = 0.694$	$\mu_{CR} = 0.101$
Diversity = 5.4884e-01	$k_t = [4, 2, 3, 3]$			
Generation 750:	Best = 4.038398e+02	FES = 75100	$\mu_F = 0.699$	$\mu_{CR} = 0.099$
Diversity = 5.7601e-01	$k_t = [1, 1, 1, 1]$			
Generation 760:	Best = 4.037024e+02	FES = 76100	$\mu_F = 0.704$	$\mu_{CR} = 0.099$
Diversity = 5.3354e-01	$k_t = [2, 4, 4, 3]$			

Generation 770:	Best = 4.035092e+02	FES = 77100	$\mu_F = 0.697$	$\mu_{CR} = 0.105$
Diversity = 5.1945e-01	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 4.034155e+02	FES = 78100	$\mu_F = 0.692$	$\mu_{CR} = 0.114$
Diversity = 4.7021e-01	$k_t = [4, 2, 3, 4]$			
Generation 790:	Best = 4.032282e+02	FES = 79100	$\mu_F = 0.703$	$\mu_{CR} = 0.102$
Diversity = 4.0461e-01	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 4.031405e+02	FES = 80100	$\mu_F = 0.720$	$\mu_{CR} = 0.096$
Diversity = 5.0016e-01	$k_t = [3, 2, 4, 3]$			
Generation 810:	Best = 4.030211e+02	FES = 81100	$\mu_F = 0.732$	$\mu_{CR} = 0.095$
Diversity = 5.1645e-01	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 4.028967e+02	FES = 82100	$\mu_F = 0.754$	$\mu_{CR} = 0.094$
Diversity = 5.0368e-01	$k_t = [3, 2, 4, 4]$			
Generation 830:	Best = 4.027310e+02	FES = 83100	$\mu_F = 0.773$	$\mu_{CR} = 0.097$
Diversity = 5.1430e-01	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 4.026752e+02	FES = 84100	$\mu_F = 0.770$	$\mu_{CR} = 0.091$
Diversity = 5.1664e-01	$k_t = [2, 2, 4, 4]$			
Generation 850:	Best = 4.026202e+02	FES = 85100	$\mu_F = 0.763$	$\mu_{CR} = 0.087$
Diversity = 4.9133e-01	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 4.025185e+02	FES = 86100	$\mu_F = 0.765$	$\mu_{CR} = 0.090$
Diversity = 4.3818e-01	$k_t = [2, 3, 4, 4]$			
Generation 870:	Best = 4.024153e+02	FES = 87100	$\mu_F = 0.758$	$\mu_{CR} = 0.093$
Diversity = 4.6133e-01	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 4.022845e+02	FES = 88100	$\mu_F = 0.774$	$\mu_{CR} = 0.091$
Diversity = 5.4853e-01	$k_t = [2, 4, 4, 2]$			
Generation 890:	Best = 4.022452e+02	FES = 89100	$\mu_F = 0.762$	$\mu_{CR} = 0.088$
Diversity = 6.2650e-01	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 4.021717e+02	FES = 90100	$\mu_F = 0.742$	$\mu_{CR} = 0.072$
Diversity = 6.4026e-01	$k_t = [4, 2, 3, 3]$			
Generation 910:	Best = 4.019879e+02	FES = 91100	$\mu_F = 0.747$	$\mu_{CR} = 0.074$
Diversity = 6.4974e-01	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 4.019754e+02	FES = 92100	$\mu_F = 0.740$	$\mu_{CR} = 0.076$
Diversity = 6.0893e-01	$k_t = [2, 4, 2, 2]$			
Generation 930:	Best = 4.019182e+02	FES = 93100	$\mu_F = 0.758$	$\mu_{CR} = 0.078$
Diversity = 6.5024e-01	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 4.018722e+02	FES = 94100	$\mu_F = 0.749$	$\mu_{CR} = 0.074$
Diversity = 6.0988e-01	$k_t = [2, 4, 2, 4]$			
Generation 950:	Best = 4.018339e+02	FES = 95100	$\mu_F = 0.748$	$\mu_{CR} = 0.071$
Diversity = 6.1287e-01	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 4.017360e+02	FES = 96100	$\mu_F = 0.753$	$\mu_{CR} = 0.071$
Diversity = 5.8312e-01	$k_t = [3, 3, 4, 2]$			
Generation 970:	Best = 4.016982e+02	FES = 97100	$\mu_F = 0.746$	$\mu_{CR} = 0.081$
Diversity = 5.5457e-01	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 4.016556e+02	FES = 98100	$\mu_F = 0.733$	$\mu_{CR} = 0.077$
Diversity = 5.3954e-01	$k_t = [2, 3, 4, 4]$			
Generation 990:	Best = 4.015880e+02	FES = 99100	$\mu_F = 0.730$	$\mu_{CR} = 0.074$
Diversity = 4.5588e-01	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 4.015677e+02	FES = 100100	$\mu_F = 0.720$	$\mu_{CR} = 0.087$
Diversity = 4.0943e-01	$k_t = [2, 3, 4, 3]$			
Generation 1010:	Best = 4.015226e+02	FES = 101100	$\mu_F = 0.736$	$\mu_{CR} = 0.082$
Diversity = 4.9564e-01	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 4.014806e+02	FES = 102100	$\mu_F = 0.729$	$\mu_{CR} = 0.081$
Diversity = 4.4227e-01	$k_t = [3, 2, 3, 4]$			
Generation 1030:	Best = 4.014461e+02	FES = 103100	$\mu_F = 0.748$	$\mu_{CR} = 0.086$
Diversity = 4.0181e-01	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 4.014150e+02 FES = 104100 μ_F = 0.751 μ_{CR} = 0.090
 Diversity = 4.7104e-01 k_t = [2, 4, 4, 3]
 Generation 1050: Best = 4.013689e+02 FES = 105100 μ_F = 0.754 μ_{CR} = 0.081
 Diversity = 4.9393e-01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 4.013349e+02 FES = 106100 μ_F = 0.730 μ_{CR} = 0.073
 Diversity = 4.5800e-01 k_t = [2, 4, 3, 2]
 Generation 1070: Best = 4.012578e+02 FES = 107100 μ_F = 0.726 μ_{CR} = 0.084
 Diversity = 5.0919e-01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 4.012555e+02 FES = 108100 μ_F = 0.725 μ_{CR} = 0.093
 Diversity = 4.5498e-01 k_t = [4, 4, 2, 3]
 Generation 1090: Best = 4.012345e+02 FES = 109100 μ_F = 0.732 μ_{CR} = 0.085
 Diversity = 5.2155e-01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 4.012071e+02 FES = 110100 μ_F = 0.728 μ_{CR} = 0.084
 Diversity = 4.8143e-01 k_t = [4, 4, 3, 3]
 Generation 1110: Best = 4.011654e+02 FES = 111100 μ_F = 0.724 μ_{CR} = 0.086
 Diversity = 3.9906e-01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 4.011467e+02 FES = 112100 μ_F = 0.706 μ_{CR} = 0.082
 Diversity = 3.8569e-01 k_t = [2, 3, 3, 3]
 Generation 1130: Best = 4.011116e+02 FES = 113100 μ_F = 0.718 μ_{CR} = 0.089
 Diversity = 4.3190e-01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 4.010921e+02 FES = 114100 μ_F = 0.730 μ_{CR} = 0.091
 Diversity = 4.3110e-01 k_t = [4, 4, 2, 3]
 Generation 1150: Best = 4.010504e+02 FES = 115100 μ_F = 0.748 μ_{CR} = 0.087
 Diversity = 4.2638e-01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 4.010240e+02 FES = 116100 μ_F = 0.753 μ_{CR} = 0.099
 Diversity = 4.4661e-01 k_t = [3, 4, 2, 3]
 Generation 1170: Best = 4.009968e+02 FES = 117100 μ_F = 0.754 μ_{CR} = 0.081
 Diversity = 4.1321e-01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 4.009875e+02 FES = 118100 μ_F = 0.741 μ_{CR} = 0.075
 Diversity = 4.0617e-01 k_t = [3, 4, 4, 3]
 Generation 1190: Best = 4.009616e+02 FES = 119100 μ_F = 0.728 μ_{CR} = 0.079
 Diversity = 4.0354e-01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 4.009479e+02 FES = 120100 μ_F = 0.722 μ_{CR} = 0.075
 Diversity = 3.9150e-01 k_t = [4, 2, 3, 2]
 Generation 1210: Best = 4.009345e+02 FES = 121100 μ_F = 0.741 μ_{CR} = 0.076
 Diversity = 4.1279e-01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 4.009232e+02 FES = 122100 μ_F = 0.748 μ_{CR} = 0.082
 Diversity = 4.0634e-01 k_t = [2, 2, 4, 4]
 Generation 1230: Best = 4.008908e+02 FES = 123100 μ_F = 0.749 μ_{CR} = 0.082
 Diversity = 3.6073e-01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 4.008888e+02 FES = 124100 μ_F = 0.753 μ_{CR} = 0.076
 Diversity = 3.3284e-01 k_t = [2, 2, 4, 3]
 Generation 1250: Best = 4.008725e+02 FES = 125100 μ_F = 0.747 μ_{CR} = 0.065
 Diversity = 3.4092e-01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 4.008623e+02 FES = 126100 μ_F = 0.743 μ_{CR} = 0.069
 Diversity = 3.3516e-01 k_t = [3, 2, 3, 3]
 Generation 1270: Best = 4.008573e+02 FES = 127100 μ_F = 0.733 μ_{CR} = 0.060
 Diversity = 2.9874e-01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 4.008449e+02 FES = 128100 μ_F = 0.746 μ_{CR} = 0.079
 Diversity = 2.9244e-01 k_t = [2, 3, 2, 3]
 Generation 1290: Best = 4.008338e+02 FES = 129100 μ_F = 0.734 μ_{CR} = 0.077
 Diversity = 2.9816e-01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 4.008275e+02 FES = 130100 μ_F = 0.716 μ_{CR} = 0.076
 Diversity = 2.8976e-01 k_t = [2, 2, 2, 2]

Generation 1310: Best = 4.008180e+02 FES = 131100 μ_F = 0.743 μ_{CR} = 0.072
 Diversity = 2.8123e-01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 4.008137e+02 FES = 132100 μ_F = 0.743 μ_{CR} = 0.074
 Diversity = 2.7474e-01 k_t = [4, 4, 2, 2]
 Generation 1330: Best = 4.008041e+02 FES = 133100 μ_F = 0.748 μ_{CR} = 0.085
 Diversity = 2.7587e-01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 4.007937e+02 FES = 134100 μ_F = 0.731 μ_{CR} = 0.071
 Diversity = 2.6717e-01 k_t = [2, 3, 2, 4]
 Generation 1350: Best = 4.007869e+02 FES = 135100 μ_F = 0.743 μ_{CR} = 0.069
 Diversity = 2.7666e-01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 4.007768e+02 FES = 136100 μ_F = 0.734 μ_{CR} = 0.066
 Diversity = 2.6247e-01 k_t = [3, 4, 4, 3]
 Generation 1370: Best = 4.007595e+02 FES = 137100 μ_F = 0.726 μ_{CR} = 0.078
 Diversity = 2.5530e-01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 4.007595e+02 FES = 138100 μ_F = 0.736 μ_{CR} = 0.080
 Diversity = 2.8148e-01 k_t = [3, 4, 2, 3]
 Generation 1390: Best = 4.007501e+02 FES = 139100 μ_F = 0.722 μ_{CR} = 0.081
 Diversity = 2.6240e-01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 4.007425e+02 FES = 140100 μ_F = 0.725 μ_{CR} = 0.079
 Diversity = 2.3179e-01 k_t = [4, 3, 2, 3]
 Generation 1410: Best = 4.007407e+02 FES = 141100 μ_F = 0.705 μ_{CR} = 0.076
 Diversity = 2.4480e-01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 4.007346e+02 FES = 142100 μ_F = 0.684 μ_{CR} = 0.089
 Diversity = 1.9566e-01 k_t = [3, 3, 3, 3]
 Generation 1430: Best = 4.007343e+02 FES = 143100 μ_F = 0.680 μ_{CR} = 0.088
 Diversity = 1.9397e-01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 4.007216e+02 FES = 144100 μ_F = 0.679 μ_{CR} = 0.081
 Diversity = 2.2761e-01 k_t = [3, 2, 3, 4]
 Generation 1450: Best = 4.007043e+02 FES = 145100 μ_F = 0.691 μ_{CR} = 0.096
 Diversity = 2.5593e-01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 4.006980e+02 FES = 146100 μ_F = 0.682 μ_{CR} = 0.070
 Diversity = 2.8200e-01 k_t = [4, 3, 3, 2]
 Generation 1470: Best = 4.006884e+02 FES = 147100 μ_F = 0.697 μ_{CR} = 0.068
 Diversity = 2.8819e-01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 4.006754e+02 FES = 148100 μ_F = 0.719 μ_{CR} = 0.070
 Diversity = 3.4142e-01 k_t = [4, 4, 3, 3]
 Generation 1490: Best = 4.006690e+02 FES = 149100 μ_F = 0.725 μ_{CR} = 0.076
 Diversity = 3.2545e-01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 4.006495e+02 FES = 150100 μ_F = 0.744 μ_{CR} = 0.072
 Diversity = 3.6897e-01 k_t = [3, 2, 3, 4]
 Generation 1510: Best = 4.006231e+02 FES = 151100 μ_F = 0.732 μ_{CR} = 0.073
 Diversity = 3.8565e-01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 4.006102e+02 FES = 152100 μ_F = 0.720 μ_{CR} = 0.065
 Diversity = 4.4662e-01 k_t = [4, 3, 2, 3]
 Generation 1530: Best = 4.005872e+02 FES = 153100 μ_F = 0.727 μ_{CR} = 0.070
 Diversity = 4.6859e-01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 4.005737e+02 FES = 154100 μ_F = 0.747 μ_{CR} = 0.072
 Diversity = 3.9358e-01 k_t = [4, 4, 2, 4]
 Generation 1550: Best = 4.005575e+02 FES = 155100 μ_F = 0.750 μ_{CR} = 0.067
 Diversity = 4.0932e-01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 4.005468e+02 FES = 156100 μ_F = 0.751 μ_{CR} = 0.074
 Diversity = 3.9274e-01 k_t = [3, 4, 4, 3]
 Generation 1570: Best = 4.005333e+02 FES = 157100 μ_F = 0.738 μ_{CR} = 0.069
 Diversity = 3.9068e-01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 4.005085e+02 FES = 158100 μ_F = 0.740 μ_{CR} = 0.067
 Diversity = 3.8115e-01 k_t = [3, 3, 2, 4]
 Generation 1590: Best = 4.004981e+02 FES = 159100 μ_F = 0.748 μ_{CR} = 0.077
 Diversity = 4.4941e-01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 4.004902e+02 FES = 160100 μ_F = 0.755 μ_{CR} = 0.085
 Diversity = 3.6339e-01 k_t = [2, 4, 2, 4]
 Generation 1610: Best = 4.004793e+02 FES = 161100 μ_F = 0.751 μ_{CR} = 0.090
 Diversity = 3.6623e-01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 4.004754e+02 FES = 162100 μ_F = 0.744 μ_{CR} = 0.078
 Diversity = 3.0821e-01 k_t = [4, 3, 3, 4]
 Generation 1630: Best = 4.004673e+02 FES = 163100 μ_F = 0.734 μ_{CR} = 0.061
 Diversity = 2.6700e-01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 4.004643e+02 FES = 164100 μ_F = 0.726 μ_{CR} = 0.060
 Diversity = 2.2624e-01 k_t = [3, 2, 3, 3]
 Generation 1650: Best = 4.004450e+02 FES = 165100 μ_F = 0.740 μ_{CR} = 0.071
 Diversity = 2.3314e-01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 4.004436e+02 FES = 166100 μ_F = 0.756 μ_{CR} = 0.077
 Diversity = 2.8145e-01 k_t = [4, 4, 3, 2]
 Generation 1670: Best = 4.004394e+02 FES = 167100 μ_F = 0.764 μ_{CR} = 0.068
 Diversity = 2.9478e-01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 4.004335e+02 FES = 168100 μ_F = 0.748 μ_{CR} = 0.068
 Diversity = 2.2615e-01 k_t = [3, 2, 2, 4]
 Generation 1690: Best = 4.004267e+02 FES = 169100 μ_F = 0.732 μ_{CR} = 0.079
 Diversity = 2.2576e-01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 4.004247e+02 FES = 170100 μ_F = 0.721 μ_{CR} = 0.090
 Diversity = 1.9911e-01 k_t = [4, 2, 2, 4]
 Generation 1710: Best = 4.004183e+02 FES = 171100 μ_F = 0.718 μ_{CR} = 0.091
 Diversity = 2.0483e-01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 4.004177e+02 FES = 172100 μ_F = 0.691 μ_{CR} = 0.083
 Diversity = 1.6883e-01 k_t = [4, 4, 2, 4]
 Generation 1730: Best = 4.004140e+02 FES = 173100 μ_F = 0.664 μ_{CR} = 0.099
 Diversity = 1.8251e-01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 4.004101e+02 FES = 174100 μ_F = 0.678 μ_{CR} = 0.098
 Diversity = 1.6348e-01 k_t = [4, 3, 4, 4]
 Generation 1750: Best = 4.004086e+02 FES = 175100 μ_F = 0.669 μ_{CR} = 0.085
 Diversity = 1.3557e-01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 4.004042e+02 FES = 176100 μ_F = 0.702 μ_{CR} = 0.089
 Diversity = 1.3805e-01 k_t = [2, 2, 3, 3]
 Generation 1770: Best = 4.004037e+02 FES = 177100 μ_F = 0.698 μ_{CR} = 0.073
 Diversity = 1.4863e-01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 4.004022e+02 FES = 178100 μ_F = 0.700 μ_{CR} = 0.071
 Diversity = 1.4602e-01 k_t = [3, 2, 3, 4]
 Generation 1790: Best = 4.004020e+02 FES = 179100 μ_F = 0.693 μ_{CR} = 0.078
 Diversity = 1.3726e-01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 4.004009e+02 FES = 180100 μ_F = 0.686 μ_{CR} = 0.066
 Diversity = 1.2699e-01 k_t = [3, 3, 2, 2]
 Generation 1810: Best = 4.003992e+02 FES = 181100 μ_F = 0.716 μ_{CR} = 0.076
 Diversity = 1.2331e-01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 4.003992e+02 FES = 182100 μ_F = 0.721 μ_{CR} = 0.081
 Diversity = 1.0460e-01 k_t = [2, 4, 3, 4]
 Generation 1830: Best = 4.003975e+02 FES = 183100 μ_F = 0.709 μ_{CR} = 0.079
 Diversity = 1.0385e-01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 4.003966e+02 FES = 184100 μ_F = 0.723 μ_{CR} = 0.078
 Diversity = 1.0391e-01 k_t = [4, 2, 4, 4]

Generation 1850: Best = 4.003963e+02 FES = 185100 μ_F = 0.728 μ_{CR} = 0.086
 Diversity = 1.0964e-01 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 4.003956e+02 FES = 186100 μ_F = 0.716 μ_{CR} = 0.073
 Diversity = 1.1152e-01 k_t = [2, 3, 4, 3]
 Generation 1870: Best = 4.003953e+02 FES = 187100 μ_F = 0.695 μ_{CR} = 0.074
 Diversity = 1.1282e-01 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 4.003929e+02 FES = 188100 μ_F = 0.711 μ_{CR} = 0.073
 Diversity = 9.2000e-02 k_t = [3, 4, 4, 4]
 Generation 1890: Best = 4.003929e+02 FES = 189100 μ_F = 0.737 μ_{CR} = 0.070
 Diversity = 1.0584e-01 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 4.003924e+02 FES = 190100 μ_F = 0.737 μ_{CR} = 0.075
 Diversity = 1.0472e-01 k_t = [3, 4, 3, 3]
 Generation 1910: Best = 4.003917e+02 FES = 191100 μ_F = 0.722 μ_{CR} = 0.081
 Diversity = 8.8733e-02 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 4.003907e+02 FES = 192100 μ_F = 0.744 μ_{CR} = 0.072
 Diversity = 8.9324e-02 k_t = [2, 3, 4, 2]
 Generation 1930: Best = 4.003901e+02 FES = 193100 μ_F = 0.754 μ_{CR} = 0.074
 Diversity = 9.0542e-02 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 4.003898e+02 FES = 194100 μ_F = 0.762 μ_{CR} = 0.070
 Diversity = 8.5611e-02 k_t = [2, 3, 3, 4]
 Generation 1950: Best = 4.003882e+02 FES = 195100 μ_F = 0.753 μ_{CR} = 0.074
 Diversity = 8.2544e-02 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 4.003868e+02 FES = 196100 μ_F = 0.736 μ_{CR} = 0.082
 Diversity = 9.5567e-02 k_t = [2, 2, 4, 3]
 Generation 1970: Best = 4.003857e+02 FES = 197100 μ_F = 0.747 μ_{CR} = 0.067
 Diversity = 1.0736e-01 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 4.003853e+02 FES = 198100 μ_F = 0.727 μ_{CR} = 0.070
 Diversity = 1.1469e-01 k_t = [4, 3, 3, 4]
 Generation 1990: Best = 4.003827e+02 FES = 199100 μ_F = 0.743 μ_{CR} = 0.079
 Diversity = 1.2538e-01 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 4.003827e+02 FES = 200100 μ_F = 0.751 μ_{CR} = 0.074
 Diversity = 1.1964e-01 k_t = [3, 3, 4, 4]
 Generation 2010: Best = 4.003825e+02 FES = 201100 μ_F = 0.722 μ_{CR} = 0.070
 Diversity = 1.1565e-01 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 4.003819e+02 FES = 202100 μ_F = 0.725 μ_{CR} = 0.064
 Diversity = 1.1010e-01 k_t = [4, 3, 3, 2]
 Generation 2030: Best = 4.003807e+02 FES = 203100 μ_F = 0.727 μ_{CR} = 0.063
 Diversity = 1.0766e-01 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 4.003795e+02 FES = 204100 μ_F = 0.722 μ_{CR} = 0.071
 Diversity = 1.0296e-01 k_t = [3, 3, 2, 3]
 Generation 2050: Best = 4.003794e+02 FES = 205100 μ_F = 0.727 μ_{CR} = 0.068
 Diversity = 1.0425e-01 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 4.003791e+02 FES = 206100 μ_F = 0.703 μ_{CR} = 0.069
 Diversity = 1.0076e-01 k_t = [3, 4, 2, 2]
 Generation 2070: Best = 4.003779e+02 FES = 207100 μ_F = 0.705 μ_{CR} = 0.074
 Diversity = 1.0952e-01 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 4.003767e+02 FES = 208100 μ_F = 0.706 μ_{CR} = 0.068
 Diversity = 1.0814e-01 k_t = [2, 2, 2, 2]
 Generation 2090: Best = 4.003749e+02 FES = 209100 μ_F = 0.697 μ_{CR} = 0.068
 Diversity = 1.1106e-01 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 4.003741e+02 FES = 210100 μ_F = 0.687 μ_{CR} = 0.077
 Diversity = 1.0093e-01 k_t = [4, 2, 4, 2]
 Generation 2110: Best = 4.003723e+02 FES = 211100 μ_F = 0.674 μ_{CR} = 0.075
 Diversity = 1.1975e-01 k_t = [1, 1, 1, 1]

Generation 2120: Best = 4.003711e+02 FES = 212100 μ_F = 0.677 μ_{CR} = 0.073
 Diversity = 1.1029e-01 k_t = [4, 3, 4, 4]
 Generation 2130: Best = 4.003711e+02 FES = 213100 μ_F = 0.661 μ_{CR} = 0.072
 Diversity = 1.0510e-01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 4.003708e+02 FES = 214100 μ_F = 0.672 μ_{CR} = 0.076
 Diversity = 1.0051e-01 k_t = [4, 4, 2, 4]
 Generation 2150: Best = 4.003696e+02 FES = 215100 μ_F = 0.666 μ_{CR} = 0.081
 Diversity = 9.8277e-02 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 4.003687e+02 FES = 216100 μ_F = 0.667 μ_{CR} = 0.072
 Diversity = 1.1651e-01 k_t = [4, 3, 2, 2]
 Generation 2170: Best = 4.003674e+02 FES = 217100 μ_F = 0.696 μ_{CR} = 0.062
 Diversity = 1.0048e-01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 4.003674e+02 FES = 218100 μ_F = 0.716 μ_{CR} = 0.067
 Diversity = 9.9623e-02 k_t = [2, 2, 4, 4]
 Generation 2190: Best = 4.003661e+02 FES = 219100 μ_F = 0.738 μ_{CR} = 0.069
 Diversity = 1.1101e-01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 4.003645e+02 FES = 220100 μ_F = 0.752 μ_{CR} = 0.071
 Diversity = 1.1654e-01 k_t = [3, 3, 3, 3]
 Generation 2210: Best = 4.003632e+02 FES = 221100 μ_F = 0.762 μ_{CR} = 0.068
 Diversity = 1.1661e-01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 4.003632e+02 FES = 222100 μ_F = 0.781 μ_{CR} = 0.074
 Diversity = 1.1717e-01 k_t = [4, 4, 4, 2]
 Generation 2230: Best = 4.003611e+02 FES = 223100 μ_F = 0.801 μ_{CR} = 0.072
 Diversity = 1.2895e-01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 4.003580e+02 FES = 224100 μ_F = 0.804 μ_{CR} = 0.067
 Diversity = 1.1689e-01 k_t = [3, 3, 3, 3]
 Generation 2250: Best = 4.003564e+02 FES = 225100 μ_F = 0.806 μ_{CR} = 0.062
 Diversity = 1.2485e-01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 4.003564e+02 FES = 226100 μ_F = 0.796 μ_{CR} = 0.079
 Diversity = 1.2132e-01 k_t = [4, 2, 4, 4]
 Generation 2270: Best = 4.003561e+02 FES = 227100 μ_F = 0.795 μ_{CR} = 0.083
 Diversity = 1.0333e-01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 4.003530e+02 FES = 228100 μ_F = 0.808 μ_{CR} = 0.075
 Diversity = 1.0699e-01 k_t = [2, 2, 4, 2]
 Generation 2290: Best = 4.003482e+02 FES = 229100 μ_F = 0.809 μ_{CR} = 0.076
 Diversity = 1.3153e-01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 4.003468e+02 FES = 230100 μ_F = 0.794 μ_{CR} = 0.085
 Diversity = 1.4692e-01 k_t = [3, 4, 3, 2]
 Generation 2310: Best = 4.003446e+02 FES = 231100 μ_F = 0.792 μ_{CR} = 0.079
 Diversity = 1.5037e-01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 4.003415e+02 FES = 232100 μ_F = 0.733 μ_{CR} = 0.087
 Diversity = 1.4628e-01 k_t = [2, 4, 2, 2]
 Generation 2330: Best = 4.003353e+02 FES = 233100 μ_F = 0.680 μ_{CR} = 0.088
 Diversity = 1.6233e-01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 4.003353e+02 FES = 234100 μ_F = 0.709 μ_{CR} = 0.066
 Diversity = 1.6935e-01 k_t = [4, 2, 4, 2]
 Generation 2350: Best = 4.003353e+02 FES = 235100 μ_F = 0.710 μ_{CR} = 0.068
 Diversity = 1.6933e-01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 4.003336e+02 FES = 236100 μ_F = 0.718 μ_{CR} = 0.082
 Diversity = 1.7092e-01 k_t = [2, 4, 4, 4]
 Generation 2370: Best = 4.003336e+02 FES = 237100 μ_F = 0.714 μ_{CR} = 0.079
 Diversity = 1.5109e-01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 4.003311e+02 FES = 238100 μ_F = 0.729 μ_{CR} = 0.073
 Diversity = 1.4188e-01 k_t = [2, 2, 2, 4]

Generation 2390: Best = 4.003304e+02 FES = 239100 μ_F = 0.722 μ_{CR} = 0.072
 Diversity = 1.2140e-01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 4.003289e+02 FES = 240100 μ_F = 0.728 μ_{CR} = 0.076
 Diversity = 1.1569e-01 k_t = [3, 3, 3, 2]
 Generation 2410: Best = 4.003281e+02 FES = 241100 μ_F = 0.729 μ_{CR} = 0.068
 Diversity = 1.2910e-01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 4.003267e+02 FES = 242100 μ_F = 0.759 μ_{CR} = 0.065
 Diversity = 1.4704e-01 k_t = [3, 3, 3, 3]
 Generation 2430: Best = 4.003261e+02 FES = 243100 μ_F = 0.745 μ_{CR} = 0.078
 Diversity = 1.3000e-01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 4.003259e+02 FES = 244100 μ_F = 0.726 μ_{CR} = 0.075
 Diversity = 1.2366e-01 k_t = [4, 2, 2, 3]
 Generation 2450: Best = 4.003250e+02 FES = 245100 μ_F = 0.743 μ_{CR} = 0.066
 Diversity = 1.2084e-01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 4.003238e+02 FES = 246100 μ_F = 0.736 μ_{CR} = 0.066
 Diversity = 1.2229e-01 k_t = [3, 4, 3, 2]
 Generation 2470: Best = 4.003228e+02 FES = 247100 μ_F = 0.743 μ_{CR} = 0.068
 Diversity = 1.1786e-01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 4.003220e+02 FES = 248100 μ_F = 0.727 μ_{CR} = 0.059
 Diversity = 1.0210e-01 k_t = [2, 4, 4, 3]
 Generation 2490: Best = 4.003214e+02 FES = 249100 μ_F = 0.709 μ_{CR} = 0.067
 Diversity = 8.9537e-02 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 4.003209e+02 FES = 250100 μ_F = 0.701 μ_{CR} = 0.076
 Diversity = 8.6247e-02 k_t = [2, 2, 2, 4]
 Generation 2510: Best = 4.003204e+02 FES = 251100 μ_F = 0.701 μ_{CR} = 0.068
 Diversity = 7.8842e-02 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 4.003198e+02 FES = 252100 μ_F = 0.680 μ_{CR} = 0.070
 Diversity = 8.3579e-02 k_t = [3, 3, 3, 3]
 Generation 2530: Best = 4.003194e+02 FES = 253100 μ_F = 0.640 μ_{CR} = 0.066
 Diversity = 6.5128e-02 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 4.003185e+02 FES = 254100 μ_F = 0.649 μ_{CR} = 0.081
 Diversity = 6.3020e-02 k_t = [4, 4, 2, 2]
 Generation 2550: Best = 4.003175e+02 FES = 255100 μ_F = 0.687 μ_{CR} = 0.078
 Diversity = 6.8086e-02 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 4.003175e+02 FES = 256100 μ_F = 0.694 μ_{CR} = 0.067
 Diversity = 7.4851e-02 k_t = [2, 3, 3, 2]
 Generation 2570: Best = 4.003173e+02 FES = 257100 μ_F = 0.686 μ_{CR} = 0.070
 Diversity = 6.4795e-02 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 4.003168e+02 FES = 258100 μ_F = 0.684 μ_{CR} = 0.059
 Diversity = 7.1252e-02 k_t = [2, 3, 4, 2]
 Generation 2590: Best = 4.003163e+02 FES = 259100 μ_F = 0.679 μ_{CR} = 0.074
 Diversity = 7.1535e-02 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 4.003152e+02 FES = 260100 μ_F = 0.696 μ_{CR} = 0.073
 Diversity = 7.3320e-02 k_t = [3, 4, 4, 4]
 Generation 2610: Best = 4.003147e+02 FES = 261100 μ_F = 0.709 μ_{CR} = 0.066
 Diversity = 6.8235e-02 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 4.003146e+02 FES = 262100 μ_F = 0.709 μ_{CR} = 0.080
 Diversity = 7.5473e-02 k_t = [2, 2, 4, 2]
 Generation 2630: Best = 4.003141e+02 FES = 263100 μ_F = 0.703 μ_{CR} = 0.061
 Diversity = 7.4313e-02 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 4.003132e+02 FES = 264100 μ_F = 0.718 μ_{CR} = 0.067
 Diversity = 5.8555e-02 k_t = [4, 3, 4, 4]
 Generation 2650: Best = 4.003128e+02 FES = 265100 μ_F = 0.734 μ_{CR} = 0.078
 Diversity = 7.2560e-02 k_t = [1, 1, 1, 1]

Generation 2660: Best = 4.003124e+02 FES = 266100 μ_F = 0.764 μ_{CR} = 0.075
 Diversity = 6.9466e-02 k_t = [2, 3, 4, 3]
 Generation 2670: Best = 4.003113e+02 FES = 267100 μ_F = 0.758 μ_{CR} = 0.070
 Diversity = 6.2356e-02 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 4.003112e+02 FES = 268100 μ_F = 0.750 μ_{CR} = 0.063
 Diversity = 6.6229e-02 k_t = [2, 2, 2, 3]
 Generation 2690: Best = 4.003109e+02 FES = 269100 μ_F = 0.763 μ_{CR} = 0.061
 Diversity = 7.6242e-02 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 4.003107e+02 FES = 270100 μ_F = 0.731 μ_{CR} = 0.076
 Diversity = 7.1839e-02 k_t = [4, 2, 2, 4]
 Generation 2710: Best = 4.003099e+02 FES = 271100 μ_F = 0.758 μ_{CR} = 0.063
 Diversity = 6.6705e-02 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 4.003090e+02 FES = 272100 μ_F = 0.775 μ_{CR} = 0.065
 Diversity = 7.0301e-02 k_t = [3, 3, 3, 4]
 Generation 2730: Best = 4.003072e+02 FES = 273100 μ_F = 0.778 μ_{CR} = 0.069
 Diversity = 9.2496e-02 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 4.003057e+02 FES = 274100 μ_F = 0.771 μ_{CR} = 0.073
 Diversity = 9.5791e-02 k_t = [4, 4, 2, 3]
 Generation 2750: Best = 4.003054e+02 FES = 275100 μ_F = 0.780 μ_{CR} = 0.071
 Diversity = 1.2343e-01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 4.003039e+02 FES = 276100 μ_F = 0.770 μ_{CR} = 0.073
 Diversity = 1.0577e-01 k_t = [3, 2, 2, 3]
 Generation 2770: Best = 4.003032e+02 FES = 277100 μ_F = 0.736 μ_{CR} = 0.072
 Diversity = 1.0438e-01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 4.003029e+02 FES = 278100 μ_F = 0.744 μ_{CR} = 0.062
 Diversity = 1.0606e-01 k_t = [4, 2, 4, 4]
 Generation 2790: Best = 4.003029e+02 FES = 279100 μ_F = 0.743 μ_{CR} = 0.059
 Diversity = 1.0243e-01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 4.003023e+02 FES = 280100 μ_F = 0.732 μ_{CR} = 0.052
 Diversity = 9.3818e-02 k_t = [3, 2, 3, 3]
 Generation 2810: Best = 4.003016e+02 FES = 281100 μ_F = 0.745 μ_{CR} = 0.066
 Diversity = 9.1441e-02 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 4.003008e+02 FES = 282100 μ_F = 0.739 μ_{CR} = 0.061
 Diversity = 9.2673e-02 k_t = [3, 3, 3, 2]
 Generation 2830: Best = 4.003006e+02 FES = 283100 μ_F = 0.721 μ_{CR} = 0.062
 Diversity = 8.4617e-02 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 4.003000e+02 FES = 284100 μ_F = 0.719 μ_{CR} = 0.065
 Diversity = 7.6356e-02 k_t = [4, 3, 2, 4]
 Generation 2850: Best = 4.002994e+02 FES = 285100 μ_F = 0.743 μ_{CR} = 0.070
 Diversity = 7.4672e-02 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 4.002994e+02 FES = 286100 μ_F = 0.768 μ_{CR} = 0.059
 Diversity = 7.3402e-02 k_t = [4, 3, 4, 2]
 Generation 2870: Best = 4.002990e+02 FES = 287100 μ_F = 0.745 μ_{CR} = 0.071
 Diversity = 7.4450e-02 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 4.002982e+02 FES = 288100 μ_F = 0.761 μ_{CR} = 0.066
 Diversity = 7.0037e-02 k_t = [3, 2, 3, 4]
 Generation 2890: Best = 4.002982e+02 FES = 289100 μ_F = 0.740 μ_{CR} = 0.062
 Diversity = 6.7619e-02 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 4.002977e+02 FES = 290100 μ_F = 0.743 μ_{CR} = 0.065
 Diversity = 7.8984e-02 k_t = [2, 3, 4, 2]
 Generation 2910: Best = 4.002977e+02 FES = 291100 μ_F = 0.734 μ_{CR} = 0.067
 Diversity = 7.0703e-02 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 4.002977e+02 FES = 292100 μ_F = 0.741 μ_{CR} = 0.070
 Diversity = 6.9516e-02 k_t = [2, 4, 4, 2]

Generation 2930: Best = 4.002972e+02 FES = 293100 μ_F = 0.759 μ_{CR} = 0.070
 Diversity = 6.5219e-02 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 4.002970e+02 FES = 294100 μ_F = 0.739 μ_{CR} = 0.074
 Diversity = 6.1802e-02 k_t = [3, 3, 2, 4]
 Generation 2950: Best = 4.002969e+02 FES = 295100 μ_F = 0.739 μ_{CR} = 0.067
 Diversity = 5.8495e-02 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 4.002967e+02 FES = 296100 μ_F = 0.748 μ_{CR} = 0.069
 Diversity = 5.8441e-02 k_t = [2, 3, 3, 2]
 Generation 2970: Best = 4.002964e+02 FES = 297100 μ_F = 0.741 μ_{CR} = 0.073
 Diversity = 6.0348e-02 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 4.002962e+02 FES = 298100 μ_F = 0.710 μ_{CR} = 0.069
 Diversity = 5.5182e-02 k_t = [2, 3, 3, 3]
 Generation 2990: Best = 4.002961e+02 FES = 299100 μ_F = 0.720 μ_{CR} = 0.071
 Diversity = 6.0128e-02 k_t = [1, 1, 1, 1]

Optimization complete.

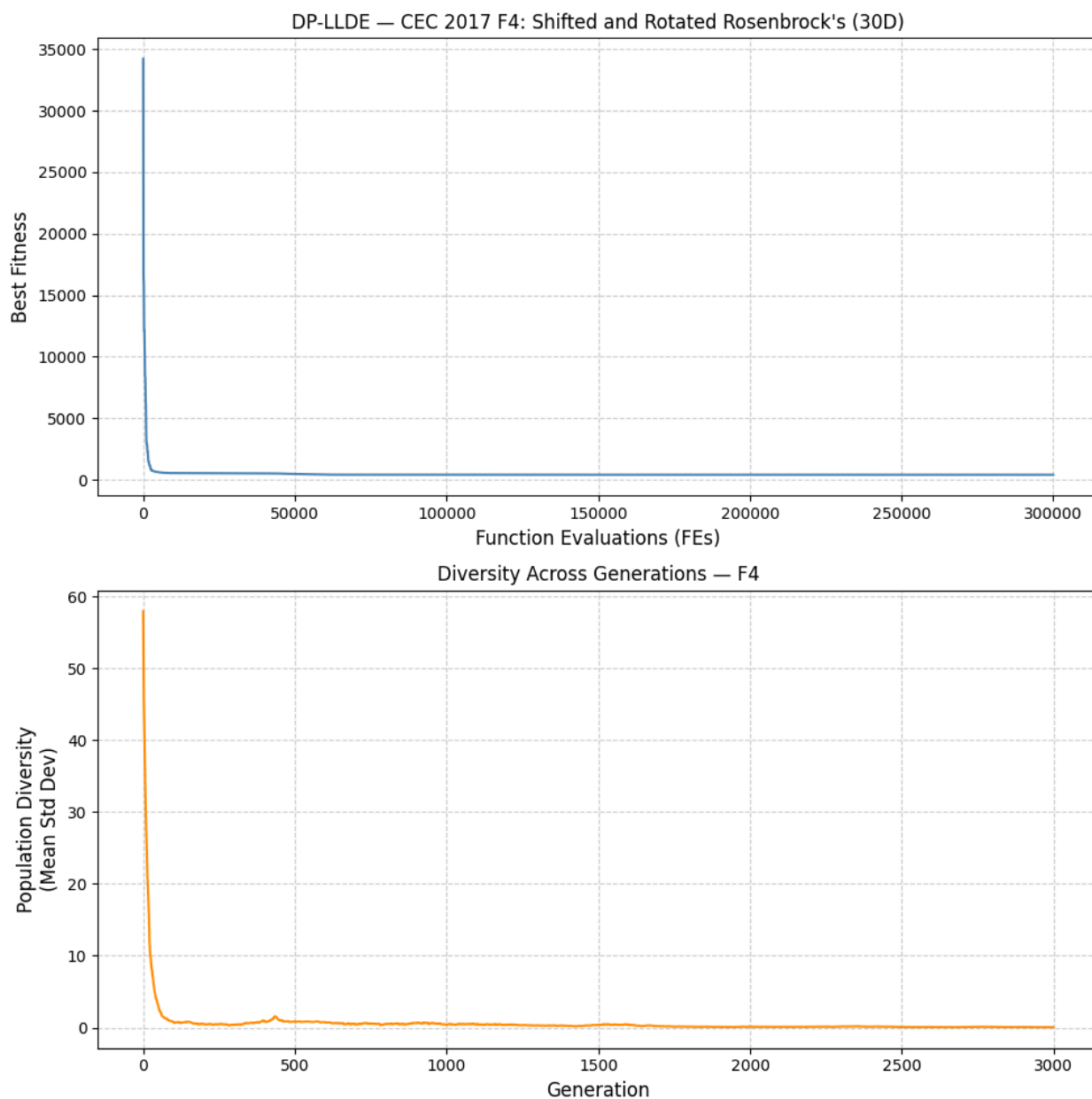
Best Fitness = 4.002958e+02

Total Function Evals = 300000

Total Successful Updates = 63579

```

=====
=
Results:
  Best Fitness      : 4.002958e+02
  Global Optimum    : 4.000000e+02
  Error (|best-optimum|) : 2.957904e-01
  Best Solution (5 dim) : [ 32.87160794    7.75909992 -55.82995448   77.7155678
19.04335138]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F4.png

```
In [ ]: # =====
# F5 – Shifted and Rotated Rastrigin
# =====
run_and_plot('F5', make_wrapper(simple.f5), 500.0, "Shifted and Rotated Rastrigin")
```

=====

=

DP-LLDE on CEC 2017 F5 – Single Run

=====

=

Function : Shifted and Rotated Rastrigin's (30D)

Bounds : [-100, 100]

Global optimum : 500.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 1.177228e+03

Generation 10: Best = 8.177005e+02 FES = 1100 $\mu_F = 0.563$ $\mu_{CR} = 0.356$

Diversity = 2.8475e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 7.318789e+02 FES = 2100 $\mu_F = 0.591$ $\mu_{CR} = 0.352$

Diversity = 2.1269e+01 $k_t = [4, 2, 4, 4]$

Generation 30: Best = 7.133249e+02 FES = 3100 $\mu_F = 0.607$ $\mu_{CR} = 0.328$

Diversity = 1.7896e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 7.133249e+02 FES = 4100 $\mu_F = 0.650$ $\mu_{CR} = 0.291$

Diversity = 1.6373e+01 $k_t = [4, 3, 4, 2]$

Generation 50: Best = 7.133249e+02 FES = 5100 $\mu_F = 0.680$ $\mu_{CR} = 0.225$

Diversity = 1.5485e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 6.938667e+02 FES = 6100 $\mu_F = 0.665$ $\mu_{CR} = 0.187$

Diversity = 1.4892e+01 $k_t = [3, 3, 4, 3]$

Generation 70: Best = 6.938667e+02 FES = 7100 $\mu_F = 0.692$ $\mu_{CR} = 0.145$

Diversity = 1.4589e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 6.938667e+02 FES = 8100 $\mu_F = 0.722$ $\mu_{CR} = 0.112$

Diversity = 1.4147e+01 $k_t = [2, 3, 4, 3]$

Generation 90: Best = 6.840546e+02 FES = 9100 $\mu_F = 0.716$ $\mu_{CR} = 0.095$

Diversity = 1.3907e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 6.699713e+02 FES = 10100 $\mu_F = 0.730$ $\mu_{CR} = 0.075$

Diversity = 1.3868e+01 $k_t = [3, 4, 4, 2]$

Generation 110: Best = 6.699713e+02 FES = 11100 $\mu_F = 0.758$ $\mu_{CR} = 0.061$

Diversity = 1.3922e+01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 6.699713e+02 FES = 12100 $\mu_F = 0.786$ $\mu_{CR} = 0.065$

Diversity = 1.4094e+01 $k_t = [3, 4, 2, 4]$

Generation 130: Best = 6.643581e+02 FES = 13100 $\mu_F = 0.812$ $\mu_{CR} = 0.070$

Diversity = 1.4170e+01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 6.559013e+02 FES = 14100 $\mu_F = 0.771$ $\mu_{CR} = 0.062$

Diversity = 1.4201e+01 $k_t = [2, 4, 2, 2]$

Generation 150: Best = 6.559013e+02 FES = 15100 $\mu_F = 0.786$ $\mu_{CR} = 0.053$

Diversity = 1.4228e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 6.549669e+02 FES = 16100 $\mu_F = 0.779$ $\mu_{CR} = 0.058$

Diversity = 1.4184e+01 $k_t = [2, 4, 3, 2]$

Generation 170: Best = 6.549669e+02 FES = 17100 $\mu_F = 0.772$ $\mu_{CR} = 0.055$

Diversity = 1.4011e+01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 6.538704e+02 FES = 18100 $\mu_F = 0.800$ $\mu_{CR} = 0.052$

Diversity = 1.3912e+01 $k_t = [3, 2, 3, 4]$

Generation 190: Best = 6.527379e+02 FES = 19100 $\mu_F = 0.704$ $\mu_{CR} = 0.062$

Diversity = 1.3887e+01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 6.397133e+02 FES = 20100 $\mu_F = 0.729$ $\mu_{CR} = 0.060$

Diversity = 1.3895e+01 $k_t = [3, 3, 3, 4]$

Generation 210: Best = 6.352001e+02 FES = 21100 $\mu_F = 0.718$ $\mu_{CR} = 0.055$

Diversity = 1.3878e+01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 6.352001e+02 FES = 22100 $\mu_F = 0.714$ $\mu_{CR} = 0.053$

Diversity = 1.3906e+01 $k_t = [3, 2, 2, 2]$

Generation	230: Best = 6.269841e+02	FES = 23100	$\mu_F = 0.577$	$\mu_{CR} = 0.060$
Diversity =	1.3904e+01	$k_t = [1, 1, 1, 1]$		
Generation	240: Best = 6.139813e+02	FES = 24100	$\mu_F = 0.593$	$\mu_{CR} = 0.059$
Diversity =	1.3832e+01	$k_t = [2, 4, 4, 3]$		
Generation	250: Best = 6.139813e+02	FES = 25100	$\mu_F = 0.611$	$\mu_{CR} = 0.050$
Diversity =	1.3825e+01	$k_t = [1, 1, 1, 1]$		
Generation	260: Best = 6.116817e+02	FES = 26100	$\mu_F = 0.607$	$\mu_{CR} = 0.052$
Diversity =	1.3803e+01	$k_t = [2, 4, 2, 2]$		
Generation	270: Best = 6.116817e+02	FES = 27100	$\mu_F = 0.576$	$\mu_{CR} = 0.039$
Diversity =	1.3803e+01	$k_t = [1, 1, 1, 1]$		
Generation	280: Best = 6.116817e+02	FES = 28100	$\mu_F = 0.638$	$\mu_{CR} = 0.045$
Diversity =	1.3826e+01	$k_t = [4, 4, 4, 4]$		
Generation	290: Best = 6.116817e+02	FES = 29100	$\mu_F = 0.582$	$\mu_{CR} = 0.047$
Diversity =	1.3813e+01	$k_t = [1, 1, 1, 1]$		
Generation	300: Best = 6.114604e+02	FES = 30100	$\mu_F = 0.604$	$\mu_{CR} = 0.046$
Diversity =	1.3776e+01	$k_t = [4, 2, 3, 2]$		
Generation	310: Best = 6.114604e+02	FES = 31100	$\mu_F = 0.669$	$\mu_{CR} = 0.056$
Diversity =	1.3757e+01	$k_t = [1, 1, 1, 1]$		
Generation	320: Best = 6.114604e+02	FES = 32100	$\mu_F = 0.607$	$\mu_{CR} = 0.056$
Diversity =	1.3752e+01	$k_t = [3, 4, 2, 3]$		
Generation	330: Best = 6.114604e+02	FES = 33100	$\mu_F = 0.615$	$\mu_{CR} = 0.060$
Diversity =	1.3745e+01	$k_t = [1, 1, 1, 1]$		
Generation	340: Best = 6.114604e+02	FES = 34100	$\mu_F = 0.562$	$\mu_{CR} = 0.043$
Diversity =	1.3726e+01	$k_t = [2, 2, 2, 2]$		
Generation	350: Best = 6.049267e+02	FES = 35100	$\mu_F = 0.582$	$\mu_{CR} = 0.045$
Diversity =	1.3695e+01	$k_t = [1, 1, 1, 1]$		
Generation	360: Best = 6.049267e+02	FES = 36100	$\mu_F = 0.484$	$\mu_{CR} = 0.064$
Diversity =	1.3708e+01	$k_t = [3, 2, 4, 3]$		
Generation	370: Best = 6.049267e+02	FES = 37100	$\mu_F = 0.502$	$\mu_{CR} = 0.055$
Diversity =	1.3681e+01	$k_t = [1, 1, 1, 1]$		
Generation	380: Best = 6.049267e+02	FES = 38100	$\mu_F = 0.468$	$\mu_{CR} = 0.045$
Diversity =	1.3605e+01	$k_t = [2, 2, 4, 3]$		
Generation	390: Best = 6.043452e+02	FES = 39100	$\mu_F = 0.567$	$\mu_{CR} = 0.049$
Diversity =	1.3577e+01	$k_t = [1, 1, 1, 1]$		
Generation	400: Best = 6.043452e+02	FES = 40100	$\mu_F = 0.567$	$\mu_{CR} = 0.059$
Diversity =	1.3509e+01	$k_t = [2, 2, 4, 4]$		
Generation	410: Best = 6.043452e+02	FES = 41100	$\mu_F = 0.581$	$\mu_{CR} = 0.065$
Diversity =	1.3525e+01	$k_t = [1, 1, 1, 1]$		
Generation	420: Best = 6.035816e+02	FES = 42100	$\mu_F = 0.551$	$\mu_{CR} = 0.057$
Diversity =	1.3500e+01	$k_t = [3, 4, 3, 3]$		
Generation	430: Best = 6.035816e+02	FES = 43100	$\mu_F = 0.558$	$\mu_{CR} = 0.063$
Diversity =	1.3509e+01	$k_t = [1, 1, 1, 1]$		
Generation	440: Best = 6.035816e+02	FES = 44100	$\mu_F = 0.486$	$\mu_{CR} = 0.050$
Diversity =	1.3494e+01	$k_t = [3, 4, 4, 2]$		
Generation	450: Best = 6.035816e+02	FES = 45100	$\mu_F = 0.529$	$\mu_{CR} = 0.049$
Diversity =	1.3445e+01	$k_t = [1, 1, 1, 1]$		
Generation	460: Best = 6.024420e+02	FES = 46100	$\mu_F = 0.520$	$\mu_{CR} = 0.049$
Diversity =	1.3454e+01	$k_t = [3, 3, 4, 3]$		
Generation	470: Best = 5.962112e+02	FES = 47100	$\mu_F = 0.536$	$\mu_{CR} = 0.050$
Diversity =	1.3407e+01	$k_t = [1, 1, 1, 1]$		
Generation	480: Best = 5.962112e+02	FES = 48100	$\mu_F = 0.498$	$\mu_{CR} = 0.050$
Diversity =	1.3388e+01	$k_t = [4, 3, 2, 3]$		
Generation	490: Best = 5.962112e+02	FES = 49100	$\mu_F = 0.499$	$\mu_{CR} = 0.040$
Diversity =	1.3372e+01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 5.949415e+02	FES = 50100	$\mu_F = 0.588$	$\mu_{CR} = 0.052$
Diversity = 1.3289e+01	k_t = [2, 4, 4, 4]		
Generation 510: Best = 5.949415e+02	FES = 51100	$\mu_F = 0.713$	$\mu_{CR} = 0.051$
Diversity = 1.3279e+01	k_t = [1, 1, 1, 1]		
Generation 520: Best = 5.949415e+02	FES = 52100	$\mu_F = 0.700$	$\mu_{CR} = 0.047$
Diversity = 1.3081e+01	k_t = [2, 3, 2, 4]		
Generation 530: Best = 5.949415e+02	FES = 53100	$\mu_F = 0.664$	$\mu_{CR} = 0.042$
Diversity = 1.3080e+01	k_t = [1, 1, 1, 1]		
Generation 540: Best = 5.899171e+02	FES = 54100	$\mu_F = 0.722$	$\mu_{CR} = 0.043$
Diversity = 1.3060e+01	k_t = [3, 3, 3, 4]		
Generation 550: Best = 5.868200e+02	FES = 55100	$\mu_F = 0.685$	$\mu_{CR} = 0.045$
Diversity = 1.2945e+01	k_t = [1, 1, 1, 1]		
Generation 560: Best = 5.858263e+02	FES = 56100	$\mu_F = 0.596$	$\mu_{CR} = 0.047$
Diversity = 1.2970e+01	k_t = [2, 4, 2, 3]		
Generation 570: Best = 5.855366e+02	FES = 57100	$\mu_F = 0.556$	$\mu_{CR} = 0.048$
Diversity = 1.2960e+01	k_t = [1, 1, 1, 1]		
Generation 580: Best = 5.835638e+02	FES = 58100	$\mu_F = 0.571$	$\mu_{CR} = 0.038$
Diversity = 1.2889e+01	k_t = [4, 2, 4, 4]		
Generation 590: Best = 5.835638e+02	FES = 59100	$\mu_F = 0.551$	$\mu_{CR} = 0.042$
Diversity = 1.2918e+01	k_t = [1, 1, 1, 1]		
Generation 600: Best = 5.772085e+02	FES = 60100	$\mu_F = 0.462$	$\mu_{CR} = 0.046$
Diversity = 1.2913e+01	k_t = [3, 4, 4, 2]		
Generation 610: Best = 5.772085e+02	FES = 61100	$\mu_F = 0.431$	$\mu_{CR} = 0.056$
Diversity = 1.2910e+01	k_t = [1, 1, 1, 1]		
Generation 620: Best = 5.772085e+02	FES = 62100	$\mu_F = 0.355$	$\mu_{CR} = 0.061$
Diversity = 1.2900e+01	k_t = [4, 3, 2, 4]		
Generation 630: Best = 5.772085e+02	FES = 63100	$\mu_F = 0.400$	$\mu_{CR} = 0.057$
Diversity = 1.2867e+01	k_t = [1, 1, 1, 1]		
Generation 640: Best = 5.772085e+02	FES = 64100	$\mu_F = 0.416$	$\mu_{CR} = 0.053$
Diversity = 1.2830e+01	k_t = [4, 2, 3, 4]		
Generation 650: Best = 5.772085e+02	FES = 65100	$\mu_F = 0.381$	$\mu_{CR} = 0.056$
Diversity = 1.2827e+01	k_t = [1, 1, 1, 1]		
Generation 660: Best = 5.772021e+02	FES = 66100	$\mu_F = 0.407$	$\mu_{CR} = 0.048$
Diversity = 1.2825e+01	k_t = [2, 2, 2, 2]		
Generation 670: Best = 5.772021e+02	FES = 67100	$\mu_F = 0.451$	$\mu_{CR} = 0.074$
Diversity = 1.2795e+01	k_t = [1, 1, 1, 1]		
Generation 680: Best = 5.772021e+02	FES = 68100	$\mu_F = 0.467$	$\mu_{CR} = 0.066$
Diversity = 1.2772e+01	k_t = [3, 2, 4, 4]		
Generation 690: Best = 5.725878e+02	FES = 69100	$\mu_F = 0.483$	$\mu_{CR} = 0.044$
Diversity = 1.2768e+01	k_t = [1, 1, 1, 1]		
Generation 700: Best = 5.725878e+02	FES = 70100	$\mu_F = 0.583$	$\mu_{CR} = 0.042$
Diversity = 1.2765e+01	k_t = [4, 4, 3, 2]		
Generation 710: Best = 5.725878e+02	FES = 71100	$\mu_F = 0.585$	$\mu_{CR} = 0.039$
Diversity = 1.2762e+01	k_t = [1, 1, 1, 1]		
Generation 720: Best = 5.725878e+02	FES = 72100	$\mu_F = 0.578$	$\mu_{CR} = 0.048$
Diversity = 1.2775e+01	k_t = [4, 3, 3, 2]		
Generation 730: Best = 5.725878e+02	FES = 73100	$\mu_F = 0.585$	$\mu_{CR} = 0.050$
Diversity = 1.2754e+01	k_t = [1, 1, 1, 1]		
Generation 740: Best = 5.725878e+02	FES = 74100	$\mu_F = 0.574$	$\mu_{CR} = 0.048$
Diversity = 1.2734e+01	k_t = [3, 2, 4, 4]		
Generation 750: Best = 5.725878e+02	FES = 75100	$\mu_F = 0.519$	$\mu_{CR} = 0.056$
Diversity = 1.2733e+01	k_t = [1, 1, 1, 1]		
Generation 760: Best = 5.725878e+02	FES = 76100	$\mu_F = 0.412$	$\mu_{CR} = 0.053$
Diversity = 1.2723e+01	k_t = [3, 3, 4, 2]		

Generation 770: Best = 5.725878e+02	FES = 77100	$\mu_F = 0.439$	$\mu_{CR} = 0.056$
Diversity = 1.2717e+01	k_t = [1, 1, 1, 1]		
Generation 780: Best = 5.711969e+02	FES = 78100	$\mu_F = 0.497$	$\mu_{CR} = 0.054$
Diversity = 1.2707e+01	k_t = [3, 3, 3, 3]		
Generation 790: Best = 5.711969e+02	FES = 79100	$\mu_F = 0.464$	$\mu_{CR} = 0.056$
Diversity = 1.2710e+01	k_t = [1, 1, 1, 1]		
Generation 800: Best = 5.708718e+02	FES = 80100	$\mu_F = 0.410$	$\mu_{CR} = 0.050$
Diversity = 1.2683e+01	k_t = [3, 2, 3, 4]		
Generation 810: Best = 5.674198e+02	FES = 81100	$\mu_F = 0.382$	$\mu_{CR} = 0.055$
Diversity = 1.2682e+01	k_t = [1, 1, 1, 1]		
Generation 820: Best = 5.672005e+02	FES = 82100	$\mu_F = 0.386$	$\mu_{CR} = 0.054$
Diversity = 1.2680e+01	k_t = [3, 3, 4, 2]		
Generation 830: Best = 5.672005e+02	FES = 83100	$\mu_F = 0.349$	$\mu_{CR} = 0.065$
Diversity = 1.2673e+01	k_t = [1, 1, 1, 1]		
Generation 840: Best = 5.655317e+02	FES = 84100	$\mu_F = 0.348$	$\mu_{CR} = 0.064$
Diversity = 1.2670e+01	k_t = [4, 3, 3, 3]		
Generation 850: Best = 5.655317e+02	FES = 85100	$\mu_F = 0.312$	$\mu_{CR} = 0.062$
Diversity = 1.2666e+01	k_t = [1, 1, 1, 1]		
Generation 860: Best = 5.652113e+02	FES = 86100	$\mu_F = 0.387$	$\mu_{CR} = 0.067$
Diversity = 1.2595e+01	k_t = [2, 4, 2, 4]		
Generation 870: Best = 5.652113e+02	FES = 87100	$\mu_F = 0.377$	$\mu_{CR} = 0.063$
Diversity = 1.2587e+01	k_t = [1, 1, 1, 1]		
Generation 880: Best = 5.652113e+02	FES = 88100	$\mu_F = 0.409$	$\mu_{CR} = 0.050$
Diversity = 1.2585e+01	k_t = [3, 4, 3, 2]		
Generation 890: Best = 5.651332e+02	FES = 89100	$\mu_F = 0.317$	$\mu_{CR} = 0.049$
Diversity = 1.2584e+01	k_t = [1, 1, 1, 1]		
Generation 900: Best = 5.634010e+02	FES = 90100	$\mu_F = 0.331$	$\mu_{CR} = 0.065$
Diversity = 1.2535e+01	k_t = [2, 2, 4, 4]		
Generation 910: Best = 5.623391e+02	FES = 91100	$\mu_F = 0.347$	$\mu_{CR} = 0.054$
Diversity = 1.2531e+01	k_t = [1, 1, 1, 1]		
Generation 920: Best = 5.623391e+02	FES = 92100	$\mu_F = 0.356$	$\mu_{CR} = 0.062$
Diversity = 1.2486e+01	k_t = [4, 3, 3, 4]		
Generation 930: Best = 5.614857e+02	FES = 93100	$\mu_F = 0.311$	$\mu_{CR} = 0.058$
Diversity = 1.2479e+01	k_t = [1, 1, 1, 1]		
Generation 940: Best = 5.614835e+02	FES = 94100	$\mu_F = 0.248$	$\mu_{CR} = 0.057$
Diversity = 1.2480e+01	k_t = [4, 3, 4, 4]		
Generation 950: Best = 5.592890e+02	FES = 95100	$\mu_F = 0.226$	$\mu_{CR} = 0.061$
Diversity = 1.2455e+01	k_t = [1, 1, 1, 1]		
Generation 960: Best = 5.588884e+02	FES = 96100	$\mu_F = 0.207$	$\mu_{CR} = 0.055$
Diversity = 1.2448e+01	k_t = [4, 4, 4, 2]		
Generation 970: Best = 5.581356e+02	FES = 97100	$\mu_F = 0.221$	$\mu_{CR} = 0.065$
Diversity = 1.2440e+01	k_t = [1, 1, 1, 1]		
Generation 980: Best = 5.580791e+02	FES = 98100	$\mu_F = 0.217$	$\mu_{CR} = 0.067$
Diversity = 1.2434e+01	k_t = [4, 4, 3, 2]		
Generation 990: Best = 5.580791e+02	FES = 99100	$\mu_F = 0.341$	$\mu_{CR} = 0.055$
Diversity = 1.2422e+01	k_t = [1, 1, 1, 1]		
Generation 1000: Best = 5.578339e+02	FES = 100100	$\mu_F = 0.371$	$\mu_{CR} = 0.059$
Diversity = 1.2422e+01	k_t = [4, 3, 2, 3]		
Generation 1010: Best = 5.578339e+02	FES = 101100	$\mu_F = 0.270$	$\mu_{CR} = 0.057$
Diversity = 1.2417e+01	k_t = [1, 1, 1, 1]		
Generation 1020: Best = 5.578339e+02	FES = 102100	$\mu_F = 0.310$	$\mu_{CR} = 0.040$
Diversity = 1.2386e+01	k_t = [3, 3, 3, 4]		
Generation 1030: Best = 5.578339e+02	FES = 103100	$\mu_F = 0.273$	$\mu_{CR} = 0.048$
Diversity = 1.2385e+01	k_t = [1, 1, 1, 1]		

Generation 1040: Best = 5.577415e+02 FES = 104100 μ_F = 0.443 μ_{CR} = 0.059
 Diversity = 1.2379e+01 k_t = [4, 2, 4, 2]
 Generation 1050: Best = 5.577415e+02 FES = 105100 μ_F = 0.535 μ_{CR} = 0.054
 Diversity = 1.2382e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 5.577415e+02 FES = 106100 μ_F = 0.604 μ_{CR} = 0.058
 Diversity = 1.2366e+01 k_t = [3, 4, 3, 3]
 Generation 1070: Best = 5.552980e+02 FES = 107100 μ_F = 0.663 μ_{CR} = 0.050
 Diversity = 1.2368e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 5.544406e+02 FES = 108100 μ_F = 0.707 μ_{CR} = 0.048
 Diversity = 1.2383e+01 k_t = [4, 3, 4, 2]
 Generation 1090: Best = 5.544406e+02 FES = 109100 μ_F = 0.695 μ_{CR} = 0.051
 Diversity = 1.2383e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 5.544406e+02 FES = 110100 μ_F = 0.525 μ_{CR} = 0.045
 Diversity = 1.2383e+01 k_t = [3, 4, 4, 2]
 Generation 1110: Best = 5.544406e+02 FES = 111100 μ_F = 0.474 μ_{CR} = 0.050
 Diversity = 1.2383e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 5.544406e+02 FES = 112100 μ_F = 0.367 μ_{CR} = 0.064
 Diversity = 1.2384e+01 k_t = [3, 3, 3, 3]
 Generation 1130: Best = 5.541826e+02 FES = 113100 μ_F = 0.292 μ_{CR} = 0.091
 Diversity = 1.2384e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 5.541826e+02 FES = 114100 μ_F = 0.312 μ_{CR} = 0.089
 Diversity = 1.2377e+01 k_t = [4, 4, 4, 3]
 Generation 1150: Best = 5.541826e+02 FES = 115100 μ_F = 0.323 μ_{CR} = 0.101
 Diversity = 1.2369e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 5.496414e+02 FES = 116100 μ_F = 0.352 μ_{CR} = 0.077
 Diversity = 1.2367e+01 k_t = [2, 3, 3, 4]
 Generation 1170: Best = 5.496414e+02 FES = 117100 μ_F = 0.327 μ_{CR} = 0.075
 Diversity = 1.2368e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 5.495839e+02 FES = 118100 μ_F = 0.425 μ_{CR} = 0.061
 Diversity = 1.2362e+01 k_t = [4, 4, 3, 3]
 Generation 1190: Best = 5.495839e+02 FES = 119100 μ_F = 0.461 μ_{CR} = 0.060
 Diversity = 1.2360e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 5.472221e+02 FES = 120100 μ_F = 0.504 μ_{CR} = 0.045
 Diversity = 1.2272e+01 k_t = [3, 2, 4, 4]
 Generation 1210: Best = 5.472221e+02 FES = 121100 μ_F = 0.520 μ_{CR} = 0.034
 Diversity = 1.2277e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 5.465130e+02 FES = 122100 μ_F = 0.594 μ_{CR} = 0.035
 Diversity = 1.2189e+01 k_t = [4, 4, 2, 3]
 Generation 1230: Best = 5.465130e+02 FES = 123100 μ_F = 0.487 μ_{CR} = 0.046
 Diversity = 1.2190e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 5.465130e+02 FES = 124100 μ_F = 0.400 μ_{CR} = 0.044
 Diversity = 1.2213e+01 k_t = [2, 4, 2, 4]
 Generation 1250: Best = 5.465130e+02 FES = 125100 μ_F = 0.354 μ_{CR} = 0.048
 Diversity = 1.2218e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 5.453180e+02 FES = 126100 μ_F = 0.427 μ_{CR} = 0.057
 Diversity = 1.2220e+01 k_t = [3, 4, 4, 3]
 Generation 1270: Best = 5.453180e+02 FES = 127100 μ_F = 0.405 μ_{CR} = 0.063
 Diversity = 1.2220e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 5.453180e+02 FES = 128100 μ_F = 0.302 μ_{CR} = 0.046
 Diversity = 1.2219e+01 k_t = [2, 3, 4, 2]
 Generation 1290: Best = 5.453180e+02 FES = 129100 μ_F = 0.320 μ_{CR} = 0.053
 Diversity = 1.2205e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 5.453180e+02 FES = 130100 μ_F = 0.211 μ_{CR} = 0.074
 Diversity = 1.2201e+01 k_t = [2, 4, 4, 3]

Generation 1310: Best = 5.453180e+02 FES = 131100 μ_F = 0.236 μ_{CR} = 0.060
 Diversity = 1.2169e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 5.453180e+02 FES = 132100 μ_F = 0.366 μ_{CR} = 0.052
 Diversity = 1.2118e+01 k_t = [3, 3, 3, 4]
 Generation 1330: Best = 5.453180e+02 FES = 133100 μ_F = 0.350 μ_{CR} = 0.043
 Diversity = 1.2106e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 5.453180e+02 FES = 134100 μ_F = 0.357 μ_{CR} = 0.056
 Diversity = 1.2105e+01 k_t = [3, 2, 4, 2]
 Generation 1350: Best = 5.446797e+02 FES = 135100 μ_F = 0.353 μ_{CR} = 0.049
 Diversity = 1.2086e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 5.446797e+02 FES = 136100 μ_F = 0.446 μ_{CR} = 0.057
 Diversity = 1.2088e+01 k_t = [3, 3, 4, 4]
 Generation 1370: Best = 5.441389e+02 FES = 137100 μ_F = 0.357 μ_{CR} = 0.058
 Diversity = 1.2087e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 5.441389e+02 FES = 138100 μ_F = 0.347 μ_{CR} = 0.066
 Diversity = 1.1997e+01 k_t = [2, 4, 4, 4]
 Generation 1390: Best = 5.441389e+02 FES = 139100 μ_F = 0.394 μ_{CR} = 0.054
 Diversity = 1.1998e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 5.441389e+02 FES = 140100 μ_F = 0.346 μ_{CR} = 0.044
 Diversity = 1.1997e+01 k_t = [3, 3, 3, 3]
 Generation 1410: Best = 5.429598e+02 FES = 141100 μ_F = 0.395 μ_{CR} = 0.040
 Diversity = 1.1999e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 5.429598e+02 FES = 142100 μ_F = 0.382 μ_{CR} = 0.054
 Diversity = 1.2001e+01 k_t = [3, 4, 3, 2]
 Generation 1430: Best = 5.429598e+02 FES = 143100 μ_F = 0.332 μ_{CR} = 0.053
 Diversity = 1.2002e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 5.423472e+02 FES = 144100 μ_F = 0.393 μ_{CR} = 0.043
 Diversity = 1.2002e+01 k_t = [2, 2, 2, 3]
 Generation 1450: Best = 5.423472e+02 FES = 145100 μ_F = 0.470 μ_{CR} = 0.033
 Diversity = 1.2002e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 5.418530e+02 FES = 146100 μ_F = 0.457 μ_{CR} = 0.035
 Diversity = 1.1957e+01 k_t = [4, 3, 3, 4]
 Generation 1470: Best = 5.418530e+02 FES = 147100 μ_F = 0.446 μ_{CR} = 0.032
 Diversity = 1.1959e+01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 5.418530e+02 FES = 148100 μ_F = 0.491 μ_{CR} = 0.046
 Diversity = 1.1984e+01 k_t = [3, 3, 3, 4]
 Generation 1490: Best = 5.417736e+02 FES = 149100 μ_F = 0.406 μ_{CR} = 0.048
 Diversity = 1.1986e+01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 5.408659e+02 FES = 150100 μ_F = 0.369 μ_{CR} = 0.049
 Diversity = 1.2046e+01 k_t = [4, 4, 4, 2]
 Generation 1510: Best = 5.408659e+02 FES = 151100 μ_F = 0.352 μ_{CR} = 0.041
 Diversity = 1.2049e+01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 5.408659e+02 FES = 152100 μ_F = 0.382 μ_{CR} = 0.050
 Diversity = 1.2037e+01 k_t = [3, 2, 3, 4]
 Generation 1530: Best = 5.408659e+02 FES = 153100 μ_F = 0.342 μ_{CR} = 0.047
 Diversity = 1.2038e+01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 5.408659e+02 FES = 154100 μ_F = 0.398 μ_{CR} = 0.037
 Diversity = 1.2028e+01 k_t = [4, 4, 2, 2]
 Generation 1550: Best = 5.408659e+02 FES = 155100 μ_F = 0.379 μ_{CR} = 0.039
 Diversity = 1.2027e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 5.408659e+02 FES = 156100 μ_F = 0.357 μ_{CR} = 0.051
 Diversity = 1.2027e+01 k_t = [4, 2, 2, 2]
 Generation 1570: Best = 5.398155e+02 FES = 157100 μ_F = 0.428 μ_{CR} = 0.050
 Diversity = 1.2023e+01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 5.398155e+02 FES = 158100 μ_F = 0.454 μ_{CR} = 0.040
 Diversity = 1.2013e+01 k_t = [4, 4, 4, 3]
 Generation 1590: Best = 5.398155e+02 FES = 159100 μ_F = 0.485 μ_{CR} = 0.049
 Diversity = 1.2013e+01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 5.398155e+02 FES = 160100 μ_F = 0.550 μ_{CR} = 0.050
 Diversity = 1.2013e+01 k_t = [4, 3, 2, 2]
 Generation 1610: Best = 5.398155e+02 FES = 161100 μ_F = 0.521 μ_{CR} = 0.043
 Diversity = 1.2015e+01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 5.398155e+02 FES = 162100 μ_F = 0.454 μ_{CR} = 0.035
 Diversity = 1.2015e+01 k_t = [4, 4, 3, 3]
 Generation 1630: Best = 5.393736e+02 FES = 163100 μ_F = 0.386 μ_{CR} = 0.035
 Diversity = 1.2017e+01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 5.393736e+02 FES = 164100 μ_F = 0.526 μ_{CR} = 0.047
 Diversity = 1.1971e+01 k_t = [4, 3, 2, 4]
 Generation 1650: Best = 5.393736e+02 FES = 165100 μ_F = 0.599 μ_{CR} = 0.042
 Diversity = 1.1971e+01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 5.393699e+02 FES = 166100 μ_F = 0.619 μ_{CR} = 0.042
 Diversity = 1.1947e+01 k_t = [4, 3, 2, 4]
 Generation 1670: Best = 5.393699e+02 FES = 167100 μ_F = 0.539 μ_{CR} = 0.036
 Diversity = 1.1948e+01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 5.377843e+02 FES = 168100 μ_F = 0.519 μ_{CR} = 0.035
 Diversity = 1.1951e+01 k_t = [2, 2, 3, 3]
 Generation 1690: Best = 5.377843e+02 FES = 169100 μ_F = 0.485 μ_{CR} = 0.039
 Diversity = 1.1953e+01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 5.377843e+02 FES = 170100 μ_F = 0.501 μ_{CR} = 0.053
 Diversity = 1.1953e+01 k_t = [3, 2, 4, 2]
 Generation 1710: Best = 5.377843e+02 FES = 171100 μ_F = 0.414 μ_{CR} = 0.064
 Diversity = 1.1956e+01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 5.377787e+02 FES = 172100 μ_F = 0.466 μ_{CR} = 0.051
 Diversity = 1.1916e+01 k_t = [2, 4, 2, 4]
 Generation 1730: Best = 5.377787e+02 FES = 173100 μ_F = 0.349 μ_{CR} = 0.046
 Diversity = 1.1917e+01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 5.377787e+02 FES = 174100 μ_F = 0.457 μ_{CR} = 0.052
 Diversity = 1.1815e+01 k_t = [3, 2, 4, 4]
 Generation 1750: Best = 5.377787e+02 FES = 175100 μ_F = 0.389 μ_{CR} = 0.051
 Diversity = 1.1815e+01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 5.377787e+02 FES = 176100 μ_F = 0.462 μ_{CR} = 0.047
 Diversity = 1.1820e+01 k_t = [3, 3, 2, 2]
 Generation 1770: Best = 5.377787e+02 FES = 177100 μ_F = 0.497 μ_{CR} = 0.044
 Diversity = 1.1826e+01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 5.377787e+02 FES = 178100 μ_F = 0.437 μ_{CR} = 0.033
 Diversity = 1.1828e+01 k_t = [3, 4, 3, 3]
 Generation 1790: Best = 5.377787e+02 FES = 179100 μ_F = 0.526 μ_{CR} = 0.028
 Diversity = 1.1832e+01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 5.377787e+02 FES = 180100 μ_F = 0.618 μ_{CR} = 0.035
 Diversity = 1.1712e+01 k_t = [2, 4, 3, 4]
 Generation 1810: Best = 5.358005e+02 FES = 181100 μ_F = 0.596 μ_{CR} = 0.035
 Diversity = 1.1717e+01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 5.358005e+02 FES = 182100 μ_F = 0.566 μ_{CR} = 0.038
 Diversity = 1.1718e+01 k_t = [3, 4, 3, 2]
 Generation 1830: Best = 5.347057e+02 FES = 183100 μ_F = 0.494 μ_{CR} = 0.040
 Diversity = 1.1708e+01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 5.347057e+02 FES = 184100 μ_F = 0.421 μ_{CR} = 0.040
 Diversity = 1.1711e+01 k_t = [2, 2, 3, 2]

Generation 1850: Best = 5.347057e+02	FES = 185100	$\mu_F = 0.416$	$\mu_{CR} = 0.038$
Diversity = 1.1741e+01	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 5.345600e+02	FES = 186100	$\mu_F = 0.315$	$\mu_{CR} = 0.059$
Diversity = 1.1744e+01	k_t = [4, 4, 3, 3]		
Generation 1870: Best = 5.344134e+02	FES = 187100	$\mu_F = 0.275$	$\mu_{CR} = 0.068$
Diversity = 1.1745e+01	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 5.341728e+02	FES = 188100	$\mu_F = 0.230$	$\mu_{CR} = 0.068$
Diversity = 1.1762e+01	k_t = [3, 3, 3, 3]		
Generation 1890: Best = 5.341482e+02	FES = 189100	$\mu_F = 0.267$	$\mu_{CR} = 0.070$
Diversity = 1.1762e+01	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 5.341482e+02	FES = 190100	$\mu_F = 0.363$	$\mu_{CR} = 0.071$
Diversity = 1.1717e+01	k_t = [3, 4, 4, 2]		
Generation 1910: Best = 5.341482e+02	FES = 191100	$\mu_F = 0.399$	$\mu_{CR} = 0.051$
Diversity = 1.1717e+01	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 5.341482e+02	FES = 192100	$\mu_F = 0.452$	$\mu_{CR} = 0.062$
Diversity = 1.1720e+01	k_t = [2, 2, 2, 2]		
Generation 1930: Best = 5.341482e+02	FES = 193100	$\mu_F = 0.375$	$\mu_{CR} = 0.058$
Diversity = 1.1721e+01	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 5.341482e+02	FES = 194100	$\mu_F = 0.439$	$\mu_{CR} = 0.049$
Diversity = 1.1639e+01	k_t = [2, 4, 2, 4]		
Generation 1950: Best = 5.341482e+02	FES = 195100	$\mu_F = 0.325$	$\mu_{CR} = 0.049$
Diversity = 1.1640e+01	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 5.341482e+02	FES = 196100	$\mu_F = 0.287$	$\mu_{CR} = 0.047$
Diversity = 1.1641e+01	k_t = [2, 2, 3, 3]		
Generation 1970: Best = 5.341482e+02	FES = 197100	$\mu_F = 0.200$	$\mu_{CR} = 0.052$
Diversity = 1.1639e+01	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 5.340954e+02	FES = 198100	$\mu_F = 0.274$	$\mu_{CR} = 0.051$
Diversity = 1.1502e+01	k_t = [2, 2, 3, 4]		
Generation 1990: Best = 5.340954e+02	FES = 199100	$\mu_F = 0.296$	$\mu_{CR} = 0.061$
Diversity = 1.1501e+01	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 5.340954e+02	FES = 200100	$\mu_F = 0.318$	$\mu_{CR} = 0.063$
Diversity = 1.1495e+01	k_t = [3, 4, 2, 4]		
Generation 2010: Best = 5.339691e+02	FES = 201100	$\mu_F = 0.343$	$\mu_{CR} = 0.065$
Diversity = 1.1495e+01	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 5.339691e+02	FES = 202100	$\mu_F = 0.287$	$\mu_{CR} = 0.068$
Diversity = 1.1495e+01	k_t = [3, 4, 2, 3]		
Generation 2030: Best = 5.339691e+02	FES = 203100	$\mu_F = 0.293$	$\mu_{CR} = 0.055$
Diversity = 1.1495e+01	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 5.339691e+02	FES = 204100	$\mu_F = 0.368$	$\mu_{CR} = 0.052$
Diversity = 1.1468e+01	k_t = [2, 2, 2, 4]		
Generation 2050: Best = 5.339691e+02	FES = 205100	$\mu_F = 0.437$	$\mu_{CR} = 0.041$
Diversity = 1.1466e+01	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 5.339691e+02	FES = 206100	$\mu_F = 0.412$	$\mu_{CR} = 0.049$
Diversity = 1.1467e+01	k_t = [3, 3, 2, 2]		
Generation 2070: Best = 5.339691e+02	FES = 207100	$\mu_F = 0.395$	$\mu_{CR} = 0.070$
Diversity = 1.1467e+01	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 5.339691e+02	FES = 208100	$\mu_F = 0.331$	$\mu_{CR} = 0.060$
Diversity = 1.1471e+01	k_t = [3, 2, 2, 2]		
Generation 2090: Best = 5.339691e+02	FES = 209100	$\mu_F = 0.333$	$\mu_{CR} = 0.058$
Diversity = 1.1445e+01	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 5.339691e+02	FES = 210100	$\mu_F = 0.353$	$\mu_{CR} = 0.062$
Diversity = 1.1412e+01	k_t = [3, 4, 2, 2]		
Generation 2110: Best = 5.339691e+02	FES = 211100	$\mu_F = 0.377$	$\mu_{CR} = 0.050$
Diversity = 1.1413e+01	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 5.339691e+02 FES = 212100 μ_F = 0.327 μ_{CR} = 0.056
 Diversity = 1.1411e+01 k_t = [3, 2, 3, 2]
 Generation 2130: Best = 5.339691e+02 FES = 213100 μ_F = 0.371 μ_{CR} = 0.058
 Diversity = 1.1412e+01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 5.339691e+02 FES = 214100 μ_F = 0.302 μ_{CR} = 0.051
 Diversity = 1.1413e+01 k_t = [2, 4, 4, 4]
 Generation 2150: Best = 5.324655e+02 FES = 215100 μ_F = 0.511 μ_{CR} = 0.053
 Diversity = 1.1410e+01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 5.324655e+02 FES = 216100 μ_F = 0.522 μ_{CR} = 0.053
 Diversity = 1.1394e+01 k_t = [2, 4, 3, 4]
 Generation 2170: Best = 5.321984e+02 FES = 217100 μ_F = 0.476 μ_{CR} = 0.051
 Diversity = 1.1388e+01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 5.321984e+02 FES = 218100 μ_F = 0.434 μ_{CR} = 0.054
 Diversity = 1.1387e+01 k_t = [4, 2, 2, 3]
 Generation 2190: Best = 5.321984e+02 FES = 219100 μ_F = 0.431 μ_{CR} = 0.051
 Diversity = 1.1387e+01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 5.321984e+02 FES = 220100 μ_F = 0.435 μ_{CR} = 0.051
 Diversity = 1.1387e+01 k_t = [2, 2, 3, 2]
 Generation 2210: Best = 5.321984e+02 FES = 221100 μ_F = 0.439 μ_{CR} = 0.055
 Diversity = 1.1387e+01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 5.321984e+02 FES = 222100 μ_F = 0.478 μ_{CR} = 0.044
 Diversity = 1.1385e+01 k_t = [3, 3, 3, 3]
 Generation 2230: Best = 5.315007e+02 FES = 223100 μ_F = 0.459 μ_{CR} = 0.032
 Diversity = 1.1381e+01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 5.315007e+02 FES = 224100 μ_F = 0.479 μ_{CR} = 0.064
 Diversity = 1.1381e+01 k_t = [4, 4, 2, 2]
 Generation 2250: Best = 5.315007e+02 FES = 225100 μ_F = 0.474 μ_{CR} = 0.050
 Diversity = 1.1382e+01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 5.315007e+02 FES = 226100 μ_F = 0.484 μ_{CR} = 0.041
 Diversity = 1.1381e+01 k_t = [2, 4, 3, 2]
 Generation 2270: Best = 5.315007e+02 FES = 227100 μ_F = 0.514 μ_{CR} = 0.046
 Diversity = 1.1384e+01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 5.315007e+02 FES = 228100 μ_F = 0.508 μ_{CR} = 0.047
 Diversity = 1.1385e+01 k_t = [2, 4, 3, 3]
 Generation 2290: Best = 5.315007e+02 FES = 229100 μ_F = 0.633 μ_{CR} = 0.041
 Diversity = 1.1380e+01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 5.315007e+02 FES = 230100 μ_F = 0.648 μ_{CR} = 0.038
 Diversity = 1.1315e+01 k_t = [3, 3, 2, 4]
 Generation 2310: Best = 5.315007e+02 FES = 231100 μ_F = 0.658 μ_{CR} = 0.038
 Diversity = 1.1315e+01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 5.315007e+02 FES = 232100 μ_F = 0.598 μ_{CR} = 0.042
 Diversity = 1.1299e+01 k_t = [4, 2, 2, 2]
 Generation 2330: Best = 5.315007e+02 FES = 233100 μ_F = 0.491 μ_{CR} = 0.043
 Diversity = 1.1299e+01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 5.315007e+02 FES = 234100 μ_F = 0.406 μ_{CR} = 0.050
 Diversity = 1.1298e+01 k_t = [2, 4, 4, 2]
 Generation 2350: Best = 5.315007e+02 FES = 235100 μ_F = 0.544 μ_{CR} = 0.051
 Diversity = 1.1287e+01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 5.315007e+02 FES = 236100 μ_F = 0.720 μ_{CR} = 0.064
 Diversity = 1.1170e+01 k_t = [2, 4, 2, 4]
 Generation 2370: Best = 5.315007e+02 FES = 237100 μ_F = 0.687 μ_{CR} = 0.066
 Diversity = 1.1119e+01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 5.315007e+02 FES = 238100 μ_F = 0.722 μ_{CR} = 0.054
 Diversity = 1.1038e+01 k_t = [2, 4, 4, 2]

Generation 2390: Best = 5.312438e+02 FES = 239100 μ_F = 0.709 μ_{CR} = 0.047
 Diversity = 1.0961e+01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 5.312438e+02 FES = 240100 μ_F = 0.703 μ_{CR} = 0.061
 Diversity = 1.0958e+01 k_t = [2, 3, 3, 2]
 Generation 2410: Best = 5.312438e+02 FES = 241100 μ_F = 0.619 μ_{CR} = 0.078
 Diversity = 1.0957e+01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 5.312438e+02 FES = 242100 μ_F = 0.587 μ_{CR} = 0.070
 Diversity = 1.0930e+01 k_t = [4, 4, 4, 2]
 Generation 2430: Best = 5.312438e+02 FES = 243100 μ_F = 0.515 μ_{CR} = 0.066
 Diversity = 1.0931e+01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 5.312438e+02 FES = 244100 μ_F = 0.506 μ_{CR} = 0.064
 Diversity = 1.0824e+01 k_t = [4, 4, 3, 4]
 Generation 2450: Best = 5.312438e+02 FES = 245100 μ_F = 0.488 μ_{CR} = 0.050
 Diversity = 1.0826e+01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 5.311220e+02 FES = 246100 μ_F = 0.498 μ_{CR} = 0.046
 Diversity = 1.0826e+01 k_t = [2, 2, 3, 3]
 Generation 2470: Best = 5.311220e+02 FES = 247100 μ_F = 0.572 μ_{CR} = 0.042
 Diversity = 1.0829e+01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 5.311220e+02 FES = 248100 μ_F = 0.530 μ_{CR} = 0.038
 Diversity = 1.0829e+01 k_t = [3, 3, 2, 4]
 Generation 2490: Best = 5.311220e+02 FES = 249100 μ_F = 0.480 μ_{CR} = 0.037
 Diversity = 1.0829e+01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 5.311220e+02 FES = 250100 μ_F = 0.505 μ_{CR} = 0.053
 Diversity = 1.0828e+01 k_t = [3, 3, 3, 2]
 Generation 2510: Best = 5.311220e+02 FES = 251100 μ_F = 0.593 μ_{CR} = 0.036
 Diversity = 1.0818e+01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 5.311220e+02 FES = 252100 μ_F = 0.587 μ_{CR} = 0.049
 Diversity = 1.0818e+01 k_t = [3, 3, 4, 2]
 Generation 2530: Best = 5.311220e+02 FES = 253100 μ_F = 0.614 μ_{CR} = 0.044
 Diversity = 1.0818e+01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 5.311220e+02 FES = 254100 μ_F = 0.647 μ_{CR} = 0.036
 Diversity = 1.0819e+01 k_t = [2, 4, 3, 3]
 Generation 2550: Best = 5.311220e+02 FES = 255100 μ_F = 0.638 μ_{CR} = 0.039
 Diversity = 1.0819e+01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 5.311220e+02 FES = 256100 μ_F = 0.563 μ_{CR} = 0.058
 Diversity = 1.0820e+01 k_t = [2, 4, 2, 3]
 Generation 2570: Best = 5.311220e+02 FES = 257100 μ_F = 0.530 μ_{CR} = 0.059
 Diversity = 1.0820e+01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 5.311220e+02 FES = 258100 μ_F = 0.564 μ_{CR} = 0.061
 Diversity = 1.0748e+01 k_t = [2, 3, 2, 4]
 Generation 2590: Best = 5.311220e+02 FES = 259100 μ_F = 0.600 μ_{CR} = 0.069
 Diversity = 1.0740e+01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 5.311220e+02 FES = 260100 μ_F = 0.629 μ_{CR} = 0.046
 Diversity = 1.0700e+01 k_t = [4, 2, 4, 4]
 Generation 2610: Best = 5.311181e+02 FES = 261100 μ_F = 0.558 μ_{CR} = 0.066
 Diversity = 1.0700e+01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 5.311181e+02 FES = 262100 μ_F = 0.576 μ_{CR} = 0.055
 Diversity = 1.0695e+01 k_t = [2, 4, 2, 2]
 Generation 2630: Best = 5.311181e+02 FES = 263100 μ_F = 0.530 μ_{CR} = 0.044
 Diversity = 1.0696e+01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 5.311181e+02 FES = 264100 μ_F = 0.442 μ_{CR} = 0.065
 Diversity = 1.0695e+01 k_t = [3, 3, 2, 4]
 Generation 2650: Best = 5.311181e+02 FES = 265100 μ_F = 0.392 μ_{CR} = 0.053
 Diversity = 1.0694e+01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 5.311181e+02 FES = 266100 μ_F = 0.385 μ_{CR} = 0.049
 Diversity = 1.0690e+01 k_t = [2, 3, 3, 2]
 Generation 2670: Best = 5.311181e+02 FES = 267100 μ_F = 0.433 μ_{CR} = 0.050
 Diversity = 1.0689e+01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 5.311181e+02 FES = 268100 μ_F = 0.440 μ_{CR} = 0.065
 Diversity = 1.0689e+01 k_t = [2, 4, 4, 2]
 Generation 2690: Best = 5.311181e+02 FES = 269100 μ_F = 0.393 μ_{CR} = 0.058
 Diversity = 1.0670e+01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 5.311181e+02 FES = 270100 μ_F = 0.304 μ_{CR} = 0.045
 Diversity = 1.0669e+01 k_t = [3, 3, 4, 2]
 Generation 2710: Best = 5.311181e+02 FES = 271100 μ_F = 0.320 μ_{CR} = 0.047
 Diversity = 1.0670e+01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 5.311181e+02 FES = 272100 μ_F = 0.344 μ_{CR} = 0.037
 Diversity = 1.0670e+01 k_t = [2, 3, 2, 2]
 Generation 2730: Best = 5.311181e+02 FES = 273100 μ_F = 0.457 μ_{CR} = 0.038
 Diversity = 1.0672e+01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 5.311181e+02 FES = 274100 μ_F = 0.473 μ_{CR} = 0.039
 Diversity = 1.0672e+01 k_t = [2, 3, 4, 3]
 Generation 2750: Best = 5.311181e+02 FES = 275100 μ_F = 0.369 μ_{CR} = 0.060
 Diversity = 1.0673e+01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 5.311181e+02 FES = 276100 μ_F = 0.294 μ_{CR} = 0.075
 Diversity = 1.0672e+01 k_t = [2, 3, 2, 3]
 Generation 2770: Best = 5.311181e+02 FES = 277100 μ_F = 0.354 μ_{CR} = 0.051
 Diversity = 1.0651e+01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 5.311181e+02 FES = 278100 μ_F = 0.440 μ_{CR} = 0.051
 Diversity = 1.0592e+01 k_t = [3, 3, 2, 4]
 Generation 2790: Best = 5.311181e+02 FES = 279100 μ_F = 0.474 μ_{CR} = 0.057
 Diversity = 1.0564e+01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 5.304453e+02 FES = 280100 μ_F = 0.443 μ_{CR} = 0.052
 Diversity = 1.0564e+01 k_t = [2, 3, 4, 2]
 Generation 2810: Best = 5.304453e+02 FES = 281100 μ_F = 0.336 μ_{CR} = 0.055
 Diversity = 1.0564e+01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 5.304453e+02 FES = 282100 μ_F = 0.396 μ_{CR} = 0.055
 Diversity = 1.0565e+01 k_t = [4, 3, 2, 2]
 Generation 2830: Best = 5.304453e+02 FES = 283100 μ_F = 0.283 μ_{CR} = 0.069
 Diversity = 1.0564e+01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 5.304453e+02 FES = 284100 μ_F = 0.362 μ_{CR} = 0.060
 Diversity = 1.0565e+01 k_t = [3, 3, 2, 2]
 Generation 2850: Best = 5.304453e+02 FES = 285100 μ_F = 0.247 μ_{CR} = 0.057
 Diversity = 1.0565e+01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 5.304453e+02 FES = 286100 μ_F = 0.153 μ_{CR} = 0.069
 Diversity = 1.0564e+01 k_t = [3, 2, 4, 2]
 Generation 2870: Best = 5.304453e+02 FES = 287100 μ_F = 0.159 μ_{CR} = 0.066
 Diversity = 1.0563e+01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 5.304453e+02 FES = 288100 μ_F = 0.212 μ_{CR} = 0.081
 Diversity = 1.0560e+01 k_t = [4, 2, 3, 2]
 Generation 2890: Best = 5.304453e+02 FES = 289100 μ_F = 0.179 μ_{CR} = 0.066
 Diversity = 1.0560e+01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 5.304453e+02 FES = 290100 μ_F = 0.279 μ_{CR} = 0.065
 Diversity = 1.0584e+01 k_t = [4, 4, 2, 2]
 Generation 2910: Best = 5.304451e+02 FES = 291100 μ_F = 0.400 μ_{CR} = 0.054
 Diversity = 1.0584e+01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 5.304451e+02 FES = 292100 μ_F = 0.411 μ_{CR} = 0.052
 Diversity = 1.0583e+01 k_t = [4, 4, 2, 2]

Generation 2930: Best = 5.304451e+02 FES = 293100 μ_F = 0.484 μ_{CR} = 0.041
 Diversity = 1.0577e+01 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 5.304451e+02 FES = 294100 μ_F = 0.452 μ_{CR} = 0.040
 Diversity = 1.0525e+01 k_t = [2, 2, 2, 4]
 Generation 2950: Best = 5.304451e+02 FES = 295100 μ_F = 0.443 μ_{CR} = 0.037
 Diversity = 1.0525e+01 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 5.304451e+02 FES = 296100 μ_F = 0.410 μ_{CR} = 0.039
 Diversity = 1.0506e+01 k_t = [4, 3, 4, 4]
 Generation 2970: Best = 5.304451e+02 FES = 297100 μ_F = 0.476 μ_{CR} = 0.041
 Diversity = 1.0505e+01 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 5.304451e+02 FES = 298100 μ_F = 0.397 μ_{CR} = 0.048
 Diversity = 1.0505e+01 k_t = [2, 3, 4, 3]
 Generation 2990: Best = 5.304451e+02 FES = 299100 μ_F = 0.442 μ_{CR} = 0.047
 Diversity = 1.0504e+01 k_t = [1, 1, 1, 1]

Optimization complete.

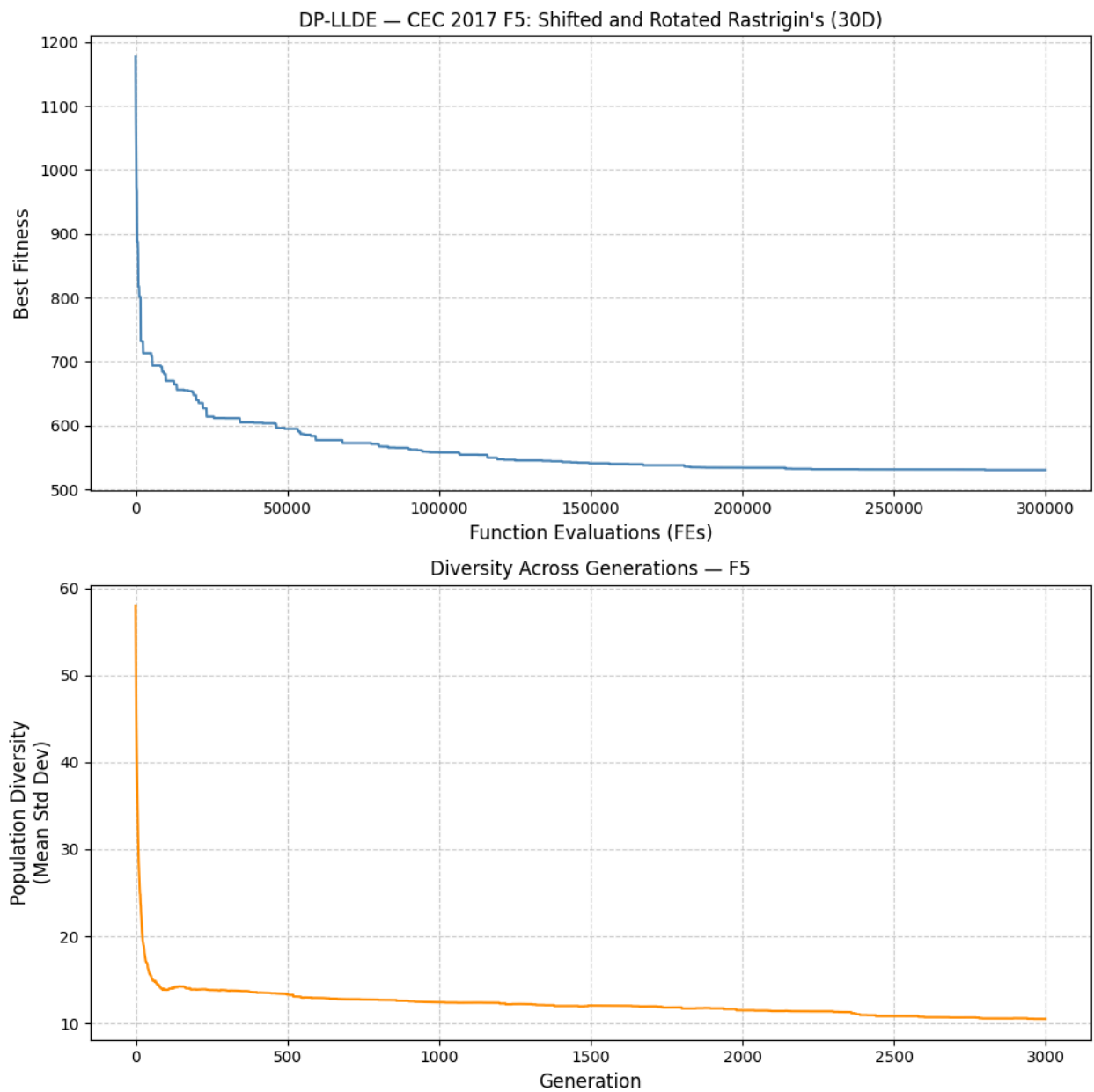
Best Fitness = 5.304451e+02

Total Function Evals = 300000

Total Successful Updates = 7949

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=
Results:
  Best Fitness      : 5.304451e+02
  Global Optimum    : 5.000000e+02
  Error (|best-optimum|) : 3.044509e+01
  Best Solution (5 dim) : [-22.81300654  57.24473027 -5.86542779 -59.28010296
19.4314962 ]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F5.png

```
In [ ]: # =====
# F6 – Shifted and Rotated Schaffer's F7
# =====
run_and_plot('F6', make_wrapper(simple.f6), 600.0, "Shifted and Rotated Schaff
```

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DP-LLDE on CEC 2017 F6 – Single Run

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Function : Shifted and Rotated Schaffer's F7 (30D)
Bounds : [-100, 100]
Global optimum : 600.0
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 7.816536e+02

Generation 10:	Best = 6.955437e+02	FES = 1100	$\mu_F = 0.560$	$\mu_{CR} = 0.357$
Diversity = 3.3200e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 6.707250e+02	FES = 2100	$\mu_F = 0.593$	$\mu_{CR} = 0.362$
Diversity = 2.2138e+01	$k_t = [4, 2, 4, 4]$			
Generation 30:	Best = 6.528993e+02	FES = 3100	$\mu_F = 0.642$	$\mu_{CR} = 0.342$
Diversity = 1.6062e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 6.402661e+02	FES = 4100	$\mu_F = 0.670$	$\mu_{CR} = 0.337$
Diversity = 1.3311e+01	$k_t = [4, 3, 4, 2]$			
Generation 50:	Best = 6.347208e+02	FES = 5100	$\mu_F = 0.705$	$\mu_{CR} = 0.342$
Diversity = 1.0919e+01	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 6.271148e+02	FES = 6100	$\mu_F = 0.727$	$\mu_{CR} = 0.327$
Diversity = 8.5051e+00	$k_t = [3, 3, 4, 3]$			
Generation 70:	Best = 6.224523e+02	FES = 7100	$\mu_F = 0.751$	$\mu_{CR} = 0.334$
Diversity = 7.3502e+00	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 6.203521e+02	FES = 8100	$\mu_F = 0.744$	$\mu_{CR} = 0.311$
Diversity = 6.3017e+00	$k_t = [2, 2, 4, 3]$			
Generation 90:	Best = 6.158553e+02	FES = 9100	$\mu_F = 0.739$	$\mu_{CR} = 0.297$
Diversity = 5.6339e+00	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 6.148193e+02	FES = 10100	$\mu_F = 0.742$	$\mu_{CR} = 0.272$
Diversity = 4.6201e+00	$k_t = [2, 3, 2, 2]$			
Generation 110:	Best = 6.121165e+02	FES = 11100	$\mu_F = 0.712$	$\mu_{CR} = 0.267$
Diversity = 3.8393e+00	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 6.116674e+02	FES = 12100	$\mu_F = 0.707$	$\mu_{CR} = 0.232$
Diversity = 3.4950e+00	$k_t = [4, 3, 3, 2]$			
Generation 130:	Best = 6.101454e+02	FES = 13100	$\mu_F = 0.699$	$\mu_{CR} = 0.226$
Diversity = 3.0482e+00	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 6.083271e+02	FES = 14100	$\mu_F = 0.718$	$\mu_{CR} = 0.214$
Diversity = 2.7091e+00	$k_t = [4, 3, 4, 4]$			
Generation 150:	Best = 6.058816e+02	FES = 15100	$\mu_F = 0.697$	$\mu_{CR} = 0.196$
Diversity = 2.3831e+00	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 6.058263e+02	FES = 16100	$\mu_F = 0.691$	$\mu_{CR} = 0.180$
Diversity = 2.0589e+00	$k_t = [2, 3, 4, 3]$			
Generation 170:	Best = 6.057337e+02	FES = 17100	$\mu_F = 0.702$	$\mu_{CR} = 0.181$
Diversity = 1.6758e+00	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 6.045586e+02	FES = 18100	$\mu_F = 0.709$	$\mu_{CR} = 0.169$
Diversity = 1.3855e+00	$k_t = [2, 3, 3, 2]$			
Generation 190:	Best = 6.045353e+02	FES = 19100	$\mu_F = 0.721$	$\mu_{CR} = 0.160$
Diversity = 1.1687e+00	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 6.038557e+02	FES = 20100	$\mu_F = 0.735$	$\mu_{CR} = 0.157$
Diversity = 1.0203e+00	$k_t = [4, 2, 2, 3]$			
Generation 210:	Best = 6.037146e+02	FES = 21100	$\mu_F = 0.715$	$\mu_{CR} = 0.159$
Diversity = 9.3803e-01	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 6.029609e+02	FES = 22100	$\mu_F = 0.721$	$\mu_{CR} = 0.153$
Diversity = 8.3994e-01	$k_t = [3, 2, 4, 4]$			

Generation	230: Best = 6.029347e+02	FES = 23100	$\mu_F = 0.703$	$\mu_{CR} = 0.148$
Diversity =	7.1464e-01	$k_t = [1, 1, 1, 1]$		
Generation	240: Best = 6.026333e+02	FES = 24100	$\mu_F = 0.702$	$\mu_{CR} = 0.142$
Diversity =	6.6238e-01	$k_t = [3, 4, 3, 3]$		
Generation	250: Best = 6.025829e+02	FES = 25100	$\mu_F = 0.688$	$\mu_{CR} = 0.125$
Diversity =	6.0403e-01	$k_t = [1, 1, 1, 1]$		
Generation	260: Best = 6.022027e+02	FES = 26100	$\mu_F = 0.674$	$\mu_{CR} = 0.108$
Diversity =	5.7459e-01	$k_t = [3, 3, 4, 3]$		
Generation	270: Best = 6.021693e+02	FES = 27100	$\mu_F = 0.718$	$\mu_{CR} = 0.103$
Diversity =	5.7758e-01	$k_t = [1, 1, 1, 1]$		
Generation	280: Best = 6.021015e+02	FES = 28100	$\mu_F = 0.725$	$\mu_{CR} = 0.105$
Diversity =	5.4915e-01	$k_t = [3, 4, 2, 3]$		
Generation	290: Best = 6.019887e+02	FES = 29100	$\mu_F = 0.723$	$\mu_{CR} = 0.106$
Diversity =	5.4677e-01	$k_t = [1, 1, 1, 1]$		
Generation	300: Best = 6.019215e+02	FES = 30100	$\mu_F = 0.681$	$\mu_{CR} = 0.099$
Diversity =	5.4402e-01	$k_t = [2, 4, 4, 2]$		
Generation	310: Best = 6.017433e+02	FES = 31100	$\mu_F = 0.693$	$\mu_{CR} = 0.090$
Diversity =	5.0094e-01	$k_t = [1, 1, 1, 1]$		
Generation	320: Best = 6.016330e+02	FES = 32100	$\mu_F = 0.683$	$\mu_{CR} = 0.098$
Diversity =	4.2818e-01	$k_t = [3, 2, 4, 3]$		
Generation	330: Best = 6.015482e+02	FES = 33100	$\mu_F = 0.675$	$\mu_{CR} = 0.094$
Diversity =	3.7523e-01	$k_t = [1, 1, 1, 1]$		
Generation	340: Best = 6.013934e+02	FES = 34100	$\mu_F = 0.641$	$\mu_{CR} = 0.102$
Diversity =	3.4822e-01	$k_t = [4, 4, 4, 2]$		
Generation	350: Best = 6.013541e+02	FES = 35100	$\mu_F = 0.640$	$\mu_{CR} = 0.111$
Diversity =	2.9543e-01	$k_t = [1, 1, 1, 1]$		
Generation	360: Best = 6.013469e+02	FES = 36100	$\mu_F = 0.666$	$\mu_{CR} = 0.109$
Diversity =	2.4079e-01	$k_t = [3, 4, 4, 4]$		
Generation	370: Best = 6.012835e+02	FES = 37100	$\mu_F = 0.662$	$\mu_{CR} = 0.104$
Diversity =	2.0295e-01	$k_t = [1, 1, 1, 1]$		
Generation	380: Best = 6.012349e+02	FES = 38100	$\mu_F = 0.683$	$\mu_{CR} = 0.097$
Diversity =	1.9426e-01	$k_t = [4, 3, 2, 3]$		
Generation	390: Best = 6.011645e+02	FES = 39100	$\mu_F = 0.664$	$\mu_{CR} = 0.098$
Diversity =	1.8595e-01	$k_t = [1, 1, 1, 1]$		
Generation	400: Best = 6.011478e+02	FES = 40100	$\mu_F = 0.670$	$\mu_{CR} = 0.092$
Diversity =	1.7504e-01	$k_t = [3, 3, 2, 2]$		
Generation	410: Best = 6.011207e+02	FES = 41100	$\mu_F = 0.670$	$\mu_{CR} = 0.096$
Diversity =	1.6920e-01	$k_t = [1, 1, 1, 1]$		
Generation	420: Best = 6.011175e+02	FES = 42100	$\mu_F = 0.686$	$\mu_{CR} = 0.089$
Diversity =	1.6060e-01	$k_t = [4, 2, 2, 3]$		
Generation	430: Best = 6.011132e+02	FES = 43100	$\mu_F = 0.699$	$\mu_{CR} = 0.078$
Diversity =	1.5776e-01	$k_t = [1, 1, 1, 1]$		
Generation	440: Best = 6.010753e+02	FES = 44100	$\mu_F = 0.664$	$\mu_{CR} = 0.073$
Diversity =	1.5299e-01	$k_t = [2, 2, 2, 3]$		
Generation	450: Best = 6.010499e+02	FES = 45100	$\mu_F = 0.645$	$\mu_{CR} = 0.067$
Diversity =	1.4617e-01	$k_t = [1, 1, 1, 1]$		
Generation	460: Best = 6.010355e+02	FES = 46100	$\mu_F = 0.632$	$\mu_{CR} = 0.077$
Diversity =	1.3378e-01	$k_t = [2, 4, 3, 3]$		
Generation	470: Best = 6.010007e+02	FES = 47100	$\mu_F = 0.627$	$\mu_{CR} = 0.087$
Diversity =	1.3135e-01	$k_t = [1, 1, 1, 1]$		
Generation	480: Best = 6.009550e+02	FES = 48100	$\mu_F = 0.641$	$\mu_{CR} = 0.085$
Diversity =	1.2633e-01	$k_t = [3, 2, 4, 2]$		
Generation	490: Best = 6.009550e+02	FES = 49100	$\mu_F = 0.636$	$\mu_{CR} = 0.082$
Diversity =	1.2354e-01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 6.009550e+02	FES = 50100	$\mu_F = 0.643$	$\mu_{CR} = 0.075$
Diversity = 1.1867e-01	$k_t = [2, 2, 4, 4]$		
Generation 510: Best = 6.009496e+02	FES = 51100	$\mu_F = 0.686$	$\mu_{CR} = 0.063$
Diversity = 1.1446e-01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 6.009298e+02	FES = 52100	$\mu_F = 0.682$	$\mu_{CR} = 0.067$
Diversity = 1.0889e-01	$k_t = [2, 2, 4, 3]$		
Generation 530: Best = 6.009295e+02	FES = 53100	$\mu_F = 0.670$	$\mu_{CR} = 0.065$
Diversity = 1.0975e-01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 6.009295e+02	FES = 54100	$\mu_F = 0.664$	$\mu_{CR} = 0.075$
Diversity = 1.0318e-01	$k_t = [3, 2, 2, 4]$		
Generation 550: Best = 6.009295e+02	FES = 55100	$\mu_F = 0.704$	$\mu_{CR} = 0.068$
Diversity = 1.0026e-01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 6.008919e+02	FES = 56100	$\mu_F = 0.699$	$\mu_{CR} = 0.080$
Diversity = 9.8054e-02	$k_t = [2, 4, 4, 3]$		
Generation 570: Best = 6.008695e+02	FES = 57100	$\mu_F = 0.674$	$\mu_{CR} = 0.090$
Diversity = 9.6576e-02	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 6.008695e+02	FES = 58100	$\mu_F = 0.654$	$\mu_{CR} = 0.068$
Diversity = 9.1531e-02	$k_t = [3, 4, 3, 4]$		
Generation 590: Best = 6.008538e+02	FES = 59100	$\mu_F = 0.672$	$\mu_{CR} = 0.063$
Diversity = 8.5977e-02	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 6.008312e+02	FES = 60100	$\mu_F = 0.699$	$\mu_{CR} = 0.068$
Diversity = 8.4716e-02	$k_t = [4, 3, 4, 4]$		
Generation 610: Best = 6.008312e+02	FES = 61100	$\mu_F = 0.742$	$\mu_{CR} = 0.066$
Diversity = 8.4248e-02	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 6.008312e+02	FES = 62100	$\mu_F = 0.713$	$\mu_{CR} = 0.066$
Diversity = 7.4166e-02	$k_t = [2, 3, 4, 4]$		
Generation 630: Best = 6.008312e+02	FES = 63100	$\mu_F = 0.717$	$\mu_{CR} = 0.072$
Diversity = 7.0518e-02	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 6.008166e+02	FES = 64100	$\mu_F = 0.766$	$\mu_{CR} = 0.077$
Diversity = 6.5616e-02	$k_t = [2, 2, 3, 4]$		
Generation 650: Best = 6.008123e+02	FES = 65100	$\mu_F = 0.752$	$\mu_{CR} = 0.070$
Diversity = 6.0700e-02	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 6.007687e+02	FES = 66100	$\mu_F = 0.728$	$\mu_{CR} = 0.078$
Diversity = 5.4803e-02	$k_t = [2, 2, 3, 4]$		
Generation 670: Best = 6.007530e+02	FES = 67100	$\mu_F = 0.758$	$\mu_{CR} = 0.084$
Diversity = 5.2030e-02	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 6.007378e+02	FES = 68100	$\mu_F = 0.708$	$\mu_{CR} = 0.079$
Diversity = 4.7106e-02	$k_t = [4, 4, 3, 4]$		
Generation 690: Best = 6.007308e+02	FES = 69100	$\mu_F = 0.696$	$\mu_{CR} = 0.076$
Diversity = 4.3065e-02	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 6.007301e+02	FES = 70100	$\mu_F = 0.721$	$\mu_{CR} = 0.078$
Diversity = 4.0270e-02	$k_t = [2, 2, 2, 2]$		
Generation 710: Best = 6.007136e+02	FES = 71100	$\mu_F = 0.731$	$\mu_{CR} = 0.077$
Diversity = 3.8898e-02	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 6.006866e+02	FES = 72100	$\mu_F = 0.716$	$\mu_{CR} = 0.076$
Diversity = 3.5890e-02	$k_t = [2, 3, 2, 3]$		
Generation 730: Best = 6.006866e+02	FES = 73100	$\mu_F = 0.736$	$\mu_{CR} = 0.082$
Diversity = 3.5240e-02	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 6.006866e+02	FES = 74100	$\mu_F = 0.731$	$\mu_{CR} = 0.094$
Diversity = 3.1782e-02	$k_t = [3, 4, 3, 4]$		
Generation 750: Best = 6.006740e+02	FES = 75100	$\mu_F = 0.745$	$\mu_{CR} = 0.091$
Diversity = 2.6912e-02	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 6.006677e+02	FES = 76100	$\mu_F = 0.747$	$\mu_{CR} = 0.089$
Diversity = 2.6674e-02	$k_t = [3, 3, 3, 2]$		

Generation 770: Best = 6.006660e+02	FES = 77100	$\mu_F = 0.746$	$\mu_{CR} = 0.096$
Diversity = 2.5030e-02	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 6.006466e+02	FES = 78100	$\mu_F = 0.726$	$\mu_{CR} = 0.085$
Diversity = 2.3175e-02	$k_t = [4, 4, 2, 4]$		
Generation 790: Best = 6.006400e+02	FES = 79100	$\mu_F = 0.673$	$\mu_{CR} = 0.092$
Diversity = 2.1578e-02	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 6.006219e+02	FES = 80100	$\mu_F = 0.709$	$\mu_{CR} = 0.083$
Diversity = 1.8790e-02	$k_t = [2, 4, 3, 4]$		
Generation 810: Best = 6.006219e+02	FES = 81100	$\mu_F = 0.735$	$\mu_{CR} = 0.083$
Diversity = 1.7415e-02	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 6.006219e+02	FES = 82100	$\mu_F = 0.734$	$\mu_{CR} = 0.089$
Diversity = 1.6069e-02	$k_t = [4, 3, 2, 3]$		
Generation 830: Best = 6.006164e+02	FES = 83100	$\mu_F = 0.728$	$\mu_{CR} = 0.078$
Diversity = 1.4155e-02	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 6.006160e+02	FES = 84100	$\mu_F = 0.725$	$\mu_{CR} = 0.075$
Diversity = 1.2981e-02	$k_t = [4, 2, 2, 4]$		
Generation 850: Best = 6.006108e+02	FES = 85100	$\mu_F = 0.703$	$\mu_{CR} = 0.083$
Diversity = 1.2656e-02	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 6.006084e+02	FES = 86100	$\mu_F = 0.738$	$\mu_{CR} = 0.081$
Diversity = 1.1501e-02	$k_t = [4, 2, 4, 4]$		
Generation 870: Best = 6.006003e+02	FES = 87100	$\mu_F = 0.728$	$\mu_{CR} = 0.068$
Diversity = 1.1190e-02	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 6.005947e+02	FES = 88100	$\mu_F = 0.755$	$\mu_{CR} = 0.072$
Diversity = 1.1057e-02	$k_t = [4, 4, 3, 4]$		
Generation 890: Best = 6.005940e+02	FES = 89100	$\mu_F = 0.747$	$\mu_{CR} = 0.069$
Diversity = 1.0913e-02	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 6.005863e+02	FES = 90100	$\mu_F = 0.736$	$\mu_{CR} = 0.074$
Diversity = 1.0799e-02	$k_t = [4, 2, 4, 3]$		
Generation 910: Best = 6.005854e+02	FES = 91100	$\mu_F = 0.751$	$\mu_{CR} = 0.068$
Diversity = 1.0783e-02	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 6.005854e+02	FES = 92100	$\mu_F = 0.763$	$\mu_{CR} = 0.075$
Diversity = 1.0289e-02	$k_t = [2, 2, 3, 3]$		
Generation 930: Best = 6.005850e+02	FES = 93100	$\mu_F = 0.771$	$\mu_{CR} = 0.071$
Diversity = 9.8932e-03	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 6.005815e+02	FES = 94100	$\mu_F = 0.757$	$\mu_{CR} = 0.062$
Diversity = 9.9408e-03	$k_t = [3, 2, 2, 2]$		
Generation 950: Best = 6.005806e+02	FES = 95100	$\mu_F = 0.785$	$\mu_{CR} = 0.055$
Diversity = 1.0027e-02	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 6.005800e+02	FES = 96100	$\mu_F = 0.804$	$\mu_{CR} = 0.064$
Diversity = 9.1904e-03	$k_t = [3, 4, 2, 4]$		
Generation 970: Best = 6.005771e+02	FES = 97100	$\mu_F = 0.834$	$\mu_{CR} = 0.059$
Diversity = 8.5721e-03	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 6.005756e+02	FES = 98100	$\mu_F = 0.808$	$\mu_{CR} = 0.054$
Diversity = 7.6683e-03	$k_t = [3, 3, 2, 4]$		
Generation 990: Best = 6.005753e+02	FES = 99100	$\mu_F = 0.776$	$\mu_{CR} = 0.061$
Diversity = 7.7389e-03	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 6.005753e+02	FES = 100100	$\mu_F = 0.764$	$\mu_{CR} = 0.061$
Diversity = 7.0899e-03	$k_t = [2, 3, 4, 4]$		
Generation 1010: Best = 6.005697e+02	FES = 101100	$\mu_F = 0.820$	$\mu_{CR} = 0.057$
Diversity = 6.7783e-03	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 6.005690e+02	FES = 102100	$\mu_F = 0.783$	$\mu_{CR} = 0.059$
Diversity = 6.4032e-03	$k_t = [4, 4, 4, 4]$		
Generation 1030: Best = 6.005690e+02	FES = 103100	$\mu_F = 0.746$	$\mu_{CR} = 0.056$
Diversity = 6.2866e-03	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 6.005687e+02 FES = 104100 μ_F = 0.783 μ_{CR} = 0.060
 Diversity = 6.0912e-03 k_t = [4, 2, 4, 4]
 Generation 1050: Best = 6.005687e+02 FES = 105100 μ_F = 0.765 μ_{CR} = 0.063
 Diversity = 5.9053e-03 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 6.005687e+02 FES = 106100 μ_F = 0.770 μ_{CR} = 0.056
 Diversity = 5.4300e-03 k_t = [2, 2, 3, 3]
 Generation 1070: Best = 6.005687e+02 FES = 107100 μ_F = 0.736 μ_{CR} = 0.058
 Diversity = 5.4458e-03 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 6.005687e+02 FES = 108100 μ_F = 0.704 μ_{CR} = 0.067
 Diversity = 5.2555e-03 k_t = [2, 2, 4, 4]
 Generation 1090: Best = 6.005687e+02 FES = 109100 μ_F = 0.668 μ_{CR} = 0.061
 Diversity = 5.0154e-03 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 6.005687e+02 FES = 110100 μ_F = 0.676 μ_{CR} = 0.069
 Diversity = 4.8446e-03 k_t = [3, 3, 2, 4]
 Generation 1110: Best = 6.005686e+02 FES = 111100 μ_F = 0.644 μ_{CR} = 0.065
 Diversity = 4.6496e-03 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 6.005665e+02 FES = 112100 μ_F = 0.661 μ_{CR} = 0.066
 Diversity = 4.5449e-03 k_t = [3, 4, 4, 4]
 Generation 1130: Best = 6.005640e+02 FES = 113100 μ_F = 0.680 μ_{CR} = 0.066
 Diversity = 4.4503e-03 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 6.005640e+02 FES = 114100 μ_F = 0.670 μ_{CR} = 0.052
 Diversity = 4.4790e-03 k_t = [3, 4, 3, 4]
 Generation 1150: Best = 6.005606e+02 FES = 115100 μ_F = 0.674 μ_{CR} = 0.052
 Diversity = 4.4136e-03 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 6.005605e+02 FES = 116100 μ_F = 0.691 μ_{CR} = 0.050
 Diversity = 4.3632e-03 k_t = [4, 4, 3, 3]
 Generation 1170: Best = 6.005605e+02 FES = 117100 μ_F = 0.699 μ_{CR} = 0.055
 Diversity = 4.3005e-03 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 6.005594e+02 FES = 118100 μ_F = 0.656 μ_{CR} = 0.058
 Diversity = 4.2509e-03 k_t = [2, 2, 3, 2]
 Generation 1190: Best = 6.005594e+02 FES = 119100 μ_F = 0.650 μ_{CR} = 0.068
 Diversity = 4.1654e-03 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 6.005594e+02 FES = 120100 μ_F = 0.675 μ_{CR} = 0.050
 Diversity = 4.1301e-03 k_t = [3, 3, 2, 3]
 Generation 1210: Best = 6.005591e+02 FES = 121100 μ_F = 0.676 μ_{CR} = 0.053
 Diversity = 4.0802e-03 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 6.005560e+02 FES = 122100 μ_F = 0.718 μ_{CR} = 0.051
 Diversity = 4.1005e-03 k_t = [3, 3, 3, 2]
 Generation 1230: Best = 6.005560e+02 FES = 123100 μ_F = 0.656 μ_{CR} = 0.056
 Diversity = 4.0544e-03 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 6.005560e+02 FES = 124100 μ_F = 0.630 μ_{CR} = 0.057
 Diversity = 4.0515e-03 k_t = [4, 4, 2, 2]
 Generation 1250: Best = 6.005560e+02 FES = 125100 μ_F = 0.600 μ_{CR} = 0.057
 Diversity = 4.0170e-03 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 6.005543e+02 FES = 126100 μ_F = 0.613 μ_{CR} = 0.063
 Diversity = 3.9877e-03 k_t = [4, 3, 2, 3]
 Generation 1270: Best = 6.005527e+02 FES = 127100 μ_F = 0.586 μ_{CR} = 0.057
 Diversity = 4.0255e-03 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 6.005527e+02 FES = 128100 μ_F = 0.567 μ_{CR} = 0.064
 Diversity = 4.0013e-03 k_t = [2, 4, 4, 2]
 Generation 1290: Best = 6.005527e+02 FES = 129100 μ_F = 0.539 μ_{CR} = 0.062
 Diversity = 4.0031e-03 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 6.005507e+02 FES = 130100 μ_F = 0.562 μ_{CR} = 0.061
 Diversity = 3.7469e-03 k_t = [3, 2, 3, 4]

Generation 1310: Best = 6.005507e+02 FES = 131100 μ_F = 0.565 μ_{CR} = 0.052
 Diversity = 3.7413e-03 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 6.005485e+02 FES = 132100 μ_F = 0.535 μ_{CR} = 0.051
 Diversity = 3.7349e-03 k_t = [2, 2, 4, 2]
 Generation 1330: Best = 6.005485e+02 FES = 133100 μ_F = 0.546 μ_{CR} = 0.052
 Diversity = 3.7278e-03 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 6.005480e+02 FES = 134100 μ_F = 0.576 μ_{CR} = 0.058
 Diversity = 3.7233e-03 k_t = [2, 2, 3, 2]
 Generation 1350: Best = 6.005479e+02 FES = 135100 μ_F = 0.589 μ_{CR} = 0.057
 Diversity = 3.7143e-03 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 6.005479e+02 FES = 136100 μ_F = 0.572 μ_{CR} = 0.048
 Diversity = 3.6312e-03 k_t = [3, 3, 3, 4]
 Generation 1370: Best = 6.005479e+02 FES = 137100 μ_F = 0.593 μ_{CR} = 0.054
 Diversity = 3.6253e-03 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 6.005474e+02 FES = 138100 μ_F = 0.541 μ_{CR} = 0.047
 Diversity = 3.6281e-03 k_t = [4, 2, 2, 3]
 Generation 1390: Best = 6.005474e+02 FES = 139100 μ_F = 0.601 μ_{CR} = 0.064
 Diversity = 3.6304e-03 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 6.005472e+02 FES = 140100 μ_F = 0.548 μ_{CR} = 0.062
 Diversity = 3.6074e-03 k_t = [2, 3, 4, 3]
 Generation 1410: Best = 6.005466e+02 FES = 141100 μ_F = 0.524 μ_{CR} = 0.049
 Diversity = 3.6148e-03 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 6.005458e+02 FES = 142100 μ_F = 0.544 μ_{CR} = 0.063
 Diversity = 3.5768e-03 k_t = [4, 3, 2, 3]
 Generation 1430: Best = 6.005456e+02 FES = 143100 μ_F = 0.495 μ_{CR} = 0.069
 Diversity = 3.5028e-03 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 6.005456e+02 FES = 144100 μ_F = 0.515 μ_{CR} = 0.058
 Diversity = 3.4813e-03 k_t = [3, 3, 4, 4]
 Generation 1450: Best = 6.005456e+02 FES = 145100 μ_F = 0.491 μ_{CR} = 0.067
 Diversity = 3.3626e-03 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 6.005456e+02 FES = 146100 μ_F = 0.478 μ_{CR} = 0.071
 Diversity = 3.3430e-03 k_t = [4, 2, 4, 4]
 Generation 1470: Best = 6.005455e+02 FES = 147100 μ_F = 0.468 μ_{CR} = 0.073
 Diversity = 3.2830e-03 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 6.005455e+02 FES = 148100 μ_F = 0.491 μ_{CR} = 0.074
 Diversity = 3.2356e-03 k_t = [3, 4, 2, 2]
 Generation 1490: Best = 6.005455e+02 FES = 149100 μ_F = 0.451 μ_{CR} = 0.068
 Diversity = 3.2300e-03 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 6.005455e+02 FES = 150100 μ_F = 0.474 μ_{CR} = 0.074
 Diversity = 3.1978e-03 k_t = [2, 2, 2, 2]
 Generation 1510: Best = 6.005455e+02 FES = 151100 μ_F = 0.434 μ_{CR} = 0.073
 Diversity = 3.1893e-03 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 6.005441e+02 FES = 152100 μ_F = 0.424 μ_{CR} = 0.076
 Diversity = 3.1555e-03 k_t = [3, 4, 4, 4]
 Generation 1530: Best = 6.005417e+02 FES = 153100 μ_F = 0.402 μ_{CR} = 0.078
 Diversity = 3.1512e-03 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 6.005413e+02 FES = 154100 μ_F = 0.449 μ_{CR} = 0.065
 Diversity = 3.1523e-03 k_t = [3, 4, 4, 3]
 Generation 1550: Best = 6.005410e+02 FES = 155100 μ_F = 0.478 μ_{CR} = 0.049
 Diversity = 3.1528e-03 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 6.005410e+02 FES = 156100 μ_F = 0.461 μ_{CR} = 0.066
 Diversity = 3.1671e-03 k_t = [2, 4, 2, 4]
 Generation 1570: Best = 6.005409e+02 FES = 157100 μ_F = 0.446 μ_{CR} = 0.056
 Diversity = 3.1706e-03 k_t = [1, 1, 1, 1]

Generation 1580: Best = 6.005409e+02 FES = 158100 μ_F = 0.427 μ_{CR} = 0.067
 Diversity = 3.1676e-03 k_t = [2, 2, 2, 2]
 Generation 1590: Best = 6.005408e+02 FES = 159100 μ_F = 0.457 μ_{CR} = 0.055
 Diversity = 3.1666e-03 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 6.005403e+02 FES = 160100 μ_F = 0.538 μ_{CR} = 0.047
 Diversity = 3.1680e-03 k_t = [3, 3, 2, 2]
 Generation 1610: Best = 6.005403e+02 FES = 161100 μ_F = 0.599 μ_{CR} = 0.044
 Diversity = 3.1697e-03 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 6.005403e+02 FES = 162100 μ_F = 0.640 μ_{CR} = 0.049
 Diversity = 3.1328e-03 k_t = [2, 4, 2, 4]
 Generation 1630: Best = 6.005399e+02 FES = 163100 μ_F = 0.641 μ_{CR} = 0.053
 Diversity = 3.1339e-03 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 6.005395e+02 FES = 164100 μ_F = 0.580 μ_{CR} = 0.049
 Diversity = 3.1105e-03 k_t = [3, 2, 2, 3]
 Generation 1650: Best = 6.005395e+02 FES = 165100 μ_F = 0.535 μ_{CR} = 0.045
 Diversity = 3.1107e-03 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 6.005395e+02 FES = 166100 μ_F = 0.533 μ_{CR} = 0.071
 Diversity = 3.1101e-03 k_t = [3, 3, 4, 3]
 Generation 1670: Best = 6.005395e+02 FES = 167100 μ_F = 0.590 μ_{CR} = 0.045
 Diversity = 3.1124e-03 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 6.005395e+02 FES = 168100 μ_F = 0.510 μ_{CR} = 0.041
 Diversity = 3.1116e-03 k_t = [3, 4, 3, 2]
 Generation 1690: Best = 6.005395e+02 FES = 169100 μ_F = 0.467 μ_{CR} = 0.056
 Diversity = 3.1120e-03 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 6.005395e+02 FES = 170100 μ_F = 0.463 μ_{CR} = 0.054
 Diversity = 3.1109e-03 k_t = [2, 3, 2, 2]
 Generation 1710: Best = 6.005395e+02 FES = 171100 μ_F = 0.466 μ_{CR} = 0.045
 Diversity = 3.1096e-03 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 6.005395e+02 FES = 172100 μ_F = 0.538 μ_{CR} = 0.041
 Diversity = 3.0780e-03 k_t = [4, 2, 4, 3]
 Generation 1730: Best = 6.005395e+02 FES = 173100 μ_F = 0.536 μ_{CR} = 0.035
 Diversity = 3.0792e-03 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 6.005391e+02 FES = 174100 μ_F = 0.556 μ_{CR} = 0.050
 Diversity = 3.0102e-03 k_t = [3, 2, 4, 4]
 Generation 1750: Best = 6.005391e+02 FES = 175100 μ_F = 0.529 μ_{CR} = 0.055
 Diversity = 2.9108e-03 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 6.005391e+02 FES = 176100 μ_F = 0.473 μ_{CR} = 0.057
 Diversity = 2.6536e-03 k_t = [3, 3, 4, 4]
 Generation 1770: Best = 6.005391e+02 FES = 177100 μ_F = 0.547 μ_{CR} = 0.066
 Diversity = 2.6140e-03 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 6.005374e+02 FES = 178100 μ_F = 0.527 μ_{CR} = 0.049
 Diversity = 2.6076e-03 k_t = [2, 4, 4, 3]
 Generation 1790: Best = 6.005374e+02 FES = 179100 μ_F = 0.529 μ_{CR} = 0.054
 Diversity = 2.6108e-03 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 6.005373e+02 FES = 180100 μ_F = 0.527 μ_{CR} = 0.049
 Diversity = 2.5758e-03 k_t = [2, 4, 3, 4]
 Generation 1810: Best = 6.005373e+02 FES = 181100 μ_F = 0.498 μ_{CR} = 0.041
 Diversity = 2.5764e-03 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 6.005373e+02 FES = 182100 μ_F = 0.511 μ_{CR} = 0.050
 Diversity = 2.5504e-03 k_t = [3, 2, 3, 2]
 Generation 1830: Best = 6.005372e+02 FES = 183100 μ_F = 0.502 μ_{CR} = 0.045
 Diversity = 2.5222e-03 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 6.005372e+02 FES = 184100 μ_F = 0.458 μ_{CR} = 0.046
 Diversity = 2.5274e-03 k_t = [4, 4, 3, 2]

Generation 1850: Best = 6.005372e+02	FES = 185100	$\mu_F = 0.417$	$\mu_{CR} = 0.056$
Diversity = 2.5221e-03	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 6.005371e+02	FES = 186100	$\mu_F = 0.422$	$\mu_{CR} = 0.053$
Diversity = 2.5148e-03	$k_t = [2, 4, 3, 2]$		
Generation 1870: Best = 6.005371e+02	FES = 187100	$\mu_F = 0.468$	$\mu_{CR} = 0.050$
Diversity = 2.4782e-03	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 6.005369e+02	FES = 188100	$\mu_F = 0.509$	$\mu_{CR} = 0.050$
Diversity = 2.4781e-03	$k_t = [3, 2, 2, 3]$		
Generation 1890: Best = 6.005368e+02	FES = 189100	$\mu_F = 0.571$	$\mu_{CR} = 0.048$
Diversity = 2.4765e-03	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 6.005365e+02	FES = 190100	$\mu_F = 0.579$	$\mu_{CR} = 0.045$
Diversity = 2.4164e-03	$k_t = [3, 2, 3, 4]$		
Generation 1910: Best = 6.005353e+02	FES = 191100	$\mu_F = 0.524$	$\mu_{CR} = 0.042$
Diversity = 2.3515e-03	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 6.005353e+02	FES = 192100	$\mu_F = 0.535$	$\mu_{CR} = 0.043$
Diversity = 2.2953e-03	$k_t = [3, 2, 4, 3]$		
Generation 1930: Best = 6.005353e+02	FES = 193100	$\mu_F = 0.495$	$\mu_{CR} = 0.055$
Diversity = 2.2530e-03	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 6.005353e+02	FES = 194100	$\mu_F = 0.500$	$\mu_{CR} = 0.070$
Diversity = 2.1767e-03	$k_t = [4, 2, 3, 2]$		
Generation 1950: Best = 6.005352e+02	FES = 195100	$\mu_F = 0.525$	$\mu_{CR} = 0.058$
Diversity = 2.1767e-03	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 6.005352e+02	FES = 196100	$\mu_F = 0.491$	$\mu_{CR} = 0.055$
Diversity = 2.1846e-03	$k_t = [3, 4, 3, 3]$		
Generation 1970: Best = 6.005345e+02	FES = 197100	$\mu_F = 0.510$	$\mu_{CR} = 0.057$
Diversity = 2.1831e-03	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 6.005344e+02	FES = 198100	$\mu_F = 0.578$	$\mu_{CR} = 0.059$
Diversity = 2.1661e-03	$k_t = [2, 2, 3, 3]$		
Generation 1990: Best = 6.005344e+02	FES = 199100	$\mu_F = 0.704$	$\mu_{CR} = 0.047$
Diversity = 2.1638e-03	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 6.005344e+02	FES = 200100	$\mu_F = 0.636$	$\mu_{CR} = 0.059$
Diversity = 2.1707e-03	$k_t = [3, 3, 3, 2]$		
Generation 2010: Best = 6.005344e+02	FES = 201100	$\mu_F = 0.552$	$\mu_{CR} = 0.048$
Diversity = 2.1677e-03	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 6.005344e+02	FES = 202100	$\mu_F = 0.493$	$\mu_{CR} = 0.044$
Diversity = 2.1681e-03	$k_t = [4, 3, 4, 2]$		
Generation 2030: Best = 6.005344e+02	FES = 203100	$\mu_F = 0.531$	$\mu_{CR} = 0.060$
Diversity = 2.1695e-03	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 6.005335e+02	FES = 204100	$\mu_F = 0.550$	$\mu_{CR} = 0.049$
Diversity = 2.1686e-03	$k_t = [4, 4, 3, 2]$		
Generation 2050: Best = 6.005333e+02	FES = 205100	$\mu_F = 0.557$	$\mu_{CR} = 0.043$
Diversity = 2.1685e-03	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 6.005333e+02	FES = 206100	$\mu_F = 0.484$	$\mu_{CR} = 0.050$
Diversity = 2.1599e-03	$k_t = [2, 3, 4, 3]$		
Generation 2070: Best = 6.005333e+02	FES = 207100	$\mu_F = 0.454$	$\mu_{CR} = 0.044$
Diversity = 2.1602e-03	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 6.005333e+02	FES = 208100	$\mu_F = 0.424$	$\mu_{CR} = 0.042$
Diversity = 2.1570e-03	$k_t = [3, 2, 4, 3]$		
Generation 2090: Best = 6.005333e+02	FES = 209100	$\mu_F = 0.458$	$\mu_{CR} = 0.055$
Diversity = 2.1561e-03	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 6.005333e+02	FES = 210100	$\mu_F = 0.425$	$\mu_{CR} = 0.047$
Diversity = 2.1572e-03	$k_t = [2, 2, 3, 3]$		
Generation 2110: Best = 6.005333e+02	FES = 211100	$\mu_F = 0.418$	$\mu_{CR} = 0.051$
Diversity = 2.1567e-03	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 6.005333e+02 FES = 212100 μ_F = 0.400 μ_{CR} = 0.050
 Diversity = 2.1574e-03 k_t = [2, 4, 2, 4]
 Generation 2130: Best = 6.005333e+02 FES = 213100 μ_F = 0.429 μ_{CR} = 0.057
 Diversity = 2.1208e-03 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 6.005333e+02 FES = 214100 μ_F = 0.434 μ_{CR} = 0.055
 Diversity = 2.1090e-03 k_t = [2, 4, 4, 4]
 Generation 2150: Best = 6.005333e+02 FES = 215100 μ_F = 0.405 μ_{CR} = 0.068
 Diversity = 2.1067e-03 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 6.005333e+02 FES = 216100 μ_F = 0.385 μ_{CR} = 0.061
 Diversity = 2.1099e-03 k_t = [3, 3, 3, 3]
 Generation 2170: Best = 6.005328e+02 FES = 217100 μ_F = 0.341 μ_{CR} = 0.059
 Diversity = 2.1091e-03 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 6.005327e+02 FES = 218100 μ_F = 0.410 μ_{CR} = 0.056
 Diversity = 1.9906e-03 k_t = [2, 3, 2, 4]
 Generation 2190: Best = 6.005326e+02 FES = 219100 μ_F = 0.433 μ_{CR} = 0.054
 Diversity = 1.9931e-03 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 6.005326e+02 FES = 220100 μ_F = 0.503 μ_{CR} = 0.057
 Diversity = 1.9891e-03 k_t = [2, 3, 3, 3]
 Generation 2210: Best = 6.005326e+02 FES = 221100 μ_F = 0.577 μ_{CR} = 0.055
 Diversity = 1.9211e-03 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 6.005324e+02 FES = 222100 μ_F = 0.649 μ_{CR} = 0.051
 Diversity = 1.7702e-03 k_t = [2, 3, 3, 4]
 Generation 2230: Best = 6.005319e+02 FES = 223100 μ_F = 0.612 μ_{CR} = 0.039
 Diversity = 1.7709e-03 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 6.005317e+02 FES = 224100 μ_F = 0.514 μ_{CR} = 0.053
 Diversity = 1.7711e-03 k_t = [4, 3, 3, 2]
 Generation 2250: Best = 6.005316e+02 FES = 225100 μ_F = 0.528 μ_{CR} = 0.048
 Diversity = 1.6750e-03 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 6.005315e+02 FES = 226100 μ_F = 0.565 μ_{CR} = 0.041
 Diversity = 1.6356e-03 k_t = [2, 4, 3, 2]
 Generation 2270: Best = 6.005315e+02 FES = 227100 μ_F = 0.587 μ_{CR} = 0.040
 Diversity = 1.6348e-03 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 6.005315e+02 FES = 228100 μ_F = 0.593 μ_{CR} = 0.058
 Diversity = 1.6301e-03 k_t = [3, 4, 4, 3]
 Generation 2290: Best = 6.005314e+02 FES = 229100 μ_F = 0.662 μ_{CR} = 0.056
 Diversity = 1.6300e-03 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 6.005313e+02 FES = 230100 μ_F = 0.494 μ_{CR} = 0.070
 Diversity = 1.6292e-03 k_t = [4, 3, 2, 2]
 Generation 2310: Best = 6.005309e+02 FES = 231100 μ_F = 0.522 μ_{CR} = 0.055
 Diversity = 1.6292e-03 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 6.005309e+02 FES = 232100 μ_F = 0.539 μ_{CR} = 0.055
 Diversity = 1.5490e-03 k_t = [3, 3, 3, 4]
 Generation 2330: Best = 6.005309e+02 FES = 233100 μ_F = 0.580 μ_{CR} = 0.048
 Diversity = 1.4163e-03 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 6.005309e+02 FES = 234100 μ_F = 0.577 μ_{CR} = 0.060
 Diversity = 1.2103e-03 k_t = [4, 2, 2, 3]
 Generation 2350: Best = 6.005309e+02 FES = 235100 μ_F = 0.613 μ_{CR} = 0.051
 Diversity = 1.1724e-03 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 6.005309e+02 FES = 236100 μ_F = 0.614 μ_{CR} = 0.061
 Diversity = 1.0995e-03 k_t = [4, 4, 2, 2]
 Generation 2370: Best = 6.005299e+02 FES = 237100 μ_F = 0.589 μ_{CR} = 0.075
 Diversity = 1.0707e-03 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 6.005299e+02 FES = 238100 μ_F = 0.628 μ_{CR} = 0.069
 Diversity = 1.0810e-03 k_t = [2, 3, 3, 3]

Generation 2390: Best = 6.005293e+02 FES = 239100 μ_F = 0.622 μ_{CR} = 0.079
 Diversity = 1.0657e-03 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 6.005293e+02 FES = 240100 μ_F = 0.632 μ_{CR} = 0.076
 Diversity = 1.0153e-03 k_t = [2, 2, 4, 3]
 Generation 2410: Best = 6.005293e+02 FES = 241100 μ_F = 0.622 μ_{CR} = 0.066
 Diversity = 1.0031e-03 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 6.005292e+02 FES = 242100 μ_F = 0.679 μ_{CR} = 0.069
 Diversity = 8.9004e-04 k_t = [2, 4, 2, 4]
 Generation 2430: Best = 6.005292e+02 FES = 243100 μ_F = 0.677 μ_{CR} = 0.056
 Diversity = 8.5310e-04 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 6.005289e+02 FES = 244100 μ_F = 0.664 μ_{CR} = 0.052
 Diversity = 8.4398e-04 k_t = [3, 3, 2, 3]
 Generation 2450: Best = 6.005279e+02 FES = 245100 μ_F = 0.662 μ_{CR} = 0.067
 Diversity = 8.3749e-04 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 6.005273e+02 FES = 246100 μ_F = 0.667 μ_{CR} = 0.073
 Diversity = 8.3152e-04 k_t = [4, 3, 3, 2]
 Generation 2470: Best = 6.005273e+02 FES = 247100 μ_F = 0.708 μ_{CR} = 0.081
 Diversity = 8.2873e-04 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 6.005256e+02 FES = 248100 μ_F = 0.700 μ_{CR} = 0.070
 Diversity = 8.3139e-04 k_t = [3, 3, 3, 3]
 Generation 2490: Best = 6.005256e+02 FES = 249100 μ_F = 0.730 μ_{CR} = 0.078
 Diversity = 7.8429e-04 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 6.005246e+02 FES = 250100 μ_F = 0.727 μ_{CR} = 0.072
 Diversity = 7.5173e-04 k_t = [3, 3, 4, 4]
 Generation 2510: Best = 6.005246e+02 FES = 251100 μ_F = 0.661 μ_{CR} = 0.061
 Diversity = 7.0263e-04 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 6.005246e+02 FES = 252100 μ_F = 0.668 μ_{CR} = 0.062
 Diversity = 6.6774e-04 k_t = [3, 3, 3, 2]
 Generation 2530: Best = 6.005242e+02 FES = 253100 μ_F = 0.680 μ_{CR} = 0.068
 Diversity = 6.5446e-04 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 6.005224e+02 FES = 254100 μ_F = 0.652 μ_{CR} = 0.072
 Diversity = 6.3356e-04 k_t = [4, 3, 3, 3]
 Generation 2550: Best = 6.005224e+02 FES = 255100 μ_F = 0.650 μ_{CR} = 0.073
 Diversity = 5.8616e-04 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 6.005203e+02 FES = 256100 μ_F = 0.649 μ_{CR} = 0.072
 Diversity = 5.5202e-04 k_t = [4, 4, 4, 2]
 Generation 2570: Best = 6.005201e+02 FES = 257100 μ_F = 0.693 μ_{CR} = 0.064
 Diversity = 4.9399e-04 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 6.005196e+02 FES = 258100 μ_F = 0.678 μ_{CR} = 0.073
 Diversity = 4.4974e-04 k_t = [4, 4, 2, 4]
 Generation 2590: Best = 6.005193e+02 FES = 259100 μ_F = 0.694 μ_{CR} = 0.071
 Diversity = 4.3098e-04 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 6.005193e+02 FES = 260100 μ_F = 0.660 μ_{CR} = 0.078
 Diversity = 3.9072e-04 k_t = [3, 2, 4, 3]
 Generation 2610: Best = 6.005193e+02 FES = 261100 μ_F = 0.655 μ_{CR} = 0.072
 Diversity = 3.5414e-04 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 6.005185e+02 FES = 262100 μ_F = 0.663 μ_{CR} = 0.076
 Diversity = 2.9244e-04 k_t = [2, 4, 3, 4]
 Generation 2630: Best = 6.005185e+02 FES = 263100 μ_F = 0.680 μ_{CR} = 0.085
 Diversity = 2.6103e-04 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 6.005185e+02 FES = 264100 μ_F = 0.690 μ_{CR} = 0.088
 Diversity = 2.2757e-04 k_t = [4, 4, 4, 2]
 Generation 2650: Best = 6.005179e+02 FES = 265100 μ_F = 0.692 μ_{CR} = 0.085
 Diversity = 2.2615e-04 k_t = [1, 1, 1, 1]

Generation 2660: Best = 6.005176e+02 FES = 266100 μ_F = 0.679 μ_{CR} = 0.083
 Diversity = 2.1805e-04 k_t = [4, 3, 4, 3]
 Generation 2670: Best = 6.005175e+02 FES = 267100 μ_F = 0.684 μ_{CR} = 0.084
 Diversity = 2.0489e-04 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 6.005175e+02 FES = 268100 μ_F = 0.672 μ_{CR} = 0.074
 Diversity = 2.0192e-04 k_t = [2, 2, 4, 2]
 Generation 2690: Best = 6.005175e+02 FES = 269100 μ_F = 0.676 μ_{CR} = 0.064
 Diversity = 1.9465e-04 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 6.005172e+02 FES = 270100 μ_F = 0.666 μ_{CR} = 0.069
 Diversity = 1.8363e-04 k_t = [3, 2, 4, 2]
 Generation 2710: Best = 6.005162e+02 FES = 271100 μ_F = 0.660 μ_{CR} = 0.071
 Diversity = 1.7840e-04 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 6.005162e+02 FES = 272100 μ_F = 0.675 μ_{CR} = 0.072
 Diversity = 1.7186e-04 k_t = [3, 3, 2, 4]
 Generation 2730: Best = 6.005161e+02 FES = 273100 μ_F = 0.686 μ_{CR} = 0.072
 Diversity = 1.6995e-04 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 6.005161e+02 FES = 274100 μ_F = 0.699 μ_{CR} = 0.063
 Diversity = 1.6953e-04 k_t = [3, 2, 3, 3]
 Generation 2750: Best = 6.005160e+02 FES = 275100 μ_F = 0.670 μ_{CR} = 0.065
 Diversity = 1.6471e-04 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 6.005157e+02 FES = 276100 μ_F = 0.671 μ_{CR} = 0.078
 Diversity = 1.6199e-04 k_t = [4, 4, 4, 2]
 Generation 2770: Best = 6.005156e+02 FES = 277100 μ_F = 0.641 μ_{CR} = 0.087
 Diversity = 1.5908e-04 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 6.005156e+02 FES = 278100 μ_F = 0.659 μ_{CR} = 0.085
 Diversity = 1.4890e-04 k_t = [2, 3, 4, 2]
 Generation 2790: Best = 6.005155e+02 FES = 279100 μ_F = 0.678 μ_{CR} = 0.078
 Diversity = 1.5207e-04 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 6.005154e+02 FES = 280100 μ_F = 0.660 μ_{CR} = 0.071
 Diversity = 1.4275e-04 k_t = [4, 3, 2, 2]
 Generation 2810: Best = 6.005154e+02 FES = 281100 μ_F = 0.679 μ_{CR} = 0.073
 Diversity = 1.3837e-04 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 6.005152e+02 FES = 282100 μ_F = 0.721 μ_{CR} = 0.063
 Diversity = 1.3603e-04 k_t = [4, 3, 2, 4]
 Generation 2830: Best = 6.005152e+02 FES = 283100 μ_F = 0.735 μ_{CR} = 0.062
 Diversity = 1.3543e-04 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 6.005148e+02 FES = 284100 μ_F = 0.701 μ_{CR} = 0.059
 Diversity = 1.3774e-04 k_t = [4, 4, 2, 4]
 Generation 2850: Best = 6.005148e+02 FES = 285100 μ_F = 0.691 μ_{CR} = 0.063
 Diversity = 1.3912e-04 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 6.005145e+02 FES = 286100 μ_F = 0.674 μ_{CR} = 0.070
 Diversity = 1.4102e-04 k_t = [4, 3, 4, 3]
 Generation 2870: Best = 6.005145e+02 FES = 287100 μ_F = 0.666 μ_{CR} = 0.059
 Diversity = 1.4272e-04 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 6.005141e+02 FES = 288100 μ_F = 0.678 μ_{CR} = 0.059
 Diversity = 1.4900e-04 k_t = [3, 4, 2, 2]
 Generation 2890: Best = 6.005137e+02 FES = 289100 μ_F = 0.682 μ_{CR} = 0.061
 Diversity = 1.4486e-04 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 6.005136e+02 FES = 290100 μ_F = 0.671 μ_{CR} = 0.056
 Diversity = 1.4862e-04 k_t = [2, 2, 3, 3]
 Generation 2910: Best = 6.005135e+02 FES = 291100 μ_F = 0.703 μ_{CR} = 0.057
 Diversity = 1.3984e-04 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 6.005135e+02 FES = 292100 μ_F = 0.688 μ_{CR} = 0.057
 Diversity = 1.1998e-04 k_t = [3, 2, 3, 3]

Generation 2930: Best = 6.005133e+02 FES = 293100 μ_F = 0.687 μ_{CR} = 0.069
 Diversity = 1.3307e-04 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 6.005131e+02 FES = 294100 μ_F = 0.714 μ_{CR} = 0.071
 Diversity = 1.7199e-04 k_t = [2, 2, 3, 2]
 Generation 2950: Best = 6.005131e+02 FES = 295100 μ_F = 0.716 μ_{CR} = 0.063
 Diversity = 1.9010e-04 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 6.005127e+02 FES = 296100 μ_F = 0.749 μ_{CR} = 0.068
 Diversity = 1.8287e-04 k_t = [4, 4, 4, 4]
 Generation 2970: Best = 6.005127e+02 FES = 297100 μ_F = 0.687 μ_{CR} = 0.058
 Diversity = 1.6125e-04 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 6.005125e+02 FES = 298100 μ_F = 0.714 μ_{CR} = 0.058
 Diversity = 1.5894e-04 k_t = [4, 2, 4, 4]
 Generation 2990: Best = 6.005122e+02 FES = 299100 μ_F = 0.710 μ_{CR} = 0.055
 Diversity = 1.4566e-04 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 6.005122e+02

Total Function Evals = 300000

Total Successful Updates = 22647

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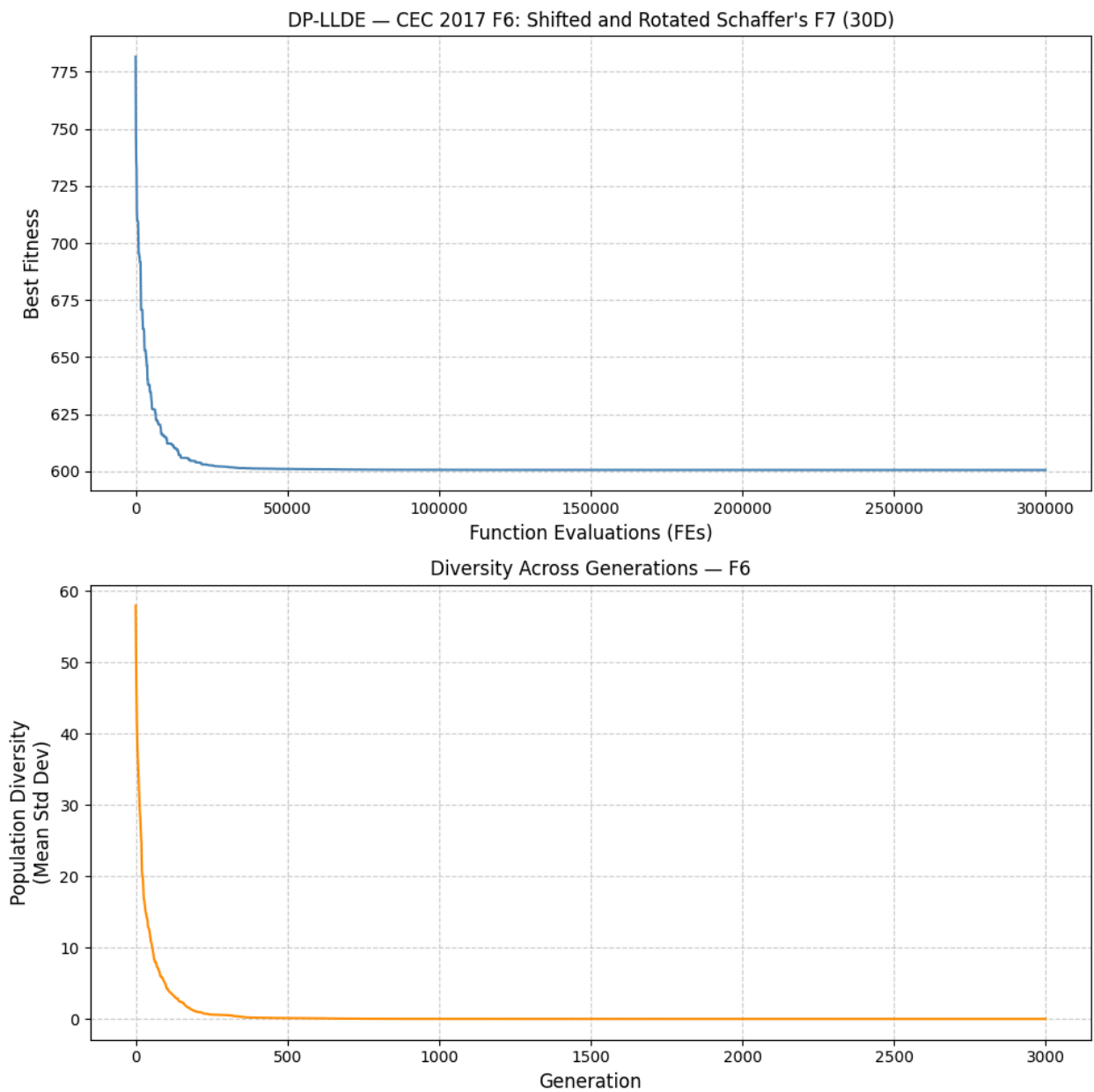
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Results:

Best Fitness : 6.005122e+02
 Global Optimum : 6.000000e+02
 Error (|best-optimum|) : 5.122056e-01
 Best Solution (5 dim) : [47.32615724 -24.65658714 -41.9919819 -55.89596257
 74.24673895]
 D^(0) (init. diversity): 5.7976e+01

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Graph saved as convergence_F6.png

```
In [ ]: # =====
# F7 — Shifted and Rotated Lunacek Bi-Rastrigin
# =====
run_and_plot('F7', make_wrapper(simple.f7), 700.0, "Shifted and Rotated Lunacek Bi-Rastrigin")
```

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DP-LLDE on CEC 2017 F7 – Single Run

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Function : Shifted and Rotated Lunacek Bi-Rastrigin's (30D)
Bounds : [-100, 100]
Global optimum : 700.0
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 3.268478e+03

Generation 10:	Best = 1.345653e+03	FES = 1100	$\mu_F = 0.540$	$\mu_{CR} = 0.359$
Diversity = 2.0364e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 1.100074e+03	FES = 2100	$\mu_F = 0.587$	$\mu_{CR} = 0.360$
Diversity = 9.7141e+00	$k_t = [4, 2, 4, 4]$			
Generation 30:	Best = 1.020078e+03	FES = 3100	$\mu_F = 0.639$	$\mu_{CR} = 0.343$
Diversity = 7.4260e+00	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 9.833655e+02	FES = 4100	$\mu_F = 0.680$	$\mu_{CR} = 0.308$
Diversity = 6.9734e+00	$k_t = [4, 3, 4, 2]$			
Generation 50:	Best = 9.751660e+02	FES = 5100	$\mu_F = 0.704$	$\mu_{CR} = 0.272$
Diversity = 6.5126e+00	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 9.678479e+02	FES = 6100	$\mu_F = 0.702$	$\mu_{CR} = 0.260$
Diversity = 6.1630e+00	$k_t = [3, 3, 2, 4]$			
Generation 70:	Best = 9.473500e+02	FES = 7100	$\mu_F = 0.725$	$\mu_{CR} = 0.192$
Diversity = 6.1107e+00	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 9.473500e+02	FES = 8100	$\mu_F = 0.730$	$\mu_{CR} = 0.139$
Diversity = 6.0832e+00	$k_t = [2, 2, 3, 2]$			
Generation 90:	Best = 9.333106e+02	FES = 9100	$\mu_F = 0.747$	$\mu_{CR} = 0.119$
Diversity = 6.0835e+00	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 9.333106e+02	FES = 10100	$\mu_F = 0.752$	$\mu_{CR} = 0.098$
Diversity = 6.0694e+00	$k_t = [4, 4, 3, 3]$			
Generation 110:	Best = 9.333106e+02	FES = 11100	$\mu_F = 0.744$	$\mu_{CR} = 0.075$
Diversity = 5.9482e+00	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 9.309193e+02	FES = 12100	$\mu_F = 0.765$	$\mu_{CR} = 0.074$
Diversity = 5.9471e+00	$k_t = [4, 2, 4, 3]$			
Generation 130:	Best = 9.309193e+02	FES = 13100	$\mu_F = 0.770$	$\mu_{CR} = 0.065$
Diversity = 5.8905e+00	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 9.135109e+02	FES = 14100	$\mu_F = 0.773$	$\mu_{CR} = 0.073$
Diversity = 5.9128e+00	$k_t = [2, 2, 2, 2]$			
Generation 150:	Best = 9.135109e+02	FES = 15100	$\mu_F = 0.770$	$\mu_{CR} = 0.064$
Diversity = 5.8254e+00	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 9.017650e+02	FES = 16100	$\mu_F = 0.769$	$\mu_{CR} = 0.054$
Diversity = 5.7861e+00	$k_t = [4, 4, 3, 3]$			
Generation 170:	Best = 9.017650e+02	FES = 17100	$\mu_F = 0.770$	$\mu_{CR} = 0.062$
Diversity = 5.7575e+00	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 8.943239e+02	FES = 18100	$\mu_F = 0.782$	$\mu_{CR} = 0.047$
Diversity = 5.6992e+00	$k_t = [2, 2, 3, 2]$			
Generation 190:	Best = 8.943239e+02	FES = 19100	$\mu_F = 0.782$	$\mu_{CR} = 0.057$
Diversity = 5.6603e+00	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 8.780059e+02	FES = 20100	$\mu_F = 0.748$	$\mu_{CR} = 0.055$
Diversity = 5.6781e+00	$k_t = [3, 2, 2, 4]$			
Generation 210:	Best = 8.780059e+02	FES = 21100	$\mu_F = 0.766$	$\mu_{CR} = 0.062$
Diversity = 5.6407e+00	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 8.780059e+02	FES = 22100	$\mu_F = 0.704$	$\mu_{CR} = 0.056$
Diversity = 5.6133e+00	$k_t = [3, 3, 2, 3]$			

Generation 230: Best = 8.780059e+02	FES = 23100	$\mu_F = 0.697$	$\mu_{CR} = 0.061$
Diversity = 5.6354e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 8.738917e+02	FES = 24100	$\mu_F = 0.688$	$\mu_{CR} = 0.065$
Diversity = 5.6464e+00	$k_t = [4, 3, 3, 2]$		
Generation 250: Best = 8.738917e+02	FES = 25100	$\mu_F = 0.614$	$\mu_{CR} = 0.059$
Diversity = 5.6133e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 8.738917e+02	FES = 26100	$\mu_F = 0.614$	$\mu_{CR} = 0.052$
Diversity = 5.5577e+00	$k_t = [2, 4, 2, 3]$		
Generation 270: Best = 8.703584e+02	FES = 27100	$\mu_F = 0.656$	$\mu_{CR} = 0.044$
Diversity = 5.5088e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 8.703584e+02	FES = 28100	$\mu_F = 0.728$	$\mu_{CR} = 0.043$
Diversity = 5.4720e+00	$k_t = [2, 4, 4, 4]$		
Generation 290: Best = 8.703584e+02	FES = 29100	$\mu_F = 0.745$	$\mu_{CR} = 0.036$
Diversity = 5.4683e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 8.589654e+02	FES = 30100	$\mu_F = 0.764$	$\mu_{CR} = 0.048$
Diversity = 5.4613e+00	$k_t = [2, 3, 3, 3]$		
Generation 310: Best = 8.589654e+02	FES = 31100	$\mu_F = 0.788$	$\mu_{CR} = 0.049$
Diversity = 5.4491e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 8.582011e+02	FES = 32100	$\mu_F = 0.763$	$\mu_{CR} = 0.061$
Diversity = 5.4491e+00	$k_t = [2, 4, 3, 3]$		
Generation 330: Best = 8.582011e+02	FES = 33100	$\mu_F = 0.797$	$\mu_{CR} = 0.048$
Diversity = 5.4479e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 8.582011e+02	FES = 34100	$\mu_F = 0.753$	$\mu_{CR} = 0.040$
Diversity = 5.4010e+00	$k_t = [3, 3, 2, 4]$		
Generation 350: Best = 8.582011e+02	FES = 35100	$\mu_F = 0.691$	$\mu_{CR} = 0.039$
Diversity = 5.3874e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 8.552138e+02	FES = 36100	$\mu_F = 0.638$	$\mu_{CR} = 0.041$
Diversity = 5.3771e+00	$k_t = [3, 4, 3, 2]$		
Generation 370: Best = 8.552138e+02	FES = 37100	$\mu_F = 0.592$	$\mu_{CR} = 0.052$
Diversity = 5.3683e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 8.552138e+02	FES = 38100	$\mu_F = 0.614$	$\mu_{CR} = 0.068$
Diversity = 5.3273e+00	$k_t = [2, 3, 3, 2]$		
Generation 390: Best = 8.552138e+02	FES = 39100	$\mu_F = 0.608$	$\mu_{CR} = 0.056$
Diversity = 5.3050e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 8.543613e+02	FES = 40100	$\mu_F = 0.665$	$\mu_{CR} = 0.056$
Diversity = 5.2346e+00	$k_t = [4, 2, 3, 4]$		
Generation 410: Best = 8.543613e+02	FES = 41100	$\mu_F = 0.709$	$\mu_{CR} = 0.045$
Diversity = 5.2376e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 8.543613e+02	FES = 42100	$\mu_F = 0.577$	$\mu_{CR} = 0.048$
Diversity = 5.2334e+00	$k_t = [4, 4, 4, 2]$		
Generation 430: Best = 8.454391e+02	FES = 43100	$\mu_F = 0.462$	$\mu_{CR} = 0.048$
Diversity = 5.2252e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 8.454391e+02	FES = 44100	$\mu_F = 0.473$	$\mu_{CR} = 0.053$
Diversity = 5.1702e+00	$k_t = [2, 4, 3, 2]$		
Generation 450: Best = 8.454391e+02	FES = 45100	$\mu_F = 0.327$	$\mu_{CR} = 0.049$
Diversity = 5.1568e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 8.454391e+02	FES = 46100	$\mu_F = 0.390$	$\mu_{CR} = 0.057$
Diversity = 5.1428e+00	$k_t = [2, 3, 4, 2]$		
Generation 470: Best = 8.454391e+02	FES = 47100	$\mu_F = 0.379$	$\mu_{CR} = 0.054$
Diversity = 5.1391e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 8.419783e+02	FES = 48100	$\mu_F = 0.309$	$\mu_{CR} = 0.051$
Diversity = 5.1310e+00	$k_t = [2, 2, 2, 3]$		
Generation 490: Best = 8.419783e+02	FES = 49100	$\mu_F = 0.366$	$\mu_{CR} = 0.055$
Diversity = 5.1245e+00	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 8.417650e+02	FES = 50100	$\mu_F = 0.494$	$\mu_{CR} = 0.063$
Diversity = 5.1202e+00	k_t = [2, 4, 2, 2]		
Generation 510: Best = 8.417650e+02	FES = 51100	$\mu_F = 0.410$	$\mu_{CR} = 0.052$
Diversity = 5.1138e+00	k_t = [1, 1, 1, 1]		
Generation 520: Best = 8.417650e+02	FES = 52100	$\mu_F = 0.351$	$\mu_{CR} = 0.047$
Diversity = 5.1064e+00	k_t = [2, 2, 4, 3]		
Generation 530: Best = 8.417650e+02	FES = 53100	$\mu_F = 0.362$	$\mu_{CR} = 0.055$
Diversity = 5.1073e+00	k_t = [1, 1, 1, 1]		
Generation 540: Best = 8.387669e+02	FES = 54100	$\mu_F = 0.363$	$\mu_{CR} = 0.055$
Diversity = 5.1049e+00	k_t = [2, 4, 3, 2]		
Generation 550: Best = 8.380497e+02	FES = 55100	$\mu_F = 0.418$	$\mu_{CR} = 0.052$
Diversity = 5.0994e+00	k_t = [1, 1, 1, 1]		
Generation 560: Best = 8.343481e+02	FES = 56100	$\mu_F = 0.388$	$\mu_{CR} = 0.050$
Diversity = 5.0928e+00	k_t = [2, 3, 2, 3]		
Generation 570: Best = 8.343481e+02	FES = 57100	$\mu_F = 0.407$	$\mu_{CR} = 0.032$
Diversity = 5.0866e+00	k_t = [1, 1, 1, 1]		
Generation 580: Best = 8.298814e+02	FES = 58100	$\mu_F = 0.406$	$\mu_{CR} = 0.048$
Diversity = 5.0889e+00	k_t = [4, 2, 2, 3]		
Generation 590: Best = 8.298814e+02	FES = 59100	$\mu_F = 0.367$	$\mu_{CR} = 0.057$
Diversity = 5.0837e+00	k_t = [1, 1, 1, 1]		
Generation 600: Best = 8.298814e+02	FES = 60100	$\mu_F = 0.395$	$\mu_{CR} = 0.050$
Diversity = 5.0786e+00	k_t = [2, 3, 2, 2]		
Generation 610: Best = 8.230976e+02	FES = 61100	$\mu_F = 0.402$	$\mu_{CR} = 0.057$
Diversity = 5.0710e+00	k_t = [1, 1, 1, 1]		
Generation 620: Best = 8.230976e+02	FES = 62100	$\mu_F = 0.364$	$\mu_{CR} = 0.065$
Diversity = 5.0837e+00	k_t = [2, 3, 3, 4]		
Generation 630: Best = 8.230976e+02	FES = 63100	$\mu_F = 0.339$	$\mu_{CR} = 0.056$
Diversity = 5.0829e+00	k_t = [1, 1, 1, 1]		
Generation 640: Best = 8.230976e+02	FES = 64100	$\mu_F = 0.384$	$\mu_{CR} = 0.060$
Diversity = 5.0688e+00	k_t = [3, 2, 2, 4]		
Generation 650: Best = 8.230976e+02	FES = 65100	$\mu_F = 0.401$	$\mu_{CR} = 0.051$
Diversity = 5.0690e+00	k_t = [1, 1, 1, 1]		
Generation 660: Best = 8.230976e+02	FES = 66100	$\mu_F = 0.414$	$\mu_{CR} = 0.054$
Diversity = 5.0699e+00	k_t = [4, 3, 3, 2]		
Generation 670: Best = 8.230976e+02	FES = 67100	$\mu_F = 0.345$	$\mu_{CR} = 0.051$
Diversity = 5.0665e+00	k_t = [1, 1, 1, 1]		
Generation 680: Best = 8.230976e+02	FES = 68100	$\mu_F = 0.377$	$\mu_{CR} = 0.044$
Diversity = 5.0639e+00	k_t = [2, 3, 2, 4]		
Generation 690: Best = 8.163335e+02	FES = 69100	$\mu_F = 0.252$	$\mu_{CR} = 0.055$
Diversity = 5.0624e+00	k_t = [1, 1, 1, 1]		
Generation 700: Best = 8.163335e+02	FES = 70100	$\mu_F = 0.302$	$\mu_{CR} = 0.063$
Diversity = 5.0574e+00	k_t = [2, 4, 4, 3]		
Generation 710: Best = 8.163335e+02	FES = 71100	$\mu_F = 0.300$	$\mu_{CR} = 0.055$
Diversity = 5.0564e+00	k_t = [1, 1, 1, 1]		
Generation 720: Best = 8.163335e+02	FES = 72100	$\mu_F = 0.327$	$\mu_{CR} = 0.079$
Diversity = 5.0499e+00	k_t = [4, 3, 4, 4]		
Generation 730: Best = 8.163335e+02	FES = 73100	$\mu_F = 0.322$	$\mu_{CR} = 0.063$
Diversity = 5.0482e+00	k_t = [1, 1, 1, 1]		
Generation 740: Best = 8.122012e+02	FES = 74100	$\mu_F = 0.286$	$\mu_{CR} = 0.063$
Diversity = 5.0423e+00	k_t = [2, 3, 2, 3]		
Generation 750: Best = 8.122012e+02	FES = 75100	$\mu_F = 0.402$	$\mu_{CR} = 0.053$
Diversity = 5.0308e+00	k_t = [1, 1, 1, 1]		
Generation 760: Best = 8.122012e+02	FES = 76100	$\mu_F = 0.490$	$\mu_{CR} = 0.060$
Diversity = 5.0339e+00	k_t = [4, 4, 2, 3]		

Generation 770: Best = 8.122012e+02	FES = 77100	$\mu_F = 0.437$	$\mu_{CR} = 0.060$
Diversity = 5.0342e+00	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 8.122012e+02	FES = 78100	$\mu_F = 0.482$	$\mu_{CR} = 0.043$
Diversity = 5.0341e+00	$k_t = [3, 4, 3, 2]$		
Generation 790: Best = 8.035896e+02	FES = 79100	$\mu_F = 0.527$	$\mu_{CR} = 0.037$
Diversity = 5.0341e+00	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 8.029259e+02	FES = 80100	$\mu_F = 0.537$	$\mu_{CR} = 0.040$
Diversity = 5.0331e+00	$k_t = [2, 2, 3, 4]$		
Generation 810: Best = 8.029259e+02	FES = 81100	$\mu_F = 0.554$	$\mu_{CR} = 0.038$
Diversity = 5.0330e+00	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 8.029259e+02	FES = 82100	$\mu_F = 0.485$	$\mu_{CR} = 0.048$
Diversity = 5.0305e+00	$k_t = [4, 2, 3, 2]$		
Generation 830: Best = 8.029259e+02	FES = 83100	$\mu_F = 0.404$	$\mu_{CR} = 0.050$
Diversity = 5.0293e+00	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 8.028586e+02	FES = 84100	$\mu_F = 0.414$	$\mu_{CR} = 0.056$
Diversity = 5.0312e+00	$k_t = [2, 2, 4, 2]$		
Generation 850: Best = 8.028586e+02	FES = 85100	$\mu_F = 0.407$	$\mu_{CR} = 0.049$
Diversity = 5.0281e+00	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 8.028586e+02	FES = 86100	$\mu_F = 0.375$	$\mu_{CR} = 0.046$
Diversity = 5.0273e+00	$k_t = [3, 3, 2, 3]$		
Generation 870: Best = 8.028586e+02	FES = 87100	$\mu_F = 0.313$	$\mu_{CR} = 0.054$
Diversity = 5.0252e+00	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 8.020879e+02	FES = 88100	$\mu_F = 0.248$	$\mu_{CR} = 0.044$
Diversity = 5.0235e+00	$k_t = [3, 2, 2, 2]$		
Generation 890: Best = 8.019738e+02	FES = 89100	$\mu_F = 0.204$	$\mu_{CR} = 0.061$
Diversity = 5.0177e+00	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 8.019738e+02	FES = 90100	$\mu_F = 0.266$	$\mu_{CR} = 0.054$
Diversity = 4.9940e+00	$k_t = [4, 4, 2, 4]$		
Generation 910: Best = 8.019738e+02	FES = 91100	$\mu_F = 0.233$	$\mu_{CR} = 0.052$
Diversity = 4.9938e+00	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 8.019738e+02	FES = 92100	$\mu_F = 0.201$	$\mu_{CR} = 0.060$
Diversity = 4.9901e+00	$k_t = [3, 2, 2, 2]$		
Generation 930: Best = 8.019738e+02	FES = 93100	$\mu_F = 0.263$	$\mu_{CR} = 0.059$
Diversity = 4.9896e+00	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 8.019738e+02	FES = 94100	$\mu_F = 0.284$	$\mu_{CR} = 0.048$
Diversity = 4.9857e+00	$k_t = [2, 2, 2, 3]$		
Generation 950: Best = 8.009876e+02	FES = 95100	$\mu_F = 0.284$	$\mu_{CR} = 0.045$
Diversity = 4.9873e+00	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 7.998464e+02	FES = 96100	$\mu_F = 0.321$	$\mu_{CR} = 0.055$
Diversity = 4.9864e+00	$k_t = [4, 3, 4, 2]$		
Generation 970: Best = 7.998464e+02	FES = 97100	$\mu_F = 0.325$	$\mu_{CR} = 0.054$
Diversity = 4.9865e+00	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 7.991541e+02	FES = 98100	$\mu_F = 0.229$	$\mu_{CR} = 0.045$
Diversity = 4.9812e+00	$k_t = [2, 4, 2, 2]$		
Generation 990: Best = 7.990334e+02	FES = 99100	$\mu_F = 0.186$	$\mu_{CR} = 0.042$
Diversity = 4.9780e+00	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 7.970683e+02	FES = 100100	$\mu_F = 0.297$	$\mu_{CR} = 0.052$
Diversity = 4.9741e+00	$k_t = [3, 3, 2, 4]$		
Generation 1010: Best = 7.970683e+02	FES = 101100	$\mu_F = 0.274$	$\mu_{CR} = 0.045$
Diversity = 4.9755e+00	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 7.970683e+02	FES = 102100	$\mu_F = 0.210$	$\mu_{CR} = 0.064$
Diversity = 4.9618e+00	$k_t = [3, 4, 4, 2]$		
Generation 1030: Best = 7.962811e+02	FES = 103100	$\mu_F = 0.143$	$\mu_{CR} = 0.068$
Diversity = 4.9557e+00	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 7.962811e+02 FES = 104100 μ_F = 0.117 μ_{CR} = 0.086
 Diversity = 4.9470e+00 k_t = [4, 3, 2, 2]
 Generation 1050: Best = 7.961435e+02 FES = 105100 μ_F = 0.116 μ_{CR} = 0.080
 Diversity = 4.9442e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 7.959849e+02 FES = 106100 μ_F = 0.107 μ_{CR} = 0.069
 Diversity = 4.9381e+00 k_t = [4, 4, 4, 3]
 Generation 1070: Best = 7.928406e+02 FES = 107100 μ_F = 0.140 μ_{CR} = 0.074
 Diversity = 4.9369e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 7.928406e+02 FES = 108100 μ_F = 0.203 μ_{CR} = 0.061
 Diversity = 4.9308e+00 k_t = [3, 4, 2, 2]
 Generation 1090: Best = 7.889000e+02 FES = 109100 μ_F = 0.181 μ_{CR} = 0.073
 Diversity = 4.9303e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 7.889000e+02 FES = 110100 μ_F = 0.255 μ_{CR} = 0.059
 Diversity = 4.9647e+00 k_t = [4, 3, 4, 4]
 Generation 1110: Best = 7.828367e+02 FES = 111100 μ_F = 0.217 μ_{CR} = 0.063
 Diversity = 4.9645e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 7.805332e+02 FES = 112100 μ_F = 0.305 μ_{CR} = 0.062
 Diversity = 4.9646e+00 k_t = [2, 3, 3, 2]
 Generation 1130: Best = 7.805332e+02 FES = 113100 μ_F = 0.223 μ_{CR} = 0.070
 Diversity = 4.9638e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 7.805332e+02 FES = 114100 μ_F = 0.184 μ_{CR} = 0.071
 Diversity = 4.9630e+00 k_t = [4, 4, 2, 2]
 Generation 1150: Best = 7.804555e+02 FES = 115100 μ_F = 0.227 μ_{CR} = 0.060
 Diversity = 4.9613e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 7.804555e+02 FES = 116100 μ_F = 0.201 μ_{CR} = 0.070
 Diversity = 4.9600e+00 k_t = [4, 2, 3, 2]
 Generation 1170: Best = 7.804555e+02 FES = 117100 μ_F = 0.206 μ_{CR} = 0.060
 Diversity = 4.9595e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 7.804555e+02 FES = 118100 μ_F = 0.310 μ_{CR} = 0.063
 Diversity = 4.9590e+00 k_t = [2, 2, 2, 4]
 Generation 1190: Best = 7.804555e+02 FES = 119100 μ_F = 0.332 μ_{CR} = 0.057
 Diversity = 4.9596e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 7.804555e+02 FES = 120100 μ_F = 0.353 μ_{CR} = 0.052
 Diversity = 4.8981e+00 k_t = [3, 2, 3, 4]
 Generation 1210: Best = 7.804555e+02 FES = 121100 μ_F = 0.237 μ_{CR} = 0.060
 Diversity = 4.8981e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 7.804555e+02 FES = 122100 μ_F = 0.255 μ_{CR} = 0.065
 Diversity = 4.8780e+00 k_t = [4, 3, 2, 4]
 Generation 1230: Best = 7.804555e+02 FES = 123100 μ_F = 0.274 μ_{CR} = 0.075
 Diversity = 4.8772e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 7.759613e+02 FES = 124100 μ_F = 0.306 μ_{CR} = 0.053
 Diversity = 4.8770e+00 k_t = [4, 4, 4, 2]
 Generation 1250: Best = 7.759613e+02 FES = 125100 μ_F = 0.200 μ_{CR} = 0.051
 Diversity = 4.8764e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 7.759283e+02 FES = 126100 μ_F = 0.165 μ_{CR} = 0.064
 Diversity = 4.8745e+00 k_t = [4, 3, 2, 3]
 Generation 1270: Best = 7.759283e+02 FES = 127100 μ_F = 0.154 μ_{CR} = 0.063
 Diversity = 4.8739e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 7.757006e+02 FES = 128100 μ_F = 0.196 μ_{CR} = 0.058
 Diversity = 4.8727e+00 k_t = [3, 2, 4, 2]
 Generation 1290: Best = 7.757006e+02 FES = 129100 μ_F = 0.216 μ_{CR} = 0.049
 Diversity = 4.8719e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 7.757006e+02 FES = 130100 μ_F = 0.311 μ_{CR} = 0.046
 Diversity = 4.8612e+00 k_t = [2, 4, 2, 4]

Generation 1310: Best = 7.757006e+02 FES = 131100 μ_F = 0.278 μ_{CR} = 0.059
 Diversity = 4.8598e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 7.757006e+02 FES = 132100 μ_F = 0.237 μ_{CR} = 0.056
 Diversity = 4.8585e+00 k_t = [2, 2, 4, 2]
 Generation 1330: Best = 7.757006e+02 FES = 133100 μ_F = 0.221 μ_{CR} = 0.055
 Diversity = 4.8543e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 7.757006e+02 FES = 134100 μ_F = 0.230 μ_{CR} = 0.047
 Diversity = 4.8282e+00 k_t = [4, 4, 3, 4]
 Generation 1350: Best = 7.742501e+02 FES = 135100 μ_F = 0.204 μ_{CR} = 0.039
 Diversity = 4.8280e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 7.742501e+02 FES = 136100 μ_F = 0.330 μ_{CR} = 0.046
 Diversity = 4.8073e+00 k_t = [4, 3, 3, 4]
 Generation 1370: Best = 7.739273e+02 FES = 137100 μ_F = 0.235 μ_{CR} = 0.040
 Diversity = 4.8072e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 7.739273e+02 FES = 138100 μ_F = 0.242 μ_{CR} = 0.044
 Diversity = 4.8057e+00 k_t = [3, 4, 4, 3]
 Generation 1390: Best = 7.739273e+02 FES = 139100 μ_F = 0.222 μ_{CR} = 0.050
 Diversity = 4.8057e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 7.739273e+02 FES = 140100 μ_F = 0.270 μ_{CR} = 0.043
 Diversity = 4.8056e+00 k_t = [4, 3, 3, 2]
 Generation 1410: Best = 7.739273e+02 FES = 141100 μ_F = 0.272 μ_{CR} = 0.066
 Diversity = 4.8025e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 7.725043e+02 FES = 142100 μ_F = 0.294 μ_{CR} = 0.075
 Diversity = 4.8028e+00 k_t = [4, 3, 3, 3]
 Generation 1430: Best = 7.725043e+02 FES = 143100 μ_F = 0.289 μ_{CR} = 0.053
 Diversity = 4.8036e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 7.725043e+02 FES = 144100 μ_F = 0.339 μ_{CR} = 0.050
 Diversity = 4.8010e+00 k_t = [3, 4, 3, 4]
 Generation 1450: Best = 7.725043e+02 FES = 145100 μ_F = 0.313 μ_{CR} = 0.052
 Diversity = 4.8017e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 7.725043e+02 FES = 146100 μ_F = 0.373 μ_{CR} = 0.052
 Diversity = 4.8241e+00 k_t = [2, 3, 3, 4]
 Generation 1470: Best = 7.725043e+02 FES = 147100 μ_F = 0.332 μ_{CR} = 0.046
 Diversity = 4.8243e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 7.725043e+02 FES = 148100 μ_F = 0.266 μ_{CR} = 0.054
 Diversity = 4.8244e+00 k_t = [3, 4, 4, 4]
 Generation 1490: Best = 7.720698e+02 FES = 149100 μ_F = 0.207 μ_{CR} = 0.048
 Diversity = 4.8242e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 7.718943e+02 FES = 150100 μ_F = 0.198 μ_{CR} = 0.054
 Diversity = 4.8244e+00 k_t = [3, 3, 2, 2]
 Generation 1510: Best = 7.718943e+02 FES = 151100 μ_F = 0.162 μ_{CR} = 0.063
 Diversity = 4.8245e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 7.718943e+02 FES = 152100 μ_F = 0.175 μ_{CR} = 0.056
 Diversity = 4.8236e+00 k_t = [3, 2, 2, 3]
 Generation 1530: Best = 7.718943e+02 FES = 153100 μ_F = 0.148 μ_{CR} = 0.062
 Diversity = 4.8209e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 7.718943e+02 FES = 154100 μ_F = 0.200 μ_{CR} = 0.076
 Diversity = 4.7861e+00 k_t = [2, 3, 4, 4]
 Generation 1550: Best = 7.718943e+02 FES = 155100 μ_F = 0.271 μ_{CR} = 0.074
 Diversity = 4.7861e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 7.716374e+02 FES = 156100 μ_F = 0.336 μ_{CR} = 0.048
 Diversity = 4.7863e+00 k_t = [4, 4, 3, 3]
 Generation 1570: Best = 7.716374e+02 FES = 157100 μ_F = 0.373 μ_{CR} = 0.046
 Diversity = 4.7851e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 7.716374e+02 FES = 158100 μ_F = 0.416 μ_{CR} = 0.041
 Diversity = 4.7820e+00 k_t = [4, 2, 4, 4]
 Generation 1590: Best = 7.716374e+02 FES = 159100 μ_F = 0.310 μ_{CR} = 0.049
 Diversity = 4.7821e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 7.716374e+02 FES = 160100 μ_F = 0.265 μ_{CR} = 0.045
 Diversity = 4.7577e+00 k_t = [2, 2, 2, 4]
 Generation 1610: Best = 7.716374e+02 FES = 161100 μ_F = 0.225 μ_{CR} = 0.055
 Diversity = 4.7561e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 7.716374e+02 FES = 162100 μ_F = 0.267 μ_{CR} = 0.051
 Diversity = 4.7245e+00 k_t = [3, 4, 2, 4]
 Generation 1630: Best = 7.716374e+02 FES = 163100 μ_F = 0.286 μ_{CR} = 0.046
 Diversity = 4.7243e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 7.714358e+02 FES = 164100 μ_F = 0.274 μ_{CR} = 0.075
 Diversity = 4.7271e+00 k_t = [4, 4, 2, 2]
 Generation 1650: Best = 7.714358e+02 FES = 165100 μ_F = 0.233 μ_{CR} = 0.053
 Diversity = 4.7272e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 7.712247e+02 FES = 166100 μ_F = 0.210 μ_{CR} = 0.077
 Diversity = 4.7273e+00 k_t = [4, 2, 2, 3]
 Generation 1670: Best = 7.706006e+02 FES = 167100 μ_F = 0.164 μ_{CR} = 0.057
 Diversity = 4.7280e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 7.705178e+02 FES = 168100 μ_F = 0.227 μ_{CR} = 0.074
 Diversity = 4.7378e+00 k_t = [4, 3, 3, 4]
 Generation 1690: Best = 7.705178e+02 FES = 169100 μ_F = 0.202 μ_{CR} = 0.076
 Diversity = 4.7382e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 7.703029e+02 FES = 170100 μ_F = 0.226 μ_{CR} = 0.064
 Diversity = 4.7378e+00 k_t = [4, 4, 3, 3]
 Generation 1710: Best = 7.703029e+02 FES = 171100 μ_F = 0.131 μ_{CR} = 0.056
 Diversity = 4.7371e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 7.701765e+02 FES = 172100 μ_F = 0.194 μ_{CR} = 0.051
 Diversity = 4.7375e+00 k_t = [4, 4, 2, 2]
 Generation 1730: Best = 7.701765e+02 FES = 173100 μ_F = 0.132 μ_{CR} = 0.052
 Diversity = 4.7373e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 7.701543e+02 FES = 174100 μ_F = 0.115 μ_{CR} = 0.075
 Diversity = 4.7365e+00 k_t = [4, 2, 3, 4]
 Generation 1750: Best = 7.697446e+02 FES = 175100 μ_F = 0.189 μ_{CR} = 0.060
 Diversity = 4.7362e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 7.697446e+02 FES = 176100 μ_F = 0.176 μ_{CR} = 0.061
 Diversity = 4.7357e+00 k_t = [2, 4, 4, 4]
 Generation 1770: Best = 7.697444e+02 FES = 177100 μ_F = 0.113 μ_{CR} = 0.072
 Diversity = 4.7334e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 7.691436e+02 FES = 178100 μ_F = 0.121 μ_{CR} = 0.064
 Diversity = 4.7338e+00 k_t = [2, 4, 2, 4]
 Generation 1790: Best = 7.691436e+02 FES = 179100 μ_F = 0.120 μ_{CR} = 0.065
 Diversity = 4.7335e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 7.688229e+02 FES = 180100 μ_F = 0.188 μ_{CR} = 0.065
 Diversity = 4.7256e+00 k_t = [4, 2, 4, 4]
 Generation 1810: Best = 7.687723e+02 FES = 181100 μ_F = 0.200 μ_{CR} = 0.057
 Diversity = 4.7258e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 7.687723e+02 FES = 182100 μ_F = 0.164 μ_{CR} = 0.056
 Diversity = 4.7261e+00 k_t = [2, 3, 3, 4]
 Generation 1830: Best = 7.680598e+02 FES = 183100 μ_F = 0.164 μ_{CR} = 0.051
 Diversity = 4.7268e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 7.680598e+02 FES = 184100 μ_F = 0.199 μ_{CR} = 0.050
 Diversity = 4.7251e+00 k_t = [4, 2, 4, 4]

Generation 1850: Best = 7.680598e+02	FES = 185100	$\mu_F = 0.183$	$\mu_{CR} = 0.061$
Diversity = 4.7239e+00	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 7.680598e+02	FES = 186100	$\mu_F = 0.216$	$\mu_{CR} = 0.063$
Diversity = 4.7237e+00	$k_t = [4, 2, 2, 3]$		
Generation 1870: Best = 7.680598e+02	FES = 187100	$\mu_F = 0.205$	$\mu_{CR} = 0.061$
Diversity = 4.7234e+00	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 7.678117e+02	FES = 188100	$\mu_F = 0.191$	$\mu_{CR} = 0.045$
Diversity = 4.7235e+00	$k_t = [4, 3, 3, 2]$		
Generation 1890: Best = 7.678117e+02	FES = 189100	$\mu_F = 0.160$	$\mu_{CR} = 0.041$
Diversity = 4.7234e+00	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 7.678117e+02	FES = 190100	$\mu_F = 0.149$	$\mu_{CR} = 0.043$
Diversity = 4.7233e+00	$k_t = [2, 4, 2, 2]$		
Generation 1910: Best = 7.678117e+02	FES = 191100	$\mu_F = 0.137$	$\mu_{CR} = 0.049$
Diversity = 4.7218e+00	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 7.678117e+02	FES = 192100	$\mu_F = 0.151$	$\mu_{CR} = 0.052$
Diversity = 4.7220e+00	$k_t = [3, 2, 4, 3]$		
Generation 1930: Best = 7.678117e+02	FES = 193100	$\mu_F = 0.177$	$\mu_{CR} = 0.051$
Diversity = 4.7195e+00	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 7.678117e+02	FES = 194100	$\mu_F = 0.225$	$\mu_{CR} = 0.055$
Diversity = 4.7194e+00	$k_t = [4, 4, 3, 3]$		
Generation 1950: Best = 7.678117e+02	FES = 195100	$\mu_F = 0.283$	$\mu_{CR} = 0.048$
Diversity = 4.7199e+00	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 7.677634e+02	FES = 196100	$\mu_F = 0.223$	$\mu_{CR} = 0.051$
Diversity = 4.7198e+00	$k_t = [4, 2, 2, 2]$		
Generation 1970: Best = 7.677634e+02	FES = 197100	$\mu_F = 0.170$	$\mu_{CR} = 0.066$
Diversity = 4.7196e+00	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 7.675302e+02	FES = 198100	$\mu_F = 0.187$	$\mu_{CR} = 0.060$
Diversity = 4.7195e+00	$k_t = [2, 4, 4, 2]$		
Generation 1990: Best = 7.675302e+02	FES = 199100	$\mu_F = 0.155$	$\mu_{CR} = 0.056$
Diversity = 4.7194e+00	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 7.675302e+02	FES = 200100	$\mu_F = 0.238$	$\mu_{CR} = 0.060$
Diversity = 4.7102e+00	$k_t = [4, 4, 4, 4]$		
Generation 2010: Best = 7.675302e+02	FES = 201100	$\mu_F = 0.157$	$\mu_{CR} = 0.112$
Diversity = 4.7104e+00	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 7.675302e+02	FES = 202100	$\mu_F = 0.305$	$\mu_{CR} = 0.083$
Diversity = 4.7111e+00	$k_t = [4, 3, 3, 2]$		
Generation 2030: Best = 7.675302e+02	FES = 203100	$\mu_F = 0.238$	$\mu_{CR} = 0.064$
Diversity = 4.7112e+00	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 7.675302e+02	FES = 204100	$\mu_F = 0.187$	$\mu_{CR} = 0.083$
Diversity = 4.7106e+00	$k_t = [2, 2, 3, 2]$		
Generation 2050: Best = 7.675302e+02	FES = 205100	$\mu_F = 0.204$	$\mu_{CR} = 0.061$
Diversity = 4.7114e+00	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 7.675302e+02	FES = 206100	$\mu_F = 0.165$	$\mu_{CR} = 0.047$
Diversity = 4.7113e+00	$k_t = [3, 4, 2, 2]$		
Generation 2070: Best = 7.675302e+02	FES = 207100	$\mu_F = 0.153$	$\mu_{CR} = 0.049$
Diversity = 4.7113e+00	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 7.675302e+02	FES = 208100	$\mu_F = 0.201$	$\mu_{CR} = 0.072$
Diversity = 4.6837e+00	$k_t = [2, 3, 2, 4]$		
Generation 2090: Best = 7.675302e+02	FES = 209100	$\mu_F = 0.215$	$\mu_{CR} = 0.064$
Diversity = 4.6841e+00	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 7.675302e+02	FES = 210100	$\mu_F = 0.210$	$\mu_{CR} = 0.041$
Diversity = 4.6844e+00	$k_t = [2, 3, 4, 3]$		
Generation 2110: Best = 7.673818e+02	FES = 211100	$\mu_F = 0.167$	$\mu_{CR} = 0.044$
Diversity = 4.6853e+00	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 7.673818e+02 FES = 212100 μ_F = 0.098 μ_{CR} = 0.040
 Diversity = 4.6848e+00 k_t = [2, 4, 2, 3]
 Generation 2130: Best = 7.673127e+02 FES = 213100 μ_F = 0.071 μ_{CR} = 0.049
 Diversity = 4.6845e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 7.673127e+02 FES = 214100 μ_F = 0.065 μ_{CR} = 0.083
 Diversity = 4.6841e+00 k_t = [3, 2, 4, 2]
 Generation 2150: Best = 7.673127e+02 FES = 215100 μ_F = 0.061 μ_{CR} = 0.082
 Diversity = 4.6836e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 7.669573e+02 FES = 216100 μ_F = 0.074 μ_{CR} = 0.091
 Diversity = 4.6827e+00 k_t = [3, 4, 4, 3]
 Generation 2170: Best = 7.669573e+02 FES = 217100 μ_F = 0.133 μ_{CR} = 0.070
 Diversity = 4.6823e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 7.669550e+02 FES = 218100 μ_F = 0.152 μ_{CR} = 0.072
 Diversity = 4.6828e+00 k_t = [2, 2, 2, 3]
 Generation 2190: Best = 7.669550e+02 FES = 219100 μ_F = 0.154 μ_{CR} = 0.075
 Diversity = 4.6825e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 7.669361e+02 FES = 220100 μ_F = 0.131 μ_{CR} = 0.068
 Diversity = 4.6824e+00 k_t = [3, 4, 4, 3]
 Generation 2210: Best = 7.668757e+02 FES = 221100 μ_F = 0.099 μ_{CR} = 0.066
 Diversity = 4.6825e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 7.668757e+02 FES = 222100 μ_F = 0.178 μ_{CR} = 0.064
 Diversity = 4.6458e+00 k_t = [2, 2, 3, 4]
 Generation 2230: Best = 7.668757e+02 FES = 223100 μ_F = 0.157 μ_{CR} = 0.064
 Diversity = 4.6459e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 7.668757e+02 FES = 224100 μ_F = 0.109 μ_{CR} = 0.077
 Diversity = 4.6457e+00 k_t = [2, 2, 2, 2]
 Generation 2250: Best = 7.668615e+02 FES = 225100 μ_F = 0.087 μ_{CR} = 0.080
 Diversity = 4.6460e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 7.668615e+02 FES = 226100 μ_F = 0.081 μ_{CR} = 0.069
 Diversity = 4.6456e+00 k_t = [3, 2, 2, 2]
 Generation 2270: Best = 7.668615e+02 FES = 227100 μ_F = 0.072 μ_{CR} = 0.074
 Diversity = 4.6466e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 7.667064e+02 FES = 228100 μ_F = 0.154 μ_{CR} = 0.081
 Diversity = 4.6213e+00 k_t = [3, 3, 2, 4]
 Generation 2290: Best = 7.666865e+02 FES = 229100 μ_F = 0.204 μ_{CR} = 0.070
 Diversity = 4.6217e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 7.666865e+02 FES = 230100 μ_F = 0.203 μ_{CR} = 0.067
 Diversity = 4.6218e+00 k_t = [4, 4, 3, 3]
 Generation 2310: Best = 7.666865e+02 FES = 231100 μ_F = 0.158 μ_{CR} = 0.049
 Diversity = 4.6217e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 7.664843e+02 FES = 232100 μ_F = 0.252 μ_{CR} = 0.063
 Diversity = 4.5554e+00 k_t = [3, 4, 2, 4]
 Generation 2330: Best = 7.664843e+02 FES = 233100 μ_F = 0.208 μ_{CR} = 0.055
 Diversity = 4.5552e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 7.664843e+02 FES = 234100 μ_F = 0.224 μ_{CR} = 0.056
 Diversity = 4.5554e+00 k_t = [3, 3, 4, 2]
 Generation 2350: Best = 7.664437e+02 FES = 235100 μ_F = 0.200 μ_{CR} = 0.055
 Diversity = 4.5554e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 7.664437e+02 FES = 236100 μ_F = 0.304 μ_{CR} = 0.045
 Diversity = 4.5644e+00 k_t = [4, 2, 2, 4]
 Generation 2370: Best = 7.664437e+02 FES = 237100 μ_F = 0.284 μ_{CR} = 0.040
 Diversity = 4.5644e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 7.664437e+02 FES = 238100 μ_F = 0.267 μ_{CR} = 0.033
 Diversity = 4.5645e+00 k_t = [3, 3, 3, 3]

Generation 2390: Best = 7.664437e+02 FES = 239100 μ_F = 0.237 μ_{CR} = 0.043
 Diversity = 4.5649e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 7.664437e+02 FES = 240100 μ_F = 0.173 μ_{CR} = 0.053
 Diversity = 4.5648e+00 k_t = [4, 3, 2, 3]
 Generation 2410: Best = 7.664171e+02 FES = 241100 μ_F = 0.181 μ_{CR} = 0.061
 Diversity = 4.5653e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 7.664171e+02 FES = 242100 μ_F = 0.250 μ_{CR} = 0.060
 Diversity = 4.5598e+00 k_t = [2, 3, 3, 4]
 Generation 2430: Best = 7.664171e+02 FES = 243100 μ_F = 0.224 μ_{CR} = 0.049
 Diversity = 4.5597e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 7.664171e+02 FES = 244100 μ_F = 0.325 μ_{CR} = 0.046
 Diversity = 4.5511e+00 k_t = [3, 2, 2, 4]
 Generation 2450: Best = 7.664171e+02 FES = 245100 μ_F = 0.258 μ_{CR} = 0.039
 Diversity = 4.5511e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 7.664171e+02 FES = 246100 μ_F = 0.217 μ_{CR} = 0.050
 Diversity = 4.5512e+00 k_t = [4, 2, 3, 2]
 Generation 2470: Best = 7.661459e+02 FES = 247100 μ_F = 0.201 μ_{CR} = 0.052
 Diversity = 4.5512e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 7.661459e+02 FES = 248100 μ_F = 0.230 μ_{CR} = 0.078
 Diversity = 4.5260e+00 k_t = [4, 3, 4, 4]
 Generation 2490: Best = 7.661254e+02 FES = 249100 μ_F = 0.166 μ_{CR} = 0.081
 Diversity = 4.5261e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 7.661084e+02 FES = 250100 μ_F = 0.133 μ_{CR} = 0.086
 Diversity = 4.5258e+00 k_t = [4, 4, 2, 3]
 Generation 2510: Best = 7.661084e+02 FES = 251100 μ_F = 0.091 μ_{CR} = 0.063
 Diversity = 4.5256e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 7.657501e+02 FES = 252100 μ_F = 0.186 μ_{CR} = 0.094
 Diversity = 4.5342e+00 k_t = [3, 3, 3, 4]
 Generation 2530: Best = 7.657501e+02 FES = 253100 μ_F = 0.156 μ_{CR} = 0.073
 Diversity = 4.5343e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 7.657501e+02 FES = 254100 μ_F = 0.163 μ_{CR} = 0.086
 Diversity = 4.5344e+00 k_t = [4, 2, 2, 2]
 Generation 2550: Best = 7.657501e+02 FES = 255100 μ_F = 0.157 μ_{CR} = 0.060
 Diversity = 4.5347e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 7.657501e+02 FES = 256100 μ_F = 0.262 μ_{CR} = 0.063
 Diversity = 4.5297e+00 k_t = [3, 2, 3, 4]
 Generation 2570: Best = 7.657501e+02 FES = 257100 μ_F = 0.182 μ_{CR} = 0.071
 Diversity = 4.5298e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 7.657501e+02 FES = 258100 μ_F = 0.186 μ_{CR} = 0.065
 Diversity = 4.5297e+00 k_t = [3, 4, 4, 3]
 Generation 2590: Best = 7.657468e+02 FES = 259100 μ_F = 0.170 μ_{CR} = 0.063
 Diversity = 4.5310e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 7.657468e+02 FES = 260100 μ_F = 0.207 μ_{CR} = 0.066
 Diversity = 4.5153e+00 k_t = [4, 3, 4, 4]
 Generation 2610: Best = 7.657468e+02 FES = 261100 μ_F = 0.143 μ_{CR} = 0.054
 Diversity = 4.5154e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 7.657468e+02 FES = 262100 μ_F = 0.128 μ_{CR} = 0.051
 Diversity = 4.5158e+00 k_t = [3, 3, 4, 2]
 Generation 2630: Best = 7.654352e+02 FES = 263100 μ_F = 0.116 μ_{CR} = 0.058
 Diversity = 4.5156e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 7.654352e+02 FES = 264100 μ_F = 0.112 μ_{CR} = 0.082
 Diversity = 4.5148e+00 k_t = [4, 4, 2, 2]
 Generation 2650: Best = 7.654352e+02 FES = 265100 μ_F = 0.133 μ_{CR} = 0.069
 Diversity = 4.5150e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 7.654352e+02 FES = 266100 μ_F = 0.202 μ_{CR} = 0.057
 Diversity = 4.4969e+00 k_t = [4, 3, 4, 4]
 Generation 2670: Best = 7.654352e+02 FES = 267100 μ_F = 0.197 μ_{CR} = 0.051
 Diversity = 4.4971e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 7.654352e+02 FES = 268100 μ_F = 0.233 μ_{CR} = 0.043
 Diversity = 4.4923e+00 k_t = [2, 2, 4, 4]
 Generation 2690: Best = 7.654352e+02 FES = 269100 μ_F = 0.185 μ_{CR} = 0.041
 Diversity = 4.4903e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 7.654352e+02 FES = 270100 μ_F = 0.169 μ_{CR} = 0.048
 Diversity = 4.4905e+00 k_t = [4, 2, 3, 2]
 Generation 2710: Best = 7.654352e+02 FES = 271100 μ_F = 0.137 μ_{CR} = 0.045
 Diversity = 4.4907e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 7.654352e+02 FES = 272100 μ_F = 0.215 μ_{CR} = 0.045
 Diversity = 4.4909e+00 k_t = [3, 4, 3, 2]
 Generation 2730: Best = 7.654090e+02 FES = 273100 μ_F = 0.169 μ_{CR} = 0.048
 Diversity = 4.4911e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 7.654090e+02 FES = 274100 μ_F = 0.252 μ_{CR} = 0.059
 Diversity = 4.4575e+00 k_t = [3, 4, 4, 4]
 Generation 2750: Best = 7.654090e+02 FES = 275100 μ_F = 0.235 μ_{CR} = 0.047
 Diversity = 4.4577e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 7.654090e+02 FES = 276100 μ_F = 0.223 μ_{CR} = 0.049
 Diversity = 4.4582e+00 k_t = [4, 3, 2, 2]
 Generation 2770: Best = 7.654090e+02 FES = 277100 μ_F = 0.215 μ_{CR} = 0.060
 Diversity = 4.4584e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 7.654090e+02 FES = 278100 μ_F = 0.290 μ_{CR} = 0.050
 Diversity = 4.4473e+00 k_t = [4, 2, 3, 4]
 Generation 2790: Best = 7.654090e+02 FES = 279100 μ_F = 0.235 μ_{CR} = 0.031
 Diversity = 4.4474e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 7.654090e+02 FES = 280100 μ_F = 0.229 μ_{CR} = 0.041
 Diversity = 4.4477e+00 k_t = [2, 2, 2, 3]
 Generation 2810: Best = 7.654090e+02 FES = 281100 μ_F = 0.147 μ_{CR} = 0.049
 Diversity = 4.4478e+00 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 7.654090e+02 FES = 282100 μ_F = 0.118 μ_{CR} = 0.054
 Diversity = 4.4478e+00 k_t = [3, 2, 3, 2]
 Generation 2830: Best = 7.653358e+02 FES = 283100 μ_F = 0.080 μ_{CR} = 0.054
 Diversity = 4.4476e+00 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 7.653358e+02 FES = 284100 μ_F = 0.121 μ_{CR} = 0.047
 Diversity = 4.4476e+00 k_t = [3, 4, 2, 3]
 Generation 2850: Best = 7.653358e+02 FES = 285100 μ_F = 0.085 μ_{CR} = 0.058
 Diversity = 4.4476e+00 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 7.653358e+02 FES = 286100 μ_F = 0.112 μ_{CR} = 0.058
 Diversity = 4.4477e+00 k_t = [3, 4, 2, 2]
 Generation 2870: Best = 7.653358e+02 FES = 287100 μ_F = 0.122 μ_{CR} = 0.057
 Diversity = 4.4475e+00 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 7.653358e+02 FES = 288100 μ_F = 0.083 μ_{CR} = 0.055
 Diversity = 4.4474e+00 k_t = [2, 3, 4, 4]
 Generation 2890: Best = 7.653358e+02 FES = 289100 μ_F = 0.071 μ_{CR} = 0.061
 Diversity = 4.4479e+00 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 7.653021e+02 FES = 290100 μ_F = 0.175 μ_{CR} = 0.065
 Diversity = 4.4555e+00 k_t = [4, 4, 3, 4]
 Generation 2910: Best = 7.653021e+02 FES = 291100 μ_F = 0.136 μ_{CR} = 0.052
 Diversity = 4.4557e+00 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 7.653021e+02 FES = 292100 μ_F = 0.112 μ_{CR} = 0.065
 Diversity = 4.4557e+00 k_t = [4, 3, 2, 4]

Generation 2930: Best = 7.653021e+02 FES = 293100 μ_F = 0.159 μ_{CR} = 0.074
 Diversity = 4.4555e+00 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 7.653021e+02 FES = 294100 μ_F = 0.195 μ_{CR} = 0.071
 Diversity = 4.4558e+00 k_t = [2, 3, 4, 2]
 Generation 2950: Best = 7.653021e+02 FES = 295100 μ_F = 0.170 μ_{CR} = 0.055
 Diversity = 4.4563e+00 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 7.653021e+02 FES = 296100 μ_F = 0.201 μ_{CR} = 0.047
 Diversity = 4.4660e+00 k_t = [2, 2, 2, 4]
 Generation 2970: Best = 7.653021e+02 FES = 297100 μ_F = 0.177 μ_{CR} = 0.053
 Diversity = 4.4659e+00 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 7.653021e+02 FES = 298100 μ_F = 0.134 μ_{CR} = 0.043
 Diversity = 4.4658e+00 k_t = [3, 2, 4, 4]
 Generation 2990: Best = 7.652807e+02 FES = 299100 μ_F = 0.110 μ_{CR} = 0.034
 Diversity = 4.4660e+00 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 7.652807e+02

Total Function Evals = 300000

Total Successful Updates = 8809

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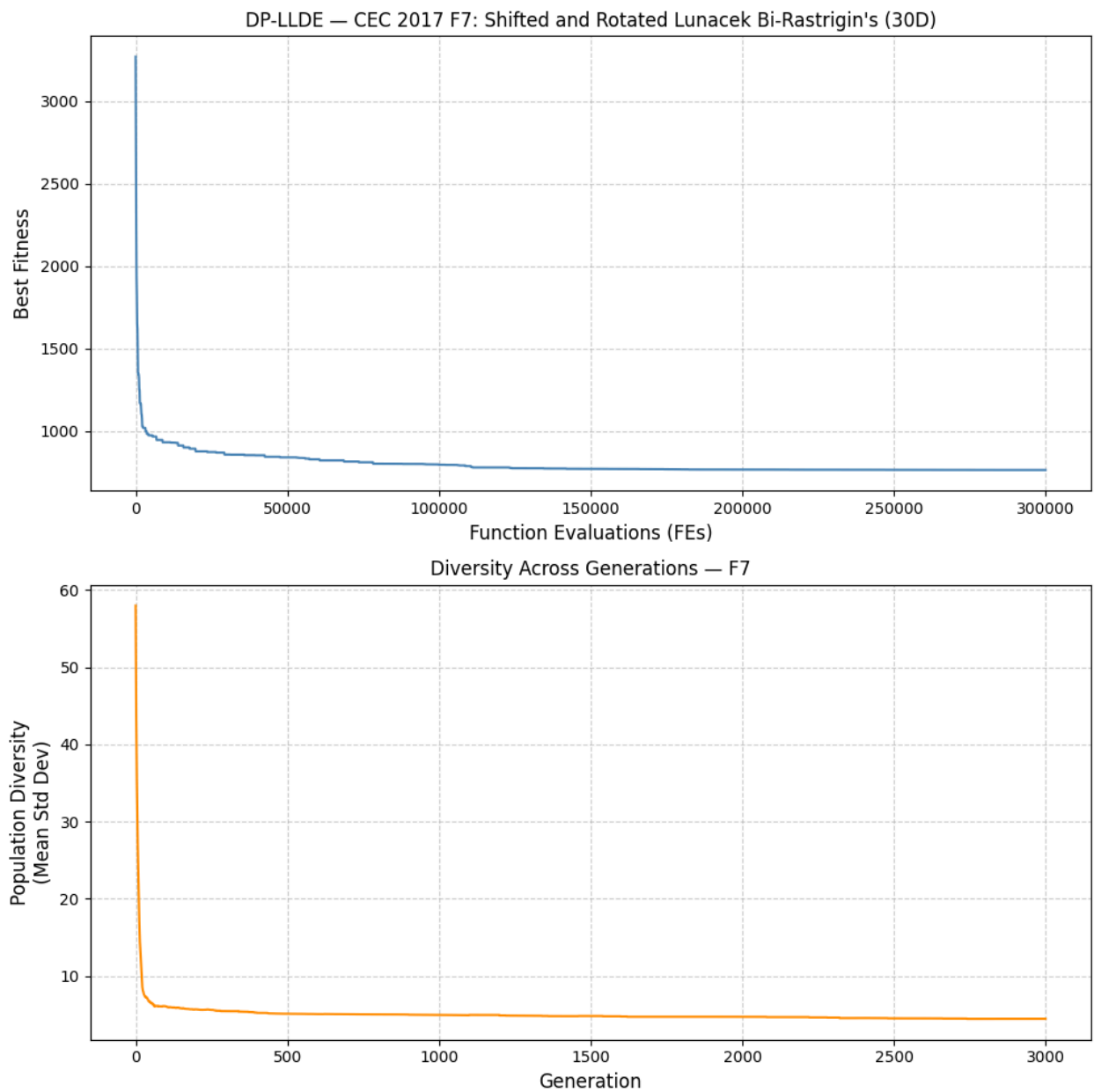
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Results:

Best Fitness : 7.652807e+02
 Global Optimum : 7.000000e+02
 Error (|best-optimum|) : 6.528069e+01
 Best Solution (5 dim) : [-16.44868226 16.33598671 -19.18291757 -2.72039523
 13.31663977]
 D^(0) (init. diversity): 5.7976e+01

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Graph saved as convergence_F7.png

```
In [ ]: # =====
# F8 — Shifted and Rotated Non-Continuous Rastrigin
# =====
run_and_plot('F8', make_wrapper(simple.f8), 800.0, "Shifted and Rotated Non-Co
```

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DP-LLDE on CEC 2017 F8 – Single Run

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Function : Shifted and Rotated Non-Continuous Rastrigin's (30D)
Bounds : [-100, 100]
Global optimum : 800.0
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 1.442304e+03

Generation 10:	Best = 1.157041e+03	FES = 1100	$\mu_F = 0.558$	$\mu_{CR} = 0.349$
Diversity = 2.9184e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 1.054876e+03	FES = 2100	$\mu_F = 0.604$	$\mu_{CR} = 0.337$
Diversity = 2.2847e+01	$k_t = [4, 4, 2, 4]$			
Generation 30:	Best = 1.054876e+03	FES = 3100	$\mu_F = 0.623$	$\mu_{CR} = 0.311$
Diversity = 1.9218e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 1.027439e+03	FES = 4100	$\mu_F = 0.650$	$\mu_{CR} = 0.279$
Diversity = 1.7499e+01	$k_t = [2, 4, 4, 2]$			
Generation 50:	Best = 1.021357e+03	FES = 5100	$\mu_F = 0.670$	$\mu_{CR} = 0.259$
Diversity = 1.6530e+01	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 1.003654e+03	FES = 6100	$\mu_F = 0.701$	$\mu_{CR} = 0.208$
Diversity = 1.6155e+01	$k_t = [3, 3, 4, 3]$			
Generation 70:	Best = 1.003654e+03	FES = 7100	$\mu_F = 0.659$	$\mu_{CR} = 0.157$
Diversity = 1.5945e+01	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 9.913331e+02	FES = 8100	$\mu_F = 0.719$	$\mu_{CR} = 0.102$
Diversity = 1.5635e+01	$k_t = [2, 2, 4, 3]$			
Generation 90:	Best = 9.868384e+02	FES = 9100	$\mu_F = 0.703$	$\mu_{CR} = 0.092$
Diversity = 1.5524e+01	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 9.865168e+02	FES = 10100	$\mu_F = 0.699$	$\mu_{CR} = 0.081$
Diversity = 1.5709e+01	$k_t = [2, 3, 4, 3]$			
Generation 110:	Best = 9.825309e+02	FES = 11100	$\mu_F = 0.702$	$\mu_{CR} = 0.077$
Diversity = 1.5671e+01	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 9.706441e+02	FES = 12100	$\mu_F = 0.740$	$\mu_{CR} = 0.061$
Diversity = 1.5539e+01	$k_t = [2, 3, 2, 3]$			
Generation 130:	Best = 9.543767e+02	FES = 13100	$\mu_F = 0.711$	$\mu_{CR} = 0.062$
Diversity = 1.5603e+01	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 9.487443e+02	FES = 14100	$\mu_F = 0.705$	$\mu_{CR} = 0.063$
Diversity = 1.5596e+01	$k_t = [3, 3, 4, 3]$			
Generation 150:	Best = 9.402734e+02	FES = 15100	$\mu_F = 0.715$	$\mu_{CR} = 0.068$
Diversity = 1.5574e+01	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 9.402734e+02	FES = 16100	$\mu_F = 0.720$	$\mu_{CR} = 0.054$
Diversity = 1.5314e+01	$k_t = [3, 3, 4, 4]$			
Generation 170:	Best = 9.396072e+02	FES = 17100	$\mu_F = 0.748$	$\mu_{CR} = 0.052$
Diversity = 1.5278e+01	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 9.396072e+02	FES = 18100	$\mu_F = 0.757$	$\mu_{CR} = 0.062$
Diversity = 1.4998e+01	$k_t = [4, 2, 2, 4]$			
Generation 190:	Best = 9.396072e+02	FES = 19100	$\mu_F = 0.755$	$\mu_{CR} = 0.058$
Diversity = 1.4955e+01	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 9.391942e+02	FES = 20100	$\mu_F = 0.755$	$\mu_{CR} = 0.060$
Diversity = 1.4859e+01	$k_t = [2, 2, 3, 4]$			
Generation 210:	Best = 9.391942e+02	FES = 21100	$\mu_F = 0.773$	$\mu_{CR} = 0.072$
Diversity = 1.4798e+01	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 9.391942e+02	FES = 22100	$\mu_F = 0.783$	$\mu_{CR} = 0.060$
Diversity = 1.4763e+01	$k_t = [4, 3, 4, 2]$			

Generation 230: Best = 9.391942e+02	FES = 23100	$\mu_F = 0.798$	$\mu_{CR} = 0.047$
Diversity = 1.4736e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 9.322486e+02	FES = 24100	$\mu_F = 0.765$	$\mu_{CR} = 0.044$
Diversity = 1.4583e+01	$k_t = [4, 2, 3, 4]$		
Generation 250: Best = 9.246623e+02	FES = 25100	$\mu_F = 0.763$	$\mu_{CR} = 0.054$
Diversity = 1.4621e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 9.246623e+02	FES = 26100	$\mu_F = 0.662$	$\mu_{CR} = 0.066$
Diversity = 1.4596e+01	$k_t = [3, 2, 2, 3]$		
Generation 270: Best = 9.173357e+02	FES = 27100	$\mu_F = 0.647$	$\mu_{CR} = 0.048$
Diversity = 1.4601e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 9.173357e+02	FES = 28100	$\mu_F = 0.598$	$\mu_{CR} = 0.058$
Diversity = 1.4640e+01	$k_t = [4, 4, 3, 4]$		
Generation 290: Best = 9.173357e+02	FES = 29100	$\mu_F = 0.636$	$\mu_{CR} = 0.053$
Diversity = 1.4715e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 9.173357e+02	FES = 30100	$\mu_F = 0.620$	$\mu_{CR} = 0.050$
Diversity = 1.4649e+01	$k_t = [2, 4, 4, 4]$		
Generation 310: Best = 9.142647e+02	FES = 31100	$\mu_F = 0.670$	$\mu_{CR} = 0.046$
Diversity = 1.4656e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 9.142647e+02	FES = 32100	$\mu_F = 0.655$	$\mu_{CR} = 0.056$
Diversity = 1.4635e+01	$k_t = [2, 4, 3, 3]$		
Generation 330: Best = 9.142647e+02	FES = 33100	$\mu_F = 0.636$	$\mu_{CR} = 0.055$
Diversity = 1.4636e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 9.136698e+02	FES = 34100	$\mu_F = 0.669$	$\mu_{CR} = 0.061$
Diversity = 1.4643e+01	$k_t = [4, 2, 2, 4]$		
Generation 350: Best = 9.136698e+02	FES = 35100	$\mu_F = 0.742$	$\mu_{CR} = 0.053$
Diversity = 1.4594e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 9.135314e+02	FES = 36100	$\mu_F = 0.735$	$\mu_{CR} = 0.046$
Diversity = 1.4589e+01	$k_t = [3, 4, 4, 2]$		
Generation 370: Best = 9.100301e+02	FES = 37100	$\mu_F = 0.755$	$\mu_{CR} = 0.055$
Diversity = 1.4575e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 9.100301e+02	FES = 38100	$\mu_F = 0.777$	$\mu_{CR} = 0.037$
Diversity = 1.4362e+01	$k_t = [4, 2, 4, 4]$		
Generation 390: Best = 9.100301e+02	FES = 39100	$\mu_F = 0.832$	$\mu_{CR} = 0.032$
Diversity = 1.4369e+01	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 9.100301e+02	FES = 40100	$\mu_F = 0.821$	$\mu_{CR} = 0.035$
Diversity = 1.4426e+01	$k_t = [3, 4, 4, 4]$		
Generation 410: Best = 9.090002e+02	FES = 41100	$\mu_F = 0.813$	$\mu_{CR} = 0.034$
Diversity = 1.4403e+01	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 9.090002e+02	FES = 42100	$\mu_F = 0.783$	$\mu_{CR} = 0.034$
Diversity = 1.4392e+01	$k_t = [3, 2, 3, 2]$		
Generation 430: Best = 9.069245e+02	FES = 43100	$\mu_F = 0.777$	$\mu_{CR} = 0.041$
Diversity = 1.4289e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 9.059362e+02	FES = 44100	$\mu_F = 0.751$	$\mu_{CR} = 0.036$
Diversity = 1.4299e+01	$k_t = [4, 3, 3, 3]$		
Generation 450: Best = 9.030717e+02	FES = 45100	$\mu_F = 0.604$	$\mu_{CR} = 0.049$
Diversity = 1.4301e+01	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 9.030717e+02	FES = 46100	$\mu_F = 0.496$	$\mu_{CR} = 0.046$
Diversity = 1.4289e+01	$k_t = [3, 2, 2, 3]$		
Generation 470: Best = 9.028604e+02	FES = 47100	$\mu_F = 0.485$	$\mu_{CR} = 0.057$
Diversity = 1.4272e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 9.028604e+02	FES = 48100	$\mu_F = 0.502$	$\mu_{CR} = 0.049$
Diversity = 1.4265e+01	$k_t = [4, 3, 2, 3]$		
Generation 490: Best = 9.022130e+02	FES = 49100	$\mu_F = 0.512$	$\mu_{CR} = 0.050$
Diversity = 1.4270e+01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 9.007339e+02	FES = 50100	$\mu_F = 0.443$	$\mu_{CR} = 0.057$
Diversity = 1.4255e+01	k_t = [4, 3, 4, 2]		
Generation 510: Best = 8.929051e+02	FES = 51100	$\mu_F = 0.409$	$\mu_{CR} = 0.046$
Diversity = 1.4217e+01	k_t = [1, 1, 1, 1]		
Generation 520: Best = 8.929051e+02	FES = 52100	$\mu_F = 0.394$	$\mu_{CR} = 0.060$
Diversity = 1.4199e+01	k_t = [3, 2, 3, 3]		
Generation 530: Best = 8.923356e+02	FES = 53100	$\mu_F = 0.302$	$\mu_{CR} = 0.057$
Diversity = 1.4172e+01	k_t = [1, 1, 1, 1]		
Generation 540: Best = 8.874556e+02	FES = 54100	$\mu_F = 0.265$	$\mu_{CR} = 0.052$
Diversity = 1.4148e+01	k_t = [4, 2, 2, 2]		
Generation 550: Best = 8.874556e+02	FES = 55100	$\mu_F = 0.258$	$\mu_{CR} = 0.074$
Diversity = 1.4130e+01	k_t = [1, 1, 1, 1]		
Generation 560: Best = 8.842694e+02	FES = 56100	$\mu_F = 0.264$	$\mu_{CR} = 0.073$
Diversity = 1.4125e+01	k_t = [3, 2, 2, 3]		
Generation 570: Best = 8.774902e+02	FES = 57100	$\mu_F = 0.282$	$\mu_{CR} = 0.073$
Diversity = 1.4119e+01	k_t = [1, 1, 1, 1]		
Generation 580: Best = 8.734136e+02	FES = 58100	$\mu_F = 0.299$	$\mu_{CR} = 0.076$
Diversity = 1.4079e+01	k_t = [4, 2, 3, 3]		
Generation 590: Best = 8.733513e+02	FES = 59100	$\mu_F = 0.387$	$\mu_{CR} = 0.069$
Diversity = 1.4067e+01	k_t = [1, 1, 1, 1]		
Generation 600: Best = 8.733513e+02	FES = 60100	$\mu_F = 0.441$	$\mu_{CR} = 0.063$
Diversity = 1.3995e+01	k_t = [4, 2, 3, 4]		
Generation 610: Best = 8.733513e+02	FES = 61100	$\mu_F = 0.477$	$\mu_{CR} = 0.068$
Diversity = 1.3991e+01	k_t = [1, 1, 1, 1]		
Generation 620: Best = 8.733513e+02	FES = 62100	$\mu_F = 0.447$	$\mu_{CR} = 0.072$
Diversity = 1.3981e+01	k_t = [3, 3, 3, 3]		
Generation 630: Best = 8.726930e+02	FES = 63100	$\mu_F = 0.455$	$\mu_{CR} = 0.049$
Diversity = 1.3960e+01	k_t = [1, 1, 1, 1]		
Generation 640: Best = 8.726930e+02	FES = 64100	$\mu_F = 0.350$	$\mu_{CR} = 0.048$
Diversity = 1.3962e+01	k_t = [2, 3, 3, 2]		
Generation 650: Best = 8.688503e+02	FES = 65100	$\mu_F = 0.418$	$\mu_{CR} = 0.062$
Diversity = 1.3945e+01	k_t = [1, 1, 1, 1]		
Generation 660: Best = 8.682508e+02	FES = 66100	$\mu_F = 0.381$	$\mu_{CR} = 0.062$
Diversity = 1.3944e+01	k_t = [3, 3, 2, 2]		
Generation 670: Best = 8.681346e+02	FES = 67100	$\mu_F = 0.376$	$\mu_{CR} = 0.061$
Diversity = 1.3915e+01	k_t = [1, 1, 1, 1]		
Generation 680: Best = 8.681346e+02	FES = 68100	$\mu_F = 0.421$	$\mu_{CR} = 0.061$
Diversity = 1.3905e+01	k_t = [4, 4, 4, 3]		
Generation 690: Best = 8.681346e+02	FES = 69100	$\mu_F = 0.302$	$\mu_{CR} = 0.062$
Diversity = 1.3897e+01	k_t = [1, 1, 1, 1]		
Generation 700: Best = 8.661978e+02	FES = 70100	$\mu_F = 0.284$	$\mu_{CR} = 0.059$
Diversity = 1.3889e+01	k_t = [3, 3, 3, 2]		
Generation 710: Best = 8.659180e+02	FES = 71100	$\mu_F = 0.338$	$\mu_{CR} = 0.063$
Diversity = 1.3887e+01	k_t = [1, 1, 1, 1]		
Generation 720: Best = 8.631997e+02	FES = 72100	$\mu_F = 0.361$	$\mu_{CR} = 0.074$
Diversity = 1.3774e+01	k_t = [4, 2, 4, 4]		
Generation 730: Best = 8.621003e+02	FES = 73100	$\mu_F = 0.414$	$\mu_{CR} = 0.061$
Diversity = 1.3755e+01	k_t = [1, 1, 1, 1]		
Generation 740: Best = 8.621003e+02	FES = 74100	$\mu_F = 0.477$	$\mu_{CR} = 0.039$
Diversity = 1.3792e+01	k_t = [2, 2, 2, 3]		
Generation 750: Best = 8.621003e+02	FES = 75100	$\mu_F = 0.393$	$\mu_{CR} = 0.046$
Diversity = 1.3789e+01	k_t = [1, 1, 1, 1]		
Generation 760: Best = 8.621003e+02	FES = 76100	$\mu_F = 0.348$	$\mu_{CR} = 0.061$
Diversity = 1.3786e+01	k_t = [3, 4, 3, 2]		

Generation 770:	Best = 8.621003e+02	FES = 77100	$\mu_F = 0.352$	$\mu_{CR} = 0.058$
Diversity = 1.3774e+01	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 8.610884e+02	FES = 78100	$\mu_F = 0.295$	$\mu_{CR} = 0.059$
Diversity = 1.3769e+01	$k_t = [3, 3, 2, 2]$			
Generation 790:	Best = 8.608194e+02	FES = 79100	$\mu_F = 0.279$	$\mu_{CR} = 0.056$
Diversity = 1.3763e+01	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 8.608194e+02	FES = 80100	$\mu_F = 0.398$	$\mu_{CR} = 0.065$
Diversity = 1.3743e+01	$k_t = [2, 2, 3, 4]$			
Generation 810:	Best = 8.608194e+02	FES = 81100	$\mu_F = 0.408$	$\mu_{CR} = 0.046$
Diversity = 1.3743e+01	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 8.608194e+02	FES = 82100	$\mu_F = 0.391$	$\mu_{CR} = 0.043$
Diversity = 1.3744e+01	$k_t = [4, 4, 3, 2]$			
Generation 830:	Best = 8.608194e+02	FES = 83100	$\mu_F = 0.398$	$\mu_{CR} = 0.056$
Diversity = 1.3721e+01	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 8.608194e+02	FES = 84100	$\mu_F = 0.395$	$\mu_{CR} = 0.075$
Diversity = 1.3717e+01	$k_t = [3, 2, 4, 2]$			
Generation 850:	Best = 8.602626e+02	FES = 85100	$\mu_F = 0.383$	$\mu_{CR} = 0.056$
Diversity = 1.3714e+01	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 8.602626e+02	FES = 86100	$\mu_F = 0.416$	$\mu_{CR} = 0.069$
Diversity = 1.3732e+01	$k_t = [4, 3, 2, 2]$			
Generation 870:	Best = 8.602626e+02	FES = 87100	$\mu_F = 0.311$	$\mu_{CR} = 0.053$
Diversity = 1.3716e+01	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 8.602626e+02	FES = 88100	$\mu_F = 0.315$	$\mu_{CR} = 0.046$
Diversity = 1.3712e+01	$k_t = [2, 3, 4, 2]$			
Generation 890:	Best = 8.602626e+02	FES = 89100	$\mu_F = 0.316$	$\mu_{CR} = 0.050$
Diversity = 1.3710e+01	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 8.597302e+02	FES = 90100	$\mu_F = 0.355$	$\mu_{CR} = 0.052$
Diversity = 1.3721e+01	$k_t = [4, 2, 3, 4]$			
Generation 910:	Best = 8.567205e+02	FES = 91100	$\mu_F = 0.348$	$\mu_{CR} = 0.048$
Diversity = 1.3719e+01	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 8.567205e+02	FES = 92100	$\mu_F = 0.288$	$\mu_{CR} = 0.060$
Diversity = 1.3709e+01	$k_t = [2, 3, 4, 4]$			
Generation 930:	Best = 8.567205e+02	FES = 93100	$\mu_F = 0.312$	$\mu_{CR} = 0.058$
Diversity = 1.3710e+01	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 8.521894e+02	FES = 94100	$\mu_F = 0.267$	$\mu_{CR} = 0.051$
Diversity = 1.3711e+01	$k_t = [3, 3, 4, 3]$			
Generation 950:	Best = 8.513937e+02	FES = 95100	$\mu_F = 0.239$	$\mu_{CR} = 0.054$
Diversity = 1.3711e+01	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 8.513221e+02	FES = 96100	$\mu_F = 0.266$	$\mu_{CR} = 0.056$
Diversity = 1.3710e+01	$k_t = [3, 4, 4, 3]$			
Generation 970:	Best = 8.513221e+02	FES = 97100	$\mu_F = 0.287$	$\mu_{CR} = 0.069$
Diversity = 1.3690e+01	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 8.508305e+02	FES = 98100	$\mu_F = 0.242$	$\mu_{CR} = 0.065$
Diversity = 1.3689e+01	$k_t = [4, 2, 2, 2]$			
Generation 990:	Best = 8.508305e+02	FES = 99100	$\mu_F = 0.247$	$\mu_{CR} = 0.062$
Diversity = 1.3693e+01	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 8.508305e+02	FES = 100100	$\mu_F = 0.348$	$\mu_{CR} = 0.054$
Diversity = 1.3680e+01	$k_t = [3, 2, 4, 4]$			
Generation 1010:	Best = 8.508305e+02	FES = 101100	$\mu_F = 0.383$	$\mu_{CR} = 0.046$
Diversity = 1.3674e+01	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 8.508305e+02	FES = 102100	$\mu_F = 0.415$	$\mu_{CR} = 0.042$
Diversity = 1.3653e+01	$k_t = [3, 2, 2, 4]$			
Generation 1030:	Best = 8.508305e+02	FES = 103100	$\mu_F = 0.404$	$\mu_{CR} = 0.043$
Diversity = 1.3658e+01	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 8.508305e+02 FES = 104100 μ_F = 0.355 μ_{CR} = 0.052
 Diversity = 1.3656e+01 k_t = [3, 3, 2, 2]
 Generation 1050: Best = 8.508305e+02 FES = 105100 μ_F = 0.306 μ_{CR} = 0.054
 Diversity = 1.3641e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 8.508305e+02 FES = 106100 μ_F = 0.344 μ_{CR} = 0.063
 Diversity = 1.3653e+01 k_t = [4, 2, 3, 3]
 Generation 1070: Best = 8.508305e+02 FES = 107100 μ_F = 0.346 μ_{CR} = 0.049
 Diversity = 1.3635e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 8.508305e+02 FES = 108100 μ_F = 0.322 μ_{CR} = 0.056
 Diversity = 1.3638e+01 k_t = [2, 3, 4, 3]
 Generation 1090: Best = 8.508305e+02 FES = 109100 μ_F = 0.253 μ_{CR} = 0.063
 Diversity = 1.3640e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 8.504636e+02 FES = 110100 μ_F = 0.326 μ_{CR} = 0.057
 Diversity = 1.3623e+01 k_t = [2, 4, 2, 3]
 Generation 1110: Best = 8.504636e+02 FES = 111100 μ_F = 0.234 μ_{CR} = 0.049
 Diversity = 1.3615e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 8.500631e+02 FES = 112100 μ_F = 0.218 μ_{CR} = 0.057
 Diversity = 1.3614e+01 k_t = [3, 2, 4, 4]
 Generation 1130: Best = 8.500631e+02 FES = 113100 μ_F = 0.192 μ_{CR} = 0.058
 Diversity = 1.3609e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 8.484350e+02 FES = 114100 μ_F = 0.239 μ_{CR} = 0.052
 Diversity = 1.3607e+01 k_t = [2, 2, 4, 2]
 Generation 1150: Best = 8.468890e+02 FES = 115100 μ_F = 0.234 μ_{CR} = 0.057
 Diversity = 1.3601e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 8.468890e+02 FES = 116100 μ_F = 0.248 μ_{CR} = 0.072
 Diversity = 1.3691e+01 k_t = [2, 2, 3, 4]
 Generation 1170: Best = 8.468890e+02 FES = 117100 μ_F = 0.206 μ_{CR} = 0.062
 Diversity = 1.3689e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 8.450408e+02 FES = 118100 μ_F = 0.204 μ_{CR} = 0.058
 Diversity = 1.3696e+01 k_t = [3, 4, 4, 3]
 Generation 1190: Best = 8.442658e+02 FES = 119100 μ_F = 0.217 μ_{CR} = 0.052
 Diversity = 1.3688e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 8.442658e+02 FES = 120100 μ_F = 0.228 μ_{CR} = 0.060
 Diversity = 1.3690e+01 k_t = [3, 4, 2, 2]
 Generation 1210: Best = 8.442658e+02 FES = 121100 μ_F = 0.228 μ_{CR} = 0.058
 Diversity = 1.3688e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 8.442658e+02 FES = 122100 μ_F = 0.205 μ_{CR} = 0.064
 Diversity = 1.3684e+01 k_t = [3, 3, 2, 2]
 Generation 1230: Best = 8.441317e+02 FES = 123100 μ_F = 0.211 μ_{CR} = 0.060
 Diversity = 1.3683e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 8.434540e+02 FES = 124100 μ_F = 0.257 μ_{CR} = 0.058
 Diversity = 1.3679e+01 k_t = [3, 4, 3, 3]
 Generation 1250: Best = 8.434540e+02 FES = 125100 μ_F = 0.298 μ_{CR} = 0.057
 Diversity = 1.3678e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 8.433775e+02 FES = 126100 μ_F = 0.337 μ_{CR} = 0.048
 Diversity = 1.3683e+01 k_t = [2, 3, 2, 4]
 Generation 1270: Best = 8.433775e+02 FES = 127100 μ_F = 0.294 μ_{CR} = 0.057
 Diversity = 1.3685e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 8.433775e+02 FES = 128100 μ_F = 0.322 μ_{CR} = 0.050
 Diversity = 1.3685e+01 k_t = [4, 3, 2, 3]
 Generation 1290: Best = 8.433775e+02 FES = 129100 μ_F = 0.332 μ_{CR} = 0.056
 Diversity = 1.3690e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 8.386087e+02 FES = 130100 μ_F = 0.353 μ_{CR} = 0.054
 Diversity = 1.3689e+01 k_t = [4, 4, 3, 3]

Generation 1310: Best = 8.376546e+02 FES = 131100 μ_F = 0.373 μ_{CR} = 0.038
 Diversity = 1.3693e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 8.375178e+02 FES = 132100 μ_F = 0.301 μ_{CR} = 0.042
 Diversity = 1.3694e+01 k_t = [2, 4, 2, 2]
 Generation 1330: Best = 8.375178e+02 FES = 133100 μ_F = 0.288 μ_{CR} = 0.046
 Diversity = 1.3695e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 8.361920e+02 FES = 134100 μ_F = 0.315 μ_{CR} = 0.058
 Diversity = 1.3732e+01 k_t = [4, 3, 3, 4]
 Generation 1350: Best = 8.359858e+02 FES = 135100 μ_F = 0.288 μ_{CR} = 0.052
 Diversity = 1.3732e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 8.359858e+02 FES = 136100 μ_F = 0.259 μ_{CR} = 0.043
 Diversity = 1.3730e+01 k_t = [3, 2, 3, 2]
 Generation 1370: Best = 8.358970e+02 FES = 137100 μ_F = 0.242 μ_{CR} = 0.045
 Diversity = 1.3730e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 8.358970e+02 FES = 138100 μ_F = 0.282 μ_{CR} = 0.054
 Diversity = 1.3730e+01 k_t = [4, 2, 3, 2]
 Generation 1390: Best = 8.358519e+02 FES = 139100 μ_F = 0.335 μ_{CR} = 0.052
 Diversity = 1.3754e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 8.358519e+02 FES = 140100 μ_F = 0.339 μ_{CR} = 0.041
 Diversity = 1.3755e+01 k_t = [2, 4, 2, 3]
 Generation 1410: Best = 8.358519e+02 FES = 141100 μ_F = 0.315 μ_{CR} = 0.036
 Diversity = 1.3751e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 8.358519e+02 FES = 142100 μ_F = 0.376 μ_{CR} = 0.060
 Diversity = 1.3779e+01 k_t = [3, 4, 3, 4]
 Generation 1430: Best = 8.357444e+02 FES = 143100 μ_F = 0.367 μ_{CR} = 0.060
 Diversity = 1.3780e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 8.357444e+02 FES = 144100 μ_F = 0.356 μ_{CR} = 0.045
 Diversity = 1.3749e+01 k_t = [4, 3, 3, 4]
 Generation 1450: Best = 8.357444e+02 FES = 145100 μ_F = 0.267 μ_{CR} = 0.055
 Diversity = 1.3749e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 8.353004e+02 FES = 146100 μ_F = 0.338 μ_{CR} = 0.048
 Diversity = 1.3789e+01 k_t = [2, 4, 3, 3]
 Generation 1470: Best = 8.353004e+02 FES = 147100 μ_F = 0.311 μ_{CR} = 0.047
 Diversity = 1.3791e+01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 8.352821e+02 FES = 148100 μ_F = 0.398 μ_{CR} = 0.037
 Diversity = 1.3798e+01 k_t = [2, 3, 3, 4]
 Generation 1490: Best = 8.351016e+02 FES = 149100 μ_F = 0.489 μ_{CR} = 0.033
 Diversity = 1.3800e+01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 8.351016e+02 FES = 150100 μ_F = 0.408 μ_{CR} = 0.031
 Diversity = 1.3799e+01 k_t = [2, 3, 3, 3]
 Generation 1510: Best = 8.351016e+02 FES = 151100 μ_F = 0.409 μ_{CR} = 0.046
 Diversity = 1.3799e+01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 8.351016e+02 FES = 152100 μ_F = 0.357 μ_{CR} = 0.045
 Diversity = 1.3808e+01 k_t = [3, 3, 4, 3]
 Generation 1530: Best = 8.351016e+02 FES = 153100 μ_F = 0.339 μ_{CR} = 0.040
 Diversity = 1.3811e+01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 8.351016e+02 FES = 154100 μ_F = 0.321 μ_{CR} = 0.047
 Diversity = 1.3767e+01 k_t = [2, 3, 4, 4]
 Generation 1550: Best = 8.351016e+02 FES = 155100 μ_F = 0.306 μ_{CR} = 0.037
 Diversity = 1.3791e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 8.351016e+02 FES = 156100 μ_F = 0.295 μ_{CR} = 0.040
 Diversity = 1.3792e+01 k_t = [2, 2, 3, 3]
 Generation 1570: Best = 8.351016e+02 FES = 157100 μ_F = 0.293 μ_{CR} = 0.048
 Diversity = 1.3793e+01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 8.351016e+02 FES = 158100 μ_F = 0.208 μ_{CR} = 0.045
 Diversity = 1.3791e+01 k_t = [2, 4, 4, 4]
 Generation 1590: Best = 8.351016e+02 FES = 159100 μ_F = 0.273 μ_{CR} = 0.032
 Diversity = 1.3793e+01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 8.351016e+02 FES = 160100 μ_F = 0.336 μ_{CR} = 0.041
 Diversity = 1.3777e+01 k_t = [4, 3, 4, 2]
 Generation 1610: Best = 8.351016e+02 FES = 161100 μ_F = 0.325 μ_{CR} = 0.066
 Diversity = 1.3777e+01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 8.351016e+02 FES = 162100 μ_F = 0.268 μ_{CR} = 0.062
 Diversity = 1.3774e+01 k_t = [3, 4, 4, 2]
 Generation 1630: Best = 8.351016e+02 FES = 163100 μ_F = 0.237 μ_{CR} = 0.065
 Diversity = 1.3773e+01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 8.351016e+02 FES = 164100 μ_F = 0.261 μ_{CR} = 0.058
 Diversity = 1.3773e+01 k_t = [2, 4, 2, 3]
 Generation 1650: Best = 8.351016e+02 FES = 165100 μ_F = 0.267 μ_{CR} = 0.056
 Diversity = 1.3774e+01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 8.351016e+02 FES = 166100 μ_F = 0.302 μ_{CR} = 0.057
 Diversity = 1.3761e+01 k_t = [3, 3, 3, 4]
 Generation 1670: Best = 8.351016e+02 FES = 167100 μ_F = 0.283 μ_{CR} = 0.056
 Diversity = 1.3765e+01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 8.351016e+02 FES = 168100 μ_F = 0.342 μ_{CR} = 0.055
 Diversity = 1.3764e+01 k_t = [2, 2, 4, 2]
 Generation 1690: Best = 8.351016e+02 FES = 169100 μ_F = 0.275 μ_{CR} = 0.060
 Diversity = 1.3766e+01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 8.351016e+02 FES = 170100 μ_F = 0.256 μ_{CR} = 0.045
 Diversity = 1.3769e+01 k_t = [3, 3, 2, 4]
 Generation 1710: Best = 8.351016e+02 FES = 171100 μ_F = 0.257 μ_{CR} = 0.042
 Diversity = 1.3769e+01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 8.351016e+02 FES = 172100 μ_F = 0.250 μ_{CR} = 0.046
 Diversity = 1.3771e+01 k_t = [2, 4, 2, 2]
 Generation 1730: Best = 8.348063e+02 FES = 173100 μ_F = 0.218 μ_{CR} = 0.051
 Diversity = 1.3770e+01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 8.348063e+02 FES = 174100 μ_F = 0.207 μ_{CR} = 0.060
 Diversity = 1.3772e+01 k_t = [3, 4, 2, 3]
 Generation 1750: Best = 8.348063e+02 FES = 175100 μ_F = 0.195 μ_{CR} = 0.067
 Diversity = 1.3765e+01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 8.347982e+02 FES = 176100 μ_F = 0.209 μ_{CR} = 0.056
 Diversity = 1.3764e+01 k_t = [2, 2, 4, 4]
 Generation 1770: Best = 8.347982e+02 FES = 177100 μ_F = 0.178 μ_{CR} = 0.059
 Diversity = 1.3763e+01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 8.347982e+02 FES = 178100 μ_F = 0.280 μ_{CR} = 0.055
 Diversity = 1.3735e+01 k_t = [4, 3, 2, 4]
 Generation 1790: Best = 8.347982e+02 FES = 179100 μ_F = 0.291 μ_{CR} = 0.046
 Diversity = 1.3733e+01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 8.347982e+02 FES = 180100 μ_F = 0.317 μ_{CR} = 0.035
 Diversity = 1.3680e+01 k_t = [2, 2, 3, 3]
 Generation 1810: Best = 8.347982e+02 FES = 181100 μ_F = 0.311 μ_{CR} = 0.038
 Diversity = 1.3680e+01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 8.347982e+02 FES = 182100 μ_F = 0.275 μ_{CR} = 0.040
 Diversity = 1.3681e+01 k_t = [3, 2, 2, 4]
 Generation 1830: Best = 8.347982e+02 FES = 183100 μ_F = 0.430 μ_{CR} = 0.034
 Diversity = 1.3682e+01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 8.347982e+02 FES = 184100 μ_F = 0.406 μ_{CR} = 0.030
 Diversity = 1.3683e+01 k_t = [3, 3, 4, 2]

Generation 1850: Best = 8.342499e+02	FES = 185100	$\mu_F = 0.368$	$\mu_{CR} = 0.046$
Diversity = 1.3602e+01	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 8.341908e+02	FES = 186100	$\mu_F = 0.304$	$\mu_{CR} = 0.052$
Diversity = 1.3603e+01	$k_t = [2, 2, 2, 3]$		
Generation 1870: Best = 8.341908e+02	FES = 187100	$\mu_F = 0.282$	$\mu_{CR} = 0.046$
Diversity = 1.3603e+01	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 8.341908e+02	FES = 188100	$\mu_F = 0.291$	$\mu_{CR} = 0.056$
Diversity = 1.3596e+01	$k_t = [2, 4, 2, 3]$		
Generation 1890: Best = 8.341908e+02	FES = 189100	$\mu_F = 0.286$	$\mu_{CR} = 0.059$
Diversity = 1.3615e+01	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 8.341395e+02	FES = 190100	$\mu_F = 0.301$	$\mu_{CR} = 0.054$
Diversity = 1.3572e+01	$k_t = [4, 2, 3, 4]$		
Generation 1910: Best = 8.341395e+02	FES = 191100	$\mu_F = 0.262$	$\mu_{CR} = 0.054$
Diversity = 1.3573e+01	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 8.341299e+02	FES = 192100	$\mu_F = 0.313$	$\mu_{CR} = 0.039$
Diversity = 1.3541e+01	$k_t = [2, 2, 4, 4]$		
Generation 1930: Best = 8.341299e+02	FES = 193100	$\mu_F = 0.359$	$\mu_{CR} = 0.056$
Diversity = 1.3534e+01	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 8.341299e+02	FES = 194100	$\mu_F = 0.346$	$\mu_{CR} = 0.043$
Diversity = 1.3530e+01	$k_t = [3, 2, 4, 2]$		
Generation 1950: Best = 8.341299e+02	FES = 195100	$\mu_F = 0.289$	$\mu_{CR} = 0.047$
Diversity = 1.3529e+01	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 8.341299e+02	FES = 196100	$\mu_F = 0.252$	$\mu_{CR} = 0.042$
Diversity = 1.3530e+01	$k_t = [3, 4, 2, 3]$		
Generation 1970: Best = 8.341299e+02	FES = 197100	$\mu_F = 0.286$	$\mu_{CR} = 0.041$
Diversity = 1.3468e+01	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 8.341299e+02	FES = 198100	$\mu_F = 0.384$	$\mu_{CR} = 0.054$
Diversity = 1.3404e+01	$k_t = [2, 2, 3, 4]$		
Generation 1990: Best = 8.341299e+02	FES = 199100	$\mu_F = 0.372$	$\mu_{CR} = 0.067$
Diversity = 1.3404e+01	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 8.341299e+02	FES = 200100	$\mu_F = 0.380$	$\mu_{CR} = 0.070$
Diversity = 1.3351e+01	$k_t = [4, 4, 2, 4]$		
Generation 2010: Best = 8.341299e+02	FES = 201100	$\mu_F = 0.431$	$\mu_{CR} = 0.056$
Diversity = 1.3355e+01	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 8.341299e+02	FES = 202100	$\mu_F = 0.305$	$\mu_{CR} = 0.063$
Diversity = 1.3355e+01	$k_t = [4, 3, 3, 3]$		
Generation 2030: Best = 8.341299e+02	FES = 203100	$\mu_F = 0.248$	$\mu_{CR} = 0.050$
Diversity = 1.3355e+01	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 8.341299e+02	FES = 204100	$\mu_F = 0.322$	$\mu_{CR} = 0.058$
Diversity = 1.3356e+01	$k_t = [3, 4, 2, 2]$		
Generation 2050: Best = 8.341299e+02	FES = 205100	$\mu_F = 0.250$	$\mu_{CR} = 0.054$
Diversity = 1.3356e+01	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 8.341299e+02	FES = 206100	$\mu_F = 0.350$	$\mu_{CR} = 0.070$
Diversity = 1.3298e+01	$k_t = [4, 2, 2, 3]$		
Generation 2070: Best = 8.341181e+02	FES = 207100	$\mu_F = 0.338$	$\mu_{CR} = 0.050$
Diversity = 1.3298e+01	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 8.339578e+02	FES = 208100	$\mu_F = 0.310$	$\mu_{CR} = 0.055$
Diversity = 1.3301e+01	$k_t = [2, 3, 4, 3]$		
Generation 2090: Best = 8.339578e+02	FES = 209100	$\mu_F = 0.256$	$\mu_{CR} = 0.043$
Diversity = 1.3303e+01	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 8.339578e+02	FES = 210100	$\mu_F = 0.221$	$\mu_{CR} = 0.042$
Diversity = 1.3304e+01	$k_t = [3, 4, 4, 2]$		
Generation 2110: Best = 8.339578e+02	FES = 211100	$\mu_F = 0.173$	$\mu_{CR} = 0.082$
Diversity = 1.3303e+01	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 8.339578e+02 FES = 212100 μ_F = 0.099 μ_{CR} = 0.111
 Diversity = 1.3298e+01 k_t = [3, 2, 4, 2]
 Generation 2130: Best = 8.339578e+02 FES = 213100 μ_F = 0.084 μ_{CR} = 0.107
 Diversity = 1.3293e+01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 8.339578e+02 FES = 214100 μ_F = 0.183 μ_{CR} = 0.098
 Diversity = 1.3293e+01 k_t = [3, 3, 3, 4]
 Generation 2150: Best = 8.339578e+02 FES = 215100 μ_F = 0.170 μ_{CR} = 0.078
 Diversity = 1.3292e+01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 8.339279e+02 FES = 216100 μ_F = 0.237 μ_{CR} = 0.069
 Diversity = 1.3300e+01 k_t = [4, 2, 3, 4]
 Generation 2170: Best = 8.339241e+02 FES = 217100 μ_F = 0.255 μ_{CR} = 0.045
 Diversity = 1.3300e+01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 8.338820e+02 FES = 218100 μ_F = 0.219 μ_{CR} = 0.053
 Diversity = 1.3300e+01 k_t = [2, 3, 4, 3]
 Generation 2190: Best = 8.338820e+02 FES = 219100 μ_F = 0.183 μ_{CR} = 0.049
 Diversity = 1.3297e+01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 8.338820e+02 FES = 220100 μ_F = 0.172 μ_{CR} = 0.048
 Diversity = 1.3294e+01 k_t = [2, 4, 3, 4]
 Generation 2210: Best = 8.338820e+02 FES = 221100 μ_F = 0.189 μ_{CR} = 0.048
 Diversity = 1.3294e+01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 8.338820e+02 FES = 222100 μ_F = 0.228 μ_{CR} = 0.071
 Diversity = 1.3278e+01 k_t = [4, 3, 3, 4]
 Generation 2230: Best = 8.338820e+02 FES = 223100 μ_F = 0.227 μ_{CR} = 0.062
 Diversity = 1.3279e+01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 8.338820e+02 FES = 224100 μ_F = 0.213 μ_{CR} = 0.047
 Diversity = 1.3279e+01 k_t = [4, 3, 2, 3]
 Generation 2250: Best = 8.338820e+02 FES = 225100 μ_F = 0.173 μ_{CR} = 0.047
 Diversity = 1.3280e+01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 8.338793e+02 FES = 226100 μ_F = 0.141 μ_{CR} = 0.045
 Diversity = 1.3280e+01 k_t = [2, 3, 2, 3]
 Generation 2270: Best = 8.338206e+02 FES = 227100 μ_F = 0.148 μ_{CR} = 0.049
 Diversity = 1.3279e+01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 8.338206e+02 FES = 228100 μ_F = 0.117 μ_{CR} = 0.065
 Diversity = 1.3279e+01 k_t = [3, 4, 4, 4]
 Generation 2290: Best = 8.338206e+02 FES = 229100 μ_F = 0.130 μ_{CR} = 0.053
 Diversity = 1.3281e+01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 8.338205e+02 FES = 230100 μ_F = 0.190 μ_{CR} = 0.054
 Diversity = 1.3282e+01 k_t = [3, 2, 2, 2]
 Generation 2310: Best = 8.338205e+02 FES = 231100 μ_F = 0.176 μ_{CR} = 0.044
 Diversity = 1.3283e+01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 8.338205e+02 FES = 232100 μ_F = 0.168 μ_{CR} = 0.052
 Diversity = 1.3281e+01 k_t = [4, 4, 2, 3]
 Generation 2330: Best = 8.338205e+02 FES = 233100 μ_F = 0.163 μ_{CR} = 0.063
 Diversity = 1.3281e+01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 8.338205e+02 FES = 234100 μ_F = 0.171 μ_{CR} = 0.043
 Diversity = 1.3281e+01 k_t = [2, 3, 3, 3]
 Generation 2350: Best = 8.338205e+02 FES = 235100 μ_F = 0.159 μ_{CR} = 0.047
 Diversity = 1.3280e+01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 8.338205e+02 FES = 236100 μ_F = 0.139 μ_{CR} = 0.055
 Diversity = 1.3278e+01 k_t = [4, 2, 2, 3]
 Generation 2370: Best = 8.338205e+02 FES = 237100 μ_F = 0.126 μ_{CR} = 0.070
 Diversity = 1.3278e+01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 8.338106e+02 FES = 238100 μ_F = 0.149 μ_{CR} = 0.064
 Diversity = 1.3278e+01 k_t = [2, 3, 3, 2]

Generation 2390: Best = 8.338106e+02 FES = 239100 μ_F = 0.220 μ_{CR} = 0.063
 Diversity = 1.3278e+01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 8.338106e+02 FES = 240100 μ_F = 0.186 μ_{CR} = 0.058
 Diversity = 1.3280e+01 k_t = [2, 3, 2, 3]
 Generation 2410: Best = 8.338106e+02 FES = 241100 μ_F = 0.204 μ_{CR} = 0.064
 Diversity = 1.3281e+01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 8.338106e+02 FES = 242100 μ_F = 0.165 μ_{CR} = 0.050
 Diversity = 1.3281e+01 k_t = [2, 2, 4, 2]
 Generation 2430: Best = 8.338106e+02 FES = 243100 μ_F = 0.146 μ_{CR} = 0.061
 Diversity = 1.3281e+01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 8.338106e+02 FES = 244100 μ_F = 0.137 μ_{CR} = 0.055
 Diversity = 1.3279e+01 k_t = [4, 3, 4, 2]
 Generation 2450: Best = 8.338106e+02 FES = 245100 μ_F = 0.108 μ_{CR} = 0.059
 Diversity = 1.3277e+01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 8.337517e+02 FES = 246100 μ_F = 0.122 μ_{CR} = 0.072
 Diversity = 1.3284e+01 k_t = [4, 3, 2, 4]
 Generation 2470: Best = 8.336764e+02 FES = 247100 μ_F = 0.109 μ_{CR} = 0.065
 Diversity = 1.3282e+01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 8.336764e+02 FES = 248100 μ_F = 0.157 μ_{CR} = 0.070
 Diversity = 1.3282e+01 k_t = [4, 2, 4, 2]
 Generation 2490: Best = 8.333912e+02 FES = 249100 μ_F = 0.094 μ_{CR} = 0.066
 Diversity = 1.3281e+01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 8.333912e+02 FES = 250100 μ_F = 0.075 μ_{CR} = 0.064
 Diversity = 1.3282e+01 k_t = [3, 2, 4, 4]
 Generation 2510: Best = 8.333912e+02 FES = 251100 μ_F = 0.144 μ_{CR} = 0.065
 Diversity = 1.3319e+01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 8.333912e+02 FES = 252100 μ_F = 0.212 μ_{CR} = 0.106
 Diversity = 1.3328e+01 k_t = [2, 2, 2, 4]
 Generation 2530: Best = 8.333564e+02 FES = 253100 μ_F = 0.204 μ_{CR} = 0.087
 Diversity = 1.3328e+01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 8.333564e+02 FES = 254100 μ_F = 0.196 μ_{CR} = 0.070
 Diversity = 1.3337e+01 k_t = [2, 3, 2, 4]
 Generation 2550: Best = 8.333564e+02 FES = 255100 μ_F = 0.105 μ_{CR} = 0.065
 Diversity = 1.3338e+01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 8.333111e+02 FES = 256100 μ_F = 0.189 μ_{CR} = 0.069
 Diversity = 1.3337e+01 k_t = [4, 2, 4, 3]
 Generation 2570: Best = 8.333111e+02 FES = 257100 μ_F = 0.166 μ_{CR} = 0.060
 Diversity = 1.3338e+01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 8.333111e+02 FES = 258100 μ_F = 0.208 μ_{CR} = 0.055
 Diversity = 1.3338e+01 k_t = [3, 3, 4, 3]
 Generation 2590: Best = 8.333111e+02 FES = 259100 μ_F = 0.260 μ_{CR} = 0.064
 Diversity = 1.3339e+01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 8.333111e+02 FES = 260100 μ_F = 0.230 μ_{CR} = 0.054
 Diversity = 1.3340e+01 k_t = [4, 4, 2, 2]
 Generation 2610: Best = 8.333111e+02 FES = 261100 μ_F = 0.179 μ_{CR} = 0.045
 Diversity = 1.3342e+01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 8.333111e+02 FES = 262100 μ_F = 0.195 μ_{CR} = 0.048
 Diversity = 1.3337e+01 k_t = [3, 4, 3, 4]
 Generation 2630: Best = 8.333111e+02 FES = 263100 μ_F = 0.099 μ_{CR} = 0.061
 Diversity = 1.3337e+01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 8.332065e+02 FES = 264100 μ_F = 0.078 μ_{CR} = 0.083
 Diversity = 1.3337e+01 k_t = [2, 2, 4, 3]
 Generation 2650: Best = 8.332065e+02 FES = 265100 μ_F = 0.076 μ_{CR} = 0.069
 Diversity = 1.3337e+01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 8.332065e+02 FES = 266100 μ_F = 0.136 μ_{CR} = 0.079
 Diversity = 1.3299e+01 k_t = [2, 2, 4, 3]
 Generation 2670: Best = 8.331028e+02 FES = 267100 μ_F = 0.143 μ_{CR} = 0.075
 Diversity = 1.3300e+01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 8.331028e+02 FES = 268100 μ_F = 0.297 μ_{CR} = 0.057
 Diversity = 1.3200e+01 k_t = [3, 2, 4, 4]
 Generation 2690: Best = 8.331028e+02 FES = 269100 μ_F = 0.306 μ_{CR} = 0.064
 Diversity = 1.3198e+01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 8.331028e+02 FES = 270100 μ_F = 0.325 μ_{CR} = 0.052
 Diversity = 1.3188e+01 k_t = [4, 4, 3, 4]
 Generation 2710: Best = 8.331028e+02 FES = 271100 μ_F = 0.288 μ_{CR} = 0.056
 Diversity = 1.3188e+01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 8.330224e+02 FES = 272100 μ_F = 0.223 μ_{CR} = 0.047
 Diversity = 1.3188e+01 k_t = [3, 3, 2, 3]
 Generation 2730: Best = 8.330224e+02 FES = 273100 μ_F = 0.277 μ_{CR} = 0.046
 Diversity = 1.3189e+01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 8.329318e+02 FES = 274100 μ_F = 0.340 μ_{CR} = 0.049
 Diversity = 1.3187e+01 k_t = [2, 3, 3, 4]
 Generation 2750: Best = 8.329318e+02 FES = 275100 μ_F = 0.282 μ_{CR} = 0.046
 Diversity = 1.3187e+01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 8.329318e+02 FES = 276100 μ_F = 0.261 μ_{CR} = 0.040
 Diversity = 1.3187e+01 k_t = [2, 2, 3, 2]
 Generation 2770: Best = 8.329318e+02 FES = 277100 μ_F = 0.273 μ_{CR} = 0.029
 Diversity = 1.3188e+01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 8.329318e+02 FES = 278100 μ_F = 0.233 μ_{CR} = 0.051
 Diversity = 1.3189e+01 k_t = [3, 4, 4, 3]
 Generation 2790: Best = 8.329318e+02 FES = 279100 μ_F = 0.168 μ_{CR} = 0.062
 Diversity = 1.3190e+01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 8.329318e+02 FES = 280100 μ_F = 0.192 μ_{CR} = 0.060
 Diversity = 1.3191e+01 k_t = [2, 3, 3, 4]
 Generation 2810: Best = 8.329318e+02 FES = 281100 μ_F = 0.220 μ_{CR} = 0.061
 Diversity = 1.3193e+01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 8.329318e+02 FES = 282100 μ_F = 0.195 μ_{CR} = 0.074
 Diversity = 1.3193e+01 k_t = [4, 2, 4, 2]
 Generation 2830: Best = 8.329197e+02 FES = 283100 μ_F = 0.162 μ_{CR} = 0.058
 Diversity = 1.3193e+01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 8.329197e+02 FES = 284100 μ_F = 0.151 μ_{CR} = 0.053
 Diversity = 1.3193e+01 k_t = [3, 3, 2, 4]
 Generation 2850: Best = 8.329197e+02 FES = 285100 μ_F = 0.143 μ_{CR} = 0.055
 Diversity = 1.3195e+01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 8.328331e+02 FES = 286100 μ_F = 0.231 μ_{CR} = 0.056
 Diversity = 1.3191e+01 k_t = [3, 4, 2, 3]
 Generation 2870: Best = 8.328331e+02 FES = 287100 μ_F = 0.173 μ_{CR} = 0.051
 Diversity = 1.3192e+01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 8.324392e+02 FES = 288100 μ_F = 0.191 μ_{CR} = 0.080
 Diversity = 1.3193e+01 k_t = [4, 4, 4, 2]
 Generation 2890: Best = 8.324392e+02 FES = 289100 μ_F = 0.172 μ_{CR} = 0.081
 Diversity = 1.3194e+01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 8.323915e+02 FES = 290100 μ_F = 0.157 μ_{CR} = 0.080
 Diversity = 1.3194e+01 k_t = [4, 4, 3, 3]
 Generation 2910: Best = 8.323915e+02 FES = 291100 μ_F = 0.149 μ_{CR} = 0.046
 Diversity = 1.3194e+01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 8.323915e+02 FES = 292100 μ_F = 0.232 μ_{CR} = 0.082
 Diversity = 1.3159e+01 k_t = [4, 2, 4, 4]

Generation 2930: Best = 8.323915e+02 FES = 293100 μ_F = 0.198 μ_{CR} = 0.064
 Diversity = 1.3158e+01 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 8.323855e+02 FES = 294100 μ_F = 0.180 μ_{CR} = 0.053
 Diversity = 1.3160e+01 k_t = [2, 2, 3, 4]
 Generation 2950: Best = 8.323295e+02 FES = 295100 μ_F = 0.127 μ_{CR} = 0.067
 Diversity = 1.3158e+01 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 8.323295e+02 FES = 296100 μ_F = 0.199 μ_{CR} = 0.069
 Diversity = 1.3125e+01 k_t = [3, 2, 4, 4]
 Generation 2970: Best = 8.323295e+02 FES = 297100 μ_F = 0.246 μ_{CR} = 0.053
 Diversity = 1.3126e+01 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 8.322158e+02 FES = 298100 μ_F = 0.268 μ_{CR} = 0.049
 Diversity = 1.3127e+01 k_t = [2, 3, 2, 2]
 Generation 2990: Best = 8.317910e+02 FES = 299100 μ_F = 0.220 μ_{CR} = 0.061
 Diversity = 1.3127e+01 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 8.317910e+02

Total Function Evals = 300000

Total Successful Updates = 8371

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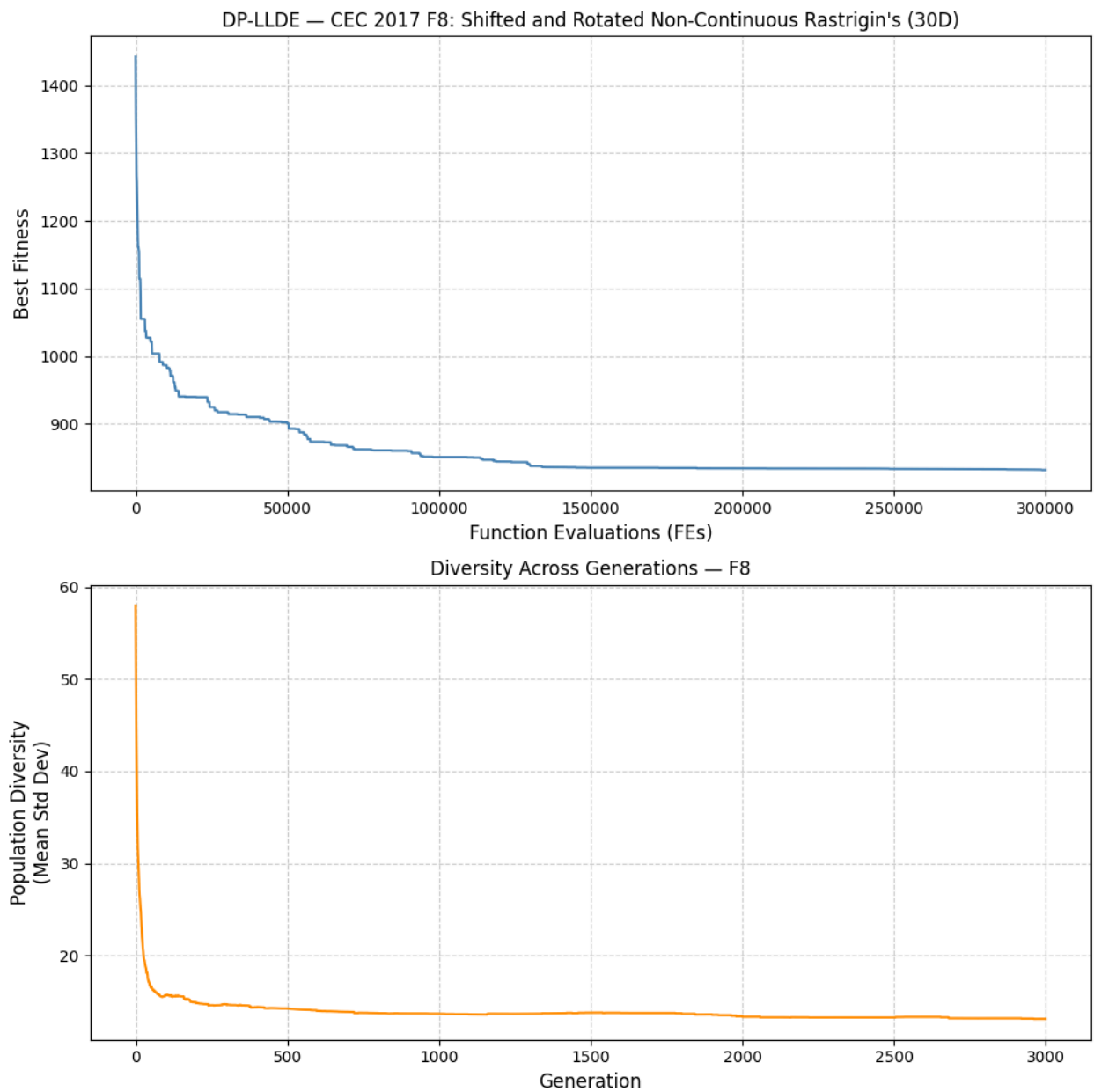
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Results:

Best Fitness : 8.317910e+02
 Global Optimum : 8.000000e+02
 Error (|best-optimum|) : 3.179096e+01
 Best Solution (5 dim) : [28.73729249 -28.7530276 7.93980899 -43.71449242
 -32.77712553]
 D^(0) (init. diversity): 5.7976e+01

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Graph saved as convergence_F8.png

```
In [ ]: # =====
# F9 - Shifted and Rotated Levy
# =====
run_and_plot('F9', make_wrapper(simple.f9), 900.0, 'Shifted and Rotated Levy')
```

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DP-LLDE on CEC 2017 F9 – Single Run

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Function : Shifted and Rotated Levy (30D)
Bounds : [-100, 100]
Global optimum : 900.0
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 2.774722e+04

Generation 10:	Best = 1.001547e+04	FES = 1100	$\mu_F = 0.561$	$\mu_{CR} = 0.354$
Diversity = 3.3861e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 4.530745e+03	FES = 2100	$\mu_F = 0.616$	$\mu_{CR} = 0.347$
Diversity = 2.0056e+01	$k_t = [4, 4, 2, 4]$			
Generation 30:	Best = 2.872079e+03	FES = 3100	$\mu_F = 0.652$	$\mu_{CR} = 0.355$
Diversity = 1.2534e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 2.197849e+03	FES = 4100	$\mu_F = 0.675$	$\mu_{CR} = 0.350$
Diversity = 9.1666e+00	$k_t = [2, 4, 4, 2]$			
Generation 50:	Best = 1.507747e+03	FES = 5100	$\mu_F = 0.695$	$\mu_{CR} = 0.369$
Diversity = 7.1897e+00	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 1.330057e+03	FES = 6100	$\mu_F = 0.702$	$\mu_{CR} = 0.370$
Diversity = 5.3191e+00	$k_t = [3, 3, 2, 4]$			
Generation 70:	Best = 1.184680e+03	FES = 7100	$\mu_F = 0.712$	$\mu_{CR} = 0.361$
Diversity = 4.0180e+00	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 1.042836e+03	FES = 8100	$\mu_F = 0.730$	$\mu_{CR} = 0.345$
Diversity = 3.1608e+00	$k_t = [2, 2, 4, 3]$			
Generation 90:	Best = 9.854623e+02	FES = 9100	$\mu_F = 0.735$	$\mu_{CR} = 0.344$
Diversity = 2.6019e+00	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 9.343031e+02	FES = 10100	$\mu_F = 0.737$	$\mu_{CR} = 0.352$
Diversity = 1.9960e+00	$k_t = [2, 2, 3, 3]$			
Generation 110:	Best = 9.172467e+02	FES = 11100	$\mu_F = 0.738$	$\mu_{CR} = 0.348$
Diversity = 1.4473e+00	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 9.094902e+02	FES = 12100	$\mu_F = 0.736$	$\mu_{CR} = 0.345$
Diversity = 1.0429e+00	$k_t = [4, 3, 3, 2]$			
Generation 130:	Best = 9.051209e+02	FES = 13100	$\mu_F = 0.714$	$\mu_{CR} = 0.335$
Diversity = 7.6304e-01	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 9.025687e+02	FES = 14100	$\mu_F = 0.701$	$\mu_{CR} = 0.338$
Diversity = 5.9315e-01	$k_t = [4, 2, 2, 3]$			
Generation 150:	Best = 9.014085e+02	FES = 15100	$\mu_F = 0.687$	$\mu_{CR} = 0.331$
Diversity = 4.3568e-01	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 9.007040e+02	FES = 16100	$\mu_F = 0.678$	$\mu_{CR} = 0.328$
Diversity = 2.6798e-01	$k_t = [3, 3, 3, 3]$			
Generation 170:	Best = 9.005506e+02	FES = 17100	$\mu_F = 0.680$	$\mu_{CR} = 0.322$
Diversity = 1.8778e-01	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 9.003994e+02	FES = 18100	$\mu_F = 0.678$	$\mu_{CR} = 0.317$
Diversity = 1.3086e-01	$k_t = [4, 4, 2, 2]$			
Generation 190:	Best = 9.003104e+02	FES = 19100	$\mu_F = 0.675$	$\mu_{CR} = 0.320$
Diversity = 8.7646e-02	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 9.002429e+02	FES = 20100	$\mu_F = 0.677$	$\mu_{CR} = 0.316$
Diversity = 5.0194e-02	$k_t = [3, 3, 2, 3]$			
Generation 210:	Best = 9.002218e+02	FES = 21100	$\mu_F = 0.678$	$\mu_{CR} = 0.322$
Diversity = 3.3295e-02	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 9.002065e+02	FES = 22100	$\mu_F = 0.683$	$\mu_{CR} = 0.320$
Diversity = 2.0215e-02	$k_t = [2, 3, 3, 3]$			

Generation	230:	Best = 9.001987e+02	FES = 23100	$\mu_F = 0.690$	$\mu_{CR} = 0.316$
Diversity	=	1.5778e-02	$k_t = [1, 1, 1, 1]$		
Generation	240:	Best = 9.001944e+02	FES = 24100	$\mu_F = 0.707$	$\mu_{CR} = 0.317$
Diversity	=	1.1444e-02	$k_t = [2, 3, 3, 2]$		
Generation	250:	Best = 9.001915e+02	FES = 25100	$\mu_F = 0.723$	$\mu_{CR} = 0.321$
Diversity	=	1.0262e-02	$k_t = [1, 1, 1, 1]$		
Generation	260:	Best = 9.001891e+02	FES = 26100	$\mu_F = 0.727$	$\mu_{CR} = 0.311$
Diversity	=	8.0712e-03	$k_t = [3, 2, 4, 4]$		
Generation	270:	Best = 9.001873e+02	FES = 27100	$\mu_F = 0.732$	$\mu_{CR} = 0.315$
Diversity	=	7.2471e-03	$k_t = [1, 1, 1, 1]$		
Generation	280:	Best = 9.001845e+02	FES = 28100	$\mu_F = 0.747$	$\mu_{CR} = 0.311$
Diversity	=	6.9250e-03	$k_t = [3, 3, 2, 3]$		
Generation	290:	Best = 9.001833e+02	FES = 29100	$\mu_F = 0.764$	$\mu_{CR} = 0.309$
Diversity	=	6.5717e-03	$k_t = [1, 1, 1, 1]$		
Generation	300:	Best = 9.001821e+02	FES = 30100	$\mu_F = 0.761$	$\mu_{CR} = 0.312$
Diversity	=	5.4538e-03	$k_t = [3, 4, 3, 4]$		
Generation	310:	Best = 9.001812e+02	FES = 31100	$\mu_F = 0.763$	$\mu_{CR} = 0.310$
Diversity	=	5.4082e-03	$k_t = [1, 1, 1, 1]$		
Generation	320:	Best = 9.001807e+02	FES = 32100	$\mu_F = 0.756$	$\mu_{CR} = 0.315$
Diversity	=	4.3520e-03	$k_t = [4, 3, 3, 2]$		
Generation	330:	Best = 9.001804e+02	FES = 33100	$\mu_F = 0.755$	$\mu_{CR} = 0.303$
Diversity	=	3.7074e-03	$k_t = [1, 1, 1, 1]$		
Generation	340:	Best = 9.001801e+02	FES = 34100	$\mu_F = 0.748$	$\mu_{CR} = 0.298$
Diversity	=	3.2291e-03	$k_t = [2, 3, 4, 2]$		
Generation	350:	Best = 9.001799e+02	FES = 35100	$\mu_F = 0.748$	$\mu_{CR} = 0.298$
Diversity	=	2.7199e-03	$k_t = [1, 1, 1, 1]$		
Generation	360:	Best = 9.001798e+02	FES = 36100	$\mu_F = 0.763$	$\mu_{CR} = 0.300$
Diversity	=	2.1878e-03	$k_t = [3, 4, 3, 4]$		
Generation	370:	Best = 9.001796e+02	FES = 37100	$\mu_F = 0.772$	$\mu_{CR} = 0.307$
Diversity	=	2.0688e-03	$k_t = [1, 1, 1, 1]$		
Generation	380:	Best = 9.001795e+02	FES = 38100	$\mu_F = 0.777$	$\mu_{CR} = 0.302$
Diversity	=	1.6591e-03	$k_t = [4, 4, 3, 3]$		
Generation	390:	Best = 9.001795e+02	FES = 39100	$\mu_F = 0.781$	$\mu_{CR} = 0.300$
Diversity	=	1.5393e-03	$k_t = [1, 1, 1, 1]$		
Generation	400:	Best = 9.001794e+02	FES = 40100	$\mu_F = 0.775$	$\mu_{CR} = 0.295$
Diversity	=	1.3717e-03	$k_t = [4, 3, 2, 3]$		
Generation	410:	Best = 9.001793e+02	FES = 41100	$\mu_F = 0.775$	$\mu_{CR} = 0.293$
Diversity	=	1.2680e-03	$k_t = [1, 1, 1, 1]$		
Generation	420:	Best = 9.001793e+02	FES = 42100	$\mu_F = 0.782$	$\mu_{CR} = 0.289$
Diversity	=	1.1054e-03	$k_t = [3, 2, 4, 2]$		
Generation	430:	Best = 9.001793e+02	FES = 43100	$\mu_F = 0.770$	$\mu_{CR} = 0.283$
Diversity	=	1.0165e-03	$k_t = [1, 1, 1, 1]$		
Generation	440:	Best = 9.001793e+02	FES = 44100	$\mu_F = 0.775$	$\mu_{CR} = 0.279$
Diversity	=	8.1786e-04	$k_t = [4, 3, 2, 3]$		
Generation	450:	Best = 9.001792e+02	FES = 45100	$\mu_F = 0.775$	$\mu_{CR} = 0.276$
Diversity	=	6.9340e-04	$k_t = [1, 1, 1, 1]$		
Generation	460:	Best = 9.001792e+02	FES = 46100	$\mu_F = 0.774$	$\mu_{CR} = 0.272$
Diversity	=	5.4876e-04	$k_t = [2, 2, 4, 4]$		
Generation	470:	Best = 9.001792e+02	FES = 47100	$\mu_F = 0.765$	$\mu_{CR} = 0.265$
Diversity	=	5.2233e-04	$k_t = [1, 1, 1, 1]$		
Generation	480:	Best = 9.001792e+02	FES = 48100	$\mu_F = 0.761$	$\mu_{CR} = 0.266$
Diversity	=	4.6838e-04	$k_t = [3, 2, 4, 3]$		
Generation	490:	Best = 9.001792e+02	FES = 49100	$\mu_F = 0.773$	$\mu_{CR} = 0.264$
Diversity	=	4.3816e-04	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 9.001792e+02	FES = 50100	$\mu_F = 0.773$	$\mu_{CR} = 0.247$
Diversity = 3.7627e-04	$k_t = [3, 3, 2, 2]$		
Generation 510: Best = 9.001792e+02	FES = 51100	$\mu_F = 0.777$	$\mu_{CR} = 0.243$
Diversity = 3.9132e-04	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 9.001792e+02	FES = 52100	$\mu_F = 0.775$	$\mu_{CR} = 0.238$
Diversity = 3.3319e-04	$k_t = [3, 3, 3, 4]$		
Generation 530: Best = 9.001792e+02	FES = 53100	$\mu_F = 0.774$	$\mu_{CR} = 0.239$
Diversity = 3.0996e-04	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 9.001792e+02	FES = 54100	$\mu_F = 0.783$	$\mu_{CR} = 0.231$
Diversity = 2.9338e-04	$k_t = [4, 4, 3, 2]$		
Generation 550: Best = 9.001792e+02	FES = 55100	$\mu_F = 0.787$	$\mu_{CR} = 0.223$
Diversity = 2.5825e-04	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 9.001792e+02	FES = 56100	$\mu_F = 0.791$	$\mu_{CR} = 0.224$
Diversity = 2.4563e-04	$k_t = [2, 4, 3, 3]$		
Generation 570: Best = 9.001792e+02	FES = 57100	$\mu_F = 0.780$	$\mu_{CR} = 0.225$
Diversity = 2.4807e-04	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 9.001792e+02	FES = 58100	$\mu_F = 0.792$	$\mu_{CR} = 0.219$
Diversity = 2.5919e-04	$k_t = [3, 3, 2, 2]$		
Generation 590: Best = 9.001792e+02	FES = 59100	$\mu_F = 0.784$	$\mu_{CR} = 0.224$
Diversity = 2.4814e-04	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 9.001792e+02	FES = 60100	$\mu_F = 0.790$	$\mu_{CR} = 0.228$
Diversity = 2.1514e-04	$k_t = [4, 4, 4, 4]$		
Generation 610: Best = 9.001792e+02	FES = 61100	$\mu_F = 0.791$	$\mu_{CR} = 0.228$
Diversity = 1.9734e-04	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 9.001792e+02	FES = 62100	$\mu_F = 0.780$	$\mu_{CR} = 0.229$
Diversity = 1.8351e-04	$k_t = [2, 3, 4, 4]$		
Generation 630: Best = 9.001792e+02	FES = 63100	$\mu_F = 0.787$	$\mu_{CR} = 0.230$
Diversity = 1.8865e-04	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 9.001792e+02	FES = 64100	$\mu_F = 0.777$	$\mu_{CR} = 0.233$
Diversity = 1.5501e-04	$k_t = [3, 2, 2, 4]$		
Generation 650: Best = 9.001792e+02	FES = 65100	$\mu_F = 0.769$	$\mu_{CR} = 0.221$
Diversity = 1.4643e-04	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 9.001792e+02	FES = 66100	$\mu_F = 0.780$	$\mu_{CR} = 0.218$
Diversity = 1.3618e-04	$k_t = [4, 4, 2, 3]$		
Generation 670: Best = 9.001792e+02	FES = 67100	$\mu_F = 0.776$	$\mu_{CR} = 0.207$
Diversity = 1.3432e-04	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 9.001792e+02	FES = 68100	$\mu_F = 0.774$	$\mu_{CR} = 0.209$
Diversity = 1.0478e-04	$k_t = [4, 4, 3, 4]$		
Generation 690: Best = 9.001792e+02	FES = 69100	$\mu_F = 0.778$	$\mu_{CR} = 0.205$
Diversity = 9.5320e-05	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 9.001792e+02	FES = 70100	$\mu_F = 0.784$	$\mu_{CR} = 0.221$
Diversity = 8.0424e-05	$k_t = [3, 3, 4, 3]$		
Generation 710: Best = 9.001792e+02	FES = 71100	$\mu_F = 0.780$	$\mu_{CR} = 0.214$
Diversity = 7.2831e-05	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 9.001792e+02	FES = 72100	$\mu_F = 0.775$	$\mu_{CR} = 0.217$
Diversity = 6.4802e-05	$k_t = [3, 3, 3, 2]$		
Generation 730: Best = 9.001792e+02	FES = 73100	$\mu_F = 0.782$	$\mu_{CR} = 0.214$
Diversity = 6.0581e-05	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 9.001792e+02	FES = 74100	$\mu_F = 0.781$	$\mu_{CR} = 0.210$
Diversity = 4.7705e-05	$k_t = [4, 2, 4, 4]$		
Generation 750: Best = 9.001792e+02	FES = 75100	$\mu_F = 0.782$	$\mu_{CR} = 0.216$
Diversity = 4.4336e-05	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 9.001792e+02	FES = 76100	$\mu_F = 0.782$	$\mu_{CR} = 0.220$
Diversity = 4.1149e-05	$k_t = [4, 3, 3, 2]$		

Generation 770: Best = 9.001792e+02	FES = 77100	$\mu_F = 0.781$	$\mu_{CR} = 0.216$
Diversity = 3.6449e-05	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 9.001792e+02	FES = 78100	$\mu_F = 0.783$	$\mu_{CR} = 0.231$
Diversity = 3.6460e-05	$k_t = [4, 4, 2, 3]$		
Generation 790: Best = 9.001792e+02	FES = 79100	$\mu_F = 0.794$	$\mu_{CR} = 0.243$
Diversity = 3.6686e-05	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 9.001792e+02	FES = 80100	$\mu_F = 0.781$	$\mu_{CR} = 0.249$
Diversity = 3.3712e-05	$k_t = [3, 3, 3, 3]$		
Generation 810: Best = 9.001792e+02	FES = 81100	$\mu_F = 0.783$	$\mu_{CR} = 0.246$
Diversity = 3.0949e-05	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 9.001792e+02	FES = 82100	$\mu_F = 0.779$	$\mu_{CR} = 0.251$
Diversity = 3.1028e-05	$k_t = [4, 2, 2, 3]$		
Generation 830: Best = 9.001792e+02	FES = 83100	$\mu_F = 0.773$	$\mu_{CR} = 0.250$
Diversity = 2.8400e-05	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 9.001792e+02	FES = 84100	$\mu_F = 0.765$	$\mu_{CR} = 0.248$
Diversity = 2.3043e-05	$k_t = [2, 2, 3, 3]$		
Generation 850: Best = 9.001792e+02	FES = 85100	$\mu_F = 0.765$	$\mu_{CR} = 0.242$
Diversity = 2.5452e-05	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 9.001792e+02	FES = 86100	$\mu_F = 0.770$	$\mu_{CR} = 0.239$
Diversity = 2.3049e-05	$k_t = [4, 3, 4, 2]$		
Generation 870: Best = 9.001792e+02	FES = 87100	$\mu_F = 0.774$	$\mu_{CR} = 0.239$
Diversity = 2.0046e-05	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 9.001792e+02	FES = 88100	$\mu_F = 0.775$	$\mu_{CR} = 0.247$
Diversity = 1.5597e-05	$k_t = [4, 4, 3, 3]$		
Generation 890: Best = 9.001792e+02	FES = 89100	$\mu_F = 0.776$	$\mu_{CR} = 0.245$
Diversity = 1.5220e-05	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 9.001792e+02	FES = 90100	$\mu_F = 0.784$	$\mu_{CR} = 0.242$
Diversity = 1.3543e-05	$k_t = [4, 3, 2, 3]$		
Generation 910: Best = 9.001792e+02	FES = 91100	$\mu_F = 0.788$	$\mu_{CR} = 0.245$
Diversity = 1.4321e-05	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 9.001792e+02	FES = 92100	$\mu_F = 0.789$	$\mu_{CR} = 0.243$
Diversity = 1.2035e-05	$k_t = [2, 4, 3, 4]$		
Generation 930: Best = 9.001792e+02	FES = 93100	$\mu_F = 0.788$	$\mu_{CR} = 0.242$
Diversity = 1.0555e-05	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 9.001792e+02	FES = 94100	$\mu_F = 0.780$	$\mu_{CR} = 0.237$
Diversity = 8.8759e-06	$k_t = [2, 4, 4, 2]$		
Generation 950: Best = 9.001792e+02	FES = 95100	$\mu_F = 0.778$	$\mu_{CR} = 0.245$
Diversity = 7.9188e-06	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 9.001792e+02	FES = 96100	$\mu_F = 0.778$	$\mu_{CR} = 0.244$
Diversity = 6.5907e-06	$k_t = [2, 4, 3, 3]$		
Generation 970: Best = 9.001792e+02	FES = 97100	$\mu_F = 0.779$	$\mu_{CR} = 0.246$
Diversity = 6.6125e-06	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 9.001792e+02	FES = 98100	$\mu_F = 0.757$	$\mu_{CR} = 0.238$
Diversity = 5.4552e-06	$k_t = [4, 2, 4, 4]$		
Generation 990: Best = 9.001792e+02	FES = 99100	$\mu_F = 0.765$	$\mu_{CR} = 0.241$
Diversity = 4.6490e-06	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 9.001792e+02	FES = 100100	$\mu_F = 0.767$	$\mu_{CR} = 0.250$
Diversity = 4.1726e-06	$k_t = [2, 4, 4, 2]$		
Generation 1010: Best = 9.001792e+02	FES = 101100	$\mu_F = 0.749$	$\mu_{CR} = 0.246$
Diversity = 4.2495e-06	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 9.001792e+02	FES = 102100	$\mu_F = 0.754$	$\mu_{CR} = 0.242$
Diversity = 3.6234e-06	$k_t = [2, 2, 4, 3]$		
Generation 1030: Best = 9.001792e+02	FES = 103100	$\mu_F = 0.756$	$\mu_{CR} = 0.240$
Diversity = 3.2635e-06	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 9.001792e+02 FES = 104100 μ_F = 0.763 μ_{CR} = 0.236
 Diversity = 2.8790e-06 k_t = [4, 2, 2, 3]
 Generation 1050: Best = 9.001792e+02 FES = 105100 μ_F = 0.776 μ_{CR} = 0.235
 Diversity = 3.0502e-06 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 9.001792e+02 FES = 106100 μ_F = 0.785 μ_{CR} = 0.238
 Diversity = 2.7457e-06 k_t = [4, 4, 2, 3]
 Generation 1070: Best = 9.001792e+02 FES = 107100 μ_F = 0.797 μ_{CR} = 0.223
 Diversity = 2.6563e-06 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 9.001792e+02 FES = 108100 μ_F = 0.788 μ_{CR} = 0.228
 Diversity = 2.5509e-06 k_t = [4, 3, 4, 2]
 Generation 1090: Best = 9.001792e+02 FES = 109100 μ_F = 0.789 μ_{CR} = 0.234
 Diversity = 2.7698e-06 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 9.001792e+02 FES = 110100 μ_F = 0.790 μ_{CR} = 0.238
 Diversity = 2.5693e-06 k_t = [4, 2, 4, 2]
 Generation 1110: Best = 9.001792e+02 FES = 111100 μ_F = 0.799 μ_{CR} = 0.230
 Diversity = 2.6106e-06 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 9.001792e+02 FES = 112100 μ_F = 0.796 μ_{CR} = 0.222
 Diversity = 2.2271e-06 k_t = [4, 4, 3, 2]
 Generation 1130: Best = 9.001792e+02 FES = 113100 μ_F = 0.778 μ_{CR} = 0.216
 Diversity = 2.0510e-06 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 9.001792e+02 FES = 114100 μ_F = 0.783 μ_{CR} = 0.217
 Diversity = 1.5826e-06 k_t = [4, 2, 3, 4]
 Generation 1150: Best = 9.001792e+02 FES = 115100 μ_F = 0.778 μ_{CR} = 0.218
 Diversity = 1.4743e-06 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 9.001792e+02 FES = 116100 μ_F = 0.776 μ_{CR} = 0.223
 Diversity = 1.2628e-06 k_t = [2, 2, 2, 4]
 Generation 1170: Best = 9.001792e+02 FES = 117100 μ_F = 0.788 μ_{CR} = 0.221
 Diversity = 1.3443e-06 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 9.001792e+02 FES = 118100 μ_F = 0.775 μ_{CR} = 0.220
 Diversity = 1.1289e-06 k_t = [2, 2, 2, 2]
 Generation 1190: Best = 9.001792e+02 FES = 119100 μ_F = 0.767 μ_{CR} = 0.216
 Diversity = 1.2233e-06 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 9.001792e+02 FES = 120100 μ_F = 0.761 μ_{CR} = 0.226
 Diversity = 1.1572e-06 k_t = [2, 3, 3, 4]
 Generation 1210: Best = 9.001792e+02 FES = 121100 μ_F = 0.766 μ_{CR} = 0.220
 Diversity = 1.0112e-06 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 9.001792e+02 FES = 122100 μ_F = 0.763 μ_{CR} = 0.223
 Diversity = 9.8525e-07 k_t = [4, 2, 4, 2]
 Generation 1230: Best = 9.001792e+02 FES = 123100 μ_F = 0.761 μ_{CR} = 0.217
 Diversity = 8.3263e-07 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 9.001792e+02 FES = 124100 μ_F = 0.764 μ_{CR} = 0.213
 Diversity = 6.3970e-07 k_t = [4, 3, 3, 4]
 Generation 1250: Best = 9.001792e+02 FES = 125100 μ_F = 0.769 μ_{CR} = 0.217
 Diversity = 5.2783e-07 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 9.001792e+02 FES = 126100 μ_F = 0.761 μ_{CR} = 0.213
 Diversity = 4.9286e-07 k_t = [2, 3, 4, 2]
 Generation 1270: Best = 9.001792e+02 FES = 127100 μ_F = 0.757 μ_{CR} = 0.207
 Diversity = 4.5829e-07 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 9.001792e+02 FES = 128100 μ_F = 0.749 μ_{CR} = 0.212
 Diversity = 3.8189e-07 k_t = [3, 4, 2, 4]
 Generation 1290: Best = 9.001792e+02 FES = 129100 μ_F = 0.751 μ_{CR} = 0.220
 Diversity = 3.6661e-07 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 9.001792e+02 FES = 130100 μ_F = 0.763 μ_{CR} = 0.219
 Diversity = 3.2431e-07 k_t = [3, 4, 2, 3]

Generation 1310: Best = 9.001792e+02 FES = 131100 μ_F = 0.767 μ_{CR} = 0.213
 Diversity = 3.2039e-07 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 9.001792e+02 FES = 132100 μ_F = 0.769 μ_{CR} = 0.208
 Diversity = 2.7540e-07 k_t = [2, 2, 3, 2]
 Generation 1330: Best = 9.001792e+02 FES = 133100 μ_F = 0.772 μ_{CR} = 0.203
 Diversity = 2.7523e-07 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 9.001792e+02 FES = 134100 μ_F = 0.777 μ_{CR} = 0.208
 Diversity = 2.8407e-07 k_t = [4, 2, 3, 3]
 Generation 1350: Best = 9.001792e+02 FES = 135100 μ_F = 0.780 μ_{CR} = 0.205
 Diversity = 2.9186e-07 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 9.001792e+02 FES = 136100 μ_F = 0.777 μ_{CR} = 0.204
 Diversity = 2.7832e-07 k_t = [3, 3, 4, 3]
 Generation 1370: Best = 9.001792e+02 FES = 137100 μ_F = 0.775 μ_{CR} = 0.194
 Diversity = 2.7633e-07 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 9.001792e+02 FES = 138100 μ_F = 0.777 μ_{CR} = 0.192
 Diversity = 2.6221e-07 k_t = [3, 3, 3, 3]
 Generation 1390: Best = 9.001792e+02 FES = 139100 μ_F = 0.771 μ_{CR} = 0.189
 Diversity = 2.4376e-07 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 9.001792e+02 FES = 140100 μ_F = 0.779 μ_{CR} = 0.198
 Diversity = 2.0166e-07 k_t = [4, 2, 4, 4]
 Generation 1410: Best = 9.001792e+02 FES = 141100 μ_F = 0.780 μ_{CR} = 0.190
 Diversity = 1.8355e-07 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 9.001792e+02 FES = 142100 μ_F = 0.782 μ_{CR} = 0.187
 Diversity = 1.9612e-07 k_t = [3, 3, 2, 2]
 Generation 1430: Best = 9.001792e+02 FES = 143100 μ_F = 0.781 μ_{CR} = 0.189
 Diversity = 2.3288e-07 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 9.001792e+02 FES = 144100 μ_F = 0.795 μ_{CR} = 0.178
 Diversity = 1.9398e-07 k_t = [3, 3, 3, 4]
 Generation 1450: Best = 9.001792e+02 FES = 145100 μ_F = 0.796 μ_{CR} = 0.182
 Diversity = 1.7004e-07 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 9.001792e+02 FES = 146100 μ_F = 0.788 μ_{CR} = 0.179
 Diversity = 1.5425e-07 k_t = [3, 2, 2, 4]
 Generation 1470: Best = 9.001792e+02 FES = 147100 μ_F = 0.785 μ_{CR} = 0.182
 Diversity = 1.7523e-07 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 9.001792e+02 FES = 148100 μ_F = 0.794 μ_{CR} = 0.173
 Diversity = 1.6359e-07 k_t = [2, 3, 2, 3]
 Generation 1490: Best = 9.001792e+02 FES = 149100 μ_F = 0.795 μ_{CR} = 0.167
 Diversity = 1.5501e-07 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 9.001792e+02 FES = 150100 μ_F = 0.792 μ_{CR} = 0.174
 Diversity = 1.2558e-07 k_t = [3, 4, 3, 2]
 Generation 1510: Best = 9.001792e+02 FES = 151100 μ_F = 0.795 μ_{CR} = 0.176
 Diversity = 1.1555e-07 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 9.001792e+02 FES = 152100 μ_F = 0.794 μ_{CR} = 0.174
 Diversity = 1.0614e-07 k_t = [4, 4, 2, 2]
 Generation 1530: Best = 9.001792e+02 FES = 153100 μ_F = 0.806 μ_{CR} = 0.170
 Diversity = 9.6233e-08 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 9.001792e+02 FES = 154100 μ_F = 0.810 μ_{CR} = 0.171
 Diversity = 8.6334e-08 k_t = [2, 3, 2, 4]
 Generation 1550: Best = 9.001792e+02 FES = 155100 μ_F = 0.816 μ_{CR} = 0.163
 Diversity = 9.0880e-08 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 9.001792e+02 FES = 156100 μ_F = 0.815 μ_{CR} = 0.160
 Diversity = 8.4733e-08 k_t = [4, 3, 2, 3]
 Generation 1570: Best = 9.001792e+02 FES = 157100 μ_F = 0.819 μ_{CR} = 0.153
 Diversity = 8.1639e-08 k_t = [1, 1, 1, 1]

Generation 1580: Best = 9.001792e+02 FES = 158100 μ_F = 0.819 μ_{CR} = 0.154
 Diversity = 7.4231e-08 k_t = [2, 3, 2, 4]
 Generation 1590: Best = 9.001792e+02 FES = 159100 μ_F = 0.821 μ_{CR} = 0.159
 Diversity = 8.1197e-08 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 9.001792e+02 FES = 160100 μ_F = 0.824 μ_{CR} = 0.158
 Diversity = 7.4914e-08 k_t = [3, 2, 2, 4]
 Generation 1610: Best = 9.001792e+02 FES = 161100 μ_F = 0.811 μ_{CR} = 0.151
 Diversity = 6.6994e-08 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 9.001792e+02 FES = 162100 μ_F = 0.798 μ_{CR} = 0.141
 Diversity = 6.2722e-08 k_t = [3, 2, 3, 2]
 Generation 1630: Best = 9.001792e+02 FES = 163100 μ_F = 0.791 μ_{CR} = 0.139
 Diversity = 6.2235e-08 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 9.001792e+02 FES = 164100 μ_F = 0.790 μ_{CR} = 0.139
 Diversity = 5.7886e-08 k_t = [4, 3, 2, 2]
 Generation 1650: Best = 9.001792e+02 FES = 165100 μ_F = 0.787 μ_{CR} = 0.139
 Diversity = 5.5208e-08 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 9.001792e+02 FES = 166100 μ_F = 0.791 μ_{CR} = 0.138
 Diversity = 5.1658e-08 k_t = [3, 4, 3, 3]
 Generation 1670: Best = 9.001792e+02 FES = 167100 μ_F = 0.797 μ_{CR} = 0.149
 Diversity = 4.9818e-08 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 9.001792e+02 FES = 168100 μ_F = 0.797 μ_{CR} = 0.148
 Diversity = 4.7069e-08 k_t = [4, 3, 4, 2]
 Generation 1690: Best = 9.001792e+02 FES = 169100 μ_F = 0.801 μ_{CR} = 0.151
 Diversity = 4.8854e-08 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 9.001792e+02 FES = 170100 μ_F = 0.806 μ_{CR} = 0.148
 Diversity = 4.6833e-08 k_t = [2, 2, 2, 4]
 Generation 1710: Best = 9.001792e+02 FES = 171100 μ_F = 0.807 μ_{CR} = 0.150
 Diversity = 4.8847e-08 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 9.001792e+02 FES = 172100 μ_F = 0.801 μ_{CR} = 0.139
 Diversity = 4.7983e-08 k_t = [4, 4, 3, 2]
 Generation 1730: Best = 9.001792e+02 FES = 173100 μ_F = 0.806 μ_{CR} = 0.137
 Diversity = 4.7684e-08 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 9.001792e+02 FES = 174100 μ_F = 0.803 μ_{CR} = 0.137
 Diversity = 4.4224e-08 k_t = [3, 2, 4, 2]
 Generation 1750: Best = 9.001792e+02 FES = 175100 μ_F = 0.806 μ_{CR} = 0.140
 Diversity = 4.3096e-08 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 9.001792e+02 FES = 176100 μ_F = 0.806 μ_{CR} = 0.135
 Diversity = 3.8741e-08 k_t = [2, 3, 2, 4]
 Generation 1770: Best = 9.001792e+02 FES = 177100 μ_F = 0.803 μ_{CR} = 0.142
 Diversity = 4.0430e-08 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 9.001792e+02 FES = 178100 μ_F = 0.804 μ_{CR} = 0.154
 Diversity = 4.0154e-08 k_t = [3, 2, 2, 4]
 Generation 1790: Best = 9.001792e+02 FES = 179100 μ_F = 0.799 μ_{CR} = 0.158
 Diversity = 4.3850e-08 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 9.001792e+02 FES = 180100 μ_F = 0.800 μ_{CR} = 0.162
 Diversity = 4.0077e-08 k_t = [4, 4, 4, 3]
 Generation 1810: Best = 9.001792e+02 FES = 181100 μ_F = 0.809 μ_{CR} = 0.164
 Diversity = 3.8239e-08 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 9.001792e+02 FES = 182100 μ_F = 0.806 μ_{CR} = 0.169
 Diversity = 3.6766e-08 k_t = [4, 3, 2, 3]
 Generation 1830: Best = 9.001792e+02 FES = 183100 μ_F = 0.808 μ_{CR} = 0.165
 Diversity = 3.7012e-08 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 9.001792e+02 FES = 184100 μ_F = 0.811 μ_{CR} = 0.161
 Diversity = 3.4510e-08 k_t = [2, 4, 4, 4]

Generation 1850: Best = 9.001792e+02	FES = 185100	$\mu_F = 0.820$	$\mu_{CR} = 0.168$
Diversity = 3.6262e-08	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 9.001792e+02	FES = 186100	$\mu_F = 0.817$	$\mu_{CR} = 0.173$
Diversity = 3.6857e-08	$k_t = [4, 2, 3, 2]$		
Generation 1870: Best = 9.001792e+02	FES = 187100	$\mu_F = 0.816$	$\mu_{CR} = 0.173$
Diversity = 3.8130e-08	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 9.001792e+02	FES = 188100	$\mu_F = 0.815$	$\mu_{CR} = 0.168$
Diversity = 3.8410e-08	$k_t = [4, 3, 4, 2]$		
Generation 1890: Best = 9.001792e+02	FES = 189100	$\mu_F = 0.818$	$\mu_{CR} = 0.168$
Diversity = 3.8647e-08	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 9.001792e+02	FES = 190100	$\mu_F = 0.824$	$\mu_{CR} = 0.162$
Diversity = 3.7264e-08	$k_t = [3, 4, 3, 3]$		
Generation 1910: Best = 9.001792e+02	FES = 191100	$\mu_F = 0.821$	$\mu_{CR} = 0.161$
Diversity = 3.8044e-08	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 9.001792e+02	FES = 192100	$\mu_F = 0.812$	$\mu_{CR} = 0.163$
Diversity = 3.6397e-08	$k_t = [4, 2, 2, 2]$		
Generation 1930: Best = 9.001792e+02	FES = 193100	$\mu_F = 0.811$	$\mu_{CR} = 0.162$
Diversity = 3.5455e-08	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 9.001792e+02	FES = 194100	$\mu_F = 0.817$	$\mu_{CR} = 0.162$
Diversity = 3.2934e-08	$k_t = [3, 4, 3, 4]$		
Generation 1950: Best = 9.001792e+02	FES = 195100	$\mu_F = 0.814$	$\mu_{CR} = 0.159$
Diversity = 3.3655e-08	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 9.001792e+02	FES = 196100	$\mu_F = 0.804$	$\mu_{CR} = 0.165$
Diversity = 3.3551e-08	$k_t = [3, 3, 3, 4]$		
Generation 1970: Best = 9.001792e+02	FES = 197100	$\mu_F = 0.805$	$\mu_{CR} = 0.165$
Diversity = 3.4965e-08	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 9.001792e+02	FES = 198100	$\mu_F = 0.804$	$\mu_{CR} = 0.162$
Diversity = 3.3985e-08	$k_t = [4, 2, 3, 4]$		
Generation 1990: Best = 9.001792e+02	FES = 199100	$\mu_F = 0.812$	$\mu_{CR} = 0.161$
Diversity = 3.3974e-08	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 9.001792e+02	FES = 200100	$\mu_F = 0.822$	$\mu_{CR} = 0.164$
Diversity = 3.3834e-08	$k_t = [3, 4, 4, 2]$		
Generation 2010: Best = 9.001792e+02	FES = 201100	$\mu_F = 0.827$	$\mu_{CR} = 0.166$
Diversity = 3.3120e-08	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 9.001792e+02	FES = 202100	$\mu_F = 0.822$	$\mu_{CR} = 0.169$
Diversity = 3.2633e-08	$k_t = [2, 2, 3, 3]$		
Generation 2030: Best = 9.001792e+02	FES = 203100	$\mu_F = 0.824$	$\mu_{CR} = 0.167$
Diversity = 3.5821e-08	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 9.001792e+02	FES = 204100	$\mu_F = 0.820$	$\mu_{CR} = 0.166$
Diversity = 3.4222e-08	$k_t = [3, 4, 4, 2]$		
Generation 2050: Best = 9.001792e+02	FES = 205100	$\mu_F = 0.831$	$\mu_{CR} = 0.174$
Diversity = 3.1394e-08	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 9.001792e+02	FES = 206100	$\mu_F = 0.828$	$\mu_{CR} = 0.167$
Diversity = 3.2190e-08	$k_t = [4, 3, 3, 2]$		
Generation 2070: Best = 9.001792e+02	FES = 207100	$\mu_F = 0.831$	$\mu_{CR} = 0.173$
Diversity = 3.3113e-08	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 9.001792e+02	FES = 208100	$\mu_F = 0.823$	$\mu_{CR} = 0.169$
Diversity = 3.0197e-08	$k_t = [2, 2, 4, 4]$		
Generation 2090: Best = 9.001792e+02	FES = 209100	$\mu_F = 0.820$	$\mu_{CR} = 0.168$
Diversity = 3.3042e-08	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 9.001792e+02	FES = 210100	$\mu_F = 0.820$	$\mu_{CR} = 0.172$
Diversity = 3.3143e-08	$k_t = [4, 4, 3, 2]$		
Generation 2110: Best = 9.001792e+02	FES = 211100	$\mu_F = 0.822$	$\mu_{CR} = 0.169$
Diversity = 3.2687e-08	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 9.001792e+02 FES = 212100 μ_F = 0.824 μ_{CR} = 0.171
 Diversity = 3.1402e-08 k_t = [4, 3, 3, 3]
 Generation 2130: Best = 9.001792e+02 FES = 213100 μ_F = 0.834 μ_{CR} = 0.177
 Diversity = 3.1121e-08 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 9.001792e+02 FES = 214100 μ_F = 0.836 μ_{CR} = 0.170
 Diversity = 2.9096e-08 k_t = [2, 4, 3, 2]
 Generation 2150: Best = 9.001792e+02 FES = 215100 μ_F = 0.840 μ_{CR} = 0.163
 Diversity = 3.0542e-08 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 9.001792e+02 FES = 216100 μ_F = 0.835 μ_{CR} = 0.166
 Diversity = 2.9731e-08 k_t = [2, 4, 3, 3]
 Generation 2170: Best = 9.001792e+02 FES = 217100 μ_F = 0.838 μ_{CR} = 0.168
 Diversity = 2.9104e-08 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 9.001792e+02 FES = 218100 μ_F = 0.839 μ_{CR} = 0.166
 Diversity = 2.9426e-08 k_t = [4, 2, 3, 4]
 Generation 2190: Best = 9.001792e+02 FES = 219100 μ_F = 0.831 μ_{CR} = 0.168
 Diversity = 2.7548e-08 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 9.001792e+02 FES = 220100 μ_F = 0.831 μ_{CR} = 0.167
 Diversity = 2.6064e-08 k_t = [3, 2, 2, 3]
 Generation 2210: Best = 9.001792e+02 FES = 221100 μ_F = 0.831 μ_{CR} = 0.165
 Diversity = 2.6630e-08 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 9.001792e+02 FES = 222100 μ_F = 0.830 μ_{CR} = 0.160
 Diversity = 2.4596e-08 k_t = [4, 2, 3, 4]
 Generation 2230: Best = 9.001792e+02 FES = 223100 μ_F = 0.829 μ_{CR} = 0.159
 Diversity = 2.6839e-08 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 9.001792e+02 FES = 224100 μ_F = 0.823 μ_{CR} = 0.164
 Diversity = 2.4884e-08 k_t = [4, 4, 3, 3]
 Generation 2250: Best = 9.001792e+02 FES = 225100 μ_F = 0.826 μ_{CR} = 0.163
 Diversity = 2.4339e-08 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 9.001792e+02 FES = 226100 μ_F = 0.825 μ_{CR} = 0.165
 Diversity = 2.5271e-08 k_t = [2, 2, 2, 4]
 Generation 2270: Best = 9.001792e+02 FES = 227100 μ_F = 0.834 μ_{CR} = 0.169
 Diversity = 2.7678e-08 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 9.001792e+02 FES = 228100 μ_F = 0.832 μ_{CR} = 0.166
 Diversity = 2.7771e-08 k_t = [2, 2, 2, 3]
 Generation 2290: Best = 9.001792e+02 FES = 229100 μ_F = 0.826 μ_{CR} = 0.167
 Diversity = 2.7179e-08 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 9.001792e+02 FES = 230100 μ_F = 0.828 μ_{CR} = 0.169
 Diversity = 2.5776e-08 k_t = [3, 4, 3, 3]
 Generation 2310: Best = 9.001792e+02 FES = 231100 μ_F = 0.829 μ_{CR} = 0.174
 Diversity = 2.5717e-08 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 9.001792e+02 FES = 232100 μ_F = 0.826 μ_{CR} = 0.170
 Diversity = 2.5505e-08 k_t = [3, 3, 2, 4]
 Generation 2330: Best = 9.001792e+02 FES = 233100 μ_F = 0.830 μ_{CR} = 0.164
 Diversity = 2.9735e-08 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 9.001792e+02 FES = 234100 μ_F = 0.833 μ_{CR} = 0.165
 Diversity = 2.7393e-08 k_t = [4, 3, 4, 2]
 Generation 2350: Best = 9.001792e+02 FES = 235100 μ_F = 0.831 μ_{CR} = 0.161
 Diversity = 2.6158e-08 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 9.001792e+02 FES = 236100 μ_F = 0.829 μ_{CR} = 0.162
 Diversity = 2.7156e-08 k_t = [4, 3, 3, 2]
 Generation 2370: Best = 9.001792e+02 FES = 237100 μ_F = 0.830 μ_{CR} = 0.161
 Diversity = 2.8232e-08 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 9.001792e+02 FES = 238100 μ_F = 0.818 μ_{CR} = 0.162
 Diversity = 2.6060e-08 k_t = [4, 4, 2, 2]

Generation 2390: Best = 9.001792e+02 FES = 239100 μ_F = 0.821 μ_{CR} = 0.166
 Diversity = 2.9225e-08 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 9.001792e+02 FES = 240100 μ_F = 0.822 μ_{CR} = 0.171
 Diversity = 3.0262e-08 k_t = [4, 4, 4, 2]
 Generation 2410: Best = 9.001792e+02 FES = 241100 μ_F = 0.830 μ_{CR} = 0.168
 Diversity = 3.1670e-08 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 9.001792e+02 FES = 242100 μ_F = 0.828 μ_{CR} = 0.166
 Diversity = 3.1346e-08 k_t = [3, 3, 2, 2]
 Generation 2430: Best = 9.001792e+02 FES = 243100 μ_F = 0.832 μ_{CR} = 0.169
 Diversity = 3.3831e-08 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 9.001792e+02 FES = 244100 μ_F = 0.821 μ_{CR} = 0.166
 Diversity = 3.3629e-08 k_t = [2, 3, 2, 4]
 Generation 2450: Best = 9.001792e+02 FES = 245100 μ_F = 0.811 μ_{CR} = 0.155
 Diversity = 3.3137e-08 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 9.001792e+02 FES = 246100 μ_F = 0.813 μ_{CR} = 0.152
 Diversity = 3.2101e-08 k_t = [2, 4, 4, 4]
 Generation 2470: Best = 9.001792e+02 FES = 247100 μ_F = 0.817 μ_{CR} = 0.148
 Diversity = 3.4392e-08 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 9.001792e+02 FES = 248100 μ_F = 0.811 μ_{CR} = 0.152
 Diversity = 3.2594e-08 k_t = [4, 2, 4, 2]
 Generation 2490: Best = 9.001792e+02 FES = 249100 μ_F = 0.813 μ_{CR} = 0.167
 Diversity = 3.0812e-08 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 9.001792e+02 FES = 250100 μ_F = 0.813 μ_{CR} = 0.166
 Diversity = 3.2171e-08 k_t = [2, 3, 2, 2]
 Generation 2510: Best = 9.001792e+02 FES = 251100 μ_F = 0.815 μ_{CR} = 0.163
 Diversity = 3.2933e-08 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 9.001792e+02 FES = 252100 μ_F = 0.819 μ_{CR} = 0.163
 Diversity = 3.3193e-08 k_t = [3, 2, 4, 2]
 Generation 2530: Best = 9.001792e+02 FES = 253100 μ_F = 0.820 μ_{CR} = 0.164
 Diversity = 3.5322e-08 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 9.001792e+02 FES = 254100 μ_F = 0.816 μ_{CR} = 0.171
 Diversity = 3.6067e-08 k_t = [2, 3, 2, 3]
 Generation 2550: Best = 9.001792e+02 FES = 255100 μ_F = 0.815 μ_{CR} = 0.173
 Diversity = 3.5082e-08 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 9.001792e+02 FES = 256100 μ_F = 0.821 μ_{CR} = 0.170
 Diversity = 3.3942e-08 k_t = [3, 3, 4, 3]
 Generation 2570: Best = 9.001792e+02 FES = 257100 μ_F = 0.826 μ_{CR} = 0.176
 Diversity = 3.7405e-08 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 9.001792e+02 FES = 258100 μ_F = 0.830 μ_{CR} = 0.174
 Diversity = 3.9332e-08 k_t = [3, 4, 2, 2]
 Generation 2590: Best = 9.001792e+02 FES = 259100 μ_F = 0.827 μ_{CR} = 0.173
 Diversity = 3.8379e-08 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 9.001792e+02 FES = 260100 μ_F = 0.830 μ_{CR} = 0.174
 Diversity = 3.8421e-08 k_t = [4, 3, 3, 4]
 Generation 2610: Best = 9.001792e+02 FES = 261100 μ_F = 0.829 μ_{CR} = 0.171
 Diversity = 3.9338e-08 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 9.001792e+02 FES = 262100 μ_F = 0.826 μ_{CR} = 0.166
 Diversity = 3.8386e-08 k_t = [4, 2, 4, 4]
 Generation 2630: Best = 9.001792e+02 FES = 263100 μ_F = 0.837 μ_{CR} = 0.169
 Diversity = 3.8747e-08 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 9.001792e+02 FES = 264100 μ_F = 0.838 μ_{CR} = 0.172
 Diversity = 3.4886e-08 k_t = [3, 2, 3, 4]
 Generation 2650: Best = 9.001792e+02 FES = 265100 μ_F = 0.834 μ_{CR} = 0.168
 Diversity = 3.7362e-08 k_t = [1, 1, 1, 1]

Generation 2660: Best = 9.001792e+02 FES = 266100 μ_F = 0.833 μ_{CR} = 0.162
 Diversity = 3.7167e-08 k_t = [3, 3, 3, 4]
 Generation 2670: Best = 9.001792e+02 FES = 267100 μ_F = 0.834 μ_{CR} = 0.159
 Diversity = 3.7466e-08 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 9.001792e+02 FES = 268100 μ_F = 0.829 μ_{CR} = 0.162
 Diversity = 3.3936e-08 k_t = [4, 3, 4, 4]
 Generation 2690: Best = 9.001792e+02 FES = 269100 μ_F = 0.832 μ_{CR} = 0.160
 Diversity = 3.3215e-08 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 9.001792e+02 FES = 270100 μ_F = 0.831 μ_{CR} = 0.153
 Diversity = 3.4892e-08 k_t = [2, 3, 3, 2]
 Generation 2710: Best = 9.001792e+02 FES = 271100 μ_F = 0.823 μ_{CR} = 0.155
 Diversity = 3.5096e-08 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 9.001792e+02 FES = 272100 μ_F = 0.816 μ_{CR} = 0.160
 Diversity = 3.4120e-08 k_t = [3, 2, 4, 2]
 Generation 2730: Best = 9.001792e+02 FES = 273100 μ_F = 0.826 μ_{CR} = 0.158
 Diversity = 3.4801e-08 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 9.001792e+02 FES = 274100 μ_F = 0.828 μ_{CR} = 0.160
 Diversity = 3.7767e-08 k_t = [2, 4, 2, 2]
 Generation 2750: Best = 9.001792e+02 FES = 275100 μ_F = 0.820 μ_{CR} = 0.158
 Diversity = 3.8750e-08 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 9.001792e+02 FES = 276100 μ_F = 0.813 μ_{CR} = 0.158
 Diversity = 3.6274e-08 k_t = [2, 4, 4, 3]
 Generation 2770: Best = 9.001792e+02 FES = 277100 μ_F = 0.811 μ_{CR} = 0.163
 Diversity = 3.9316e-08 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 9.001792e+02 FES = 278100 μ_F = 0.818 μ_{CR} = 0.165
 Diversity = 3.9671e-08 k_t = [3, 2, 4, 3]
 Generation 2790: Best = 9.001792e+02 FES = 279100 μ_F = 0.815 μ_{CR} = 0.173
 Diversity = 4.2775e-08 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 9.001792e+02 FES = 280100 μ_F = 0.812 μ_{CR} = 0.169
 Diversity = 4.1166e-08 k_t = [3, 4, 3, 2]
 Generation 2810: Best = 9.001792e+02 FES = 281100 μ_F = 0.813 μ_{CR} = 0.167
 Diversity = 3.9606e-08 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 9.001792e+02 FES = 282100 μ_F = 0.813 μ_{CR} = 0.169
 Diversity = 3.8613e-08 k_t = [2, 3, 3, 2]
 Generation 2830: Best = 9.001792e+02 FES = 283100 μ_F = 0.820 μ_{CR} = 0.167
 Diversity = 3.9926e-08 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 9.001792e+02 FES = 284100 μ_F = 0.816 μ_{CR} = 0.165
 Diversity = 3.8877e-08 k_t = [2, 2, 3, 3]
 Generation 2850: Best = 9.001792e+02 FES = 285100 μ_F = 0.815 μ_{CR} = 0.158
 Diversity = 4.1114e-08 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 9.001792e+02 FES = 286100 μ_F = 0.819 μ_{CR} = 0.161
 Diversity = 4.2745e-08 k_t = [3, 2, 4, 4]
 Generation 2870: Best = 9.001792e+02 FES = 287100 μ_F = 0.821 μ_{CR} = 0.159
 Diversity = 4.5526e-08 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 9.001792e+02 FES = 288100 μ_F = 0.826 μ_{CR} = 0.164
 Diversity = 4.7827e-08 k_t = [4, 4, 4, 3]
 Generation 2890: Best = 9.001792e+02 FES = 289100 μ_F = 0.834 μ_{CR} = 0.167
 Diversity = 5.1620e-08 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 9.001792e+02 FES = 290100 μ_F = 0.833 μ_{CR} = 0.165
 Diversity = 4.8531e-08 k_t = [3, 3, 3, 4]
 Generation 2910: Best = 9.001792e+02 FES = 291100 μ_F = 0.835 μ_{CR} = 0.172
 Diversity = 5.5757e-08 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 9.001792e+02 FES = 292100 μ_F = 0.841 μ_{CR} = 0.171
 Diversity = 5.6971e-08 k_t = [4, 2, 3, 3]

Generation 2930: Best = 9.001792e+02 FES = 293100 μ_F = 0.840 μ_{CR} = 0.174
 Diversity = 6.4821e-08 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 9.001792e+02 FES = 294100 μ_F = 0.840 μ_{CR} = 0.178
 Diversity = 6.5844e-08 k_t = [3, 4, 4, 2]
 Generation 2950: Best = 9.001792e+02 FES = 295100 μ_F = 0.842 μ_{CR} = 0.175
 Diversity = 7.6579e-08 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 9.001792e+02 FES = 296100 μ_F = 0.839 μ_{CR} = 0.172
 Diversity = 8.4964e-08 k_t = [3, 3, 3, 2]
 Generation 2970: Best = 9.001792e+02 FES = 297100 μ_F = 0.839 μ_{CR} = 0.164
 Diversity = 8.5698e-08 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 9.001792e+02 FES = 298100 μ_F = 0.829 μ_{CR} = 0.166
 Diversity = 8.4441e-08 k_t = [2, 4, 2, 2]
 Generation 2990: Best = 9.001792e+02 FES = 299100 μ_F = 0.828 μ_{CR} = 0.167
 Diversity = 8.5803e-08 k_t = [1, 1, 1, 1]

Optimization complete.

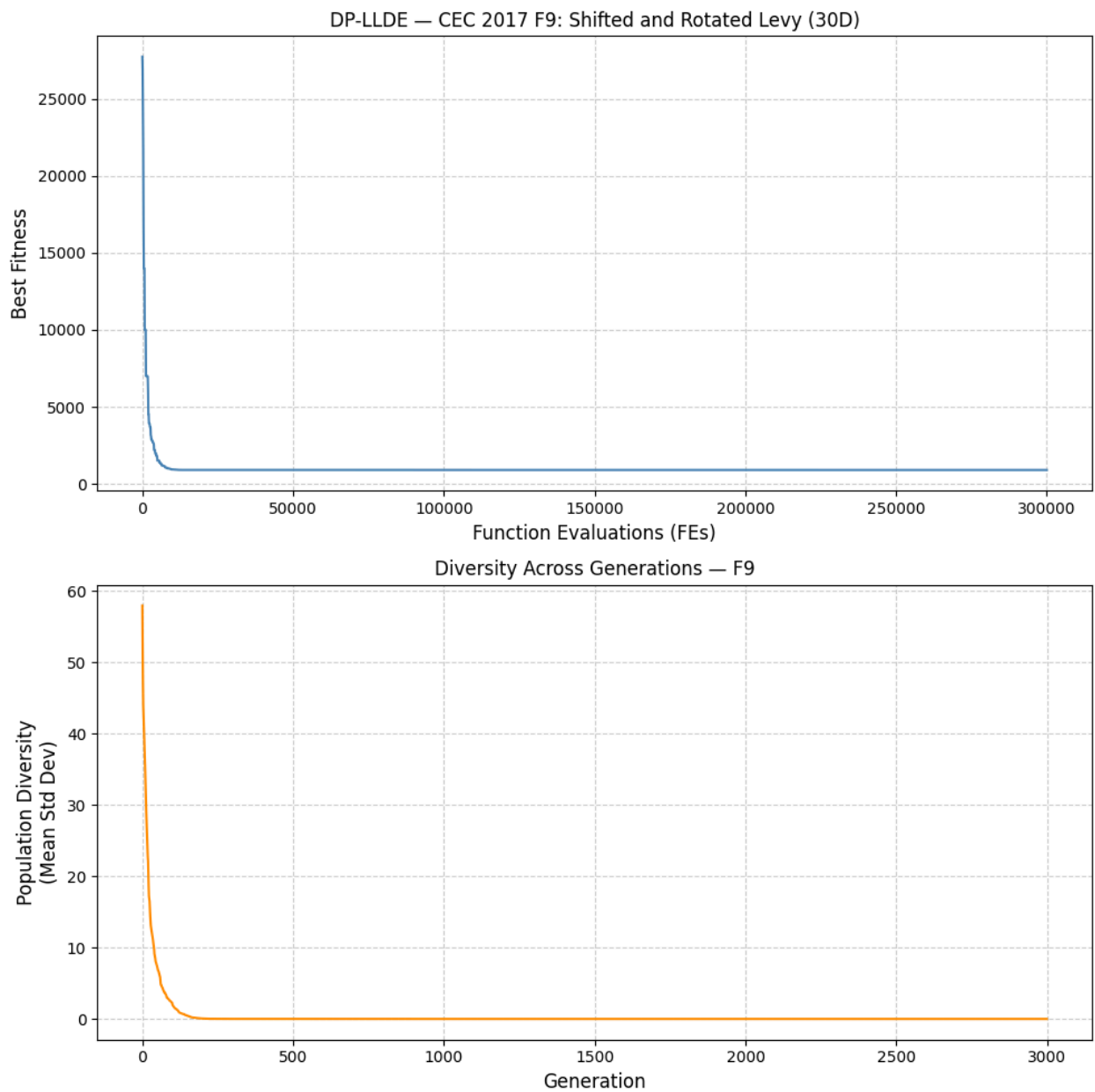
Best Fitness = 9.001792e+02

Total Function Evals = 300000

Total Successful Updates = 153698

```

=====
=
Results:
  Best Fitness      : 9.001792e+02
  Global Optimum    : 9.000000e+02
  Error (|best-optimum|) : 1.791842e-01
  Best Solution (5 dim) : [-22.7714942    1.25950773   21.68061469   51.93725513
-16.81644129]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F9.png

```
In [ ]: # =====
# F10 - Shifted and Rotated Schwefel
# =====
run_and_plot('F10', make_wrapper(simple.f10), 1000.0, "Shifted and Rotated Sch
```

=====

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DP-LLDE on CEC 2017 F10 – Single Run

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Function : Shifted and Rotated Schwefel's (30D)

Bounds : [-100, 100]

Global optimum : 1000.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 9.881110e+03

Generation 10: Best = 9.818516e+03 FES = 1100 $\mu_F = 0.578$ $\mu_{CR} = 0.333$

Diversity = 4.9890e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 9.556862e+03 FES = 2100 $\mu_F = 0.583$ $\mu_{CR} = 0.338$

Diversity = 4.8677e+01 $k_t = [4, 3, 4, 3]$

Generation 30: Best = 9.214234e+03 FES = 3100 $\mu_F = 0.676$ $\mu_{CR} = 0.263$

Diversity = 4.7647e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 9.013522e+03 FES = 4100 $\mu_F = 0.678$ $\mu_{CR} = 0.196$

Diversity = 4.6975e+01 $k_t = [3, 4, 2, 2]$

Generation 50: Best = 8.603907e+03 FES = 5100 $\mu_F = 0.660$ $\mu_{CR} = 0.160$

Diversity = 4.6603e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 8.603907e+03 FES = 6100 $\mu_F = 0.722$ $\mu_{CR} = 0.139$

Diversity = 4.6366e+01 $k_t = [4, 2, 2, 4]$

Generation 70: Best = 7.839157e+03 FES = 7100 $\mu_F = 0.723$ $\mu_{CR} = 0.114$

Diversity = 4.6250e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 7.839157e+03 FES = 8100 $\mu_F = 0.757$ $\mu_{CR} = 0.096$

Diversity = 4.6275e+01 $k_t = [3, 3, 4, 2]$

Generation 90: Best = 7.839157e+03 FES = 9100 $\mu_F = 0.784$ $\mu_{CR} = 0.071$

Diversity = 4.5940e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 7.839157e+03 FES = 10100 $\mu_F = 0.789$ $\mu_{CR} = 0.057$

Diversity = 4.6041e+01 $k_t = [2, 4, 3, 3]$

Generation 110: Best = 7.839157e+03 FES = 11100 $\mu_F = 0.804$ $\mu_{CR} = 0.056$

Diversity = 4.6022e+01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 7.839157e+03 FES = 12100 $\mu_F = 0.796$ $\mu_{CR} = 0.054$

Diversity = 4.6508e+01 $k_t = [3, 3, 2, 3]$

Generation 130: Best = 7.818788e+03 FES = 13100 $\mu_F = 0.829$ $\mu_{CR} = 0.050$

Diversity = 4.6735e+01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 6.608789e+03 FES = 14100 $\mu_F = 0.848$ $\mu_{CR} = 0.056$

Diversity = 4.6265e+01 $k_t = [2, 2, 4, 4]$

Generation 150: Best = 6.608789e+03 FES = 15100 $\mu_F = 0.803$ $\mu_{CR} = 0.057$

Diversity = 4.6321e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 6.550950e+03 FES = 16100 $\mu_F = 0.796$ $\mu_{CR} = 0.044$

Diversity = 4.6230e+01 $k_t = [4, 4, 2, 3]$

Generation 170: Best = 6.550950e+03 FES = 17100 $\mu_F = 0.785$ $\mu_{CR} = 0.065$

Diversity = 4.6339e+01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 6.550950e+03 FES = 18100 $\mu_F = 0.812$ $\mu_{CR} = 0.063$

Diversity = 4.6505e+01 $k_t = [2, 3, 2, 3]$

Generation 190: Best = 6.550950e+03 FES = 19100 $\mu_F = 0.816$ $\mu_{CR} = 0.062$

Diversity = 4.6436e+01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 6.372743e+03 FES = 20100 $\mu_F = 0.789$ $\mu_{CR} = 0.048$

Diversity = 4.6447e+01 $k_t = [4, 2, 2, 3]$

Generation 210: Best = 6.307938e+03 FES = 21100 $\mu_F = 0.812$ $\mu_{CR} = 0.046$

Diversity = 4.6346e+01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 6.307938e+03 FES = 22100 $\mu_F = 0.744$ $\mu_{CR} = 0.032$

Diversity = 4.6299e+01 $k_t = [3, 3, 2, 3]$

Generation 230: Best = 6.307938e+03	FES = 23100	$\mu_F = 0.750$	$\mu_{CR} = 0.035$
Diversity = 4.6169e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 6.307938e+03	FES = 24100	$\mu_F = 0.763$	$\mu_{CR} = 0.037$
Diversity = 4.6117e+01	$k_t = [3, 3, 3, 4]$		
Generation 250: Best = 6.307938e+03	FES = 25100	$\mu_F = 0.721$	$\mu_{CR} = 0.041$
Diversity = 4.6203e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 6.307938e+03	FES = 26100	$\mu_F = 0.662$	$\mu_{CR} = 0.040$
Diversity = 4.6342e+01	$k_t = [3, 2, 2, 3]$		
Generation 270: Best = 6.307938e+03	FES = 27100	$\mu_F = 0.704$	$\mu_{CR} = 0.054$
Diversity = 4.6401e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 6.305341e+03	FES = 28100	$\mu_F = 0.738$	$\mu_{CR} = 0.043$
Diversity = 4.6414e+01	$k_t = [3, 2, 2, 2]$		
Generation 290: Best = 6.305341e+03	FES = 29100	$\mu_F = 0.675$	$\mu_{CR} = 0.049$
Diversity = 4.6388e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 6.305341e+03	FES = 30100	$\mu_F = 0.665$	$\mu_{CR} = 0.050$
Diversity = 4.6419e+01	$k_t = [2, 3, 4, 3]$		
Generation 310: Best = 6.305341e+03	FES = 31100	$\mu_F = 0.654$	$\mu_{CR} = 0.042$
Diversity = 4.6437e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 6.305341e+03	FES = 32100	$\mu_F = 0.643$	$\mu_{CR} = 0.045$
Diversity = 4.6365e+01	$k_t = [2, 2, 4, 3]$		
Generation 330: Best = 6.305341e+03	FES = 33100	$\mu_F = 0.637$	$\mu_{CR} = 0.046$
Diversity = 4.6700e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 6.305341e+03	FES = 34100	$\mu_F = 0.595$	$\mu_{CR} = 0.044$
Diversity = 4.6793e+01	$k_t = [4, 3, 4, 4]$		
Generation 350: Best = 6.305341e+03	FES = 35100	$\mu_F = 0.624$	$\mu_{CR} = 0.052$
Diversity = 4.6563e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 6.215401e+03	FES = 36100	$\mu_F = 0.738$	$\mu_{CR} = 0.038$
Diversity = 4.6550e+01	$k_t = [3, 4, 2, 3]$		
Generation 370: Best = 6.215401e+03	FES = 37100	$\mu_F = 0.667$	$\mu_{CR} = 0.042$
Diversity = 4.6546e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 6.215401e+03	FES = 38100	$\mu_F = 0.734$	$\mu_{CR} = 0.043$
Diversity = 4.6554e+01	$k_t = [3, 4, 4, 3]$		
Generation 390: Best = 6.215401e+03	FES = 39100	$\mu_F = 0.753$	$\mu_{CR} = 0.040$
Diversity = 4.6533e+01	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 6.026970e+03	FES = 40100	$\mu_F = 0.686$	$\mu_{CR} = 0.037$
Diversity = 4.6482e+01	$k_t = [3, 3, 3, 2]$		
Generation 410: Best = 6.026970e+03	FES = 41100	$\mu_F = 0.746$	$\mu_{CR} = 0.033$
Diversity = 4.6544e+01	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 5.963861e+03	FES = 42100	$\mu_F = 0.755$	$\mu_{CR} = 0.045$
Diversity = 4.6526e+01	$k_t = [4, 3, 2, 2]$		
Generation 430: Best = 5.963861e+03	FES = 43100	$\mu_F = 0.714$	$\mu_{CR} = 0.040$
Diversity = 4.6335e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 5.963861e+03	FES = 44100	$\mu_F = 0.593$	$\mu_{CR} = 0.054$
Diversity = 4.6264e+01	$k_t = [4, 2, 4, 2]$		
Generation 450: Best = 5.963861e+03	FES = 45100	$\mu_F = 0.645$	$\mu_{CR} = 0.049$
Diversity = 4.6386e+01	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 5.751004e+03	FES = 46100	$\mu_F = 0.533$	$\mu_{CR} = 0.056$
Diversity = 4.6356e+01	$k_t = [2, 3, 4, 2]$		
Generation 470: Best = 5.751004e+03	FES = 47100	$\mu_F = 0.481$	$\mu_{CR} = 0.043$
Diversity = 4.6219e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 5.336825e+03	FES = 48100	$\mu_F = 0.489$	$\mu_{CR} = 0.054$
Diversity = 4.6219e+01	$k_t = [4, 2, 4, 2]$		
Generation 490: Best = 5.336825e+03	FES = 49100	$\mu_F = 0.503$	$\mu_{CR} = 0.043$
Diversity = 4.6411e+01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 5.336825e+03	FES = 50100	$\mu_F = 0.517$	$\mu_{CR} = 0.036$
Diversity = 4.6429e+01	k_t = [4, 3, 4, 3]		
Generation 510: Best = 5.336825e+03	FES = 51100	$\mu_F = 0.591$	$\mu_{CR} = 0.045$
Diversity = 4.6490e+01	k_t = [1, 1, 1, 1]		
Generation 520: Best = 5.336825e+03	FES = 52100	$\mu_F = 0.600$	$\mu_{CR} = 0.038$
Diversity = 4.6445e+01	k_t = [4, 2, 4, 2]		
Generation 530: Best = 5.336825e+03	FES = 53100	$\mu_F = 0.644$	$\mu_{CR} = 0.042$
Diversity = 4.6089e+01	k_t = [1, 1, 1, 1]		
Generation 540: Best = 5.336825e+03	FES = 54100	$\mu_F = 0.583$	$\mu_{CR} = 0.043$
Diversity = 4.6075e+01	k_t = [2, 3, 2, 3]		
Generation 550: Best = 5.336825e+03	FES = 55100	$\mu_F = 0.648$	$\mu_{CR} = 0.037$
Diversity = 4.6128e+01	k_t = [1, 1, 1, 1]		
Generation 560: Best = 5.336825e+03	FES = 56100	$\mu_F = 0.638$	$\mu_{CR} = 0.036$
Diversity = 4.6150e+01	k_t = [3, 2, 3, 2]		
Generation 570: Best = 5.336825e+03	FES = 57100	$\mu_F = 0.514$	$\mu_{CR} = 0.035$
Diversity = 4.6159e+01	k_t = [1, 1, 1, 1]		
Generation 580: Best = 5.336825e+03	FES = 58100	$\mu_F = 0.594$	$\mu_{CR} = 0.040$
Diversity = 4.6384e+01	k_t = [4, 4, 3, 4]		
Generation 590: Best = 5.336825e+03	FES = 59100	$\mu_F = 0.598$	$\mu_{CR} = 0.057$
Diversity = 4.6407e+01	k_t = [1, 1, 1, 1]		
Generation 600: Best = 5.336825e+03	FES = 60100	$\mu_F = 0.576$	$\mu_{CR} = 0.058$
Diversity = 4.6408e+01	k_t = [4, 3, 2, 3]		
Generation 610: Best = 5.336825e+03	FES = 61100	$\mu_F = 0.592$	$\mu_{CR} = 0.044$
Diversity = 4.6411e+01	k_t = [1, 1, 1, 1]		
Generation 620: Best = 5.336825e+03	FES = 62100	$\mu_F = 0.633$	$\mu_{CR} = 0.049$
Diversity = 4.6416e+01	k_t = [3, 2, 3, 4]		
Generation 630: Best = 5.336825e+03	FES = 63100	$\mu_F = 0.602$	$\mu_{CR} = 0.042$
Diversity = 4.6382e+01	k_t = [1, 1, 1, 1]		
Generation 640: Best = 5.336825e+03	FES = 64100	$\mu_F = 0.592$	$\mu_{CR} = 0.036$
Diversity = 4.6346e+01	k_t = [3, 3, 4, 2]		
Generation 650: Best = 5.336825e+03	FES = 65100	$\mu_F = 0.645$	$\mu_{CR} = 0.030$
Diversity = 4.6337e+01	k_t = [1, 1, 1, 1]		
Generation 660: Best = 5.336825e+03	FES = 66100	$\mu_F = 0.637$	$\mu_{CR} = 0.043$
Diversity = 4.6220e+01	k_t = [2, 2, 2, 2]		
Generation 670: Best = 5.336825e+03	FES = 67100	$\mu_F = 0.486$	$\mu_{CR} = 0.038$
Diversity = 4.6212e+01	k_t = [1, 1, 1, 1]		
Generation 680: Best = 5.336825e+03	FES = 68100	$\mu_F = 0.509$	$\mu_{CR} = 0.042$
Diversity = 4.5968e+01	k_t = [4, 2, 4, 4]		
Generation 690: Best = 5.336825e+03	FES = 69100	$\mu_F = 0.512$	$\mu_{CR} = 0.047$
Diversity = 4.5884e+01	k_t = [1, 1, 1, 1]		
Generation 700: Best = 5.336825e+03	FES = 70100	$\mu_F = 0.604$	$\mu_{CR} = 0.048$
Diversity = 4.5641e+01	k_t = [3, 2, 4, 4]		
Generation 710: Best = 5.336825e+03	FES = 71100	$\mu_F = 0.665$	$\mu_{CR} = 0.041$
Diversity = 4.5640e+01	k_t = [1, 1, 1, 1]		
Generation 720: Best = 5.336825e+03	FES = 72100	$\mu_F = 0.563$	$\mu_{CR} = 0.050$
Diversity = 4.5631e+01	k_t = [4, 2, 3, 2]		
Generation 730: Best = 5.336825e+03	FES = 73100	$\mu_F = 0.608$	$\mu_{CR} = 0.049$
Diversity = 4.5592e+01	k_t = [1, 1, 1, 1]		
Generation 740: Best = 5.336825e+03	FES = 74100	$\mu_F = 0.513$	$\mu_{CR} = 0.033$
Diversity = 4.5565e+01	k_t = [2, 3, 2, 2]		
Generation 750: Best = 5.336825e+03	FES = 75100	$\mu_F = 0.408$	$\mu_{CR} = 0.046$
Diversity = 4.5572e+01	k_t = [1, 1, 1, 1]		
Generation 760: Best = 5.336825e+03	FES = 76100	$\mu_F = 0.486$	$\mu_{CR} = 0.049$
Diversity = 4.5501e+01	k_t = [4, 3, 3, 4]		

Generation 770: Best = 5.336825e+03	FES = 77100	$\mu_F = 0.408$	$\mu_{CR} = 0.051$
Diversity = 4.5494e+01	k_t = [1, 1, 1, 1]		
Generation 780: Best = 5.336825e+03	FES = 78100	$\mu_F = 0.456$	$\mu_{CR} = 0.045$
Diversity = 4.5527e+01	k_t = [3, 4, 3, 2]		
Generation 790: Best = 5.276293e+03	FES = 79100	$\mu_F = 0.326$	$\mu_{CR} = 0.051$
Diversity = 4.5527e+01	k_t = [1, 1, 1, 1]		
Generation 800: Best = 5.276293e+03	FES = 80100	$\mu_F = 0.338$	$\mu_{CR} = 0.051$
Diversity = 4.5489e+01	k_t = [3, 3, 2, 2]		
Generation 810: Best = 5.276293e+03	FES = 81100	$\mu_F = 0.289$	$\mu_{CR} = 0.056$
Diversity = 4.5470e+01	k_t = [1, 1, 1, 1]		
Generation 820: Best = 5.276293e+03	FES = 82100	$\mu_F = 0.382$	$\mu_{CR} = 0.049$
Diversity = 4.5476e+01	k_t = [4, 4, 2, 3]		
Generation 830: Best = 5.276293e+03	FES = 83100	$\mu_F = 0.378$	$\mu_{CR} = 0.041$
Diversity = 4.5457e+01	k_t = [1, 1, 1, 1]		
Generation 840: Best = 5.276293e+03	FES = 84100	$\mu_F = 0.338$	$\mu_{CR} = 0.044$
Diversity = 4.5429e+01	k_t = [3, 4, 4, 2]		
Generation 850: Best = 5.276293e+03	FES = 85100	$\mu_F = 0.315$	$\mu_{CR} = 0.039$
Diversity = 4.5413e+01	k_t = [1, 1, 1, 1]		
Generation 860: Best = 5.249948e+03	FES = 86100	$\mu_F = 0.318$	$\mu_{CR} = 0.044$
Diversity = 4.5327e+01	k_t = [4, 4, 3, 3]		
Generation 870: Best = 5.249948e+03	FES = 87100	$\mu_F = 0.282$	$\mu_{CR} = 0.053$
Diversity = 4.5280e+01	k_t = [1, 1, 1, 1]		
Generation 880: Best = 5.249948e+03	FES = 88100	$\mu_F = 0.310$	$\mu_{CR} = 0.064$
Diversity = 4.5268e+01	k_t = [4, 4, 3, 2]		
Generation 890: Best = 5.249948e+03	FES = 89100	$\mu_F = 0.320$	$\mu_{CR} = 0.054$
Diversity = 4.5264e+01	k_t = [1, 1, 1, 1]		
Generation 900: Best = 5.249948e+03	FES = 90100	$\mu_F = 0.300$	$\mu_{CR} = 0.046$
Diversity = 4.5260e+01	k_t = [2, 4, 4, 2]		
Generation 910: Best = 5.249948e+03	FES = 91100	$\mu_F = 0.318$	$\mu_{CR} = 0.037$
Diversity = 4.5244e+01	k_t = [1, 1, 1, 1]		
Generation 920: Best = 5.217443e+03	FES = 92100	$\mu_F = 0.263$	$\mu_{CR} = 0.083$
Diversity = 4.5238e+01	k_t = [3, 2, 2, 3]		
Generation 930: Best = 5.073005e+03	FES = 93100	$\mu_F = 0.242$	$\mu_{CR} = 0.079$
Diversity = 4.5228e+01	k_t = [1, 1, 1, 1]		
Generation 940: Best = 5.073005e+03	FES = 94100	$\mu_F = 0.162$	$\mu_{CR} = 0.067$
Diversity = 4.5190e+01	k_t = [3, 2, 4, 2]		
Generation 950: Best = 5.073005e+03	FES = 95100	$\mu_F = 0.156$	$\mu_{CR} = 0.076$
Diversity = 4.5162e+01	k_t = [1, 1, 1, 1]		
Generation 960: Best = 5.073005e+03	FES = 96100	$\mu_F = 0.262$	$\mu_{CR} = 0.056$
Diversity = 4.5327e+01	k_t = [3, 2, 3, 4]		
Generation 970: Best = 5.034703e+03	FES = 97100	$\mu_F = 0.310$	$\mu_{CR} = 0.049$
Diversity = 4.5322e+01	k_t = [1, 1, 1, 1]		
Generation 980: Best = 5.034703e+03	FES = 98100	$\mu_F = 0.298$	$\mu_{CR} = 0.065$
Diversity = 4.5343e+01	k_t = [2, 2, 2, 4]		
Generation 990: Best = 4.850421e+03	FES = 99100	$\mu_F = 0.207$	$\mu_{CR} = 0.057$
Diversity = 4.5326e+01	k_t = [1, 1, 1, 1]		
Generation 1000: Best = 4.850421e+03	FES = 100100	$\mu_F = 0.253$	$\mu_{CR} = 0.074$
Diversity = 4.5657e+01	k_t = [4, 2, 3, 4]		
Generation 1010: Best = 4.850421e+03	FES = 101100	$\mu_F = 0.269$	$\mu_{CR} = 0.065$
Diversity = 4.5540e+01	k_t = [1, 1, 1, 1]		
Generation 1020: Best = 4.850421e+03	FES = 102100	$\mu_F = 0.239$	$\mu_{CR} = 0.069$
Diversity = 4.5554e+01	k_t = [4, 3, 2, 3]		
Generation 1030: Best = 4.845147e+03	FES = 103100	$\mu_F = 0.265$	$\mu_{CR} = 0.059$
Diversity = 4.5554e+01	k_t = [1, 1, 1, 1]		

Generation 1040: Best = 4.803625e+03 FES = 104100 μ_F = 0.177 μ_{CR} = 0.062
 Diversity = 4.5542e+01 k_t = [2, 2, 4, 3]
 Generation 1050: Best = 4.803625e+03 FES = 105100 μ_F = 0.207 μ_{CR} = 0.057
 Diversity = 4.5467e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 4.791111e+03 FES = 106100 μ_F = 0.225 μ_{CR} = 0.045
 Diversity = 4.5425e+01 k_t = [3, 3, 4, 4]
 Generation 1070: Best = 4.791111e+03 FES = 107100 μ_F = 0.178 μ_{CR} = 0.047
 Diversity = 4.5411e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 4.791111e+03 FES = 108100 μ_F = 0.204 μ_{CR} = 0.048
 Diversity = 4.5409e+01 k_t = [4, 2, 2, 2]
 Generation 1090: Best = 4.621995e+03 FES = 109100 μ_F = 0.242 μ_{CR} = 0.036
 Diversity = 4.5401e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 4.621995e+03 FES = 110100 μ_F = 0.319 μ_{CR} = 0.034
 Diversity = 4.5443e+01 k_t = [2, 4, 4, 2]
 Generation 1110: Best = 4.619799e+03 FES = 111100 μ_F = 0.222 μ_{CR} = 0.052
 Diversity = 4.5427e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 4.428488e+03 FES = 112100 μ_F = 0.160 μ_{CR} = 0.046
 Diversity = 4.5421e+01 k_t = [2, 3, 3, 3]
 Generation 1130: Best = 4.428488e+03 FES = 113100 μ_F = 0.134 μ_{CR} = 0.046
 Diversity = 4.5416e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 4.391884e+03 FES = 114100 μ_F = 0.196 μ_{CR} = 0.056
 Diversity = 4.5403e+01 k_t = [2, 3, 3, 2]
 Generation 1150: Best = 4.386890e+03 FES = 115100 μ_F = 0.166 μ_{CR} = 0.049
 Diversity = 4.5400e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 4.360549e+03 FES = 116100 μ_F = 0.178 μ_{CR} = 0.081
 Diversity = 4.5407e+01 k_t = [4, 2, 4, 4]
 Generation 1170: Best = 4.360549e+03 FES = 117100 μ_F = 0.189 μ_{CR} = 0.082
 Diversity = 4.5456e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 4.360549e+03 FES = 118100 μ_F = 0.163 μ_{CR} = 0.076
 Diversity = 4.5438e+01 k_t = [2, 3, 2, 3]
 Generation 1190: Best = 4.360549e+03 FES = 119100 μ_F = 0.166 μ_{CR} = 0.074
 Diversity = 4.5432e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 4.360549e+03 FES = 120100 μ_F = 0.113 μ_{CR} = 0.086
 Diversity = 4.5426e+01 k_t = [3, 3, 3, 2]
 Generation 1210: Best = 4.270784e+03 FES = 121100 μ_F = 0.149 μ_{CR} = 0.064
 Diversity = 4.5432e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 4.248865e+03 FES = 122100 μ_F = 0.111 μ_{CR} = 0.067
 Diversity = 4.5417e+01 k_t = [3, 3, 2, 3]
 Generation 1230: Best = 4.209754e+03 FES = 123100 μ_F = 0.146 μ_{CR} = 0.063
 Diversity = 4.5413e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 4.209754e+03 FES = 124100 μ_F = 0.147 μ_{CR} = 0.070
 Diversity = 4.5413e+01 k_t = [3, 3, 3, 2]
 Generation 1250: Best = 4.209754e+03 FES = 125100 μ_F = 0.131 μ_{CR} = 0.065
 Diversity = 4.5401e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 4.198369e+03 FES = 126100 μ_F = 0.150 μ_{CR} = 0.062
 Diversity = 4.5397e+01 k_t = [3, 4, 4, 4]
 Generation 1270: Best = 4.198369e+03 FES = 127100 μ_F = 0.237 μ_{CR} = 0.058
 Diversity = 4.5409e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 4.198369e+03 FES = 128100 μ_F = 0.232 μ_{CR} = 0.043
 Diversity = 4.5407e+01 k_t = [3, 3, 2, 3]
 Generation 1290: Best = 4.148522e+03 FES = 129100 μ_F = 0.223 μ_{CR} = 0.048
 Diversity = 4.5411e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 4.148522e+03 FES = 130100 μ_F = 0.231 μ_{CR} = 0.060
 Diversity = 4.5408e+01 k_t = [4, 4, 3, 3]

Generation 1310: Best = 4.148522e+03 FES = 131100 μ_F = 0.230 μ_{CR} = 0.051
 Diversity = 4.5407e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 4.148522e+03 FES = 132100 μ_F = 0.263 μ_{CR} = 0.041
 Diversity = 4.5338e+01 k_t = [3, 3, 2, 4]
 Generation 1330: Best = 4.148522e+03 FES = 133100 μ_F = 0.184 μ_{CR} = 0.051
 Diversity = 4.5323e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 4.148522e+03 FES = 134100 μ_F = 0.165 μ_{CR} = 0.052
 Diversity = 4.5332e+01 k_t = [4, 2, 3, 2]
 Generation 1350: Best = 4.148522e+03 FES = 135100 μ_F = 0.142 μ_{CR} = 0.053
 Diversity = 4.5321e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 4.148522e+03 FES = 136100 μ_F = 0.195 μ_{CR} = 0.062
 Diversity = 4.5304e+01 k_t = [3, 2, 4, 4]
 Generation 1370: Best = 4.057657e+03 FES = 137100 μ_F = 0.172 μ_{CR} = 0.060
 Diversity = 4.5290e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 4.055759e+03 FES = 138100 μ_F = 0.140 μ_{CR} = 0.055
 Diversity = 4.5273e+01 k_t = [3, 2, 4, 3]
 Generation 1390: Best = 4.055759e+03 FES = 139100 μ_F = 0.234 μ_{CR} = 0.052
 Diversity = 4.5282e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 4.053458e+03 FES = 140100 μ_F = 0.258 μ_{CR} = 0.055
 Diversity = 4.5276e+01 k_t = [3, 4, 2, 2]
 Generation 1410: Best = 4.053040e+03 FES = 141100 μ_F = 0.222 μ_{CR} = 0.050
 Diversity = 4.5277e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 4.033116e+03 FES = 142100 μ_F = 0.161 μ_{CR} = 0.053
 Diversity = 4.5275e+01 k_t = [2, 3, 4, 4]
 Generation 1430: Best = 4.033116e+03 FES = 143100 μ_F = 0.124 μ_{CR} = 0.072
 Diversity = 4.5257e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 4.031562e+03 FES = 144100 μ_F = 0.218 μ_{CR} = 0.072
 Diversity = 4.5245e+01 k_t = [2, 2, 4, 3]
 Generation 1450: Best = 4.003095e+03 FES = 145100 μ_F = 0.136 μ_{CR} = 0.060
 Diversity = 4.5245e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 4.003095e+03 FES = 146100 μ_F = 0.092 μ_{CR} = 0.058
 Diversity = 4.5245e+01 k_t = [3, 4, 4, 3]
 Generation 1470: Best = 3.992537e+03 FES = 147100 μ_F = 0.094 μ_{CR} = 0.064
 Diversity = 4.5240e+01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 3.783627e+03 FES = 148100 μ_F = 0.133 μ_{CR} = 0.067
 Diversity = 4.5239e+01 k_t = [3, 3, 2, 4]
 Generation 1490: Best = 3.783627e+03 FES = 149100 μ_F = 0.124 μ_{CR} = 0.064
 Diversity = 4.5234e+01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 3.783627e+03 FES = 150100 μ_F = 0.108 μ_{CR} = 0.065
 Diversity = 4.5221e+01 k_t = [3, 3, 4, 2]
 Generation 1510: Best = 3.783627e+03 FES = 151100 μ_F = 0.194 μ_{CR} = 0.063
 Diversity = 4.5216e+01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 3.783627e+03 FES = 152100 μ_F = 0.143 μ_{CR} = 0.059
 Diversity = 4.5214e+01 k_t = [3, 4, 3, 3]
 Generation 1530: Best = 3.740063e+03 FES = 153100 μ_F = 0.194 μ_{CR} = 0.049
 Diversity = 4.5215e+01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 3.740063e+03 FES = 154100 μ_F = 0.205 μ_{CR} = 0.056
 Diversity = 4.5226e+01 k_t = [4, 2, 3, 4]
 Generation 1550: Best = 3.740063e+03 FES = 155100 μ_F = 0.177 μ_{CR} = 0.072
 Diversity = 4.5218e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 3.711072e+03 FES = 156100 μ_F = 0.220 μ_{CR} = 0.065
 Diversity = 4.5255e+01 k_t = [4, 2, 4, 4]
 Generation 1570: Best = 3.711072e+03 FES = 157100 μ_F = 0.202 μ_{CR} = 0.053
 Diversity = 4.5254e+01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 3.711072e+03 FES = 158100 μ_F = 0.160 μ_{CR} = 0.075
 Diversity = 4.5232e+01 k_t = [3, 2, 2, 2]
 Generation 1590: Best = 3.711072e+03 FES = 159100 μ_F = 0.220 μ_{CR} = 0.053
 Diversity = 4.5231e+01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 3.711072e+03 FES = 160100 μ_F = 0.194 μ_{CR} = 0.055
 Diversity = 4.5232e+01 k_t = [2, 4, 3, 2]
 Generation 1610: Best = 3.711072e+03 FES = 161100 μ_F = 0.153 μ_{CR} = 0.045
 Diversity = 4.5235e+01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 3.711072e+03 FES = 162100 μ_F = 0.134 μ_{CR} = 0.059
 Diversity = 4.5236e+01 k_t = [3, 2, 3, 2]
 Generation 1630: Best = 3.711072e+03 FES = 163100 μ_F = 0.125 μ_{CR} = 0.050
 Diversity = 4.5239e+01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 3.609583e+03 FES = 164100 μ_F = 0.220 μ_{CR} = 0.050
 Diversity = 4.5264e+01 k_t = [4, 2, 3, 3]
 Generation 1650: Best = 3.609583e+03 FES = 165100 μ_F = 0.375 μ_{CR} = 0.052
 Diversity = 4.5173e+01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 3.600509e+03 FES = 166100 μ_F = 0.358 μ_{CR} = 0.047
 Diversity = 4.5171e+01 k_t = [2, 2, 4, 2]
 Generation 1670: Best = 3.600509e+03 FES = 167100 μ_F = 0.324 μ_{CR} = 0.041
 Diversity = 4.5209e+01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 3.600509e+03 FES = 168100 μ_F = 0.266 μ_{CR} = 0.072
 Diversity = 4.5189e+01 k_t = [2, 2, 2, 2]
 Generation 1690: Best = 3.600509e+03 FES = 169100 μ_F = 0.228 μ_{CR} = 0.085
 Diversity = 4.5183e+01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 3.600509e+03 FES = 170100 μ_F = 0.258 μ_{CR} = 0.066
 Diversity = 4.5185e+01 k_t = [3, 2, 3, 2]
 Generation 1710: Best = 3.600509e+03 FES = 171100 μ_F = 0.352 μ_{CR} = 0.064
 Diversity = 4.5187e+01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 3.600509e+03 FES = 172100 μ_F = 0.303 μ_{CR} = 0.058
 Diversity = 4.5185e+01 k_t = [3, 4, 2, 3]
 Generation 1730: Best = 3.600509e+03 FES = 173100 μ_F = 0.287 μ_{CR} = 0.052
 Diversity = 4.5189e+01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 3.600509e+03 FES = 174100 μ_F = 0.258 μ_{CR} = 0.051
 Diversity = 4.5186e+01 k_t = [3, 4, 3, 2]
 Generation 1750: Best = 3.596937e+03 FES = 175100 μ_F = 0.241 μ_{CR} = 0.064
 Diversity = 4.5181e+01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 3.596937e+03 FES = 176100 μ_F = 0.216 μ_{CR} = 0.048
 Diversity = 4.5181e+01 k_t = [2, 3, 2, 2]
 Generation 1770: Best = 3.596937e+03 FES = 177100 μ_F = 0.174 μ_{CR} = 0.040
 Diversity = 4.5184e+01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 3.577553e+03 FES = 178100 μ_F = 0.133 μ_{CR} = 0.060
 Diversity = 4.5176e+01 k_t = [4, 3, 3, 4]
 Generation 1790: Best = 3.577553e+03 FES = 179100 μ_F = 0.190 μ_{CR} = 0.067
 Diversity = 4.5120e+01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 3.577553e+03 FES = 180100 μ_F = 0.255 μ_{CR} = 0.055
 Diversity = 4.5240e+01 k_t = [3, 3, 3, 4]
 Generation 1810: Best = 3.577553e+03 FES = 181100 μ_F = 0.206 μ_{CR} = 0.049
 Diversity = 4.5229e+01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 3.556204e+03 FES = 182100 μ_F = 0.195 μ_{CR} = 0.072
 Diversity = 4.5231e+01 k_t = [3, 2, 4, 3]
 Generation 1830: Best = 3.556204e+03 FES = 183100 μ_F = 0.161 μ_{CR} = 0.058
 Diversity = 4.5233e+01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 3.555200e+03 FES = 184100 μ_F = 0.129 μ_{CR} = 0.055
 Diversity = 4.5229e+01 k_t = [3, 3, 4, 4]

Generation 1850: Best = 3.555200e+03 FES = 185100 μ_F = 0.204 μ_{CR} = 0.050
 Diversity = 4.5231e+01 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 3.496204e+03 FES = 186100 μ_F = 0.212 μ_{CR} = 0.055
 Diversity = 4.5231e+01 k_t = [3, 4, 2, 3]
 Generation 1870: Best = 3.480646e+03 FES = 187100 μ_F = 0.235 μ_{CR} = 0.050
 Diversity = 4.5229e+01 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 3.480646e+03 FES = 188100 μ_F = 0.246 μ_{CR} = 0.040
 Diversity = 4.5234e+01 k_t = [4, 3, 4, 2]
 Generation 1890: Best = 3.480646e+03 FES = 189100 μ_F = 0.256 μ_{CR} = 0.051
 Diversity = 4.5235e+01 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 3.480646e+03 FES = 190100 μ_F = 0.234 μ_{CR} = 0.056
 Diversity = 4.5288e+01 k_t = [4, 3, 4, 4]
 Generation 1910: Best = 3.480646e+03 FES = 191100 μ_F = 0.329 μ_{CR} = 0.046
 Diversity = 4.5288e+01 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 3.480646e+03 FES = 192100 μ_F = 0.394 μ_{CR} = 0.050
 Diversity = 4.5287e+01 k_t = [4, 2, 4, 2]
 Generation 1930: Best = 3.480646e+03 FES = 193100 μ_F = 0.346 μ_{CR} = 0.046
 Diversity = 4.5287e+01 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 3.480646e+03 FES = 194100 μ_F = 0.283 μ_{CR} = 0.047
 Diversity = 4.5284e+01 k_t = [4, 3, 3, 2]
 Generation 1950: Best = 3.480646e+03 FES = 195100 μ_F = 0.253 μ_{CR} = 0.041
 Diversity = 4.5279e+01 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 3.480646e+03 FES = 196100 μ_F = 0.213 μ_{CR} = 0.055
 Diversity = 4.5271e+01 k_t = [3, 4, 3, 4]
 Generation 1970: Best = 3.480646e+03 FES = 197100 μ_F = 0.208 μ_{CR} = 0.049
 Diversity = 4.5268e+01 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 3.470230e+03 FES = 198100 μ_F = 0.233 μ_{CR} = 0.068
 Diversity = 4.5264e+01 k_t = [4, 2, 4, 2]
 Generation 1990: Best = 3.449127e+03 FES = 199100 μ_F = 0.241 μ_{CR} = 0.058
 Diversity = 4.5262e+01 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 3.448186e+03 FES = 200100 μ_F = 0.197 μ_{CR} = 0.065
 Diversity = 4.5260e+01 k_t = [4, 2, 2, 3]
 Generation 2010: Best = 3.448186e+03 FES = 201100 μ_F = 0.151 μ_{CR} = 0.075
 Diversity = 4.5257e+01 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 3.431605e+03 FES = 202100 μ_F = 0.173 μ_{CR} = 0.084
 Diversity = 4.5277e+01 k_t = [4, 3, 3, 2]
 Generation 2030: Best = 3.431605e+03 FES = 203100 μ_F = 0.145 μ_{CR} = 0.056
 Diversity = 4.5276e+01 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 3.431605e+03 FES = 204100 μ_F = 0.131 μ_{CR} = 0.064
 Diversity = 4.5274e+01 k_t = [4, 2, 4, 2]
 Generation 2050: Best = 3.431605e+03 FES = 205100 μ_F = 0.147 μ_{CR} = 0.053
 Diversity = 4.5275e+01 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 3.431605e+03 FES = 206100 μ_F = 0.299 μ_{CR} = 0.057
 Diversity = 4.5426e+01 k_t = [4, 4, 2, 4]
 Generation 2070: Best = 3.431605e+03 FES = 207100 μ_F = 0.294 μ_{CR} = 0.056
 Diversity = 4.5449e+01 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 3.375774e+03 FES = 208100 μ_F = 0.266 μ_{CR} = 0.068
 Diversity = 4.5444e+01 k_t = [2, 3, 4, 3]
 Generation 2090: Best = 3.375774e+03 FES = 209100 μ_F = 0.209 μ_{CR} = 0.079
 Diversity = 4.5442e+01 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 3.375774e+03 FES = 210100 μ_F = 0.249 μ_{CR} = 0.067
 Diversity = 4.5447e+01 k_t = [3, 4, 3, 3]
 Generation 2110: Best = 3.375774e+03 FES = 211100 μ_F = 0.322 μ_{CR} = 0.056
 Diversity = 4.5475e+01 k_t = [1, 1, 1, 1]

Generation 2120: Best = 3.375774e+03 FES = 212100 μ_F = 0.360 μ_{CR} = 0.047
 Diversity = 4.5478e+01 k_t = [2, 4, 4, 4]
 Generation 2130: Best = 3.375774e+03 FES = 213100 μ_F = 0.405 μ_{CR} = 0.048
 Diversity = 4.5475e+01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 3.375774e+03 FES = 214100 μ_F = 0.367 μ_{CR} = 0.051
 Diversity = 4.5469e+01 k_t = [2, 4, 3, 2]
 Generation 2150: Best = 3.361601e+03 FES = 215100 μ_F = 0.325 μ_{CR} = 0.063
 Diversity = 4.5469e+01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 3.361601e+03 FES = 216100 μ_F = 0.366 μ_{CR} = 0.054
 Diversity = 4.5259e+01 k_t = [3, 3, 4, 4]
 Generation 2170: Best = 3.361601e+03 FES = 217100 μ_F = 0.324 μ_{CR} = 0.050
 Diversity = 4.5257e+01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 3.361601e+03 FES = 218100 μ_F = 0.417 μ_{CR} = 0.050
 Diversity = 4.5276e+01 k_t = [2, 4, 2, 3]
 Generation 2190: Best = 3.361601e+03 FES = 219100 μ_F = 0.391 μ_{CR} = 0.059
 Diversity = 4.5274e+01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 3.361601e+03 FES = 220100 μ_F = 0.361 μ_{CR} = 0.053
 Diversity = 4.5275e+01 k_t = [4, 4, 4, 3]
 Generation 2210: Best = 3.358150e+03 FES = 221100 μ_F = 0.321 μ_{CR} = 0.050
 Diversity = 4.5259e+01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 3.358150e+03 FES = 222100 μ_F = 0.414 μ_{CR} = 0.039
 Diversity = 4.5438e+01 k_t = [2, 3, 2, 4]
 Generation 2230: Best = 3.358150e+03 FES = 223100 μ_F = 0.375 μ_{CR} = 0.077
 Diversity = 4.5427e+01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 3.356651e+03 FES = 224100 μ_F = 0.357 μ_{CR} = 0.076
 Diversity = 4.5424e+01 k_t = [4, 2, 3, 4]
 Generation 2250: Best = 3.356614e+03 FES = 225100 μ_F = 0.374 μ_{CR} = 0.069
 Diversity = 4.5440e+01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 3.301518e+03 FES = 226100 μ_F = 0.447 μ_{CR} = 0.069
 Diversity = 4.5370e+01 k_t = [4, 2, 4, 4]
 Generation 2270: Best = 3.301518e+03 FES = 227100 μ_F = 0.448 μ_{CR} = 0.051
 Diversity = 4.5369e+01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 3.254387e+03 FES = 228100 μ_F = 0.475 μ_{CR} = 0.068
 Diversity = 4.5370e+01 k_t = [4, 3, 2, 2]
 Generation 2290: Best = 3.254387e+03 FES = 229100 μ_F = 0.315 μ_{CR} = 0.077
 Diversity = 4.5327e+01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 3.254387e+03 FES = 230100 μ_F = 0.283 μ_{CR} = 0.083
 Diversity = 4.5221e+01 k_t = [4, 2, 4, 4]
 Generation 2310: Best = 3.254387e+03 FES = 231100 μ_F = 0.169 μ_{CR} = 0.073
 Diversity = 4.5215e+01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 3.254387e+03 FES = 232100 μ_F = 0.167 μ_{CR} = 0.088
 Diversity = 4.5215e+01 k_t = [2, 3, 3, 2]
 Generation 2330: Best = 3.254387e+03 FES = 233100 μ_F = 0.169 μ_{CR} = 0.085
 Diversity = 4.5210e+01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 3.254387e+03 FES = 234100 μ_F = 0.223 μ_{CR} = 0.073
 Diversity = 4.5222e+01 k_t = [4, 3, 3, 2]
 Generation 2350: Best = 3.254387e+03 FES = 235100 μ_F = 0.232 μ_{CR} = 0.076
 Diversity = 4.5223e+01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 3.254387e+03 FES = 236100 μ_F = 0.205 μ_{CR} = 0.071
 Diversity = 4.5221e+01 k_t = [2, 4, 4, 3]
 Generation 2370: Best = 3.254257e+03 FES = 237100 μ_F = 0.189 μ_{CR} = 0.089
 Diversity = 4.5221e+01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 3.254257e+03 FES = 238100 μ_F = 0.257 μ_{CR} = 0.081
 Diversity = 4.5158e+01 k_t = [3, 4, 3, 2]

Generation 2390: Best = 3.060455e+03 FES = 239100 μ_F = 0.347 μ_{CR} = 0.080
 Diversity = 4.5157e+01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 2.993752e+03 FES = 240100 μ_F = 0.378 μ_{CR} = 0.071
 Diversity = 4.5063e+01 k_t = [3, 4, 2, 3]
 Generation 2410: Best = 2.993752e+03 FES = 241100 μ_F = 0.425 μ_{CR} = 0.061
 Diversity = 4.5066e+01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 2.993752e+03 FES = 242100 μ_F = 0.475 μ_{CR} = 0.069
 Diversity = 4.5051e+01 k_t = [3, 3, 4, 4]
 Generation 2430: Best = 2.993752e+03 FES = 243100 μ_F = 0.511 μ_{CR} = 0.064
 Diversity = 4.5050e+01 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 2.986993e+03 FES = 244100 μ_F = 0.497 μ_{CR} = 0.074
 Diversity = 4.5096e+01 k_t = [3, 3, 2, 4]
 Generation 2450: Best = 2.949778e+03 FES = 245100 μ_F = 0.538 μ_{CR} = 0.062
 Diversity = 4.5096e+01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 2.949673e+03 FES = 246100 μ_F = 0.455 μ_{CR} = 0.062
 Diversity = 4.5095e+01 k_t = [4, 3, 3, 2]
 Generation 2470: Best = 2.949074e+03 FES = 247100 μ_F = 0.438 μ_{CR} = 0.059
 Diversity = 4.5098e+01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 2.937059e+03 FES = 248100 μ_F = 0.319 μ_{CR} = 0.083
 Diversity = 4.5096e+01 k_t = [2, 3, 2, 3]
 Generation 2490: Best = 2.937059e+03 FES = 249100 μ_F = 0.433 μ_{CR} = 0.071
 Diversity = 4.5060e+01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 2.882529e+03 FES = 250100 μ_F = 0.426 μ_{CR} = 0.050
 Diversity = 4.4992e+01 k_t = [3, 3, 4, 4]
 Generation 2510: Best = 2.873276e+03 FES = 251100 μ_F = 0.539 μ_{CR} = 0.053
 Diversity = 4.4895e+01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 2.873276e+03 FES = 252100 μ_F = 0.578 μ_{CR} = 0.049
 Diversity = 4.4857e+01 k_t = [2, 4, 2, 3]
 Generation 2530: Best = 2.873276e+03 FES = 253100 μ_F = 0.664 μ_{CR} = 0.043
 Diversity = 4.4797e+01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 2.873276e+03 FES = 254100 μ_F = 0.620 μ_{CR} = 0.041
 Diversity = 4.4796e+01 k_t = [4, 2, 3, 2]
 Generation 2550: Best = 2.850642e+03 FES = 255100 μ_F = 0.633 μ_{CR} = 0.038
 Diversity = 4.4801e+01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 2.850642e+03 FES = 256100 μ_F = 0.579 μ_{CR} = 0.037
 Diversity = 4.4904e+01 k_t = [4, 3, 4, 3]
 Generation 2570: Best = 2.850642e+03 FES = 257100 μ_F = 0.576 μ_{CR} = 0.047
 Diversity = 4.4922e+01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 2.850642e+03 FES = 258100 μ_F = 0.575 μ_{CR} = 0.063
 Diversity = 4.4953e+01 k_t = [2, 4, 2, 2]
 Generation 2590: Best = 2.850642e+03 FES = 259100 μ_F = 0.561 μ_{CR} = 0.054
 Diversity = 4.4953e+01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 2.850642e+03 FES = 260100 μ_F = 0.543 μ_{CR} = 0.056
 Diversity = 4.4955e+01 k_t = [2, 4, 2, 2]
 Generation 2610: Best = 2.840731e+03 FES = 261100 μ_F = 0.556 μ_{CR} = 0.060
 Diversity = 4.4947e+01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 2.840731e+03 FES = 262100 μ_F = 0.598 μ_{CR} = 0.054
 Diversity = 4.4909e+01 k_t = [2, 2, 3, 4]
 Generation 2630: Best = 2.840731e+03 FES = 263100 μ_F = 0.603 μ_{CR} = 0.061
 Diversity = 4.4910e+01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 2.840731e+03 FES = 264100 μ_F = 0.637 μ_{CR} = 0.067
 Diversity = 4.4799e+01 k_t = [4, 2, 2, 4]
 Generation 2650: Best = 2.839586e+03 FES = 265100 μ_F = 0.644 μ_{CR} = 0.067
 Diversity = 4.4799e+01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 2.839586e+03 FES = 266100 μ_F = 0.636 μ_{CR} = 0.066
 Diversity = 4.4771e+01 k_t = [3, 3, 4, 3]
 Generation 2670: Best = 2.839586e+03 FES = 267100 μ_F = 0.573 μ_{CR} = 0.075
 Diversity = 4.4771e+01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 2.839586e+03 FES = 268100 μ_F = 0.602 μ_{CR} = 0.069
 Diversity = 4.4991e+01 k_t = [4, 2, 3, 4]
 Generation 2690: Best = 2.839586e+03 FES = 269100 μ_F = 0.571 μ_{CR} = 0.054
 Diversity = 4.4992e+01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 2.839586e+03 FES = 270100 μ_F = 0.593 μ_{CR} = 0.050
 Diversity = 4.4992e+01 k_t = [3, 3, 2, 4]
 Generation 2710: Best = 2.839586e+03 FES = 271100 μ_F = 0.629 μ_{CR} = 0.046
 Diversity = 4.4976e+01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 2.833272e+03 FES = 272100 μ_F = 0.664 μ_{CR} = 0.048
 Diversity = 4.5090e+01 k_t = [4, 3, 4, 4]
 Generation 2730: Best = 2.833272e+03 FES = 273100 μ_F = 0.719 μ_{CR} = 0.049
 Diversity = 4.5147e+01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 2.833272e+03 FES = 274100 μ_F = 0.766 μ_{CR} = 0.052
 Diversity = 4.5114e+01 k_t = [3, 2, 3, 4]
 Generation 2750: Best = 2.787069e+03 FES = 275100 μ_F = 0.759 μ_{CR} = 0.056
 Diversity = 4.5120e+01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 2.787069e+03 FES = 276100 μ_F = 0.780 μ_{CR} = 0.058
 Diversity = 4.5120e+01 k_t = [2, 4, 3, 2]
 Generation 2770: Best = 2.780053e+03 FES = 277100 μ_F = 0.753 μ_{CR} = 0.057
 Diversity = 4.5119e+01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 2.780053e+03 FES = 278100 μ_F = 0.690 μ_{CR} = 0.060
 Diversity = 4.5120e+01 k_t = [4, 2, 3, 2]
 Generation 2790: Best = 2.780053e+03 FES = 279100 μ_F = 0.724 μ_{CR} = 0.065
 Diversity = 4.5120e+01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 2.780053e+03 FES = 280100 μ_F = 0.789 μ_{CR} = 0.058
 Diversity = 4.5093e+01 k_t = [2, 3, 3, 2]
 Generation 2810: Best = 2.780053e+03 FES = 281100 μ_F = 0.816 μ_{CR} = 0.053
 Diversity = 4.5086e+01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 2.779672e+03 FES = 282100 μ_F = 0.726 μ_{CR} = 0.038
 Diversity = 4.5086e+01 k_t = [2, 3, 2, 2]
 Generation 2830: Best = 2.777593e+03 FES = 283100 μ_F = 0.725 μ_{CR} = 0.037
 Diversity = 4.5120e+01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 2.767092e+03 FES = 284100 μ_F = 0.724 μ_{CR} = 0.049
 Diversity = 4.5121e+01 k_t = [4, 2, 2, 2]
 Generation 2850: Best = 2.767092e+03 FES = 285100 μ_F = 0.759 μ_{CR} = 0.042
 Diversity = 4.5184e+01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 2.767092e+03 FES = 286100 μ_F = 0.773 μ_{CR} = 0.046
 Diversity = 4.5119e+01 k_t = [3, 2, 3, 2]
 Generation 2870: Best = 2.757875e+03 FES = 287100 μ_F = 0.633 μ_{CR} = 0.070
 Diversity = 4.5118e+01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 2.757875e+03 FES = 288100 μ_F = 0.685 μ_{CR} = 0.057
 Diversity = 4.5099e+01 k_t = [3, 3, 3, 4]
 Generation 2890: Best = 2.741584e+03 FES = 289100 μ_F = 0.683 μ_{CR} = 0.061
 Diversity = 4.5100e+01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 2.741584e+03 FES = 290100 μ_F = 0.569 μ_{CR} = 0.065
 Diversity = 4.5120e+01 k_t = [3, 3, 3, 3]
 Generation 2910: Best = 2.726168e+03 FES = 291100 μ_F = 0.622 μ_{CR} = 0.060
 Diversity = 4.5027e+01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 2.726168e+03 FES = 292100 μ_F = 0.598 μ_{CR} = 0.058
 Diversity = 4.5026e+01 k_t = [2, 4, 4, 2]

Generation 2930: Best = 2.724368e+03 FES = 293100 μ_F = 0.601 μ_{CR} = 0.061
 Diversity = 4.5026e+01 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 2.723839e+03 FES = 294100 μ_F = 0.562 μ_{CR} = 0.062
 Diversity = 4.5016e+01 k_t = [2, 4, 4, 3]
 Generation 2950: Best = 2.719206e+03 FES = 295100 μ_F = 0.587 μ_{CR} = 0.091
 Diversity = 4.5007e+01 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 2.709234e+03 FES = 296100 μ_F = 0.600 μ_{CR} = 0.098
 Diversity = 4.5008e+01 k_t = [3, 3, 4, 2]
 Generation 2970: Best = 2.703158e+03 FES = 297100 μ_F = 0.653 μ_{CR} = 0.080
 Diversity = 4.5005e+01 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 2.701312e+03 FES = 298100 μ_F = 0.577 μ_{CR} = 0.075
 Diversity = 4.4997e+01 k_t = [4, 4, 4, 4]
 Generation 2990: Best = 2.701312e+03 FES = 299100 μ_F = 0.623 μ_{CR} = 0.056
 Diversity = 4.4952e+01 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 2.701312e+03

Total Function Evals = 300000

Total Successful Updates = 5177

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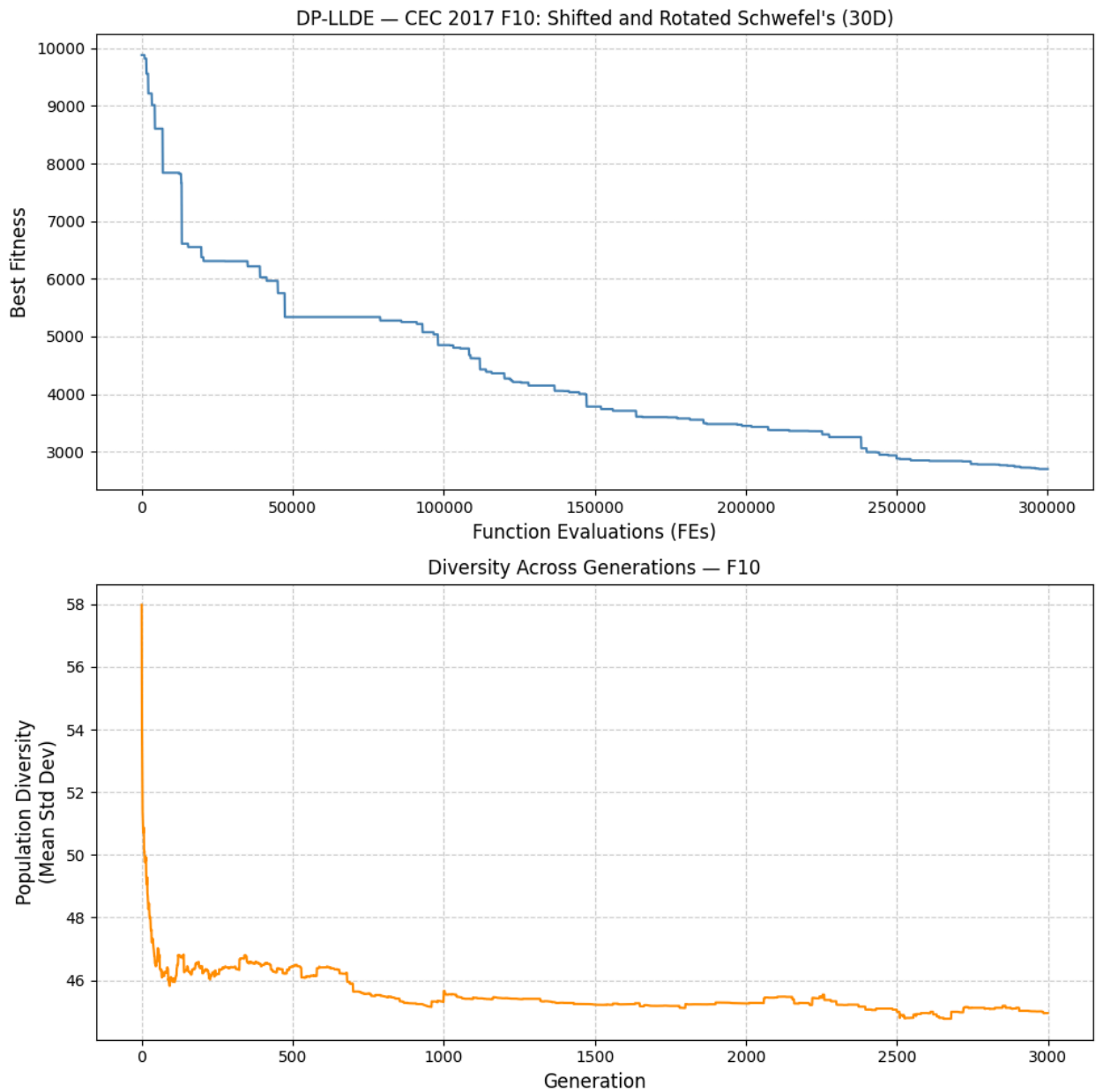
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Results:

Best Fitness : 2.701312e+03
 Global Optimum : 1.000000e+03
 Error (|best-optimum|) : 1.701312e+03
 Best Solution (5 dim) : [-11.15924289 95.30235889 63.21259387 99.91512187
 12.73974296]
 D^(0) (init. diversity): 5.7976e+01

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Graph saved as convergence_F10.png

6.3 Hybrid Functions (F11-F20)

```
In [ ]: # =====
# F11 — Hybrid Function 1 (Zakharov; Rosenbrock; Rastrigin)
# =====
run_and_plot('F11', make_wrapper(hybrid.f11), 1100.0, 'Hybrid: Zakharov; Rosenbrock; Rastrigin')
```

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DP-LLDE on CEC 2017 F11 – Single Run

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Function : Hybrid: Zakharov; Rosenbrock; Rastrigin (30D)

Bounds : [-100, 100]

Global optimum : 1100.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 1.525880e+04

Generation 10: Best = 7.724076e+03 FES = 1100 $\mu_F = 0.574$ $\mu_{CR} = 0.337$

Diversity = 4.2822e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 3.429033e+03 FES = 2100 $\mu_F = 0.602$ $\mu_{CR} = 0.320$

Diversity = 2.7801e+01 $k_t = [4, 3, 4, 3]$

Generation 30: Best = 2.425102e+03 FES = 3100 $\mu_F = 0.627$ $\mu_{CR} = 0.293$

Diversity = 1.9289e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 1.792592e+03 FES = 4100 $\mu_F = 0.655$ $\mu_{CR} = 0.274$

Diversity = 1.5359e+01 $k_t = [4, 3, 4, 2]$

Generation 50: Best = 1.536701e+03 FES = 5100 $\mu_F = 0.670$ $\mu_{CR} = 0.259$

Diversity = 1.1717e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 1.429263e+03 FES = 6100 $\mu_F = 0.690$ $\mu_{CR} = 0.240$

Diversity = 9.5903e+00 $k_t = [2, 4, 4, 3]$

Generation 70: Best = 1.338048e+03 FES = 7100 $\mu_F = 0.699$ $\mu_{CR} = 0.219$

Diversity = 8.8546e+00 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 1.314989e+03 FES = 8100 $\mu_F = 0.699$ $\mu_{CR} = 0.207$

Diversity = 8.1980e+00 $k_t = [4, 2, 4, 2]$

Generation 90: Best = 1.287959e+03 FES = 9100 $\mu_F = 0.704$ $\mu_{CR} = 0.180$

Diversity = 7.5453e+00 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 1.279311e+03 FES = 10100 $\mu_F = 0.700$ $\mu_{CR} = 0.158$

Diversity = 7.1951e+00 $k_t = [4, 2, 2, 2]$

Generation 110: Best = 1.266242e+03 FES = 11100 $\mu_F = 0.729$ $\mu_{CR} = 0.139$

Diversity = 6.7975e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 1.251539e+03 FES = 12100 $\mu_F = 0.746$ $\mu_{CR} = 0.121$

Diversity = 6.2683e+00 $k_t = [4, 4, 3, 4]$

Generation 130: Best = 1.240022e+03 FES = 13100 $\mu_F = 0.750$ $\mu_{CR} = 0.108$

Diversity = 6.0737e+00 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 1.225821e+03 FES = 14100 $\mu_F = 0.754$ $\mu_{CR} = 0.108$

Diversity = 5.7994e+00 $k_t = [4, 3, 4, 3]$

Generation 150: Best = 1.225049e+03 FES = 15100 $\mu_F = 0.751$ $\mu_{CR} = 0.093$

Diversity = 5.7033e+00 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 1.218158e+03 FES = 16100 $\mu_F = 0.752$ $\mu_{CR} = 0.085$

Diversity = 5.7493e+00 $k_t = [4, 3, 4, 3]$

Generation 170: Best = 1.210880e+03 FES = 17100 $\mu_F = 0.737$ $\mu_{CR} = 0.079$

Diversity = 5.8115e+00 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 1.208435e+03 FES = 18100 $\mu_F = 0.731$ $\mu_{CR} = 0.073$

Diversity = 5.8071e+00 $k_t = [2, 3, 2, 3]$

Generation 190: Best = 1.206697e+03 FES = 19100 $\mu_F = 0.726$ $\mu_{CR} = 0.084$

Diversity = 5.5482e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 1.202682e+03 FES = 20100 $\mu_F = 0.739$ $\mu_{CR} = 0.078$

Diversity = 5.4065e+00 $k_t = [2, 4, 2, 4]$

Generation 210: Best = 1.190412e+03 FES = 21100 $\mu_F = 0.726$ $\mu_{CR} = 0.077$

Diversity = 5.3479e+00 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 1.189411e+03 FES = 22100 $\mu_F = 0.750$ $\mu_{CR} = 0.081$

Diversity = 5.3937e+00 $k_t = [3, 3, 2, 2]$

Generation 230: Best = 1.189411e+03	FES = 23100	$\mu_F = 0.749$	$\mu_{CR} = 0.075$
Diversity = 5.3530e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 1.186940e+03	FES = 24100	$\mu_F = 0.768$	$\mu_{CR} = 0.077$
Diversity = 5.2928e+00	$k_t = [2, 2, 4, 3]$		
Generation 250: Best = 1.186940e+03	FES = 25100	$\mu_F = 0.785$	$\mu_{CR} = 0.070$
Diversity = 5.2129e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.184671e+03	FES = 26100	$\mu_F = 0.799$	$\mu_{CR} = 0.076$
Diversity = 5.2761e+00	$k_t = [3, 4, 3, 3]$		
Generation 270: Best = 1.181880e+03	FES = 27100	$\mu_F = 0.800$	$\mu_{CR} = 0.080$
Diversity = 5.3510e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.180585e+03	FES = 28100	$\mu_F = 0.818$	$\mu_{CR} = 0.076$
Diversity = 5.1822e+00	$k_t = [4, 2, 4, 4]$		
Generation 290: Best = 1.179306e+03	FES = 29100	$\mu_F = 0.818$	$\mu_{CR} = 0.073$
Diversity = 5.1475e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.174182e+03	FES = 30100	$\mu_F = 0.838$	$\mu_{CR} = 0.071$
Diversity = 5.0419e+00	$k_t = [2, 4, 4, 3]$		
Generation 310: Best = 1.174047e+03	FES = 31100	$\mu_F = 0.824$	$\mu_{CR} = 0.083$
Diversity = 4.9857e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.173990e+03	FES = 32100	$\mu_F = 0.792$	$\mu_{CR} = 0.076$
Diversity = 4.8227e+00	$k_t = [3, 4, 4, 4]$		
Generation 330: Best = 1.168676e+03	FES = 33100	$\mu_F = 0.808$	$\mu_{CR} = 0.083$
Diversity = 4.8373e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.163040e+03	FES = 34100	$\mu_F = 0.795$	$\mu_{CR} = 0.079$
Diversity = 4.6693e+00	$k_t = [3, 3, 3, 4]$		
Generation 350: Best = 1.163040e+03	FES = 35100	$\mu_F = 0.811$	$\mu_{CR} = 0.077$
Diversity = 4.7141e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.161779e+03	FES = 36100	$\mu_F = 0.810$	$\mu_{CR} = 0.072$
Diversity = 4.5789e+00	$k_t = [4, 3, 4, 4]$		
Generation 370: Best = 1.160366e+03	FES = 37100	$\mu_F = 0.804$	$\mu_{CR} = 0.076$
Diversity = 4.5276e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 1.158267e+03	FES = 38100	$\mu_F = 0.792$	$\mu_{CR} = 0.070$
Diversity = 4.5330e+00	$k_t = [4, 2, 2, 2]$		
Generation 390: Best = 1.158267e+03	FES = 39100	$\mu_F = 0.825$	$\mu_{CR} = 0.069$
Diversity = 4.5324e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 1.155889e+03	FES = 40100	$\mu_F = 0.817$	$\mu_{CR} = 0.068$
Diversity = 4.4983e+00	$k_t = [2, 4, 3, 3]$		
Generation 410: Best = 1.154356e+03	FES = 41100	$\mu_F = 0.798$	$\mu_{CR} = 0.073$
Diversity = 4.5014e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 1.154282e+03	FES = 42100	$\mu_F = 0.796$	$\mu_{CR} = 0.073$
Diversity = 4.4843e+00	$k_t = [3, 3, 4, 2]$		
Generation 430: Best = 1.153818e+03	FES = 43100	$\mu_F = 0.797$	$\mu_{CR} = 0.082$
Diversity = 4.5075e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 1.153775e+03	FES = 44100	$\mu_F = 0.800$	$\mu_{CR} = 0.082$
Diversity = 4.4202e+00	$k_t = [3, 2, 3, 4]$		
Generation 450: Best = 1.151384e+03	FES = 45100	$\mu_F = 0.808$	$\mu_{CR} = 0.078$
Diversity = 4.3602e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 1.150026e+03	FES = 46100	$\mu_F = 0.781$	$\mu_{CR} = 0.075$
Diversity = 4.3030e+00	$k_t = [2, 3, 3, 3]$		
Generation 470: Best = 1.149802e+03	FES = 47100	$\mu_F = 0.763$	$\mu_{CR} = 0.078$
Diversity = 4.3326e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.149066e+03	FES = 48100	$\mu_F = 0.783$	$\mu_{CR} = 0.077$
Diversity = 4.2890e+00	$k_t = [4, 4, 4, 3]$		
Generation 490: Best = 1.147245e+03	FES = 49100	$\mu_F = 0.797$	$\mu_{CR} = 0.082$
Diversity = 4.1873e+00	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 1.147163e+03	FES = 50100	$\mu_F = 0.795$	$\mu_{CR} = 0.082$
Diversity = 4.1232e+00	$k_t = [3, 3, 2, 2]$		
Generation 510: Best = 1.147163e+03	FES = 51100	$\mu_F = 0.818$	$\mu_{CR} = 0.088$
Diversity = 4.0193e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.146122e+03	FES = 52100	$\mu_F = 0.837$	$\mu_{CR} = 0.097$
Diversity = 3.8215e+00	$k_t = [4, 3, 4, 3]$		
Generation 530: Best = 1.144796e+03	FES = 53100	$\mu_F = 0.832$	$\mu_{CR} = 0.094$
Diversity = 3.7643e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.144717e+03	FES = 54100	$\mu_F = 0.832$	$\mu_{CR} = 0.094$
Diversity = 3.5143e+00	$k_t = [3, 3, 2, 3]$		
Generation 550: Best = 1.144428e+03	FES = 55100	$\mu_F = 0.837$	$\mu_{CR} = 0.081$
Diversity = 3.4246e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.144428e+03	FES = 56100	$\mu_F = 0.851$	$\mu_{CR} = 0.092$
Diversity = 3.0052e+00	$k_t = [3, 2, 2, 4]$		
Generation 570: Best = 1.144048e+03	FES = 57100	$\mu_F = 0.859$	$\mu_{CR} = 0.098$
Diversity = 2.3772e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.143822e+03	FES = 58100	$\mu_F = 0.857$	$\mu_{CR} = 0.102$
Diversity = 1.5225e+00	$k_t = [3, 4, 2, 4]$		
Generation 590: Best = 1.142140e+03	FES = 59100	$\mu_F = 0.844$	$\mu_{CR} = 0.101$
Diversity = 1.3217e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.141810e+03	FES = 60100	$\mu_F = 0.832$	$\mu_{CR} = 0.105$
Diversity = 1.1374e+00	$k_t = [3, 4, 4, 2]$		
Generation 610: Best = 1.141313e+03	FES = 61100	$\mu_F = 0.816$	$\mu_{CR} = 0.117$
Diversity = 9.7705e-01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.140841e+03	FES = 62100	$\mu_F = 0.802$	$\mu_{CR} = 0.132$
Diversity = 9.0630e-01	$k_t = [2, 2, 3, 2]$		
Generation 630: Best = 1.140314e+03	FES = 63100	$\mu_F = 0.785$	$\mu_{CR} = 0.144$
Diversity = 8.4187e-01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.139711e+03	FES = 64100	$\mu_F = 0.770$	$\mu_{CR} = 0.152$
Diversity = 4.0612e-01	$k_t = [3, 2, 3, 2]$		
Generation 650: Best = 1.139332e+03	FES = 65100	$\mu_F = 0.763$	$\mu_{CR} = 0.141$
Diversity = 3.5556e-01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.139083e+03	FES = 66100	$\mu_F = 0.745$	$\mu_{CR} = 0.134$
Diversity = 2.9639e-01	$k_t = [4, 4, 4, 2]$		
Generation 670: Best = 1.138731e+03	FES = 67100	$\mu_F = 0.747$	$\mu_{CR} = 0.138$
Diversity = 2.5730e-01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.138621e+03	FES = 68100	$\mu_F = 0.738$	$\mu_{CR} = 0.147$
Diversity = 2.3153e-01	$k_t = [4, 2, 3, 2]$		
Generation 690: Best = 1.138520e+03	FES = 69100	$\mu_F = 0.719$	$\mu_{CR} = 0.153$
Diversity = 2.1333e-01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.138486e+03	FES = 70100	$\mu_F = 0.724$	$\mu_{CR} = 0.149$
Diversity = 1.9448e-01	$k_t = [2, 3, 3, 2]$		
Generation 710: Best = 1.138419e+03	FES = 71100	$\mu_F = 0.714$	$\mu_{CR} = 0.132$
Diversity = 1.9790e-01	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 1.138395e+03	FES = 72100	$\mu_F = 0.716$	$\mu_{CR} = 0.129$
Diversity = 1.7260e-01	$k_t = [3, 3, 3, 4]$		
Generation 730: Best = 1.138343e+03	FES = 73100	$\mu_F = 0.714$	$\mu_{CR} = 0.123$
Diversity = 1.6691e-01	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 1.138314e+03	FES = 74100	$\mu_F = 0.713$	$\mu_{CR} = 0.121$
Diversity = 1.3964e-01	$k_t = [4, 4, 2, 4]$		
Generation 750: Best = 1.138287e+03	FES = 75100	$\mu_F = 0.709$	$\mu_{CR} = 0.121$
Diversity = 1.3278e-01	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 1.138268e+03	FES = 76100	$\mu_F = 0.714$	$\mu_{CR} = 0.122$
Diversity = 1.2640e-01	$k_t = [2, 3, 2, 3]$		

Generation 770: Best = 1.138259e+03 FES = 77100 μ_F = 0.725 μ_{CR} = 0.109
 Diversity = 1.2224e-01 k_t = [1, 1, 1, 1]
 Generation 780: Best = 1.138247e+03 FES = 78100 μ_F = 0.721 μ_{CR} = 0.101
 Diversity = 1.0984e-01 k_t = [2, 3, 4, 4]
 Generation 790: Best = 1.138238e+03 FES = 79100 μ_F = 0.712 μ_{CR} = 0.101
 Diversity = 1.1651e-01 k_t = [1, 1, 1, 1]
 Generation 800: Best = 1.138210e+03 FES = 80100 μ_F = 0.719 μ_{CR} = 0.104
 Diversity = 1.3380e-01 k_t = [2, 2, 3, 3]
 Generation 810: Best = 1.138202e+03 FES = 81100 μ_F = 0.737 μ_{CR} = 0.101
 Diversity = 1.2463e-01 k_t = [1, 1, 1, 1]
 Generation 820: Best = 1.138181e+03 FES = 82100 μ_F = 0.751 μ_{CR} = 0.101
 Diversity = 1.2289e-01 k_t = [4, 2, 2, 4]
 Generation 830: Best = 1.138164e+03 FES = 83100 μ_F = 0.769 μ_{CR} = 0.114
 Diversity = 1.1852e-01 k_t = [1, 1, 1, 1]
 Generation 840: Best = 1.138159e+03 FES = 84100 μ_F = 0.758 μ_{CR} = 0.119
 Diversity = 1.2958e-01 k_t = [2, 4, 4, 2]
 Generation 850: Best = 1.138149e+03 FES = 85100 μ_F = 0.742 μ_{CR} = 0.106
 Diversity = 1.5659e-01 k_t = [1, 1, 1, 1]
 Generation 860: Best = 1.138128e+03 FES = 86100 μ_F = 0.756 μ_{CR} = 0.103
 Diversity = 1.4295e-01 k_t = [3, 3, 2, 3]
 Generation 870: Best = 1.138107e+03 FES = 87100 μ_F = 0.757 μ_{CR} = 0.087
 Diversity = 1.6080e-01 k_t = [1, 1, 1, 1]
 Generation 880: Best = 1.138100e+03 FES = 88100 μ_F = 0.755 μ_{CR} = 0.088
 Diversity = 1.6652e-01 k_t = [2, 2, 3, 4]
 Generation 890: Best = 1.138093e+03 FES = 89100 μ_F = 0.760 μ_{CR} = 0.083
 Diversity = 1.5261e-01 k_t = [1, 1, 1, 1]
 Generation 900: Best = 1.138079e+03 FES = 90100 μ_F = 0.743 μ_{CR} = 0.084
 Diversity = 1.5340e-01 k_t = [4, 4, 4, 2]
 Generation 910: Best = 1.138072e+03 FES = 91100 μ_F = 0.731 μ_{CR} = 0.085
 Diversity = 1.6608e-01 k_t = [1, 1, 1, 1]
 Generation 920: Best = 1.138060e+03 FES = 92100 μ_F = 0.734 μ_{CR} = 0.094
 Diversity = 1.3122e-01 k_t = [4, 2, 2, 4]
 Generation 930: Best = 1.138045e+03 FES = 93100 μ_F = 0.728 μ_{CR} = 0.110
 Diversity = 1.2562e-01 k_t = [1, 1, 1, 1]
 Generation 940: Best = 1.138039e+03 FES = 94100 μ_F = 0.736 μ_{CR} = 0.111
 Diversity = 1.3448e-01 k_t = [4, 4, 4, 2]
 Generation 950: Best = 1.138023e+03 FES = 95100 μ_F = 0.747 μ_{CR} = 0.097
 Diversity = 1.5752e-01 k_t = [1, 1, 1, 1]
 Generation 960: Best = 1.138016e+03 FES = 96100 μ_F = 0.749 μ_{CR} = 0.099
 Diversity = 1.1830e-01 k_t = [4, 4, 4, 3]
 Generation 970: Best = 1.138009e+03 FES = 97100 μ_F = 0.741 μ_{CR} = 0.096
 Diversity = 1.4498e-01 k_t = [1, 1, 1, 1]
 Generation 980: Best = 1.138004e+03 FES = 98100 μ_F = 0.740 μ_{CR} = 0.101
 Diversity = 1.3812e-01 k_t = [3, 3, 2, 3]
 Generation 990: Best = 1.137999e+03 FES = 99100 μ_F = 0.737 μ_{CR} = 0.086
 Diversity = 1.3092e-01 k_t = [1, 1, 1, 1]
 Generation 1000: Best = 1.137997e+03 FES = 100100 μ_F = 0.732 μ_{CR} = 0.091
 Diversity = 1.2185e-01 k_t = [2, 4, 4, 3]
 Generation 1010: Best = 1.137993e+03 FES = 101100 μ_F = 0.739 μ_{CR} = 0.084
 Diversity = 1.2763e-01 k_t = [1, 1, 1, 1]
 Generation 1020: Best = 1.137986e+03 FES = 102100 μ_F = 0.734 μ_{CR} = 0.087
 Diversity = 1.2735e-01 k_t = [3, 3, 3, 4]
 Generation 1030: Best = 1.137979e+03 FES = 103100 μ_F = 0.758 μ_{CR} = 0.086
 Diversity = 1.3311e-01 k_t = [1, 1, 1, 1]

Generation 1040: Best = 1.137971e+03 FES = 104100 μ_F = 0.760 μ_{CR} = 0.080
 Diversity = 1.1620e-01 k_t = [2, 2, 4, 4]
 Generation 1050: Best = 1.137966e+03 FES = 105100 μ_F = 0.766 μ_{CR} = 0.091
 Diversity = 1.0807e-01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 1.137959e+03 FES = 106100 μ_F = 0.758 μ_{CR} = 0.093
 Diversity = 1.0831e-01 k_t = [3, 2, 4, 3]
 Generation 1070: Best = 1.137954e+03 FES = 107100 μ_F = 0.745 μ_{CR} = 0.098
 Diversity = 1.2347e-01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 1.137953e+03 FES = 108100 μ_F = 0.762 μ_{CR} = 0.091
 Diversity = 1.1426e-01 k_t = [3, 3, 3, 3]
 Generation 1090: Best = 1.137946e+03 FES = 109100 μ_F = 0.767 μ_{CR} = 0.091
 Diversity = 1.1332e-01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.137938e+03 FES = 110100 μ_F = 0.767 μ_{CR} = 0.095
 Diversity = 1.3169e-01 k_t = [2, 2, 2, 2]
 Generation 1110: Best = 1.137925e+03 FES = 111100 μ_F = 0.772 μ_{CR} = 0.085
 Diversity = 1.3365e-01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.137925e+03 FES = 112100 μ_F = 0.788 μ_{CR} = 0.085
 Diversity = 1.3772e-01 k_t = [4, 2, 4, 4]
 Generation 1130: Best = 1.137923e+03 FES = 113100 μ_F = 0.799 μ_{CR} = 0.073
 Diversity = 1.4894e-01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.137920e+03 FES = 114100 μ_F = 0.801 μ_{CR} = 0.081
 Diversity = 1.4311e-01 k_t = [2, 3, 4, 4]
 Generation 1150: Best = 1.137920e+03 FES = 115100 μ_F = 0.795 μ_{CR} = 0.084
 Diversity = 1.2483e-01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.137917e+03 FES = 116100 μ_F = 0.773 μ_{CR} = 0.086
 Diversity = 1.2546e-01 k_t = [3, 2, 4, 4]
 Generation 1170: Best = 1.137914e+03 FES = 117100 μ_F = 0.777 μ_{CR} = 0.084
 Diversity = 1.1758e-01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.137912e+03 FES = 118100 μ_F = 0.782 μ_{CR} = 0.074
 Diversity = 1.0093e-01 k_t = [4, 4, 3, 3]
 Generation 1190: Best = 1.137911e+03 FES = 119100 μ_F = 0.763 μ_{CR} = 0.080
 Diversity = 9.8551e-02 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.137910e+03 FES = 120100 μ_F = 0.755 μ_{CR} = 0.085
 Diversity = 9.0418e-02 k_t = [2, 4, 4, 2]
 Generation 1210: Best = 1.137910e+03 FES = 121100 μ_F = 0.748 μ_{CR} = 0.092
 Diversity = 9.1892e-02 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.137907e+03 FES = 122100 μ_F = 0.774 μ_{CR} = 0.081
 Diversity = 8.8276e-02 k_t = [2, 2, 3, 2]
 Generation 1230: Best = 1.137906e+03 FES = 123100 μ_F = 0.779 μ_{CR} = 0.079
 Diversity = 8.6346e-02 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.137905e+03 FES = 124100 μ_F = 0.787 μ_{CR} = 0.084
 Diversity = 8.6617e-02 k_t = [3, 3, 2, 4]
 Generation 1250: Best = 1.137902e+03 FES = 125100 μ_F = 0.796 μ_{CR} = 0.083
 Diversity = 7.5143e-02 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.137901e+03 FES = 126100 μ_F = 0.789 μ_{CR} = 0.079
 Diversity = 7.1597e-02 k_t = [2, 3, 2, 4]
 Generation 1270: Best = 1.137897e+03 FES = 127100 μ_F = 0.798 μ_{CR} = 0.083
 Diversity = 6.1352e-02 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.137897e+03 FES = 128100 μ_F = 0.799 μ_{CR} = 0.080
 Diversity = 7.1184e-02 k_t = [3, 4, 3, 4]
 Generation 1290: Best = 1.137896e+03 FES = 129100 μ_F = 0.796 μ_{CR} = 0.082
 Diversity = 6.3993e-02 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.137895e+03 FES = 130100 μ_F = 0.790 μ_{CR} = 0.089
 Diversity = 6.0009e-02 k_t = [3, 3, 3, 4]

Generation 1310: Best = 1.137894e+03 FES = 131100 μ_F = 0.774 μ_{CR} = 0.090
 Diversity = 5.8647e-02 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.137893e+03 FES = 132100 μ_F = 0.763 μ_{CR} = 0.091
 Diversity = 6.1287e-02 k_t = [2, 2, 2, 2]
 Generation 1330: Best = 1.137892e+03 FES = 133100 μ_F = 0.767 μ_{CR} = 0.092
 Diversity = 6.1462e-02 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.137892e+03 FES = 134100 μ_F = 0.761 μ_{CR} = 0.086
 Diversity = 5.3475e-02 k_t = [2, 3, 3, 4]
 Generation 1350: Best = 1.137890e+03 FES = 135100 μ_F = 0.757 μ_{CR} = 0.082
 Diversity = 5.0469e-02 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.137890e+03 FES = 136100 μ_F = 0.753 μ_{CR} = 0.084
 Diversity = 5.6561e-02 k_t = [4, 4, 2, 2]
 Generation 1370: Best = 1.137889e+03 FES = 137100 μ_F = 0.765 μ_{CR} = 0.083
 Diversity = 5.6481e-02 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.137889e+03 FES = 138100 μ_F = 0.764 μ_{CR} = 0.075
 Diversity = 5.0148e-02 k_t = [2, 3, 2, 4]
 Generation 1390: Best = 1.137888e+03 FES = 139100 μ_F = 0.771 μ_{CR} = 0.084
 Diversity = 4.1305e-02 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.137888e+03 FES = 140100 μ_F = 0.769 μ_{CR} = 0.105
 Diversity = 4.2947e-02 k_t = [3, 4, 3, 2]
 Generation 1410: Best = 1.137887e+03 FES = 141100 μ_F = 0.764 μ_{CR} = 0.104
 Diversity = 4.7002e-02 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.137886e+03 FES = 142100 μ_F = 0.754 μ_{CR} = 0.096
 Diversity = 4.0490e-02 k_t = [4, 2, 3, 4]
 Generation 1430: Best = 1.137886e+03 FES = 143100 μ_F = 0.765 μ_{CR} = 0.095
 Diversity = 4.0234e-02 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.137885e+03 FES = 144100 μ_F = 0.766 μ_{CR} = 0.093
 Diversity = 4.2932e-02 k_t = [2, 4, 2, 3]
 Generation 1450: Best = 1.137885e+03 FES = 145100 μ_F = 0.740 μ_{CR} = 0.097
 Diversity = 4.2903e-02 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.137885e+03 FES = 146100 μ_F = 0.726 μ_{CR} = 0.092
 Diversity = 4.2369e-02 k_t = [2, 2, 4, 3]
 Generation 1470: Best = 1.137884e+03 FES = 147100 μ_F = 0.725 μ_{CR} = 0.085
 Diversity = 3.6712e-02 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.137884e+03 FES = 148100 μ_F = 0.724 μ_{CR} = 0.091
 Diversity = 3.8232e-02 k_t = [3, 3, 3, 2]
 Generation 1490: Best = 1.137884e+03 FES = 149100 μ_F = 0.735 μ_{CR} = 0.088
 Diversity = 3.6148e-02 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.137883e+03 FES = 150100 μ_F = 0.722 μ_{CR} = 0.084
 Diversity = 3.5707e-02 k_t = [3, 4, 2, 2]
 Generation 1510: Best = 1.137883e+03 FES = 151100 μ_F = 0.736 μ_{CR} = 0.080
 Diversity = 3.2765e-02 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.137883e+03 FES = 152100 μ_F = 0.731 μ_{CR} = 0.085
 Diversity = 2.9335e-02 k_t = [4, 4, 3, 2]
 Generation 1530: Best = 1.137883e+03 FES = 153100 μ_F = 0.730 μ_{CR} = 0.078
 Diversity = 2.8704e-02 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.137882e+03 FES = 154100 μ_F = 0.737 μ_{CR} = 0.084
 Diversity = 2.8304e-02 k_t = [4, 4, 3, 2]
 Generation 1550: Best = 1.137882e+03 FES = 155100 μ_F = 0.763 μ_{CR} = 0.079
 Diversity = 2.6496e-02 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.137882e+03 FES = 156100 μ_F = 0.777 μ_{CR} = 0.085
 Diversity = 2.7796e-02 k_t = [3, 2, 3, 3]
 Generation 1570: Best = 1.137881e+03 FES = 157100 μ_F = 0.759 μ_{CR} = 0.088
 Diversity = 2.7798e-02 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.137881e+03 FES = 158100 μ_F = 0.748 μ_{CR} = 0.092
 Diversity = 2.8113e-02 k_t = [4, 2, 2, 2]
 Generation 1590: Best = 1.137881e+03 FES = 159100 μ_F = 0.748 μ_{CR} = 0.086
 Diversity = 2.6550e-02 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.137881e+03 FES = 160100 μ_F = 0.753 μ_{CR} = 0.089
 Diversity = 2.5801e-02 k_t = [2, 4, 2, 3]
 Generation 1610: Best = 1.137881e+03 FES = 161100 μ_F = 0.765 μ_{CR} = 0.096
 Diversity = 2.3752e-02 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.137881e+03 FES = 162100 μ_F = 0.772 μ_{CR} = 0.095
 Diversity = 2.3786e-02 k_t = [3, 4, 2, 4]
 Generation 1630: Best = 1.137880e+03 FES = 163100 μ_F = 0.776 μ_{CR} = 0.098
 Diversity = 2.3839e-02 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.137880e+03 FES = 164100 μ_F = 0.784 μ_{CR} = 0.084
 Diversity = 2.5115e-02 k_t = [2, 3, 3, 2]
 Generation 1650: Best = 1.137880e+03 FES = 165100 μ_F = 0.807 μ_{CR} = 0.087
 Diversity = 3.0223e-02 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.137879e+03 FES = 166100 μ_F = 0.814 μ_{CR} = 0.093
 Diversity = 2.7538e-02 k_t = [4, 3, 3, 4]
 Generation 1670: Best = 1.137879e+03 FES = 167100 μ_F = 0.810 μ_{CR} = 0.086
 Diversity = 2.7387e-02 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.137879e+03 FES = 168100 μ_F = 0.806 μ_{CR} = 0.090
 Diversity = 2.5652e-02 k_t = [4, 2, 3, 4]
 Generation 1690: Best = 1.137879e+03 FES = 169100 μ_F = 0.817 μ_{CR} = 0.083
 Diversity = 2.8500e-02 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.137878e+03 FES = 170100 μ_F = 0.800 μ_{CR} = 0.078
 Diversity = 2.6322e-02 k_t = [2, 4, 4, 4]
 Generation 1710: Best = 1.137878e+03 FES = 171100 μ_F = 0.798 μ_{CR} = 0.082
 Diversity = 3.0369e-02 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.137878e+03 FES = 172100 μ_F = 0.801 μ_{CR} = 0.077
 Diversity = 3.0007e-02 k_t = [4, 3, 4, 2]
 Generation 1730: Best = 1.137878e+03 FES = 173100 μ_F = 0.799 μ_{CR} = 0.082
 Diversity = 3.6779e-02 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.137877e+03 FES = 174100 μ_F = 0.805 μ_{CR} = 0.084
 Diversity = 3.4608e-02 k_t = [3, 3, 3, 3]
 Generation 1750: Best = 1.137877e+03 FES = 175100 μ_F = 0.815 μ_{CR} = 0.092
 Diversity = 3.2306e-02 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.137877e+03 FES = 176100 μ_F = 0.828 μ_{CR} = 0.096
 Diversity = 3.1833e-02 k_t = [3, 4, 3, 2]
 Generation 1770: Best = 1.137877e+03 FES = 177100 μ_F = 0.826 μ_{CR} = 0.087
 Diversity = 2.9437e-02 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.137877e+03 FES = 178100 μ_F = 0.824 μ_{CR} = 0.081
 Diversity = 2.6857e-02 k_t = [3, 4, 3, 2]
 Generation 1790: Best = 1.137876e+03 FES = 179100 μ_F = 0.808 μ_{CR} = 0.084
 Diversity = 2.6703e-02 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 1.137876e+03 FES = 180100 μ_F = 0.797 μ_{CR} = 0.083
 Diversity = 2.5479e-02 k_t = [2, 2, 3, 2]
 Generation 1810: Best = 1.137876e+03 FES = 181100 μ_F = 0.778 μ_{CR} = 0.100
 Diversity = 2.2063e-02 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 1.137876e+03 FES = 182100 μ_F = 0.780 μ_{CR} = 0.096
 Diversity = 2.1502e-02 k_t = [4, 4, 4, 4]
 Generation 1830: Best = 1.137876e+03 FES = 183100 μ_F = 0.777 μ_{CR} = 0.101
 Diversity = 2.1020e-02 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 1.137876e+03 FES = 184100 μ_F = 0.779 μ_{CR} = 0.101
 Diversity = 2.1493e-02 k_t = [3, 2, 2, 2]

Generation 1850: Best = 1.137876e+03 FES = 185100 μ_F = 0.801 μ_{CR} = 0.097
 Diversity = 2.1515e-02 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 1.137876e+03 FES = 186100 μ_F = 0.813 μ_{CR} = 0.094
 Diversity = 2.0235e-02 k_t = [2, 3, 4, 2]
 Generation 1870: Best = 1.137875e+03 FES = 187100 μ_F = 0.800 μ_{CR} = 0.084
 Diversity = 2.0151e-02 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 1.137875e+03 FES = 188100 μ_F = 0.804 μ_{CR} = 0.090
 Diversity = 2.7556e-02 k_t = [4, 2, 3, 2]
 Generation 1890: Best = 1.137874e+03 FES = 189100 μ_F = 0.805 μ_{CR} = 0.097
 Diversity = 3.2614e-02 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 1.137873e+03 FES = 190100 μ_F = 0.789 μ_{CR} = 0.090
 Diversity = 3.6177e-02 k_t = [2, 4, 2, 2]
 Generation 1910: Best = 1.137873e+03 FES = 191100 μ_F = 0.796 μ_{CR} = 0.091
 Diversity = 3.7005e-02 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 1.137873e+03 FES = 192100 μ_F = 0.797 μ_{CR} = 0.091
 Diversity = 3.5124e-02 k_t = [3, 3, 4, 3]
 Generation 1930: Best = 1.137873e+03 FES = 193100 μ_F = 0.800 μ_{CR} = 0.084
 Diversity = 3.6472e-02 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 1.137873e+03 FES = 194100 μ_F = 0.788 μ_{CR} = 0.080
 Diversity = 3.0575e-02 k_t = [2, 2, 3, 3]
 Generation 1950: Best = 1.137873e+03 FES = 195100 μ_F = 0.776 μ_{CR} = 0.087
 Diversity = 2.4147e-02 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 1.137873e+03 FES = 196100 μ_F = 0.772 μ_{CR} = 0.085
 Diversity = 2.1169e-02 k_t = [2, 3, 4, 2]
 Generation 1970: Best = 1.137873e+03 FES = 197100 μ_F = 0.769 μ_{CR} = 0.081
 Diversity = 2.0814e-02 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 1.137872e+03 FES = 198100 μ_F = 0.763 μ_{CR} = 0.088
 Diversity = 1.6835e-02 k_t = [3, 4, 2, 2]
 Generation 1990: Best = 1.137872e+03 FES = 199100 μ_F = 0.744 μ_{CR} = 0.095
 Diversity = 1.5741e-02 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 1.137872e+03 FES = 200100 μ_F = 0.744 μ_{CR} = 0.092
 Diversity = 1.3868e-02 k_t = [2, 3, 4, 2]
 Generation 2010: Best = 1.137872e+03 FES = 201100 μ_F = 0.752 μ_{CR} = 0.088
 Diversity = 1.4247e-02 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 1.137872e+03 FES = 202100 μ_F = 0.780 μ_{CR} = 0.088
 Diversity = 1.4772e-02 k_t = [3, 3, 4, 3]
 Generation 2030: Best = 1.137872e+03 FES = 203100 μ_F = 0.775 μ_{CR} = 0.088
 Diversity = 1.5828e-02 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 1.137872e+03 FES = 204100 μ_F = 0.788 μ_{CR} = 0.087
 Diversity = 1.5273e-02 k_t = [4, 3, 2, 3]
 Generation 2050: Best = 1.137872e+03 FES = 205100 μ_F = 0.796 μ_{CR} = 0.084
 Diversity = 1.7871e-02 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 1.137872e+03 FES = 206100 μ_F = 0.796 μ_{CR} = 0.092
 Diversity = 1.6910e-02 k_t = [4, 3, 2, 2]
 Generation 2070: Best = 1.137872e+03 FES = 207100 μ_F = 0.789 μ_{CR} = 0.085
 Diversity = 1.7971e-02 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 1.137872e+03 FES = 208100 μ_F = 0.784 μ_{CR} = 0.084
 Diversity = 1.4505e-02 k_t = [3, 3, 4, 4]
 Generation 2090: Best = 1.137872e+03 FES = 209100 μ_F = 0.780 μ_{CR} = 0.082
 Diversity = 1.2429e-02 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 1.137871e+03 FES = 210100 μ_F = 0.760 μ_{CR} = 0.086
 Diversity = 1.2100e-02 k_t = [3, 3, 3, 4]
 Generation 2110: Best = 1.137871e+03 FES = 211100 μ_F = 0.765 μ_{CR} = 0.085
 Diversity = 1.1656e-02 k_t = [1, 1, 1, 1]

Generation 2120: Best = 1.137871e+03 FES = 212100 μ_F = 0.764 μ_{CR} = 0.081
 Diversity = 1.1272e-02 k_t = [2, 4, 4, 4]
 Generation 2130: Best = 1.137871e+03 FES = 213100 μ_F = 0.786 μ_{CR} = 0.077
 Diversity = 1.0807e-02 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.137871e+03 FES = 214100 μ_F = 0.784 μ_{CR} = 0.083
 Diversity = 1.1471e-02 k_t = [4, 3, 2, 2]
 Generation 2150: Best = 1.137871e+03 FES = 215100 μ_F = 0.770 μ_{CR} = 0.093
 Diversity = 1.1657e-02 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.137871e+03 FES = 216100 μ_F = 0.777 μ_{CR} = 0.099
 Diversity = 1.1480e-02 k_t = [3, 2, 3, 3]
 Generation 2170: Best = 1.137871e+03 FES = 217100 μ_F = 0.775 μ_{CR} = 0.100
 Diversity = 1.0759e-02 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.137871e+03 FES = 218100 μ_F = 0.768 μ_{CR} = 0.097
 Diversity = 1.1179e-02 k_t = [4, 4, 3, 2]
 Generation 2190: Best = 1.137871e+03 FES = 219100 μ_F = 0.781 μ_{CR} = 0.099
 Diversity = 1.2390e-02 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 1.137871e+03 FES = 220100 μ_F = 0.782 μ_{CR} = 0.098
 Diversity = 1.4233e-02 k_t = [4, 2, 2, 2]
 Generation 2210: Best = 1.137871e+03 FES = 221100 μ_F = 0.761 μ_{CR} = 0.095
 Diversity = 1.7278e-02 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 1.137871e+03 FES = 222100 μ_F = 0.771 μ_{CR} = 0.099
 Diversity = 1.5075e-02 k_t = [4, 2, 4, 2]
 Generation 2230: Best = 1.137871e+03 FES = 223100 μ_F = 0.767 μ_{CR} = 0.110
 Diversity = 1.5853e-02 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.137871e+03 FES = 224100 μ_F = 0.765 μ_{CR} = 0.104
 Diversity = 1.3946e-02 k_t = [4, 4, 4, 2]
 Generation 2250: Best = 1.137871e+03 FES = 225100 μ_F = 0.772 μ_{CR} = 0.085
 Diversity = 1.3252e-02 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.137871e+03 FES = 226100 μ_F = 0.777 μ_{CR} = 0.090
 Diversity = 1.1508e-02 k_t = [4, 4, 4, 3]
 Generation 2270: Best = 1.137871e+03 FES = 227100 μ_F = 0.771 μ_{CR} = 0.095
 Diversity = 1.2317e-02 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.137871e+03 FES = 228100 μ_F = 0.778 μ_{CR} = 0.093
 Diversity = 1.1097e-02 k_t = [4, 2, 3, 3]
 Generation 2290: Best = 1.137870e+03 FES = 229100 μ_F = 0.783 μ_{CR} = 0.089
 Diversity = 1.1328e-02 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.137870e+03 FES = 230100 μ_F = 0.786 μ_{CR} = 0.093
 Diversity = 1.1250e-02 k_t = [2, 2, 4, 3]
 Generation 2310: Best = 1.137870e+03 FES = 231100 μ_F = 0.780 μ_{CR} = 0.089
 Diversity = 1.3749e-02 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.137870e+03 FES = 232100 μ_F = 0.766 μ_{CR} = 0.087
 Diversity = 1.2106e-02 k_t = [3, 2, 3, 3]
 Generation 2330: Best = 1.137870e+03 FES = 233100 μ_F = 0.768 μ_{CR} = 0.086
 Diversity = 1.2053e-02 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.137870e+03 FES = 234100 μ_F = 0.772 μ_{CR} = 0.088
 Diversity = 1.2044e-02 k_t = [4, 3, 3, 3]
 Generation 2350: Best = 1.137870e+03 FES = 235100 μ_F = 0.785 μ_{CR} = 0.080
 Diversity = 1.1791e-02 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.137870e+03 FES = 236100 μ_F = 0.769 μ_{CR} = 0.084
 Diversity = 1.0410e-02 k_t = [3, 3, 2, 4]
 Generation 2370: Best = 1.137870e+03 FES = 237100 μ_F = 0.777 μ_{CR} = 0.081
 Diversity = 1.1284e-02 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.137870e+03 FES = 238100 μ_F = 0.786 μ_{CR} = 0.088
 Diversity = 1.1232e-02 k_t = [2, 4, 4, 2]

Generation 2390: Best = 1.137870e+03 FES = 239100 μ_F = 0.789 μ_{CR} = 0.087
 Diversity = 1.3423e-02 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.137870e+03 FES = 240100 μ_F = 0.799 μ_{CR} = 0.089
 Diversity = 1.3590e-02 k_t = [4, 2, 4, 4]
 Generation 2410: Best = 1.137870e+03 FES = 241100 μ_F = 0.810 μ_{CR} = 0.087
 Diversity = 1.3778e-02 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.137870e+03 FES = 242100 μ_F = 0.786 μ_{CR} = 0.091
 Diversity = 1.1675e-02 k_t = [2, 2, 3, 4]
 Generation 2430: Best = 1.137870e+03 FES = 243100 μ_F = 0.776 μ_{CR} = 0.090
 Diversity = 1.0301e-02 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.137870e+03 FES = 244100 μ_F = 0.780 μ_{CR} = 0.083
 Diversity = 1.0020e-02 k_t = [3, 3, 2, 3]
 Generation 2450: Best = 1.137870e+03 FES = 245100 μ_F = 0.780 μ_{CR} = 0.087
 Diversity = 1.1459e-02 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.137870e+03 FES = 246100 μ_F = 0.788 μ_{CR} = 0.091
 Diversity = 1.1738e-02 k_t = [2, 2, 3, 4]
 Generation 2470: Best = 1.137869e+03 FES = 247100 μ_F = 0.782 μ_{CR} = 0.088
 Diversity = 1.1446e-02 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.137869e+03 FES = 248100 μ_F = 0.757 μ_{CR} = 0.096
 Diversity = 1.0915e-02 k_t = [2, 2, 4, 2]
 Generation 2490: Best = 1.137869e+03 FES = 249100 μ_F = 0.780 μ_{CR} = 0.101
 Diversity = 1.1639e-02 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.137869e+03 FES = 250100 μ_F = 0.779 μ_{CR} = 0.094
 Diversity = 1.1996e-02 k_t = [3, 3, 3, 4]
 Generation 2510: Best = 1.137869e+03 FES = 251100 μ_F = 0.781 μ_{CR} = 0.087
 Diversity = 1.1779e-02 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.137869e+03 FES = 252100 μ_F = 0.785 μ_{CR} = 0.091
 Diversity = 1.2223e-02 k_t = [3, 3, 3, 3]
 Generation 2530: Best = 1.137869e+03 FES = 253100 μ_F = 0.781 μ_{CR} = 0.098
 Diversity = 1.1959e-02 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.137869e+03 FES = 254100 μ_F = 0.779 μ_{CR} = 0.093
 Diversity = 1.0874e-02 k_t = [2, 3, 4, 4]
 Generation 2550: Best = 1.137869e+03 FES = 255100 μ_F = 0.786 μ_{CR} = 0.082
 Diversity = 1.1432e-02 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.137869e+03 FES = 256100 μ_F = 0.787 μ_{CR} = 0.080
 Diversity = 1.2343e-02 k_t = [4, 2, 2, 3]
 Generation 2570: Best = 1.137869e+03 FES = 257100 μ_F = 0.803 μ_{CR} = 0.080
 Diversity = 1.5001e-02 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.137869e+03 FES = 258100 μ_F = 0.797 μ_{CR} = 0.082
 Diversity = 1.5193e-02 k_t = [2, 4, 2, 2]
 Generation 2590: Best = 1.137868e+03 FES = 259100 μ_F = 0.799 μ_{CR} = 0.084
 Diversity = 1.7976e-02 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.137868e+03 FES = 260100 μ_F = 0.807 μ_{CR} = 0.098
 Diversity = 1.8827e-02 k_t = [2, 3, 4, 3]
 Generation 2610: Best = 1.137868e+03 FES = 261100 μ_F = 0.797 μ_{CR} = 0.089
 Diversity = 1.9041e-02 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.137868e+03 FES = 262100 μ_F = 0.801 μ_{CR} = 0.088
 Diversity = 2.0990e-02 k_t = [2, 3, 4, 4]
 Generation 2630: Best = 1.137868e+03 FES = 263100 μ_F = 0.793 μ_{CR} = 0.090
 Diversity = 2.2215e-02 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.137868e+03 FES = 264100 μ_F = 0.782 μ_{CR} = 0.087
 Diversity = 2.0802e-02 k_t = [3, 2, 4, 3]
 Generation 2650: Best = 1.137868e+03 FES = 265100 μ_F = 0.767 μ_{CR} = 0.092
 Diversity = 1.8448e-02 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.137868e+03 FES = 266100 μ_F = 0.757 μ_{CR} = 0.096
 Diversity = 1.7474e-02 k_t = [4, 4, 2, 3]
 Generation 2670: Best = 1.137867e+03 FES = 267100 μ_F = 0.762 μ_{CR} = 0.092
 Diversity = 1.7950e-02 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.137867e+03 FES = 268100 μ_F = 0.782 μ_{CR} = 0.088
 Diversity = 1.6761e-02 k_t = [2, 2, 2, 3]
 Generation 2690: Best = 1.137867e+03 FES = 269100 μ_F = 0.791 μ_{CR} = 0.085
 Diversity = 1.7672e-02 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.137867e+03 FES = 270100 μ_F = 0.781 μ_{CR} = 0.086
 Diversity = 1.4057e-02 k_t = [2, 2, 3, 3]
 Generation 2710: Best = 1.137867e+03 FES = 271100 μ_F = 0.770 μ_{CR} = 0.085
 Diversity = 1.2789e-02 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.137867e+03 FES = 272100 μ_F = 0.760 μ_{CR} = 0.091
 Diversity = 1.0822e-02 k_t = [4, 2, 2, 4]
 Generation 2730: Best = 1.137867e+03 FES = 273100 μ_F = 0.771 μ_{CR} = 0.084
 Diversity = 9.5660e-03 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.137867e+03 FES = 274100 μ_F = 0.768 μ_{CR} = 0.098
 Diversity = 9.6812e-03 k_t = [4, 4, 3, 2]
 Generation 2750: Best = 1.137867e+03 FES = 275100 μ_F = 0.783 μ_{CR} = 0.087
 Diversity = 9.1104e-03 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.137867e+03 FES = 276100 μ_F = 0.780 μ_{CR} = 0.084
 Diversity = 8.1486e-03 k_t = [3, 3, 4, 2]
 Generation 2770: Best = 1.137867e+03 FES = 277100 μ_F = 0.784 μ_{CR} = 0.089
 Diversity = 8.3424e-03 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.137867e+03 FES = 278100 μ_F = 0.784 μ_{CR} = 0.080
 Diversity = 8.8717e-03 k_t = [3, 4, 3, 2]
 Generation 2790: Best = 1.137867e+03 FES = 279100 μ_F = 0.784 μ_{CR} = 0.073
 Diversity = 9.2002e-03 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.137867e+03 FES = 280100 μ_F = 0.796 μ_{CR} = 0.077
 Diversity = 8.6491e-03 k_t = [3, 4, 4, 3]
 Generation 2810: Best = 1.137867e+03 FES = 281100 μ_F = 0.773 μ_{CR} = 0.075
 Diversity = 8.5767e-03 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.137867e+03 FES = 282100 μ_F = 0.768 μ_{CR} = 0.089
 Diversity = 9.4262e-03 k_t = [2, 3, 4, 3]
 Generation 2830: Best = 1.137867e+03 FES = 283100 μ_F = 0.768 μ_{CR} = 0.086
 Diversity = 8.9335e-03 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.137867e+03 FES = 284100 μ_F = 0.798 μ_{CR} = 0.089
 Diversity = 8.8526e-03 k_t = [2, 4, 3, 2]
 Generation 2850: Best = 1.137867e+03 FES = 285100 μ_F = 0.797 μ_{CR} = 0.083
 Diversity = 8.7911e-03 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.137867e+03 FES = 286100 μ_F = 0.788 μ_{CR} = 0.082
 Diversity = 9.5170e-03 k_t = [4, 4, 2, 2]
 Generation 2870: Best = 1.137867e+03 FES = 287100 μ_F = 0.775 μ_{CR} = 0.078
 Diversity = 8.9626e-03 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.137867e+03 FES = 288100 μ_F = 0.779 μ_{CR} = 0.080
 Diversity = 7.5404e-03 k_t = [3, 2, 3, 4]
 Generation 2890: Best = 1.137867e+03 FES = 289100 μ_F = 0.769 μ_{CR} = 0.083
 Diversity = 7.7234e-03 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.137867e+03 FES = 290100 μ_F = 0.779 μ_{CR} = 0.075
 Diversity = 6.6536e-03 k_t = [3, 4, 4, 2]
 Generation 2910: Best = 1.137867e+03 FES = 291100 μ_F = 0.768 μ_{CR} = 0.076
 Diversity = 6.6006e-03 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.137867e+03 FES = 292100 μ_F = 0.764 μ_{CR} = 0.078
 Diversity = 6.4361e-03 k_t = [4, 2, 3, 3]

```

Generation 2930: Best = 1.137867e+03  FES = 293100   $\mu_F$  = 0.788   $\mu_{CR}$  = 0.073
Diversity = 6.5558e-03  k_t = [1, 1, 1, 1]
Generation 2940: Best = 1.137867e+03  FES = 294100   $\mu_F$  = 0.797   $\mu_{CR}$  = 0.089
Diversity = 7.4959e-03  k_t = [3, 4, 3, 2]
Generation 2950: Best = 1.137867e+03  FES = 295100   $\mu_F$  = 0.786   $\mu_{CR}$  = 0.094
Diversity = 7.7179e-03  k_t = [1, 1, 1, 1]
Generation 2960: Best = 1.137867e+03  FES = 296100   $\mu_F$  = 0.776   $\mu_{CR}$  = 0.081
Diversity = 6.8666e-03  k_t = [4, 4, 4, 4]
Generation 2970: Best = 1.137867e+03  FES = 297100   $\mu_F$  = 0.785   $\mu_{CR}$  = 0.082
Diversity = 6.5785e-03  k_t = [1, 1, 1, 1]
Generation 2980: Best = 1.137867e+03  FES = 298100   $\mu_F$  = 0.797   $\mu_{CR}$  = 0.073
Diversity = 6.0322e-03  k_t = [4, 4, 3, 2]
Generation 2990: Best = 1.137867e+03  FES = 299100   $\mu_F$  = 0.802   $\mu_{CR}$  = 0.079
Diversity = 5.5686e-03  k_t = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 1.137867e+03

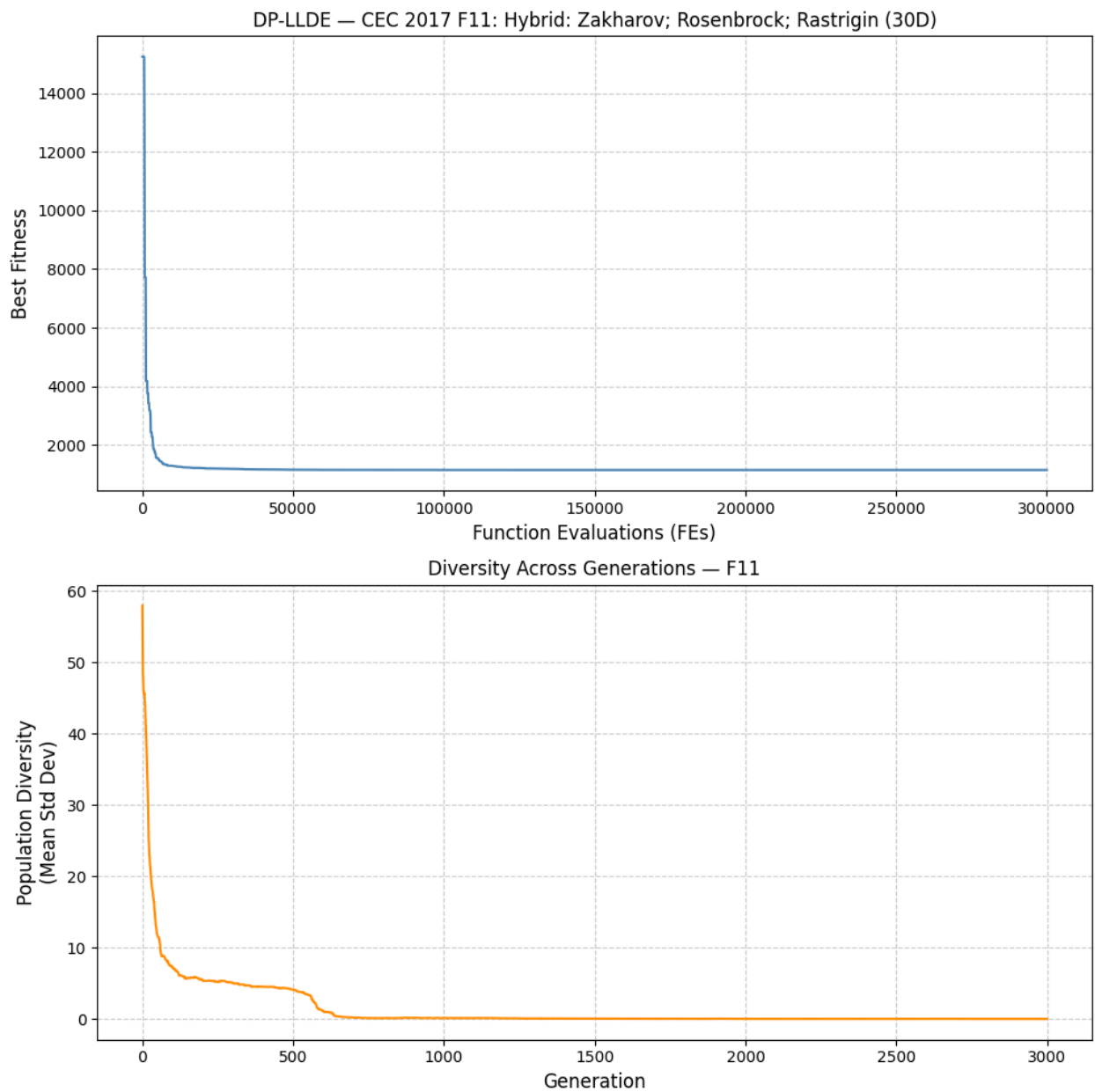
Total Function Evals = 300000

Total Successful Updates = 59917

```

=====
=
Results:
  Best Fitness      : 1.137867e+03
  Global Optimum    : 1.100000e+03
  Error (|best-optimum|) : 3.786682e+01
  Best Solution (5 dim) : [-21.38276561 -15.17113539 -24.1380122  1.65228685
17.92636505]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F11.png

```
In [ ]: # =====
# F12 - Hybrid Function 2 (Elliptic; Schwefel; Bent Cigar)
# =====
run_and_plot('F12', make_wrapper(hybrid.f12), 1200.0, 'Hybrid: Elliptic; Schwe
```


=====

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DP-LLDE on CEC 2017 F12 – Single Run

=====

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Function : Hybrid: Elliptic; Schwefel; Bent Cigar (30D)
Bounds : [-100, 100]
Global optimum : 1200.0
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 1.806776e+11
Generation 10: Best = 2.604417e+10 FES = 1100 $\mu_F = 0.574$ $\mu_{CR} = 0.360$
Diversity = 2.6848e+01 $k_t = [1, 1, 1, 1]$
Generation 20: Best = 4.215199e+09 FES = 2100 $\mu_F = 0.617$ $\mu_{CR} = 0.360$
Diversity = 1.6658e+01 $k_t = [4, 2, 4, 4]$
Generation 30: Best = 9.918870e+08 FES = 3100 $\mu_F = 0.635$ $\mu_{CR} = 0.351$
Diversity = 1.1610e+01 $k_t = [1, 1, 1, 1]$
Generation 40: Best = 2.259906e+08 FES = 4100 $\mu_F = 0.667$ $\mu_{CR} = 0.349$
Diversity = 9.9725e+00 $k_t = [4, 3, 4, 2]$
Generation 50: Best = 5.823218e+07 FES = 5100 $\mu_F = 0.691$ $\mu_{CR} = 0.360$
Diversity = 9.4236e+00 $k_t = [1, 1, 1, 1]$
Generation 60: Best = 1.958944e+07 FES = 6100 $\mu_F = 0.687$ $\mu_{CR} = 0.357$
Diversity = 7.5276e+00 $k_t = [3, 3, 4, 2]$
Generation 70: Best = 1.024216e+07 FES = 7100 $\mu_F = 0.678$ $\mu_{CR} = 0.353$
Diversity = 6.1452e+00 $k_t = [1, 1, 1, 1]$
Generation 80: Best = 5.001807e+06 FES = 8100 $\mu_F = 0.689$ $\mu_{CR} = 0.340$
Diversity = 5.2721e+00 $k_t = [2, 4, 4, 3]$
Generation 90: Best = 3.013104e+06 FES = 9100 $\mu_F = 0.686$ $\mu_{CR} = 0.347$
Diversity = 4.0428e+00 $k_t = [1, 1, 1, 1]$
Generation 100: Best = 1.762083e+06 FES = 10100 $\mu_F = 0.680$ $\mu_{CR} = 0.342$
Diversity = 3.2583e+00 $k_t = [2, 2, 4, 4]$
Generation 110: Best = 1.278368e+06 FES = 11100 $\mu_F = 0.703$ $\mu_{CR} = 0.322$
Diversity = 2.9333e+00 $k_t = [1, 1, 1, 1]$
Generation 120: Best = 1.130402e+06 FES = 12100 $\mu_F = 0.726$ $\mu_{CR} = 0.316$
Diversity = 2.9072e+00 $k_t = [3, 2, 3, 4]$
Generation 130: Best = 9.769110e+05 FES = 13100 $\mu_F = 0.725$ $\mu_{CR} = 0.292$
Diversity = 2.4920e+00 $k_t = [1, 1, 1, 1]$
Generation 140: Best = 8.829869e+05 FES = 14100 $\mu_F = 0.711$ $\mu_{CR} = 0.247$
Diversity = 2.4419e+00 $k_t = [2, 3, 2, 4]$
Generation 150: Best = 8.500888e+05 FES = 15100 $\mu_F = 0.727$ $\mu_{CR} = 0.209$
Diversity = 2.3269e+00 $k_t = [1, 1, 1, 1]$
Generation 160: Best = 7.850598e+05 FES = 16100 $\mu_F = 0.744$ $\mu_{CR} = 0.168$
Diversity = 2.5702e+00 $k_t = [3, 4, 2, 3]$
Generation 170: Best = 7.502501e+05 FES = 17100 $\mu_F = 0.752$ $\mu_{CR} = 0.158$
Diversity = 2.4833e+00 $k_t = [1, 1, 1, 1]$
Generation 180: Best = 7.251061e+05 FES = 18100 $\mu_F = 0.756$ $\mu_{CR} = 0.139$
Diversity = 2.3293e+00 $k_t = [3, 3, 4, 2]$
Generation 190: Best = 7.020135e+05 FES = 19100 $\mu_F = 0.777$ $\mu_{CR} = 0.111$
Diversity = 2.6305e+00 $k_t = [1, 1, 1, 1]$
Generation 200: Best = 6.852334e+05 FES = 20100 $\mu_F = 0.784$ $\mu_{CR} = 0.094$
Diversity = 2.5896e+00 $k_t = [3, 4, 3, 3]$
Generation 210: Best = 6.749216e+05 FES = 21100 $\mu_F = 0.795$ $\mu_{CR} = 0.082$
Diversity = 2.6839e+00 $k_t = [1, 1, 1, 1]$
Generation 220: Best = 6.613683e+05 FES = 22100 $\mu_F = 0.780$ $\mu_{CR} = 0.084$
Diversity = 2.5829e+00 $k_t = [4, 2, 4, 4]$

Generation 230: Best = 6.374415e+05	FES = 23100	$\mu_F = 0.779$	$\mu_{CR} = 0.087$
Diversity = 2.3324e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 6.346136e+05	FES = 24100	$\mu_F = 0.782$	$\mu_{CR} = 0.085$
Diversity = 2.7492e+00	$k_t = [2, 4, 3, 3]$		
Generation 250: Best = 6.246214e+05	FES = 25100	$\mu_F = 0.774$	$\mu_{CR} = 0.085$
Diversity = 2.4892e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 6.133708e+05	FES = 26100	$\mu_F = 0.774$	$\mu_{CR} = 0.090$
Diversity = 2.5859e+00	$k_t = [3, 4, 3, 4]$		
Generation 270: Best = 6.057402e+05	FES = 27100	$\mu_F = 0.781$	$\mu_{CR} = 0.077$
Diversity = 2.5483e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 5.947876e+05	FES = 28100	$\mu_F = 0.782$	$\mu_{CR} = 0.091$
Diversity = 2.8395e+00	$k_t = [2, 2, 3, 3]$		
Generation 290: Best = 5.882816e+05	FES = 29100	$\mu_F = 0.789$	$\mu_{CR} = 0.083$
Diversity = 2.9086e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 5.846759e+05	FES = 30100	$\mu_F = 0.791$	$\mu_{CR} = 0.081$
Diversity = 2.9692e+00	$k_t = [3, 3, 4, 2]$		
Generation 310: Best = 5.774104e+05	FES = 31100	$\mu_F = 0.792$	$\mu_{CR} = 0.072$
Diversity = 3.0645e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 5.697137e+05	FES = 32100	$\mu_F = 0.805$	$\mu_{CR} = 0.088$
Diversity = 3.4360e+00	$k_t = [3, 3, 2, 4]$		
Generation 330: Best = 5.619573e+05	FES = 33100	$\mu_F = 0.805$	$\mu_{CR} = 0.093$
Diversity = 3.5273e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 5.570602e+05	FES = 34100	$\mu_F = 0.789$	$\mu_{CR} = 0.098$
Diversity = 3.4827e+00	$k_t = [2, 3, 4, 3]$		
Generation 350: Best = 5.521114e+05	FES = 35100	$\mu_F = 0.790$	$\mu_{CR} = 0.088$
Diversity = 3.6475e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 5.492615e+05	FES = 36100	$\mu_F = 0.809$	$\mu_{CR} = 0.080$
Diversity = 3.3841e+00	$k_t = [2, 3, 3, 4]$		
Generation 370: Best = 5.437666e+05	FES = 37100	$\mu_F = 0.799$	$\mu_{CR} = 0.078$
Diversity = 3.7160e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 5.395744e+05	FES = 38100	$\mu_F = 0.790$	$\mu_{CR} = 0.089$
Diversity = 4.1633e+00	$k_t = [3, 4, 2, 4]$		
Generation 390: Best = 5.326015e+05	FES = 39100	$\mu_F = 0.789$	$\mu_{CR} = 0.091$
Diversity = 3.5645e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 5.274191e+05	FES = 40100	$\mu_F = 0.793$	$\mu_{CR} = 0.103$
Diversity = 3.8053e+00	$k_t = [3, 3, 3, 3]$		
Generation 410: Best = 5.134275e+05	FES = 41100	$\mu_F = 0.793$	$\mu_{CR} = 0.093$
Diversity = 4.3118e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 5.039875e+05	FES = 42100	$\mu_F = 0.803$	$\mu_{CR} = 0.095$
Diversity = 4.2988e+00	$k_t = [3, 3, 4, 2]$		
Generation 430: Best = 4.895208e+05	FES = 43100	$\mu_F = 0.806$	$\mu_{CR} = 0.106$
Diversity = 4.7298e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 4.818277e+05	FES = 44100	$\mu_F = 0.817$	$\mu_{CR} = 0.091$
Diversity = 5.0011e+00	$k_t = [3, 3, 2, 3]$		
Generation 450: Best = 4.751440e+05	FES = 45100	$\mu_F = 0.804$	$\mu_{CR} = 0.084$
Diversity = 5.1031e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 4.737029e+05	FES = 46100	$\mu_F = 0.794$	$\mu_{CR} = 0.079$
Diversity = 4.9892e+00	$k_t = [4, 2, 3, 2]$		
Generation 470: Best = 4.701994e+05	FES = 47100	$\mu_F = 0.784$	$\mu_{CR} = 0.084$
Diversity = 4.7976e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 4.639175e+05	FES = 48100	$\mu_F = 0.771$	$\mu_{CR} = 0.081$
Diversity = 4.5998e+00	$k_t = [3, 2, 4, 2]$		
Generation 490: Best = 4.598144e+05	FES = 49100	$\mu_F = 0.776$	$\mu_{CR} = 0.078$
Diversity = 4.6903e+00	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 4.552353e+05	FES = 50100	$\mu_F = 0.782$	$\mu_{CR} = 0.073$
Diversity = 4.1842e+00	k_t = [3, 3, 4, 3]		
Generation 510: Best = 4.522506e+05	FES = 51100	$\mu_F = 0.784$	$\mu_{CR} = 0.081$
Diversity = 3.8163e+00	k_t = [1, 1, 1, 1]		
Generation 520: Best = 4.477298e+05	FES = 52100	$\mu_F = 0.801$	$\mu_{CR} = 0.077$
Diversity = 3.9193e+00	k_t = [2, 3, 4, 3]		
Generation 530: Best = 4.443718e+05	FES = 53100	$\mu_F = 0.786$	$\mu_{CR} = 0.077$
Diversity = 3.6010e+00	k_t = [1, 1, 1, 1]		
Generation 540: Best = 4.390971e+05	FES = 54100	$\mu_F = 0.791$	$\mu_{CR} = 0.076$
Diversity = 3.4786e+00	k_t = [3, 4, 4, 4]		
Generation 550: Best = 4.361084e+05	FES = 55100	$\mu_F = 0.790$	$\mu_{CR} = 0.082$
Diversity = 3.2511e+00	k_t = [1, 1, 1, 1]		
Generation 560: Best = 4.321489e+05	FES = 56100	$\mu_F = 0.800$	$\mu_{CR} = 0.078$
Diversity = 3.7126e+00	k_t = [2, 4, 2, 4]		
Generation 570: Best = 4.281018e+05	FES = 57100	$\mu_F = 0.801$	$\mu_{CR} = 0.091$
Diversity = 4.3791e+00	k_t = [1, 1, 1, 1]		
Generation 580: Best = 4.175051e+05	FES = 58100	$\mu_F = 0.805$	$\mu_{CR} = 0.094$
Diversity = 5.0626e+00	k_t = [3, 4, 4, 2]		
Generation 590: Best = 4.147474e+05	FES = 59100	$\mu_F = 0.809$	$\mu_{CR} = 0.100$
Diversity = 5.8337e+00	k_t = [1, 1, 1, 1]		
Generation 600: Best = 4.097018e+05	FES = 60100	$\mu_F = 0.805$	$\mu_{CR} = 0.090$
Diversity = 5.8838e+00	k_t = [4, 2, 2, 3]		
Generation 610: Best = 4.037316e+05	FES = 61100	$\mu_F = 0.799$	$\mu_{CR} = 0.083$
Diversity = 6.5695e+00	k_t = [1, 1, 1, 1]		
Generation 620: Best = 3.949106e+05	FES = 62100	$\mu_F = 0.783$	$\mu_{CR} = 0.079$
Diversity = 6.7721e+00	k_t = [2, 3, 2, 4]		
Generation 630: Best = 3.896398e+05	FES = 63100	$\mu_F = 0.802$	$\mu_{CR} = 0.088$
Diversity = 6.7538e+00	k_t = [1, 1, 1, 1]		
Generation 640: Best = 3.860403e+05	FES = 64100	$\mu_F = 0.812$	$\mu_{CR} = 0.097$
Diversity = 7.3279e+00	k_t = [3, 3, 4, 2]		
Generation 650: Best = 3.822557e+05	FES = 65100	$\mu_F = 0.797$	$\mu_{CR} = 0.088$
Diversity = 8.6041e+00	k_t = [1, 1, 1, 1]		
Generation 660: Best = 3.791522e+05	FES = 66100	$\mu_F = 0.797$	$\mu_{CR} = 0.091$
Diversity = 8.9836e+00	k_t = [2, 2, 4, 2]		
Generation 670: Best = 3.745216e+05	FES = 67100	$\mu_F = 0.790$	$\mu_{CR} = 0.089$
Diversity = 9.4415e+00	k_t = [1, 1, 1, 1]		
Generation 680: Best = 3.711120e+05	FES = 68100	$\mu_F = 0.794$	$\mu_{CR} = 0.086$
Diversity = 8.3900e+00	k_t = [4, 3, 3, 4]		
Generation 690: Best = 3.672434e+05	FES = 69100	$\mu_F = 0.803$	$\mu_{CR} = 0.083$
Diversity = 7.6152e+00	k_t = [1, 1, 1, 1]		
Generation 700: Best = 3.654748e+05	FES = 70100	$\mu_F = 0.799$	$\mu_{CR} = 0.087$
Diversity = 6.9791e+00	k_t = [4, 3, 4, 3]		
Generation 710: Best = 3.608332e+05	FES = 71100	$\mu_F = 0.803$	$\mu_{CR} = 0.081$
Diversity = 6.6231e+00	k_t = [1, 1, 1, 1]		
Generation 720: Best = 3.594705e+05	FES = 72100	$\mu_F = 0.814$	$\mu_{CR} = 0.078$
Diversity = 6.9893e+00	k_t = [3, 2, 4, 2]		
Generation 730: Best = 3.564111e+05	FES = 73100	$\mu_F = 0.798$	$\mu_{CR} = 0.077$
Diversity = 8.2281e+00	k_t = [1, 1, 1, 1]		
Generation 740: Best = 3.520174e+05	FES = 74100	$\mu_F = 0.793$	$\mu_{CR} = 0.085$
Diversity = 8.4989e+00	k_t = [2, 3, 2, 3]		
Generation 750: Best = 3.496520e+05	FES = 75100	$\mu_F = 0.804$	$\mu_{CR} = 0.084$
Diversity = 8.9278e+00	k_t = [1, 1, 1, 1]		
Generation 760: Best = 3.447015e+05	FES = 76100	$\mu_F = 0.807$	$\mu_{CR} = 0.084$
Diversity = 8.6329e+00	k_t = [2, 4, 3, 4]		

Generation 770: Best = 3.408461e+05 FES = 77100 μ_F = 0.807 μ_{CR} = 0.079
 Diversity = 9.0478e+00 k_t = [1, 1, 1, 1]
 Generation 780: Best = 3.394373e+05 FES = 78100 μ_F = 0.797 μ_{CR} = 0.079
 Diversity = 8.8661e+00 k_t = [3, 2, 3, 4]
 Generation 790: Best = 3.361967e+05 FES = 79100 μ_F = 0.794 μ_{CR} = 0.080
 Diversity = 9.5227e+00 k_t = [1, 1, 1, 1]
 Generation 800: Best = 3.328934e+05 FES = 80100 μ_F = 0.798 μ_{CR} = 0.075
 Diversity = 9.4928e+00 k_t = [3, 2, 3, 3]
 Generation 810: Best = 3.310919e+05 FES = 81100 μ_F = 0.807 μ_{CR} = 0.073
 Diversity = 9.8945e+00 k_t = [1, 1, 1, 1]
 Generation 820: Best = 3.260936e+05 FES = 82100 μ_F = 0.795 μ_{CR} = 0.082
 Diversity = 8.1992e+00 k_t = [4, 2, 2, 3]
 Generation 830: Best = 3.243792e+05 FES = 83100 μ_F = 0.800 μ_{CR} = 0.080
 Diversity = 7.4187e+00 k_t = [1, 1, 1, 1]
 Generation 840: Best = 3.202434e+05 FES = 84100 μ_F = 0.800 μ_{CR} = 0.088
 Diversity = 7.4268e+00 k_t = [2, 3, 4, 3]
 Generation 850: Best = 3.167857e+05 FES = 85100 μ_F = 0.803 μ_{CR} = 0.080
 Diversity = 6.8473e+00 k_t = [1, 1, 1, 1]
 Generation 860: Best = 3.146407e+05 FES = 86100 μ_F = 0.798 μ_{CR} = 0.076
 Diversity = 6.1276e+00 k_t = [4, 4, 4, 2]
 Generation 870: Best = 3.130118e+05 FES = 87100 μ_F = 0.791 μ_{CR} = 0.066
 Diversity = 5.9350e+00 k_t = [1, 1, 1, 1]
 Generation 880: Best = 3.106770e+05 FES = 88100 μ_F = 0.785 μ_{CR} = 0.071
 Diversity = 5.0042e+00 k_t = [4, 4, 3, 3]
 Generation 890: Best = 3.086469e+05 FES = 89100 μ_F = 0.786 μ_{CR} = 0.072
 Diversity = 4.8127e+00 k_t = [1, 1, 1, 1]
 Generation 900: Best = 3.048075e+05 FES = 90100 μ_F = 0.797 μ_{CR} = 0.070
 Diversity = 4.9216e+00 k_t = [2, 4, 4, 2]
 Generation 910: Best = 3.034882e+05 FES = 91100 μ_F = 0.804 μ_{CR} = 0.071
 Diversity = 5.0130e+00 k_t = [1, 1, 1, 1]
 Generation 920: Best = 2.950663e+05 FES = 92100 μ_F = 0.813 μ_{CR} = 0.077
 Diversity = 4.6616e+00 k_t = [2, 4, 3, 2]
 Generation 930: Best = 2.920727e+05 FES = 93100 μ_F = 0.819 μ_{CR} = 0.088
 Diversity = 5.3777e+00 k_t = [1, 1, 1, 1]
 Generation 940: Best = 2.889502e+05 FES = 94100 μ_F = 0.825 μ_{CR} = 0.086
 Diversity = 5.3278e+00 k_t = [4, 2, 2, 4]
 Generation 950: Best = 2.858452e+05 FES = 95100 μ_F = 0.812 μ_{CR} = 0.091
 Diversity = 5.1884e+00 k_t = [1, 1, 1, 1]
 Generation 960: Best = 2.818572e+05 FES = 96100 μ_F = 0.808 μ_{CR} = 0.088
 Diversity = 5.0378e+00 k_t = [3, 3, 2, 4]
 Generation 970: Best = 2.776410e+05 FES = 97100 μ_F = 0.798 μ_{CR} = 0.086
 Diversity = 4.7966e+00 k_t = [1, 1, 1, 1]
 Generation 980: Best = 2.754721e+05 FES = 98100 μ_F = 0.813 μ_{CR} = 0.085
 Diversity = 5.1192e+00 k_t = [2, 4, 2, 2]
 Generation 990: Best = 2.730730e+05 FES = 99100 μ_F = 0.816 μ_{CR} = 0.082
 Diversity = 5.3905e+00 k_t = [1, 1, 1, 1]
 Generation 1000: Best = 2.715824e+05 FES = 100100 μ_F = 0.825 μ_{CR} = 0.069
 Diversity = 6.0898e+00 k_t = [3, 4, 3, 4]
 Generation 1010: Best = 2.686477e+05 FES = 101100 μ_F = 0.824 μ_{CR} = 0.068
 Diversity = 6.4437e+00 k_t = [1, 1, 1, 1]
 Generation 1020: Best = 2.662596e+05 FES = 102100 μ_F = 0.824 μ_{CR} = 0.072
 Diversity = 7.3563e+00 k_t = [4, 3, 3, 2]
 Generation 1030: Best = 2.628793e+05 FES = 103100 μ_F = 0.819 μ_{CR} = 0.065
 Diversity = 7.0946e+00 k_t = [1, 1, 1, 1]

Generation 1040: Best = 2.616962e+05 FES = 104100 μ_F = 0.816 μ_{CR} = 0.072
 Diversity = 6.3556e+00 k_t = [3, 4, 3, 2]
 Generation 1050: Best = 2.586359e+05 FES = 105100 μ_F = 0.812 μ_{CR} = 0.075
 Diversity = 6.4106e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 2.525928e+05 FES = 106100 μ_F = 0.812 μ_{CR} = 0.074
 Diversity = 6.7974e+00 k_t = [2, 4, 2, 2]
 Generation 1070: Best = 2.513068e+05 FES = 107100 μ_F = 0.813 μ_{CR} = 0.075
 Diversity = 5.5779e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 2.461677e+05 FES = 108100 μ_F = 0.813 μ_{CR} = 0.072
 Diversity = 5.9174e+00 k_t = [4, 4, 2, 2]
 Generation 1090: Best = 2.427334e+05 FES = 109100 μ_F = 0.807 μ_{CR} = 0.077
 Diversity = 5.8995e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 2.378534e+05 FES = 110100 μ_F = 0.799 μ_{CR} = 0.084
 Diversity = 6.0280e+00 k_t = [3, 4, 3, 4]
 Generation 1110: Best = 2.351535e+05 FES = 111100 μ_F = 0.796 μ_{CR} = 0.077
 Diversity = 5.0411e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 2.322462e+05 FES = 112100 μ_F = 0.814 μ_{CR} = 0.082
 Diversity = 5.0073e+00 k_t = [2, 3, 4, 3]
 Generation 1130: Best = 2.300927e+05 FES = 113100 μ_F = 0.821 μ_{CR} = 0.077
 Diversity = 4.4140e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 2.256545e+05 FES = 114100 μ_F = 0.827 μ_{CR} = 0.079
 Diversity = 4.5238e+00 k_t = [3, 4, 3, 2]
 Generation 1150: Best = 2.219785e+05 FES = 115100 μ_F = 0.826 μ_{CR} = 0.079
 Diversity = 4.2206e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 2.200760e+05 FES = 116100 μ_F = 0.845 μ_{CR} = 0.079
 Diversity = 5.4556e+00 k_t = [2, 2, 2, 4]
 Generation 1170: Best = 2.157066e+05 FES = 117100 μ_F = 0.834 μ_{CR} = 0.081
 Diversity = 4.8103e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 2.127490e+05 FES = 118100 μ_F = 0.814 μ_{CR} = 0.081
 Diversity = 4.6526e+00 k_t = [2, 2, 4, 4]
 Generation 1190: Best = 2.100011e+05 FES = 119100 μ_F = 0.819 μ_{CR} = 0.075
 Diversity = 4.7828e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 2.078021e+05 FES = 120100 μ_F = 0.817 μ_{CR} = 0.080
 Diversity = 4.5197e+00 k_t = [4, 4, 2, 2]
 Generation 1210: Best = 2.054643e+05 FES = 121100 μ_F = 0.817 μ_{CR} = 0.081
 Diversity = 4.9714e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 2.033114e+05 FES = 122100 μ_F = 0.817 μ_{CR} = 0.078
 Diversity = 5.1164e+00 k_t = [4, 2, 4, 2]
 Generation 1230: Best = 1.991304e+05 FES = 123100 μ_F = 0.815 μ_{CR} = 0.081
 Diversity = 5.5423e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.971325e+05 FES = 124100 μ_F = 0.820 μ_{CR} = 0.084
 Diversity = 5.0749e+00 k_t = [4, 2, 2, 3]
 Generation 1250: Best = 1.914610e+05 FES = 125100 μ_F = 0.829 μ_{CR} = 0.090
 Diversity = 5.3592e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.881072e+05 FES = 126100 μ_F = 0.811 μ_{CR} = 0.087
 Diversity = 5.9402e+00 k_t = [2, 2, 2, 3]
 Generation 1270: Best = 1.861648e+05 FES = 127100 μ_F = 0.805 μ_{CR} = 0.079
 Diversity = 5.4816e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.825574e+05 FES = 128100 μ_F = 0.805 μ_{CR} = 0.077
 Diversity = 5.2658e+00 k_t = [2, 2, 3, 3]
 Generation 1290: Best = 1.814476e+05 FES = 129100 μ_F = 0.803 μ_{CR} = 0.084
 Diversity = 5.7653e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.774080e+05 FES = 130100 μ_F = 0.796 μ_{CR} = 0.083
 Diversity = 5.8502e+00 k_t = [3, 3, 2, 3]

Generation 1310: Best = 1.771067e+05 FES = 131100 μ_F = 0.802 μ_{CR} = 0.067
 Diversity = 5.2376e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.758438e+05 FES = 132100 μ_F = 0.784 μ_{CR} = 0.071
 Diversity = 5.0321e+00 k_t = [3, 3, 2, 4]
 Generation 1330: Best = 1.748251e+05 FES = 133100 μ_F = 0.807 μ_{CR} = 0.076
 Diversity = 4.8427e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.716516e+05 FES = 134100 μ_F = 0.797 μ_{CR} = 0.075
 Diversity = 4.6038e+00 k_t = [2, 3, 2, 4]
 Generation 1350: Best = 1.709970e+05 FES = 135100 μ_F = 0.796 μ_{CR} = 0.078
 Diversity = 4.3299e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.689278e+05 FES = 136100 μ_F = 0.793 μ_{CR} = 0.080
 Diversity = 3.8378e+00 k_t = [2, 4, 2, 4]
 Generation 1370: Best = 1.675885e+05 FES = 137100 μ_F = 0.796 μ_{CR} = 0.077
 Diversity = 3.4042e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.662892e+05 FES = 138100 μ_F = 0.801 μ_{CR} = 0.075
 Diversity = 3.7014e+00 k_t = [3, 3, 3, 4]
 Generation 1390: Best = 1.653386e+05 FES = 139100 μ_F = 0.801 μ_{CR} = 0.083
 Diversity = 3.4186e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.638330e+05 FES = 140100 μ_F = 0.809 μ_{CR} = 0.087
 Diversity = 4.1636e+00 k_t = [4, 3, 3, 2]
 Generation 1410: Best = 1.629257e+05 FES = 141100 μ_F = 0.806 μ_{CR} = 0.096
 Diversity = 4.3819e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.615092e+05 FES = 142100 μ_F = 0.800 μ_{CR} = 0.089
 Diversity = 4.6436e+00 k_t = [3, 2, 4, 3]
 Generation 1430: Best = 1.605885e+05 FES = 143100 μ_F = 0.789 μ_{CR} = 0.084
 Diversity = 4.4338e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.584602e+05 FES = 144100 μ_F = 0.788 μ_{CR} = 0.080
 Diversity = 4.4573e+00 k_t = [2, 2, 4, 4]
 Generation 1450: Best = 1.575869e+05 FES = 145100 μ_F = 0.798 μ_{CR} = 0.083
 Diversity = 4.6047e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.547203e+05 FES = 146100 μ_F = 0.792 μ_{CR} = 0.076
 Diversity = 4.5282e+00 k_t = [4, 2, 2, 3]
 Generation 1470: Best = 1.529216e+05 FES = 147100 μ_F = 0.802 μ_{CR} = 0.080
 Diversity = 4.5722e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.502428e+05 FES = 148100 μ_F = 0.804 μ_{CR} = 0.078
 Diversity = 4.4025e+00 k_t = [4, 3, 2, 4]
 Generation 1490: Best = 1.479564e+05 FES = 149100 μ_F = 0.801 μ_{CR} = 0.083
 Diversity = 4.0095e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.432497e+05 FES = 150100 μ_F = 0.811 μ_{CR} = 0.077
 Diversity = 4.5511e+00 k_t = [2, 4, 4, 2]
 Generation 1510: Best = 1.416306e+05 FES = 151100 μ_F = 0.818 μ_{CR} = 0.078
 Diversity = 4.8679e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.383244e+05 FES = 152100 μ_F = 0.822 μ_{CR} = 0.083
 Diversity = 5.2700e+00 k_t = [4, 3, 2, 4]
 Generation 1530: Best = 1.340740e+05 FES = 153100 μ_F = 0.816 μ_{CR} = 0.090
 Diversity = 4.8865e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.325981e+05 FES = 154100 μ_F = 0.823 μ_{CR} = 0.088
 Diversity = 5.4935e+00 k_t = [3, 4, 3, 2]
 Generation 1550: Best = 1.308028e+05 FES = 155100 μ_F = 0.823 μ_{CR} = 0.080
 Diversity = 5.7744e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.284138e+05 FES = 156100 μ_F = 0.813 μ_{CR} = 0.079
 Diversity = 6.3246e+00 k_t = [2, 4, 2, 2]
 Generation 1570: Best = 1.266598e+05 FES = 157100 μ_F = 0.816 μ_{CR} = 0.068
 Diversity = 6.8493e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.244081e+05 FES = 158100 μ_F = 0.821 μ_{CR} = 0.069
 Diversity = 6.9322e+00 k_t = [2, 2, 3, 2]
 Generation 1590: Best = 1.212988e+05 FES = 159100 μ_F = 0.836 μ_{CR} = 0.062
 Diversity = 7.2156e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.176693e+05 FES = 160100 μ_F = 0.808 μ_{CR} = 0.065
 Diversity = 6.9816e+00 k_t = [3, 4, 4, 4]
 Generation 1610: Best = 1.158609e+05 FES = 161100 μ_F = 0.814 μ_{CR} = 0.065
 Diversity = 6.6722e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.142848e+05 FES = 162100 μ_F = 0.818 μ_{CR} = 0.068
 Diversity = 6.7894e+00 k_t = [3, 4, 4, 4]
 Generation 1630: Best = 1.121874e+05 FES = 163100 μ_F = 0.821 μ_{CR} = 0.077
 Diversity = 7.2346e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.104717e+05 FES = 164100 μ_F = 0.812 μ_{CR} = 0.069
 Diversity = 7.0161e+00 k_t = [4, 2, 4, 4]
 Generation 1650: Best = 1.087662e+05 FES = 165100 μ_F = 0.822 μ_{CR} = 0.076
 Diversity = 7.0093e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.079032e+05 FES = 166100 μ_F = 0.831 μ_{CR} = 0.065
 Diversity = 6.1532e+00 k_t = [2, 4, 2, 4]
 Generation 1670: Best = 1.058551e+05 FES = 167100 μ_F = 0.826 μ_{CR} = 0.074
 Diversity = 5.8584e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.046973e+05 FES = 168100 μ_F = 0.815 μ_{CR} = 0.078
 Diversity = 5.8456e+00 k_t = [3, 2, 2, 4]
 Generation 1690: Best = 1.030654e+05 FES = 169100 μ_F = 0.792 μ_{CR} = 0.074
 Diversity = 5.7154e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.008235e+05 FES = 170100 μ_F = 0.791 μ_{CR} = 0.073
 Diversity = 6.4380e+00 k_t = [2, 3, 2, 4]
 Generation 1710: Best = 9.850231e+04 FES = 171100 μ_F = 0.786 μ_{CR} = 0.068
 Diversity = 6.4701e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 9.805323e+04 FES = 172100 μ_F = 0.779 μ_{CR} = 0.074
 Diversity = 6.4017e+00 k_t = [3, 2, 4, 2]
 Generation 1730: Best = 9.662042e+04 FES = 173100 μ_F = 0.776 μ_{CR} = 0.064
 Diversity = 6.1027e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 9.573535e+04 FES = 174100 μ_F = 0.796 μ_{CR} = 0.074
 Diversity = 6.0732e+00 k_t = [2, 2, 3, 4]
 Generation 1750: Best = 9.190081e+04 FES = 175100 μ_F = 0.801 μ_{CR} = 0.077
 Diversity = 6.3264e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 9.135271e+04 FES = 176100 μ_F = 0.811 μ_{CR} = 0.069
 Diversity = 6.6935e+00 k_t = [3, 3, 4, 2]
 Generation 1770: Best = 8.886349e+04 FES = 177100 μ_F = 0.813 μ_{CR} = 0.071
 Diversity = 7.0799e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 8.815846e+04 FES = 178100 μ_F = 0.810 μ_{CR} = 0.077
 Diversity = 6.3812e+00 k_t = [2, 4, 4, 3]
 Generation 1790: Best = 8.685695e+04 FES = 179100 μ_F = 0.820 μ_{CR} = 0.070
 Diversity = 6.0145e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 8.590948e+04 FES = 180100 μ_F = 0.820 μ_{CR} = 0.075
 Diversity = 6.5078e+00 k_t = [4, 2, 4, 4]
 Generation 1810: Best = 8.245706e+04 FES = 181100 μ_F = 0.809 μ_{CR} = 0.057
 Diversity = 6.6200e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 8.245706e+04 FES = 182100 μ_F = 0.817 μ_{CR} = 0.059
 Diversity = 6.8990e+00 k_t = [4, 4, 2, 3]
 Generation 1830: Best = 8.165449e+04 FES = 183100 μ_F = 0.821 μ_{CR} = 0.067
 Diversity = 7.4817e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 7.993913e+04 FES = 184100 μ_F = 0.825 μ_{CR} = 0.068
 Diversity = 7.7306e+00 k_t = [2, 4, 4, 2]

Generation 1850: Best = 7.890881e+04	FES = 185100	$\mu_F = 0.832$	$\mu_{CR} = 0.067$
Diversity = 8.4848e+00	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 7.747291e+04	FES = 186100	$\mu_F = 0.811$	$\mu_{CR} = 0.073$
Diversity = 8.4017e+00	$k_t = [4, 4, 3, 2]$		
Generation 1870: Best = 7.684311e+04	FES = 187100	$\mu_F = 0.817$	$\mu_{CR} = 0.061$
Diversity = 7.8511e+00	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 7.600477e+04	FES = 188100	$\mu_F = 0.815$	$\mu_{CR} = 0.064$
Diversity = 7.3050e+00	$k_t = [4, 3, 2, 2]$		
Generation 1890: Best = 7.503601e+04	FES = 189100	$\mu_F = 0.800$	$\mu_{CR} = 0.073$
Diversity = 6.7735e+00	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 7.423875e+04	FES = 190100	$\mu_F = 0.793$	$\mu_{CR} = 0.063$
Diversity = 6.4733e+00	$k_t = [2, 4, 3, 4]$		
Generation 1910: Best = 7.253415e+04	FES = 191100	$\mu_F = 0.806$	$\mu_{CR} = 0.074$
Diversity = 7.1692e+00	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 7.115486e+04	FES = 192100	$\mu_F = 0.800$	$\mu_{CR} = 0.076$
Diversity = 6.1160e+00	$k_t = [3, 2, 4, 2]$		
Generation 1930: Best = 6.911502e+04	FES = 193100	$\mu_F = 0.810$	$\mu_{CR} = 0.075$
Diversity = 5.8235e+00	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 6.726800e+04	FES = 194100	$\mu_F = 0.820$	$\mu_{CR} = 0.069$
Diversity = 5.5606e+00	$k_t = [3, 4, 2, 3]$		
Generation 1950: Best = 6.550515e+04	FES = 195100	$\mu_F = 0.816$	$\mu_{CR} = 0.075$
Diversity = 6.4568e+00	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 6.298275e+04	FES = 196100	$\mu_F = 0.800$	$\mu_{CR} = 0.073$
Diversity = 6.7029e+00	$k_t = [4, 2, 3, 2]$		
Generation 1970: Best = 6.256227e+04	FES = 197100	$\mu_F = 0.812$	$\mu_{CR} = 0.079$
Diversity = 6.4230e+00	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 6.089604e+04	FES = 198100	$\mu_F = 0.815$	$\mu_{CR} = 0.081$
Diversity = 6.3321e+00	$k_t = [3, 3, 2, 4]$		
Generation 1990: Best = 5.954707e+04	FES = 199100	$\mu_F = 0.816$	$\mu_{CR} = 0.072$
Diversity = 5.6409e+00	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 5.761378e+04	FES = 200100	$\mu_F = 0.837$	$\mu_{CR} = 0.066$
Diversity = 5.8566e+00	$k_t = [3, 2, 3, 3]$		
Generation 2010: Best = 5.437384e+04	FES = 201100	$\mu_F = 0.841$	$\mu_{CR} = 0.080$
Diversity = 6.5873e+00	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 5.297302e+04	FES = 202100	$\mu_F = 0.838$	$\mu_{CR} = 0.081$
Diversity = 5.9961e+00	$k_t = [3, 4, 4, 3]$		
Generation 2030: Best = 4.997793e+04	FES = 203100	$\mu_F = 0.840$	$\mu_{CR} = 0.070$
Diversity = 5.4981e+00	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 4.628814e+04	FES = 204100	$\mu_F = 0.830$	$\mu_{CR} = 0.072$
Diversity = 5.4023e+00	$k_t = [2, 3, 4, 3]$		
Generation 2050: Best = 4.481450e+04	FES = 205100	$\mu_F = 0.832$	$\mu_{CR} = 0.088$
Diversity = 5.7321e+00	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 4.342294e+04	FES = 206100	$\mu_F = 0.828$	$\mu_{CR} = 0.085$
Diversity = 5.4149e+00	$k_t = [4, 4, 4, 3]$		
Generation 2070: Best = 4.158936e+04	FES = 207100	$\mu_F = 0.822$	$\mu_{CR} = 0.078$
Diversity = 5.1558e+00	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 3.935276e+04	FES = 208100	$\mu_F = 0.822$	$\mu_{CR} = 0.071$
Diversity = 5.3230e+00	$k_t = [4, 4, 4, 2]$		
Generation 2090: Best = 3.856146e+04	FES = 209100	$\mu_F = 0.810$	$\mu_{CR} = 0.064$
Diversity = 5.1985e+00	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 3.716904e+04	FES = 210100	$\mu_F = 0.812$	$\mu_{CR} = 0.058$
Diversity = 5.3253e+00	$k_t = [2, 4, 4, 3]$		
Generation 2110: Best = 3.532203e+04	FES = 211100	$\mu_F = 0.820$	$\mu_{CR} = 0.064$
Diversity = 5.4783e+00	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 3.398319e+04 FES = 212100 μ_F = 0.835 μ_{CR} = 0.065
 Diversity = 5.6367e+00 k_t = [4, 4, 4, 4]
 Generation 2130: Best = 3.261919e+04 FES = 213100 μ_F = 0.830 μ_{CR} = 0.071
 Diversity = 5.0521e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 3.134747e+04 FES = 214100 μ_F = 0.816 μ_{CR} = 0.077
 Diversity = 5.1944e+00 k_t = [4, 2, 3, 4]
 Generation 2150: Best = 3.081805e+04 FES = 215100 μ_F = 0.826 μ_{CR} = 0.076
 Diversity = 5.4815e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 2.970693e+04 FES = 216100 μ_F = 0.833 μ_{CR} = 0.059
 Diversity = 5.2449e+00 k_t = [4, 3, 3, 3]
 Generation 2170: Best = 2.867553e+04 FES = 217100 μ_F = 0.836 μ_{CR} = 0.061
 Diversity = 5.2957e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 2.827517e+04 FES = 218100 μ_F = 0.827 μ_{CR} = 0.065
 Diversity = 4.9595e+00 k_t = [4, 4, 3, 3]
 Generation 2190: Best = 2.782816e+04 FES = 219100 μ_F = 0.820 μ_{CR} = 0.061
 Diversity = 4.3837e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 2.674636e+04 FES = 220100 μ_F = 0.837 μ_{CR} = 0.063
 Diversity = 4.0836e+00 k_t = [4, 2, 3, 4]
 Generation 2210: Best = 2.626665e+04 FES = 221100 μ_F = 0.819 μ_{CR} = 0.059
 Diversity = 3.7919e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 2.562189e+04 FES = 222100 μ_F = 0.808 μ_{CR} = 0.057
 Diversity = 3.8875e+00 k_t = [2, 2, 2, 2]
 Generation 2230: Best = 2.504164e+04 FES = 223100 μ_F = 0.807 μ_{CR} = 0.066
 Diversity = 3.9435e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 2.457930e+04 FES = 224100 μ_F = 0.799 μ_{CR} = 0.067
 Diversity = 3.6223e+00 k_t = [2, 2, 4, 3]
 Generation 2250: Best = 2.320739e+04 FES = 225100 μ_F = 0.792 μ_{CR} = 0.066
 Diversity = 3.3968e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 2.216746e+04 FES = 226100 μ_F = 0.804 μ_{CR} = 0.064
 Diversity = 3.1522e+00 k_t = [2, 4, 4, 3]
 Generation 2270: Best = 2.156382e+04 FES = 227100 μ_F = 0.814 μ_{CR} = 0.062
 Diversity = 3.0721e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 2.043612e+04 FES = 228100 μ_F = 0.808 μ_{CR} = 0.068
 Diversity = 2.8551e+00 k_t = [3, 4, 2, 2]
 Generation 2290: Best = 1.963021e+04 FES = 229100 μ_F = 0.815 μ_{CR} = 0.068
 Diversity = 2.9516e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.881570e+04 FES = 230100 μ_F = 0.820 μ_{CR} = 0.065
 Diversity = 3.2636e+00 k_t = [3, 2, 3, 4]
 Generation 2310: Best = 1.803742e+04 FES = 231100 μ_F = 0.825 μ_{CR} = 0.072
 Diversity = 3.8436e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.743295e+04 FES = 232100 μ_F = 0.802 μ_{CR} = 0.068
 Diversity = 3.7542e+00 k_t = [4, 2, 2, 2]
 Generation 2330: Best = 1.734157e+04 FES = 233100 μ_F = 0.815 μ_{CR} = 0.066
 Diversity = 3.9431e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.694456e+04 FES = 234100 μ_F = 0.818 μ_{CR} = 0.065
 Diversity = 4.2707e+00 k_t = [4, 3, 3, 2]
 Generation 2350: Best = 1.634514e+04 FES = 235100 μ_F = 0.818 μ_{CR} = 0.064
 Diversity = 4.0702e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.604386e+04 FES = 236100 μ_F = 0.791 μ_{CR} = 0.062
 Diversity = 3.9069e+00 k_t = [2, 2, 4, 4]
 Generation 2370: Best = 1.577827e+04 FES = 237100 μ_F = 0.782 μ_{CR} = 0.065
 Diversity = 3.4709e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.570303e+04 FES = 238100 μ_F = 0.781 μ_{CR} = 0.062
 Diversity = 3.3586e+00 k_t = [2, 4, 3, 4]

Generation 2390: Best = 1.528749e+04 FES = 239100 μ_F = 0.775 μ_{CR} = 0.056
 Diversity = 3.1723e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.519260e+04 FES = 240100 μ_F = 0.762 μ_{CR} = 0.060
 Diversity = 3.1183e+00 k_t = [4, 4, 4, 3]
 Generation 2410: Best = 1.480470e+04 FES = 241100 μ_F = 0.752 μ_{CR} = 0.055
 Diversity = 3.0226e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.472475e+04 FES = 242100 μ_F = 0.763 μ_{CR} = 0.052
 Diversity = 2.9114e+00 k_t = [2, 2, 2, 3]
 Generation 2430: Best = 1.471727e+04 FES = 243100 μ_F = 0.777 μ_{CR} = 0.058
 Diversity = 3.0613e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.462346e+04 FES = 244100 μ_F = 0.776 μ_{CR} = 0.064
 Diversity = 2.8319e+00 k_t = [2, 3, 2, 3]
 Generation 2450: Best = 1.452667e+04 FES = 245100 μ_F = 0.783 μ_{CR} = 0.061
 Diversity = 2.5744e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.427794e+04 FES = 246100 μ_F = 0.800 μ_{CR} = 0.067
 Diversity = 2.2027e+00 k_t = [3, 3, 4, 3]
 Generation 2470: Best = 1.402345e+04 FES = 247100 μ_F = 0.778 μ_{CR} = 0.060
 Diversity = 2.2029e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.386998e+04 FES = 248100 μ_F = 0.811 μ_{CR} = 0.058
 Diversity = 2.2937e+00 k_t = [3, 3, 2, 2]
 Generation 2490: Best = 1.368799e+04 FES = 249100 μ_F = 0.805 μ_{CR} = 0.057
 Diversity = 2.4858e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.368799e+04 FES = 250100 μ_F = 0.822 μ_{CR} = 0.065
 Diversity = 2.3927e+00 k_t = [2, 3, 3, 4]
 Generation 2510: Best = 1.368069e+04 FES = 251100 μ_F = 0.798 μ_{CR} = 0.069
 Diversity = 2.2501e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.368028e+04 FES = 252100 μ_F = 0.813 μ_{CR} = 0.057
 Diversity = 2.2982e+00 k_t = [2, 3, 4, 4]
 Generation 2530: Best = 1.362060e+04 FES = 253100 μ_F = 0.814 μ_{CR} = 0.058
 Diversity = 2.2214e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.338076e+04 FES = 254100 μ_F = 0.831 μ_{CR} = 0.054
 Diversity = 2.2677e+00 k_t = [4, 2, 4, 2]
 Generation 2550: Best = 1.338076e+04 FES = 255100 μ_F = 0.827 μ_{CR} = 0.052
 Diversity = 2.3077e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.333309e+04 FES = 256100 μ_F = 0.824 μ_{CR} = 0.047
 Diversity = 2.3205e+00 k_t = [4, 2, 2, 3]
 Generation 2570: Best = 1.322911e+04 FES = 257100 μ_F = 0.795 μ_{CR} = 0.052
 Diversity = 2.3799e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.322911e+04 FES = 258100 μ_F = 0.786 μ_{CR} = 0.055
 Diversity = 2.3364e+00 k_t = [2, 4, 4, 4]
 Generation 2590: Best = 1.322911e+04 FES = 259100 μ_F = 0.760 μ_{CR} = 0.051
 Diversity = 2.3538e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.322629e+04 FES = 260100 μ_F = 0.772 μ_{CR} = 0.051
 Diversity = 2.4398e+00 k_t = [2, 3, 4, 2]
 Generation 2610: Best = 1.314966e+04 FES = 261100 μ_F = 0.745 μ_{CR} = 0.057
 Diversity = 2.5035e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.306679e+04 FES = 262100 μ_F = 0.746 μ_{CR} = 0.073
 Diversity = 2.5235e+00 k_t = [4, 3, 2, 3]
 Generation 2630: Best = 1.297246e+04 FES = 263100 μ_F = 0.740 μ_{CR} = 0.060
 Diversity = 2.5191e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.294387e+04 FES = 264100 μ_F = 0.767 μ_{CR} = 0.062
 Diversity = 2.4780e+00 k_t = [3, 2, 2, 3]
 Generation 2650: Best = 1.290321e+04 FES = 265100 μ_F = 0.770 μ_{CR} = 0.064
 Diversity = 2.0584e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.280589e+04 FES = 266100 μ_F = 0.769 μ_{CR} = 0.075
 Diversity = 1.6535e+00 k_t = [4, 2, 3, 3]
 Generation 2670: Best = 1.271786e+04 FES = 267100 μ_F = 0.733 μ_{CR} = 0.072
 Diversity = 1.2872e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.267213e+04 FES = 268100 μ_F = 0.732 μ_{CR} = 0.068
 Diversity = 9.6887e-01 k_t = [3, 4, 2, 3]
 Generation 2690: Best = 1.267213e+04 FES = 269100 μ_F = 0.760 μ_{CR} = 0.064
 Diversity = 9.3210e-01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.266462e+04 FES = 270100 μ_F = 0.766 μ_{CR} = 0.078
 Diversity = 7.6155e-01 k_t = [4, 3, 2, 2]
 Generation 2710: Best = 1.251329e+04 FES = 271100 μ_F = 0.762 μ_{CR} = 0.076
 Diversity = 6.0630e-01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.247452e+04 FES = 272100 μ_F = 0.748 μ_{CR} = 0.074
 Diversity = 4.6290e-01 k_t = [2, 2, 4, 3]
 Generation 2730: Best = 1.241022e+04 FES = 273100 μ_F = 0.756 μ_{CR} = 0.071
 Diversity = 4.2800e-01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.234703e+04 FES = 274100 μ_F = 0.736 μ_{CR} = 0.076
 Diversity = 3.8708e-01 k_t = [2, 4, 2, 2]
 Generation 2750: Best = 1.234691e+04 FES = 275100 μ_F = 0.735 μ_{CR} = 0.071
 Diversity = 3.6051e-01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.227090e+04 FES = 276100 μ_F = 0.748 μ_{CR} = 0.069
 Diversity = 3.5024e-01 k_t = [4, 4, 2, 3]
 Generation 2770: Best = 1.220602e+04 FES = 277100 μ_F = 0.735 μ_{CR} = 0.067
 Diversity = 3.2688e-01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.220231e+04 FES = 278100 μ_F = 0.736 μ_{CR} = 0.072
 Diversity = 2.6598e-01 k_t = [3, 4, 3, 4]
 Generation 2790: Best = 1.218496e+04 FES = 279100 μ_F = 0.727 μ_{CR} = 0.070
 Diversity = 2.3415e-01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.217133e+04 FES = 280100 μ_F = 0.739 μ_{CR} = 0.066
 Diversity = 2.0324e-01 k_t = [3, 4, 4, 3]
 Generation 2810: Best = 1.206577e+04 FES = 281100 μ_F = 0.755 μ_{CR} = 0.070
 Diversity = 2.1008e-01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.200173e+04 FES = 282100 μ_F = 0.786 μ_{CR} = 0.070
 Diversity = 2.2282e-01 k_t = [2, 2, 4, 2]
 Generation 2830: Best = 1.199566e+04 FES = 283100 μ_F = 0.792 μ_{CR} = 0.070
 Diversity = 2.6114e-01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.197687e+04 FES = 284100 μ_F = 0.793 μ_{CR} = 0.071
 Diversity = 2.5327e-01 k_t = [2, 2, 3, 3]
 Generation 2850: Best = 1.185317e+04 FES = 285100 μ_F = 0.798 μ_{CR} = 0.078
 Diversity = 2.3927e-01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.183938e+04 FES = 286100 μ_F = 0.785 μ_{CR} = 0.073
 Diversity = 2.6063e-01 k_t = [2, 4, 4, 2]
 Generation 2870: Best = 1.183929e+04 FES = 287100 μ_F = 0.783 μ_{CR} = 0.072
 Diversity = 2.7756e-01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.173437e+04 FES = 288100 μ_F = 0.756 μ_{CR} = 0.070
 Diversity = 2.4707e-01 k_t = [3, 2, 4, 4]
 Generation 2890: Best = 1.173437e+04 FES = 289100 μ_F = 0.784 μ_{CR} = 0.068
 Diversity = 2.5664e-01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.173437e+04 FES = 290100 μ_F = 0.792 μ_{CR} = 0.071
 Diversity = 2.6351e-01 k_t = [2, 4, 2, 2]
 Generation 2910: Best = 1.173203e+04 FES = 291100 μ_F = 0.807 μ_{CR} = 0.076
 Diversity = 2.5119e-01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.171356e+04 FES = 292100 μ_F = 0.793 μ_{CR} = 0.073
 Diversity = 2.2432e-01 k_t = [2, 2, 2, 4]

Generation 2930: Best = 1.171248e+04 FES = 293100 μ_F = 0.778 μ_{CR} = 0.062
 Diversity = 1.8341e-01 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 1.168737e+04 FES = 294100 μ_F = 0.799 μ_{CR} = 0.076
 Diversity = 1.7497e-01 k_t = [3, 2, 4, 3]
 Generation 2950: Best = 1.168009e+04 FES = 295100 μ_F = 0.799 μ_{CR} = 0.080
 Diversity = 1.8032e-01 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 1.167899e+04 FES = 296100 μ_F = 0.782 μ_{CR} = 0.079
 Diversity = 1.7410e-01 k_t = [3, 2, 4, 4]
 Generation 2970: Best = 1.163886e+04 FES = 297100 μ_F = 0.795 μ_{CR} = 0.077
 Diversity = 1.6508e-01 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 1.163378e+04 FES = 298100 μ_F = 0.791 μ_{CR} = 0.071
 Diversity = 1.8741e-01 k_t = [3, 4, 3, 4]
 Generation 2990: Best = 1.163271e+04 FES = 299100 μ_F = 0.787 μ_{CR} = 0.074
 Diversity = 1.7595e-01 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 1.162609e+04

Total Function Evals = 300000

Total Successful Updates = 66373

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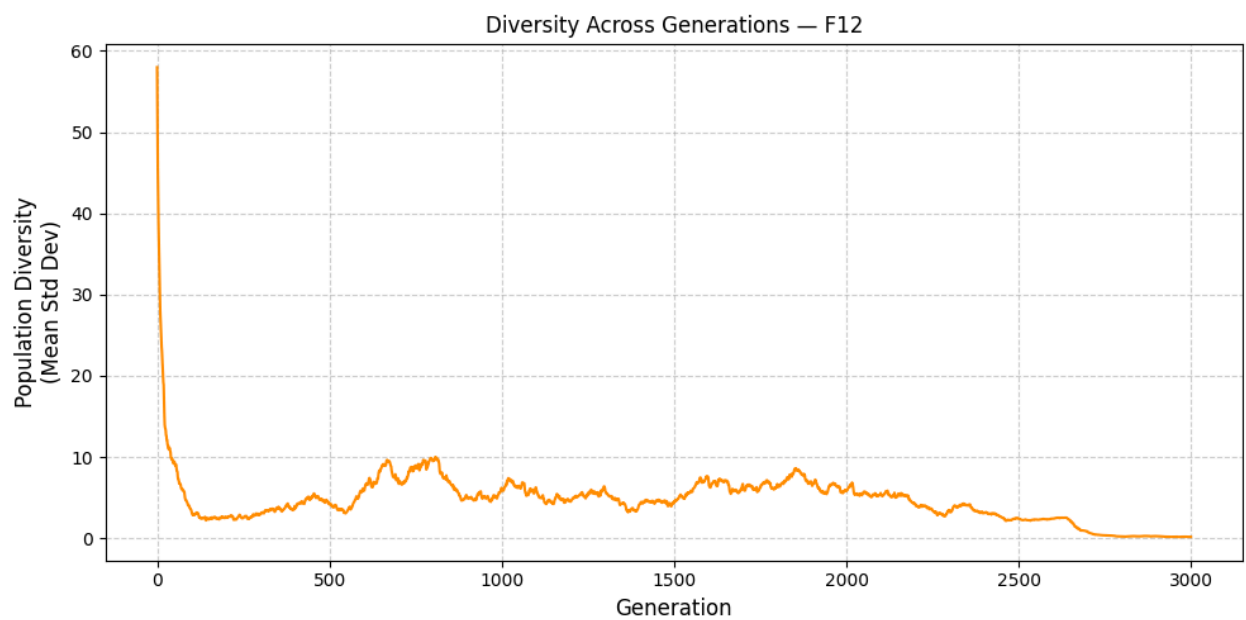
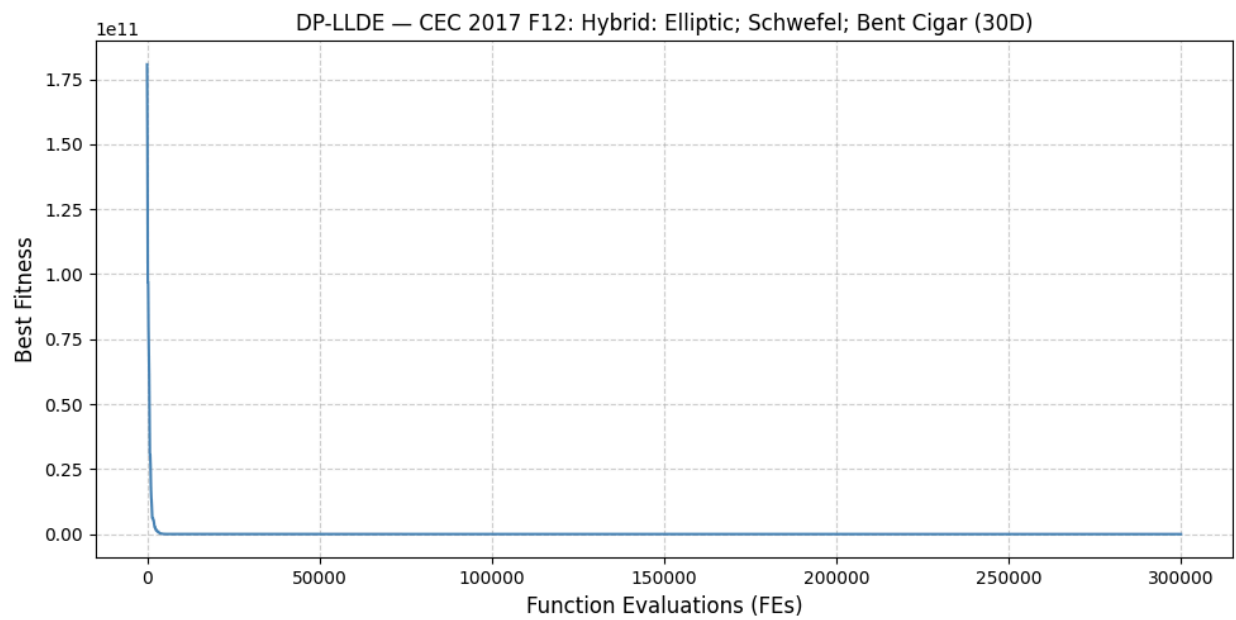
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Results:

Best Fitness : 1.162609e+04
 Global Optimum : 1.200000e+03
 Error (|best-optimum|) : 1.042609e+04
 Best Solution (5 dim) : [50.04121371 98.05224369 -74.73566009 17.18439568
 -25.82523221]
 D^(0) (init. diversity): 5.7976e+01

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Graph saved as convergence_F12.png

```
In [ ]: # =====
# F13 – Hybrid Function 3 (Bent Cigar; Rosenbrock; Bi-Rastrigin)
# =====
run_and_plot('F13', make_wrapper(hybrid.f13), 1300.0, 'Hybrid: Bent Cigar; Ros
```

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DP-LLDE on CEC 2017 F13 – Single Run

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Function : Hybrid: Bent Cigar; Rosenbrock; Bi-Rastrigin (30D)
Bounds : [-100, 100]
Global optimum : 1300.0
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 1.700763e+11

Generation 10:	Best = 6.052474e+09	FES = 1100	$\mu_F = 0.577$	$\mu_{CR} = 0.355$
Diversity = 3.1632e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 8.296867e+08	FES = 2100	$\mu_F = 0.623$	$\mu_{CR} = 0.359$
Diversity = 2.1293e+01	$k_t = [4, 4, 2, 4]$			
Generation 30:	Best = 1.352904e+08	FES = 3100	$\mu_F = 0.647$	$\mu_{CR} = 0.362$
Diversity = 1.7665e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 1.543205e+07	FES = 4100	$\mu_F = 0.667$	$\mu_{CR} = 0.347$
Diversity = 1.5848e+01	$k_t = [2, 3, 3, 2]$			
Generation 50:	Best = 1.324521e+07	FES = 5100	$\mu_F = 0.668$	$\mu_{CR} = 0.338$
Diversity = 1.5597e+01	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 4.072451e+06	FES = 6100	$\mu_F = 0.676$	$\mu_{CR} = 0.313$
Diversity = 1.3464e+01	$k_t = [3, 3, 2, 3]$			
Generation 70:	Best = 1.149842e+06	FES = 7100	$\mu_F = 0.697$	$\mu_{CR} = 0.266$
Diversity = 1.1364e+01	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 3.269239e+05	FES = 8100	$\mu_F = 0.729$	$\mu_{CR} = 0.193$
Diversity = 1.0845e+01	$k_t = [2, 3, 4, 4]$			
Generation 90:	Best = 1.488990e+05	FES = 9100	$\mu_F = 0.733$	$\mu_{CR} = 0.150$
Diversity = 9.4039e+00	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 7.205033e+04	FES = 10100	$\mu_F = 0.745$	$\mu_{CR} = 0.108$
Diversity = 9.1796e+00	$k_t = [2, 2, 2, 2]$			
Generation 110:	Best = 4.294196e+04	FES = 11100	$\mu_F = 0.761$	$\mu_{CR} = 0.091$
Diversity = 7.8184e+00	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 2.920621e+04	FES = 12100	$\mu_F = 0.771$	$\mu_{CR} = 0.089$
Diversity = 7.9908e+00	$k_t = [3, 3, 2, 4]$			
Generation 130:	Best = 2.294219e+04	FES = 13100	$\mu_F = 0.778$	$\mu_{CR} = 0.080$
Diversity = 7.4604e+00	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 2.014577e+04	FES = 14100	$\mu_F = 0.793$	$\mu_{CR} = 0.072$
Diversity = 6.7890e+00	$k_t = [2, 4, 3, 2]$			
Generation 150:	Best = 1.610842e+04	FES = 15100	$\mu_F = 0.803$	$\mu_{CR} = 0.074$
Diversity = 6.7368e+00	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 1.423190e+04	FES = 16100	$\mu_F = 0.805$	$\mu_{CR} = 0.076$
Diversity = 6.4067e+00	$k_t = [4, 4, 4, 3]$			
Generation 170:	Best = 1.251095e+04	FES = 17100	$\mu_F = 0.799$	$\mu_{CR} = 0.072$
Diversity = 6.4900e+00	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 1.136936e+04	FES = 18100	$\mu_F = 0.816$	$\mu_{CR} = 0.072$
Diversity = 6.5082e+00	$k_t = [4, 2, 3, 2]$			
Generation 190:	Best = 1.041801e+04	FES = 19100	$\mu_F = 0.807$	$\mu_{CR} = 0.069$
Diversity = 6.4654e+00	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 9.857199e+03	FES = 20100	$\mu_F = 0.803$	$\mu_{CR} = 0.069$
Diversity = 5.7551e+00	$k_t = [4, 2, 2, 4]$			
Generation 210:	Best = 9.264783e+03	FES = 21100	$\mu_F = 0.807$	$\mu_{CR} = 0.074$
Diversity = 5.5150e+00	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 8.807605e+03	FES = 22100	$\mu_F = 0.805$	$\mu_{CR} = 0.072$
Diversity = 5.3653e+00	$k_t = [3, 3, 4, 4]$			

Generation 230: Best = 8.186366e+03	FES = 23100	$\mu_F = 0.805$	$\mu_{CR} = 0.074$
Diversity = 5.1145e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 7.731557e+03	FES = 24100	$\mu_F = 0.809$	$\mu_{CR} = 0.068$
Diversity = 4.6755e+00	$k_t = [2, 3, 4, 3]$		
Generation 250: Best = 7.256537e+03	FES = 25100	$\mu_F = 0.822$	$\mu_{CR} = 0.078$
Diversity = 4.8613e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 6.827323e+03	FES = 26100	$\mu_F = 0.811$	$\mu_{CR} = 0.074$
Diversity = 4.7829e+00	$k_t = [2, 2, 3, 3]$		
Generation 270: Best = 6.418759e+03	FES = 27100	$\mu_F = 0.816$	$\mu_{CR} = 0.075$
Diversity = 5.1927e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 6.286175e+03	FES = 28100	$\mu_F = 0.794$	$\mu_{CR} = 0.073$
Diversity = 5.0586e+00	$k_t = [4, 2, 2, 4]$		
Generation 290: Best = 5.897296e+03	FES = 29100	$\mu_F = 0.804$	$\mu_{CR} = 0.066$
Diversity = 4.8837e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 5.506730e+03	FES = 30100	$\mu_F = 0.804$	$\mu_{CR} = 0.073$
Diversity = 4.9797e+00	$k_t = [2, 2, 4, 4]$		
Generation 310: Best = 5.022227e+03	FES = 31100	$\mu_F = 0.808$	$\mu_{CR} = 0.067$
Diversity = 5.3277e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 4.678513e+03	FES = 32100	$\mu_F = 0.809$	$\mu_{CR} = 0.065$
Diversity = 5.6262e+00	$k_t = [3, 3, 3, 4]$		
Generation 330: Best = 4.416654e+03	FES = 33100	$\mu_F = 0.816$	$\mu_{CR} = 0.069$
Diversity = 4.7579e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 4.171152e+03	FES = 34100	$\mu_F = 0.821$	$\mu_{CR} = 0.065$
Diversity = 4.7508e+00	$k_t = [3, 4, 2, 2]$		
Generation 350: Best = 4.050327e+03	FES = 35100	$\mu_F = 0.810$	$\mu_{CR} = 0.067$
Diversity = 4.4918e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 3.868846e+03	FES = 36100	$\mu_F = 0.804$	$\mu_{CR} = 0.072$
Diversity = 4.0968e+00	$k_t = [2, 3, 2, 3]$		
Generation 370: Best = 3.791819e+03	FES = 37100	$\mu_F = 0.807$	$\mu_{CR} = 0.070$
Diversity = 4.1716e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 3.709545e+03	FES = 38100	$\mu_F = 0.812$	$\mu_{CR} = 0.070$
Diversity = 3.9898e+00	$k_t = [2, 3, 3, 3]$		
Generation 390: Best = 3.515898e+03	FES = 39100	$\mu_F = 0.817$	$\mu_{CR} = 0.064$
Diversity = 3.6149e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 3.443110e+03	FES = 40100	$\mu_F = 0.827$	$\mu_{CR} = 0.067$
Diversity = 3.5477e+00	$k_t = [2, 3, 2, 4]$		
Generation 410: Best = 3.322529e+03	FES = 41100	$\mu_F = 0.827$	$\mu_{CR} = 0.067$
Diversity = 3.4590e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 3.239193e+03	FES = 42100	$\mu_F = 0.821$	$\mu_{CR} = 0.068$
Diversity = 3.5230e+00	$k_t = [3, 4, 3, 2]$		
Generation 430: Best = 3.097920e+03	FES = 43100	$\mu_F = 0.814$	$\mu_{CR} = 0.065$
Diversity = 3.5419e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 3.060916e+03	FES = 44100	$\mu_F = 0.814$	$\mu_{CR} = 0.056$
Diversity = 3.4453e+00	$k_t = [3, 3, 3, 4]$		
Generation 450: Best = 2.984372e+03	FES = 45100	$\mu_F = 0.809$	$\mu_{CR} = 0.062$
Diversity = 2.9640e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 2.944592e+03	FES = 46100	$\mu_F = 0.795$	$\mu_{CR} = 0.062$
Diversity = 2.6570e+00	$k_t = [3, 4, 2, 4]$		
Generation 470: Best = 2.845686e+03	FES = 47100	$\mu_F = 0.793$	$\mu_{CR} = 0.071$
Diversity = 2.6514e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.802303e+03	FES = 48100	$\mu_F = 0.790$	$\mu_{CR} = 0.066$
Diversity = 2.7605e+00	$k_t = [4, 4, 2, 2]$		
Generation 490: Best = 2.747882e+03	FES = 49100	$\mu_F = 0.814$	$\mu_{CR} = 0.059$
Diversity = 2.7518e+00	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 2.701549e+03	FES = 50100	$\mu_F = 0.795$	$\mu_{CR} = 0.061$
Diversity = 2.5965e+00	$k_t = [2, 4, 4, 3]$		
Generation 510: Best = 2.701549e+03	FES = 51100	$\mu_F = 0.794$	$\mu_{CR} = 0.069$
Diversity = 2.6349e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.621483e+03	FES = 52100	$\mu_F = 0.785$	$\mu_{CR} = 0.061$
Diversity = 2.7189e+00	$k_t = [2, 3, 3, 3]$		
Generation 530: Best = 2.559641e+03	FES = 53100	$\mu_F = 0.791$	$\mu_{CR} = 0.059$
Diversity = 2.6541e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.533365e+03	FES = 54100	$\mu_F = 0.769$	$\mu_{CR} = 0.066$
Diversity = 2.4156e+00	$k_t = [2, 4, 4, 4]$		
Generation 550: Best = 2.464067e+03	FES = 55100	$\mu_F = 0.760$	$\mu_{CR} = 0.054$
Diversity = 2.0886e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.440652e+03	FES = 56100	$\mu_F = 0.763$	$\mu_{CR} = 0.051$
Diversity = 1.9239e+00	$k_t = [4, 4, 3, 3]$		
Generation 570: Best = 2.370763e+03	FES = 57100	$\mu_F = 0.757$	$\mu_{CR} = 0.052$
Diversity = 1.8545e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.358430e+03	FES = 58100	$\mu_F = 0.754$	$\mu_{CR} = 0.053$
Diversity = 1.6940e+00	$k_t = [2, 3, 3, 3]$		
Generation 590: Best = 2.340577e+03	FES = 59100	$\mu_F = 0.760$	$\mu_{CR} = 0.056$
Diversity = 1.6256e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.335729e+03	FES = 60100	$\mu_F = 0.780$	$\mu_{CR} = 0.054$
Diversity = 1.5647e+00	$k_t = [3, 2, 2, 2]$		
Generation 610: Best = 2.313392e+03	FES = 61100	$\mu_F = 0.789$	$\mu_{CR} = 0.046$
Diversity = 1.5605e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.296043e+03	FES = 62100	$\mu_F = 0.788$	$\mu_{CR} = 0.046$
Diversity = 1.4344e+00	$k_t = [4, 4, 3, 3]$		
Generation 630: Best = 2.231252e+03	FES = 63100	$\mu_F = 0.785$	$\mu_{CR} = 0.060$
Diversity = 1.5277e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.204819e+03	FES = 64100	$\mu_F = 0.799$	$\mu_{CR} = 0.049$
Diversity = 1.7683e+00	$k_t = [2, 4, 3, 4]$		
Generation 650: Best = 2.204447e+03	FES = 65100	$\mu_F = 0.815$	$\mu_{CR} = 0.058$
Diversity = 1.8518e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.190766e+03	FES = 66100	$\mu_F = 0.841$	$\mu_{CR} = 0.053$
Diversity = 1.8974e+00	$k_t = [2, 2, 3, 4]$		
Generation 670: Best = 2.186643e+03	FES = 67100	$\mu_F = 0.824$	$\mu_{CR} = 0.049$
Diversity = 1.8576e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.181781e+03	FES = 68100	$\mu_F = 0.796$	$\mu_{CR} = 0.053$
Diversity = 1.7353e+00	$k_t = [3, 3, 3, 2]$		
Generation 690: Best = 2.151513e+03	FES = 69100	$\mu_F = 0.799$	$\mu_{CR} = 0.061$
Diversity = 1.7690e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.151513e+03	FES = 70100	$\mu_F = 0.844$	$\mu_{CR} = 0.064$
Diversity = 1.6862e+00	$k_t = [4, 2, 4, 3]$		
Generation 710: Best = 2.144805e+03	FES = 71100	$\mu_F = 0.817$	$\mu_{CR} = 0.056$
Diversity = 1.7059e+00	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 2.131803e+03	FES = 72100	$\mu_F = 0.845$	$\mu_{CR} = 0.056$
Diversity = 1.6838e+00	$k_t = [3, 3, 3, 4]$		
Generation 730: Best = 2.113802e+03	FES = 73100	$\mu_F = 0.833$	$\mu_{CR} = 0.060$
Diversity = 1.6746e+00	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 2.098777e+03	FES = 74100	$\mu_F = 0.845$	$\mu_{CR} = 0.061$
Diversity = 1.6852e+00	$k_t = [3, 2, 2, 3]$		
Generation 750: Best = 2.097296e+03	FES = 75100	$\mu_F = 0.847$	$\mu_{CR} = 0.061$
Diversity = 1.7906e+00	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 2.052714e+03	FES = 76100	$\mu_F = 0.816$	$\mu_{CR} = 0.061$
Diversity = 1.7450e+00	$k_t = [2, 4, 3, 4]$		

Generation 770:	Best = 2.052714e+03	FES = 77100	$\mu_F = 0.814$	$\mu_{CR} = 0.055$
Diversity = 1.7347e+00	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 2.049262e+03	FES = 78100	$\mu_F = 0.802$	$\mu_{CR} = 0.049$
Diversity = 1.6377e+00	$k_t = [2, 4, 3, 4]$			
Generation 790:	Best = 2.030988e+03	FES = 79100	$\mu_F = 0.792$	$\mu_{CR} = 0.061$
Diversity = 1.6328e+00	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 2.028481e+03	FES = 80100	$\mu_F = 0.806$	$\mu_{CR} = 0.053$
Diversity = 1.7139e+00	$k_t = [2, 3, 3, 2]$			
Generation 810:	Best = 2.026242e+03	FES = 81100	$\mu_F = 0.811$	$\mu_{CR} = 0.058$
Diversity = 1.8301e+00	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 2.014009e+03	FES = 82100	$\mu_F = 0.813$	$\mu_{CR} = 0.056$
Diversity = 1.8572e+00	$k_t = [4, 4, 2, 3]$			
Generation 830:	Best = 2.008340e+03	FES = 83100	$\mu_F = 0.808$	$\mu_{CR} = 0.061$
Diversity = 1.9243e+00	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 1.989073e+03	FES = 84100	$\mu_F = 0.796$	$\mu_{CR} = 0.058$
Diversity = 1.9094e+00	$k_t = [4, 3, 4, 4]$			
Generation 850:	Best = 1.961768e+03	FES = 85100	$\mu_F = 0.779$	$\mu_{CR} = 0.059$
Diversity = 1.9201e+00	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 1.959073e+03	FES = 86100	$\mu_F = 0.773$	$\mu_{CR} = 0.054$
Diversity = 1.8423e+00	$k_t = [2, 4, 2, 4]$			
Generation 870:	Best = 1.959073e+03	FES = 87100	$\mu_F = 0.796$	$\mu_{CR} = 0.056$
Diversity = 1.8181e+00	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 1.943791e+03	FES = 88100	$\mu_F = 0.815$	$\mu_{CR} = 0.057$
Diversity = 1.9981e+00	$k_t = [4, 3, 3, 3]$			
Generation 890:	Best = 1.907730e+03	FES = 89100	$\mu_F = 0.828$	$\mu_{CR} = 0.055$
Diversity = 2.0790e+00	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 1.864751e+03	FES = 90100	$\mu_F = 0.843$	$\mu_{CR} = 0.060$
Diversity = 2.1607e+00	$k_t = [2, 2, 2, 4]$			
Generation 910:	Best = 1.864751e+03	FES = 91100	$\mu_F = 0.826$	$\mu_{CR} = 0.061$
Diversity = 2.1651e+00	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 1.864751e+03	FES = 92100	$\mu_F = 0.831$	$\mu_{CR} = 0.059$
Diversity = 2.0516e+00	$k_t = [3, 3, 2, 4]$			
Generation 930:	Best = 1.847036e+03	FES = 93100	$\mu_F = 0.813$	$\mu_{CR} = 0.063$
Diversity = 2.0085e+00	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 1.847036e+03	FES = 94100	$\mu_F = 0.803$	$\mu_{CR} = 0.067$
Diversity = 1.9558e+00	$k_t = [4, 4, 2, 2]$			
Generation 950:	Best = 1.801768e+03	FES = 95100	$\mu_F = 0.806$	$\mu_{CR} = 0.064$
Diversity = 1.9157e+00	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 1.799409e+03	FES = 96100	$\mu_F = 0.816$	$\mu_{CR} = 0.054$
Diversity = 2.0477e+00	$k_t = [4, 3, 2, 3]$			
Generation 970:	Best = 1.799409e+03	FES = 97100	$\mu_F = 0.824$	$\mu_{CR} = 0.054$
Diversity = 1.9282e+00	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 1.769435e+03	FES = 98100	$\mu_F = 0.825$	$\mu_{CR} = 0.070$
Diversity = 1.8904e+00	$k_t = [4, 2, 4, 4]$			
Generation 990:	Best = 1.767172e+03	FES = 99100	$\mu_F = 0.813$	$\mu_{CR} = 0.063$
Diversity = 1.7515e+00	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 1.759229e+03	FES = 100100	$\mu_F = 0.787$	$\mu_{CR} = 0.057$
Diversity = 1.7304e+00	$k_t = [4, 2, 4, 4]$			
Generation 1010:	Best = 1.742871e+03	FES = 101100	$\mu_F = 0.743$	$\mu_{CR} = 0.064$
Diversity = 1.6584e+00	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 1.742871e+03	FES = 102100	$\mu_F = 0.777$	$\mu_{CR} = 0.059$
Diversity = 1.6388e+00	$k_t = [4, 4, 4, 3]$			
Generation 1030:	Best = 1.712475e+03	FES = 103100	$\mu_F = 0.783$	$\mu_{CR} = 0.051$
Diversity = 1.6294e+00	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 1.712300e+03 FES = 104100 μ_F = 0.781 μ_{CR} = 0.046
 Diversity = 1.6042e+00 k_t = [3, 4, 4, 3]
 Generation 1050: Best = 1.703942e+03 FES = 105100 μ_F = 0.778 μ_{CR} = 0.044
 Diversity = 1.5575e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 1.701878e+03 FES = 106100 μ_F = 0.805 μ_{CR} = 0.049
 Diversity = 1.5619e+00 k_t = [4, 4, 3, 4]
 Generation 1070: Best = 1.679594e+03 FES = 107100 μ_F = 0.821 μ_{CR} = 0.054
 Diversity = 1.6055e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 1.661229e+03 FES = 108100 μ_F = 0.809 μ_{CR} = 0.052
 Diversity = 1.7041e+00 k_t = [4, 2, 2, 2]
 Generation 1090: Best = 1.627942e+03 FES = 109100 μ_F = 0.813 μ_{CR} = 0.058
 Diversity = 2.0579e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.618010e+03 FES = 110100 μ_F = 0.818 μ_{CR} = 0.056
 Diversity = 1.9503e+00 k_t = [4, 4, 4, 4]
 Generation 1110: Best = 1.603497e+03 FES = 111100 μ_F = 0.851 μ_{CR} = 0.062
 Diversity = 1.8531e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.594703e+03 FES = 112100 μ_F = 0.847 μ_{CR} = 0.059
 Diversity = 1.6984e+00 k_t = [2, 3, 3, 3]
 Generation 1130: Best = 1.586654e+03 FES = 113100 μ_F = 0.836 μ_{CR} = 0.056
 Diversity = 1.7293e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.575081e+03 FES = 114100 μ_F = 0.831 μ_{CR} = 0.056
 Diversity = 1.8174e+00 k_t = [4, 3, 4, 2]
 Generation 1150: Best = 1.552485e+03 FES = 115100 μ_F = 0.831 μ_{CR} = 0.056
 Diversity = 1.9516e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.552297e+03 FES = 116100 μ_F = 0.844 μ_{CR} = 0.065
 Diversity = 1.9531e+00 k_t = [2, 4, 3, 4]
 Generation 1170: Best = 1.546494e+03 FES = 117100 μ_F = 0.815 μ_{CR} = 0.056
 Diversity = 1.8610e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.534544e+03 FES = 118100 μ_F = 0.810 μ_{CR} = 0.054
 Diversity = 1.7998e+00 k_t = [4, 2, 2, 4]
 Generation 1190: Best = 1.525810e+03 FES = 119100 μ_F = 0.821 μ_{CR} = 0.051
 Diversity = 1.7721e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.493173e+03 FES = 120100 μ_F = 0.828 μ_{CR} = 0.050
 Diversity = 1.8600e+00 k_t = [3, 3, 4, 2]
 Generation 1210: Best = 1.480913e+03 FES = 121100 μ_F = 0.800 μ_{CR} = 0.052
 Diversity = 1.8778e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.480913e+03 FES = 122100 μ_F = 0.787 μ_{CR} = 0.054
 Diversity = 1.7816e+00 k_t = [3, 3, 4, 3]
 Generation 1230: Best = 1.480473e+03 FES = 123100 μ_F = 0.787 μ_{CR} = 0.051
 Diversity = 1.6836e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.469603e+03 FES = 124100 μ_F = 0.790 μ_{CR} = 0.053
 Diversity = 1.5653e+00 k_t = [4, 2, 3, 4]
 Generation 1250: Best = 1.459695e+03 FES = 125100 μ_F = 0.781 μ_{CR} = 0.057
 Diversity = 1.5238e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.457352e+03 FES = 126100 μ_F = 0.784 μ_{CR} = 0.063
 Diversity = 1.4815e+00 k_t = [4, 4, 3, 4]
 Generation 1270: Best = 1.457352e+03 FES = 127100 μ_F = 0.769 μ_{CR} = 0.051
 Diversity = 1.4474e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.438539e+03 FES = 128100 μ_F = 0.771 μ_{CR} = 0.050
 Diversity = 1.4313e+00 k_t = [3, 3, 3, 2]
 Generation 1290: Best = 1.438539e+03 FES = 129100 μ_F = 0.779 μ_{CR} = 0.052
 Diversity = 1.4243e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.426145e+03 FES = 130100 μ_F = 0.797 μ_{CR} = 0.053
 Diversity = 1.3916e+00 k_t = [4, 2, 4, 3]

Generation 1310: Best = 1.426145e+03 FES = 131100 μ_F = 0.795 μ_{CR} = 0.056
 Diversity = 1.4011e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.426145e+03 FES = 132100 μ_F = 0.774 μ_{CR} = 0.051
 Diversity = 1.3767e+00 k_t = [4, 2, 4, 2]
 Generation 1330: Best = 1.426095e+03 FES = 133100 μ_F = 0.791 μ_{CR} = 0.051
 Diversity = 1.3774e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.426095e+03 FES = 134100 μ_F = 0.795 μ_{CR} = 0.049
 Diversity = 1.4298e+00 k_t = [3, 2, 2, 3]
 Generation 1350: Best = 1.426095e+03 FES = 135100 μ_F = 0.820 μ_{CR} = 0.055
 Diversity = 1.4015e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.426095e+03 FES = 136100 μ_F = 0.808 μ_{CR} = 0.044
 Diversity = 1.4177e+00 k_t = [2, 2, 2, 2]
 Generation 1370: Best = 1.425649e+03 FES = 137100 μ_F = 0.832 μ_{CR} = 0.048
 Diversity = 1.4148e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.425649e+03 FES = 138100 μ_F = 0.824 μ_{CR} = 0.055
 Diversity = 1.4202e+00 k_t = [4, 2, 2, 3]
 Generation 1390: Best = 1.425633e+03 FES = 139100 μ_F = 0.830 μ_{CR} = 0.064
 Diversity = 1.4051e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.425633e+03 FES = 140100 μ_F = 0.847 μ_{CR} = 0.047
 Diversity = 1.3837e+00 k_t = [4, 3, 3, 2]
 Generation 1410: Best = 1.425633e+03 FES = 141100 μ_F = 0.836 μ_{CR} = 0.040
 Diversity = 1.3959e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.424537e+03 FES = 142100 μ_F = 0.796 μ_{CR} = 0.046
 Diversity = 1.3964e+00 k_t = [2, 2, 4, 3]
 Generation 1430: Best = 1.424537e+03 FES = 143100 μ_F = 0.808 μ_{CR} = 0.040
 Diversity = 1.3945e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.424537e+03 FES = 144100 μ_F = 0.823 μ_{CR} = 0.044
 Diversity = 1.3616e+00 k_t = [3, 2, 2, 3]
 Generation 1450: Best = 1.424537e+03 FES = 145100 μ_F = 0.778 μ_{CR} = 0.052
 Diversity = 1.3509e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.424537e+03 FES = 146100 μ_F = 0.793 μ_{CR} = 0.048
 Diversity = 1.3471e+00 k_t = [2, 2, 4, 2]
 Generation 1470: Best = 1.424537e+03 FES = 147100 μ_F = 0.805 μ_{CR} = 0.045
 Diversity = 1.3333e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.424403e+03 FES = 148100 μ_F = 0.737 μ_{CR} = 0.049
 Diversity = 1.3280e+00 k_t = [3, 2, 4, 3]
 Generation 1490: Best = 1.424403e+03 FES = 149100 μ_F = 0.761 μ_{CR} = 0.036
 Diversity = 1.3215e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.422743e+03 FES = 150100 μ_F = 0.694 μ_{CR} = 0.047
 Diversity = 1.3069e+00 k_t = [2, 2, 2, 4]
 Generation 1510: Best = 1.416697e+03 FES = 151100 μ_F = 0.645 μ_{CR} = 0.043
 Diversity = 1.3018e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.415879e+03 FES = 152100 μ_F = 0.679 μ_{CR} = 0.045
 Diversity = 1.2999e+00 k_t = [3, 4, 3, 3]
 Generation 1530: Best = 1.412999e+03 FES = 153100 μ_F = 0.617 μ_{CR} = 0.049
 Diversity = 1.2774e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.412766e+03 FES = 154100 μ_F = 0.695 μ_{CR} = 0.047
 Diversity = 1.2643e+00 k_t = [2, 4, 2, 4]
 Generation 1550: Best = 1.412766e+03 FES = 155100 μ_F = 0.728 μ_{CR} = 0.057
 Diversity = 1.2509e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.412215e+03 FES = 156100 μ_F = 0.688 μ_{CR} = 0.043
 Diversity = 1.2532e+00 k_t = [2, 3, 3, 3]
 Generation 1570: Best = 1.409845e+03 FES = 157100 μ_F = 0.712 μ_{CR} = 0.039
 Diversity = 1.2533e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.409845e+03 FES = 158100 μ_F = 0.738 μ_{CR} = 0.039
 Diversity = 1.2363e+00 k_t = [2, 4, 4, 3]
 Generation 1590: Best = 1.409845e+03 FES = 159100 μ_F = 0.707 μ_{CR} = 0.050
 Diversity = 1.2309e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.409845e+03 FES = 160100 μ_F = 0.696 μ_{CR} = 0.066
 Diversity = 1.2289e+00 k_t = [4, 3, 4, 2]
 Generation 1610: Best = 1.409845e+03 FES = 161100 μ_F = 0.669 μ_{CR} = 0.065
 Diversity = 1.2282e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.409845e+03 FES = 162100 μ_F = 0.691 μ_{CR} = 0.063
 Diversity = 1.2154e+00 k_t = [4, 4, 4, 4]
 Generation 1630: Best = 1.407727e+03 FES = 163100 μ_F = 0.720 μ_{CR} = 0.048
 Diversity = 1.2120e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.407727e+03 FES = 164100 μ_F = 0.744 μ_{CR} = 0.048
 Diversity = 1.1998e+00 k_t = [3, 4, 4, 2]
 Generation 1650: Best = 1.407727e+03 FES = 165100 μ_F = 0.750 μ_{CR} = 0.045
 Diversity = 1.2036e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.407727e+03 FES = 166100 μ_F = 0.788 μ_{CR} = 0.044
 Diversity = 1.2070e+00 k_t = [3, 2, 3, 3]
 Generation 1670: Best = 1.404736e+03 FES = 167100 μ_F = 0.790 μ_{CR} = 0.045
 Diversity = 1.2102e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.404736e+03 FES = 168100 μ_F = 0.816 μ_{CR} = 0.038
 Diversity = 1.2130e+00 k_t = [3, 4, 4, 2]
 Generation 1690: Best = 1.403853e+03 FES = 169100 μ_F = 0.782 μ_{CR} = 0.036
 Diversity = 1.2010e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.399954e+03 FES = 170100 μ_F = 0.774 μ_{CR} = 0.050
 Diversity = 1.1989e+00 k_t = [2, 3, 3, 3]
 Generation 1710: Best = 1.399954e+03 FES = 171100 μ_F = 0.746 μ_{CR} = 0.044
 Diversity = 1.1948e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.399954e+03 FES = 172100 μ_F = 0.701 μ_{CR} = 0.042
 Diversity = 1.2093e+00 k_t = [3, 3, 4, 3]
 Generation 1730: Best = 1.399954e+03 FES = 173100 μ_F = 0.713 μ_{CR} = 0.040
 Diversity = 1.2062e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.399954e+03 FES = 174100 μ_F = 0.681 μ_{CR} = 0.045
 Diversity = 1.2045e+00 k_t = [3, 2, 4, 2]
 Generation 1750: Best = 1.399213e+03 FES = 175100 μ_F = 0.694 μ_{CR} = 0.049
 Diversity = 1.1988e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.399213e+03 FES = 176100 μ_F = 0.742 μ_{CR} = 0.062
 Diversity = 1.1908e+00 k_t = [4, 2, 2, 4]
 Generation 1770: Best = 1.399190e+03 FES = 177100 μ_F = 0.735 μ_{CR} = 0.049
 Diversity = 1.1900e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.399190e+03 FES = 178100 μ_F = 0.743 μ_{CR} = 0.048
 Diversity = 1.1609e+00 k_t = [2, 4, 3, 2]
 Generation 1790: Best = 1.379988e+03 FES = 179100 μ_F = 0.689 μ_{CR} = 0.045
 Diversity = 1.1611e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 1.379988e+03 FES = 180100 μ_F = 0.781 μ_{CR} = 0.045
 Diversity = 1.1581e+00 k_t = [2, 2, 4, 2]
 Generation 1810: Best = 1.379988e+03 FES = 181100 μ_F = 0.812 μ_{CR} = 0.039
 Diversity = 1.1620e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 1.379988e+03 FES = 182100 μ_F = 0.743 μ_{CR} = 0.033
 Diversity = 1.1539e+00 k_t = [4, 2, 2, 4]
 Generation 1830: Best = 1.379988e+03 FES = 183100 μ_F = 0.751 μ_{CR} = 0.038
 Diversity = 1.1530e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 1.379940e+03 FES = 184100 μ_F = 0.688 μ_{CR} = 0.038
 Diversity = 1.1559e+00 k_t = [4, 4, 3, 2]

Generation 1850: Best = 1.379917e+03 FES = 185100 μ_F = 0.653 μ_{CR} = 0.042
 Diversity = 1.1601e+00 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 1.379917e+03 FES = 186100 μ_F = 0.680 μ_{CR} = 0.040
 Diversity = 1.1573e+00 k_t = [4, 2, 4, 3]
 Generation 1870: Best = 1.379862e+03 FES = 187100 μ_F = 0.662 μ_{CR} = 0.048
 Diversity = 1.1534e+00 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 1.379828e+03 FES = 188100 μ_F = 0.658 μ_{CR} = 0.039
 Diversity = 1.1510e+00 k_t = [3, 2, 2, 4]
 Generation 1890: Best = 1.379828e+03 FES = 189100 μ_F = 0.702 μ_{CR} = 0.045
 Diversity = 1.1484e+00 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 1.379828e+03 FES = 190100 μ_F = 0.731 μ_{CR} = 0.041
 Diversity = 1.1448e+00 k_t = [3, 3, 4, 2]
 Generation 1910: Best = 1.378174e+03 FES = 191100 μ_F = 0.717 μ_{CR} = 0.041
 Diversity = 1.1474e+00 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 1.378174e+03 FES = 192100 μ_F = 0.747 μ_{CR} = 0.039
 Diversity = 1.1139e+00 k_t = [3, 4, 2, 4]
 Generation 1930: Best = 1.378174e+03 FES = 193100 μ_F = 0.768 μ_{CR} = 0.045
 Diversity = 1.1083e+00 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 1.378174e+03 FES = 194100 μ_F = 0.808 μ_{CR} = 0.037
 Diversity = 1.1033e+00 k_t = [3, 3, 4, 3]
 Generation 1950: Best = 1.378174e+03 FES = 195100 μ_F = 0.779 μ_{CR} = 0.047
 Diversity = 1.0972e+00 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 1.378174e+03 FES = 196100 μ_F = 0.809 μ_{CR} = 0.046
 Diversity = 1.0906e+00 k_t = [3, 4, 4, 4]
 Generation 1970: Best = 1.378174e+03 FES = 197100 μ_F = 0.802 μ_{CR} = 0.034
 Diversity = 1.0895e+00 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 1.378174e+03 FES = 198100 μ_F = 0.798 μ_{CR} = 0.034
 Diversity = 1.0860e+00 k_t = [2, 4, 4, 3]
 Generation 1990: Best = 1.378174e+03 FES = 199100 μ_F = 0.784 μ_{CR} = 0.042
 Diversity = 1.0833e+00 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 1.378174e+03 FES = 200100 μ_F = 0.732 μ_{CR} = 0.040
 Diversity = 1.0811e+00 k_t = [2, 2, 3, 2]
 Generation 2010: Best = 1.378174e+03 FES = 201100 μ_F = 0.701 μ_{CR} = 0.052
 Diversity = 1.0815e+00 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 1.378174e+03 FES = 202100 μ_F = 0.712 μ_{CR} = 0.047
 Diversity = 1.0890e+00 k_t = [2, 2, 2, 3]
 Generation 2030: Best = 1.378174e+03 FES = 203100 μ_F = 0.784 μ_{CR} = 0.046
 Diversity = 1.0885e+00 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 1.378174e+03 FES = 204100 μ_F = 0.791 μ_{CR} = 0.036
 Diversity = 1.0830e+00 k_t = [2, 2, 4, 2]
 Generation 2050: Best = 1.378163e+03 FES = 205100 μ_F = 0.740 μ_{CR} = 0.032
 Diversity = 1.0847e+00 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 1.378163e+03 FES = 206100 μ_F = 0.803 μ_{CR} = 0.040
 Diversity = 1.0846e+00 k_t = [4, 2, 2, 3]
 Generation 2070: Best = 1.378163e+03 FES = 207100 μ_F = 0.756 μ_{CR} = 0.052
 Diversity = 1.0800e+00 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 1.378163e+03 FES = 208100 μ_F = 0.800 μ_{CR} = 0.051
 Diversity = 1.0662e+00 k_t = [4, 2, 4, 3]
 Generation 2090: Best = 1.378163e+03 FES = 209100 μ_F = 0.749 μ_{CR} = 0.035
 Diversity = 1.0584e+00 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 1.378163e+03 FES = 210100 μ_F = 0.691 μ_{CR} = 0.044
 Diversity = 1.0578e+00 k_t = [3, 4, 4, 3]
 Generation 2110: Best = 1.378163e+03 FES = 211100 μ_F = 0.720 μ_{CR} = 0.043
 Diversity = 1.0587e+00 k_t = [1, 1, 1, 1]

Generation 2120: Best = 1.378163e+03 FES = 212100 μ_F = 0.777 μ_{CR} = 0.043
 Diversity = 1.0545e+00 k_t = [3, 2, 3, 2]
 Generation 2130: Best = 1.375624e+03 FES = 213100 μ_F = 0.792 μ_{CR} = 0.037
 Diversity = 1.0548e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.375624e+03 FES = 214100 μ_F = 0.735 μ_{CR} = 0.040
 Diversity = 1.0512e+00 k_t = [4, 2, 3, 3]
 Generation 2150: Best = 1.375598e+03 FES = 215100 μ_F = 0.741 μ_{CR} = 0.036
 Diversity = 1.0491e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.375172e+03 FES = 216100 μ_F = 0.729 μ_{CR} = 0.035
 Diversity = 1.0538e+00 k_t = [2, 4, 3, 2]
 Generation 2170: Best = 1.375172e+03 FES = 217100 μ_F = 0.723 μ_{CR} = 0.034
 Diversity = 1.0532e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.375172e+03 FES = 218100 μ_F = 0.736 μ_{CR} = 0.032
 Diversity = 1.0420e+00 k_t = [2, 2, 3, 4]
 Generation 2190: Best = 1.374766e+03 FES = 219100 μ_F = 0.716 μ_{CR} = 0.043
 Diversity = 1.0522e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 1.374766e+03 FES = 220100 μ_F = 0.773 μ_{CR} = 0.049
 Diversity = 1.0499e+00 k_t = [2, 4, 4, 2]
 Generation 2210: Best = 1.374760e+03 FES = 221100 μ_F = 0.796 μ_{CR} = 0.039
 Diversity = 1.0504e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 1.374760e+03 FES = 222100 μ_F = 0.727 μ_{CR} = 0.039
 Diversity = 1.0485e+00 k_t = [3, 3, 4, 3]
 Generation 2230: Best = 1.374760e+03 FES = 223100 μ_F = 0.684 μ_{CR} = 0.048
 Diversity = 1.0487e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.374760e+03 FES = 224100 μ_F = 0.655 μ_{CR} = 0.041
 Diversity = 1.0472e+00 k_t = [4, 3, 3, 3]
 Generation 2250: Best = 1.374760e+03 FES = 225100 μ_F = 0.560 μ_{CR} = 0.051
 Diversity = 1.0480e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.374760e+03 FES = 226100 μ_F = 0.579 μ_{CR} = 0.037
 Diversity = 1.0467e+00 k_t = [4, 2, 4, 4]
 Generation 2270: Best = 1.374425e+03 FES = 227100 μ_F = 0.612 μ_{CR} = 0.040
 Diversity = 1.0445e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.373870e+03 FES = 228100 μ_F = 0.702 μ_{CR} = 0.047
 Diversity = 1.0352e+00 k_t = [2, 2, 2, 4]
 Generation 2290: Best = 1.373870e+03 FES = 229100 μ_F = 0.688 μ_{CR} = 0.044
 Diversity = 1.0289e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.371217e+03 FES = 230100 μ_F = 0.760 μ_{CR} = 0.046
 Diversity = 1.0270e+00 k_t = [4, 2, 2, 3]
 Generation 2310: Best = 1.370552e+03 FES = 231100 μ_F = 0.781 μ_{CR} = 0.048
 Diversity = 1.0263e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.370552e+03 FES = 232100 μ_F = 0.785 μ_{CR} = 0.063
 Diversity = 1.0264e+00 k_t = [4, 4, 3, 2]
 Generation 2330: Best = 1.370487e+03 FES = 233100 μ_F = 0.780 μ_{CR} = 0.051
 Diversity = 1.0259e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.369791e+03 FES = 234100 μ_F = 0.706 μ_{CR} = 0.063
 Diversity = 1.0254e+00 k_t = [2, 4, 4, 3]
 Generation 2350: Best = 1.369791e+03 FES = 235100 μ_F = 0.736 μ_{CR} = 0.060
 Diversity = 1.0166e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.369761e+03 FES = 236100 μ_F = 0.734 μ_{CR} = 0.051
 Diversity = 1.0142e+00 k_t = [3, 4, 4, 4]
 Generation 2370: Best = 1.368844e+03 FES = 237100 μ_F = 0.753 μ_{CR} = 0.044
 Diversity = 1.0142e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.368844e+03 FES = 238100 μ_F = 0.748 μ_{CR} = 0.041
 Diversity = 1.0135e+00 k_t = [3, 3, 3, 2]

Generation 2390: Best = 1.368844e+03 FES = 239100 μ_F = 0.803 μ_{CR} = 0.046
 Diversity = 1.0117e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.368844e+03 FES = 240100 μ_F = 0.834 μ_{CR} = 0.049
 Diversity = 1.0067e+00 k_t = [2, 2, 2, 3]
 Generation 2410: Best = 1.368844e+03 FES = 241100 μ_F = 0.880 μ_{CR} = 0.046
 Diversity = 1.0068e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.368844e+03 FES = 242100 μ_F = 0.864 μ_{CR} = 0.048
 Diversity = 1.0137e+00 k_t = [2, 3, 2, 3]
 Generation 2430: Best = 1.368838e+03 FES = 243100 μ_F = 0.758 μ_{CR} = 0.039
 Diversity = 1.0092e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.368838e+03 FES = 244100 μ_F = 0.735 μ_{CR} = 0.039
 Diversity = 9.9465e-01 k_t = [2, 3, 4, 4]
 Generation 2450: Best = 1.368838e+03 FES = 245100 μ_F = 0.655 μ_{CR} = 0.043
 Diversity = 9.9278e-01 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.368838e+03 FES = 246100 μ_F = 0.683 μ_{CR} = 0.044
 Diversity = 9.7359e-01 k_t = [4, 4, 2, 4]
 Generation 2470: Best = 1.368838e+03 FES = 247100 μ_F = 0.711 μ_{CR} = 0.039
 Diversity = 9.6070e-01 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.368838e+03 FES = 248100 μ_F = 0.682 μ_{CR} = 0.038
 Diversity = 9.5998e-01 k_t = [2, 2, 2, 2]
 Generation 2490: Best = 1.368838e+03 FES = 249100 μ_F = 0.724 μ_{CR} = 0.040
 Diversity = 9.6052e-01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.368424e+03 FES = 250100 μ_F = 0.795 μ_{CR} = 0.046
 Diversity = 9.6384e-01 k_t = [4, 3, 3, 4]
 Generation 2510: Best = 1.368424e+03 FES = 251100 μ_F = 0.754 μ_{CR} = 0.045
 Diversity = 9.6485e-01 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.368424e+03 FES = 252100 μ_F = 0.745 μ_{CR} = 0.046
 Diversity = 9.6356e-01 k_t = [3, 4, 4, 3]
 Generation 2530: Best = 1.368377e+03 FES = 253100 μ_F = 0.654 μ_{CR} = 0.037
 Diversity = 9.6355e-01 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.366315e+03 FES = 254100 μ_F = 0.598 μ_{CR} = 0.041
 Diversity = 9.5765e-01 k_t = [2, 4, 2, 4]
 Generation 2550: Best = 1.366315e+03 FES = 255100 μ_F = 0.575 μ_{CR} = 0.036
 Diversity = 9.4858e-01 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.366315e+03 FES = 256100 μ_F = 0.543 μ_{CR} = 0.045
 Diversity = 9.4345e-01 k_t = [2, 3, 3, 4]
 Generation 2570: Best = 1.366315e+03 FES = 257100 μ_F = 0.552 μ_{CR} = 0.054
 Diversity = 9.4176e-01 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.366006e+03 FES = 258100 μ_F = 0.623 μ_{CR} = 0.052
 Diversity = 9.3861e-01 k_t = [4, 3, 3, 2]
 Generation 2590: Best = 1.366006e+03 FES = 259100 μ_F = 0.647 μ_{CR} = 0.043
 Diversity = 9.3796e-01 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.366006e+03 FES = 260100 μ_F = 0.655 μ_{CR} = 0.047
 Diversity = 9.3765e-01 k_t = [4, 3, 4, 2]
 Generation 2610: Best = 1.366006e+03 FES = 261100 μ_F = 0.703 μ_{CR} = 0.047
 Diversity = 9.3333e-01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.366006e+03 FES = 262100 μ_F = 0.709 μ_{CR} = 0.037
 Diversity = 9.2883e-01 k_t = [4, 2, 3, 4]
 Generation 2630: Best = 1.366006e+03 FES = 263100 μ_F = 0.748 μ_{CR} = 0.039
 Diversity = 9.2785e-01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.366006e+03 FES = 264100 μ_F = 0.722 μ_{CR} = 0.046
 Diversity = 9.1470e-01 k_t = [3, 2, 4, 4]
 Generation 2650: Best = 1.366006e+03 FES = 265100 μ_F = 0.754 μ_{CR} = 0.049
 Diversity = 9.1374e-01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.366006e+03 FES = 266100 μ_F = 0.798 μ_{CR} = 0.047
 Diversity = 9.1407e-01 k_t = [4, 3, 2, 2]
 Generation 2670: Best = 1.366006e+03 FES = 267100 μ_F = 0.819 μ_{CR} = 0.041
 Diversity = 9.1414e-01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.366006e+03 FES = 268100 μ_F = 0.846 μ_{CR} = 0.038
 Diversity = 8.7355e-01 k_t = [4, 3, 2, 4]
 Generation 2690: Best = 1.366006e+03 FES = 269100 μ_F = 0.862 μ_{CR} = 0.030
 Diversity = 8.7531e-01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.366006e+03 FES = 270100 μ_F = 0.885 μ_{CR} = 0.040
 Diversity = 8.7764e-01 k_t = [3, 2, 3, 4]
 Generation 2710: Best = 1.366006e+03 FES = 271100 μ_F = 0.835 μ_{CR} = 0.038
 Diversity = 8.7150e-01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.362039e+03 FES = 272100 μ_F = 0.851 μ_{CR} = 0.038
 Diversity = 8.6784e-01 k_t = [3, 3, 4, 2]
 Generation 2730: Best = 1.362039e+03 FES = 273100 μ_F = 0.755 μ_{CR} = 0.035
 Diversity = 8.6769e-01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.362039e+03 FES = 274100 μ_F = 0.786 μ_{CR} = 0.038
 Diversity = 8.6376e-01 k_t = [3, 3, 3, 4]
 Generation 2750: Best = 1.362039e+03 FES = 275100 μ_F = 0.703 μ_{CR} = 0.047
 Diversity = 8.5504e-01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.362039e+03 FES = 276100 μ_F = 0.739 μ_{CR} = 0.045
 Diversity = 8.3590e-01 k_t = [3, 3, 3, 4]
 Generation 2770: Best = 1.362039e+03 FES = 277100 μ_F = 0.744 μ_{CR} = 0.043
 Diversity = 8.3396e-01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.360618e+03 FES = 278100 μ_F = 0.777 μ_{CR} = 0.041
 Diversity = 8.2954e-01 k_t = [4, 3, 3, 4]
 Generation 2790: Best = 1.360618e+03 FES = 279100 μ_F = 0.719 μ_{CR} = 0.048
 Diversity = 8.2928e-01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.360618e+03 FES = 280100 μ_F = 0.764 μ_{CR} = 0.038
 Diversity = 8.2886e-01 k_t = [4, 4, 2, 2]
 Generation 2810: Best = 1.360603e+03 FES = 281100 μ_F = 0.773 μ_{CR} = 0.033
 Diversity = 8.2887e-01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.360544e+03 FES = 282100 μ_F = 0.787 μ_{CR} = 0.037
 Diversity = 7.9441e-01 k_t = [3, 3, 3, 4]
 Generation 2830: Best = 1.360544e+03 FES = 283100 μ_F = 0.802 μ_{CR} = 0.044
 Diversity = 7.9554e-01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.360544e+03 FES = 284100 μ_F = 0.703 μ_{CR} = 0.047
 Diversity = 7.8144e-01 k_t = [2, 2, 4, 4]
 Generation 2850: Best = 1.360544e+03 FES = 285100 μ_F = 0.753 μ_{CR} = 0.038
 Diversity = 7.8074e-01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.360544e+03 FES = 286100 μ_F = 0.763 μ_{CR} = 0.040
 Diversity = 7.6924e-01 k_t = [3, 2, 4, 4]
 Generation 2870: Best = 1.360516e+03 FES = 287100 μ_F = 0.792 μ_{CR} = 0.046
 Diversity = 7.6908e-01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.360516e+03 FES = 288100 μ_F = 0.787 μ_{CR} = 0.040
 Diversity = 7.6847e-01 k_t = [2, 3, 3, 2]
 Generation 2890: Best = 1.360516e+03 FES = 289100 μ_F = 0.814 μ_{CR} = 0.036
 Diversity = 7.6808e-01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.360516e+03 FES = 290100 μ_F = 0.815 μ_{CR} = 0.039
 Diversity = 7.5619e-01 k_t = [2, 4, 3, 2]
 Generation 2910: Best = 1.359817e+03 FES = 291100 μ_F = 0.767 μ_{CR} = 0.051
 Diversity = 7.5579e-01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.359558e+03 FES = 292100 μ_F = 0.775 μ_{CR} = 0.054
 Diversity = 7.5618e-01 k_t = [4, 3, 3, 3]

Generation 2930: Best = 1.359558e+03 FES = 293100 μ_F = 0.753 μ_{CR} = 0.042
 Diversity = 7.5469e-01 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 1.359557e+03 FES = 294100 μ_F = 0.749 μ_{CR} = 0.048
 Diversity = 7.4815e-01 k_t = [4, 4, 2, 3]
 Generation 2950: Best = 1.359557e+03 FES = 295100 μ_F = 0.812 μ_{CR} = 0.041
 Diversity = 7.3787e-01 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 1.358050e+03 FES = 296100 μ_F = 0.811 μ_{CR} = 0.049
 Diversity = 7.3645e-01 k_t = [3, 3, 4, 3]
 Generation 2970: Best = 1.358050e+03 FES = 297100 μ_F = 0.746 μ_{CR} = 0.049
 Diversity = 7.3676e-01 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 1.358050e+03 FES = 298100 μ_F = 0.734 μ_{CR} = 0.051
 Diversity = 7.3359e-01 k_t = [2, 3, 4, 2]
 Generation 2990: Best = 1.358050e+03 FES = 299100 μ_F = 0.688 μ_{CR} = 0.053
 Diversity = 7.3279e-01 k_t = [1, 1, 1, 1]

Optimization complete.

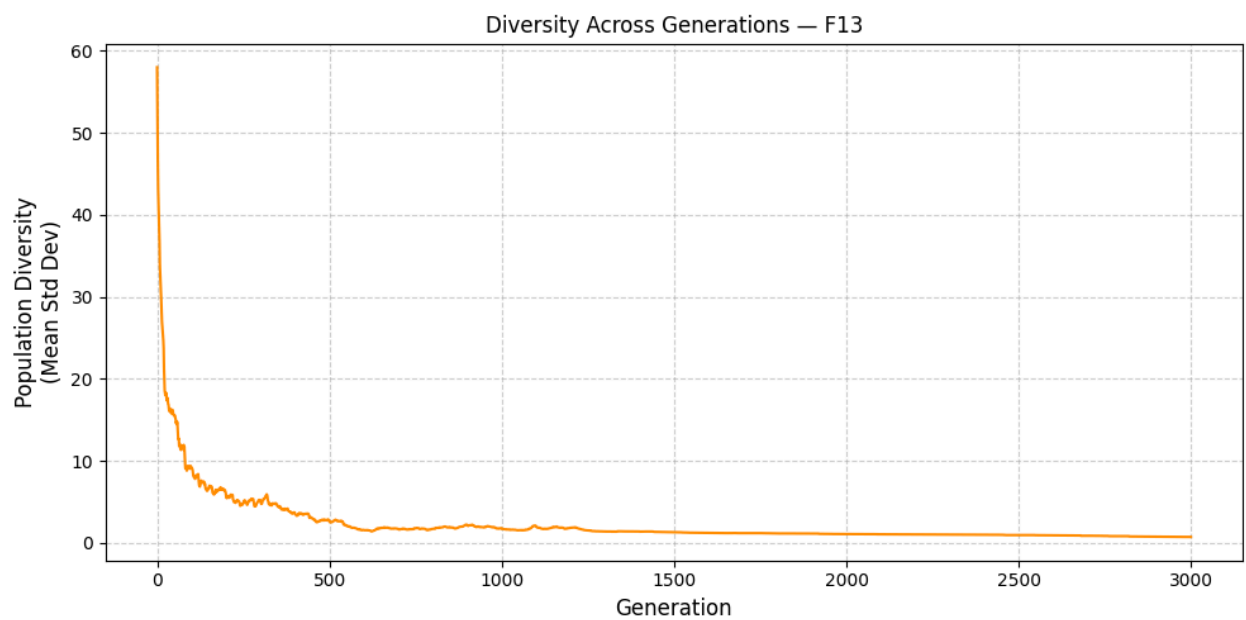
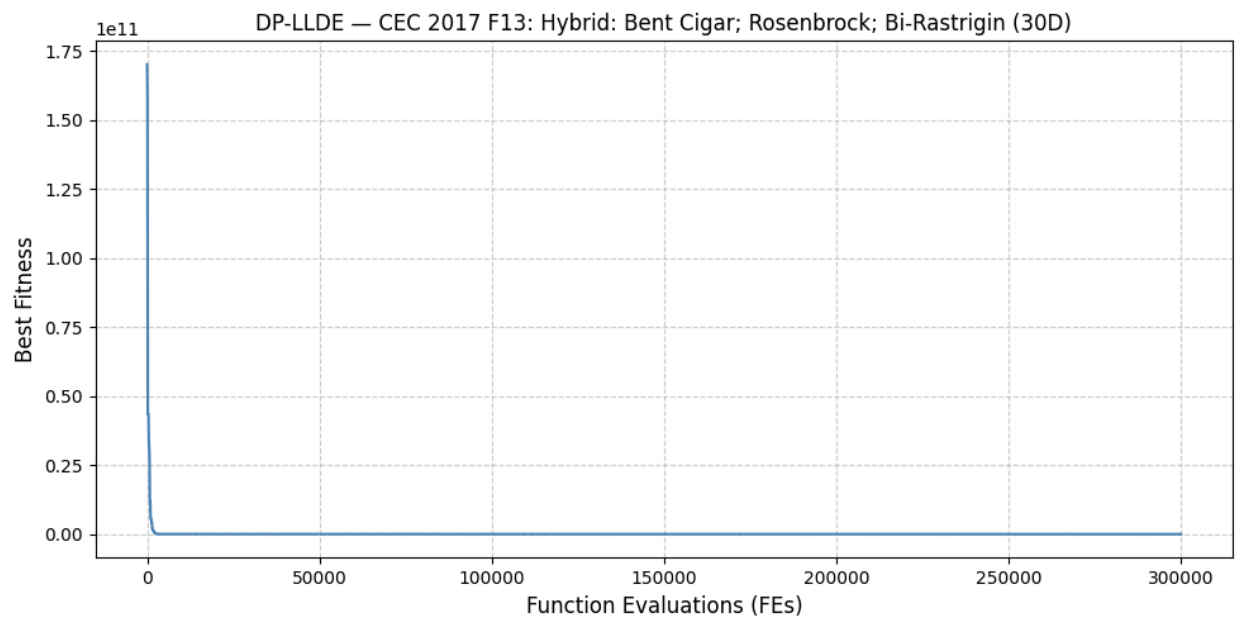
Best Fitness = 1.358050e+03

Total Function Evals = 300000

Total Successful Updates = 27721

```

=====
=
Results:
  Best Fitness      : 1.358050e+03
  Global Optimum    : 1.300000e+03
  Error (|best-optimum|) : 5.804976e+01
  Best Solution (5 dim) : [ 2.2830033 -76.61900223 5.04324535 -14.46708761
37.94433759]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F13.png

```
In [ ]: # =====
# F14 – Hybrid Function 4 (Elliptic; Ackley; Schaffer; Rastrigin)
# =====
run_and_plot('F14', make_wrapper(hybrid.f14), 1400.0, 'Hybrid: Elliptic; Ackle
```

=====

=

DP-LLDE on CEC 2017 F14 – Single Run

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Function : Hybrid: Elliptic; Ackley; Schaffer; Rastrigin (30D)

Bounds : [-100, 100]

Global optimum : 1400.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 3.706702e+07

Generation 10: Best = 1.965078e+06 FES = 1100 $\mu_F = 0.572$ $\mu_{CR} = 0.340$

Diversity = 4.3504e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 1.634271e+05 FES = 2100 $\mu_F = 0.623$ $\mu_{CR} = 0.303$

Diversity = 3.5217e+01 $k_t = [4, 3, 4, 3]$

Generation 30: Best = 1.633563e+05 FES = 3100 $\mu_F = 0.665$ $\mu_{CR} = 0.250$

Diversity = 3.0257e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 1.970422e+04 FES = 4100 $\mu_F = 0.707$ $\mu_{CR} = 0.188$

Diversity = 2.9020e+01 $k_t = [3, 4, 2, 2]$

Generation 50: Best = 1.612298e+04 FES = 5100 $\mu_F = 0.726$ $\mu_{CR} = 0.154$

Diversity = 2.8537e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 5.262157e+03 FES = 6100 $\mu_F = 0.749$ $\mu_{CR} = 0.125$

Diversity = 2.7500e+01 $k_t = [3, 2, 4, 4]$

Generation 70: Best = 2.633994e+03 FES = 7100 $\mu_F = 0.757$ $\mu_{CR} = 0.110$

Diversity = 2.4088e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 2.197039e+03 FES = 8100 $\mu_F = 0.765$ $\mu_{CR} = 0.089$

Diversity = 2.4220e+01 $k_t = [3, 4, 2, 3]$

Generation 90: Best = 1.874382e+03 FES = 9100 $\mu_F = 0.778$ $\mu_{CR} = 0.080$

Diversity = 2.3814e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 1.799742e+03 FES = 10100 $\mu_F = 0.789$ $\mu_{CR} = 0.081$

Diversity = 2.1969e+01 $k_t = [3, 4, 4, 4]$

Generation 110: Best = 1.660833e+03 FES = 11100 $\mu_F = 0.803$ $\mu_{CR} = 0.082$

Diversity = 2.0538e+01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 1.636007e+03 FES = 12100 $\mu_F = 0.795$ $\mu_{CR} = 0.078$

Diversity = 1.9928e+01 $k_t = [2, 2, 3, 3]$

Generation 130: Best = 1.553368e+03 FES = 13100 $\mu_F = 0.791$ $\mu_{CR} = 0.082$

Diversity = 1.8187e+01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 1.553368e+03 FES = 14100 $\mu_F = 0.779$ $\mu_{CR} = 0.080$

Diversity = 1.5714e+01 $k_t = [3, 2, 3, 3]$

Generation 150: Best = 1.553368e+03 FES = 15100 $\mu_F = 0.768$ $\mu_{CR} = 0.070$

Diversity = 1.4403e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 1.516320e+03 FES = 16100 $\mu_F = 0.762$ $\mu_{CR} = 0.068$

Diversity = 1.3398e+01 $k_t = [3, 4, 3, 3]$

Generation 170: Best = 1.516320e+03 FES = 17100 $\mu_F = 0.751$ $\mu_{CR} = 0.061$

Diversity = 1.2540e+01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 1.516320e+03 FES = 18100 $\mu_F = 0.762$ $\mu_{CR} = 0.065$

Diversity = 1.1481e+01 $k_t = [3, 2, 3, 3]$

Generation 190: Best = 1.516320e+03 FES = 19100 $\mu_F = 0.780$ $\mu_{CR} = 0.063$

Diversity = 1.1553e+01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 1.508895e+03 FES = 20100 $\mu_F = 0.782$ $\mu_{CR} = 0.055$

Diversity = 1.1400e+01 $k_t = [2, 2, 3, 3]$

Generation 210: Best = 1.508895e+03 FES = 21100 $\mu_F = 0.795$ $\mu_{CR} = 0.057$

Diversity = 1.1508e+01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 1.507410e+03 FES = 22100 $\mu_F = 0.805$ $\mu_{CR} = 0.057$

Diversity = 1.1354e+01 $k_t = [4, 2, 3, 2]$

Generation 230: Best = 1.500389e+03	FES = 23100	$\mu_F = 0.793$	$\mu_{CR} = 0.058$
Diversity = 1.1380e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 1.500389e+03	FES = 24100	$\mu_F = 0.805$	$\mu_{CR} = 0.057$
Diversity = 1.1238e+01	$k_t = [4, 2, 3, 2]$		
Generation 250: Best = 1.498462e+03	FES = 25100	$\mu_F = 0.794$	$\mu_{CR} = 0.057$
Diversity = 1.1362e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.496436e+03	FES = 26100	$\mu_F = 0.823$	$\mu_{CR} = 0.060$
Diversity = 1.1018e+01	$k_t = [2, 2, 2, 3]$		
Generation 270: Best = 1.496436e+03	FES = 27100	$\mu_F = 0.841$	$\mu_{CR} = 0.050$
Diversity = 1.0996e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.494346e+03	FES = 28100	$\mu_F = 0.834$	$\mu_{CR} = 0.055$
Diversity = 1.0792e+01	$k_t = [3, 2, 2, 2]$		
Generation 290: Best = 1.494346e+03	FES = 29100	$\mu_F = 0.801$	$\mu_{CR} = 0.049$
Diversity = 1.0650e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.494346e+03	FES = 30100	$\mu_F = 0.807$	$\mu_{CR} = 0.052$
Diversity = 1.0615e+01	$k_t = [4, 2, 2, 2]$		
Generation 310: Best = 1.490909e+03	FES = 31100	$\mu_F = 0.824$	$\mu_{CR} = 0.061$
Diversity = 1.0585e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.490909e+03	FES = 32100	$\mu_F = 0.826$	$\mu_{CR} = 0.057$
Diversity = 1.0470e+01	$k_t = [4, 4, 4, 3]$		
Generation 330: Best = 1.490909e+03	FES = 33100	$\mu_F = 0.792$	$\mu_{CR} = 0.051$
Diversity = 1.0419e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.490624e+03	FES = 34100	$\mu_F = 0.788$	$\mu_{CR} = 0.045$
Diversity = 1.0427e+01	$k_t = [3, 2, 3, 3]$		
Generation 350: Best = 1.487536e+03	FES = 35100	$\mu_F = 0.805$	$\mu_{CR} = 0.041$
Diversity = 1.0312e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.484863e+03	FES = 36100	$\mu_F = 0.815$	$\mu_{CR} = 0.033$
Diversity = 1.0241e+01	$k_t = [3, 4, 4, 4]$		
Generation 370: Best = 1.482532e+03	FES = 37100	$\mu_F = 0.768$	$\mu_{CR} = 0.037$
Diversity = 1.0206e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 1.471799e+03	FES = 38100	$\mu_F = 0.811$	$\mu_{CR} = 0.049$
Diversity = 9.9204e+00	$k_t = [4, 3, 3, 4]$		
Generation 390: Best = 1.471799e+03	FES = 39100	$\mu_F = 0.836$	$\mu_{CR} = 0.045$
Diversity = 9.9615e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 1.471799e+03	FES = 40100	$\mu_F = 0.724$	$\mu_{CR} = 0.055$
Diversity = 9.8987e+00	$k_t = [3, 2, 3, 2]$		
Generation 410: Best = 1.471799e+03	FES = 41100	$\mu_F = 0.635$	$\mu_{CR} = 0.056$
Diversity = 9.9625e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 1.471799e+03	FES = 42100	$\mu_F = 0.666$	$\mu_{CR} = 0.054$
Diversity = 1.0017e+01	$k_t = [3, 2, 3, 3]$		
Generation 430: Best = 1.471799e+03	FES = 43100	$\mu_F = 0.631$	$\mu_{CR} = 0.062$
Diversity = 1.0035e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 1.471783e+03	FES = 44100	$\mu_F = 0.606$	$\mu_{CR} = 0.056$
Diversity = 1.0020e+01	$k_t = [3, 2, 4, 2]$		
Generation 450: Best = 1.471627e+03	FES = 45100	$\mu_F = 0.646$	$\mu_{CR} = 0.057$
Diversity = 9.8461e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 1.471627e+03	FES = 46100	$\mu_F = 0.670$	$\mu_{CR} = 0.061$
Diversity = 9.7933e+00	$k_t = [2, 3, 4, 3]$		
Generation 470: Best = 1.471627e+03	FES = 47100	$\mu_F = 0.697$	$\mu_{CR} = 0.057$
Diversity = 9.7892e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.471627e+03	FES = 48100	$\mu_F = 0.703$	$\mu_{CR} = 0.048$
Diversity = 9.7045e+00	$k_t = [2, 3, 4, 2]$		
Generation 490: Best = 1.471627e+03	FES = 49100	$\mu_F = 0.748$	$\mu_{CR} = 0.055$
Diversity = 9.7898e+00	$k_t = [1, 1, 1, 1]$		

Generation 500:	Best = 1.471627e+03	FES = 50100	$\mu_F = 0.704$	$\mu_{CR} = 0.056$
Diversity = 9.6829e+00	$k_t = [2, 2, 4, 2]$			
Generation 510:	Best = 1.468960e+03	FES = 51100	$\mu_F = 0.721$	$\mu_{CR} = 0.053$
Diversity = 9.6421e+00	$k_t = [1, 1, 1, 1]$			
Generation 520:	Best = 1.468960e+03	FES = 52100	$\mu_F = 0.776$	$\mu_{CR} = 0.045$
Diversity = 9.6622e+00	$k_t = [4, 4, 4, 2]$			
Generation 530:	Best = 1.468742e+03	FES = 53100	$\mu_F = 0.742$	$\mu_{CR} = 0.054$
Diversity = 9.6995e+00	$k_t = [1, 1, 1, 1]$			
Generation 540:	Best = 1.466885e+03	FES = 54100	$\mu_F = 0.708$	$\mu_{CR} = 0.038$
Diversity = 9.6889e+00	$k_t = [2, 3, 4, 4]$			
Generation 550:	Best = 1.466778e+03	FES = 55100	$\mu_F = 0.707$	$\mu_{CR} = 0.042$
Diversity = 9.6927e+00	$k_t = [1, 1, 1, 1]$			
Generation 560:	Best = 1.466778e+03	FES = 56100	$\mu_F = 0.709$	$\mu_{CR} = 0.052$
Diversity = 9.7225e+00	$k_t = [2, 3, 4, 2]$			
Generation 570:	Best = 1.466778e+03	FES = 57100	$\mu_F = 0.756$	$\mu_{CR} = 0.052$
Diversity = 9.7074e+00	$k_t = [1, 1, 1, 1]$			
Generation 580:	Best = 1.466778e+03	FES = 58100	$\mu_F = 0.767$	$\mu_{CR} = 0.045$
Diversity = 9.6589e+00	$k_t = [2, 3, 4, 2]$			
Generation 590:	Best = 1.466778e+03	FES = 59100	$\mu_F = 0.831$	$\mu_{CR} = 0.050$
Diversity = 9.6728e+00	$k_t = [1, 1, 1, 1]$			
Generation 600:	Best = 1.466778e+03	FES = 60100	$\mu_F = 0.820$	$\mu_{CR} = 0.051$
Diversity = 9.6616e+00	$k_t = [4, 3, 4, 3]$			
Generation 610:	Best = 1.466778e+03	FES = 61100	$\mu_F = 0.816$	$\mu_{CR} = 0.056$
Diversity = 9.6875e+00	$k_t = [1, 1, 1, 1]$			
Generation 620:	Best = 1.464592e+03	FES = 62100	$\mu_F = 0.809$	$\mu_{CR} = 0.045$
Diversity = 9.6555e+00	$k_t = [4, 3, 4, 2]$			
Generation 630:	Best = 1.464592e+03	FES = 63100	$\mu_F = 0.733$	$\mu_{CR} = 0.042$
Diversity = 9.6279e+00	$k_t = [1, 1, 1, 1]$			
Generation 640:	Best = 1.464592e+03	FES = 64100	$\mu_F = 0.775$	$\mu_{CR} = 0.052$
Diversity = 9.5445e+00	$k_t = [4, 4, 3, 2]$			
Generation 650:	Best = 1.464592e+03	FES = 65100	$\mu_F = 0.737$	$\mu_{CR} = 0.043$
Diversity = 9.5631e+00	$k_t = [1, 1, 1, 1]$			
Generation 660:	Best = 1.464592e+03	FES = 66100	$\mu_F = 0.734$	$\mu_{CR} = 0.050$
Diversity = 9.5010e+00	$k_t = [2, 2, 2, 3]$			
Generation 670:	Best = 1.464592e+03	FES = 67100	$\mu_F = 0.762$	$\mu_{CR} = 0.053$
Diversity = 9.5186e+00	$k_t = [1, 1, 1, 1]$			
Generation 680:	Best = 1.464592e+03	FES = 68100	$\mu_F = 0.726$	$\mu_{CR} = 0.050$
Diversity = 9.5197e+00	$k_t = [4, 3, 4, 2]$			
Generation 690:	Best = 1.464592e+03	FES = 69100	$\mu_F = 0.763$	$\mu_{CR} = 0.053$
Diversity = 9.4335e+00	$k_t = [1, 1, 1, 1]$			
Generation 700:	Best = 1.463382e+03	FES = 70100	$\mu_F = 0.775$	$\mu_{CR} = 0.049$
Diversity = 9.3484e+00	$k_t = [2, 3, 3, 4]$			
Generation 710:	Best = 1.462740e+03	FES = 71100	$\mu_F = 0.779$	$\mu_{CR} = 0.039$
Diversity = 9.4477e+00	$k_t = [1, 1, 1, 1]$			
Generation 720:	Best = 1.461101e+03	FES = 72100	$\mu_F = 0.774$	$\mu_{CR} = 0.035$
Diversity = 9.3592e+00	$k_t = [2, 3, 4, 4]$			
Generation 730:	Best = 1.459956e+03	FES = 73100	$\mu_F = 0.831$	$\mu_{CR} = 0.038$
Diversity = 9.3152e+00	$k_t = [1, 1, 1, 1]$			
Generation 740:	Best = 1.459956e+03	FES = 74100	$\mu_F = 0.775$	$\mu_{CR} = 0.051$
Diversity = 9.2925e+00	$k_t = [2, 2, 3, 2]$			
Generation 750:	Best = 1.459956e+03	FES = 75100	$\mu_F = 0.784$	$\mu_{CR} = 0.045$
Diversity = 9.2581e+00	$k_t = [1, 1, 1, 1]$			
Generation 760:	Best = 1.459492e+03	FES = 76100	$\mu_F = 0.779$	$\mu_{CR} = 0.046$
Diversity = 9.2771e+00	$k_t = [2, 2, 4, 2]$			

Generation 770:	Best = 1.459492e+03	FES = 77100	$\mu_F = 0.782$	$\mu_{CR} = 0.050$
Diversity = 9.2774e+00	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 1.459492e+03	FES = 78100	$\mu_F = 0.794$	$\mu_{CR} = 0.047$
Diversity = 9.1413e+00	$k_t = [3, 2, 2, 4]$			
Generation 790:	Best = 1.459492e+03	FES = 79100	$\mu_F = 0.825$	$\mu_{CR} = 0.053$
Diversity = 9.0610e+00	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 1.459492e+03	FES = 80100	$\mu_F = 0.861$	$\mu_{CR} = 0.055$
Diversity = 8.9416e+00	$k_t = [4, 4, 2, 4]$			
Generation 810:	Best = 1.459492e+03	FES = 81100	$\mu_F = 0.790$	$\mu_{CR} = 0.047$
Diversity = 8.9875e+00	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 1.459492e+03	FES = 82100	$\mu_F = 0.807$	$\mu_{CR} = 0.073$
Diversity = 8.8598e+00	$k_t = [4, 4, 2, 4]$			
Generation 830:	Best = 1.456696e+03	FES = 83100	$\mu_F = 0.831$	$\mu_{CR} = 0.058$
Diversity = 8.9587e+00	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 1.456696e+03	FES = 84100	$\mu_F = 0.837$	$\mu_{CR} = 0.049$
Diversity = 8.9940e+00	$k_t = [4, 2, 2, 3]$			
Generation 850:	Best = 1.456696e+03	FES = 85100	$\mu_F = 0.798$	$\mu_{CR} = 0.053$
Diversity = 9.0182e+00	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 1.456218e+03	FES = 86100	$\mu_F = 0.706$	$\mu_{CR} = 0.048$
Diversity = 9.0638e+00	$k_t = [2, 2, 2, 2]$			
Generation 870:	Best = 1.456218e+03	FES = 87100	$\mu_F = 0.646$	$\mu_{CR} = 0.038$
Diversity = 9.0856e+00	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 1.455663e+03	FES = 88100	$\mu_F = 0.640$	$\mu_{CR} = 0.033$
Diversity = 9.0995e+00	$k_t = [3, 3, 2, 2]$			
Generation 890:	Best = 1.455663e+03	FES = 89100	$\mu_F = 0.596$	$\mu_{CR} = 0.035$
Diversity = 9.0745e+00	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 1.454462e+03	FES = 90100	$\mu_F = 0.623$	$\mu_{CR} = 0.029$
Diversity = 9.0767e+00	$k_t = [3, 3, 3, 2]$			
Generation 910:	Best = 1.454462e+03	FES = 91100	$\mu_F = 0.606$	$\mu_{CR} = 0.050$
Diversity = 9.0121e+00	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 1.454462e+03	FES = 92100	$\mu_F = 0.669$	$\mu_{CR} = 0.046$
Diversity = 8.9781e+00	$k_t = [3, 2, 4, 2]$			
Generation 930:	Best = 1.454369e+03	FES = 93100	$\mu_F = 0.697$	$\mu_{CR} = 0.041$
Diversity = 8.9842e+00	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 1.454369e+03	FES = 94100	$\mu_F = 0.651$	$\mu_{CR} = 0.040$
Diversity = 8.9096e+00	$k_t = [3, 2, 4, 2]$			
Generation 950:	Best = 1.454369e+03	FES = 95100	$\mu_F = 0.622$	$\mu_{CR} = 0.045$
Diversity = 8.9050e+00	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 1.454369e+03	FES = 96100	$\mu_F = 0.607$	$\mu_{CR} = 0.052$
Diversity = 8.8331e+00	$k_t = [2, 3, 2, 3]$			
Generation 970:	Best = 1.452791e+03	FES = 97100	$\mu_F = 0.671$	$\mu_{CR} = 0.053$
Diversity = 8.7950e+00	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 1.451636e+03	FES = 98100	$\mu_F = 0.706$	$\mu_{CR} = 0.053$
Diversity = 8.7365e+00	$k_t = [4, 3, 4, 4]$			
Generation 990:	Best = 1.451093e+03	FES = 99100	$\mu_F = 0.664$	$\mu_{CR} = 0.064$
Diversity = 8.7127e+00	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 1.450662e+03	FES = 100100	$\mu_F = 0.688$	$\mu_{CR} = 0.063$
Diversity = 8.5423e+00	$k_t = [2, 4, 4, 4]$			
Generation 1010:	Best = 1.450662e+03	FES = 101100	$\mu_F = 0.805$	$\mu_{CR} = 0.066$
Diversity = 8.5505e+00	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 1.450662e+03	FES = 102100	$\mu_F = 0.770$	$\mu_{CR} = 0.070$
Diversity = 8.5453e+00	$k_t = [4, 3, 2, 3]$			
Generation 1030:	Best = 1.449102e+03	FES = 103100	$\mu_F = 0.793$	$\mu_{CR} = 0.064$
Diversity = 8.4152e+00	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 1.448264e+03 FES = 104100 μ_F = 0.808 μ_{CR} = 0.069
 Diversity = 8.4748e+00 k_t = [3, 4, 4, 2]
 Generation 1050: Best = 1.448264e+03 FES = 105100 μ_F = 0.796 μ_{CR} = 0.059
 Diversity = 8.4266e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 1.448264e+03 FES = 106100 μ_F = 0.827 μ_{CR} = 0.057
 Diversity = 8.2574e+00 k_t = [4, 3, 3, 4]
 Generation 1070: Best = 1.448264e+03 FES = 107100 μ_F = 0.842 μ_{CR} = 0.039
 Diversity = 8.2996e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 1.448264e+03 FES = 108100 μ_F = 0.849 μ_{CR} = 0.066
 Diversity = 8.2166e+00 k_t = [2, 2, 4, 3]
 Generation 1090: Best = 1.447862e+03 FES = 109100 μ_F = 0.823 μ_{CR} = 0.070
 Diversity = 8.1429e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.447862e+03 FES = 110100 μ_F = 0.819 μ_{CR} = 0.084
 Diversity = 8.1815e+00 k_t = [4, 3, 4, 2]
 Generation 1110: Best = 1.447862e+03 FES = 111100 μ_F = 0.849 μ_{CR} = 0.064
 Diversity = 8.1328e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.447849e+03 FES = 112100 μ_F = 0.847 μ_{CR} = 0.070
 Diversity = 8.1182e+00 k_t = [2, 4, 4, 2]
 Generation 1130: Best = 1.447849e+03 FES = 113100 μ_F = 0.835 μ_{CR} = 0.062
 Diversity = 8.0665e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.447849e+03 FES = 114100 μ_F = 0.806 μ_{CR} = 0.058
 Diversity = 8.0660e+00 k_t = [4, 4, 4, 4]
 Generation 1150: Best = 1.447849e+03 FES = 115100 μ_F = 0.814 μ_{CR} = 0.064
 Diversity = 7.9844e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.447459e+03 FES = 116100 μ_F = 0.814 μ_{CR} = 0.061
 Diversity = 7.9686e+00 k_t = [4, 3, 4, 4]
 Generation 1170: Best = 1.447131e+03 FES = 117100 μ_F = 0.773 μ_{CR} = 0.064
 Diversity = 8.0669e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.447131e+03 FES = 118100 μ_F = 0.802 μ_{CR} = 0.058
 Diversity = 7.8361e+00 k_t = [2, 3, 2, 4]
 Generation 1190: Best = 1.447131e+03 FES = 119100 μ_F = 0.828 μ_{CR} = 0.065
 Diversity = 7.7062e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.447131e+03 FES = 120100 μ_F = 0.847 μ_{CR} = 0.057
 Diversity = 7.5972e+00 k_t = [2, 3, 2, 2]
 Generation 1210: Best = 1.447131e+03 FES = 121100 μ_F = 0.860 μ_{CR} = 0.053
 Diversity = 7.4765e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.447131e+03 FES = 122100 μ_F = 0.896 μ_{CR} = 0.061
 Diversity = 7.1123e+00 k_t = [2, 4, 3, 4]
 Generation 1230: Best = 1.446872e+03 FES = 123100 μ_F = 0.903 μ_{CR} = 0.045
 Diversity = 6.8212e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.446277e+03 FES = 124100 μ_F = 0.887 μ_{CR} = 0.051
 Diversity = 6.2631e+00 k_t = [3, 2, 3, 4]
 Generation 1250: Best = 1.446272e+03 FES = 125100 μ_F = 0.892 μ_{CR} = 0.056
 Diversity = 6.0221e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.446245e+03 FES = 126100 μ_F = 0.875 μ_{CR} = 0.060
 Diversity = 5.9223e+00 k_t = [2, 3, 2, 2]
 Generation 1270: Best = 1.446245e+03 FES = 127100 μ_F = 0.841 μ_{CR} = 0.063
 Diversity = 5.8010e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.445625e+03 FES = 128100 μ_F = 0.815 μ_{CR} = 0.060
 Diversity = 5.6235e+00 k_t = [3, 2, 2, 4]
 Generation 1290: Best = 1.445156e+03 FES = 129100 μ_F = 0.826 μ_{CR} = 0.068
 Diversity = 5.5540e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.445088e+03 FES = 130100 μ_F = 0.794 μ_{CR} = 0.073
 Diversity = 5.4257e+00 k_t = [3, 3, 3, 3]

Generation 1310: Best = 1.444996e+03 FES = 131100 μ_F = 0.805 μ_{CR} = 0.058
 Diversity = 5.4591e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.444996e+03 FES = 132100 μ_F = 0.846 μ_{CR} = 0.061
 Diversity = 5.4133e+00 k_t = [2, 2, 2, 2]
 Generation 1330: Best = 1.444776e+03 FES = 133100 μ_F = 0.829 μ_{CR} = 0.066
 Diversity = 5.3620e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.444402e+03 FES = 134100 μ_F = 0.839 μ_{CR} = 0.074
 Diversity = 5.1230e+00 k_t = [3, 4, 4, 4]
 Generation 1350: Best = 1.444112e+03 FES = 135100 μ_F = 0.827 μ_{CR} = 0.073
 Diversity = 4.4320e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.443936e+03 FES = 136100 μ_F = 0.793 μ_{CR} = 0.076
 Diversity = 4.0601e+00 k_t = [4, 2, 2, 2]
 Generation 1370: Best = 1.443737e+03 FES = 137100 μ_F = 0.794 μ_{CR} = 0.074
 Diversity = 3.5145e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.443723e+03 FES = 138100 μ_F = 0.818 μ_{CR} = 0.073
 Diversity = 2.6556e+00 k_t = [2, 3, 3, 4]
 Generation 1390: Best = 1.443321e+03 FES = 139100 μ_F = 0.821 μ_{CR} = 0.070
 Diversity = 2.6253e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.443316e+03 FES = 140100 μ_F = 0.768 μ_{CR} = 0.076
 Diversity = 2.5127e+00 k_t = [3, 2, 3, 2]
 Generation 1410: Best = 1.442861e+03 FES = 141100 μ_F = 0.774 μ_{CR} = 0.064
 Diversity = 2.4737e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.442861e+03 FES = 142100 μ_F = 0.775 μ_{CR} = 0.068
 Diversity = 2.3841e+00 k_t = [3, 3, 2, 4]
 Generation 1430: Best = 1.442848e+03 FES = 143100 μ_F = 0.761 μ_{CR} = 0.058
 Diversity = 2.2922e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.442728e+03 FES = 144100 μ_F = 0.758 μ_{CR} = 0.060
 Diversity = 2.2970e+00 k_t = [3, 2, 4, 2]
 Generation 1450: Best = 1.442635e+03 FES = 145100 μ_F = 0.753 μ_{CR} = 0.056
 Diversity = 2.4555e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.442635e+03 FES = 146100 μ_F = 0.744 μ_{CR} = 0.061
 Diversity = 2.4677e+00 k_t = [2, 2, 2, 3]
 Generation 1470: Best = 1.442629e+03 FES = 147100 μ_F = 0.759 μ_{CR} = 0.060
 Diversity = 2.4611e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.442615e+03 FES = 148100 μ_F = 0.765 μ_{CR} = 0.066
 Diversity = 2.3088e+00 k_t = [3, 3, 4, 4]
 Generation 1490: Best = 1.442478e+03 FES = 149100 μ_F = 0.791 μ_{CR} = 0.067
 Diversity = 2.3087e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.442469e+03 FES = 150100 μ_F = 0.812 μ_{CR} = 0.068
 Diversity = 2.2824e+00 k_t = [3, 4, 4, 3]
 Generation 1510: Best = 1.442462e+03 FES = 151100 μ_F = 0.792 μ_{CR} = 0.059
 Diversity = 2.2902e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.442462e+03 FES = 152100 μ_F = 0.782 μ_{CR} = 0.055
 Diversity = 2.2370e+00 k_t = [2, 2, 3, 4]
 Generation 1530: Best = 1.442462e+03 FES = 153100 μ_F = 0.769 μ_{CR} = 0.054
 Diversity = 2.1635e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.442340e+03 FES = 154100 μ_F = 0.781 μ_{CR} = 0.051
 Diversity = 2.2376e+00 k_t = [2, 2, 4, 4]
 Generation 1550: Best = 1.442323e+03 FES = 155100 μ_F = 0.783 μ_{CR} = 0.047
 Diversity = 2.2351e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.442320e+03 FES = 156100 μ_F = 0.793 μ_{CR} = 0.049
 Diversity = 2.2343e+00 k_t = [3, 2, 2, 4]
 Generation 1570: Best = 1.442289e+03 FES = 157100 μ_F = 0.816 μ_{CR} = 0.048
 Diversity = 2.2318e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.442237e+03 FES = 158100 μ_F = 0.854 μ_{CR} = 0.055
 Diversity = 2.2527e+00 k_t = [3, 3, 4, 2]
 Generation 1590: Best = 1.442172e+03 FES = 159100 μ_F = 0.828 μ_{CR} = 0.056
 Diversity = 2.2663e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.442059e+03 FES = 160100 μ_F = 0.835 μ_{CR} = 0.056
 Diversity = 2.2172e+00 k_t = [2, 3, 3, 2]
 Generation 1610: Best = 1.442058e+03 FES = 161100 μ_F = 0.810 μ_{CR} = 0.049
 Diversity = 2.1573e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.442058e+03 FES = 162100 μ_F = 0.800 μ_{CR} = 0.051
 Diversity = 2.1731e+00 k_t = [3, 2, 3, 4]
 Generation 1630: Best = 1.442011e+03 FES = 163100 μ_F = 0.812 μ_{CR} = 0.050
 Diversity = 2.1803e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.442011e+03 FES = 164100 μ_F = 0.805 μ_{CR} = 0.060
 Diversity = 2.2470e+00 k_t = [3, 4, 4, 2]
 Generation 1650: Best = 1.442011e+03 FES = 165100 μ_F = 0.812 μ_{CR} = 0.061
 Diversity = 2.2441e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.442011e+03 FES = 166100 μ_F = 0.825 μ_{CR} = 0.054
 Diversity = 2.3065e+00 k_t = [3, 2, 3, 4]
 Generation 1670: Best = 1.442011e+03 FES = 167100 μ_F = 0.783 μ_{CR} = 0.049
 Diversity = 2.3091e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.442011e+03 FES = 168100 μ_F = 0.762 μ_{CR} = 0.060
 Diversity = 2.3959e+00 k_t = [3, 4, 4, 4]
 Generation 1690: Best = 1.442011e+03 FES = 169100 μ_F = 0.758 μ_{CR} = 0.056
 Diversity = 2.3894e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.442001e+03 FES = 170100 μ_F = 0.740 μ_{CR} = 0.059
 Diversity = 2.3478e+00 k_t = [2, 2, 3, 4]
 Generation 1710: Best = 1.442001e+03 FES = 171100 μ_F = 0.746 μ_{CR} = 0.064
 Diversity = 2.4288e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.441960e+03 FES = 172100 μ_F = 0.728 μ_{CR} = 0.056
 Diversity = 2.4452e+00 k_t = [2, 4, 4, 4]
 Generation 1730: Best = 1.441960e+03 FES = 173100 μ_F = 0.727 μ_{CR} = 0.056
 Diversity = 2.4273e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.441949e+03 FES = 174100 μ_F = 0.712 μ_{CR} = 0.050
 Diversity = 2.3852e+00 k_t = [2, 2, 2, 2]
 Generation 1750: Best = 1.441949e+03 FES = 175100 μ_F = 0.706 μ_{CR} = 0.052
 Diversity = 2.3920e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.441943e+03 FES = 176100 μ_F = 0.729 μ_{CR} = 0.045
 Diversity = 2.4094e+00 k_t = [3, 3, 4, 2]
 Generation 1770: Best = 1.441903e+03 FES = 177100 μ_F = 0.758 μ_{CR} = 0.046
 Diversity = 2.4415e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.441903e+03 FES = 178100 μ_F = 0.765 μ_{CR} = 0.068
 Diversity = 2.4565e+00 k_t = [3, 4, 4, 3]
 Generation 1790: Best = 1.441869e+03 FES = 179100 μ_F = 0.732 μ_{CR} = 0.057
 Diversity = 2.5547e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 1.441861e+03 FES = 180100 μ_F = 0.778 μ_{CR} = 0.060
 Diversity = 2.6384e+00 k_t = [4, 4, 2, 4]
 Generation 1810: Best = 1.441861e+03 FES = 181100 μ_F = 0.778 μ_{CR} = 0.052
 Diversity = 2.6191e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 1.441861e+03 FES = 182100 μ_F = 0.753 μ_{CR} = 0.059
 Diversity = 2.6357e+00 k_t = [2, 3, 4, 3]
 Generation 1830: Best = 1.441861e+03 FES = 183100 μ_F = 0.756 μ_{CR} = 0.046
 Diversity = 2.6240e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 1.441860e+03 FES = 184100 μ_F = 0.772 μ_{CR} = 0.051
 Diversity = 2.6460e+00 k_t = [4, 3, 4, 3]

Generation 1850: Best = 1.441860e+03 FES = 185100 μ_F = 0.765 μ_{CR} = 0.060
 Diversity = 2.6632e+00 k_t = [1, 1, 1, 1]
 Generation 1860: Best = 1.441860e+03 FES = 186100 μ_F = 0.780 μ_{CR} = 0.049
 Diversity = 2.6958e+00 k_t = [3, 4, 2, 2]
 Generation 1870: Best = 1.441860e+03 FES = 187100 μ_F = 0.811 μ_{CR} = 0.049
 Diversity = 2.7588e+00 k_t = [1, 1, 1, 1]
 Generation 1880: Best = 1.441860e+03 FES = 188100 μ_F = 0.795 μ_{CR} = 0.059
 Diversity = 2.8478e+00 k_t = [3, 3, 3, 3]
 Generation 1890: Best = 1.441858e+03 FES = 189100 μ_F = 0.814 μ_{CR} = 0.065
 Diversity = 3.0208e+00 k_t = [1, 1, 1, 1]
 Generation 1900: Best = 1.441539e+03 FES = 190100 μ_F = 0.834 μ_{CR} = 0.057
 Diversity = 3.3162e+00 k_t = [3, 3, 2, 3]
 Generation 1910: Best = 1.441468e+03 FES = 191100 μ_F = 0.819 μ_{CR} = 0.053
 Diversity = 3.4091e+00 k_t = [1, 1, 1, 1]
 Generation 1920: Best = 1.441337e+03 FES = 192100 μ_F = 0.820 μ_{CR} = 0.056
 Diversity = 3.4403e+00 k_t = [3, 4, 3, 4]
 Generation 1930: Best = 1.441337e+03 FES = 193100 μ_F = 0.818 μ_{CR} = 0.054
 Diversity = 3.5811e+00 k_t = [1, 1, 1, 1]
 Generation 1940: Best = 1.441334e+03 FES = 194100 μ_F = 0.787 μ_{CR} = 0.060
 Diversity = 3.5749e+00 k_t = [3, 2, 3, 3]
 Generation 1950: Best = 1.441333e+03 FES = 195100 μ_F = 0.820 μ_{CR} = 0.054
 Diversity = 3.8519e+00 k_t = [1, 1, 1, 1]
 Generation 1960: Best = 1.441301e+03 FES = 196100 μ_F = 0.820 μ_{CR} = 0.056
 Diversity = 3.9436e+00 k_t = [2, 3, 3, 2]
 Generation 1970: Best = 1.441211e+03 FES = 197100 μ_F = 0.818 μ_{CR} = 0.055
 Diversity = 3.9870e+00 k_t = [1, 1, 1, 1]
 Generation 1980: Best = 1.441211e+03 FES = 198100 μ_F = 0.805 μ_{CR} = 0.046
 Diversity = 3.9267e+00 k_t = [4, 2, 2, 4]
 Generation 1990: Best = 1.441191e+03 FES = 199100 μ_F = 0.793 μ_{CR} = 0.050
 Diversity = 3.9889e+00 k_t = [1, 1, 1, 1]
 Generation 2000: Best = 1.441191e+03 FES = 200100 μ_F = 0.812 μ_{CR} = 0.044
 Diversity = 4.2224e+00 k_t = [3, 4, 2, 4]
 Generation 2010: Best = 1.441175e+03 FES = 201100 μ_F = 0.801 μ_{CR} = 0.045
 Diversity = 4.4072e+00 k_t = [1, 1, 1, 1]
 Generation 2020: Best = 1.441175e+03 FES = 202100 μ_F = 0.800 μ_{CR} = 0.059
 Diversity = 4.6446e+00 k_t = [2, 3, 4, 3]
 Generation 2030: Best = 1.441175e+03 FES = 203100 μ_F = 0.805 μ_{CR} = 0.057
 Diversity = 4.8900e+00 k_t = [1, 1, 1, 1]
 Generation 2040: Best = 1.441142e+03 FES = 204100 μ_F = 0.832 μ_{CR} = 0.053
 Diversity = 4.9403e+00 k_t = [4, 4, 2, 2]
 Generation 2050: Best = 1.441126e+03 FES = 205100 μ_F = 0.821 μ_{CR} = 0.060
 Diversity = 4.9916e+00 k_t = [1, 1, 1, 1]
 Generation 2060: Best = 1.441070e+03 FES = 206100 μ_F = 0.815 μ_{CR} = 0.058
 Diversity = 5.2044e+00 k_t = [4, 4, 3, 4]
 Generation 2070: Best = 1.440800e+03 FES = 207100 μ_F = 0.827 μ_{CR} = 0.055
 Diversity = 5.1862e+00 k_t = [1, 1, 1, 1]
 Generation 2080: Best = 1.440790e+03 FES = 208100 μ_F = 0.812 μ_{CR} = 0.054
 Diversity = 5.1514e+00 k_t = [3, 4, 2, 4]
 Generation 2090: Best = 1.440790e+03 FES = 209100 μ_F = 0.812 μ_{CR} = 0.045
 Diversity = 5.1432e+00 k_t = [1, 1, 1, 1]
 Generation 2100: Best = 1.440783e+03 FES = 210100 μ_F = 0.840 μ_{CR} = 0.055
 Diversity = 5.1413e+00 k_t = [2, 2, 3, 4]
 Generation 2110: Best = 1.440721e+03 FES = 211100 μ_F = 0.843 μ_{CR} = 0.058
 Diversity = 5.1515e+00 k_t = [1, 1, 1, 1]

Generation 2120: Best = 1.440721e+03 FES = 212100 μ_F = 0.841 μ_{CR} = 0.062
 Diversity = 5.1428e+00 k_t = [4, 4, 4, 3]
 Generation 2130: Best = 1.440721e+03 FES = 213100 μ_F = 0.839 μ_{CR} = 0.060
 Diversity = 5.0412e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.440721e+03 FES = 214100 μ_F = 0.849 μ_{CR} = 0.058
 Diversity = 4.9632e+00 k_t = [4, 3, 4, 3]
 Generation 2150: Best = 1.440721e+03 FES = 215100 μ_F = 0.842 μ_{CR} = 0.059
 Diversity = 4.9906e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.440721e+03 FES = 216100 μ_F = 0.834 μ_{CR} = 0.066
 Diversity = 4.8659e+00 k_t = [2, 4, 3, 2]
 Generation 2170: Best = 1.440701e+03 FES = 217100 μ_F = 0.806 μ_{CR} = 0.064
 Diversity = 4.7795e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.440698e+03 FES = 218100 μ_F = 0.850 μ_{CR} = 0.060
 Diversity = 4.6412e+00 k_t = [3, 3, 4, 4]
 Generation 2190: Best = 1.440501e+03 FES = 219100 μ_F = 0.851 μ_{CR} = 0.056
 Diversity = 4.7619e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 1.440495e+03 FES = 220100 μ_F = 0.856 μ_{CR} = 0.071
 Diversity = 4.6825e+00 k_t = [2, 4, 2, 4]
 Generation 2210: Best = 1.440495e+03 FES = 221100 μ_F = 0.862 μ_{CR} = 0.059
 Diversity = 4.6394e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 1.440495e+03 FES = 222100 μ_F = 0.860 μ_{CR} = 0.046
 Diversity = 4.6583e+00 k_t = [4, 2, 2, 3]
 Generation 2230: Best = 1.440495e+03 FES = 223100 μ_F = 0.857 μ_{CR} = 0.050
 Diversity = 4.6687e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.440495e+03 FES = 224100 μ_F = 0.879 μ_{CR} = 0.049
 Diversity = 4.7057e+00 k_t = [2, 2, 2, 3]
 Generation 2250: Best = 1.440489e+03 FES = 225100 μ_F = 0.868 μ_{CR} = 0.051
 Diversity = 4.7799e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.440478e+03 FES = 226100 μ_F = 0.878 μ_{CR} = 0.062
 Diversity = 4.8042e+00 k_t = [2, 4, 2, 4]
 Generation 2270: Best = 1.440478e+03 FES = 227100 μ_F = 0.886 μ_{CR} = 0.050
 Diversity = 4.8311e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.440434e+03 FES = 228100 μ_F = 0.870 μ_{CR} = 0.058
 Diversity = 4.8102e+00 k_t = [3, 2, 4, 2]
 Generation 2290: Best = 1.440434e+03 FES = 229100 μ_F = 0.882 μ_{CR} = 0.052
 Diversity = 4.7441e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.440434e+03 FES = 230100 μ_F = 0.844 μ_{CR} = 0.062
 Diversity = 4.6895e+00 k_t = [2, 4, 2, 4]
 Generation 2310: Best = 1.440434e+03 FES = 231100 μ_F = 0.794 μ_{CR} = 0.054
 Diversity = 4.7479e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.440434e+03 FES = 232100 μ_F = 0.793 μ_{CR} = 0.056
 Diversity = 4.7346e+00 k_t = [4, 3, 4, 3]
 Generation 2330: Best = 1.440434e+03 FES = 233100 μ_F = 0.789 μ_{CR} = 0.050
 Diversity = 4.7439e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.440434e+03 FES = 234100 μ_F = 0.780 μ_{CR} = 0.054
 Diversity = 4.7825e+00 k_t = [4, 4, 2, 4]
 Generation 2350: Best = 1.440432e+03 FES = 235100 μ_F = 0.809 μ_{CR} = 0.052
 Diversity = 4.8379e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.440419e+03 FES = 236100 μ_F = 0.808 μ_{CR} = 0.049
 Diversity = 4.9588e+00 k_t = [3, 3, 4, 4]
 Generation 2370: Best = 1.440419e+03 FES = 237100 μ_F = 0.790 μ_{CR} = 0.049
 Diversity = 4.9738e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.440419e+03 FES = 238100 μ_F = 0.802 μ_{CR} = 0.057
 Diversity = 4.9735e+00 k_t = [3, 2, 2, 3]

Generation 2390: Best = 1.440419e+03 FES = 239100 μ_F = 0.816 μ_{CR} = 0.055
 Diversity = 4.9891e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.440419e+03 FES = 240100 μ_F = 0.810 μ_{CR} = 0.055
 Diversity = 5.0472e+00 k_t = [4, 3, 4, 3]
 Generation 2410: Best = 1.440419e+03 FES = 241100 μ_F = 0.822 μ_{CR} = 0.065
 Diversity = 5.1221e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.440342e+03 FES = 242100 μ_F = 0.838 μ_{CR} = 0.061
 Diversity = 5.1035e+00 k_t = [4, 4, 3, 2]
 Generation 2430: Best = 1.440342e+03 FES = 243100 μ_F = 0.842 μ_{CR} = 0.053
 Diversity = 5.0921e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.440341e+03 FES = 244100 μ_F = 0.783 μ_{CR} = 0.053
 Diversity = 5.0796e+00 k_t = [4, 2, 4, 2]
 Generation 2450: Best = 1.440339e+03 FES = 245100 μ_F = 0.830 μ_{CR} = 0.056
 Diversity = 5.0471e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.440339e+03 FES = 246100 μ_F = 0.845 μ_{CR} = 0.046
 Diversity = 5.0885e+00 k_t = [2, 3, 2, 2]
 Generation 2470: Best = 1.440339e+03 FES = 247100 μ_F = 0.863 μ_{CR} = 0.047
 Diversity = 5.0807e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.440334e+03 FES = 248100 μ_F = 0.855 μ_{CR} = 0.058
 Diversity = 5.0697e+00 k_t = [3, 3, 2, 3]
 Generation 2490: Best = 1.440334e+03 FES = 249100 μ_F = 0.843 μ_{CR} = 0.049
 Diversity = 5.1392e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.440334e+03 FES = 250100 μ_F = 0.831 μ_{CR} = 0.049
 Diversity = 5.1792e+00 k_t = [3, 4, 2, 4]
 Generation 2510: Best = 1.440332e+03 FES = 251100 μ_F = 0.841 μ_{CR} = 0.047
 Diversity = 5.1641e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.440332e+03 FES = 252100 μ_F = 0.855 μ_{CR} = 0.046
 Diversity = 5.1609e+00 k_t = [2, 3, 4, 2]
 Generation 2530: Best = 1.440332e+03 FES = 253100 μ_F = 0.819 μ_{CR} = 0.054
 Diversity = 5.2660e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.440317e+03 FES = 254100 μ_F = 0.839 μ_{CR} = 0.061
 Diversity = 5.2820e+00 k_t = [2, 3, 2, 2]
 Generation 2550: Best = 1.440317e+03 FES = 255100 μ_F = 0.802 μ_{CR} = 0.058
 Diversity = 5.2612e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.440299e+03 FES = 256100 μ_F = 0.765 μ_{CR} = 0.061
 Diversity = 5.2458e+00 k_t = [3, 3, 4, 3]
 Generation 2570: Best = 1.440299e+03 FES = 257100 μ_F = 0.794 μ_{CR} = 0.043
 Diversity = 5.2712e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.440217e+03 FES = 258100 μ_F = 0.823 μ_{CR} = 0.048
 Diversity = 5.2528e+00 k_t = [4, 3, 2, 2]
 Generation 2590: Best = 1.440217e+03 FES = 259100 μ_F = 0.851 μ_{CR} = 0.052
 Diversity = 5.2940e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.440212e+03 FES = 260100 μ_F = 0.842 μ_{CR} = 0.046
 Diversity = 5.3479e+00 k_t = [2, 4, 2, 4]
 Generation 2610: Best = 1.440212e+03 FES = 261100 μ_F = 0.762 μ_{CR} = 0.040
 Diversity = 5.3497e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.440212e+03 FES = 262100 μ_F = 0.766 μ_{CR} = 0.043
 Diversity = 5.3223e+00 k_t = [4, 4, 3, 2]
 Generation 2630: Best = 1.440203e+03 FES = 263100 μ_F = 0.771 μ_{CR} = 0.057
 Diversity = 5.3222e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.440203e+03 FES = 264100 μ_F = 0.811 μ_{CR} = 0.057
 Diversity = 5.3059e+00 k_t = [4, 4, 3, 4]
 Generation 2650: Best = 1.440203e+03 FES = 265100 μ_F = 0.795 μ_{CR} = 0.055
 Diversity = 5.3144e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.440203e+03 FES = 266100 μ_F = 0.777 μ_{CR} = 0.057
 Diversity = 5.3217e+00 k_t = [4, 3, 2, 4]
 Generation 2670: Best = 1.440202e+03 FES = 267100 μ_F = 0.764 μ_{CR} = 0.057
 Diversity = 5.2957e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.440198e+03 FES = 268100 μ_F = 0.752 μ_{CR} = 0.055
 Diversity = 5.2538e+00 k_t = [3, 4, 3, 2]
 Generation 2690: Best = 1.440192e+03 FES = 269100 μ_F = 0.760 μ_{CR} = 0.044
 Diversity = 5.2410e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.440187e+03 FES = 270100 μ_F = 0.730 μ_{CR} = 0.054
 Diversity = 5.2428e+00 k_t = [3, 3, 4, 2]
 Generation 2710: Best = 1.440184e+03 FES = 271100 μ_F = 0.746 μ_{CR} = 0.058
 Diversity = 5.2628e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.440182e+03 FES = 272100 μ_F = 0.774 μ_{CR} = 0.053
 Diversity = 5.2626e+00 k_t = [3, 4, 3, 2]
 Generation 2730: Best = 1.440182e+03 FES = 273100 μ_F = 0.800 μ_{CR} = 0.057
 Diversity = 5.2643e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.440182e+03 FES = 274100 μ_F = 0.828 μ_{CR} = 0.060
 Diversity = 5.2697e+00 k_t = [3, 4, 4, 3]
 Generation 2750: Best = 1.440156e+03 FES = 275100 μ_F = 0.815 μ_{CR} = 0.051
 Diversity = 5.2693e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.440156e+03 FES = 276100 μ_F = 0.798 μ_{CR} = 0.060
 Diversity = 5.2639e+00 k_t = [3, 4, 2, 3]
 Generation 2770: Best = 1.440156e+03 FES = 277100 μ_F = 0.775 μ_{CR} = 0.049
 Diversity = 5.2369e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.440156e+03 FES = 278100 μ_F = 0.823 μ_{CR} = 0.050
 Diversity = 5.2368e+00 k_t = [3, 2, 4, 3]
 Generation 2790: Best = 1.440156e+03 FES = 279100 μ_F = 0.805 μ_{CR} = 0.050
 Diversity = 5.2366e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.440152e+03 FES = 280100 μ_F = 0.805 μ_{CR} = 0.053
 Diversity = 5.3205e+00 k_t = [3, 4, 2, 4]
 Generation 2810: Best = 1.440152e+03 FES = 281100 μ_F = 0.791 μ_{CR} = 0.046
 Diversity = 5.3205e+00 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.440152e+03 FES = 282100 μ_F = 0.821 μ_{CR} = 0.053
 Diversity = 5.3405e+00 k_t = [4, 4, 2, 2]
 Generation 2830: Best = 1.440146e+03 FES = 283100 μ_F = 0.854 μ_{CR} = 0.049
 Diversity = 5.3370e+00 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.440146e+03 FES = 284100 μ_F = 0.856 μ_{CR} = 0.052
 Diversity = 5.3611e+00 k_t = [2, 4, 4, 4]
 Generation 2850: Best = 1.440146e+03 FES = 285100 μ_F = 0.850 μ_{CR} = 0.043
 Diversity = 5.3465e+00 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.440146e+03 FES = 286100 μ_F = 0.828 μ_{CR} = 0.041
 Diversity = 5.3866e+00 k_t = [3, 3, 3, 4]
 Generation 2870: Best = 1.440144e+03 FES = 287100 μ_F = 0.851 μ_{CR} = 0.042
 Diversity = 5.3864e+00 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.440144e+03 FES = 288100 μ_F = 0.856 μ_{CR} = 0.056
 Diversity = 5.3862e+00 k_t = [3, 3, 4, 2]
 Generation 2890: Best = 1.440144e+03 FES = 289100 μ_F = 0.858 μ_{CR} = 0.061
 Diversity = 5.3867e+00 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.440144e+03 FES = 290100 μ_F = 0.886 μ_{CR} = 0.063
 Diversity = 5.4074e+00 k_t = [2, 2, 4, 3]
 Generation 2910: Best = 1.440144e+03 FES = 291100 μ_F = 0.851 μ_{CR} = 0.056
 Diversity = 5.4354e+00 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.440144e+03 FES = 292100 μ_F = 0.828 μ_{CR} = 0.058
 Diversity = 5.4353e+00 k_t = [2, 2, 3, 2]

```

Generation 2930: Best = 1.440144e+03  FES = 293100   $\mu_F$  = 0.822   $\mu_{CR}$  = 0.050
Diversity = 5.4310e+00  k_t = [1, 1, 1, 1]
Generation 2940: Best = 1.440144e+03  FES = 294100   $\mu_F$  = 0.790   $\mu_{CR}$  = 0.047
Diversity = 5.4206e+00  k_t = [2, 2, 2, 2]
Generation 2950: Best = 1.440144e+03  FES = 295100   $\mu_F$  = 0.797   $\mu_{CR}$  = 0.050
Diversity = 5.4411e+00  k_t = [1, 1, 1, 1]
Generation 2960: Best = 1.440144e+03  FES = 296100   $\mu_F$  = 0.815   $\mu_{CR}$  = 0.052
Diversity = 5.4244e+00  k_t = [3, 2, 3, 4]
Generation 2970: Best = 1.440144e+03  FES = 297100   $\mu_F$  = 0.854   $\mu_{CR}$  = 0.041
Diversity = 5.4202e+00  k_t = [1, 1, 1, 1]
Generation 2980: Best = 1.440144e+03  FES = 298100   $\mu_F$  = 0.832   $\mu_{CR}$  = 0.046
Diversity = 5.4346e+00  k_t = [4, 3, 4, 2]
Generation 2990: Best = 1.439910e+03  FES = 299100   $\mu_F$  = 0.850   $\mu_{CR}$  = 0.048
Diversity = 5.4139e+00  k_t = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 1.439882e+03

Total Function Evals = 300000

Total Successful Updates = 19239

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Results:

Best Fitness : 1.439882e+03

Global Optimum : 1.400000e+03

Error (|best-optimum|) : 3.988158e+01

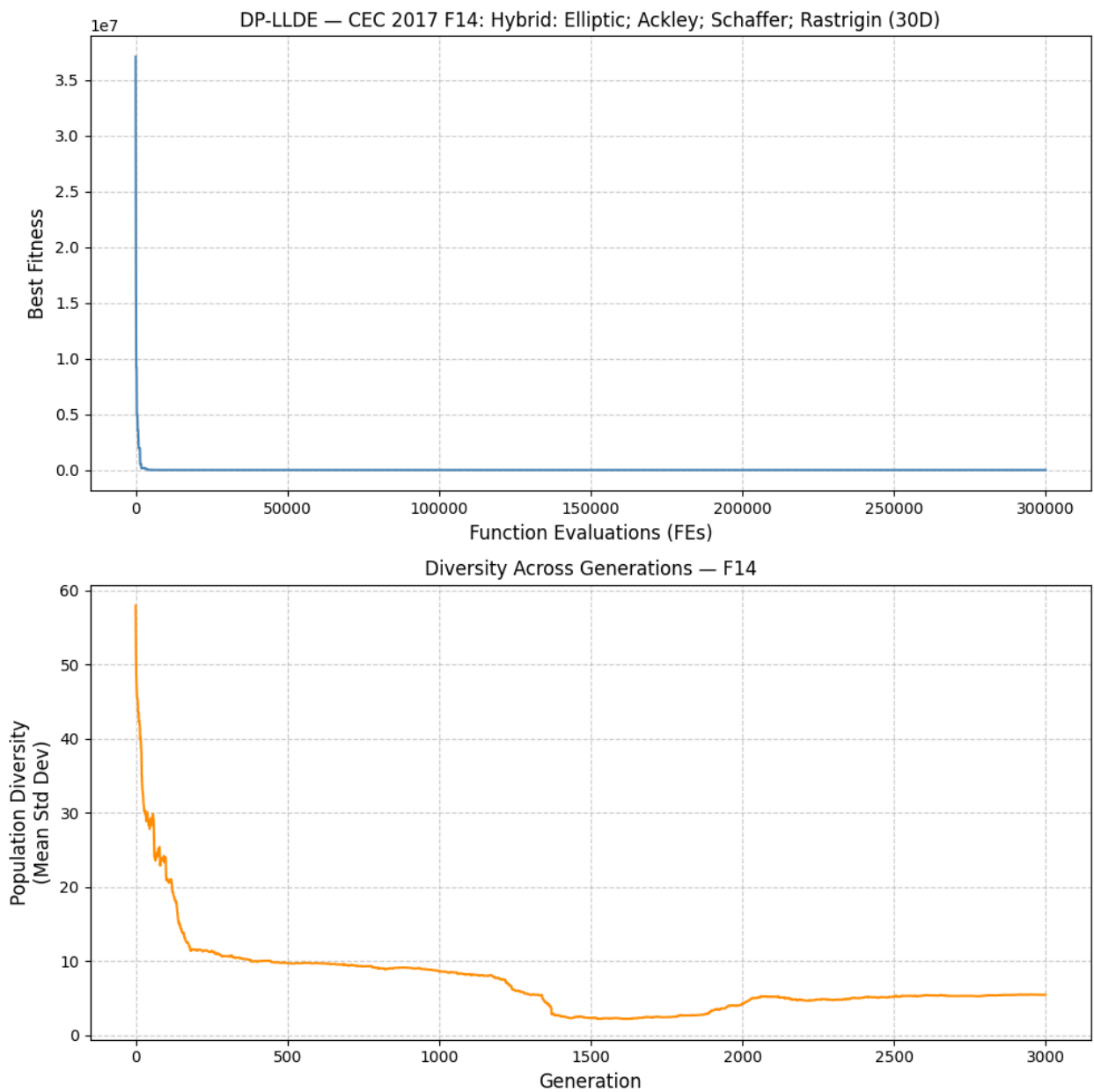
Best Solution (5 dim) : [-36.60829552 -83.71761601 88.91404064 79.13461496
61.88289946]

D^(0) (init. diversity): 5.7976e+01

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Graph saved as convergence_F14.png

```
In [ ]: # =====
# F15 – Hybrid Function 5 (Bent Cigar; HGBat; Rastrigin; Rosenbrock)
# =====
run_and_plot('F15', make_wrapper(hybrid.f15), 1500.0, 'Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock')
```

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DP-LLDE on CEC 2017 F15 – Single Run

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Function : Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock (30D)
Bounds : [-100, 100]
Global optimum : 1500.0
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 4.037230e+10
Generation 10: Best = 6.531062e+08 FES = 1100 $\mu_F = 0.564$ $\mu_{CR} = 0.352$
Diversity = 3.4097e+01 $k_t = [1, 1, 1, 1]$
Generation 20: Best = 1.099479e+08 FES = 2100 $\mu_F = 0.620$ $\mu_{CR} = 0.350$
Diversity = 2.8090e+01 $k_t = [4, 4, 2, 4]$
Generation 30: Best = 7.558369e+06 FES = 3100 $\mu_F = 0.664$ $\mu_{CR} = 0.315$
Diversity = 2.2415e+01 $k_t = [1, 1, 1, 1]$
Generation 40: Best = 6.674627e+05 FES = 4100 $\mu_F = 0.688$ $\mu_{CR} = 0.280$
Diversity = 2.2713e+01 $k_t = [4, 3, 4, 2]$
Generation 50: Best = 6.674627e+05 FES = 5100 $\mu_F = 0.691$ $\mu_{CR} = 0.240$
Diversity = 2.1100e+01 $k_t = [1, 1, 1, 1]$
Generation 60: Best = 1.650706e+05 FES = 6100 $\mu_F = 0.712$ $\mu_{CR} = 0.184$
Diversity = 2.0089e+01 $k_t = [3, 4, 2, 2]$
Generation 70: Best = 4.007748e+04 FES = 7100 $\mu_F = 0.729$ $\mu_{CR} = 0.152$
Diversity = 2.0809e+01 $k_t = [1, 1, 1, 1]$
Generation 80: Best = 2.676579e+04 FES = 8100 $\mu_F = 0.761$ $\mu_{CR} = 0.129$
Diversity = 1.9938e+01 $k_t = [3, 3, 2, 3]$
Generation 90: Best = 1.733009e+04 FES = 9100 $\mu_F = 0.777$ $\mu_{CR} = 0.115$
Diversity = 1.8576e+01 $k_t = [1, 1, 1, 1]$
Generation 100: Best = 8.219759e+03 FES = 10100 $\mu_F = 0.777$ $\mu_{CR} = 0.102$
Diversity = 1.6666e+01 $k_t = [3, 4, 4, 3]$
Generation 110: Best = 6.225315e+03 FES = 11100 $\mu_F = 0.799$ $\mu_{CR} = 0.094$
Diversity = 1.6092e+01 $k_t = [1, 1, 1, 1]$
Generation 120: Best = 4.688493e+03 FES = 12100 $\mu_F = 0.806$ $\mu_{CR} = 0.089$
Diversity = 1.5467e+01 $k_t = [4, 3, 2, 3]$
Generation 130: Best = 3.237345e+03 FES = 13100 $\mu_F = 0.807$ $\mu_{CR} = 0.095$
Diversity = 1.4915e+01 $k_t = [1, 1, 1, 1]$
Generation 140: Best = 2.371916e+03 FES = 14100 $\mu_F = 0.797$ $\mu_{CR} = 0.095$
Diversity = 1.4173e+01 $k_t = [3, 4, 4, 2]$
Generation 150: Best = 2.295022e+03 FES = 15100 $\mu_F = 0.789$ $\mu_{CR} = 0.089$
Diversity = 1.3643e+01 $k_t = [1, 1, 1, 1]$
Generation 160: Best = 2.132004e+03 FES = 16100 $\mu_F = 0.781$ $\mu_{CR} = 0.087$
Diversity = 1.2708e+01 $k_t = [3, 3, 3, 2]$
Generation 170: Best = 1.954817e+03 FES = 17100 $\mu_F = 0.786$ $\mu_{CR} = 0.084$
Diversity = 1.1074e+01 $k_t = [1, 1, 1, 1]$
Generation 180: Best = 1.862926e+03 FES = 18100 $\mu_F = 0.791$ $\mu_{CR} = 0.088$
Diversity = 9.8038e+00 $k_t = [4, 2, 4, 3]$
Generation 190: Best = 1.835152e+03 FES = 19100 $\mu_F = 0.777$ $\mu_{CR} = 0.084$
Diversity = 8.7399e+00 $k_t = [1, 1, 1, 1]$
Generation 200: Best = 1.784447e+03 FES = 20100 $\mu_F = 0.784$ $\mu_{CR} = 0.081$
Diversity = 7.8831e+00 $k_t = [4, 3, 2, 4]$
Generation 210: Best = 1.721423e+03 FES = 21100 $\mu_F = 0.789$ $\mu_{CR} = 0.080$
Diversity = 7.2695e+00 $k_t = [1, 1, 1, 1]$
Generation 220: Best = 1.698490e+03 FES = 22100 $\mu_F = 0.776$ $\mu_{CR} = 0.071$
Diversity = 6.8494e+00 $k_t = [3, 4, 3, 3]$

Generation 230: Best = 1.693710e+03	FES = 23100	$\mu_F = 0.767$	$\mu_{CR} = 0.082$
Diversity = 6.6399e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 1.680056e+03	FES = 24100	$\mu_F = 0.771$	$\mu_{CR} = 0.078$
Diversity = 6.4687e+00	$k_t = [4, 2, 3, 4]$		
Generation 250: Best = 1.673168e+03	FES = 25100	$\mu_F = 0.767$	$\mu_{CR} = 0.066$
Diversity = 5.9228e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 1.646268e+03	FES = 26100	$\mu_F = 0.777$	$\mu_{CR} = 0.059$
Diversity = 5.8453e+00	$k_t = [4, 4, 4, 4]$		
Generation 270: Best = 1.646268e+03	FES = 27100	$\mu_F = 0.746$	$\mu_{CR} = 0.057$
Diversity = 5.5947e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.643353e+03	FES = 28100	$\mu_F = 0.754$	$\mu_{CR} = 0.055$
Diversity = 5.3445e+00	$k_t = [2, 4, 4, 3]$		
Generation 290: Best = 1.633915e+03	FES = 29100	$\mu_F = 0.774$	$\mu_{CR} = 0.068$
Diversity = 5.2410e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.633915e+03	FES = 30100	$\mu_F = 0.765$	$\mu_{CR} = 0.070$
Diversity = 5.0504e+00	$k_t = [2, 2, 4, 4]$		
Generation 310: Best = 1.633915e+03	FES = 31100	$\mu_F = 0.790$	$\mu_{CR} = 0.056$
Diversity = 4.9276e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.623949e+03	FES = 32100	$\mu_F = 0.822$	$\mu_{CR} = 0.052$
Diversity = 4.9076e+00	$k_t = [3, 3, 2, 2]$		
Generation 330: Best = 1.623949e+03	FES = 33100	$\mu_F = 0.830$	$\mu_{CR} = 0.058$
Diversity = 4.9434e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.623949e+03	FES = 34100	$\mu_F = 0.827$	$\mu_{CR} = 0.054$
Diversity = 4.9642e+00	$k_t = [3, 3, 4, 2]$		
Generation 350: Best = 1.623949e+03	FES = 35100	$\mu_F = 0.821$	$\mu_{CR} = 0.046$
Diversity = 4.9204e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.623949e+03	FES = 36100	$\mu_F = 0.832$	$\mu_{CR} = 0.049$
Diversity = 4.8687e+00	$k_t = [2, 4, 3, 2]$		
Generation 370: Best = 1.623375e+03	FES = 37100	$\mu_F = 0.816$	$\mu_{CR} = 0.050$
Diversity = 4.8532e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 1.619964e+03	FES = 38100	$\mu_F = 0.804$	$\mu_{CR} = 0.049$
Diversity = 4.7726e+00	$k_t = [4, 2, 2, 3]$		
Generation 390: Best = 1.615560e+03	FES = 39100	$\mu_F = 0.798$	$\mu_{CR} = 0.058$
Diversity = 4.7342e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 1.615040e+03	FES = 40100	$\mu_F = 0.758$	$\mu_{CR} = 0.056$
Diversity = 4.7173e+00	$k_t = [2, 2, 3, 4]$		
Generation 410: Best = 1.613719e+03	FES = 41100	$\mu_F = 0.739$	$\mu_{CR} = 0.049$
Diversity = 4.6603e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 1.612105e+03	FES = 42100	$\mu_F = 0.766$	$\mu_{CR} = 0.048$
Diversity = 4.6296e+00	$k_t = [3, 2, 3, 2]$		
Generation 430: Best = 1.610884e+03	FES = 43100	$\mu_F = 0.775$	$\mu_{CR} = 0.049$
Diversity = 4.5372e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 1.610884e+03	FES = 44100	$\mu_F = 0.781$	$\mu_{CR} = 0.053$
Diversity = 4.4650e+00	$k_t = [4, 4, 4, 3]$		
Generation 450: Best = 1.608734e+03	FES = 45100	$\mu_F = 0.811$	$\mu_{CR} = 0.054$
Diversity = 4.3678e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 1.608734e+03	FES = 46100	$\mu_F = 0.799$	$\mu_{CR} = 0.056$
Diversity = 4.3577e+00	$k_t = [3, 4, 2, 3]$		
Generation 470: Best = 1.608025e+03	FES = 47100	$\mu_F = 0.838$	$\mu_{CR} = 0.058$
Diversity = 4.3225e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 1.607342e+03	FES = 48100	$\mu_F = 0.819$	$\mu_{CR} = 0.060$
Diversity = 4.3184e+00	$k_t = [2, 2, 2, 2]$		
Generation 490: Best = 1.605363e+03	FES = 49100	$\mu_F = 0.859$	$\mu_{CR} = 0.052$
Diversity = 4.3299e+00	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 1.601539e+03	FES = 50100	$\mu_F = 0.785$	$\mu_{CR} = 0.054$
Diversity = 4.3170e+00	$k_t = [3, 4, 2, 2]$		
Generation 510: Best = 1.601539e+03	FES = 51100	$\mu_F = 0.773$	$\mu_{CR} = 0.056$
Diversity = 4.2860e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 1.590056e+03	FES = 52100	$\mu_F = 0.768$	$\mu_{CR} = 0.055$
Diversity = 4.2669e+00	$k_t = [4, 2, 3, 3]$		
Generation 530: Best = 1.590056e+03	FES = 53100	$\mu_F = 0.772$	$\mu_{CR} = 0.058$
Diversity = 4.2472e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 1.590056e+03	FES = 54100	$\mu_F = 0.746$	$\mu_{CR} = 0.054$
Diversity = 4.1601e+00	$k_t = [3, 4, 4, 4]$		
Generation 550: Best = 1.589188e+03	FES = 55100	$\mu_F = 0.774$	$\mu_{CR} = 0.056$
Diversity = 4.1119e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 1.588560e+03	FES = 56100	$\mu_F = 0.763$	$\mu_{CR} = 0.053$
Diversity = 4.0682e+00	$k_t = [2, 2, 3, 2]$		
Generation 570: Best = 1.588560e+03	FES = 57100	$\mu_F = 0.793$	$\mu_{CR} = 0.052$
Diversity = 4.0428e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 1.588385e+03	FES = 58100	$\mu_F = 0.789$	$\mu_{CR} = 0.050$
Diversity = 3.9189e+00	$k_t = [4, 3, 3, 4]$		
Generation 590: Best = 1.588343e+03	FES = 59100	$\mu_F = 0.819$	$\mu_{CR} = 0.054$
Diversity = 3.9327e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 1.588122e+03	FES = 60100	$\mu_F = 0.797$	$\mu_{CR} = 0.058$
Diversity = 3.9354e+00	$k_t = [2, 4, 3, 3]$		
Generation 610: Best = 1.588122e+03	FES = 61100	$\mu_F = 0.806$	$\mu_{CR} = 0.053$
Diversity = 3.9359e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 1.587140e+03	FES = 62100	$\mu_F = 0.814$	$\mu_{CR} = 0.067$
Diversity = 3.9305e+00	$k_t = [4, 2, 4, 4]$		
Generation 630: Best = 1.586195e+03	FES = 63100	$\mu_F = 0.801$	$\mu_{CR} = 0.062$
Diversity = 3.8914e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 1.586195e+03	FES = 64100	$\mu_F = 0.824$	$\mu_{CR} = 0.054$
Diversity = 3.7198e+00	$k_t = [3, 2, 3, 4]$		
Generation 650: Best = 1.586195e+03	FES = 65100	$\mu_F = 0.834$	$\mu_{CR} = 0.053$
Diversity = 3.6478e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 1.586195e+03	FES = 66100	$\mu_F = 0.833$	$\mu_{CR} = 0.070$
Diversity = 3.6473e+00	$k_t = [4, 2, 4, 3]$		
Generation 670: Best = 1.585357e+03	FES = 67100	$\mu_F = 0.851$	$\mu_{CR} = 0.055$
Diversity = 3.6463e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 1.585357e+03	FES = 68100	$\mu_F = 0.844$	$\mu_{CR} = 0.050$
Diversity = 3.5474e+00	$k_t = [3, 4, 2, 4]$		
Generation 690: Best = 1.585357e+03	FES = 69100	$\mu_F = 0.815$	$\mu_{CR} = 0.060$
Diversity = 3.5257e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 1.585357e+03	FES = 70100	$\mu_F = 0.790$	$\mu_{CR} = 0.057$
Diversity = 3.5183e+00	$k_t = [4, 3, 4, 2]$		
Generation 710: Best = 1.583434e+03	FES = 71100	$\mu_F = 0.810$	$\mu_{CR} = 0.051$
Diversity = 3.4736e+00	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 1.583434e+03	FES = 72100	$\mu_F = 0.857$	$\mu_{CR} = 0.049$
Diversity = 3.3558e+00	$k_t = [3, 2, 3, 4]$		
Generation 730: Best = 1.583434e+03	FES = 73100	$\mu_F = 0.856$	$\mu_{CR} = 0.051$
Diversity = 3.3709e+00	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 1.582740e+03	FES = 74100	$\mu_F = 0.859$	$\mu_{CR} = 0.070$
Diversity = 3.3531e+00	$k_t = [4, 3, 4, 3]$		
Generation 750: Best = 1.582740e+03	FES = 75100	$\mu_F = 0.861$	$\mu_{CR} = 0.066$
Diversity = 3.3014e+00	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 1.582740e+03	FES = 76100	$\mu_F = 0.858$	$\mu_{CR} = 0.061$
Diversity = 3.2049e+00	$k_t = [4, 3, 2, 3]$		

Generation 770: Best = 1.582225e+03 FES = 77100 μ_F = 0.848 μ_{CR} = 0.054
 Diversity = 3.1538e+00 k_t = [1, 1, 1, 1]
 Generation 780: Best = 1.580757e+03 FES = 78100 μ_F = 0.844 μ_{CR} = 0.057
 Diversity = 3.1283e+00 k_t = [2, 2, 4, 4]
 Generation 790: Best = 1.580739e+03 FES = 79100 μ_F = 0.827 μ_{CR} = 0.062
 Diversity = 3.0374e+00 k_t = [1, 1, 1, 1]
 Generation 800: Best = 1.580739e+03 FES = 80100 μ_F = 0.852 μ_{CR} = 0.060
 Diversity = 2.9998e+00 k_t = [2, 4, 2, 3]
 Generation 810: Best = 1.580739e+03 FES = 81100 μ_F = 0.852 μ_{CR} = 0.057
 Diversity = 2.9981e+00 k_t = [1, 1, 1, 1]
 Generation 820: Best = 1.580672e+03 FES = 82100 μ_F = 0.870 μ_{CR} = 0.066
 Diversity = 2.9109e+00 k_t = [4, 3, 2, 3]
 Generation 830: Best = 1.580672e+03 FES = 83100 μ_F = 0.877 μ_{CR} = 0.064
 Diversity = 2.8901e+00 k_t = [1, 1, 1, 1]
 Generation 840: Best = 1.580429e+03 FES = 84100 μ_F = 0.908 μ_{CR} = 0.060
 Diversity = 2.8276e+00 k_t = [2, 3, 4, 4]
 Generation 850: Best = 1.579839e+03 FES = 85100 μ_F = 0.882 μ_{CR} = 0.065
 Diversity = 2.7196e+00 k_t = [1, 1, 1, 1]
 Generation 860: Best = 1.579580e+03 FES = 86100 μ_F = 0.876 μ_{CR} = 0.068
 Diversity = 2.6211e+00 k_t = [4, 4, 4, 2]
 Generation 870: Best = 1.579429e+03 FES = 87100 μ_F = 0.867 μ_{CR} = 0.067
 Diversity = 2.3490e+00 k_t = [1, 1, 1, 1]
 Generation 880: Best = 1.579140e+03 FES = 88100 μ_F = 0.873 μ_{CR} = 0.065
 Diversity = 1.8967e+00 k_t = [2, 4, 2, 4]
 Generation 890: Best = 1.578626e+03 FES = 89100 μ_F = 0.859 μ_{CR} = 0.073
 Diversity = 1.3368e+00 k_t = [1, 1, 1, 1]
 Generation 900: Best = 1.577855e+03 FES = 90100 μ_F = 0.848 μ_{CR} = 0.070
 Diversity = 1.0592e+00 k_t = [3, 3, 3, 3]
 Generation 910: Best = 1.577177e+03 FES = 91100 μ_F = 0.810 μ_{CR} = 0.073
 Diversity = 7.2083e-01 k_t = [1, 1, 1, 1]
 Generation 920: Best = 1.576681e+03 FES = 92100 μ_F = 0.786 μ_{CR} = 0.072
 Diversity = 6.0936e-01 k_t = [2, 4, 4, 3]
 Generation 930: Best = 1.576189e+03 FES = 93100 μ_F = 0.787 μ_{CR} = 0.080
 Diversity = 5.9330e-01 k_t = [1, 1, 1, 1]
 Generation 940: Best = 1.575865e+03 FES = 94100 μ_F = 0.777 μ_{CR} = 0.082
 Diversity = 5.7931e-01 k_t = [2, 4, 2, 3]
 Generation 950: Best = 1.575814e+03 FES = 95100 μ_F = 0.761 μ_{CR} = 0.076
 Diversity = 5.2828e-01 k_t = [1, 1, 1, 1]
 Generation 960: Best = 1.575430e+03 FES = 96100 μ_F = 0.770 μ_{CR} = 0.073
 Diversity = 4.8780e-01 k_t = [2, 4, 3, 3]
 Generation 970: Best = 1.575290e+03 FES = 97100 μ_F = 0.754 μ_{CR} = 0.075
 Diversity = 4.7334e-01 k_t = [1, 1, 1, 1]
 Generation 980: Best = 1.575217e+03 FES = 98100 μ_F = 0.744 μ_{CR} = 0.076
 Diversity = 4.5417e-01 k_t = [2, 4, 3, 2]
 Generation 990: Best = 1.575182e+03 FES = 99100 μ_F = 0.749 μ_{CR} = 0.077
 Diversity = 4.4638e-01 k_t = [1, 1, 1, 1]
 Generation 1000: Best = 1.575129e+03 FES = 100100 μ_F = 0.773 μ_{CR} = 0.076
 Diversity = 4.4392e-01 k_t = [3, 4, 2, 4]
 Generation 1010: Best = 1.574954e+03 FES = 101100 μ_F = 0.777 μ_{CR} = 0.068
 Diversity = 4.3778e-01 k_t = [1, 1, 1, 1]
 Generation 1020: Best = 1.574927e+03 FES = 102100 μ_F = 0.779 μ_{CR} = 0.073
 Diversity = 4.2981e-01 k_t = [4, 3, 2, 3]
 Generation 1030: Best = 1.574859e+03 FES = 103100 μ_F = 0.784 μ_{CR} = 0.078
 Diversity = 4.2987e-01 k_t = [1, 1, 1, 1]

Generation 1040: Best = 1.574763e+03 FES = 104100 μ_F = 0.793 μ_{CR} = 0.077
 Diversity = 4.2487e-01 k_t = [4, 3, 4, 3]
 Generation 1050: Best = 1.574657e+03 FES = 105100 μ_F = 0.804 μ_{CR} = 0.077
 Diversity = 4.3640e-01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 1.574657e+03 FES = 106100 μ_F = 0.810 μ_{CR} = 0.071
 Diversity = 4.4444e-01 k_t = [2, 4, 3, 3]
 Generation 1070: Best = 1.574575e+03 FES = 107100 μ_F = 0.802 μ_{CR} = 0.070
 Diversity = 4.4933e-01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 1.574510e+03 FES = 108100 μ_F = 0.794 μ_{CR} = 0.069
 Diversity = 4.3976e-01 k_t = [2, 4, 3, 3]
 Generation 1090: Best = 1.574443e+03 FES = 109100 μ_F = 0.793 μ_{CR} = 0.073
 Diversity = 4.3644e-01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 1.574266e+03 FES = 110100 μ_F = 0.793 μ_{CR} = 0.075
 Diversity = 4.3513e-01 k_t = [4, 2, 3, 2]
 Generation 1110: Best = 1.574222e+03 FES = 111100 μ_F = 0.794 μ_{CR} = 0.080
 Diversity = 4.2205e-01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.574193e+03 FES = 112100 μ_F = 0.780 μ_{CR} = 0.076
 Diversity = 4.2079e-01 k_t = [4, 4, 2, 2]
 Generation 1130: Best = 1.574027e+03 FES = 113100 μ_F = 0.766 μ_{CR} = 0.072
 Diversity = 4.0730e-01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.574004e+03 FES = 114100 μ_F = 0.774 μ_{CR} = 0.076
 Diversity = 3.8068e-01 k_t = [2, 2, 2, 4]
 Generation 1150: Best = 1.573930e+03 FES = 115100 μ_F = 0.777 μ_{CR} = 0.067
 Diversity = 3.8035e-01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.573885e+03 FES = 116100 μ_F = 0.762 μ_{CR} = 0.065
 Diversity = 3.7020e-01 k_t = [4, 3, 2, 4]
 Generation 1170: Best = 1.573862e+03 FES = 117100 μ_F = 0.752 μ_{CR} = 0.063
 Diversity = 3.4349e-01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.573855e+03 FES = 118100 μ_F = 0.756 μ_{CR} = 0.065
 Diversity = 3.4385e-01 k_t = [4, 4, 4, 2]
 Generation 1190: Best = 1.573835e+03 FES = 119100 μ_F = 0.728 μ_{CR} = 0.068
 Diversity = 3.3683e-01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.573798e+03 FES = 120100 μ_F = 0.761 μ_{CR} = 0.074
 Diversity = 3.3614e-01 k_t = [2, 4, 3, 2]
 Generation 1210: Best = 1.573787e+03 FES = 121100 μ_F = 0.768 μ_{CR} = 0.068
 Diversity = 3.3879e-01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.573783e+03 FES = 122100 μ_F = 0.791 μ_{CR} = 0.063
 Diversity = 3.4031e-01 k_t = [3, 3, 3, 2]
 Generation 1230: Best = 1.573748e+03 FES = 123100 μ_F = 0.798 μ_{CR} = 0.075
 Diversity = 3.4932e-01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.573700e+03 FES = 124100 μ_F = 0.783 μ_{CR} = 0.077
 Diversity = 3.5886e-01 k_t = [2, 3, 2, 2]
 Generation 1250: Best = 1.573671e+03 FES = 125100 μ_F = 0.795 μ_{CR} = 0.074
 Diversity = 3.7007e-01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.573650e+03 FES = 126100 μ_F = 0.779 μ_{CR} = 0.071
 Diversity = 3.4354e-01 k_t = [3, 3, 2, 4]
 Generation 1270: Best = 1.573610e+03 FES = 127100 μ_F = 0.775 μ_{CR} = 0.067
 Diversity = 3.1017e-01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.573562e+03 FES = 128100 μ_F = 0.793 μ_{CR} = 0.070
 Diversity = 3.0644e-01 k_t = [4, 4, 3, 3]
 Generation 1290: Best = 1.573522e+03 FES = 129100 μ_F = 0.802 μ_{CR} = 0.077
 Diversity = 2.9816e-01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.573474e+03 FES = 130100 μ_F = 0.782 μ_{CR} = 0.066
 Diversity = 3.2184e-01 k_t = [4, 3, 3, 2]

Generation 1310: Best = 1.573444e+03 FES = 131100 μ_F = 0.794 μ_{CR} = 0.069
 Diversity = 3.3224e-01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.573415e+03 FES = 132100 μ_F = 0.806 μ_{CR} = 0.070
 Diversity = 3.2683e-01 k_t = [3, 2, 2, 3]
 Generation 1330: Best = 1.573411e+03 FES = 133100 μ_F = 0.809 μ_{CR} = 0.068
 Diversity = 3.2475e-01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.573374e+03 FES = 134100 μ_F = 0.795 μ_{CR} = 0.076
 Diversity = 3.4028e-01 k_t = [2, 3, 3, 2]
 Generation 1350: Best = 1.573364e+03 FES = 135100 μ_F = 0.792 μ_{CR} = 0.066
 Diversity = 3.5120e-01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.573349e+03 FES = 136100 μ_F = 0.784 μ_{CR} = 0.067
 Diversity = 3.0898e-01 k_t = [2, 2, 4, 4]
 Generation 1370: Best = 1.573320e+03 FES = 137100 μ_F = 0.781 μ_{CR} = 0.063
 Diversity = 2.8553e-01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.573300e+03 FES = 138100 μ_F = 0.795 μ_{CR} = 0.066
 Diversity = 2.7051e-01 k_t = [4, 4, 4, 3]
 Generation 1390: Best = 1.573293e+03 FES = 139100 μ_F = 0.782 μ_{CR} = 0.068
 Diversity = 2.6984e-01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.573257e+03 FES = 140100 μ_F = 0.776 μ_{CR} = 0.067
 Diversity = 2.5861e-01 k_t = [3, 4, 2, 2]
 Generation 1410: Best = 1.573236e+03 FES = 141100 μ_F = 0.756 μ_{CR} = 0.071
 Diversity = 2.5441e-01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.573235e+03 FES = 142100 μ_F = 0.749 μ_{CR} = 0.064
 Diversity = 2.2904e-01 k_t = [4, 2, 2, 4]
 Generation 1430: Best = 1.573221e+03 FES = 143100 μ_F = 0.750 μ_{CR} = 0.070
 Diversity = 2.1207e-01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.573206e+03 FES = 144100 μ_F = 0.761 μ_{CR} = 0.071
 Diversity = 1.9456e-01 k_t = [4, 2, 4, 4]
 Generation 1450: Best = 1.573201e+03 FES = 145100 μ_F = 0.774 μ_{CR} = 0.059
 Diversity = 1.8615e-01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.573188e+03 FES = 146100 μ_F = 0.776 μ_{CR} = 0.058
 Diversity = 1.9423e-01 k_t = [4, 3, 3, 3]
 Generation 1470: Best = 1.573183e+03 FES = 147100 μ_F = 0.781 μ_{CR} = 0.062
 Diversity = 1.9904e-01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.573175e+03 FES = 148100 μ_F = 0.796 μ_{CR} = 0.078
 Diversity = 1.9455e-01 k_t = [3, 4, 3, 4]
 Generation 1490: Best = 1.573175e+03 FES = 149100 μ_F = 0.822 μ_{CR} = 0.068
 Diversity = 1.8936e-01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.573171e+03 FES = 150100 μ_F = 0.808 μ_{CR} = 0.063
 Diversity = 1.9296e-01 k_t = [4, 3, 2, 2]
 Generation 1510: Best = 1.573159e+03 FES = 151100 μ_F = 0.802 μ_{CR} = 0.062
 Diversity = 1.9512e-01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.573140e+03 FES = 152100 μ_F = 0.809 μ_{CR} = 0.070
 Diversity = 1.8727e-01 k_t = [4, 4, 4, 4]
 Generation 1530: Best = 1.573133e+03 FES = 153100 μ_F = 0.832 μ_{CR} = 0.063
 Diversity = 1.7912e-01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.573131e+03 FES = 154100 μ_F = 0.818 μ_{CR} = 0.076
 Diversity = 1.6577e-01 k_t = [2, 2, 3, 4]
 Generation 1550: Best = 1.573128e+03 FES = 155100 μ_F = 0.805 μ_{CR} = 0.066
 Diversity = 1.7297e-01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.573126e+03 FES = 156100 μ_F = 0.798 μ_{CR} = 0.062
 Diversity = 1.7827e-01 k_t = [3, 3, 2, 3]
 Generation 1570: Best = 1.573115e+03 FES = 157100 μ_F = 0.785 μ_{CR} = 0.065
 Diversity = 1.7689e-01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.573115e+03 FES = 158100 μ_F = 0.801 μ_{CR} = 0.066
 Diversity = 1.7483e-01 k_t = [4, 2, 2, 4]
 Generation 1590: Best = 1.573109e+03 FES = 159100 μ_F = 0.811 μ_{CR} = 0.064
 Diversity = 1.7832e-01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.573103e+03 FES = 160100 μ_F = 0.800 μ_{CR} = 0.066
 Diversity = 1.7711e-01 k_t = [3, 2, 3, 4]
 Generation 1610: Best = 1.573084e+03 FES = 161100 μ_F = 0.800 μ_{CR} = 0.068
 Diversity = 1.6734e-01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.573070e+03 FES = 162100 μ_F = 0.818 μ_{CR} = 0.069
 Diversity = 1.7007e-01 k_t = [3, 3, 3, 2]
 Generation 1630: Best = 1.573047e+03 FES = 163100 μ_F = 0.813 μ_{CR} = 0.071
 Diversity = 1.6852e-01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.573041e+03 FES = 164100 μ_F = 0.803 μ_{CR} = 0.075
 Diversity = 1.7611e-01 k_t = [2, 2, 4, 4]
 Generation 1650: Best = 1.573041e+03 FES = 165100 μ_F = 0.812 μ_{CR} = 0.066
 Diversity = 1.6785e-01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.573033e+03 FES = 166100 μ_F = 0.816 μ_{CR} = 0.070
 Diversity = 1.6551e-01 k_t = [3, 3, 2, 3]
 Generation 1670: Best = 1.573031e+03 FES = 167100 μ_F = 0.815 μ_{CR} = 0.064
 Diversity = 1.6204e-01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.573021e+03 FES = 168100 μ_F = 0.823 μ_{CR} = 0.057
 Diversity = 1.4834e-01 k_t = [4, 3, 2, 4]
 Generation 1690: Best = 1.573021e+03 FES = 169100 μ_F = 0.786 μ_{CR} = 0.060
 Diversity = 1.4759e-01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.573005e+03 FES = 170100 μ_F = 0.775 μ_{CR} = 0.067
 Diversity = 1.4634e-01 k_t = [3, 3, 3, 2]
 Generation 1710: Best = 1.573000e+03 FES = 171100 μ_F = 0.764 μ_{CR} = 0.070
 Diversity = 1.4641e-01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.572987e+03 FES = 172100 μ_F = 0.748 μ_{CR} = 0.066
 Diversity = 1.4830e-01 k_t = [2, 4, 3, 3]
 Generation 1730: Best = 1.572972e+03 FES = 173100 μ_F = 0.741 μ_{CR} = 0.064
 Diversity = 1.4582e-01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.572972e+03 FES = 174100 μ_F = 0.761 μ_{CR} = 0.068
 Diversity = 1.4757e-01 k_t = [4, 4, 3, 3]
 Generation 1750: Best = 1.572954e+03 FES = 175100 μ_F = 0.773 μ_{CR} = 0.079
 Diversity = 1.4565e-01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.572946e+03 FES = 176100 μ_F = 0.770 μ_{CR} = 0.081
 Diversity = 1.4078e-01 k_t = [3, 3, 2, 3]
 Generation 1770: Best = 1.572944e+03 FES = 177100 μ_F = 0.792 μ_{CR} = 0.072
 Diversity = 1.5355e-01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.572938e+03 FES = 178100 μ_F = 0.799 μ_{CR} = 0.065
 Diversity = 1.3836e-01 k_t = [3, 3, 3, 3]
 Generation 1790: Best = 1.572923e+03 FES = 179100 μ_F = 0.785 μ_{CR} = 0.079
 Diversity = 1.5341e-01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 1.572918e+03 FES = 180100 μ_F = 0.774 μ_{CR} = 0.073
 Diversity = 1.4153e-01 k_t = [2, 3, 2, 3]
 Generation 1810: Best = 1.572911e+03 FES = 181100 μ_F = 0.778 μ_{CR} = 0.077
 Diversity = 1.4099e-01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 1.572911e+03 FES = 182100 μ_F = 0.783 μ_{CR} = 0.071
 Diversity = 1.3610e-01 k_t = [2, 3, 3, 4]
 Generation 1830: Best = 1.572900e+03 FES = 183100 μ_F = 0.780 μ_{CR} = 0.069
 Diversity = 1.2307e-01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 1.572890e+03 FES = 184100 μ_F = 0.793 μ_{CR} = 0.062
 Diversity = 1.2364e-01 k_t = [3, 3, 2, 4]

Generation 1850: Best = 1.572887e+03	FES = 185100	$\mu_F = 0.796$	$\mu_{CR} = 0.059$
Diversity = 1.3531e-01	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 1.572883e+03	FES = 186100	$\mu_F = 0.800$	$\mu_{CR} = 0.060$
Diversity = 1.2414e-01	k_t = [4, 3, 3, 3]		
Generation 1870: Best = 1.572879e+03	FES = 187100	$\mu_F = 0.797$	$\mu_{CR} = 0.063$
Diversity = 1.2677e-01	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 1.572866e+03	FES = 188100	$\mu_F = 0.791$	$\mu_{CR} = 0.064$
Diversity = 1.2502e-01	k_t = [4, 4, 4, 4]		
Generation 1890: Best = 1.572857e+03	FES = 189100	$\mu_F = 0.795$	$\mu_{CR} = 0.069$
Diversity = 1.2705e-01	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 1.572847e+03	FES = 190100	$\mu_F = 0.797$	$\mu_{CR} = 0.069$
Diversity = 1.3470e-01	k_t = [2, 4, 4, 2]		
Generation 1910: Best = 1.572843e+03	FES = 191100	$\mu_F = 0.794$	$\mu_{CR} = 0.073$
Diversity = 1.5112e-01	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 1.572835e+03	FES = 192100	$\mu_F = 0.788$	$\mu_{CR} = 0.068$
Diversity = 1.3060e-01	k_t = [3, 4, 2, 3]		
Generation 1930: Best = 1.572823e+03	FES = 193100	$\mu_F = 0.785$	$\mu_{CR} = 0.067$
Diversity = 1.2628e-01	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 1.572820e+03	FES = 194100	$\mu_F = 0.785$	$\mu_{CR} = 0.079$
Diversity = 1.2498e-01	k_t = [3, 4, 2, 4]		
Generation 1950: Best = 1.572809e+03	FES = 195100	$\mu_F = 0.793$	$\mu_{CR} = 0.074$
Diversity = 1.2066e-01	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 1.572809e+03	FES = 196100	$\mu_F = 0.799$	$\mu_{CR} = 0.065$
Diversity = 1.2197e-01	k_t = [3, 3, 3, 2]		
Generation 1970: Best = 1.572799e+03	FES = 197100	$\mu_F = 0.795$	$\mu_{CR} = 0.068$
Diversity = 1.2395e-01	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 1.572799e+03	FES = 198100	$\mu_F = 0.803$	$\mu_{CR} = 0.060$
Diversity = 1.1188e-01	k_t = [3, 4, 4, 3]		
Generation 1990: Best = 1.572799e+03	FES = 199100	$\mu_F = 0.803$	$\mu_{CR} = 0.065$
Diversity = 1.1145e-01	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 1.572793e+03	FES = 200100	$\mu_F = 0.800$	$\mu_{CR} = 0.069$
Diversity = 1.1133e-01	k_t = [4, 2, 2, 4]		
Generation 2010: Best = 1.572785e+03	FES = 201100	$\mu_F = 0.792$	$\mu_{CR} = 0.061$
Diversity = 1.1276e-01	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 1.572774e+03	FES = 202100	$\mu_F = 0.802$	$\mu_{CR} = 0.064$
Diversity = 1.1811e-01	k_t = [3, 2, 4, 2]		
Generation 2030: Best = 1.572770e+03	FES = 203100	$\mu_F = 0.812$	$\mu_{CR} = 0.066$
Diversity = 1.2964e-01	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 1.572769e+03	FES = 204100	$\mu_F = 0.789$	$\mu_{CR} = 0.067$
Diversity = 1.3742e-01	k_t = [3, 2, 3, 2]		
Generation 2050: Best = 1.572760e+03	FES = 205100	$\mu_F = 0.809$	$\mu_{CR} = 0.062$
Diversity = 1.3588e-01	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 1.572760e+03	FES = 206100	$\mu_F = 0.827$	$\mu_{CR} = 0.069$
Diversity = 1.3249e-01	k_t = [2, 4, 4, 2]		
Generation 2070: Best = 1.572760e+03	FES = 207100	$\mu_F = 0.808$	$\mu_{CR} = 0.058$
Diversity = 1.3013e-01	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 1.572756e+03	FES = 208100	$\mu_F = 0.814$	$\mu_{CR} = 0.058$
Diversity = 1.2870e-01	k_t = [3, 2, 3, 2]		
Generation 2090: Best = 1.572751e+03	FES = 209100	$\mu_F = 0.829$	$\mu_{CR} = 0.064$
Diversity = 1.2463e-01	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 1.572751e+03	FES = 210100	$\mu_F = 0.816$	$\mu_{CR} = 0.069$
Diversity = 1.1899e-01	k_t = [3, 2, 4, 4]		
Generation 2110: Best = 1.572740e+03	FES = 211100	$\mu_F = 0.828$	$\mu_{CR} = 0.071$
Diversity = 1.1416e-01	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 1.572738e+03 FES = 212100 μ_F = 0.834 μ_{CR} = 0.074
 Diversity = 1.1843e-01 k_t = [3, 2, 4, 2]
 Generation 2130: Best = 1.572738e+03 FES = 213100 μ_F = 0.834 μ_{CR} = 0.067
 Diversity = 1.1806e-01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.572732e+03 FES = 214100 μ_F = 0.841 μ_{CR} = 0.067
 Diversity = 1.1935e-01 k_t = [2, 2, 2, 4]
 Generation 2150: Best = 1.572732e+03 FES = 215100 μ_F = 0.838 μ_{CR} = 0.069
 Diversity = 1.2197e-01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.572725e+03 FES = 216100 μ_F = 0.838 μ_{CR} = 0.068
 Diversity = 1.2175e-01 k_t = [3, 4, 4, 2]
 Generation 2170: Best = 1.572722e+03 FES = 217100 μ_F = 0.854 μ_{CR} = 0.069
 Diversity = 1.1980e-01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.572709e+03 FES = 218100 μ_F = 0.835 μ_{CR} = 0.067
 Diversity = 1.1558e-01 k_t = [4, 2, 2, 4]
 Generation 2190: Best = 1.572709e+03 FES = 219100 μ_F = 0.826 μ_{CR} = 0.062
 Diversity = 1.1451e-01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 1.572704e+03 FES = 220100 μ_F = 0.846 μ_{CR} = 0.071
 Diversity = 1.2294e-01 k_t = [2, 4, 4, 3]
 Generation 2210: Best = 1.572704e+03 FES = 221100 μ_F = 0.831 μ_{CR} = 0.067
 Diversity = 1.3594e-01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 1.572701e+03 FES = 222100 μ_F = 0.818 μ_{CR} = 0.066
 Diversity = 1.3397e-01 k_t = [4, 3, 4, 4]
 Generation 2230: Best = 1.572696e+03 FES = 223100 μ_F = 0.813 μ_{CR} = 0.066
 Diversity = 1.1882e-01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.572690e+03 FES = 224100 μ_F = 0.807 μ_{CR} = 0.069
 Diversity = 1.1874e-01 k_t = [4, 4, 4, 4]
 Generation 2250: Best = 1.572689e+03 FES = 225100 μ_F = 0.790 μ_{CR} = 0.065
 Diversity = 1.0696e-01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.572679e+03 FES = 226100 μ_F = 0.775 μ_{CR} = 0.064
 Diversity = 1.0235e-01 k_t = [3, 2, 3, 3]
 Generation 2270: Best = 1.572677e+03 FES = 227100 μ_F = 0.778 μ_{CR} = 0.064
 Diversity = 1.0380e-01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.572674e+03 FES = 228100 μ_F = 0.810 μ_{CR} = 0.061
 Diversity = 9.9431e-02 k_t = [3, 3, 4, 4]
 Generation 2290: Best = 1.572668e+03 FES = 229100 μ_F = 0.790 μ_{CR} = 0.063
 Diversity = 9.4725e-02 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.572668e+03 FES = 230100 μ_F = 0.792 μ_{CR} = 0.060
 Diversity = 9.7320e-02 k_t = [2, 2, 4, 3]
 Generation 2310: Best = 1.572661e+03 FES = 231100 μ_F = 0.783 μ_{CR} = 0.066
 Diversity = 1.0658e-01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.572649e+03 FES = 232100 μ_F = 0.790 μ_{CR} = 0.066
 Diversity = 1.0603e-01 k_t = [3, 3, 4, 3]
 Generation 2330: Best = 1.572649e+03 FES = 233100 μ_F = 0.766 μ_{CR} = 0.069
 Diversity = 1.0615e-01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.572645e+03 FES = 234100 μ_F = 0.755 μ_{CR} = 0.073
 Diversity = 1.0876e-01 k_t = [3, 3, 3, 2]
 Generation 2350: Best = 1.572642e+03 FES = 235100 μ_F = 0.766 μ_{CR} = 0.067
 Diversity = 1.1203e-01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.572641e+03 FES = 236100 μ_F = 0.774 μ_{CR} = 0.060
 Diversity = 9.8709e-02 k_t = [3, 3, 4, 2]
 Generation 2370: Best = 1.572637e+03 FES = 237100 μ_F = 0.771 μ_{CR} = 0.058
 Diversity = 9.2164e-02 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.572637e+03 FES = 238100 μ_F = 0.786 μ_{CR} = 0.067
 Diversity = 9.1307e-02 k_t = [2, 2, 4, 4]

Generation 2390: Best = 1.572635e+03 FES = 239100 μ_F = 0.750 μ_{CR} = 0.062
 Diversity = 8.9368e-02 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.572631e+03 FES = 240100 μ_F = 0.781 μ_{CR} = 0.057
 Diversity = 8.3888e-02 k_t = [2, 3, 3, 3]
 Generation 2410: Best = 1.572631e+03 FES = 241100 μ_F = 0.789 μ_{CR} = 0.059
 Diversity = 8.2388e-02 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.572629e+03 FES = 242100 μ_F = 0.783 μ_{CR} = 0.061
 Diversity = 7.7471e-02 k_t = [3, 2, 3, 4]
 Generation 2430: Best = 1.572625e+03 FES = 243100 μ_F = 0.793 μ_{CR} = 0.063
 Diversity = 7.2254e-02 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.572624e+03 FES = 244100 μ_F = 0.787 μ_{CR} = 0.060
 Diversity = 6.7330e-02 k_t = [3, 2, 4, 3]
 Generation 2450: Best = 1.572624e+03 FES = 245100 μ_F = 0.789 μ_{CR} = 0.062
 Diversity = 6.7603e-02 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.572622e+03 FES = 246100 μ_F = 0.786 μ_{CR} = 0.063
 Diversity = 6.4088e-02 k_t = [4, 4, 3, 3]
 Generation 2470: Best = 1.572620e+03 FES = 247100 μ_F = 0.782 μ_{CR} = 0.063
 Diversity = 6.2338e-02 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.572618e+03 FES = 248100 μ_F = 0.792 μ_{CR} = 0.074
 Diversity = 5.8580e-02 k_t = [2, 4, 2, 2]
 Generation 2490: Best = 1.572616e+03 FES = 249100 μ_F = 0.791 μ_{CR} = 0.064
 Diversity = 5.5742e-02 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.572615e+03 FES = 250100 μ_F = 0.809 μ_{CR} = 0.071
 Diversity = 5.3089e-02 k_t = [2, 2, 2, 4]
 Generation 2510: Best = 1.572612e+03 FES = 251100 μ_F = 0.818 μ_{CR} = 0.062
 Diversity = 5.1815e-02 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.572607e+03 FES = 252100 μ_F = 0.812 μ_{CR} = 0.061
 Diversity = 5.3544e-02 k_t = [4, 4, 4, 3]
 Generation 2530: Best = 1.572605e+03 FES = 253100 μ_F = 0.826 μ_{CR} = 0.065
 Diversity = 5.3647e-02 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.572603e+03 FES = 254100 μ_F = 0.795 μ_{CR} = 0.067
 Diversity = 5.1785e-02 k_t = [3, 4, 2, 4]
 Generation 2550: Best = 1.572603e+03 FES = 255100 μ_F = 0.795 μ_{CR} = 0.067
 Diversity = 5.2020e-02 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.572598e+03 FES = 256100 μ_F = 0.797 μ_{CR} = 0.067
 Diversity = 4.7900e-02 k_t = [2, 2, 2, 3]
 Generation 2570: Best = 1.572596e+03 FES = 257100 μ_F = 0.787 μ_{CR} = 0.071
 Diversity = 5.0889e-02 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.572591e+03 FES = 258100 μ_F = 0.782 μ_{CR} = 0.068
 Diversity = 5.4005e-02 k_t = [3, 3, 3, 2]
 Generation 2590: Best = 1.572589e+03 FES = 259100 μ_F = 0.776 μ_{CR} = 0.065
 Diversity = 5.5831e-02 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.572588e+03 FES = 260100 μ_F = 0.785 μ_{CR} = 0.067
 Diversity = 5.3173e-02 k_t = [4, 2, 4, 4]
 Generation 2610: Best = 1.572587e+03 FES = 261100 μ_F = 0.760 μ_{CR} = 0.063
 Diversity = 5.3326e-02 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.572586e+03 FES = 262100 μ_F = 0.769 μ_{CR} = 0.071
 Diversity = 5.5341e-02 k_t = [4, 4, 4, 3]
 Generation 2630: Best = 1.572585e+03 FES = 263100 μ_F = 0.773 μ_{CR} = 0.071
 Diversity = 5.1517e-02 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.572583e+03 FES = 264100 μ_F = 0.765 μ_{CR} = 0.068
 Diversity = 5.1339e-02 k_t = [3, 3, 2, 3]
 Generation 2650: Best = 1.572583e+03 FES = 265100 μ_F = 0.769 μ_{CR} = 0.062
 Diversity = 5.1548e-02 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.572579e+03 FES = 266100 μ_F = 0.816 μ_{CR} = 0.063
 Diversity = 4.9940e-02 k_t = [4, 3, 3, 3]
 Generation 2670: Best = 1.572578e+03 FES = 267100 μ_F = 0.816 μ_{CR} = 0.064
 Diversity = 5.6316e-02 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.572578e+03 FES = 268100 μ_F = 0.815 μ_{CR} = 0.059
 Diversity = 5.5350e-02 k_t = [3, 3, 3, 4]
 Generation 2690: Best = 1.572578e+03 FES = 269100 μ_F = 0.815 μ_{CR} = 0.066
 Diversity = 5.4741e-02 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.572574e+03 FES = 270100 μ_F = 0.808 μ_{CR} = 0.063
 Diversity = 5.4985e-02 k_t = [3, 2, 4, 3]
 Generation 2710: Best = 1.572574e+03 FES = 271100 μ_F = 0.826 μ_{CR} = 0.063
 Diversity = 5.2876e-02 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.572574e+03 FES = 272100 μ_F = 0.843 μ_{CR} = 0.065
 Diversity = 4.9821e-02 k_t = [3, 3, 3, 4]
 Generation 2730: Best = 1.572574e+03 FES = 273100 μ_F = 0.831 μ_{CR} = 0.074
 Diversity = 4.8620e-02 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.572572e+03 FES = 274100 μ_F = 0.845 μ_{CR} = 0.066
 Diversity = 4.9257e-02 k_t = [3, 4, 2, 2]
 Generation 2750: Best = 1.572572e+03 FES = 275100 μ_F = 0.845 μ_{CR} = 0.067
 Diversity = 4.8061e-02 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.572572e+03 FES = 276100 μ_F = 0.852 μ_{CR} = 0.065
 Diversity = 4.7501e-02 k_t = [2, 2, 4, 2]
 Generation 2770: Best = 1.572569e+03 FES = 277100 μ_F = 0.874 μ_{CR} = 0.064
 Diversity = 4.7504e-02 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.572568e+03 FES = 278100 μ_F = 0.858 μ_{CR} = 0.063
 Diversity = 4.6690e-02 k_t = [4, 2, 2, 2]
 Generation 2790: Best = 1.572568e+03 FES = 279100 μ_F = 0.859 μ_{CR} = 0.070
 Diversity = 4.6605e-02 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.572566e+03 FES = 280100 μ_F = 0.849 μ_{CR} = 0.069
 Diversity = 4.6959e-02 k_t = [2, 2, 2, 2]
 Generation 2810: Best = 1.572566e+03 FES = 281100 μ_F = 0.859 μ_{CR} = 0.064
 Diversity = 4.7098e-02 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.572565e+03 FES = 282100 μ_F = 0.867 μ_{CR} = 0.064
 Diversity = 4.6858e-02 k_t = [4, 3, 3, 4]
 Generation 2830: Best = 1.572565e+03 FES = 283100 μ_F = 0.869 μ_{CR} = 0.064
 Diversity = 4.8236e-02 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.572564e+03 FES = 284100 μ_F = 0.860 μ_{CR} = 0.070
 Diversity = 4.7804e-02 k_t = [3, 2, 2, 2]
 Generation 2850: Best = 1.572561e+03 FES = 285100 μ_F = 0.853 μ_{CR} = 0.068
 Diversity = 4.8066e-02 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.572560e+03 FES = 286100 μ_F = 0.833 μ_{CR} = 0.064
 Diversity = 5.2323e-02 k_t = [3, 3, 4, 2]
 Generation 2870: Best = 1.572557e+03 FES = 287100 μ_F = 0.822 μ_{CR} = 0.069
 Diversity = 5.3575e-02 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.572554e+03 FES = 288100 μ_F = 0.822 μ_{CR} = 0.077
 Diversity = 4.7832e-02 k_t = [3, 2, 2, 4]
 Generation 2890: Best = 1.572547e+03 FES = 289100 μ_F = 0.825 μ_{CR} = 0.076
 Diversity = 5.2126e-02 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.572547e+03 FES = 290100 μ_F = 0.814 μ_{CR} = 0.076
 Diversity = 5.5111e-02 k_t = [3, 2, 2, 4]
 Generation 2910: Best = 1.572544e+03 FES = 291100 μ_F = 0.814 μ_{CR} = 0.075
 Diversity = 6.0150e-02 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.572540e+03 FES = 292100 μ_F = 0.787 μ_{CR} = 0.082
 Diversity = 5.7946e-02 k_t = [2, 2, 4, 4]

Generation 2930: Best = 1.572538e+03 FES = 293100 μ_F = 0.778 μ_{CR} = 0.082
 Diversity = 5.5798e-02 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 1.572533e+03 FES = 294100 μ_F = 0.785 μ_{CR} = 0.085
 Diversity = 5.8774e-02 k_t = [2, 3, 3, 3]
 Generation 2950: Best = 1.572527e+03 FES = 295100 μ_F = 0.799 μ_{CR} = 0.087
 Diversity = 5.4458e-02 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 1.572526e+03 FES = 296100 μ_F = 0.800 μ_{CR} = 0.077
 Diversity = 5.4368e-02 k_t = [4, 2, 4, 3]
 Generation 2970: Best = 1.572526e+03 FES = 297100 μ_F = 0.810 μ_{CR} = 0.078
 Diversity = 5.2419e-02 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 1.572526e+03 FES = 298100 μ_F = 0.822 μ_{CR} = 0.074
 Diversity = 5.3336e-02 k_t = [2, 4, 4, 3]
 Generation 2990: Best = 1.572524e+03 FES = 299100 μ_F = 0.814 μ_{CR} = 0.073
 Diversity = 5.3639e-02 k_t = [1, 1, 1, 1]

Optimization complete.

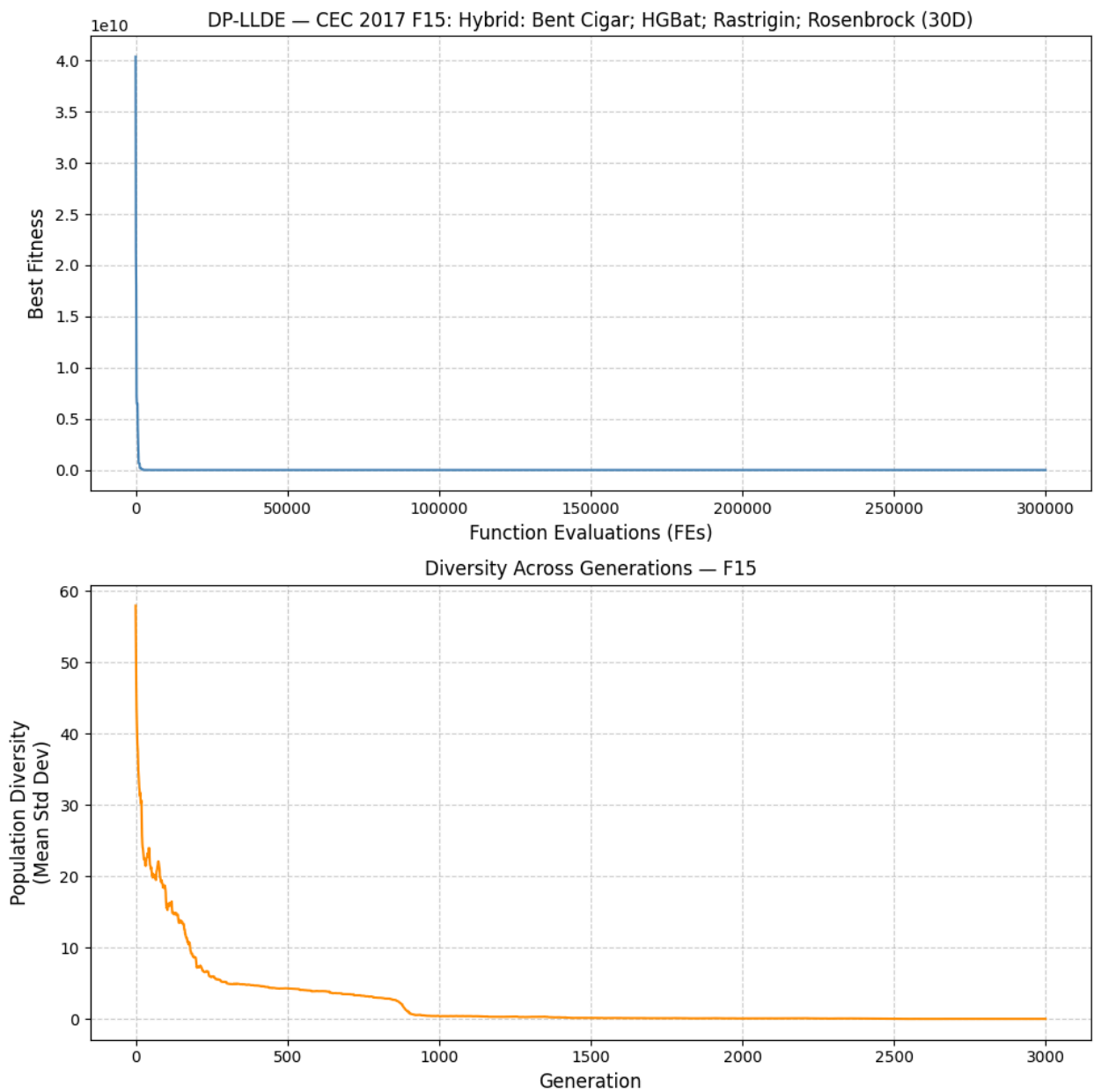
Best Fitness = 1.572524e+03

Total Function Evals = 300000

Total Successful Updates = 41358

```

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=
Results:
  Best Fitness      : 1.572524e+03
  Global Optimum    : 1.500000e+03
  Error (|best-optimum|) : 7.252393e+01
  Best Solution (5 dim) : [ 5.07361709  1.38099486  45.18697996 -87.89623486
77.82915691]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F15.png

```
In [ ]: # =====
# F16 – Hybrid Function 6 (Schaffer; HGBat; Rosenbrock; Schwefel)
# =====
run_and_plot('F16', make_wrapper(hybrid.f16), 1600.0, 'Hybrid: Schaffer; HGBat
```

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DP-LLDE on CEC 2017 F16 – Single Run

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Function : Hybrid: Schaffer; HGBat; Rosenbrock; Schwefel (30D)

Bounds : [-100, 100]

Global optimum : 1600.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 8.664946e+03

Generation 10: Best = 4.560186e+03 FES = 1100 $\mu_F = 0.584$ $\mu_{CR} = 0.361$

Diversity = 3.5445e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 3.895189e+03 FES = 2100 $\mu_F = 0.615$ $\mu_{CR} = 0.352$

Diversity = 2.9310e+01 $k_t = [4, 2, 4, 4]$

Generation 30: Best = 3.746545e+03 FES = 3100 $\mu_F = 0.642$ $\mu_{CR} = 0.335$

Diversity = 2.7494e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 3.746545e+03 FES = 4100 $\mu_F = 0.656$ $\mu_{CR} = 0.315$

Diversity = 2.6165e+01 $k_t = [4, 3, 4, 2]$

Generation 50: Best = 3.647710e+03 FES = 5100 $\mu_F = 0.703$ $\mu_{CR} = 0.278$

Diversity = 2.5164e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 3.460418e+03 FES = 6100 $\mu_F = 0.747$ $\mu_{CR} = 0.202$

Diversity = 2.5662e+01 $k_t = [2, 4, 2, 3]$

Generation 70: Best = 3.409365e+03 FES = 7100 $\mu_F = 0.750$ $\mu_{CR} = 0.135$

Diversity = 2.5821e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 3.394991e+03 FES = 8100 $\mu_F = 0.770$ $\mu_{CR} = 0.099$

Diversity = 2.6012e+01 $k_t = [4, 4, 2, 3]$

Generation 90: Best = 3.014736e+03 FES = 9100 $\mu_F = 0.783$ $\mu_{CR} = 0.078$

Diversity = 2.5885e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 3.011462e+03 FES = 10100 $\mu_F = 0.793$ $\mu_{CR} = 0.074$

Diversity = 2.5649e+01 $k_t = [4, 4, 4, 2]$

Generation 110: Best = 2.982613e+03 FES = 11100 $\mu_F = 0.784$ $\mu_{CR} = 0.070$

Diversity = 2.5204e+01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 2.958862e+03 FES = 12100 $\mu_F = 0.799$ $\mu_{CR} = 0.067$

Diversity = 2.5470e+01 $k_t = [3, 3, 4, 3]$

Generation 130: Best = 2.958862e+03 FES = 13100 $\mu_F = 0.822$ $\mu_{CR} = 0.069$

Diversity = 2.5621e+01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 2.945634e+03 FES = 14100 $\mu_F = 0.812$ $\mu_{CR} = 0.059$

Diversity = 2.5150e+01 $k_t = [3, 2, 2, 4]$

Generation 150: Best = 2.931762e+03 FES = 15100 $\mu_F = 0.816$ $\mu_{CR} = 0.061$

Diversity = 2.5069e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 2.931762e+03 FES = 16100 $\mu_F = 0.845$ $\mu_{CR} = 0.059$

Diversity = 2.4713e+01 $k_t = [4, 4, 4, 4]$

Generation 170: Best = 2.894529e+03 FES = 17100 $\mu_F = 0.818$ $\mu_{CR} = 0.061$

Diversity = 2.4486e+01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 2.894529e+03 FES = 18100 $\mu_F = 0.830$ $\mu_{CR} = 0.062$

Diversity = 2.4491e+01 $k_t = [2, 3, 2, 4]$

Generation 190: Best = 2.889103e+03 FES = 19100 $\mu_F = 0.826$ $\mu_{CR} = 0.070$

Diversity = 2.4588e+01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 2.884245e+03 FES = 20100 $\mu_F = 0.807$ $\mu_{CR} = 0.070$

Diversity = 2.4397e+01 $k_t = [3, 4, 4, 3]$

Generation 210: Best = 2.829593e+03 FES = 21100 $\mu_F = 0.796$ $\mu_{CR} = 0.071$

Diversity = 2.4362e+01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 2.815508e+03 FES = 22100 $\mu_F = 0.792$ $\mu_{CR} = 0.066$

Diversity = 2.4498e+01 $k_t = [3, 3, 2, 3]$

Generation 230: Best = 2.815508e+03	FES = 23100	$\mu_F = 0.815$	$\mu_{CR} = 0.068$
Diversity = 2.4418e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 2.815508e+03	FES = 24100	$\mu_F = 0.780$	$\mu_{CR} = 0.087$
Diversity = 2.4492e+01	$k_t = [2, 3, 4, 3]$		
Generation 250: Best = 2.777434e+03	FES = 25100	$\mu_F = 0.800$	$\mu_{CR} = 0.072$
Diversity = 2.4321e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 2.777434e+03	FES = 26100	$\mu_F = 0.749$	$\mu_{CR} = 0.070$
Diversity = 2.4057e+01	$k_t = [2, 4, 4, 4]$		
Generation 270: Best = 2.777431e+03	FES = 27100	$\mu_F = 0.753$	$\mu_{CR} = 0.060$
Diversity = 2.3981e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 2.766752e+03	FES = 28100	$\mu_F = 0.766$	$\mu_{CR} = 0.058$
Diversity = 2.3951e+01	$k_t = [4, 2, 2, 2]$		
Generation 290: Best = 2.761875e+03	FES = 29100	$\mu_F = 0.793$	$\mu_{CR} = 0.065$
Diversity = 2.3739e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 2.761875e+03	FES = 30100	$\mu_F = 0.825$	$\mu_{CR} = 0.063$
Diversity = 2.3510e+01	$k_t = [4, 3, 3, 4]$		
Generation 310: Best = 2.761875e+03	FES = 31100	$\mu_F = 0.827$	$\mu_{CR} = 0.062$
Diversity = 2.3497e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 2.761875e+03	FES = 32100	$\mu_F = 0.820$	$\mu_{CR} = 0.068$
Diversity = 2.3113e+01	$k_t = [4, 3, 2, 4]$		
Generation 330: Best = 2.637944e+03	FES = 33100	$\mu_F = 0.795$	$\mu_{CR} = 0.074$
Diversity = 2.3207e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 2.607255e+03	FES = 34100	$\mu_F = 0.800$	$\mu_{CR} = 0.065$
Diversity = 2.3318e+01	$k_t = [3, 3, 4, 4]$		
Generation 350: Best = 2.607255e+03	FES = 35100	$\mu_F = 0.812$	$\mu_{CR} = 0.062$
Diversity = 2.3270e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 2.607251e+03	FES = 36100	$\mu_F = 0.817$	$\mu_{CR} = 0.057$
Diversity = 2.3100e+01	$k_t = [2, 3, 4, 3]$		
Generation 370: Best = 2.587060e+03	FES = 37100	$\mu_F = 0.822$	$\mu_{CR} = 0.061$
Diversity = 2.3143e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 2.584787e+03	FES = 38100	$\mu_F = 0.830$	$\mu_{CR} = 0.064$
Diversity = 2.2749e+01	$k_t = [2, 2, 3, 4]$		
Generation 390: Best = 2.584787e+03	FES = 39100	$\mu_F = 0.820$	$\mu_{CR} = 0.067$
Diversity = 2.2645e+01	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 2.580710e+03	FES = 40100	$\mu_F = 0.825$	$\mu_{CR} = 0.068$
Diversity = 2.2492e+01	$k_t = [3, 3, 2, 2]$		
Generation 410: Best = 2.580463e+03	FES = 41100	$\mu_F = 0.819$	$\mu_{CR} = 0.060$
Diversity = 2.2600e+01	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 2.580446e+03	FES = 42100	$\mu_F = 0.771$	$\mu_{CR} = 0.059$
Diversity = 2.2336e+01	$k_t = [4, 4, 2, 2]$		
Generation 430: Best = 2.580446e+03	FES = 43100	$\mu_F = 0.805$	$\mu_{CR} = 0.050$
Diversity = 2.2258e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 2.576631e+03	FES = 44100	$\mu_F = 0.801$	$\mu_{CR} = 0.061$
Diversity = 2.1987e+01	$k_t = [3, 4, 3, 3]$		
Generation 450: Best = 2.576631e+03	FES = 45100	$\mu_F = 0.848$	$\mu_{CR} = 0.055$
Diversity = 2.1920e+01	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 2.576469e+03	FES = 46100	$\mu_F = 0.845$	$\mu_{CR} = 0.047$
Diversity = 2.1886e+01	$k_t = [3, 3, 2, 3]$		
Generation 470: Best = 2.562629e+03	FES = 47100	$\mu_F = 0.805$	$\mu_{CR} = 0.056$
Diversity = 2.1826e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.532609e+03	FES = 48100	$\mu_F = 0.780$	$\mu_{CR} = 0.058$
Diversity = 2.1869e+01	$k_t = [2, 2, 2, 2]$		
Generation 490: Best = 2.532609e+03	FES = 49100	$\mu_F = 0.811$	$\mu_{CR} = 0.062$
Diversity = 2.1806e+01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 2.532609e+03	FES = 50100	$\mu_F = 0.829$	$\mu_{CR} = 0.064$
Diversity = 2.1734e+01	k_t = [3, 4, 2, 3]		
Generation 510: Best = 2.532609e+03	FES = 51100	$\mu_F = 0.832$	$\mu_{CR} = 0.071$
Diversity = 2.1733e+01	k_t = [1, 1, 1, 1]		
Generation 520: Best = 2.532609e+03	FES = 52100	$\mu_F = 0.871$	$\mu_{CR} = 0.056$
Diversity = 2.1714e+01	k_t = [3, 2, 4, 2]		
Generation 530: Best = 2.531725e+03	FES = 53100	$\mu_F = 0.873$	$\mu_{CR} = 0.051$
Diversity = 2.1904e+01	k_t = [1, 1, 1, 1]		
Generation 540: Best = 2.531725e+03	FES = 54100	$\mu_F = 0.841$	$\mu_{CR} = 0.056$
Diversity = 2.1882e+01	k_t = [4, 4, 4, 3]		
Generation 550: Best = 2.531526e+03	FES = 55100	$\mu_F = 0.861$	$\mu_{CR} = 0.060$
Diversity = 2.2049e+01	k_t = [1, 1, 1, 1]		
Generation 560: Best = 2.478625e+03	FES = 56100	$\mu_F = 0.814$	$\mu_{CR} = 0.064$
Diversity = 2.2125e+01	k_t = [4, 3, 2, 2]		
Generation 570: Best = 2.467643e+03	FES = 57100	$\mu_F = 0.805$	$\mu_{CR} = 0.065$
Diversity = 2.1890e+01	k_t = [1, 1, 1, 1]		
Generation 580: Best = 2.465912e+03	FES = 58100	$\mu_F = 0.812$	$\mu_{CR} = 0.064$
Diversity = 2.1896e+01	k_t = [4, 3, 3, 4]		
Generation 590: Best = 2.403861e+03	FES = 59100	$\mu_F = 0.790$	$\mu_{CR} = 0.073$
Diversity = 2.1965e+01	k_t = [1, 1, 1, 1]		
Generation 600: Best = 2.403245e+03	FES = 60100	$\mu_F = 0.794$	$\mu_{CR} = 0.064$
Diversity = 2.1765e+01	k_t = [3, 2, 2, 3]		
Generation 610: Best = 2.400857e+03	FES = 61100	$\mu_F = 0.763$	$\mu_{CR} = 0.064$
Diversity = 2.1822e+01	k_t = [1, 1, 1, 1]		
Generation 620: Best = 2.395380e+03	FES = 62100	$\mu_F = 0.733$	$\mu_{CR} = 0.064$
Diversity = 2.1267e+01	k_t = [2, 2, 2, 4]		
Generation 630: Best = 2.395322e+03	FES = 63100	$\mu_F = 0.715$	$\mu_{CR} = 0.063$
Diversity = 2.1172e+01	k_t = [1, 1, 1, 1]		
Generation 640: Best = 2.395322e+03	FES = 64100	$\mu_F = 0.712$	$\mu_{CR} = 0.062$
Diversity = 2.1044e+01	k_t = [3, 4, 4, 3]		
Generation 650: Best = 2.360713e+03	FES = 65100	$\mu_F = 0.755$	$\mu_{CR} = 0.058$
Diversity = 2.1049e+01	k_t = [1, 1, 1, 1]		
Generation 660: Best = 2.357602e+03	FES = 66100	$\mu_F = 0.788$	$\mu_{CR} = 0.064$
Diversity = 2.0913e+01	k_t = [2, 4, 4, 4]		
Generation 670: Best = 2.357602e+03	FES = 67100	$\mu_F = 0.790$	$\mu_{CR} = 0.068$
Diversity = 2.0946e+01	k_t = [1, 1, 1, 1]		
Generation 680: Best = 2.357602e+03	FES = 68100	$\mu_F = 0.808$	$\mu_{CR} = 0.063$
Diversity = 2.0847e+01	k_t = [3, 2, 4, 3]		
Generation 690: Best = 2.357517e+03	FES = 69100	$\mu_F = 0.809$	$\mu_{CR} = 0.062$
Diversity = 2.0925e+01	k_t = [1, 1, 1, 1]		
Generation 700: Best = 2.357517e+03	FES = 70100	$\mu_F = 0.810$	$\mu_{CR} = 0.080$
Diversity = 2.0932e+01	k_t = [4, 4, 4, 3]		
Generation 710: Best = 2.352854e+03	FES = 71100	$\mu_F = 0.780$	$\mu_{CR} = 0.067$
Diversity = 2.0861e+01	k_t = [1, 1, 1, 1]		
Generation 720: Best = 2.352854e+03	FES = 72100	$\mu_F = 0.770$	$\mu_{CR} = 0.067$
Diversity = 2.0672e+01	k_t = [3, 4, 3, 4]		
Generation 730: Best = 2.340977e+03	FES = 73100	$\mu_F = 0.808$	$\mu_{CR} = 0.066$
Diversity = 2.0641e+01	k_t = [1, 1, 1, 1]		
Generation 740: Best = 2.340977e+03	FES = 74100	$\mu_F = 0.839$	$\mu_{CR} = 0.070$
Diversity = 2.0675e+01	k_t = [3, 3, 3, 3]		
Generation 750: Best = 2.340977e+03	FES = 75100	$\mu_F = 0.827$	$\mu_{CR} = 0.075$
Diversity = 2.0637e+01	k_t = [1, 1, 1, 1]		
Generation 760: Best = 2.335853e+03	FES = 76100	$\mu_F = 0.796$	$\mu_{CR} = 0.071$
Diversity = 2.0596e+01	k_t = [4, 2, 2, 3]		

Generation 770:	Best = 2.335429e+03	FES = 77100	$\mu_F = 0.808$	$\mu_{CR} = 0.055$
Diversity = 2.0552e+01	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 2.333125e+03	FES = 78100	$\mu_F = 0.782$	$\mu_{CR} = 0.067$
Diversity = 2.0205e+01	$k_t = [4, 2, 2, 4]$			
Generation 790:	Best = 2.315278e+03	FES = 79100	$\mu_F = 0.787$	$\mu_{CR} = 0.058$
Diversity = 2.0113e+01	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 2.315278e+03	FES = 80100	$\mu_F = 0.827$	$\mu_{CR} = 0.056$
Diversity = 1.9843e+01	$k_t = [4, 2, 3, 4]$			
Generation 810:	Best = 2.315278e+03	FES = 81100	$\mu_F = 0.816$	$\mu_{CR} = 0.067$
Diversity = 1.9769e+01	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 2.310497e+03	FES = 82100	$\mu_F = 0.837$	$\mu_{CR} = 0.073$
Diversity = 1.9438e+01	$k_t = [4, 3, 2, 4]$			
Generation 830:	Best = 2.310497e+03	FES = 83100	$\mu_F = 0.829$	$\mu_{CR} = 0.071$
Diversity = 1.9298e+01	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 2.310497e+03	FES = 84100	$\mu_F = 0.806$	$\mu_{CR} = 0.067$
Diversity = 1.9380e+01	$k_t = [2, 2, 4, 2]$			
Generation 850:	Best = 2.310284e+03	FES = 85100	$\mu_F = 0.801$	$\mu_{CR} = 0.059$
Diversity = 1.9446e+01	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 2.310284e+03	FES = 86100	$\mu_F = 0.809$	$\mu_{CR} = 0.056$
Diversity = 1.9224e+01	$k_t = [3, 4, 3, 4]$			
Generation 870:	Best = 2.309095e+03	FES = 87100	$\mu_F = 0.800$	$\mu_{CR} = 0.061$
Diversity = 1.9089e+01	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 2.309094e+03	FES = 88100	$\mu_F = 0.799$	$\mu_{CR} = 0.064$
Diversity = 1.9127e+01	$k_t = [4, 3, 3, 3]$			
Generation 890:	Best = 2.246297e+03	FES = 89100	$\mu_F = 0.775$	$\mu_{CR} = 0.056$
Diversity = 1.9090e+01	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 2.246290e+03	FES = 90100	$\mu_F = 0.796$	$\mu_{CR} = 0.066$
Diversity = 1.8919e+01	$k_t = [2, 4, 2, 4]$			
Generation 910:	Best = 2.245455e+03	FES = 91100	$\mu_F = 0.780$	$\mu_{CR} = 0.063$
Diversity = 1.9043e+01	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 2.244888e+03	FES = 92100	$\mu_F = 0.783$	$\mu_{CR} = 0.074$
Diversity = 1.8987e+01	$k_t = [4, 3, 4, 4]$			
Generation 930:	Best = 2.244888e+03	FES = 93100	$\mu_F = 0.790$	$\mu_{CR} = 0.067$
Diversity = 1.8969e+01	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 2.244012e+03	FES = 94100	$\mu_F = 0.797$	$\mu_{CR} = 0.069$
Diversity = 1.8847e+01	$k_t = [4, 4, 2, 2]$			
Generation 950:	Best = 2.244012e+03	FES = 95100	$\mu_F = 0.802$	$\mu_{CR} = 0.072$
Diversity = 1.8870e+01	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 2.244012e+03	FES = 96100	$\mu_F = 0.812$	$\mu_{CR} = 0.075$
Diversity = 1.8761e+01	$k_t = [3, 4, 2, 3]$			
Generation 970:	Best = 2.243526e+03	FES = 97100	$\mu_F = 0.850$	$\mu_{CR} = 0.072$
Diversity = 1.8734e+01	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 2.242815e+03	FES = 98100	$\mu_F = 0.858$	$\mu_{CR} = 0.068$
Diversity = 1.8376e+01	$k_t = [3, 2, 4, 4]$			
Generation 990:	Best = 2.242815e+03	FES = 99100	$\mu_F = 0.869$	$\mu_{CR} = 0.069$
Diversity = 1.8090e+01	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 2.242815e+03	FES = 100100	$\mu_F = 0.859$	$\mu_{CR} = 0.064$
Diversity = 1.7540e+01	$k_t = [3, 4, 2, 3]$			
Generation 1010:	Best = 2.239890e+03	FES = 101100	$\mu_F = 0.863$	$\mu_{CR} = 0.068$
Diversity = 1.7114e+01	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 2.229576e+03	FES = 102100	$\mu_F = 0.847$	$\mu_{CR} = 0.073$
Diversity = 1.6730e+01	$k_t = [3, 2, 3, 2]$			
Generation 1030:	Best = 2.229576e+03	FES = 103100	$\mu_F = 0.873$	$\mu_{CR} = 0.087$
Diversity = 1.6645e+01	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 2.159013e+03 FES = 104100 μ_F = 0.900 μ_{CR} = 0.098
 Diversity = 1.6428e+01 k_t = [3, 3, 4, 3]
 Generation 1050: Best = 2.159013e+03 FES = 105100 μ_F = 0.898 μ_{CR} = 0.086
 Diversity = 1.6488e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 2.155782e+03 FES = 106100 μ_F = 0.907 μ_{CR} = 0.078
 Diversity = 1.6359e+01 k_t = [4, 2, 4, 3]
 Generation 1070: Best = 2.146005e+03 FES = 107100 μ_F = 0.889 μ_{CR} = 0.081
 Diversity = 1.6360e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 2.133359e+03 FES = 108100 μ_F = 0.892 μ_{CR} = 0.084
 Diversity = 1.5947e+01 k_t = [4, 4, 4, 3]
 Generation 1090: Best = 2.081625e+03 FES = 109100 μ_F = 0.883 μ_{CR} = 0.096
 Diversity = 1.5676e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 2.018146e+03 FES = 110100 μ_F = 0.891 μ_{CR} = 0.113
 Diversity = 1.5191e+01 k_t = [3, 2, 3, 2]
 Generation 1110: Best = 2.015363e+03 FES = 111100 μ_F = 0.889 μ_{CR} = 0.116
 Diversity = 1.4240e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 1.968363e+03 FES = 112100 μ_F = 0.889 μ_{CR} = 0.127
 Diversity = 1.1169e+01 k_t = [3, 3, 3, 2]
 Generation 1130: Best = 1.924256e+03 FES = 113100 μ_F = 0.874 μ_{CR} = 0.125
 Diversity = 9.4259e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 1.889025e+03 FES = 114100 μ_F = 0.854 μ_{CR} = 0.127
 Diversity = 8.9437e+00 k_t = [3, 4, 3, 3]
 Generation 1150: Best = 1.887556e+03 FES = 115100 μ_F = 0.851 μ_{CR} = 0.125
 Diversity = 9.7946e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 1.882509e+03 FES = 116100 μ_F = 0.850 μ_{CR} = 0.141
 Diversity = 9.7975e+00 k_t = [4, 2, 4, 3]
 Generation 1170: Best = 1.868200e+03 FES = 117100 μ_F = 0.832 μ_{CR} = 0.150
 Diversity = 8.4837e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 1.866302e+03 FES = 118100 μ_F = 0.808 μ_{CR} = 0.150
 Diversity = 7.4282e+00 k_t = [2, 3, 2, 2]
 Generation 1190: Best = 1.863239e+03 FES = 119100 μ_F = 0.791 μ_{CR} = 0.156
 Diversity = 6.9160e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 1.862333e+03 FES = 120100 μ_F = 0.765 μ_{CR} = 0.165
 Diversity = 6.4240e+00 k_t = [2, 4, 4, 3]
 Generation 1210: Best = 1.861620e+03 FES = 121100 μ_F = 0.755 μ_{CR} = 0.159
 Diversity = 5.7293e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 1.861148e+03 FES = 122100 μ_F = 0.740 μ_{CR} = 0.174
 Diversity = 4.7590e+00 k_t = [4, 4, 3, 4]
 Generation 1230: Best = 1.861022e+03 FES = 123100 μ_F = 0.733 μ_{CR} = 0.158
 Diversity = 4.3695e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 1.860599e+03 FES = 124100 μ_F = 0.707 μ_{CR} = 0.152
 Diversity = 3.8040e+00 k_t = [4, 3, 4, 3]
 Generation 1250: Best = 1.860448e+03 FES = 125100 μ_F = 0.707 μ_{CR} = 0.140
 Diversity = 3.4548e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 1.860356e+03 FES = 126100 μ_F = 0.697 μ_{CR} = 0.126
 Diversity = 2.9955e+00 k_t = [2, 4, 4, 2]
 Generation 1270: Best = 1.860356e+03 FES = 127100 μ_F = 0.720 μ_{CR} = 0.111
 Diversity = 2.8274e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 1.860255e+03 FES = 128100 μ_F = 0.753 μ_{CR} = 0.102
 Diversity = 2.6168e+00 k_t = [3, 4, 4, 3]
 Generation 1290: Best = 1.860199e+03 FES = 129100 μ_F = 0.772 μ_{CR} = 0.096
 Diversity = 2.4784e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 1.860197e+03 FES = 130100 μ_F = 0.760 μ_{CR} = 0.101
 Diversity = 2.4301e+00 k_t = [4, 4, 3, 2]

Generation 1310: Best = 1.860190e+03 FES = 131100 μ_F = 0.790 μ_{CR} = 0.087
 Diversity = 2.3812e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.860183e+03 FES = 132100 μ_F = 0.780 μ_{CR} = 0.100
 Diversity = 2.2877e+00 k_t = [2, 2, 2, 2]
 Generation 1330: Best = 1.860162e+03 FES = 133100 μ_F = 0.772 μ_{CR} = 0.088
 Diversity = 2.1179e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.860107e+03 FES = 134100 μ_F = 0.754 μ_{CR} = 0.100
 Diversity = 2.1263e+00 k_t = [2, 2, 2, 2]
 Generation 1350: Best = 1.860104e+03 FES = 135100 μ_F = 0.773 μ_{CR} = 0.074
 Diversity = 2.1873e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.860084e+03 FES = 136100 μ_F = 0.741 μ_{CR} = 0.083
 Diversity = 2.1190e+00 k_t = [3, 4, 2, 2]
 Generation 1370: Best = 1.860084e+03 FES = 137100 μ_F = 0.756 μ_{CR} = 0.077
 Diversity = 2.1026e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.860084e+03 FES = 138100 μ_F = 0.733 μ_{CR} = 0.085
 Diversity = 1.9507e+00 k_t = [4, 3, 4, 4]
 Generation 1390: Best = 1.860084e+03 FES = 139100 μ_F = 0.761 μ_{CR} = 0.080
 Diversity = 1.9124e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.860077e+03 FES = 140100 μ_F = 0.729 μ_{CR} = 0.078
 Diversity = 1.8940e+00 k_t = [2, 4, 2, 2]
 Generation 1410: Best = 1.860075e+03 FES = 141100 μ_F = 0.704 μ_{CR} = 0.075
 Diversity = 1.8946e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.860072e+03 FES = 142100 μ_F = 0.704 μ_{CR} = 0.070
 Diversity = 1.9049e+00 k_t = [3, 2, 3, 2]
 Generation 1430: Best = 1.860072e+03 FES = 143100 μ_F = 0.739 μ_{CR} = 0.067
 Diversity = 1.8797e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.860060e+03 FES = 144100 μ_F = 0.760 μ_{CR} = 0.055
 Diversity = 1.8621e+00 k_t = [3, 4, 3, 3]
 Generation 1450: Best = 1.860060e+03 FES = 145100 μ_F = 0.743 μ_{CR} = 0.068
 Diversity = 1.8235e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.860053e+03 FES = 146100 μ_F = 0.749 μ_{CR} = 0.065
 Diversity = 1.8300e+00 k_t = [4, 2, 2, 2]
 Generation 1470: Best = 1.860048e+03 FES = 147100 μ_F = 0.751 μ_{CR} = 0.060
 Diversity = 1.8136e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.860045e+03 FES = 148100 μ_F = 0.769 μ_{CR} = 0.066
 Diversity = 1.8006e+00 k_t = [2, 3, 4, 2]
 Generation 1490: Best = 1.860039e+03 FES = 149100 μ_F = 0.769 μ_{CR} = 0.060
 Diversity = 1.7975e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.860039e+03 FES = 150100 μ_F = 0.778 μ_{CR} = 0.067
 Diversity = 1.7664e+00 k_t = [3, 3, 3, 4]
 Generation 1510: Best = 1.860031e+03 FES = 151100 μ_F = 0.770 μ_{CR} = 0.065
 Diversity = 1.7614e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.860030e+03 FES = 152100 μ_F = 0.790 μ_{CR} = 0.063
 Diversity = 1.7897e+00 k_t = [3, 4, 2, 3]
 Generation 1530: Best = 1.860026e+03 FES = 153100 μ_F = 0.779 μ_{CR} = 0.069
 Diversity = 1.8014e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.860025e+03 FES = 154100 μ_F = 0.759 μ_{CR} = 0.058
 Diversity = 1.8011e+00 k_t = [3, 4, 4, 3]
 Generation 1550: Best = 1.860025e+03 FES = 155100 μ_F = 0.745 μ_{CR} = 0.060
 Diversity = 1.7824e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.860023e+03 FES = 156100 μ_F = 0.747 μ_{CR} = 0.058
 Diversity = 1.7934e+00 k_t = [4, 3, 2, 2]
 Generation 1570: Best = 1.860023e+03 FES = 157100 μ_F = 0.753 μ_{CR} = 0.059
 Diversity = 1.7801e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 1.860023e+03 FES = 158100 μ_F = 0.745 μ_{CR} = 0.057
 Diversity = 1.7663e+00 k_t = [2, 2, 2, 2]
 Generation 1590: Best = 1.860022e+03 FES = 159100 μ_F = 0.721 μ_{CR} = 0.063
 Diversity = 1.8118e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 1.860022e+03 FES = 160100 μ_F = 0.754 μ_{CR} = 0.059
 Diversity = 1.7984e+00 k_t = [2, 3, 4, 4]
 Generation 1610: Best = 1.860020e+03 FES = 161100 μ_F = 0.771 μ_{CR} = 0.059
 Diversity = 1.7828e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 1.860019e+03 FES = 162100 μ_F = 0.806 μ_{CR} = 0.058
 Diversity = 1.8105e+00 k_t = [2, 3, 4, 4]
 Generation 1630: Best = 1.860019e+03 FES = 163100 μ_F = 0.806 μ_{CR} = 0.061
 Diversity = 1.7873e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 1.860019e+03 FES = 164100 μ_F = 0.770 μ_{CR} = 0.056
 Diversity = 1.7867e+00 k_t = [3, 4, 4, 2]
 Generation 1650: Best = 1.860019e+03 FES = 165100 μ_F = 0.770 μ_{CR} = 0.062
 Diversity = 1.7850e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 1.860018e+03 FES = 166100 μ_F = 0.747 μ_{CR} = 0.062
 Diversity = 1.7878e+00 k_t = [3, 4, 4, 3]
 Generation 1670: Best = 1.860017e+03 FES = 167100 μ_F = 0.733 μ_{CR} = 0.060
 Diversity = 1.7838e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 1.860017e+03 FES = 168100 μ_F = 0.738 μ_{CR} = 0.067
 Diversity = 1.7852e+00 k_t = [3, 2, 3, 2]
 Generation 1690: Best = 1.860017e+03 FES = 169100 μ_F = 0.762 μ_{CR} = 0.063
 Diversity = 1.7888e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 1.860010e+03 FES = 170100 μ_F = 0.763 μ_{CR} = 0.070
 Diversity = 1.7720e+00 k_t = [3, 4, 4, 4]
 Generation 1710: Best = 1.860010e+03 FES = 171100 μ_F = 0.753 μ_{CR} = 0.062
 Diversity = 1.7545e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 1.860009e+03 FES = 172100 μ_F = 0.780 μ_{CR} = 0.067
 Diversity = 1.7610e+00 k_t = [3, 2, 3, 2]
 Generation 1730: Best = 1.860008e+03 FES = 173100 μ_F = 0.766 μ_{CR} = 0.066
 Diversity = 1.7735e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 1.860008e+03 FES = 174100 μ_F = 0.772 μ_{CR} = 0.062
 Diversity = 1.7786e+00 k_t = [3, 4, 2, 4]
 Generation 1750: Best = 1.860008e+03 FES = 175100 μ_F = 0.796 μ_{CR} = 0.057
 Diversity = 1.7773e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 1.860008e+03 FES = 176100 μ_F = 0.781 μ_{CR} = 0.053
 Diversity = 1.7845e+00 k_t = [4, 3, 2, 2]
 Generation 1770: Best = 1.860008e+03 FES = 177100 μ_F = 0.770 μ_{CR} = 0.056
 Diversity = 1.8014e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 1.860008e+03 FES = 178100 μ_F = 0.780 μ_{CR} = 0.062
 Diversity = 1.8251e+00 k_t = [3, 3, 3, 4]
 Generation 1790: Best = 1.860008e+03 FES = 179100 μ_F = 0.789 μ_{CR} = 0.059
 Diversity = 1.8221e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 1.860008e+03 FES = 180100 μ_F = 0.796 μ_{CR} = 0.058
 Diversity = 1.7905e+00 k_t = [3, 2, 4, 4]
 Generation 1810: Best = 1.860008e+03 FES = 181100 μ_F = 0.801 μ_{CR} = 0.055
 Diversity = 1.7895e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 1.860007e+03 FES = 182100 μ_F = 0.808 μ_{CR} = 0.050
 Diversity = 1.7893e+00 k_t = [4, 4, 2, 3]
 Generation 1830: Best = 1.860004e+03 FES = 183100 μ_F = 0.782 μ_{CR} = 0.061
 Diversity = 1.7969e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 1.860004e+03 FES = 184100 μ_F = 0.820 μ_{CR} = 0.061
 Diversity = 1.7911e+00 k_t = [2, 2, 4, 3]

Generation 1850: Best = 1.860004e+03	FES = 185100	$\mu_F = 0.772$	$\mu_{CR} = 0.054$
Diversity = 1.7910e+00	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 1.860003e+03	FES = 186100	$\mu_F = 0.788$	$\mu_{CR} = 0.055$
Diversity = 1.7910e+00	k_t = [3, 3, 4, 3]		
Generation 1870: Best = 1.860003e+03	FES = 187100	$\mu_F = 0.758$	$\mu_{CR} = 0.058$
Diversity = 1.7921e+00	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 1.860003e+03	FES = 188100	$\mu_F = 0.749$	$\mu_{CR} = 0.054$
Diversity = 1.7752e+00	k_t = [2, 2, 4, 2]		
Generation 1890: Best = 1.860003e+03	FES = 189100	$\mu_F = 0.763$	$\mu_{CR} = 0.055$
Diversity = 1.7748e+00	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 1.860003e+03	FES = 190100	$\mu_F = 0.763$	$\mu_{CR} = 0.056$
Diversity = 1.7747e+00	k_t = [3, 2, 4, 2]		
Generation 1910: Best = 1.860002e+03	FES = 191100	$\mu_F = 0.749$	$\mu_{CR} = 0.048$
Diversity = 1.7738e+00	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 1.860002e+03	FES = 192100	$\mu_F = 0.753$	$\mu_{CR} = 0.047$
Diversity = 1.7790e+00	k_t = [3, 3, 4, 3]		
Generation 1930: Best = 1.860002e+03	FES = 193100	$\mu_F = 0.744$	$\mu_{CR} = 0.054$
Diversity = 1.7838e+00	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 1.860002e+03	FES = 194100	$\mu_F = 0.789$	$\mu_{CR} = 0.056$
Diversity = 1.7470e+00	k_t = [2, 4, 3, 4]		
Generation 1950: Best = 1.860002e+03	FES = 195100	$\mu_F = 0.795$	$\mu_{CR} = 0.055$
Diversity = 1.7475e+00	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 1.860002e+03	FES = 196100	$\mu_F = 0.811$	$\mu_{CR} = 0.049$
Diversity = 1.7282e+00	k_t = [4, 4, 2, 4]		
Generation 1970: Best = 1.860002e+03	FES = 197100	$\mu_F = 0.823$	$\mu_{CR} = 0.049$
Diversity = 1.7284e+00	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 1.860002e+03	FES = 198100	$\mu_F = 0.764$	$\mu_{CR} = 0.050$
Diversity = 1.6992e+00	k_t = [3, 3, 3, 3]		
Generation 1990: Best = 1.860002e+03	FES = 199100	$\mu_F = 0.758$	$\mu_{CR} = 0.058$
Diversity = 1.6988e+00	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 1.860002e+03	FES = 200100	$\mu_F = 0.766$	$\mu_{CR} = 0.068$
Diversity = 1.7157e+00	k_t = [2, 2, 4, 3]		
Generation 2010: Best = 1.860002e+03	FES = 201100	$\mu_F = 0.806$	$\mu_{CR} = 0.060$
Diversity = 1.7073e+00	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 1.860002e+03	FES = 202100	$\mu_F = 0.824$	$\mu_{CR} = 0.054$
Diversity = 1.6802e+00	k_t = [3, 3, 4, 4]		
Generation 2030: Best = 1.860002e+03	FES = 203100	$\mu_F = 0.801$	$\mu_{CR} = 0.056$
Diversity = 1.6727e+00	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 1.860002e+03	FES = 204100	$\mu_F = 0.801$	$\mu_{CR} = 0.058$
Diversity = 1.6755e+00	k_t = [4, 2, 2, 2]		
Generation 2050: Best = 1.860001e+03	FES = 205100	$\mu_F = 0.806$	$\mu_{CR} = 0.059$
Diversity = 1.6753e+00	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 1.860001e+03	FES = 206100	$\mu_F = 0.792$	$\mu_{CR} = 0.055$
Diversity = 1.6832e+00	k_t = [2, 3, 4, 4]		
Generation 2070: Best = 1.860000e+03	FES = 207100	$\mu_F = 0.825$	$\mu_{CR} = 0.045$
Diversity = 1.6831e+00	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 1.859999e+03	FES = 208100	$\mu_F = 0.839$	$\mu_{CR} = 0.049$
Diversity = 1.6695e+00	k_t = [3, 2, 2, 4]		
Generation 2090: Best = 1.859999e+03	FES = 209100	$\mu_F = 0.817$	$\mu_{CR} = 0.051$
Diversity = 1.6759e+00	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 1.859998e+03	FES = 210100	$\mu_F = 0.817$	$\mu_{CR} = 0.053$
Diversity = 1.6758e+00	k_t = [4, 3, 4, 2]		
Generation 2110: Best = 1.859998e+03	FES = 211100	$\mu_F = 0.786$	$\mu_{CR} = 0.055$
Diversity = 1.6745e+00	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 1.859997e+03 FES = 212100 μ_F = 0.751 μ_{CR} = 0.052
 Diversity = 1.6741e+00 k_t = [4, 4, 2, 2]
 Generation 2130: Best = 1.859997e+03 FES = 213100 μ_F = 0.768 μ_{CR} = 0.050
 Diversity = 1.6739e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 1.859997e+03 FES = 214100 μ_F = 0.800 μ_{CR} = 0.049
 Diversity = 1.6628e+00 k_t = [2, 3, 4, 2]
 Generation 2150: Best = 1.859997e+03 FES = 215100 μ_F = 0.779 μ_{CR} = 0.055
 Diversity = 1.6491e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 1.859997e+03 FES = 216100 μ_F = 0.770 μ_{CR} = 0.052
 Diversity = 1.6382e+00 k_t = [2, 3, 4, 2]
 Generation 2170: Best = 1.859997e+03 FES = 217100 μ_F = 0.748 μ_{CR} = 0.062
 Diversity = 1.6380e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 1.859997e+03 FES = 218100 μ_F = 0.772 μ_{CR} = 0.053
 Diversity = 1.6382e+00 k_t = [3, 4, 2, 2]
 Generation 2190: Best = 1.859997e+03 FES = 219100 μ_F = 0.784 μ_{CR} = 0.056
 Diversity = 1.6370e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 1.859996e+03 FES = 220100 μ_F = 0.807 μ_{CR} = 0.063
 Diversity = 1.6521e+00 k_t = [2, 3, 2, 4]
 Generation 2210: Best = 1.859995e+03 FES = 221100 μ_F = 0.826 μ_{CR} = 0.060
 Diversity = 1.6378e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 1.859995e+03 FES = 222100 μ_F = 0.831 μ_{CR} = 0.054
 Diversity = 1.6204e+00 k_t = [4, 4, 2, 4]
 Generation 2230: Best = 1.859995e+03 FES = 223100 μ_F = 0.830 μ_{CR} = 0.048
 Diversity = 1.6208e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 1.859995e+03 FES = 224100 μ_F = 0.843 μ_{CR} = 0.056
 Diversity = 1.6107e+00 k_t = [2, 2, 2, 3]
 Generation 2250: Best = 1.859993e+03 FES = 225100 μ_F = 0.820 μ_{CR} = 0.058
 Diversity = 1.6108e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 1.859993e+03 FES = 226100 μ_F = 0.811 μ_{CR} = 0.050
 Diversity = 1.6448e+00 k_t = [4, 3, 3, 4]
 Generation 2270: Best = 1.859993e+03 FES = 227100 μ_F = 0.808 μ_{CR} = 0.048
 Diversity = 1.6447e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 1.859989e+03 FES = 228100 μ_F = 0.808 μ_{CR} = 0.053
 Diversity = 1.6447e+00 k_t = [3, 2, 4, 3]
 Generation 2290: Best = 1.859989e+03 FES = 229100 μ_F = 0.785 μ_{CR} = 0.052
 Diversity = 1.6446e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 1.859989e+03 FES = 230100 μ_F = 0.771 μ_{CR} = 0.059
 Diversity = 1.6446e+00 k_t = [3, 3, 3, 3]
 Generation 2310: Best = 1.859989e+03 FES = 231100 μ_F = 0.785 μ_{CR} = 0.052
 Diversity = 1.6446e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 1.859989e+03 FES = 232100 μ_F = 0.786 μ_{CR} = 0.061
 Diversity = 1.6447e+00 k_t = [4, 4, 2, 3]
 Generation 2330: Best = 1.859989e+03 FES = 233100 μ_F = 0.815 μ_{CR} = 0.063
 Diversity = 1.6448e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 1.859989e+03 FES = 234100 μ_F = 0.801 μ_{CR} = 0.061
 Diversity = 1.6448e+00 k_t = [3, 4, 3, 3]
 Generation 2350: Best = 1.859989e+03 FES = 235100 μ_F = 0.810 μ_{CR} = 0.056
 Diversity = 1.6449e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 1.859989e+03 FES = 236100 μ_F = 0.844 μ_{CR} = 0.054
 Diversity = 1.6451e+00 k_t = [2, 3, 4, 2]
 Generation 2370: Best = 1.859989e+03 FES = 237100 μ_F = 0.857 μ_{CR} = 0.055
 Diversity = 1.6464e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 1.859989e+03 FES = 238100 μ_F = 0.847 μ_{CR} = 0.063
 Diversity = 1.6466e+00 k_t = [2, 2, 4, 3]

Generation 2390: Best = 1.859982e+03 FES = 239100 μ_F = 0.864 μ_{CR} = 0.060
 Diversity = 1.6502e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 1.859982e+03 FES = 240100 μ_F = 0.864 μ_{CR} = 0.067
 Diversity = 1.6727e+00 k_t = [3, 2, 4, 4]
 Generation 2410: Best = 1.859981e+03 FES = 241100 μ_F = 0.839 μ_{CR} = 0.057
 Diversity = 1.6728e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 1.859981e+03 FES = 242100 μ_F = 0.819 μ_{CR} = 0.047
 Diversity = 1.6522e+00 k_t = [2, 2, 3, 4]
 Generation 2430: Best = 1.859981e+03 FES = 243100 μ_F = 0.824 μ_{CR} = 0.052
 Diversity = 1.6523e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 1.859981e+03 FES = 244100 μ_F = 0.837 μ_{CR} = 0.056
 Diversity = 1.6773e+00 k_t = [4, 3, 4, 4]
 Generation 2450: Best = 1.859981e+03 FES = 245100 μ_F = 0.834 μ_{CR} = 0.059
 Diversity = 1.6774e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 1.859981e+03 FES = 246100 μ_F = 0.831 μ_{CR} = 0.062
 Diversity = 1.6775e+00 k_t = [2, 4, 4, 3]
 Generation 2470: Best = 1.859981e+03 FES = 247100 μ_F = 0.828 μ_{CR} = 0.056
 Diversity = 1.6777e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 1.859981e+03 FES = 248100 μ_F = 0.804 μ_{CR} = 0.053
 Diversity = 1.6777e+00 k_t = [3, 4, 3, 3]
 Generation 2490: Best = 1.859981e+03 FES = 249100 μ_F = 0.784 μ_{CR} = 0.042
 Diversity = 1.6777e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 1.859981e+03 FES = 250100 μ_F = 0.779 μ_{CR} = 0.054
 Diversity = 1.6778e+00 k_t = [3, 2, 2, 2]
 Generation 2510: Best = 1.859981e+03 FES = 251100 μ_F = 0.772 μ_{CR} = 0.057
 Diversity = 1.6773e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 1.859981e+03 FES = 252100 μ_F = 0.797 μ_{CR} = 0.053
 Diversity = 1.6774e+00 k_t = [4, 2, 4, 2]
 Generation 2530: Best = 1.859981e+03 FES = 253100 μ_F = 0.812 μ_{CR} = 0.059
 Diversity = 1.6775e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 1.859981e+03 FES = 254100 μ_F = 0.815 μ_{CR} = 0.064
 Diversity = 1.6804e+00 k_t = [2, 4, 3, 4]
 Generation 2550: Best = 1.859981e+03 FES = 255100 μ_F = 0.815 μ_{CR} = 0.062
 Diversity = 1.6804e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 1.859980e+03 FES = 256100 μ_F = 0.806 μ_{CR} = 0.053
 Diversity = 1.6814e+00 k_t = [4, 4, 2, 4]
 Generation 2570: Best = 1.859980e+03 FES = 257100 μ_F = 0.820 μ_{CR} = 0.052
 Diversity = 1.6815e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 1.859980e+03 FES = 258100 μ_F = 0.825 μ_{CR} = 0.056
 Diversity = 1.6816e+00 k_t = [3, 3, 3, 2]
 Generation 2590: Best = 1.859977e+03 FES = 259100 μ_F = 0.806 μ_{CR} = 0.053
 Diversity = 1.6814e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 1.859977e+03 FES = 260100 μ_F = 0.822 μ_{CR} = 0.042
 Diversity = 1.6970e+00 k_t = [4, 4, 3, 4]
 Generation 2610: Best = 1.859977e+03 FES = 261100 μ_F = 0.820 μ_{CR} = 0.054
 Diversity = 1.6970e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 1.859977e+03 FES = 262100 μ_F = 0.815 μ_{CR} = 0.054
 Diversity = 1.6970e+00 k_t = [2, 2, 4, 3]
 Generation 2630: Best = 1.859977e+03 FES = 263100 μ_F = 0.798 μ_{CR} = 0.058
 Diversity = 1.6971e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 1.859977e+03 FES = 264100 μ_F = 0.837 μ_{CR} = 0.061
 Diversity = 1.6972e+00 k_t = [4, 3, 2, 3]
 Generation 2650: Best = 1.859977e+03 FES = 265100 μ_F = 0.835 μ_{CR} = 0.058
 Diversity = 1.6951e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 1.859977e+03 FES = 266100 μ_F = 0.840 μ_{CR} = 0.050
 Diversity = 1.6925e+00 k_t = [3, 3, 3, 2]
 Generation 2670: Best = 1.859977e+03 FES = 267100 μ_F = 0.829 μ_{CR} = 0.050
 Diversity = 1.6925e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 1.859977e+03 FES = 268100 μ_F = 0.845 μ_{CR} = 0.061
 Diversity = 1.6922e+00 k_t = [3, 3, 3, 4]
 Generation 2690: Best = 1.859955e+03 FES = 269100 μ_F = 0.831 μ_{CR} = 0.055
 Diversity = 1.6921e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 1.859955e+03 FES = 270100 μ_F = 0.815 μ_{CR} = 0.054
 Diversity = 1.6922e+00 k_t = [2, 4, 3, 2]
 Generation 2710: Best = 1.859954e+03 FES = 271100 μ_F = 0.802 μ_{CR} = 0.059
 Diversity = 1.6923e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 1.859954e+03 FES = 272100 μ_F = 0.803 μ_{CR} = 0.054
 Diversity = 1.6928e+00 k_t = [3, 3, 2, 3]
 Generation 2730: Best = 1.859954e+03 FES = 273100 μ_F = 0.809 μ_{CR} = 0.055
 Diversity = 1.6928e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 1.859954e+03 FES = 274100 μ_F = 0.798 μ_{CR} = 0.059
 Diversity = 1.6928e+00 k_t = [2, 3, 2, 3]
 Generation 2750: Best = 1.859954e+03 FES = 275100 μ_F = 0.784 μ_{CR} = 0.052
 Diversity = 1.6928e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 1.859954e+03 FES = 276100 μ_F = 0.790 μ_{CR} = 0.063
 Diversity = 1.6952e+00 k_t = [3, 3, 4, 2]
 Generation 2770: Best = 1.859954e+03 FES = 277100 μ_F = 0.729 μ_{CR} = 0.067
 Diversity = 1.6953e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 1.859954e+03 FES = 278100 μ_F = 0.758 μ_{CR} = 0.074
 Diversity = 1.6953e+00 k_t = [4, 2, 2, 3]
 Generation 2790: Best = 1.859954e+03 FES = 279100 μ_F = 0.770 μ_{CR} = 0.062
 Diversity = 1.6953e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 1.859954e+03 FES = 280100 μ_F = 0.788 μ_{CR} = 0.066
 Diversity = 1.7232e+00 k_t = [4, 3, 2, 4]
 Generation 2810: Best = 1.859954e+03 FES = 281100 μ_F = 0.820 μ_{CR} = 0.060
 Diversity = 1.7152e+00 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 1.859954e+03 FES = 282100 μ_F = 0.816 μ_{CR} = 0.060
 Diversity = 1.7152e+00 k_t = [3, 2, 3, 2]
 Generation 2830: Best = 1.859954e+03 FES = 283100 μ_F = 0.831 μ_{CR} = 0.058
 Diversity = 1.7153e+00 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 1.859954e+03 FES = 284100 μ_F = 0.850 μ_{CR} = 0.059
 Diversity = 1.7173e+00 k_t = [3, 2, 2, 2]
 Generation 2850: Best = 1.859954e+03 FES = 285100 μ_F = 0.842 μ_{CR} = 0.057
 Diversity = 1.7173e+00 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 1.859954e+03 FES = 286100 μ_F = 0.856 μ_{CR} = 0.058
 Diversity = 1.7174e+00 k_t = [4, 2, 3, 2]
 Generation 2870: Best = 1.859954e+03 FES = 287100 μ_F = 0.843 μ_{CR} = 0.063
 Diversity = 1.7175e+00 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 1.859954e+03 FES = 288100 μ_F = 0.836 μ_{CR} = 0.068
 Diversity = 1.7176e+00 k_t = [3, 4, 3, 3]
 Generation 2890: Best = 1.859954e+03 FES = 289100 μ_F = 0.835 μ_{CR} = 0.055
 Diversity = 1.7251e+00 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 1.859954e+03 FES = 290100 μ_F = 0.819 μ_{CR} = 0.051
 Diversity = 1.7251e+00 k_t = [4, 3, 4, 2]
 Generation 2910: Best = 1.859954e+03 FES = 291100 μ_F = 0.839 μ_{CR} = 0.056
 Diversity = 1.7252e+00 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 1.859954e+03 FES = 292100 μ_F = 0.843 μ_{CR} = 0.057
 Diversity = 1.7276e+00 k_t = [2, 3, 4, 4]

Generation 2930: Best = 1.859954e+03 FES = 293100 μ_F = 0.853 μ_{CR} = 0.056
 Diversity = 1.7276e+00 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 1.859954e+03 FES = 294100 μ_F = 0.834 μ_{CR} = 0.065
 Diversity = 1.7277e+00 k_t = [4, 3, 4, 2]
 Generation 2950: Best = 1.859954e+03 FES = 295100 μ_F = 0.808 μ_{CR} = 0.057
 Diversity = 1.7277e+00 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 1.859954e+03 FES = 296100 μ_F = 0.805 μ_{CR} = 0.064
 Diversity = 1.7276e+00 k_t = [2, 4, 4, 3]
 Generation 2970: Best = 1.859954e+03 FES = 297100 μ_F = 0.791 μ_{CR} = 0.054
 Diversity = 1.7276e+00 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 1.859954e+03 FES = 298100 μ_F = 0.785 μ_{CR} = 0.051
 Diversity = 1.7277e+00 k_t = [4, 2, 4, 2]
 Generation 2990: Best = 1.859954e+03 FES = 299100 μ_F = 0.782 μ_{CR} = 0.055
 Diversity = 1.7277e+00 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 1.859954e+03

Total Function Evals = 300000

Total Successful Updates = 23645

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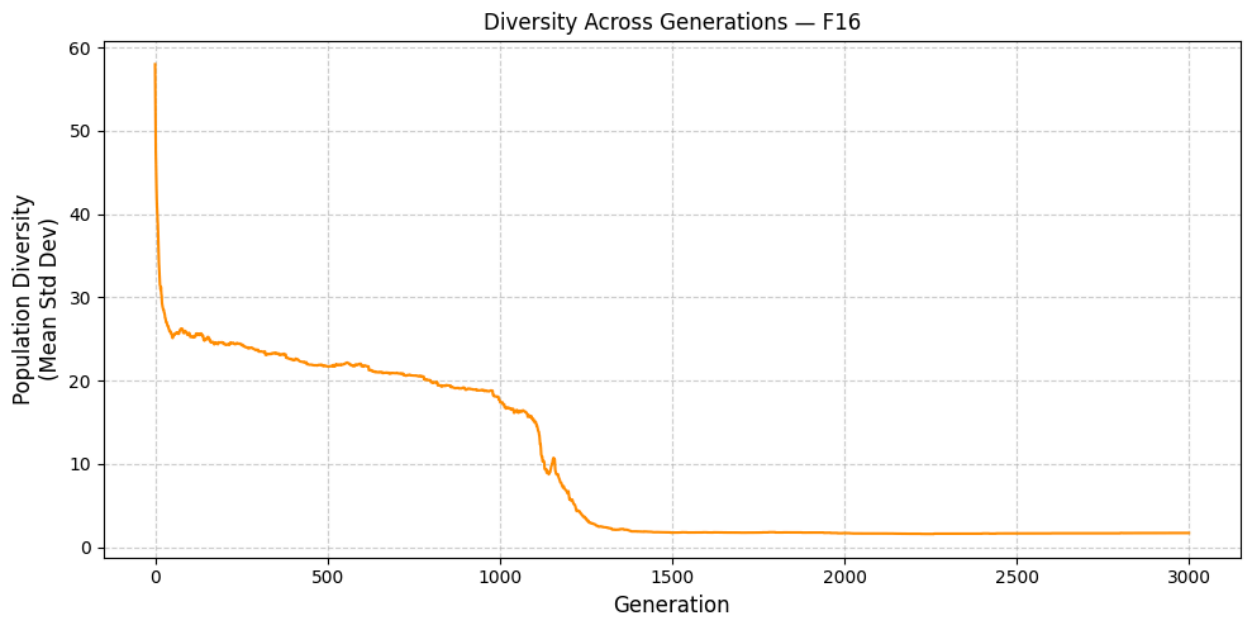
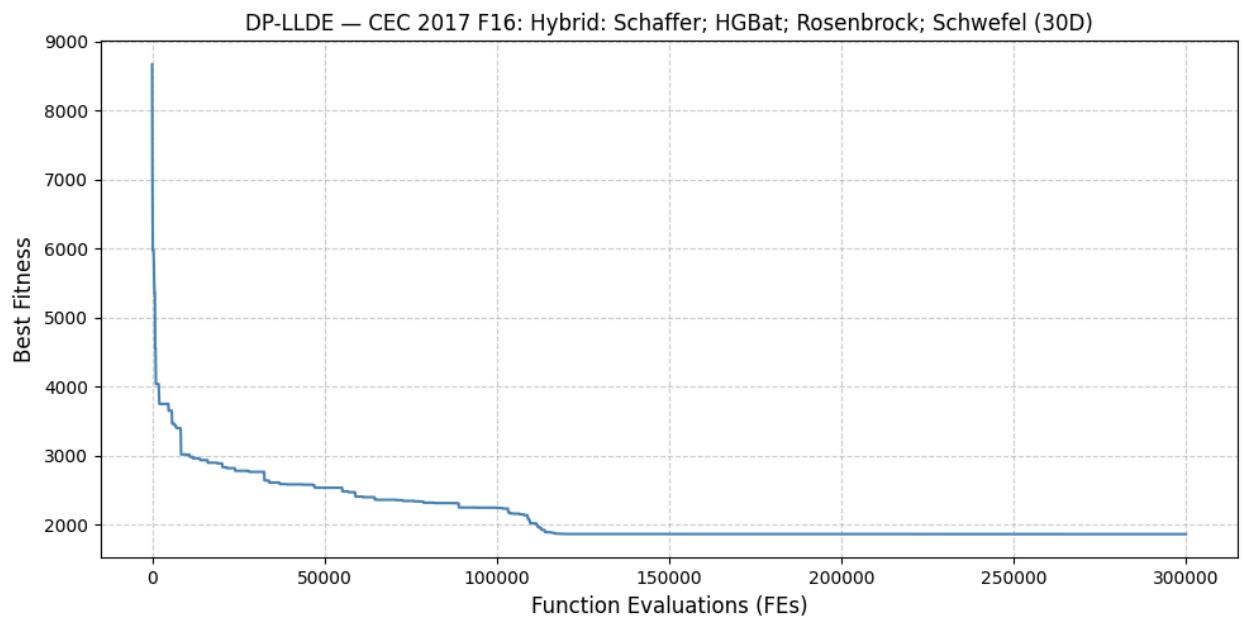
=

Results:

Best Fitness : 1.859954e+03
 Global Optimum : 1.600000e+03
 Error (|best-optimum|) : 2.599537e+02
 Best Solution (5 dim) : [44.70649093 53.855763 78.65181831 -8.05960277
 -21.61264028]
 D^(0) (init. diversity): 5.7976e+01

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Graph saved as convergence_F16.png

```
In [ ]: # =====
# F17 – Hybrid Function 7 (Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Ra
# =====
run_and_plot('F17', make_wrapper(hybrid.f17), 1700.0, 'Hybrid: Katsuura; Ackle
```

```

=====
=
DP-LLDE on CEC 2017 F17 – Single Run
=====
=
Function      : Hybrid: Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastr
igin (30D)
Bounds       : [-100, 100]
Global optimum : 1700.0
Initial  $D^*(0) = 5.7976e+01$  | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 4.634765e+03
Generation  10: Best = 3.209299e+03 FES =  1100  $\mu_F = 0.572$   $\mu_{CR} = 0.354$ 
Diversity = 3.4514e+01  $k_t = [1, 1, 1, 1]$ 
Generation  20: Best = 2.599169e+03 FES =  2100  $\mu_F = 0.627$   $\mu_{CR} = 0.316$ 
Diversity = 3.1439e+01  $k_t = [4, 3, 4, 3]$ 
Generation  30: Best = 2.599169e+03 FES =  3100  $\mu_F = 0.671$   $\mu_{CR} = 0.268$ 
Diversity = 3.0647e+01  $k_t = [1, 1, 1, 1]$ 
Generation  40: Best = 2.455955e+03 FES =  4100  $\mu_F = 0.660$   $\mu_{CR} = 0.236$ 
Diversity = 2.9463e+01  $k_t = [3, 4, 2, 2]$ 
Generation  50: Best = 2.427968e+03 FES =  5100  $\mu_F = 0.712$   $\mu_{CR} = 0.182$ 
Diversity = 2.9521e+01  $k_t = [1, 1, 1, 1]$ 
Generation  60: Best = 2.390597e+03 FES =  6100  $\mu_F = 0.743$   $\mu_{CR} = 0.152$ 
Diversity = 2.9522e+01  $k_t = [2, 2, 3, 3]$ 
Generation  70: Best = 2.320507e+03 FES =  7100  $\mu_F = 0.761$   $\mu_{CR} = 0.120$ 
Diversity = 2.9449e+01  $k_t = [1, 1, 1, 1]$ 
Generation  80: Best = 2.319483e+03 FES =  8100  $\mu_F = 0.799$   $\mu_{CR} = 0.108$ 
Diversity = 2.9375e+01  $k_t = [4, 2, 3, 4]$ 
Generation  90: Best = 2.265757e+03 FES =  9100  $\mu_F = 0.793$   $\mu_{CR} = 0.103$ 
Diversity = 2.9388e+01  $k_t = [1, 1, 1, 1]$ 
Generation 100: Best = 2.256918e+03 FES = 10100  $\mu_F = 0.765$   $\mu_{CR} = 0.094$ 
Diversity = 2.8907e+01  $k_t = [3, 3, 4, 2]$ 
Generation 110: Best = 2.246627e+03 FES = 11100  $\mu_F = 0.799$   $\mu_{CR} = 0.096$ 
Diversity = 2.9646e+01  $k_t = [1, 1, 1, 1]$ 
Generation 120: Best = 2.226897e+03 FES = 12100  $\mu_F = 0.817$   $\mu_{CR} = 0.085$ 
Diversity = 2.9547e+01  $k_t = [2, 4, 3, 4]$ 
Generation 130: Best = 2.211158e+03 FES = 13100  $\mu_F = 0.820$   $\mu_{CR} = 0.074$ 
Diversity = 2.9579e+01  $k_t = [1, 1, 1, 1]$ 
Generation 140: Best = 2.211158e+03 FES = 14100  $\mu_F = 0.845$   $\mu_{CR} = 0.070$ 
Diversity = 2.9674e+01  $k_t = [2, 2, 2, 3]$ 
Generation 150: Best = 2.121848e+03 FES = 15100  $\mu_F = 0.803$   $\mu_{CR} = 0.058$ 
Diversity = 2.9800e+01  $k_t = [1, 1, 1, 1]$ 
Generation 160: Best = 2.068825e+03 FES = 16100  $\mu_F = 0.799$   $\mu_{CR} = 0.067$ 
Diversity = 2.9879e+01  $k_t = [2, 3, 2, 3]$ 
Generation 170: Best = 2.057105e+03 FES = 17100  $\mu_F = 0.774$   $\mu_{CR} = 0.074$ 
Diversity = 2.9804e+01  $k_t = [1, 1, 1, 1]$ 
Generation 180: Best = 2.057105e+03 FES = 18100  $\mu_F = 0.785$   $\mu_{CR} = 0.070$ 
Diversity = 2.9616e+01  $k_t = [2, 3, 2, 4]$ 
Generation 190: Best = 2.050285e+03 FES = 19100  $\mu_F = 0.776$   $\mu_{CR} = 0.081$ 
Diversity = 2.9993e+01  $k_t = [1, 1, 1, 1]$ 
Generation 200: Best = 2.050285e+03 FES = 20100  $\mu_F = 0.813$   $\mu_{CR} = 0.076$ 
Diversity = 3.0221e+01  $k_t = [4, 4, 3, 2]$ 
Generation 210: Best = 2.050285e+03 FES = 21100  $\mu_F = 0.809$   $\mu_{CR} = 0.070$ 
Diversity = 3.0237e+01  $k_t = [1, 1, 1, 1]$ 
Generation 220: Best = 2.045283e+03 FES = 22100  $\mu_F = 0.833$   $\mu_{CR} = 0.074$ 

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Diversity = 3.0580e+01	k_t = [4, 3, 3, 2]				
Generation 230: Best = 2.045283e+03	FES = 23100	$\mu_F = 0.805$	$\mu_{CR} = 0.081$		
Diversity = 3.0544e+01	k_t = [1, 1, 1, 1]				
Generation 240: Best = 2.039244e+03	FES = 24100	$\mu_F = 0.840$	$\mu_{CR} = 0.087$		
Diversity = 3.0567e+01	k_t = [2, 3, 3, 3]				
Generation 250: Best = 2.039240e+03	FES = 25100	$\mu_F = 0.844$	$\mu_{CR} = 0.074$		
Diversity = 3.0418e+01	k_t = [1, 1, 1, 1]				
Generation 260: Best = 2.039240e+03	FES = 26100	$\mu_F = 0.868$	$\mu_{CR} = 0.055$		
Diversity = 3.0146e+01	k_t = [3, 4, 3, 4]				
Generation 270: Best = 1.989752e+03	FES = 27100	$\mu_F = 0.850$	$\mu_{CR} = 0.061$		
Diversity = 3.0162e+01	k_t = [1, 1, 1, 1]				
Generation 280: Best = 1.989752e+03	FES = 28100	$\mu_F = 0.826$	$\mu_{CR} = 0.059$		
Diversity = 3.0044e+01	k_t = [4, 2, 4, 3]				
Generation 290: Best = 1.989752e+03	FES = 29100	$\mu_F = 0.827$	$\mu_{CR} = 0.061$		
Diversity = 3.0067e+01	k_t = [1, 1, 1, 1]				
Generation 300: Best = 1.989752e+03	FES = 30100	$\mu_F = 0.851$	$\mu_{CR} = 0.072$		
Diversity = 3.0219e+01	k_t = [4, 4, 2, 2]				
Generation 310: Best = 1.982278e+03	FES = 31100	$\mu_F = 0.876$	$\mu_{CR} = 0.068$		
Diversity = 3.0436e+01	k_t = [1, 1, 1, 1]				
Generation 320: Best = 1.937900e+03	FES = 32100	$\mu_F = 0.875$	$\mu_{CR} = 0.060$		
Diversity = 3.0637e+01	k_t = [2, 3, 2, 2]				
Generation 330: Best = 1.937693e+03	FES = 33100	$\mu_F = 0.876$	$\mu_{CR} = 0.061$		
Diversity = 3.0612e+01	k_t = [1, 1, 1, 1]				
Generation 340: Best = 1.937693e+03	FES = 34100	$\mu_F = 0.890$	$\mu_{CR} = 0.058$		
Diversity = 3.0574e+01	k_t = [3, 3, 4, 2]				
Generation 350: Best = 1.937693e+03	FES = 35100	$\mu_F = 0.892$	$\mu_{CR} = 0.044$		
Diversity = 3.0524e+01	k_t = [1, 1, 1, 1]				
Generation 360: Best = 1.912344e+03	FES = 36100	$\mu_F = 0.882$	$\mu_{CR} = 0.071$		
Diversity = 3.0574e+01	k_t = [3, 3, 3, 2]				
Generation 370: Best = 1.906317e+03	FES = 37100	$\mu_F = 0.890$	$\mu_{CR} = 0.067$		
Diversity = 3.0543e+01	k_t = [1, 1, 1, 1]				
Generation 380: Best = 1.906317e+03	FES = 38100	$\mu_F = 0.902$	$\mu_{CR} = 0.064$		
Diversity = 3.0036e+01	k_t = [4, 2, 2, 4]				
Generation 390: Best = 1.903104e+03	FES = 39100	$\mu_F = 0.911$	$\mu_{CR} = 0.064$		
Diversity = 3.0104e+01	k_t = [1, 1, 1, 1]				
Generation 400: Best = 1.903104e+03	FES = 40100	$\mu_F = 0.894$	$\mu_{CR} = 0.063$		
Diversity = 3.0006e+01	k_t = [4, 2, 4, 3]				
Generation 410: Best = 1.903104e+03	FES = 41100	$\mu_F = 0.885$	$\mu_{CR} = 0.089$		
Diversity = 3.0192e+01	k_t = [1, 1, 1, 1]				
Generation 420: Best = 1.903104e+03	FES = 42100	$\mu_F = 0.872$	$\mu_{CR} = 0.088$		
Diversity = 3.0253e+01	k_t = [2, 3, 4, 2]				
Generation 430: Best = 1.903104e+03	FES = 43100	$\mu_F = 0.864$	$\mu_{CR} = 0.085$		
Diversity = 3.0162e+01	k_t = [1, 1, 1, 1]				
Generation 440: Best = 1.902877e+03	FES = 44100	$\mu_F = 0.844$	$\mu_{CR} = 0.070$		
Diversity = 2.9790e+01	k_t = [3, 3, 3, 4]				
Generation 450: Best = 1.902877e+03	FES = 45100	$\mu_F = 0.856$	$\mu_{CR} = 0.061$		
Diversity = 2.9702e+01	k_t = [1, 1, 1, 1]				
Generation 460: Best = 1.902877e+03	FES = 46100	$\mu_F = 0.887$	$\mu_{CR} = 0.060$		
Diversity = 2.9834e+01	k_t = [3, 3, 3, 4]				
Generation 470: Best = 1.870298e+03	FES = 47100	$\mu_F = 0.861$	$\mu_{CR} = 0.072$		
Diversity = 2.9792e+01	k_t = [1, 1, 1, 1]				
Generation 480: Best = 1.870169e+03	FES = 48100	$\mu_F = 0.864$	$\mu_{CR} = 0.073$		
Diversity = 2.9660e+01	k_t = [3, 4, 3, 3]				
Generation 490: Best = 1.870169e+03	FES = 49100	$\mu_F = 0.878$	$\mu_{CR} = 0.079$		

Diversity = 2.9588e+01	k_t = [1, 1, 1, 1]				
Generation 500: Best = 1.870169e+03	FES = 50100	$\mu_F = 0.887$	$\mu_{CR} = 0.067$		
Diversity = 2.9048e+01	k_t = [3, 3, 4, 4]				
Generation 510: Best = 1.870169e+03	FES = 51100	$\mu_F = 0.890$	$\mu_{CR} = 0.083$		
Diversity = 2.9140e+01	k_t = [1, 1, 1, 1]				
Generation 520: Best = 1.868732e+03	FES = 52100	$\mu_F = 0.885$	$\mu_{CR} = 0.068$		
Diversity = 2.9076e+01	k_t = [3, 4, 4, 4]				
Generation 530: Best = 1.866599e+03	FES = 53100	$\mu_F = 0.884$	$\mu_{CR} = 0.063$		
Diversity = 2.9111e+01	k_t = [1, 1, 1, 1]				
Generation 540: Best = 1.849357e+03	FES = 54100	$\mu_F = 0.861$	$\mu_{CR} = 0.062$		
Diversity = 2.9168e+01	k_t = [2, 4, 4, 3]				
Generation 550: Best = 1.849357e+03	FES = 55100	$\mu_F = 0.819$	$\mu_{CR} = 0.050$		
Diversity = 2.9227e+01	k_t = [1, 1, 1, 1]				
Generation 560: Best = 1.842241e+03	FES = 56100	$\mu_F = 0.833$	$\mu_{CR} = 0.054$		
Diversity = 2.9256e+01	k_t = [3, 4, 4, 2]				
Generation 570: Best = 1.842241e+03	FES = 57100	$\mu_F = 0.836$	$\mu_{CR} = 0.075$		
Diversity = 2.9314e+01	k_t = [1, 1, 1, 1]				
Generation 580: Best = 1.842241e+03	FES = 58100	$\mu_F = 0.816$	$\mu_{CR} = 0.083$		
Diversity = 2.9378e+01	k_t = [3, 3, 3, 3]				
Generation 590: Best = 1.842241e+03	FES = 59100	$\mu_F = 0.779$	$\mu_{CR} = 0.082$		
Diversity = 2.9146e+01	k_t = [1, 1, 1, 1]				
Generation 600: Best = 1.813559e+03	FES = 60100	$\mu_F = 0.830$	$\mu_{CR} = 0.079$		
Diversity = 2.9044e+01	k_t = [4, 3, 2, 2]				
Generation 610: Best = 1.813559e+03	FES = 61100	$\mu_F = 0.848$	$\mu_{CR} = 0.064$		
Diversity = 2.9206e+01	k_t = [1, 1, 1, 1]				
Generation 620: Best = 1.813559e+03	FES = 62100	$\mu_F = 0.864$	$\mu_{CR} = 0.067$		
Diversity = 2.8678e+01	k_t = [3, 2, 3, 4]				
Generation 630: Best = 1.813559e+03	FES = 63100	$\mu_F = 0.853$	$\mu_{CR} = 0.069$		
Diversity = 2.8543e+01	k_t = [1, 1, 1, 1]				
Generation 640: Best = 1.804635e+03	FES = 64100	$\mu_F = 0.865$	$\mu_{CR} = 0.078$		
Diversity = 2.8031e+01	k_t = [2, 4, 3, 4]				
Generation 650: Best = 1.804635e+03	FES = 65100	$\mu_F = 0.852$	$\mu_{CR} = 0.059$		
Diversity = 2.8252e+01	k_t = [1, 1, 1, 1]				
Generation 660: Best = 1.804635e+03	FES = 66100	$\mu_F = 0.843$	$\mu_{CR} = 0.084$		
Diversity = 2.7870e+01	k_t = [4, 3, 4, 4]				
Generation 670: Best = 1.804635e+03	FES = 67100	$\mu_F = 0.819$	$\mu_{CR} = 0.080$		
Diversity = 2.7843e+01	k_t = [1, 1, 1, 1]				
Generation 680: Best = 1.804635e+03	FES = 68100	$\mu_F = 0.846$	$\mu_{CR} = 0.072$		
Diversity = 2.7992e+01	k_t = [2, 3, 3, 2]				
Generation 690: Best = 1.804635e+03	FES = 69100	$\mu_F = 0.846$	$\mu_{CR} = 0.074$		
Diversity = 2.7768e+01	k_t = [1, 1, 1, 1]				
Generation 700: Best = 1.804635e+03	FES = 70100	$\mu_F = 0.879$	$\mu_{CR} = 0.058$		
Diversity = 2.7082e+01	k_t = [4, 4, 4, 4]				
Generation 710: Best = 1.804635e+03	FES = 71100	$\mu_F = 0.857$	$\mu_{CR} = 0.057$		
Diversity = 2.7127e+01	k_t = [1, 1, 1, 1]				
Generation 720: Best = 1.804635e+03	FES = 72100	$\mu_F = 0.857$	$\mu_{CR} = 0.061$		
Diversity = 2.7223e+01	k_t = [4, 2, 2, 2]				
Generation 730: Best = 1.804635e+03	FES = 73100	$\mu_F = 0.851$	$\mu_{CR} = 0.066$		
Diversity = 2.7362e+01	k_t = [1, 1, 1, 1]				
Generation 740: Best = 1.799587e+03	FES = 74100	$\mu_F = 0.877$	$\mu_{CR} = 0.051$		
Diversity = 2.7459e+01	k_t = [2, 3, 4, 3]				
Generation 750: Best = 1.799587e+03	FES = 75100	$\mu_F = 0.880$	$\mu_{CR} = 0.056$		
Diversity = 2.7363e+01	k_t = [1, 1, 1, 1]				
Generation 760: Best = 1.799587e+03	FES = 76100	$\mu_F = 0.860$	$\mu_{CR} = 0.052$		

Diversity = 2.7327e+01	k_t = [3, 3, 4, 2]				
Generation 770: Best = 1.799587e+03	FES = 77100	$\mu_F = 0.837$	$\mu_{CR} = 0.056$		
Diversity = 2.7573e+01	k_t = [1, 1, 1, 1]				
Generation 780: Best = 1.799587e+03	FES = 78100	$\mu_F = 0.882$	$\mu_{CR} = 0.056$		
Diversity = 2.7004e+01	k_t = [4, 3, 4, 4]				
Generation 790: Best = 1.799587e+03	FES = 79100	$\mu_F = 0.874$	$\mu_{CR} = 0.044$		
Diversity = 2.6663e+01	k_t = [1, 1, 1, 1]				
Generation 800: Best = 1.799587e+03	FES = 80100	$\mu_F = 0.800$	$\mu_{CR} = 0.071$		
Diversity = 2.6560e+01	k_t = [3, 2, 3, 3]				
Generation 810: Best = 1.796508e+03	FES = 81100	$\mu_F = 0.799$	$\mu_{CR} = 0.054$		
Diversity = 2.6536e+01	k_t = [1, 1, 1, 1]				
Generation 820: Best = 1.793305e+03	FES = 82100	$\mu_F = 0.811$	$\mu_{CR} = 0.072$		
Diversity = 2.6325e+01	k_t = [4, 4, 3, 4]				
Generation 830: Best = 1.789981e+03	FES = 83100	$\mu_F = 0.799$	$\mu_{CR} = 0.060$		
Diversity = 2.5806e+01	k_t = [1, 1, 1, 1]				
Generation 840: Best = 1.789981e+03	FES = 84100	$\mu_F = 0.822$	$\mu_{CR} = 0.054$		
Diversity = 2.5525e+01	k_t = [3, 2, 2, 4]				
Generation 850: Best = 1.789981e+03	FES = 85100	$\mu_F = 0.836$	$\mu_{CR} = 0.060$		
Diversity = 2.5385e+01	k_t = [1, 1, 1, 1]				
Generation 860: Best = 1.788987e+03	FES = 86100	$\mu_F = 0.821$	$\mu_{CR} = 0.073$		
Diversity = 2.5554e+01	k_t = [3, 4, 2, 3]				
Generation 870: Best = 1.788579e+03	FES = 87100	$\mu_F = 0.809$	$\mu_{CR} = 0.054$		
Diversity = 2.5360e+01	k_t = [1, 1, 1, 1]				
Generation 880: Best = 1.788311e+03	FES = 88100	$\mu_F = 0.823$	$\mu_{CR} = 0.045$		
Diversity = 2.5291e+01	k_t = [2, 4, 3, 2]				
Generation 890: Best = 1.788311e+03	FES = 89100	$\mu_F = 0.825$	$\mu_{CR} = 0.050$		
Diversity = 2.5561e+01	k_t = [1, 1, 1, 1]				
Generation 900: Best = 1.788311e+03	FES = 90100	$\mu_F = 0.856$	$\mu_{CR} = 0.052$		
Diversity = 2.5547e+01	k_t = [3, 3, 3, 3]				
Generation 910: Best = 1.788311e+03	FES = 91100	$\mu_F = 0.854$	$\mu_{CR} = 0.062$		
Diversity = 2.5385e+01	k_t = [1, 1, 1, 1]				
Generation 920: Best = 1.788311e+03	FES = 92100	$\mu_F = 0.882$	$\mu_{CR} = 0.058$		
Diversity = 2.5302e+01	k_t = [2, 3, 3, 2]				
Generation 930: Best = 1.784999e+03	FES = 93100	$\mu_F = 0.837$	$\mu_{CR} = 0.057$		
Diversity = 2.5281e+01	k_t = [1, 1, 1, 1]				
Generation 940: Best = 1.784038e+03	FES = 94100	$\mu_F = 0.791$	$\mu_{CR} = 0.068$		
Diversity = 2.5212e+01	k_t = [4, 4, 4, 2]				
Generation 950: Best = 1.781143e+03	FES = 95100	$\mu_F = 0.806$	$\mu_{CR} = 0.072$		
Diversity = 2.5129e+01	k_t = [1, 1, 1, 1]				
Generation 960: Best = 1.781076e+03	FES = 96100	$\mu_F = 0.817$	$\mu_{CR} = 0.064$		
Diversity = 2.4746e+01	k_t = [4, 2, 2, 4]				
Generation 970: Best = 1.781076e+03	FES = 97100	$\mu_F = 0.786$	$\mu_{CR} = 0.066$		
Diversity = 2.4561e+01	k_t = [1, 1, 1, 1]				
Generation 980: Best = 1.778603e+03	FES = 98100	$\mu_F = 0.791$	$\mu_{CR} = 0.055$		
Diversity = 2.4189e+01	k_t = [3, 3, 3, 2]				
Generation 990: Best = 1.778603e+03	FES = 99100	$\mu_F = 0.835$	$\mu_{CR} = 0.066$		
Diversity = 2.4173e+01	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 1.778504e+03	FES = 100100	$\mu_F = 0.773$	$\mu_{CR} = 0.062$		
Diversity = 2.4207e+01	k_t = [3, 3, 2, 2]				
Generation 1010: Best = 1.777196e+03	FES = 101100	$\mu_F = 0.727$	$\mu_{CR} = 0.067$		
Diversity = 2.3983e+01	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 1.774712e+03	FES = 102100	$\mu_F = 0.761$	$\mu_{CR} = 0.051$		
Diversity = 2.3590e+01	k_t = [2, 3, 4, 4]				
Generation 1030: Best = 1.774712e+03	FES = 103100	$\mu_F = 0.800$	$\mu_{CR} = 0.050$		

Diversity = 2.3610e+01	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 1.772307e+03	FES = 104100	$\mu_F = 0.820$	$\mu_{CR} = 0.060$	
Diversity = 2.3583e+01	k_t = [4, 2, 2, 2]			
Generation 1050: Best = 1.770802e+03	FES = 105100	$\mu_F = 0.812$	$\mu_{CR} = 0.067$	
Diversity = 2.3597e+01	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 1.770802e+03	FES = 106100	$\mu_F = 0.835$	$\mu_{CR} = 0.075$	
Diversity = 2.3547e+01	k_t = [4, 3, 2, 2]			
Generation 1070: Best = 1.770243e+03	FES = 107100	$\mu_F = 0.806$	$\mu_{CR} = 0.063$	
Diversity = 2.3282e+01	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 1.770243e+03	FES = 108100	$\mu_F = 0.848$	$\mu_{CR} = 0.052$	
Diversity = 2.2739e+01	k_t = [2, 2, 2, 3]			
Generation 1090: Best = 1.766173e+03	FES = 109100	$\mu_F = 0.873$	$\mu_{CR} = 0.062$	
Diversity = 2.2733e+01	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 1.765436e+03	FES = 110100	$\mu_F = 0.886$	$\mu_{CR} = 0.050$	
Diversity = 2.2576e+01	k_t = [2, 4, 2, 2]			
Generation 1110: Best = 1.765436e+03	FES = 111100	$\mu_F = 0.893$	$\mu_{CR} = 0.050$	
Diversity = 2.2507e+01	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 1.764912e+03	FES = 112100	$\mu_F = 0.876$	$\mu_{CR} = 0.060$	
Diversity = 2.2359e+01	k_t = [3, 4, 3, 4]			
Generation 1130: Best = 1.764912e+03	FES = 113100	$\mu_F = 0.864$	$\mu_{CR} = 0.053$	
Diversity = 2.2319e+01	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 1.764657e+03	FES = 114100	$\mu_F = 0.879$	$\mu_{CR} = 0.060$	
Diversity = 2.1555e+01	k_t = [2, 4, 3, 4]			
Generation 1150: Best = 1.763537e+03	FES = 115100	$\mu_F = 0.900$	$\mu_{CR} = 0.050$	
Diversity = 2.1202e+01	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 1.761838e+03	FES = 116100	$\mu_F = 0.890$	$\mu_{CR} = 0.069$	
Diversity = 2.1218e+01	k_t = [2, 2, 4, 2]			
Generation 1170: Best = 1.761838e+03	FES = 117100	$\mu_F = 0.863$	$\mu_{CR} = 0.065$	
Diversity = 2.1156e+01	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 1.760967e+03	FES = 118100	$\mu_F = 0.855$	$\mu_{CR} = 0.061$	
Diversity = 2.1317e+01	k_t = [4, 4, 2, 2]			
Generation 1190: Best = 1.760966e+03	FES = 119100	$\mu_F = 0.854$	$\mu_{CR} = 0.058$	
Diversity = 2.0949e+01	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 1.760780e+03	FES = 120100	$\mu_F = 0.843$	$\mu_{CR} = 0.057$	
Diversity = 2.0580e+01	k_t = [3, 4, 3, 2]			
Generation 1210: Best = 1.760769e+03	FES = 121100	$\mu_F = 0.826$	$\mu_{CR} = 0.067$	
Diversity = 2.0161e+01	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 1.760356e+03	FES = 122100	$\mu_F = 0.834$	$\mu_{CR} = 0.079$	
Diversity = 1.9696e+01	k_t = [3, 2, 2, 2]			
Generation 1230: Best = 1.757890e+03	FES = 123100	$\mu_F = 0.818$	$\mu_{CR} = 0.069$	
Diversity = 1.9595e+01	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 1.752477e+03	FES = 124100	$\mu_F = 0.846$	$\mu_{CR} = 0.056$	
Diversity = 1.9500e+01	k_t = [2, 3, 2, 3]			
Generation 1250: Best = 1.752477e+03	FES = 125100	$\mu_F = 0.841$	$\mu_{CR} = 0.048$	
Diversity = 1.9279e+01	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 1.751869e+03	FES = 126100	$\mu_F = 0.832$	$\mu_{CR} = 0.046$	
Diversity = 1.8990e+01	k_t = [2, 4, 2, 3]			
Generation 1270: Best = 1.751601e+03	FES = 127100	$\mu_F = 0.825$	$\mu_{CR} = 0.054$	
Diversity = 1.8905e+01	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 1.751508e+03	FES = 128100	$\mu_F = 0.830$	$\mu_{CR} = 0.069$	
Diversity = 1.8753e+01	k_t = [3, 2, 4, 4]			
Generation 1290: Best = 1.751508e+03	FES = 129100	$\mu_F = 0.796$	$\mu_{CR} = 0.076$	
Diversity = 1.8617e+01	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 1.751508e+03	FES = 130100	$\mu_F = 0.801$	$\mu_{CR} = 0.074$	

Diversity = 1.8583e+01	k_t = [3, 3, 3, 2]			
Generation 1310: Best = 1.751508e+03	FES = 131100	$\mu_F = 0.816$	$\mu_{CR} = 0.067$	
Diversity = 1.8382e+01	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 1.751508e+03	FES = 132100	$\mu_F = 0.826$	$\mu_{CR} = 0.051$	
Diversity = 1.8258e+01	k_t = [2, 3, 4, 2]			
Generation 1330: Best = 1.751508e+03	FES = 133100	$\mu_F = 0.864$	$\mu_{CR} = 0.070$	
Diversity = 1.7999e+01	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 1.751508e+03	FES = 134100	$\mu_F = 0.873$	$\mu_{CR} = 0.065$	
Diversity = 1.7783e+01	k_t = [2, 4, 3, 3]			
Generation 1350: Best = 1.751508e+03	FES = 135100	$\mu_F = 0.860$	$\mu_{CR} = 0.059$	
Diversity = 1.7626e+01	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 1.751508e+03	FES = 136100	$\mu_F = 0.863$	$\mu_{CR} = 0.071$	
Diversity = 1.6318e+01	k_t = [2, 4, 3, 4]			
Generation 1370: Best = 1.751447e+03	FES = 137100	$\mu_F = 0.875$	$\mu_{CR} = 0.070$	
Diversity = 1.6278e+01	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 1.751433e+03	FES = 138100	$\mu_F = 0.876$	$\mu_{CR} = 0.066$	
Diversity = 1.6030e+01	k_t = [4, 4, 3, 3]			
Generation 1390: Best = 1.751378e+03	FES = 139100	$\mu_F = 0.859$	$\mu_{CR} = 0.062$	
Diversity = 1.5681e+01	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 1.750761e+03	FES = 140100	$\mu_F = 0.840$	$\mu_{CR} = 0.052$	
Diversity = 1.5708e+01	k_t = [4, 4, 2, 4]			
Generation 1410: Best = 1.749437e+03	FES = 141100	$\mu_F = 0.808$	$\mu_{CR} = 0.057$	
Diversity = 1.5523e+01	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 1.749417e+03	FES = 142100	$\mu_F = 0.830$	$\mu_{CR} = 0.053$	
Diversity = 1.5515e+01	k_t = [2, 4, 3, 2]			
Generation 1430: Best = 1.749417e+03	FES = 143100	$\mu_F = 0.845$	$\mu_{CR} = 0.050$	
Diversity = 1.5517e+01	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 1.749417e+03	FES = 144100	$\mu_F = 0.823$	$\mu_{CR} = 0.048$	
Diversity = 1.5229e+01	k_t = [4, 3, 2, 3]			
Generation 1450: Best = 1.749417e+03	FES = 145100	$\mu_F = 0.806$	$\mu_{CR} = 0.043$	
Diversity = 1.5242e+01	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 1.749417e+03	FES = 146100	$\mu_F = 0.827$	$\mu_{CR} = 0.038$	
Diversity = 1.5143e+01	k_t = [2, 4, 2, 2]			
Generation 1470: Best = 1.749388e+03	FES = 147100	$\mu_F = 0.770$	$\mu_{CR} = 0.047$	
Diversity = 1.5126e+01	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 1.749388e+03	FES = 148100	$\mu_F = 0.736$	$\mu_{CR} = 0.051$	
Diversity = 1.4981e+01	k_t = [4, 4, 4, 3]			
Generation 1490: Best = 1.749388e+03	FES = 149100	$\mu_F = 0.801$	$\mu_{CR} = 0.056$	
Diversity = 1.5035e+01	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 1.749388e+03	FES = 150100	$\mu_F = 0.840$	$\mu_{CR} = 0.043$	
Diversity = 1.5170e+01	k_t = [3, 4, 2, 3]			
Generation 1510: Best = 1.749388e+03	FES = 151100	$\mu_F = 0.777$	$\mu_{CR} = 0.042$	
Diversity = 1.5100e+01	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 1.749388e+03	FES = 152100	$\mu_F = 0.817$	$\mu_{CR} = 0.038$	
Diversity = 1.5103e+01	k_t = [3, 3, 3, 3]			
Generation 1530: Best = 1.749120e+03	FES = 153100	$\mu_F = 0.866$	$\mu_{CR} = 0.045$	
Diversity = 1.5169e+01	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 1.748396e+03	FES = 154100	$\mu_F = 0.876$	$\mu_{CR} = 0.047$	
Diversity = 1.5076e+01	k_t = [4, 4, 2, 4]			
Generation 1550: Best = 1.745275e+03	FES = 155100	$\mu_F = 0.873$	$\mu_{CR} = 0.057$	
Diversity = 1.4682e+01	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 1.745275e+03	FES = 156100	$\mu_F = 0.844$	$\mu_{CR} = 0.052$	
Diversity = 1.4617e+01	k_t = [4, 2, 4, 2]			
Generation 1570: Best = 1.745275e+03	FES = 157100	$\mu_F = 0.868$	$\mu_{CR} = 0.041$	

Diversity = 1.4619e+01	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 1.745057e+03	FES = 158100	$\mu_F = 0.844$	$\mu_{CR} = 0.045$	
Diversity = 1.4620e+01	k_t = [2, 2, 3, 2]			
Generation 1590: Best = 1.745053e+03	FES = 159100	$\mu_F = 0.817$	$\mu_{CR} = 0.043$	
Diversity = 1.4514e+01	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 1.745053e+03	FES = 160100	$\mu_F = 0.827$	$\mu_{CR} = 0.045$	
Diversity = 1.4486e+01	k_t = [4, 2, 2, 3]			
Generation 1610: Best = 1.745053e+03	FES = 161100	$\mu_F = 0.806$	$\mu_{CR} = 0.048$	
Diversity = 1.4555e+01	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 1.745053e+03	FES = 162100	$\mu_F = 0.802$	$\mu_{CR} = 0.042$	
Diversity = 1.4536e+01	k_t = [2, 4, 3, 4]			
Generation 1630: Best = 1.745053e+03	FES = 163100	$\mu_F = 0.803$	$\mu_{CR} = 0.047$	
Diversity = 1.4638e+01	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 1.745053e+03	FES = 164100	$\mu_F = 0.821$	$\mu_{CR} = 0.053$	
Diversity = 1.4713e+01	k_t = [3, 4, 4, 3]			
Generation 1650: Best = 1.744733e+03	FES = 165100	$\mu_F = 0.806$	$\mu_{CR} = 0.048$	
Diversity = 1.4714e+01	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 1.744306e+03	FES = 166100	$\mu_F = 0.841$	$\mu_{CR} = 0.046$	
Diversity = 1.4859e+01	k_t = [2, 4, 3, 4]			
Generation 1670: Best = 1.744306e+03	FES = 167100	$\mu_F = 0.831$	$\mu_{CR} = 0.046$	
Diversity = 1.4892e+01	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 1.744250e+03	FES = 168100	$\mu_F = 0.850$	$\mu_{CR} = 0.049$	
Diversity = 1.4615e+01	k_t = [2, 4, 2, 4]			
Generation 1690: Best = 1.744250e+03	FES = 169100	$\mu_F = 0.871$	$\mu_{CR} = 0.043$	
Diversity = 1.4634e+01	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 1.744043e+03	FES = 170100	$\mu_F = 0.841$	$\mu_{CR} = 0.051$	
Diversity = 1.4663e+01	k_t = [2, 3, 3, 2]			
Generation 1710: Best = 1.744043e+03	FES = 171100	$\mu_F = 0.848$	$\mu_{CR} = 0.046$	
Diversity = 1.4651e+01	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 1.743582e+03	FES = 172100	$\mu_F = 0.848$	$\mu_{CR} = 0.041$	
Diversity = 1.4711e+01	k_t = [3, 2, 3, 3]			
Generation 1730: Best = 1.743582e+03	FES = 173100	$\mu_F = 0.842$	$\mu_{CR} = 0.037$	
Diversity = 1.4733e+01	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 1.743582e+03	FES = 174100	$\mu_F = 0.849$	$\mu_{CR} = 0.042$	
Diversity = 1.4727e+01	k_t = [4, 3, 3, 2]			
Generation 1750: Best = 1.743422e+03	FES = 175100	$\mu_F = 0.852$	$\mu_{CR} = 0.057$	
Diversity = 1.4729e+01	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 1.742778e+03	FES = 176100	$\mu_F = 0.823$	$\mu_{CR} = 0.054$	
Diversity = 1.4716e+01	k_t = [2, 3, 2, 4]			
Generation 1770: Best = 1.742778e+03	FES = 177100	$\mu_F = 0.807$	$\mu_{CR} = 0.051$	
Diversity = 1.4731e+01	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 1.742778e+03	FES = 178100	$\mu_F = 0.806$	$\mu_{CR} = 0.051$	
Diversity = 1.4636e+01	k_t = [3, 2, 3, 3]			
Generation 1790: Best = 1.742778e+03	FES = 179100	$\mu_F = 0.847$	$\mu_{CR} = 0.045$	
Diversity = 1.4607e+01	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 1.742778e+03	FES = 180100	$\mu_F = 0.824$	$\mu_{CR} = 0.049$	
Diversity = 1.4554e+01	k_t = [3, 2, 3, 2]			
Generation 1810: Best = 1.742778e+03	FES = 181100	$\mu_F = 0.827$	$\mu_{CR} = 0.049$	
Diversity = 1.4715e+01	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 1.742517e+03	FES = 182100	$\mu_F = 0.817$	$\mu_{CR} = 0.044$	
Diversity = 1.4693e+01	k_t = [3, 3, 2, 4]			
Generation 1830: Best = 1.742220e+03	FES = 183100	$\mu_F = 0.817$	$\mu_{CR} = 0.058$	
Diversity = 1.4944e+01	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 1.742220e+03	FES = 184100	$\mu_F = 0.870$	$\mu_{CR} = 0.057$	

Diversity = 1.4698e+01	k_t = [4, 2, 3, 4]			
Generation 1850: Best = 1.742220e+03	FES = 185100	$\mu_F = 0.843$	$\mu_{CR} = 0.045$	
Diversity = 1.4722e+01	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 1.742220e+03	FES = 186100	$\mu_F = 0.863$	$\mu_{CR} = 0.054$	
Diversity = 1.4789e+01	k_t = [3, 4, 3, 2]			
Generation 1870: Best = 1.742220e+03	FES = 187100	$\mu_F = 0.823$	$\mu_{CR} = 0.058$	
Diversity = 1.4797e+01	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 1.742220e+03	FES = 188100	$\mu_F = 0.831$	$\mu_{CR} = 0.056$	
Diversity = 1.4825e+01	k_t = [2, 3, 3, 4]			
Generation 1890: Best = 1.742220e+03	FES = 189100	$\mu_F = 0.850$	$\mu_{CR} = 0.045$	
Diversity = 1.4864e+01	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 1.742220e+03	FES = 190100	$\mu_F = 0.877$	$\mu_{CR} = 0.049$	
Diversity = 1.4767e+01	k_t = [3, 4, 4, 4]			
Generation 1910: Best = 1.742094e+03	FES = 191100	$\mu_F = 0.880$	$\mu_{CR} = 0.054$	
Diversity = 1.4777e+01	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 1.742094e+03	FES = 192100	$\mu_F = 0.873$	$\mu_{CR} = 0.059$	
Diversity = 1.4795e+01	k_t = [4, 3, 4, 3]			
Generation 1930: Best = 1.742094e+03	FES = 193100	$\mu_F = 0.842$	$\mu_{CR} = 0.054$	
Diversity = 1.4838e+01	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 1.742094e+03	FES = 194100	$\mu_F = 0.865$	$\mu_{CR} = 0.055$	
Diversity = 1.5061e+01	k_t = [4, 2, 3, 2]			
Generation 1950: Best = 1.742094e+03	FES = 195100	$\mu_F = 0.876$	$\mu_{CR} = 0.042$	
Diversity = 1.5054e+01	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 1.742094e+03	FES = 196100	$\mu_F = 0.911$	$\mu_{CR} = 0.046$	
Diversity = 1.5023e+01	k_t = [2, 4, 3, 4]			
Generation 1970: Best = 1.742094e+03	FES = 197100	$\mu_F = 0.878$	$\mu_{CR} = 0.048$	
Diversity = 1.5106e+01	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 1.742006e+03	FES = 198100	$\mu_F = 0.870$	$\mu_{CR} = 0.038$	
Diversity = 1.5307e+01	k_t = [3, 3, 3, 2]			
Generation 1990: Best = 1.742005e+03	FES = 199100	$\mu_F = 0.862$	$\mu_{CR} = 0.057$	
Diversity = 1.5226e+01	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 1.741705e+03	FES = 200100	$\mu_F = 0.848$	$\mu_{CR} = 0.059$	
Diversity = 1.5268e+01	k_t = [3, 4, 4, 4]			
Generation 2010: Best = 1.741571e+03	FES = 201100	$\mu_F = 0.848$	$\mu_{CR} = 0.047$	
Diversity = 1.5276e+01	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 1.740489e+03	FES = 202100	$\mu_F = 0.829$	$\mu_{CR} = 0.045$	
Diversity = 1.5355e+01	k_t = [2, 2, 4, 3]			
Generation 2030: Best = 1.740489e+03	FES = 203100	$\mu_F = 0.798$	$\mu_{CR} = 0.047$	
Diversity = 1.5440e+01	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 1.740489e+03	FES = 204100	$\mu_F = 0.785$	$\mu_{CR} = 0.053$	
Diversity = 1.5397e+01	k_t = [4, 2, 2, 3]			
Generation 2050: Best = 1.740489e+03	FES = 205100	$\mu_F = 0.806$	$\mu_{CR} = 0.053$	
Diversity = 1.5371e+01	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 1.739577e+03	FES = 206100	$\mu_F = 0.799$	$\mu_{CR} = 0.057$	
Diversity = 1.5391e+01	k_t = [4, 3, 3, 3]			
Generation 2070: Best = 1.739576e+03	FES = 207100	$\mu_F = 0.795$	$\mu_{CR} = 0.045$	
Diversity = 1.5376e+01	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 1.739576e+03	FES = 208100	$\mu_F = 0.787$	$\mu_{CR} = 0.042$	
Diversity = 1.5413e+01	k_t = [3, 4, 3, 3]			
Generation 2090: Best = 1.739576e+03	FES = 209100	$\mu_F = 0.738$	$\mu_{CR} = 0.037$	
Diversity = 1.5387e+01	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 1.739576e+03	FES = 210100	$\mu_F = 0.759$	$\mu_{CR} = 0.047$	
Diversity = 1.5428e+01	k_t = [3, 2, 3, 3]			
Generation 2110: Best = 1.739028e+03	FES = 211100	$\mu_F = 0.768$	$\mu_{CR} = 0.045$	

Diversity = 1.5386e+01	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 1.739028e+03	FES = 212100	$\mu_F = 0.816$	$\mu_{CR} = 0.054$	
Diversity = 1.5379e+01	k_t = [3, 4, 4, 2]			
Generation 2130: Best = 1.739028e+03	FES = 213100	$\mu_F = 0.796$	$\mu_{CR} = 0.050$	
Diversity = 1.5483e+01	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 1.738653e+03	FES = 214100	$\mu_F = 0.786$	$\mu_{CR} = 0.053$	
Diversity = 1.5496e+01	k_t = [3, 4, 4, 2]			
Generation 2150: Best = 1.738653e+03	FES = 215100	$\mu_F = 0.779$	$\mu_{CR} = 0.040$	
Diversity = 1.5560e+01	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 1.738653e+03	FES = 216100	$\mu_F = 0.752$	$\mu_{CR} = 0.039$	
Diversity = 1.5464e+01	k_t = [2, 3, 2, 3]			
Generation 2170: Best = 1.738653e+03	FES = 217100	$\mu_F = 0.750$	$\mu_{CR} = 0.033$	
Diversity = 1.5366e+01	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 1.738653e+03	FES = 218100	$\mu_F = 0.722$	$\mu_{CR} = 0.042$	
Diversity = 1.5562e+01	k_t = [4, 3, 3, 3]			
Generation 2190: Best = 1.738653e+03	FES = 219100	$\mu_F = 0.784$	$\mu_{CR} = 0.056$	
Diversity = 1.5567e+01	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 1.738653e+03	FES = 220100	$\mu_F = 0.839$	$\mu_{CR} = 0.053$	
Diversity = 1.5485e+01	k_t = [3, 2, 3, 4]			
Generation 2210: Best = 1.738653e+03	FES = 221100	$\mu_F = 0.835$	$\mu_{CR} = 0.054$	
Diversity = 1.5412e+01	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 1.738653e+03	FES = 222100	$\mu_F = 0.862$	$\mu_{CR} = 0.066$	
Diversity = 1.5371e+01	k_t = [2, 4, 2, 2]			
Generation 2230: Best = 1.738653e+03	FES = 223100	$\mu_F = 0.859$	$\mu_{CR} = 0.057$	
Diversity = 1.5571e+01	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 1.738653e+03	FES = 224100	$\mu_F = 0.894$	$\mu_{CR} = 0.048$	
Diversity = 1.5475e+01	k_t = [3, 4, 2, 3]			
Generation 2250: Best = 1.738504e+03	FES = 225100	$\mu_F = 0.877$	$\mu_{CR} = 0.051$	
Diversity = 1.5499e+01	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 1.737911e+03	FES = 226100	$\mu_F = 0.871$	$\mu_{CR} = 0.045$	
Diversity = 1.5442e+01	k_t = [2, 4, 3, 2]			
Generation 2270: Best = 1.736942e+03	FES = 227100	$\mu_F = 0.879$	$\mu_{CR} = 0.046$	
Diversity = 1.5403e+01	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 1.736942e+03	FES = 228100	$\mu_F = 0.906$	$\mu_{CR} = 0.058$	
Diversity = 1.5233e+01	k_t = [4, 2, 2, 2]			
Generation 2290: Best = 1.736942e+03	FES = 229100	$\mu_F = 0.918$	$\mu_{CR} = 0.055$	
Diversity = 1.5236e+01	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 1.736942e+03	FES = 230100	$\mu_F = 0.906$	$\mu_{CR} = 0.052$	
Diversity = 1.5091e+01	k_t = [4, 3, 2, 3]			
Generation 2310: Best = 1.736942e+03	FES = 231100	$\mu_F = 0.901$	$\mu_{CR} = 0.054$	
Diversity = 1.5169e+01	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 1.736938e+03	FES = 232100	$\mu_F = 0.861$	$\mu_{CR} = 0.051$	
Diversity = 1.5183e+01	k_t = [4, 2, 4, 2]			
Generation 2330: Best = 1.736938e+03	FES = 233100	$\mu_F = 0.894$	$\mu_{CR} = 0.049$	
Diversity = 1.4982e+01	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 1.736937e+03	FES = 234100	$\mu_F = 0.891$	$\mu_{CR} = 0.057$	
Diversity = 1.4746e+01	k_t = [3, 3, 2, 4]			
Generation 2350: Best = 1.736937e+03	FES = 235100	$\mu_F = 0.868$	$\mu_{CR} = 0.055$	
Diversity = 1.4485e+01	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 1.736937e+03	FES = 236100	$\mu_F = 0.878$	$\mu_{CR} = 0.052$	
Diversity = 1.4331e+01	k_t = [3, 2, 2, 3]			
Generation 2370: Best = 1.736937e+03	FES = 237100	$\mu_F = 0.888$	$\mu_{CR} = 0.050$	
Diversity = 1.4212e+01	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 1.736937e+03	FES = 238100	$\mu_F = 0.909$	$\mu_{CR} = 0.054$	

Diversity = 1.3306e+01	k_t = [3, 3, 2, 4]			
Generation 2390: Best = 1.736936e+03	FES = 239100	$\mu_F = 0.902$	$\mu_{CR} = 0.048$	
Diversity = 1.2957e+01	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 1.736936e+03	FES = 240100	$\mu_F = 0.880$	$\mu_{CR} = 0.061$	
Diversity = 1.2406e+01	k_t = [3, 4, 2, 4]			
Generation 2410: Best = 1.736935e+03	FES = 241100	$\mu_F = 0.862$	$\mu_{CR} = 0.067$	
Diversity = 1.2239e+01	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 1.736935e+03	FES = 242100	$\mu_F = 0.842$	$\mu_{CR} = 0.074$	
Diversity = 1.2119e+01	k_t = [4, 2, 4, 2]			
Generation 2430: Best = 1.736696e+03	FES = 243100	$\mu_F = 0.809$	$\mu_{CR} = 0.071$	
Diversity = 1.2080e+01	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 1.736688e+03	FES = 244100	$\mu_F = 0.838$	$\mu_{CR} = 0.067$	
Diversity = 1.1426e+01	k_t = [4, 2, 4, 4]			
Generation 2450: Best = 1.736688e+03	FES = 245100	$\mu_F = 0.830$	$\mu_{CR} = 0.065$	
Diversity = 1.1253e+01	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 1.736688e+03	FES = 246100	$\mu_F = 0.847$	$\mu_{CR} = 0.065$	
Diversity = 1.0969e+01	k_t = [3, 3, 4, 4]			
Generation 2470: Best = 1.736688e+03	FES = 247100	$\mu_F = 0.850$	$\mu_{CR} = 0.069$	
Diversity = 1.0430e+01	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 1.736688e+03	FES = 248100	$\mu_F = 0.846$	$\mu_{CR} = 0.066$	
Diversity = 1.0423e+01	k_t = [3, 4, 4, 3]			
Generation 2490: Best = 1.736688e+03	FES = 249100	$\mu_F = 0.806$	$\mu_{CR} = 0.066$	
Diversity = 1.0391e+01	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 1.736687e+03	FES = 250100	$\mu_F = 0.788$	$\mu_{CR} = 0.063$	
Diversity = 9.7522e+00	k_t = [4, 3, 3, 4]			
Generation 2510: Best = 1.736687e+03	FES = 251100	$\mu_F = 0.813$	$\mu_{CR} = 0.065$	
Diversity = 9.7339e+00	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 1.736538e+03	FES = 252100	$\mu_F = 0.806$	$\mu_{CR} = 0.061$	
Diversity = 9.8014e+00	k_t = [2, 3, 4, 2]			
Generation 2530: Best = 1.736523e+03	FES = 253100	$\mu_F = 0.819$	$\mu_{CR} = 0.057$	
Diversity = 9.7200e+00	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 1.736523e+03	FES = 254100	$\mu_F = 0.825$	$\mu_{CR} = 0.056$	
Diversity = 9.7246e+00	k_t = [2, 2, 4, 2]			
Generation 2550: Best = 1.736405e+03	FES = 255100	$\mu_F = 0.799$	$\mu_{CR} = 0.060$	
Diversity = 9.5868e+00	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 1.736405e+03	FES = 256100	$\mu_F = 0.825$	$\mu_{CR} = 0.069$	
Diversity = 9.2926e+00	k_t = [2, 4, 2, 4]			
Generation 2570: Best = 1.736405e+03	FES = 257100	$\mu_F = 0.813$	$\mu_{CR} = 0.072$	
Diversity = 9.3017e+00	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 1.736080e+03	FES = 258100	$\mu_F = 0.838$	$\mu_{CR} = 0.061$	
Diversity = 9.3413e+00	k_t = [2, 3, 4, 4]			
Generation 2590: Best = 1.736080e+03	FES = 259100	$\mu_F = 0.847$	$\mu_{CR} = 0.061$	
Diversity = 9.3824e+00	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 1.736080e+03	FES = 260100	$\mu_F = 0.857$	$\mu_{CR} = 0.063$	
Diversity = 9.3780e+00	k_t = [3, 2, 2, 2]			
Generation 2610: Best = 1.736080e+03	FES = 261100	$\mu_F = 0.870$	$\mu_{CR} = 0.066$	
Diversity = 9.3671e+00	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 1.736076e+03	FES = 262100	$\mu_F = 0.865$	$\mu_{CR} = 0.059$	
Diversity = 9.3388e+00	k_t = [2, 4, 3, 4]			
Generation 2630: Best = 1.735949e+03	FES = 263100	$\mu_F = 0.845$	$\mu_{CR} = 0.062$	
Diversity = 9.4056e+00	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 1.734295e+03	FES = 264100	$\mu_F = 0.838$	$\mu_{CR} = 0.057$	
Diversity = 9.3377e+00	k_t = [3, 4, 3, 2]			
Generation 2650: Best = 1.734295e+03	FES = 265100	$\mu_F = 0.849$	$\mu_{CR} = 0.055$	

Diversity = 9.3162e+00	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 1.734295e+03	FES = 266100	$\mu_F = 0.858$	$\mu_{CR} = 0.061$	
Diversity = 9.2925e+00	k_t = [4, 2, 4, 2]			
Generation 2670: Best = 1.734295e+03	FES = 267100	$\mu_F = 0.861$	$\mu_{CR} = 0.056$	
Diversity = 9.3444e+00	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 1.734280e+03	FES = 268100	$\mu_F = 0.856$	$\mu_{CR} = 0.047$	
Diversity = 9.3140e+00	k_t = [2, 3, 3, 4]			
Generation 2690: Best = 1.734280e+03	FES = 269100	$\mu_F = 0.829$	$\mu_{CR} = 0.050$	
Diversity = 9.3315e+00	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 1.734228e+03	FES = 270100	$\mu_F = 0.829$	$\mu_{CR} = 0.064$	
Diversity = 9.3432e+00	k_t = [3, 4, 2, 2]			
Generation 2710: Best = 1.733550e+03	FES = 271100	$\mu_F = 0.833$	$\mu_{CR} = 0.060$	
Diversity = 9.2719e+00	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 1.733548e+03	FES = 272100	$\mu_F = 0.829$	$\mu_{CR} = 0.051$	
Diversity = 9.3103e+00	k_t = [4, 4, 2, 3]			
Generation 2730: Best = 1.733466e+03	FES = 273100	$\mu_F = 0.844$	$\mu_{CR} = 0.057$	
Diversity = 9.3820e+00	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 1.733461e+03	FES = 274100	$\mu_F = 0.851$	$\mu_{CR} = 0.057$	
Diversity = 9.0383e+00	k_t = [2, 2, 2, 4]			
Generation 2750: Best = 1.733461e+03	FES = 275100	$\mu_F = 0.841$	$\mu_{CR} = 0.062$	
Diversity = 9.0828e+00	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 1.733152e+03	FES = 276100	$\mu_F = 0.828$	$\mu_{CR} = 0.064$	
Diversity = 9.0390e+00	k_t = [3, 3, 4, 2]			
Generation 2770: Best = 1.733152e+03	FES = 277100	$\mu_F = 0.832$	$\mu_{CR} = 0.053$	
Diversity = 9.1703e+00	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 1.732899e+03	FES = 278100	$\mu_F = 0.846$	$\mu_{CR} = 0.054$	
Diversity = 9.0987e+00	k_t = [4, 3, 4, 3]			
Generation 2790: Best = 1.732892e+03	FES = 279100	$\mu_F = 0.852$	$\mu_{CR} = 0.049$	
Diversity = 9.0961e+00	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 1.732311e+03	FES = 280100	$\mu_F = 0.859$	$\mu_{CR} = 0.062$	
Diversity = 9.1094e+00	k_t = [2, 4, 3, 3]			
Generation 2810: Best = 1.732311e+03	FES = 281100	$\mu_F = 0.846$	$\mu_{CR} = 0.051$	
Diversity = 9.1599e+00	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 1.732309e+03	FES = 282100	$\mu_F = 0.847$	$\mu_{CR} = 0.051$	
Diversity = 9.2074e+00	k_t = [3, 3, 3, 2]			
Generation 2830: Best = 1.732248e+03	FES = 283100	$\mu_F = 0.834$	$\mu_{CR} = 0.046$	
Diversity = 9.2249e+00	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 1.731894e+03	FES = 284100	$\mu_F = 0.842$	$\mu_{CR} = 0.057$	
Diversity = 9.0791e+00	k_t = [4, 2, 3, 3]			
Generation 2850: Best = 1.730076e+03	FES = 285100	$\mu_F = 0.812$	$\mu_{CR} = 0.067$	
Diversity = 9.0968e+00	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 1.729153e+03	FES = 286100	$\mu_F = 0.794$	$\mu_{CR} = 0.060$	
Diversity = 9.0445e+00	k_t = [4, 2, 3, 3]			
Generation 2870: Best = 1.729153e+03	FES = 287100	$\mu_F = 0.772$	$\mu_{CR} = 0.059$	
Diversity = 9.1432e+00	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 1.729153e+03	FES = 288100	$\mu_F = 0.757$	$\mu_{CR} = 0.057$	
Diversity = 9.2224e+00	k_t = [2, 4, 4, 3]			
Generation 2890: Best = 1.728988e+03	FES = 289100	$\mu_F = 0.791$	$\mu_{CR} = 0.054$	
Diversity = 9.2211e+00	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 1.728988e+03	FES = 290100	$\mu_F = 0.806$	$\mu_{CR} = 0.058$	
Diversity = 9.2374e+00	k_t = [4, 2, 3, 2]			
Generation 2910: Best = 1.728988e+03	FES = 291100	$\mu_F = 0.787$	$\mu_{CR} = 0.057$	
Diversity = 9.2642e+00	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 1.728988e+03	FES = 292100	$\mu_F = 0.798$	$\mu_{CR} = 0.060$	

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Diversity = 9.2259e+00  k_t = [4, 2, 3, 3]
Generation 2930: Best = 1.728988e+03  FES = 293100   $\mu_F$  = 0.830   $\mu_{CR}$  = 0.061
Diversity = 9.2175e+00  k_t = [1, 1, 1, 1]
Generation 2940: Best = 1.728988e+03  FES = 294100   $\mu_F$  = 0.809   $\mu_{CR}$  = 0.054
Diversity = 9.1128e+00  k_t = [4, 4, 3, 3]
Generation 2950: Best = 1.727895e+03  FES = 295100   $\mu_F$  = 0.838   $\mu_{CR}$  = 0.053
Diversity = 9.1295e+00  k_t = [1, 1, 1, 1]
Generation 2960: Best = 1.727894e+03  FES = 296100   $\mu_F$  = 0.807   $\mu_{CR}$  = 0.056
Diversity = 9.1014e+00  k_t = [3, 3, 3, 2]
Generation 2970: Best = 1.727894e+03  FES = 297100   $\mu_F$  = 0.789   $\mu_{CR}$  = 0.052
Diversity = 9.1472e+00  k_t = [1, 1, 1, 1]
Generation 2980: Best = 1.727870e+03  FES = 298100   $\mu_F$  = 0.792   $\mu_{CR}$  = 0.057
Diversity = 9.1460e+00  k_t = [3, 4, 3, 3]
Generation 2990: Best = 1.725904e+03  FES = 299100   $\mu_F$  = 0.791   $\mu_{CR}$  = 0.066
Diversity = 9.1429e+00  k_t = [1, 1, 1, 1]

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Optimization complete.

Best Fitness = 1.725904e+03

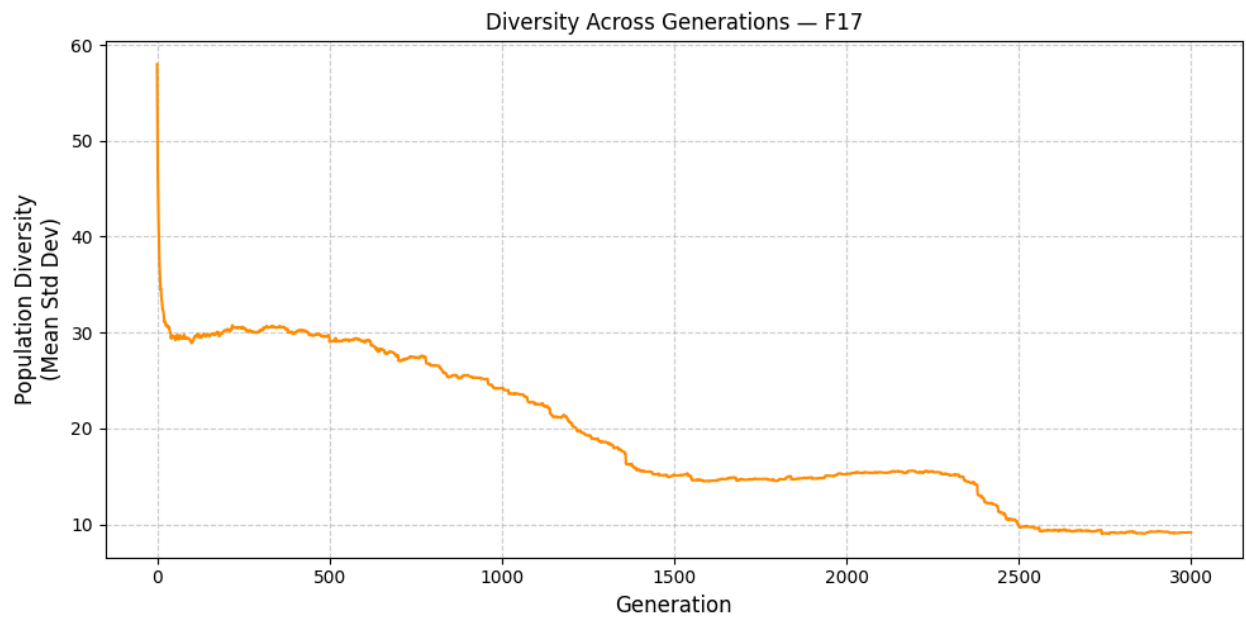
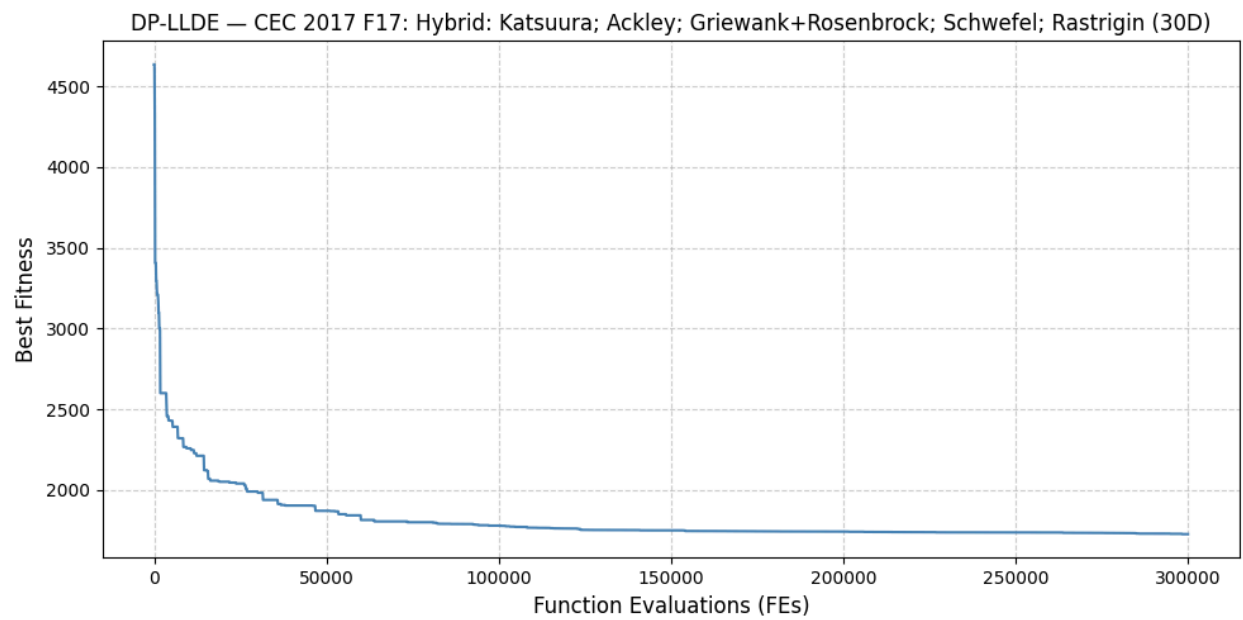
Total Function Evals = 300000

Total Successful Updates = 15532

```

=====
=
Results:
  Best Fitness      : 1.725904e+03
  Global Optimum    : 1.700000e+03
  Error (|best-optimum|) : 2.590427e+01
  Best Solution (5 dim) : [ 25.24974195  19.56096452 -11.62938497  19.1569042
-32.81123299]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F17.png

```
In [ ]: # =====
# F18 – Hybrid Function 8 (Elliptic; Ackley; Rastrigin; HGBat; Discus)
# =====
run_and_plot('F18', make_wrapper(hybrid.f18), 1800.0, 'Hybrid: Elliptic; Ackle
```

```

=====
=
DP-LLDE on CEC 2017 F18 – Single Run
=====
=
Function      : Hybrid: Elliptic; Ackley; Rastrigin; HGBat; Discus (30D)
Bounds       : [-100, 100]
Global optimum : 1800.0
Initial D^(0) = 5.7976e+01 | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 3.522341e+08
Generation  10: Best = 2.214072e+07 FES =  1100   $\mu_F$  = 0.560   $\mu_{CR}$  = 0.344
Diversity = 4.1923e+01 k_t = [1, 1, 1, 1]
Generation  20: Best = 6.612029e+06 FES =  2100   $\mu_F$  = 0.605   $\mu_{CR}$  = 0.314
Diversity = 2.9997e+01 k_t = [4, 4, 2, 4]
Generation  30: Best = 1.370786e+06 FES =  3100   $\mu_F$  = 0.650   $\mu_{CR}$  = 0.232
Diversity = 2.7002e+01 k_t = [1, 1, 1, 1]
Generation  40: Best = 5.827566e+05 FES =  4100   $\mu_F$  = 0.688   $\mu_{CR}$  = 0.152
Diversity = 2.3755e+01 k_t = [3, 4, 2, 2]
Generation  50: Best = 2.716468e+05 FES =  5100   $\mu_F$  = 0.713   $\mu_{CR}$  = 0.104
Diversity = 2.3080e+01 k_t = [1, 1, 1, 1]
Generation  60: Best = 8.751768e+04 FES =  6100   $\mu_F$  = 0.722   $\mu_{CR}$  = 0.078
Diversity = 2.2393e+01 k_t = [3, 3, 3, 2]
Generation  70: Best = 6.031028e+04 FES =  7100   $\mu_F$  = 0.732   $\mu_{CR}$  = 0.069
Diversity = 2.1771e+01 k_t = [1, 1, 1, 1]
Generation  80: Best = 3.727704e+04 FES =  8100   $\mu_F$  = 0.751   $\mu_{CR}$  = 0.057
Diversity = 2.1062e+01 k_t = [4, 2, 4, 3]
Generation  90: Best = 2.607230e+04 FES =  9100   $\mu_F$  = 0.767   $\mu_{CR}$  = 0.061
Diversity = 1.7949e+01 k_t = [1, 1, 1, 1]
Generation 100: Best = 2.001066e+04 FES = 10100   $\mu_F$  = 0.773   $\mu_{CR}$  = 0.055
Diversity = 1.4805e+01 k_t = [2, 4, 2, 4]
Generation 110: Best = 1.635624e+04 FES = 11100   $\mu_F$  = 0.794   $\mu_{CR}$  = 0.051
Diversity = 1.2482e+01 k_t = [1, 1, 1, 1]
Generation 120: Best = 1.451373e+04 FES = 12100   $\mu_F$  = 0.806   $\mu_{CR}$  = 0.048
Diversity = 1.3077e+01 k_t = [3, 4, 3, 2]
Generation 130: Best = 1.189743e+04 FES = 13100   $\mu_F$  = 0.801   $\mu_{CR}$  = 0.053
Diversity = 1.2787e+01 k_t = [1, 1, 1, 1]
Generation 140: Best = 9.839177e+03 FES = 14100   $\mu_F$  = 0.798   $\mu_{CR}$  = 0.058
Diversity = 1.2443e+01 k_t = [4, 4, 3, 3]
Generation 150: Best = 9.115643e+03 FES = 15100   $\mu_F$  = 0.810   $\mu_{CR}$  = 0.057
Diversity = 1.2481e+01 k_t = [1, 1, 1, 1]
Generation 160: Best = 8.251418e+03 FES = 16100   $\mu_F$  = 0.814   $\mu_{CR}$  = 0.055
Diversity = 1.1843e+01 k_t = [3, 4, 3, 3]
Generation 170: Best = 7.614096e+03 FES = 17100   $\mu_F$  = 0.807   $\mu_{CR}$  = 0.059
Diversity = 1.1885e+01 k_t = [1, 1, 1, 1]
Generation 180: Best = 6.982897e+03 FES = 18100   $\mu_F$  = 0.807   $\mu_{CR}$  = 0.060
Diversity = 1.1558e+01 k_t = [4, 4, 3, 2]
Generation 190: Best = 6.758541e+03 FES = 19100   $\mu_F$  = 0.824   $\mu_{CR}$  = 0.066
Diversity = 1.3715e+01 k_t = [1, 1, 1, 1]
Generation 200: Best = 6.547464e+03 FES = 20100   $\mu_F$  = 0.819   $\mu_{CR}$  = 0.063
Diversity = 1.4329e+01 k_t = [3, 3, 2, 4]
Generation 210: Best = 6.318998e+03 FES = 21100   $\mu_F$  = 0.826   $\mu_{CR}$  = 0.057
Diversity = 1.5703e+01 k_t = [1, 1, 1, 1]
Generation 220: Best = 6.191535e+03 FES = 22100   $\mu_F$  = 0.815   $\mu_{CR}$  = 0.054
Diversity = 1.4064e+01 k_t = [3, 2, 4, 3]

```

Generation	230:	Best = 6.153364e+03	FES = 23100	$\mu_F = 0.824$	$\mu_{CR} = 0.057$
Diversity =	1.2329e+01	k_t = [1, 1, 1, 1]			
Generation	240:	Best = 6.074617e+03	FES = 24100	$\mu_F = 0.834$	$\mu_{CR} = 0.056$
Diversity =	1.2192e+01	k_t = [4, 2, 2, 2]			
Generation	250:	Best = 5.978226e+03	FES = 25100	$\mu_F = 0.821$	$\mu_{CR} = 0.060$
Diversity =	1.2483e+01	k_t = [1, 1, 1, 1]			
Generation	260:	Best = 5.931282e+03	FES = 26100	$\mu_F = 0.808$	$\mu_{CR} = 0.055$
Diversity =	1.1152e+01	k_t = [2, 2, 2, 3]			
Generation	270:	Best = 5.892001e+03	FES = 27100	$\mu_F = 0.810$	$\mu_{CR} = 0.052$
Diversity =	1.0449e+01	k_t = [1, 1, 1, 1]			
Generation	280:	Best = 5.856612e+03	FES = 28100	$\mu_F = 0.811$	$\mu_{CR} = 0.055$
Diversity =	1.0399e+01	k_t = [3, 3, 3, 3]			
Generation	290:	Best = 5.835483e+03	FES = 29100	$\mu_F = 0.798$	$\mu_{CR} = 0.061$
Diversity =	1.0311e+01	k_t = [1, 1, 1, 1]			
Generation	300:	Best = 5.761557e+03	FES = 30100	$\mu_F = 0.803$	$\mu_{CR} = 0.054$
Diversity =	9.4467e+00	k_t = [4, 2, 2, 3]			
Generation	310:	Best = 5.754827e+03	FES = 31100	$\mu_F = 0.813$	$\mu_{CR} = 0.061$
Diversity =	9.3740e+00	k_t = [1, 1, 1, 1]			
Generation	320:	Best = 5.713746e+03	FES = 32100	$\mu_F = 0.820$	$\mu_{CR} = 0.058$
Diversity =	9.8099e+00	k_t = [2, 3, 3, 2]			
Generation	330:	Best = 5.671897e+03	FES = 33100	$\mu_F = 0.819$	$\mu_{CR} = 0.055$
Diversity =	1.0204e+01	k_t = [1, 1, 1, 1]			
Generation	340:	Best = 5.636043e+03	FES = 34100	$\mu_F = 0.827$	$\mu_{CR} = 0.062$
Diversity =	1.0823e+01	k_t = [2, 2, 2, 2]			
Generation	350:	Best = 5.602177e+03	FES = 35100	$\mu_F = 0.829$	$\mu_{CR} = 0.065$
Diversity =	1.0833e+01	k_t = [1, 1, 1, 1]			
Generation	360:	Best = 5.545807e+03	FES = 36100	$\mu_F = 0.821$	$\mu_{CR} = 0.056$
Diversity =	1.0414e+01	k_t = [3, 2, 4, 3]			
Generation	370:	Best = 5.528045e+03	FES = 37100	$\mu_F = 0.819$	$\mu_{CR} = 0.050$
Diversity =	1.0795e+01	k_t = [1, 1, 1, 1]			
Generation	380:	Best = 5.496385e+03	FES = 38100	$\mu_F = 0.816$	$\mu_{CR} = 0.055$
Diversity =	1.1198e+01	k_t = [2, 2, 4, 4]			
Generation	390:	Best = 5.445755e+03	FES = 39100	$\mu_F = 0.817$	$\mu_{CR} = 0.062$
Diversity =	1.1712e+01	k_t = [1, 1, 1, 1]			
Generation	400:	Best = 5.418918e+03	FES = 40100	$\mu_F = 0.842$	$\mu_{CR} = 0.066$
Diversity =	1.2036e+01	k_t = [3, 3, 2, 3]			
Generation	410:	Best = 5.399651e+03	FES = 41100	$\mu_F = 0.848$	$\mu_{CR} = 0.062$
Diversity =	1.2269e+01	k_t = [1, 1, 1, 1]			
Generation	420:	Best = 5.376561e+03	FES = 42100	$\mu_F = 0.835$	$\mu_{CR} = 0.065$
Diversity =	1.2460e+01	k_t = [4, 2, 3, 2]			
Generation	430:	Best = 5.316430e+03	FES = 43100	$\mu_F = 0.842$	$\mu_{CR} = 0.056$
Diversity =	1.2314e+01	k_t = [1, 1, 1, 1]			
Generation	440:	Best = 5.311507e+03	FES = 44100	$\mu_F = 0.823$	$\mu_{CR} = 0.050$
Diversity =	1.2394e+01	k_t = [2, 2, 2, 4]			
Generation	450:	Best = 5.277137e+03	FES = 45100	$\mu_F = 0.827$	$\mu_{CR} = 0.048$
Diversity =	1.1511e+01	k_t = [1, 1, 1, 1]			
Generation	460:	Best = 5.252126e+03	FES = 46100	$\mu_F = 0.805$	$\mu_{CR} = 0.056$
Diversity =	1.0194e+01	k_t = [3, 4, 2, 4]			
Generation	470:	Best = 5.219792e+03	FES = 47100	$\mu_F = 0.789$	$\mu_{CR} = 0.054$
Diversity =	1.0084e+01	k_t = [1, 1, 1, 1]			
Generation	480:	Best = 5.188827e+03	FES = 48100	$\mu_F = 0.822$	$\mu_{CR} = 0.050$
Diversity =	1.0109e+01	k_t = [2, 2, 2, 2]			
Generation	490:	Best = 5.188436e+03	FES = 49100	$\mu_F = 0.821$	$\mu_{CR} = 0.053$
Diversity =	1.0137e+01	k_t = [1, 1, 1, 1]			

Generation 500: Best = 5.134381e+03	FES = 50100	$\mu_F = 0.826$	$\mu_{CR} = 0.050$
Diversity = 1.0224e+01	k_t = [3, 3, 4, 3]		
Generation 510: Best = 5.109644e+03	FES = 51100	$\mu_F = 0.826$	$\mu_{CR} = 0.051$
Diversity = 9.9549e+00	k_t = [1, 1, 1, 1]		
Generation 520: Best = 5.033850e+03	FES = 52100	$\mu_F = 0.843$	$\mu_{CR} = 0.055$
Diversity = 1.0610e+01	k_t = [2, 4, 4, 3]		
Generation 530: Best = 4.914559e+03	FES = 53100	$\mu_F = 0.858$	$\mu_{CR} = 0.058$
Diversity = 1.0900e+01	k_t = [1, 1, 1, 1]		
Generation 540: Best = 4.896658e+03	FES = 54100	$\mu_F = 0.852$	$\mu_{CR} = 0.058$
Diversity = 1.2281e+01	k_t = [2, 2, 3, 4]		
Generation 550: Best = 4.827858e+03	FES = 55100	$\mu_F = 0.830$	$\mu_{CR} = 0.051$
Diversity = 1.1448e+01	k_t = [1, 1, 1, 1]		
Generation 560: Best = 4.729462e+03	FES = 56100	$\mu_F = 0.816$	$\mu_{CR} = 0.051$
Diversity = 1.1048e+01	k_t = [2, 2, 3, 3]		
Generation 570: Best = 4.633109e+03	FES = 57100	$\mu_F = 0.833$	$\mu_{CR} = 0.053$
Diversity = 1.1886e+01	k_t = [1, 1, 1, 1]		
Generation 580: Best = 4.603972e+03	FES = 58100	$\mu_F = 0.829$	$\mu_{CR} = 0.054$
Diversity = 1.1056e+01	k_t = [4, 2, 3, 3]		
Generation 590: Best = 4.576457e+03	FES = 59100	$\mu_F = 0.834$	$\mu_{CR} = 0.049$
Diversity = 1.0171e+01	k_t = [1, 1, 1, 1]		
Generation 600: Best = 4.491600e+03	FES = 60100	$\mu_F = 0.847$	$\mu_{CR} = 0.060$
Diversity = 1.0285e+01	k_t = [3, 2, 3, 4]		
Generation 610: Best = 4.419884e+03	FES = 61100	$\mu_F = 0.832$	$\mu_{CR} = 0.056$
Diversity = 9.9580e+00	k_t = [1, 1, 1, 1]		
Generation 620: Best = 4.409223e+03	FES = 62100	$\mu_F = 0.825$	$\mu_{CR} = 0.058$
Diversity = 1.0673e+01	k_t = [4, 3, 4, 2]		
Generation 630: Best = 4.303346e+03	FES = 63100	$\mu_F = 0.823$	$\mu_{CR} = 0.069$
Diversity = 1.0937e+01	k_t = [1, 1, 1, 1]		
Generation 640: Best = 4.300297e+03	FES = 64100	$\mu_F = 0.818$	$\mu_{CR} = 0.060$
Diversity = 1.0705e+01	k_t = [2, 3, 4, 4]		
Generation 650: Best = 4.284434e+03	FES = 65100	$\mu_F = 0.837$	$\mu_{CR} = 0.064$
Diversity = 1.0616e+01	k_t = [1, 1, 1, 1]		
Generation 660: Best = 4.223394e+03	FES = 66100	$\mu_F = 0.832$	$\mu_{CR} = 0.056$
Diversity = 1.0673e+01	k_t = [3, 2, 2, 2]		
Generation 670: Best = 4.198664e+03	FES = 67100	$\mu_F = 0.833$	$\mu_{CR} = 0.048$
Diversity = 1.0993e+01	k_t = [1, 1, 1, 1]		
Generation 680: Best = 4.178319e+03	FES = 68100	$\mu_F = 0.834$	$\mu_{CR} = 0.059$
Diversity = 1.0469e+01	k_t = [2, 2, 2, 4]		
Generation 690: Best = 4.131584e+03	FES = 69100	$\mu_F = 0.830$	$\mu_{CR} = 0.053$
Diversity = 1.0171e+01	k_t = [1, 1, 1, 1]		
Generation 700: Best = 4.130953e+03	FES = 70100	$\mu_F = 0.831$	$\mu_{CR} = 0.058$
Diversity = 1.0217e+01	k_t = [2, 2, 4, 2]		
Generation 710: Best = 4.096043e+03	FES = 71100	$\mu_F = 0.815$	$\mu_{CR} = 0.059$
Diversity = 1.0412e+01	k_t = [1, 1, 1, 1]		
Generation 720: Best = 4.046952e+03	FES = 72100	$\mu_F = 0.809$	$\mu_{CR} = 0.053$
Diversity = 1.0922e+01	k_t = [2, 2, 4, 2]		
Generation 730: Best = 4.042915e+03	FES = 73100	$\mu_F = 0.800$	$\mu_{CR} = 0.053$
Diversity = 1.0174e+01	k_t = [1, 1, 1, 1]		
Generation 740: Best = 4.042478e+03	FES = 74100	$\mu_F = 0.801$	$\mu_{CR} = 0.056$
Diversity = 9.6785e+00	k_t = [4, 4, 3, 4]		
Generation 750: Best = 4.025599e+03	FES = 75100	$\mu_F = 0.811$	$\mu_{CR} = 0.065$
Diversity = 9.3601e+00	k_t = [1, 1, 1, 1]		
Generation 760: Best = 4.025599e+03	FES = 76100	$\mu_F = 0.791$	$\mu_{CR} = 0.057$
Diversity = 9.0951e+00	k_t = [4, 3, 4, 3]		

Generation 770:	Best = 4.020008e+03	FES = 77100	$\mu_F = 0.785$	$\mu_{CR} = 0.047$
Diversity = 8.8323e+00	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 4.008381e+03	FES = 78100	$\mu_F = 0.757$	$\mu_{CR} = 0.055$
Diversity = 8.9115e+00	$k_t = [3, 4, 4, 3]$			
Generation 790:	Best = 4.008381e+03	FES = 79100	$\mu_F = 0.753$	$\mu_{CR} = 0.055$
Diversity = 9.2353e+00	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 4.008381e+03	FES = 80100	$\mu_F = 0.741$	$\mu_{CR} = 0.050$
Diversity = 9.1154e+00	$k_t = [4, 2, 2, 3]$			
Generation 810:	Best = 4.005714e+03	FES = 81100	$\mu_F = 0.775$	$\mu_{CR} = 0.048$
Diversity = 8.8726e+00	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 4.005714e+03	FES = 82100	$\mu_F = 0.803$	$\mu_{CR} = 0.047$
Diversity = 8.9083e+00	$k_t = [3, 3, 4, 2]$			
Generation 830:	Best = 4.001967e+03	FES = 83100	$\mu_F = 0.800$	$\mu_{CR} = 0.047$
Diversity = 8.7483e+00	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 3.999564e+03	FES = 84100	$\mu_F = 0.818$	$\mu_{CR} = 0.051$
Diversity = 8.5377e+00	$k_t = [3, 4, 2, 3]$			
Generation 850:	Best = 3.999564e+03	FES = 85100	$\mu_F = 0.796$	$\mu_{CR} = 0.048$
Diversity = 8.4610e+00	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 3.996723e+03	FES = 86100	$\mu_F = 0.813$	$\mu_{CR} = 0.043$
Diversity = 8.2484e+00	$k_t = [2, 4, 4, 4]$			
Generation 870:	Best = 3.992177e+03	FES = 87100	$\mu_F = 0.804$	$\mu_{CR} = 0.049$
Diversity = 8.3358e+00	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 3.992177e+03	FES = 88100	$\mu_F = 0.813$	$\mu_{CR} = 0.054$
Diversity = 8.3179e+00	$k_t = [4, 2, 2, 4]$			
Generation 890:	Best = 3.991381e+03	FES = 89100	$\mu_F = 0.841$	$\mu_{CR} = 0.051$
Diversity = 8.1654e+00	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 3.981842e+03	FES = 90100	$\mu_F = 0.826$	$\mu_{CR} = 0.047$
Diversity = 8.1868e+00	$k_t = [4, 4, 4, 3]$			
Generation 910:	Best = 3.980319e+03	FES = 91100	$\mu_F = 0.827$	$\mu_{CR} = 0.046$
Diversity = 7.9894e+00	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 3.972655e+03	FES = 92100	$\mu_F = 0.842$	$\mu_{CR} = 0.050$
Diversity = 7.8614e+00	$k_t = [2, 3, 4, 3]$			
Generation 930:	Best = 3.971841e+03	FES = 93100	$\mu_F = 0.852$	$\mu_{CR} = 0.046$
Diversity = 8.0293e+00	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 3.970913e+03	FES = 94100	$\mu_F = 0.862$	$\mu_{CR} = 0.046$
Diversity = 8.3694e+00	$k_t = [2, 2, 3, 2]$			
Generation 950:	Best = 3.964002e+03	FES = 95100	$\mu_F = 0.870$	$\mu_{CR} = 0.048$
Diversity = 8.8959e+00	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 3.911254e+03	FES = 96100	$\mu_F = 0.874$	$\mu_{CR} = 0.056$
Diversity = 9.6509e+00	$k_t = [3, 2, 3, 4]$			
Generation 970:	Best = 3.888462e+03	FES = 97100	$\mu_F = 0.866$	$\mu_{CR} = 0.059$
Diversity = 9.1473e+00	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 3.872879e+03	FES = 98100	$\mu_F = 0.861$	$\mu_{CR} = 0.048$
Diversity = 9.1240e+00	$k_t = [3, 2, 4, 2]$			
Generation 990:	Best = 3.852439e+03	FES = 99100	$\mu_F = 0.853$	$\mu_{CR} = 0.061$
Diversity = 1.0065e+01	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 3.821679e+03	FES = 100100	$\mu_F = 0.848$	$\mu_{CR} = 0.072$
Diversity = 1.0525e+01	$k_t = [4, 3, 3, 3]$			
Generation 1010:	Best = 3.770590e+03	FES = 101100	$\mu_F = 0.831$	$\mu_{CR} = 0.063$
Diversity = 1.0705e+01	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 3.770590e+03	FES = 102100	$\mu_F = 0.835$	$\mu_{CR} = 0.065$
Diversity = 1.0519e+01	$k_t = [3, 4, 3, 4]$			
Generation 1030:	Best = 3.749017e+03	FES = 103100	$\mu_F = 0.831$	$\mu_{CR} = 0.050$
Diversity = 1.0448e+01	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 3.720328e+03 FES = 104100 μ_F = 0.829 μ_{CR} = 0.050
 Diversity = 1.0286e+01 k_t = [3, 4, 3, 2]
 Generation 1050: Best = 3.700682e+03 FES = 105100 μ_F = 0.820 μ_{CR} = 0.070
 Diversity = 1.0421e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 3.649357e+03 FES = 106100 μ_F = 0.827 μ_{CR} = 0.062
 Diversity = 1.0942e+01 k_t = [4, 2, 3, 3]
 Generation 1070: Best = 3.625623e+03 FES = 107100 μ_F = 0.828 μ_{CR} = 0.055
 Diversity = 1.1585e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 3.594491e+03 FES = 108100 μ_F = 0.826 μ_{CR} = 0.056
 Diversity = 1.2266e+01 k_t = [3, 2, 4, 2]
 Generation 1090: Best = 3.575714e+03 FES = 109100 μ_F = 0.828 μ_{CR} = 0.052
 Diversity = 1.2046e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 3.569501e+03 FES = 110100 μ_F = 0.821 μ_{CR} = 0.046
 Diversity = 1.1810e+01 k_t = [3, 3, 4, 4]
 Generation 1110: Best = 3.542327e+03 FES = 111100 μ_F = 0.820 μ_{CR} = 0.048
 Diversity = 1.1596e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 3.512721e+03 FES = 112100 μ_F = 0.823 μ_{CR} = 0.049
 Diversity = 1.1533e+01 k_t = [2, 2, 4, 4]
 Generation 1130: Best = 3.510156e+03 FES = 113100 μ_F = 0.821 μ_{CR} = 0.050
 Diversity = 1.1534e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 3.503296e+03 FES = 114100 μ_F = 0.834 μ_{CR} = 0.051
 Diversity = 1.1394e+01 k_t = [2, 3, 3, 2]
 Generation 1150: Best = 3.491915e+03 FES = 115100 μ_F = 0.844 μ_{CR} = 0.050
 Diversity = 1.1626e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 3.478815e+03 FES = 116100 μ_F = 0.809 μ_{CR} = 0.061
 Diversity = 1.1689e+01 k_t = [2, 3, 3, 3]
 Generation 1170: Best = 3.472825e+03 FES = 117100 μ_F = 0.811 μ_{CR} = 0.064
 Diversity = 1.1838e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 3.433571e+03 FES = 118100 μ_F = 0.817 μ_{CR} = 0.059
 Diversity = 1.1988e+01 k_t = [4, 2, 4, 3]
 Generation 1190: Best = 3.386039e+03 FES = 119100 μ_F = 0.825 μ_{CR} = 0.056
 Diversity = 1.1662e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 3.374166e+03 FES = 120100 μ_F = 0.819 μ_{CR} = 0.055
 Diversity = 1.0482e+01 k_t = [3, 4, 4, 2]
 Generation 1210: Best = 3.337515e+03 FES = 121100 μ_F = 0.826 μ_{CR} = 0.051
 Diversity = 1.0509e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 3.290963e+03 FES = 122100 μ_F = 0.822 μ_{CR} = 0.049
 Diversity = 1.0731e+01 k_t = [3, 3, 2, 3]
 Generation 1230: Best = 3.243908e+03 FES = 123100 μ_F = 0.832 μ_{CR} = 0.049
 Diversity = 1.0680e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 3.243908e+03 FES = 124100 μ_F = 0.827 μ_{CR} = 0.050
 Diversity = 9.9631e+00 k_t = [3, 4, 2, 2]
 Generation 1250: Best = 3.227392e+03 FES = 125100 μ_F = 0.828 μ_{CR} = 0.050
 Diversity = 1.0291e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 3.197292e+03 FES = 126100 μ_F = 0.838 μ_{CR} = 0.052
 Diversity = 1.0037e+01 k_t = [3, 3, 4, 3]
 Generation 1270: Best = 3.177949e+03 FES = 127100 μ_F = 0.842 μ_{CR} = 0.052
 Diversity = 1.0153e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 3.150115e+03 FES = 128100 μ_F = 0.825 μ_{CR} = 0.057
 Diversity = 1.0107e+01 k_t = [2, 2, 4, 2]
 Generation 1290: Best = 3.146987e+03 FES = 129100 μ_F = 0.810 μ_{CR} = 0.059
 Diversity = 1.0023e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 3.108023e+03 FES = 130100 μ_F = 0.802 μ_{CR} = 0.051
 Diversity = 9.8056e+00 k_t = [3, 3, 2, 3]

Generation 1310: Best = 3.069378e+03 FES = 131100 μ_F = 0.798 μ_{CR} = 0.048
 Diversity = 9.6782e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 3.068616e+03 FES = 132100 μ_F = 0.809 μ_{CR} = 0.051
 Diversity = 9.5832e+00 k_t = [3, 4, 2, 4]
 Generation 1330: Best = 3.041234e+03 FES = 133100 μ_F = 0.830 μ_{CR} = 0.063
 Diversity = 1.0498e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 2.975142e+03 FES = 134100 μ_F = 0.805 μ_{CR} = 0.056
 Diversity = 1.0080e+01 k_t = [2, 3, 2, 4]
 Generation 1350: Best = 2.973114e+03 FES = 135100 μ_F = 0.806 μ_{CR} = 0.061
 Diversity = 1.0092e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 2.946279e+03 FES = 136100 μ_F = 0.818 μ_{CR} = 0.060
 Diversity = 9.9177e+00 k_t = [2, 4, 3, 2]
 Generation 1370: Best = 2.911854e+03 FES = 137100 μ_F = 0.826 μ_{CR} = 0.062
 Diversity = 1.0207e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 2.869772e+03 FES = 138100 μ_F = 0.831 μ_{CR} = 0.059
 Diversity = 9.6894e+00 k_t = [4, 2, 3, 4]
 Generation 1390: Best = 2.848102e+03 FES = 139100 μ_F = 0.833 μ_{CR} = 0.064
 Diversity = 8.9563e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 2.809398e+03 FES = 140100 μ_F = 0.831 μ_{CR} = 0.058
 Diversity = 9.0961e+00 k_t = [3, 4, 2, 2]
 Generation 1410: Best = 2.762079e+03 FES = 141100 μ_F = 0.833 μ_{CR} = 0.061
 Diversity = 9.7250e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 2.726662e+03 FES = 142100 μ_F = 0.819 μ_{CR} = 0.054
 Diversity = 9.4524e+00 k_t = [3, 3, 2, 2]
 Generation 1430: Best = 2.694935e+03 FES = 143100 μ_F = 0.804 μ_{CR} = 0.059
 Diversity = 9.2205e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 2.689084e+03 FES = 144100 μ_F = 0.803 μ_{CR} = 0.054
 Diversity = 8.5351e+00 k_t = [2, 3, 3, 3]
 Generation 1450: Best = 2.689084e+03 FES = 145100 μ_F = 0.813 μ_{CR} = 0.047
 Diversity = 8.3386e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 2.686630e+03 FES = 146100 μ_F = 0.831 μ_{CR} = 0.047
 Diversity = 8.0832e+00 k_t = [2, 4, 2, 4]
 Generation 1470: Best = 2.686630e+03 FES = 147100 μ_F = 0.825 μ_{CR} = 0.042
 Diversity = 8.1915e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 2.682256e+03 FES = 148100 μ_F = 0.810 μ_{CR} = 0.050
 Diversity = 7.8449e+00 k_t = [4, 2, 4, 2]
 Generation 1490: Best = 2.672916e+03 FES = 149100 μ_F = 0.827 μ_{CR} = 0.053
 Diversity = 7.9045e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 2.665783e+03 FES = 150100 μ_F = 0.822 μ_{CR} = 0.051
 Diversity = 7.2995e+00 k_t = [2, 3, 3, 4]
 Generation 1510: Best = 2.665783e+03 FES = 151100 μ_F = 0.836 μ_{CR} = 0.047
 Diversity = 7.4627e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 2.637657e+03 FES = 152100 μ_F = 0.842 μ_{CR} = 0.053
 Diversity = 8.0477e+00 k_t = [4, 3, 3, 3]
 Generation 1530: Best = 2.629303e+03 FES = 153100 μ_F = 0.855 μ_{CR} = 0.054
 Diversity = 9.1673e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 2.606716e+03 FES = 154100 μ_F = 0.826 μ_{CR} = 0.047
 Diversity = 9.8537e+00 k_t = [2, 4, 4, 2]
 Generation 1550: Best = 2.572975e+03 FES = 155100 μ_F = 0.820 μ_{CR} = 0.057
 Diversity = 1.0589e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 2.563409e+03 FES = 156100 μ_F = 0.816 μ_{CR} = 0.047
 Diversity = 1.0785e+01 k_t = [4, 4, 3, 2]
 Generation 1570: Best = 2.518803e+03 FES = 157100 μ_F = 0.820 μ_{CR} = 0.052
 Diversity = 1.1051e+01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 2.507381e+03 FES = 158100 μ_F = 0.825 μ_{CR} = 0.056
 Diversity = 1.1163e+01 k_t = [4, 4, 3, 3]
 Generation 1590: Best = 2.471984e+03 FES = 159100 μ_F = 0.820 μ_{CR} = 0.060
 Diversity = 1.1567e+01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 2.464679e+03 FES = 160100 μ_F = 0.817 μ_{CR} = 0.051
 Diversity = 1.1740e+01 k_t = [2, 4, 3, 4]
 Generation 1610: Best = 2.432119e+03 FES = 161100 μ_F = 0.812 μ_{CR} = 0.051
 Diversity = 1.1211e+01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 2.425639e+03 FES = 162100 μ_F = 0.800 μ_{CR} = 0.050
 Diversity = 1.0991e+01 k_t = [4, 3, 4, 3]
 Generation 1630: Best = 2.424441e+03 FES = 163100 μ_F = 0.809 μ_{CR} = 0.048
 Diversity = 1.0792e+01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 2.417333e+03 FES = 164100 μ_F = 0.801 μ_{CR} = 0.051
 Diversity = 9.8452e+00 k_t = [2, 3, 3, 4]
 Generation 1650: Best = 2.413609e+03 FES = 165100 μ_F = 0.794 μ_{CR} = 0.052
 Diversity = 9.5745e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 2.402731e+03 FES = 166100 μ_F = 0.823 μ_{CR} = 0.057
 Diversity = 9.7589e+00 k_t = [2, 3, 3, 4]
 Generation 1670: Best = 2.401471e+03 FES = 167100 μ_F = 0.816 μ_{CR} = 0.050
 Diversity = 9.3769e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 2.397852e+03 FES = 168100 μ_F = 0.831 μ_{CR} = 0.051
 Diversity = 9.8443e+00 k_t = [3, 4, 4, 3]
 Generation 1690: Best = 2.393995e+03 FES = 169100 μ_F = 0.839 μ_{CR} = 0.050
 Diversity = 9.6883e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 2.391030e+03 FES = 170100 μ_F = 0.821 μ_{CR} = 0.046
 Diversity = 9.6186e+00 k_t = [2, 4, 4, 2]
 Generation 1710: Best = 2.390459e+03 FES = 171100 μ_F = 0.811 μ_{CR} = 0.041
 Diversity = 9.4696e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 2.385852e+03 FES = 172100 μ_F = 0.833 μ_{CR} = 0.042
 Diversity = 9.3981e+00 k_t = [2, 3, 3, 3]
 Generation 1730: Best = 2.385852e+03 FES = 173100 μ_F = 0.850 μ_{CR} = 0.039
 Diversity = 9.7564e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 2.373694e+03 FES = 174100 μ_F = 0.859 μ_{CR} = 0.046
 Diversity = 9.9543e+00 k_t = [4, 3, 2, 4]
 Generation 1750: Best = 2.373611e+03 FES = 175100 μ_F = 0.849 μ_{CR} = 0.051
 Diversity = 9.9060e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 2.373611e+03 FES = 176100 μ_F = 0.827 μ_{CR} = 0.051
 Diversity = 9.9440e+00 k_t = [2, 3, 3, 2]
 Generation 1770: Best = 2.373611e+03 FES = 177100 μ_F = 0.827 μ_{CR} = 0.057
 Diversity = 9.9596e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 2.373611e+03 FES = 178100 μ_F = 0.824 μ_{CR} = 0.055
 Diversity = 9.9520e+00 k_t = [3, 4, 3, 4]
 Generation 1790: Best = 2.370727e+03 FES = 179100 μ_F = 0.817 μ_{CR} = 0.044
 Diversity = 9.9331e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 2.370727e+03 FES = 180100 μ_F = 0.801 μ_{CR} = 0.044
 Diversity = 9.8504e+00 k_t = [3, 2, 4, 2]
 Generation 1810: Best = 2.370572e+03 FES = 181100 μ_F = 0.787 μ_{CR} = 0.049
 Diversity = 9.7137e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 2.370358e+03 FES = 182100 μ_F = 0.760 μ_{CR} = 0.046
 Diversity = 9.5689e+00 k_t = [4, 3, 3, 2]
 Generation 1830: Best = 2.364001e+03 FES = 183100 μ_F = 0.709 μ_{CR} = 0.047
 Diversity = 9.5953e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 2.362410e+03 FES = 184100 μ_F = 0.663 μ_{CR} = 0.049
 Diversity = 9.7535e+00 k_t = [2, 3, 3, 3]

Generation 1850: Best = 2.362239e+03	FES = 185100	$\mu_F = 0.721$	$\mu_{CR} = 0.050$
Diversity = 9.7483e+00	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 2.361627e+03	FES = 186100	$\mu_F = 0.724$	$\mu_{CR} = 0.049$
Diversity = 9.7055e+00	k_t = [4, 3, 4, 2]		
Generation 1870: Best = 2.361627e+03	FES = 187100	$\mu_F = 0.692$	$\mu_{CR} = 0.049$
Diversity = 9.7276e+00	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 2.361627e+03	FES = 188100	$\mu_F = 0.714$	$\mu_{CR} = 0.050$
Diversity = 9.6882e+00	k_t = [4, 2, 3, 4]		
Generation 1890: Best = 2.361530e+03	FES = 189100	$\mu_F = 0.770$	$\mu_{CR} = 0.044$
Diversity = 9.9571e+00	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 2.361530e+03	FES = 190100	$\mu_F = 0.760$	$\mu_{CR} = 0.040$
Diversity = 9.9851e+00	k_t = [2, 3, 2, 2]		
Generation 1910: Best = 2.360629e+03	FES = 191100	$\mu_F = 0.763$	$\mu_{CR} = 0.041$
Diversity = 1.0091e+01	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 2.360565e+03	FES = 192100	$\mu_F = 0.769$	$\mu_{CR} = 0.044$
Diversity = 9.9773e+00	k_t = [3, 2, 2, 2]		
Generation 1930: Best = 2.339306e+03	FES = 193100	$\mu_F = 0.799$	$\mu_{CR} = 0.040$
Diversity = 1.0380e+01	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 2.338443e+03	FES = 194100	$\mu_F = 0.832$	$\mu_{CR} = 0.050$
Diversity = 1.0503e+01	k_t = [2, 2, 2, 4]		
Generation 1950: Best = 2.337608e+03	FES = 195100	$\mu_F = 0.868$	$\mu_{CR} = 0.053$
Diversity = 1.1139e+01	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 2.331496e+03	FES = 196100	$\mu_F = 0.867$	$\mu_{CR} = 0.053$
Diversity = 1.1587e+01	k_t = [2, 2, 2, 3]		
Generation 1970: Best = 2.331496e+03	FES = 197100	$\mu_F = 0.860$	$\mu_{CR} = 0.045$
Diversity = 1.2138e+01	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 2.330575e+03	FES = 198100	$\mu_F = 0.884$	$\mu_{CR} = 0.046$
Diversity = 1.2683e+01	k_t = [4, 3, 3, 2]		
Generation 1990: Best = 2.319896e+03	FES = 199100	$\mu_F = 0.862$	$\mu_{CR} = 0.049$
Diversity = 1.2464e+01	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 2.298710e+03	FES = 200100	$\mu_F = 0.860$	$\mu_{CR} = 0.039$
Diversity = 1.1980e+01	k_t = [2, 2, 3, 4]		
Generation 2010: Best = 2.297895e+03	FES = 201100	$\mu_F = 0.859$	$\mu_{CR} = 0.045$
Diversity = 1.1451e+01	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 2.276949e+03	FES = 202100	$\mu_F = 0.854$	$\mu_{CR} = 0.059$
Diversity = 1.0056e+01	k_t = [4, 2, 4, 4]		
Generation 2030: Best = 2.271972e+03	FES = 203100	$\mu_F = 0.854$	$\mu_{CR} = 0.054$
Diversity = 8.8550e+00	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 2.263519e+03	FES = 204100	$\mu_F = 0.850$	$\mu_{CR} = 0.058$
Diversity = 8.2275e+00	k_t = [4, 3, 3, 2]		
Generation 2050: Best = 2.258077e+03	FES = 205100	$\mu_F = 0.853$	$\mu_{CR} = 0.046$
Diversity = 8.3849e+00	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 2.252350e+03	FES = 206100	$\mu_F = 0.880$	$\mu_{CR} = 0.049$
Diversity = 8.7533e+00	k_t = [2, 3, 2, 4]		
Generation 2070: Best = 2.242718e+03	FES = 207100	$\mu_F = 0.868$	$\mu_{CR} = 0.062$
Diversity = 9.3508e+00	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 2.242718e+03	FES = 208100	$\mu_F = 0.859$	$\mu_{CR} = 0.057$
Diversity = 9.0303e+00	k_t = [4, 2, 3, 3]		
Generation 2090: Best = 2.236525e+03	FES = 209100	$\mu_F = 0.846$	$\mu_{CR} = 0.055$
Diversity = 9.6498e+00	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 2.231814e+03	FES = 210100	$\mu_F = 0.830$	$\mu_{CR} = 0.053$
Diversity = 9.2399e+00	k_t = [3, 3, 4, 4]		
Generation 2110: Best = 2.225801e+03	FES = 211100	$\mu_F = 0.834$	$\mu_{CR} = 0.056$
Diversity = 9.1969e+00	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 2.216725e+03 FES = 212100 μ_F = 0.840 μ_{CR} = 0.045
 Diversity = 9.2410e+00 k_t = [2, 3, 3, 4]
 Generation 2130: Best = 2.215205e+03 FES = 213100 μ_F = 0.828 μ_{CR} = 0.045
 Diversity = 9.2498e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 2.214475e+03 FES = 214100 μ_F = 0.827 μ_{CR} = 0.042
 Diversity = 9.5064e+00 k_t = [4, 4, 2, 2]
 Generation 2150: Best = 2.208328e+03 FES = 215100 μ_F = 0.826 μ_{CR} = 0.040
 Diversity = 9.3322e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 2.208328e+03 FES = 216100 μ_F = 0.834 μ_{CR} = 0.041
 Diversity = 9.4618e+00 k_t = [4, 4, 2, 4]
 Generation 2170: Best = 2.205507e+03 FES = 217100 μ_F = 0.829 μ_{CR} = 0.047
 Diversity = 9.3278e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 2.205061e+03 FES = 218100 μ_F = 0.833 μ_{CR} = 0.051
 Diversity = 9.0953e+00 k_t = [2, 3, 2, 3]
 Generation 2190: Best = 2.204130e+03 FES = 219100 μ_F = 0.844 μ_{CR} = 0.051
 Diversity = 9.2672e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 2.204130e+03 FES = 220100 μ_F = 0.795 μ_{CR} = 0.044
 Diversity = 9.0871e+00 k_t = [2, 4, 4, 2]
 Generation 2210: Best = 2.204130e+03 FES = 221100 μ_F = 0.795 μ_{CR} = 0.039
 Diversity = 8.9780e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 2.204130e+03 FES = 222100 μ_F = 0.800 μ_{CR} = 0.046
 Diversity = 8.6897e+00 k_t = [2, 4, 2, 3]
 Generation 2230: Best = 2.204130e+03 FES = 223100 μ_F = 0.850 μ_{CR} = 0.057
 Diversity = 8.7102e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 2.199151e+03 FES = 224100 μ_F = 0.849 μ_{CR} = 0.060
 Diversity = 8.8875e+00 k_t = [2, 2, 3, 4]
 Generation 2250: Best = 2.197517e+03 FES = 225100 μ_F = 0.849 μ_{CR} = 0.047
 Diversity = 8.8175e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 2.196953e+03 FES = 226100 μ_F = 0.857 μ_{CR} = 0.041
 Diversity = 8.9524e+00 k_t = [2, 4, 2, 2]
 Generation 2270: Best = 2.196782e+03 FES = 227100 μ_F = 0.842 μ_{CR} = 0.047
 Diversity = 9.0480e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 2.196782e+03 FES = 228100 μ_F = 0.843 μ_{CR} = 0.053
 Diversity = 8.9289e+00 k_t = [3, 4, 4, 2]
 Generation 2290: Best = 2.196032e+03 FES = 229100 μ_F = 0.826 μ_{CR} = 0.075
 Diversity = 8.9672e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 2.195979e+03 FES = 230100 μ_F = 0.838 μ_{CR} = 0.056
 Diversity = 8.7530e+00 k_t = [2, 4, 4, 4]
 Generation 2310: Best = 2.195032e+03 FES = 231100 μ_F = 0.798 μ_{CR} = 0.046
 Diversity = 8.7608e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 2.195032e+03 FES = 232100 μ_F = 0.802 μ_{CR} = 0.046
 Diversity = 8.6811e+00 k_t = [2, 3, 4, 3]
 Generation 2330: Best = 2.193776e+03 FES = 233100 μ_F = 0.785 μ_{CR} = 0.043
 Diversity = 8.5571e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 2.193776e+03 FES = 234100 μ_F = 0.788 μ_{CR} = 0.043
 Diversity = 8.4378e+00 k_t = [2, 3, 4, 4]
 Generation 2350: Best = 2.192724e+03 FES = 235100 μ_F = 0.808 μ_{CR} = 0.047
 Diversity = 8.4753e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 2.192724e+03 FES = 236100 μ_F = 0.789 μ_{CR} = 0.049
 Diversity = 8.4074e+00 k_t = [4, 3, 2, 4]
 Generation 2370: Best = 2.190181e+03 FES = 237100 μ_F = 0.796 μ_{CR} = 0.045
 Diversity = 8.2975e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 2.189850e+03 FES = 238100 μ_F = 0.809 μ_{CR} = 0.041
 Diversity = 8.3257e+00 k_t = [4, 3, 4, 3]

Generation 2390: Best = 2.189530e+03 FES = 239100 μ_F = 0.807 μ_{CR} = 0.039
 Diversity = 8.3398e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 2.189530e+03 FES = 240100 μ_F = 0.794 μ_{CR} = 0.044
 Diversity = 8.1116e+00 k_t = [2, 3, 4, 2]
 Generation 2410: Best = 2.187936e+03 FES = 241100 μ_F = 0.829 μ_{CR} = 0.053
 Diversity = 8.4252e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 2.185632e+03 FES = 242100 μ_F = 0.827 μ_{CR} = 0.048
 Diversity = 8.3464e+00 k_t = [3, 2, 3, 2]
 Generation 2430: Best = 2.185632e+03 FES = 243100 μ_F = 0.854 μ_{CR} = 0.049
 Diversity = 8.3793e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 2.185632e+03 FES = 244100 μ_F = 0.866 μ_{CR} = 0.048
 Diversity = 8.0802e+00 k_t = [2, 4, 2, 4]
 Generation 2450: Best = 2.185556e+03 FES = 245100 μ_F = 0.864 μ_{CR} = 0.046
 Diversity = 8.0141e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 2.183649e+03 FES = 246100 μ_F = 0.838 μ_{CR} = 0.048
 Diversity = 8.0023e+00 k_t = [2, 4, 3, 2]
 Generation 2470: Best = 2.182831e+03 FES = 247100 μ_F = 0.841 μ_{CR} = 0.047
 Diversity = 7.6526e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 2.179709e+03 FES = 248100 μ_F = 0.843 μ_{CR} = 0.054
 Diversity = 6.1048e+00 k_t = [2, 3, 2, 2]
 Generation 2490: Best = 2.177155e+03 FES = 249100 μ_F = 0.835 μ_{CR} = 0.059
 Diversity = 5.6521e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 2.173264e+03 FES = 250100 μ_F = 0.821 μ_{CR} = 0.066
 Diversity = 4.9577e+00 k_t = [3, 4, 2, 2]
 Generation 2510: Best = 2.169020e+03 FES = 251100 μ_F = 0.826 μ_{CR} = 0.058
 Diversity = 4.7346e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 2.165802e+03 FES = 252100 μ_F = 0.814 μ_{CR} = 0.054
 Diversity = 4.0090e+00 k_t = [2, 4, 3, 4]
 Generation 2530: Best = 2.162680e+03 FES = 253100 μ_F = 0.810 μ_{CR} = 0.052
 Diversity = 3.5291e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 2.160930e+03 FES = 254100 μ_F = 0.821 μ_{CR} = 0.059
 Diversity = 3.3136e+00 k_t = [2, 4, 3, 2]
 Generation 2550: Best = 2.158602e+03 FES = 255100 μ_F = 0.825 μ_{CR} = 0.054
 Diversity = 3.4337e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 2.157760e+03 FES = 256100 μ_F = 0.817 μ_{CR} = 0.053
 Diversity = 3.5628e+00 k_t = [2, 4, 4, 3]
 Generation 2570: Best = 2.156533e+03 FES = 257100 μ_F = 0.823 μ_{CR} = 0.052
 Diversity = 3.3893e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 2.155560e+03 FES = 258100 μ_F = 0.820 μ_{CR} = 0.056
 Diversity = 3.2940e+00 k_t = [4, 3, 4, 4]
 Generation 2590: Best = 2.153142e+03 FES = 259100 μ_F = 0.815 μ_{CR} = 0.059
 Diversity = 3.2140e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 2.153063e+03 FES = 260100 μ_F = 0.817 μ_{CR} = 0.056
 Diversity = 3.7619e+00 k_t = [4, 2, 2, 2]
 Generation 2610: Best = 2.151379e+03 FES = 261100 μ_F = 0.816 μ_{CR} = 0.058
 Diversity = 4.0016e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 2.150721e+03 FES = 262100 μ_F = 0.812 μ_{CR} = 0.049
 Diversity = 3.5829e+00 k_t = [4, 4, 2, 2]
 Generation 2630: Best = 2.150049e+03 FES = 263100 μ_F = 0.804 μ_{CR} = 0.045
 Diversity = 3.8681e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 2.149634e+03 FES = 264100 μ_F = 0.812 μ_{CR} = 0.050
 Diversity = 3.8448e+00 k_t = [2, 4, 2, 4]
 Generation 2650: Best = 2.148779e+03 FES = 265100 μ_F = 0.789 μ_{CR} = 0.052
 Diversity = 3.2845e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 2.147867e+03 FES = 266100 μ_F = 0.790 μ_{CR} = 0.046
 Diversity = 3.2954e+00 k_t = [4, 3, 3, 2]
 Generation 2670: Best = 2.147305e+03 FES = 267100 μ_F = 0.785 μ_{CR} = 0.045
 Diversity = 3.4528e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 2.146345e+03 FES = 268100 μ_F = 0.812 μ_{CR} = 0.051
 Diversity = 3.7431e+00 k_t = [2, 2, 4, 2]
 Generation 2690: Best = 2.144895e+03 FES = 269100 μ_F = 0.821 μ_{CR} = 0.061
 Diversity = 4.1184e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 2.144362e+03 FES = 270100 μ_F = 0.826 μ_{CR} = 0.059
 Diversity = 4.1906e+00 k_t = [4, 3, 2, 2]
 Generation 2710: Best = 2.143035e+03 FES = 271100 μ_F = 0.830 μ_{CR} = 0.061
 Diversity = 4.2539e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 2.142328e+03 FES = 272100 μ_F = 0.820 μ_{CR} = 0.056
 Diversity = 4.2150e+00 k_t = [2, 2, 3, 2]
 Generation 2730: Best = 2.141444e+03 FES = 273100 μ_F = 0.815 μ_{CR} = 0.051
 Diversity = 4.4631e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 2.140367e+03 FES = 274100 μ_F = 0.819 μ_{CR} = 0.049
 Diversity = 4.3901e+00 k_t = [3, 3, 2, 4]
 Generation 2750: Best = 2.140089e+03 FES = 275100 μ_F = 0.822 μ_{CR} = 0.051
 Diversity = 3.6461e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 2.138436e+03 FES = 276100 μ_F = 0.832 μ_{CR} = 0.054
 Diversity = 3.8397e+00 k_t = [2, 2, 4, 3]
 Generation 2770: Best = 2.137744e+03 FES = 277100 μ_F = 0.841 μ_{CR} = 0.064
 Diversity = 3.6532e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 2.136517e+03 FES = 278100 μ_F = 0.841 μ_{CR} = 0.056
 Diversity = 3.7063e+00 k_t = [2, 4, 3, 4]
 Generation 2790: Best = 2.135295e+03 FES = 279100 μ_F = 0.843 μ_{CR} = 0.050
 Diversity = 3.5400e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 2.134010e+03 FES = 280100 μ_F = 0.808 μ_{CR} = 0.053
 Diversity = 3.2793e+00 k_t = [2, 3, 4, 3]
 Generation 2810: Best = 2.133229e+03 FES = 281100 μ_F = 0.812 μ_{CR} = 0.059
 Diversity = 3.2046e+00 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 2.133023e+03 FES = 282100 μ_F = 0.812 μ_{CR} = 0.061
 Diversity = 3.4342e+00 k_t = [2, 4, 2, 2]
 Generation 2830: Best = 2.131477e+03 FES = 283100 μ_F = 0.818 μ_{CR} = 0.060
 Diversity = 3.6447e+00 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 2.130790e+03 FES = 284100 μ_F = 0.816 μ_{CR} = 0.053
 Diversity = 4.1059e+00 k_t = [4, 4, 4, 2]
 Generation 2850: Best = 2.129628e+03 FES = 285100 μ_F = 0.801 μ_{CR} = 0.059
 Diversity = 4.4192e+00 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 2.128312e+03 FES = 286100 μ_F = 0.805 μ_{CR} = 0.057
 Diversity = 4.8332e+00 k_t = [4, 3, 3, 3]
 Generation 2870: Best = 2.128168e+03 FES = 287100 μ_F = 0.800 μ_{CR} = 0.067
 Diversity = 4.7769e+00 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 2.127032e+03 FES = 288100 μ_F = 0.818 μ_{CR} = 0.054
 Diversity = 4.4192e+00 k_t = [3, 2, 3, 2]
 Generation 2890: Best = 2.125966e+03 FES = 289100 μ_F = 0.825 μ_{CR} = 0.054
 Diversity = 5.0587e+00 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 2.125109e+03 FES = 290100 μ_F = 0.827 μ_{CR} = 0.063
 Diversity = 5.1872e+00 k_t = [2, 3, 3, 3]
 Generation 2910: Best = 2.124531e+03 FES = 291100 μ_F = 0.804 μ_{CR} = 0.052
 Diversity = 5.3223e+00 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 2.123223e+03 FES = 292100 μ_F = 0.801 μ_{CR} = 0.057
 Diversity = 5.6466e+00 k_t = [2, 3, 4, 2]

Generation 2930: Best = 2.121764e+03 FES = 293100 μ_F = 0.824 μ_{CR} = 0.059
 Diversity = 6.5955e+00 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 2.120841e+03 FES = 294100 μ_F = 0.819 μ_{CR} = 0.065
 Diversity = 6.6716e+00 k_t = [4, 2, 2, 2]
 Generation 2950: Best = 2.119659e+03 FES = 295100 μ_F = 0.806 μ_{CR} = 0.055
 Diversity = 6.7483e+00 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 2.118487e+03 FES = 296100 μ_F = 0.801 μ_{CR} = 0.060
 Diversity = 6.6655e+00 k_t = [4, 3, 4, 3]
 Generation 2970: Best = 2.118487e+03 FES = 297100 μ_F = 0.809 μ_{CR} = 0.058
 Diversity = 6.6188e+00 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 2.117609e+03 FES = 298100 μ_F = 0.798 μ_{CR} = 0.056
 Diversity = 5.5436e+00 k_t = [2, 3, 3, 4]
 Generation 2990: Best = 2.116863e+03 FES = 299100 μ_F = 0.810 μ_{CR} = 0.050
 Diversity = 5.2060e+00 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 2.115914e+03

Total Function Evals = 300000

Total Successful Updates = 43531

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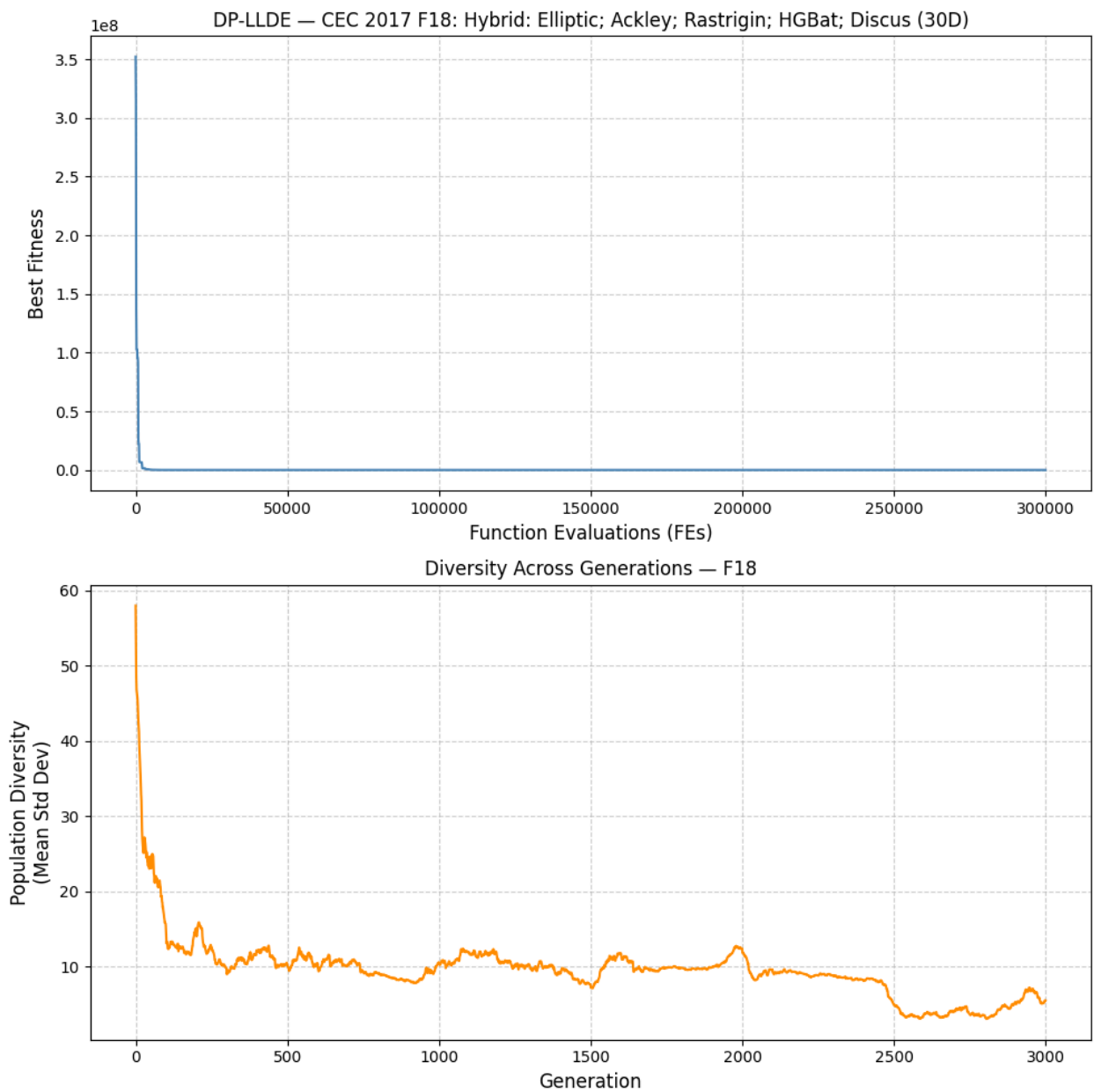
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Results:

Best Fitness : 2.115914e+03
 Global Optimum : 1.800000e+03
 Error (|best-optimum|) : 3.159139e+02
 Best Solution (5 dim) : [22.68226224 10.12060245 15.6120562 -39.32752257
 1.50764354]
 D^(0) (init. diversity): 5.7976e+01

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Graph saved as convergence_F18.png

```
In [ ]: # =====
# F19 – Hybrid Function 9 (Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierst
# =====
run_and_plot('F19', make_wrapper(hybrid.f19), 1900.0, 'Hybrid: Bent Cigar; Ras
```

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DP-LLDE on CEC 2017 F19 – Single Run

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Function : Hybrid: Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstras
s; Schaffer (30D)

Bounds : [-100, 100]

Global optimum : 1900.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 7.567444e+10

Generation 10: Best = 1.882078e+09	FES = 1100	$\mu_F = 0.577$	$\mu_{CR} = 0.354$
Diversity = 3.4711e+01	$k_t = [1, 1, 1, 1]$		
Generation 20: Best = 9.586931e+07	FES = 2100	$\mu_F = 0.617$	$\mu_{CR} = 0.346$
Diversity = 2.6788e+01	$k_t = [4, 4, 2, 4]$		
Generation 30: Best = 1.366870e+07	FES = 3100	$\mu_F = 0.656$	$\mu_{CR} = 0.335$
Diversity = 2.1293e+01	$k_t = [1, 1, 1, 1]$		
Generation 40: Best = 3.378541e+06	FES = 4100	$\mu_F = 0.682$	$\mu_{CR} = 0.303$
Diversity = 1.8449e+01	$k_t = [2, 3, 3, 2]$		
Generation 50: Best = 9.660792e+05	FES = 5100	$\mu_F = 0.698$	$\mu_{CR} = 0.262$
Diversity = 1.7843e+01	$k_t = [1, 1, 1, 1]$		
Generation 60: Best = 2.340890e+05	FES = 6100	$\mu_F = 0.722$	$\mu_{CR} = 0.229$
Diversity = 1.8289e+01	$k_t = [3, 3, 2, 3]$		
Generation 70: Best = 1.378342e+05	FES = 7100	$\mu_F = 0.740$	$\mu_{CR} = 0.186$
Diversity = 1.7422e+01	$k_t = [1, 1, 1, 1]$		
Generation 80: Best = 3.880156e+04	FES = 8100	$\mu_F = 0.763$	$\mu_{CR} = 0.144$
Diversity = 1.7624e+01	$k_t = [3, 3, 3, 3]$		
Generation 90: Best = 1.348220e+04	FES = 9100	$\mu_F = 0.780$	$\mu_{CR} = 0.126$
Diversity = 1.6991e+01	$k_t = [1, 1, 1, 1]$		
Generation 100: Best = 5.912579e+03	FES = 10100	$\mu_F = 0.787$	$\mu_{CR} = 0.109$
Diversity = 1.6149e+01	$k_t = [3, 2, 4, 2]$		
Generation 110: Best = 3.560284e+03	FES = 11100	$\mu_F = 0.796$	$\mu_{CR} = 0.091$
Diversity = 1.7524e+01	$k_t = [1, 1, 1, 1]$		
Generation 120: Best = 2.504115e+03	FES = 12100	$\mu_F = 0.802$	$\mu_{CR} = 0.085$
Diversity = 1.7292e+01	$k_t = [4, 2, 4, 4]$		
Generation 130: Best = 2.287971e+03	FES = 13100	$\mu_F = 0.802$	$\mu_{CR} = 0.083$
Diversity = 1.5366e+01	$k_t = [1, 1, 1, 1]$		
Generation 140: Best = 2.211245e+03	FES = 14100	$\mu_F = 0.801$	$\mu_{CR} = 0.077$
Diversity = 1.3849e+01	$k_t = [4, 4, 3, 3]$		
Generation 150: Best = 2.082323e+03	FES = 15100	$\mu_F = 0.807$	$\mu_{CR} = 0.077$
Diversity = 1.4547e+01	$k_t = [1, 1, 1, 1]$		
Generation 160: Best = 2.043944e+03	FES = 16100	$\mu_F = 0.810$	$\mu_{CR} = 0.074$
Diversity = 1.4708e+01	$k_t = [2, 3, 4, 3]$		
Generation 170: Best = 1.996152e+03	FES = 17100	$\mu_F = 0.814$	$\mu_{CR} = 0.074$
Diversity = 1.3592e+01	$k_t = [1, 1, 1, 1]$		
Generation 180: Best = 1.995429e+03	FES = 18100	$\mu_F = 0.810$	$\mu_{CR} = 0.076$
Diversity = 1.3430e+01	$k_t = [2, 4, 2, 3]$		
Generation 190: Best = 1.973166e+03	FES = 19100	$\mu_F = 0.796$	$\mu_{CR} = 0.078$
Diversity = 1.3036e+01	$k_t = [1, 1, 1, 1]$		
Generation 200: Best = 1.973166e+03	FES = 20100	$\mu_F = 0.781$	$\mu_{CR} = 0.069$
Diversity = 1.3040e+01	$k_t = [3, 4, 2, 2]$		
Generation 210: Best = 1.966586e+03	FES = 21100	$\mu_F = 0.769$	$\mu_{CR} = 0.064$
Diversity = 1.2412e+01	$k_t = [1, 1, 1, 1]$		
Generation 220: Best = 1.966437e+03	FES = 22100	$\mu_F = 0.786$	$\mu_{CR} = 0.061$

Diversity = 1.1878e+01	k_t = [2, 2, 4, 4]				
Generation 230: Best = 1.959492e+03	FES = 23100	$\mu_F = 0.812$	$\mu_{CR} = 0.056$		
Diversity = 1.2129e+01	k_t = [1, 1, 1, 1]				
Generation 240: Best = 1.959492e+03	FES = 24100	$\mu_F = 0.830$	$\mu_{CR} = 0.062$		
Diversity = 1.1540e+01	k_t = [3, 3, 4, 4]				
Generation 250: Best = 1.954226e+03	FES = 25100	$\mu_F = 0.847$	$\mu_{CR} = 0.076$		
Diversity = 1.1646e+01	k_t = [1, 1, 1, 1]				
Generation 260: Best = 1.954226e+03	FES = 26100	$\mu_F = 0.853$	$\mu_{CR} = 0.080$		
Diversity = 1.1977e+01	k_t = [2, 2, 3, 4]				
Generation 270: Best = 1.951508e+03	FES = 27100	$\mu_F = 0.826$	$\mu_{CR} = 0.066$		
Diversity = 1.1975e+01	k_t = [1, 1, 1, 1]				
Generation 280: Best = 1.950283e+03	FES = 28100	$\mu_F = 0.837$	$\mu_{CR} = 0.066$		
Diversity = 1.1799e+01	k_t = [4, 3, 4, 3]				
Generation 290: Best = 1.946781e+03	FES = 29100	$\mu_F = 0.848$	$\mu_{CR} = 0.056$		
Diversity = 1.1689e+01	k_t = [1, 1, 1, 1]				
Generation 300: Best = 1.946270e+03	FES = 30100	$\mu_F = 0.856$	$\mu_{CR} = 0.060$		
Diversity = 1.1884e+01	k_t = [2, 2, 4, 2]				
Generation 310: Best = 1.945940e+03	FES = 31100	$\mu_F = 0.846$	$\mu_{CR} = 0.051$		
Diversity = 1.1935e+01	k_t = [1, 1, 1, 1]				
Generation 320: Best = 1.945518e+03	FES = 32100	$\mu_F = 0.838$	$\mu_{CR} = 0.056$		
Diversity = 1.2072e+01	k_t = [4, 2, 2, 4]				
Generation 330: Best = 1.945518e+03	FES = 33100	$\mu_F = 0.831$	$\mu_{CR} = 0.058$		
Diversity = 1.1924e+01	k_t = [1, 1, 1, 1]				
Generation 340: Best = 1.943553e+03	FES = 34100	$\mu_F = 0.847$	$\mu_{CR} = 0.061$		
Diversity = 1.1768e+01	k_t = [2, 2, 4, 2]				
Generation 350: Best = 1.939670e+03	FES = 35100	$\mu_F = 0.866$	$\mu_{CR} = 0.060$		
Diversity = 1.1964e+01	k_t = [1, 1, 1, 1]				
Generation 360: Best = 1.939670e+03	FES = 36100	$\mu_F = 0.866$	$\mu_{CR} = 0.053$		
Diversity = 1.2049e+01	k_t = [4, 3, 4, 4]				
Generation 370: Best = 1.938614e+03	FES = 37100	$\mu_F = 0.854$	$\mu_{CR} = 0.054$		
Diversity = 1.2063e+01	k_t = [1, 1, 1, 1]				
Generation 380: Best = 1.938614e+03	FES = 38100	$\mu_F = 0.887$	$\mu_{CR} = 0.058$		
Diversity = 1.2009e+01	k_t = [2, 2, 2, 4]				
Generation 390: Best = 1.938609e+03	FES = 39100	$\mu_F = 0.885$	$\mu_{CR} = 0.057$		
Diversity = 1.2110e+01	k_t = [1, 1, 1, 1]				
Generation 400: Best = 1.938609e+03	FES = 40100	$\mu_F = 0.855$	$\mu_{CR} = 0.051$		
Diversity = 1.2053e+01	k_t = [2, 4, 2, 3]				
Generation 410: Best = 1.938215e+03	FES = 41100	$\mu_F = 0.865$	$\mu_{CR} = 0.048$		
Diversity = 1.2091e+01	k_t = [1, 1, 1, 1]				
Generation 420: Best = 1.937798e+03	FES = 42100	$\mu_F = 0.883$	$\mu_{CR} = 0.058$		
Diversity = 1.1968e+01	k_t = [4, 3, 3, 2]				
Generation 430: Best = 1.934320e+03	FES = 43100	$\mu_F = 0.903$	$\mu_{CR} = 0.057$		
Diversity = 1.2020e+01	k_t = [1, 1, 1, 1]				
Generation 440: Best = 1.933358e+03	FES = 44100	$\mu_F = 0.893$	$\mu_{CR} = 0.050$		
Diversity = 1.1933e+01	k_t = [2, 2, 3, 2]				
Generation 450: Best = 1.933358e+03	FES = 45100	$\mu_F = 0.919$	$\mu_{CR} = 0.056$		
Diversity = 1.2084e+01	k_t = [1, 1, 1, 1]				
Generation 460: Best = 1.933358e+03	FES = 46100	$\mu_F = 0.907$	$\mu_{CR} = 0.051$		
Diversity = 1.1978e+01	k_t = [3, 2, 4, 2]				
Generation 470: Best = 1.931359e+03	FES = 47100	$\mu_F = 0.901$	$\mu_{CR} = 0.057$		
Diversity = 1.2086e+01	k_t = [1, 1, 1, 1]				
Generation 480: Best = 1.931359e+03	FES = 48100	$\mu_F = 0.879$	$\mu_{CR} = 0.051$		
Diversity = 1.2276e+01	k_t = [3, 4, 3, 2]				
Generation 490: Best = 1.931324e+03	FES = 49100	$\mu_F = 0.821$	$\mu_{CR} = 0.054$		

Diversity = 1.2513e+01	k_t = [1, 1, 1, 1]					
Generation 500: Best = 1.930800e+03	FES = 50100	$\mu_F = 0.833$	$\mu_{CR} = 0.051$			
Diversity = 1.2659e+01	k_t = [3, 4, 2, 4]					
Generation 510: Best = 1.930800e+03	FES = 51100	$\mu_F = 0.848$	$\mu_{CR} = 0.054$			
Diversity = 1.2734e+01	k_t = [1, 1, 1, 1]					
Generation 520: Best = 1.930800e+03	FES = 52100	$\mu_F = 0.882$	$\mu_{CR} = 0.054$			
Diversity = 1.2934e+01	k_t = [2, 2, 2, 3]					
Generation 530: Best = 1.930800e+03	FES = 53100	$\mu_F = 0.891$	$\mu_{CR} = 0.051$			
Diversity = 1.3036e+01	k_t = [1, 1, 1, 1]					
Generation 540: Best = 1.930800e+03	FES = 54100	$\mu_F = 0.863$	$\mu_{CR} = 0.063$			
Diversity = 1.3068e+01	k_t = [4, 4, 3, 3]					
Generation 550: Best = 1.929821e+03	FES = 55100	$\mu_F = 0.881$	$\mu_{CR} = 0.054$			
Diversity = 1.3073e+01	k_t = [1, 1, 1, 1]					
Generation 560: Best = 1.929821e+03	FES = 56100	$\mu_F = 0.891$	$\mu_{CR} = 0.052$			
Diversity = 1.2987e+01	k_t = [2, 2, 3, 2]					
Generation 570: Best = 1.929821e+03	FES = 57100	$\mu_F = 0.786$	$\mu_{CR} = 0.050$			
Diversity = 1.2937e+01	k_t = [1, 1, 1, 1]					
Generation 580: Best = 1.929706e+03	FES = 58100	$\mu_F = 0.806$	$\mu_{CR} = 0.039$			
Diversity = 1.2912e+01	k_t = [2, 2, 4, 3]					
Generation 590: Best = 1.929705e+03	FES = 59100	$\mu_F = 0.767$	$\mu_{CR} = 0.037$			
Diversity = 1.2970e+01	k_t = [1, 1, 1, 1]					
Generation 600: Best = 1.929705e+03	FES = 60100	$\mu_F = 0.759$	$\mu_{CR} = 0.049$			
Diversity = 1.3032e+01	k_t = [4, 3, 4, 2]					
Generation 610: Best = 1.929611e+03	FES = 61100	$\mu_F = 0.765$	$\mu_{CR} = 0.057$			
Diversity = 1.3067e+01	k_t = [1, 1, 1, 1]					
Generation 620: Best = 1.928437e+03	FES = 62100	$\mu_F = 0.758$	$\mu_{CR} = 0.063$			
Diversity = 1.2799e+01	k_t = [4, 3, 4, 4]					
Generation 630: Best = 1.928364e+03	FES = 63100	$\mu_F = 0.736$	$\mu_{CR} = 0.046$			
Diversity = 1.2764e+01	k_t = [1, 1, 1, 1]					
Generation 640: Best = 1.928364e+03	FES = 64100	$\mu_F = 0.725$	$\mu_{CR} = 0.056$			
Diversity = 1.2803e+01	k_t = [3, 3, 3, 2]					
Generation 650: Best = 1.926624e+03	FES = 65100	$\mu_F = 0.717$	$\mu_{CR} = 0.040$			
Diversity = 1.2815e+01	k_t = [1, 1, 1, 1]					
Generation 660: Best = 1.926052e+03	FES = 66100	$\mu_F = 0.756$	$\mu_{CR} = 0.035$			
Diversity = 1.2776e+01	k_t = [2, 3, 2, 3]					
Generation 670: Best = 1.925867e+03	FES = 67100	$\mu_F = 0.777$	$\mu_{CR} = 0.042$			
Diversity = 1.2796e+01	k_t = [1, 1, 1, 1]					
Generation 680: Best = 1.925867e+03	FES = 68100	$\mu_F = 0.834$	$\mu_{CR} = 0.050$			
Diversity = 1.2662e+01	k_t = [4, 4, 2, 4]					
Generation 690: Best = 1.923819e+03	FES = 69100	$\mu_F = 0.834$	$\mu_{CR} = 0.042$			
Diversity = 1.2620e+01	k_t = [1, 1, 1, 1]					
Generation 700: Best = 1.923819e+03	FES = 70100	$\mu_F = 0.782$	$\mu_{CR} = 0.052$			
Diversity = 1.2574e+01	k_t = [2, 3, 2, 2]					
Generation 710: Best = 1.923819e+03	FES = 71100	$\mu_F = 0.839$	$\mu_{CR} = 0.049$			
Diversity = 1.2579e+01	k_t = [1, 1, 1, 1]					
Generation 720: Best = 1.923819e+03	FES = 72100	$\mu_F = 0.815$	$\mu_{CR} = 0.053$			
Diversity = 1.2607e+01	k_t = [2, 3, 2, 2]					
Generation 730: Best = 1.923819e+03	FES = 73100	$\mu_F = 0.746$	$\mu_{CR} = 0.038$			
Diversity = 1.2633e+01	k_t = [1, 1, 1, 1]					
Generation 740: Best = 1.923819e+03	FES = 74100	$\mu_F = 0.724$	$\mu_{CR} = 0.037$			
Diversity = 1.2634e+01	k_t = [2, 3, 3, 3]					
Generation 750: Best = 1.923614e+03	FES = 75100	$\mu_F = 0.709$	$\mu_{CR} = 0.033$			
Diversity = 1.2691e+01	k_t = [1, 1, 1, 1]					
Generation 760: Best = 1.923614e+03	FES = 76100	$\mu_F = 0.779$	$\mu_{CR} = 0.048$			

Diversity = 1.2465e+01	k_t = [4, 3, 4, 4]				
Generation 770: Best = 1.923614e+03	FES = 77100	$\mu_F = 0.725$	$\mu_{CR} = 0.055$		
Diversity = 1.2491e+01	k_t = [1, 1, 1, 1]				
Generation 780: Best = 1.923338e+03	FES = 78100	$\mu_F = 0.769$	$\mu_{CR} = 0.046$		
Diversity = 1.2357e+01	k_t = [3, 4, 4, 4]				
Generation 790: Best = 1.923338e+03	FES = 79100	$\mu_F = 0.807$	$\mu_{CR} = 0.051$		
Diversity = 1.2352e+01	k_t = [1, 1, 1, 1]				
Generation 800: Best = 1.923338e+03	FES = 80100	$\mu_F = 0.834$	$\mu_{CR} = 0.055$		
Diversity = 1.2373e+01	k_t = [3, 2, 2, 4]				
Generation 810: Best = 1.922709e+03	FES = 81100	$\mu_F = 0.740$	$\mu_{CR} = 0.052$		
Diversity = 1.2323e+01	k_t = [1, 1, 1, 1]				
Generation 820: Best = 1.922709e+03	FES = 82100	$\mu_F = 0.745$	$\mu_{CR} = 0.049$		
Diversity = 1.2378e+01	k_t = [4, 4, 2, 4]				
Generation 830: Best = 1.922709e+03	FES = 83100	$\mu_F = 0.788$	$\mu_{CR} = 0.047$		
Diversity = 1.2372e+01	k_t = [1, 1, 1, 1]				
Generation 840: Best = 1.921776e+03	FES = 84100	$\mu_F = 0.815$	$\mu_{CR} = 0.049$		
Diversity = 1.2176e+01	k_t = [4, 4, 4, 4]				
Generation 850: Best = 1.921776e+03	FES = 85100	$\mu_F = 0.776$	$\mu_{CR} = 0.044$		
Diversity = 1.2100e+01	k_t = [1, 1, 1, 1]				
Generation 860: Best = 1.921776e+03	FES = 86100	$\mu_F = 0.821$	$\mu_{CR} = 0.050$		
Diversity = 1.2105e+01	k_t = [4, 2, 2, 3]				
Generation 870: Best = 1.921716e+03	FES = 87100	$\mu_F = 0.794$	$\mu_{CR} = 0.048$		
Diversity = 1.2067e+01	k_t = [1, 1, 1, 1]				
Generation 880: Best = 1.921575e+03	FES = 88100	$\mu_F = 0.832$	$\mu_{CR} = 0.059$		
Diversity = 1.2060e+01	k_t = [4, 4, 4, 4]				
Generation 890: Best = 1.921575e+03	FES = 89100	$\mu_F = 0.834$	$\mu_{CR} = 0.048$		
Diversity = 1.1997e+01	k_t = [1, 1, 1, 1]				
Generation 900: Best = 1.921575e+03	FES = 90100	$\mu_F = 0.861$	$\mu_{CR} = 0.046$		
Diversity = 1.1996e+01	k_t = [2, 3, 2, 3]				
Generation 910: Best = 1.921575e+03	FES = 91100	$\mu_F = 0.774$	$\mu_{CR} = 0.036$		
Diversity = 1.1932e+01	k_t = [1, 1, 1, 1]				
Generation 920: Best = 1.921575e+03	FES = 92100	$\mu_F = 0.850$	$\mu_{CR} = 0.041$		
Diversity = 1.1836e+01	k_t = [3, 4, 4, 4]				
Generation 930: Best = 1.921575e+03	FES = 93100	$\mu_F = 0.803$	$\mu_{CR} = 0.051$		
Diversity = 1.1891e+01	k_t = [1, 1, 1, 1]				
Generation 940: Best = 1.921575e+03	FES = 94100	$\mu_F = 0.784$	$\mu_{CR} = 0.055$		
Diversity = 1.2002e+01	k_t = [2, 2, 3, 2]				
Generation 950: Best = 1.921513e+03	FES = 95100	$\mu_F = 0.831$	$\mu_{CR} = 0.047$		
Diversity = 1.2005e+01	k_t = [1, 1, 1, 1]				
Generation 960: Best = 1.921513e+03	FES = 96100	$\mu_F = 0.843$	$\mu_{CR} = 0.047$		
Diversity = 1.1956e+01	k_t = [4, 4, 4, 4]				
Generation 970: Best = 1.921513e+03	FES = 97100	$\mu_F = 0.835$	$\mu_{CR} = 0.046$		
Diversity = 1.1844e+01	k_t = [1, 1, 1, 1]				
Generation 980: Best = 1.921513e+03	FES = 98100	$\mu_F = 0.798$	$\mu_{CR} = 0.049$		
Diversity = 1.1881e+01	k_t = [4, 4, 3, 3]				
Generation 990: Best = 1.921513e+03	FES = 99100	$\mu_F = 0.803$	$\mu_{CR} = 0.036$		
Diversity = 1.1922e+01	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 1.921513e+03	FES = 100100	$\mu_F = 0.758$	$\mu_{CR} = 0.045$		
Diversity = 1.1827e+01	k_t = [2, 3, 4, 4]				
Generation 1010: Best = 1.921513e+03	FES = 101100	$\mu_F = 0.753$	$\mu_{CR} = 0.042$		
Diversity = 1.1823e+01	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 1.920704e+03	FES = 102100	$\mu_F = 0.736$	$\mu_{CR} = 0.049$		
Diversity = 1.1882e+01	k_t = [4, 2, 4, 4]				
Generation 1030: Best = 1.920704e+03	FES = 103100	$\mu_F = 0.735$	$\mu_{CR} = 0.046$		

Diversity = 1.1965e+01	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 1.920704e+03	FES = 104100	$\mu_F = 0.728$	$\mu_{CR} = 0.048$	
Diversity = 1.1897e+01	k_t = [4, 2, 2, 3]			
Generation 1050: Best = 1.920704e+03	FES = 105100	$\mu_F = 0.758$	$\mu_{CR} = 0.043$	
Diversity = 1.1883e+01	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 1.920704e+03	FES = 106100	$\mu_F = 0.738$	$\mu_{CR} = 0.055$	
Diversity = 1.1901e+01	k_t = [3, 3, 4, 4]			
Generation 1070: Best = 1.920704e+03	FES = 107100	$\mu_F = 0.699$	$\mu_{CR} = 0.045$	
Diversity = 1.1806e+01	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 1.920596e+03	FES = 108100	$\mu_F = 0.749$	$\mu_{CR} = 0.055$	
Diversity = 1.1751e+01	k_t = [4, 3, 4, 3]			
Generation 1090: Best = 1.920529e+03	FES = 109100	$\mu_F = 0.801$	$\mu_{CR} = 0.055$	
Diversity = 1.1906e+01	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 1.920529e+03	FES = 110100	$\mu_F = 0.860$	$\mu_{CR} = 0.044$	
Diversity = 1.1895e+01	k_t = [2, 3, 3, 3]			
Generation 1110: Best = 1.920529e+03	FES = 111100	$\mu_F = 0.808$	$\mu_{CR} = 0.040$	
Diversity = 1.1863e+01	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 1.920529e+03	FES = 112100	$\mu_F = 0.854$	$\mu_{CR} = 0.039$	
Diversity = 1.1679e+01	k_t = [4, 2, 4, 4]			
Generation 1130: Best = 1.919905e+03	FES = 113100	$\mu_F = 0.818$	$\mu_{CR} = 0.039$	
Diversity = 1.1709e+01	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 1.919905e+03	FES = 114100	$\mu_F = 0.825$	$\mu_{CR} = 0.055$	
Diversity = 1.1766e+01	k_t = [2, 2, 3, 2]			
Generation 1150: Best = 1.919905e+03	FES = 115100	$\mu_F = 0.789$	$\mu_{CR} = 0.051$	
Diversity = 1.1802e+01	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 1.919905e+03	FES = 116100	$\mu_F = 0.710$	$\mu_{CR} = 0.056$	
Diversity = 1.1742e+01	k_t = [3, 4, 3, 4]			
Generation 1170: Best = 1.919905e+03	FES = 117100	$\mu_F = 0.810$	$\mu_{CR} = 0.048$	
Diversity = 1.1709e+01	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 1.919905e+03	FES = 118100	$\mu_F = 0.737$	$\mu_{CR} = 0.063$	
Diversity = 1.1683e+01	k_t = [2, 3, 3, 3]			
Generation 1190: Best = 1.919905e+03	FES = 119100	$\mu_F = 0.712$	$\mu_{CR} = 0.047$	
Diversity = 1.1608e+01	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 1.916801e+03	FES = 120100	$\mu_F = 0.719$	$\mu_{CR} = 0.046$	
Diversity = 1.1572e+01	k_t = [4, 2, 3, 2]			
Generation 1210: Best = 1.916801e+03	FES = 121100	$\mu_F = 0.770$	$\mu_{CR} = 0.056$	
Diversity = 1.1594e+01	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 1.916801e+03	FES = 122100	$\mu_F = 0.780$	$\mu_{CR} = 0.055$	
Diversity = 1.1601e+01	k_t = [3, 2, 2, 3]			
Generation 1230: Best = 1.916801e+03	FES = 123100	$\mu_F = 0.693$	$\mu_{CR} = 0.055$	
Diversity = 1.1680e+01	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 1.916801e+03	FES = 124100	$\mu_F = 0.636$	$\mu_{CR} = 0.056$	
Diversity = 1.1623e+01	k_t = [3, 4, 3, 4]			
Generation 1250: Best = 1.916798e+03	FES = 125100	$\mu_F = 0.602$	$\mu_{CR} = 0.053$	
Diversity = 1.1525e+01	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 1.916798e+03	FES = 126100	$\mu_F = 0.648$	$\mu_{CR} = 0.051$	
Diversity = 1.1169e+01	k_t = [3, 2, 4, 4]			
Generation 1270: Best = 1.916798e+03	FES = 127100	$\mu_F = 0.708$	$\mu_{CR} = 0.055$	
Diversity = 1.1127e+01	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 1.916798e+03	FES = 128100	$\mu_F = 0.703$	$\mu_{CR} = 0.049$	
Diversity = 1.1135e+01	k_t = [2, 2, 4, 3]			
Generation 1290: Best = 1.916798e+03	FES = 129100	$\mu_F = 0.660$	$\mu_{CR} = 0.056$	
Diversity = 1.1116e+01	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 1.916798e+03	FES = 130100	$\mu_F = 0.688$	$\mu_{CR} = 0.042$	

Diversity = 1.1120e+01 k_t = [3, 4, 3, 2]
 Generation 1310: Best = 1.916798e+03 FES = 131100 μ_F = 0.729 μ_{CR} = 0.056
 Diversity = 1.1080e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 1.916798e+03 FES = 132100 μ_F = 0.743 μ_{CR} = 0.049
 Diversity = 1.1036e+01 k_t = [4, 4, 2, 2]
 Generation 1330: Best = 1.916798e+03 FES = 133100 μ_F = 0.707 μ_{CR} = 0.054
 Diversity = 1.1069e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 1.916798e+03 FES = 134100 μ_F = 0.734 μ_{CR} = 0.048
 Diversity = 1.0933e+01 k_t = [2, 4, 3, 4]
 Generation 1350: Best = 1.916798e+03 FES = 135100 μ_F = 0.776 μ_{CR} = 0.041
 Diversity = 1.0825e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 1.916760e+03 FES = 136100 μ_F = 0.802 μ_{CR} = 0.044
 Diversity = 1.0746e+01 k_t = [2, 2, 4, 4]
 Generation 1370: Best = 1.916760e+03 FES = 137100 μ_F = 0.813 μ_{CR} = 0.049
 Diversity = 1.0697e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 1.916760e+03 FES = 138100 μ_F = 0.843 μ_{CR} = 0.046
 Diversity = 1.0519e+01 k_t = [3, 4, 4, 4]
 Generation 1390: Best = 1.916760e+03 FES = 139100 μ_F = 0.809 μ_{CR} = 0.052
 Diversity = 1.0440e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 1.916760e+03 FES = 140100 μ_F = 0.815 μ_{CR} = 0.039
 Diversity = 1.0448e+01 k_t = [2, 3, 2, 2]
 Generation 1410: Best = 1.916760e+03 FES = 141100 μ_F = 0.774 μ_{CR} = 0.043
 Diversity = 1.0353e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 1.916760e+03 FES = 142100 μ_F = 0.740 μ_{CR} = 0.049
 Diversity = 1.0328e+01 k_t = [2, 4, 4, 4]
 Generation 1430: Best = 1.916377e+03 FES = 143100 μ_F = 0.722 μ_{CR} = 0.046
 Diversity = 1.0256e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 1.916377e+03 FES = 144100 μ_F = 0.633 μ_{CR} = 0.050
 Diversity = 1.0199e+01 k_t = [3, 2, 2, 2]
 Generation 1450: Best = 1.916323e+03 FES = 145100 μ_F = 0.689 μ_{CR} = 0.052
 Diversity = 1.0173e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 1.916323e+03 FES = 146100 μ_F = 0.733 μ_{CR} = 0.056
 Diversity = 1.0179e+01 k_t = [3, 2, 3, 3]
 Generation 1470: Best = 1.916323e+03 FES = 147100 μ_F = 0.715 μ_{CR} = 0.046
 Diversity = 1.0154e+01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 1.916323e+03 FES = 148100 μ_F = 0.679 μ_{CR} = 0.048
 Diversity = 1.0104e+01 k_t = [3, 3, 4, 3]
 Generation 1490: Best = 1.916323e+03 FES = 149100 μ_F = 0.710 μ_{CR} = 0.042
 Diversity = 1.0095e+01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 1.916323e+03 FES = 150100 μ_F = 0.741 μ_{CR} = 0.041
 Diversity = 1.0065e+01 k_t = [4, 3, 3, 4]
 Generation 1510: Best = 1.916323e+03 FES = 151100 μ_F = 0.761 μ_{CR} = 0.044
 Diversity = 1.0075e+01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 1.916323e+03 FES = 152100 μ_F = 0.765 μ_{CR} = 0.052
 Diversity = 1.0014e+01 k_t = [3, 2, 2, 2]
 Generation 1530: Best = 1.916323e+03 FES = 153100 μ_F = 0.718 μ_{CR} = 0.054
 Diversity = 9.9691e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 1.916217e+03 FES = 154100 μ_F = 0.761 μ_{CR} = 0.061
 Diversity = 1.0016e+01 k_t = [4, 4, 4, 2]
 Generation 1550: Best = 1.916217e+03 FES = 155100 μ_F = 0.765 μ_{CR} = 0.047
 Diversity = 1.0044e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 1.916217e+03 FES = 156100 μ_F = 0.835 μ_{CR} = 0.047
 Diversity = 1.0049e+01 k_t = [4, 4, 3, 3]
 Generation 1570: Best = 1.916217e+03 FES = 157100 μ_F = 0.714 μ_{CR} = 0.042

Diversity = 1.0077e+01	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 1.916217e+03	FES = 158100	$\mu_F = 0.750$	$\mu_{CR} = 0.041$	
Diversity = 1.0034e+01	k_t = [3, 4, 2, 3]			
Generation 1590: Best = 1.916217e+03	FES = 159100	$\mu_F = 0.810$	$\mu_{CR} = 0.035$	
Diversity = 9.8936e+00	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 1.915827e+03	FES = 160100	$\mu_F = 0.762$	$\mu_{CR} = 0.036$	
Diversity = 9.5891e+00	k_t = [3, 2, 4, 4]			
Generation 1610: Best = 1.915808e+03	FES = 161100	$\mu_F = 0.752$	$\mu_{CR} = 0.037$	
Diversity = 9.4923e+00	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 1.915772e+03	FES = 162100	$\mu_F = 0.672$	$\mu_{CR} = 0.038$	
Diversity = 9.5293e+00	k_t = [2, 2, 4, 4]			
Generation 1630: Best = 1.915772e+03	FES = 163100	$\mu_F = 0.750$	$\mu_{CR} = 0.045$	
Diversity = 9.5111e+00	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 1.915696e+03	FES = 164100	$\mu_F = 0.809$	$\mu_{CR} = 0.043$	
Diversity = 9.2409e+00	k_t = [3, 3, 3, 4]			
Generation 1650: Best = 1.915696e+03	FES = 165100	$\mu_F = 0.740$	$\mu_{CR} = 0.045$	
Diversity = 9.2425e+00	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 1.915696e+03	FES = 166100	$\mu_F = 0.805$	$\mu_{CR} = 0.048$	
Diversity = 8.9250e+00	k_t = [2, 4, 2, 4]			
Generation 1670: Best = 1.915696e+03	FES = 167100	$\mu_F = 0.742$	$\mu_{CR} = 0.047$	
Diversity = 8.8576e+00	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 1.915678e+03	FES = 168100	$\mu_F = 0.787$	$\mu_{CR} = 0.067$	
Diversity = 8.8144e+00	k_t = [3, 2, 4, 4]			
Generation 1690: Best = 1.915678e+03	FES = 169100	$\mu_F = 0.739$	$\mu_{CR} = 0.054$	
Diversity = 8.7921e+00	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 1.915522e+03	FES = 170100	$\mu_F = 0.692$	$\mu_{CR} = 0.046$	
Diversity = 8.7481e+00	k_t = [4, 4, 2, 4]			
Generation 1710: Best = 1.915513e+03	FES = 171100	$\mu_F = 0.781$	$\mu_{CR} = 0.063$	
Diversity = 8.7655e+00	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 1.915441e+03	FES = 172100	$\mu_F = 0.797$	$\mu_{CR} = 0.052$	
Diversity = 8.8767e+00	k_t = [3, 4, 4, 2]			
Generation 1730: Best = 1.915441e+03	FES = 173100	$\mu_F = 0.781$	$\mu_{CR} = 0.049$	
Diversity = 8.8288e+00	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 1.915253e+03	FES = 174100	$\mu_F = 0.804$	$\mu_{CR} = 0.057$	
Diversity = 8.8950e+00	k_t = [3, 3, 4, 2]			
Generation 1750: Best = 1.915163e+03	FES = 175100	$\mu_F = 0.786$	$\mu_{CR} = 0.045$	
Diversity = 8.8490e+00	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 1.914933e+03	FES = 176100	$\mu_F = 0.795$	$\mu_{CR} = 0.048$	
Diversity = 8.7715e+00	k_t = [4, 3, 4, 4]			
Generation 1770: Best = 1.914933e+03	FES = 177100	$\mu_F = 0.775$	$\mu_{CR} = 0.048$	
Diversity = 8.7519e+00	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 1.914933e+03	FES = 178100	$\mu_F = 0.721$	$\mu_{CR} = 0.050$	
Diversity = 8.7465e+00	k_t = [2, 2, 3, 3]			
Generation 1790: Best = 1.914933e+03	FES = 179100	$\mu_F = 0.655$	$\mu_{CR} = 0.051$	
Diversity = 8.6881e+00	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 1.914806e+03	FES = 180100	$\mu_F = 0.661$	$\mu_{CR} = 0.042$	
Diversity = 8.6824e+00	k_t = [3, 4, 4, 2]			
Generation 1810: Best = 1.914799e+03	FES = 181100	$\mu_F = 0.683$	$\mu_{CR} = 0.038$	
Diversity = 8.6294e+00	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 1.914799e+03	FES = 182100	$\mu_F = 0.704$	$\mu_{CR} = 0.038$	
Diversity = 8.6489e+00	k_t = [2, 3, 3, 3]			
Generation 1830: Best = 1.914799e+03	FES = 183100	$\mu_F = 0.682$	$\mu_{CR} = 0.043$	
Diversity = 8.6381e+00	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 1.914799e+03	FES = 184100	$\mu_F = 0.627$	$\mu_{CR} = 0.033$	

Diversity = 8.5949e+00	k_t = [2, 4, 4, 4]			
Generation 1850: Best = 1.914799e+03	FES = 185100	$\mu_F = 0.671$	$\mu_{CR} = 0.049$	
Diversity = 8.5955e+00	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 1.914799e+03	FES = 186100	$\mu_F = 0.705$	$\mu_{CR} = 0.051$	
Diversity = 8.6125e+00	k_t = [2, 4, 4, 3]			
Generation 1870: Best = 1.914799e+03	FES = 187100	$\mu_F = 0.718$	$\mu_{CR} = 0.054$	
Diversity = 8.5520e+00	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 1.914794e+03	FES = 188100	$\mu_F = 0.720$	$\mu_{CR} = 0.055$	
Diversity = 8.4687e+00	k_t = [4, 2, 3, 4]			
Generation 1890: Best = 1.914794e+03	FES = 189100	$\mu_F = 0.652$	$\mu_{CR} = 0.045$	
Diversity = 8.4690e+00	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 1.914794e+03	FES = 190100	$\mu_F = 0.646$	$\mu_{CR} = 0.053$	
Diversity = 8.4546e+00	k_t = [4, 2, 2, 2]			
Generation 1910: Best = 1.914794e+03	FES = 191100	$\mu_F = 0.728$	$\mu_{CR} = 0.041$	
Diversity = 8.4131e+00	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 1.914701e+03	FES = 192100	$\mu_F = 0.637$	$\mu_{CR} = 0.057$	
Diversity = 8.3712e+00	k_t = [4, 4, 2, 3]			
Generation 1930: Best = 1.914701e+03	FES = 193100	$\mu_F = 0.697$	$\mu_{CR} = 0.051$	
Diversity = 8.3405e+00	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 1.914701e+03	FES = 194100	$\mu_F = 0.725$	$\mu_{CR} = 0.046$	
Diversity = 8.2680e+00	k_t = [3, 2, 4, 2]			
Generation 1950: Best = 1.914701e+03	FES = 195100	$\mu_F = 0.755$	$\mu_{CR} = 0.053$	
Diversity = 8.1783e+00	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 1.914701e+03	FES = 196100	$\mu_F = 0.740$	$\mu_{CR} = 0.042$	
Diversity = 8.1915e+00	k_t = [2, 2, 3, 2]			
Generation 1970: Best = 1.914683e+03	FES = 197100	$\mu_F = 0.729$	$\mu_{CR} = 0.043$	
Diversity = 8.1109e+00	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 1.914683e+03	FES = 198100	$\mu_F = 0.647$	$\mu_{CR} = 0.043$	
Diversity = 8.0342e+00	k_t = [2, 3, 4, 2]			
Generation 1990: Best = 1.914683e+03	FES = 199100	$\mu_F = 0.682$	$\mu_{CR} = 0.045$	
Diversity = 8.0204e+00	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 1.913717e+03	FES = 200100	$\mu_F = 0.739$	$\mu_{CR} = 0.051$	
Diversity = 8.1076e+00	k_t = [4, 2, 3, 2]			
Generation 2010: Best = 1.913717e+03	FES = 201100	$\mu_F = 0.752$	$\mu_{CR} = 0.060$	
Diversity = 8.1220e+00	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 1.913717e+03	FES = 202100	$\mu_F = 0.771$	$\mu_{CR} = 0.055$	
Diversity = 8.1123e+00	k_t = [4, 4, 2, 4]			
Generation 2030: Best = 1.913717e+03	FES = 203100	$\mu_F = 0.792$	$\mu_{CR} = 0.054$	
Diversity = 8.0751e+00	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 1.913717e+03	FES = 204100	$\mu_F = 0.823$	$\mu_{CR} = 0.060$	
Diversity = 8.1862e+00	k_t = [4, 4, 4, 4]			
Generation 2050: Best = 1.913717e+03	FES = 205100	$\mu_F = 0.824$	$\mu_{CR} = 0.057$	
Diversity = 8.1715e+00	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 1.913717e+03	FES = 206100	$\mu_F = 0.761$	$\mu_{CR} = 0.050$	
Diversity = 8.1989e+00	k_t = [3, 2, 4, 3]			
Generation 2070: Best = 1.913717e+03	FES = 207100	$\mu_F = 0.795$	$\mu_{CR} = 0.046$	
Diversity = 8.2033e+00	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 1.913447e+03	FES = 208100	$\mu_F = 0.775$	$\mu_{CR} = 0.038$	
Diversity = 8.1495e+00	k_t = [2, 4, 4, 4]			
Generation 2090: Best = 1.913447e+03	FES = 209100	$\mu_F = 0.784$	$\mu_{CR} = 0.054$	
Diversity = 8.1608e+00	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 1.913447e+03	FES = 210100	$\mu_F = 0.825$	$\mu_{CR} = 0.048$	
Diversity = 8.0390e+00	k_t = [3, 3, 2, 2]			
Generation 2110: Best = 1.913280e+03	FES = 211100	$\mu_F = 0.809$	$\mu_{CR} = 0.061$	

Diversity = 8.0687e+00	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 1.913259e+03	FES = 212100	$\mu_F = 0.762$	$\mu_{CR} = 0.056$	
Diversity = 8.0729e+00	k_t = [4, 4, 2, 3]			
Generation 2130: Best = 1.913234e+03	FES = 213100	$\mu_F = 0.740$	$\mu_{CR} = 0.056$	
Diversity = 8.1370e+00	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 1.913176e+03	FES = 214100	$\mu_F = 0.702$	$\mu_{CR} = 0.056$	
Diversity = 8.1343e+00	k_t = [2, 2, 4, 2]			
Generation 2150: Best = 1.913174e+03	FES = 215100	$\mu_F = 0.690$	$\mu_{CR} = 0.042$	
Diversity = 8.0349e+00	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 1.913174e+03	FES = 216100	$\mu_F = 0.673$	$\mu_{CR} = 0.042$	
Diversity = 8.0107e+00	k_t = [4, 3, 2, 2]			
Generation 2170: Best = 1.913174e+03	FES = 217100	$\mu_F = 0.707$	$\mu_{CR} = 0.044$	
Diversity = 8.0221e+00	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 1.913174e+03	FES = 218100	$\mu_F = 0.741$	$\mu_{CR} = 0.046$	
Diversity = 7.8940e+00	k_t = [2, 3, 4, 3]			
Generation 2190: Best = 1.913174e+03	FES = 219100	$\mu_F = 0.785$	$\mu_{CR} = 0.042$	
Diversity = 7.8153e+00	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 1.913168e+03	FES = 220100	$\mu_F = 0.780$	$\mu_{CR} = 0.046$	
Diversity = 7.8870e+00	k_t = [3, 4, 3, 3]			
Generation 2210: Best = 1.913168e+03	FES = 221100	$\mu_F = 0.680$	$\mu_{CR} = 0.043$	
Diversity = 7.8977e+00	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 1.913160e+03	FES = 222100	$\mu_F = 0.695$	$\mu_{CR} = 0.036$	
Diversity = 7.8952e+00	k_t = [2, 3, 3, 2]			
Generation 2230: Best = 1.913087e+03	FES = 223100	$\mu_F = 0.677$	$\mu_{CR} = 0.047$	
Diversity = 7.9202e+00	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 1.913087e+03	FES = 224100	$\mu_F = 0.654$	$\mu_{CR} = 0.048$	
Diversity = 8.0859e+00	k_t = [4, 2, 3, 2]			
Generation 2250: Best = 1.913087e+03	FES = 225100	$\mu_F = 0.614$	$\mu_{CR} = 0.058$	
Diversity = 8.0877e+00	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 1.913087e+03	FES = 226100	$\mu_F = 0.655$	$\mu_{CR} = 0.062$	
Diversity = 7.9557e+00	k_t = [4, 4, 3, 2]			
Generation 2270: Best = 1.913087e+03	FES = 227100	$\mu_F = 0.666$	$\mu_{CR} = 0.053$	
Diversity = 7.9689e+00	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 1.913087e+03	FES = 228100	$\mu_F = 0.601$	$\mu_{CR} = 0.049$	
Diversity = 8.0231e+00	k_t = [4, 4, 2, 2]			
Generation 2290: Best = 1.913087e+03	FES = 229100	$\mu_F = 0.607$	$\mu_{CR} = 0.053$	
Diversity = 8.0296e+00	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 1.913087e+03	FES = 230100	$\mu_F = 0.678$	$\mu_{CR} = 0.041$	
Diversity = 8.0056e+00	k_t = [4, 3, 4, 4]			
Generation 2310: Best = 1.913087e+03	FES = 231100	$\mu_F = 0.597$	$\mu_{CR} = 0.052$	
Diversity = 7.9705e+00	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 1.913087e+03	FES = 232100	$\mu_F = 0.659$	$\mu_{CR} = 0.049$	
Diversity = 7.9676e+00	k_t = [3, 3, 3, 2]			
Generation 2330: Best = 1.913087e+03	FES = 233100	$\mu_F = 0.733$	$\mu_{CR} = 0.056$	
Diversity = 7.8949e+00	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 1.913087e+03	FES = 234100	$\mu_F = 0.812$	$\mu_{CR} = 0.055$	
Diversity = 8.0343e+00	k_t = [4, 2, 3, 4]			
Generation 2350: Best = 1.913087e+03	FES = 235100	$\mu_F = 0.709$	$\mu_{CR} = 0.055$	
Diversity = 8.0145e+00	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 1.913079e+03	FES = 236100	$\mu_F = 0.726$	$\mu_{CR} = 0.065$	
Diversity = 8.0539e+00	k_t = [4, 3, 2, 4]			
Generation 2370: Best = 1.913079e+03	FES = 237100	$\mu_F = 0.674$	$\mu_{CR} = 0.055$	
Diversity = 8.0521e+00	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 1.913079e+03	FES = 238100	$\mu_F = 0.563$	$\mu_{CR} = 0.066$	

Diversity = 8.0550e+00	k_t = [2, 4, 3, 2]			
Generation 2390: Best = 1.913069e+03	FES = 239100	$\mu_F = 0.646$	$\mu_{CR} = 0.060$	
Diversity = 8.0213e+00	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 1.913048e+03	FES = 240100	$\mu_F = 0.619$	$\mu_{CR} = 0.048$	
Diversity = 8.0240e+00	k_t = [4, 3, 4, 4]			
Generation 2410: Best = 1.912364e+03	FES = 241100	$\mu_F = 0.544$	$\mu_{CR} = 0.050$	
Diversity = 7.9799e+00	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 1.912364e+03	FES = 242100	$\mu_F = 0.610$	$\mu_{CR} = 0.052$	
Diversity = 7.9785e+00	k_t = [3, 4, 3, 3]			
Generation 2430: Best = 1.912364e+03	FES = 243100	$\mu_F = 0.680$	$\mu_{CR} = 0.060$	
Diversity = 8.0470e+00	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 1.912364e+03	FES = 244100	$\mu_F = 0.734$	$\mu_{CR} = 0.067$	
Diversity = 8.0791e+00	k_t = [3, 3, 4, 2]			
Generation 2450: Best = 1.912289e+03	FES = 245100	$\mu_F = 0.702$	$\mu_{CR} = 0.054$	
Diversity = 8.0785e+00	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 1.912289e+03	FES = 246100	$\mu_F = 0.663$	$\mu_{CR} = 0.048$	
Diversity = 8.0922e+00	k_t = [3, 3, 3, 2]			
Generation 2470: Best = 1.912289e+03	FES = 247100	$\mu_F = 0.650$	$\mu_{CR} = 0.067$	
Diversity = 8.1242e+00	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 1.912289e+03	FES = 248100	$\mu_F = 0.671$	$\mu_{CR} = 0.062$	
Diversity = 8.1209e+00	k_t = [4, 3, 2, 3]			
Generation 2490: Best = 1.912289e+03	FES = 249100	$\mu_F = 0.612$	$\mu_{CR} = 0.049$	
Diversity = 8.1216e+00	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 1.912289e+03	FES = 250100	$\mu_F = 0.671$	$\mu_{CR} = 0.049$	
Diversity = 8.1295e+00	k_t = [4, 3, 3, 2]			
Generation 2510: Best = 1.912289e+03	FES = 251100	$\mu_F = 0.714$	$\mu_{CR} = 0.046$	
Diversity = 8.1126e+00	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 1.912231e+03	FES = 252100	$\mu_F = 0.704$	$\mu_{CR} = 0.044$	
Diversity = 8.1078e+00	k_t = [2, 2, 4, 2]			
Generation 2530: Best = 1.912231e+03	FES = 253100	$\mu_F = 0.683$	$\mu_{CR} = 0.038$	
Diversity = 8.0803e+00	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 1.911937e+03	FES = 254100	$\mu_F = 0.789$	$\mu_{CR} = 0.042$	
Diversity = 8.1159e+00	k_t = [4, 2, 4, 4]			
Generation 2550: Best = 1.911937e+03	FES = 255100	$\mu_F = 0.772$	$\mu_{CR} = 0.041$	
Diversity = 8.1118e+00	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 1.911937e+03	FES = 256100	$\mu_F = 0.787$	$\mu_{CR} = 0.036$	
Diversity = 8.1900e+00	k_t = [2, 2, 4, 2]			
Generation 2570: Best = 1.911900e+03	FES = 257100	$\mu_F = 0.778$	$\mu_{CR} = 0.046$	
Diversity = 8.1684e+00	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 1.911900e+03	FES = 258100	$\mu_F = 0.841$	$\mu_{CR} = 0.049$	
Diversity = 8.1330e+00	k_t = [3, 3, 3, 2]			
Generation 2590: Best = 1.911900e+03	FES = 259100	$\mu_F = 0.764$	$\mu_{CR} = 0.039$	
Diversity = 8.1642e+00	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 1.911900e+03	FES = 260100	$\mu_F = 0.720$	$\mu_{CR} = 0.041$	
Diversity = 8.1418e+00	k_t = [3, 4, 4, 2]			
Generation 2610: Best = 1.911900e+03	FES = 261100	$\mu_F = 0.685$	$\mu_{CR} = 0.036$	
Diversity = 8.1525e+00	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 1.911890e+03	FES = 262100	$\mu_F = 0.700$	$\mu_{CR} = 0.039$	
Diversity = 8.1362e+00	k_t = [3, 2, 3, 4]			
Generation 2630: Best = 1.911890e+03	FES = 263100	$\mu_F = 0.652$	$\mu_{CR} = 0.045$	
Diversity = 8.1597e+00	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 1.911890e+03	FES = 264100	$\mu_F = 0.660$	$\mu_{CR} = 0.041$	
Diversity = 8.1623e+00	k_t = [3, 4, 4, 2]			
Generation 2650: Best = 1.911619e+03	FES = 265100	$\mu_F = 0.712$	$\mu_{CR} = 0.045$	

Diversity = 8.1706e+00	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 1.911609e+03	FES = 266100	$\mu_F = 0.736$	$\mu_{CR} = 0.046$	
Diversity = 8.0689e+00	k_t = [2, 4, 3, 4]			
Generation 2670: Best = 1.911609e+03	FES = 267100	$\mu_F = 0.734$	$\mu_{CR} = 0.042$	
Diversity = 8.0716e+00	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 1.911609e+03	FES = 268100	$\mu_F = 0.759$	$\mu_{CR} = 0.044$	
Diversity = 8.0650e+00	k_t = [4, 3, 3, 4]			
Generation 2690: Best = 1.911563e+03	FES = 269100	$\mu_F = 0.780$	$\mu_{CR} = 0.044$	
Diversity = 8.1628e+00	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 1.911563e+03	FES = 270100	$\mu_F = 0.726$	$\mu_{CR} = 0.039$	
Diversity = 8.1597e+00	k_t = [3, 4, 2, 3]			
Generation 2710: Best = 1.911563e+03	FES = 271100	$\mu_F = 0.767$	$\mu_{CR} = 0.039$	
Diversity = 8.1713e+00	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 1.911538e+03	FES = 272100	$\mu_F = 0.806$	$\mu_{CR} = 0.040$	
Diversity = 8.1580e+00	k_t = [4, 4, 2, 2]			
Generation 2730: Best = 1.911538e+03	FES = 273100	$\mu_F = 0.791$	$\mu_{CR} = 0.041$	
Diversity = 8.1946e+00	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 1.911536e+03	FES = 274100	$\mu_F = 0.745$	$\mu_{CR} = 0.047$	
Diversity = 8.2001e+00	k_t = [3, 4, 3, 2]			
Generation 2750: Best = 1.911536e+03	FES = 275100	$\mu_F = 0.774$	$\mu_{CR} = 0.041$	
Diversity = 8.2037e+00	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 1.911536e+03	FES = 276100	$\mu_F = 0.759$	$\mu_{CR} = 0.039$	
Diversity = 8.2273e+00	k_t = [4, 3, 2, 2]			
Generation 2770: Best = 1.911532e+03	FES = 277100	$\mu_F = 0.781$	$\mu_{CR} = 0.044$	
Diversity = 8.2784e+00	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 1.911532e+03	FES = 278100	$\mu_F = 0.793$	$\mu_{CR} = 0.053$	
Diversity = 8.2653e+00	k_t = [4, 2, 3, 2]			
Generation 2790: Best = 1.911532e+03	FES = 279100	$\mu_F = 0.709$	$\mu_{CR} = 0.049$	
Diversity = 8.2654e+00	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 1.911510e+03	FES = 280100	$\mu_F = 0.704$	$\mu_{CR} = 0.048$	
Diversity = 8.2710e+00	k_t = [4, 4, 4, 2]			
Generation 2810: Best = 1.911510e+03	FES = 281100	$\mu_F = 0.714$	$\mu_{CR} = 0.043$	
Diversity = 8.2776e+00	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 1.911510e+03	FES = 282100	$\mu_F = 0.728$	$\mu_{CR} = 0.042$	
Diversity = 8.2563e+00	k_t = [4, 4, 2, 3]			
Generation 2830: Best = 1.911510e+03	FES = 283100	$\mu_F = 0.735$	$\mu_{CR} = 0.035$	
Diversity = 8.2270e+00	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 1.911510e+03	FES = 284100	$\mu_F = 0.712$	$\mu_{CR} = 0.029$	
Diversity = 8.2457e+00	k_t = [2, 3, 4, 2]			
Generation 2850: Best = 1.911510e+03	FES = 285100	$\mu_F = 0.719$	$\mu_{CR} = 0.026$	
Diversity = 8.2387e+00	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 1.911510e+03	FES = 286100	$\mu_F = 0.744$	$\mu_{CR} = 0.044$	
Diversity = 8.2125e+00	k_t = [4, 3, 4, 2]			
Generation 2870: Best = 1.911492e+03	FES = 287100	$\mu_F = 0.745$	$\mu_{CR} = 0.051$	
Diversity = 8.2129e+00	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 1.911492e+03	FES = 288100	$\mu_F = 0.752$	$\mu_{CR} = 0.048$	
Diversity = 8.2517e+00	k_t = [2, 4, 2, 4]			
Generation 2890: Best = 1.911478e+03	FES = 289100	$\mu_F = 0.797$	$\mu_{CR} = 0.038$	
Diversity = 8.2575e+00	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 1.911478e+03	FES = 290100	$\mu_F = 0.813$	$\mu_{CR} = 0.054$	
Diversity = 8.1196e+00	k_t = [4, 4, 4, 4]			
Generation 2910: Best = 1.911478e+03	FES = 291100	$\mu_F = 0.797$	$\mu_{CR} = 0.058$	
Diversity = 8.0955e+00	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 1.911478e+03	FES = 292100	$\mu_F = 0.817$	$\mu_{CR} = 0.058$	

```

Diversity = 8.1267e+00 k_t = [2, 2, 3, 3]
Generation 2930: Best = 1.911477e+03 FES = 293100  $\mu_F$  = 0.855  $\mu_{CR}$  = 0.058
Diversity = 7.9603e+00 k_t = [1, 1, 1, 1]
Generation 2940: Best = 1.911477e+03 FES = 294100  $\mu_F$  = 0.810  $\mu_{CR}$  = 0.049
Diversity = 7.9608e+00 k_t = [3, 3, 4, 3]
Generation 2950: Best = 1.911415e+03 FES = 295100  $\mu_F$  = 0.799  $\mu_{CR}$  = 0.052
Diversity = 7.8997e+00 k_t = [1, 1, 1, 1]
Generation 2960: Best = 1.911365e+03 FES = 296100  $\mu_F$  = 0.835  $\mu_{CR}$  = 0.041
Diversity = 7.5389e+00 k_t = [2, 4, 4, 4]
Generation 2970: Best = 1.911364e+03 FES = 297100  $\mu_F$  = 0.842  $\mu_{CR}$  = 0.044
Diversity = 7.2781e+00 k_t = [1, 1, 1, 1]
Generation 2980: Best = 1.911364e+03 FES = 298100  $\mu_F$  = 0.865  $\mu_{CR}$  = 0.045
Diversity = 7.1897e+00 k_t = [3, 3, 3, 3]
Generation 2990: Best = 1.911364e+03 FES = 299100  $\mu_F$  = 0.860  $\mu_{CR}$  = 0.035
Diversity = 7.0664e+00 k_t = [1, 1, 1, 1]

```

Optimization complete.

```

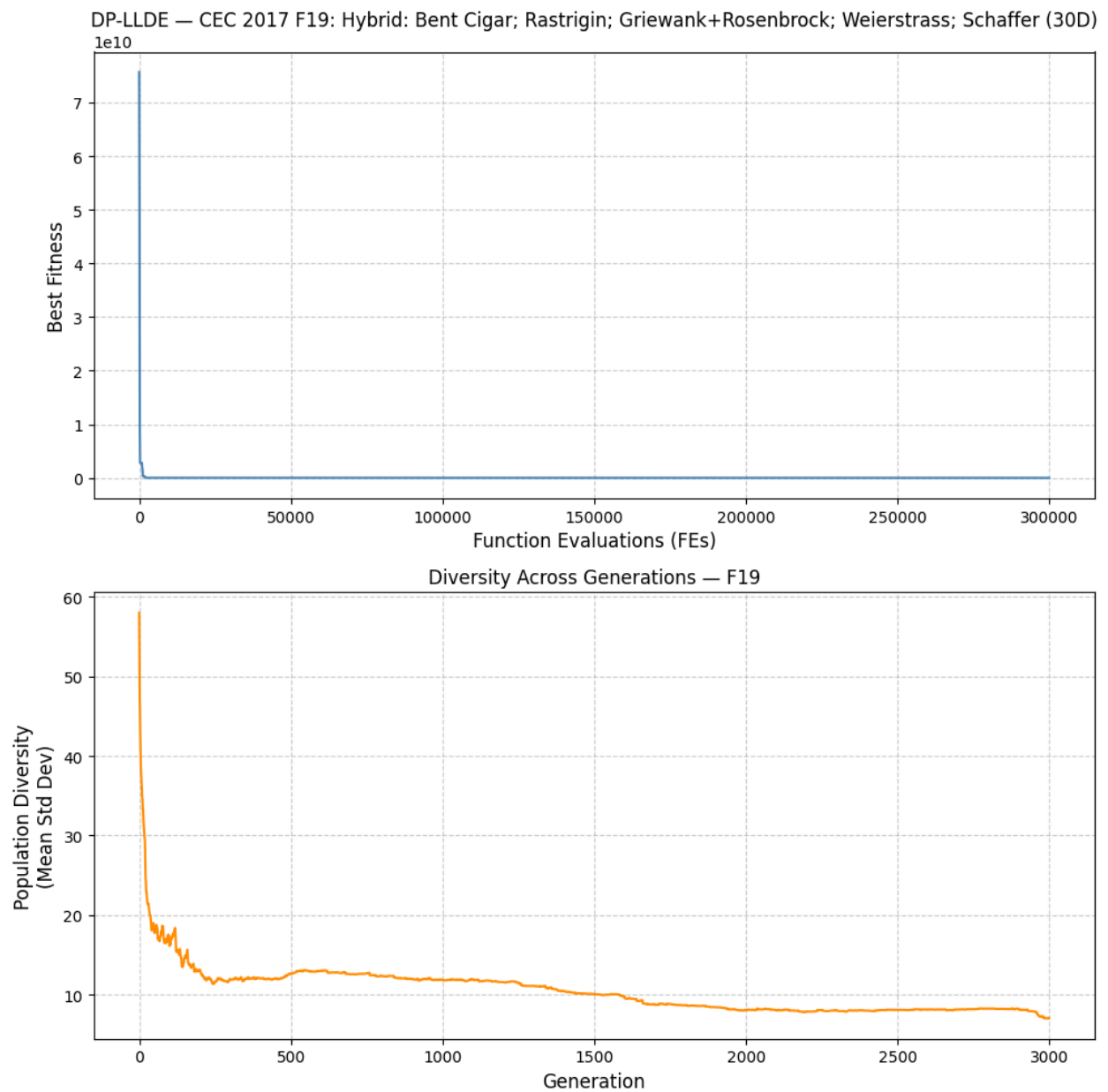
Best Fitness           = 1.911364e+03
Total Function Evals   = 300000
Total Successful Updates = 14312

```

```

=====
=
Results:
  Best Fitness           : 1.911364e+03
  Global Optimum         : 1.900000e+03
  Error (|best-optimum|) : 1.136359e+01
  Best Solution (5 dim)  : [-12.64554111 -21.46584548 -17.51068235 -50.657953
-5.77545232]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F19.png

```
In [ ]: # =====
# F20 — Hybrid Function 10 (HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; S
# =====
run_and_plot('F20', make_wrapper(hybrid.f20), 2000.0, 'Hybrid: HappyCat; Katsu
```


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DP-LLDE on CEC 2017 F20 – Single Run

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Function : Hybrid: HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer (30D)

Bounds : [-100, 100]

Global optimum : 2000.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 3.694893e+03

Generation 10: Best = 3.161492e+03 FES = 1100 $\mu_F = 0.587$ $\mu_{CR} = 0.324$

Diversity = 4.5234e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 3.150299e+03 FES = 2100 $\mu_F = 0.607$ $\mu_{CR} = 0.300$

Diversity = 4.2083e+01 $k_t = [4, 3, 4, 3]$

Generation 30: Best = 2.767477e+03 FES = 3100 $\mu_F = 0.667$ $\mu_{CR} = 0.244$

Diversity = 4.1052e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 2.767477e+03 FES = 4100 $\mu_F = 0.706$ $\mu_{CR} = 0.175$

Diversity = 4.0735e+01 $k_t = [3, 3, 4, 4]$

Generation 50: Best = 2.767477e+03 FES = 5100 $\mu_F = 0.719$ $\mu_{CR} = 0.139$

Diversity = 3.9567e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 2.765930e+03 FES = 6100 $\mu_F = 0.738$ $\mu_{CR} = 0.130$

Diversity = 4.0259e+01 $k_t = [3, 2, 4, 4]$

Generation 70: Best = 2.755781e+03 FES = 7100 $\mu_F = 0.740$ $\mu_{CR} = 0.106$

Diversity = 3.8932e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 2.754036e+03 FES = 8100 $\mu_F = 0.777$ $\mu_{CR} = 0.088$

Diversity = 3.8929e+01 $k_t = [3, 4, 3, 4]$

Generation 90: Best = 2.753105e+03 FES = 9100 $\mu_F = 0.814$ $\mu_{CR} = 0.091$

Diversity = 3.8769e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 2.564207e+03 FES = 10100 $\mu_F = 0.798$ $\mu_{CR} = 0.088$

Diversity = 3.7305e+01 $k_t = [4, 3, 3, 4]$

Generation 110: Best = 2.550755e+03 FES = 11100 $\mu_F = 0.774$ $\mu_{CR} = 0.085$

Diversity = 3.7094e+01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 2.538010e+03 FES = 12100 $\mu_F = 0.805$ $\mu_{CR} = 0.085$

Diversity = 3.6512e+01 $k_t = [2, 4, 2, 3]$

Generation 130: Best = 2.504883e+03 FES = 13100 $\mu_F = 0.812$ $\mu_{CR} = 0.067$

Diversity = 3.6651e+01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 2.504883e+03 FES = 14100 $\mu_F = 0.827$ $\mu_{CR} = 0.068$

Diversity = 3.6658e+01 $k_t = [3, 3, 4, 3]$

Generation 150: Best = 2.504354e+03 FES = 15100 $\mu_F = 0.842$ $\mu_{CR} = 0.077$

Diversity = 3.6569e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 2.504354e+03 FES = 16100 $\mu_F = 0.859$ $\mu_{CR} = 0.081$

Diversity = 3.6222e+01 $k_t = [2, 4, 4, 3]$

Generation 170: Best = 2.504347e+03 FES = 17100 $\mu_F = 0.854$ $\mu_{CR} = 0.075$

Diversity = 3.6365e+01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 2.468920e+03 FES = 18100 $\mu_F = 0.851$ $\mu_{CR} = 0.073$

Diversity = 3.5905e+01 $k_t = [2, 3, 2, 4]$

Generation 190: Best = 2.453948e+03 FES = 19100 $\mu_F = 0.847$ $\mu_{CR} = 0.076$

Diversity = 3.5927e+01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 2.422749e+03 FES = 20100 $\mu_F = 0.856$ $\mu_{CR} = 0.072$

Diversity = 3.5936e+01 $k_t = [3, 4, 2, 3]$

Generation 210: Best = 2.413150e+03 FES = 21100 $\mu_F = 0.863$ $\mu_{CR} = 0.073$

Diversity = 3.5647e+01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 2.407106e+03 FES = 22100 $\mu_F = 0.864$ $\mu_{CR} = 0.083$

Diversity = 3.5521e+01	k_t = [2, 2, 3, 2]				
Generation 230: Best = 2.301304e+03	FES = 23100	$\mu_F = 0.850$	$\mu_{CR} = 0.072$		
Diversity = 3.5707e+01	k_t = [1, 1, 1, 1]				
Generation 240: Best = 2.301304e+03	FES = 24100	$\mu_F = 0.848$	$\mu_{CR} = 0.070$		
Diversity = 3.5271e+01	k_t = [3, 3, 4, 4]				
Generation 250: Best = 2.268411e+03	FES = 25100	$\mu_F = 0.841$	$\mu_{CR} = 0.070$		
Diversity = 3.5017e+01	k_t = [1, 1, 1, 1]				
Generation 260: Best = 2.247732e+03	FES = 26100	$\mu_F = 0.835$	$\mu_{CR} = 0.067$		
Diversity = 3.4345e+01	k_t = [2, 3, 4, 4]				
Generation 270: Best = 2.247732e+03	FES = 27100	$\mu_F = 0.827$	$\mu_{CR} = 0.063$		
Diversity = 3.3810e+01	k_t = [1, 1, 1, 1]				
Generation 280: Best = 2.247569e+03	FES = 28100	$\mu_F = 0.844$	$\mu_{CR} = 0.081$		
Diversity = 3.4045e+01	k_t = [4, 3, 4, 3]				
Generation 290: Best = 2.247569e+03	FES = 29100	$\mu_F = 0.808$	$\mu_{CR} = 0.069$		
Diversity = 3.3759e+01	k_t = [1, 1, 1, 1]				
Generation 300: Best = 2.246621e+03	FES = 30100	$\mu_F = 0.849$	$\mu_{CR} = 0.076$		
Diversity = 3.3259e+01	k_t = [3, 3, 2, 4]				
Generation 310: Best = 2.246621e+03	FES = 31100	$\mu_F = 0.842$	$\mu_{CR} = 0.073$		
Diversity = 3.3224e+01	k_t = [1, 1, 1, 1]				
Generation 320: Best = 2.194646e+03	FES = 32100	$\mu_F = 0.877$	$\mu_{CR} = 0.066$		
Diversity = 3.3031e+01	k_t = [4, 4, 4, 3]				
Generation 330: Best = 2.194460e+03	FES = 33100	$\mu_F = 0.879$	$\mu_{CR} = 0.071$		
Diversity = 3.2887e+01	k_t = [1, 1, 1, 1]				
Generation 340: Best = 2.194460e+03	FES = 34100	$\mu_F = 0.901$	$\mu_{CR} = 0.064$		
Diversity = 3.2686e+01	k_t = [2, 4, 3, 2]				
Generation 350: Best = 2.193838e+03	FES = 35100	$\mu_F = 0.875$	$\mu_{CR} = 0.070$		
Diversity = 3.2629e+01	k_t = [1, 1, 1, 1]				
Generation 360: Best = 2.173466e+03	FES = 36100	$\mu_F = 0.879$	$\mu_{CR} = 0.070$		
Diversity = 3.2559e+01	k_t = [2, 3, 3, 2]				
Generation 370: Best = 2.173382e+03	FES = 37100	$\mu_F = 0.837$	$\mu_{CR} = 0.063$		
Diversity = 3.2581e+01	k_t = [1, 1, 1, 1]				
Generation 380: Best = 2.162205e+03	FES = 38100	$\mu_F = 0.855$	$\mu_{CR} = 0.062$		
Diversity = 3.2854e+01	k_t = [3, 4, 3, 2]				
Generation 390: Best = 2.162205e+03	FES = 39100	$\mu_F = 0.836$	$\mu_{CR} = 0.072$		
Diversity = 3.2868e+01	k_t = [1, 1, 1, 1]				
Generation 400: Best = 2.162205e+03	FES = 40100	$\mu_F = 0.856$	$\mu_{CR} = 0.073$		
Diversity = 3.2892e+01	k_t = [2, 2, 3, 2]				
Generation 410: Best = 2.162205e+03	FES = 41100	$\mu_F = 0.836$	$\mu_{CR} = 0.075$		
Diversity = 3.2782e+01	k_t = [1, 1, 1, 1]				
Generation 420: Best = 2.161496e+03	FES = 42100	$\mu_F = 0.860$	$\mu_{CR} = 0.071$		
Diversity = 3.2782e+01	k_t = [2, 3, 4, 2]				
Generation 430: Best = 2.161496e+03	FES = 43100	$\mu_F = 0.866$	$\mu_{CR} = 0.080$		
Diversity = 3.2865e+01	k_t = [1, 1, 1, 1]				
Generation 440: Best = 2.161496e+03	FES = 44100	$\mu_F = 0.880$	$\mu_{CR} = 0.065$		
Diversity = 3.2845e+01	k_t = [3, 2, 3, 3]				
Generation 450: Best = 2.158929e+03	FES = 45100	$\mu_F = 0.890$	$\mu_{CR} = 0.057$		
Diversity = 3.2800e+01	k_t = [1, 1, 1, 1]				
Generation 460: Best = 2.128467e+03	FES = 46100	$\mu_F = 0.890$	$\mu_{CR} = 0.059$		
Diversity = 3.2322e+01	k_t = [3, 3, 4, 4]				
Generation 470: Best = 2.128467e+03	FES = 47100	$\mu_F = 0.898$	$\mu_{CR} = 0.064$		
Diversity = 3.2312e+01	k_t = [1, 1, 1, 1]				
Generation 480: Best = 2.119940e+03	FES = 48100	$\mu_F = 0.902$	$\mu_{CR} = 0.062$		
Diversity = 3.1548e+01	k_t = [4, 2, 2, 4]				
Generation 490: Best = 2.119940e+03	FES = 49100	$\mu_F = 0.898$	$\mu_{CR} = 0.070$		

Diversity = 3.1354e+01	k_t = [1, 1, 1, 1]					
Generation 500: Best = 2.119940e+03	FES = 50100	$\mu_F = 0.904$	$\mu_{CR} = 0.071$			
Diversity = 3.1149e+01	k_t = [4, 3, 2, 2]					
Generation 510: Best = 2.119578e+03	FES = 51100	$\mu_F = 0.873$	$\mu_{CR} = 0.072$			
Diversity = 3.1512e+01	k_t = [1, 1, 1, 1]					
Generation 520: Best = 2.118765e+03	FES = 52100	$\mu_F = 0.882$	$\mu_{CR} = 0.077$			
Diversity = 3.0813e+01	k_t = [2, 2, 2, 4]					
Generation 530: Best = 2.118765e+03	FES = 53100	$\mu_F = 0.883$	$\mu_{CR} = 0.057$			
Diversity = 3.0779e+01	k_t = [1, 1, 1, 1]					
Generation 540: Best = 2.115875e+03	FES = 54100	$\mu_F = 0.829$	$\mu_{CR} = 0.066$			
Diversity = 3.0806e+01	k_t = [2, 3, 4, 2]					
Generation 550: Best = 2.115875e+03	FES = 55100	$\mu_F = 0.844$	$\mu_{CR} = 0.065$			
Diversity = 3.0904e+01	k_t = [1, 1, 1, 1]					
Generation 560: Best = 2.115875e+03	FES = 56100	$\mu_F = 0.849$	$\mu_{CR} = 0.051$			
Diversity = 3.1093e+01	k_t = [3, 2, 2, 3]					
Generation 570: Best = 2.107360e+03	FES = 57100	$\mu_F = 0.874$	$\mu_{CR} = 0.057$			
Diversity = 3.1300e+01	k_t = [1, 1, 1, 1]					
Generation 580: Best = 2.106620e+03	FES = 58100	$\mu_F = 0.840$	$\mu_{CR} = 0.059$			
Diversity = 3.0994e+01	k_t = [4, 4, 4, 3]					
Generation 590: Best = 2.106620e+03	FES = 59100	$\mu_F = 0.836$	$\mu_{CR} = 0.050$			
Diversity = 3.0808e+01	k_t = [1, 1, 1, 1]					
Generation 600: Best = 2.106620e+03	FES = 60100	$\mu_F = 0.840$	$\mu_{CR} = 0.062$			
Diversity = 3.0437e+01	k_t = [3, 4, 4, 3]					
Generation 610: Best = 2.102159e+03	FES = 61100	$\mu_F = 0.853$	$\mu_{CR} = 0.078$			
Diversity = 3.0298e+01	k_t = [1, 1, 1, 1]					
Generation 620: Best = 2.102159e+03	FES = 62100	$\mu_F = 0.863$	$\mu_{CR} = 0.070$			
Diversity = 3.0110e+01	k_t = [3, 2, 2, 2]					
Generation 630: Best = 2.102135e+03	FES = 63100	$\mu_F = 0.872$	$\mu_{CR} = 0.066$			
Diversity = 2.9980e+01	k_t = [1, 1, 1, 1]					
Generation 640: Best = 2.102061e+03	FES = 64100	$\mu_F = 0.903$	$\mu_{CR} = 0.063$			
Diversity = 2.9716e+01	k_t = [3, 4, 4, 4]					
Generation 650: Best = 2.102061e+03	FES = 65100	$\mu_F = 0.923$	$\mu_{CR} = 0.067$			
Diversity = 2.9480e+01	k_t = [1, 1, 1, 1]					
Generation 660: Best = 2.102061e+03	FES = 66100	$\mu_F = 0.894$	$\mu_{CR} = 0.061$			
Diversity = 2.9769e+01	k_t = [4, 3, 4, 3]					
Generation 670: Best = 2.101237e+03	FES = 67100	$\mu_F = 0.890$	$\mu_{CR} = 0.059$			
Diversity = 3.0592e+01	k_t = [1, 1, 1, 1]					
Generation 680: Best = 2.086240e+03	FES = 68100	$\mu_F = 0.884$	$\mu_{CR} = 0.061$			
Diversity = 3.0860e+01	k_t = [4, 3, 2, 3]					
Generation 690: Best = 2.085769e+03	FES = 69100	$\mu_F = 0.889$	$\mu_{CR} = 0.050$			
Diversity = 3.0728e+01	k_t = [1, 1, 1, 1]					
Generation 700: Best = 2.085769e+03	FES = 70100	$\mu_F = 0.872$	$\mu_{CR} = 0.063$			
Diversity = 3.0495e+01	k_t = [3, 4, 4, 2]					
Generation 710: Best = 2.085095e+03	FES = 71100	$\mu_F = 0.849$	$\mu_{CR} = 0.064$			
Diversity = 3.0368e+01	k_t = [1, 1, 1, 1]					
Generation 720: Best = 2.079618e+03	FES = 72100	$\mu_F = 0.848$	$\mu_{CR} = 0.060$			
Diversity = 3.0119e+01	k_t = [2, 2, 3, 4]					
Generation 730: Best = 2.074843e+03	FES = 73100	$\mu_F = 0.860$	$\mu_{CR} = 0.064$			
Diversity = 2.9864e+01	k_t = [1, 1, 1, 1]					
Generation 740: Best = 2.074566e+03	FES = 74100	$\mu_F = 0.863$	$\mu_{CR} = 0.073$			
Diversity = 2.9820e+01	k_t = [2, 2, 3, 4]					
Generation 750: Best = 2.073672e+03	FES = 75100	$\mu_F = 0.870$	$\mu_{CR} = 0.069$			
Diversity = 3.0202e+01	k_t = [1, 1, 1, 1]					
Generation 760: Best = 2.073672e+03	FES = 76100	$\mu_F = 0.888$	$\mu_{CR} = 0.071$			

Diversity = 2.9830e+01	k_t = [2, 4, 3, 3]				
Generation 770: Best = 2.073672e+03	FES = 77100	$\mu_F = 0.906$	$\mu_{CR} = 0.076$		
Diversity = 2.9566e+01	k_t = [1, 1, 1, 1]				
Generation 780: Best = 2.073672e+03	FES = 78100	$\mu_F = 0.878$	$\mu_{CR} = 0.080$		
Diversity = 2.9280e+01	k_t = [4, 4, 2, 2]				
Generation 790: Best = 2.071782e+03	FES = 79100	$\mu_F = 0.884$	$\mu_{CR} = 0.076$		
Diversity = 2.8744e+01	k_t = [1, 1, 1, 1]				
Generation 800: Best = 2.071782e+03	FES = 80100	$\mu_F = 0.895$	$\mu_{CR} = 0.087$		
Diversity = 2.8389e+01	k_t = [4, 4, 2, 3]				
Generation 810: Best = 2.071256e+03	FES = 81100	$\mu_F = 0.877$	$\mu_{CR} = 0.081$		
Diversity = 2.6746e+01	k_t = [1, 1, 1, 1]				
Generation 820: Best = 2.069146e+03	FES = 82100	$\mu_F = 0.894$	$\mu_{CR} = 0.104$		
Diversity = 2.4860e+01	k_t = [2, 4, 4, 4]				
Generation 830: Best = 2.058750e+03	FES = 83100	$\mu_F = 0.889$	$\mu_{CR} = 0.094$		
Diversity = 2.4110e+01	k_t = [1, 1, 1, 1]				
Generation 840: Best = 2.058750e+03	FES = 84100	$\mu_F = 0.867$	$\mu_{CR} = 0.085$		
Diversity = 2.3170e+01	k_t = [3, 3, 3, 2]				
Generation 850: Best = 2.058197e+03	FES = 85100	$\mu_F = 0.860$	$\mu_{CR} = 0.082$		
Diversity = 2.3193e+01	k_t = [1, 1, 1, 1]				
Generation 860: Best = 2.058197e+03	FES = 86100	$\mu_F = 0.843$	$\mu_{CR} = 0.081$		
Diversity = 2.2293e+01	k_t = [2, 4, 3, 2]				
Generation 870: Best = 2.057062e+03	FES = 87100	$\mu_F = 0.822$	$\mu_{CR} = 0.086$		
Diversity = 2.1995e+01	k_t = [1, 1, 1, 1]				
Generation 880: Best = 2.055849e+03	FES = 88100	$\mu_F = 0.817$	$\mu_{CR} = 0.084$		
Diversity = 2.1473e+01	k_t = [3, 2, 2, 3]				
Generation 890: Best = 2.055705e+03	FES = 89100	$\mu_F = 0.799$	$\mu_{CR} = 0.072$		
Diversity = 2.1379e+01	k_t = [1, 1, 1, 1]				
Generation 900: Best = 2.050213e+03	FES = 90100	$\mu_F = 0.795$	$\mu_{CR} = 0.067$		
Diversity = 2.1184e+01	k_t = [2, 4, 4, 3]				
Generation 910: Best = 2.048200e+03	FES = 91100	$\mu_F = 0.815$	$\mu_{CR} = 0.074$		
Diversity = 2.1214e+01	k_t = [1, 1, 1, 1]				
Generation 920: Best = 2.046921e+03	FES = 92100	$\mu_F = 0.835$	$\mu_{CR} = 0.075$		
Diversity = 2.0885e+01	k_t = [4, 2, 2, 4]				
Generation 930: Best = 2.045569e+03	FES = 93100	$\mu_F = 0.827$	$\mu_{CR} = 0.080$		
Diversity = 2.0812e+01	k_t = [1, 1, 1, 1]				
Generation 940: Best = 2.045569e+03	FES = 94100	$\mu_F = 0.830$	$\mu_{CR} = 0.078$		
Diversity = 2.0785e+01	k_t = [3, 4, 3, 4]				
Generation 950: Best = 2.045569e+03	FES = 95100	$\mu_F = 0.849$	$\mu_{CR} = 0.077$		
Diversity = 2.0638e+01	k_t = [1, 1, 1, 1]				
Generation 960: Best = 2.045569e+03	FES = 96100	$\mu_F = 0.865$	$\mu_{CR} = 0.075$		
Diversity = 2.0507e+01	k_t = [3, 3, 3, 3]				
Generation 970: Best = 2.045569e+03	FES = 97100	$\mu_F = 0.854$	$\mu_{CR} = 0.069$		
Diversity = 2.0219e+01	k_t = [1, 1, 1, 1]				
Generation 980: Best = 2.044525e+03	FES = 98100	$\mu_F = 0.850$	$\mu_{CR} = 0.065$		
Diversity = 2.0216e+01	k_t = [2, 2, 2, 3]				
Generation 990: Best = 2.044525e+03	FES = 99100	$\mu_F = 0.836$	$\mu_{CR} = 0.055$		
Diversity = 2.0237e+01	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 2.044525e+03	FES = 100100	$\mu_F = 0.821$	$\mu_{CR} = 0.066$		
Diversity = 2.0140e+01	k_t = [3, 2, 2, 2]				
Generation 1010: Best = 2.044525e+03	FES = 101100	$\mu_F = 0.787$	$\mu_{CR} = 0.068$		
Diversity = 1.9932e+01	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 2.044323e+03	FES = 102100	$\mu_F = 0.795$	$\mu_{CR} = 0.066$		
Diversity = 1.9610e+01	k_t = [2, 4, 4, 4]				
Generation 1030: Best = 2.043713e+03	FES = 103100	$\mu_F = 0.773$	$\mu_{CR} = 0.051$		

Diversity = 1.9275e+01	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 2.043713e+03	FES = 104100	$\mu_F = 0.762$	$\mu_{CR} = 0.068$	
Diversity = 1.9276e+01	k_t = [2, 4, 2, 4]			
Generation 1050: Best = 2.043713e+03	FES = 105100	$\mu_F = 0.780$	$\mu_{CR} = 0.074$	
Diversity = 1.9032e+01	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 2.043713e+03	FES = 106100	$\mu_F = 0.763$	$\mu_{CR} = 0.063$	
Diversity = 1.8953e+01	k_t = [2, 2, 4, 3]			
Generation 1070: Best = 2.043416e+03	FES = 107100	$\mu_F = 0.781$	$\mu_{CR} = 0.052$	
Diversity = 1.8749e+01	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 2.042185e+03	FES = 108100	$\mu_F = 0.789$	$\mu_{CR} = 0.056$	
Diversity = 1.8765e+01	k_t = [2, 2, 2, 3]			
Generation 1090: Best = 2.041825e+03	FES = 109100	$\mu_F = 0.792$	$\mu_{CR} = 0.066$	
Diversity = 1.8792e+01	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 2.041770e+03	FES = 110100	$\mu_F = 0.774$	$\mu_{CR} = 0.062$	
Diversity = 1.8804e+01	k_t = [2, 2, 4, 3]			
Generation 1110: Best = 2.041770e+03	FES = 111100	$\mu_F = 0.816$	$\mu_{CR} = 0.060$	
Diversity = 1.8879e+01	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 2.038079e+03	FES = 112100	$\mu_F = 0.829$	$\mu_{CR} = 0.063$	
Diversity = 1.8762e+01	k_t = [3, 4, 4, 2]			
Generation 1130: Best = 2.038079e+03	FES = 113100	$\mu_F = 0.751$	$\mu_{CR} = 0.059$	
Diversity = 1.8555e+01	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 2.038079e+03	FES = 114100	$\mu_F = 0.679$	$\mu_{CR} = 0.067$	
Diversity = 1.8379e+01	k_t = [3, 3, 3, 3]			
Generation 1150: Best = 2.037925e+03	FES = 115100	$\mu_F = 0.684$	$\mu_{CR} = 0.075$	
Diversity = 1.8349e+01	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 2.037925e+03	FES = 116100	$\mu_F = 0.708$	$\mu_{CR} = 0.060$	
Diversity = 1.8281e+01	k_t = [4, 2, 4, 4]			
Generation 1170: Best = 2.037925e+03	FES = 117100	$\mu_F = 0.750$	$\mu_{CR} = 0.067$	
Diversity = 1.8195e+01	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 2.037925e+03	FES = 118100	$\mu_F = 0.760$	$\mu_{CR} = 0.062$	
Diversity = 1.8154e+01	k_t = [2, 4, 4, 3]			
Generation 1190: Best = 2.037925e+03	FES = 119100	$\mu_F = 0.736$	$\mu_{CR} = 0.053$	
Diversity = 1.8006e+01	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 2.037872e+03	FES = 120100	$\mu_F = 0.766$	$\mu_{CR} = 0.063$	
Diversity = 1.7998e+01	k_t = [2, 4, 2, 2]			
Generation 1210: Best = 2.037699e+03	FES = 121100	$\mu_F = 0.714$	$\mu_{CR} = 0.054$	
Diversity = 1.7949e+01	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 2.037503e+03	FES = 122100	$\mu_F = 0.745$	$\mu_{CR} = 0.049$	
Diversity = 1.7654e+01	k_t = [3, 4, 2, 4]			
Generation 1230: Best = 2.037503e+03	FES = 123100	$\mu_F = 0.754$	$\mu_{CR} = 0.052$	
Diversity = 1.7463e+01	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 2.037503e+03	FES = 124100	$\mu_F = 0.779$	$\mu_{CR} = 0.059$	
Diversity = 1.7431e+01	k_t = [3, 4, 3, 3]			
Generation 1250: Best = 2.037503e+03	FES = 125100	$\mu_F = 0.750$	$\mu_{CR} = 0.065$	
Diversity = 1.7370e+01	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 2.037503e+03	FES = 126100	$\mu_F = 0.779$	$\mu_{CR} = 0.063$	
Diversity = 1.7358e+01	k_t = [4, 2, 3, 4]			
Generation 1270: Best = 2.037439e+03	FES = 127100	$\mu_F = 0.785$	$\mu_{CR} = 0.061$	
Diversity = 1.7216e+01	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 2.037439e+03	FES = 128100	$\mu_F = 0.821$	$\mu_{CR} = 0.057$	
Diversity = 1.7280e+01	k_t = [3, 4, 4, 2]			
Generation 1290: Best = 2.036569e+03	FES = 129100	$\mu_F = 0.827$	$\mu_{CR} = 0.051$	
Diversity = 1.7191e+01	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 2.036548e+03	FES = 130100	$\mu_F = 0.862$	$\mu_{CR} = 0.050$	

Diversity = 1.7286e+01	k_t = [4, 2, 2, 3]			
Generation 1310: Best = 2.036269e+03	FES = 131100	$\mu_F = 0.802$	$\mu_{CR} = 0.057$	
Diversity = 1.7273e+01	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 2.036225e+03	FES = 132100	$\mu_F = 0.785$	$\mu_{CR} = 0.063$	
Diversity = 1.7283e+01	k_t = [3, 3, 4, 4]			
Generation 1330: Best = 2.036225e+03	FES = 133100	$\mu_F = 0.741$	$\mu_{CR} = 0.053$	
Diversity = 1.7272e+01	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 2.036035e+03	FES = 134100	$\mu_F = 0.687$	$\mu_{CR} = 0.048$	
Diversity = 1.7280e+01	k_t = [2, 4, 3, 3]			
Generation 1350: Best = 2.035902e+03	FES = 135100	$\mu_F = 0.676$	$\mu_{CR} = 0.066$	
Diversity = 1.7234e+01	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 2.035765e+03	FES = 136100	$\mu_F = 0.699$	$\mu_{CR} = 0.070$	
Diversity = 1.7235e+01	k_t = [3, 2, 2, 2]			
Generation 1370: Best = 2.035765e+03	FES = 137100	$\mu_F = 0.750$	$\mu_{CR} = 0.068$	
Diversity = 1.7226e+01	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 2.034924e+03	FES = 138100	$\mu_F = 0.784$	$\mu_{CR} = 0.060$	
Diversity = 1.7144e+01	k_t = [2, 4, 4, 2]			
Generation 1390: Best = 2.034924e+03	FES = 139100	$\mu_F = 0.809$	$\mu_{CR} = 0.063$	
Diversity = 1.7080e+01	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 2.034556e+03	FES = 140100	$\mu_F = 0.799$	$\mu_{CR} = 0.050$	
Diversity = 1.7015e+01	k_t = [3, 2, 4, 4]			
Generation 1410: Best = 2.034528e+03	FES = 141100	$\mu_F = 0.797$	$\mu_{CR} = 0.039$	
Diversity = 1.6976e+01	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 2.034485e+03	FES = 142100	$\mu_F = 0.774$	$\mu_{CR} = 0.051$	
Diversity = 1.6926e+01	k_t = [2, 4, 3, 2]			
Generation 1430: Best = 2.034297e+03	FES = 143100	$\mu_F = 0.794$	$\mu_{CR} = 0.055$	
Diversity = 1.6933e+01	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 2.033839e+03	FES = 144100	$\mu_F = 0.768$	$\mu_{CR} = 0.041$	
Diversity = 1.6919e+01	k_t = [2, 3, 3, 2]			
Generation 1450: Best = 2.033755e+03	FES = 145100	$\mu_F = 0.805$	$\mu_{CR} = 0.044$	
Diversity = 1.6941e+01	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 2.033755e+03	FES = 146100	$\mu_F = 0.793$	$\mu_{CR} = 0.036$	
Diversity = 1.6944e+01	k_t = [2, 3, 3, 3]			
Generation 1470: Best = 2.033755e+03	FES = 147100	$\mu_F = 0.774$	$\mu_{CR} = 0.035$	
Diversity = 1.6862e+01	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 2.033578e+03	FES = 148100	$\mu_F = 0.762$	$\mu_{CR} = 0.046$	
Diversity = 1.6953e+01	k_t = [2, 3, 4, 4]			
Generation 1490: Best = 2.033571e+03	FES = 149100	$\mu_F = 0.748$	$\mu_{CR} = 0.043$	
Diversity = 1.6993e+01	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 2.032349e+03	FES = 150100	$\mu_F = 0.736$	$\mu_{CR} = 0.038$	
Diversity = 1.7077e+01	k_t = [2, 4, 4, 4]			
Generation 1510: Best = 2.031909e+03	FES = 151100	$\mu_F = 0.789$	$\mu_{CR} = 0.047$	
Diversity = 1.7066e+01	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 2.031909e+03	FES = 152100	$\mu_F = 0.775$	$\mu_{CR} = 0.048$	
Diversity = 1.6981e+01	k_t = [2, 2, 4, 2]			
Generation 1530: Best = 2.031907e+03	FES = 153100	$\mu_F = 0.709$	$\mu_{CR} = 0.038$	
Diversity = 1.6992e+01	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 2.031907e+03	FES = 154100	$\mu_F = 0.777$	$\mu_{CR} = 0.050$	
Diversity = 1.6761e+01	k_t = [3, 3, 4, 4]			
Generation 1550: Best = 2.031907e+03	FES = 155100	$\mu_F = 0.771$	$\mu_{CR} = 0.045$	
Diversity = 1.6678e+01	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 2.030585e+03	FES = 156100	$\mu_F = 0.752$	$\mu_{CR} = 0.055$	
Diversity = 1.6571e+01	k_t = [4, 3, 4, 2]			
Generation 1570: Best = 2.030426e+03	FES = 157100	$\mu_F = 0.804$	$\mu_{CR} = 0.051$	

Diversity = 1.6584e+01	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 2.030422e+03	FES = 158100	$\mu_F = 0.795$	$\mu_{CR} = 0.045$	
Diversity = 1.6380e+01	k_t = [2, 4, 4, 4]			
Generation 1590: Best = 2.030173e+03	FES = 159100	$\mu_F = 0.727$	$\mu_{CR} = 0.043$	
Diversity = 1.6443e+01	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 2.030173e+03	FES = 160100	$\mu_F = 0.723$	$\mu_{CR} = 0.051$	
Diversity = 1.6492e+01	k_t = [4, 3, 2, 3]			
Generation 1610: Best = 2.029798e+03	FES = 161100	$\mu_F = 0.721$	$\mu_{CR} = 0.056$	
Diversity = 1.6510e+01	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 2.029798e+03	FES = 162100	$\mu_F = 0.735$	$\mu_{CR} = 0.063$	
Diversity = 1.6686e+01	k_t = [2, 4, 4, 2]			
Generation 1630: Best = 2.029798e+03	FES = 163100	$\mu_F = 0.732$	$\mu_{CR} = 0.058$	
Diversity = 1.6653e+01	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 2.029798e+03	FES = 164100	$\mu_F = 0.779$	$\mu_{CR} = 0.049$	
Diversity = 1.6654e+01	k_t = [2, 3, 2, 3]			
Generation 1650: Best = 2.029798e+03	FES = 165100	$\mu_F = 0.788$	$\mu_{CR} = 0.057$	
Diversity = 1.6647e+01	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 2.029798e+03	FES = 166100	$\mu_F = 0.779$	$\mu_{CR} = 0.052$	
Diversity = 1.6654e+01	k_t = [4, 4, 4, 2]			
Generation 1670: Best = 2.029798e+03	FES = 167100	$\mu_F = 0.763$	$\mu_{CR} = 0.056$	
Diversity = 1.6661e+01	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 2.029789e+03	FES = 168100	$\mu_F = 0.822$	$\mu_{CR} = 0.048$	
Diversity = 1.6673e+01	k_t = [2, 3, 2, 2]			
Generation 1690: Best = 2.029745e+03	FES = 169100	$\mu_F = 0.811$	$\mu_{CR} = 0.052$	
Diversity = 1.6716e+01	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 2.029742e+03	FES = 170100	$\mu_F = 0.862$	$\mu_{CR} = 0.053$	
Diversity = 1.6527e+01	k_t = [4, 4, 3, 4]			
Generation 1710: Best = 2.029742e+03	FES = 171100	$\mu_F = 0.839$	$\mu_{CR} = 0.050$	
Diversity = 1.6494e+01	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 2.029740e+03	FES = 172100	$\mu_F = 0.873$	$\mu_{CR} = 0.042$	
Diversity = 1.6226e+01	k_t = [2, 3, 2, 4]			
Generation 1730: Best = 2.029740e+03	FES = 173100	$\mu_F = 0.824$	$\mu_{CR} = 0.043$	
Diversity = 1.6232e+01	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 2.029736e+03	FES = 174100	$\mu_F = 0.772$	$\mu_{CR} = 0.045$	
Diversity = 1.6282e+01	k_t = [3, 3, 2, 2]			
Generation 1750: Best = 2.029429e+03	FES = 175100	$\mu_F = 0.760$	$\mu_{CR} = 0.058$	
Diversity = 1.6289e+01	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 2.029429e+03	FES = 176100	$\mu_F = 0.735$	$\mu_{CR} = 0.054$	
Diversity = 1.6268e+01	k_t = [2, 4, 4, 2]			
Generation 1770: Best = 2.029428e+03	FES = 177100	$\mu_F = 0.737$	$\mu_{CR} = 0.052$	
Diversity = 1.6270e+01	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 2.029428e+03	FES = 178100	$\mu_F = 0.778$	$\mu_{CR} = 0.047$	
Diversity = 1.6241e+01	k_t = [3, 3, 2, 4]			
Generation 1790: Best = 2.029428e+03	FES = 179100	$\mu_F = 0.784$	$\mu_{CR} = 0.040$	
Diversity = 1.6223e+01	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 2.029428e+03	FES = 180100	$\mu_F = 0.796$	$\mu_{CR} = 0.045$	
Diversity = 1.6110e+01	k_t = [4, 2, 3, 4]			
Generation 1810: Best = 2.029297e+03	FES = 181100	$\mu_F = 0.807$	$\mu_{CR} = 0.048$	
Diversity = 1.6100e+01	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 2.029297e+03	FES = 182100	$\mu_F = 0.853$	$\mu_{CR} = 0.045$	
Diversity = 1.6187e+01	k_t = [3, 3, 4, 4]			
Generation 1830: Best = 2.029297e+03	FES = 183100	$\mu_F = 0.845$	$\mu_{CR} = 0.047$	
Diversity = 1.6154e+01	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 2.029061e+03	FES = 184100	$\mu_F = 0.841$	$\mu_{CR} = 0.046$	

Diversity = 1.6172e+01	k_t = [2, 2, 3, 2]			
Generation 1850: Best = 2.029061e+03	FES = 185100	$\mu_F = 0.839$	$\mu_{CR} = 0.050$	
Diversity = 1.6185e+01	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 2.029061e+03	FES = 186100	$\mu_F = 0.836$	$\mu_{CR} = 0.054$	
Diversity = 1.6126e+01	k_t = [3, 4, 4, 3]			
Generation 1870: Best = 2.029061e+03	FES = 187100	$\mu_F = 0.828$	$\mu_{CR} = 0.050$	
Diversity = 1.6163e+01	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 2.029061e+03	FES = 188100	$\mu_F = 0.851$	$\mu_{CR} = 0.043$	
Diversity = 1.6085e+01	k_t = [3, 4, 2, 2]			
Generation 1890: Best = 2.029061e+03	FES = 189100	$\mu_F = 0.831$	$\mu_{CR} = 0.065$	
Diversity = 1.6105e+01	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 2.029061e+03	FES = 190100	$\mu_F = 0.844$	$\mu_{CR} = 0.067$	
Diversity = 1.6013e+01	k_t = [4, 2, 3, 4]			
Generation 1910: Best = 2.029061e+03	FES = 191100	$\mu_F = 0.845$	$\mu_{CR} = 0.068$	
Diversity = 1.5896e+01	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 2.029061e+03	FES = 192100	$\mu_F = 0.856$	$\mu_{CR} = 0.061$	
Diversity = 1.5804e+01	k_t = [2, 2, 4, 3]			
Generation 1930: Best = 2.029061e+03	FES = 193100	$\mu_F = 0.874$	$\mu_{CR} = 0.058$	
Diversity = 1.5800e+01	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 2.028463e+03	FES = 194100	$\mu_F = 0.898$	$\mu_{CR} = 0.059$	
Diversity = 1.5548e+01	k_t = [2, 2, 3, 4]			
Generation 1950: Best = 2.028437e+03	FES = 195100	$\mu_F = 0.898$	$\mu_{CR} = 0.050$	
Diversity = 1.5658e+01	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 2.028327e+03	FES = 196100	$\mu_F = 0.890$	$\mu_{CR} = 0.043$	
Diversity = 1.5518e+01	k_t = [4, 2, 2, 3]			
Generation 1970: Best = 2.028327e+03	FES = 197100	$\mu_F = 0.896$	$\mu_{CR} = 0.057$	
Diversity = 1.5537e+01	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 2.028327e+03	FES = 198100	$\mu_F = 0.895$	$\mu_{CR} = 0.053$	
Diversity = 1.5554e+01	k_t = [2, 4, 3, 3]			
Generation 1990: Best = 2.028327e+03	FES = 199100	$\mu_F = 0.908$	$\mu_{CR} = 0.066$	
Diversity = 1.5473e+01	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 2.028303e+03	FES = 200100	$\mu_F = 0.901$	$\mu_{CR} = 0.058$	
Diversity = 1.5372e+01	k_t = [4, 4, 4, 4]			
Generation 2010: Best = 2.028280e+03	FES = 201100	$\mu_F = 0.902$	$\mu_{CR} = 0.050$	
Diversity = 1.5401e+01	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 2.028259e+03	FES = 202100	$\mu_F = 0.905$	$\mu_{CR} = 0.058$	
Diversity = 1.5054e+01	k_t = [3, 4, 3, 3]			
Generation 2030: Best = 2.028241e+03	FES = 203100	$\mu_F = 0.897$	$\mu_{CR} = 0.057$	
Diversity = 1.5337e+01	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 2.028241e+03	FES = 204100	$\mu_F = 0.906$	$\mu_{CR} = 0.069$	
Diversity = 1.5580e+01	k_t = [3, 3, 2, 3]			
Generation 2050: Best = 2.028241e+03	FES = 205100	$\mu_F = 0.903$	$\mu_{CR} = 0.073$	
Diversity = 1.5516e+01	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 2.028241e+03	FES = 206100	$\mu_F = 0.875$	$\mu_{CR} = 0.072$	
Diversity = 1.5542e+01	k_t = [2, 4, 3, 4]			
Generation 2070: Best = 2.028046e+03	FES = 207100	$\mu_F = 0.877$	$\mu_{CR} = 0.085$	
Diversity = 1.5612e+01	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 2.027907e+03	FES = 208100	$\mu_F = 0.889$	$\mu_{CR} = 0.079$	
Diversity = 1.5399e+01	k_t = [2, 4, 3, 3]			
Generation 2090: Best = 2.027896e+03	FES = 209100	$\mu_F = 0.893$	$\mu_{CR} = 0.077$	
Diversity = 1.5574e+01	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 2.027896e+03	FES = 210100	$\mu_F = 0.888$	$\mu_{CR} = 0.078$	
Diversity = 1.5409e+01	k_t = [2, 3, 4, 2]			
Generation 2110: Best = 2.027895e+03	FES = 211100	$\mu_F = 0.882$	$\mu_{CR} = 0.073$	

Diversity = 1.5415e+01	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 2.027895e+03	FES = 212100	$\mu_F = 0.907$	$\mu_{CR} = 0.067$	
Diversity = 1.5005e+01	k_t = [3, 4, 2, 4]			
Generation 2130: Best = 2.027630e+03	FES = 213100	$\mu_F = 0.876$	$\mu_{CR} = 0.079$	
Diversity = 1.4348e+01	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 2.027630e+03	FES = 214100	$\mu_F = 0.875$	$\mu_{CR} = 0.071$	
Diversity = 1.3808e+01	k_t = [2, 4, 3, 2]			
Generation 2150: Best = 2.027628e+03	FES = 215100	$\mu_F = 0.878$	$\mu_{CR} = 0.065$	
Diversity = 1.2805e+01	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 2.027575e+03	FES = 216100	$\mu_F = 0.873$	$\mu_{CR} = 0.067$	
Diversity = 1.2033e+01	k_t = [4, 2, 4, 4]			
Generation 2170: Best = 2.027371e+03	FES = 217100	$\mu_F = 0.865$	$\mu_{CR} = 0.074$	
Diversity = 1.1642e+01	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 2.026813e+03	FES = 218100	$\mu_F = 0.877$	$\mu_{CR} = 0.077$	
Diversity = 1.1708e+01	k_t = [2, 4, 4, 2]			
Generation 2190: Best = 2.026786e+03	FES = 219100	$\mu_F = 0.867$	$\mu_{CR} = 0.081$	
Diversity = 1.1932e+01	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 2.026781e+03	FES = 220100	$\mu_F = 0.872$	$\mu_{CR} = 0.074$	
Diversity = 1.1394e+01	k_t = [4, 3, 2, 2]			
Generation 2210: Best = 2.026656e+03	FES = 221100	$\mu_F = 0.865$	$\mu_{CR} = 0.085$	
Diversity = 1.1068e+01	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 2.026435e+03	FES = 222100	$\mu_F = 0.851$	$\mu_{CR} = 0.075$	
Diversity = 1.1123e+01	k_t = [2, 3, 4, 4]			
Generation 2230: Best = 2.026431e+03	FES = 223100	$\mu_F = 0.879$	$\mu_{CR} = 0.067$	
Diversity = 1.1140e+01	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 2.026369e+03	FES = 224100	$\mu_F = 0.876$	$\mu_{CR} = 0.079$	
Diversity = 1.0962e+01	k_t = [3, 3, 4, 3]			
Generation 2250: Best = 2.026312e+03	FES = 225100	$\mu_F = 0.865$	$\mu_{CR} = 0.069$	
Diversity = 1.0998e+01	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 2.026312e+03	FES = 226100	$\mu_F = 0.851$	$\mu_{CR} = 0.068$	
Diversity = 1.0775e+01	k_t = [3, 3, 2, 4]			
Generation 2270: Best = 2.026312e+03	FES = 227100	$\mu_F = 0.854$	$\mu_{CR} = 0.064$	
Diversity = 1.0719e+01	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 2.026305e+03	FES = 228100	$\mu_F = 0.835$	$\mu_{CR} = 0.068$	
Diversity = 1.0525e+01	k_t = [3, 2, 3, 3]			
Generation 2290: Best = 2.026150e+03	FES = 229100	$\mu_F = 0.833$	$\mu_{CR} = 0.074$	
Diversity = 1.0718e+01	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 2.026148e+03	FES = 230100	$\mu_F = 0.847$	$\mu_{CR} = 0.072$	
Diversity = 1.0543e+01	k_t = [3, 4, 4, 3]			
Generation 2310: Best = 2.026148e+03	FES = 231100	$\mu_F = 0.851$	$\mu_{CR} = 0.070$	
Diversity = 1.0674e+01	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 2.025931e+03	FES = 232100	$\mu_F = 0.848$	$\mu_{CR} = 0.070$	
Diversity = 1.0745e+01	k_t = [4, 4, 3, 3]			
Generation 2330: Best = 2.025907e+03	FES = 233100	$\mu_F = 0.856$	$\mu_{CR} = 0.069$	
Diversity = 1.0678e+01	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 2.025905e+03	FES = 234100	$\mu_F = 0.841$	$\mu_{CR} = 0.068$	
Diversity = 1.0390e+01	k_t = [3, 3, 2, 2]			
Generation 2350: Best = 2.025904e+03	FES = 235100	$\mu_F = 0.846$	$\mu_{CR} = 0.075$	
Diversity = 1.0276e+01	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 2.025880e+03	FES = 236100	$\mu_F = 0.805$	$\mu_{CR} = 0.073$	
Diversity = 1.0321e+01	k_t = [3, 3, 2, 2]			
Generation 2370: Best = 2.025880e+03	FES = 237100	$\mu_F = 0.790$	$\mu_{CR} = 0.068$	
Diversity = 1.0296e+01	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 2.025879e+03	FES = 238100	$\mu_F = 0.792$	$\mu_{CR} = 0.066$	

Diversity = 9.9301e+00	k_t = [3, 3, 4, 4]			
Generation 2390: Best = 2.025755e+03	FES = 239100	$\mu_F = 0.774$	$\mu_{CR} = 0.065$	
Diversity = 9.9400e+00	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 2.025751e+03	FES = 240100	$\mu_F = 0.802$	$\mu_{CR} = 0.062$	
Diversity = 1.0018e+01	k_t = [2, 2, 2, 2]			
Generation 2410: Best = 2.025751e+03	FES = 241100	$\mu_F = 0.805$	$\mu_{CR} = 0.082$	
Diversity = 1.0025e+01	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 2.025698e+03	FES = 242100	$\mu_F = 0.827$	$\mu_{CR} = 0.081$	
Diversity = 1.0048e+01	k_t = [3, 4, 3, 2]			
Generation 2430: Best = 2.025698e+03	FES = 243100	$\mu_F = 0.812$	$\mu_{CR} = 0.071$	
Diversity = 1.0022e+01	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 2.025698e+03	FES = 244100	$\mu_F = 0.838$	$\mu_{CR} = 0.064$	
Diversity = 9.8036e+00	k_t = [3, 4, 3, 2]			
Generation 2450: Best = 2.025658e+03	FES = 245100	$\mu_F = 0.846$	$\mu_{CR} = 0.055$	
Diversity = 9.7697e+00	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 2.025589e+03	FES = 246100	$\mu_F = 0.830$	$\mu_{CR} = 0.063$	
Diversity = 9.8390e+00	k_t = [3, 2, 4, 3]			
Generation 2470: Best = 2.025589e+03	FES = 247100	$\mu_F = 0.822$	$\mu_{CR} = 0.055$	
Diversity = 9.9016e+00	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 2.025589e+03	FES = 248100	$\mu_F = 0.831$	$\mu_{CR} = 0.057$	
Diversity = 9.9427e+00	k_t = [3, 4, 3, 4]			
Generation 2490: Best = 2.025588e+03	FES = 249100	$\mu_F = 0.824$	$\mu_{CR} = 0.066$	
Diversity = 9.8840e+00	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 2.025526e+03	FES = 250100	$\mu_F = 0.852$	$\mu_{CR} = 0.065$	
Diversity = 9.7675e+00	k_t = [4, 3, 4, 2]			
Generation 2510: Best = 2.025525e+03	FES = 251100	$\mu_F = 0.860$	$\mu_{CR} = 0.063$	
Diversity = 9.8062e+00	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 2.025525e+03	FES = 252100	$\mu_F = 0.823$	$\mu_{CR} = 0.066$	
Diversity = 9.8493e+00	k_t = [2, 3, 4, 4]			
Generation 2530: Best = 2.025410e+03	FES = 253100	$\mu_F = 0.815$	$\mu_{CR} = 0.064$	
Diversity = 9.7675e+00	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 2.025348e+03	FES = 254100	$\mu_F = 0.818$	$\mu_{CR} = 0.064$	
Diversity = 9.7855e+00	k_t = [2, 2, 2, 2]			
Generation 2550: Best = 2.025348e+03	FES = 255100	$\mu_F = 0.829$	$\mu_{CR} = 0.055$	
Diversity = 9.7249e+00	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 2.025347e+03	FES = 256100	$\mu_F = 0.825$	$\mu_{CR} = 0.058$	
Diversity = 9.7484e+00	k_t = [2, 4, 4, 3]			
Generation 2570: Best = 2.025347e+03	FES = 257100	$\mu_F = 0.830$	$\mu_{CR} = 0.060$	
Diversity = 9.6934e+00	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 2.025345e+03	FES = 258100	$\mu_F = 0.847$	$\mu_{CR} = 0.060$	
Diversity = 9.6617e+00	k_t = [4, 4, 4, 3]			
Generation 2590: Best = 2.025345e+03	FES = 259100	$\mu_F = 0.866$	$\mu_{CR} = 0.052$	
Diversity = 9.6664e+00	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 2.025345e+03	FES = 260100	$\mu_F = 0.853$	$\mu_{CR} = 0.058$	
Diversity = 9.6808e+00	k_t = [4, 4, 2, 3]			
Generation 2610: Best = 2.025344e+03	FES = 261100	$\mu_F = 0.864$	$\mu_{CR} = 0.059$	
Diversity = 9.7906e+00	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 2.025344e+03	FES = 262100	$\mu_F = 0.861$	$\mu_{CR} = 0.063$	
Diversity = 9.7768e+00	k_t = [4, 4, 3, 3]			
Generation 2630: Best = 2.025314e+03	FES = 263100	$\mu_F = 0.856$	$\mu_{CR} = 0.060$	
Diversity = 9.7525e+00	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 2.025313e+03	FES = 264100	$\mu_F = 0.864$	$\mu_{CR} = 0.057$	
Diversity = 9.7125e+00	k_t = [4, 2, 3, 4]			
Generation 2650: Best = 2.025310e+03	FES = 265100	$\mu_F = 0.864$	$\mu_{CR} = 0.055$	

Diversity = 9.6384e+00	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 2.025310e+03	FES = 266100	$\mu_F = 0.865$	$\mu_{CR} = 0.062$	
Diversity = 9.6448e+00	k_t = [2, 2, 3, 2]			
Generation 2670: Best = 2.025310e+03	FES = 267100	$\mu_F = 0.868$	$\mu_{CR} = 0.061$	
Diversity = 9.6025e+00	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 2.025310e+03	FES = 268100	$\mu_F = 0.866$	$\mu_{CR} = 0.066$	
Diversity = 9.6220e+00	k_t = [4, 4, 4, 3]			
Generation 2690: Best = 2.025309e+03	FES = 269100	$\mu_F = 0.882$	$\mu_{CR} = 0.055$	
Diversity = 9.6217e+00	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 2.025309e+03	FES = 270100	$\mu_F = 0.891$	$\mu_{CR} = 0.052$	
Diversity = 9.6147e+00	k_t = [3, 2, 4, 2]			
Generation 2710: Best = 2.025309e+03	FES = 271100	$\mu_F = 0.866$	$\mu_{CR} = 0.049$	
Diversity = 9.6116e+00	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 2.025309e+03	FES = 272100	$\mu_F = 0.875$	$\mu_{CR} = 0.049$	
Diversity = 9.7126e+00	k_t = [4, 4, 2, 4]			
Generation 2730: Best = 2.025166e+03	FES = 273100	$\mu_F = 0.851$	$\mu_{CR} = 0.056$	
Diversity = 9.7123e+00	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 2.025166e+03	FES = 274100	$\mu_F = 0.840$	$\mu_{CR} = 0.058$	
Diversity = 9.6770e+00	k_t = [3, 4, 2, 2]			
Generation 2750: Best = 2.025166e+03	FES = 275100	$\mu_F = 0.825$	$\mu_{CR} = 0.060$	
Diversity = 9.6308e+00	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 2.025166e+03	FES = 276100	$\mu_F = 0.808$	$\mu_{CR} = 0.046$	
Diversity = 9.5830e+00	k_t = [2, 3, 4, 3]			
Generation 2770: Best = 2.025166e+03	FES = 277100	$\mu_F = 0.818$	$\mu_{CR} = 0.059$	
Diversity = 9.5923e+00	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 2.025166e+03	FES = 278100	$\mu_F = 0.826$	$\mu_{CR} = 0.059$	
Diversity = 9.5773e+00	k_t = [4, 2, 2, 2]			
Generation 2790: Best = 2.025166e+03	FES = 279100	$\mu_F = 0.806$	$\mu_{CR} = 0.060$	
Diversity = 9.4762e+00	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 2.025165e+03	FES = 280100	$\mu_F = 0.822$	$\mu_{CR} = 0.064$	
Diversity = 9.4749e+00	k_t = [4, 2, 4, 2]			
Generation 2810: Best = 2.025159e+03	FES = 281100	$\mu_F = 0.827$	$\mu_{CR} = 0.069$	
Diversity = 9.4711e+00	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 2.025159e+03	FES = 282100	$\mu_F = 0.808$	$\mu_{CR} = 0.060$	
Diversity = 9.4250e+00	k_t = [4, 4, 2, 3]			
Generation 2830: Best = 2.025155e+03	FES = 283100	$\mu_F = 0.823$	$\mu_{CR} = 0.047$	
Diversity = 9.4373e+00	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 2.025155e+03	FES = 284100	$\mu_F = 0.756$	$\mu_{CR} = 0.057$	
Diversity = 9.2767e+00	k_t = [4, 4, 4, 4]			
Generation 2850: Best = 2.025138e+03	FES = 285100	$\mu_F = 0.778$	$\mu_{CR} = 0.051$	
Diversity = 9.3540e+00	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 2.025138e+03	FES = 286100	$\mu_F = 0.778$	$\mu_{CR} = 0.052$	
Diversity = 9.2889e+00	k_t = [2, 2, 2, 4]			
Generation 2870: Best = 2.025138e+03	FES = 287100	$\mu_F = 0.815$	$\mu_{CR} = 0.050$	
Diversity = 9.2536e+00	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 2.025138e+03	FES = 288100	$\mu_F = 0.817$	$\mu_{CR} = 0.049$	
Diversity = 9.2698e+00	k_t = [2, 2, 2, 2]			
Generation 2890: Best = 2.025138e+03	FES = 289100	$\mu_F = 0.799$	$\mu_{CR} = 0.050$	
Diversity = 9.2698e+00	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 2.025138e+03	FES = 290100	$\mu_F = 0.825$	$\mu_{CR} = 0.062$	
Diversity = 9.3155e+00	k_t = [3, 2, 2, 4]			
Generation 2910: Best = 2.025138e+03	FES = 291100	$\mu_F = 0.811$	$\mu_{CR} = 0.052$	
Diversity = 9.2768e+00	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 2.025138e+03	FES = 292100	$\mu_F = 0.778$	$\mu_{CR} = 0.054$	

```

Diversity = 9.1762e+00 k_t = [3, 2, 4, 4]
Generation 2930: Best = 2.025138e+03 FES = 293100  $\mu_F$  = 0.821  $\mu_{CR}$  = 0.061
Diversity = 9.1767e+00 k_t = [1, 1, 1, 1]
Generation 2940: Best = 2.025119e+03 FES = 294100  $\mu_F$  = 0.848  $\mu_{CR}$  = 0.064
Diversity = 9.0718e+00 k_t = [3, 2, 3, 4]
Generation 2950: Best = 2.025104e+03 FES = 295100  $\mu_F$  = 0.828  $\mu_{CR}$  = 0.051
Diversity = 8.9997e+00 k_t = [1, 1, 1, 1]
Generation 2960: Best = 2.025104e+03 FES = 296100  $\mu_F$  = 0.838  $\mu_{CR}$  = 0.046
Diversity = 8.9996e+00 k_t = [4, 4, 2, 3]
Generation 2970: Best = 2.025104e+03 FES = 297100  $\mu_F$  = 0.849  $\mu_{CR}$  = 0.052
Diversity = 8.9995e+00 k_t = [1, 1, 1, 1]
Generation 2980: Best = 2.025104e+03 FES = 298100  $\mu_F$  = 0.847  $\mu_{CR}$  = 0.058
Diversity = 8.8732e+00 k_t = [4, 4, 3, 4]
Generation 2990: Best = 2.025104e+03 FES = 299100  $\mu_F$  = 0.861  $\mu_{CR}$  = 0.047
Diversity = 8.8732e+00 k_t = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 2.025104e+03

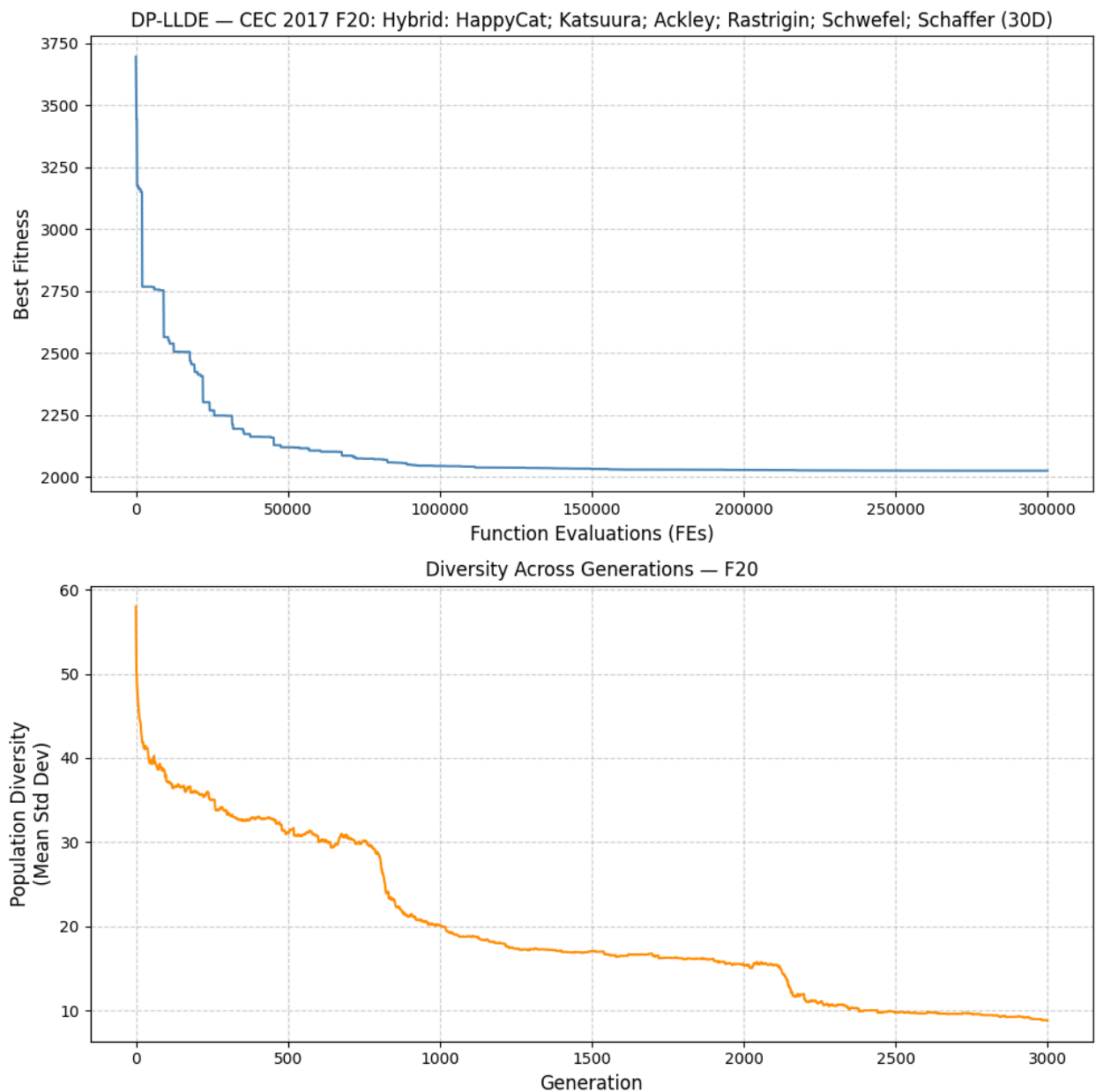
Total Function Evals = 300000

Total Successful Updates = 18159

```

=====
=
Results:
  Best Fitness      : 2.025104e+03
  Global Optimum    : 2.000000e+03
  Error (|best-optimum|) : 2.510370e+01
  Best Solution (5 dim) : [ 29.7101045 -85.90379598 -23.30284983  60.25308011
56.35320174]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F20.png

6.4 Composite Functions (F21-F30)

```
In [ ]: # =====
# F21 — Composition Function 1 (Rosenbrock; Elliptic; Rastrigin)
# =====
run_and_plot('F21', make_wrapper(composition.f21), 2100.0, 'Composition: Roser
```

=====

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DP-LLDE on CEC 2017 F21 – Single Run

=====

=

Function : Composition: Rosenbrock; Elliptic; Rastrigin (30D)
Bounds : [-100, 100]
Global optimum : 2100.0
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 2.946639e+03
Generation 10: Best = 2.637368e+03 FES = 1100 $\mu_F = 0.579$ $\mu_{CR} = 0.353$
Diversity = 3.2831e+01 $k_t = [1, 1, 1, 1]$
Generation 20: Best = 2.584986e+03 FES = 2100 $\mu_F = 0.629$ $\mu_{CR} = 0.348$
Diversity = 2.4371e+01 $k_t = [4, 4, 2, 4]$
Generation 30: Best = 2.543141e+03 FES = 3100 $\mu_F = 0.663$ $\mu_{CR} = 0.312$
Diversity = 2.1270e+01 $k_t = [1, 1, 1, 1]$
Generation 40: Best = 2.513852e+03 FES = 4100 $\mu_F = 0.665$ $\mu_{CR} = 0.279$
Diversity = 1.9748e+01 $k_t = [4, 3, 4, 2]$
Generation 50: Best = 2.513852e+03 FES = 5100 $\mu_F = 0.699$ $\mu_{CR} = 0.227$
Diversity = 1.8962e+01 $k_t = [1, 1, 1, 1]$
Generation 60: Best = 2.513316e+03 FES = 6100 $\mu_F = 0.725$ $\mu_{CR} = 0.193$
Diversity = 1.8360e+01 $k_t = [3, 4, 2, 2]$
Generation 70: Best = 2.506574e+03 FES = 7100 $\mu_F = 0.691$ $\mu_{CR} = 0.158$
Diversity = 1.7797e+01 $k_t = [1, 1, 1, 1]$
Generation 80: Best = 2.482642e+03 FES = 8100 $\mu_F = 0.752$ $\mu_{CR} = 0.123$
Diversity = 1.7281e+01 $k_t = [4, 4, 3, 4]$
Generation 90: Best = 2.482642e+03 FES = 9100 $\mu_F = 0.746$ $\mu_{CR} = 0.098$
Diversity = 1.6940e+01 $k_t = [1, 1, 1, 1]$
Generation 100: Best = 2.469222e+03 FES = 10100 $\mu_F = 0.757$ $\mu_{CR} = 0.091$
Diversity = 1.7001e+01 $k_t = [3, 4, 4, 3]$
Generation 110: Best = 2.464784e+03 FES = 11100 $\mu_F = 0.737$ $\mu_{CR} = 0.071$
Diversity = 1.6803e+01 $k_t = [1, 1, 1, 1]$
Generation 120: Best = 2.464784e+03 FES = 12100 $\mu_F = 0.719$ $\mu_{CR} = 0.068$
Diversity = 1.6565e+01 $k_t = [2, 2, 4, 4]$
Generation 130: Best = 2.458746e+03 FES = 13100 $\mu_F = 0.689$ $\mu_{CR} = 0.059$
Diversity = 1.6445e+01 $k_t = [1, 1, 1, 1]$
Generation 140: Best = 2.458746e+03 FES = 14100 $\mu_F = 0.729$ $\mu_{CR} = 0.061$
Diversity = 1.6478e+01 $k_t = [2, 4, 2, 2]$
Generation 150: Best = 2.451180e+03 FES = 15100 $\mu_F = 0.705$ $\mu_{CR} = 0.064$
Diversity = 1.6220e+01 $k_t = [1, 1, 1, 1]$
Generation 160: Best = 2.451180e+03 FES = 16100 $\mu_F = 0.719$ $\mu_{CR} = 0.057$
Diversity = 1.6305e+01 $k_t = [3, 2, 3, 4]$
Generation 170: Best = 2.439891e+03 FES = 17100 $\mu_F = 0.680$ $\mu_{CR} = 0.059$
Diversity = 1.6274e+01 $k_t = [1, 1, 1, 1]$
Generation 180: Best = 2.429825e+03 FES = 18100 $\mu_F = 0.626$ $\mu_{CR} = 0.065$
Diversity = 1.6211e+01 $k_t = [3, 4, 3, 3]$
Generation 190: Best = 2.423526e+03 FES = 19100 $\mu_F = 0.609$ $\mu_{CR} = 0.089$
Diversity = 1.6044e+01 $k_t = [1, 1, 1, 1]$
Generation 200: Best = 2.423526e+03 FES = 20100 $\mu_F = 0.621$ $\mu_{CR} = 0.074$
Diversity = 1.5987e+01 $k_t = [3, 2, 3, 2]$
Generation 210: Best = 2.423526e+03 FES = 21100 $\mu_F = 0.599$ $\mu_{CR} = 0.060$
Diversity = 1.5891e+01 $k_t = [1, 1, 1, 1]$
Generation 220: Best = 2.423526e+03 FES = 22100 $\mu_F = 0.639$ $\mu_{CR} = 0.057$
Diversity = 1.5952e+01 $k_t = [3, 2, 2, 2]$

Generation 230: Best = 2.416893e+03	FES = 23100	$\mu_F = 0.644$	$\mu_{CR} = 0.050$
Diversity = 1.5859e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 2.416893e+03	FES = 24100	$\mu_F = 0.576$	$\mu_{CR} = 0.064$
Diversity = 1.5796e+01	$k_t = [2, 4, 2, 3]$		
Generation 250: Best = 2.414451e+03	FES = 25100	$\mu_F = 0.573$	$\mu_{CR} = 0.060$
Diversity = 1.5719e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 2.402288e+03	FES = 26100	$\mu_F = 0.568$	$\mu_{CR} = 0.070$
Diversity = 1.5745e+01	$k_t = [2, 4, 3, 4]$		
Generation 270: Best = 2.402288e+03	FES = 27100	$\mu_F = 0.601$	$\mu_{CR} = 0.067$
Diversity = 1.5574e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 2.402288e+03	FES = 28100	$\mu_F = 0.650$	$\mu_{CR} = 0.063$
Diversity = 1.5483e+01	$k_t = [4, 3, 2, 2]$		
Generation 290: Best = 2.402288e+03	FES = 29100	$\mu_F = 0.644$	$\mu_{CR} = 0.059$
Diversity = 1.5469e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 2.402288e+03	FES = 30100	$\mu_F = 0.624$	$\mu_{CR} = 0.062$
Diversity = 1.5421e+01	$k_t = [3, 3, 3, 3]$		
Generation 310: Best = 2.402288e+03	FES = 31100	$\mu_F = 0.565$	$\mu_{CR} = 0.068$
Diversity = 1.5383e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 2.401791e+03	FES = 32100	$\mu_F = 0.593$	$\mu_{CR} = 0.066$
Diversity = 1.5359e+01	$k_t = [2, 2, 4, 3]$		
Generation 330: Best = 2.401791e+03	FES = 33100	$\mu_F = 0.604$	$\mu_{CR} = 0.057$
Diversity = 1.5230e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 2.401791e+03	FES = 34100	$\mu_F = 0.583$	$\mu_{CR} = 0.044$
Diversity = 1.5158e+01	$k_t = [4, 2, 4, 4]$		
Generation 350: Best = 2.394620e+03	FES = 35100	$\mu_F = 0.581$	$\mu_{CR} = 0.038$
Diversity = 1.5124e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 2.392336e+03	FES = 36100	$\mu_F = 0.539$	$\mu_{CR} = 0.053$
Diversity = 1.5134e+01	$k_t = [3, 4, 2, 2]$		
Generation 370: Best = 2.390851e+03	FES = 37100	$\mu_F = 0.512$	$\mu_{CR} = 0.060$
Diversity = 1.5068e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 2.390851e+03	FES = 38100	$\mu_F = 0.487$	$\mu_{CR} = 0.053$
Diversity = 1.4936e+01	$k_t = [4, 4, 3, 2]$		
Generation 390: Best = 2.390851e+03	FES = 39100	$\mu_F = 0.512$	$\mu_{CR} = 0.059$
Diversity = 1.4862e+01	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 2.390851e+03	FES = 40100	$\mu_F = 0.541$	$\mu_{CR} = 0.056$
Diversity = 1.4998e+01	$k_t = [4, 3, 3, 3]$		
Generation 410: Best = 2.378663e+03	FES = 41100	$\mu_F = 0.530$	$\mu_{CR} = 0.048$
Diversity = 1.4920e+01	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 2.378663e+03	FES = 42100	$\mu_F = 0.510$	$\mu_{CR} = 0.044$
Diversity = 1.4873e+01	$k_t = [4, 4, 2, 2]$		
Generation 430: Best = 2.378663e+03	FES = 43100	$\mu_F = 0.515$	$\mu_{CR} = 0.044$
Diversity = 1.4865e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 2.378663e+03	FES = 44100	$\mu_F = 0.573$	$\mu_{CR} = 0.048$
Diversity = 1.4802e+01	$k_t = [2, 3, 2, 4]$		
Generation 450: Best = 2.374011e+03	FES = 45100	$\mu_F = 0.567$	$\mu_{CR} = 0.045$
Diversity = 1.4835e+01	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 2.374011e+03	FES = 46100	$\mu_F = 0.545$	$\mu_{CR} = 0.052$
Diversity = 1.4793e+01	$k_t = [2, 3, 4, 2]$		
Generation 470: Best = 2.374011e+03	FES = 47100	$\mu_F = 0.613$	$\mu_{CR} = 0.048$
Diversity = 1.4779e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.369110e+03	FES = 48100	$\mu_F = 0.598$	$\mu_{CR} = 0.062$
Diversity = 1.4768e+01	$k_t = [4, 3, 2, 2]$		
Generation 490: Best = 2.369056e+03	FES = 49100	$\mu_F = 0.569$	$\mu_{CR} = 0.052$
Diversity = 1.4763e+01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 2.369056e+03	FES = 50100	$\mu_F = 0.547$	$\mu_{CR} = 0.045$
Diversity = 1.4750e+01	k_t = [3, 4, 2, 2]		
Generation 510: Best = 2.369056e+03	FES = 51100	$\mu_F = 0.533$	$\mu_{CR} = 0.056$
Diversity = 1.4733e+01	k_t = [1, 1, 1, 1]		
Generation 520: Best = 2.369056e+03	FES = 52100	$\mu_F = 0.524$	$\mu_{CR} = 0.051$
Diversity = 1.4715e+01	k_t = [2, 2, 4, 3]		
Generation 530: Best = 2.369056e+03	FES = 53100	$\mu_F = 0.445$	$\mu_{CR} = 0.063$
Diversity = 1.4704e+01	k_t = [1, 1, 1, 1]		
Generation 540: Best = 2.369056e+03	FES = 54100	$\mu_F = 0.373$	$\mu_{CR} = 0.058$
Diversity = 1.4666e+01	k_t = [4, 3, 4, 2]		
Generation 550: Best = 2.369056e+03	FES = 55100	$\mu_F = 0.353$	$\mu_{CR} = 0.050$
Diversity = 1.4646e+01	k_t = [1, 1, 1, 1]		
Generation 560: Best = 2.369056e+03	FES = 56100	$\mu_F = 0.309$	$\mu_{CR} = 0.071$
Diversity = 1.4629e+01	k_t = [2, 2, 3, 2]		
Generation 570: Best = 2.369056e+03	FES = 57100	$\mu_F = 0.298$	$\mu_{CR} = 0.065$
Diversity = 1.4605e+01	k_t = [1, 1, 1, 1]		
Generation 580: Best = 2.369056e+03	FES = 58100	$\mu_F = 0.384$	$\mu_{CR} = 0.065$
Diversity = 1.4547e+01	k_t = [3, 4, 4, 4]		
Generation 590: Best = 2.367575e+03	FES = 59100	$\mu_F = 0.342$	$\mu_{CR} = 0.062$
Diversity = 1.4529e+01	k_t = [1, 1, 1, 1]		
Generation 600: Best = 2.360305e+03	FES = 60100	$\mu_F = 0.408$	$\mu_{CR} = 0.057$
Diversity = 1.4490e+01	k_t = [3, 2, 2, 2]		
Generation 610: Best = 2.360305e+03	FES = 61100	$\mu_F = 0.349$	$\mu_{CR} = 0.059$
Diversity = 1.4461e+01	k_t = [1, 1, 1, 1]		
Generation 620: Best = 2.360024e+03	FES = 62100	$\mu_F = 0.351$	$\mu_{CR} = 0.056$
Diversity = 1.4454e+01	k_t = [3, 4, 2, 2]		
Generation 630: Best = 2.360024e+03	FES = 63100	$\mu_F = 0.327$	$\mu_{CR} = 0.066$
Diversity = 1.4437e+01	k_t = [1, 1, 1, 1]		
Generation 640: Best = 2.360024e+03	FES = 64100	$\mu_F = 0.306$	$\mu_{CR} = 0.066$
Diversity = 1.4427e+01	k_t = [4, 2, 4, 3]		
Generation 650: Best = 2.358589e+03	FES = 65100	$\mu_F = 0.333$	$\mu_{CR} = 0.056$
Diversity = 1.4418e+01	k_t = [1, 1, 1, 1]		
Generation 660: Best = 2.358589e+03	FES = 66100	$\mu_F = 0.329$	$\mu_{CR} = 0.062$
Diversity = 1.4421e+01	k_t = [2, 4, 4, 3]		
Generation 670: Best = 2.357768e+03	FES = 67100	$\mu_F = 0.300$	$\mu_{CR} = 0.060$
Diversity = 1.4397e+01	k_t = [1, 1, 1, 1]		
Generation 680: Best = 2.357768e+03	FES = 68100	$\mu_F = 0.352$	$\mu_{CR} = 0.063$
Diversity = 1.4451e+01	k_t = [2, 2, 3, 4]		
Generation 690: Best = 2.357768e+03	FES = 69100	$\mu_F = 0.334$	$\mu_{CR} = 0.049$
Diversity = 1.4441e+01	k_t = [1, 1, 1, 1]		
Generation 700: Best = 2.357768e+03	FES = 70100	$\mu_F = 0.370$	$\mu_{CR} = 0.065$
Diversity = 1.4439e+01	k_t = [3, 2, 4, 3]		
Generation 710: Best = 2.357649e+03	FES = 71100	$\mu_F = 0.339$	$\mu_{CR} = 0.059$
Diversity = 1.4419e+01	k_t = [1, 1, 1, 1]		
Generation 720: Best = 2.357649e+03	FES = 72100	$\mu_F = 0.375$	$\mu_{CR} = 0.041$
Diversity = 1.4414e+01	k_t = [3, 2, 2, 2]		
Generation 730: Best = 2.357649e+03	FES = 73100	$\mu_F = 0.318$	$\mu_{CR} = 0.035$
Diversity = 1.4409e+01	k_t = [1, 1, 1, 1]		
Generation 740: Best = 2.357604e+03	FES = 74100	$\mu_F = 0.447$	$\mu_{CR} = 0.044$
Diversity = 1.4425e+01	k_t = [3, 4, 3, 4]		
Generation 750: Best = 2.357604e+03	FES = 75100	$\mu_F = 0.413$	$\mu_{CR} = 0.044$
Diversity = 1.4410e+01	k_t = [1, 1, 1, 1]		
Generation 760: Best = 2.357604e+03	FES = 76100	$\mu_F = 0.337$	$\mu_{CR} = 0.051$
Diversity = 1.4413e+01	k_t = [3, 3, 3, 2]		

Generation 770: Best = 2.357557e+03	FES = 77100	$\mu_F = 0.336$	$\mu_{CR} = 0.053$
Diversity = 1.4378e+01	k_t = [1, 1, 1, 1]		
Generation 780: Best = 2.356694e+03	FES = 78100	$\mu_F = 0.443$	$\mu_{CR} = 0.057$
Diversity = 1.4386e+01	k_t = [4, 2, 3, 3]		
Generation 790: Best = 2.356694e+03	FES = 79100	$\mu_F = 0.346$	$\mu_{CR} = 0.078$
Diversity = 1.4374e+01	k_t = [1, 1, 1, 1]		
Generation 800: Best = 2.356694e+03	FES = 80100	$\mu_F = 0.320$	$\mu_{CR} = 0.065$
Diversity = 1.4365e+01	k_t = [3, 3, 4, 2]		
Generation 810: Best = 2.355749e+03	FES = 81100	$\mu_F = 0.354$	$\mu_{CR} = 0.051$
Diversity = 1.4358e+01	k_t = [1, 1, 1, 1]		
Generation 820: Best = 2.355749e+03	FES = 82100	$\mu_F = 0.425$	$\mu_{CR} = 0.045$
Diversity = 1.4178e+01	k_t = [2, 2, 3, 4]		
Generation 830: Best = 2.355749e+03	FES = 83100	$\mu_F = 0.501$	$\mu_{CR} = 0.038$
Diversity = 1.4181e+01	k_t = [1, 1, 1, 1]		
Generation 840: Best = 2.354601e+03	FES = 84100	$\mu_F = 0.555$	$\mu_{CR} = 0.045$
Diversity = 1.4069e+01	k_t = [2, 2, 2, 4]		
Generation 850: Best = 2.354601e+03	FES = 85100	$\mu_F = 0.454$	$\mu_{CR} = 0.051$
Diversity = 1.4076e+01	k_t = [1, 1, 1, 1]		
Generation 860: Best = 2.354601e+03	FES = 86100	$\mu_F = 0.498$	$\mu_{CR} = 0.046$
Diversity = 1.4043e+01	k_t = [3, 4, 3, 4]		
Generation 870: Best = 2.354601e+03	FES = 87100	$\mu_F = 0.537$	$\mu_{CR} = 0.041$
Diversity = 1.4077e+01	k_t = [1, 1, 1, 1]		
Generation 880: Best = 2.354601e+03	FES = 88100	$\mu_F = 0.502$	$\mu_{CR} = 0.043$
Diversity = 1.4117e+01	k_t = [4, 3, 4, 4]		
Generation 890: Best = 2.354601e+03	FES = 89100	$\mu_F = 0.399$	$\mu_{CR} = 0.051$
Diversity = 1.4055e+01	k_t = [1, 1, 1, 1]		
Generation 900: Best = 2.354074e+03	FES = 90100	$\mu_F = 0.387$	$\mu_{CR} = 0.054$
Diversity = 1.4063e+01	k_t = [3, 4, 4, 2]		
Generation 910: Best = 2.354074e+03	FES = 91100	$\mu_F = 0.413$	$\mu_{CR} = 0.049$
Diversity = 1.4084e+01	k_t = [1, 1, 1, 1]		
Generation 920: Best = 2.354074e+03	FES = 92100	$\mu_F = 0.459$	$\mu_{CR} = 0.047$
Diversity = 1.3976e+01	k_t = [3, 3, 2, 4]		
Generation 930: Best = 2.352643e+03	FES = 93100	$\mu_F = 0.355$	$\mu_{CR} = 0.063$
Diversity = 1.3974e+01	k_t = [1, 1, 1, 1]		
Generation 940: Best = 2.351537e+03	FES = 94100	$\mu_F = 0.317$	$\mu_{CR} = 0.065$
Diversity = 1.3971e+01	k_t = [4, 2, 2, 3]		
Generation 950: Best = 2.351537e+03	FES = 95100	$\mu_F = 0.304$	$\mu_{CR} = 0.058$
Diversity = 1.3969e+01	k_t = [1, 1, 1, 1]		
Generation 960: Best = 2.350068e+03	FES = 96100	$\mu_F = 0.290$	$\mu_{CR} = 0.066$
Diversity = 1.3967e+01	k_t = [2, 2, 2, 3]		
Generation 970: Best = 2.349859e+03	FES = 97100	$\mu_F = 0.331$	$\mu_{CR} = 0.063$
Diversity = 1.3951e+01	k_t = [1, 1, 1, 1]		
Generation 980: Best = 2.349733e+03	FES = 98100	$\mu_F = 0.313$	$\mu_{CR} = 0.084$
Diversity = 1.3916e+01	k_t = [2, 3, 3, 4]		
Generation 990: Best = 2.348685e+03	FES = 99100	$\mu_F = 0.236$	$\mu_{CR} = 0.076$
Diversity = 1.3909e+01	k_t = [1, 1, 1, 1]		
Generation 1000: Best = 2.348685e+03	FES = 100100	$\mu_F = 0.234$	$\mu_{CR} = 0.075$
Diversity = 1.3904e+01	k_t = [4, 4, 3, 3]		
Generation 1010: Best = 2.348685e+03	FES = 101100	$\mu_F = 0.367$	$\mu_{CR} = 0.074$
Diversity = 1.3877e+01	k_t = [1, 1, 1, 1]		
Generation 1020: Best = 2.348565e+03	FES = 102100	$\mu_F = 0.353$	$\mu_{CR} = 0.051$
Diversity = 1.3877e+01	k_t = [3, 3, 3, 3]		
Generation 1030: Best = 2.348565e+03	FES = 103100	$\mu_F = 0.357$	$\mu_{CR} = 0.059$
Diversity = 1.3881e+01	k_t = [1, 1, 1, 1]		

Generation 1040: Best = 2.348203e+03 FES = 104100 μ_F = 0.326 μ_{CR} = 0.067
 Diversity = 1.3894e+01 k_t = [4, 3, 2, 2]
 Generation 1050: Best = 2.348203e+03 FES = 105100 μ_F = 0.343 μ_{CR} = 0.079
 Diversity = 1.3889e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 2.346827e+03 FES = 106100 μ_F = 0.318 μ_{CR} = 0.071
 Diversity = 1.3931e+01 k_t = [4, 2, 4, 4]
 Generation 1070: Best = 2.345512e+03 FES = 107100 μ_F = 0.312 μ_{CR} = 0.054
 Diversity = 1.3892e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 2.345512e+03 FES = 108100 μ_F = 0.261 μ_{CR} = 0.056
 Diversity = 1.3892e+01 k_t = [3, 4, 2, 3]
 Generation 1090: Best = 2.345512e+03 FES = 109100 μ_F = 0.316 μ_{CR} = 0.056
 Diversity = 1.3857e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 2.345512e+03 FES = 110100 μ_F = 0.284 μ_{CR} = 0.057
 Diversity = 1.3855e+01 k_t = [4, 3, 2, 3]
 Generation 1110: Best = 2.345512e+03 FES = 111100 μ_F = 0.265 μ_{CR} = 0.058
 Diversity = 1.3832e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 2.345512e+03 FES = 112100 μ_F = 0.354 μ_{CR} = 0.048
 Diversity = 1.3754e+01 k_t = [3, 2, 4, 4]
 Generation 1130: Best = 2.345512e+03 FES = 113100 μ_F = 0.297 μ_{CR} = 0.048
 Diversity = 1.3752e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 2.345512e+03 FES = 114100 μ_F = 0.306 μ_{CR} = 0.058
 Diversity = 1.3757e+01 k_t = [2, 4, 2, 4]
 Generation 1150: Best = 2.345512e+03 FES = 115100 μ_F = 0.255 μ_{CR} = 0.060
 Diversity = 1.3751e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 2.345512e+03 FES = 116100 μ_F = 0.296 μ_{CR} = 0.050
 Diversity = 1.3787e+01 k_t = [2, 4, 4, 4]
 Generation 1170: Best = 2.344953e+03 FES = 117100 μ_F = 0.351 μ_{CR} = 0.051
 Diversity = 1.3785e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 2.344953e+03 FES = 118100 μ_F = 0.297 μ_{CR} = 0.052
 Diversity = 1.3819e+01 k_t = [3, 4, 3, 2]
 Generation 1190: Best = 2.344953e+03 FES = 119100 μ_F = 0.389 μ_{CR} = 0.052
 Diversity = 1.3819e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 2.342382e+03 FES = 120100 μ_F = 0.337 μ_{CR} = 0.067
 Diversity = 1.3812e+01 k_t = [4, 2, 3, 3]
 Generation 1210: Best = 2.342338e+03 FES = 121100 μ_F = 0.282 μ_{CR} = 0.057
 Diversity = 1.3809e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 2.342252e+03 FES = 122100 μ_F = 0.366 μ_{CR} = 0.045
 Diversity = 1.3813e+01 k_t = [2, 3, 3, 3]
 Generation 1230: Best = 2.342252e+03 FES = 123100 μ_F = 0.389 μ_{CR} = 0.040
 Diversity = 1.3807e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 2.341586e+03 FES = 124100 μ_F = 0.545 μ_{CR} = 0.051
 Diversity = 1.3807e+01 k_t = [3, 4, 3, 4]
 Generation 1250: Best = 2.341586e+03 FES = 125100 μ_F = 0.510 μ_{CR} = 0.054
 Diversity = 1.3812e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 2.341586e+03 FES = 126100 μ_F = 0.445 μ_{CR} = 0.055
 Diversity = 1.3812e+01 k_t = [2, 3, 2, 2]
 Generation 1270: Best = 2.341586e+03 FES = 127100 μ_F = 0.560 μ_{CR} = 0.051
 Diversity = 1.3776e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 2.341586e+03 FES = 128100 μ_F = 0.503 μ_{CR} = 0.047
 Diversity = 1.3774e+01 k_t = [3, 4, 4, 3]
 Generation 1290: Best = 2.341586e+03 FES = 129100 μ_F = 0.487 μ_{CR} = 0.044
 Diversity = 1.3776e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 2.341586e+03 FES = 130100 μ_F = 0.372 μ_{CR} = 0.046
 Diversity = 1.3777e+01 k_t = [2, 2, 2, 2]

Generation 1310: Best = 2.341586e+03 FES = 131100 μ_F = 0.385 μ_{CR} = 0.055
 Diversity = 1.3779e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 2.341586e+03 FES = 132100 μ_F = 0.382 μ_{CR} = 0.047
 Diversity = 1.3778e+01 k_t = [3, 2, 3, 2]
 Generation 1330: Best = 2.341433e+03 FES = 133100 μ_F = 0.420 μ_{CR} = 0.059
 Diversity = 1.3769e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 2.341433e+03 FES = 134100 μ_F = 0.373 μ_{CR} = 0.049
 Diversity = 1.3713e+01 k_t = [2, 2, 4, 4]
 Generation 1350: Best = 2.341433e+03 FES = 135100 μ_F = 0.305 μ_{CR} = 0.043
 Diversity = 1.3708e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 2.341433e+03 FES = 136100 μ_F = 0.326 μ_{CR} = 0.061
 Diversity = 1.3707e+01 k_t = [4, 3, 3, 4]
 Generation 1370: Best = 2.341311e+03 FES = 137100 μ_F = 0.316 μ_{CR} = 0.056
 Diversity = 1.3707e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 2.341311e+03 FES = 138100 μ_F = 0.351 μ_{CR} = 0.062
 Diversity = 1.3704e+01 k_t = [3, 2, 3, 2]
 Generation 1390: Best = 2.341311e+03 FES = 139100 μ_F = 0.353 μ_{CR} = 0.056
 Diversity = 1.3707e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 2.341311e+03 FES = 140100 μ_F = 0.321 μ_{CR} = 0.057
 Diversity = 1.3705e+01 k_t = [3, 2, 4, 2]
 Generation 1410: Best = 2.341119e+03 FES = 141100 μ_F = 0.284 μ_{CR} = 0.049
 Diversity = 1.3704e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 2.341119e+03 FES = 142100 μ_F = 0.303 μ_{CR} = 0.054
 Diversity = 1.3698e+01 k_t = [4, 4, 2, 2]
 Generation 1430: Best = 2.341099e+03 FES = 143100 μ_F = 0.306 μ_{CR} = 0.049
 Diversity = 1.3693e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 2.341099e+03 FES = 144100 μ_F = 0.450 μ_{CR} = 0.060
 Diversity = 1.3502e+01 k_t = [4, 3, 3, 4]
 Generation 1450: Best = 2.341099e+03 FES = 145100 μ_F = 0.460 μ_{CR} = 0.048
 Diversity = 1.3476e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 2.340766e+03 FES = 146100 μ_F = 0.439 μ_{CR} = 0.052
 Diversity = 1.3378e+01 k_t = [4, 3, 2, 4]
 Generation 1470: Best = 2.340766e+03 FES = 147100 μ_F = 0.459 μ_{CR} = 0.045
 Diversity = 1.3379e+01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 2.340766e+03 FES = 148100 μ_F = 0.550 μ_{CR} = 0.040
 Diversity = 1.3350e+01 k_t = [2, 2, 4, 3]
 Generation 1490: Best = 2.339937e+03 FES = 149100 μ_F = 0.632 μ_{CR} = 0.041
 Diversity = 1.3265e+01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 2.339937e+03 FES = 150100 μ_F = 0.615 μ_{CR} = 0.049
 Diversity = 1.3180e+01 k_t = [2, 3, 2, 4]
 Generation 1510: Best = 2.339937e+03 FES = 151100 μ_F = 0.575 μ_{CR} = 0.050
 Diversity = 1.3179e+01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 2.339937e+03 FES = 152100 μ_F = 0.555 μ_{CR} = 0.068
 Diversity = 1.3158e+01 k_t = [4, 3, 2, 4]
 Generation 1530: Best = 2.339933e+03 FES = 153100 μ_F = 0.477 μ_{CR} = 0.064
 Diversity = 1.3157e+01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 2.339754e+03 FES = 154100 μ_F = 0.432 μ_{CR} = 0.094
 Diversity = 1.3154e+01 k_t = [3, 3, 3, 3]
 Generation 1550: Best = 2.339754e+03 FES = 155100 μ_F = 0.457 μ_{CR} = 0.070
 Diversity = 1.3154e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 2.339711e+03 FES = 156100 μ_F = 0.525 μ_{CR} = 0.062
 Diversity = 1.3153e+01 k_t = [3, 2, 3, 3]
 Generation 1570: Best = 2.339600e+03 FES = 157100 μ_F = 0.472 μ_{CR} = 0.059
 Diversity = 1.3152e+01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 2.339600e+03 FES = 158100 μ_F = 0.435 μ_{CR} = 0.060
 Diversity = 1.3154e+01 k_t = [4, 4, 3, 2]
 Generation 1590: Best = 2.339600e+03 FES = 159100 μ_F = 0.492 μ_{CR} = 0.065
 Diversity = 1.3156e+01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 2.339600e+03 FES = 160100 μ_F = 0.500 μ_{CR} = 0.065
 Diversity = 1.3154e+01 k_t = [4, 4, 2, 2]
 Generation 1610: Best = 2.338893e+03 FES = 161100 μ_F = 0.577 μ_{CR} = 0.067
 Diversity = 1.3157e+01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 2.338888e+03 FES = 162100 μ_F = 0.617 μ_{CR} = 0.047
 Diversity = 1.3169e+01 k_t = [3, 3, 2, 4]
 Generation 1630: Best = 2.338888e+03 FES = 163100 μ_F = 0.541 μ_{CR} = 0.056
 Diversity = 1.3169e+01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 2.338466e+03 FES = 164100 μ_F = 0.599 μ_{CR} = 0.056
 Diversity = 1.3165e+01 k_t = [4, 4, 3, 3]
 Generation 1650: Best = 2.338466e+03 FES = 165100 μ_F = 0.532 μ_{CR} = 0.063
 Diversity = 1.3098e+01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 2.338426e+03 FES = 166100 μ_F = 0.546 μ_{CR} = 0.068
 Diversity = 1.3096e+01 k_t = [4, 4, 3, 2]
 Generation 1670: Best = 2.338408e+03 FES = 167100 μ_F = 0.545 μ_{CR} = 0.061
 Diversity = 1.3097e+01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 2.337360e+03 FES = 168100 μ_F = 0.485 μ_{CR} = 0.078
 Diversity = 1.2995e+01 k_t = [4, 3, 4, 4]
 Generation 1690: Best = 2.337360e+03 FES = 169100 μ_F = 0.496 μ_{CR} = 0.052
 Diversity = 1.2994e+01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 2.337360e+03 FES = 170100 μ_F = 0.518 μ_{CR} = 0.063
 Diversity = 1.2864e+01 k_t = [4, 2, 3, 4]
 Generation 1710: Best = 2.337302e+03 FES = 171100 μ_F = 0.621 μ_{CR} = 0.058
 Diversity = 1.2810e+01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 2.337302e+03 FES = 172100 μ_F = 0.548 μ_{CR} = 0.055
 Diversity = 1.2810e+01 k_t = [3, 3, 3, 3]
 Generation 1730: Best = 2.337302e+03 FES = 173100 μ_F = 0.604 μ_{CR} = 0.047
 Diversity = 1.2810e+01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 2.337056e+03 FES = 174100 μ_F = 0.618 μ_{CR} = 0.050
 Diversity = 1.2732e+01 k_t = [3, 4, 2, 4]
 Generation 1750: Best = 2.337036e+03 FES = 175100 μ_F = 0.592 μ_{CR} = 0.051
 Diversity = 1.2498e+01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 2.337036e+03 FES = 176100 μ_F = 0.569 μ_{CR} = 0.068
 Diversity = 1.2500e+01 k_t = [2, 2, 2, 2]
 Generation 1770: Best = 2.337036e+03 FES = 177100 μ_F = 0.527 μ_{CR} = 0.066
 Diversity = 1.2503e+01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 2.336914e+03 FES = 178100 μ_F = 0.560 μ_{CR} = 0.043
 Diversity = 1.2503e+01 k_t = [3, 3, 3, 3]
 Generation 1790: Best = 2.336908e+03 FES = 179100 μ_F = 0.518 μ_{CR} = 0.044
 Diversity = 1.2435e+01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 2.336908e+03 FES = 180100 μ_F = 0.531 μ_{CR} = 0.049
 Diversity = 1.2433e+01 k_t = [3, 3, 3, 2]
 Generation 1810: Best = 2.336908e+03 FES = 181100 μ_F = 0.526 μ_{CR} = 0.044
 Diversity = 1.2406e+01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 2.336908e+03 FES = 182100 μ_F = 0.534 μ_{CR} = 0.065
 Diversity = 1.2400e+01 k_t = [4, 3, 2, 3]
 Generation 1830: Best = 2.336908e+03 FES = 183100 μ_F = 0.529 μ_{CR} = 0.060
 Diversity = 1.2366e+01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 2.336908e+03 FES = 184100 μ_F = 0.493 μ_{CR} = 0.057
 Diversity = 1.2283e+01 k_t = [2, 3, 2, 3]

Generation 1850: Best = 2.336908e+03	FES = 185100	$\mu_F = 0.449$	$\mu_{CR} = 0.051$
Diversity = 1.2255e+01	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 2.336908e+03	FES = 186100	$\mu_F = 0.401$	$\mu_{CR} = 0.081$
Diversity = 1.2184e+01	k_t = [2, 4, 3, 3]		
Generation 1870: Best = 2.336908e+03	FES = 187100	$\mu_F = 0.391$	$\mu_{CR} = 0.076$
Diversity = 1.2182e+01	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 2.336908e+03	FES = 188100	$\mu_F = 0.487$	$\mu_{CR} = 0.077$
Diversity = 1.2225e+01	k_t = [2, 4, 2, 2]		
Generation 1890: Best = 2.336127e+03	FES = 189100	$\mu_F = 0.555$	$\mu_{CR} = 0.061$
Diversity = 1.2216e+01	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 2.336127e+03	FES = 190100	$\mu_F = 0.551$	$\mu_{CR} = 0.072$
Diversity = 1.2175e+01	k_t = [3, 4, 3, 4]		
Generation 1910: Best = 2.336122e+03	FES = 191100	$\mu_F = 0.512$	$\mu_{CR} = 0.085$
Diversity = 1.2173e+01	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 2.336122e+03	FES = 192100	$\mu_F = 0.525$	$\mu_{CR} = 0.085$
Diversity = 1.2037e+01	k_t = [3, 2, 4, 4]		
Generation 1930: Best = 2.336122e+03	FES = 193100	$\mu_F = 0.447$	$\mu_{CR} = 0.087$
Diversity = 1.2036e+01	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 2.335951e+03	FES = 194100	$\mu_F = 0.465$	$\mu_{CR} = 0.072$
Diversity = 1.2037e+01	k_t = [4, 4, 4, 3]		
Generation 1950: Best = 2.335951e+03	FES = 195100	$\mu_F = 0.447$	$\mu_{CR} = 0.080$
Diversity = 1.2036e+01	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 2.335951e+03	FES = 196100	$\mu_F = 0.436$	$\mu_{CR} = 0.105$
Diversity = 1.2037e+01	k_t = [2, 4, 3, 3]		
Generation 1970: Best = 2.335951e+03	FES = 197100	$\mu_F = 0.486$	$\mu_{CR} = 0.115$
Diversity = 1.2036e+01	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 2.335951e+03	FES = 198100	$\mu_F = 0.479$	$\mu_{CR} = 0.098$
Diversity = 1.1979e+01	k_t = [2, 2, 4, 4]		
Generation 1990: Best = 2.335951e+03	FES = 199100	$\mu_F = 0.475$	$\mu_{CR} = 0.092$
Diversity = 1.1980e+01	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 2.335951e+03	FES = 200100	$\mu_F = 0.600$	$\mu_{CR} = 0.093$
Diversity = 1.1895e+01	k_t = [3, 4, 4, 2]		
Generation 2010: Best = 2.335951e+03	FES = 201100	$\mu_F = 0.632$	$\mu_{CR} = 0.099$
Diversity = 1.1894e+01	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 2.335951e+03	FES = 202100	$\mu_F = 0.642$	$\mu_{CR} = 0.094$
Diversity = 1.1893e+01	k_t = [3, 4, 3, 2]		
Generation 2030: Best = 2.335951e+03	FES = 203100	$\mu_F = 0.719$	$\mu_{CR} = 0.103$
Diversity = 1.1879e+01	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 2.335951e+03	FES = 204100	$\mu_F = 0.753$	$\mu_{CR} = 0.096$
Diversity = 1.1883e+01	k_t = [3, 2, 3, 2]		
Generation 2050: Best = 2.335951e+03	FES = 205100	$\mu_F = 0.765$	$\mu_{CR} = 0.106$
Diversity = 1.1893e+01	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 2.335951e+03	FES = 206100	$\mu_F = 0.759$	$\mu_{CR} = 0.102$
Diversity = 1.1569e+01	k_t = [3, 3, 4, 4]		
Generation 2070: Best = 2.335925e+03	FES = 207100	$\mu_F = 0.813$	$\mu_{CR} = 0.099$
Diversity = 1.1607e+01	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 2.335925e+03	FES = 208100	$\mu_F = 0.801$	$\mu_{CR} = 0.116$
Diversity = 1.1583e+01	k_t = [4, 2, 2, 2]		
Generation 2090: Best = 2.335925e+03	FES = 209100	$\mu_F = 0.764$	$\mu_{CR} = 0.116$
Diversity = 1.1591e+01	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 2.335925e+03	FES = 210100	$\mu_F = 0.763$	$\mu_{CR} = 0.125$
Diversity = 1.1666e+01	k_t = [4, 4, 3, 3]		
Generation 2110: Best = 2.335925e+03	FES = 211100	$\mu_F = 0.781$	$\mu_{CR} = 0.132$
Diversity = 1.1622e+01	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 2.335439e+03 FES = 212100 μ_F = 0.766 μ_{CR} = 0.145
 Diversity = 1.1524e+01 k_t = [4, 2, 2, 3]
 Generation 2130: Best = 2.335369e+03 FES = 213100 μ_F = 0.751 μ_{CR} = 0.148
 Diversity = 1.1467e+01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 2.335306e+03 FES = 214100 μ_F = 0.797 μ_{CR} = 0.154
 Diversity = 1.1283e+01 k_t = [3, 3, 4, 4]
 Generation 2150: Best = 2.335306e+03 FES = 215100 μ_F = 0.823 μ_{CR} = 0.169
 Diversity = 1.1245e+01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 2.335306e+03 FES = 216100 μ_F = 0.824 μ_{CR} = 0.139
 Diversity = 1.1174e+01 k_t = [4, 4, 4, 2]
 Generation 2170: Best = 2.335279e+03 FES = 217100 μ_F = 0.856 μ_{CR} = 0.152
 Diversity = 1.0644e+01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 2.335279e+03 FES = 218100 μ_F = 0.885 μ_{CR} = 0.156
 Diversity = 9.6498e+00 k_t = [3, 3, 3, 3]
 Generation 2190: Best = 2.334979e+03 FES = 219100 μ_F = 0.891 μ_{CR} = 0.153
 Diversity = 7.8988e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 2.334734e+03 FES = 220100 μ_F = 0.890 μ_{CR} = 0.164
 Diversity = 4.8090e+00 k_t = [4, 3, 2, 3]
 Generation 2210: Best = 2.334678e+03 FES = 221100 μ_F = 0.863 μ_{CR} = 0.173
 Diversity = 3.9329e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 2.334330e+03 FES = 222100 μ_F = 0.829 μ_{CR} = 0.151
 Diversity = 3.7550e+00 k_t = [4, 2, 2, 2]
 Generation 2230: Best = 2.334275e+03 FES = 223100 μ_F = 0.827 μ_{CR} = 0.155
 Diversity = 3.7279e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 2.334275e+03 FES = 224100 μ_F = 0.794 μ_{CR} = 0.152
 Diversity = 4.1139e+00 k_t = [3, 2, 3, 4]
 Generation 2250: Best = 2.334191e+03 FES = 225100 μ_F = 0.785 μ_{CR} = 0.167
 Diversity = 4.0980e+00 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 2.334079e+03 FES = 226100 μ_F = 0.719 μ_{CR} = 0.161
 Diversity = 4.0624e+00 k_t = [2, 4, 3, 4]
 Generation 2270: Best = 2.334079e+03 FES = 227100 μ_F = 0.756 μ_{CR} = 0.131
 Diversity = 4.0619e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 2.334079e+03 FES = 228100 μ_F = 0.730 μ_{CR} = 0.124
 Diversity = 4.0530e+00 k_t = [3, 2, 2, 4]
 Generation 2290: Best = 2.333965e+03 FES = 229100 μ_F = 0.724 μ_{CR} = 0.165
 Diversity = 4.0458e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 2.333965e+03 FES = 230100 μ_F = 0.720 μ_{CR} = 0.172
 Diversity = 4.0525e+00 k_t = [2, 2, 3, 2]
 Generation 2310: Best = 2.333860e+03 FES = 231100 μ_F = 0.703 μ_{CR} = 0.182
 Diversity = 4.0566e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 2.333591e+03 FES = 232100 μ_F = 0.634 μ_{CR} = 0.180
 Diversity = 3.6079e+00 k_t = [4, 2, 3, 4]
 Generation 2330: Best = 2.333591e+03 FES = 233100 μ_F = 0.691 μ_{CR} = 0.203
 Diversity = 3.6092e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 2.333551e+03 FES = 234100 μ_F = 0.627 μ_{CR} = 0.183
 Diversity = 3.7939e+00 k_t = [3, 3, 4, 3]
 Generation 2350: Best = 2.333545e+03 FES = 235100 μ_F = 0.639 μ_{CR} = 0.159
 Diversity = 3.8005e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 2.333221e+03 FES = 236100 μ_F = 0.583 μ_{CR} = 0.170
 Diversity = 3.7889e+00 k_t = [4, 3, 2, 2]
 Generation 2370: Best = 2.333221e+03 FES = 237100 μ_F = 0.612 μ_{CR} = 0.184
 Diversity = 3.9692e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 2.332683e+03 FES = 238100 μ_F = 0.641 μ_{CR} = 0.180
 Diversity = 3.9664e+00 k_t = [4, 2, 3, 3]

Generation 2390: Best = 2.332683e+03 FES = 239100 μ_F = 0.643 μ_{CR} = 0.187
 Diversity = 4.1273e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 2.332683e+03 FES = 240100 μ_F = 0.643 μ_{CR} = 0.200
 Diversity = 4.2791e+00 k_t = [3, 3, 4, 2]
 Generation 2410: Best = 2.332683e+03 FES = 241100 μ_F = 0.655 μ_{CR} = 0.181
 Diversity = 4.2741e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 2.332357e+03 FES = 242100 μ_F = 0.620 μ_{CR} = 0.168
 Diversity = 4.2683e+00 k_t = [3, 4, 3, 3]
 Generation 2430: Best = 2.331989e+03 FES = 243100 μ_F = 0.653 μ_{CR} = 0.165
 Diversity = 4.4096e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 2.331910e+03 FES = 244100 μ_F = 0.643 μ_{CR} = 0.171
 Diversity = 4.5202e+00 k_t = [2, 2, 2, 4]
 Generation 2450: Best = 2.331565e+03 FES = 245100 μ_F = 0.550 μ_{CR} = 0.174
 Diversity = 4.6491e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 2.331565e+03 FES = 246100 μ_F = 0.573 μ_{CR} = 0.180
 Diversity = 4.8876e+00 k_t = [2, 4, 2, 4]
 Generation 2470: Best = 2.331222e+03 FES = 247100 μ_F = 0.576 μ_{CR} = 0.179
 Diversity = 5.3108e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 2.330992e+03 FES = 248100 μ_F = 0.648 μ_{CR} = 0.203
 Diversity = 5.5826e+00 k_t = [2, 2, 3, 3]
 Generation 2490: Best = 2.330440e+03 FES = 249100 μ_F = 0.680 μ_{CR} = 0.194
 Diversity = 5.6910e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 2.330136e+03 FES = 250100 μ_F = 0.717 μ_{CR} = 0.187
 Diversity = 5.7726e+00 k_t = [4, 3, 3, 3]
 Generation 2510: Best = 2.329837e+03 FES = 251100 μ_F = 0.743 μ_{CR} = 0.208
 Diversity = 5.7945e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 2.329771e+03 FES = 252100 μ_F = 0.784 μ_{CR} = 0.192
 Diversity = 6.1042e+00 k_t = [2, 4, 3, 4]
 Generation 2530: Best = 2.329591e+03 FES = 253100 μ_F = 0.803 μ_{CR} = 0.181
 Diversity = 6.2824e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 2.329515e+03 FES = 254100 μ_F = 0.818 μ_{CR} = 0.193
 Diversity = 6.3433e+00 k_t = [3, 2, 2, 3]
 Generation 2550: Best = 2.329358e+03 FES = 255100 μ_F = 0.821 μ_{CR} = 0.192
 Diversity = 6.3418e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 2.329263e+03 FES = 256100 μ_F = 0.825 μ_{CR} = 0.197
 Diversity = 6.1150e+00 k_t = [4, 4, 2, 3]
 Generation 2570: Best = 2.329124e+03 FES = 257100 μ_F = 0.826 μ_{CR} = 0.198
 Diversity = 4.8986e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 2.329083e+03 FES = 258100 μ_F = 0.820 μ_{CR} = 0.202
 Diversity = 2.3136e+00 k_t = [2, 2, 4, 3]
 Generation 2590: Best = 2.329077e+03 FES = 259100 μ_F = 0.817 μ_{CR} = 0.199
 Diversity = 1.3883e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 2.329057e+03 FES = 260100 μ_F = 0.818 μ_{CR} = 0.192
 Diversity = 1.3299e+00 k_t = [2, 2, 4, 4]
 Generation 2610: Best = 2.329039e+03 FES = 261100 μ_F = 0.811 μ_{CR} = 0.184
 Diversity = 1.7049e-01 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 2.329027e+03 FES = 262100 μ_F = 0.804 μ_{CR} = 0.179
 Diversity = 1.3433e-01 k_t = [4, 2, 4, 3]
 Generation 2630: Best = 2.329020e+03 FES = 263100 μ_F = 0.782 μ_{CR} = 0.182
 Diversity = 1.0245e-01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 2.329013e+03 FES = 264100 μ_F = 0.781 μ_{CR} = 0.177
 Diversity = 7.6071e-02 k_t = [3, 3, 3, 4]
 Generation 2650: Best = 2.329009e+03 FES = 265100 μ_F = 0.771 μ_{CR} = 0.173
 Diversity = 6.3956e-02 k_t = [1, 1, 1, 1]

Generation 2660: Best = 2.329006e+03 FES = 266100 μ_F = 0.765 μ_{CR} = 0.175
 Diversity = 5.0937e-02 k_t = [4, 2, 2, 3]
 Generation 2670: Best = 2.329005e+03 FES = 267100 μ_F = 0.775 μ_{CR} = 0.184
 Diversity = 5.0214e-02 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 2.329004e+03 FES = 268100 μ_F = 0.790 μ_{CR} = 0.173
 Diversity = 3.2549e-02 k_t = [4, 2, 3, 4]
 Generation 2690: Best = 2.329004e+03 FES = 269100 μ_F = 0.778 μ_{CR} = 0.170
 Diversity = 2.4949e-02 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 2.329003e+03 FES = 270100 μ_F = 0.793 μ_{CR} = 0.177
 Diversity = 2.1443e-02 k_t = [3, 3, 3, 4]
 Generation 2710: Best = 2.329003e+03 FES = 271100 μ_F = 0.800 μ_{CR} = 0.168
 Diversity = 1.7207e-02 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 2.329003e+03 FES = 272100 μ_F = 0.786 μ_{CR} = 0.172
 Diversity = 1.2747e-02 k_t = [3, 4, 2, 4]
 Generation 2730: Best = 2.329003e+03 FES = 273100 μ_F = 0.783 μ_{CR} = 0.167
 Diversity = 1.0692e-02 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 2.329003e+03 FES = 274100 μ_F = 0.764 μ_{CR} = 0.171
 Diversity = 8.3970e-03 k_t = [3, 2, 2, 4]
 Generation 2750: Best = 2.329003e+03 FES = 275100 μ_F = 0.764 μ_{CR} = 0.160
 Diversity = 7.7530e-03 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 2.329003e+03 FES = 276100 μ_F = 0.759 μ_{CR} = 0.168
 Diversity = 6.1023e-03 k_t = [2, 3, 2, 3]
 Generation 2770: Best = 2.329003e+03 FES = 277100 μ_F = 0.756 μ_{CR} = 0.169
 Diversity = 5.1391e-03 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 2.329003e+03 FES = 278100 μ_F = 0.759 μ_{CR} = 0.168
 Diversity = 3.8610e-03 k_t = [4, 4, 4, 3]
 Generation 2790: Best = 2.329003e+03 FES = 279100 μ_F = 0.754 μ_{CR} = 0.177
 Diversity = 2.8447e-03 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 2.329003e+03 FES = 280100 μ_F = 0.741 μ_{CR} = 0.175
 Diversity = 2.5447e-03 k_t = [3, 2, 2, 3]
 Generation 2810: Best = 2.329003e+03 FES = 281100 μ_F = 0.735 μ_{CR} = 0.175
 Diversity = 2.2840e-03 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 2.329003e+03 FES = 282100 μ_F = 0.751 μ_{CR} = 0.164
 Diversity = 1.6478e-03 k_t = [2, 4, 3, 2]
 Generation 2830: Best = 2.329003e+03 FES = 283100 μ_F = 0.747 μ_{CR} = 0.164
 Diversity = 1.4425e-03 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 2.329003e+03 FES = 284100 μ_F = 0.751 μ_{CR} = 0.163
 Diversity = 1.1333e-03 k_t = [4, 4, 4, 3]
 Generation 2850: Best = 2.329003e+03 FES = 285100 μ_F = 0.759 μ_{CR} = 0.163
 Diversity = 9.4782e-04 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 2.329003e+03 FES = 286100 μ_F = 0.760 μ_{CR} = 0.166
 Diversity = 8.3387e-04 k_t = [3, 4, 4, 3]
 Generation 2870: Best = 2.329003e+03 FES = 287100 μ_F = 0.762 μ_{CR} = 0.157
 Diversity = 6.7438e-04 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 2.329003e+03 FES = 288100 μ_F = 0.760 μ_{CR} = 0.158
 Diversity = 5.5511e-04 k_t = [3, 2, 3, 4]
 Generation 2890: Best = 2.329003e+03 FES = 289100 μ_F = 0.759 μ_{CR} = 0.164
 Diversity = 4.6188e-04 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 2.329003e+03 FES = 290100 μ_F = 0.761 μ_{CR} = 0.168
 Diversity = 3.9904e-04 k_t = [3, 4, 4, 4]
 Generation 2910: Best = 2.329003e+03 FES = 291100 μ_F = 0.763 μ_{CR} = 0.174
 Diversity = 3.3139e-04 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 2.329003e+03 FES = 292100 μ_F = 0.775 μ_{CR} = 0.174
 Diversity = 2.8923e-04 k_t = [4, 4, 4, 4]

Generation 2930: Best = 2.329003e+03 FES = 293100 μ_F = 0.769 μ_{CR} = 0.169
 Diversity = 2.5127e-04 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 2.329003e+03 FES = 294100 μ_F = 0.775 μ_{CR} = 0.175
 Diversity = 2.1648e-04 k_t = [4, 4, 3, 2]
 Generation 2950: Best = 2.329003e+03 FES = 295100 μ_F = 0.784 μ_{CR} = 0.178
 Diversity = 2.2165e-04 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 2.329003e+03 FES = 296100 μ_F = 0.776 μ_{CR} = 0.182
 Diversity = 1.9559e-04 k_t = [3, 4, 2, 2]
 Generation 2970: Best = 2.329003e+03 FES = 297100 μ_F = 0.774 μ_{CR} = 0.176
 Diversity = 1.8190e-04 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 2.329003e+03 FES = 298100 μ_F = 0.780 μ_{CR} = 0.176
 Diversity = 1.6606e-04 k_t = [4, 2, 3, 3]
 Generation 2990: Best = 2.329003e+03 FES = 299100 μ_F = 0.780 μ_{CR} = 0.184
 Diversity = 1.5288e-04 k_t = [1, 1, 1, 1]

Optimization complete.

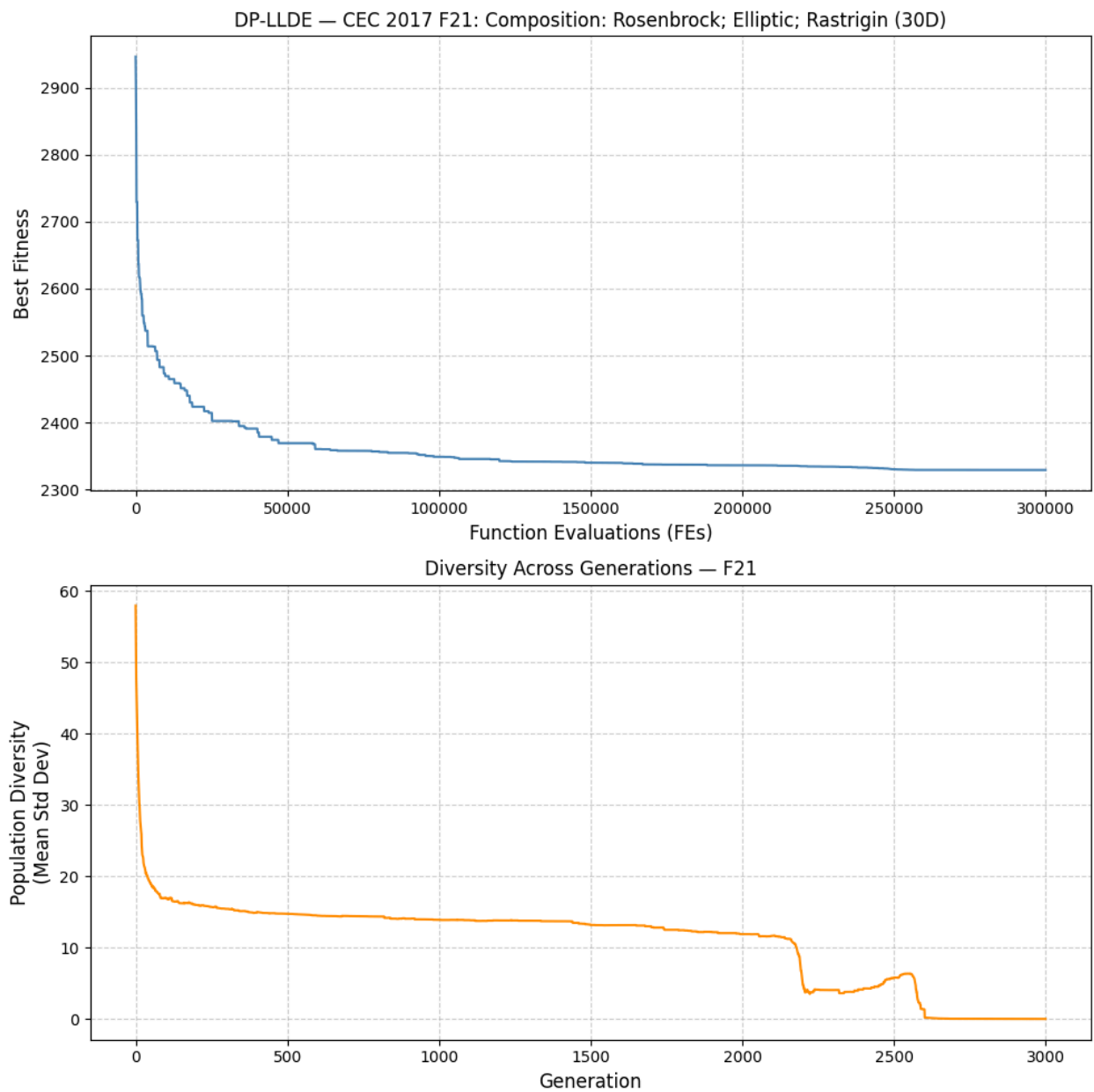
Best Fitness = 2.329003e+03

Total Function Evals = 300000

Total Successful Updates = 26846

```

=====
=
Results:
  Best Fitness      : 2.329003e+03
  Global Optimum    : 2.100000e+03
  Error (|best-optimum|) : 2.290028e+02
  Best Solution (5 dim) : [-28.49680138 -12.9733929  72.74275106  11.82683228
52.60066877]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F21.png

```
In [ ]: # =====
# F22 – Composition Function 2 (Rastrigin; Griewank; Schwefel)
# =====
run_and_plot('F22', make_wrapper(composition.f22), 2200.0, 'Composition: Rastr
```

=====

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DP-LLDE on CEC 2017 F22 – Single Run

=====

=

Function : Composition: Rastrigin; Griewank; Schwefel (30D)
Bounds : [-100, 100]
Global optimum : 2200.0
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 1.151639e+04
Generation 10: Best = 4.315370e+03 FES = 1100 $\mu_F = 0.564$ $\mu_{CR} = 0.349$
Diversity = 3.9273e+01 $k_t = [1, 1, 1, 1]$
Generation 20: Best = 3.541841e+03 FES = 2100 $\mu_F = 0.616$ $\mu_{CR} = 0.349$
Diversity = 1.4995e+01 $k_t = [4, 4, 2, 4]$
Generation 30: Best = 2.734035e+03 FES = 3100 $\mu_F = 0.646$ $\mu_{CR} = 0.354$
Diversity = 7.0032e+00 $k_t = [1, 1, 1, 1]$
Generation 40: Best = 2.519533e+03 FES = 4100 $\mu_F = 0.674$ $\mu_{CR} = 0.354$
Diversity = 3.9056e+00 $k_t = [2, 4, 4, 2]$
Generation 50: Best = 2.411690e+03 FES = 5100 $\mu_F = 0.699$ $\mu_{CR} = 0.356$
Diversity = 2.5751e+00 $k_t = [1, 1, 1, 1]$
Generation 60: Best = 2.364489e+03 FES = 6100 $\mu_F = 0.704$ $\mu_{CR} = 0.365$
Diversity = 1.6279e+00 $k_t = [3, 3, 2, 4]$
Generation 70: Best = 2.345123e+03 FES = 7100 $\mu_F = 0.718$ $\mu_{CR} = 0.368$
Diversity = 1.0316e+00 $k_t = [1, 1, 1, 1]$
Generation 80: Best = 2.329467e+03 FES = 8100 $\mu_F = 0.730$ $\mu_{CR} = 0.379$
Diversity = 7.2325e-01 $k_t = [2, 2, 4, 3]$
Generation 90: Best = 2.322013e+03 FES = 9100 $\mu_F = 0.712$ $\mu_{CR} = 0.379$
Diversity = 6.2950e-01 $k_t = [1, 1, 1, 1]$
Generation 100: Best = 2.317982e+03 FES = 10100 $\mu_F = 0.709$ $\mu_{CR} = 0.377$
Diversity = 4.4582e-01 $k_t = [2, 2, 3, 3]$
Generation 110: Best = 2.315816e+03 FES = 11100 $\mu_F = 0.684$ $\mu_{CR} = 0.375$
Diversity = 3.8176e-01 $k_t = [1, 1, 1, 1]$
Generation 120: Best = 2.312288e+03 FES = 12100 $\mu_F = 0.674$ $\mu_{CR} = 0.351$
Diversity = 3.2890e-01 $k_t = [4, 3, 3, 2]$
Generation 130: Best = 2.310700e+03 FES = 13100 $\mu_F = 0.660$ $\mu_{CR} = 0.329$
Diversity = 2.9296e-01 $k_t = [1, 1, 1, 1]$
Generation 140: Best = 2.308266e+03 FES = 14100 $\mu_F = 0.634$ $\mu_{CR} = 0.318$
Diversity = 2.2876e-01 $k_t = [3, 4, 4, 4]$
Generation 150: Best = 2.307379e+03 FES = 15100 $\mu_F = 0.646$ $\mu_{CR} = 0.310$
Diversity = 1.7561e-01 $k_t = [1, 1, 1, 1]$
Generation 160: Best = 2.306610e+03 FES = 16100 $\mu_F = 0.635$ $\mu_{CR} = 0.296$
Diversity = 1.2844e-01 $k_t = [3, 4, 3, 2]$
Generation 170: Best = 2.306091e+03 FES = 17100 $\mu_F = 0.629$ $\mu_{CR} = 0.276$
Diversity = 9.7589e-02 $k_t = [1, 1, 1, 1]$
Generation 180: Best = 2.305656e+03 FES = 18100 $\mu_F = 0.623$ $\mu_{CR} = 0.277$
Diversity = 7.0133e-02 $k_t = [4, 3, 4, 2]$
Generation 190: Best = 2.305343e+03 FES = 19100 $\mu_F = 0.613$ $\mu_{CR} = 0.273$
Diversity = 5.2515e-02 $k_t = [1, 1, 1, 1]$
Generation 200: Best = 2.305152e+03 FES = 20100 $\mu_F = 0.624$ $\mu_{CR} = 0.274$
Diversity = 3.5419e-02 $k_t = [3, 4, 3, 3]$
Generation 210: Best = 2.305002e+03 FES = 21100 $\mu_F = 0.645$ $\mu_{CR} = 0.275$
Diversity = 2.6810e-02 $k_t = [1, 1, 1, 1]$
Generation 220: Best = 2.304886e+03 FES = 22100 $\mu_F = 0.668$ $\mu_{CR} = 0.276$
Diversity = 2.0219e-02 $k_t = [2, 2, 2, 4]$

Generation	230:	Best = 2.304824e+03	FES = 23100	$\mu_F = 0.680$	$\mu_{CR} = 0.275$
Diversity	= 1.9044e-02	$k_t = [1, 1, 1, 1]$			
Generation	240:	Best = 2.304748e+03	FES = 24100	$\mu_F = 0.694$	$\mu_{CR} = 0.274$
Diversity	= 1.5457e-02	$k_t = [2, 4, 4, 4]$			
Generation	250:	Best = 2.304670e+03	FES = 25100	$\mu_F = 0.711$	$\mu_{CR} = 0.268$
Diversity	= 1.3410e-02	$k_t = [1, 1, 1, 1]$			
Generation	260:	Best = 2.304573e+03	FES = 26100	$\mu_F = 0.730$	$\mu_{CR} = 0.266$
Diversity	= 1.3920e-02	$k_t = [4, 3, 2, 4]$			
Generation	270:	Best = 2.304508e+03	FES = 27100	$\mu_F = 0.750$	$\mu_{CR} = 0.264$
Diversity	= 1.4254e-02	$k_t = [1, 1, 1, 1]$			
Generation	280:	Best = 2.304441e+03	FES = 28100	$\mu_F = 0.757$	$\mu_{CR} = 0.261$
Diversity	= 1.3790e-02	$k_t = [3, 2, 3, 2]$			
Generation	290:	Best = 2.304397e+03	FES = 29100	$\mu_F = 0.768$	$\mu_{CR} = 0.261$
Diversity	= 1.3396e-02	$k_t = [1, 1, 1, 1]$			
Generation	300:	Best = 2.304316e+03	FES = 30100	$\mu_F = 0.773$	$\mu_{CR} = 0.272$
Diversity	= 1.2618e-02	$k_t = [3, 2, 4, 4]$			
Generation	310:	Best = 2.304264e+03	FES = 31100	$\mu_F = 0.772$	$\mu_{CR} = 0.269$
Diversity	= 1.4922e-02	$k_t = [1, 1, 1, 1]$			
Generation	320:	Best = 2.304201e+03	FES = 32100	$\mu_F = 0.772$	$\mu_{CR} = 0.266$
Diversity	= 1.4726e-02	$k_t = [3, 4, 2, 4]$			
Generation	330:	Best = 2.304173e+03	FES = 33100	$\mu_F = 0.775$	$\mu_{CR} = 0.270$
Diversity	= 1.3784e-02	$k_t = [1, 1, 1, 1]$			
Generation	340:	Best = 2.304130e+03	FES = 34100	$\mu_F = 0.768$	$\mu_{CR} = 0.272$
Diversity	= 1.2478e-02	$k_t = [3, 3, 2, 2]$			
Generation	350:	Best = 2.304100e+03	FES = 35100	$\mu_F = 0.768$	$\mu_{CR} = 0.261$
Diversity	= 1.1293e-02	$k_t = [1, 1, 1, 1]$			
Generation	360:	Best = 2.304066e+03	FES = 36100	$\mu_F = 0.768$	$\mu_{CR} = 0.257$
Diversity	= 1.0058e-02	$k_t = [2, 2, 3, 4]$			
Generation	370:	Best = 2.304037e+03	FES = 37100	$\mu_F = 0.772$	$\mu_{CR} = 0.259$
Diversity	= 1.0141e-02	$k_t = [1, 1, 1, 1]$			
Generation	380:	Best = 2.303996e+03	FES = 38100	$\mu_F = 0.771$	$\mu_{CR} = 0.260$
Diversity	= 1.0035e-02	$k_t = [3, 4, 3, 2]$			
Generation	390:	Best = 2.303962e+03	FES = 39100	$\mu_F = 0.784$	$\mu_{CR} = 0.264$
Diversity	= 1.0857e-02	$k_t = [1, 1, 1, 1]$			
Generation	400:	Best = 2.303925e+03	FES = 40100	$\mu_F = 0.776$	$\mu_{CR} = 0.258$
Diversity	= 1.0801e-02	$k_t = [4, 3, 2, 3]$			
Generation	410:	Best = 2.303902e+03	FES = 41100	$\mu_F = 0.766$	$\mu_{CR} = 0.265$
Diversity	= 1.0509e-02	$k_t = [1, 1, 1, 1]$			
Generation	420:	Best = 2.303879e+03	FES = 42100	$\mu_F = 0.769$	$\mu_{CR} = 0.259$
Diversity	= 9.2556e-03	$k_t = [2, 2, 3, 4]$			
Generation	430:	Best = 2.303849e+03	FES = 43100	$\mu_F = 0.781$	$\mu_{CR} = 0.255$
Diversity	= 9.5806e-03	$k_t = [1, 1, 1, 1]$			
Generation	440:	Best = 2.303833e+03	FES = 44100	$\mu_F = 0.775$	$\mu_{CR} = 0.251$
Diversity	= 8.8900e-03	$k_t = [4, 4, 3, 3]$			
Generation	450:	Best = 2.303815e+03	FES = 45100	$\mu_F = 0.773$	$\mu_{CR} = 0.259$
Diversity	= 8.9923e-03	$k_t = [1, 1, 1, 1]$			
Generation	460:	Best = 2.303798e+03	FES = 46100	$\mu_F = 0.765$	$\mu_{CR} = 0.253$
Diversity	= 8.5039e-03	$k_t = [2, 2, 3, 2]$			
Generation	470:	Best = 2.303789e+03	FES = 47100	$\mu_F = 0.765$	$\mu_{CR} = 0.246$
Diversity	= 8.0599e-03	$k_t = [1, 1, 1, 1]$			
Generation	480:	Best = 2.303775e+03	FES = 48100	$\mu_F = 0.756$	$\mu_{CR} = 0.243$
Diversity	= 7.6863e-03	$k_t = [4, 4, 2, 2]$			
Generation	490:	Best = 2.303764e+03	FES = 49100	$\mu_F = 0.756$	$\mu_{CR} = 0.242$
Diversity	= 7.2630e-03	$k_t = [1, 1, 1, 1]$			

Generation 500: Best = 2.303750e+03	FES = 50100	$\mu_F = 0.753$	$\mu_{CR} = 0.249$
Diversity = 6.8912e-03	$k_t = [2, 2, 4, 3]$		
Generation 510: Best = 2.303744e+03	FES = 51100	$\mu_F = 0.761$	$\mu_{CR} = 0.245$
Diversity = 6.6245e-03	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.303737e+03	FES = 52100	$\mu_F = 0.755$	$\mu_{CR} = 0.245$
Diversity = 5.8908e-03	$k_t = [4, 3, 3, 2]$		
Generation 530: Best = 2.303728e+03	FES = 53100	$\mu_F = 0.760$	$\mu_{CR} = 0.246$
Diversity = 5.6526e-03	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.303722e+03	FES = 54100	$\mu_F = 0.761$	$\mu_{CR} = 0.243$
Diversity = 4.9036e-03	$k_t = [3, 4, 2, 4]$		
Generation 550: Best = 2.303721e+03	FES = 55100	$\mu_F = 0.752$	$\mu_{CR} = 0.241$
Diversity = 4.8515e-03	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.303715e+03	FES = 56100	$\mu_F = 0.754$	$\mu_{CR} = 0.246$
Diversity = 4.5966e-03	$k_t = [2, 3, 2, 3]$		
Generation 570: Best = 2.303712e+03	FES = 57100	$\mu_F = 0.741$	$\mu_{CR} = 0.252$
Diversity = 4.3360e-03	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.303708e+03	FES = 58100	$\mu_F = 0.744$	$\mu_{CR} = 0.251$
Diversity = 4.0079e-03	$k_t = [4, 3, 3, 2]$		
Generation 590: Best = 2.303703e+03	FES = 59100	$\mu_F = 0.753$	$\mu_{CR} = 0.261$
Diversity = 4.1273e-03	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.303701e+03	FES = 60100	$\mu_F = 0.750$	$\mu_{CR} = 0.264$
Diversity = 3.5073e-03	$k_t = [4, 4, 4, 4]$		
Generation 610: Best = 2.303697e+03	FES = 61100	$\mu_F = 0.757$	$\mu_{CR} = 0.260$
Diversity = 3.4578e-03	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.303694e+03	FES = 62100	$\mu_F = 0.752$	$\mu_{CR} = 0.254$
Diversity = 3.4877e-03	$k_t = [3, 3, 2, 2]$		
Generation 630: Best = 2.303690e+03	FES = 63100	$\mu_F = 0.751$	$\mu_{CR} = 0.255$
Diversity = 3.4709e-03	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.303689e+03	FES = 64100	$\mu_F = 0.744$	$\mu_{CR} = 0.244$
Diversity = 3.1937e-03	$k_t = [2, 4, 2, 4]$		
Generation 650: Best = 2.303687e+03	FES = 65100	$\mu_F = 0.732$	$\mu_{CR} = 0.241$
Diversity = 2.9236e-03	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.303685e+03	FES = 66100	$\mu_F = 0.739$	$\mu_{CR} = 0.240$
Diversity = 2.6935e-03	$k_t = [4, 4, 3, 4]$		
Generation 670: Best = 2.303683e+03	FES = 67100	$\mu_F = 0.735$	$\mu_{CR} = 0.232$
Diversity = 2.5899e-03	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.303681e+03	FES = 68100	$\mu_F = 0.745$	$\mu_{CR} = 0.229$
Diversity = 2.4728e-03	$k_t = [3, 4, 4, 2]$		
Generation 690: Best = 2.303680e+03	FES = 69100	$\mu_F = 0.750$	$\mu_{CR} = 0.227$
Diversity = 2.3272e-03	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.303678e+03	FES = 70100	$\mu_F = 0.763$	$\mu_{CR} = 0.217$
Diversity = 2.3644e-03	$k_t = [3, 3, 2, 2]$		
Generation 710: Best = 2.303676e+03	FES = 71100	$\mu_F = 0.763$	$\mu_{CR} = 0.221$
Diversity = 2.5702e-03	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 2.303674e+03	FES = 72100	$\mu_F = 0.766$	$\mu_{CR} = 0.222$
Diversity = 2.3843e-03	$k_t = [3, 2, 4, 4]$		
Generation 730: Best = 2.303673e+03	FES = 73100	$\mu_F = 0.763$	$\mu_{CR} = 0.216$
Diversity = 2.3707e-03	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 2.303671e+03	FES = 74100	$\mu_F = 0.764$	$\mu_{CR} = 0.208$
Diversity = 2.3192e-03	$k_t = [2, 3, 4, 2]$		
Generation 750: Best = 2.303670e+03	FES = 75100	$\mu_F = 0.768$	$\mu_{CR} = 0.216$
Diversity = 2.5083e-03	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 2.303669e+03	FES = 76100	$\mu_F = 0.761$	$\mu_{CR} = 0.212$
Diversity = 2.5043e-03	$k_t = [2, 4, 4, 2]$		

Generation 770: Best = 2.303667e+03	FES = 77100	$\mu_F = 0.759$	$\mu_{CR} = 0.212$
Diversity = 2.3407e-03	$k_t = [1, 1, 1, 1]$		
Generation 780: Best = 2.303666e+03	FES = 78100	$\mu_F = 0.765$	$\mu_{CR} = 0.214$
Diversity = 2.4528e-03	$k_t = [2, 3, 3, 3]$		
Generation 790: Best = 2.303665e+03	FES = 79100	$\mu_F = 0.746$	$\mu_{CR} = 0.221$
Diversity = 2.3044e-03	$k_t = [1, 1, 1, 1]$		
Generation 800: Best = 2.303665e+03	FES = 80100	$\mu_F = 0.730$	$\mu_{CR} = 0.227$
Diversity = 2.0531e-03	$k_t = [3, 4, 4, 2]$		
Generation 810: Best = 2.303664e+03	FES = 81100	$\mu_F = 0.737$	$\mu_{CR} = 0.222$
Diversity = 1.9727e-03	$k_t = [1, 1, 1, 1]$		
Generation 820: Best = 2.303664e+03	FES = 82100	$\mu_F = 0.720$	$\mu_{CR} = 0.215$
Diversity = 1.7196e-03	$k_t = [4, 3, 4, 2]$		
Generation 830: Best = 2.303663e+03	FES = 83100	$\mu_F = 0.717$	$\mu_{CR} = 0.218$
Diversity = 1.5408e-03	$k_t = [1, 1, 1, 1]$		
Generation 840: Best = 2.303663e+03	FES = 84100	$\mu_F = 0.724$	$\mu_{CR} = 0.211$
Diversity = 1.3259e-03	$k_t = [3, 4, 4, 2]$		
Generation 850: Best = 2.303663e+03	FES = 85100	$\mu_F = 0.726$	$\mu_{CR} = 0.213$
Diversity = 1.2028e-03	$k_t = [1, 1, 1, 1]$		
Generation 860: Best = 2.303663e+03	FES = 86100	$\mu_F = 0.741$	$\mu_{CR} = 0.214$
Diversity = 1.0569e-03	$k_t = [3, 2, 2, 3]$		
Generation 870: Best = 2.303662e+03	FES = 87100	$\mu_F = 0.742$	$\mu_{CR} = 0.219$
Diversity = 1.0576e-03	$k_t = [1, 1, 1, 1]$		
Generation 880: Best = 2.303662e+03	FES = 88100	$\mu_F = 0.751$	$\mu_{CR} = 0.213$
Diversity = 9.7106e-04	$k_t = [2, 2, 4, 3]$		
Generation 890: Best = 2.303662e+03	FES = 89100	$\mu_F = 0.763$	$\mu_{CR} = 0.215$
Diversity = 1.0309e-03	$k_t = [1, 1, 1, 1]$		
Generation 900: Best = 2.303661e+03	FES = 90100	$\mu_F = 0.770$	$\mu_{CR} = 0.221$
Diversity = 1.0831e-03	$k_t = [3, 2, 2, 2]$		
Generation 910: Best = 2.303660e+03	FES = 91100	$\mu_F = 0.781$	$\mu_{CR} = 0.226$
Diversity = 1.4222e-03	$k_t = [1, 1, 1, 1]$		
Generation 920: Best = 2.303659e+03	FES = 92100	$\mu_F = 0.780$	$\mu_{CR} = 0.239$
Diversity = 1.6484e-03	$k_t = [4, 2, 4, 2]$		
Generation 930: Best = 2.303657e+03	FES = 93100	$\mu_F = 0.799$	$\mu_{CR} = 0.244$
Diversity = 2.0598e-03	$k_t = [1, 1, 1, 1]$		
Generation 940: Best = 2.303656e+03	FES = 94100	$\mu_F = 0.800$	$\mu_{CR} = 0.246$
Diversity = 2.3700e-03	$k_t = [2, 4, 4, 2]$		
Generation 950: Best = 2.303654e+03	FES = 95100	$\mu_F = 0.800$	$\mu_{CR} = 0.253$
Diversity = 2.4956e-03	$k_t = [1, 1, 1, 1]$		
Generation 960: Best = 2.303651e+03	FES = 96100	$\mu_F = 0.796$	$\mu_{CR} = 0.257$
Diversity = 2.3618e-03	$k_t = [4, 3, 4, 4]$		
Generation 970: Best = 2.303650e+03	FES = 97100	$\mu_F = 0.781$	$\mu_{CR} = 0.252$
Diversity = 2.2501e-03	$k_t = [1, 1, 1, 1]$		
Generation 980: Best = 2.303646e+03	FES = 98100	$\mu_F = 0.780$	$\mu_{CR} = 0.256$
Diversity = 2.2306e-03	$k_t = [2, 4, 2, 2]$		
Generation 990: Best = 2.303645e+03	FES = 99100	$\mu_F = 0.794$	$\mu_{CR} = 0.260$
Diversity = 2.4634e-03	$k_t = [1, 1, 1, 1]$		
Generation 1000: Best = 2.303643e+03	FES = 100100	$\mu_F = 0.788$	$\mu_{CR} = 0.263$
Diversity = 2.2158e-03	$k_t = [2, 3, 4, 4]$		
Generation 1010: Best = 2.303642e+03	FES = 101100	$\mu_F = 0.780$	$\mu_{CR} = 0.257$
Diversity = 2.3619e-03	$k_t = [1, 1, 1, 1]$		
Generation 1020: Best = 2.303641e+03	FES = 102100	$\mu_F = 0.776$	$\mu_{CR} = 0.252$
Diversity = 2.0930e-03	$k_t = [4, 2, 4, 4]$		
Generation 1030: Best = 2.303639e+03	FES = 103100	$\mu_F = 0.774$	$\mu_{CR} = 0.260$
Diversity = 2.2575e-03	$k_t = [1, 1, 1, 1]$		

Generation 1040: Best = 2.303638e+03 FES = 104100 μ_F = 0.776 μ_{CR} = 0.267
 Diversity = 2.1083e-03 k_t = [3, 3, 3, 2]
 Generation 1050: Best = 2.303637e+03 FES = 105100 μ_F = 0.774 μ_{CR} = 0.276
 Diversity = 1.9901e-03 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 2.303636e+03 FES = 106100 μ_F = 0.766 μ_{CR} = 0.260
 Diversity = 1.8782e-03 k_t = [2, 3, 2, 3]
 Generation 1070: Best = 2.303636e+03 FES = 107100 μ_F = 0.761 μ_{CR} = 0.259
 Diversity = 1.9001e-03 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 2.303635e+03 FES = 108100 μ_F = 0.750 μ_{CR} = 0.251
 Diversity = 1.5407e-03 k_t = [3, 3, 3, 3]
 Generation 1090: Best = 2.303635e+03 FES = 109100 μ_F = 0.748 μ_{CR} = 0.245
 Diversity = 1.5345e-03 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 2.303634e+03 FES = 110100 μ_F = 0.740 μ_{CR} = 0.253
 Diversity = 1.4069e-03 k_t = [3, 4, 2, 4]
 Generation 1110: Best = 2.303633e+03 FES = 111100 μ_F = 0.748 μ_{CR} = 0.248
 Diversity = 1.3923e-03 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 2.303632e+03 FES = 112100 μ_F = 0.751 μ_{CR} = 0.249
 Diversity = 1.6382e-03 k_t = [3, 3, 3, 2]
 Generation 1130: Best = 2.303632e+03 FES = 113100 μ_F = 0.747 μ_{CR} = 0.247
 Diversity = 1.5759e-03 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 2.303632e+03 FES = 114100 μ_F = 0.748 μ_{CR} = 0.237
 Diversity = 1.3396e-03 k_t = [2, 4, 4, 2]
 Generation 1150: Best = 2.303631e+03 FES = 115100 μ_F = 0.741 μ_{CR} = 0.238
 Diversity = 1.3686e-03 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 2.303630e+03 FES = 116100 μ_F = 0.752 μ_{CR} = 0.238
 Diversity = 1.3454e-03 k_t = [4, 4, 4, 2]
 Generation 1170: Best = 2.303630e+03 FES = 117100 μ_F = 0.756 μ_{CR} = 0.237
 Diversity = 1.2565e-03 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 2.303629e+03 FES = 118100 μ_F = 0.767 μ_{CR} = 0.242
 Diversity = 1.1499e-03 k_t = [4, 2, 3, 4]
 Generation 1190: Best = 2.303629e+03 FES = 119100 μ_F = 0.763 μ_{CR} = 0.250
 Diversity = 1.2362e-03 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 2.303629e+03 FES = 120100 μ_F = 0.756 μ_{CR} = 0.246
 Diversity = 1.1771e-03 k_t = [3, 3, 2, 4]
 Generation 1210: Best = 2.303628e+03 FES = 121100 μ_F = 0.751 μ_{CR} = 0.246
 Diversity = 1.0939e-03 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 2.303628e+03 FES = 122100 μ_F = 0.738 μ_{CR} = 0.240
 Diversity = 9.5302e-04 k_t = [4, 3, 4, 3]
 Generation 1230: Best = 2.303628e+03 FES = 123100 μ_F = 0.728 μ_{CR} = 0.244
 Diversity = 8.1991e-04 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 2.303628e+03 FES = 124100 μ_F = 0.724 μ_{CR} = 0.252
 Diversity = 7.2352e-04 k_t = [2, 2, 3, 3]
 Generation 1250: Best = 2.303628e+03 FES = 125100 μ_F = 0.731 μ_{CR} = 0.246
 Diversity = 7.3397e-04 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 2.303628e+03 FES = 126100 μ_F = 0.745 μ_{CR} = 0.252
 Diversity = 7.7406e-04 k_t = [2, 3, 2, 3]
 Generation 1270: Best = 2.303627e+03 FES = 127100 μ_F = 0.760 μ_{CR} = 0.246
 Diversity = 8.0354e-04 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 2.303627e+03 FES = 128100 μ_F = 0.759 μ_{CR} = 0.247
 Diversity = 7.6096e-04 k_t = [4, 3, 4, 2]
 Generation 1290: Best = 2.303627e+03 FES = 129100 μ_F = 0.748 μ_{CR} = 0.251
 Diversity = 7.1963e-04 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 2.303627e+03 FES = 130100 μ_F = 0.759 μ_{CR} = 0.249
 Diversity = 7.8629e-04 k_t = [4, 3, 3, 2]

Generation 1310: Best = 2.303627e+03 FES = 131100 μ_F = 0.773 μ_{CR} = 0.253
 Diversity = 8.0517e-04 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 2.303627e+03 FES = 132100 μ_F = 0.768 μ_{CR} = 0.248
 Diversity = 7.3199e-04 k_t = [2, 4, 3, 3]
 Generation 1330: Best = 2.303626e+03 FES = 133100 μ_F = 0.770 μ_{CR} = 0.251
 Diversity = 7.3924e-04 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 2.303626e+03 FES = 134100 μ_F = 0.780 μ_{CR} = 0.256
 Diversity = 7.3448e-04 k_t = [2, 2, 4, 2]
 Generation 1350: Best = 2.303626e+03 FES = 135100 μ_F = 0.776 μ_{CR} = 0.258
 Diversity = 7.2512e-04 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 2.303626e+03 FES = 136100 μ_F = 0.778 μ_{CR} = 0.256
 Diversity = 7.4462e-04 k_t = [3, 3, 4, 2]
 Generation 1370: Best = 2.303626e+03 FES = 137100 μ_F = 0.776 μ_{CR} = 0.259
 Diversity = 6.9785e-04 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 2.303626e+03 FES = 138100 μ_F = 0.758 μ_{CR} = 0.250
 Diversity = 6.4883e-04 k_t = [4, 2, 4, 3]
 Generation 1390: Best = 2.303626e+03 FES = 139100 μ_F = 0.759 μ_{CR} = 0.248
 Diversity = 6.3084e-04 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 2.303626e+03 FES = 140100 μ_F = 0.753 μ_{CR} = 0.246
 Diversity = 5.7173e-04 k_t = [2, 2, 4, 4]
 Generation 1410: Best = 2.303626e+03 FES = 141100 μ_F = 0.743 μ_{CR} = 0.245
 Diversity = 5.4061e-04 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 2.303625e+03 FES = 142100 μ_F = 0.745 μ_{CR} = 0.244
 Diversity = 5.2545e-04 k_t = [3, 2, 2, 2]
 Generation 1430: Best = 2.303625e+03 FES = 143100 μ_F = 0.755 μ_{CR} = 0.239
 Diversity = 5.0464e-04 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 2.303625e+03 FES = 144100 μ_F = 0.760 μ_{CR} = 0.252
 Diversity = 5.2244e-04 k_t = [4, 4, 2, 4]
 Generation 1450: Best = 2.303625e+03 FES = 145100 μ_F = 0.766 μ_{CR} = 0.249
 Diversity = 4.6486e-04 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 2.303625e+03 FES = 146100 μ_F = 0.757 μ_{CR} = 0.250
 Diversity = 4.6703e-04 k_t = [2, 4, 2, 4]
 Generation 1470: Best = 2.303625e+03 FES = 147100 μ_F = 0.760 μ_{CR} = 0.258
 Diversity = 5.4049e-04 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 2.303625e+03 FES = 148100 μ_F = 0.755 μ_{CR} = 0.250
 Diversity = 4.6705e-04 k_t = [4, 2, 3, 3]
 Generation 1490: Best = 2.303625e+03 FES = 149100 μ_F = 0.749 μ_{CR} = 0.251
 Diversity = 4.3209e-04 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 2.303625e+03 FES = 150100 μ_F = 0.757 μ_{CR} = 0.258
 Diversity = 3.9705e-04 k_t = [3, 3, 3, 2]
 Generation 1510: Best = 2.303625e+03 FES = 151100 μ_F = 0.749 μ_{CR} = 0.263
 Diversity = 3.8358e-04 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 2.303625e+03 FES = 152100 μ_F = 0.753 μ_{CR} = 0.266
 Diversity = 3.3609e-04 k_t = [2, 2, 2, 4]
 Generation 1530: Best = 2.303625e+03 FES = 153100 μ_F = 0.759 μ_{CR} = 0.265
 Diversity = 3.1487e-04 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 2.303625e+03 FES = 154100 μ_F = 0.758 μ_{CR} = 0.266
 Diversity = 3.1919e-04 k_t = [2, 4, 4, 2]
 Generation 1550: Best = 2.303625e+03 FES = 155100 μ_F = 0.759 μ_{CR} = 0.267
 Diversity = 3.1556e-04 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 2.303625e+03 FES = 156100 μ_F = 0.771 μ_{CR} = 0.272
 Diversity = 3.3107e-04 k_t = [2, 4, 3, 3]
 Generation 1570: Best = 2.303625e+03 FES = 157100 μ_F = 0.758 μ_{CR} = 0.269
 Diversity = 3.1744e-04 k_t = [1, 1, 1, 1]

Generation 1580: Best = 2.303625e+03 FES = 158100 μ_F = 0.757 μ_{CR} = 0.246
 Diversity = 2.8201e-04 k_t = [4, 4, 3, 4]
 Generation 1590: Best = 2.303625e+03 FES = 159100 μ_F = 0.767 μ_{CR} = 0.247
 Diversity = 2.7994e-04 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 2.303624e+03 FES = 160100 μ_F = 0.766 μ_{CR} = 0.255
 Diversity = 2.9101e-04 k_t = [3, 2, 3, 3]
 Generation 1610: Best = 2.303624e+03 FES = 161100 μ_F = 0.767 μ_{CR} = 0.245
 Diversity = 2.9190e-04 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 2.303624e+03 FES = 162100 μ_F = 0.763 μ_{CR} = 0.239
 Diversity = 2.6850e-04 k_t = [2, 2, 4, 3]
 Generation 1630: Best = 2.303624e+03 FES = 163100 μ_F = 0.763 μ_{CR} = 0.236
 Diversity = 2.6210e-04 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 2.303624e+03 FES = 164100 μ_F = 0.771 μ_{CR} = 0.245
 Diversity = 2.7322e-04 k_t = [4, 2, 4, 2]
 Generation 1650: Best = 2.303624e+03 FES = 165100 μ_F = 0.779 μ_{CR} = 0.245
 Diversity = 2.9488e-04 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 2.303624e+03 FES = 166100 μ_F = 0.797 μ_{CR} = 0.247
 Diversity = 3.0913e-04 k_t = [3, 3, 4, 3]
 Generation 1670: Best = 2.303624e+03 FES = 167100 μ_F = 0.783 μ_{CR} = 0.241
 Diversity = 3.2743e-04 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 2.303624e+03 FES = 168100 μ_F = 0.788 μ_{CR} = 0.244
 Diversity = 3.8183e-04 k_t = [3, 3, 4, 2]
 Generation 1690: Best = 2.303624e+03 FES = 169100 μ_F = 0.797 μ_{CR} = 0.256
 Diversity = 4.7719e-04 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 2.303624e+03 FES = 170100 μ_F = 0.794 μ_{CR} = 0.256
 Diversity = 5.1494e-04 k_t = [4, 3, 3, 4]
 Generation 1710: Best = 2.303624e+03 FES = 171100 μ_F = 0.807 μ_{CR} = 0.256
 Diversity = 5.4645e-04 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 2.303624e+03 FES = 172100 μ_F = 0.785 μ_{CR} = 0.261
 Diversity = 4.9228e-04 k_t = [3, 3, 3, 2]
 Generation 1730: Best = 2.303624e+03 FES = 173100 μ_F = 0.776 μ_{CR} = 0.251
 Diversity = 5.1205e-04 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 2.303624e+03 FES = 174100 μ_F = 0.759 μ_{CR} = 0.255
 Diversity = 4.7947e-04 k_t = [3, 4, 2, 2]
 Generation 1750: Best = 2.303624e+03 FES = 175100 μ_F = 0.762 μ_{CR} = 0.260
 Diversity = 4.7906e-04 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 2.303624e+03 FES = 176100 μ_F = 0.759 μ_{CR} = 0.258
 Diversity = 4.5843e-04 k_t = [4, 3, 2, 4]
 Generation 1770: Best = 2.303623e+03 FES = 177100 μ_F = 0.760 μ_{CR} = 0.258
 Diversity = 4.6058e-04 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 2.303623e+03 FES = 178100 μ_F = 0.748 μ_{CR} = 0.268
 Diversity = 4.6337e-04 k_t = [4, 3, 3, 3]
 Generation 1790: Best = 2.303623e+03 FES = 179100 μ_F = 0.764 μ_{CR} = 0.275
 Diversity = 5.1033e-04 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 2.303623e+03 FES = 180100 μ_F = 0.761 μ_{CR} = 0.281
 Diversity = 4.9917e-04 k_t = [2, 2, 2, 2]
 Generation 1810: Best = 2.303623e+03 FES = 181100 μ_F = 0.759 μ_{CR} = 0.280
 Diversity = 5.8383e-04 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 2.303623e+03 FES = 182100 μ_F = 0.769 μ_{CR} = 0.278
 Diversity = 6.5991e-04 k_t = [3, 3, 4, 2]
 Generation 1830: Best = 2.303623e+03 FES = 183100 μ_F = 0.774 μ_{CR} = 0.284
 Diversity = 6.9886e-04 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 2.303622e+03 FES = 184100 μ_F = 0.772 μ_{CR} = 0.285
 Diversity = 6.4104e-04 k_t = [2, 4, 4, 3]

Generation 1850: Best = 2.303622e+03	FES = 185100	$\mu_F = 0.784$	$\mu_{CR} = 0.282$
Diversity = 7.2220e-04	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 2.303622e+03	FES = 186100	$\mu_F = 0.795$	$\mu_{CR} = 0.283$
Diversity = 8.5609e-04	$k_t = [4, 4, 3, 2]$		
Generation 1870: Best = 2.303622e+03	FES = 187100	$\mu_F = 0.782$	$\mu_{CR} = 0.273$
Diversity = 8.1414e-04	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 2.303621e+03	FES = 188100	$\mu_F = 0.787$	$\mu_{CR} = 0.262$
Diversity = 7.8313e-04	$k_t = [3, 4, 2, 2]$		
Generation 1890: Best = 2.303621e+03	FES = 189100	$\mu_F = 0.786$	$\mu_{CR} = 0.259$
Diversity = 6.7603e-04	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 2.303621e+03	FES = 190100	$\mu_F = 0.768$	$\mu_{CR} = 0.256$
Diversity = 5.6917e-04	$k_t = [2, 4, 2, 3]$		
Generation 1910: Best = 2.303621e+03	FES = 191100	$\mu_F = 0.762$	$\mu_{CR} = 0.248$
Diversity = 4.9699e-04	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 2.303621e+03	FES = 192100	$\mu_F = 0.754$	$\mu_{CR} = 0.240$
Diversity = 4.2806e-04	$k_t = [2, 2, 3, 3]$		
Generation 1930: Best = 2.303621e+03	FES = 193100	$\mu_F = 0.748$	$\mu_{CR} = 0.239$
Diversity = 4.7419e-04	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 2.303621e+03	FES = 194100	$\mu_F = 0.760$	$\mu_{CR} = 0.239$
Diversity = 5.0793e-04	$k_t = [2, 3, 2, 2]$		
Generation 1950: Best = 2.303621e+03	FES = 195100	$\mu_F = 0.757$	$\mu_{CR} = 0.243$
Diversity = 5.4006e-04	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 2.303621e+03	FES = 196100	$\mu_F = 0.750$	$\mu_{CR} = 0.250$
Diversity = 5.3306e-04	$k_t = [3, 2, 2, 2]$		
Generation 1970: Best = 2.303621e+03	FES = 197100	$\mu_F = 0.739$	$\mu_{CR} = 0.249$
Diversity = 4.5611e-04	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 2.303621e+03	FES = 198100	$\mu_F = 0.739$	$\mu_{CR} = 0.240$
Diversity = 3.7233e-04	$k_t = [4, 4, 4, 3]$		
Generation 1990: Best = 2.303621e+03	FES = 199100	$\mu_F = 0.748$	$\mu_{CR} = 0.244$
Diversity = 3.3059e-04	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 2.303621e+03	FES = 200100	$\mu_F = 0.743$	$\mu_{CR} = 0.234$
Diversity = 2.9126e-04	$k_t = [3, 3, 3, 4]$		
Generation 2010: Best = 2.303621e+03	FES = 201100	$\mu_F = 0.750$	$\mu_{CR} = 0.239$
Diversity = 2.8011e-04	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 2.303621e+03	FES = 202100	$\mu_F = 0.756$	$\mu_{CR} = 0.236$
Diversity = 2.7585e-04	$k_t = [3, 4, 4, 2]$		
Generation 2030: Best = 2.303621e+03	FES = 203100	$\mu_F = 0.755$	$\mu_{CR} = 0.242$
Diversity = 2.5994e-04	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 2.303621e+03	FES = 204100	$\mu_F = 0.767$	$\mu_{CR} = 0.251$
Diversity = 2.3521e-04	$k_t = [4, 2, 3, 4]$		
Generation 2050: Best = 2.303621e+03	FES = 205100	$\mu_F = 0.773$	$\mu_{CR} = 0.251$
Diversity = 2.2849e-04	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 2.303621e+03	FES = 206100	$\mu_F = 0.766$	$\mu_{CR} = 0.252$
Diversity = 2.2667e-04	$k_t = [2, 3, 3, 4]$		
Generation 2070: Best = 2.303621e+03	FES = 207100	$\mu_F = 0.769$	$\mu_{CR} = 0.254$
Diversity = 2.2001e-04	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 2.303621e+03	FES = 208100	$\mu_F = 0.764$	$\mu_{CR} = 0.249$
Diversity = 2.0974e-04	$k_t = [3, 4, 3, 2]$		
Generation 2090: Best = 2.303621e+03	FES = 209100	$\mu_F = 0.770$	$\mu_{CR} = 0.251$
Diversity = 2.1583e-04	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 2.303621e+03	FES = 210100	$\mu_F = 0.768$	$\mu_{CR} = 0.251$
Diversity = 1.8079e-04	$k_t = [3, 2, 4, 3]$		
Generation 2110: Best = 2.303621e+03	FES = 211100	$\mu_F = 0.769$	$\mu_{CR} = 0.256$
Diversity = 1.9435e-04	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 2.303621e+03 FES = 212100 μ_F = 0.764 μ_{CR} = 0.249
 Diversity = 1.6927e-04 k_t = [4, 3, 4, 3]
 Generation 2130: Best = 2.303621e+03 FES = 213100 μ_F = 0.754 μ_{CR} = 0.245
 Diversity = 1.5653e-04 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 2.303621e+03 FES = 214100 μ_F = 0.751 μ_{CR} = 0.246
 Diversity = 1.3280e-04 k_t = [2, 3, 2, 3]
 Generation 2150: Best = 2.303621e+03 FES = 215100 μ_F = 0.745 μ_{CR} = 0.234
 Diversity = 1.3713e-04 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 2.303621e+03 FES = 216100 μ_F = 0.749 μ_{CR} = 0.248
 Diversity = 1.1434e-04 k_t = [4, 4, 4, 3]
 Generation 2170: Best = 2.303621e+03 FES = 217100 μ_F = 0.765 μ_{CR} = 0.253
 Diversity = 1.1984e-04 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 2.303621e+03 FES = 218100 μ_F = 0.766 μ_{CR} = 0.267
 Diversity = 1.2343e-04 k_t = [4, 3, 3, 4]
 Generation 2190: Best = 2.303621e+03 FES = 219100 μ_F = 0.767 μ_{CR} = 0.263
 Diversity = 1.3104e-04 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 2.303621e+03 FES = 220100 μ_F = 0.764 μ_{CR} = 0.269
 Diversity = 1.3448e-04 k_t = [3, 2, 3, 3]
 Generation 2210: Best = 2.303621e+03 FES = 221100 μ_F = 0.768 μ_{CR} = 0.261
 Diversity = 1.5665e-04 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 2.303621e+03 FES = 222100 μ_F = 0.768 μ_{CR} = 0.264
 Diversity = 1.4943e-04 k_t = [3, 2, 2, 3]
 Generation 2230: Best = 2.303621e+03 FES = 223100 μ_F = 0.766 μ_{CR} = 0.258
 Diversity = 1.4468e-04 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 2.303621e+03 FES = 224100 μ_F = 0.769 μ_{CR} = 0.258
 Diversity = 1.2915e-04 k_t = [3, 4, 3, 4]
 Generation 2250: Best = 2.303621e+03 FES = 225100 μ_F = 0.768 μ_{CR} = 0.240
 Diversity = 1.2109e-04 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 2.303621e+03 FES = 226100 μ_F = 0.772 μ_{CR} = 0.234
 Diversity = 1.1477e-04 k_t = [4, 3, 3, 2]
 Generation 2270: Best = 2.303621e+03 FES = 227100 μ_F = 0.772 μ_{CR} = 0.238
 Diversity = 1.0901e-04 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 2.303621e+03 FES = 228100 μ_F = 0.778 μ_{CR} = 0.235
 Diversity = 9.7487e-05 k_t = [4, 4, 3, 4]
 Generation 2290: Best = 2.303621e+03 FES = 229100 μ_F = 0.776 μ_{CR} = 0.243
 Diversity = 9.1225e-05 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 2.303621e+03 FES = 230100 μ_F = 0.776 μ_{CR} = 0.244
 Diversity = 9.4510e-05 k_t = [2, 3, 4, 2]
 Generation 2310: Best = 2.303621e+03 FES = 231100 μ_F = 0.767 μ_{CR} = 0.240
 Diversity = 9.9071e-05 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 2.303621e+03 FES = 232100 μ_F = 0.771 μ_{CR} = 0.248
 Diversity = 9.0080e-05 k_t = [2, 3, 2, 4]
 Generation 2330: Best = 2.303621e+03 FES = 233100 μ_F = 0.777 μ_{CR} = 0.237
 Diversity = 9.1618e-05 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 2.303621e+03 FES = 234100 μ_F = 0.764 μ_{CR} = 0.229
 Diversity = 8.9197e-05 k_t = [2, 3, 4, 2]
 Generation 2350: Best = 2.303621e+03 FES = 235100 μ_F = 0.754 μ_{CR} = 0.233
 Diversity = 9.0820e-05 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 2.303621e+03 FES = 236100 μ_F = 0.751 μ_{CR} = 0.226
 Diversity = 7.9952e-05 k_t = [3, 2, 4, 3]
 Generation 2370: Best = 2.303621e+03 FES = 237100 μ_F = 0.746 μ_{CR} = 0.228
 Diversity = 8.3874e-05 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 2.303621e+03 FES = 238100 μ_F = 0.755 μ_{CR} = 0.227
 Diversity = 7.7155e-05 k_t = [2, 4, 4, 2]

Generation 2390: Best = 2.303621e+03 FES = 239100 μ_F = 0.761 μ_{CR} = 0.224
 Diversity = 7.6119e-05 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 2.303621e+03 FES = 240100 μ_F = 0.750 μ_{CR} = 0.236
 Diversity = 7.3585e-05 k_t = [2, 2, 3, 2]
 Generation 2410: Best = 2.303621e+03 FES = 241100 μ_F = 0.739 μ_{CR} = 0.225
 Diversity = 7.3987e-05 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 2.303621e+03 FES = 242100 μ_F = 0.747 μ_{CR} = 0.224
 Diversity = 7.1347e-05 k_t = [2, 2, 3, 3]
 Generation 2430: Best = 2.303621e+03 FES = 243100 μ_F = 0.746 μ_{CR} = 0.222
 Diversity = 6.8625e-05 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 2.303621e+03 FES = 244100 μ_F = 0.769 μ_{CR} = 0.223
 Diversity = 6.6891e-05 k_t = [3, 4, 4, 2]
 Generation 2450: Best = 2.303621e+03 FES = 245100 μ_F = 0.773 μ_{CR} = 0.227
 Diversity = 6.9402e-05 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 2.303621e+03 FES = 246100 μ_F = 0.778 μ_{CR} = 0.235
 Diversity = 7.9344e-05 k_t = [4, 4, 4, 2]
 Generation 2470: Best = 2.303621e+03 FES = 247100 μ_F = 0.787 μ_{CR} = 0.225
 Diversity = 7.4009e-05 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 2.303621e+03 FES = 248100 μ_F = 0.775 μ_{CR} = 0.235
 Diversity = 7.7791e-05 k_t = [2, 2, 2, 4]
 Generation 2490: Best = 2.303621e+03 FES = 249100 μ_F = 0.780 μ_{CR} = 0.240
 Diversity = 7.8396e-05 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 2.303621e+03 FES = 250100 μ_F = 0.786 μ_{CR} = 0.238
 Diversity = 7.0527e-05 k_t = [2, 4, 4, 4]
 Generation 2510: Best = 2.303621e+03 FES = 251100 μ_F = 0.789 μ_{CR} = 0.231
 Diversity = 7.1683e-05 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 2.303621e+03 FES = 252100 μ_F = 0.785 μ_{CR} = 0.232
 Diversity = 6.7022e-05 k_t = [2, 3, 4, 4]
 Generation 2530: Best = 2.303621e+03 FES = 253100 μ_F = 0.788 μ_{CR} = 0.227
 Diversity = 6.8073e-05 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 2.303621e+03 FES = 254100 μ_F = 0.785 μ_{CR} = 0.220
 Diversity = 5.9673e-05 k_t = [2, 2, 2, 3]
 Generation 2550: Best = 2.303621e+03 FES = 255100 μ_F = 0.768 μ_{CR} = 0.212
 Diversity = 5.5513e-05 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 2.303621e+03 FES = 256100 μ_F = 0.750 μ_{CR} = 0.213
 Diversity = 4.7443e-05 k_t = [4, 4, 3, 4]
 Generation 2570: Best = 2.303621e+03 FES = 257100 μ_F = 0.749 μ_{CR} = 0.210
 Diversity = 4.3039e-05 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 2.303621e+03 FES = 258100 μ_F = 0.761 μ_{CR} = 0.215
 Diversity = 3.9364e-05 k_t = [2, 4, 4, 3]
 Generation 2590: Best = 2.303621e+03 FES = 259100 μ_F = 0.748 μ_{CR} = 0.212
 Diversity = 4.1134e-05 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 2.303621e+03 FES = 260100 μ_F = 0.749 μ_{CR} = 0.205
 Diversity = 3.5393e-05 k_t = [3, 4, 3, 3]
 Generation 2610: Best = 2.303621e+03 FES = 261100 μ_F = 0.749 μ_{CR} = 0.210
 Diversity = 3.7208e-05 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 2.303621e+03 FES = 262100 μ_F = 0.749 μ_{CR} = 0.208
 Diversity = 3.5372e-05 k_t = [4, 3, 4, 2]
 Generation 2630: Best = 2.303621e+03 FES = 263100 μ_F = 0.759 μ_{CR} = 0.208
 Diversity = 3.4345e-05 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 2.303621e+03 FES = 264100 μ_F = 0.746 μ_{CR} = 0.213
 Diversity = 2.9124e-05 k_t = [4, 4, 3, 4]
 Generation 2650: Best = 2.303621e+03 FES = 265100 μ_F = 0.749 μ_{CR} = 0.203
 Diversity = 2.7839e-05 k_t = [1, 1, 1, 1]

Generation 2660: Best = 2.303621e+03 FES = 266100 μ_F = 0.736 μ_{CR} = 0.201
 Diversity = 2.4186e-05 k_t = [3, 2, 4, 4]
 Generation 2670: Best = 2.303621e+03 FES = 267100 μ_F = 0.733 μ_{CR} = 0.199
 Diversity = 2.0584e-05 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 2.303621e+03 FES = 268100 μ_F = 0.747 μ_{CR} = 0.195
 Diversity = 1.9878e-05 k_t = [4, 4, 4, 3]
 Generation 2690: Best = 2.303621e+03 FES = 269100 μ_F = 0.743 μ_{CR} = 0.201
 Diversity = 2.1025e-05 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 2.303621e+03 FES = 270100 μ_F = 0.749 μ_{CR} = 0.195
 Diversity = 2.1450e-05 k_t = [3, 2, 2, 2]
 Generation 2710: Best = 2.303621e+03 FES = 271100 μ_F = 0.760 μ_{CR} = 0.191
 Diversity = 2.0256e-05 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 2.303621e+03 FES = 272100 μ_F = 0.774 μ_{CR} = 0.204
 Diversity = 1.9198e-05 k_t = [2, 4, 4, 4]
 Generation 2730: Best = 2.303621e+03 FES = 273100 μ_F = 0.788 μ_{CR} = 0.208
 Diversity = 2.3198e-05 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 2.303621e+03 FES = 274100 μ_F = 0.790 μ_{CR} = 0.213
 Diversity = 2.2354e-05 k_t = [4, 3, 3, 4]
 Generation 2750: Best = 2.303621e+03 FES = 275100 μ_F = 0.782 μ_{CR} = 0.224
 Diversity = 2.1822e-05 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 2.303621e+03 FES = 276100 μ_F = 0.770 μ_{CR} = 0.224
 Diversity = 1.9425e-05 k_t = [4, 4, 2, 4]
 Generation 2770: Best = 2.303621e+03 FES = 277100 μ_F = 0.778 μ_{CR} = 0.221
 Diversity = 1.7981e-05 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 2.303621e+03 FES = 278100 μ_F = 0.778 μ_{CR} = 0.210
 Diversity = 1.6886e-05 k_t = [4, 2, 2, 4]
 Generation 2790: Best = 2.303621e+03 FES = 279100 μ_F = 0.783 μ_{CR} = 0.212
 Diversity = 1.6374e-05 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 2.303621e+03 FES = 280100 μ_F = 0.776 μ_{CR} = 0.208
 Diversity = 1.5023e-05 k_t = [3, 4, 4, 3]
 Generation 2810: Best = 2.303621e+03 FES = 281100 μ_F = 0.776 μ_{CR} = 0.206
 Diversity = 1.4335e-05 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 2.303621e+03 FES = 282100 μ_F = 0.762 μ_{CR} = 0.201
 Diversity = 1.3466e-05 k_t = [4, 3, 2, 3]
 Generation 2830: Best = 2.303621e+03 FES = 283100 μ_F = 0.772 μ_{CR} = 0.203
 Diversity = 1.3977e-05 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 2.303621e+03 FES = 284100 μ_F = 0.784 μ_{CR} = 0.198
 Diversity = 1.4088e-05 k_t = [2, 4, 4, 3]
 Generation 2850: Best = 2.303621e+03 FES = 285100 μ_F = 0.788 μ_{CR} = 0.203
 Diversity = 1.4794e-05 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 2.303621e+03 FES = 286100 μ_F = 0.797 μ_{CR} = 0.197
 Diversity = 1.3328e-05 k_t = [3, 3, 3, 4]
 Generation 2870: Best = 2.303621e+03 FES = 287100 μ_F = 0.786 μ_{CR} = 0.204
 Diversity = 1.4219e-05 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 2.303621e+03 FES = 288100 μ_F = 0.782 μ_{CR} = 0.207
 Diversity = 1.3625e-05 k_t = [2, 4, 3, 2]
 Generation 2890: Best = 2.303621e+03 FES = 289100 μ_F = 0.779 μ_{CR} = 0.206
 Diversity = 1.3340e-05 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 2.303621e+03 FES = 290100 μ_F = 0.766 μ_{CR} = 0.214
 Diversity = 1.1640e-05 k_t = [4, 3, 4, 3]
 Generation 2910: Best = 2.303621e+03 FES = 291100 μ_F = 0.769 μ_{CR} = 0.215
 Diversity = 1.0856e-05 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 2.303621e+03 FES = 292100 μ_F = 0.770 μ_{CR} = 0.219
 Diversity = 1.0325e-05 k_t = [3, 4, 4, 2]

Generation 2930: Best = 2.303621e+03 FES = 293100 μ_F = 0.780 μ_{CR} = 0.218
 Diversity = 1.0373e-05 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 2.303621e+03 FES = 294100 μ_F = 0.783 μ_{CR} = 0.213
 Diversity = 1.0826e-05 k_t = [3, 4, 2, 2]
 Generation 2950: Best = 2.303621e+03 FES = 295100 μ_F = 0.772 μ_{CR} = 0.210
 Diversity = 9.5514e-06 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 2.303621e+03 FES = 296100 μ_F = 0.783 μ_{CR} = 0.209
 Diversity = 8.9639e-06 k_t = [3, 4, 3, 4]
 Generation 2970: Best = 2.303621e+03 FES = 297100 μ_F = 0.784 μ_{CR} = 0.208
 Diversity = 9.2123e-06 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 2.303621e+03 FES = 298100 μ_F = 0.788 μ_{CR} = 0.207
 Diversity = 8.7134e-06 k_t = [2, 3, 4, 4]
 Generation 2990: Best = 2.303621e+03 FES = 299100 μ_F = 0.778 μ_{CR} = 0.218
 Diversity = 8.2657e-06 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 2.303621e+03

Total Function Evals = 300000

Total Successful Updates = 124998

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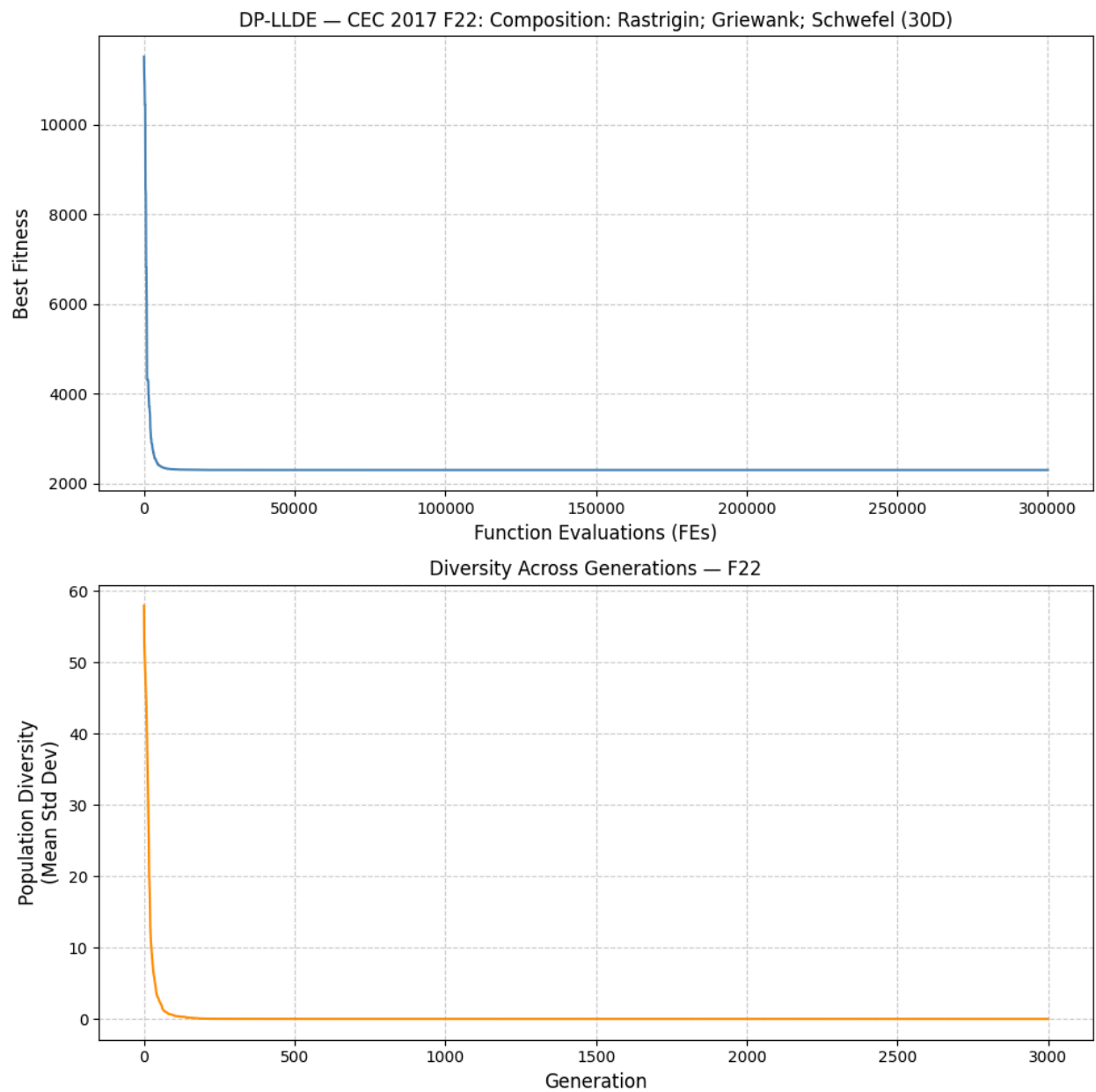
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Results:

Best Fitness : 2.303621e+03
 Global Optimum : 2.200000e+03
 Error (|best-optimum|) : 1.036205e+02
 Best Solution (5 dim) : [-52.38595423 11.069395 23.05202413 34.66206306
 30.30338657]
 D^(0) (init. diversity): 5.7976e+01

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Graph saved as convergence_F22.png

```
In [ ]: # =====  
# F23 — Composition Function 3 (Rosenbrock; Ackley; Schwefel; Rastrigin)  
# =====  
run_and_plot('F23', make_wrapper(composition.f23), 2300.0, 'Composition: Rosen
```

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DP-LLDE on CEC 2017 F23 – Single Run

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Function : Composition: Rosenbrock; Ackley; Schwefel; Rastrigin (30D)
Bounds : [-100, 100]
Global optimum : 2300.0
Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 4.071208e+03

Generation 10:	Best = 3.237922e+03	FES = 1100	$\mu_F = 0.598$	$\mu_{CR} = 0.355$
Diversity = 3.8950e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 3.011578e+03	FES = 2100	$\mu_F = 0.662$	$\mu_{CR} = 0.351$
Diversity = 2.4750e+01	$k_t = [4, 4, 2, 4]$			
Generation 30:	Best = 2.898037e+03	FES = 3100	$\mu_F = 0.694$	$\mu_{CR} = 0.342$
Diversity = 1.9815e+01	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 2.898037e+03	FES = 4100	$\mu_F = 0.717$	$\mu_{CR} = 0.324$
Diversity = 1.8123e+01	$k_t = [2, 4, 4, 2]$			
Generation 50:	Best = 2.882252e+03	FES = 5100	$\mu_F = 0.755$	$\mu_{CR} = 0.311$
Diversity = 1.7327e+01	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 2.830102e+03	FES = 6100	$\mu_F = 0.757$	$\mu_{CR} = 0.264$
Diversity = 1.6684e+01	$k_t = [3, 3, 4, 2]$			
Generation 70:	Best = 2.830102e+03	FES = 7100	$\mu_F = 0.761$	$\mu_{CR} = 0.237$
Diversity = 1.6108e+01	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 2.830102e+03	FES = 8100	$\mu_F = 0.769$	$\mu_{CR} = 0.212$
Diversity = 1.5277e+01	$k_t = [2, 4, 4, 4]$			
Generation 90:	Best = 2.830102e+03	FES = 9100	$\mu_F = 0.779$	$\mu_{CR} = 0.179$
Diversity = 1.5099e+01	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 2.830102e+03	FES = 10100	$\mu_F = 0.722$	$\mu_{CR} = 0.132$
Diversity = 1.4827e+01	$k_t = [4, 3, 3, 3]$			
Generation 110:	Best = 2.827628e+03	FES = 11100	$\mu_F = 0.718$	$\mu_{CR} = 0.093$
Diversity = 1.4714e+01	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 2.810658e+03	FES = 12100	$\mu_F = 0.689$	$\mu_{CR} = 0.092$
Diversity = 1.4356e+01	$k_t = [3, 3, 3, 2]$			
Generation 130:	Best = 2.810658e+03	FES = 13100	$\mu_F = 0.701$	$\mu_{CR} = 0.079$
Diversity = 1.4286e+01	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 2.810658e+03	FES = 14100	$\mu_F = 0.647$	$\mu_{CR} = 0.076$
Diversity = 1.4076e+01	$k_t = [3, 2, 4, 3]$			
Generation 150:	Best = 2.810658e+03	FES = 15100	$\mu_F = 0.641$	$\mu_{CR} = 0.060$
Diversity = 1.3980e+01	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 2.807180e+03	FES = 16100	$\mu_F = 0.640$	$\mu_{CR} = 0.058$
Diversity = 1.3821e+01	$k_t = [3, 4, 3, 4]$			
Generation 170:	Best = 2.802309e+03	FES = 17100	$\mu_F = 0.650$	$\mu_{CR} = 0.054$
Diversity = 1.3696e+01	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 2.798406e+03	FES = 18100	$\mu_F = 0.713$	$\mu_{CR} = 0.046$
Diversity = 1.3390e+01	$k_t = [3, 4, 3, 4]$			
Generation 190:	Best = 2.798406e+03	FES = 19100	$\mu_F = 0.689$	$\mu_{CR} = 0.054$
Diversity = 1.3380e+01	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 2.795036e+03	FES = 20100	$\mu_F = 0.743$	$\mu_{CR} = 0.057$
Diversity = 1.3323e+01	$k_t = [3, 4, 2, 3]$			
Generation 210:	Best = 2.792730e+03	FES = 21100	$\mu_F = 0.741$	$\mu_{CR} = 0.060$
Diversity = 1.3147e+01	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 2.767433e+03	FES = 22100	$\mu_F = 0.709$	$\mu_{CR} = 0.057$
Diversity = 1.3130e+01	$k_t = [4, 3, 2, 2]$			

Generation 230: Best = 2.767433e+03	FES = 23100	$\mu_F = 0.724$	$\mu_{CR} = 0.064$
Diversity = 1.3112e+01	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 2.767433e+03	FES = 24100	$\mu_F = 0.747$	$\mu_{CR} = 0.055$
Diversity = 1.2973e+01	$k_t = [4, 3, 2, 4]$		
Generation 250: Best = 2.767433e+03	FES = 25100	$\mu_F = 0.791$	$\mu_{CR} = 0.054$
Diversity = 1.2959e+01	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 2.767433e+03	FES = 26100	$\mu_F = 0.732$	$\mu_{CR} = 0.055$
Diversity = 1.2934e+01	$k_t = [4, 2, 3, 3]$		
Generation 270: Best = 2.767433e+03	FES = 27100	$\mu_F = 0.670$	$\mu_{CR} = 0.063$
Diversity = 1.2844e+01	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 2.767433e+03	FES = 28100	$\mu_F = 0.695$	$\mu_{CR} = 0.050$
Diversity = 1.2810e+01	$k_t = [3, 3, 4, 3]$		
Generation 290: Best = 2.767433e+03	FES = 29100	$\mu_F = 0.685$	$\mu_{CR} = 0.053$
Diversity = 1.2816e+01	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 2.767433e+03	FES = 30100	$\mu_F = 0.676$	$\mu_{CR} = 0.048$
Diversity = 1.2831e+01	$k_t = [4, 4, 2, 3]$		
Generation 310: Best = 2.767433e+03	FES = 31100	$\mu_F = 0.689$	$\mu_{CR} = 0.058$
Diversity = 1.2798e+01	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 2.767433e+03	FES = 32100	$\mu_F = 0.704$	$\mu_{CR} = 0.057$
Diversity = 1.2792e+01	$k_t = [4, 3, 3, 2]$		
Generation 330: Best = 2.763727e+03	FES = 33100	$\mu_F = 0.668$	$\mu_{CR} = 0.060$
Diversity = 1.2790e+01	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 2.763727e+03	FES = 34100	$\mu_F = 0.680$	$\mu_{CR} = 0.051$
Diversity = 1.2784e+01	$k_t = [3, 4, 3, 3]$		
Generation 350: Best = 2.763727e+03	FES = 35100	$\mu_F = 0.666$	$\mu_{CR} = 0.053$
Diversity = 1.2784e+01	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 2.757203e+03	FES = 36100	$\mu_F = 0.652$	$\mu_{CR} = 0.059$
Diversity = 1.2767e+01	$k_t = [4, 2, 3, 3]$		
Generation 370: Best = 2.752242e+03	FES = 37100	$\mu_F = 0.647$	$\mu_{CR} = 0.057$
Diversity = 1.2765e+01	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 2.752242e+03	FES = 38100	$\mu_F = 0.673$	$\mu_{CR} = 0.053$
Diversity = 1.2761e+01	$k_t = [2, 3, 2, 2]$		
Generation 390: Best = 2.748829e+03	FES = 39100	$\mu_F = 0.653$	$\mu_{CR} = 0.059$
Diversity = 1.2761e+01	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 2.746743e+03	FES = 40100	$\mu_F = 0.666$	$\mu_{CR} = 0.054$
Diversity = 1.2747e+01	$k_t = [4, 4, 4, 2]$		
Generation 410: Best = 2.746673e+03	FES = 41100	$\mu_F = 0.693$	$\mu_{CR} = 0.053$
Diversity = 1.2723e+01	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 2.746673e+03	FES = 42100	$\mu_F = 0.697$	$\mu_{CR} = 0.052$
Diversity = 1.2721e+01	$k_t = [2, 3, 2, 2]$		
Generation 430: Best = 2.746673e+03	FES = 43100	$\mu_F = 0.676$	$\mu_{CR} = 0.045$
Diversity = 1.2716e+01	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 2.746673e+03	FES = 44100	$\mu_F = 0.670$	$\mu_{CR} = 0.040$
Diversity = 1.2683e+01	$k_t = [2, 4, 3, 3]$		
Generation 450: Best = 2.745216e+03	FES = 45100	$\mu_F = 0.715$	$\mu_{CR} = 0.039$
Diversity = 1.2684e+01	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 2.744241e+03	FES = 46100	$\mu_F = 0.695$	$\mu_{CR} = 0.048$
Diversity = 1.2693e+01	$k_t = [4, 3, 3, 3]$		
Generation 470: Best = 2.744241e+03	FES = 47100	$\mu_F = 0.695$	$\mu_{CR} = 0.055$
Diversity = 1.2671e+01	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 2.744241e+03	FES = 48100	$\mu_F = 0.666$	$\mu_{CR} = 0.048$
Diversity = 1.2576e+01	$k_t = [4, 4, 4, 4]$		
Generation 490: Best = 2.741216e+03	FES = 49100	$\mu_F = 0.625$	$\mu_{CR} = 0.052$
Diversity = 1.2572e+01	$k_t = [1, 1, 1, 1]$		

Generation 500: Best = 2.741216e+03	FES = 50100	$\mu_F = 0.511$	$\mu_{CR} = 0.043$
Diversity = 1.2568e+01	$k_t = [2, 2, 4, 2]$		
Generation 510: Best = 2.741216e+03	FES = 51100	$\mu_F = 0.496$	$\mu_{CR} = 0.047$
Diversity = 1.2581e+01	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.741216e+03	FES = 52100	$\mu_F = 0.548$	$\mu_{CR} = 0.061$
Diversity = 1.2586e+01	$k_t = [4, 3, 2, 4]$		
Generation 530: Best = 2.741216e+03	FES = 53100	$\mu_F = 0.545$	$\mu_{CR} = 0.050$
Diversity = 1.2618e+01	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.741216e+03	FES = 54100	$\mu_F = 0.559$	$\mu_{CR} = 0.049$
Diversity = 1.2594e+01	$k_t = [2, 4, 2, 3]$		
Generation 550: Best = 2.733487e+03	FES = 55100	$\mu_F = 0.477$	$\mu_{CR} = 0.059$
Diversity = 1.2606e+01	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.733487e+03	FES = 56100	$\mu_F = 0.465$	$\mu_{CR} = 0.052$
Diversity = 1.2611e+01	$k_t = [3, 3, 2, 2]$		
Generation 570: Best = 2.732846e+03	FES = 57100	$\mu_F = 0.464$	$\mu_{CR} = 0.051$
Diversity = 1.2603e+01	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.732383e+03	FES = 58100	$\mu_F = 0.496$	$\mu_{CR} = 0.065$
Diversity = 1.2612e+01	$k_t = [4, 2, 2, 4]$		
Generation 590: Best = 2.728651e+03	FES = 59100	$\mu_F = 0.526$	$\mu_{CR} = 0.057$
Diversity = 1.2607e+01	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.728651e+03	FES = 60100	$\mu_F = 0.506$	$\mu_{CR} = 0.056$
Diversity = 1.2588e+01	$k_t = [2, 2, 3, 2]$		
Generation 610: Best = 2.726126e+03	FES = 61100	$\mu_F = 0.432$	$\mu_{CR} = 0.072$
Diversity = 1.2525e+01	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.726126e+03	FES = 62100	$\mu_F = 0.290$	$\mu_{CR} = 0.080$
Diversity = 1.2505e+01	$k_t = [2, 4, 2, 3]$		
Generation 630: Best = 2.725749e+03	FES = 63100	$\mu_F = 0.254$	$\mu_{CR} = 0.082$
Diversity = 1.2474e+01	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.719663e+03	FES = 64100	$\mu_F = 0.248$	$\mu_{CR} = 0.082$
Diversity = 1.2441e+01	$k_t = [3, 2, 4, 2]$		
Generation 650: Best = 2.714091e+03	FES = 65100	$\mu_F = 0.255$	$\mu_{CR} = 0.080$
Diversity = 1.2423e+01	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.714091e+03	FES = 66100	$\mu_F = 0.253$	$\mu_{CR} = 0.082$
Diversity = 1.2386e+01	$k_t = [4, 4, 3, 3]$		
Generation 670: Best = 2.714091e+03	FES = 67100	$\mu_F = 0.319$	$\mu_{CR} = 0.093$
Diversity = 1.2364e+01	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.712717e+03	FES = 68100	$\mu_F = 0.438$	$\mu_{CR} = 0.074$
Diversity = 1.2414e+01	$k_t = [3, 2, 2, 4]$		
Generation 690: Best = 2.709971e+03	FES = 69100	$\mu_F = 0.417$	$\mu_{CR} = 0.055$
Diversity = 1.2421e+01	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.709971e+03	FES = 70100	$\mu_F = 0.466$	$\mu_{CR} = 0.055$
Diversity = 1.2464e+01	$k_t = [3, 3, 3, 4]$		
Generation 710: Best = 2.709971e+03	FES = 71100	$\mu_F = 0.530$	$\mu_{CR} = 0.046$
Diversity = 1.2463e+01	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 2.709971e+03	FES = 72100	$\mu_F = 0.471$	$\mu_{CR} = 0.054$
Diversity = 1.2495e+01	$k_t = [4, 3, 2, 4]$		
Generation 730: Best = 2.709971e+03	FES = 73100	$\mu_F = 0.514$	$\mu_{CR} = 0.053$
Diversity = 1.2448e+01	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 2.709971e+03	FES = 74100	$\mu_F = 0.521$	$\mu_{CR} = 0.034$
Diversity = 1.2447e+01	$k_t = [4, 4, 4, 2]$		
Generation 750: Best = 2.709971e+03	FES = 75100	$\mu_F = 0.523$	$\mu_{CR} = 0.046$
Diversity = 1.2442e+01	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 2.707983e+03	FES = 76100	$\mu_F = 0.572$	$\mu_{CR} = 0.041$
Diversity = 1.2440e+01	$k_t = [2, 3, 3, 2]$		

Generation 770:	Best = 2.707983e+03	FES = 77100	$\mu_F = 0.417$	$\mu_{CR} = 0.060$
Diversity = 1.2440e+01	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 2.707983e+03	FES = 78100	$\mu_F = 0.419$	$\mu_{CR} = 0.054$
Diversity = 1.2412e+01	$k_t = [3, 4, 3, 2]$			
Generation 790:	Best = 2.706791e+03	FES = 79100	$\mu_F = 0.398$	$\mu_{CR} = 0.038$
Diversity = 1.2418e+01	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 2.706791e+03	FES = 80100	$\mu_F = 0.455$	$\mu_{CR} = 0.057$
Diversity = 1.2387e+01	$k_t = [2, 3, 2, 4]$			
Generation 810:	Best = 2.706078e+03	FES = 81100	$\mu_F = 0.433$	$\mu_{CR} = 0.055$
Diversity = 1.2381e+01	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 2.706078e+03	FES = 82100	$\mu_F = 0.403$	$\mu_{CR} = 0.063$
Diversity = 1.2383e+01	$k_t = [2, 3, 2, 2]$			
Generation 830:	Best = 2.705781e+03	FES = 83100	$\mu_F = 0.447$	$\mu_{CR} = 0.055$
Diversity = 1.2383e+01	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 2.705781e+03	FES = 84100	$\mu_F = 0.485$	$\mu_{CR} = 0.063$
Diversity = 1.2319e+01	$k_t = [4, 4, 4, 4]$			
Generation 850:	Best = 2.703549e+03	FES = 85100	$\mu_F = 0.558$	$\mu_{CR} = 0.050$
Diversity = 1.2313e+01	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 2.703549e+03	FES = 86100	$\mu_F = 0.501$	$\mu_{CR} = 0.043$
Diversity = 1.2345e+01	$k_t = [2, 4, 3, 4]$			
Generation 870:	Best = 2.703549e+03	FES = 87100	$\mu_F = 0.509$	$\mu_{CR} = 0.039$
Diversity = 1.2342e+01	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 2.703549e+03	FES = 88100	$\mu_F = 0.452$	$\mu_{CR} = 0.049$
Diversity = 1.2335e+01	$k_t = [2, 3, 2, 4]$			
Generation 890:	Best = 2.703148e+03	FES = 89100	$\mu_F = 0.373$	$\mu_{CR} = 0.044$
Diversity = 1.2333e+01	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 2.702305e+03	FES = 90100	$\mu_F = 0.408$	$\mu_{CR} = 0.058$
Diversity = 1.2331e+01	$k_t = [2, 4, 2, 4]$			
Generation 910:	Best = 2.702305e+03	FES = 91100	$\mu_F = 0.331$	$\mu_{CR} = 0.055$
Diversity = 1.2317e+01	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 2.701184e+03	FES = 92100	$\mu_F = 0.337$	$\mu_{CR} = 0.054$
Diversity = 1.2296e+01	$k_t = [2, 2, 2, 3]$			
Generation 930:	Best = 2.701184e+03	FES = 93100	$\mu_F = 0.306$	$\mu_{CR} = 0.067$
Diversity = 1.2282e+01	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 2.701184e+03	FES = 94100	$\mu_F = 0.324$	$\mu_{CR} = 0.069$
Diversity = 1.2288e+01	$k_t = [2, 3, 3, 2]$			
Generation 950:	Best = 2.699353e+03	FES = 95100	$\mu_F = 0.285$	$\mu_{CR} = 0.069$
Diversity = 1.2285e+01	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 2.698552e+03	FES = 96100	$\mu_F = 0.293$	$\mu_{CR} = 0.078$
Diversity = 1.2290e+01	$k_t = [3, 4, 4, 3]$			
Generation 970:	Best = 2.698531e+03	FES = 97100	$\mu_F = 0.379$	$\mu_{CR} = 0.057$
Diversity = 1.2290e+01	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 2.698531e+03	FES = 98100	$\mu_F = 0.496$	$\mu_{CR} = 0.052$
Diversity = 1.2245e+01	$k_t = [2, 3, 3, 4]$			
Generation 990:	Best = 2.698531e+03	FES = 99100	$\mu_F = 0.527$	$\mu_{CR} = 0.049$
Diversity = 1.2223e+01	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 2.698531e+03	FES = 100100	$\mu_F = 0.618$	$\mu_{CR} = 0.037$
Diversity = 1.2191e+01	$k_t = [3, 2, 3, 3]$			
Generation 1010:	Best = 2.698531e+03	FES = 101100	$\mu_F = 0.651$	$\mu_{CR} = 0.039$
Diversity = 1.2193e+01	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 2.698119e+03	FES = 102100	$\mu_F = 0.711$	$\mu_{CR} = 0.041$
Diversity = 1.2089e+01	$k_t = [2, 4, 2, 4]$			
Generation 1030:	Best = 2.698119e+03	FES = 103100	$\mu_F = 0.585$	$\mu_{CR} = 0.037$
Diversity = 1.2091e+01	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 2.698119e+03 FES = 104100 μ_F = 0.564 μ_{CR} = 0.043
 Diversity = 1.2096e+01 k_t = [4, 2, 4, 2]
 Generation 1050: Best = 2.698119e+03 FES = 105100 μ_F = 0.598 μ_{CR} = 0.051
 Diversity = 1.1988e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 2.696097e+03 FES = 106100 μ_F = 0.618 μ_{CR} = 0.061
 Diversity = 1.1988e+01 k_t = [4, 4, 4, 2]
 Generation 1070: Best = 2.696097e+03 FES = 107100 μ_F = 0.633 μ_{CR} = 0.050
 Diversity = 1.1989e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 2.696097e+03 FES = 108100 μ_F = 0.619 μ_{CR} = 0.042
 Diversity = 1.2001e+01 k_t = [3, 3, 3, 2]
 Generation 1090: Best = 2.696073e+03 FES = 109100 μ_F = 0.664 μ_{CR} = 0.037
 Diversity = 1.2000e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 2.694784e+03 FES = 110100 μ_F = 0.654 μ_{CR} = 0.046
 Diversity = 1.1995e+01 k_t = [4, 4, 3, 4]
 Generation 1110: Best = 2.694784e+03 FES = 111100 μ_F = 0.560 μ_{CR} = 0.051
 Diversity = 1.1994e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 2.694498e+03 FES = 112100 μ_F = 0.397 μ_{CR} = 0.065
 Diversity = 1.1997e+01 k_t = [2, 2, 2, 3]
 Generation 1130: Best = 2.693738e+03 FES = 113100 μ_F = 0.334 μ_{CR} = 0.068
 Diversity = 1.1996e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 2.692509e+03 FES = 114100 μ_F = 0.349 μ_{CR} = 0.069
 Diversity = 1.1989e+01 k_t = [3, 2, 2, 2]
 Generation 1150: Best = 2.692509e+03 FES = 115100 μ_F = 0.319 μ_{CR} = 0.066
 Diversity = 1.1987e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 2.691954e+03 FES = 116100 μ_F = 0.294 μ_{CR} = 0.066
 Diversity = 1.1986e+01 k_t = [4, 3, 4, 4]
 Generation 1170: Best = 2.691954e+03 FES = 117100 μ_F = 0.303 μ_{CR} = 0.055
 Diversity = 1.1983e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 2.690681e+03 FES = 118100 μ_F = 0.298 μ_{CR} = 0.059
 Diversity = 1.1987e+01 k_t = [2, 2, 3, 3]
 Generation 1190: Best = 2.687644e+03 FES = 119100 μ_F = 0.215 μ_{CR} = 0.083
 Diversity = 1.1977e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 2.687644e+03 FES = 120100 μ_F = 0.223 μ_{CR} = 0.072
 Diversity = 1.1979e+01 k_t = [4, 2, 3, 2]
 Generation 1210: Best = 2.685782e+03 FES = 121100 μ_F = 0.236 μ_{CR} = 0.065
 Diversity = 1.1975e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 2.685155e+03 FES = 122100 μ_F = 0.316 μ_{CR} = 0.067
 Diversity = 1.1929e+01 k_t = [4, 4, 2, 4]
 Generation 1230: Best = 2.685155e+03 FES = 123100 μ_F = 0.389 μ_{CR} = 0.049
 Diversity = 1.1924e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 2.684173e+03 FES = 124100 μ_F = 0.442 μ_{CR} = 0.043
 Diversity = 1.1904e+01 k_t = [4, 3, 3, 3]
 Generation 1250: Best = 2.684173e+03 FES = 125100 μ_F = 0.444 μ_{CR} = 0.054
 Diversity = 1.1906e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 2.683949e+03 FES = 126100 μ_F = 0.401 μ_{CR} = 0.065
 Diversity = 1.1948e+01 k_t = [4, 4, 2, 4]
 Generation 1270: Best = 2.683949e+03 FES = 127100 μ_F = 0.338 μ_{CR} = 0.070
 Diversity = 1.1942e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 2.683933e+03 FES = 128100 μ_F = 0.302 μ_{CR} = 0.062
 Diversity = 1.1881e+01 k_t = [4, 3, 3, 2]
 Generation 1290: Best = 2.683933e+03 FES = 129100 μ_F = 0.300 μ_{CR} = 0.058
 Diversity = 1.1882e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 2.683687e+03 FES = 130100 μ_F = 0.329 μ_{CR} = 0.054
 Diversity = 1.1833e+01 k_t = [3, 3, 4, 2]

Generation 1310: Best = 2.683461e+03 FES = 131100 μ_F = 0.267 μ_{CR} = 0.060
 Diversity = 1.1826e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 2.683461e+03 FES = 132100 μ_F = 0.304 μ_{CR} = 0.085
 Diversity = 1.1824e+01 k_t = [4, 2, 4, 2]
 Generation 1330: Best = 2.683335e+03 FES = 133100 μ_F = 0.277 μ_{CR} = 0.069
 Diversity = 1.1826e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 2.683335e+03 FES = 134100 μ_F = 0.295 μ_{CR} = 0.068
 Diversity = 1.1744e+01 k_t = [4, 2, 3, 4]
 Generation 1350: Best = 2.682975e+03 FES = 135100 μ_F = 0.314 μ_{CR} = 0.081
 Diversity = 1.1745e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 2.682975e+03 FES = 136100 μ_F = 0.280 μ_{CR} = 0.058
 Diversity = 1.1737e+01 k_t = [3, 3, 2, 2]
 Generation 1370: Best = 2.682975e+03 FES = 137100 μ_F = 0.318 μ_{CR} = 0.046
 Diversity = 1.1740e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 2.682975e+03 FES = 138100 μ_F = 0.381 μ_{CR} = 0.048
 Diversity = 1.1785e+01 k_t = [3, 3, 4, 4]
 Generation 1390: Best = 2.682835e+03 FES = 139100 μ_F = 0.441 μ_{CR} = 0.047
 Diversity = 1.1797e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 2.682578e+03 FES = 140100 μ_F = 0.419 μ_{CR} = 0.042
 Diversity = 1.1690e+01 k_t = [4, 3, 4, 4]
 Generation 1410: Best = 2.680502e+03 FES = 141100 μ_F = 0.362 μ_{CR} = 0.040
 Diversity = 1.1689e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 2.679742e+03 FES = 142100 μ_F = 0.355 μ_{CR} = 0.037
 Diversity = 1.1686e+01 k_t = [2, 3, 4, 2]
 Generation 1430: Best = 2.679742e+03 FES = 143100 μ_F = 0.340 μ_{CR} = 0.052
 Diversity = 1.1688e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 2.679270e+03 FES = 144100 μ_F = 0.388 μ_{CR} = 0.054
 Diversity = 1.1688e+01 k_t = [3, 3, 2, 4]
 Generation 1450: Best = 2.679270e+03 FES = 145100 μ_F = 0.452 μ_{CR} = 0.054
 Diversity = 1.1689e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 2.679270e+03 FES = 146100 μ_F = 0.419 μ_{CR} = 0.060
 Diversity = 1.1691e+01 k_t = [2, 2, 4, 2]
 Generation 1470: Best = 2.679242e+03 FES = 147100 μ_F = 0.454 μ_{CR} = 0.070
 Diversity = 1.1692e+01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 2.677673e+03 FES = 148100 μ_F = 0.434 μ_{CR} = 0.053
 Diversity = 1.1693e+01 k_t = [4, 2, 3, 3]
 Generation 1490: Best = 2.677673e+03 FES = 149100 μ_F = 0.415 μ_{CR} = 0.054
 Diversity = 1.1699e+01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 2.677292e+03 FES = 150100 μ_F = 0.395 μ_{CR} = 0.058
 Diversity = 1.1701e+01 k_t = [4, 3, 4, 3]
 Generation 1510: Best = 2.677292e+03 FES = 151100 μ_F = 0.420 μ_{CR} = 0.054
 Diversity = 1.1677e+01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 2.676832e+03 FES = 152100 μ_F = 0.487 μ_{CR} = 0.059
 Diversity = 1.1678e+01 k_t = [4, 4, 2, 2]
 Generation 1530: Best = 2.676832e+03 FES = 153100 μ_F = 0.411 μ_{CR} = 0.051
 Diversity = 1.1678e+01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 2.676628e+03 FES = 154100 μ_F = 0.430 μ_{CR} = 0.053
 Diversity = 1.1668e+01 k_t = [3, 4, 2, 2]
 Generation 1550: Best = 2.676628e+03 FES = 155100 μ_F = 0.315 μ_{CR} = 0.043
 Diversity = 1.1668e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 2.676628e+03 FES = 156100 μ_F = 0.400 μ_{CR} = 0.055
 Diversity = 1.1672e+01 k_t = [2, 3, 4, 2]
 Generation 1570: Best = 2.676470e+03 FES = 157100 μ_F = 0.410 μ_{CR} = 0.067
 Diversity = 1.1645e+01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 2.676470e+03 FES = 158100 μ_F = 0.514 μ_{CR} = 0.058
 Diversity = 1.1565e+01 k_t = [2, 4, 4, 4]
 Generation 1590: Best = 2.675823e+03 FES = 159100 μ_F = 0.501 μ_{CR} = 0.042
 Diversity = 1.1567e+01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 2.675823e+03 FES = 160100 μ_F = 0.519 μ_{CR} = 0.048
 Diversity = 1.1581e+01 k_t = [2, 4, 4, 2]
 Generation 1610: Best = 2.675823e+03 FES = 161100 μ_F = 0.404 μ_{CR} = 0.044
 Diversity = 1.1581e+01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 2.675823e+03 FES = 162100 μ_F = 0.384 μ_{CR} = 0.032
 Diversity = 1.1582e+01 k_t = [4, 2, 3, 2]
 Generation 1630: Best = 2.675823e+03 FES = 163100 μ_F = 0.349 μ_{CR} = 0.069
 Diversity = 1.1580e+01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 2.675823e+03 FES = 164100 μ_F = 0.288 μ_{CR} = 0.061
 Diversity = 1.1579e+01 k_t = [2, 3, 3, 2]
 Generation 1650: Best = 2.675823e+03 FES = 165100 μ_F = 0.371 μ_{CR} = 0.048
 Diversity = 1.1583e+01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 2.675823e+03 FES = 166100 μ_F = 0.296 μ_{CR} = 0.062
 Diversity = 1.1577e+01 k_t = [3, 2, 2, 3]
 Generation 1670: Best = 2.675734e+03 FES = 167100 μ_F = 0.279 μ_{CR} = 0.057
 Diversity = 1.1577e+01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 2.675617e+03 FES = 168100 μ_F = 0.244 μ_{CR} = 0.078
 Diversity = 1.1572e+01 k_t = [4, 4, 4, 2]
 Generation 1690: Best = 2.675617e+03 FES = 169100 μ_F = 0.336 μ_{CR} = 0.061
 Diversity = 1.1591e+01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 2.675617e+03 FES = 170100 μ_F = 0.364 μ_{CR} = 0.052
 Diversity = 1.1588e+01 k_t = [4, 2, 2, 2]
 Generation 1710: Best = 2.675617e+03 FES = 171100 μ_F = 0.342 μ_{CR} = 0.041
 Diversity = 1.1589e+01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 2.675524e+03 FES = 172100 μ_F = 0.356 μ_{CR} = 0.045
 Diversity = 1.1587e+01 k_t = [3, 4, 3, 4]
 Generation 1730: Best = 2.675504e+03 FES = 173100 μ_F = 0.357 μ_{CR} = 0.056
 Diversity = 1.1582e+01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 2.675457e+03 FES = 174100 μ_F = 0.428 μ_{CR} = 0.052
 Diversity = 1.1597e+01 k_t = [4, 4, 2, 3]
 Generation 1750: Best = 2.675457e+03 FES = 175100 μ_F = 0.530 μ_{CR} = 0.061
 Diversity = 1.1574e+01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 2.675457e+03 FES = 176100 μ_F = 0.514 μ_{CR} = 0.065
 Diversity = 1.1577e+01 k_t = [3, 4, 2, 3]
 Generation 1770: Best = 2.675457e+03 FES = 177100 μ_F = 0.594 μ_{CR} = 0.061
 Diversity = 1.1524e+01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 2.675428e+03 FES = 178100 μ_F = 0.583 μ_{CR} = 0.050
 Diversity = 1.1527e+01 k_t = [2, 2, 2, 3]
 Generation 1790: Best = 2.675428e+03 FES = 179100 μ_F = 0.517 μ_{CR} = 0.050
 Diversity = 1.1478e+01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 2.675428e+03 FES = 180100 μ_F = 0.499 μ_{CR} = 0.051
 Diversity = 1.1484e+01 k_t = [4, 2, 2, 3]
 Generation 1810: Best = 2.675428e+03 FES = 181100 μ_F = 0.506 μ_{CR} = 0.046
 Diversity = 1.1423e+01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 2.675428e+03 FES = 182100 μ_F = 0.516 μ_{CR} = 0.040
 Diversity = 1.1399e+01 k_t = [4, 4, 3, 4]
 Generation 1830: Best = 2.675428e+03 FES = 183100 μ_F = 0.565 μ_{CR} = 0.047
 Diversity = 1.1401e+01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 2.675428e+03 FES = 184100 μ_F = 0.490 μ_{CR} = 0.045
 Diversity = 1.1397e+01 k_t = [2, 3, 3, 4]

Generation 1850: Best = 2.675332e+03	FES = 185100	$\mu_F = 0.524$	$\mu_{CR} = 0.053$
Diversity = 1.1396e+01	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 2.675304e+03	FES = 186100	$\mu_F = 0.635$	$\mu_{CR} = 0.050$
Diversity = 1.1378e+01	$k_t = [2, 2, 4, 4]$		
Generation 1870: Best = 2.675304e+03	FES = 187100	$\mu_F = 0.584$	$\mu_{CR} = 0.050$
Diversity = 1.1378e+01	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 2.675261e+03	FES = 188100	$\mu_F = 0.645$	$\mu_{CR} = 0.053$
Diversity = 1.1268e+01	$k_t = [3, 2, 4, 4]$		
Generation 1890: Best = 2.675261e+03	FES = 189100	$\mu_F = 0.576$	$\mu_{CR} = 0.075$
Diversity = 1.1273e+01	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 2.675261e+03	FES = 190100	$\mu_F = 0.503$	$\mu_{CR} = 0.062$
Diversity = 1.1290e+01	$k_t = [3, 3, 4, 4]$		
Generation 1910: Best = 2.675024e+03	FES = 191100	$\mu_F = 0.522$	$\mu_{CR} = 0.058$
Diversity = 1.1233e+01	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 2.675024e+03	FES = 192100	$\mu_F = 0.556$	$\mu_{CR} = 0.057$
Diversity = 1.1233e+01	$k_t = [4, 2, 3, 3]$		
Generation 1930: Best = 2.675024e+03	FES = 193100	$\mu_F = 0.655$	$\mu_{CR} = 0.048$
Diversity = 1.1230e+01	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 2.675024e+03	FES = 194100	$\mu_F = 0.679$	$\mu_{CR} = 0.045$
Diversity = 1.1122e+01	$k_t = [4, 3, 2, 3]$		
Generation 1950: Best = 2.675024e+03	FES = 195100	$\mu_F = 0.615$	$\mu_{CR} = 0.046$
Diversity = 1.1122e+01	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 2.674971e+03	FES = 196100	$\mu_F = 0.607$	$\mu_{CR} = 0.046$
Diversity = 1.1125e+01	$k_t = [3, 3, 3, 2]$		
Generation 1970: Best = 2.674971e+03	FES = 197100	$\mu_F = 0.624$	$\mu_{CR} = 0.043$
Diversity = 1.1094e+01	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 2.674971e+03	FES = 198100	$\mu_F = 0.663$	$\mu_{CR} = 0.054$
Diversity = 1.1095e+01	$k_t = [3, 4, 3, 3]$		
Generation 1990: Best = 2.674947e+03	FES = 199100	$\mu_F = 0.686$	$\mu_{CR} = 0.066$
Diversity = 1.1069e+01	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 2.674947e+03	FES = 200100	$\mu_F = 0.619$	$\mu_{CR} = 0.063$
Diversity = 1.1070e+01	$k_t = [2, 4, 4, 3]$		
Generation 2010: Best = 2.674947e+03	FES = 201100	$\mu_F = 0.610$	$\mu_{CR} = 0.050$
Diversity = 1.1070e+01	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 2.674947e+03	FES = 202100	$\mu_F = 0.545$	$\mu_{CR} = 0.067$
Diversity = 1.1074e+01	$k_t = [4, 2, 2, 3]$		
Generation 2030: Best = 2.674947e+03	FES = 203100	$\mu_F = 0.520$	$\mu_{CR} = 0.048$
Diversity = 1.1078e+01	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 2.674947e+03	FES = 204100	$\mu_F = 0.506$	$\mu_{CR} = 0.062$
Diversity = 1.1072e+01	$k_t = [2, 3, 4, 3]$		
Generation 2050: Best = 2.674848e+03	FES = 205100	$\mu_F = 0.554$	$\mu_{CR} = 0.060$
Diversity = 1.1076e+01	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 2.674848e+03	FES = 206100	$\mu_F = 0.560$	$\mu_{CR} = 0.070$
Diversity = 1.1009e+01	$k_t = [3, 2, 3, 3]$		
Generation 2070: Best = 2.674745e+03	FES = 207100	$\mu_F = 0.651$	$\mu_{CR} = 0.076$
Diversity = 1.0980e+01	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 2.674744e+03	FES = 208100	$\mu_F = 0.627$	$\mu_{CR} = 0.080$
Diversity = 1.0978e+01	$k_t = [3, 4, 3, 3]$		
Generation 2090: Best = 2.674744e+03	FES = 209100	$\mu_F = 0.619$	$\mu_{CR} = 0.093$
Diversity = 1.0981e+01	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 2.674744e+03	FES = 210100	$\mu_F = 0.590$	$\mu_{CR} = 0.058$
Diversity = 1.0982e+01	$k_t = [4, 3, 3, 3]$		
Generation 2110: Best = 2.674744e+03	FES = 211100	$\mu_F = 0.492$	$\mu_{CR} = 0.074$
Diversity = 1.0982e+01	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 2.674744e+03 FES = 212100 μ_F = 0.568 μ_{CR} = 0.070
 Diversity = 1.0875e+01 k_t = [2, 2, 2, 4]
 Generation 2130: Best = 2.674728e+03 FES = 213100 μ_F = 0.525 μ_{CR} = 0.065
 Diversity = 1.0865e+01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 2.674677e+03 FES = 214100 μ_F = 0.434 μ_{CR} = 0.061
 Diversity = 1.0863e+01 k_t = [3, 2, 4, 3]
 Generation 2150: Best = 2.674677e+03 FES = 215100 μ_F = 0.457 μ_{CR} = 0.068
 Diversity = 1.0866e+01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 2.674622e+03 FES = 216100 μ_F = 0.605 μ_{CR} = 0.062
 Diversity = 1.0771e+01 k_t = [4, 2, 3, 4]
 Generation 2170: Best = 2.674471e+03 FES = 217100 μ_F = 0.475 μ_{CR} = 0.052
 Diversity = 1.0775e+01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 2.674434e+03 FES = 218100 μ_F = 0.474 μ_{CR} = 0.058
 Diversity = 1.0775e+01 k_t = [4, 4, 2, 3]
 Generation 2190: Best = 2.674434e+03 FES = 219100 μ_F = 0.416 μ_{CR} = 0.053
 Diversity = 1.0775e+01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 2.674400e+03 FES = 220100 μ_F = 0.381 μ_{CR} = 0.056
 Diversity = 1.0777e+01 k_t = [3, 4, 4, 3]
 Generation 2210: Best = 2.674400e+03 FES = 221100 μ_F = 0.440 μ_{CR} = 0.059
 Diversity = 1.0730e+01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 2.674400e+03 FES = 222100 μ_F = 0.461 μ_{CR} = 0.069
 Diversity = 1.0731e+01 k_t = [3, 3, 2, 4]
 Generation 2230: Best = 2.674400e+03 FES = 223100 μ_F = 0.429 μ_{CR} = 0.054
 Diversity = 1.0651e+01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 2.674312e+03 FES = 224100 μ_F = 0.405 μ_{CR} = 0.060
 Diversity = 1.0646e+01 k_t = [3, 3, 2, 2]
 Generation 2250: Best = 2.674312e+03 FES = 225100 μ_F = 0.391 μ_{CR} = 0.059
 Diversity = 1.0643e+01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 2.674170e+03 FES = 226100 μ_F = 0.415 μ_{CR} = 0.059
 Diversity = 1.0645e+01 k_t = [4, 3, 3, 4]
 Generation 2270: Best = 2.674170e+03 FES = 227100 μ_F = 0.446 μ_{CR} = 0.058
 Diversity = 1.0617e+01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 2.674170e+03 FES = 228100 μ_F = 0.493 μ_{CR} = 0.065
 Diversity = 1.0583e+01 k_t = [2, 2, 4, 2]
 Generation 2290: Best = 2.674071e+03 FES = 229100 μ_F = 0.602 μ_{CR} = 0.059
 Diversity = 1.0542e+01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 2.674019e+03 FES = 230100 μ_F = 0.632 μ_{CR} = 0.056
 Diversity = 1.0467e+01 k_t = [2, 2, 2, 2]
 Generation 2310: Best = 2.674002e+03 FES = 231100 μ_F = 0.621 μ_{CR} = 0.057
 Diversity = 1.0470e+01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 2.674002e+03 FES = 232100 μ_F = 0.595 μ_{CR} = 0.052
 Diversity = 1.0359e+01 k_t = [4, 2, 3, 2]
 Generation 2330: Best = 2.674002e+03 FES = 233100 μ_F = 0.555 μ_{CR} = 0.065
 Diversity = 1.0355e+01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 2.673988e+03 FES = 234100 μ_F = 0.572 μ_{CR} = 0.070
 Diversity = 1.0319e+01 k_t = [2, 4, 2, 3]
 Generation 2350: Best = 2.673988e+03 FES = 235100 μ_F = 0.527 μ_{CR} = 0.052
 Diversity = 1.0204e+01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 2.673988e+03 FES = 236100 μ_F = 0.525 μ_{CR} = 0.051
 Diversity = 1.0132e+01 k_t = [2, 3, 3, 4]
 Generation 2370: Best = 2.673974e+03 FES = 237100 μ_F = 0.580 μ_{CR} = 0.051
 Diversity = 1.0131e+01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 2.673959e+03 FES = 238100 μ_F = 0.543 μ_{CR} = 0.064
 Diversity = 1.0131e+01 k_t = [2, 3, 2, 2]

Generation 2390: Best = 2.673897e+03 FES = 239100 μ_F = 0.587 μ_{CR} = 0.062
 Diversity = 1.0023e+01 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 2.673897e+03 FES = 240100 μ_F = 0.513 μ_{CR} = 0.054
 Diversity = 1.0023e+01 k_t = [4, 2, 3, 2]
 Generation 2410: Best = 2.673878e+03 FES = 241100 μ_F = 0.492 μ_{CR} = 0.046
 Diversity = 1.0024e+01 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 2.673862e+03 FES = 242100 μ_F = 0.520 μ_{CR} = 0.056
 Diversity = 9.9635e+00 k_t = [3, 2, 4, 4]
 Generation 2430: Best = 2.673797e+03 FES = 243100 μ_F = 0.528 μ_{CR} = 0.056
 Diversity = 9.9646e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 2.673788e+03 FES = 244100 μ_F = 0.606 μ_{CR} = 0.051
 Diversity = 9.9393e+00 k_t = [2, 3, 2, 2]
 Generation 2450: Best = 2.673721e+03 FES = 245100 μ_F = 0.621 μ_{CR} = 0.066
 Diversity = 9.9135e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 2.673721e+03 FES = 246100 μ_F = 0.654 μ_{CR} = 0.080
 Diversity = 9.9158e+00 k_t = [3, 4, 4, 2]
 Generation 2470: Best = 2.673719e+03 FES = 247100 μ_F = 0.651 μ_{CR} = 0.061
 Diversity = 9.9142e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 2.673719e+03 FES = 248100 μ_F = 0.589 μ_{CR} = 0.072
 Diversity = 9.8675e+00 k_t = [3, 4, 4, 2]
 Generation 2490: Best = 2.673719e+03 FES = 249100 μ_F = 0.679 μ_{CR} = 0.078
 Diversity = 9.6657e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 2.673717e+03 FES = 250100 μ_F = 0.707 μ_{CR} = 0.087
 Diversity = 9.5763e+00 k_t = [3, 4, 4, 4]
 Generation 2510: Best = 2.673712e+03 FES = 251100 μ_F = 0.717 μ_{CR} = 0.064
 Diversity = 9.3575e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 2.673712e+03 FES = 252100 μ_F = 0.765 μ_{CR} = 0.077
 Diversity = 9.1905e+00 k_t = [2, 4, 3, 3]
 Generation 2530: Best = 2.673712e+03 FES = 253100 μ_F = 0.821 μ_{CR} = 0.070
 Diversity = 9.0185e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 2.673712e+03 FES = 254100 μ_F = 0.817 μ_{CR} = 0.065
 Diversity = 8.9052e+00 k_t = [3, 4, 4, 2]
 Generation 2550: Best = 2.673712e+03 FES = 255100 μ_F = 0.818 μ_{CR} = 0.089
 Diversity = 8.7280e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 2.673670e+03 FES = 256100 μ_F = 0.842 μ_{CR} = 0.082
 Diversity = 8.3617e+00 k_t = [4, 2, 2, 3]
 Generation 2570: Best = 2.673670e+03 FES = 257100 μ_F = 0.786 μ_{CR} = 0.071
 Diversity = 8.2387e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 2.673670e+03 FES = 258100 μ_F = 0.774 μ_{CR} = 0.067
 Diversity = 8.1618e+00 k_t = [2, 4, 2, 2]
 Generation 2590: Best = 2.673668e+03 FES = 259100 μ_F = 0.771 μ_{CR} = 0.082
 Diversity = 8.1193e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 2.673619e+03 FES = 260100 μ_F = 0.818 μ_{CR} = 0.075
 Diversity = 7.9998e+00 k_t = [4, 2, 2, 3]
 Generation 2610: Best = 2.673619e+03 FES = 261100 μ_F = 0.752 μ_{CR} = 0.093
 Diversity = 7.8572e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 2.673601e+03 FES = 262100 μ_F = 0.753 μ_{CR} = 0.093
 Diversity = 7.6850e+00 k_t = [2, 2, 4, 2]
 Generation 2630: Best = 2.673601e+03 FES = 263100 μ_F = 0.764 μ_{CR} = 0.086
 Diversity = 7.5423e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 2.673601e+03 FES = 264100 μ_F = 0.783 μ_{CR} = 0.078
 Diversity = 7.1756e+00 k_t = [2, 3, 4, 2]
 Generation 2650: Best = 2.673584e+03 FES = 265100 μ_F = 0.741 μ_{CR} = 0.074
 Diversity = 6.9179e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 2.673584e+03 FES = 266100 μ_F = 0.727 μ_{CR} = 0.084
 Diversity = 6.6469e+00 k_t = [3, 4, 3, 2]
 Generation 2670: Best = 2.673584e+03 FES = 267100 μ_F = 0.759 μ_{CR} = 0.079
 Diversity = 6.4563e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 2.673584e+03 FES = 268100 μ_F = 0.814 μ_{CR} = 0.069
 Diversity = 6.1360e+00 k_t = [3, 2, 4, 2]
 Generation 2690: Best = 2.673584e+03 FES = 269100 μ_F = 0.804 μ_{CR} = 0.062
 Diversity = 5.2918e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 2.673584e+03 FES = 270100 μ_F = 0.823 μ_{CR} = 0.066
 Diversity = 3.7005e+00 k_t = [4, 2, 4, 4]
 Generation 2710: Best = 2.673584e+03 FES = 271100 μ_F = 0.831 μ_{CR} = 0.080
 Diversity = 2.6505e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 2.673584e+03 FES = 272100 μ_F = 0.828 μ_{CR} = 0.066
 Diversity = 2.3375e+00 k_t = [3, 2, 2, 4]
 Generation 2730: Best = 2.673570e+03 FES = 273100 μ_F = 0.812 μ_{CR} = 0.080
 Diversity = 2.3420e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 2.673515e+03 FES = 274100 μ_F = 0.788 μ_{CR} = 0.101
 Diversity = 2.3585e+00 k_t = [3, 2, 3, 3]
 Generation 2750: Best = 2.673490e+03 FES = 275100 μ_F = 0.803 μ_{CR} = 0.109
 Diversity = 2.3489e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 2.673465e+03 FES = 276100 μ_F = 0.787 μ_{CR} = 0.109
 Diversity = 2.2022e+00 k_t = [3, 3, 4, 3]
 Generation 2770: Best = 2.673445e+03 FES = 277100 μ_F = 0.781 μ_{CR} = 0.130
 Diversity = 1.9875e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 2.673423e+03 FES = 278100 μ_F = 0.787 μ_{CR} = 0.139
 Diversity = 1.5673e+00 k_t = [4, 2, 3, 3]
 Generation 2790: Best = 2.673420e+03 FES = 279100 μ_F = 0.798 μ_{CR} = 0.156
 Diversity = 1.3246e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 2.673405e+03 FES = 280100 μ_F = 0.787 μ_{CR} = 0.168
 Diversity = 7.9469e-01 k_t = [2, 4, 3, 4]
 Generation 2810: Best = 2.673396e+03 FES = 281100 μ_F = 0.804 μ_{CR} = 0.173
 Diversity = 4.8279e-01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 2.673382e+03 FES = 282100 μ_F = 0.798 μ_{CR} = 0.177
 Diversity = 4.9564e-02 k_t = [4, 2, 4, 3]
 Generation 2830: Best = 2.673374e+03 FES = 283100 μ_F = 0.767 μ_{CR} = 0.178
 Diversity = 3.2104e-02 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 2.673368e+03 FES = 284100 μ_F = 0.760 μ_{CR} = 0.172
 Diversity = 2.1294e-02 k_t = [2, 2, 4, 4]
 Generation 2850: Best = 2.673365e+03 FES = 285100 μ_F = 0.749 μ_{CR} = 0.175
 Diversity = 1.4621e-02 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 2.673364e+03 FES = 286100 μ_F = 0.730 μ_{CR} = 0.174
 Diversity = 1.0492e-02 k_t = [2, 3, 2, 3]
 Generation 2870: Best = 2.673363e+03 FES = 287100 μ_F = 0.735 μ_{CR} = 0.176
 Diversity = 7.6077e-03 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 2.673363e+03 FES = 288100 μ_F = 0.726 μ_{CR} = 0.179
 Diversity = 5.4968e-03 k_t = [2, 3, 2, 2]
 Generation 2890: Best = 2.673362e+03 FES = 289100 μ_F = 0.714 μ_{CR} = 0.188
 Diversity = 4.1911e-03 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 2.673362e+03 FES = 290100 μ_F = 0.714 μ_{CR} = 0.196
 Diversity = 3.0759e-03 k_t = [2, 4, 3, 2]
 Generation 2910: Best = 2.673362e+03 FES = 291100 μ_F = 0.717 μ_{CR} = 0.199
 Diversity = 2.4799e-03 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 2.673362e+03 FES = 292100 μ_F = 0.716 μ_{CR} = 0.198
 Diversity = 1.7279e-03 k_t = [3, 4, 4, 4]

Generation 2930: Best = 2.673362e+03 FES = 293100 μ_F = 0.731 μ_{CR} = 0.209
 Diversity = 1.3186e-03 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 2.673362e+03 FES = 294100 μ_F = 0.739 μ_{CR} = 0.214
 Diversity = 9.9556e-04 k_t = [2, 4, 4, 3]
 Generation 2950: Best = 2.673362e+03 FES = 295100 μ_F = 0.742 μ_{CR} = 0.221
 Diversity = 8.8793e-04 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 2.673362e+03 FES = 296100 μ_F = 0.744 μ_{CR} = 0.227
 Diversity = 7.6393e-04 k_t = [4, 3, 2, 3]
 Generation 2970: Best = 2.673362e+03 FES = 297100 μ_F = 0.750 μ_{CR} = 0.238
 Diversity = 6.4766e-04 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 2.673362e+03 FES = 298100 μ_F = 0.746 μ_{CR} = 0.242
 Diversity = 5.1041e-04 k_t = [3, 4, 2, 4]
 Generation 2990: Best = 2.673362e+03 FES = 299100 μ_F = 0.753 μ_{CR} = 0.247
 Diversity = 4.2779e-04 k_t = [1, 1, 1, 1]

Optimization complete.

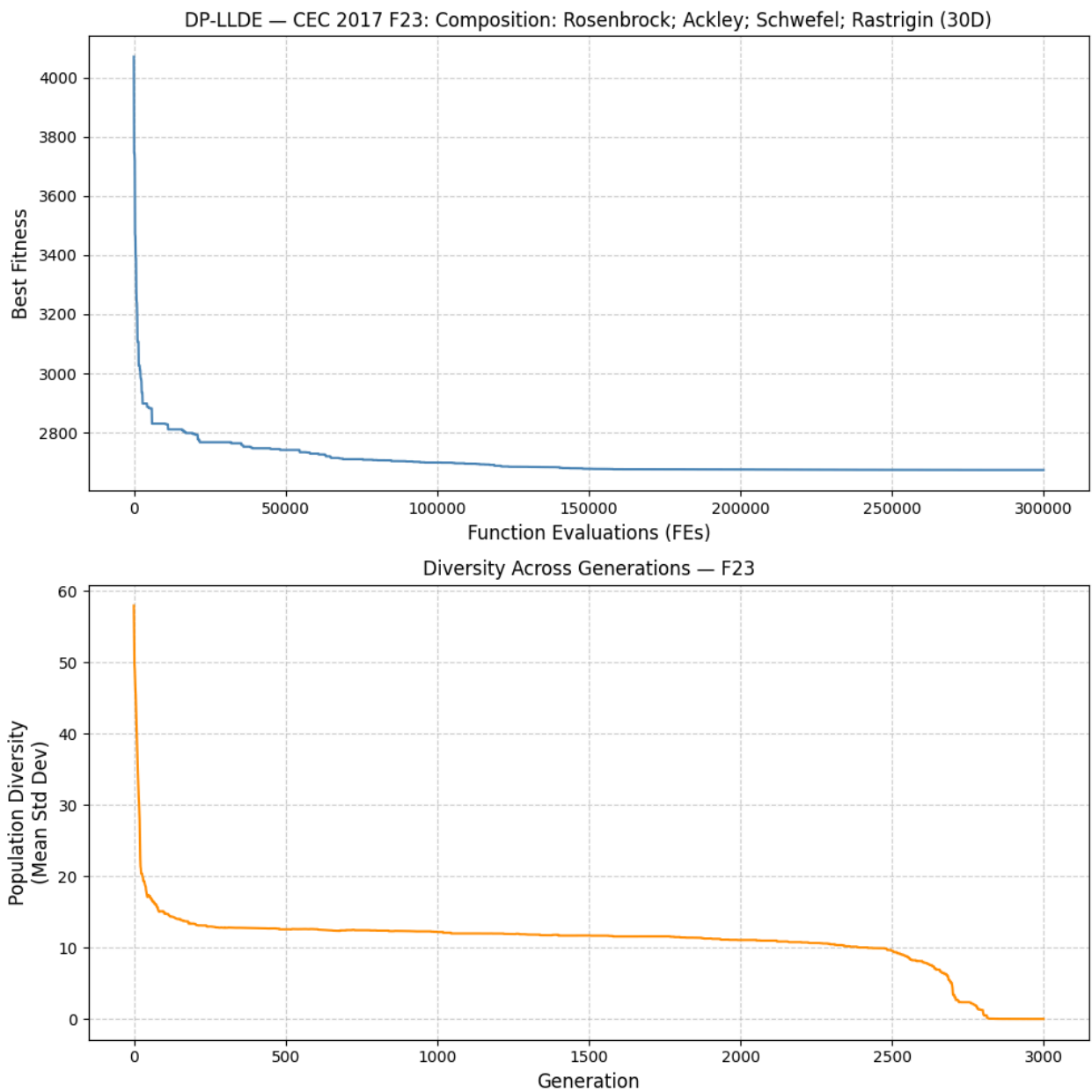
Best Fitness = 2.673362e+03

Total Function Evals = 300000

Total Successful Updates = 20031

```

=====
=
Results:
  Best Fitness      : 2.673362e+03
  Global Optimum    : 2.300000e+03
  Error (|best-optimum|) : 3.733618e+02
  Best Solution (5 dim) : [-87.58021416 -78.94449312 -49.04893293  68.27311277
-78.3824595 ]
  D^(0) (init. diversity): 5.7976e+01
=====
=
  
```



Graph saved as convergence_F23.png

```
In [ ]: # =====
# F24 — Composition Function 4 (Ackley; Elliptic; Griewank; Rastrigin)
# =====
run_and_plot('F24', make_wrapper(composition.f24), 2400.0, 'Composition: Ackle
```

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DP-LLDE on CEC 2017 F24 – Single Run

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=

Function : Composition: Ackley; Elliptic; Griewank; Rastrigin (30D)

Bounds : [-100, 100]

Global optimum : 2400.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 3.931320e+03

Generation 10: Best = 3.381428e+03 FES = 1100 $\mu_F = 0.587$ $\mu_{CR} = 0.356$

Diversity = 3.5905e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 3.120124e+03 FES = 2100 $\mu_F = 0.649$ $\mu_{CR} = 0.356$

Diversity = 2.2373e+01 $k_t = [4, 2, 4, 4]$

Generation 30: Best = 3.074589e+03 FES = 3100 $\mu_F = 0.657$ $\mu_{CR} = 0.351$

Diversity = 1.6798e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 3.061114e+03 FES = 4100 $\mu_F = 0.672$ $\mu_{CR} = 0.313$

Diversity = 1.5189e+01 $k_t = [4, 2, 4, 4]$

Generation 50: Best = 3.032845e+03 FES = 5100 $\mu_F = 0.647$ $\mu_{CR} = 0.306$

Diversity = 1.4665e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 3.032845e+03 FES = 6100 $\mu_F = 0.656$ $\mu_{CR} = 0.305$

Diversity = 1.3816e+01 $k_t = [3, 3, 2, 4]$

Generation 70: Best = 3.032845e+03 FES = 7100 $\mu_F = 0.648$ $\mu_{CR} = 0.244$

Diversity = 1.3416e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 2.999826e+03 FES = 8100 $\mu_F = 0.588$ $\mu_{CR} = 0.215$

Diversity = 1.3352e+01 $k_t = [4, 3, 4, 4]$

Generation 90: Best = 2.999826e+03 FES = 9100 $\mu_F = 0.567$ $\mu_{CR} = 0.186$

Diversity = 1.2990e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 2.965355e+03 FES = 10100 $\mu_F = 0.560$ $\mu_{CR} = 0.150$

Diversity = 1.2426e+01 $k_t = [2, 2, 2, 3]$

Generation 110: Best = 2.965355e+03 FES = 11100 $\mu_F = 0.538$ $\mu_{CR} = 0.124$

Diversity = 1.2194e+01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 2.963998e+03 FES = 12100 $\mu_F = 0.541$ $\mu_{CR} = 0.107$

Diversity = 1.1682e+01 $k_t = [4, 4, 3, 4]$

Generation 130: Best = 2.963998e+03 FES = 13100 $\mu_F = 0.561$ $\mu_{CR} = 0.091$

Diversity = 1.1635e+01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 2.963998e+03 FES = 14100 $\mu_F = 0.562$ $\mu_{CR} = 0.075$

Diversity = 1.1402e+01 $k_t = [2, 2, 2, 3]$

Generation 150: Best = 2.940025e+03 FES = 15100 $\mu_F = 0.547$ $\mu_{CR} = 0.067$

Diversity = 1.1179e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 2.936160e+03 FES = 16100 $\mu_F = 0.567$ $\mu_{CR} = 0.065$

Diversity = 1.1004e+01 $k_t = [2, 2, 3, 4]$

Generation 170: Best = 2.935717e+03 FES = 17100 $\mu_F = 0.549$ $\mu_{CR} = 0.068$

Diversity = 1.0893e+01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 2.932771e+03 FES = 18100 $\mu_F = 0.549$ $\mu_{CR} = 0.072$

Diversity = 1.0748e+01 $k_t = [4, 4, 4, 3]$

Generation 190: Best = 2.927521e+03 FES = 19100 $\mu_F = 0.565$ $\mu_{CR} = 0.066$

Diversity = 1.0721e+01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 2.924808e+03 FES = 20100 $\mu_F = 0.542$ $\mu_{CR} = 0.071$

Diversity = 1.0632e+01 $k_t = [2, 2, 3, 4]$

Generation 210: Best = 2.915944e+03 FES = 21100 $\mu_F = 0.556$ $\mu_{CR} = 0.068$

Diversity = 1.0547e+01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 2.914088e+03 FES = 22100 $\mu_F = 0.546$ $\mu_{CR} = 0.063$

Diversity = 1.0495e+01 $k_t = [3, 2, 4, 2]$

Generation 230: Best = 2.913788e+03	FES = 23100	$\mu_F = 0.561$	$\mu_{CR} = 0.060$
Diversity = 1.0425e+01	k_t = [1, 1, 1, 1]		
Generation 240: Best = 2.907619e+03	FES = 24100	$\mu_F = 0.574$	$\mu_{CR} = 0.057$
Diversity = 1.0402e+01	k_t = [3, 3, 4, 2]		
Generation 250: Best = 2.905573e+03	FES = 25100	$\mu_F = 0.472$	$\mu_{CR} = 0.061$
Diversity = 1.0341e+01	k_t = [1, 1, 1, 1]		
Generation 260: Best = 2.902931e+03	FES = 26100	$\mu_F = 0.476$	$\mu_{CR} = 0.077$
Diversity = 1.0294e+01	k_t = [4, 4, 2, 2]		
Generation 270: Best = 2.902931e+03	FES = 27100	$\mu_F = 0.458$	$\mu_{CR} = 0.065$
Diversity = 1.0289e+01	k_t = [1, 1, 1, 1]		
Generation 280: Best = 2.902116e+03	FES = 28100	$\mu_F = 0.451$	$\mu_{CR} = 0.057$
Diversity = 1.0238e+01	k_t = [3, 3, 4, 3]		
Generation 290: Best = 2.900456e+03	FES = 29100	$\mu_F = 0.452$	$\mu_{CR} = 0.054$
Diversity = 1.0224e+01	k_t = [1, 1, 1, 1]		
Generation 300: Best = 2.900456e+03	FES = 30100	$\mu_F = 0.506$	$\mu_{CR} = 0.062$
Diversity = 1.0213e+01	k_t = [2, 3, 4, 3]		
Generation 310: Best = 2.900456e+03	FES = 31100	$\mu_F = 0.524$	$\mu_{CR} = 0.053$
Diversity = 1.0222e+01	k_t = [1, 1, 1, 1]		
Generation 320: Best = 2.900456e+03	FES = 32100	$\mu_F = 0.514$	$\mu_{CR} = 0.052$
Diversity = 1.0111e+01	k_t = [2, 3, 4, 2]		
Generation 330: Best = 2.897220e+03	FES = 33100	$\mu_F = 0.448$	$\mu_{CR} = 0.059$
Diversity = 1.0103e+01	k_t = [1, 1, 1, 1]		
Generation 340: Best = 2.889743e+03	FES = 34100	$\mu_F = 0.453$	$\mu_{CR} = 0.069$
Diversity = 1.0089e+01	k_t = [3, 4, 2, 3]		
Generation 350: Best = 2.889743e+03	FES = 35100	$\mu_F = 0.484$	$\mu_{CR} = 0.068$
Diversity = 1.0075e+01	k_t = [1, 1, 1, 1]		
Generation 360: Best = 2.884927e+03	FES = 36100	$\mu_F = 0.506$	$\mu_{CR} = 0.064$
Diversity = 9.9454e+00	k_t = [4, 2, 3, 4]		
Generation 370: Best = 2.883484e+03	FES = 37100	$\mu_F = 0.578$	$\mu_{CR} = 0.065$
Diversity = 9.9416e+00	k_t = [1, 1, 1, 1]		
Generation 380: Best = 2.882941e+03	FES = 38100	$\mu_F = 0.607$	$\mu_{CR} = 0.063$
Diversity = 9.8959e+00	k_t = [3, 3, 3, 2]		
Generation 390: Best = 2.882616e+03	FES = 39100	$\mu_F = 0.550$	$\mu_{CR} = 0.062$
Diversity = 9.8876e+00	k_t = [1, 1, 1, 1]		
Generation 400: Best = 2.878594e+03	FES = 40100	$\mu_F = 0.485$	$\mu_{CR} = 0.048$
Diversity = 9.8859e+00	k_t = [3, 3, 3, 3]		
Generation 410: Best = 2.876511e+03	FES = 41100	$\mu_F = 0.478$	$\mu_{CR} = 0.057$
Diversity = 9.8759e+00	k_t = [1, 1, 1, 1]		
Generation 420: Best = 2.875305e+03	FES = 42100	$\mu_F = 0.444$	$\mu_{CR} = 0.061$
Diversity = 9.8856e+00	k_t = [2, 4, 4, 3]		
Generation 430: Best = 2.872687e+03	FES = 43100	$\mu_F = 0.424$	$\mu_{CR} = 0.069$
Diversity = 9.8712e+00	k_t = [1, 1, 1, 1]		
Generation 440: Best = 2.870009e+03	FES = 44100	$\mu_F = 0.432$	$\mu_{CR} = 0.065$
Diversity = 9.8310e+00	k_t = [3, 4, 3, 3]		
Generation 450: Best = 2.867320e+03	FES = 45100	$\mu_F = 0.410$	$\mu_{CR} = 0.057$
Diversity = 9.8283e+00	k_t = [1, 1, 1, 1]		
Generation 460: Best = 2.858464e+03	FES = 46100	$\mu_F = 0.373$	$\mu_{CR} = 0.081$
Diversity = 9.7919e+00	k_t = [2, 3, 4, 2]		
Generation 470: Best = 2.858260e+03	FES = 47100	$\mu_F = 0.392$	$\mu_{CR} = 0.062$
Diversity = 9.7750e+00	k_t = [1, 1, 1, 1]		
Generation 480: Best = 2.857767e+03	FES = 48100	$\mu_F = 0.384$	$\mu_{CR} = 0.067$
Diversity = 9.7749e+00	k_t = [2, 2, 3, 3]		
Generation 490: Best = 2.857767e+03	FES = 49100	$\mu_F = 0.434$	$\mu_{CR} = 0.061$
Diversity = 9.7543e+00	k_t = [1, 1, 1, 1]		

Generation 500: Best = 2.853982e+03	FES = 50100	$\mu_F = 0.493$	$\mu_{CR} = 0.069$
Diversity = 9.7190e+00	$k_t = [4, 2, 2, 4]$		
Generation 510: Best = 2.853982e+03	FES = 51100	$\mu_F = 0.516$	$\mu_{CR} = 0.056$
Diversity = 9.6948e+00	$k_t = [1, 1, 1, 1]$		
Generation 520: Best = 2.851445e+03	FES = 52100	$\mu_F = 0.540$	$\mu_{CR} = 0.071$
Diversity = 9.6097e+00	$k_t = [4, 3, 2, 3]$		
Generation 530: Best = 2.851445e+03	FES = 53100	$\mu_F = 0.641$	$\mu_{CR} = 0.051$
Diversity = 9.5936e+00	$k_t = [1, 1, 1, 1]$		
Generation 540: Best = 2.851445e+03	FES = 54100	$\mu_F = 0.620$	$\mu_{CR} = 0.046$
Diversity = 9.5395e+00	$k_t = [2, 2, 4, 4]$		
Generation 550: Best = 2.850872e+03	FES = 55100	$\mu_F = 0.630$	$\mu_{CR} = 0.047$
Diversity = 9.5483e+00	$k_t = [1, 1, 1, 1]$		
Generation 560: Best = 2.850872e+03	FES = 56100	$\mu_F = 0.652$	$\mu_{CR} = 0.054$
Diversity = 9.5659e+00	$k_t = [3, 2, 4, 3]$		
Generation 570: Best = 2.850360e+03	FES = 57100	$\mu_F = 0.485$	$\mu_{CR} = 0.071$
Diversity = 9.5869e+00	$k_t = [1, 1, 1, 1]$		
Generation 580: Best = 2.850360e+03	FES = 58100	$\mu_F = 0.599$	$\mu_{CR} = 0.054$
Diversity = 9.5323e+00	$k_t = [2, 4, 4, 4]$		
Generation 590: Best = 2.850360e+03	FES = 59100	$\mu_F = 0.530$	$\mu_{CR} = 0.054$
Diversity = 9.5439e+00	$k_t = [1, 1, 1, 1]$		
Generation 600: Best = 2.850203e+03	FES = 60100	$\mu_F = 0.582$	$\mu_{CR} = 0.050$
Diversity = 9.5674e+00	$k_t = [4, 4, 3, 4]$		
Generation 610: Best = 2.850203e+03	FES = 61100	$\mu_F = 0.655$	$\mu_{CR} = 0.041$
Diversity = 9.5957e+00	$k_t = [1, 1, 1, 1]$		
Generation 620: Best = 2.850094e+03	FES = 62100	$\mu_F = 0.675$	$\mu_{CR} = 0.036$
Diversity = 9.6039e+00	$k_t = [2, 2, 2, 3]$		
Generation 630: Best = 2.850094e+03	FES = 63100	$\mu_F = 0.670$	$\mu_{CR} = 0.035$
Diversity = 9.6119e+00	$k_t = [1, 1, 1, 1]$		
Generation 640: Best = 2.849949e+03	FES = 64100	$\mu_F = 0.669$	$\mu_{CR} = 0.039$
Diversity = 9.4579e+00	$k_t = [4, 2, 4, 2]$		
Generation 650: Best = 2.849949e+03	FES = 65100	$\mu_F = 0.697$	$\mu_{CR} = 0.044$
Diversity = 9.4431e+00	$k_t = [1, 1, 1, 1]$		
Generation 660: Best = 2.849949e+03	FES = 66100	$\mu_F = 0.717$	$\mu_{CR} = 0.046$
Diversity = 9.4062e+00	$k_t = [4, 3, 4, 4]$		
Generation 670: Best = 2.849947e+03	FES = 67100	$\mu_F = 0.717$	$\mu_{CR} = 0.044$
Diversity = 9.3961e+00	$k_t = [1, 1, 1, 1]$		
Generation 680: Best = 2.849908e+03	FES = 68100	$\mu_F = 0.696$	$\mu_{CR} = 0.043$
Diversity = 9.4028e+00	$k_t = [2, 4, 2, 3]$		
Generation 690: Best = 2.849908e+03	FES = 69100	$\mu_F = 0.691$	$\mu_{CR} = 0.053$
Diversity = 9.4035e+00	$k_t = [1, 1, 1, 1]$		
Generation 700: Best = 2.849908e+03	FES = 70100	$\mu_F = 0.648$	$\mu_{CR} = 0.053$
Diversity = 9.3842e+00	$k_t = [2, 4, 2, 3]$		
Generation 710: Best = 2.849908e+03	FES = 71100	$\mu_F = 0.721$	$\mu_{CR} = 0.048$
Diversity = 9.3559e+00	$k_t = [1, 1, 1, 1]$		
Generation 720: Best = 2.849908e+03	FES = 72100	$\mu_F = 0.675$	$\mu_{CR} = 0.048$
Diversity = 9.3504e+00	$k_t = [4, 4, 4, 4]$		
Generation 730: Best = 2.849385e+03	FES = 73100	$\mu_F = 0.539$	$\mu_{CR} = 0.057$
Diversity = 9.3538e+00	$k_t = [1, 1, 1, 1]$		
Generation 740: Best = 2.849385e+03	FES = 74100	$\mu_F = 0.543$	$\mu_{CR} = 0.056$
Diversity = 9.3431e+00	$k_t = [4, 3, 4, 4]$		
Generation 750: Best = 2.849385e+03	FES = 75100	$\mu_F = 0.483$	$\mu_{CR} = 0.053$
Diversity = 9.3538e+00	$k_t = [1, 1, 1, 1]$		
Generation 760: Best = 2.849385e+03	FES = 76100	$\mu_F = 0.519$	$\mu_{CR} = 0.050$
Diversity = 9.3317e+00	$k_t = [2, 4, 3, 4]$		

Generation 770:	Best = 2.849385e+03	FES = 77100	$\mu_F = 0.541$	$\mu_{CR} = 0.047$
Diversity = 9.3380e+00	$k_t = [1, 1, 1, 1]$			
Generation 780:	Best = 2.847066e+03	FES = 78100	$\mu_F = 0.465$	$\mu_{CR} = 0.055$
Diversity = 9.3370e+00	$k_t = [3, 4, 2, 3]$			
Generation 790:	Best = 2.846716e+03	FES = 79100	$\mu_F = 0.529$	$\mu_{CR} = 0.054$
Diversity = 9.2829e+00	$k_t = [1, 1, 1, 1]$			
Generation 800:	Best = 2.845992e+03	FES = 80100	$\mu_F = 0.553$	$\mu_{CR} = 0.051$
Diversity = 9.2804e+00	$k_t = [4, 3, 4, 4]$			
Generation 810:	Best = 2.845992e+03	FES = 81100	$\mu_F = 0.538$	$\mu_{CR} = 0.044$
Diversity = 9.2883e+00	$k_t = [1, 1, 1, 1]$			
Generation 820:	Best = 2.844564e+03	FES = 82100	$\mu_F = 0.555$	$\mu_{CR} = 0.055$
Diversity = 9.2894e+00	$k_t = [2, 4, 2, 3]$			
Generation 830:	Best = 2.844499e+03	FES = 83100	$\mu_F = 0.575$	$\mu_{CR} = 0.049$
Diversity = 9.3013e+00	$k_t = [1, 1, 1, 1]$			
Generation 840:	Best = 2.844499e+03	FES = 84100	$\mu_F = 0.584$	$\mu_{CR} = 0.063$
Diversity = 9.2876e+00	$k_t = [3, 2, 4, 2]$			
Generation 850:	Best = 2.844499e+03	FES = 85100	$\mu_F = 0.555$	$\mu_{CR} = 0.049$
Diversity = 9.2744e+00	$k_t = [1, 1, 1, 1]$			
Generation 860:	Best = 2.844499e+03	FES = 86100	$\mu_F = 0.584$	$\mu_{CR} = 0.041$
Diversity = 9.2832e+00	$k_t = [3, 2, 3, 2]$			
Generation 870:	Best = 2.844499e+03	FES = 87100	$\mu_F = 0.485$	$\mu_{CR} = 0.044$
Diversity = 9.2827e+00	$k_t = [1, 1, 1, 1]$			
Generation 880:	Best = 2.844474e+03	FES = 88100	$\mu_F = 0.533$	$\mu_{CR} = 0.047$
Diversity = 9.1997e+00	$k_t = [4, 4, 2, 4]$			
Generation 890:	Best = 2.843817e+03	FES = 89100	$\mu_F = 0.561$	$\mu_{CR} = 0.058$
Diversity = 9.1941e+00	$k_t = [1, 1, 1, 1]$			
Generation 900:	Best = 2.843817e+03	FES = 90100	$\mu_F = 0.623$	$\mu_{CR} = 0.059$
Diversity = 9.1708e+00	$k_t = [3, 2, 3, 4]$			
Generation 910:	Best = 2.843817e+03	FES = 91100	$\mu_F = 0.496$	$\mu_{CR} = 0.068$
Diversity = 9.1732e+00	$k_t = [1, 1, 1, 1]$			
Generation 920:	Best = 2.843817e+03	FES = 92100	$\mu_F = 0.460$	$\mu_{CR} = 0.055$
Diversity = 9.1727e+00	$k_t = [2, 4, 3, 2]$			
Generation 930:	Best = 2.843211e+03	FES = 93100	$\mu_F = 0.521$	$\mu_{CR} = 0.056$
Diversity = 9.2088e+00	$k_t = [1, 1, 1, 1]$			
Generation 940:	Best = 2.842184e+03	FES = 94100	$\mu_F = 0.535$	$\mu_{CR} = 0.068$
Diversity = 9.1222e+00	$k_t = [3, 4, 4, 4]$			
Generation 950:	Best = 2.842184e+03	FES = 95100	$\mu_F = 0.477$	$\mu_{CR} = 0.070$
Diversity = 9.1229e+00	$k_t = [1, 1, 1, 1]$			
Generation 960:	Best = 2.842184e+03	FES = 96100	$\mu_F = 0.535$	$\mu_{CR} = 0.060$
Diversity = 9.0761e+00	$k_t = [3, 3, 3, 4]$			
Generation 970:	Best = 2.842184e+03	FES = 97100	$\mu_F = 0.529$	$\mu_{CR} = 0.052$
Diversity = 9.0373e+00	$k_t = [1, 1, 1, 1]$			
Generation 980:	Best = 2.842153e+03	FES = 98100	$\mu_F = 0.557$	$\mu_{CR} = 0.048$
Diversity = 9.0221e+00	$k_t = [2, 2, 4, 4]$			
Generation 990:	Best = 2.842153e+03	FES = 99100	$\mu_F = 0.487$	$\mu_{CR} = 0.065$
Diversity = 9.0217e+00	$k_t = [1, 1, 1, 1]$			
Generation 1000:	Best = 2.842153e+03	FES = 100100	$\mu_F = 0.490$	$\mu_{CR} = 0.070$
Diversity = 9.0221e+00	$k_t = [4, 3, 3, 2]$			
Generation 1010:	Best = 2.841592e+03	FES = 101100	$\mu_F = 0.415$	$\mu_{CR} = 0.068$
Diversity = 9.0185e+00	$k_t = [1, 1, 1, 1]$			
Generation 1020:	Best = 2.839726e+03	FES = 102100	$\mu_F = 0.456$	$\mu_{CR} = 0.055$
Diversity = 9.0201e+00	$k_t = [4, 4, 3, 3]$			
Generation 1030:	Best = 2.839188e+03	FES = 103100	$\mu_F = 0.465$	$\mu_{CR} = 0.056$
Diversity = 9.0174e+00	$k_t = [1, 1, 1, 1]$			

Generation 1040: Best = 2.839188e+03 FES = 104100 μ_F = 0.523 μ_{CR} = 0.054
 Diversity = 8.9881e+00 k_t = [3, 3, 2, 4]
 Generation 1050: Best = 2.839188e+03 FES = 105100 μ_F = 0.546 μ_{CR} = 0.041
 Diversity = 8.9876e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 2.839188e+03 FES = 106100 μ_F = 0.553 μ_{CR} = 0.057
 Diversity = 8.9308e+00 k_t = [2, 2, 4, 4]
 Generation 1070: Best = 2.839102e+03 FES = 107100 μ_F = 0.465 μ_{CR} = 0.067
 Diversity = 8.9309e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 2.839008e+03 FES = 108100 μ_F = 0.357 μ_{CR} = 0.087
 Diversity = 8.9302e+00 k_t = [2, 2, 4, 3]
 Generation 1090: Best = 2.839008e+03 FES = 109100 μ_F = 0.301 μ_{CR} = 0.091
 Diversity = 8.9364e+00 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 2.838867e+03 FES = 110100 μ_F = 0.400 μ_{CR} = 0.075
 Diversity = 8.8642e+00 k_t = [4, 4, 2, 3]
 Generation 1110: Best = 2.838867e+03 FES = 111100 μ_F = 0.315 μ_{CR} = 0.074
 Diversity = 8.8550e+00 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 2.838320e+03 FES = 112100 μ_F = 0.269 μ_{CR} = 0.073
 Diversity = 8.8510e+00 k_t = [3, 3, 3, 2]
 Generation 1130: Best = 2.837843e+03 FES = 113100 μ_F = 0.280 μ_{CR} = 0.068
 Diversity = 8.8491e+00 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 2.837843e+03 FES = 114100 μ_F = 0.298 μ_{CR} = 0.068
 Diversity = 8.8447e+00 k_t = [2, 3, 2, 2]
 Generation 1150: Best = 2.837841e+03 FES = 115100 μ_F = 0.375 μ_{CR} = 0.062
 Diversity = 8.7610e+00 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 2.837841e+03 FES = 116100 μ_F = 0.393 μ_{CR} = 0.058
 Diversity = 8.7587e+00 k_t = [3, 2, 4, 3]
 Generation 1170: Best = 2.837841e+03 FES = 117100 μ_F = 0.398 μ_{CR} = 0.061
 Diversity = 8.7601e+00 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 2.837579e+03 FES = 118100 μ_F = 0.343 μ_{CR} = 0.093
 Diversity = 8.7501e+00 k_t = [3, 2, 4, 2]
 Generation 1190: Best = 2.837401e+03 FES = 119100 μ_F = 0.310 μ_{CR} = 0.088
 Diversity = 8.7528e+00 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 2.837401e+03 FES = 120100 μ_F = 0.342 μ_{CR} = 0.078
 Diversity = 8.7068e+00 k_t = [3, 2, 4, 4]
 Generation 1210: Best = 2.837401e+03 FES = 121100 μ_F = 0.279 μ_{CR} = 0.065
 Diversity = 8.7074e+00 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 2.837333e+03 FES = 122100 μ_F = 0.288 μ_{CR} = 0.071
 Diversity = 8.7043e+00 k_t = [2, 2, 4, 2]
 Generation 1230: Best = 2.837333e+03 FES = 123100 μ_F = 0.268 μ_{CR} = 0.090
 Diversity = 8.6558e+00 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 2.837333e+03 FES = 124100 μ_F = 0.389 μ_{CR} = 0.091
 Diversity = 8.5292e+00 k_t = [3, 4, 2, 4]
 Generation 1250: Best = 2.837333e+03 FES = 125100 μ_F = 0.459 μ_{CR} = 0.068
 Diversity = 8.5268e+00 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 2.837333e+03 FES = 126100 μ_F = 0.397 μ_{CR} = 0.070
 Diversity = 8.5281e+00 k_t = [3, 4, 4, 2]
 Generation 1270: Best = 2.837333e+03 FES = 127100 μ_F = 0.415 μ_{CR} = 0.056
 Diversity = 8.4872e+00 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 2.837293e+03 FES = 128100 μ_F = 0.390 μ_{CR} = 0.056
 Diversity = 8.4870e+00 k_t = [4, 4, 3, 2]
 Generation 1290: Best = 2.837293e+03 FES = 129100 μ_F = 0.398 μ_{CR} = 0.054
 Diversity = 8.4531e+00 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 2.837293e+03 FES = 130100 μ_F = 0.351 μ_{CR} = 0.051
 Diversity = 8.4590e+00 k_t = [2, 2, 2, 3]

Generation 1310: Best = 2.837293e+03 FES = 131100 μ_F = 0.390 μ_{CR} = 0.038
 Diversity = 8.4595e+00 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 2.837272e+03 FES = 132100 μ_F = 0.397 μ_{CR} = 0.058
 Diversity = 8.4505e+00 k_t = [3, 4, 2, 4]
 Generation 1330: Best = 2.837272e+03 FES = 133100 μ_F = 0.416 μ_{CR} = 0.055
 Diversity = 8.4601e+00 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 2.836515e+03 FES = 134100 μ_F = 0.465 μ_{CR} = 0.063
 Diversity = 8.4123e+00 k_t = [4, 2, 4, 2]
 Generation 1350: Best = 2.836515e+03 FES = 135100 μ_F = 0.402 μ_{CR} = 0.065
 Diversity = 8.3181e+00 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 2.836504e+03 FES = 136100 μ_F = 0.433 μ_{CR} = 0.071
 Diversity = 8.3154e+00 k_t = [4, 4, 4, 3]
 Generation 1370: Best = 2.836504e+03 FES = 137100 μ_F = 0.458 μ_{CR} = 0.049
 Diversity = 8.3146e+00 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 2.836504e+03 FES = 138100 μ_F = 0.495 μ_{CR} = 0.056
 Diversity = 8.3191e+00 k_t = [4, 3, 4, 3]
 Generation 1390: Best = 2.836504e+03 FES = 139100 μ_F = 0.462 μ_{CR} = 0.044
 Diversity = 8.3220e+00 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 2.836018e+03 FES = 140100 μ_F = 0.404 μ_{CR} = 0.061
 Diversity = 8.3173e+00 k_t = [4, 3, 3, 2]
 Generation 1410: Best = 2.836000e+03 FES = 141100 μ_F = 0.336 μ_{CR} = 0.052
 Diversity = 8.3147e+00 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 2.835997e+03 FES = 142100 μ_F = 0.336 μ_{CR} = 0.054
 Diversity = 8.3192e+00 k_t = [3, 2, 3, 3]
 Generation 1430: Best = 2.835997e+03 FES = 143100 μ_F = 0.314 μ_{CR} = 0.057
 Diversity = 8.3273e+00 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 2.835914e+03 FES = 144100 μ_F = 0.375 μ_{CR} = 0.072
 Diversity = 8.2915e+00 k_t = [3, 2, 4, 4]
 Generation 1450: Best = 2.835914e+03 FES = 145100 μ_F = 0.336 μ_{CR} = 0.067
 Diversity = 8.2922e+00 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 2.835888e+03 FES = 146100 μ_F = 0.297 μ_{CR} = 0.071
 Diversity = 8.2366e+00 k_t = [4, 3, 4, 4]
 Generation 1470: Best = 2.835833e+03 FES = 147100 μ_F = 0.309 μ_{CR} = 0.071
 Diversity = 8.2358e+00 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 2.835792e+03 FES = 148100 μ_F = 0.288 μ_{CR} = 0.068
 Diversity = 8.2364e+00 k_t = [4, 3, 2, 2]
 Generation 1490: Best = 2.835732e+03 FES = 149100 μ_F = 0.364 μ_{CR} = 0.072
 Diversity = 8.2339e+00 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 2.835712e+03 FES = 150100 μ_F = 0.391 μ_{CR} = 0.077
 Diversity = 8.2352e+00 k_t = [4, 3, 3, 2]
 Generation 1510: Best = 2.835712e+03 FES = 151100 μ_F = 0.468 μ_{CR} = 0.064
 Diversity = 8.2332e+00 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 2.835712e+03 FES = 152100 μ_F = 0.537 μ_{CR} = 0.054
 Diversity = 8.1352e+00 k_t = [3, 4, 2, 2]
 Generation 1530: Best = 2.835712e+03 FES = 153100 μ_F = 0.539 μ_{CR} = 0.039
 Diversity = 8.0625e+00 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 2.835699e+03 FES = 154100 μ_F = 0.513 μ_{CR} = 0.048
 Diversity = 8.0231e+00 k_t = [4, 2, 4, 3]
 Generation 1550: Best = 2.835699e+03 FES = 155100 μ_F = 0.573 μ_{CR} = 0.051
 Diversity = 7.9565e+00 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 2.835699e+03 FES = 156100 μ_F = 0.611 μ_{CR} = 0.055
 Diversity = 7.9575e+00 k_t = [3, 2, 2, 2]
 Generation 1570: Best = 2.835699e+03 FES = 157100 μ_F = 0.583 μ_{CR} = 0.051
 Diversity = 7.9459e+00 k_t = [1, 1, 1, 1]

Generation 1580: Best = 2.835699e+03 FES = 158100 μ_F = 0.615 μ_{CR} = 0.047
 Diversity = 7.9245e+00 k_t = [3, 4, 4, 3]
 Generation 1590: Best = 2.835699e+03 FES = 159100 μ_F = 0.546 μ_{CR} = 0.051
 Diversity = 7.8548e+00 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 2.835686e+03 FES = 160100 μ_F = 0.521 μ_{CR} = 0.043
 Diversity = 7.8304e+00 k_t = [3, 3, 2, 2]
 Generation 1610: Best = 2.835686e+03 FES = 161100 μ_F = 0.606 μ_{CR} = 0.061
 Diversity = 7.8104e+00 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 2.835578e+03 FES = 162100 μ_F = 0.569 μ_{CR} = 0.075
 Diversity = 7.6493e+00 k_t = [4, 3, 4, 4]
 Generation 1630: Best = 2.835578e+03 FES = 163100 μ_F = 0.564 μ_{CR} = 0.077
 Diversity = 7.6492e+00 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 2.835578e+03 FES = 164100 μ_F = 0.517 μ_{CR} = 0.086
 Diversity = 7.6418e+00 k_t = [4, 4, 2, 2]
 Generation 1650: Best = 2.835578e+03 FES = 165100 μ_F = 0.458 μ_{CR} = 0.071
 Diversity = 7.5795e+00 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 2.835578e+03 FES = 166100 μ_F = 0.472 μ_{CR} = 0.059
 Diversity = 7.4352e+00 k_t = [3, 2, 4, 4]
 Generation 1670: Best = 2.835578e+03 FES = 167100 μ_F = 0.385 μ_{CR} = 0.056
 Diversity = 7.4333e+00 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 2.835552e+03 FES = 168100 μ_F = 0.356 μ_{CR} = 0.077
 Diversity = 7.3966e+00 k_t = [3, 3, 2, 4]
 Generation 1690: Best = 2.835552e+03 FES = 169100 μ_F = 0.333 μ_{CR} = 0.074
 Diversity = 7.3948e+00 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 2.835552e+03 FES = 170100 μ_F = 0.288 μ_{CR} = 0.078
 Diversity = 7.3963e+00 k_t = [3, 3, 2, 3]
 Generation 1710: Best = 2.835552e+03 FES = 171100 μ_F = 0.329 μ_{CR} = 0.064
 Diversity = 7.3968e+00 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 2.835492e+03 FES = 172100 μ_F = 0.366 μ_{CR} = 0.058
 Diversity = 7.3364e+00 k_t = [4, 2, 3, 3]
 Generation 1730: Best = 2.835492e+03 FES = 173100 μ_F = 0.389 μ_{CR} = 0.052
 Diversity = 7.3354e+00 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 2.835492e+03 FES = 174100 μ_F = 0.399 μ_{CR} = 0.059
 Diversity = 7.3349e+00 k_t = [3, 2, 3, 2]
 Generation 1750: Best = 2.835492e+03 FES = 175100 μ_F = 0.415 μ_{CR} = 0.046
 Diversity = 7.2252e+00 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 2.835492e+03 FES = 176100 μ_F = 0.313 μ_{CR} = 0.060
 Diversity = 7.2251e+00 k_t = [4, 3, 4, 2]
 Generation 1770: Best = 2.835492e+03 FES = 177100 μ_F = 0.273 μ_{CR} = 0.055
 Diversity = 7.2252e+00 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 2.835480e+03 FES = 178100 μ_F = 0.309 μ_{CR} = 0.100
 Diversity = 7.2217e+00 k_t = [4, 4, 4, 2]
 Generation 1790: Best = 2.835480e+03 FES = 179100 μ_F = 0.347 μ_{CR} = 0.086
 Diversity = 7.2241e+00 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 2.835480e+03 FES = 180100 μ_F = 0.395 μ_{CR} = 0.071
 Diversity = 7.1471e+00 k_t = [2, 2, 3, 4]
 Generation 1810: Best = 2.835480e+03 FES = 181100 μ_F = 0.319 μ_{CR} = 0.065
 Diversity = 7.1469e+00 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 2.835480e+03 FES = 182100 μ_F = 0.278 μ_{CR} = 0.069
 Diversity = 7.1483e+00 k_t = [2, 2, 3, 3]
 Generation 1830: Best = 2.835480e+03 FES = 183100 μ_F = 0.275 μ_{CR} = 0.060
 Diversity = 7.1476e+00 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 2.835436e+03 FES = 184100 μ_F = 0.292 μ_{CR} = 0.078
 Diversity = 7.1435e+00 k_t = [2, 3, 4, 3]

Generation 1850: Best = 2.835436e+03	FES = 185100	$\mu_F = 0.254$	$\mu_{CR} = 0.075$
Diversity = 7.1428e+00	$k_t = [1, 1, 1, 1]$		
Generation 1860: Best = 2.835406e+03	FES = 186100	$\mu_F = 0.323$	$\mu_{CR} = 0.098$
Diversity = 7.1380e+00	$k_t = [4, 3, 2, 2]$		
Generation 1870: Best = 2.835374e+03	FES = 187100	$\mu_F = 0.432$	$\mu_{CR} = 0.125$
Diversity = 7.1356e+00	$k_t = [1, 1, 1, 1]$		
Generation 1880: Best = 2.835314e+03	FES = 188100	$\mu_F = 0.521$	$\mu_{CR} = 0.139$
Diversity = 6.8317e+00	$k_t = [2, 4, 4, 4]$		
Generation 1890: Best = 2.835282e+03	FES = 189100	$\mu_F = 0.559$	$\mu_{CR} = 0.140$
Diversity = 6.7103e+00	$k_t = [1, 1, 1, 1]$		
Generation 1900: Best = 2.835260e+03	FES = 190100	$\mu_F = 0.581$	$\mu_{CR} = 0.142$
Diversity = 6.6633e+00	$k_t = [2, 2, 3, 2]$		
Generation 1910: Best = 2.835234e+03	FES = 191100	$\mu_F = 0.586$	$\mu_{CR} = 0.156$
Diversity = 6.5902e+00	$k_t = [1, 1, 1, 1]$		
Generation 1920: Best = 2.835221e+03	FES = 192100	$\mu_F = 0.625$	$\mu_{CR} = 0.156$
Diversity = 6.5208e+00	$k_t = [3, 2, 2, 2]$		
Generation 1930: Best = 2.835209e+03	FES = 193100	$\mu_F = 0.634$	$\mu_{CR} = 0.164$
Diversity = 6.5205e+00	$k_t = [1, 1, 1, 1]$		
Generation 1940: Best = 2.835205e+03	FES = 194100	$\mu_F = 0.675$	$\mu_{CR} = 0.162$
Diversity = 6.4398e+00	$k_t = [4, 3, 4, 2]$		
Generation 1950: Best = 2.835202e+03	FES = 195100	$\mu_F = 0.697$	$\mu_{CR} = 0.168$
Diversity = 6.3458e+00	$k_t = [1, 1, 1, 1]$		
Generation 1960: Best = 2.835201e+03	FES = 196100	$\mu_F = 0.718$	$\mu_{CR} = 0.174$
Diversity = 6.0148e+00	$k_t = [3, 2, 4, 4]$		
Generation 1970: Best = 2.835199e+03	FES = 197100	$\mu_F = 0.755$	$\mu_{CR} = 0.181$
Diversity = 5.9172e+00	$k_t = [1, 1, 1, 1]$		
Generation 1980: Best = 2.835199e+03	FES = 198100	$\mu_F = 0.743$	$\mu_{CR} = 0.189$
Diversity = 5.9097e+00	$k_t = [4, 2, 2, 3]$		
Generation 1990: Best = 2.835198e+03	FES = 199100	$\mu_F = 0.745$	$\mu_{CR} = 0.173$
Diversity = 5.9105e+00	$k_t = [1, 1, 1, 1]$		
Generation 2000: Best = 2.835198e+03	FES = 200100	$\mu_F = 0.771$	$\mu_{CR} = 0.169$
Diversity = 5.7335e+00	$k_t = [3, 3, 4, 4]$		
Generation 2010: Best = 2.835198e+03	FES = 201100	$\mu_F = 0.762$	$\mu_{CR} = 0.157$
Diversity = 5.7186e+00	$k_t = [1, 1, 1, 1]$		
Generation 2020: Best = 2.835197e+03	FES = 202100	$\mu_F = 0.769$	$\mu_{CR} = 0.169$
Diversity = 5.5346e+00	$k_t = [3, 4, 3, 3]$		
Generation 2030: Best = 2.835197e+03	FES = 203100	$\mu_F = 0.773$	$\mu_{CR} = 0.186$
Diversity = 5.4493e+00	$k_t = [1, 1, 1, 1]$		
Generation 2040: Best = 2.835197e+03	FES = 204100	$\mu_F = 0.778$	$\mu_{CR} = 0.161$
Diversity = 4.8852e+00	$k_t = [4, 2, 2, 4]$		
Generation 2050: Best = 2.835197e+03	FES = 205100	$\mu_F = 0.756$	$\mu_{CR} = 0.154$
Diversity = 4.8852e+00	$k_t = [1, 1, 1, 1]$		
Generation 2060: Best = 2.835197e+03	FES = 206100	$\mu_F = 0.767$	$\mu_{CR} = 0.167$
Diversity = 4.6044e+00	$k_t = [4, 4, 4, 4]$		
Generation 2070: Best = 2.835197e+03	FES = 207100	$\mu_F = 0.763$	$\mu_{CR} = 0.173$
Diversity = 4.5083e+00	$k_t = [1, 1, 1, 1]$		
Generation 2080: Best = 2.835197e+03	FES = 208100	$\mu_F = 0.775$	$\mu_{CR} = 0.187$
Diversity = 4.3837e+00	$k_t = [2, 4, 2, 2]$		
Generation 2090: Best = 2.835197e+03	FES = 209100	$\mu_F = 0.783$	$\mu_{CR} = 0.180$
Diversity = 4.3822e+00	$k_t = [1, 1, 1, 1]$		
Generation 2100: Best = 2.835197e+03	FES = 210100	$\mu_F = 0.794$	$\mu_{CR} = 0.180$
Diversity = 4.3794e+00	$k_t = [2, 4, 4, 2]$		
Generation 2110: Best = 2.835197e+03	FES = 211100	$\mu_F = 0.793$	$\mu_{CR} = 0.185$
Diversity = 4.1435e+00	$k_t = [1, 1, 1, 1]$		

Generation 2120: Best = 2.835197e+03 FES = 212100 μ_F = 0.788 μ_{CR} = 0.185
 Diversity = 4.1428e+00 k_t = [4, 4, 4, 2]
 Generation 2130: Best = 2.835197e+03 FES = 213100 μ_F = 0.792 μ_{CR} = 0.189
 Diversity = 3.5240e+00 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 2.835197e+03 FES = 214100 μ_F = 0.780 μ_{CR} = 0.175
 Diversity = 3.4648e+00 k_t = [4, 2, 2, 2]
 Generation 2150: Best = 2.835197e+03 FES = 215100 μ_F = 0.788 μ_{CR} = 0.183
 Diversity = 3.3556e+00 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 2.835197e+03 FES = 216100 μ_F = 0.777 μ_{CR} = 0.186
 Diversity = 2.9670e+00 k_t = [3, 3, 2, 4]
 Generation 2170: Best = 2.835197e+03 FES = 217100 μ_F = 0.783 μ_{CR} = 0.186
 Diversity = 2.0255e+00 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 2.835197e+03 FES = 218100 μ_F = 0.785 μ_{CR} = 0.185
 Diversity = 1.6466e+00 k_t = [4, 2, 4, 2]
 Generation 2190: Best = 2.835197e+03 FES = 219100 μ_F = 0.781 μ_{CR} = 0.198
 Diversity = 1.6463e+00 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 2.835197e+03 FES = 220100 μ_F = 0.780 μ_{CR} = 0.201
 Diversity = 1.6461e+00 k_t = [3, 3, 4, 3]
 Generation 2210: Best = 2.835197e+03 FES = 221100 μ_F = 0.797 μ_{CR} = 0.195
 Diversity = 1.5523e+00 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 2.835197e+03 FES = 222100 μ_F = 0.798 μ_{CR} = 0.199
 Diversity = 1.0931e+00 k_t = [2, 4, 4, 4]
 Generation 2230: Best = 2.835197e+03 FES = 223100 μ_F = 0.788 μ_{CR} = 0.194
 Diversity = 1.0931e+00 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 2.835197e+03 FES = 224100 μ_F = 0.800 μ_{CR} = 0.186
 Diversity = 5.0607e-01 k_t = [4, 2, 4, 4]
 Generation 2250: Best = 2.835197e+03 FES = 225100 μ_F = 0.805 μ_{CR} = 0.196
 Diversity = 5.0609e-01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 2.835197e+03 FES = 226100 μ_F = 0.792 μ_{CR} = 0.198
 Diversity = 5.0609e-01 k_t = [2, 3, 3, 2]
 Generation 2270: Best = 2.835197e+03 FES = 227100 μ_F = 0.784 μ_{CR} = 0.200
 Diversity = 5.0609e-01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 2.835197e+03 FES = 228100 μ_F = 0.766 μ_{CR} = 0.199
 Diversity = 2.0967e-03 k_t = [3, 2, 2, 2]
 Generation 2290: Best = 2.835197e+03 FES = 229100 μ_F = 0.770 μ_{CR} = 0.204
 Diversity = 9.4780e-05 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 2.835197e+03 FES = 230100 μ_F = 0.757 μ_{CR} = 0.213
 Diversity = 2.3411e-05 k_t = [4, 3, 2, 4]
 Generation 2310: Best = 2.835197e+03 FES = 231100 μ_F = 0.731 μ_{CR} = 0.212
 Diversity = 1.5823e-05 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 2.835197e+03 FES = 232100 μ_F = 0.713 μ_{CR} = 0.217
 Diversity = 1.0807e-05 k_t = [3, 3, 3, 4]
 Generation 2330: Best = 2.835197e+03 FES = 233100 μ_F = 0.693 μ_{CR} = 0.218
 Diversity = 7.4404e-06 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 2.835197e+03 FES = 234100 μ_F = 0.684 μ_{CR} = 0.220
 Diversity = 4.6487e-06 k_t = [4, 3, 4, 4]
 Generation 2350: Best = 2.835197e+03 FES = 235100 μ_F = 0.675 μ_{CR} = 0.224
 Diversity = 2.8805e-06 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 2.835197e+03 FES = 236100 μ_F = 0.677 μ_{CR} = 0.228
 Diversity = 1.9718e-06 k_t = [3, 2, 3, 2]
 Generation 2370: Best = 2.835197e+03 FES = 237100 μ_F = 0.679 μ_{CR} = 0.228
 Diversity = 1.5767e-06 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 2.835197e+03 FES = 238100 μ_F = 0.686 μ_{CR} = 0.230
 Diversity = 9.9816e-07 k_t = [4, 4, 4, 4]

Generation 2390: Best = 2.835197e+03 FES = 239100 μ_F = 0.674 μ_{CR} = 0.234
 Diversity = 7.3294e-07 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 2.835197e+03 FES = 240100 μ_F = 0.674 μ_{CR} = 0.225
 Diversity = 5.1025e-07 k_t = [2, 2, 4, 4]
 Generation 2410: Best = 2.835197e+03 FES = 241100 μ_F = 0.682 μ_{CR} = 0.224
 Diversity = 3.9756e-07 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 2.835197e+03 FES = 242100 μ_F = 0.691 μ_{CR} = 0.223
 Diversity = 2.8406e-07 k_t = [3, 4, 2, 4]
 Generation 2430: Best = 2.835197e+03 FES = 243100 μ_F = 0.702 μ_{CR} = 0.220
 Diversity = 2.1476e-07 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 2.835197e+03 FES = 244100 μ_F = 0.715 μ_{CR} = 0.219
 Diversity = 1.8599e-07 k_t = [2, 3, 2, 3]
 Generation 2450: Best = 2.835197e+03 FES = 245100 μ_F = 0.727 μ_{CR} = 0.208
 Diversity = 1.7753e-07 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 2.835197e+03 FES = 246100 μ_F = 0.724 μ_{CR} = 0.208
 Diversity = 1.5474e-07 k_t = [2, 4, 3, 3]
 Generation 2470: Best = 2.835197e+03 FES = 247100 μ_F = 0.718 μ_{CR} = 0.191
 Diversity = 1.5045e-07 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 2.835197e+03 FES = 248100 μ_F = 0.725 μ_{CR} = 0.182
 Diversity = 1.3766e-07 k_t = [2, 3, 2, 4]
 Generation 2490: Best = 2.835197e+03 FES = 249100 μ_F = 0.735 μ_{CR} = 0.181
 Diversity = 1.4432e-07 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 2.835197e+03 FES = 250100 μ_F = 0.741 μ_{CR} = 0.185
 Diversity = 1.3611e-07 k_t = [3, 3, 2, 3]
 Generation 2510: Best = 2.835197e+03 FES = 251100 μ_F = 0.754 μ_{CR} = 0.180
 Diversity = 1.4550e-07 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 2.835197e+03 FES = 252100 μ_F = 0.768 μ_{CR} = 0.184
 Diversity = 1.3030e-07 k_t = [2, 3, 4, 4]
 Generation 2530: Best = 2.835197e+03 FES = 253100 μ_F = 0.777 μ_{CR} = 0.182
 Diversity = 1.3749e-07 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 2.835197e+03 FES = 254100 μ_F = 0.781 μ_{CR} = 0.187
 Diversity = 1.3156e-07 k_t = [4, 4, 2, 2]
 Generation 2550: Best = 2.835197e+03 FES = 255100 μ_F = 0.787 μ_{CR} = 0.190
 Diversity = 1.3395e-07 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 2.835197e+03 FES = 256100 μ_F = 0.795 μ_{CR} = 0.185
 Diversity = 1.2688e-07 k_t = [4, 4, 3, 3]
 Generation 2570: Best = 2.835197e+03 FES = 257100 μ_F = 0.794 μ_{CR} = 0.179
 Diversity = 1.2896e-07 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 2.835197e+03 FES = 258100 μ_F = 0.803 μ_{CR} = 0.171
 Diversity = 1.1478e-07 k_t = [3, 2, 2, 4]
 Generation 2590: Best = 2.835197e+03 FES = 259100 μ_F = 0.800 μ_{CR} = 0.164
 Diversity = 1.1486e-07 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 2.835197e+03 FES = 260100 μ_F = 0.813 μ_{CR} = 0.153
 Diversity = 1.0719e-07 k_t = [3, 4, 2, 2]
 Generation 2610: Best = 2.835197e+03 FES = 261100 μ_F = 0.804 μ_{CR} = 0.145
 Diversity = 1.0307e-07 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 2.835197e+03 FES = 262100 μ_F = 0.803 μ_{CR} = 0.146
 Diversity = 9.6940e-08 k_t = [3, 3, 3, 3]
 Generation 2630: Best = 2.835197e+03 FES = 263100 μ_F = 0.808 μ_{CR} = 0.145
 Diversity = 9.6507e-08 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 2.835197e+03 FES = 264100 μ_F = 0.799 μ_{CR} = 0.135
 Diversity = 9.4028e-08 k_t = [3, 2, 3, 3]
 Generation 2650: Best = 2.835197e+03 FES = 265100 μ_F = 0.797 μ_{CR} = 0.134
 Diversity = 9.3316e-08 k_t = [1, 1, 1, 1]

Generation 2660: Best = 2.835197e+03 FES = 266100 μ_F = 0.806 μ_{CR} = 0.132
 Diversity = 9.1551e-08 k_t = [3, 2, 3, 4]
 Generation 2670: Best = 2.835197e+03 FES = 267100 μ_F = 0.807 μ_{CR} = 0.148
 Diversity = 9.2053e-08 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 2.835197e+03 FES = 268100 μ_F = 0.807 μ_{CR} = 0.142
 Diversity = 9.0501e-08 k_t = [2, 4, 2, 2]
 Generation 2690: Best = 2.835197e+03 FES = 269100 μ_F = 0.802 μ_{CR} = 0.129
 Diversity = 9.0728e-08 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 2.835197e+03 FES = 270100 μ_F = 0.791 μ_{CR} = 0.128
 Diversity = 8.8041e-08 k_t = [4, 2, 4, 3]
 Generation 2710: Best = 2.835197e+03 FES = 271100 μ_F = 0.802 μ_{CR} = 0.131
 Diversity = 8.8703e-08 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 2.835197e+03 FES = 272100 μ_F = 0.815 μ_{CR} = 0.129
 Diversity = 8.7643e-08 k_t = [2, 4, 4, 4]
 Generation 2730: Best = 2.835197e+03 FES = 273100 μ_F = 0.818 μ_{CR} = 0.131
 Diversity = 8.8535e-08 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 2.835197e+03 FES = 274100 μ_F = 0.813 μ_{CR} = 0.138
 Diversity = 8.7892e-08 k_t = [4, 3, 3, 2]
 Generation 2750: Best = 2.835197e+03 FES = 275100 μ_F = 0.823 μ_{CR} = 0.134
 Diversity = 8.8017e-08 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 2.835197e+03 FES = 276100 μ_F = 0.824 μ_{CR} = 0.151
 Diversity = 8.6895e-08 k_t = [4, 4, 4, 3]
 Generation 2770: Best = 2.835197e+03 FES = 277100 μ_F = 0.827 μ_{CR} = 0.146
 Diversity = 8.5629e-08 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 2.835197e+03 FES = 278100 μ_F = 0.812 μ_{CR} = 0.146
 Diversity = 8.6215e-08 k_t = [4, 4, 4, 3]
 Generation 2790: Best = 2.835197e+03 FES = 279100 μ_F = 0.820 μ_{CR} = 0.143
 Diversity = 8.8704e-08 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 2.835197e+03 FES = 280100 μ_F = 0.830 μ_{CR} = 0.141
 Diversity = 8.9455e-08 k_t = [2, 2, 4, 2]
 Generation 2810: Best = 2.835197e+03 FES = 281100 μ_F = 0.826 μ_{CR} = 0.147
 Diversity = 9.3366e-08 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 2.835197e+03 FES = 282100 μ_F = 0.817 μ_{CR} = 0.142
 Diversity = 9.4909e-08 k_t = [4, 3, 3, 2]
 Generation 2830: Best = 2.835197e+03 FES = 283100 μ_F = 0.818 μ_{CR} = 0.140
 Diversity = 9.6773e-08 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 2.835197e+03 FES = 284100 μ_F = 0.811 μ_{CR} = 0.145
 Diversity = 9.2783e-08 k_t = [3, 2, 4, 4]
 Generation 2850: Best = 2.835197e+03 FES = 285100 μ_F = 0.796 μ_{CR} = 0.148
 Diversity = 9.3325e-08 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 2.835197e+03 FES = 286100 μ_F = 0.796 μ_{CR} = 0.145
 Diversity = 9.2372e-08 k_t = [2, 3, 3, 3]
 Generation 2870: Best = 2.835197e+03 FES = 287100 μ_F = 0.797 μ_{CR} = 0.149
 Diversity = 9.3826e-08 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 2.835197e+03 FES = 288100 μ_F = 0.795 μ_{CR} = 0.145
 Diversity = 9.5985e-08 k_t = [4, 3, 2, 3]
 Generation 2890: Best = 2.835197e+03 FES = 289100 μ_F = 0.779 μ_{CR} = 0.148
 Diversity = 9.6648e-08 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 2.835197e+03 FES = 290100 μ_F = 0.771 μ_{CR} = 0.150
 Diversity = 9.1541e-08 k_t = [2, 3, 2, 4]
 Generation 2910: Best = 2.835197e+03 FES = 291100 μ_F = 0.792 μ_{CR} = 0.151
 Diversity = 9.1987e-08 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 2.835197e+03 FES = 292100 μ_F = 0.797 μ_{CR} = 0.153
 Diversity = 9.2776e-08 k_t = [4, 4, 2, 4]

Generation 2930: Best = 2.835197e+03 FES = 293100 μ_F = 0.796 μ_{CR} = 0.153
 Diversity = 9.4372e-08 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 2.835197e+03 FES = 294100 μ_F = 0.797 μ_{CR} = 0.151
 Diversity = 9.3138e-08 k_t = [2, 4, 3, 3]
 Generation 2950: Best = 2.835197e+03 FES = 295100 μ_F = 0.809 μ_{CR} = 0.153
 Diversity = 9.5022e-08 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 2.835197e+03 FES = 296100 μ_F = 0.800 μ_{CR} = 0.150
 Diversity = 9.1856e-08 k_t = [2, 3, 2, 2]
 Generation 2970: Best = 2.835197e+03 FES = 297100 μ_F = 0.794 μ_{CR} = 0.143
 Diversity = 9.0726e-08 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 2.835197e+03 FES = 298100 μ_F = 0.805 μ_{CR} = 0.137
 Diversity = 8.7680e-08 k_t = [3, 3, 4, 3]
 Generation 2990: Best = 2.835197e+03 FES = 299100 μ_F = 0.806 μ_{CR} = 0.140
 Diversity = 8.6174e-08 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 2.835197e+03

Total Function Evals = 300000

Total Successful Updates = 43732

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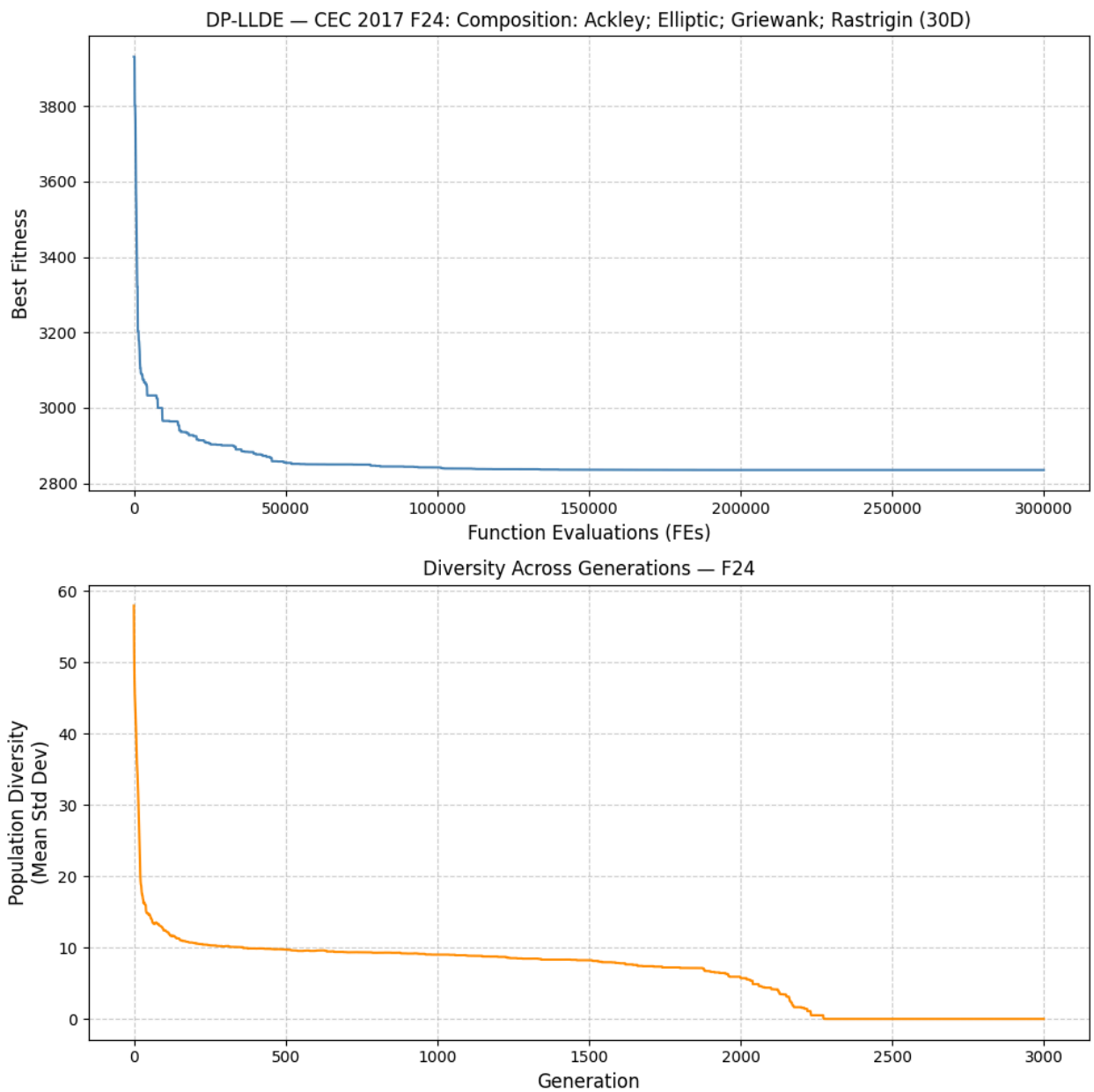
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Results:

Best Fitness : 2.835197e+03
 Global Optimum : 2.400000e+03
 Error (|best-optimum|) : 4.351970e+02
 Best Solution (5 dim) : [73.63766764 74.58664256 -81.04431928 -85.55783612
 84.53398065]
 D^(0) (init. diversity): 5.7976e+01

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Graph saved as convergence_F24.png

```
In [ ]: # =====
# F25 — Composition Function 5 (Rastrigin; HappyCat; Ackley; Discus; Rosenbrock)
# =====
run_and_plot('F25', make_wrapper(composition.f25), 2500.0, 'Composition: Rastr
```

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DP-LLDE on CEC 2017 F25 – Single Run

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Function : Composition: Rastrigin; HappyCat; Ackley; Discus; Rosenbrock (30D)

Bounds : [-100, 100]

Global optimum : 2500.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 2.562141e+04

Generation 10:	Best = 4.162292e+03	FES = 1100	$\mu_F = 0.550$	$\mu_{CR} = 0.353$
Diversity = 2.4672e+01	$k_t = [1, 1, 1, 1]$			
Generation 20:	Best = 3.300223e+03	FES = 2100	$\mu_F = 0.593$	$\mu_{CR} = 0.358$
Diversity = 1.2166e+01	$k_t = [4, 4, 2, 4]$			
Generation 30:	Best = 3.080578e+03	FES = 3100	$\mu_F = 0.623$	$\mu_{CR} = 0.366$
Diversity = 6.9477e+00	$k_t = [1, 1, 1, 1]$			
Generation 40:	Best = 3.046493e+03	FES = 4100	$\mu_F = 0.651$	$\mu_{CR} = 0.375$
Diversity = 4.4208e+00	$k_t = [2, 4, 4, 2]$			
Generation 50:	Best = 3.020967e+03	FES = 5100	$\mu_F = 0.671$	$\mu_{CR} = 0.387$
Diversity = 2.8394e+00	$k_t = [1, 1, 1, 1]$			
Generation 60:	Best = 3.001220e+03	FES = 6100	$\mu_F = 0.688$	$\mu_{CR} = 0.402$
Diversity = 1.9738e+00	$k_t = [3, 3, 2, 4]$			
Generation 70:	Best = 2.989291e+03	FES = 7100	$\mu_F = 0.716$	$\mu_{CR} = 0.406$
Diversity = 1.6482e+00	$k_t = [1, 1, 1, 1]$			
Generation 80:	Best = 2.979362e+03	FES = 8100	$\mu_F = 0.745$	$\mu_{CR} = 0.420$
Diversity = 1.4595e+00	$k_t = [2, 2, 4, 3]$			
Generation 90:	Best = 2.970736e+03	FES = 9100	$\mu_F = 0.744$	$\mu_{CR} = 0.424$
Diversity = 1.3915e+00	$k_t = [1, 1, 1, 1]$			
Generation 100:	Best = 2.959996e+03	FES = 10100	$\mu_F = 0.758$	$\mu_{CR} = 0.426$
Diversity = 1.5040e+00	$k_t = [2, 2, 3, 3]$			
Generation 110:	Best = 2.949469e+03	FES = 11100	$\mu_F = 0.773$	$\mu_{CR} = 0.425$
Diversity = 1.6433e+00	$k_t = [1, 1, 1, 1]$			
Generation 120:	Best = 2.939184e+03	FES = 12100	$\mu_F = 0.783$	$\mu_{CR} = 0.424$
Diversity = 1.6760e+00	$k_t = [4, 3, 3, 2]$			
Generation 130:	Best = 2.932271e+03	FES = 13100	$\mu_F = 0.791$	$\mu_{CR} = 0.426$
Diversity = 1.5076e+00	$k_t = [1, 1, 1, 1]$			
Generation 140:	Best = 2.926043e+03	FES = 14100	$\mu_F = 0.788$	$\mu_{CR} = 0.428$
Diversity = 1.3000e+00	$k_t = [4, 3, 4, 4]$			
Generation 150:	Best = 2.921818e+03	FES = 15100	$\mu_F = 0.795$	$\mu_{CR} = 0.429$
Diversity = 1.0710e+00	$k_t = [1, 1, 1, 1]$			
Generation 160:	Best = 2.918878e+03	FES = 16100	$\mu_F = 0.785$	$\mu_{CR} = 0.425$
Diversity = 1.0277e+00	$k_t = [4, 3, 2, 4]$			
Generation 170:	Best = 2.915986e+03	FES = 17100	$\mu_F = 0.784$	$\mu_{CR} = 0.434$
Diversity = 9.4656e-01	$k_t = [1, 1, 1, 1]$			
Generation 180:	Best = 2.913696e+03	FES = 18100	$\mu_F = 0.777$	$\mu_{CR} = 0.445$
Diversity = 8.9564e-01	$k_t = [2, 4, 2, 3]$			
Generation 190:	Best = 2.911226e+03	FES = 19100	$\mu_F = 0.775$	$\mu_{CR} = 0.451$
Diversity = 7.9781e-01	$k_t = [1, 1, 1, 1]$			
Generation 200:	Best = 2.909431e+03	FES = 20100	$\mu_F = 0.765$	$\mu_{CR} = 0.455$
Diversity = 6.3738e-01	$k_t = [3, 4, 2, 2]$			
Generation 210:	Best = 2.908422e+03	FES = 21100	$\mu_F = 0.774$	$\mu_{CR} = 0.457$
Diversity = 6.0459e-01	$k_t = [1, 1, 1, 1]$			
Generation 220:	Best = 2.907248e+03	FES = 22100	$\mu_F = 0.773$	$\mu_{CR} = 0.453$

Diversity = 5.4980e-01	k_t = [2, 2, 2, 2]				
Generation 230: Best = 2.906624e+03	FES = 23100	$\mu_F = 0.773$	$\mu_{CR} = 0.459$		
Diversity = 5.2699e-01	k_t = [1, 1, 1, 1]				
Generation 240: Best = 2.905482e+03	FES = 24100	$\mu_F = 0.771$	$\mu_{CR} = 0.460$		
Diversity = 4.9284e-01	k_t = [2, 3, 3, 2]				
Generation 250: Best = 2.904675e+03	FES = 25100	$\mu_F = 0.779$	$\mu_{CR} = 0.466$		
Diversity = 4.0133e-01	k_t = [1, 1, 1, 1]				
Generation 260: Best = 2.904119e+03	FES = 26100	$\mu_F = 0.766$	$\mu_{CR} = 0.454$		
Diversity = 3.7601e-01	k_t = [2, 3, 3, 3]				
Generation 270: Best = 2.903818e+03	FES = 27100	$\mu_F = 0.731$	$\mu_{CR} = 0.459$		
Diversity = 3.7507e-01	k_t = [1, 1, 1, 1]				
Generation 280: Best = 2.903379e+03	FES = 28100	$\mu_F = 0.731$	$\mu_{CR} = 0.449$		
Diversity = 3.1439e-01	k_t = [3, 2, 4, 3]				
Generation 290: Best = 2.903103e+03	FES = 29100	$\mu_F = 0.715$	$\mu_{CR} = 0.452$		
Diversity = 2.9603e-01	k_t = [1, 1, 1, 1]				
Generation 300: Best = 2.902745e+03	FES = 30100	$\mu_F = 0.711$	$\mu_{CR} = 0.455$		
Diversity = 2.5106e-01	k_t = [2, 2, 2, 4]				
Generation 310: Best = 2.902531e+03	FES = 31100	$\mu_F = 0.707$	$\mu_{CR} = 0.456$		
Diversity = 2.4262e-01	k_t = [1, 1, 1, 1]				
Generation 320: Best = 2.902202e+03	FES = 32100	$\mu_F = 0.702$	$\mu_{CR} = 0.459$		
Diversity = 2.2251e-01	k_t = [3, 3, 2, 2]				
Generation 330: Best = 2.901979e+03	FES = 33100	$\mu_F = 0.696$	$\mu_{CR} = 0.453$		
Diversity = 1.9105e-01	k_t = [1, 1, 1, 1]				
Generation 340: Best = 2.901762e+03	FES = 34100	$\mu_F = 0.700$	$\mu_{CR} = 0.444$		
Diversity = 1.4135e-01	k_t = [2, 2, 3, 4]				
Generation 350: Best = 2.901668e+03	FES = 35100	$\mu_F = 0.693$	$\mu_{CR} = 0.439$		
Diversity = 9.9988e-02	k_t = [1, 1, 1, 1]				
Generation 360: Best = 2.901602e+03	FES = 36100	$\mu_F = 0.698$	$\mu_{CR} = 0.435$		
Diversity = 7.1593e-02	k_t = [4, 4, 3, 3]				
Generation 370: Best = 2.901540e+03	FES = 37100	$\mu_F = 0.683$	$\mu_{CR} = 0.429$		
Diversity = 5.4590e-02	k_t = [1, 1, 1, 1]				
Generation 380: Best = 2.901509e+03	FES = 38100	$\mu_F = 0.681$	$\mu_{CR} = 0.422$		
Diversity = 3.9104e-02	k_t = [2, 3, 4, 3]				
Generation 390: Best = 2.901466e+03	FES = 39100	$\mu_F = 0.682$	$\mu_{CR} = 0.424$		
Diversity = 3.4518e-02	k_t = [1, 1, 1, 1]				
Generation 400: Best = 2.901447e+03	FES = 40100	$\mu_F = 0.682$	$\mu_{CR} = 0.414$		
Diversity = 2.5151e-02	k_t = [4, 4, 3, 4]				
Generation 410: Best = 2.901432e+03	FES = 41100	$\mu_F = 0.685$	$\mu_{CR} = 0.409$		
Diversity = 1.8028e-02	k_t = [1, 1, 1, 1]				
Generation 420: Best = 2.901425e+03	FES = 42100	$\mu_F = 0.698$	$\mu_{CR} = 0.414$		
Diversity = 1.3900e-02	k_t = [2, 2, 4, 3]				
Generation 430: Best = 2.901419e+03	FES = 43100	$\mu_F = 0.712$	$\mu_{CR} = 0.415$		
Diversity = 1.1391e-02	k_t = [1, 1, 1, 1]				
Generation 440: Best = 2.901416e+03	FES = 44100	$\mu_F = 0.725$	$\mu_{CR} = 0.409$		
Diversity = 8.4489e-03	k_t = [2, 3, 2, 3]				
Generation 450: Best = 2.901414e+03	FES = 45100	$\mu_F = 0.720$	$\mu_{CR} = 0.418$		
Diversity = 7.1127e-03	k_t = [1, 1, 1, 1]				
Generation 460: Best = 2.901412e+03	FES = 46100	$\mu_F = 0.728$	$\mu_{CR} = 0.412$		
Diversity = 6.2411e-03	k_t = [4, 2, 4, 2]				
Generation 470: Best = 2.901411e+03	FES = 47100	$\mu_F = 0.722$	$\mu_{CR} = 0.402$		
Diversity = 5.6175e-03	k_t = [1, 1, 1, 1]				
Generation 480: Best = 2.901409e+03	FES = 48100	$\mu_F = 0.732$	$\mu_{CR} = 0.401$		
Diversity = 5.1309e-03	k_t = [4, 2, 2, 4]				
Generation 490: Best = 2.901408e+03	FES = 49100	$\mu_F = 0.746$	$\mu_{CR} = 0.399$		

Diversity = 5.4952e-03	k_t = [1, 1, 1, 1]					
Generation 500: Best = 2.901408e+03	FES = 50100	$\mu_F = 0.732$	$\mu_{CR} = 0.404$			
Diversity = 4.4414e-03	k_t = [2, 3, 4, 4]					
Generation 510: Best = 2.901407e+03	FES = 51100	$\mu_F = 0.732$	$\mu_{CR} = 0.399$			
Diversity = 4.5055e-03	k_t = [1, 1, 1, 1]					
Generation 520: Best = 2.901406e+03	FES = 52100	$\mu_F = 0.734$	$\mu_{CR} = 0.406$			
Diversity = 4.4942e-03	k_t = [3, 3, 4, 3]					
Generation 530: Best = 2.901406e+03	FES = 53100	$\mu_F = 0.736$	$\mu_{CR} = 0.402$			
Diversity = 4.3660e-03	k_t = [1, 1, 1, 1]					
Generation 540: Best = 2.901405e+03	FES = 54100	$\mu_F = 0.757$	$\mu_{CR} = 0.391$			
Diversity = 3.8041e-03	k_t = [4, 2, 2, 4]					
Generation 550: Best = 2.901404e+03	FES = 55100	$\mu_F = 0.768$	$\mu_{CR} = 0.392$			
Diversity = 3.7888e-03	k_t = [1, 1, 1, 1]					
Generation 560: Best = 2.901399e+03	FES = 56100	$\mu_F = 0.757$	$\mu_{CR} = 0.394$			
Diversity = 6.7277e-03	k_t = [3, 4, 3, 2]					
Generation 570: Best = 2.901336e+03	FES = 57100	$\mu_F = 0.777$	$\mu_{CR} = 0.385$			
Diversity = 1.7292e-02	k_t = [1, 1, 1, 1]					
Generation 580: Best = 2.901325e+03	FES = 58100	$\mu_F = 0.790$	$\mu_{CR} = 0.380$			
Diversity = 1.1443e-02	k_t = [4, 2, 2, 3]					
Generation 590: Best = 2.901313e+03	FES = 59100	$\mu_F = 0.794$	$\mu_{CR} = 0.369$			
Diversity = 1.1335e-02	k_t = [1, 1, 1, 1]					
Generation 600: Best = 2.901299e+03	FES = 60100	$\mu_F = 0.807$	$\mu_{CR} = 0.361$			
Diversity = 1.3479e-02	k_t = [3, 3, 3, 4]					
Generation 610: Best = 2.901279e+03	FES = 61100	$\mu_F = 0.814$	$\mu_{CR} = 0.363$			
Diversity = 1.7864e-02	k_t = [1, 1, 1, 1]					
Generation 620: Best = 2.901267e+03	FES = 62100	$\mu_F = 0.806$	$\mu_{CR} = 0.360$			
Diversity = 1.7003e-02	k_t = [2, 3, 4, 3]					
Generation 630: Best = 2.901255e+03	FES = 63100	$\mu_F = 0.796$	$\mu_{CR} = 0.355$			
Diversity = 1.8982e-02	k_t = [1, 1, 1, 1]					
Generation 640: Best = 2.901248e+03	FES = 64100	$\mu_F = 0.771$	$\mu_{CR} = 0.354$			
Diversity = 1.7969e-02	k_t = [4, 2, 4, 3]					
Generation 650: Best = 2.901239e+03	FES = 65100	$\mu_F = 0.765$	$\mu_{CR} = 0.346$			
Diversity = 1.7348e-02	k_t = [1, 1, 1, 1]					
Generation 660: Best = 2.901181e+03	FES = 66100	$\mu_F = 0.757$	$\mu_{CR} = 0.323$			
Diversity = 2.9476e-02	k_t = [2, 2, 4, 4]					
Generation 670: Best = 2.901130e+03	FES = 67100	$\mu_F = 0.775$	$\mu_{CR} = 0.326$			
Diversity = 3.8364e-02	k_t = [1, 1, 1, 1]					
Generation 680: Best = 2.901112e+03	FES = 68100	$\mu_F = 0.770$	$\mu_{CR} = 0.317$			
Diversity = 2.6798e-02	k_t = [4, 2, 4, 4]					
Generation 690: Best = 2.901095e+03	FES = 69100	$\mu_F = 0.765$	$\mu_{CR} = 0.309$			
Diversity = 2.3828e-02	k_t = [1, 1, 1, 1]					
Generation 700: Best = 2.901038e+03	FES = 70100	$\mu_F = 0.760$	$\mu_{CR} = 0.313$			
Diversity = 4.2115e-02	k_t = [4, 2, 2, 2]					
Generation 710: Best = 2.900997e+03	FES = 71100	$\mu_F = 0.755$	$\mu_{CR} = 0.309$			
Diversity = 3.5586e-02	k_t = [1, 1, 1, 1]					
Generation 720: Best = 2.900968e+03	FES = 72100	$\mu_F = 0.755$	$\mu_{CR} = 0.314$			
Diversity = 2.9074e-02	k_t = [3, 4, 3, 3]					
Generation 730: Best = 2.900929e+03	FES = 73100	$\mu_F = 0.752$	$\mu_{CR} = 0.315$			
Diversity = 3.7319e-02	k_t = [1, 1, 1, 1]					
Generation 740: Best = 2.900901e+03	FES = 74100	$\mu_F = 0.745$	$\mu_{CR} = 0.312$			
Diversity = 2.8587e-02	k_t = [3, 4, 2, 2]					
Generation 750: Best = 2.900893e+03	FES = 75100	$\mu_F = 0.736$	$\mu_{CR} = 0.310$			
Diversity = 2.3455e-02	k_t = [1, 1, 1, 1]					
Generation 760: Best = 2.900887e+03	FES = 76100	$\mu_F = 0.707$	$\mu_{CR} = 0.311$			

Diversity = 1.5986e-02	k_t = [3, 3, 2, 3]				
Generation 770: Best = 2.900885e+03	FES = 77100	$\mu_F = 0.718$	$\mu_{CR} = 0.313$		
Diversity = 1.2768e-02	k_t = [1, 1, 1, 1]				
Generation 780: Best = 2.900883e+03	FES = 78100	$\mu_F = 0.710$	$\mu_{CR} = 0.316$		
Diversity = 9.2613e-03	k_t = [2, 4, 3, 4]				
Generation 790: Best = 2.900882e+03	FES = 79100	$\mu_F = 0.703$	$\mu_{CR} = 0.319$		
Diversity = 7.6458e-03	k_t = [1, 1, 1, 1]				
Generation 800: Best = 2.900881e+03	FES = 80100	$\mu_F = 0.710$	$\mu_{CR} = 0.312$		
Diversity = 5.9672e-03	k_t = [2, 4, 3, 2]				
Generation 810: Best = 2.900880e+03	FES = 81100	$\mu_F = 0.691$	$\mu_{CR} = 0.308$		
Diversity = 4.8331e-03	k_t = [1, 1, 1, 1]				
Generation 820: Best = 2.900880e+03	FES = 82100	$\mu_F = 0.682$	$\mu_{CR} = 0.312$		
Diversity = 3.8349e-03	k_t = [4, 2, 3, 2]				
Generation 830: Best = 2.900879e+03	FES = 83100	$\mu_F = 0.704$	$\mu_{CR} = 0.309$		
Diversity = 2.9610e-03	k_t = [1, 1, 1, 1]				
Generation 840: Best = 2.900879e+03	FES = 84100	$\mu_F = 0.698$	$\mu_{CR} = 0.310$		
Diversity = 2.2826e-03	k_t = [3, 4, 2, 2]				
Generation 850: Best = 2.900879e+03	FES = 85100	$\mu_F = 0.711$	$\mu_{CR} = 0.304$		
Diversity = 1.8438e-03	k_t = [1, 1, 1, 1]				
Generation 860: Best = 2.900879e+03	FES = 86100	$\mu_F = 0.714$	$\mu_{CR} = 0.311$		
Diversity = 1.4120e-03	k_t = [2, 4, 4, 3]				
Generation 870: Best = 2.900879e+03	FES = 87100	$\mu_F = 0.710$	$\mu_{CR} = 0.303$		
Diversity = 1.2120e-03	k_t = [1, 1, 1, 1]				
Generation 880: Best = 2.900879e+03	FES = 88100	$\mu_F = 0.712$	$\mu_{CR} = 0.306$		
Diversity = 1.1320e-03	k_t = [4, 2, 2, 2]				
Generation 890: Best = 2.900879e+03	FES = 89100	$\mu_F = 0.718$	$\mu_{CR} = 0.308$		
Diversity = 9.5812e-04	k_t = [1, 1, 1, 1]				
Generation 900: Best = 2.900879e+03	FES = 90100	$\mu_F = 0.730$	$\mu_{CR} = 0.305$		
Diversity = 8.6891e-04	k_t = [2, 2, 2, 2]				
Generation 910: Best = 2.900879e+03	FES = 91100	$\mu_F = 0.744$	$\mu_{CR} = 0.308$		
Diversity = 9.0502e-04	k_t = [1, 1, 1, 1]				
Generation 920: Best = 2.900879e+03	FES = 92100	$\mu_F = 0.746$	$\mu_{CR} = 0.295$		
Diversity = 8.9997e-04	k_t = [3, 3, 2, 2]				
Generation 930: Best = 2.900879e+03	FES = 93100	$\mu_F = 0.737$	$\mu_{CR} = 0.284$		
Diversity = 8.2971e-04	k_t = [1, 1, 1, 1]				
Generation 940: Best = 2.900878e+03	FES = 94100	$\mu_F = 0.729$	$\mu_{CR} = 0.283$		
Diversity = 6.5498e-04	k_t = [4, 3, 3, 2]				
Generation 950: Best = 2.900878e+03	FES = 95100	$\mu_F = 0.736$	$\mu_{CR} = 0.281$		
Diversity = 5.9517e-04	k_t = [1, 1, 1, 1]				
Generation 960: Best = 2.900878e+03	FES = 96100	$\mu_F = 0.730$	$\mu_{CR} = 0.279$		
Diversity = 4.8567e-04	k_t = [4, 3, 3, 3]				
Generation 970: Best = 2.900878e+03	FES = 97100	$\mu_F = 0.723$	$\mu_{CR} = 0.277$		
Diversity = 4.2715e-04	k_t = [1, 1, 1, 1]				
Generation 980: Best = 2.900878e+03	FES = 98100	$\mu_F = 0.723$	$\mu_{CR} = 0.283$		
Diversity = 3.9567e-04	k_t = [2, 4, 2, 2]				
Generation 990: Best = 2.900878e+03	FES = 99100	$\mu_F = 0.730$	$\mu_{CR} = 0.287$		
Diversity = 3.6609e-04	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 2.900878e+03	FES = 100100	$\mu_F = 0.728$	$\mu_{CR} = 0.289$		
Diversity = 2.9106e-04	k_t = [3, 3, 2, 4]				
Generation 1010: Best = 2.900878e+03	FES = 101100	$\mu_F = 0.730$	$\mu_{CR} = 0.281$		
Diversity = 2.5397e-04	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 2.900878e+03	FES = 102100	$\mu_F = 0.742$	$\mu_{CR} = 0.281$		
Diversity = 2.3975e-04	k_t = [4, 3, 4, 2]				
Generation 1030: Best = 2.900878e+03	FES = 103100	$\mu_F = 0.741$	$\mu_{CR} = 0.270$		

Diversity = 2.0241e-04	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 2.900878e+03	FES = 104100	$\mu_F = 0.743$	$\mu_{CR} = 0.259$	
Diversity = 1.6826e-04	k_t = [3, 2, 4, 3]			
Generation 1050: Best = 2.900878e+03	FES = 105100	$\mu_F = 0.754$	$\mu_{CR} = 0.260$	
Diversity = 1.5871e-04	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 2.900878e+03	FES = 106100	$\mu_F = 0.758$	$\mu_{CR} = 0.250$	
Diversity = 1.4752e-04	k_t = [3, 3, 3, 2]			
Generation 1070: Best = 2.900878e+03	FES = 107100	$\mu_F = 0.768$	$\mu_{CR} = 0.251$	
Diversity = 1.4288e-04	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 2.900878e+03	FES = 108100	$\mu_F = 0.775$	$\mu_{CR} = 0.240$	
Diversity = 1.3807e-04	k_t = [2, 3, 3, 2]			
Generation 1090: Best = 2.900878e+03	FES = 109100	$\mu_F = 0.774$	$\mu_{CR} = 0.241$	
Diversity = 1.3835e-04	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 2.900878e+03	FES = 110100	$\mu_F = 0.770$	$\mu_{CR} = 0.242$	
Diversity = 1.2701e-04	k_t = [2, 2, 2, 4]			
Generation 1110: Best = 2.900878e+03	FES = 111100	$\mu_F = 0.769$	$\mu_{CR} = 0.243$	
Diversity = 1.3809e-04	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 2.900878e+03	FES = 112100	$\mu_F = 0.765$	$\mu_{CR} = 0.241$	
Diversity = 1.1831e-04	k_t = [3, 3, 3, 3]			
Generation 1130: Best = 2.900878e+03	FES = 113100	$\mu_F = 0.769$	$\mu_{CR} = 0.249$	
Diversity = 1.1477e-04	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 2.900878e+03	FES = 114100	$\mu_F = 0.755$	$\mu_{CR} = 0.240$	
Diversity = 1.0398e-04	k_t = [4, 2, 3, 2]			
Generation 1150: Best = 2.900878e+03	FES = 115100	$\mu_F = 0.772$	$\mu_{CR} = 0.238$	
Diversity = 1.1533e-04	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 2.900878e+03	FES = 116100	$\mu_F = 0.765$	$\mu_{CR} = 0.240$	
Diversity = 1.2133e-04	k_t = [3, 4, 3, 3]			
Generation 1170: Best = 2.900878e+03	FES = 117100	$\mu_F = 0.775$	$\mu_{CR} = 0.244$	
Diversity = 1.3411e-04	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 2.900878e+03	FES = 118100	$\mu_F = 0.784$	$\mu_{CR} = 0.249$	
Diversity = 1.2836e-04	k_t = [3, 4, 2, 4]			
Generation 1190: Best = 2.900878e+03	FES = 119100	$\mu_F = 0.783$	$\mu_{CR} = 0.247$	
Diversity = 1.2707e-04	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 2.900878e+03	FES = 120100	$\mu_F = 0.787$	$\mu_{CR} = 0.247$	
Diversity = 1.1269e-04	k_t = [3, 3, 4, 4]			
Generation 1210: Best = 2.900878e+03	FES = 121100	$\mu_F = 0.798$	$\mu_{CR} = 0.242$	
Diversity = 1.1654e-04	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 2.900878e+03	FES = 122100	$\mu_F = 0.791$	$\mu_{CR} = 0.232$	
Diversity = 1.1102e-04	k_t = [2, 3, 4, 4]			
Generation 1230: Best = 2.900878e+03	FES = 123100	$\mu_F = 0.789$	$\mu_{CR} = 0.230$	
Diversity = 1.1517e-04	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 2.900878e+03	FES = 124100	$\mu_F = 0.784$	$\mu_{CR} = 0.229$	
Diversity = 1.0172e-04	k_t = [3, 4, 4, 2]			
Generation 1250: Best = 2.900878e+03	FES = 125100	$\mu_F = 0.784$	$\mu_{CR} = 0.227$	
Diversity = 9.0306e-05	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 2.900878e+03	FES = 126100	$\mu_F = 0.783$	$\mu_{CR} = 0.230$	
Diversity = 8.0299e-05	k_t = [3, 3, 2, 3]			
Generation 1270: Best = 2.900878e+03	FES = 127100	$\mu_F = 0.765$	$\mu_{CR} = 0.225$	
Diversity = 8.1400e-05	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 2.900878e+03	FES = 128100	$\mu_F = 0.758$	$\mu_{CR} = 0.216$	
Diversity = 7.0962e-05	k_t = [3, 3, 2, 3]			
Generation 1290: Best = 2.900878e+03	FES = 129100	$\mu_F = 0.751$	$\mu_{CR} = 0.218$	
Diversity = 7.3632e-05	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 2.900878e+03	FES = 130100	$\mu_F = 0.756$	$\mu_{CR} = 0.213$	

Diversity = 6.7342e-05	k_t = [3, 2, 2, 3]			
Generation 1310: Best = 2.900878e+03	FES = 131100	$\mu_F = 0.751$	$\mu_{CR} = 0.217$	
Diversity = 6.6350e-05	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 2.900878e+03	FES = 132100	$\mu_F = 0.743$	$\mu_{CR} = 0.223$	
Diversity = 6.3311e-05	k_t = [2, 3, 4, 2]			
Generation 1330: Best = 2.900878e+03	FES = 133100	$\mu_F = 0.758$	$\mu_{CR} = 0.222$	
Diversity = 6.4376e-05	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 2.900878e+03	FES = 134100	$\mu_F = 0.762$	$\mu_{CR} = 0.234$	
Diversity = 6.1035e-05	k_t = [3, 2, 2, 2]			
Generation 1350: Best = 2.900878e+03	FES = 135100	$\mu_F = 0.767$	$\mu_{CR} = 0.239$	
Diversity = 6.1406e-05	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 2.900878e+03	FES = 136100	$\mu_F = 0.773$	$\mu_{CR} = 0.253$	
Diversity = 5.4662e-05	k_t = [4, 2, 4, 4]			
Generation 1370: Best = 2.900878e+03	FES = 137100	$\mu_F = 0.774$	$\mu_{CR} = 0.251$	
Diversity = 5.0035e-05	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 2.900878e+03	FES = 138100	$\mu_F = 0.773$	$\mu_{CR} = 0.248$	
Diversity = 4.1482e-05	k_t = [3, 4, 3, 4]			
Generation 1390: Best = 2.900878e+03	FES = 139100	$\mu_F = 0.777$	$\mu_{CR} = 0.251$	
Diversity = 4.1750e-05	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 2.900878e+03	FES = 140100	$\mu_F = 0.759$	$\mu_{CR} = 0.249$	
Diversity = 4.1768e-05	k_t = [4, 3, 3, 2]			
Generation 1410: Best = 2.900878e+03	FES = 141100	$\mu_F = 0.766$	$\mu_{CR} = 0.252$	
Diversity = 4.1941e-05	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 2.900878e+03	FES = 142100	$\mu_F = 0.765$	$\mu_{CR} = 0.249$	
Diversity = 3.7115e-05	k_t = [3, 3, 4, 3]			
Generation 1430: Best = 2.900878e+03	FES = 143100	$\mu_F = 0.769$	$\mu_{CR} = 0.245$	
Diversity = 3.2995e-05	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 2.900878e+03	FES = 144100	$\mu_F = 0.760$	$\mu_{CR} = 0.244$	
Diversity = 2.8908e-05	k_t = [2, 2, 2, 3]			
Generation 1450: Best = 2.900878e+03	FES = 145100	$\mu_F = 0.761$	$\mu_{CR} = 0.249$	
Diversity = 3.0526e-05	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 2.900878e+03	FES = 146100	$\mu_F = 0.763$	$\mu_{CR} = 0.249$	
Diversity = 2.7578e-05	k_t = [4, 4, 2, 2]			
Generation 1470: Best = 2.900878e+03	FES = 147100	$\mu_F = 0.765$	$\mu_{CR} = 0.254$	
Diversity = 2.3747e-05	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 2.900878e+03	FES = 148100	$\mu_F = 0.775$	$\mu_{CR} = 0.255$	
Diversity = 2.1514e-05	k_t = [4, 3, 4, 4]			
Generation 1490: Best = 2.900878e+03	FES = 149100	$\mu_F = 0.771$	$\mu_{CR} = 0.259$	
Diversity = 1.8121e-05	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 2.900878e+03	FES = 150100	$\mu_F = 0.757$	$\mu_{CR} = 0.253$	
Diversity = 1.5478e-05	k_t = [2, 2, 3, 4]			
Generation 1510: Best = 2.900878e+03	FES = 151100	$\mu_F = 0.764$	$\mu_{CR} = 0.254$	
Diversity = 1.4783e-05	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 2.900878e+03	FES = 152100	$\mu_F = 0.757$	$\mu_{CR} = 0.256$	
Diversity = 1.2367e-05	k_t = [4, 2, 4, 3]			
Generation 1530: Best = 2.900878e+03	FES = 153100	$\mu_F = 0.757$	$\mu_{CR} = 0.257$	
Diversity = 1.1785e-05	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 2.900878e+03	FES = 154100	$\mu_F = 0.748$	$\mu_{CR} = 0.259$	
Diversity = 1.1304e-05	k_t = [4, 3, 4, 2]			
Generation 1550: Best = 2.900878e+03	FES = 155100	$\mu_F = 0.757$	$\mu_{CR} = 0.257$	
Diversity = 1.1425e-05	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 2.900878e+03	FES = 156100	$\mu_F = 0.767$	$\mu_{CR} = 0.261$	
Diversity = 1.2494e-05	k_t = [3, 4, 2, 2]			
Generation 1570: Best = 2.900878e+03	FES = 157100	$\mu_F = 0.781$	$\mu_{CR} = 0.257$	

Diversity = 1.3527e-05	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 2.900878e+03	FES = 158100	$\mu_F = 0.766$	$\mu_{CR} = 0.255$	
Diversity = 1.4482e-05	k_t = [2, 2, 3, 2]			
Generation 1590: Best = 2.900878e+03	FES = 159100	$\mu_F = 0.763$	$\mu_{CR} = 0.250$	
Diversity = 1.6774e-05	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 2.900878e+03	FES = 160100	$\mu_F = 0.769$	$\mu_{CR} = 0.252$	
Diversity = 1.5168e-05	k_t = [3, 2, 3, 4]			
Generation 1610: Best = 2.900878e+03	FES = 161100	$\mu_F = 0.757$	$\mu_{CR} = 0.249$	
Diversity = 1.5102e-05	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 2.900878e+03	FES = 162100	$\mu_F = 0.750$	$\mu_{CR} = 0.254$	
Diversity = 1.3678e-05	k_t = [2, 2, 3, 4]			
Generation 1630: Best = 2.900878e+03	FES = 163100	$\mu_F = 0.750$	$\mu_{CR} = 0.253$	
Diversity = 1.1691e-05	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 2.900878e+03	FES = 164100	$\mu_F = 0.754$	$\mu_{CR} = 0.246$	
Diversity = 1.0041e-05	k_t = [4, 4, 4, 4]			
Generation 1650: Best = 2.900878e+03	FES = 165100	$\mu_F = 0.770$	$\mu_{CR} = 0.235$	
Diversity = 1.0431e-05	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 2.900878e+03	FES = 166100	$\mu_F = 0.779$	$\mu_{CR} = 0.240$	
Diversity = 9.8676e-06	k_t = [3, 4, 4, 4]			
Generation 1670: Best = 2.900878e+03	FES = 167100	$\mu_F = 0.781$	$\mu_{CR} = 0.243$	
Diversity = 9.6110e-06	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 2.900878e+03	FES = 168100	$\mu_F = 0.783$	$\mu_{CR} = 0.238$	
Diversity = 9.0462e-06	k_t = [4, 2, 3, 4]			
Generation 1690: Best = 2.900878e+03	FES = 169100	$\mu_F = 0.776$	$\mu_{CR} = 0.239$	
Diversity = 8.4354e-06	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 2.900878e+03	FES = 170100	$\mu_F = 0.768$	$\mu_{CR} = 0.236$	
Diversity = 6.7586e-06	k_t = [4, 3, 3, 3]			
Generation 1710: Best = 2.900878e+03	FES = 171100	$\mu_F = 0.768$	$\mu_{CR} = 0.241$	
Diversity = 6.4877e-06	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 2.900878e+03	FES = 172100	$\mu_F = 0.753$	$\mu_{CR} = 0.232$	
Diversity = 6.0997e-06	k_t = [3, 2, 2, 2]			
Generation 1730: Best = 2.900878e+03	FES = 173100	$\mu_F = 0.739$	$\mu_{CR} = 0.226$	
Diversity = 5.5846e-06	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 2.900878e+03	FES = 174100	$\mu_F = 0.746$	$\mu_{CR} = 0.223$	
Diversity = 4.5337e-06	k_t = [3, 4, 4, 4]			
Generation 1750: Best = 2.900878e+03	FES = 175100	$\mu_F = 0.766$	$\mu_{CR} = 0.226$	
Diversity = 4.2685e-06	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 2.900878e+03	FES = 176100	$\mu_F = 0.768$	$\mu_{CR} = 0.221$	
Diversity = 4.1943e-06	k_t = [4, 3, 3, 3]			
Generation 1770: Best = 2.900878e+03	FES = 177100	$\mu_F = 0.773$	$\mu_{CR} = 0.212$	
Diversity = 4.1958e-06	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 2.900878e+03	FES = 178100	$\mu_F = 0.767$	$\mu_{CR} = 0.220$	
Diversity = 4.1162e-06	k_t = [2, 2, 3, 4]			
Generation 1790: Best = 2.900878e+03	FES = 179100	$\mu_F = 0.775$	$\mu_{CR} = 0.216$	
Diversity = 3.9055e-06	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 2.900878e+03	FES = 180100	$\mu_F = 0.776$	$\mu_{CR} = 0.221$	
Diversity = 3.5218e-06	k_t = [4, 3, 2, 2]			
Generation 1810: Best = 2.900878e+03	FES = 181100	$\mu_F = 0.769$	$\mu_{CR} = 0.219$	
Diversity = 3.3374e-06	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 2.900878e+03	FES = 182100	$\mu_F = 0.778$	$\mu_{CR} = 0.218$	
Diversity = 3.0018e-06	k_t = [2, 3, 2, 3]			
Generation 1830: Best = 2.900878e+03	FES = 183100	$\mu_F = 0.771$	$\mu_{CR} = 0.215$	
Diversity = 2.7511e-06	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 2.900878e+03	FES = 184100	$\mu_F = 0.762$	$\mu_{CR} = 0.213$	

Diversity = 2.1632e-06	k_t = [2, 4, 2, 4]			
Generation 1850: Best = 2.900878e+03	FES = 185100	$\mu_F = 0.763$	$\mu_{CR} = 0.211$	
Diversity = 2.3289e-06	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 2.900878e+03	FES = 186100	$\mu_F = 0.762$	$\mu_{CR} = 0.214$	
Diversity = 1.9717e-06	k_t = [3, 2, 2, 4]			
Generation 1870: Best = 2.900878e+03	FES = 187100	$\mu_F = 0.763$	$\mu_{CR} = 0.213$	
Diversity = 1.8665e-06	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 2.900878e+03	FES = 188100	$\mu_F = 0.762$	$\mu_{CR} = 0.205$	
Diversity = 1.6889e-06	k_t = [4, 3, 4, 3]			
Generation 1890: Best = 2.900878e+03	FES = 189100	$\mu_F = 0.771$	$\mu_{CR} = 0.207$	
Diversity = 1.5658e-06	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 2.900878e+03	FES = 190100	$\mu_F = 0.777$	$\mu_{CR} = 0.208$	
Diversity = 1.3817e-06	k_t = [2, 3, 4, 4]			
Generation 1910: Best = 2.900878e+03	FES = 191100	$\mu_F = 0.776$	$\mu_{CR} = 0.222$	
Diversity = 1.3316e-06	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 2.900878e+03	FES = 192100	$\mu_F = 0.774$	$\mu_{CR} = 0.231$	
Diversity = 1.2231e-06	k_t = [3, 4, 3, 2]			
Generation 1930: Best = 2.900878e+03	FES = 193100	$\mu_F = 0.783$	$\mu_{CR} = 0.241$	
Diversity = 1.3171e-06	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 2.900878e+03	FES = 194100	$\mu_F = 0.777$	$\mu_{CR} = 0.244$	
Diversity = 1.3081e-06	k_t = [3, 2, 3, 3]			
Generation 1950: Best = 2.900878e+03	FES = 195100	$\mu_F = 0.782$	$\mu_{CR} = 0.242$	
Diversity = 1.3039e-06	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 2.900878e+03	FES = 196100	$\mu_F = 0.781$	$\mu_{CR} = 0.248$	
Diversity = 1.4005e-06	k_t = [3, 3, 4, 2]			
Generation 1970: Best = 2.900878e+03	FES = 197100	$\mu_F = 0.778$	$\mu_{CR} = 0.254$	
Diversity = 1.6095e-06	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 2.900878e+03	FES = 198100	$\mu_F = 0.777$	$\mu_{CR} = 0.249$	
Diversity = 1.3807e-06	k_t = [4, 4, 3, 2]			
Generation 1990: Best = 2.900878e+03	FES = 199100	$\mu_F = 0.776$	$\mu_{CR} = 0.253$	
Diversity = 1.3142e-06	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 2.900878e+03	FES = 200100	$\mu_F = 0.764$	$\mu_{CR} = 0.250$	
Diversity = 1.2005e-06	k_t = [2, 2, 4, 3]			
Generation 2010: Best = 2.900878e+03	FES = 201100	$\mu_F = 0.771$	$\mu_{CR} = 0.250$	
Diversity = 1.2431e-06	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 2.900878e+03	FES = 202100	$\mu_F = 0.777$	$\mu_{CR} = 0.253$	
Diversity = 1.0775e-06	k_t = [3, 3, 4, 3]			
Generation 2030: Best = 2.900878e+03	FES = 203100	$\mu_F = 0.777$	$\mu_{CR} = 0.248$	
Diversity = 9.9802e-07	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 2.900878e+03	FES = 204100	$\mu_F = 0.776$	$\mu_{CR} = 0.243$	
Diversity = 1.0012e-06	k_t = [2, 2, 3, 3]			
Generation 2050: Best = 2.900878e+03	FES = 205100	$\mu_F = 0.772$	$\mu_{CR} = 0.248$	
Diversity = 1.1462e-06	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 2.900878e+03	FES = 206100	$\mu_F = 0.776$	$\mu_{CR} = 0.251$	
Diversity = 1.0142e-06	k_t = [2, 3, 3, 3]			
Generation 2070: Best = 2.900878e+03	FES = 207100	$\mu_F = 0.775$	$\mu_{CR} = 0.242$	
Diversity = 1.0599e-06	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 2.900878e+03	FES = 208100	$\mu_F = 0.788$	$\mu_{CR} = 0.239$	
Diversity = 1.2222e-06	k_t = [2, 2, 3, 3]			
Generation 2090: Best = 2.900878e+03	FES = 209100	$\mu_F = 0.779$	$\mu_{CR} = 0.239$	
Diversity = 1.2970e-06	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 2.900878e+03	FES = 210100	$\mu_F = 0.796$	$\mu_{CR} = 0.236$	
Diversity = 1.2695e-06	k_t = [4, 2, 3, 4]			
Generation 2110: Best = 2.900878e+03	FES = 211100	$\mu_F = 0.792$	$\mu_{CR} = 0.230$	

Diversity = 1.2944e-06	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 2.900878e+03	FES = 212100	$\mu_F = 0.788$	$\mu_{CR} = 0.228$	
Diversity = 1.2564e-06	k_t = [4, 4, 2, 3]			
Generation 2130: Best = 2.900878e+03	FES = 213100	$\mu_F = 0.803$	$\mu_{CR} = 0.234$	
Diversity = 1.2872e-06	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 2.900878e+03	FES = 214100	$\mu_F = 0.799$	$\mu_{CR} = 0.224$	
Diversity = 1.1724e-06	k_t = [4, 4, 4, 4]			
Generation 2150: Best = 2.900878e+03	FES = 215100	$\mu_F = 0.796$	$\mu_{CR} = 0.219$	
Diversity = 1.2447e-06	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 2.900878e+03	FES = 216100	$\mu_F = 0.799$	$\mu_{CR} = 0.221$	
Diversity = 1.3130e-06	k_t = [4, 4, 3, 2]			
Generation 2170: Best = 2.900878e+03	FES = 217100	$\mu_F = 0.789$	$\mu_{CR} = 0.217$	
Diversity = 1.1691e-06	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 2.900878e+03	FES = 218100	$\mu_F = 0.780$	$\mu_{CR} = 0.219$	
Diversity = 1.0509e-06	k_t = [2, 3, 3, 3]			
Generation 2190: Best = 2.900878e+03	FES = 219100	$\mu_F = 0.772$	$\mu_{CR} = 0.211$	
Diversity = 1.1083e-06	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 2.900878e+03	FES = 220100	$\mu_F = 0.772$	$\mu_{CR} = 0.224$	
Diversity = 9.1644e-07	k_t = [4, 4, 2, 4]			
Generation 2210: Best = 2.900878e+03	FES = 221100	$\mu_F = 0.779$	$\mu_{CR} = 0.223$	
Diversity = 8.1878e-07	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 2.900878e+03	FES = 222100	$\mu_F = 0.763$	$\mu_{CR} = 0.220$	
Diversity = 6.9788e-07	k_t = [3, 2, 4, 4]			
Generation 2230: Best = 2.900878e+03	FES = 223100	$\mu_F = 0.766$	$\mu_{CR} = 0.215$	
Diversity = 6.3479e-07	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 2.900878e+03	FES = 224100	$\mu_F = 0.755$	$\mu_{CR} = 0.215$	
Diversity = 5.4131e-07	k_t = [3, 2, 2, 4]			
Generation 2250: Best = 2.900878e+03	FES = 225100	$\mu_F = 0.764$	$\mu_{CR} = 0.208$	
Diversity = 5.9426e-07	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 2.900878e+03	FES = 226100	$\mu_F = 0.766$	$\mu_{CR} = 0.202$	
Diversity = 6.0545e-07	k_t = [4, 2, 3, 2]			
Generation 2270: Best = 2.900878e+03	FES = 227100	$\mu_F = 0.766$	$\mu_{CR} = 0.191$	
Diversity = 5.9126e-07	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 2.900878e+03	FES = 228100	$\mu_F = 0.775$	$\mu_{CR} = 0.191$	
Diversity = 5.8793e-07	k_t = [4, 4, 3, 3]			
Generation 2290: Best = 2.900878e+03	FES = 229100	$\mu_F = 0.782$	$\mu_{CR} = 0.193$	
Diversity = 6.1751e-07	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 2.900878e+03	FES = 230100	$\mu_F = 0.784$	$\mu_{CR} = 0.188$	
Diversity = 6.1789e-07	k_t = [2, 4, 4, 2]			
Generation 2310: Best = 2.900878e+03	FES = 231100	$\mu_F = 0.782$	$\mu_{CR} = 0.197$	
Diversity = 6.9182e-07	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 2.900878e+03	FES = 232100	$\mu_F = 0.790$	$\mu_{CR} = 0.206$	
Diversity = 7.8558e-07	k_t = [2, 4, 2, 4]			
Generation 2330: Best = 2.900878e+03	FES = 233100	$\mu_F = 0.794$	$\mu_{CR} = 0.212$	
Diversity = 8.9491e-07	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 2.900878e+03	FES = 234100	$\mu_F = 0.806$	$\mu_{CR} = 0.211$	
Diversity = 9.6337e-07	k_t = [2, 2, 4, 2]			
Generation 2350: Best = 2.900878e+03	FES = 235100	$\mu_F = 0.814$	$\mu_{CR} = 0.209$	
Diversity = 1.1347e-06	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 2.900878e+03	FES = 236100	$\mu_F = 0.817$	$\mu_{CR} = 0.213$	
Diversity = 1.2973e-06	k_t = [3, 2, 2, 2]			
Generation 2370: Best = 2.900878e+03	FES = 237100	$\mu_F = 0.799$	$\mu_{CR} = 0.210$	
Diversity = 1.2497e-06	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 2.900878e+03	FES = 238100	$\mu_F = 0.799$	$\mu_{CR} = 0.210$	

Diversity = 1.2580e-06	k_t = [4, 2, 2, 4]			
Generation 2390: Best = 2.900878e+03	FES = 239100	$\mu_F = 0.791$	$\mu_{CR} = 0.201$	
Diversity = 1.2937e-06	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 2.900878e+03	FES = 240100	$\mu_F = 0.789$	$\mu_{CR} = 0.197$	
Diversity = 1.2889e-06	k_t = [4, 4, 2, 2]			
Generation 2410: Best = 2.900878e+03	FES = 241100	$\mu_F = 0.785$	$\mu_{CR} = 0.212$	
Diversity = 1.2565e-06	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 2.900878e+03	FES = 242100	$\mu_F = 0.788$	$\mu_{CR} = 0.216$	
Diversity = 1.3123e-06	k_t = [2, 4, 4, 3]			
Generation 2430: Best = 2.900878e+03	FES = 243100	$\mu_F = 0.790$	$\mu_{CR} = 0.228$	
Diversity = 1.4019e-06	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 2.900878e+03	FES = 244100	$\mu_F = 0.799$	$\mu_{CR} = 0.219$	
Diversity = 1.3683e-06	k_t = [4, 3, 2, 4]			
Generation 2450: Best = 2.900878e+03	FES = 245100	$\mu_F = 0.797$	$\mu_{CR} = 0.219$	
Diversity = 1.6480e-06	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 2.900878e+03	FES = 246100	$\mu_F = 0.796$	$\mu_{CR} = 0.221$	
Diversity = 1.5768e-06	k_t = [3, 3, 3, 3]			
Generation 2470: Best = 2.900878e+03	FES = 247100	$\mu_F = 0.780$	$\mu_{CR} = 0.223$	
Diversity = 1.6821e-06	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 2.900878e+03	FES = 248100	$\mu_F = 0.777$	$\mu_{CR} = 0.214$	
Diversity = 1.4496e-06	k_t = [3, 4, 4, 3]			
Generation 2490: Best = 2.900878e+03	FES = 249100	$\mu_F = 0.766$	$\mu_{CR} = 0.225$	
Diversity = 1.4398e-06	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 2.900878e+03	FES = 250100	$\mu_F = 0.769$	$\mu_{CR} = 0.220$	
Diversity = 1.4081e-06	k_t = [3, 2, 2, 4]			
Generation 2510: Best = 2.900878e+03	FES = 251100	$\mu_F = 0.767$	$\mu_{CR} = 0.224$	
Diversity = 1.4791e-06	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 2.900878e+03	FES = 252100	$\mu_F = 0.785$	$\mu_{CR} = 0.225$	
Diversity = 1.5250e-06	k_t = [2, 2, 4, 3]			
Generation 2530: Best = 2.900878e+03	FES = 253100	$\mu_F = 0.783$	$\mu_{CR} = 0.231$	
Diversity = 1.7650e-06	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 2.900878e+03	FES = 254100	$\mu_F = 0.788$	$\mu_{CR} = 0.228$	
Diversity = 1.6649e-06	k_t = [4, 3, 3, 4]			
Generation 2550: Best = 2.900878e+03	FES = 255100	$\mu_F = 0.796$	$\mu_{CR} = 0.230$	
Diversity = 1.8555e-06	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 2.900878e+03	FES = 256100	$\mu_F = 0.791$	$\mu_{CR} = 0.227$	
Diversity = 1.8948e-06	k_t = [2, 4, 4, 4]			
Generation 2570: Best = 2.900878e+03	FES = 257100	$\mu_F = 0.789$	$\mu_{CR} = 0.232$	
Diversity = 2.2009e-06	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 2.900878e+03	FES = 258100	$\mu_F = 0.793$	$\mu_{CR} = 0.240$	
Diversity = 2.0210e-06	k_t = [2, 2, 4, 4]			
Generation 2590: Best = 2.900878e+03	FES = 259100	$\mu_F = 0.769$	$\mu_{CR} = 0.239$	
Diversity = 1.9265e-06	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 2.900878e+03	FES = 260100	$\mu_F = 0.771$	$\mu_{CR} = 0.243$	
Diversity = 1.6631e-06	k_t = [2, 3, 4, 4]			
Generation 2610: Best = 2.900878e+03	FES = 261100	$\mu_F = 0.770$	$\mu_{CR} = 0.238$	
Diversity = 1.5439e-06	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 2.900878e+03	FES = 262100	$\mu_F = 0.773$	$\mu_{CR} = 0.240$	
Diversity = 1.4425e-06	k_t = [2, 3, 4, 4]			
Generation 2630: Best = 2.900878e+03	FES = 263100	$\mu_F = 0.771$	$\mu_{CR} = 0.249$	
Diversity = 1.5436e-06	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 2.900878e+03	FES = 264100	$\mu_F = 0.767$	$\mu_{CR} = 0.250$	
Diversity = 1.5493e-06	k_t = [4, 2, 4, 4]			
Generation 2650: Best = 2.900878e+03	FES = 265100	$\mu_F = 0.773$	$\mu_{CR} = 0.248$	

Diversity = 1.5897e-06	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 2.900878e+03	FES = 266100	$\mu_F = 0.773$	$\mu_{CR} = 0.251$	
Diversity = 1.7563e-06	k_t = [4, 3, 2, 3]			
Generation 2670: Best = 2.900878e+03	FES = 267100	$\mu_F = 0.771$	$\mu_{CR} = 0.250$	
Diversity = 1.8520e-06	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 2.900878e+03	FES = 268100	$\mu_F = 0.768$	$\mu_{CR} = 0.249$	
Diversity = 1.6148e-06	k_t = [3, 2, 3, 4]			
Generation 2690: Best = 2.900878e+03	FES = 269100	$\mu_F = 0.770$	$\mu_{CR} = 0.247$	
Diversity = 1.6366e-06	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 2.900878e+03	FES = 270100	$\mu_F = 0.759$	$\mu_{CR} = 0.250$	
Diversity = 1.7895e-06	k_t = [3, 3, 3, 2]			
Generation 2710: Best = 2.900878e+03	FES = 271100	$\mu_F = 0.763$	$\mu_{CR} = 0.246$	
Diversity = 1.5941e-06	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 2.900878e+03	FES = 272100	$\mu_F = 0.760$	$\mu_{CR} = 0.240$	
Diversity = 1.3385e-06	k_t = [4, 2, 4, 3]			
Generation 2730: Best = 2.900878e+03	FES = 273100	$\mu_F = 0.758$	$\mu_{CR} = 0.236$	
Diversity = 1.2238e-06	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 2.900878e+03	FES = 274100	$\mu_F = 0.772$	$\mu_{CR} = 0.234$	
Diversity = 1.3368e-06	k_t = [2, 2, 2, 2]			
Generation 2750: Best = 2.900878e+03	FES = 275100	$\mu_F = 0.778$	$\mu_{CR} = 0.239$	
Diversity = 1.5090e-06	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 2.900878e+03	FES = 276100	$\mu_F = 0.795$	$\mu_{CR} = 0.243$	
Diversity = 1.5467e-06	k_t = [4, 2, 2, 4]			
Generation 2770: Best = 2.900878e+03	FES = 277100	$\mu_F = 0.794$	$\mu_{CR} = 0.243$	
Diversity = 1.7191e-06	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 2.900878e+03	FES = 278100	$\mu_F = 0.808$	$\mu_{CR} = 0.243$	
Diversity = 1.5938e-06	k_t = [2, 2, 3, 3]			
Generation 2790: Best = 2.900878e+03	FES = 279100	$\mu_F = 0.802$	$\mu_{CR} = 0.260$	
Diversity = 1.6430e-06	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 2.900878e+03	FES = 280100	$\mu_F = 0.791$	$\mu_{CR} = 0.263$	
Diversity = 1.3550e-06	k_t = [3, 4, 4, 3]			
Generation 2810: Best = 2.900878e+03	FES = 281100	$\mu_F = 0.793$	$\mu_{CR} = 0.269$	
Diversity = 1.5285e-06	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 2.900878e+03	FES = 282100	$\mu_F = 0.789$	$\mu_{CR} = 0.271$	
Diversity = 1.6082e-06	k_t = [2, 3, 4, 2]			
Generation 2830: Best = 2.900878e+03	FES = 283100	$\mu_F = 0.787$	$\mu_{CR} = 0.261$	
Diversity = 1.4728e-06	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 2.900878e+03	FES = 284100	$\mu_F = 0.785$	$\mu_{CR} = 0.249$	
Diversity = 1.4328e-06	k_t = [3, 4, 4, 4]			
Generation 2850: Best = 2.900878e+03	FES = 285100	$\mu_F = 0.781$	$\mu_{CR} = 0.248$	
Diversity = 1.4256e-06	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 2.900878e+03	FES = 286100	$\mu_F = 0.780$	$\mu_{CR} = 0.258$	
Diversity = 1.4117e-06	k_t = [2, 4, 4, 3]			
Generation 2870: Best = 2.900878e+03	FES = 287100	$\mu_F = 0.782$	$\mu_{CR} = 0.266$	
Diversity = 1.3369e-06	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 2.900878e+03	FES = 288100	$\mu_F = 0.787$	$\mu_{CR} = 0.254$	
Diversity = 1.2989e-06	k_t = [3, 4, 4, 3]			
Generation 2890: Best = 2.900878e+03	FES = 289100	$\mu_F = 0.790$	$\mu_{CR} = 0.251$	
Diversity = 1.3284e-06	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 2.900878e+03	FES = 290100	$\mu_F = 0.784$	$\mu_{CR} = 0.245$	
Diversity = 1.2306e-06	k_t = [3, 3, 4, 3]			
Generation 2910: Best = 2.900878e+03	FES = 291100	$\mu_F = 0.773$	$\mu_{CR} = 0.252$	
Diversity = 1.2897e-06	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 2.900878e+03	FES = 292100	$\mu_F = 0.777$	$\mu_{CR} = 0.247$	

```

Diversity = 1.2172e-06  k_t = [4, 3, 3, 3]
Generation 2930: Best = 2.900878e+03  FES = 293100   $\mu_F$  = 0.778   $\mu_{CR}$  = 0.253
Diversity = 1.3112e-06  k_t = [1, 1, 1, 1]
Generation 2940: Best = 2.900878e+03  FES = 294100   $\mu_F$  = 0.771   $\mu_{CR}$  = 0.253
Diversity = 1.2530e-06  k_t = [3, 3, 4, 3]
Generation 2950: Best = 2.900878e+03  FES = 295100   $\mu_F$  = 0.772   $\mu_{CR}$  = 0.251
Diversity = 1.4795e-06  k_t = [1, 1, 1, 1]
Generation 2960: Best = 2.900878e+03  FES = 296100   $\mu_F$  = 0.779   $\mu_{CR}$  = 0.246
Diversity = 1.3865e-06  k_t = [4, 3, 4, 4]
Generation 2970: Best = 2.900878e+03  FES = 297100   $\mu_F$  = 0.782   $\mu_{CR}$  = 0.247
Diversity = 1.2649e-06  k_t = [1, 1, 1, 1]
Generation 2980: Best = 2.900878e+03  FES = 298100   $\mu_F$  = 0.776   $\mu_{CR}$  = 0.238
Diversity = 1.1773e-06  k_t = [4, 4, 3, 4]
Generation 2990: Best = 2.900878e+03  FES = 299100   $\mu_F$  = 0.776   $\mu_{CR}$  = 0.247
Diversity = 1.1141e-06  k_t = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 2.900878e+03

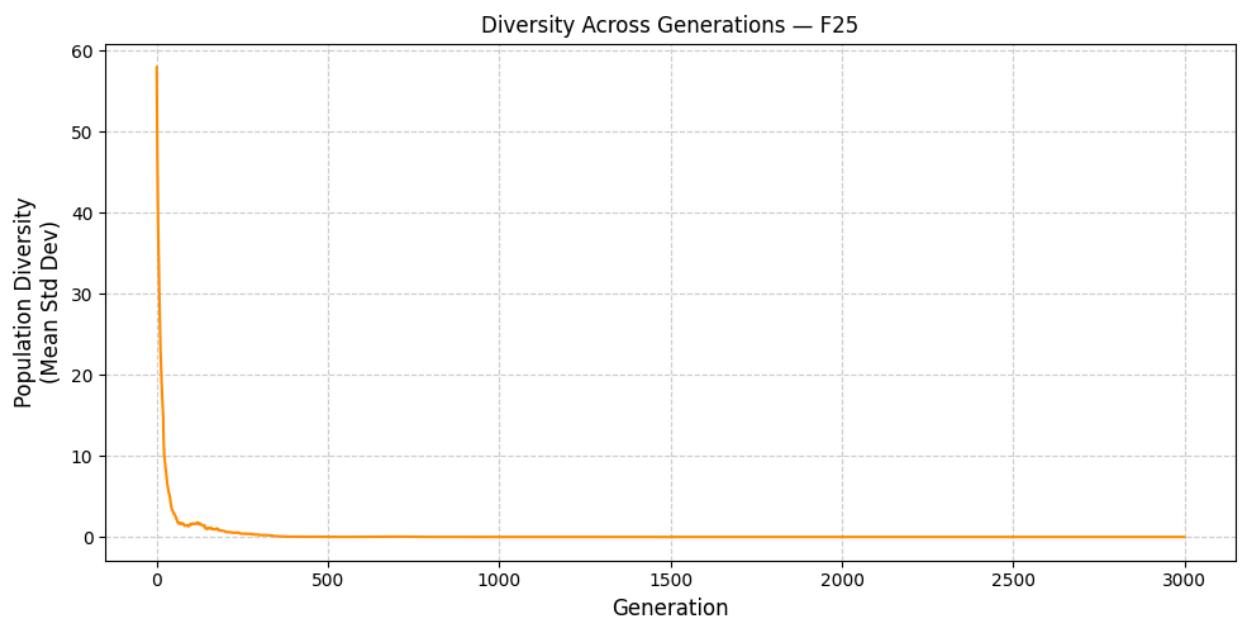
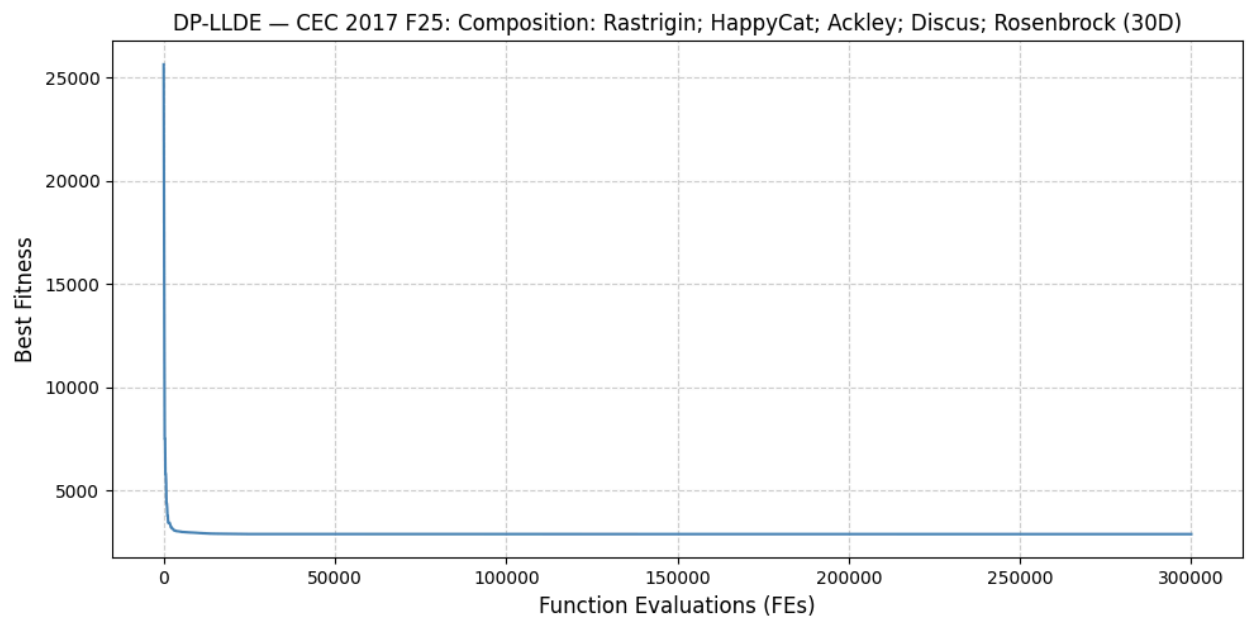
Total Function Evals = 300000

Total Successful Updates = 132338

```

=====
=
Results:
  Best Fitness      : 2.900878e+03
  Global Optimum    : 2.500000e+03
  Error (|best-optimum|) : 4.008784e+02
  Best Solution (5 dim) : [ 32.60604669  29.84606219  -1.42030297   8.87776247
-27.26734245]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F25.png

```
In [ ]: # =====
# F26 – Composition Function 6 (Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin)
# =====
run_and_plot('F26', make_wrapper(composition.f26), 2600.0, 'Composition: Schaf
```

=====

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DP-LLDE on CEC 2017 F26 – Single Run

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=

Function : Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin (30D)

Bounds : [-100, 100]

Global optimum : 2600.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 1.514240e+04

Generation 10: Best = 8.967790e+03	FES = 1100	$\mu_F = 0.564$	$\mu_{CR} = 0.359$
Diversity = 3.0983e+01	$k_t = [1, 1, 1, 1]$		
Generation 20: Best = 7.262001e+03	FES = 2100	$\mu_F = 0.619$	$\mu_{CR} = 0.355$
Diversity = 2.0492e+01	$k_t = [4, 2, 4, 4]$		
Generation 30: Best = 6.601064e+03	FES = 3100	$\mu_F = 0.646$	$\mu_{CR} = 0.347$
Diversity = 1.4591e+01	$k_t = [1, 1, 1, 1]$		
Generation 40: Best = 6.438475e+03	FES = 4100	$\mu_F = 0.665$	$\mu_{CR} = 0.333$
Diversity = 1.3490e+01	$k_t = [4, 3, 4, 2]$		
Generation 50: Best = 6.113691e+03	FES = 5100	$\mu_F = 0.693$	$\mu_{CR} = 0.304$
Diversity = 1.2698e+01	$k_t = [1, 1, 1, 1]$		
Generation 60: Best = 6.113691e+03	FES = 6100	$\mu_F = 0.708$	$\mu_{CR} = 0.248$
Diversity = 1.2289e+01	$k_t = [3, 3, 2, 4]$		
Generation 70: Best = 5.857334e+03	FES = 7100	$\mu_F = 0.705$	$\mu_{CR} = 0.192$
Diversity = 1.1845e+01	$k_t = [1, 1, 1, 1]$		
Generation 80: Best = 5.853489e+03	FES = 8100	$\mu_F = 0.670$	$\mu_{CR} = 0.156$
Diversity = 1.1582e+01	$k_t = [4, 2, 2, 2]$		
Generation 90: Best = 5.640891e+03	FES = 9100	$\mu_F = 0.694$	$\mu_{CR} = 0.133$
Diversity = 1.1531e+01	$k_t = [1, 1, 1, 1]$		
Generation 100: Best = 5.640891e+03	FES = 10100	$\mu_F = 0.736$	$\mu_{CR} = 0.115$
Diversity = 1.1279e+01	$k_t = [4, 2, 3, 3]$		
Generation 110: Best = 5.347205e+03	FES = 11100	$\mu_F = 0.686$	$\mu_{CR} = 0.098$
Diversity = 1.1216e+01	$k_t = [1, 1, 1, 1]$		
Generation 120: Best = 5.347205e+03	FES = 12100	$\mu_F = 0.732$	$\mu_{CR} = 0.076$
Diversity = 1.1257e+01	$k_t = [4, 4, 3, 4]$		
Generation 130: Best = 5.347205e+03	FES = 13100	$\mu_F = 0.737$	$\mu_{CR} = 0.079$
Diversity = 1.1225e+01	$k_t = [1, 1, 1, 1]$		
Generation 140: Best = 5.347205e+03	FES = 14100	$\mu_F = 0.717$	$\mu_{CR} = 0.077$
Diversity = 1.1182e+01	$k_t = [3, 2, 4, 2]$		
Generation 150: Best = 5.258755e+03	FES = 15100	$\mu_F = 0.722$	$\mu_{CR} = 0.063$
Diversity = 1.1216e+01	$k_t = [1, 1, 1, 1]$		
Generation 160: Best = 5.198234e+03	FES = 16100	$\mu_F = 0.737$	$\mu_{CR} = 0.064$
Diversity = 1.1090e+01	$k_t = [3, 4, 3, 4]$		
Generation 170: Best = 5.128487e+03	FES = 17100	$\mu_F = 0.708$	$\mu_{CR} = 0.072$
Diversity = 1.0915e+01	$k_t = [1, 1, 1, 1]$		
Generation 180: Best = 5.128487e+03	FES = 18100	$\mu_F = 0.680$	$\mu_{CR} = 0.070$
Diversity = 1.0920e+01	$k_t = [2, 2, 4, 2]$		
Generation 190: Best = 5.079921e+03	FES = 19100	$\mu_F = 0.682$	$\mu_{CR} = 0.066$
Diversity = 1.0778e+01	$k_t = [1, 1, 1, 1]$		
Generation 200: Best = 4.908016e+03	FES = 20100	$\mu_F = 0.682$	$\mu_{CR} = 0.069$
Diversity = 1.0735e+01	$k_t = [3, 3, 4, 3]$		
Generation 210: Best = 4.772056e+03	FES = 21100	$\mu_F = 0.715$	$\mu_{CR} = 0.064$
Diversity = 1.0631e+01	$k_t = [1, 1, 1, 1]$		
Generation 220: Best = 4.769305e+03	FES = 22100	$\mu_F = 0.591$	$\mu_{CR} = 0.065$

Diversity = 1.0479e+01	k_t = [4, 4, 4, 4]				
Generation 230: Best = 4.715312e+03	FES = 23100	$\mu_F = 0.580$	$\mu_{CR} = 0.072$		
Diversity = 1.0438e+01	k_t = [1, 1, 1, 1]				
Generation 240: Best = 4.715312e+03	FES = 24100	$\mu_F = 0.547$	$\mu_{CR} = 0.080$		
Diversity = 1.0360e+01	k_t = [3, 2, 3, 4]				
Generation 250: Best = 4.715312e+03	FES = 25100	$\mu_F = 0.542$	$\mu_{CR} = 0.078$		
Diversity = 1.0288e+01	k_t = [1, 1, 1, 1]				
Generation 260: Best = 4.691137e+03	FES = 26100	$\mu_F = 0.571$	$\mu_{CR} = 0.066$		
Diversity = 1.0090e+01	k_t = [2, 2, 3, 4]				
Generation 270: Best = 4.691137e+03	FES = 27100	$\mu_F = 0.574$	$\mu_{CR} = 0.059$		
Diversity = 9.9966e+00	k_t = [1, 1, 1, 1]				
Generation 280: Best = 4.691137e+03	FES = 28100	$\mu_F = 0.603$	$\mu_{CR} = 0.062$		
Diversity = 9.9827e+00	k_t = [2, 4, 4, 2]				
Generation 290: Best = 4.691137e+03	FES = 29100	$\mu_F = 0.594$	$\mu_{CR} = 0.055$		
Diversity = 9.9217e+00	k_t = [1, 1, 1, 1]				
Generation 300: Best = 4.691137e+03	FES = 30100	$\mu_F = 0.588$	$\mu_{CR} = 0.073$		
Diversity = 9.9270e+00	k_t = [4, 2, 2, 3]				
Generation 310: Best = 4.636996e+03	FES = 31100	$\mu_F = 0.611$	$\mu_{CR} = 0.066$		
Diversity = 9.8932e+00	k_t = [1, 1, 1, 1]				
Generation 320: Best = 4.636996e+03	FES = 32100	$\mu_F = 0.605$	$\mu_{CR} = 0.069$		
Diversity = 9.8586e+00	k_t = [2, 2, 4, 3]				
Generation 330: Best = 4.636996e+03	FES = 33100	$\mu_F = 0.616$	$\mu_{CR} = 0.054$		
Diversity = 9.8522e+00	k_t = [1, 1, 1, 1]				
Generation 340: Best = 4.636996e+03	FES = 34100	$\mu_F = 0.663$	$\mu_{CR} = 0.058$		
Diversity = 9.8561e+00	k_t = [4, 3, 3, 2]				
Generation 350: Best = 4.636996e+03	FES = 35100	$\mu_F = 0.680$	$\mu_{CR} = 0.063$		
Diversity = 9.8121e+00	k_t = [1, 1, 1, 1]				
Generation 360: Best = 4.607340e+03	FES = 36100	$\mu_F = 0.733$	$\mu_{CR} = 0.062$		
Diversity = 9.7148e+00	k_t = [2, 2, 2, 4]				
Generation 370: Best = 4.531053e+03	FES = 37100	$\mu_F = 0.737$	$\mu_{CR} = 0.047$		
Diversity = 9.6982e+00	k_t = [1, 1, 1, 1]				
Generation 380: Best = 4.531053e+03	FES = 38100	$\mu_F = 0.695$	$\mu_{CR} = 0.051$		
Diversity = 9.6508e+00	k_t = [4, 3, 4, 4]				
Generation 390: Best = 4.526661e+03	FES = 39100	$\mu_F = 0.711$	$\mu_{CR} = 0.053$		
Diversity = 9.5971e+00	k_t = [1, 1, 1, 1]				
Generation 400: Best = 4.519300e+03	FES = 40100	$\mu_F = 0.639$	$\mu_{CR} = 0.063$		
Diversity = 9.4726e+00	k_t = [3, 3, 3, 4]				
Generation 410: Best = 4.489172e+03	FES = 41100	$\mu_F = 0.681$	$\mu_{CR} = 0.070$		
Diversity = 9.4640e+00	k_t = [1, 1, 1, 1]				
Generation 420: Best = 4.487463e+03	FES = 42100	$\mu_F = 0.675$	$\mu_{CR} = 0.060$		
Diversity = 9.4624e+00	k_t = [3, 4, 4, 2]				
Generation 430: Best = 4.487463e+03	FES = 43100	$\mu_F = 0.690$	$\mu_{CR} = 0.058$		
Diversity = 9.4591e+00	k_t = [1, 1, 1, 1]				
Generation 440: Best = 4.482550e+03	FES = 44100	$\mu_F = 0.651$	$\mu_{CR} = 0.052$		
Diversity = 9.4585e+00	k_t = [3, 4, 3, 3]				
Generation 450: Best = 4.482550e+03	FES = 45100	$\mu_F = 0.682$	$\mu_{CR} = 0.048$		
Diversity = 9.4374e+00	k_t = [1, 1, 1, 1]				
Generation 460: Best = 4.477470e+03	FES = 46100	$\mu_F = 0.703$	$\mu_{CR} = 0.047$		
Diversity = 9.3325e+00	k_t = [4, 4, 2, 4]				
Generation 470: Best = 4.465274e+03	FES = 47100	$\mu_F = 0.734$	$\mu_{CR} = 0.052$		
Diversity = 9.3146e+00	k_t = [1, 1, 1, 1]				
Generation 480: Best = 4.465274e+03	FES = 48100	$\mu_F = 0.692$	$\mu_{CR} = 0.049$		
Diversity = 9.2932e+00	k_t = [2, 4, 4, 2]				
Generation 490: Best = 4.465274e+03	FES = 49100	$\mu_F = 0.701$	$\mu_{CR} = 0.054$		

Diversity = 9.2888e+00	k_t = [1, 1, 1, 1]					
Generation 500: Best = 4.465176e+03	FES = 50100	$\mu_F = 0.640$	$\mu_{CR} = 0.040$			
Diversity = 9.2969e+00	k_t = [3, 2, 3, 2]					
Generation 510: Best = 4.447154e+03	FES = 51100	$\mu_F = 0.644$	$\mu_{CR} = 0.037$			
Diversity = 9.2908e+00	k_t = [1, 1, 1, 1]					
Generation 520: Best = 4.403996e+03	FES = 52100	$\mu_F = 0.676$	$\mu_{CR} = 0.045$			
Diversity = 9.2204e+00	k_t = [3, 2, 2, 4]					
Generation 530: Best = 4.402544e+03	FES = 53100	$\mu_F = 0.726$	$\mu_{CR} = 0.040$			
Diversity = 9.1398e+00	k_t = [1, 1, 1, 1]					
Generation 540: Best = 4.364953e+03	FES = 54100	$\mu_F = 0.694$	$\mu_{CR} = 0.046$			
Diversity = 8.9260e+00	k_t = [3, 3, 4, 4]					
Generation 550: Best = 4.364753e+03	FES = 55100	$\mu_F = 0.679$	$\mu_{CR} = 0.047$			
Diversity = 8.9404e+00	k_t = [1, 1, 1, 1]					
Generation 560: Best = 4.364753e+03	FES = 56100	$\mu_F = 0.702$	$\mu_{CR} = 0.049$			
Diversity = 8.9136e+00	k_t = [2, 2, 4, 2]					
Generation 570: Best = 4.364282e+03	FES = 57100	$\mu_F = 0.603$	$\mu_{CR} = 0.047$			
Diversity = 8.8333e+00	k_t = [1, 1, 1, 1]					
Generation 580: Best = 4.357308e+03	FES = 58100	$\mu_F = 0.607$	$\mu_{CR} = 0.053$			
Diversity = 8.7771e+00	k_t = [3, 2, 3, 4]					
Generation 590: Best = 4.357308e+03	FES = 59100	$\mu_F = 0.621$	$\mu_{CR} = 0.054$			
Diversity = 8.7703e+00	k_t = [1, 1, 1, 1]					
Generation 600: Best = 4.321151e+03	FES = 60100	$\mu_F = 0.624$	$\mu_{CR} = 0.052$			
Diversity = 8.7663e+00	k_t = [4, 3, 4, 2]					
Generation 610: Best = 4.296801e+03	FES = 61100	$\mu_F = 0.597$	$\mu_{CR} = 0.055$			
Diversity = 8.7666e+00	k_t = [1, 1, 1, 1]					
Generation 620: Best = 4.296801e+03	FES = 62100	$\mu_F = 0.659$	$\mu_{CR} = 0.068$			
Diversity = 8.7633e+00	k_t = [4, 4, 2, 2]					
Generation 630: Best = 4.296801e+03	FES = 63100	$\mu_F = 0.623$	$\mu_{CR} = 0.053$			
Diversity = 8.7305e+00	k_t = [1, 1, 1, 1]					
Generation 640: Best = 4.296801e+03	FES = 64100	$\mu_F = 0.677$	$\mu_{CR} = 0.046$			
Diversity = 8.5360e+00	k_t = [3, 3, 3, 4]					
Generation 650: Best = 4.296801e+03	FES = 65100	$\mu_F = 0.652$	$\mu_{CR} = 0.054$			
Diversity = 8.5305e+00	k_t = [1, 1, 1, 1]					
Generation 660: Best = 4.296801e+03	FES = 66100	$\mu_F = 0.738$	$\mu_{CR} = 0.052$			
Diversity = 8.3870e+00	k_t = [3, 2, 2, 4]					
Generation 670: Best = 4.296095e+03	FES = 67100	$\mu_F = 0.693$	$\mu_{CR} = 0.051$			
Diversity = 8.3890e+00	k_t = [1, 1, 1, 1]					
Generation 680: Best = 4.274274e+03	FES = 68100	$\mu_F = 0.677$	$\mu_{CR} = 0.055$			
Diversity = 8.2356e+00	k_t = [4, 3, 4, 3]					
Generation 690: Best = 4.274274e+03	FES = 69100	$\mu_F = 0.674$	$\mu_{CR} = 0.050$			
Diversity = 8.2338e+00	k_t = [1, 1, 1, 1]					
Generation 700: Best = 4.249572e+03	FES = 70100	$\mu_F = 0.675$	$\mu_{CR} = 0.047$			
Diversity = 8.2367e+00	k_t = [2, 2, 3, 2]					
Generation 710: Best = 4.185915e+03	FES = 71100	$\mu_F = 0.695$	$\mu_{CR} = 0.046$			
Diversity = 8.2307e+00	k_t = [1, 1, 1, 1]					
Generation 720: Best = 4.185915e+03	FES = 72100	$\mu_F = 0.695$	$\mu_{CR} = 0.050$			
Diversity = 8.2152e+00	k_t = [2, 3, 3, 2]					
Generation 730: Best = 4.153789e+03	FES = 73100	$\mu_F = 0.674$	$\mu_{CR} = 0.051$			
Diversity = 8.2078e+00	k_t = [1, 1, 1, 1]					
Generation 740: Best = 4.153789e+03	FES = 74100	$\mu_F = 0.668$	$\mu_{CR} = 0.050$			
Diversity = 8.1202e+00	k_t = [4, 2, 4, 2]					
Generation 750: Best = 4.153789e+03	FES = 75100	$\mu_F = 0.626$	$\mu_{CR} = 0.049$			
Diversity = 8.1217e+00	k_t = [1, 1, 1, 1]					
Generation 760: Best = 4.136038e+03	FES = 76100	$\mu_F = 0.600$	$\mu_{CR} = 0.061$			

Diversity = 8.1248e+00	k_t = [3, 2, 4, 3]				
Generation 770: Best = 4.136038e+03	FES = 77100	$\mu_F = 0.608$	$\mu_{CR} = 0.047$		
Diversity = 8.0804e+00	k_t = [1, 1, 1, 1]				
Generation 780: Best = 4.136038e+03	FES = 78100	$\mu_F = 0.555$	$\mu_{CR} = 0.036$		
Diversity = 8.0793e+00	k_t = [3, 4, 3, 4]				
Generation 790: Best = 4.136038e+03	FES = 79100	$\mu_F = 0.539$	$\mu_{CR} = 0.052$		
Diversity = 8.0744e+00	k_t = [1, 1, 1, 1]				
Generation 800: Best = 4.136038e+03	FES = 80100	$\mu_F = 0.518$	$\mu_{CR} = 0.053$		
Diversity = 8.0713e+00	k_t = [3, 2, 3, 3]				
Generation 810: Best = 4.133134e+03	FES = 81100	$\mu_F = 0.543$	$\mu_{CR} = 0.046$		
Diversity = 8.0581e+00	k_t = [1, 1, 1, 1]				
Generation 820: Best = 4.133134e+03	FES = 82100	$\mu_F = 0.562$	$\mu_{CR} = 0.066$		
Diversity = 8.0634e+00	k_t = [3, 3, 4, 2]				
Generation 830: Best = 4.131247e+03	FES = 83100	$\mu_F = 0.559$	$\mu_{CR} = 0.048$		
Diversity = 8.0619e+00	k_t = [1, 1, 1, 1]				
Generation 840: Best = 4.131247e+03	FES = 84100	$\mu_F = 0.599$	$\mu_{CR} = 0.048$		
Diversity = 8.0587e+00	k_t = [3, 2, 4, 4]				
Generation 850: Best = 4.131247e+03	FES = 85100	$\mu_F = 0.550$	$\mu_{CR} = 0.048$		
Diversity = 8.0587e+00	k_t = [1, 1, 1, 1]				
Generation 860: Best = 4.116578e+03	FES = 86100	$\mu_F = 0.459$	$\mu_{CR} = 0.052$		
Diversity = 8.0586e+00	k_t = [4, 3, 2, 2]				
Generation 870: Best = 4.108041e+03	FES = 87100	$\mu_F = 0.468$	$\mu_{CR} = 0.054$		
Diversity = 8.0590e+00	k_t = [1, 1, 1, 1]				
Generation 880: Best = 4.099194e+03	FES = 88100	$\mu_F = 0.417$	$\mu_{CR} = 0.062$		
Diversity = 8.0443e+00	k_t = [2, 3, 2, 2]				
Generation 890: Best = 4.090617e+03	FES = 89100	$\mu_F = 0.500$	$\mu_{CR} = 0.048$		
Diversity = 8.0108e+00	k_t = [1, 1, 1, 1]				
Generation 900: Best = 4.088921e+03	FES = 90100	$\mu_F = 0.535$	$\mu_{CR} = 0.052$		
Diversity = 7.8886e+00	k_t = [2, 3, 4, 4]				
Generation 910: Best = 4.077329e+03	FES = 91100	$\mu_F = 0.626$	$\mu_{CR} = 0.049$		
Diversity = 7.8374e+00	k_t = [1, 1, 1, 1]				
Generation 920: Best = 4.043205e+03	FES = 92100	$\mu_F = 0.621$	$\mu_{CR} = 0.046$		
Diversity = 7.8388e+00	k_t = [3, 2, 2, 2]				
Generation 930: Best = 4.043205e+03	FES = 93100	$\mu_F = 0.642$	$\mu_{CR} = 0.040$		
Diversity = 7.8380e+00	k_t = [1, 1, 1, 1]				
Generation 940: Best = 4.042026e+03	FES = 94100	$\mu_F = 0.592$	$\mu_{CR} = 0.050$		
Diversity = 7.8613e+00	k_t = [3, 2, 3, 4]				
Generation 950: Best = 4.042026e+03	FES = 95100	$\mu_F = 0.554$	$\mu_{CR} = 0.039$		
Diversity = 7.8673e+00	k_t = [1, 1, 1, 1]				
Generation 960: Best = 4.042026e+03	FES = 96100	$\mu_F = 0.617$	$\mu_{CR} = 0.044$		
Diversity = 7.8327e+00	k_t = [4, 2, 4, 4]				
Generation 970: Best = 4.042026e+03	FES = 97100	$\mu_F = 0.551$	$\mu_{CR} = 0.043$		
Diversity = 7.8221e+00	k_t = [1, 1, 1, 1]				
Generation 980: Best = 4.042026e+03	FES = 98100	$\mu_F = 0.512$	$\mu_{CR} = 0.050$		
Diversity = 7.7916e+00	k_t = [2, 2, 3, 3]				
Generation 990: Best = 4.036101e+03	FES = 99100	$\mu_F = 0.536$	$\mu_{CR} = 0.040$		
Diversity = 7.7904e+00	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 4.035307e+03	FES = 100100	$\mu_F = 0.563$	$\mu_{CR} = 0.039$		
Diversity = 7.7849e+00	k_t = [3, 4, 4, 2]				
Generation 1010: Best = 4.012093e+03	FES = 101100	$\mu_F = 0.613$	$\mu_{CR} = 0.044$		
Diversity = 7.7872e+00	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 3.999392e+03	FES = 102100	$\mu_F = 0.569$	$\mu_{CR} = 0.053$		
Diversity = 7.7816e+00	k_t = [3, 3, 3, 3]				
Generation 1030: Best = 3.999392e+03	FES = 103100	$\mu_F = 0.529$	$\mu_{CR} = 0.055$		

Diversity = 7.7804e+00	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 3.999392e+03	FES = 104100	$\mu_F = 0.583$	$\mu_{CR} = 0.062$	
Diversity = 7.7637e+00	k_t = [4, 2, 2, 4]			
Generation 1050: Best = 3.993744e+03	FES = 105100	$\mu_F = 0.491$	$\mu_{CR} = 0.047$	
Diversity = 7.7615e+00	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 3.980659e+03	FES = 106100	$\mu_F = 0.499$	$\mu_{CR} = 0.067$	
Diversity = 7.7600e+00	k_t = [4, 2, 3, 2]			
Generation 1070: Best = 3.980399e+03	FES = 107100	$\mu_F = 0.535$	$\mu_{CR} = 0.049$	
Diversity = 7.7605e+00	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 3.974125e+03	FES = 108100	$\mu_F = 0.474$	$\mu_{CR} = 0.057$	
Diversity = 7.7580e+00	k_t = [2, 3, 4, 2]			
Generation 1090: Best = 3.974125e+03	FES = 109100	$\mu_F = 0.502$	$\mu_{CR} = 0.049$	
Diversity = 7.7582e+00	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 3.968046e+03	FES = 110100	$\mu_F = 0.418$	$\mu_{CR} = 0.044$	
Diversity = 7.7572e+00	k_t = [4, 2, 4, 3]			
Generation 1110: Best = 3.968046e+03	FES = 111100	$\mu_F = 0.335$	$\mu_{CR} = 0.048$	
Diversity = 7.7550e+00	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 3.968046e+03	FES = 112100	$\mu_F = 0.391$	$\mu_{CR} = 0.072$	
Diversity = 7.7518e+00	k_t = [2, 2, 2, 2]			
Generation 1130: Best = 3.968046e+03	FES = 113100	$\mu_F = 0.371$	$\mu_{CR} = 0.055$	
Diversity = 7.7502e+00	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 3.968037e+03	FES = 114100	$\mu_F = 0.362$	$\mu_{CR} = 0.056$	
Diversity = 7.7488e+00	k_t = [2, 2, 4, 3]			
Generation 1150: Best = 3.968037e+03	FES = 115100	$\mu_F = 0.354$	$\mu_{CR} = 0.047$	
Diversity = 7.7476e+00	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 3.968037e+03	FES = 116100	$\mu_F = 0.373$	$\mu_{CR} = 0.058$	
Diversity = 7.7476e+00	k_t = [3, 3, 3, 3]			
Generation 1170: Best = 3.967616e+03	FES = 117100	$\mu_F = 0.297$	$\mu_{CR} = 0.083$	
Diversity = 7.7460e+00	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 3.950837e+03	FES = 118100	$\mu_F = 0.308$	$\mu_{CR} = 0.063$	
Diversity = 7.7447e+00	k_t = [2, 4, 2, 2]			
Generation 1190: Best = 3.950837e+03	FES = 119100	$\mu_F = 0.321$	$\mu_{CR} = 0.070$	
Diversity = 7.7474e+00	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 3.950837e+03	FES = 120100	$\mu_F = 0.395$	$\mu_{CR} = 0.067$	
Diversity = 7.7432e+00	k_t = [4, 2, 3, 2]			
Generation 1210: Best = 3.949017e+03	FES = 121100	$\mu_F = 0.433$	$\mu_{CR} = 0.052$	
Diversity = 7.7363e+00	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 3.948277e+03	FES = 122100	$\mu_F = 0.469$	$\mu_{CR} = 0.060$	
Diversity = 7.6916e+00	k_t = [2, 3, 3, 4]			
Generation 1230: Best = 3.937193e+03	FES = 123100	$\mu_F = 0.449$	$\mu_{CR} = 0.059$	
Diversity = 7.6879e+00	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 3.937000e+03	FES = 124100	$\mu_F = 0.441$	$\mu_{CR} = 0.060$	
Diversity = 7.6510e+00	k_t = [4, 2, 3, 4]			
Generation 1250: Best = 3.937000e+03	FES = 125100	$\mu_F = 0.494$	$\mu_{CR} = 0.065$	
Diversity = 7.6505e+00	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 3.936339e+03	FES = 126100	$\mu_F = 0.564$	$\mu_{CR} = 0.068$	
Diversity = 7.6485e+00	k_t = [4, 4, 2, 4]			
Generation 1270: Best = 3.935116e+03	FES = 127100	$\mu_F = 0.468$	$\mu_{CR} = 0.063$	
Diversity = 7.6484e+00	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 3.935116e+03	FES = 128100	$\mu_F = 0.480$	$\mu_{CR} = 0.055$	
Diversity = 7.6576e+00	k_t = [4, 2, 3, 3]			
Generation 1290: Best = 3.935116e+03	FES = 129100	$\mu_F = 0.472$	$\mu_{CR} = 0.046$	
Diversity = 7.6586e+00	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 3.929577e+03	FES = 130100	$\mu_F = 0.548$	$\mu_{CR} = 0.038$	

Diversity = 7.6516e+00	k_t = [2, 4, 3, 4]			
Generation 1310: Best = 3.929577e+03	FES = 131100	$\mu_F = 0.618$	$\mu_{CR} = 0.039$	
Diversity = 7.6504e+00	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 3.929577e+03	FES = 132100	$\mu_F = 0.662$	$\mu_{CR} = 0.057$	
Diversity = 7.6527e+00	k_t = [3, 4, 2, 2]			
Generation 1330: Best = 3.929577e+03	FES = 133100	$\mu_F = 0.658$	$\mu_{CR} = 0.046$	
Diversity = 7.5990e+00	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 3.929577e+03	FES = 134100	$\mu_F = 0.629$	$\mu_{CR} = 0.047$	
Diversity = 7.6178e+00	k_t = [3, 4, 3, 4]			
Generation 1350: Best = 3.903604e+03	FES = 135100	$\mu_F = 0.660$	$\mu_{CR} = 0.062$	
Diversity = 7.6014e+00	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 3.901300e+03	FES = 136100	$\mu_F = 0.698$	$\mu_{CR} = 0.046$	
Diversity = 7.5082e+00	k_t = [4, 3, 3, 4]			
Generation 1370: Best = 3.901300e+03	FES = 137100	$\mu_F = 0.764$	$\mu_{CR} = 0.056$	
Diversity = 7.5164e+00	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 3.901300e+03	FES = 138100	$\mu_F = 0.756$	$\mu_{CR} = 0.049$	
Diversity = 7.5389e+00	k_t = [4, 2, 4, 3]			
Generation 1390: Best = 3.900764e+03	FES = 139100	$\mu_F = 0.651$	$\mu_{CR} = 0.048$	
Diversity = 7.5390e+00	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 3.900764e+03	FES = 140100	$\mu_F = 0.694$	$\mu_{CR} = 0.050$	
Diversity = 7.5466e+00	k_t = [3, 2, 2, 2]			
Generation 1410: Best = 3.900764e+03	FES = 141100	$\mu_F = 0.606$	$\mu_{CR} = 0.042$	
Diversity = 7.4883e+00	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 3.900764e+03	FES = 142100	$\mu_F = 0.654$	$\mu_{CR} = 0.038$	
Diversity = 7.5015e+00	k_t = [2, 3, 2, 4]			
Generation 1430: Best = 3.896479e+03	FES = 143100	$\mu_F = 0.563$	$\mu_{CR} = 0.039$	
Diversity = 7.5127e+00	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 3.896479e+03	FES = 144100	$\mu_F = 0.582$	$\mu_{CR} = 0.044$	
Diversity = 7.5220e+00	k_t = [4, 4, 4, 4]			
Generation 1450: Best = 3.896479e+03	FES = 145100	$\mu_F = 0.548$	$\mu_{CR} = 0.053$	
Diversity = 7.5260e+00	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 3.881473e+03	FES = 146100	$\mu_F = 0.544$	$\mu_{CR} = 0.053$	
Diversity = 7.5058e+00	k_t = [4, 3, 3, 2]			
Generation 1470: Best = 3.881326e+03	FES = 147100	$\mu_F = 0.631$	$\mu_{CR} = 0.044$	
Diversity = 7.5063e+00	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 3.879079e+03	FES = 148100	$\mu_F = 0.574$	$\mu_{CR} = 0.046$	
Diversity = 7.4939e+00	k_t = [4, 3, 3, 2]			
Generation 1490: Best = 3.877945e+03	FES = 149100	$\mu_F = 0.532$	$\mu_{CR} = 0.043$	
Diversity = 7.4421e+00	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 3.877909e+03	FES = 150100	$\mu_F = 0.576$	$\mu_{CR} = 0.049$	
Diversity = 7.4430e+00	k_t = [4, 2, 3, 3]			
Generation 1510: Best = 3.877909e+03	FES = 151100	$\mu_F = 0.580$	$\mu_{CR} = 0.044$	
Diversity = 7.4437e+00	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 3.868896e+03	FES = 152100	$\mu_F = 0.469$	$\mu_{CR} = 0.037$	
Diversity = 7.4425e+00	k_t = [3, 2, 4, 3]			
Generation 1530: Best = 3.868064e+03	FES = 153100	$\mu_F = 0.602$	$\mu_{CR} = 0.040$	
Diversity = 7.4087e+00	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 3.867608e+03	FES = 154100	$\mu_F = 0.638$	$\mu_{CR} = 0.036$	
Diversity = 7.3965e+00	k_t = [3, 2, 3, 2]			
Generation 1550: Best = 3.865539e+03	FES = 155100	$\mu_F = 0.715$	$\mu_{CR} = 0.028$	
Diversity = 7.3700e+00	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 3.865539e+03	FES = 156100	$\mu_F = 0.704$	$\mu_{CR} = 0.035$	
Diversity = 7.3659e+00	k_t = [2, 2, 4, 3]			
Generation 1570: Best = 3.863562e+03	FES = 157100	$\mu_F = 0.647$	$\mu_{CR} = 0.040$	

Diversity = 7.3657e+00	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 3.862404e+03	FES = 158100	$\mu_F = 0.623$	$\mu_{CR} = 0.065$	
Diversity = 7.3766e+00	k_t = [4, 3, 3, 4]			
Generation 1590: Best = 3.862404e+03	FES = 159100	$\mu_F = 0.615$	$\mu_{CR} = 0.047$	
Diversity = 7.3760e+00	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 3.861694e+03	FES = 160100	$\mu_F = 0.559$	$\mu_{CR} = 0.044$	
Diversity = 7.3770e+00	k_t = [2, 2, 4, 2]			
Generation 1610: Best = 3.861694e+03	FES = 161100	$\mu_F = 0.518$	$\mu_{CR} = 0.054$	
Diversity = 7.3868e+00	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 3.861694e+03	FES = 162100	$\mu_F = 0.437$	$\mu_{CR} = 0.061$	
Diversity = 7.3873e+00	k_t = [3, 3, 4, 3]			
Generation 1630: Best = 3.861694e+03	FES = 163100	$\mu_F = 0.375$	$\mu_{CR} = 0.068$	
Diversity = 7.3879e+00	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 3.860705e+03	FES = 164100	$\mu_F = 0.420$	$\mu_{CR} = 0.064$	
Diversity = 7.3862e+00	k_t = [4, 3, 2, 3]			
Generation 1650: Best = 3.860705e+03	FES = 165100	$\mu_F = 0.479$	$\mu_{CR} = 0.062$	
Diversity = 7.3854e+00	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 3.858924e+03	FES = 166100	$\mu_F = 0.449$	$\mu_{CR} = 0.066$	
Diversity = 7.3841e+00	k_t = [4, 4, 4, 3]			
Generation 1670: Best = 3.858924e+03	FES = 167100	$\mu_F = 0.426$	$\mu_{CR} = 0.059$	
Diversity = 7.3836e+00	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 3.858924e+03	FES = 168100	$\mu_F = 0.532$	$\mu_{CR} = 0.066$	
Diversity = 7.3620e+00	k_t = [3, 2, 4, 4]			
Generation 1690: Best = 3.858316e+03	FES = 169100	$\mu_F = 0.527$	$\mu_{CR} = 0.052$	
Diversity = 7.3637e+00	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 3.858316e+03	FES = 170100	$\mu_F = 0.410$	$\mu_{CR} = 0.049$	
Diversity = 7.3408e+00	k_t = [3, 3, 2, 2]			
Generation 1710: Best = 3.858316e+03	FES = 171100	$\mu_F = 0.377$	$\mu_{CR} = 0.057$	
Diversity = 7.3423e+00	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 3.858316e+03	FES = 172100	$\mu_F = 0.324$	$\mu_{CR} = 0.072$	
Diversity = 7.3393e+00	k_t = [2, 3, 3, 3]			
Generation 1730: Best = 3.858316e+03	FES = 173100	$\mu_F = 0.354$	$\mu_{CR} = 0.057$	
Diversity = 7.3407e+00	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 3.858316e+03	FES = 174100	$\mu_F = 0.324$	$\mu_{CR} = 0.061$	
Diversity = 7.3476e+00	k_t = [4, 2, 2, 2]			
Generation 1750: Best = 3.858198e+03	FES = 175100	$\mu_F = 0.334$	$\mu_{CR} = 0.062$	
Diversity = 7.3492e+00	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 3.858198e+03	FES = 176100	$\mu_F = 0.346$	$\mu_{CR} = 0.069$	
Diversity = 7.3448e+00	k_t = [4, 3, 3, 2]			
Generation 1770: Best = 3.858198e+03	FES = 177100	$\mu_F = 0.350$	$\mu_{CR} = 0.067$	
Diversity = 7.3435e+00	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 3.858198e+03	FES = 178100	$\mu_F = 0.415$	$\mu_{CR} = 0.060$	
Diversity = 7.3413e+00	k_t = [4, 2, 4, 3]			
Generation 1790: Best = 3.858198e+03	FES = 179100	$\mu_F = 0.353$	$\mu_{CR} = 0.065$	
Diversity = 7.3395e+00	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 3.856262e+03	FES = 180100	$\mu_F = 0.297$	$\mu_{CR} = 0.055$	
Diversity = 7.3368e+00	k_t = [2, 4, 2, 3]			
Generation 1810: Best = 3.855436e+03	FES = 181100	$\mu_F = 0.271$	$\mu_{CR} = 0.064$	
Diversity = 7.3420e+00	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 3.855436e+03	FES = 182100	$\mu_F = 0.314$	$\mu_{CR} = 0.067$	
Diversity = 7.3016e+00	k_t = [3, 4, 4, 4]			
Generation 1830: Best = 3.855436e+03	FES = 183100	$\mu_F = 0.316$	$\mu_{CR} = 0.067$	
Diversity = 7.3027e+00	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 3.855436e+03	FES = 184100	$\mu_F = 0.306$	$\mu_{CR} = 0.067$	

Diversity = 7.2984e+00	k_t = [3, 3, 4, 4]			
Generation 1850: Best = 3.855434e+03	FES = 185100	$\mu_F = 0.305$	$\mu_{CR} = 0.069$	
Diversity = 7.2616e+00	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 3.855434e+03	FES = 186100	$\mu_F = 0.401$	$\mu_{CR} = 0.065$	
Diversity = 7.2586e+00	k_t = [3, 4, 2, 2]			
Generation 1870: Best = 3.855434e+03	FES = 187100	$\mu_F = 0.446$	$\mu_{CR} = 0.059$	
Diversity = 7.2558e+00	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 3.855434e+03	FES = 188100	$\mu_F = 0.495$	$\mu_{CR} = 0.052$	
Diversity = 7.2556e+00	k_t = [2, 3, 4, 4]			
Generation 1890: Best = 3.855434e+03	FES = 189100	$\mu_F = 0.443$	$\mu_{CR} = 0.049$	
Diversity = 7.2563e+00	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 3.855434e+03	FES = 190100	$\mu_F = 0.383$	$\mu_{CR} = 0.044$	
Diversity = 7.2596e+00	k_t = [2, 4, 3, 2]			
Generation 1910: Best = 3.855434e+03	FES = 191100	$\mu_F = 0.407$	$\mu_{CR} = 0.045$	
Diversity = 7.2592e+00	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 3.850089e+03	FES = 192100	$\mu_F = 0.341$	$\mu_{CR} = 0.061$	
Diversity = 7.2812e+00	k_t = [3, 4, 3, 4]			
Generation 1930: Best = 3.850014e+03	FES = 193100	$\mu_F = 0.273$	$\mu_{CR} = 0.071$	
Diversity = 7.2815e+00	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 3.850014e+03	FES = 194100	$\mu_F = 0.257$	$\mu_{CR} = 0.064$	
Diversity = 7.2846e+00	k_t = [2, 3, 3, 2]			
Generation 1950: Best = 3.849549e+03	FES = 195100	$\mu_F = 0.348$	$\mu_{CR} = 0.062$	
Diversity = 7.2840e+00	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 3.849549e+03	FES = 196100	$\mu_F = 0.436$	$\mu_{CR} = 0.065$	
Diversity = 7.2312e+00	k_t = [4, 4, 2, 4]			
Generation 1970: Best = 3.849549e+03	FES = 197100	$\mu_F = 0.433$	$\mu_{CR} = 0.073$	
Diversity = 7.2358e+00	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 3.849549e+03	FES = 198100	$\mu_F = 0.414$	$\mu_{CR} = 0.080$	
Diversity = 7.2369e+00	k_t = [4, 4, 4, 2]			
Generation 1990: Best = 3.849549e+03	FES = 199100	$\mu_F = 0.378$	$\mu_{CR} = 0.073$	
Diversity = 7.2345e+00	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 3.849549e+03	FES = 200100	$\mu_F = 0.425$	$\mu_{CR} = 0.069$	
Diversity = 7.2484e+00	k_t = [3, 4, 2, 4]			
Generation 2010: Best = 3.849549e+03	FES = 201100	$\mu_F = 0.372$	$\mu_{CR} = 0.053$	
Diversity = 7.2456e+00	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 3.849549e+03	FES = 202100	$\mu_F = 0.410$	$\mu_{CR} = 0.055$	
Diversity = 7.2370e+00	k_t = [3, 4, 3, 3]			
Generation 2030: Best = 3.849549e+03	FES = 203100	$\mu_F = 0.360$	$\mu_{CR} = 0.044$	
Diversity = 7.2404e+00	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 3.849507e+03	FES = 204100	$\mu_F = 0.344$	$\mu_{CR} = 0.043$	
Diversity = 7.2390e+00	k_t = [4, 2, 2, 3]			
Generation 2050: Best = 3.849507e+03	FES = 205100	$\mu_F = 0.318$	$\mu_{CR} = 0.042$	
Diversity = 7.2397e+00	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 3.847920e+03	FES = 206100	$\mu_F = 0.341$	$\mu_{CR} = 0.053$	
Diversity = 7.2395e+00	k_t = [3, 4, 3, 3]			
Generation 2070: Best = 3.847920e+03	FES = 207100	$\mu_F = 0.291$	$\mu_{CR} = 0.069$	
Diversity = 7.2402e+00	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 3.847920e+03	FES = 208100	$\mu_F = 0.237$	$\mu_{CR} = 0.071$	
Diversity = 7.2393e+00	k_t = [4, 2, 4, 2]			
Generation 2090: Best = 3.847920e+03	FES = 209100	$\mu_F = 0.234$	$\mu_{CR} = 0.066$	
Diversity = 7.2406e+00	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 3.847920e+03	FES = 210100	$\mu_F = 0.280$	$\mu_{CR} = 0.068$	
Diversity = 7.2373e+00	k_t = [3, 3, 3, 3]			
Generation 2110: Best = 3.847840e+03	FES = 211100	$\mu_F = 0.282$	$\mu_{CR} = 0.063$	

Diversity = 7.2372e+00	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 3.847840e+03	FES = 212100	$\mu_F = 0.295$	$\mu_{CR} = 0.046$	
Diversity = 7.2288e+00	k_t = [2, 4, 2, 2]			
Generation 2130: Best = 3.847497e+03	FES = 213100	$\mu_F = 0.259$	$\mu_{CR} = 0.052$	
Diversity = 7.2293e+00	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 3.847383e+03	FES = 214100	$\mu_F = 0.267$	$\mu_{CR} = 0.069$	
Diversity = 7.2289e+00	k_t = [3, 4, 4, 3]			
Generation 2150: Best = 3.847383e+03	FES = 215100	$\mu_F = 0.332$	$\mu_{CR} = 0.048$	
Diversity = 7.1887e+00	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 3.847383e+03	FES = 216100	$\mu_F = 0.338$	$\mu_{CR} = 0.062$	
Diversity = 7.1849e+00	k_t = [4, 3, 3, 3]			
Generation 2170: Best = 3.847383e+03	FES = 217100	$\mu_F = 0.309$	$\mu_{CR} = 0.057$	
Diversity = 7.1850e+00	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 3.847267e+03	FES = 218100	$\mu_F = 0.368$	$\mu_{CR} = 0.065$	
Diversity = 7.1882e+00	k_t = [4, 2, 2, 3]			
Generation 2190: Best = 3.847267e+03	FES = 219100	$\mu_F = 0.407$	$\mu_{CR} = 0.049$	
Diversity = 7.1897e+00	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 3.847007e+03	FES = 220100	$\mu_F = 0.376$	$\mu_{CR} = 0.051$	
Diversity = 7.1890e+00	k_t = [2, 3, 3, 3]			
Generation 2210: Best = 3.847007e+03	FES = 221100	$\mu_F = 0.458$	$\mu_{CR} = 0.048$	
Diversity = 7.1881e+00	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 3.846870e+03	FES = 222100	$\mu_F = 0.473$	$\mu_{CR} = 0.044$	
Diversity = 7.1636e+00	k_t = [4, 4, 4, 2]			
Generation 2230: Best = 3.846870e+03	FES = 223100	$\mu_F = 0.557$	$\mu_{CR} = 0.044$	
Diversity = 7.1630e+00	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 3.846870e+03	FES = 224100	$\mu_F = 0.521$	$\mu_{CR} = 0.041$	
Diversity = 7.1630e+00	k_t = [3, 2, 2, 2]			
Generation 2250: Best = 3.846870e+03	FES = 225100	$\mu_F = 0.482$	$\mu_{CR} = 0.047$	
Diversity = 7.1625e+00	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 3.846870e+03	FES = 226100	$\mu_F = 0.567$	$\mu_{CR} = 0.055$	
Diversity = 7.1222e+00	k_t = [3, 3, 2, 2]			
Generation 2270: Best = 3.846870e+03	FES = 227100	$\mu_F = 0.522$	$\mu_{CR} = 0.059$	
Diversity = 7.1231e+00	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 3.846820e+03	FES = 228100	$\mu_F = 0.548$	$\mu_{CR} = 0.056$	
Diversity = 7.1121e+00	k_t = [2, 4, 2, 3]			
Generation 2290: Best = 3.846261e+03	FES = 229100	$\mu_F = 0.462$	$\mu_{CR} = 0.063$	
Diversity = 7.1111e+00	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 3.846261e+03	FES = 230100	$\mu_F = 0.491$	$\mu_{CR} = 0.058$	
Diversity = 7.1117e+00	k_t = [4, 4, 2, 2]			
Generation 2310: Best = 3.844921e+03	FES = 231100	$\mu_F = 0.552$	$\mu_{CR} = 0.070$	
Diversity = 7.0927e+00	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 3.844921e+03	FES = 232100	$\mu_F = 0.553$	$\mu_{CR} = 0.056$	
Diversity = 7.0929e+00	k_t = [3, 3, 2, 3]			
Generation 2330: Best = 3.844921e+03	FES = 233100	$\mu_F = 0.538$	$\mu_{CR} = 0.052$	
Diversity = 7.0594e+00	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 3.843352e+03	FES = 234100	$\mu_F = 0.606$	$\mu_{CR} = 0.053$	
Diversity = 7.0335e+00	k_t = [4, 3, 2, 4]			
Generation 2350: Best = 3.843352e+03	FES = 235100	$\mu_F = 0.491$	$\mu_{CR} = 0.058$	
Diversity = 7.0330e+00	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 3.843352e+03	FES = 236100	$\mu_F = 0.554$	$\mu_{CR} = 0.049$	
Diversity = 7.0126e+00	k_t = [2, 3, 4, 3]			
Generation 2370: Best = 3.843352e+03	FES = 237100	$\mu_F = 0.526$	$\mu_{CR} = 0.071$	
Diversity = 7.0138e+00	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 3.843352e+03	FES = 238100	$\mu_F = 0.562$	$\mu_{CR} = 0.080$	

Diversity = 6.9853e+00	k_t = [3, 3, 3, 2]			
Generation 2390: Best = 3.841270e+03	FES = 239100	$\mu_F = 0.599$	$\mu_{CR} = 0.071$	
Diversity = 6.9865e+00	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 3.841185e+03	FES = 240100	$\mu_F = 0.619$	$\mu_{CR} = 0.068$	
Diversity = 6.8526e+00	k_t = [3, 3, 4, 4]			
Generation 2410: Best = 3.840742e+03	FES = 241100	$\mu_F = 0.616$	$\mu_{CR} = 0.084$	
Diversity = 6.8187e+00	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 3.837815e+03	FES = 242100	$\mu_F = 0.573$	$\mu_{CR} = 0.129$	
Diversity = 6.6560e+00	k_t = [4, 3, 3, 4]			
Generation 2430: Best = 3.828458e+03	FES = 243100	$\mu_F = 0.559$	$\mu_{CR} = 0.134$	
Diversity = 6.6630e+00	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 3.822834e+03	FES = 244100	$\mu_F = 0.514$	$\mu_{CR} = 0.127$	
Diversity = 6.6628e+00	k_t = [4, 2, 2, 4]			
Generation 2450: Best = 3.822834e+03	FES = 245100	$\mu_F = 0.539$	$\mu_{CR} = 0.113$	
Diversity = 6.5966e+00	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 3.821631e+03	FES = 246100	$\mu_F = 0.562$	$\mu_{CR} = 0.114$	
Diversity = 6.5995e+00	k_t = [2, 2, 2, 2]			
Generation 2470: Best = 3.821439e+03	FES = 247100	$\mu_F = 0.534$	$\mu_{CR} = 0.095$	
Diversity = 6.5617e+00	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 3.819939e+03	FES = 248100	$\mu_F = 0.606$	$\mu_{CR} = 0.106$	
Diversity = 6.5638e+00	k_t = [3, 4, 3, 2]			
Generation 2490: Best = 3.819939e+03	FES = 249100	$\mu_F = 0.647$	$\mu_{CR} = 0.098$	
Diversity = 6.4955e+00	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 3.813881e+03	FES = 250100	$\mu_F = 0.679$	$\mu_{CR} = 0.119$	
Diversity = 6.4986e+00	k_t = [3, 3, 3, 3]			
Generation 2510: Best = 3.813688e+03	FES = 251100	$\mu_F = 0.721$	$\mu_{CR} = 0.108$	
Diversity = 6.4951e+00	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 3.813688e+03	FES = 252100	$\mu_F = 0.749$	$\mu_{CR} = 0.104$	
Diversity = 6.4964e+00	k_t = [4, 4, 4, 2]			
Generation 2530: Best = 3.813688e+03	FES = 253100	$\mu_F = 0.743$	$\mu_{CR} = 0.106$	
Diversity = 6.4625e+00	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 3.812791e+03	FES = 254100	$\mu_F = 0.754$	$\mu_{CR} = 0.107$	
Diversity = 6.4182e+00	k_t = [3, 2, 3, 3]			
Generation 2550: Best = 3.812325e+03	FES = 255100	$\mu_F = 0.774$	$\mu_{CR} = 0.079$	
Diversity = 6.3480e+00	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 3.811772e+03	FES = 256100	$\mu_F = 0.776$	$\mu_{CR} = 0.093$	
Diversity = 6.2552e+00	k_t = [2, 4, 3, 2]			
Generation 2570: Best = 3.811717e+03	FES = 257100	$\mu_F = 0.808$	$\mu_{CR} = 0.094$	
Diversity = 6.0936e+00	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 3.811642e+03	FES = 258100	$\mu_F = 0.824$	$\mu_{CR} = 0.093$	
Diversity = 6.0976e+00	k_t = [4, 4, 3, 3]			
Generation 2590: Best = 3.811642e+03	FES = 259100	$\mu_F = 0.820$	$\mu_{CR} = 0.090$	
Diversity = 5.9945e+00	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 3.811489e+03	FES = 260100	$\mu_F = 0.838$	$\mu_{CR} = 0.105$	
Diversity = 5.8734e+00	k_t = [2, 4, 3, 2]			
Generation 2610: Best = 3.811305e+03	FES = 261100	$\mu_F = 0.850$	$\mu_{CR} = 0.105$	
Diversity = 5.7769e+00	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 3.811254e+03	FES = 262100	$\mu_F = 0.859$	$\mu_{CR} = 0.128$	
Diversity = 5.1786e+00	k_t = [2, 4, 2, 3]			
Generation 2630: Best = 3.811210e+03	FES = 263100	$\mu_F = 0.837$	$\mu_{CR} = 0.136$	
Diversity = 4.3612e+00	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 3.810879e+03	FES = 264100	$\mu_F = 0.859$	$\mu_{CR} = 0.134$	
Diversity = 2.0064e+00	k_t = [2, 3, 4, 4]			
Generation 2650: Best = 3.810872e+03	FES = 265100	$\mu_F = 0.845$	$\mu_{CR} = 0.123$	

Diversity = 1.6304e+00	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 3.810825e+03	FES = 266100	$\mu_F = 0.840$	$\mu_{CR} = 0.120$	
Diversity = 4.4706e-01	k_t = [2, 3, 4, 3]			
Generation 2670: Best = 3.810380e+03	FES = 267100	$\mu_F = 0.817$	$\mu_{CR} = 0.130$	
Diversity = 4.8843e-01	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 3.808481e+03	FES = 268100	$\mu_F = 0.829$	$\mu_{CR} = 0.135$	
Diversity = 4.6133e-01	k_t = [4, 3, 4, 2]			
Generation 2690: Best = 3.808003e+03	FES = 269100	$\mu_F = 0.812$	$\mu_{CR} = 0.143$	
Diversity = 4.2196e-01	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 3.807923e+03	FES = 270100	$\mu_F = 0.797$	$\mu_{CR} = 0.139$	
Diversity = 3.6963e-01	k_t = [3, 2, 3, 3]			
Generation 2710: Best = 3.807759e+03	FES = 271100	$\mu_F = 0.792$	$\mu_{CR} = 0.138$	
Diversity = 3.3343e-01	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 3.807587e+03	FES = 272100	$\mu_F = 0.771$	$\mu_{CR} = 0.134$	
Diversity = 3.1214e-01	k_t = [2, 2, 2, 3]			
Generation 2730: Best = 3.807513e+03	FES = 273100	$\mu_F = 0.764$	$\mu_{CR} = 0.126$	
Diversity = 3.2350e-01	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 3.807365e+03	FES = 274100	$\mu_F = 0.768$	$\mu_{CR} = 0.131$	
Diversity = 3.1813e-01	k_t = [3, 2, 4, 2]			
Generation 2750: Best = 3.807306e+03	FES = 275100	$\mu_F = 0.771$	$\mu_{CR} = 0.134$	
Diversity = 3.2043e-01	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 3.807224e+03	FES = 276100	$\mu_F = 0.780$	$\mu_{CR} = 0.137$	
Diversity = 3.1042e-01	k_t = [3, 2, 4, 2]			
Generation 2770: Best = 3.807192e+03	FES = 277100	$\mu_F = 0.791$	$\mu_{CR} = 0.133$	
Diversity = 3.0235e-01	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 3.807149e+03	FES = 278100	$\mu_F = 0.785$	$\mu_{CR} = 0.132$	
Diversity = 2.8775e-01	k_t = [4, 3, 3, 2]			
Generation 2790: Best = 3.807129e+03	FES = 279100	$\mu_F = 0.797$	$\mu_{CR} = 0.129$	
Diversity = 2.8114e-01	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 3.807115e+03	FES = 280100	$\mu_F = 0.785$	$\mu_{CR} = 0.126$	
Diversity = 2.6883e-01	k_t = [2, 3, 2, 4]			
Generation 2810: Best = 3.807089e+03	FES = 281100	$\mu_F = 0.787$	$\mu_{CR} = 0.145$	
Diversity = 2.2299e-01	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 3.807086e+03	FES = 282100	$\mu_F = 0.783$	$\mu_{CR} = 0.156$	
Diversity = 2.0070e-01	k_t = [4, 3, 2, 4]			
Generation 2830: Best = 3.807079e+03	FES = 283100	$\mu_F = 0.787$	$\mu_{CR} = 0.165$	
Diversity = 1.2452e-01	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 3.807074e+03	FES = 284100	$\mu_F = 0.799$	$\mu_{CR} = 0.174$	
Diversity = 4.3333e-02	k_t = [4, 3, 2, 3]			
Generation 2850: Best = 3.807072e+03	FES = 285100	$\mu_F = 0.806$	$\mu_{CR} = 0.173$	
Diversity = 1.5056e-02	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 3.807068e+03	FES = 286100	$\mu_F = 0.800$	$\mu_{CR} = 0.178$	
Diversity = 7.1715e-03	k_t = [3, 4, 4, 3]			
Generation 2870: Best = 3.807067e+03	FES = 287100	$\mu_F = 0.786$	$\mu_{CR} = 0.172$	
Diversity = 5.2847e-03	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 3.807065e+03	FES = 288100	$\mu_F = 0.765$	$\mu_{CR} = 0.178$	
Diversity = 3.9502e-03	k_t = [2, 4, 3, 3]			
Generation 2890: Best = 3.807064e+03	FES = 289100	$\mu_F = 0.765$	$\mu_{CR} = 0.182$	
Diversity = 2.9532e-03	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 3.807064e+03	FES = 290100	$\mu_F = 0.754$	$\mu_{CR} = 0.187$	
Diversity = 2.0368e-03	k_t = [4, 2, 3, 4]			
Generation 2910: Best = 3.807064e+03	FES = 291100	$\mu_F = 0.760$	$\mu_{CR} = 0.189$	
Diversity = 1.6994e-03	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 3.807064e+03	FES = 292100	$\mu_F = 0.760$	$\mu_{CR} = 0.181$	

```

Diversity = 1.2374e-03 k_t = [2, 2, 2, 3]
Generation 2930: Best = 3.807064e+03 FES = 293100  $\mu_F$  = 0.755  $\mu_{CR}$  = 0.189
Diversity = 9.4744e-04 k_t = [1, 1, 1, 1]
Generation 2940: Best = 3.807064e+03 FES = 294100  $\mu_F$  = 0.753  $\mu_{CR}$  = 0.195
Diversity = 7.8281e-04 k_t = [3, 3, 2, 2]
Generation 2950: Best = 3.807063e+03 FES = 295100  $\mu_F$  = 0.742  $\mu_{CR}$  = 0.206
Diversity = 6.0602e-04 k_t = [1, 1, 1, 1]
Generation 2960: Best = 3.807063e+03 FES = 296100  $\mu_F$  = 0.743  $\mu_{CR}$  = 0.208
Diversity = 3.9547e-04 k_t = [4, 4, 4, 3]
Generation 2970: Best = 3.807063e+03 FES = 297100  $\mu_F$  = 0.750  $\mu_{CR}$  = 0.199
Diversity = 3.2922e-04 k_t = [1, 1, 1, 1]
Generation 2980: Best = 3.807063e+03 FES = 298100  $\mu_F$  = 0.748  $\mu_{CR}$  = 0.204
Diversity = 2.4474e-04 k_t = [3, 3, 3, 2]
Generation 2990: Best = 3.807063e+03 FES = 299100  $\mu_F$  = 0.731  $\mu_{CR}$  = 0.207
Diversity = 1.6431e-04 k_t = [1, 1, 1, 1]

```

Optimization complete.

```

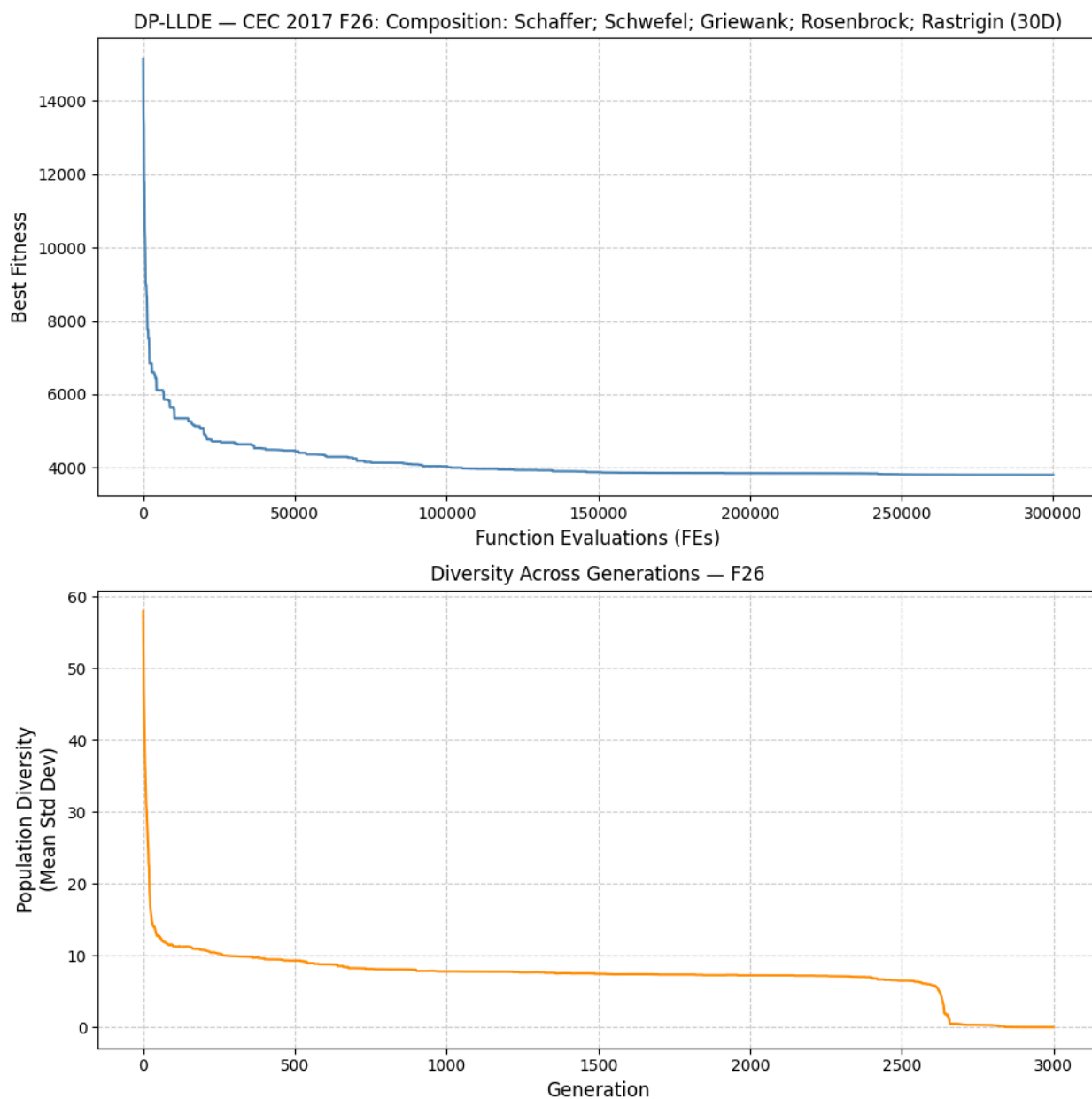
Best Fitness          = 3.807063e+03
Total Function Evals  = 300000
Total Successful Updates = 22497

```

```

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=
Results:
  Best Fitness          : 3.807063e+03
  Global Optimum        : 2.600000e+03
  Error (|best-optimum|) : 1.207063e+03
  Best Solution (5 dim) : [ 5.44244677 -72.14260577 -23.02607607 -19.3243763
-56.9879274 ]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F26.png

```
In [ ]: # =====
# F27 — Composition Function 7 (HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic)
# =====
run_and_plot('F27', make_wrapper(composition.f27), 2700.0, 'Composition: HGBat')
```

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DP-LLDE on CEC 2017 F27 – Single Run

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Function : Composition: HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer (30D)

Bounds : [-100, 100]

Global optimum : 2700.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 4.906147e+03

Generation 10: Best = 3.637261e+03 FES = 1100 $\mu_F = 0.588$ $\mu_{CR} = 0.361$

Diversity = 4.3437e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 3.461336e+03 FES = 2100 $\mu_F = 0.646$ $\mu_{CR} = 0.364$

Diversity = 2.9707e+01 $k_t = [4, 2, 4, 4]$

Generation 30: Best = 3.354980e+03 FES = 3100 $\mu_F = 0.672$ $\mu_{CR} = 0.349$

Diversity = 2.3959e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 3.327014e+03 FES = 4100 $\mu_F = 0.702$ $\mu_{CR} = 0.348$

Diversity = 1.9799e+01 $k_t = [4, 2, 4, 4]$

Generation 50: Best = 3.295557e+03 FES = 5100 $\mu_F = 0.732$ $\mu_{CR} = 0.337$

Diversity = 1.6901e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 3.277412e+03 FES = 6100 $\mu_F = 0.743$ $\mu_{CR} = 0.336$

Diversity = 1.3984e+01 $k_t = [3, 3, 2, 4]$

Generation 70: Best = 3.253560e+03 FES = 7100 $\mu_F = 0.759$ $\mu_{CR} = 0.336$

Diversity = 1.2289e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 3.239423e+03 FES = 8100 $\mu_F = 0.774$ $\mu_{CR} = 0.317$

Diversity = 1.1322e+01 $k_t = [2, 4, 4, 4]$

Generation 90: Best = 3.239423e+03 FES = 9100 $\mu_F = 0.781$ $\mu_{CR} = 0.303$

Diversity = 1.0657e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 3.233576e+03 FES = 10100 $\mu_F = 0.779$ $\mu_{CR} = 0.261$

Diversity = 1.0052e+01 $k_t = [4, 3, 4, 2]$

Generation 110: Best = 3.231428e+03 FES = 11100 $\mu_F = 0.766$ $\mu_{CR} = 0.242$

Diversity = 9.6849e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 3.225371e+03 FES = 12100 $\mu_F = 0.794$ $\mu_{CR} = 0.211$

Diversity = 9.4258e+00 $k_t = [4, 2, 3, 3]$

Generation 130: Best = 3.225371e+03 FES = 13100 $\mu_F = 0.785$ $\mu_{CR} = 0.165$

Diversity = 8.9921e+00 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 3.221896e+03 FES = 14100 $\mu_F = 0.775$ $\mu_{CR} = 0.130$

Diversity = 8.8394e+00 $k_t = [2, 4, 2, 2]$

Generation 150: Best = 3.221233e+03 FES = 15100 $\mu_F = 0.789$ $\mu_{CR} = 0.110$

Diversity = 8.9216e+00 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 3.220010e+03 FES = 16100 $\mu_F = 0.779$ $\mu_{CR} = 0.097$

Diversity = 8.7631e+00 $k_t = [4, 4, 3, 3]$

Generation 170: Best = 3.214224e+03 FES = 17100 $\mu_F = 0.791$ $\mu_{CR} = 0.087$

Diversity = 8.7303e+00 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 3.213657e+03 FES = 18100 $\mu_F = 0.764$ $\mu_{CR} = 0.081$

Diversity = 8.6952e+00 $k_t = [3, 4, 4, 2]$

Generation 190: Best = 3.213012e+03 FES = 19100 $\mu_F = 0.752$ $\mu_{CR} = 0.080$

Diversity = 8.6083e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 3.213012e+03 FES = 20100 $\mu_F = 0.764$ $\mu_{CR} = 0.087$

Diversity = 8.5338e+00 $k_t = [2, 3, 3, 2]$

Generation 210: Best = 3.213012e+03 FES = 21100 $\mu_F = 0.784$ $\mu_{CR} = 0.085$

Diversity = 8.3412e+00 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 3.209852e+03 FES = 22100 $\mu_F = 0.778$ $\mu_{CR} = 0.085$

Diversity = 8.2170e+00	k_t = [2, 3, 3, 4]				
Generation 230: Best = 3.209647e+03	FES = 23100	$\mu_F = 0.789$	$\mu_{CR} = 0.073$		
Diversity = 8.2218e+00	k_t = [1, 1, 1, 1]				
Generation 240: Best = 3.208754e+03	FES = 24100	$\mu_F = 0.806$	$\mu_{CR} = 0.070$		
Diversity = 8.1331e+00	k_t = [4, 2, 3, 4]				
Generation 250: Best = 3.208415e+03	FES = 25100	$\mu_F = 0.819$	$\mu_{CR} = 0.065$		
Diversity = 8.0989e+00	k_t = [1, 1, 1, 1]				
Generation 260: Best = 3.208333e+03	FES = 26100	$\mu_F = 0.836$	$\mu_{CR} = 0.066$		
Diversity = 7.5894e+00	k_t = [2, 4, 4, 4]				
Generation 270: Best = 3.208333e+03	FES = 27100	$\mu_F = 0.865$	$\mu_{CR} = 0.064$		
Diversity = 7.3563e+00	k_t = [1, 1, 1, 1]				
Generation 280: Best = 3.208256e+03	FES = 28100	$\mu_F = 0.836$	$\mu_{CR} = 0.082$		
Diversity = 7.2675e+00	k_t = [2, 3, 2, 2]				
Generation 290: Best = 3.208031e+03	FES = 29100	$\mu_F = 0.831$	$\mu_{CR} = 0.085$		
Diversity = 6.9631e+00	k_t = [1, 1, 1, 1]				
Generation 300: Best = 3.206266e+03	FES = 30100	$\mu_F = 0.820$	$\mu_{CR} = 0.081$		
Diversity = 6.4310e+00	k_t = [4, 3, 3, 3]				
Generation 310: Best = 3.205798e+03	FES = 31100	$\mu_F = 0.810$	$\mu_{CR} = 0.085$		
Diversity = 5.4998e+00	k_t = [1, 1, 1, 1]				
Generation 320: Best = 3.205220e+03	FES = 32100	$\mu_F = 0.788$	$\mu_{CR} = 0.095$		
Diversity = 4.8243e+00	k_t = [4, 3, 2, 4]				
Generation 330: Best = 3.203560e+03	FES = 33100	$\mu_F = 0.785$	$\mu_{CR} = 0.110$		
Diversity = 3.7071e+00	k_t = [1, 1, 1, 1]				
Generation 340: Best = 3.203381e+03	FES = 34100	$\mu_F = 0.776$	$\mu_{CR} = 0.116$		
Diversity = 2.5668e+00	k_t = [2, 2, 4, 2]				
Generation 350: Best = 3.202414e+03	FES = 35100	$\mu_F = 0.770$	$\mu_{CR} = 0.121$		
Diversity = 2.2225e+00	k_t = [1, 1, 1, 1]				
Generation 360: Best = 3.202240e+03	FES = 36100	$\mu_F = 0.756$	$\mu_{CR} = 0.118$		
Diversity = 1.5180e+00	k_t = [3, 4, 2, 3]				
Generation 370: Best = 3.201759e+03	FES = 37100	$\mu_F = 0.748$	$\mu_{CR} = 0.117$		
Diversity = 1.2602e+00	k_t = [1, 1, 1, 1]				
Generation 380: Best = 3.200552e+03	FES = 38100	$\mu_F = 0.752$	$\mu_{CR} = 0.122$		
Diversity = 9.1190e-01	k_t = [2, 3, 4, 4]				
Generation 390: Best = 3.200506e+03	FES = 39100	$\mu_F = 0.761$	$\mu_{CR} = 0.119$		
Diversity = 7.4094e-01	k_t = [1, 1, 1, 1]				
Generation 400: Best = 3.200272e+03	FES = 40100	$\mu_F = 0.750$	$\mu_{CR} = 0.112$		
Diversity = 7.1830e-01	k_t = [2, 3, 4, 3]				
Generation 410: Best = 3.200021e+03	FES = 41100	$\mu_F = 0.737$	$\mu_{CR} = 0.104$		
Diversity = 6.9523e-01	k_t = [1, 1, 1, 1]				
Generation 420: Best = 3.199891e+03	FES = 42100	$\mu_F = 0.709$	$\mu_{CR} = 0.105$		
Diversity = 6.6349e-01	k_t = [4, 2, 2, 2]				
Generation 430: Best = 3.199762e+03	FES = 43100	$\mu_F = 0.709$	$\mu_{CR} = 0.097$		
Diversity = 6.4502e-01	k_t = [1, 1, 1, 1]				
Generation 440: Best = 3.199676e+03	FES = 44100	$\mu_F = 0.719$	$\mu_{CR} = 0.101$		
Diversity = 5.8552e-01	k_t = [4, 3, 3, 4]				
Generation 450: Best = 3.199550e+03	FES = 45100	$\mu_F = 0.704$	$\mu_{CR} = 0.103$		
Diversity = 3.8134e-01	k_t = [1, 1, 1, 1]				
Generation 460: Best = 3.199389e+03	FES = 46100	$\mu_F = 0.682$	$\mu_{CR} = 0.107$		
Diversity = 3.1221e-01	k_t = [3, 4, 4, 2]				
Generation 470: Best = 3.199219e+03	FES = 47100	$\mu_F = 0.689$	$\mu_{CR} = 0.106$		
Diversity = 2.4264e-01	k_t = [1, 1, 1, 1]				
Generation 480: Best = 3.199188e+03	FES = 48100	$\mu_F = 0.676$	$\mu_{CR} = 0.104$		
Diversity = 1.8835e-01	k_t = [2, 4, 4, 3]				
Generation 490: Best = 3.199081e+03	FES = 49100	$\mu_F = 0.678$	$\mu_{CR} = 0.108$		

Diversity = 1.5507e-01	k_t = [1, 1, 1, 1]					
Generation 500: Best = 3.199069e+03	FES = 50100	$\mu_F = 0.685$	$\mu_{CR} = 0.101$			
Diversity = 1.2441e-01	k_t = [2, 2, 2, 3]					
Generation 510: Best = 3.199060e+03	FES = 51100	$\mu_F = 0.696$	$\mu_{CR} = 0.088$			
Diversity = 1.0975e-01	k_t = [1, 1, 1, 1]					
Generation 520: Best = 3.199042e+03	FES = 52100	$\mu_F = 0.712$	$\mu_{CR} = 0.081$			
Diversity = 8.9455e-02	k_t = [2, 2, 3, 3]					
Generation 530: Best = 3.199024e+03	FES = 53100	$\mu_F = 0.700$	$\mu_{CR} = 0.073$			
Diversity = 8.0799e-02	k_t = [1, 1, 1, 1]					
Generation 540: Best = 3.199007e+03	FES = 54100	$\mu_F = 0.702$	$\mu_{CR} = 0.074$			
Diversity = 7.5791e-02	k_t = [4, 2, 3, 2]					
Generation 550: Best = 3.198992e+03	FES = 55100	$\mu_F = 0.671$	$\mu_{CR} = 0.082$			
Diversity = 6.8947e-02	k_t = [1, 1, 1, 1]					
Generation 560: Best = 3.198989e+03	FES = 56100	$\mu_F = 0.677$	$\mu_{CR} = 0.083$			
Diversity = 5.6360e-02	k_t = [2, 4, 2, 4]					
Generation 570: Best = 3.198986e+03	FES = 57100	$\mu_F = 0.666$	$\mu_{CR} = 0.080$			
Diversity = 4.9642e-02	k_t = [1, 1, 1, 1]					
Generation 580: Best = 3.198981e+03	FES = 58100	$\mu_F = 0.675$	$\mu_{CR} = 0.080$			
Diversity = 4.4480e-02	k_t = [3, 2, 2, 3]					
Generation 590: Best = 3.198976e+03	FES = 59100	$\mu_F = 0.700$	$\mu_{CR} = 0.078$			
Diversity = 4.2847e-02	k_t = [1, 1, 1, 1]					
Generation 600: Best = 3.198971e+03	FES = 60100	$\mu_F = 0.691$	$\mu_{CR} = 0.075$			
Diversity = 3.9688e-02	k_t = [4, 2, 4, 2]					
Generation 610: Best = 3.198967e+03	FES = 61100	$\mu_F = 0.712$	$\mu_{CR} = 0.072$			
Diversity = 3.6634e-02	k_t = [1, 1, 1, 1]					
Generation 620: Best = 3.198963e+03	FES = 62100	$\mu_F = 0.712$	$\mu_{CR} = 0.077$			
Diversity = 3.4916e-02	k_t = [2, 2, 2, 3]					
Generation 630: Best = 3.198961e+03	FES = 63100	$\mu_F = 0.704$	$\mu_{CR} = 0.087$			
Diversity = 3.2578e-02	k_t = [1, 1, 1, 1]					
Generation 640: Best = 3.198954e+03	FES = 64100	$\mu_F = 0.712$	$\mu_{CR} = 0.082$			
Diversity = 2.8482e-02	k_t = [4, 3, 3, 3]					
Generation 650: Best = 3.198950e+03	FES = 65100	$\mu_F = 0.711$	$\mu_{CR} = 0.077$			
Diversity = 2.7642e-02	k_t = [1, 1, 1, 1]					
Generation 660: Best = 3.198937e+03	FES = 66100	$\mu_F = 0.730$	$\mu_{CR} = 0.081$			
Diversity = 3.0315e-02	k_t = [2, 4, 3, 3]					
Generation 670: Best = 3.198933e+03	FES = 67100	$\mu_F = 0.741$	$\mu_{CR} = 0.076$			
Diversity = 3.4424e-02	k_t = [1, 1, 1, 1]					
Generation 680: Best = 3.198926e+03	FES = 68100	$\mu_F = 0.756$	$\mu_{CR} = 0.073$			
Diversity = 3.3028e-02	k_t = [2, 4, 2, 2]					
Generation 690: Best = 3.198921e+03	FES = 69100	$\mu_F = 0.746$	$\mu_{CR} = 0.080$			
Diversity = 3.1740e-02	k_t = [1, 1, 1, 1]					
Generation 700: Best = 3.198920e+03	FES = 70100	$\mu_F = 0.739$	$\mu_{CR} = 0.083$			
Diversity = 2.9318e-02	k_t = [4, 4, 4, 2]					
Generation 710: Best = 3.198914e+03	FES = 71100	$\mu_F = 0.724$	$\mu_{CR} = 0.076$			
Diversity = 3.0818e-02	k_t = [1, 1, 1, 1]					
Generation 720: Best = 3.198912e+03	FES = 72100	$\mu_F = 0.741$	$\mu_{CR} = 0.087$			
Diversity = 3.0288e-02	k_t = [3, 2, 2, 2]					
Generation 730: Best = 3.198911e+03	FES = 73100	$\mu_F = 0.741$	$\mu_{CR} = 0.076$			
Diversity = 3.0059e-02	k_t = [1, 1, 1, 1]					
Generation 740: Best = 3.198911e+03	FES = 74100	$\mu_F = 0.741$	$\mu_{CR} = 0.077$			
Diversity = 2.8071e-02	k_t = [3, 4, 3, 2]					
Generation 750: Best = 3.198910e+03	FES = 75100	$\mu_F = 0.738$	$\mu_{CR} = 0.070$			
Diversity = 2.7285e-02	k_t = [1, 1, 1, 1]					
Generation 760: Best = 3.198905e+03	FES = 76100	$\mu_F = 0.738$	$\mu_{CR} = 0.073$			

Diversity = 2.5895e-02	k_t = [4, 4, 3, 3]				
Generation 770: Best = 3.198905e+03	FES = 77100	$\mu_F = 0.730$	$\mu_{CR} = 0.069$		
Diversity = 2.4710e-02	k_t = [1, 1, 1, 1]				
Generation 780: Best = 3.198902e+03	FES = 78100	$\mu_F = 0.749$	$\mu_{CR} = 0.065$		
Diversity = 2.4462e-02	k_t = [3, 3, 2, 2]				
Generation 790: Best = 3.198902e+03	FES = 79100	$\mu_F = 0.748$	$\mu_{CR} = 0.070$		
Diversity = 2.4726e-02	k_t = [1, 1, 1, 1]				
Generation 800: Best = 3.198897e+03	FES = 80100	$\mu_F = 0.766$	$\mu_{CR} = 0.075$		
Diversity = 2.1959e-02	k_t = [4, 3, 3, 2]				
Generation 810: Best = 3.198896e+03	FES = 81100	$\mu_F = 0.786$	$\mu_{CR} = 0.076$		
Diversity = 2.1780e-02	k_t = [1, 1, 1, 1]				
Generation 820: Best = 3.198895e+03	FES = 82100	$\mu_F = 0.774$	$\mu_{CR} = 0.081$		
Diversity = 2.2404e-02	k_t = [4, 3, 3, 2]				
Generation 830: Best = 3.198893e+03	FES = 83100	$\mu_F = 0.787$	$\mu_{CR} = 0.083$		
Diversity = 2.2979e-02	k_t = [1, 1, 1, 1]				
Generation 840: Best = 3.198892e+03	FES = 84100	$\mu_F = 0.794$	$\mu_{CR} = 0.071$		
Diversity = 2.0475e-02	k_t = [3, 3, 4, 4]				
Generation 850: Best = 3.198890e+03	FES = 85100	$\mu_F = 0.788$	$\mu_{CR} = 0.070$		
Diversity = 2.0011e-02	k_t = [1, 1, 1, 1]				
Generation 860: Best = 3.198890e+03	FES = 86100	$\mu_F = 0.778$	$\mu_{CR} = 0.073$		
Diversity = 1.8959e-02	k_t = [4, 4, 4, 3]				
Generation 870: Best = 3.198887e+03	FES = 87100	$\mu_F = 0.771$	$\mu_{CR} = 0.069$		
Diversity = 1.8263e-02	k_t = [1, 1, 1, 1]				
Generation 880: Best = 3.198886e+03	FES = 88100	$\mu_F = 0.762$	$\mu_{CR} = 0.076$		
Diversity = 1.9293e-02	k_t = [2, 3, 2, 2]				
Generation 890: Best = 3.198885e+03	FES = 89100	$\mu_F = 0.770$	$\mu_{CR} = 0.071$		
Diversity = 1.9044e-02	k_t = [1, 1, 1, 1]				
Generation 900: Best = 3.198882e+03	FES = 90100	$\mu_F = 0.776$	$\mu_{CR} = 0.071$		
Diversity = 1.7474e-02	k_t = [2, 3, 3, 3]				
Generation 910: Best = 3.198882e+03	FES = 91100	$\mu_F = 0.770$	$\mu_{CR} = 0.074$		
Diversity = 1.8906e-02	k_t = [1, 1, 1, 1]				
Generation 920: Best = 3.198881e+03	FES = 92100	$\mu_F = 0.794$	$\mu_{CR} = 0.068$		
Diversity = 1.8521e-02	k_t = [3, 4, 3, 3]				
Generation 930: Best = 3.198880e+03	FES = 93100	$\mu_F = 0.802$	$\mu_{CR} = 0.073$		
Diversity = 1.8388e-02	k_t = [1, 1, 1, 1]				
Generation 940: Best = 3.198879e+03	FES = 94100	$\mu_F = 0.803$	$\mu_{CR} = 0.084$		
Diversity = 1.6886e-02	k_t = [3, 3, 2, 3]				
Generation 950: Best = 3.198879e+03	FES = 95100	$\mu_F = 0.789$	$\mu_{CR} = 0.079$		
Diversity = 1.6544e-02	k_t = [1, 1, 1, 1]				
Generation 960: Best = 3.198879e+03	FES = 96100	$\mu_F = 0.793$	$\mu_{CR} = 0.073$		
Diversity = 1.5337e-02	k_t = [2, 4, 3, 2]				
Generation 970: Best = 3.198879e+03	FES = 97100	$\mu_F = 0.786$	$\mu_{CR} = 0.064$		
Diversity = 1.4443e-02	k_t = [1, 1, 1, 1]				
Generation 980: Best = 3.198878e+03	FES = 98100	$\mu_F = 0.766$	$\mu_{CR} = 0.068$		
Diversity = 1.2760e-02	k_t = [2, 3, 4, 3]				
Generation 990: Best = 3.198877e+03	FES = 99100	$\mu_F = 0.763$	$\mu_{CR} = 0.064$		
Diversity = 1.2100e-02	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 3.198877e+03	FES = 100100	$\mu_F = 0.745$	$\mu_{CR} = 0.069$		
Diversity = 1.1389e-02	k_t = [2, 4, 4, 3]				
Generation 1010: Best = 3.198877e+03	FES = 101100	$\mu_F = 0.752$	$\mu_{CR} = 0.068$		
Diversity = 1.0523e-02	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 3.198877e+03	FES = 102100	$\mu_F = 0.744$	$\mu_{CR} = 0.069$		
Diversity = 1.0160e-02	k_t = [3, 3, 2, 2]				
Generation 1030: Best = 3.198876e+03	FES = 103100	$\mu_F = 0.742$	$\mu_{CR} = 0.075$		

Diversity = 9.8031e-03	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 3.198876e+03	FES = 104100	$\mu_F = 0.740$	$\mu_{CR} = 0.064$	
Diversity = 9.2713e-03	k_t = [3, 3, 4, 2]			
Generation 1050: Best = 3.198875e+03	FES = 105100	$\mu_F = 0.728$	$\mu_{CR} = 0.061$	
Diversity = 9.1370e-03	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 3.198875e+03	FES = 106100	$\mu_F = 0.731$	$\mu_{CR} = 0.064$	
Diversity = 8.9344e-03	k_t = [3, 2, 3, 2]			
Generation 1070: Best = 3.198875e+03	FES = 107100	$\mu_F = 0.706$	$\mu_{CR} = 0.058$	
Diversity = 8.8064e-03	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 3.198875e+03	FES = 108100	$\mu_F = 0.706$	$\mu_{CR} = 0.057$	
Diversity = 8.3642e-03	k_t = [2, 4, 3, 4]			
Generation 1090: Best = 3.198875e+03	FES = 109100	$\mu_F = 0.726$	$\mu_{CR} = 0.062$	
Diversity = 8.2375e-03	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 3.198875e+03	FES = 110100	$\mu_F = 0.723$	$\mu_{CR} = 0.056$	
Diversity = 7.3115e-03	k_t = [2, 2, 2, 2]			
Generation 1110: Best = 3.198875e+03	FES = 111100	$\mu_F = 0.756$	$\mu_{CR} = 0.067$	
Diversity = 7.3062e-03	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 3.198875e+03	FES = 112100	$\mu_F = 0.758$	$\mu_{CR} = 0.067$	
Diversity = 7.0948e-03	k_t = [3, 3, 3, 2]			
Generation 1130: Best = 3.198875e+03	FES = 113100	$\mu_F = 0.765$	$\mu_{CR} = 0.060$	
Diversity = 6.6127e-03	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 3.198875e+03	FES = 114100	$\mu_F = 0.769$	$\mu_{CR} = 0.061$	
Diversity = 6.2480e-03	k_t = [4, 3, 4, 2]			
Generation 1150: Best = 3.198874e+03	FES = 115100	$\mu_F = 0.786$	$\mu_{CR} = 0.065$	
Diversity = 6.4656e-03	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 3.198874e+03	FES = 116100	$\mu_F = 0.767$	$\mu_{CR} = 0.060$	
Diversity = 6.1733e-03	k_t = [3, 3, 4, 4]			
Generation 1170: Best = 3.198874e+03	FES = 117100	$\mu_F = 0.757$	$\mu_{CR} = 0.060$	
Diversity = 5.5969e-03	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 3.198874e+03	FES = 118100	$\mu_F = 0.757$	$\mu_{CR} = 0.054$	
Diversity = 4.9989e-03	k_t = [2, 2, 2, 4]			
Generation 1190: Best = 3.198874e+03	FES = 119100	$\mu_F = 0.740$	$\mu_{CR} = 0.062$	
Diversity = 4.6054e-03	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 3.198874e+03	FES = 120100	$\mu_F = 0.744$	$\mu_{CR} = 0.057$	
Diversity = 4.6944e-03	k_t = [4, 3, 2, 4]			
Generation 1210: Best = 3.198874e+03	FES = 121100	$\mu_F = 0.750$	$\mu_{CR} = 0.064$	
Diversity = 4.4985e-03	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 3.198873e+03	FES = 122100	$\mu_F = 0.763$	$\mu_{CR} = 0.072$	
Diversity = 5.0031e-03	k_t = [2, 4, 2, 4]			
Generation 1230: Best = 3.198873e+03	FES = 123100	$\mu_F = 0.771$	$\mu_{CR} = 0.075$	
Diversity = 5.0862e-03	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 3.198873e+03	FES = 124100	$\mu_F = 0.773$	$\mu_{CR} = 0.077$	
Diversity = 5.0848e-03	k_t = [2, 3, 2, 3]			
Generation 1250: Best = 3.198873e+03	FES = 125100	$\mu_F = 0.779$	$\mu_{CR} = 0.062$	
Diversity = 5.4168e-03	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 3.198873e+03	FES = 126100	$\mu_F = 0.775$	$\mu_{CR} = 0.064$	
Diversity = 5.1588e-03	k_t = [3, 3, 2, 4]			
Generation 1270: Best = 3.198873e+03	FES = 127100	$\mu_F = 0.779$	$\mu_{CR} = 0.060$	
Diversity = 4.6683e-03	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 3.198873e+03	FES = 128100	$\mu_F = 0.779$	$\mu_{CR} = 0.057$	
Diversity = 4.2538e-03	k_t = [3, 3, 3, 3]			
Generation 1290: Best = 3.198873e+03	FES = 129100	$\mu_F = 0.775$	$\mu_{CR} = 0.060$	
Diversity = 4.0671e-03	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 3.198873e+03	FES = 130100	$\mu_F = 0.767$	$\mu_{CR} = 0.049$	

Diversity = 3.8597e-03	k_t = [4, 2, 4, 3]			
Generation 1310: Best = 3.198873e+03	FES = 131100	$\mu_F = 0.762$	$\mu_{CR} = 0.052$	
Diversity = 4.4057e-03	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 3.198872e+03	FES = 132100	$\mu_F = 0.776$	$\mu_{CR} = 0.064$	
Diversity = 4.5517e-03	k_t = [3, 4, 3, 2]			
Generation 1330: Best = 3.198872e+03	FES = 133100	$\mu_F = 0.763$	$\mu_{CR} = 0.059$	
Diversity = 4.4856e-03	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 3.198872e+03	FES = 134100	$\mu_F = 0.752$	$\mu_{CR} = 0.062$	
Diversity = 4.4023e-03	k_t = [4, 2, 4, 3]			
Generation 1350: Best = 3.198872e+03	FES = 135100	$\mu_F = 0.779$	$\mu_{CR} = 0.056$	
Diversity = 4.3924e-03	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 3.198872e+03	FES = 136100	$\mu_F = 0.780$	$\mu_{CR} = 0.063$	
Diversity = 4.5019e-03	k_t = [3, 2, 3, 3]			
Generation 1370: Best = 3.198872e+03	FES = 137100	$\mu_F = 0.778$	$\mu_{CR} = 0.055$	
Diversity = 4.1025e-03	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 3.198872e+03	FES = 138100	$\mu_F = 0.785$	$\mu_{CR} = 0.062$	
Diversity = 3.5938e-03	k_t = [3, 4, 3, 4]			
Generation 1390: Best = 3.198872e+03	FES = 139100	$\mu_F = 0.775$	$\mu_{CR} = 0.069$	
Diversity = 3.3633e-03	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 3.198872e+03	FES = 140100	$\mu_F = 0.764$	$\mu_{CR} = 0.064$	
Diversity = 3.1310e-03	k_t = [4, 3, 3, 3]			
Generation 1410: Best = 3.198872e+03	FES = 141100	$\mu_F = 0.769$	$\mu_{CR} = 0.064$	
Diversity = 2.9976e-03	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 3.198872e+03	FES = 142100	$\mu_F = 0.745$	$\mu_{CR} = 0.058$	
Diversity = 2.8883e-03	k_t = [3, 2, 2, 2]			
Generation 1430: Best = 3.198872e+03	FES = 143100	$\mu_F = 0.723$	$\mu_{CR} = 0.061$	
Diversity = 2.4969e-03	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 3.198872e+03	FES = 144100	$\mu_F = 0.729$	$\mu_{CR} = 0.064$	
Diversity = 2.2149e-03	k_t = [3, 2, 3, 4]			
Generation 1450: Best = 3.198872e+03	FES = 145100	$\mu_F = 0.735$	$\mu_{CR} = 0.058$	
Diversity = 2.1375e-03	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 3.198872e+03	FES = 146100	$\mu_F = 0.738$	$\mu_{CR} = 0.060$	
Diversity = 2.0301e-03	k_t = [2, 3, 3, 3]			
Generation 1470: Best = 3.198872e+03	FES = 147100	$\mu_F = 0.759$	$\mu_{CR} = 0.062$	
Diversity = 2.0396e-03	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 3.198872e+03	FES = 148100	$\mu_F = 0.745$	$\mu_{CR} = 0.058$	
Diversity = 1.7159e-03	k_t = [2, 2, 2, 4]			
Generation 1490: Best = 3.198872e+03	FES = 149100	$\mu_F = 0.750$	$\mu_{CR} = 0.064$	
Diversity = 1.9407e-03	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 3.198872e+03	FES = 150100	$\mu_F = 0.763$	$\mu_{CR} = 0.054$	
Diversity = 1.9519e-03	k_t = [3, 4, 4, 3]			
Generation 1510: Best = 3.198872e+03	FES = 151100	$\mu_F = 0.760$	$\mu_{CR} = 0.061$	
Diversity = 1.9925e-03	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 3.198872e+03	FES = 152100	$\mu_F = 0.756$	$\mu_{CR} = 0.066$	
Diversity = 1.8925e-03	k_t = [3, 2, 2, 2]			
Generation 1530: Best = 3.198872e+03	FES = 153100	$\mu_F = 0.729$	$\mu_{CR} = 0.064$	
Diversity = 2.0188e-03	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 3.198872e+03	FES = 154100	$\mu_F = 0.732$	$\mu_{CR} = 0.067$	
Diversity = 2.0224e-03	k_t = [4, 2, 2, 2]			
Generation 1550: Best = 3.198872e+03	FES = 155100	$\mu_F = 0.744$	$\mu_{CR} = 0.065$	
Diversity = 1.9053e-03	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 3.198872e+03	FES = 156100	$\mu_F = 0.750$	$\mu_{CR} = 0.065$	
Diversity = 1.7709e-03	k_t = [2, 2, 2, 3]			
Generation 1570: Best = 3.198872e+03	FES = 157100	$\mu_F = 0.757$	$\mu_{CR} = 0.065$	

Diversity = 1.8032e-03	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 3.198872e+03	FES = 158100	$\mu_F = 0.780$	$\mu_{CR} = 0.055$	
Diversity = 1.7332e-03	k_t = [4, 4, 4, 4]			
Generation 1590: Best = 3.198872e+03	FES = 159100	$\mu_F = 0.765$	$\mu_{CR} = 0.060$	
Diversity = 1.7607e-03	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 3.198872e+03	FES = 160100	$\mu_F = 0.776$	$\mu_{CR} = 0.064$	
Diversity = 1.6865e-03	k_t = [4, 4, 3, 3]			
Generation 1610: Best = 3.198872e+03	FES = 161100	$\mu_F = 0.780$	$\mu_{CR} = 0.063$	
Diversity = 1.8664e-03	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 3.198872e+03	FES = 162100	$\mu_F = 0.764$	$\mu_{CR} = 0.077$	
Diversity = 1.7561e-03	k_t = [2, 2, 4, 3]			
Generation 1630: Best = 3.198871e+03	FES = 163100	$\mu_F = 0.773$	$\mu_{CR} = 0.072$	
Diversity = 1.8340e-03	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 3.198871e+03	FES = 164100	$\mu_F = 0.765$	$\mu_{CR} = 0.067$	
Diversity = 2.0267e-03	k_t = [3, 2, 2, 2]			
Generation 1650: Best = 3.198871e+03	FES = 165100	$\mu_F = 0.754$	$\mu_{CR} = 0.071$	
Diversity = 2.1500e-03	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 3.198871e+03	FES = 166100	$\mu_F = 0.754$	$\mu_{CR} = 0.077$	
Diversity = 1.9971e-03	k_t = [3, 3, 3, 4]			
Generation 1670: Best = 3.198871e+03	FES = 167100	$\mu_F = 0.764$	$\mu_{CR} = 0.078$	
Diversity = 2.0194e-03	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 3.198871e+03	FES = 168100	$\mu_F = 0.775$	$\mu_{CR} = 0.070$	
Diversity = 1.9052e-03	k_t = [3, 4, 4, 3]			
Generation 1690: Best = 3.198871e+03	FES = 169100	$\mu_F = 0.775$	$\mu_{CR} = 0.068$	
Diversity = 1.8300e-03	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 3.198871e+03	FES = 170100	$\mu_F = 0.771$	$\mu_{CR} = 0.063$	
Diversity = 1.7514e-03	k_t = [3, 3, 2, 3]			
Generation 1710: Best = 3.198871e+03	FES = 171100	$\mu_F = 0.771$	$\mu_{CR} = 0.059$	
Diversity = 1.6420e-03	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 3.198871e+03	FES = 172100	$\mu_F = 0.744$	$\mu_{CR} = 0.063$	
Diversity = 1.4842e-03	k_t = [4, 2, 2, 4]			
Generation 1730: Best = 3.198871e+03	FES = 173100	$\mu_F = 0.754$	$\mu_{CR} = 0.060$	
Diversity = 1.4542e-03	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 3.198871e+03	FES = 174100	$\mu_F = 0.739$	$\mu_{CR} = 0.058$	
Diversity = 1.4134e-03	k_t = [3, 3, 3, 2]			
Generation 1750: Best = 3.198871e+03	FES = 175100	$\mu_F = 0.733$	$\mu_{CR} = 0.068$	
Diversity = 1.3605e-03	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 3.198871e+03	FES = 176100	$\mu_F = 0.747$	$\mu_{CR} = 0.066$	
Diversity = 1.2646e-03	k_t = [4, 3, 3, 3]			
Generation 1770: Best = 3.198871e+03	FES = 177100	$\mu_F = 0.743$	$\mu_{CR} = 0.058$	
Diversity = 1.3781e-03	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 3.198871e+03	FES = 178100	$\mu_F = 0.722$	$\mu_{CR} = 0.056$	
Diversity = 1.1758e-03	k_t = [2, 4, 3, 4]			
Generation 1790: Best = 3.198871e+03	FES = 179100	$\mu_F = 0.737$	$\mu_{CR} = 0.062$	
Diversity = 1.1873e-03	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 3.198871e+03	FES = 180100	$\mu_F = 0.732$	$\mu_{CR} = 0.058$	
Diversity = 1.1253e-03	k_t = [3, 2, 4, 3]			
Generation 1810: Best = 3.198871e+03	FES = 181100	$\mu_F = 0.747$	$\mu_{CR} = 0.056$	
Diversity = 1.1247e-03	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 3.198871e+03	FES = 182100	$\mu_F = 0.748$	$\mu_{CR} = 0.068$	
Diversity = 9.8327e-04	k_t = [4, 3, 2, 3]			
Generation 1830: Best = 3.198871e+03	FES = 183100	$\mu_F = 0.756$	$\mu_{CR} = 0.062$	
Diversity = 9.3605e-04	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 3.198871e+03	FES = 184100	$\mu_F = 0.756$	$\mu_{CR} = 0.063$	

Diversity = 9.3964e-04	k_t = [3, 3, 4, 2]			
Generation 1850: Best = 3.198871e+03	FES = 185100	$\mu_F = 0.754$	$\mu_{CR} = 0.073$	
Diversity = 9.1612e-04	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 3.198871e+03	FES = 186100	$\mu_F = 0.758$	$\mu_{CR} = 0.061$	
Diversity = 9.1251e-04	k_t = [2, 4, 4, 3]			
Generation 1870: Best = 3.198871e+03	FES = 187100	$\mu_F = 0.747$	$\mu_{CR} = 0.060$	
Diversity = 8.3700e-04	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 3.198871e+03	FES = 188100	$\mu_F = 0.759$	$\mu_{CR} = 0.065$	
Diversity = 8.6559e-04	k_t = [3, 2, 4, 3]			
Generation 1890: Best = 3.198871e+03	FES = 189100	$\mu_F = 0.751$	$\mu_{CR} = 0.063$	
Diversity = 8.4712e-04	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 3.198871e+03	FES = 190100	$\mu_F = 0.747$	$\mu_{CR} = 0.062$	
Diversity = 7.8407e-04	k_t = [2, 2, 4, 4]			
Generation 1910: Best = 3.198871e+03	FES = 191100	$\mu_F = 0.724$	$\mu_{CR} = 0.058$	
Diversity = 7.1991e-04	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 3.198871e+03	FES = 192100	$\mu_F = 0.728$	$\mu_{CR} = 0.068$	
Diversity = 7.3012e-04	k_t = [2, 4, 3, 2]			
Generation 1930: Best = 3.198871e+03	FES = 193100	$\mu_F = 0.746$	$\mu_{CR} = 0.065$	
Diversity = 7.1616e-04	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 3.198871e+03	FES = 194100	$\mu_F = 0.746$	$\mu_{CR} = 0.061$	
Diversity = 7.1527e-04	k_t = [2, 2, 4, 2]			
Generation 1950: Best = 3.198871e+03	FES = 195100	$\mu_F = 0.737$	$\mu_{CR} = 0.070$	
Diversity = 7.8034e-04	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 3.198871e+03	FES = 196100	$\mu_F = 0.732$	$\mu_{CR} = 0.059$	
Diversity = 7.6549e-04	k_t = [4, 4, 2, 4]			
Generation 1970: Best = 3.198871e+03	FES = 197100	$\mu_F = 0.734$	$\mu_{CR} = 0.058$	
Diversity = 7.5036e-04	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 3.198871e+03	FES = 198100	$\mu_F = 0.738$	$\mu_{CR} = 0.061$	
Diversity = 7.3387e-04	k_t = [3, 3, 2, 2]			
Generation 1990: Best = 3.198871e+03	FES = 199100	$\mu_F = 0.759$	$\mu_{CR} = 0.061$	
Diversity = 8.7921e-04	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 3.198871e+03	FES = 200100	$\mu_F = 0.756$	$\mu_{CR} = 0.071$	
Diversity = 8.5533e-04	k_t = [3, 2, 3, 3]			
Generation 2010: Best = 3.198871e+03	FES = 201100	$\mu_F = 0.782$	$\mu_{CR} = 0.063$	
Diversity = 9.3576e-04	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 3.198871e+03	FES = 202100	$\mu_F = 0.755$	$\mu_{CR} = 0.059$	
Diversity = 8.3648e-04	k_t = [3, 2, 3, 4]			
Generation 2030: Best = 3.198871e+03	FES = 203100	$\mu_F = 0.770$	$\mu_{CR} = 0.060$	
Diversity = 9.1730e-04	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 3.198871e+03	FES = 204100	$\mu_F = 0.787$	$\mu_{CR} = 0.061$	
Diversity = 1.0794e-03	k_t = [2, 2, 3, 2]			
Generation 2050: Best = 3.198871e+03	FES = 205100	$\mu_F = 0.801$	$\mu_{CR} = 0.069$	
Diversity = 1.2375e-03	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 3.198871e+03	FES = 206100	$\mu_F = 0.798$	$\mu_{CR} = 0.065$	
Diversity = 1.1881e-03	k_t = [3, 4, 4, 2]			
Generation 2070: Best = 3.198871e+03	FES = 207100	$\mu_F = 0.788$	$\mu_{CR} = 0.066$	
Diversity = 1.2644e-03	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 3.198871e+03	FES = 208100	$\mu_F = 0.798$	$\mu_{CR} = 0.063$	
Diversity = 1.3159e-03	k_t = [2, 3, 3, 2]			
Generation 2090: Best = 3.198871e+03	FES = 209100	$\mu_F = 0.769$	$\mu_{CR} = 0.061$	
Diversity = 1.2867e-03	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 3.198871e+03	FES = 210100	$\mu_F = 0.774$	$\mu_{CR} = 0.060$	
Diversity = 1.2039e-03	k_t = [3, 4, 3, 3]			
Generation 2110: Best = 3.198871e+03	FES = 211100	$\mu_F = 0.779$	$\mu_{CR} = 0.054$	

Diversity = 1.0944e-03	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 3.198871e+03	FES = 212100	$\mu_F = 0.794$	$\mu_{CR} = 0.051$	
Diversity = 9.9604e-04	k_t = [3, 4, 2, 3]			
Generation 2130: Best = 3.198871e+03	FES = 213100	$\mu_F = 0.779$	$\mu_{CR} = 0.056$	
Diversity = 9.3653e-04	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 3.198871e+03	FES = 214100	$\mu_F = 0.769$	$\mu_{CR} = 0.051$	
Diversity = 8.3873e-04	k_t = [2, 4, 4, 4]			
Generation 2150: Best = 3.198871e+03	FES = 215100	$\mu_F = 0.762$	$\mu_{CR} = 0.053$	
Diversity = 8.4473e-04	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 3.198871e+03	FES = 216100	$\mu_F = 0.757$	$\mu_{CR} = 0.062$	
Diversity = 7.7222e-04	k_t = [2, 4, 4, 4]			
Generation 2170: Best = 3.198871e+03	FES = 217100	$\mu_F = 0.741$	$\mu_{CR} = 0.061$	
Diversity = 7.3244e-04	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 3.198871e+03	FES = 218100	$\mu_F = 0.721$	$\mu_{CR} = 0.065$	
Diversity = 7.8901e-04	k_t = [3, 2, 3, 2]			
Generation 2190: Best = 3.198871e+03	FES = 219100	$\mu_F = 0.734$	$\mu_{CR} = 0.075$	
Diversity = 7.6719e-04	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 3.198871e+03	FES = 220100	$\mu_F = 0.731$	$\mu_{CR} = 0.065$	
Diversity = 6.7740e-04	k_t = [3, 2, 3, 4]			
Generation 2210: Best = 3.198871e+03	FES = 221100	$\mu_F = 0.758$	$\mu_{CR} = 0.062$	
Diversity = 6.1652e-04	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 3.198871e+03	FES = 222100	$\mu_F = 0.753$	$\mu_{CR} = 0.058$	
Diversity = 6.5750e-04	k_t = [3, 2, 2, 2]			
Generation 2230: Best = 3.198871e+03	FES = 223100	$\mu_F = 0.738$	$\mu_{CR} = 0.059$	
Diversity = 6.4854e-04	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 3.198871e+03	FES = 224100	$\mu_F = 0.740$	$\mu_{CR} = 0.060$	
Diversity = 6.0761e-04	k_t = [2, 2, 3, 4]			
Generation 2250: Best = 3.198871e+03	FES = 225100	$\mu_F = 0.756$	$\mu_{CR} = 0.064$	
Diversity = 6.7065e-04	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 3.198871e+03	FES = 226100	$\mu_F = 0.779$	$\mu_{CR} = 0.067$	
Diversity = 6.5223e-04	k_t = [2, 3, 3, 2]			
Generation 2270: Best = 3.198871e+03	FES = 227100	$\mu_F = 0.777$	$\mu_{CR} = 0.067$	
Diversity = 6.2228e-04	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 3.198871e+03	FES = 228100	$\mu_F = 0.769$	$\mu_{CR} = 0.062$	
Diversity = 5.5453e-04	k_t = [4, 4, 3, 3]			
Generation 2290: Best = 3.198871e+03	FES = 229100	$\mu_F = 0.767$	$\mu_{CR} = 0.061$	
Diversity = 5.5297e-04	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 3.198871e+03	FES = 230100	$\mu_F = 0.774$	$\mu_{CR} = 0.068$	
Diversity = 5.3192e-04	k_t = [3, 4, 2, 4]			
Generation 2310: Best = 3.198871e+03	FES = 231100	$\mu_F = 0.777$	$\mu_{CR} = 0.059$	
Diversity = 5.0255e-04	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 3.198871e+03	FES = 232100	$\mu_F = 0.752$	$\mu_{CR} = 0.050$	
Diversity = 5.0648e-04	k_t = [3, 4, 2, 3]			
Generation 2330: Best = 3.198871e+03	FES = 233100	$\mu_F = 0.754$	$\mu_{CR} = 0.055$	
Diversity = 5.0424e-04	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 3.198871e+03	FES = 234100	$\mu_F = 0.779$	$\mu_{CR} = 0.059$	
Diversity = 4.9440e-04	k_t = [3, 2, 3, 4]			
Generation 2350: Best = 3.198871e+03	FES = 235100	$\mu_F = 0.756$	$\mu_{CR} = 0.066$	
Diversity = 4.9987e-04	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 3.198871e+03	FES = 236100	$\mu_F = 0.756$	$\mu_{CR} = 0.057$	
Diversity = 4.7603e-04	k_t = [2, 3, 3, 3]			
Generation 2370: Best = 3.198871e+03	FES = 237100	$\mu_F = 0.769$	$\mu_{CR} = 0.060$	
Diversity = 4.5305e-04	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 3.198871e+03	FES = 238100	$\mu_F = 0.773$	$\mu_{CR} = 0.059$	

Diversity = 4.6772e-04	k_t = [2, 4, 3, 3]			
Generation 2390: Best = 3.198871e+03	FES = 239100	$\mu_F = 0.782$	$\mu_{CR} = 0.060$	
Diversity = 5.0912e-04	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 3.198871e+03	FES = 240100	$\mu_F = 0.777$	$\mu_{CR} = 0.060$	
Diversity = 5.8623e-04	k_t = [2, 3, 4, 3]			
Generation 2410: Best = 3.198871e+03	FES = 241100	$\mu_F = 0.792$	$\mu_{CR} = 0.063$	
Diversity = 6.7357e-04	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 3.198871e+03	FES = 242100	$\mu_F = 0.803$	$\mu_{CR} = 0.064$	
Diversity = 6.6027e-04	k_t = [4, 2, 2, 4]			
Generation 2430: Best = 3.198871e+03	FES = 243100	$\mu_F = 0.802$	$\mu_{CR} = 0.064$	
Diversity = 6.8693e-04	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 3.198871e+03	FES = 244100	$\mu_F = 0.789$	$\mu_{CR} = 0.064$	
Diversity = 6.9065e-04	k_t = [4, 2, 4, 4]			
Generation 2450: Best = 3.198871e+03	FES = 245100	$\mu_F = 0.784$	$\mu_{CR} = 0.059$	
Diversity = 6.8039e-04	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 3.198871e+03	FES = 246100	$\mu_F = 0.772$	$\mu_{CR} = 0.055$	
Diversity = 5.8921e-04	k_t = [2, 3, 4, 4]			
Generation 2470: Best = 3.198871e+03	FES = 247100	$\mu_F = 0.793$	$\mu_{CR} = 0.060$	
Diversity = 5.9770e-04	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 3.198871e+03	FES = 248100	$\mu_F = 0.786$	$\mu_{CR} = 0.053$	
Diversity = 6.2101e-04	k_t = [3, 2, 2, 4]			
Generation 2490: Best = 3.198871e+03	FES = 249100	$\mu_F = 0.785$	$\mu_{CR} = 0.060$	
Diversity = 6.6876e-04	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 3.198871e+03	FES = 250100	$\mu_F = 0.783$	$\mu_{CR} = 0.059$	
Diversity = 6.5527e-04	k_t = [2, 4, 3, 4]			
Generation 2510: Best = 3.198871e+03	FES = 251100	$\mu_F = 0.797$	$\mu_{CR} = 0.066$	
Diversity = 6.3850e-04	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 3.198871e+03	FES = 252100	$\mu_F = 0.809$	$\mu_{CR} = 0.064$	
Diversity = 5.9620e-04	k_t = [4, 3, 2, 4]			
Generation 2530: Best = 3.198871e+03	FES = 253100	$\mu_F = 0.797$	$\mu_{CR} = 0.059$	
Diversity = 5.4926e-04	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 3.198871e+03	FES = 254100	$\mu_F = 0.788$	$\mu_{CR} = 0.061$	
Diversity = 5.8448e-04	k_t = [4, 3, 2, 2]			
Generation 2550: Best = 3.198871e+03	FES = 255100	$\mu_F = 0.771$	$\mu_{CR} = 0.067$	
Diversity = 5.3818e-04	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 3.198871e+03	FES = 256100	$\mu_F = 0.767$	$\mu_{CR} = 0.066$	
Diversity = 4.9538e-04	k_t = [2, 3, 3, 4]			
Generation 2570: Best = 3.198871e+03	FES = 257100	$\mu_F = 0.752$	$\mu_{CR} = 0.072$	
Diversity = 4.1569e-04	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 3.198871e+03	FES = 258100	$\mu_F = 0.755$	$\mu_{CR} = 0.063$	
Diversity = 3.9245e-04	k_t = [3, 4, 4, 3]			
Generation 2590: Best = 3.198871e+03	FES = 259100	$\mu_F = 0.740$	$\mu_{CR} = 0.064$	
Diversity = 3.8916e-04	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 3.198871e+03	FES = 260100	$\mu_F = 0.736$	$\mu_{CR} = 0.065$	
Diversity = 3.9239e-04	k_t = [4, 3, 3, 3]			
Generation 2610: Best = 3.198871e+03	FES = 261100	$\mu_F = 0.732$	$\mu_{CR} = 0.059$	
Diversity = 3.8673e-04	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 3.198871e+03	FES = 262100	$\mu_F = 0.738$	$\mu_{CR} = 0.057$	
Diversity = 3.9851e-04	k_t = [3, 2, 2, 2]			
Generation 2630: Best = 3.198871e+03	FES = 263100	$\mu_F = 0.752$	$\mu_{CR} = 0.059$	
Diversity = 4.1874e-04	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 3.198871e+03	FES = 264100	$\mu_F = 0.761$	$\mu_{CR} = 0.058$	
Diversity = 4.2026e-04	k_t = [2, 4, 3, 4]			
Generation 2650: Best = 3.198871e+03	FES = 265100	$\mu_F = 0.756$	$\mu_{CR} = 0.055$	

Diversity = 4.0093e-04	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 3.198871e+03	FES = 266100	$\mu_F = 0.752$	$\mu_{CR} = 0.059$	
Diversity = 3.7614e-04	k_t = [3, 3, 4, 4]			
Generation 2670: Best = 3.198871e+03	FES = 267100	$\mu_F = 0.768$	$\mu_{CR} = 0.059$	
Diversity = 3.7088e-04	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 3.198871e+03	FES = 268100	$\mu_F = 0.771$	$\mu_{CR} = 0.056$	
Diversity = 3.5658e-04	k_t = [3, 4, 3, 4]			
Generation 2690: Best = 3.198871e+03	FES = 269100	$\mu_F = 0.767$	$\mu_{CR} = 0.055$	
Diversity = 3.5461e-04	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 3.198871e+03	FES = 270100	$\mu_F = 0.776$	$\mu_{CR} = 0.053$	
Diversity = 3.5404e-04	k_t = [4, 4, 3, 4]			
Generation 2710: Best = 3.198871e+03	FES = 271100	$\mu_F = 0.780$	$\mu_{CR} = 0.053$	
Diversity = 3.6318e-04	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 3.198871e+03	FES = 272100	$\mu_F = 0.777$	$\mu_{CR} = 0.057$	
Diversity = 3.4509e-04	k_t = [4, 2, 2, 4]			
Generation 2730: Best = 3.198871e+03	FES = 273100	$\mu_F = 0.772$	$\mu_{CR} = 0.053$	
Diversity = 3.1322e-04	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 3.198871e+03	FES = 274100	$\mu_F = 0.770$	$\mu_{CR} = 0.052$	
Diversity = 3.1128e-04	k_t = [4, 4, 3, 4]			
Generation 2750: Best = 3.198871e+03	FES = 275100	$\mu_F = 0.776$	$\mu_{CR} = 0.061$	
Diversity = 2.8194e-04	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 3.198871e+03	FES = 276100	$\mu_F = 0.754$	$\mu_{CR} = 0.058$	
Diversity = 2.7073e-04	k_t = [3, 2, 3, 4]			
Generation 2770: Best = 3.198871e+03	FES = 277100	$\mu_F = 0.760$	$\mu_{CR} = 0.062$	
Diversity = 2.8435e-04	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 3.198871e+03	FES = 278100	$\mu_F = 0.770$	$\mu_{CR} = 0.065$	
Diversity = 3.0182e-04	k_t = [2, 4, 2, 2]			
Generation 2790: Best = 3.198871e+03	FES = 279100	$\mu_F = 0.767$	$\mu_{CR} = 0.053$	
Diversity = 3.1336e-04	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 3.198871e+03	FES = 280100	$\mu_F = 0.768$	$\mu_{CR} = 0.072$	
Diversity = 2.9266e-04	k_t = [4, 2, 3, 3]			
Generation 2810: Best = 3.198871e+03	FES = 281100	$\mu_F = 0.803$	$\mu_{CR} = 0.066$	
Diversity = 2.6860e-04	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 3.198871e+03	FES = 282100	$\mu_F = 0.804$	$\mu_{CR} = 0.060$	
Diversity = 2.4918e-04	k_t = [3, 2, 2, 4]			
Generation 2830: Best = 3.198871e+03	FES = 283100	$\mu_F = 0.794$	$\mu_{CR} = 0.068$	
Diversity = 2.5294e-04	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 3.198871e+03	FES = 284100	$\mu_F = 0.782$	$\mu_{CR} = 0.069$	
Diversity = 2.3538e-04	k_t = [4, 2, 2, 3]			
Generation 2850: Best = 3.198871e+03	FES = 285100	$\mu_F = 0.793$	$\mu_{CR} = 0.063$	
Diversity = 2.6139e-04	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 3.198871e+03	FES = 286100	$\mu_F = 0.791$	$\mu_{CR} = 0.069$	
Diversity = 2.5679e-04	k_t = [4, 2, 3, 4]			
Generation 2870: Best = 3.198871e+03	FES = 287100	$\mu_F = 0.780$	$\mu_{CR} = 0.055$	
Diversity = 2.4393e-04	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 3.198871e+03	FES = 288100	$\mu_F = 0.787$	$\mu_{CR} = 0.053$	
Diversity = 2.2505e-04	k_t = [4, 3, 2, 3]			
Generation 2890: Best = 3.198871e+03	FES = 289100	$\mu_F = 0.783$	$\mu_{CR} = 0.062$	
Diversity = 2.1764e-04	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 3.198871e+03	FES = 290100	$\mu_F = 0.792$	$\mu_{CR} = 0.066$	
Diversity = 2.0192e-04	k_t = [3, 2, 4, 2]			
Generation 2910: Best = 3.198871e+03	FES = 291100	$\mu_F = 0.804$	$\mu_{CR} = 0.063$	
Diversity = 2.0933e-04	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 3.198871e+03	FES = 292100	$\mu_F = 0.795$	$\mu_{CR} = 0.069$	

```

Diversity = 2.1404e-04  k_t = [4, 2, 3, 3]
Generation 2930: Best = 3.198871e+03  FES = 293100   $\mu_F$  = 0.818   $\mu_{CR}$  = 0.058
Diversity = 2.2892e-04  k_t = [1, 1, 1, 1]
Generation 2940: Best = 3.198871e+03  FES = 294100   $\mu_F$  = 0.798   $\mu_{CR}$  = 0.059
Diversity = 2.3434e-04  k_t = [2, 3, 2, 3]
Generation 2950: Best = 3.198871e+03  FES = 295100   $\mu_F$  = 0.806   $\mu_{CR}$  = 0.053
Diversity = 2.4345e-04  k_t = [1, 1, 1, 1]
Generation 2960: Best = 3.198871e+03  FES = 296100   $\mu_F$  = 0.808   $\mu_{CR}$  = 0.060
Diversity = 2.4486e-04  k_t = [2, 4, 4, 2]
Generation 2970: Best = 3.198871e+03  FES = 297100   $\mu_F$  = 0.805   $\mu_{CR}$  = 0.058
Diversity = 2.4476e-04  k_t = [1, 1, 1, 1]
Generation 2980: Best = 3.198871e+03  FES = 298100   $\mu_F$  = 0.803   $\mu_{CR}$  = 0.055
Diversity = 2.5510e-04  k_t = [2, 3, 2, 2]
Generation 2990: Best = 3.198871e+03  FES = 299100   $\mu_F$  = 0.801   $\mu_{CR}$  = 0.058
Diversity = 2.5119e-04  k_t = [1, 1, 1, 1]

```

Optimization complete.

```

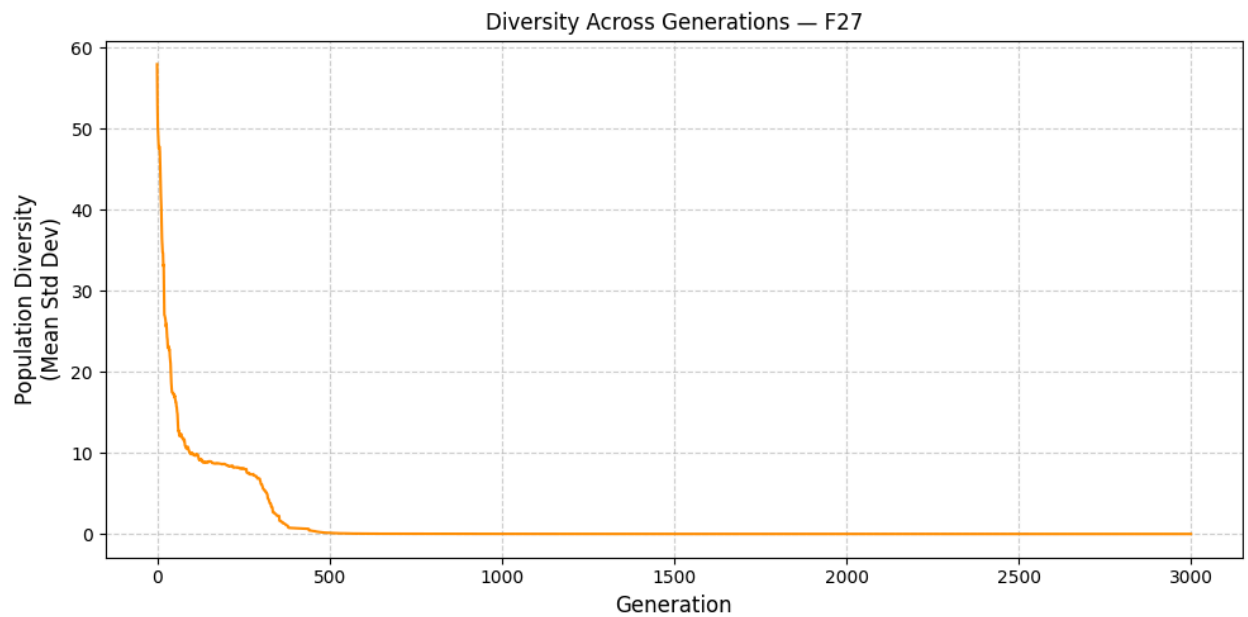
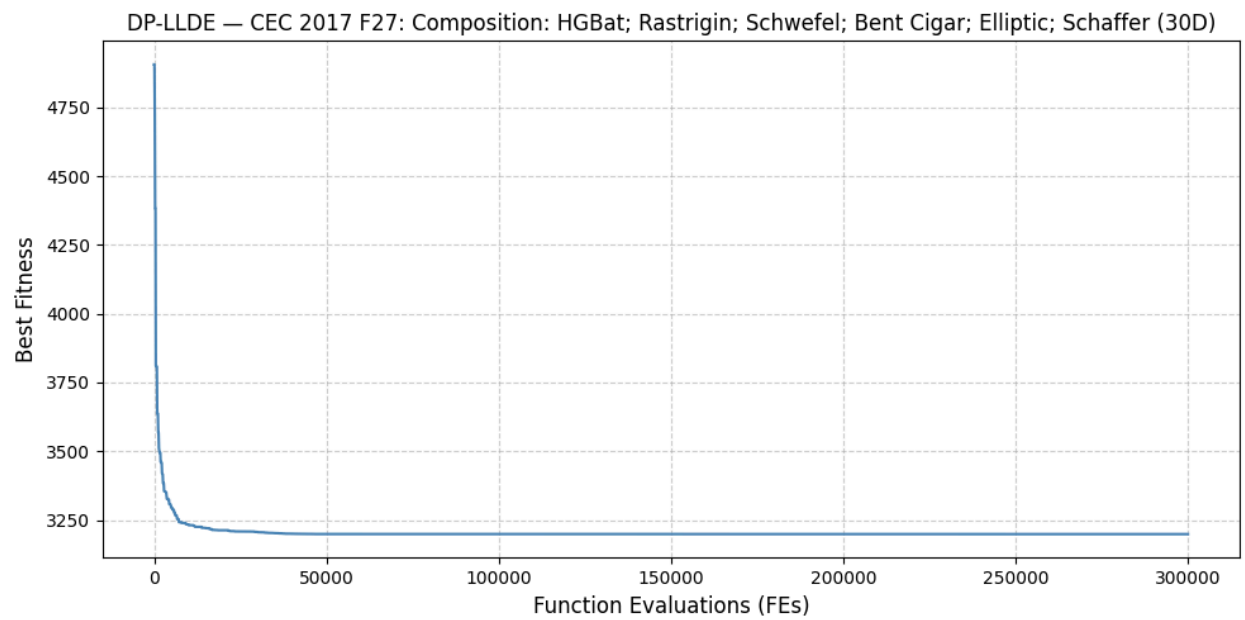
Best Fitness          = 3.198871e+03
Total Function Evals  = 300000
Total Successful Updates = 48624

```

```

=====
=
Results:
  Best Fitness          : 3.198871e+03
  Global Optimum        : 2.700000e+03
  Error (|best-optimum|) : 4.988710e+02
  Best Solution (5 dim) : [-93.88062019 -97.55738822 -99.99999883 -93.4520853
-26.43573942]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F27.png

```
In [ ]: # =====
# F28 — Composition Function 8 (Ackley; Griewank; Discus; Rosenbrock; HappyCat)
# =====
run_and_plot('F28', make_wrapper(composition.f28), 2800.0, 'Composition: Ackle
```


=====

=

DP-LLDE on CEC 2017 F28 – Single Run

=====

=

Function : Composition: Ackley; Griewank; Discus; Rosenbrock; HappyCat; S
chaffer (30D)

Bounds : [-100, 100]

Global optimum : 2800.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 1.208847e+04

Generation 10: Best = 4.921261e+03 FES = 1100 $\mu_F = 0.573$ $\mu_{CR} = 0.352$

Diversity = 2.7690e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 4.149457e+03 FES = 2100 $\mu_F = 0.611$ $\mu_{CR} = 0.360$

Diversity = 1.4529e+01 $k_t = [4, 4, 2, 4]$

Generation 30: Best = 3.704405e+03 FES = 3100 $\mu_F = 0.653$ $\mu_{CR} = 0.364$

Diversity = 8.8190e+00 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 3.549901e+03 FES = 4100 $\mu_F = 0.681$ $\mu_{CR} = 0.365$

Diversity = 5.7459e+00 $k_t = [2, 3, 3, 2]$

Generation 50: Best = 3.474070e+03 FES = 5100 $\mu_F = 0.694$ $\mu_{CR} = 0.370$

Diversity = 4.5904e+00 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 3.418783e+03 FES = 6100 $\mu_F = 0.709$ $\mu_{CR} = 0.381$

Diversity = 3.6937e+00 $k_t = [3, 2, 2, 2]$

Generation 70: Best = 3.386603e+03 FES = 7100 $\mu_F = 0.721$ $\mu_{CR} = 0.390$

Diversity = 3.1926e+00 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 3.362688e+03 FES = 8100 $\mu_F = 0.740$ $\mu_{CR} = 0.401$

Diversity = 2.4797e+00 $k_t = [2, 4, 4, 4]$

Generation 90: Best = 3.345001e+03 FES = 9100 $\mu_F = 0.762$ $\mu_{CR} = 0.408$

Diversity = 1.9550e+00 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 3.329939e+03 FES = 10100 $\mu_F = 0.767$ $\mu_{CR} = 0.411$

Diversity = 1.7324e+00 $k_t = [2, 2, 4, 4]$

Generation 110: Best = 3.315462e+03 FES = 11100 $\mu_F = 0.772$ $\mu_{CR} = 0.415$

Diversity = 1.9193e+00 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 3.301843e+03 FES = 12100 $\mu_F = 0.790$ $\mu_{CR} = 0.423$

Diversity = 2.1005e+00 $k_t = [3, 2, 3, 4]$

Generation 130: Best = 3.285879e+03 FES = 13100 $\mu_F = 0.791$ $\mu_{CR} = 0.423$

Diversity = 1.9738e+00 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 3.275560e+03 FES = 14100 $\mu_F = 0.790$ $\mu_{CR} = 0.419$

Diversity = 1.4927e+00 $k_t = [4, 2, 2, 2]$

Generation 150: Best = 3.266134e+03 FES = 15100 $\mu_F = 0.790$ $\mu_{CR} = 0.423$

Diversity = 1.4202e+00 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 3.254696e+03 FES = 16100 $\mu_F = 0.782$ $\mu_{CR} = 0.439$

Diversity = 1.1806e+00 $k_t = [3, 3, 3, 3]$

Generation 170: Best = 3.248122e+03 FES = 17100 $\mu_F = 0.789$ $\mu_{CR} = 0.445$

Diversity = 1.1897e+00 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 3.242916e+03 FES = 18100 $\mu_F = 0.783$ $\mu_{CR} = 0.453$

Diversity = 1.0790e+00 $k_t = [3, 4, 2, 3]$

Generation 190: Best = 3.237890e+03 FES = 19100 $\mu_F = 0.786$ $\mu_{CR} = 0.456$

Diversity = 1.0314e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 3.231485e+03 FES = 20100 $\mu_F = 0.786$ $\mu_{CR} = 0.463$

Diversity = 9.6509e-01 $k_t = [3, 4, 2, 4]$

Generation 210: Best = 3.227917e+03 FES = 21100 $\mu_F = 0.783$ $\mu_{CR} = 0.464$

Diversity = 9.2208e-01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 3.222495e+03 FES = 22100 $\mu_F = 0.774$ $\mu_{CR} = 0.476$

Diversity = 8.9912e-01	k_t = [4, 4, 4, 2]				
Generation 230: Best = 3.220330e+03	FES = 23100	$\mu_F = 0.767$	$\mu_{CR} = 0.475$		
Diversity = 7.8780e-01	k_t = [1, 1, 1, 1]				
Generation 240: Best = 3.218069e+03	FES = 24100	$\mu_F = 0.761$	$\mu_{CR} = 0.473$		
Diversity = 7.2667e-01	k_t = [4, 2, 4, 2]				
Generation 250: Best = 3.216697e+03	FES = 25100	$\mu_F = 0.754$	$\mu_{CR} = 0.466$		
Diversity = 6.6310e-01	k_t = [1, 1, 1, 1]				
Generation 260: Best = 3.215889e+03	FES = 26100	$\mu_F = 0.732$	$\mu_{CR} = 0.471$		
Diversity = 4.7206e-01	k_t = [2, 2, 4, 4]				
Generation 270: Best = 3.215407e+03	FES = 27100	$\mu_F = 0.730$	$\mu_{CR} = 0.476$		
Diversity = 4.2933e-01	k_t = [1, 1, 1, 1]				
Generation 280: Best = 3.214860e+03	FES = 28100	$\mu_F = 0.748$	$\mu_{CR} = 0.482$		
Diversity = 3.2518e-01	k_t = [2, 4, 3, 4]				
Generation 290: Best = 3.214507e+03	FES = 29100	$\mu_F = 0.741$	$\mu_{CR} = 0.483$		
Diversity = 3.0762e-01	k_t = [1, 1, 1, 1]				
Generation 300: Best = 3.214276e+03	FES = 30100	$\mu_F = 0.734$	$\mu_{CR} = 0.477$		
Diversity = 2.4542e-01	k_t = [4, 3, 2, 3]				
Generation 310: Best = 3.214132e+03	FES = 31100	$\mu_F = 0.730$	$\mu_{CR} = 0.481$		
Diversity = 2.0523e-01	k_t = [1, 1, 1, 1]				
Generation 320: Best = 3.214056e+03	FES = 32100	$\mu_F = 0.721$	$\mu_{CR} = 0.475$		
Diversity = 1.6351e-01	k_t = [2, 4, 2, 3]				
Generation 330: Best = 3.213943e+03	FES = 33100	$\mu_F = 0.731$	$\mu_{CR} = 0.476$		
Diversity = 1.4512e-01	k_t = [1, 1, 1, 1]				
Generation 340: Best = 3.213879e+03	FES = 34100	$\mu_F = 0.730$	$\mu_{CR} = 0.470$		
Diversity = 1.1853e-01	k_t = [2, 2, 3, 4]				
Generation 350: Best = 3.213734e+03	FES = 35100	$\mu_F = 0.742$	$\mu_{CR} = 0.483$		
Diversity = 1.3744e-01	k_t = [1, 1, 1, 1]				
Generation 360: Best = 3.213669e+03	FES = 36100	$\mu_F = 0.748$	$\mu_{CR} = 0.482$		
Diversity = 1.3630e-01	k_t = [4, 4, 3, 3]				
Generation 370: Best = 3.213550e+03	FES = 37100	$\mu_F = 0.755$	$\mu_{CR} = 0.484$		
Diversity = 1.3550e-01	k_t = [1, 1, 1, 1]				
Generation 380: Best = 3.213475e+03	FES = 38100	$\mu_F = 0.770$	$\mu_{CR} = 0.489$		
Diversity = 1.3148e-01	k_t = [4, 2, 3, 4]				
Generation 390: Best = 3.213416e+03	FES = 39100	$\mu_F = 0.767$	$\mu_{CR} = 0.486$		
Diversity = 1.2086e-01	k_t = [1, 1, 1, 1]				
Generation 400: Best = 3.213369e+03	FES = 40100	$\mu_F = 0.748$	$\mu_{CR} = 0.493$		
Diversity = 1.0282e-01	k_t = [2, 4, 2, 3]				
Generation 410: Best = 3.213327e+03	FES = 41100	$\mu_F = 0.736$	$\mu_{CR} = 0.502$		
Diversity = 1.0052e-01	k_t = [1, 1, 1, 1]				
Generation 420: Best = 3.213277e+03	FES = 42100	$\mu_F = 0.738$	$\mu_{CR} = 0.498$		
Diversity = 9.4849e-02	k_t = [4, 2, 3, 2]				
Generation 430: Best = 3.213234e+03	FES = 43100	$\mu_F = 0.732$	$\mu_{CR} = 0.508$		
Diversity = 9.0773e-02	k_t = [1, 1, 1, 1]				
Generation 440: Best = 3.213195e+03	FES = 44100	$\mu_F = 0.728$	$\mu_{CR} = 0.511$		
Diversity = 6.9595e-02	k_t = [3, 3, 2, 4]				
Generation 450: Best = 3.213167e+03	FES = 45100	$\mu_F = 0.725$	$\mu_{CR} = 0.517$		
Diversity = 6.6080e-02	k_t = [1, 1, 1, 1]				
Generation 460: Best = 3.213146e+03	FES = 46100	$\mu_F = 0.727$	$\mu_{CR} = 0.518$		
Diversity = 6.2710e-02	k_t = [2, 3, 4, 2]				
Generation 470: Best = 3.213125e+03	FES = 47100	$\mu_F = 0.727$	$\mu_{CR} = 0.519$		
Diversity = 6.6557e-02	k_t = [1, 1, 1, 1]				
Generation 480: Best = 3.213103e+03	FES = 48100	$\mu_F = 0.719$	$\mu_{CR} = 0.520$		
Diversity = 6.4456e-02	k_t = [3, 2, 3, 2]				
Generation 490: Best = 3.213082e+03	FES = 49100	$\mu_F = 0.728$	$\mu_{CR} = 0.526$		

Diversity = 6.8841e-02	k_t = [1, 1, 1, 1]					
Generation 500: Best = 3.213051e+03	FES = 50100	$\mu_F = 0.737$	$\mu_{CR} = 0.527$			
Diversity = 6.0586e-02	k_t = [4, 4, 2, 4]					
Generation 510: Best = 3.213025e+03	FES = 51100	$\mu_F = 0.761$	$\mu_{CR} = 0.536$			
Diversity = 7.7010e-02	k_t = [1, 1, 1, 1]					
Generation 520: Best = 3.212986e+03	FES = 52100	$\mu_F = 0.755$	$\mu_{CR} = 0.539$			
Diversity = 7.3100e-02	k_t = [4, 4, 2, 3]					
Generation 530: Best = 3.212965e+03	FES = 53100	$\mu_F = 0.756$	$\mu_{CR} = 0.532$			
Diversity = 7.6896e-02	k_t = [1, 1, 1, 1]					
Generation 540: Best = 3.212942e+03	FES = 54100	$\mu_F = 0.755$	$\mu_{CR} = 0.527$			
Diversity = 6.8378e-02	k_t = [2, 4, 2, 2]					
Generation 550: Best = 3.212928e+03	FES = 55100	$\mu_F = 0.744$	$\mu_{CR} = 0.527$			
Diversity = 6.2100e-02	k_t = [1, 1, 1, 1]					
Generation 560: Best = 3.212911e+03	FES = 56100	$\mu_F = 0.751$	$\mu_{CR} = 0.528$			
Diversity = 5.1609e-02	k_t = [4, 2, 3, 3]					
Generation 570: Best = 3.212895e+03	FES = 57100	$\mu_F = 0.740$	$\mu_{CR} = 0.526$			
Diversity = 5.1173e-02	k_t = [1, 1, 1, 1]					
Generation 580: Best = 3.212879e+03	FES = 58100	$\mu_F = 0.737$	$\mu_{CR} = 0.515$			
Diversity = 4.5302e-02	k_t = [3, 2, 4, 3]					
Generation 590: Best = 3.212865e+03	FES = 59100	$\mu_F = 0.751$	$\mu_{CR} = 0.499$			
Diversity = 4.6917e-02	k_t = [1, 1, 1, 1]					
Generation 600: Best = 3.212848e+03	FES = 60100	$\mu_F = 0.773$	$\mu_{CR} = 0.492$			
Diversity = 4.6652e-02	k_t = [4, 2, 2, 4]					
Generation 610: Best = 3.212832e+03	FES = 61100	$\mu_F = 0.773$	$\mu_{CR} = 0.493$			
Diversity = 5.1236e-02	k_t = [1, 1, 1, 1]					
Generation 620: Best = 3.212817e+03	FES = 62100	$\mu_F = 0.768$	$\mu_{CR} = 0.497$			
Diversity = 5.3252e-02	k_t = [3, 4, 4, 2]					
Generation 630: Best = 3.212800e+03	FES = 63100	$\mu_F = 0.771$	$\mu_{CR} = 0.507$			
Diversity = 5.7604e-02	k_t = [1, 1, 1, 1]					
Generation 640: Best = 3.212783e+03	FES = 64100	$\mu_F = 0.774$	$\mu_{CR} = 0.497$			
Diversity = 5.9500e-02	k_t = [2, 4, 4, 2]					
Generation 650: Best = 3.212763e+03	FES = 65100	$\mu_F = 0.776$	$\mu_{CR} = 0.504$			
Diversity = 6.5807e-02	k_t = [1, 1, 1, 1]					
Generation 660: Best = 3.212745e+03	FES = 66100	$\mu_F = 0.768$	$\mu_{CR} = 0.497$			
Diversity = 6.4268e-02	k_t = [3, 4, 4, 4]					
Generation 670: Best = 3.212717e+03	FES = 67100	$\mu_F = 0.770$	$\mu_{CR} = 0.498$			
Diversity = 6.4630e-02	k_t = [1, 1, 1, 1]					
Generation 680: Best = 3.212697e+03	FES = 68100	$\mu_F = 0.772$	$\mu_{CR} = 0.497$			
Diversity = 6.6305e-02	k_t = [2, 4, 2, 4]					
Generation 690: Best = 3.212670e+03	FES = 69100	$\mu_F = 0.766$	$\mu_{CR} = 0.510$			
Diversity = 5.9564e-02	k_t = [1, 1, 1, 1]					
Generation 700: Best = 3.212653e+03	FES = 70100	$\mu_F = 0.768$	$\mu_{CR} = 0.505$			
Diversity = 5.4819e-02	k_t = [3, 3, 4, 3]					
Generation 710: Best = 3.212632e+03	FES = 71100	$\mu_F = 0.771$	$\mu_{CR} = 0.512$			
Diversity = 5.5854e-02	k_t = [1, 1, 1, 1]					
Generation 720: Best = 3.212610e+03	FES = 72100	$\mu_F = 0.767$	$\mu_{CR} = 0.500$			
Diversity = 5.1586e-02	k_t = [4, 4, 4, 3]					
Generation 730: Best = 3.212592e+03	FES = 73100	$\mu_F = 0.764$	$\mu_{CR} = 0.498$			
Diversity = 5.2637e-02	k_t = [1, 1, 1, 1]					
Generation 740: Best = 3.212578e+03	FES = 74100	$\mu_F = 0.779$	$\mu_{CR} = 0.497$			
Diversity = 5.3930e-02	k_t = [3, 3, 4, 4]					
Generation 750: Best = 3.212565e+03	FES = 75100	$\mu_F = 0.776$	$\mu_{CR} = 0.491$			
Diversity = 4.6946e-02	k_t = [1, 1, 1, 1]					
Generation 760: Best = 3.212550e+03	FES = 76100	$\mu_F = 0.768$	$\mu_{CR} = 0.492$			

Diversity = 4.5289e-02	k_t = [4, 4, 4, 3]				
Generation 770: Best = 3.212537e+03	FES = 77100	$\mu_F = 0.756$	$\mu_{CR} = 0.500$		
Diversity = 5.2018e-02	k_t = [1, 1, 1, 1]				
Generation 780: Best = 3.212526e+03	FES = 78100	$\mu_F = 0.752$	$\mu_{CR} = 0.502$		
Diversity = 4.2514e-02	k_t = [4, 2, 4, 3]				
Generation 790: Best = 3.212519e+03	FES = 79100	$\mu_F = 0.748$	$\mu_{CR} = 0.509$		
Diversity = 4.2488e-02	k_t = [1, 1, 1, 1]				
Generation 800: Best = 3.212511e+03	FES = 80100	$\mu_F = 0.739$	$\mu_{CR} = 0.509$		
Diversity = 3.6332e-02	k_t = [4, 2, 4, 4]				
Generation 810: Best = 3.212500e+03	FES = 81100	$\mu_F = 0.745$	$\mu_{CR} = 0.511$		
Diversity = 3.8722e-02	k_t = [1, 1, 1, 1]				
Generation 820: Best = 3.212489e+03	FES = 82100	$\mu_F = 0.740$	$\mu_{CR} = 0.515$		
Diversity = 3.8433e-02	k_t = [3, 4, 3, 4]				
Generation 830: Best = 3.212478e+03	FES = 83100	$\mu_F = 0.746$	$\mu_{CR} = 0.514$		
Diversity = 3.9007e-02	k_t = [1, 1, 1, 1]				
Generation 840: Best = 3.212473e+03	FES = 84100	$\mu_F = 0.734$	$\mu_{CR} = 0.514$		
Diversity = 3.2862e-02	k_t = [2, 4, 4, 2]				
Generation 850: Best = 3.212467e+03	FES = 85100	$\mu_F = 0.738$	$\mu_{CR} = 0.516$		
Diversity = 2.9939e-02	k_t = [1, 1, 1, 1]				
Generation 860: Best = 3.212460e+03	FES = 86100	$\mu_F = 0.754$	$\mu_{CR} = 0.505$		
Diversity = 3.1258e-02	k_t = [2, 4, 2, 2]				
Generation 870: Best = 3.212455e+03	FES = 87100	$\mu_F = 0.753$	$\mu_{CR} = 0.506$		
Diversity = 3.3448e-02	k_t = [1, 1, 1, 1]				
Generation 880: Best = 3.212449e+03	FES = 88100	$\mu_F = 0.761$	$\mu_{CR} = 0.507$		
Diversity = 3.4356e-02	k_t = [2, 3, 2, 2]				
Generation 890: Best = 3.212441e+03	FES = 89100	$\mu_F = 0.760$	$\mu_{CR} = 0.506$		
Diversity = 4.1529e-02	k_t = [1, 1, 1, 1]				
Generation 900: Best = 3.212433e+03	FES = 90100	$\mu_F = 0.763$	$\mu_{CR} = 0.502$		
Diversity = 3.5133e-02	k_t = [2, 4, 4, 2]				
Generation 910: Best = 3.212425e+03	FES = 91100	$\mu_F = 0.756$	$\mu_{CR} = 0.518$		
Diversity = 3.6500e-02	k_t = [1, 1, 1, 1]				
Generation 920: Best = 3.212419e+03	FES = 92100	$\mu_F = 0.750$	$\mu_{CR} = 0.525$		
Diversity = 3.2194e-02	k_t = [4, 3, 4, 3]				
Generation 930: Best = 3.212407e+03	FES = 93100	$\mu_F = 0.762$	$\mu_{CR} = 0.528$		
Diversity = 3.8735e-02	k_t = [1, 1, 1, 1]				
Generation 940: Best = 3.212398e+03	FES = 94100	$\mu_F = 0.770$	$\mu_{CR} = 0.540$		
Diversity = 3.7028e-02	k_t = [3, 2, 4, 2]				
Generation 950: Best = 3.212391e+03	FES = 95100	$\mu_F = 0.782$	$\mu_{CR} = 0.550$		
Diversity = 4.1138e-02	k_t = [1, 1, 1, 1]				
Generation 960: Best = 3.212377e+03	FES = 96100	$\mu_F = 0.789$	$\mu_{CR} = 0.545$		
Diversity = 3.9333e-02	k_t = [3, 4, 3, 4]				
Generation 970: Best = 3.212368e+03	FES = 97100	$\mu_F = 0.779$	$\mu_{CR} = 0.557$		
Diversity = 4.9603e-02	k_t = [1, 1, 1, 1]				
Generation 980: Best = 3.212355e+03	FES = 98100	$\mu_F = 0.760$	$\mu_{CR} = 0.560$		
Diversity = 4.4447e-02	k_t = [3, 3, 3, 4]				
Generation 990: Best = 3.212344e+03	FES = 99100	$\mu_F = 0.766$	$\mu_{CR} = 0.557$		
Diversity = 4.6879e-02	k_t = [1, 1, 1, 1]				
Generation 1000: Best = 3.212334e+03	FES = 100100	$\mu_F = 0.760$	$\mu_{CR} = 0.554$		
Diversity = 3.9276e-02	k_t = [4, 4, 4, 4]				
Generation 1010: Best = 3.212324e+03	FES = 101100	$\mu_F = 0.771$	$\mu_{CR} = 0.554$		
Diversity = 3.9596e-02	k_t = [1, 1, 1, 1]				
Generation 1020: Best = 3.212321e+03	FES = 102100	$\mu_F = 0.748$	$\mu_{CR} = 0.555$		
Diversity = 3.4910e-02	k_t = [2, 4, 2, 2]				
Generation 1030: Best = 3.212311e+03	FES = 103100	$\mu_F = 0.745$	$\mu_{CR} = 0.556$		

Diversity = 3.4992e-02	k_t = [1, 1, 1, 1]			
Generation 1040: Best = 3.212306e+03	FES = 104100	$\mu_F = 0.730$	$\mu_{CR} = 0.557$	
Diversity = 3.1202e-02	k_t = [3, 3, 2, 3]			
Generation 1050: Best = 3.212299e+03	FES = 105100	$\mu_F = 0.742$	$\mu_{CR} = 0.557$	
Diversity = 3.0746e-02	k_t = [1, 1, 1, 1]			
Generation 1060: Best = 3.212294e+03	FES = 106100	$\mu_F = 0.746$	$\mu_{CR} = 0.554$	
Diversity = 2.5558e-02	k_t = [2, 2, 2, 3]			
Generation 1070: Best = 3.212289e+03	FES = 107100	$\mu_F = 0.755$	$\mu_{CR} = 0.553$	
Diversity = 2.6336e-02	k_t = [1, 1, 1, 1]			
Generation 1080: Best = 3.212286e+03	FES = 108100	$\mu_F = 0.768$	$\mu_{CR} = 0.554$	
Diversity = 2.5957e-02	k_t = [2, 3, 3, 2]			
Generation 1090: Best = 3.212279e+03	FES = 109100	$\mu_F = 0.768$	$\mu_{CR} = 0.552$	
Diversity = 2.6620e-02	k_t = [1, 1, 1, 1]			
Generation 1100: Best = 3.212275e+03	FES = 110100	$\mu_F = 0.763$	$\mu_{CR} = 0.563$	
Diversity = 2.5308e-02	k_t = [2, 4, 4, 2]			
Generation 1110: Best = 3.212271e+03	FES = 111100	$\mu_F = 0.757$	$\mu_{CR} = 0.560$	
Diversity = 2.6795e-02	k_t = [1, 1, 1, 1]			
Generation 1120: Best = 3.212267e+03	FES = 112100	$\mu_F = 0.756$	$\mu_{CR} = 0.553$	
Diversity = 2.0884e-02	k_t = [4, 3, 4, 4]			
Generation 1130: Best = 3.212263e+03	FES = 113100	$\mu_F = 0.773$	$\mu_{CR} = 0.549$	
Diversity = 2.4576e-02	k_t = [1, 1, 1, 1]			
Generation 1140: Best = 3.212260e+03	FES = 114100	$\mu_F = 0.770$	$\mu_{CR} = 0.539$	
Diversity = 2.1786e-02	k_t = [2, 4, 4, 4]			
Generation 1150: Best = 3.212256e+03	FES = 115100	$\mu_F = 0.760$	$\mu_{CR} = 0.550$	
Diversity = 2.3469e-02	k_t = [1, 1, 1, 1]			
Generation 1160: Best = 3.212251e+03	FES = 116100	$\mu_F = 0.782$	$\mu_{CR} = 0.548$	
Diversity = 2.7551e-02	k_t = [2, 3, 4, 4]			
Generation 1170: Best = 3.212243e+03	FES = 117100	$\mu_F = 0.791$	$\mu_{CR} = 0.549$	
Diversity = 3.6197e-02	k_t = [1, 1, 1, 1]			
Generation 1180: Best = 3.212228e+03	FES = 118100	$\mu_F = 0.794$	$\mu_{CR} = 0.555$	
Diversity = 3.7576e-02	k_t = [2, 2, 4, 4]			
Generation 1190: Best = 3.212209e+03	FES = 119100	$\mu_F = 0.791$	$\mu_{CR} = 0.549$	
Diversity = 4.9040e-02	k_t = [1, 1, 1, 1]			
Generation 1200: Best = 3.212188e+03	FES = 120100	$\mu_F = 0.776$	$\mu_{CR} = 0.558$	
Diversity = 4.8280e-02	k_t = [4, 4, 3, 4]			
Generation 1210: Best = 3.212163e+03	FES = 121100	$\mu_F = 0.780$	$\mu_{CR} = 0.558$	
Diversity = 5.9805e-02	k_t = [1, 1, 1, 1]			
Generation 1220: Best = 3.212146e+03	FES = 122100	$\mu_F = 0.762$	$\mu_{CR} = 0.556$	
Diversity = 5.6570e-02	k_t = [3, 2, 3, 4]			
Generation 1230: Best = 3.212139e+03	FES = 123100	$\mu_F = 0.755$	$\mu_{CR} = 0.556$	
Diversity = 5.2510e-02	k_t = [1, 1, 1, 1]			
Generation 1240: Best = 3.212125e+03	FES = 124100	$\mu_F = 0.749$	$\mu_{CR} = 0.566$	
Diversity = 5.0223e-02	k_t = [4, 4, 3, 3]			
Generation 1250: Best = 3.212109e+03	FES = 125100	$\mu_F = 0.768$	$\mu_{CR} = 0.567$	
Diversity = 5.3724e-02	k_t = [1, 1, 1, 1]			
Generation 1260: Best = 3.212095e+03	FES = 126100	$\mu_F = 0.764$	$\mu_{CR} = 0.560$	
Diversity = 5.6530e-02	k_t = [4, 3, 2, 2]			
Generation 1270: Best = 3.212080e+03	FES = 127100	$\mu_F = 0.771$	$\mu_{CR} = 0.560$	
Diversity = 5.1783e-02	k_t = [1, 1, 1, 1]			
Generation 1280: Best = 3.212068e+03	FES = 128100	$\mu_F = 0.767$	$\mu_{CR} = 0.559$	
Diversity = 4.9163e-02	k_t = [2, 4, 2, 3]			
Generation 1290: Best = 3.212051e+03	FES = 129100	$\mu_F = 0.763$	$\mu_{CR} = 0.559$	
Diversity = 5.6574e-02	k_t = [1, 1, 1, 1]			
Generation 1300: Best = 3.212035e+03	FES = 130100	$\mu_F = 0.750$	$\mu_{CR} = 0.559$	

Diversity = 5.3799e-02	k_t = [4, 2, 3, 2]			
Generation 1310: Best = 3.212027e+03	FES = 131100	$\mu_F = 0.743$	$\mu_{CR} = 0.550$	
Diversity = 4.7899e-02	k_t = [1, 1, 1, 1]			
Generation 1320: Best = 3.212020e+03	FES = 132100	$\mu_F = 0.747$	$\mu_{CR} = 0.549$	
Diversity = 3.8723e-02	k_t = [2, 2, 4, 3]			
Generation 1330: Best = 3.212011e+03	FES = 133100	$\mu_F = 0.753$	$\mu_{CR} = 0.558$	
Diversity = 3.8371e-02	k_t = [1, 1, 1, 1]			
Generation 1340: Best = 3.212005e+03	FES = 134100	$\mu_F = 0.753$	$\mu_{CR} = 0.556$	
Diversity = 3.8198e-02	k_t = [3, 3, 4, 3]			
Generation 1350: Best = 3.211994e+03	FES = 135100	$\mu_F = 0.748$	$\mu_{CR} = 0.559$	
Diversity = 3.9472e-02	k_t = [1, 1, 1, 1]			
Generation 1360: Best = 3.211986e+03	FES = 136100	$\mu_F = 0.757$	$\mu_{CR} = 0.565$	
Diversity = 3.9373e-02	k_t = [3, 3, 4, 2]			
Generation 1370: Best = 3.211980e+03	FES = 137100	$\mu_F = 0.746$	$\mu_{CR} = 0.569$	
Diversity = 4.3064e-02	k_t = [1, 1, 1, 1]			
Generation 1380: Best = 3.211972e+03	FES = 138100	$\mu_F = 0.743$	$\mu_{CR} = 0.569$	
Diversity = 4.0688e-02	k_t = [4, 4, 3, 2]			
Generation 1390: Best = 3.211966e+03	FES = 139100	$\mu_F = 0.736$	$\mu_{CR} = 0.569$	
Diversity = 3.8593e-02	k_t = [1, 1, 1, 1]			
Generation 1400: Best = 3.211957e+03	FES = 140100	$\mu_F = 0.746$	$\mu_{CR} = 0.566$	
Diversity = 3.9306e-02	k_t = [3, 4, 4, 2]			
Generation 1410: Best = 3.211952e+03	FES = 141100	$\mu_F = 0.738$	$\mu_{CR} = 0.570$	
Diversity = 3.3808e-02	k_t = [1, 1, 1, 1]			
Generation 1420: Best = 3.211945e+03	FES = 142100	$\mu_F = 0.745$	$\mu_{CR} = 0.558$	
Diversity = 3.5290e-02	k_t = [3, 3, 2, 2]			
Generation 1430: Best = 3.211936e+03	FES = 143100	$\mu_F = 0.739$	$\mu_{CR} = 0.562$	
Diversity = 4.4854e-02	k_t = [1, 1, 1, 1]			
Generation 1440: Best = 3.211929e+03	FES = 144100	$\mu_F = 0.737$	$\mu_{CR} = 0.558$	
Diversity = 3.7118e-02	k_t = [4, 2, 2, 2]			
Generation 1450: Best = 3.211920e+03	FES = 145100	$\mu_F = 0.746$	$\mu_{CR} = 0.546$	
Diversity = 4.1865e-02	k_t = [1, 1, 1, 1]			
Generation 1460: Best = 3.211910e+03	FES = 146100	$\mu_F = 0.742$	$\mu_{CR} = 0.532$	
Diversity = 4.2358e-02	k_t = [4, 2, 2, 2]			
Generation 1470: Best = 3.211902e+03	FES = 147100	$\mu_F = 0.748$	$\mu_{CR} = 0.538$	
Diversity = 4.1617e-02	k_t = [1, 1, 1, 1]			
Generation 1480: Best = 3.211898e+03	FES = 148100	$\mu_F = 0.733$	$\mu_{CR} = 0.541$	
Diversity = 3.7163e-02	k_t = [3, 3, 3, 2]			
Generation 1490: Best = 3.211893e+03	FES = 149100	$\mu_F = 0.725$	$\mu_{CR} = 0.541$	
Diversity = 3.3001e-02	k_t = [1, 1, 1, 1]			
Generation 1500: Best = 3.211887e+03	FES = 150100	$\mu_F = 0.727$	$\mu_{CR} = 0.541$	
Diversity = 2.8946e-02	k_t = [4, 2, 3, 3]			
Generation 1510: Best = 3.211882e+03	FES = 151100	$\mu_F = 0.737$	$\mu_{CR} = 0.536$	
Diversity = 2.7327e-02	k_t = [1, 1, 1, 1]			
Generation 1520: Best = 3.211876e+03	FES = 152100	$\mu_F = 0.753$	$\mu_{CR} = 0.532$	
Diversity = 2.6915e-02	k_t = [3, 2, 4, 4]			
Generation 1530: Best = 3.211871e+03	FES = 153100	$\mu_F = 0.762$	$\mu_{CR} = 0.529$	
Diversity = 2.9721e-02	k_t = [1, 1, 1, 1]			
Generation 1540: Best = 3.211866e+03	FES = 154100	$\mu_F = 0.764$	$\mu_{CR} = 0.524$	
Diversity = 2.7773e-02	k_t = [4, 3, 2, 2]			
Generation 1550: Best = 3.211861e+03	FES = 155100	$\mu_F = 0.764$	$\mu_{CR} = 0.519$	
Diversity = 3.0677e-02	k_t = [1, 1, 1, 1]			
Generation 1560: Best = 3.211855e+03	FES = 156100	$\mu_F = 0.760$	$\mu_{CR} = 0.509$	
Diversity = 3.0529e-02	k_t = [4, 2, 4, 3]			
Generation 1570: Best = 3.211849e+03	FES = 157100	$\mu_F = 0.768$	$\mu_{CR} = 0.502$	

Diversity = 2.8996e-02	k_t = [1, 1, 1, 1]			
Generation 1580: Best = 3.211843e+03	FES = 158100	$\mu_F = 0.767$	$\mu_{CR} = 0.497$	
Diversity = 2.8230e-02	k_t = [2, 3, 2, 3]			
Generation 1590: Best = 3.211836e+03	FES = 159100	$\mu_F = 0.772$	$\mu_{CR} = 0.490$	
Diversity = 3.3769e-02	k_t = [1, 1, 1, 1]			
Generation 1600: Best = 3.211829e+03	FES = 160100	$\mu_F = 0.778$	$\mu_{CR} = 0.488$	
Diversity = 3.8522e-02	k_t = [2, 3, 3, 2]			
Generation 1610: Best = 3.211824e+03	FES = 161100	$\mu_F = 0.774$	$\mu_{CR} = 0.475$	
Diversity = 4.0021e-02	k_t = [1, 1, 1, 1]			
Generation 1620: Best = 3.211815e+03	FES = 162100	$\mu_F = 0.778$	$\mu_{CR} = 0.467$	
Diversity = 4.3895e-02	k_t = [2, 2, 3, 2]			
Generation 1630: Best = 3.211807e+03	FES = 163100	$\mu_F = 0.746$	$\mu_{CR} = 0.458$	
Diversity = 4.3461e-02	k_t = [1, 1, 1, 1]			
Generation 1640: Best = 3.211796e+03	FES = 164100	$\mu_F = 0.743$	$\mu_{CR} = 0.459$	
Diversity = 4.3727e-02	k_t = [2, 4, 2, 2]			
Generation 1650: Best = 3.211785e+03	FES = 165100	$\mu_F = 0.736$	$\mu_{CR} = 0.463$	
Diversity = 4.9669e-02	k_t = [1, 1, 1, 1]			
Generation 1660: Best = 3.211772e+03	FES = 166100	$\mu_F = 0.739$	$\mu_{CR} = 0.464$	
Diversity = 4.1992e-02	k_t = [4, 4, 3, 4]			
Generation 1670: Best = 3.211763e+03	FES = 167100	$\mu_F = 0.751$	$\mu_{CR} = 0.461$	
Diversity = 3.8662e-02	k_t = [1, 1, 1, 1]			
Generation 1680: Best = 3.211755e+03	FES = 168100	$\mu_F = 0.754$	$\mu_{CR} = 0.464$	
Diversity = 3.5156e-02	k_t = [4, 3, 4, 2]			
Generation 1690: Best = 3.211749e+03	FES = 169100	$\mu_F = 0.764$	$\mu_{CR} = 0.459$	
Diversity = 3.3723e-02	k_t = [1, 1, 1, 1]			
Generation 1700: Best = 3.211738e+03	FES = 170100	$\mu_F = 0.771$	$\mu_{CR} = 0.459$	
Diversity = 3.7958e-02	k_t = [4, 2, 3, 3]			
Generation 1710: Best = 3.211736e+03	FES = 171100	$\mu_F = 0.768$	$\mu_{CR} = 0.452$	
Diversity = 3.8945e-02	k_t = [1, 1, 1, 1]			
Generation 1720: Best = 3.211730e+03	FES = 172100	$\mu_F = 0.769$	$\mu_{CR} = 0.445$	
Diversity = 3.4034e-02	k_t = [3, 4, 4, 4]			
Generation 1730: Best = 3.211723e+03	FES = 173100	$\mu_F = 0.765$	$\mu_{CR} = 0.442$	
Diversity = 3.7925e-02	k_t = [1, 1, 1, 1]			
Generation 1740: Best = 3.211719e+03	FES = 174100	$\mu_F = 0.765$	$\mu_{CR} = 0.436$	
Diversity = 3.7147e-02	k_t = [2, 4, 4, 2]			
Generation 1750: Best = 3.211715e+03	FES = 175100	$\mu_F = 0.764$	$\mu_{CR} = 0.428$	
Diversity = 3.4105e-02	k_t = [1, 1, 1, 1]			
Generation 1760: Best = 3.211708e+03	FES = 176100	$\mu_F = 0.763$	$\mu_{CR} = 0.415$	
Diversity = 3.4330e-02	k_t = [4, 2, 2, 2]			
Generation 1770: Best = 3.211703e+03	FES = 177100	$\mu_F = 0.758$	$\mu_{CR} = 0.411$	
Diversity = 3.1392e-02	k_t = [1, 1, 1, 1]			
Generation 1780: Best = 3.211694e+03	FES = 178100	$\mu_F = 0.764$	$\mu_{CR} = 0.407$	
Diversity = 2.9961e-02	k_t = [4, 2, 3, 3]			
Generation 1790: Best = 3.211687e+03	FES = 179100	$\mu_F = 0.768$	$\mu_{CR} = 0.403$	
Diversity = 3.8403e-02	k_t = [1, 1, 1, 1]			
Generation 1800: Best = 3.211675e+03	FES = 180100	$\mu_F = 0.773$	$\mu_{CR} = 0.416$	
Diversity = 3.9462e-02	k_t = [3, 3, 2, 2]			
Generation 1810: Best = 3.211663e+03	FES = 181100	$\mu_F = 0.772$	$\mu_{CR} = 0.417$	
Diversity = 4.5094e-02	k_t = [1, 1, 1, 1]			
Generation 1820: Best = 3.211658e+03	FES = 182100	$\mu_F = 0.767$	$\mu_{CR} = 0.413$	
Diversity = 4.2089e-02	k_t = [2, 3, 4, 2]			
Generation 1830: Best = 3.211649e+03	FES = 183100	$\mu_F = 0.768$	$\mu_{CR} = 0.403$	
Diversity = 4.1577e-02	k_t = [1, 1, 1, 1]			
Generation 1840: Best = 3.211638e+03	FES = 184100	$\mu_F = 0.775$	$\mu_{CR} = 0.406$	

Diversity = 3.9010e-02	k_t = [2, 2, 2, 4]			
Generation 1850: Best = 3.211627e+03	FES = 185100	$\mu_F = 0.782$	$\mu_{CR} = 0.401$	
Diversity = 5.1612e-02	k_t = [1, 1, 1, 1]			
Generation 1860: Best = 3.211612e+03	FES = 186100	$\mu_F = 0.760$	$\mu_{CR} = 0.401$	
Diversity = 5.2359e-02	k_t = [4, 3, 4, 3]			
Generation 1870: Best = 3.211605e+03	FES = 187100	$\mu_F = 0.765$	$\mu_{CR} = 0.400$	
Diversity = 5.3059e-02	k_t = [1, 1, 1, 1]			
Generation 1880: Best = 3.211596e+03	FES = 188100	$\mu_F = 0.766$	$\mu_{CR} = 0.395$	
Diversity = 4.3972e-02	k_t = [4, 2, 4, 4]			
Generation 1890: Best = 3.211586e+03	FES = 189100	$\mu_F = 0.755$	$\mu_{CR} = 0.406$	
Diversity = 4.7283e-02	k_t = [1, 1, 1, 1]			
Generation 1900: Best = 3.211580e+03	FES = 190100	$\mu_F = 0.755$	$\mu_{CR} = 0.405$	
Diversity = 4.0087e-02	k_t = [4, 2, 4, 3]			
Generation 1910: Best = 3.211574e+03	FES = 191100	$\mu_F = 0.747$	$\mu_{CR} = 0.404$	
Diversity = 4.2636e-02	k_t = [1, 1, 1, 1]			
Generation 1920: Best = 3.211565e+03	FES = 192100	$\mu_F = 0.745$	$\mu_{CR} = 0.412$	
Diversity = 3.6325e-02	k_t = [2, 3, 4, 4]			
Generation 1930: Best = 3.211557e+03	FES = 193100	$\mu_F = 0.753$	$\mu_{CR} = 0.405$	
Diversity = 3.6379e-02	k_t = [1, 1, 1, 1]			
Generation 1940: Best = 3.211548e+03	FES = 194100	$\mu_F = 0.751$	$\mu_{CR} = 0.395$	
Diversity = 3.9076e-02	k_t = [4, 3, 2, 2]			
Generation 1950: Best = 3.211541e+03	FES = 195100	$\mu_F = 0.748$	$\mu_{CR} = 0.395$	
Diversity = 4.2425e-02	k_t = [1, 1, 1, 1]			
Generation 1960: Best = 3.211528e+03	FES = 196100	$\mu_F = 0.759$	$\mu_{CR} = 0.390$	
Diversity = 3.9567e-02	k_t = [4, 4, 3, 3]			
Generation 1970: Best = 3.211515e+03	FES = 197100	$\mu_F = 0.777$	$\mu_{CR} = 0.387$	
Diversity = 4.9437e-02	k_t = [1, 1, 1, 1]			
Generation 1980: Best = 3.211504e+03	FES = 198100	$\mu_F = 0.782$	$\mu_{CR} = 0.387$	
Diversity = 4.6827e-02	k_t = [2, 3, 3, 4]			
Generation 1990: Best = 3.211487e+03	FES = 199100	$\mu_F = 0.788$	$\mu_{CR} = 0.385$	
Diversity = 4.7916e-02	k_t = [1, 1, 1, 1]			
Generation 2000: Best = 3.211466e+03	FES = 200100	$\mu_F = 0.784$	$\mu_{CR} = 0.382$	
Diversity = 5.2584e-02	k_t = [4, 3, 3, 3]			
Generation 2010: Best = 3.211458e+03	FES = 201100	$\mu_F = 0.784$	$\mu_{CR} = 0.381$	
Diversity = 5.7524e-02	k_t = [1, 1, 1, 1]			
Generation 2020: Best = 3.211442e+03	FES = 202100	$\mu_F = 0.780$	$\mu_{CR} = 0.373$	
Diversity = 5.1834e-02	k_t = [3, 4, 3, 4]			
Generation 2030: Best = 3.211433e+03	FES = 203100	$\mu_F = 0.780$	$\mu_{CR} = 0.373$	
Diversity = 4.6896e-02	k_t = [1, 1, 1, 1]			
Generation 2040: Best = 3.211419e+03	FES = 204100	$\mu_F = 0.768$	$\mu_{CR} = 0.368$	
Diversity = 4.1263e-02	k_t = [3, 2, 4, 4]			
Generation 2050: Best = 3.211406e+03	FES = 205100	$\mu_F = 0.787$	$\mu_{CR} = 0.373$	
Diversity = 5.1591e-02	k_t = [1, 1, 1, 1]			
Generation 2060: Best = 3.211394e+03	FES = 206100	$\mu_F = 0.786$	$\mu_{CR} = 0.375$	
Diversity = 5.8129e-02	k_t = [2, 3, 3, 2]			
Generation 2070: Best = 3.211378e+03	FES = 207100	$\mu_F = 0.757$	$\mu_{CR} = 0.377$	
Diversity = 6.0893e-02	k_t = [1, 1, 1, 1]			
Generation 2080: Best = 3.211363e+03	FES = 208100	$\mu_F = 0.758$	$\mu_{CR} = 0.374$	
Diversity = 6.0353e-02	k_t = [2, 3, 2, 4]			
Generation 2090: Best = 3.211346e+03	FES = 209100	$\mu_F = 0.752$	$\mu_{CR} = 0.382$	
Diversity = 6.0135e-02	k_t = [1, 1, 1, 1]			
Generation 2100: Best = 3.211334e+03	FES = 210100	$\mu_F = 0.753$	$\mu_{CR} = 0.374$	
Diversity = 5.4599e-02	k_t = [4, 3, 4, 4]			
Generation 2110: Best = 3.211321e+03	FES = 211100	$\mu_F = 0.760$	$\mu_{CR} = 0.367$	

Diversity = 5.4139e-02	k_t = [1, 1, 1, 1]			
Generation 2120: Best = 3.211290e+03	FES = 212100	$\mu_F = 0.774$	$\mu_{CR} = 0.357$	
Diversity = 5.8114e-02	k_t = [3, 3, 2, 4]			
Generation 2130: Best = 3.211270e+03	FES = 213100	$\mu_F = 0.779$	$\mu_{CR} = 0.350$	
Diversity = 7.7834e-02	k_t = [1, 1, 1, 1]			
Generation 2140: Best = 3.211248e+03	FES = 214100	$\mu_F = 0.774$	$\mu_{CR} = 0.360$	
Diversity = 6.9264e-02	k_t = [2, 4, 3, 3]			
Generation 2150: Best = 3.211232e+03	FES = 215100	$\mu_F = 0.767$	$\mu_{CR} = 0.358$	
Diversity = 6.8824e-02	k_t = [1, 1, 1, 1]			
Generation 2160: Best = 3.211216e+03	FES = 216100	$\mu_F = 0.774$	$\mu_{CR} = 0.356$	
Diversity = 5.9210e-02	k_t = [3, 3, 3, 3]			
Generation 2170: Best = 3.211200e+03	FES = 217100	$\mu_F = 0.776$	$\mu_{CR} = 0.345$	
Diversity = 6.2695e-02	k_t = [1, 1, 1, 1]			
Generation 2180: Best = 3.211190e+03	FES = 218100	$\mu_F = 0.778$	$\mu_{CR} = 0.343$	
Diversity = 5.3516e-02	k_t = [4, 2, 3, 4]			
Generation 2190: Best = 3.211173e+03	FES = 219100	$\mu_F = 0.788$	$\mu_{CR} = 0.336$	
Diversity = 5.3385e-02	k_t = [1, 1, 1, 1]			
Generation 2200: Best = 3.211156e+03	FES = 220100	$\mu_F = 0.785$	$\mu_{CR} = 0.342$	
Diversity = 5.5340e-02	k_t = [4, 2, 2, 2]			
Generation 2210: Best = 3.211147e+03	FES = 221100	$\mu_F = 0.782$	$\mu_{CR} = 0.334$	
Diversity = 5.6179e-02	k_t = [1, 1, 1, 1]			
Generation 2220: Best = 3.211139e+03	FES = 222100	$\mu_F = 0.768$	$\mu_{CR} = 0.326$	
Diversity = 5.0143e-02	k_t = [4, 2, 4, 4]			
Generation 2230: Best = 3.211124e+03	FES = 223100	$\mu_F = 0.773$	$\mu_{CR} = 0.320$	
Diversity = 4.7232e-02	k_t = [1, 1, 1, 1]			
Generation 2240: Best = 3.211115e+03	FES = 224100	$\mu_F = 0.770$	$\mu_{CR} = 0.315$	
Diversity = 5.0685e-02	k_t = [3, 2, 3, 3]			
Generation 2250: Best = 3.211096e+03	FES = 225100	$\mu_F = 0.762$	$\mu_{CR} = 0.321$	
Diversity = 5.3384e-02	k_t = [1, 1, 1, 1]			
Generation 2260: Best = 3.211081e+03	FES = 226100	$\mu_F = 0.768$	$\mu_{CR} = 0.311$	
Diversity = 5.4357e-02	k_t = [3, 4, 2, 2]			
Generation 2270: Best = 3.211074e+03	FES = 227100	$\mu_F = 0.783$	$\mu_{CR} = 0.301$	
Diversity = 5.9220e-02	k_t = [1, 1, 1, 1]			
Generation 2280: Best = 3.211057e+03	FES = 228100	$\mu_F = 0.790$	$\mu_{CR} = 0.283$	
Diversity = 5.1868e-02	k_t = [3, 3, 3, 3]			
Generation 2290: Best = 3.211047e+03	FES = 229100	$\mu_F = 0.780$	$\mu_{CR} = 0.279$	
Diversity = 5.2531e-02	k_t = [1, 1, 1, 1]			
Generation 2300: Best = 3.211028e+03	FES = 230100	$\mu_F = 0.798$	$\mu_{CR} = 0.272$	
Diversity = 5.4956e-02	k_t = [4, 2, 4, 4]			
Generation 2310: Best = 3.211012e+03	FES = 231100	$\mu_F = 0.808$	$\mu_{CR} = 0.272$	
Diversity = 5.9291e-02	k_t = [1, 1, 1, 1]			
Generation 2320: Best = 3.210981e+03	FES = 232100	$\mu_F = 0.807$	$\mu_{CR} = 0.273$	
Diversity = 7.1790e-02	k_t = [2, 4, 2, 2]			
Generation 2330: Best = 3.210945e+03	FES = 233100	$\mu_F = 0.799$	$\mu_{CR} = 0.268$	
Diversity = 8.8038e-02	k_t = [1, 1, 1, 1]			
Generation 2340: Best = 3.210928e+03	FES = 234100	$\mu_F = 0.801$	$\mu_{CR} = 0.250$	
Diversity = 8.8968e-02	k_t = [2, 4, 2, 4]			
Generation 2350: Best = 3.210911e+03	FES = 235100	$\mu_F = 0.800$	$\mu_{CR} = 0.242$	
Diversity = 8.3576e-02	k_t = [1, 1, 1, 1]			
Generation 2360: Best = 3.210888e+03	FES = 236100	$\mu_F = 0.792$	$\mu_{CR} = 0.236$	
Diversity = 7.6916e-02	k_t = [2, 4, 4, 3]			
Generation 2370: Best = 3.210869e+03	FES = 237100	$\mu_F = 0.794$	$\mu_{CR} = 0.240$	
Diversity = 8.0078e-02	k_t = [1, 1, 1, 1]			
Generation 2380: Best = 3.210859e+03	FES = 238100	$\mu_F = 0.797$	$\mu_{CR} = 0.242$	

Diversity = 7.6001e-02	k_t = [3, 4, 3, 2]			
Generation 2390: Best = 3.210836e+03	FES = 239100	$\mu_F = 0.796$	$\mu_{CR} = 0.243$	
Diversity = 7.4757e-02	k_t = [1, 1, 1, 1]			
Generation 2400: Best = 3.210818e+03	FES = 240100	$\mu_F = 0.796$	$\mu_{CR} = 0.243$	
Diversity = 7.7039e-02	k_t = [3, 4, 2, 2]			
Generation 2410: Best = 3.210796e+03	FES = 241100	$\mu_F = 0.781$	$\mu_{CR} = 0.232$	
Diversity = 8.2328e-02	k_t = [1, 1, 1, 1]			
Generation 2420: Best = 3.210780e+03	FES = 242100	$\mu_F = 0.770$	$\mu_{CR} = 0.224$	
Diversity = 8.3082e-02	k_t = [4, 3, 3, 3]			
Generation 2430: Best = 3.210754e+03	FES = 243100	$\mu_F = 0.774$	$\mu_{CR} = 0.222$	
Diversity = 9.0268e-02	k_t = [1, 1, 1, 1]			
Generation 2440: Best = 3.210732e+03	FES = 244100	$\mu_F = 0.780$	$\mu_{CR} = 0.210$	
Diversity = 9.3967e-02	k_t = [2, 4, 3, 2]			
Generation 2450: Best = 3.210700e+03	FES = 245100	$\mu_F = 0.788$	$\mu_{CR} = 0.213$	
Diversity = 8.8424e-02	k_t = [1, 1, 1, 1]			
Generation 2460: Best = 3.210684e+03	FES = 246100	$\mu_F = 0.776$	$\mu_{CR} = 0.206$	
Diversity = 8.0878e-02	k_t = [3, 3, 3, 3]			
Generation 2470: Best = 3.210663e+03	FES = 247100	$\mu_F = 0.776$	$\mu_{CR} = 0.204$	
Diversity = 9.4044e-02	k_t = [1, 1, 1, 1]			
Generation 2480: Best = 3.210636e+03	FES = 248100	$\mu_F = 0.779$	$\mu_{CR} = 0.200$	
Diversity = 8.7797e-02	k_t = [3, 4, 4, 3]			
Generation 2490: Best = 3.210620e+03	FES = 249100	$\mu_F = 0.772$	$\mu_{CR} = 0.209$	
Diversity = 8.9678e-02	k_t = [1, 1, 1, 1]			
Generation 2500: Best = 3.210596e+03	FES = 250100	$\mu_F = 0.779$	$\mu_{CR} = 0.211$	
Diversity = 8.7857e-02	k_t = [3, 2, 2, 4]			
Generation 2510: Best = 3.210581e+03	FES = 251100	$\mu_F = 0.779$	$\mu_{CR} = 0.199$	
Diversity = 9.5884e-02	k_t = [1, 1, 1, 1]			
Generation 2520: Best = 3.210566e+03	FES = 252100	$\mu_F = 0.778$	$\mu_{CR} = 0.191$	
Diversity = 8.5895e-02	k_t = [3, 4, 3, 3]			
Generation 2530: Best = 3.210549e+03	FES = 253100	$\mu_F = 0.788$	$\mu_{CR} = 0.193$	
Diversity = 8.8415e-02	k_t = [1, 1, 1, 1]			
Generation 2540: Best = 3.210527e+03	FES = 254100	$\mu_F = 0.777$	$\mu_{CR} = 0.192$	
Diversity = 8.0648e-02	k_t = [3, 3, 2, 4]			
Generation 2550: Best = 3.210504e+03	FES = 255100	$\mu_F = 0.779$	$\mu_{CR} = 0.183$	
Diversity = 8.5239e-02	k_t = [1, 1, 1, 1]			
Generation 2560: Best = 3.210481e+03	FES = 256100	$\mu_F = 0.772$	$\mu_{CR} = 0.173$	
Diversity = 7.6189e-02	k_t = [3, 2, 3, 3]			
Generation 2570: Best = 3.210455e+03	FES = 257100	$\mu_F = 0.768$	$\mu_{CR} = 0.170$	
Diversity = 7.9366e-02	k_t = [1, 1, 1, 1]			
Generation 2580: Best = 3.210444e+03	FES = 258100	$\mu_F = 0.776$	$\mu_{CR} = 0.173$	
Diversity = 7.7041e-02	k_t = [3, 3, 3, 2]			
Generation 2590: Best = 3.210432e+03	FES = 259100	$\mu_F = 0.780$	$\mu_{CR} = 0.182$	
Diversity = 8.3687e-02	k_t = [1, 1, 1, 1]			
Generation 2600: Best = 3.210412e+03	FES = 260100	$\mu_F = 0.793$	$\mu_{CR} = 0.179$	
Diversity = 7.8636e-02	k_t = [4, 3, 2, 2]			
Generation 2610: Best = 3.210392e+03	FES = 261100	$\mu_F = 0.792$	$\mu_{CR} = 0.180$	
Diversity = 7.7320e-02	k_t = [1, 1, 1, 1]			
Generation 2620: Best = 3.210377e+03	FES = 262100	$\mu_F = 0.784$	$\mu_{CR} = 0.179$	
Diversity = 6.8659e-02	k_t = [2, 2, 4, 4]			
Generation 2630: Best = 3.210364e+03	FES = 263100	$\mu_F = 0.773$	$\mu_{CR} = 0.184$	
Diversity = 7.0653e-02	k_t = [1, 1, 1, 1]			
Generation 2640: Best = 3.210348e+03	FES = 264100	$\mu_F = 0.774$	$\mu_{CR} = 0.184$	
Diversity = 7.3818e-02	k_t = [3, 3, 4, 2]			
Generation 2650: Best = 3.210330e+03	FES = 265100	$\mu_F = 0.778$	$\mu_{CR} = 0.182$	

Diversity = 8.1586e-02	k_t = [1, 1, 1, 1]			
Generation 2660: Best = 3.210312e+03	FES = 266100	$\mu_F = 0.789$	$\mu_{CR} = 0.189$	
Diversity = 7.2514e-02	k_t = [4, 4, 2, 4]			
Generation 2670: Best = 3.210286e+03	FES = 267100	$\mu_F = 0.785$	$\mu_{CR} = 0.196$	
Diversity = 7.0173e-02	k_t = [1, 1, 1, 1]			
Generation 2680: Best = 3.210285e+03	FES = 268100	$\mu_F = 0.788$	$\mu_{CR} = 0.198$	
Diversity = 6.4953e-02	k_t = [3, 2, 3, 3]			
Generation 2690: Best = 3.210254e+03	FES = 269100	$\mu_F = 0.800$	$\mu_{CR} = 0.194$	
Diversity = 7.3228e-02	k_t = [1, 1, 1, 1]			
Generation 2700: Best = 3.210237e+03	FES = 270100	$\mu_F = 0.806$	$\mu_{CR} = 0.191$	
Diversity = 7.9412e-02	k_t = [3, 4, 4, 4]			
Generation 2710: Best = 3.210217e+03	FES = 271100	$\mu_F = 0.791$	$\mu_{CR} = 0.196$	
Diversity = 7.7283e-02	k_t = [1, 1, 1, 1]			
Generation 2720: Best = 3.210202e+03	FES = 272100	$\mu_F = 0.785$	$\mu_{CR} = 0.203$	
Diversity = 7.0928e-02	k_t = [4, 3, 3, 4]			
Generation 2730: Best = 3.210186e+03	FES = 273100	$\mu_F = 0.783$	$\mu_{CR} = 0.190$	
Diversity = 6.9818e-02	k_t = [1, 1, 1, 1]			
Generation 2740: Best = 3.210170e+03	FES = 274100	$\mu_F = 0.785$	$\mu_{CR} = 0.193$	
Diversity = 6.9902e-02	k_t = [3, 2, 3, 2]			
Generation 2750: Best = 3.210156e+03	FES = 275100	$\mu_F = 0.790$	$\mu_{CR} = 0.187$	
Diversity = 7.4460e-02	k_t = [1, 1, 1, 1]			
Generation 2760: Best = 3.210137e+03	FES = 276100	$\mu_F = 0.795$	$\mu_{CR} = 0.182$	
Diversity = 7.1360e-02	k_t = [4, 2, 3, 2]			
Generation 2770: Best = 3.210121e+03	FES = 277100	$\mu_F = 0.789$	$\mu_{CR} = 0.182$	
Diversity = 6.9952e-02	k_t = [1, 1, 1, 1]			
Generation 2780: Best = 3.210093e+03	FES = 278100	$\mu_F = 0.780$	$\mu_{CR} = 0.186$	
Diversity = 7.0972e-02	k_t = [3, 3, 2, 2]			
Generation 2790: Best = 3.210067e+03	FES = 279100	$\mu_F = 0.773$	$\mu_{CR} = 0.193$	
Diversity = 8.3072e-02	k_t = [1, 1, 1, 1]			
Generation 2800: Best = 3.210029e+03	FES = 280100	$\mu_F = 0.773$	$\mu_{CR} = 0.198$	
Diversity = 7.9080e-02	k_t = [4, 4, 3, 4]			
Generation 2810: Best = 3.210012e+03	FES = 281100	$\mu_F = 0.771$	$\mu_{CR} = 0.198$	
Diversity = 7.7319e-02	k_t = [1, 1, 1, 1]			
Generation 2820: Best = 3.209987e+03	FES = 282100	$\mu_F = 0.780$	$\mu_{CR} = 0.206$	
Diversity = 7.9492e-02	k_t = [4, 4, 4, 3]			
Generation 2830: Best = 3.209973e+03	FES = 283100	$\mu_F = 0.782$	$\mu_{CR} = 0.195$	
Diversity = 7.6381e-02	k_t = [1, 1, 1, 1]			
Generation 2840: Best = 3.209954e+03	FES = 284100	$\mu_F = 0.776$	$\mu_{CR} = 0.197$	
Diversity = 6.8470e-02	k_t = [3, 3, 3, 4]			
Generation 2850: Best = 3.209927e+03	FES = 285100	$\mu_F = 0.789$	$\mu_{CR} = 0.197$	
Diversity = 7.2571e-02	k_t = [1, 1, 1, 1]			
Generation 2860: Best = 3.209899e+03	FES = 286100	$\mu_F = 0.772$	$\mu_{CR} = 0.197$	
Diversity = 8.8319e-02	k_t = [2, 2, 3, 2]			
Generation 2870: Best = 3.209878e+03	FES = 287100	$\mu_F = 0.772$	$\mu_{CR} = 0.196$	
Diversity = 9.0312e-02	k_t = [1, 1, 1, 1]			
Generation 2880: Best = 3.209850e+03	FES = 288100	$\mu_F = 0.770$	$\mu_{CR} = 0.195$	
Diversity = 7.4087e-02	k_t = [3, 3, 3, 4]			
Generation 2890: Best = 3.209799e+03	FES = 289100	$\mu_F = 0.780$	$\mu_{CR} = 0.196$	
Diversity = 9.3998e-02	k_t = [1, 1, 1, 1]			
Generation 2900: Best = 3.209790e+03	FES = 290100	$\mu_F = 0.794$	$\mu_{CR} = 0.191$	
Diversity = 8.7550e-02	k_t = [2, 2, 3, 4]			
Generation 2910: Best = 3.209771e+03	FES = 291100	$\mu_F = 0.782$	$\mu_{CR} = 0.183$	
Diversity = 8.2977e-02	k_t = [1, 1, 1, 1]			
Generation 2920: Best = 3.209745e+03	FES = 292100	$\mu_F = 0.791$	$\mu_{CR} = 0.175$	

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Diversity = 8.3181e-02  k_t = [2, 3, 4, 3]
Generation 2930: Best = 3.209722e+03  FES = 293100   $\mu_F$  = 0.784   $\mu_{CR}$  = 0.174
Diversity = 8.5532e-02  k_t = [1, 1, 1, 1]
Generation 2940: Best = 3.209705e+03  FES = 294100   $\mu_F$  = 0.783   $\mu_{CR}$  = 0.181
Diversity = 8.4943e-02  k_t = [3, 2, 4, 3]
Generation 2950: Best = 3.209681e+03  FES = 295100   $\mu_F$  = 0.782   $\mu_{CR}$  = 0.186
Diversity = 8.9576e-02  k_t = [1, 1, 1, 1]
Generation 2960: Best = 3.209659e+03  FES = 296100   $\mu_F$  = 0.782   $\mu_{CR}$  = 0.177
Diversity = 8.4445e-02  k_t = [4, 4, 3, 2]
Generation 2970: Best = 3.209634e+03  FES = 297100   $\mu_F$  = 0.787   $\mu_{CR}$  = 0.181
Diversity = 8.8730e-02  k_t = [1, 1, 1, 1]
Generation 2980: Best = 3.209615e+03  FES = 298100   $\mu_F$  = 0.803   $\mu_{CR}$  = 0.177
Diversity = 8.4453e-02  k_t = [4, 2, 4, 4]
Generation 2990: Best = 3.209596e+03  FES = 299100   $\mu_F$  = 0.794   $\mu_{CR}$  = 0.174
Diversity = 7.5233e-02  k_t = [1, 1, 1, 1]

```

Optimization complete.

Best Fitness = 3.209576e+03

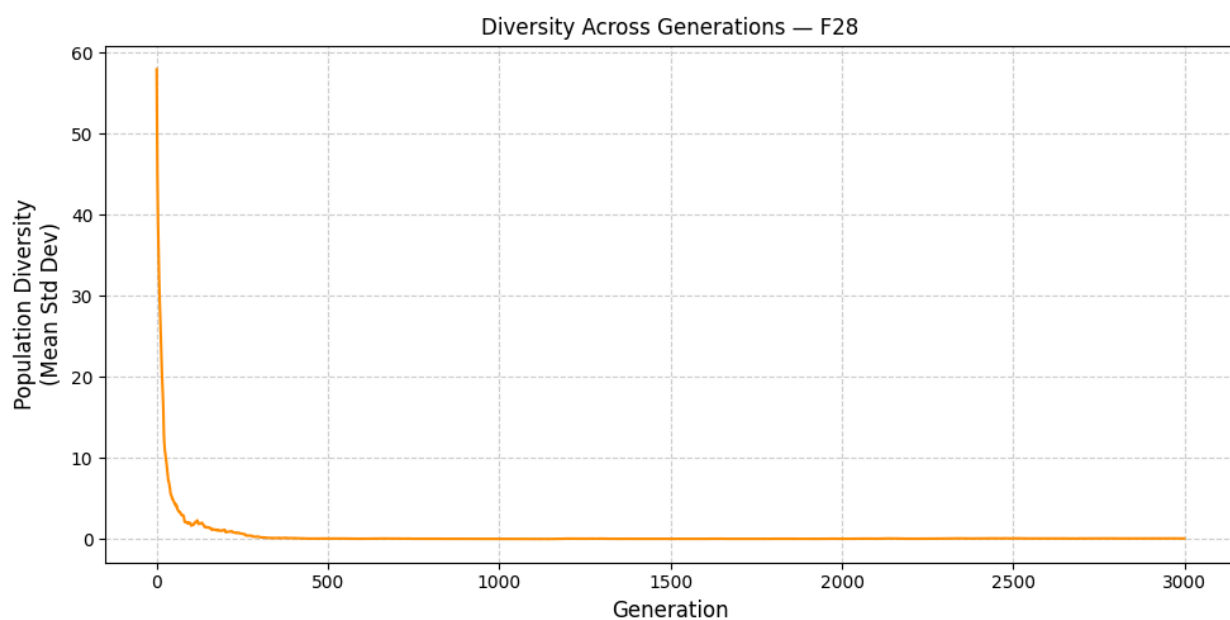
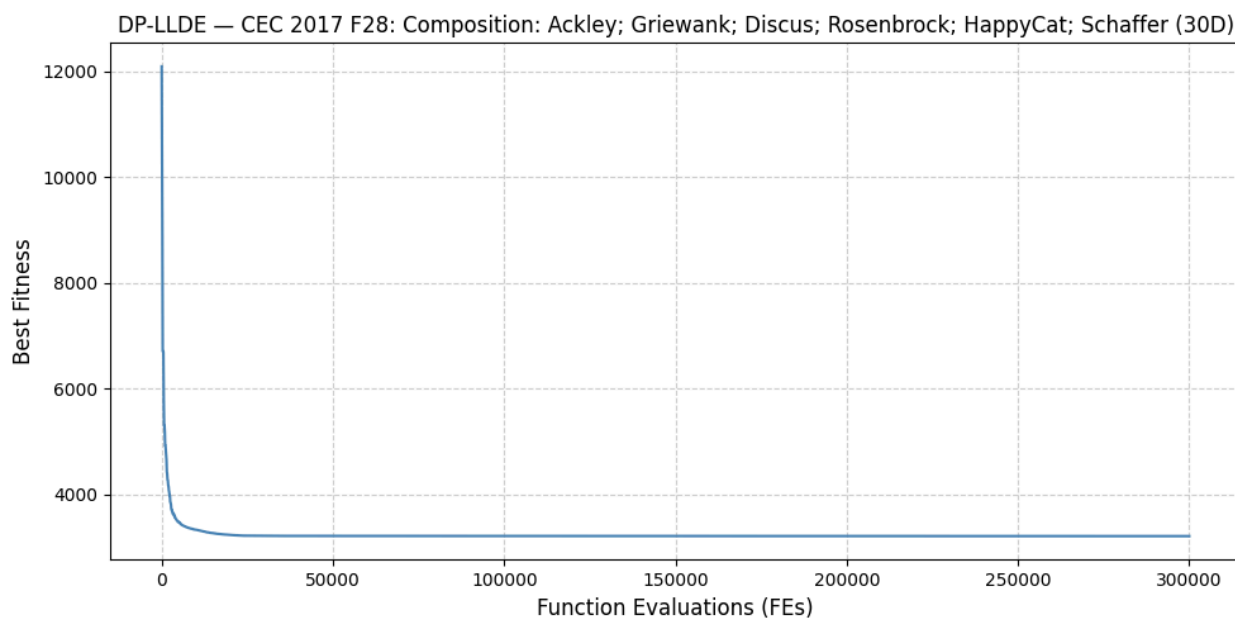
Total Function Evals = 300000

Total Successful Updates = 126736

```

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=
Results:
  Best Fitness      : 3.209576e+03
  Global Optimum    : 2.800000e+03
  Error (|best-optimum|) : 4.095760e+02
  Best Solution (5 dim) : [-30.17388777  11.09359286 -50.10059846  78.06238299
-48.75165092]
  D^(0) (init. diversity): 5.7976e+01
=====
=

```



Graph saved as convergence_F28.png

```
In [ ]: # =====
# F29 — Composition Function 9 (F15; F16; F17)
# =====
run_and_plot('F29', make_wrapper(composition.f29), 2900.0, 'Composition: F15;
```

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DP-LLDE on CEC 2017 F29 – Single Run

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=

Function : Composition: F15; F16; F17 (30D)

Bounds : [-100, 100]

Global optimum : 2900.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 8.977257e+03

Generation 10: Best = 6.156956e+03 FES = 1100 $\mu_F = 0.569$ $\mu_{CR} = 0.350$

Diversity = 3.7904e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 4.981505e+03 FES = 2100 $\mu_F = 0.627$ $\mu_{CR} = 0.353$

Diversity = 3.0650e+01 $k_t = [4, 4, 2, 4]$

Generation 30: Best = 4.752434e+03 FES = 3100 $\mu_F = 0.656$ $\mu_{CR} = 0.333$

Diversity = 2.7523e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 4.725643e+03 FES = 4100 $\mu_F = 0.666$ $\mu_{CR} = 0.324$

Diversity = 2.6158e+01 $k_t = [2, 4, 4, 2]$

Generation 50: Best = 4.702871e+03 FES = 5100 $\mu_F = 0.652$ $\mu_{CR} = 0.290$

Diversity = 2.5812e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 4.460413e+03 FES = 6100 $\mu_F = 0.709$ $\mu_{CR} = 0.260$

Diversity = 2.6327e+01 $k_t = [3, 3, 4, 3]$

Generation 70: Best = 4.270929e+03 FES = 7100 $\mu_F = 0.734$ $\mu_{CR} = 0.228$

Diversity = 2.5966e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 4.270929e+03 FES = 8100 $\mu_F = 0.719$ $\mu_{CR} = 0.180$

Diversity = 2.5682e+01 $k_t = [2, 4, 4, 4]$

Generation 90: Best = 4.270929e+03 FES = 9100 $\mu_F = 0.729$ $\mu_{CR} = 0.138$

Diversity = 2.5317e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 4.149944e+03 FES = 10100 $\mu_F = 0.762$ $\mu_{CR} = 0.120$

Diversity = 2.4846e+01 $k_t = [4, 3, 3, 3]$

Generation 110: Best = 3.994303e+03 FES = 11100 $\mu_F = 0.803$ $\mu_{CR} = 0.091$

Diversity = 2.5108e+01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 3.994303e+03 FES = 12100 $\mu_F = 0.818$ $\mu_{CR} = 0.081$

Diversity = 2.4851e+01 $k_t = [4, 4, 3, 4]$

Generation 130: Best = 3.988739e+03 FES = 13100 $\mu_F = 0.819$ $\mu_{CR} = 0.072$

Diversity = 2.4695e+01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 3.842418e+03 FES = 14100 $\mu_F = 0.809$ $\mu_{CR} = 0.067$

Diversity = 2.4714e+01 $k_t = [2, 4, 3, 2]$

Generation 150: Best = 3.814519e+03 FES = 15100 $\mu_F = 0.811$ $\mu_{CR} = 0.064$

Diversity = 2.5113e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 3.797954e+03 FES = 16100 $\mu_F = 0.810$ $\mu_{CR} = 0.062$

Diversity = 2.5076e+01 $k_t = [3, 2, 3, 3]$

Generation 170: Best = 3.744958e+03 FES = 17100 $\mu_F = 0.845$ $\mu_{CR} = 0.074$

Diversity = 2.5030e+01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 3.733267e+03 FES = 18100 $\mu_F = 0.847$ $\mu_{CR} = 0.070$

Diversity = 2.5444e+01 $k_t = [4, 4, 3, 4]$

Generation 190: Best = 3.705826e+03 FES = 19100 $\mu_F = 0.856$ $\mu_{CR} = 0.071$

Diversity = 2.5430e+01 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 3.677357e+03 FES = 20100 $\mu_F = 0.852$ $\mu_{CR} = 0.067$

Diversity = 2.5324e+01 $k_t = [4, 2, 4, 2]$

Generation 210: Best = 3.677357e+03 FES = 21100 $\mu_F = 0.869$ $\mu_{CR} = 0.065$

Diversity = 2.5438e+01 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 3.668241e+03 FES = 22100 $\mu_F = 0.872$ $\mu_{CR} = 0.065$

Diversity = 2.5211e+01 $k_t = [4, 4, 4, 2]$

Generation 230: Best = 3.635085e+03	FES = 23100	$\mu_F = 0.836$	$\mu_{CR} = 0.063$
Diversity = 2.5212e+01	k_t = [1, 1, 1, 1]		
Generation 240: Best = 3.616611e+03	FES = 24100	$\mu_F = 0.857$	$\mu_{CR} = 0.071$
Diversity = 2.5255e+01	k_t = [3, 4, 2, 3]		
Generation 250: Best = 3.616611e+03	FES = 25100	$\mu_F = 0.876$	$\mu_{CR} = 0.063$
Diversity = 2.5121e+01	k_t = [1, 1, 1, 1]		
Generation 260: Best = 3.616611e+03	FES = 26100	$\mu_F = 0.875$	$\mu_{CR} = 0.055$
Diversity = 2.4838e+01	k_t = [3, 3, 4, 2]		
Generation 270: Best = 3.596720e+03	FES = 27100	$\mu_F = 0.871$	$\mu_{CR} = 0.053$
Diversity = 2.4632e+01	k_t = [1, 1, 1, 1]		
Generation 280: Best = 3.595806e+03	FES = 28100	$\mu_F = 0.810$	$\mu_{CR} = 0.071$
Diversity = 2.4473e+01	k_t = [4, 4, 3, 3]		
Generation 290: Best = 3.595806e+03	FES = 29100	$\mu_F = 0.813$	$\mu_{CR} = 0.062$
Diversity = 2.4342e+01	k_t = [1, 1, 1, 1]		
Generation 300: Best = 3.595806e+03	FES = 30100	$\mu_F = 0.826$	$\mu_{CR} = 0.056$
Diversity = 2.4086e+01	k_t = [2, 2, 3, 4]		
Generation 310: Best = 3.595670e+03	FES = 31100	$\mu_F = 0.842$	$\mu_{CR} = 0.053$
Diversity = 2.3728e+01	k_t = [1, 1, 1, 1]		
Generation 320: Best = 3.595670e+03	FES = 32100	$\mu_F = 0.844$	$\mu_{CR} = 0.050$
Diversity = 2.3557e+01	k_t = [4, 3, 4, 4]		
Generation 330: Best = 3.595670e+03	FES = 33100	$\mu_F = 0.834$	$\mu_{CR} = 0.050$
Diversity = 2.3383e+01	k_t = [1, 1, 1, 1]		
Generation 340: Best = 3.595670e+03	FES = 34100	$\mu_F = 0.851$	$\mu_{CR} = 0.048$
Diversity = 2.3177e+01	k_t = [2, 4, 3, 2]		
Generation 350: Best = 3.595670e+03	FES = 35100	$\mu_F = 0.793$	$\mu_{CR} = 0.040$
Diversity = 2.3309e+01	k_t = [1, 1, 1, 1]		
Generation 360: Best = 3.582525e+03	FES = 36100	$\mu_F = 0.774$	$\mu_{CR} = 0.050$
Diversity = 2.3105e+01	k_t = [4, 4, 2, 2]		
Generation 370: Best = 3.581315e+03	FES = 37100	$\mu_F = 0.736$	$\mu_{CR} = 0.047$
Diversity = 2.3049e+01	k_t = [1, 1, 1, 1]		
Generation 380: Best = 3.558651e+03	FES = 38100	$\mu_F = 0.768$	$\mu_{CR} = 0.055$
Diversity = 2.2844e+01	k_t = [4, 3, 2, 4]		
Generation 390: Best = 3.537726e+03	FES = 39100	$\mu_F = 0.750$	$\mu_{CR} = 0.056$
Diversity = 2.2722e+01	k_t = [1, 1, 1, 1]		
Generation 400: Best = 3.537726e+03	FES = 40100	$\mu_F = 0.728$	$\mu_{CR} = 0.065$
Diversity = 2.2521e+01	k_t = [4, 2, 3, 4]		
Generation 410: Best = 3.503919e+03	FES = 41100	$\mu_F = 0.774$	$\mu_{CR} = 0.055$
Diversity = 2.2205e+01	k_t = [1, 1, 1, 1]		
Generation 420: Best = 3.503919e+03	FES = 42100	$\mu_F = 0.775$	$\mu_{CR} = 0.052$
Diversity = 2.1980e+01	k_t = [2, 4, 2, 2]		
Generation 430: Best = 3.503919e+03	FES = 43100	$\mu_F = 0.795$	$\mu_{CR} = 0.048$
Diversity = 2.1851e+01	k_t = [1, 1, 1, 1]		
Generation 440: Best = 3.503919e+03	FES = 44100	$\mu_F = 0.769$	$\mu_{CR} = 0.052$
Diversity = 2.1973e+01	k_t = [2, 3, 4, 2]		
Generation 450: Best = 3.503919e+03	FES = 45100	$\mu_F = 0.776$	$\mu_{CR} = 0.056$
Diversity = 2.1960e+01	k_t = [1, 1, 1, 1]		
Generation 460: Best = 3.486177e+03	FES = 46100	$\mu_F = 0.751$	$\mu_{CR} = 0.050$
Diversity = 2.1859e+01	k_t = [3, 4, 3, 4]		
Generation 470: Best = 3.486177e+03	FES = 47100	$\mu_F = 0.721$	$\mu_{CR} = 0.052$
Diversity = 2.1606e+01	k_t = [1, 1, 1, 1]		
Generation 480: Best = 3.486177e+03	FES = 48100	$\mu_F = 0.728$	$\mu_{CR} = 0.051$
Diversity = 2.1492e+01	k_t = [4, 2, 3, 4]		
Generation 490: Best = 3.486177e+03	FES = 49100	$\mu_F = 0.768$	$\mu_{CR} = 0.054$
Diversity = 2.1504e+01	k_t = [1, 1, 1, 1]		

Generation 500: Best = 3.486177e+03	FES = 50100	$\mu_F = 0.817$	$\mu_{CR} = 0.057$
Diversity = 2.1315e+01	k_t = [3, 4, 2, 4]		
Generation 510: Best = 3.484883e+03	FES = 51100	$\mu_F = 0.790$	$\mu_{CR} = 0.069$
Diversity = 2.1339e+01	k_t = [1, 1, 1, 1]		
Generation 520: Best = 3.484883e+03	FES = 52100	$\mu_F = 0.831$	$\mu_{CR} = 0.065$
Diversity = 2.1216e+01	k_t = [2, 4, 2, 2]		
Generation 530: Best = 3.481034e+03	FES = 53100	$\mu_F = 0.806$	$\mu_{CR} = 0.064$
Diversity = 2.1176e+01	k_t = [1, 1, 1, 1]		
Generation 540: Best = 3.473691e+03	FES = 54100	$\mu_F = 0.824$	$\mu_{CR} = 0.044$
Diversity = 2.0992e+01	k_t = [3, 4, 4, 4]		
Generation 550: Best = 3.473691e+03	FES = 55100	$\mu_F = 0.857$	$\mu_{CR} = 0.048$
Diversity = 2.0585e+01	k_t = [1, 1, 1, 1]		
Generation 560: Best = 3.473691e+03	FES = 56100	$\mu_F = 0.856$	$\mu_{CR} = 0.048$
Diversity = 2.0503e+01	k_t = [2, 4, 2, 2]		
Generation 570: Best = 3.473691e+03	FES = 57100	$\mu_F = 0.849$	$\mu_{CR} = 0.056$
Diversity = 2.0403e+01	k_t = [1, 1, 1, 1]		
Generation 580: Best = 3.473691e+03	FES = 58100	$\mu_F = 0.850$	$\mu_{CR} = 0.065$
Diversity = 2.0033e+01	k_t = [4, 3, 3, 4]		
Generation 590: Best = 3.471329e+03	FES = 59100	$\mu_F = 0.872$	$\mu_{CR} = 0.061$
Diversity = 1.9987e+01	k_t = [1, 1, 1, 1]		
Generation 600: Best = 3.471329e+03	FES = 60100	$\mu_F = 0.858$	$\mu_{CR} = 0.060$
Diversity = 1.9939e+01	k_t = [4, 2, 3, 2]		
Generation 610: Best = 3.471329e+03	FES = 61100	$\mu_F = 0.871$	$\mu_{CR} = 0.047$
Diversity = 1.9885e+01	k_t = [1, 1, 1, 1]		
Generation 620: Best = 3.465116e+03	FES = 62100	$\mu_F = 0.881$	$\mu_{CR} = 0.043$
Diversity = 1.9632e+01	k_t = [3, 3, 2, 3]		
Generation 630: Best = 3.465116e+03	FES = 63100	$\mu_F = 0.901$	$\mu_{CR} = 0.045$
Diversity = 1.9177e+01	k_t = [1, 1, 1, 1]		
Generation 640: Best = 3.465116e+03	FES = 64100	$\mu_F = 0.897$	$\mu_{CR} = 0.038$
Diversity = 1.8254e+01	k_t = [2, 2, 3, 4]		
Generation 650: Best = 3.461090e+03	FES = 65100	$\mu_F = 0.882$	$\mu_{CR} = 0.042$
Diversity = 1.8275e+01	k_t = [1, 1, 1, 1]		
Generation 660: Best = 3.459635e+03	FES = 66100	$\mu_F = 0.886$	$\mu_{CR} = 0.053$
Diversity = 1.7873e+01	k_t = [2, 2, 4, 4]		
Generation 670: Best = 3.459025e+03	FES = 67100	$\mu_F = 0.889$	$\mu_{CR} = 0.056$
Diversity = 1.7504e+01	k_t = [1, 1, 1, 1]		
Generation 680: Best = 3.458851e+03	FES = 68100	$\mu_F = 0.872$	$\mu_{CR} = 0.057$
Diversity = 1.7424e+01	k_t = [4, 4, 4, 2]		
Generation 690: Best = 3.458200e+03	FES = 69100	$\mu_F = 0.844$	$\mu_{CR} = 0.047$
Diversity = 1.7055e+01	k_t = [1, 1, 1, 1]		
Generation 700: Best = 3.458200e+03	FES = 70100	$\mu_F = 0.808$	$\mu_{CR} = 0.043$
Diversity = 1.6496e+01	k_t = [3, 4, 2, 4]		
Generation 710: Best = 3.458200e+03	FES = 71100	$\mu_F = 0.760$	$\mu_{CR} = 0.043$
Diversity = 1.6067e+01	k_t = [1, 1, 1, 1]		
Generation 720: Best = 3.453266e+03	FES = 72100	$\mu_F = 0.814$	$\mu_{CR} = 0.038$
Diversity = 1.6004e+01	k_t = [2, 3, 4, 3]		
Generation 730: Best = 3.453266e+03	FES = 73100	$\mu_F = 0.828$	$\mu_{CR} = 0.051$
Diversity = 1.5671e+01	k_t = [1, 1, 1, 1]		
Generation 740: Best = 3.452985e+03	FES = 74100	$\mu_F = 0.799$	$\mu_{CR} = 0.049$
Diversity = 1.5655e+01	k_t = [4, 3, 2, 2]		
Generation 750: Best = 3.452985e+03	FES = 75100	$\mu_F = 0.813$	$\mu_{CR} = 0.051$
Diversity = 1.5570e+01	k_t = [1, 1, 1, 1]		
Generation 760: Best = 3.450809e+03	FES = 76100	$\mu_F = 0.766$	$\mu_{CR} = 0.049$
Diversity = 1.4831e+01	k_t = [2, 3, 2, 4]		

Generation 770: Best = 3.450778e+03 FES = 77100 μ_F = 0.764 μ_{CR} = 0.053
 Diversity = 1.4699e+01 k_t = [1, 1, 1, 1]
 Generation 780: Best = 3.446547e+03 FES = 78100 μ_F = 0.790 μ_{CR} = 0.062
 Diversity = 1.4433e+01 k_t = [4, 4, 2, 4]
 Generation 790: Best = 3.446547e+03 FES = 79100 μ_F = 0.833 μ_{CR} = 0.067
 Diversity = 1.4143e+01 k_t = [1, 1, 1, 1]
 Generation 800: Best = 3.446547e+03 FES = 80100 μ_F = 0.861 μ_{CR} = 0.056
 Diversity = 1.4136e+01 k_t = [2, 3, 4, 2]
 Generation 810: Best = 3.441992e+03 FES = 81100 μ_F = 0.811 μ_{CR} = 0.058
 Diversity = 1.4102e+01 k_t = [1, 1, 1, 1]
 Generation 820: Best = 3.436237e+03 FES = 82100 μ_F = 0.789 μ_{CR} = 0.058
 Diversity = 1.3960e+01 k_t = [3, 2, 2, 2]
 Generation 830: Best = 3.420404e+03 FES = 83100 μ_F = 0.714 μ_{CR} = 0.059
 Diversity = 1.3991e+01 k_t = [1, 1, 1, 1]
 Generation 840: Best = 3.420404e+03 FES = 84100 μ_F = 0.715 μ_{CR} = 0.047
 Diversity = 1.3891e+01 k_t = [3, 3, 2, 2]
 Generation 850: Best = 3.420404e+03 FES = 85100 μ_F = 0.709 μ_{CR} = 0.045
 Diversity = 1.3787e+01 k_t = [1, 1, 1, 1]
 Generation 860: Best = 3.420404e+03 FES = 86100 μ_F = 0.700 μ_{CR} = 0.049
 Diversity = 1.3587e+01 k_t = [3, 4, 4, 2]
 Generation 870: Best = 3.420404e+03 FES = 87100 μ_F = 0.758 μ_{CR} = 0.042
 Diversity = 1.3518e+01 k_t = [1, 1, 1, 1]
 Generation 880: Best = 3.420404e+03 FES = 88100 μ_F = 0.745 μ_{CR} = 0.050
 Diversity = 1.3616e+01 k_t = [2, 2, 2, 3]
 Generation 890: Best = 3.419476e+03 FES = 89100 μ_F = 0.694 μ_{CR} = 0.052
 Diversity = 1.3430e+01 k_t = [1, 1, 1, 1]
 Generation 900: Best = 3.417965e+03 FES = 90100 μ_F = 0.630 μ_{CR} = 0.068
 Diversity = 1.3434e+01 k_t = [4, 4, 4, 2]
 Generation 910: Best = 3.411659e+03 FES = 91100 μ_F = 0.701 μ_{CR} = 0.059
 Diversity = 1.3399e+01 k_t = [1, 1, 1, 1]
 Generation 920: Best = 3.411659e+03 FES = 92100 μ_F = 0.732 μ_{CR} = 0.062
 Diversity = 1.3317e+01 k_t = [4, 2, 3, 4]
 Generation 930: Best = 3.408971e+03 FES = 93100 μ_F = 0.696 μ_{CR} = 0.045
 Diversity = 1.3319e+01 k_t = [1, 1, 1, 1]
 Generation 940: Best = 3.408971e+03 FES = 94100 μ_F = 0.740 μ_{CR} = 0.050
 Diversity = 1.3301e+01 k_t = [2, 2, 3, 2]
 Generation 950: Best = 3.408971e+03 FES = 95100 μ_F = 0.703 μ_{CR} = 0.041
 Diversity = 1.3244e+01 k_t = [1, 1, 1, 1]
 Generation 960: Best = 3.407717e+03 FES = 96100 μ_F = 0.715 μ_{CR} = 0.048
 Diversity = 1.3243e+01 k_t = [3, 4, 4, 3]
 Generation 970: Best = 3.401652e+03 FES = 97100 μ_F = 0.780 μ_{CR} = 0.031
 Diversity = 1.3261e+01 k_t = [1, 1, 1, 1]
 Generation 980: Best = 3.398365e+03 FES = 98100 μ_F = 0.836 μ_{CR} = 0.041
 Diversity = 1.3179e+01 k_t = [3, 2, 3, 3]
 Generation 990: Best = 3.398365e+03 FES = 99100 μ_F = 0.816 μ_{CR} = 0.041
 Diversity = 1.3300e+01 k_t = [1, 1, 1, 1]
 Generation 1000: Best = 3.398365e+03 FES = 100100 μ_F = 0.847 μ_{CR} = 0.048
 Diversity = 1.3127e+01 k_t = [4, 2, 4, 4]
 Generation 1010: Best = 3.398365e+03 FES = 101100 μ_F = 0.827 μ_{CR} = 0.048
 Diversity = 1.3029e+01 k_t = [1, 1, 1, 1]
 Generation 1020: Best = 3.393184e+03 FES = 102100 μ_F = 0.686 μ_{CR} = 0.064
 Diversity = 1.2979e+01 k_t = [2, 3, 4, 4]
 Generation 1030: Best = 3.392991e+03 FES = 103100 μ_F = 0.719 μ_{CR} = 0.042
 Diversity = 1.2963e+01 k_t = [1, 1, 1, 1]

Generation 1040: Best = 3.392991e+03 FES = 104100 μ_F = 0.728 μ_{CR} = 0.065
 Diversity = 1.2908e+01 k_t = [4, 3, 4, 2]
 Generation 1050: Best = 3.392991e+03 FES = 105100 μ_F = 0.771 μ_{CR} = 0.048
 Diversity = 1.2876e+01 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 3.392367e+03 FES = 106100 μ_F = 0.725 μ_{CR} = 0.044
 Diversity = 1.2840e+01 k_t = [3, 4, 2, 4]
 Generation 1070: Best = 3.392367e+03 FES = 107100 μ_F = 0.693 μ_{CR} = 0.037
 Diversity = 1.2828e+01 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 3.392367e+03 FES = 108100 μ_F = 0.629 μ_{CR} = 0.045
 Diversity = 1.2820e+01 k_t = [3, 3, 3, 3]
 Generation 1090: Best = 3.392367e+03 FES = 109100 μ_F = 0.634 μ_{CR} = 0.040
 Diversity = 1.2765e+01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 3.391006e+03 FES = 110100 μ_F = 0.678 μ_{CR} = 0.039
 Diversity = 1.2751e+01 k_t = [2, 4, 4, 3]
 Generation 1110: Best = 3.390223e+03 FES = 111100 μ_F = 0.613 μ_{CR} = 0.043
 Diversity = 1.2729e+01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 3.390223e+03 FES = 112100 μ_F = 0.569 μ_{CR} = 0.053
 Diversity = 1.2708e+01 k_t = [3, 3, 4, 3]
 Generation 1130: Best = 3.390223e+03 FES = 113100 μ_F = 0.606 μ_{CR} = 0.045
 Diversity = 1.2696e+01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 3.390223e+03 FES = 114100 μ_F = 0.595 μ_{CR} = 0.077
 Diversity = 1.2654e+01 k_t = [2, 2, 2, 4]
 Generation 1150: Best = 3.388187e+03 FES = 115100 μ_F = 0.590 μ_{CR} = 0.061
 Diversity = 1.2567e+01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 3.385444e+03 FES = 116100 μ_F = 0.560 μ_{CR} = 0.050
 Diversity = 1.2568e+01 k_t = [4, 3, 3, 2]
 Generation 1170: Best = 3.385444e+03 FES = 117100 μ_F = 0.534 μ_{CR} = 0.062
 Diversity = 1.2547e+01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 3.385268e+03 FES = 118100 μ_F = 0.550 μ_{CR} = 0.073
 Diversity = 1.2446e+01 k_t = [3, 3, 3, 4]
 Generation 1190: Best = 3.385268e+03 FES = 119100 μ_F = 0.619 μ_{CR} = 0.054
 Diversity = 1.2329e+01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 3.384423e+03 FES = 120100 μ_F = 0.597 μ_{CR} = 0.052
 Diversity = 1.2335e+01 k_t = [2, 4, 3, 4]
 Generation 1210: Best = 3.382000e+03 FES = 121100 μ_F = 0.498 μ_{CR} = 0.062
 Diversity = 1.2267e+01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 3.382000e+03 FES = 122100 μ_F = 0.431 μ_{CR} = 0.052
 Diversity = 1.2219e+01 k_t = [4, 2, 3, 2]
 Generation 1230: Best = 3.381675e+03 FES = 123100 μ_F = 0.430 μ_{CR} = 0.041
 Diversity = 1.2216e+01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 3.381675e+03 FES = 124100 μ_F = 0.463 μ_{CR} = 0.039
 Diversity = 1.2207e+01 k_t = [4, 2, 3, 3]
 Generation 1250: Best = 3.381675e+03 FES = 125100 μ_F = 0.476 μ_{CR} = 0.047
 Diversity = 1.2196e+01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 3.378867e+03 FES = 126100 μ_F = 0.501 μ_{CR} = 0.045
 Diversity = 1.2090e+01 k_t = [4, 2, 4, 2]
 Generation 1270: Best = 3.374798e+03 FES = 127100 μ_F = 0.449 μ_{CR} = 0.050
 Diversity = 1.2021e+01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 3.371491e+03 FES = 128100 μ_F = 0.467 μ_{CR} = 0.050
 Diversity = 1.1976e+01 k_t = [3, 2, 4, 2]
 Generation 1290: Best = 3.371343e+03 FES = 129100 μ_F = 0.478 μ_{CR} = 0.060
 Diversity = 1.1928e+01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 3.371343e+03 FES = 130100 μ_F = 0.486 μ_{CR} = 0.067
 Diversity = 1.1993e+01 k_t = [4, 2, 3, 2]

Generation 1310: Best = 3.371319e+03 FES = 131100 μ_F = 0.410 μ_{CR} = 0.051
 Diversity = 1.1974e+01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 3.371319e+03 FES = 132100 μ_F = 0.409 μ_{CR} = 0.040
 Diversity = 1.1959e+01 k_t = [4, 4, 2, 2]
 Generation 1330: Best = 3.371319e+03 FES = 133100 μ_F = 0.399 μ_{CR} = 0.041
 Diversity = 1.1951e+01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 3.368960e+03 FES = 134100 μ_F = 0.389 μ_{CR} = 0.054
 Diversity = 1.1939e+01 k_t = [2, 3, 4, 2]
 Generation 1350: Best = 3.367961e+03 FES = 135100 μ_F = 0.380 μ_{CR} = 0.063
 Diversity = 1.1924e+01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 3.363291e+03 FES = 136100 μ_F = 0.412 μ_{CR} = 0.061
 Diversity = 1.1902e+01 k_t = [4, 4, 3, 3]
 Generation 1370: Best = 3.363291e+03 FES = 137100 μ_F = 0.400 μ_{CR} = 0.048
 Diversity = 1.1878e+01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 3.359764e+03 FES = 138100 μ_F = 0.457 μ_{CR} = 0.048
 Diversity = 1.1908e+01 k_t = [3, 3, 3, 3]
 Generation 1390: Best = 3.359764e+03 FES = 139100 μ_F = 0.464 μ_{CR} = 0.050
 Diversity = 1.1887e+01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 3.359764e+03 FES = 140100 μ_F = 0.516 μ_{CR} = 0.053
 Diversity = 1.1873e+01 k_t = [2, 3, 2, 3]
 Generation 1410: Best = 3.358199e+03 FES = 141100 μ_F = 0.507 μ_{CR} = 0.048
 Diversity = 1.1849e+01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 3.358120e+03 FES = 142100 μ_F = 0.581 μ_{CR} = 0.044
 Diversity = 1.1776e+01 k_t = [4, 4, 2, 2]
 Generation 1430: Best = 3.358120e+03 FES = 143100 μ_F = 0.572 μ_{CR} = 0.043
 Diversity = 1.1779e+01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 3.358120e+03 FES = 144100 μ_F = 0.593 μ_{CR} = 0.048
 Diversity = 1.1598e+01 k_t = [3, 2, 3, 4]
 Generation 1450: Best = 3.358120e+03 FES = 145100 μ_F = 0.550 μ_{CR} = 0.053
 Diversity = 1.1510e+01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 3.358059e+03 FES = 146100 μ_F = 0.479 μ_{CR} = 0.054
 Diversity = 1.1507e+01 k_t = [3, 2, 3, 2]
 Generation 1470: Best = 3.358059e+03 FES = 147100 μ_F = 0.376 μ_{CR} = 0.062
 Diversity = 1.1499e+01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 3.357045e+03 FES = 148100 μ_F = 0.340 μ_{CR} = 0.068
 Diversity = 1.1559e+01 k_t = [4, 3, 2, 2]
 Generation 1490: Best = 3.357045e+03 FES = 149100 μ_F = 0.287 μ_{CR} = 0.064
 Diversity = 1.1545e+01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 3.357045e+03 FES = 150100 μ_F = 0.514 μ_{CR} = 0.050
 Diversity = 1.1464e+01 k_t = [4, 2, 3, 4]
 Generation 1510: Best = 3.356159e+03 FES = 151100 μ_F = 0.501 μ_{CR} = 0.060
 Diversity = 1.1410e+01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 3.356159e+03 FES = 152100 μ_F = 0.394 μ_{CR} = 0.061
 Diversity = 1.1406e+01 k_t = [4, 2, 3, 2]
 Generation 1530: Best = 3.356159e+03 FES = 153100 μ_F = 0.425 μ_{CR} = 0.063
 Diversity = 1.1287e+01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 3.356134e+03 FES = 154100 μ_F = 0.480 μ_{CR} = 0.052
 Diversity = 1.1301e+01 k_t = [3, 2, 3, 3]
 Generation 1550: Best = 3.356134e+03 FES = 155100 μ_F = 0.438 μ_{CR} = 0.048
 Diversity = 1.1298e+01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 3.356134e+03 FES = 156100 μ_F = 0.436 μ_{CR} = 0.046
 Diversity = 1.1285e+01 k_t = [3, 3, 2, 3]
 Generation 1570: Best = 3.355037e+03 FES = 157100 μ_F = 0.396 μ_{CR} = 0.051
 Diversity = 1.1281e+01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 3.355037e+03 FES = 158100 μ_F = 0.351 μ_{CR} = 0.050
 Diversity = 1.1272e+01 k_t = [3, 4, 2, 3]
 Generation 1590: Best = 3.355037e+03 FES = 159100 μ_F = 0.360 μ_{CR} = 0.054
 Diversity = 1.1273e+01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 3.354445e+03 FES = 160100 μ_F = 0.347 μ_{CR} = 0.065
 Diversity = 1.1267e+01 k_t = [3, 3, 4, 2]
 Generation 1610: Best = 3.354442e+03 FES = 161100 μ_F = 0.334 μ_{CR} = 0.063
 Diversity = 1.1257e+01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 3.354442e+03 FES = 162100 μ_F = 0.378 μ_{CR} = 0.043
 Diversity = 1.1236e+01 k_t = [4, 4, 2, 3]
 Generation 1630: Best = 3.354442e+03 FES = 163100 μ_F = 0.469 μ_{CR} = 0.042
 Diversity = 1.1160e+01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 3.354100e+03 FES = 164100 μ_F = 0.455 μ_{CR} = 0.038
 Diversity = 1.1172e+01 k_t = [3, 3, 4, 2]
 Generation 1650: Best = 3.354100e+03 FES = 165100 μ_F = 0.418 μ_{CR} = 0.036
 Diversity = 1.1187e+01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 3.354100e+03 FES = 166100 μ_F = 0.425 μ_{CR} = 0.049
 Diversity = 1.1184e+01 k_t = [4, 3, 2, 4]
 Generation 1670: Best = 3.354100e+03 FES = 167100 μ_F = 0.430 μ_{CR} = 0.038
 Diversity = 1.1162e+01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 3.354100e+03 FES = 168100 μ_F = 0.425 μ_{CR} = 0.047
 Diversity = 1.1055e+01 k_t = [2, 4, 4, 4]
 Generation 1690: Best = 3.354100e+03 FES = 169100 μ_F = 0.485 μ_{CR} = 0.042
 Diversity = 1.1039e+01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 3.354100e+03 FES = 170100 μ_F = 0.580 μ_{CR} = 0.046
 Diversity = 1.1034e+01 k_t = [2, 3, 2, 2]
 Generation 1710: Best = 3.354100e+03 FES = 171100 μ_F = 0.571 μ_{CR} = 0.044
 Diversity = 1.1033e+01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 3.354100e+03 FES = 172100 μ_F = 0.625 μ_{CR} = 0.036
 Diversity = 1.0945e+01 k_t = [3, 4, 4, 4]
 Generation 1730: Best = 3.354100e+03 FES = 173100 μ_F = 0.612 μ_{CR} = 0.042
 Diversity = 1.0916e+01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 3.354100e+03 FES = 174100 μ_F = 0.619 μ_{CR} = 0.041
 Diversity = 1.0897e+01 k_t = [3, 3, 4, 4]
 Generation 1750: Best = 3.354100e+03 FES = 175100 μ_F = 0.566 μ_{CR} = 0.044
 Diversity = 1.0860e+01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 3.354100e+03 FES = 176100 μ_F = 0.572 μ_{CR} = 0.042
 Diversity = 1.0865e+01 k_t = [4, 3, 3, 2]
 Generation 1770: Best = 3.354100e+03 FES = 177100 μ_F = 0.578 μ_{CR} = 0.042
 Diversity = 1.0775e+01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 3.354100e+03 FES = 178100 μ_F = 0.564 μ_{CR} = 0.036
 Diversity = 1.0774e+01 k_t = [2, 3, 3, 3]
 Generation 1790: Best = 3.354100e+03 FES = 179100 μ_F = 0.592 μ_{CR} = 0.035
 Diversity = 1.0731e+01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 3.354100e+03 FES = 180100 μ_F = 0.530 μ_{CR} = 0.035
 Diversity = 1.0721e+01 k_t = [2, 3, 3, 2]
 Generation 1810: Best = 3.353698e+03 FES = 181100 μ_F = 0.665 μ_{CR} = 0.038
 Diversity = 1.0718e+01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 3.353698e+03 FES = 182100 μ_F = 0.587 μ_{CR} = 0.043
 Diversity = 1.0718e+01 k_t = [2, 2, 2, 3]
 Generation 1830: Best = 3.353698e+03 FES = 183100 μ_F = 0.579 μ_{CR} = 0.046
 Diversity = 1.0717e+01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 3.352946e+03 FES = 184100 μ_F = 0.593 μ_{CR} = 0.040
 Diversity = 1.0790e+01 k_t = [3, 2, 2, 2]

Generation 1850: Best = 3.352946e+03	FES = 185100	$\mu_F = 0.549$	$\mu_{CR} = 0.032$
Diversity = 1.0789e+01	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 3.352946e+03	FES = 186100	$\mu_F = 0.498$	$\mu_{CR} = 0.032$
Diversity = 1.0788e+01	k_t = [3, 2, 4, 2]		
Generation 1870: Best = 3.352946e+03	FES = 187100	$\mu_F = 0.473$	$\mu_{CR} = 0.030$
Diversity = 1.0792e+01	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 3.352823e+03	FES = 188100	$\mu_F = 0.565$	$\mu_{CR} = 0.046$
Diversity = 1.0742e+01	k_t = [2, 4, 2, 4]		
Generation 1890: Best = 3.352823e+03	FES = 189100	$\mu_F = 0.596$	$\mu_{CR} = 0.040$
Diversity = 1.0738e+01	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 3.352823e+03	FES = 190100	$\mu_F = 0.710$	$\mu_{CR} = 0.038$
Diversity = 1.0704e+01	k_t = [2, 2, 4, 4]		
Generation 1910: Best = 3.352823e+03	FES = 191100	$\mu_F = 0.669$	$\mu_{CR} = 0.037$
Diversity = 1.0666e+01	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 3.352108e+03	FES = 192100	$\mu_F = 0.656$	$\mu_{CR} = 0.038$
Diversity = 1.0663e+01	k_t = [3, 4, 2, 3]		
Generation 1930: Best = 3.352108e+03	FES = 193100	$\mu_F = 0.654$	$\mu_{CR} = 0.038$
Diversity = 1.0607e+01	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 3.352108e+03	FES = 194100	$\mu_F = 0.647$	$\mu_{CR} = 0.045$
Diversity = 1.0495e+01	k_t = [4, 3, 4, 4]		
Generation 1950: Best = 3.352108e+03	FES = 195100	$\mu_F = 0.593$	$\mu_{CR} = 0.052$
Diversity = 1.0427e+01	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 3.352108e+03	FES = 196100	$\mu_F = 0.515$	$\mu_{CR} = 0.051$
Diversity = 1.0444e+01	k_t = [2, 3, 2, 3]		
Generation 1970: Best = 3.352087e+03	FES = 197100	$\mu_F = 0.570$	$\mu_{CR} = 0.052$
Diversity = 1.0436e+01	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 3.352087e+03	FES = 198100	$\mu_F = 0.513$	$\mu_{CR} = 0.058$
Diversity = 1.0435e+01	k_t = [2, 2, 4, 4]		
Generation 1990: Best = 3.352087e+03	FES = 199100	$\mu_F = 0.622$	$\mu_{CR} = 0.048$
Diversity = 1.0384e+01	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 3.351075e+03	FES = 200100	$\mu_F = 0.590$	$\mu_{CR} = 0.039$
Diversity = 1.0277e+01	k_t = [4, 4, 4, 2]		
Generation 2010: Best = 3.351075e+03	FES = 201100	$\mu_F = 0.595$	$\mu_{CR} = 0.052$
Diversity = 1.0265e+01	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 3.351075e+03	FES = 202100	$\mu_F = 0.545$	$\mu_{CR} = 0.068$
Diversity = 1.0196e+01	k_t = [2, 2, 3, 4]		
Generation 2030: Best = 3.351075e+03	FES = 203100	$\mu_F = 0.658$	$\mu_{CR} = 0.064$
Diversity = 1.0186e+01	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 3.351075e+03	FES = 204100	$\mu_F = 0.582$	$\mu_{CR} = 0.074$
Diversity = 1.0127e+01	k_t = [3, 3, 4, 3]		
Generation 2050: Best = 3.351075e+03	FES = 205100	$\mu_F = 0.591$	$\mu_{CR} = 0.081$
Diversity = 1.0125e+01	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 3.351075e+03	FES = 206100	$\mu_F = 0.533$	$\mu_{CR} = 0.066$
Diversity = 1.0113e+01	k_t = [4, 3, 3, 2]		
Generation 2070: Best = 3.350752e+03	FES = 207100	$\mu_F = 0.451$	$\mu_{CR} = 0.054$
Diversity = 1.0102e+01	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 3.350752e+03	FES = 208100	$\mu_F = 0.382$	$\mu_{CR} = 0.051$
Diversity = 1.0122e+01	k_t = [3, 3, 2, 4]		
Generation 2090: Best = 3.350752e+03	FES = 209100	$\mu_F = 0.453$	$\mu_{CR} = 0.046$
Diversity = 1.0116e+01	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 3.350752e+03	FES = 210100	$\mu_F = 0.468$	$\mu_{CR} = 0.045$
Diversity = 1.0150e+01	k_t = [3, 4, 4, 4]		
Generation 2110: Best = 3.350752e+03	FES = 211100	$\mu_F = 0.418$	$\mu_{CR} = 0.038$
Diversity = 1.0150e+01	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 3.350145e+03 FES = 212100 μ_F = 0.347 μ_{CR} = 0.062
 Diversity = 1.0144e+01 k_t = [3, 3, 3, 3]
 Generation 2130: Best = 3.350145e+03 FES = 213100 μ_F = 0.332 μ_{CR} = 0.041
 Diversity = 1.0138e+01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 3.349206e+03 FES = 214100 μ_F = 0.292 μ_{CR} = 0.050
 Diversity = 1.0134e+01 k_t = [2, 4, 4, 2]
 Generation 2150: Best = 3.349206e+03 FES = 215100 μ_F = 0.277 μ_{CR} = 0.046
 Diversity = 1.0121e+01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 3.349206e+03 FES = 216100 μ_F = 0.326 μ_{CR} = 0.056
 Diversity = 1.0119e+01 k_t = [4, 3, 3, 4]
 Generation 2170: Best = 3.349206e+03 FES = 217100 μ_F = 0.293 μ_{CR} = 0.059
 Diversity = 1.0115e+01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 3.349206e+03 FES = 218100 μ_F = 0.376 μ_{CR} = 0.058
 Diversity = 1.0114e+01 k_t = [3, 4, 4, 4]
 Generation 2190: Best = 3.349206e+03 FES = 219100 μ_F = 0.339 μ_{CR} = 0.055
 Diversity = 1.0116e+01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 3.348975e+03 FES = 220100 μ_F = 0.320 μ_{CR} = 0.061
 Diversity = 1.0130e+01 k_t = [2, 4, 3, 3]
 Generation 2210: Best = 3.348975e+03 FES = 221100 μ_F = 0.290 μ_{CR} = 0.074
 Diversity = 1.0129e+01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 3.348955e+03 FES = 222100 μ_F = 0.310 μ_{CR} = 0.066
 Diversity = 1.0129e+01 k_t = [4, 2, 4, 2]
 Generation 2230: Best = 3.347911e+03 FES = 223100 μ_F = 0.256 μ_{CR} = 0.079
 Diversity = 1.0129e+01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 3.346681e+03 FES = 224100 μ_F = 0.311 μ_{CR} = 0.072
 Diversity = 1.0083e+01 k_t = [3, 2, 2, 4]
 Generation 2250: Best = 3.346681e+03 FES = 225100 μ_F = 0.399 μ_{CR} = 0.052
 Diversity = 1.0011e+01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 3.346681e+03 FES = 226100 μ_F = 0.463 μ_{CR} = 0.051
 Diversity = 9.9883e+00 k_t = [3, 4, 4, 4]
 Generation 2270: Best = 3.346681e+03 FES = 227100 μ_F = 0.451 μ_{CR} = 0.057
 Diversity = 9.9888e+00 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 3.346681e+03 FES = 228100 μ_F = 0.440 μ_{CR} = 0.039
 Diversity = 9.9122e+00 k_t = [3, 4, 3, 4]
 Generation 2290: Best = 3.346354e+03 FES = 229100 μ_F = 0.359 μ_{CR} = 0.040
 Diversity = 9.9096e+00 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 3.346354e+03 FES = 230100 μ_F = 0.469 μ_{CR} = 0.038
 Diversity = 9.9075e+00 k_t = [4, 4, 4, 2]
 Generation 2310: Best = 3.346354e+03 FES = 231100 μ_F = 0.481 μ_{CR} = 0.037
 Diversity = 9.9102e+00 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 3.346354e+03 FES = 232100 μ_F = 0.428 μ_{CR} = 0.033
 Diversity = 9.9623e+00 k_t = [3, 3, 4, 4]
 Generation 2330: Best = 3.346052e+03 FES = 233100 μ_F = 0.454 μ_{CR} = 0.054
 Diversity = 9.9609e+00 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 3.346052e+03 FES = 234100 μ_F = 0.409 μ_{CR} = 0.067
 Diversity = 9.9539e+00 k_t = [4, 2, 4, 2]
 Generation 2350: Best = 3.346052e+03 FES = 235100 μ_F = 0.265 μ_{CR} = 0.055
 Diversity = 9.9489e+00 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 3.345414e+03 FES = 236100 μ_F = 0.307 μ_{CR} = 0.055
 Diversity = 9.9257e+00 k_t = [2, 4, 2, 4]
 Generation 2370: Best = 3.345414e+03 FES = 237100 μ_F = 0.233 μ_{CR} = 0.063
 Diversity = 9.9222e+00 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 3.345414e+03 FES = 238100 μ_F = 0.276 μ_{CR} = 0.061
 Diversity = 9.8937e+00 k_t = [2, 3, 4, 4]

Generation 2390: Best = 3.345388e+03 FES = 239100 μ_F = 0.166 μ_{CR} = 0.062
 Diversity = 9.8940e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 3.345388e+03 FES = 240100 μ_F = 0.210 μ_{CR} = 0.051
 Diversity = 9.7546e+00 k_t = [3, 4, 2, 4]
 Generation 2410: Best = 3.345388e+03 FES = 241100 μ_F = 0.189 μ_{CR} = 0.047
 Diversity = 9.7482e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 3.345388e+03 FES = 242100 μ_F = 0.278 μ_{CR} = 0.068
 Diversity = 9.3715e+00 k_t = [2, 4, 3, 4]
 Generation 2430: Best = 3.345388e+03 FES = 243100 μ_F = 0.292 μ_{CR} = 0.056
 Diversity = 9.3697e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 3.345188e+03 FES = 244100 μ_F = 0.305 μ_{CR} = 0.053
 Diversity = 9.3669e+00 k_t = [3, 3, 2, 3]
 Generation 2450: Best = 3.345188e+03 FES = 245100 μ_F = 0.286 μ_{CR} = 0.051
 Diversity = 9.3611e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 3.345188e+03 FES = 246100 μ_F = 0.292 μ_{CR} = 0.055
 Diversity = 9.3325e+00 k_t = [4, 3, 3, 2]
 Generation 2470: Best = 3.345188e+03 FES = 247100 μ_F = 0.292 μ_{CR} = 0.051
 Diversity = 9.3285e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 3.344978e+03 FES = 248100 μ_F = 0.291 μ_{CR} = 0.054
 Diversity = 9.3257e+00 k_t = [2, 4, 3, 3]
 Generation 2490: Best = 3.343859e+03 FES = 249100 μ_F = 0.401 μ_{CR} = 0.057
 Diversity = 9.3257e+00 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 3.343859e+03 FES = 250100 μ_F = 0.334 μ_{CR} = 0.094
 Diversity = 9.3222e+00 k_t = [2, 4, 4, 3]
 Generation 2510: Best = 3.343859e+03 FES = 251100 μ_F = 0.327 μ_{CR} = 0.069
 Diversity = 9.3196e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 3.343859e+03 FES = 252100 μ_F = 0.339 μ_{CR} = 0.053
 Diversity = 9.3219e+00 k_t = [2, 2, 4, 3]
 Generation 2530: Best = 3.343486e+03 FES = 253100 μ_F = 0.304 μ_{CR} = 0.056
 Diversity = 9.3175e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 3.343486e+03 FES = 254100 μ_F = 0.254 μ_{CR} = 0.048
 Diversity = 9.3171e+00 k_t = [2, 4, 3, 3]
 Generation 2550: Best = 3.343453e+03 FES = 255100 μ_F = 0.229 μ_{CR} = 0.051
 Diversity = 9.3148e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 3.343452e+03 FES = 256100 μ_F = 0.203 μ_{CR} = 0.053
 Diversity = 9.3125e+00 k_t = [3, 3, 2, 3]
 Generation 2570: Best = 3.343250e+03 FES = 257100 μ_F = 0.178 μ_{CR} = 0.054
 Diversity = 9.3130e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 3.343191e+03 FES = 258100 μ_F = 0.290 μ_{CR} = 0.050
 Diversity = 9.3151e+00 k_t = [3, 3, 2, 4]
 Generation 2590: Best = 3.343097e+03 FES = 259100 μ_F = 0.265 μ_{CR} = 0.042
 Diversity = 9.3140e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 3.343097e+03 FES = 260100 μ_F = 0.291 μ_{CR} = 0.037
 Diversity = 9.3130e+00 k_t = [2, 4, 2, 3]
 Generation 2610: Best = 3.342955e+03 FES = 261100 μ_F = 0.326 μ_{CR} = 0.037
 Diversity = 9.3090e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 3.342955e+03 FES = 262100 μ_F = 0.337 μ_{CR} = 0.044
 Diversity = 9.3089e+00 k_t = [4, 4, 2, 3]
 Generation 2630: Best = 3.342955e+03 FES = 263100 μ_F = 0.305 μ_{CR} = 0.067
 Diversity = 9.3070e+00 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 3.342955e+03 FES = 264100 μ_F = 0.312 μ_{CR} = 0.058
 Diversity = 9.2737e+00 k_t = [4, 2, 3, 2]
 Generation 2650: Best = 3.342918e+03 FES = 265100 μ_F = 0.353 μ_{CR} = 0.062
 Diversity = 9.2417e+00 k_t = [1, 1, 1, 1]

Generation 2660: Best = 3.342859e+03 FES = 266100 μ_F = 0.347 μ_{CR} = 0.054
 Diversity = 9.2421e+00 k_t = [3, 4, 4, 2]
 Generation 2670: Best = 3.342612e+03 FES = 267100 μ_F = 0.366 μ_{CR} = 0.041
 Diversity = 9.2376e+00 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 3.342611e+03 FES = 268100 μ_F = 0.391 μ_{CR} = 0.044
 Diversity = 9.2389e+00 k_t = [2, 4, 3, 4]
 Generation 2690: Best = 3.342304e+03 FES = 269100 μ_F = 0.342 μ_{CR} = 0.037
 Diversity = 9.2373e+00 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 3.342304e+03 FES = 270100 μ_F = 0.425 μ_{CR} = 0.045
 Diversity = 9.3039e+00 k_t = [3, 3, 2, 4]
 Generation 2710: Best = 3.342304e+03 FES = 271100 μ_F = 0.444 μ_{CR} = 0.047
 Diversity = 9.2939e+00 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 3.342304e+03 FES = 272100 μ_F = 0.464 μ_{CR} = 0.056
 Diversity = 9.3115e+00 k_t = [2, 4, 2, 4]
 Generation 2730: Best = 3.342304e+03 FES = 273100 μ_F = 0.426 μ_{CR} = 0.054
 Diversity = 9.3104e+00 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 3.342304e+03 FES = 274100 μ_F = 0.390 μ_{CR} = 0.040
 Diversity = 9.3106e+00 k_t = [3, 4, 4, 3]
 Generation 2750: Best = 3.342303e+03 FES = 275100 μ_F = 0.357 μ_{CR} = 0.049
 Diversity = 9.3099e+00 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 3.342206e+03 FES = 276100 μ_F = 0.306 μ_{CR} = 0.044
 Diversity = 9.3088e+00 k_t = [4, 4, 4, 2]
 Generation 2770: Best = 3.342027e+03 FES = 277100 μ_F = 0.266 μ_{CR} = 0.056
 Diversity = 9.3056e+00 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 3.342027e+03 FES = 278100 μ_F = 0.356 μ_{CR} = 0.058
 Diversity = 9.2803e+00 k_t = [4, 3, 4, 4]
 Generation 2790: Best = 3.342027e+03 FES = 279100 μ_F = 0.300 μ_{CR} = 0.070
 Diversity = 9.2776e+00 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 3.342027e+03 FES = 280100 μ_F = 0.345 μ_{CR} = 0.055
 Diversity = 9.2759e+00 k_t = [2, 2, 4, 2]
 Generation 2810: Best = 3.342027e+03 FES = 281100 μ_F = 0.304 μ_{CR} = 0.054
 Diversity = 9.2757e+00 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 3.342014e+03 FES = 282100 μ_F = 0.419 μ_{CR} = 0.054
 Diversity = 9.2479e+00 k_t = [4, 3, 3, 4]
 Generation 2830: Best = 3.342014e+03 FES = 283100 μ_F = 0.346 μ_{CR} = 0.049
 Diversity = 9.2473e+00 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 3.342014e+03 FES = 284100 μ_F = 0.334 μ_{CR} = 0.054
 Diversity = 9.2126e+00 k_t = [4, 3, 4, 3]
 Generation 2850: Best = 3.342014e+03 FES = 285100 μ_F = 0.286 μ_{CR} = 0.061
 Diversity = 9.2139e+00 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 3.341842e+03 FES = 286100 μ_F = 0.294 μ_{CR} = 0.068
 Diversity = 9.2142e+00 k_t = [3, 3, 4, 4]
 Generation 2870: Best = 3.341842e+03 FES = 287100 μ_F = 0.332 μ_{CR} = 0.053
 Diversity = 9.2025e+00 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 3.341842e+03 FES = 288100 μ_F = 0.353 μ_{CR} = 0.057
 Diversity = 9.1102e+00 k_t = [2, 3, 4, 4]
 Generation 2890: Best = 3.341842e+03 FES = 289100 μ_F = 0.295 μ_{CR} = 0.051
 Diversity = 9.1108e+00 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 3.341828e+03 FES = 290100 μ_F = 0.283 μ_{CR} = 0.058
 Diversity = 9.1088e+00 k_t = [4, 2, 4, 4]
 Generation 2910: Best = 3.341828e+03 FES = 291100 μ_F = 0.337 μ_{CR} = 0.050
 Diversity = 9.1089e+00 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 3.341762e+03 FES = 292100 μ_F = 0.354 μ_{CR} = 0.055
 Diversity = 9.1028e+00 k_t = [3, 4, 3, 2]

Generation 2930: Best = 3.341444e+03 FES = 293100 μ_F = 0.348 μ_{CR} = 0.050
 Diversity = 9.1024e+00 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 3.341444e+03 FES = 294100 μ_F = 0.355 μ_{CR} = 0.055
 Diversity = 9.1034e+00 k_t = [3, 2, 4, 3]
 Generation 2950: Best = 3.341236e+03 FES = 295100 μ_F = 0.269 μ_{CR} = 0.055
 Diversity = 9.1027e+00 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 3.341236e+03 FES = 296100 μ_F = 0.220 μ_{CR} = 0.064
 Diversity = 9.1032e+00 k_t = [3, 2, 3, 3]
 Generation 2970: Best = 3.341236e+03 FES = 297100 μ_F = 0.294 μ_{CR} = 0.064
 Diversity = 9.0944e+00 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 3.341232e+03 FES = 298100 μ_F = 0.302 μ_{CR} = 0.052
 Diversity = 9.0408e+00 k_t = [3, 3, 4, 3]
 Generation 2990: Best = 3.341206e+03 FES = 299100 μ_F = 0.231 μ_{CR} = 0.066
 Diversity = 9.0372e+00 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 3.341206e+03

Total Function Evals = 300000

Total Successful Updates = 9811

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Results:

Best Fitness : 3.341206e+03

Global Optimum : 2.900000e+03

Error (|best-optimum|) : 4.412057e+02

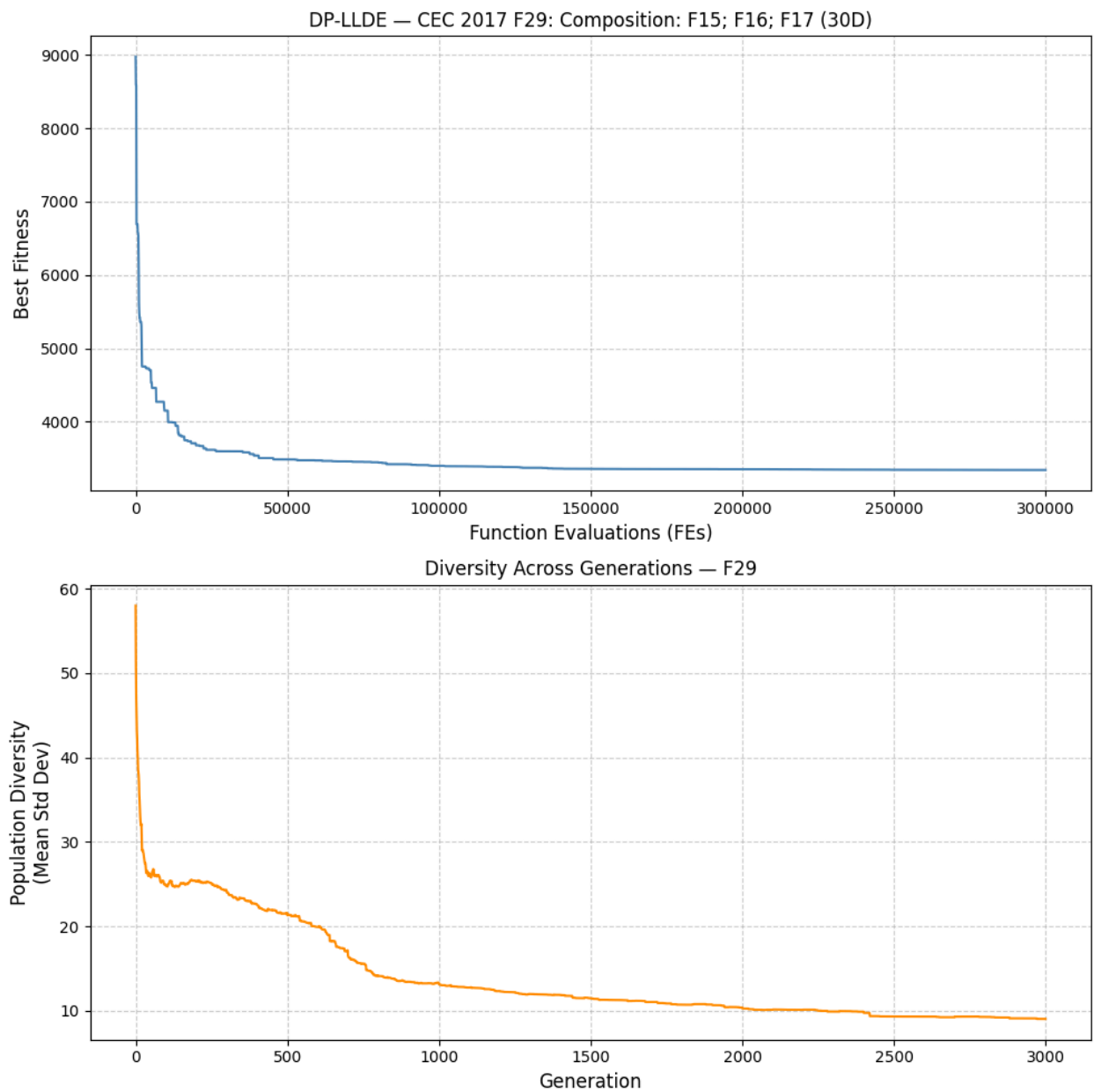
Best Solution (5 dim) : [-60.62955305 -85.47719899 89.07047976 100.

46.41573936]

$D^{(0)}$ (init. diversity): 5.7976e+01

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Graph saved as convergence_F29.png

```
In [ ]: # =====  
# F30 — Composition Function 10 (F15; F18; F19)  
# =====  
run_and_plot('F30', make_wrapper(composition.f30), 3000.0, 'Composition: F15;
```

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DP-LLDE on CEC 2017 F30 – Single Run

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Function : Composition: F15; F18; F19 (30D)

Bounds : [-100, 100]

Global optimum : 3000.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 3.426498e+10

Generation 10: Best = 2.056310e+09 FES = 1100 $\mu_F = 0.572$ $\mu_{CR} = 0.355$

Diversity = 3.2928e+01 $k_t = [1, 1, 1, 1]$

Generation 20: Best = 1.944171e+08 FES = 2100 $\mu_F = 0.631$ $\mu_{CR} = 0.362$

Diversity = 2.4959e+01 $k_t = [4, 4, 2, 4]$

Generation 30: Best = 2.428780e+07 FES = 3100 $\mu_F = 0.654$ $\mu_{CR} = 0.352$

Diversity = 2.2259e+01 $k_t = [1, 1, 1, 1]$

Generation 40: Best = 1.510815e+07 FES = 4100 $\mu_F = 0.675$ $\mu_{CR} = 0.355$

Diversity = 2.1324e+01 $k_t = [2, 3, 3, 2]$

Generation 50: Best = 8.409341e+06 FES = 5100 $\mu_F = 0.681$ $\mu_{CR} = 0.310$

Diversity = 1.9761e+01 $k_t = [1, 1, 1, 1]$

Generation 60: Best = 5.753876e+06 FES = 6100 $\mu_F = 0.690$ $\mu_{CR} = 0.245$

Diversity = 1.8027e+01 $k_t = [3, 3, 2, 3]$

Generation 70: Best = 3.290559e+06 FES = 7100 $\mu_F = 0.710$ $\mu_{CR} = 0.196$

Diversity = 1.7941e+01 $k_t = [1, 1, 1, 1]$

Generation 80: Best = 2.416741e+06 FES = 8100 $\mu_F = 0.722$ $\mu_{CR} = 0.147$

Diversity = 1.6524e+01 $k_t = [4, 2, 4, 2]$

Generation 90: Best = 1.590109e+06 FES = 9100 $\mu_F = 0.737$ $\mu_{CR} = 0.127$

Diversity = 1.5850e+01 $k_t = [1, 1, 1, 1]$

Generation 100: Best = 8.045611e+05 FES = 10100 $\mu_F = 0.761$ $\mu_{CR} = 0.098$

Diversity = 1.6446e+01 $k_t = [4, 4, 2, 2]$

Generation 110: Best = 3.401167e+05 FES = 11100 $\mu_F = 0.764$ $\mu_{CR} = 0.082$

Diversity = 1.4636e+01 $k_t = [1, 1, 1, 1]$

Generation 120: Best = 1.787710e+05 FES = 12100 $\mu_F = 0.766$ $\mu_{CR} = 0.079$

Diversity = 1.5274e+01 $k_t = [3, 3, 2, 2]$

Generation 130: Best = 1.787710e+05 FES = 13100 $\mu_F = 0.748$ $\mu_{CR} = 0.076$

Diversity = 1.3264e+01 $k_t = [1, 1, 1, 1]$

Generation 140: Best = 1.478121e+05 FES = 14100 $\mu_F = 0.748$ $\mu_{CR} = 0.067$

Diversity = 1.1995e+01 $k_t = [3, 3, 3, 3]$

Generation 150: Best = 1.177284e+05 FES = 15100 $\mu_F = 0.758$ $\mu_{CR} = 0.070$

Diversity = 1.2291e+01 $k_t = [1, 1, 1, 1]$

Generation 160: Best = 8.345561e+04 FES = 16100 $\mu_F = 0.751$ $\mu_{CR} = 0.062$

Diversity = 1.0980e+01 $k_t = [2, 4, 4, 2]$

Generation 170: Best = 5.732413e+04 FES = 17100 $\mu_F = 0.754$ $\mu_{CR} = 0.065$

Diversity = 1.0987e+01 $k_t = [1, 1, 1, 1]$

Generation 180: Best = 4.477230e+04 FES = 18100 $\mu_F = 0.755$ $\mu_{CR} = 0.058$

Diversity = 1.1469e+01 $k_t = [2, 2, 2, 4]$

Generation 190: Best = 4.043489e+04 FES = 19100 $\mu_F = 0.771$ $\mu_{CR} = 0.060$

Diversity = 9.6204e+00 $k_t = [1, 1, 1, 1]$

Generation 200: Best = 3.242316e+04 FES = 20100 $\mu_F = 0.768$ $\mu_{CR} = 0.055$

Diversity = 8.4234e+00 $k_t = [2, 4, 4, 3]$

Generation 210: Best = 2.955611e+04 FES = 21100 $\mu_F = 0.787$ $\mu_{CR} = 0.051$

Diversity = 7.8448e+00 $k_t = [1, 1, 1, 1]$

Generation 220: Best = 2.807089e+04 FES = 22100 $\mu_F = 0.794$ $\mu_{CR} = 0.047$

Diversity = 7.4847e+00 $k_t = [4, 4, 3, 2]$

Generation 230: Best = 2.610687e+04	FES = 23100	$\mu_F = 0.790$	$\mu_{CR} = 0.048$
Diversity = 8.0767e+00	$k_t = [1, 1, 1, 1]$		
Generation 240: Best = 2.438544e+04	FES = 24100	$\mu_F = 0.795$	$\mu_{CR} = 0.046$
Diversity = 7.5433e+00	$k_t = [2, 3, 4, 3]$		
Generation 250: Best = 2.362530e+04	FES = 25100	$\mu_F = 0.809$	$\mu_{CR} = 0.050$
Diversity = 7.8211e+00	$k_t = [1, 1, 1, 1]$		
Generation 260: Best = 2.164434e+04	FES = 26100	$\mu_F = 0.801$	$\mu_{CR} = 0.054$
Diversity = 8.2559e+00	$k_t = [4, 2, 4, 4]$		
Generation 270: Best = 2.087614e+04	FES = 27100	$\mu_F = 0.803$	$\mu_{CR} = 0.049$
Diversity = 8.1095e+00	$k_t = [1, 1, 1, 1]$		
Generation 280: Best = 1.930646e+04	FES = 28100	$\mu_F = 0.811$	$\mu_{CR} = 0.059$
Diversity = 8.2223e+00	$k_t = [4, 4, 3, 4]$		
Generation 290: Best = 1.782026e+04	FES = 29100	$\mu_F = 0.819$	$\mu_{CR} = 0.053$
Diversity = 7.7859e+00	$k_t = [1, 1, 1, 1]$		
Generation 300: Best = 1.676754e+04	FES = 30100	$\mu_F = 0.817$	$\mu_{CR} = 0.051$
Diversity = 7.2600e+00	$k_t = [4, 2, 2, 2]$		
Generation 310: Best = 1.471312e+04	FES = 31100	$\mu_F = 0.812$	$\mu_{CR} = 0.051$
Diversity = 7.1343e+00	$k_t = [1, 1, 1, 1]$		
Generation 320: Best = 1.388832e+04	FES = 32100	$\mu_F = 0.816$	$\mu_{CR} = 0.045$
Diversity = 6.5682e+00	$k_t = [4, 3, 3, 3]$		
Generation 330: Best = 1.388832e+04	FES = 33100	$\mu_F = 0.804$	$\mu_{CR} = 0.050$
Diversity = 6.5037e+00	$k_t = [1, 1, 1, 1]$		
Generation 340: Best = 1.351638e+04	FES = 34100	$\mu_F = 0.812$	$\mu_{CR} = 0.042$
Diversity = 5.7982e+00	$k_t = [3, 2, 3, 3]$		
Generation 350: Best = 1.314842e+04	FES = 35100	$\mu_F = 0.812$	$\mu_{CR} = 0.041$
Diversity = 5.4797e+00	$k_t = [1, 1, 1, 1]$		
Generation 360: Best = 1.236745e+04	FES = 36100	$\mu_F = 0.812$	$\mu_{CR} = 0.044$
Diversity = 4.7172e+00	$k_t = [3, 4, 4, 4]$		
Generation 370: Best = 1.221474e+04	FES = 37100	$\mu_F = 0.823$	$\mu_{CR} = 0.041$
Diversity = 4.3424e+00	$k_t = [1, 1, 1, 1]$		
Generation 380: Best = 1.172626e+04	FES = 38100	$\mu_F = 0.820$	$\mu_{CR} = 0.054$
Diversity = 4.1083e+00	$k_t = [4, 3, 3, 4]$		
Generation 390: Best = 1.150731e+04	FES = 39100	$\mu_F = 0.825$	$\mu_{CR} = 0.057$
Diversity = 4.1439e+00	$k_t = [1, 1, 1, 1]$		
Generation 400: Best = 1.150731e+04	FES = 40100	$\mu_F = 0.823$	$\mu_{CR} = 0.050$
Diversity = 4.3040e+00	$k_t = [4, 2, 2, 4]$		
Generation 410: Best = 1.120934e+04	FES = 41100	$\mu_F = 0.832$	$\mu_{CR} = 0.053$
Diversity = 3.9095e+00	$k_t = [1, 1, 1, 1]$		
Generation 420: Best = 1.097807e+04	FES = 42100	$\mu_F = 0.831$	$\mu_{CR} = 0.052$
Diversity = 3.6919e+00	$k_t = [2, 3, 3, 3]$		
Generation 430: Best = 1.082564e+04	FES = 43100	$\mu_F = 0.814$	$\mu_{CR} = 0.046$
Diversity = 3.7116e+00	$k_t = [1, 1, 1, 1]$		
Generation 440: Best = 1.054257e+04	FES = 44100	$\mu_F = 0.812$	$\mu_{CR} = 0.042$
Diversity = 3.6503e+00	$k_t = [4, 4, 4, 3]$		
Generation 450: Best = 1.029280e+04	FES = 45100	$\mu_F = 0.813$	$\mu_{CR} = 0.051$
Diversity = 3.4144e+00	$k_t = [1, 1, 1, 1]$		
Generation 460: Best = 1.009882e+04	FES = 46100	$\mu_F = 0.807$	$\mu_{CR} = 0.062$
Diversity = 2.8485e+00	$k_t = [4, 3, 3, 4]$		
Generation 470: Best = 9.912332e+03	FES = 47100	$\mu_F = 0.825$	$\mu_{CR} = 0.058$
Diversity = 2.4981e+00	$k_t = [1, 1, 1, 1]$		
Generation 480: Best = 9.754859e+03	FES = 48100	$\mu_F = 0.822$	$\mu_{CR} = 0.056$
Diversity = 2.3729e+00	$k_t = [4, 3, 3, 3]$		
Generation 490: Best = 9.681994e+03	FES = 49100	$\mu_F = 0.810$	$\mu_{CR} = 0.047$
Diversity = 2.1882e+00	$k_t = [1, 1, 1, 1]$		

Generation 500:	Best = 9.498293e+03	FES = 50100	$\mu_F = 0.800$	$\mu_{CR} = 0.041$
Diversity = 1.8534e+00	$k_t = [3, 2, 3, 3]$			
Generation 510:	Best = 9.336648e+03	FES = 51100	$\mu_F = 0.799$	$\mu_{CR} = 0.050$
Diversity = 1.8095e+00	$k_t = [1, 1, 1, 1]$			
Generation 520:	Best = 9.271779e+03	FES = 52100	$\mu_F = 0.809$	$\mu_{CR} = 0.072$
Diversity = 1.6873e+00	$k_t = [2, 2, 2, 4]$			
Generation 530:	Best = 9.206889e+03	FES = 53100	$\mu_F = 0.800$	$\mu_{CR} = 0.058$
Diversity = 1.7116e+00	$k_t = [1, 1, 1, 1]$			
Generation 540:	Best = 9.135567e+03	FES = 54100	$\mu_F = 0.792$	$\mu_{CR} = 0.057$
Diversity = 1.3394e+00	$k_t = [4, 3, 3, 3]$			
Generation 550:	Best = 9.053041e+03	FES = 55100	$\mu_F = 0.780$	$\mu_{CR} = 0.059$
Diversity = 1.2908e+00	$k_t = [1, 1, 1, 1]$			
Generation 560:	Best = 8.955945e+03	FES = 56100	$\mu_F = 0.772$	$\mu_{CR} = 0.054$
Diversity = 1.1821e+00	$k_t = [2, 4, 2, 3]$			
Generation 570:	Best = 8.919811e+03	FES = 57100	$\mu_F = 0.778$	$\mu_{CR} = 0.050$
Diversity = 1.1896e+00	$k_t = [1, 1, 1, 1]$			
Generation 580:	Best = 8.890490e+03	FES = 58100	$\mu_F = 0.783$	$\mu_{CR} = 0.043$
Diversity = 1.1161e+00	$k_t = [4, 2, 2, 2]$			
Generation 590:	Best = 8.846611e+03	FES = 59100	$\mu_F = 0.781$	$\mu_{CR} = 0.045$
Diversity = 1.0640e+00	$k_t = [1, 1, 1, 1]$			
Generation 600:	Best = 8.820630e+03	FES = 60100	$\mu_F = 0.787$	$\mu_{CR} = 0.046$
Diversity = 1.0170e+00	$k_t = [4, 3, 4, 2]$			
Generation 610:	Best = 8.802285e+03	FES = 61100	$\mu_F = 0.775$	$\mu_{CR} = 0.053$
Diversity = 1.0006e+00	$k_t = [1, 1, 1, 1]$			
Generation 620:	Best = 8.758918e+03	FES = 62100	$\mu_F = 0.762$	$\mu_{CR} = 0.066$
Diversity = 1.1452e+00	$k_t = [4, 4, 4, 2]$			
Generation 630:	Best = 8.730260e+03	FES = 63100	$\mu_F = 0.761$	$\mu_{CR} = 0.060$
Diversity = 1.2704e+00	$k_t = [1, 1, 1, 1]$			
Generation 640:	Best = 8.705952e+03	FES = 64100	$\mu_F = 0.780$	$\mu_{CR} = 0.049$
Diversity = 1.1627e+00	$k_t = [2, 4, 3, 2]$			
Generation 650:	Best = 8.697655e+03	FES = 65100	$\mu_F = 0.781$	$\mu_{CR} = 0.045$
Diversity = 1.2408e+00	$k_t = [1, 1, 1, 1]$			
Generation 660:	Best = 8.664189e+03	FES = 66100	$\mu_F = 0.782$	$\mu_{CR} = 0.053$
Diversity = 1.0216e+00	$k_t = [3, 2, 2, 4]$			
Generation 670:	Best = 8.657382e+03	FES = 67100	$\mu_F = 0.775$	$\mu_{CR} = 0.055$
Diversity = 1.0815e+00	$k_t = [1, 1, 1, 1]$			
Generation 680:	Best = 8.629819e+03	FES = 68100	$\mu_F = 0.768$	$\mu_{CR} = 0.055$
Diversity = 9.3192e-01	$k_t = [3, 4, 4, 3]$			
Generation 690:	Best = 8.612662e+03	FES = 69100	$\mu_F = 0.776$	$\mu_{CR} = 0.052$
Diversity = 1.0271e+00	$k_t = [1, 1, 1, 1]$			
Generation 700:	Best = 8.589125e+03	FES = 70100	$\mu_F = 0.783$	$\mu_{CR} = 0.058$
Diversity = 1.0080e+00	$k_t = [4, 4, 4, 4]$			
Generation 710:	Best = 8.575662e+03	FES = 71100	$\mu_F = 0.765$	$\mu_{CR} = 0.053$
Diversity = 8.7315e-01	$k_t = [1, 1, 1, 1]$			
Generation 720:	Best = 8.562316e+03	FES = 72100	$\mu_F = 0.768$	$\mu_{CR} = 0.047$
Diversity = 8.3641e-01	$k_t = [2, 4, 2, 4]$			
Generation 730:	Best = 8.553939e+03	FES = 73100	$\mu_F = 0.786$	$\mu_{CR} = 0.064$
Diversity = 9.0471e-01	$k_t = [1, 1, 1, 1]$			
Generation 740:	Best = 8.525846e+03	FES = 74100	$\mu_F = 0.781$	$\mu_{CR} = 0.051$
Diversity = 9.2655e-01	$k_t = [2, 3, 3, 3]$			
Generation 750:	Best = 8.505129e+03	FES = 75100	$\mu_F = 0.803$	$\mu_{CR} = 0.049$
Diversity = 1.1523e+00	$k_t = [1, 1, 1, 1]$			
Generation 760:	Best = 8.502494e+03	FES = 76100	$\mu_F = 0.816$	$\mu_{CR} = 0.051$
Diversity = 1.1303e+00	$k_t = [2, 3, 2, 3]$			

Generation 770: Best = 8.485124e+03 FES = 77100 μ_F = 0.792 μ_{CR} = 0.051
 Diversity = 1.0453e+00 k_t = [1, 1, 1, 1]
 Generation 780: Best = 8.480089e+03 FES = 78100 μ_F = 0.788 μ_{CR} = 0.047
 Diversity = 9.8761e-01 k_t = [3, 3, 2, 2]
 Generation 790: Best = 8.464490e+03 FES = 79100 μ_F = 0.776 μ_{CR} = 0.045
 Diversity = 1.0397e+00 k_t = [1, 1, 1, 1]
 Generation 800: Best = 8.443891e+03 FES = 80100 μ_F = 0.785 μ_{CR} = 0.056
 Diversity = 1.0732e+00 k_t = [4, 4, 3, 2]
 Generation 810: Best = 8.431245e+03 FES = 81100 μ_F = 0.795 μ_{CR} = 0.048
 Diversity = 1.0909e+00 k_t = [1, 1, 1, 1]
 Generation 820: Best = 8.412839e+03 FES = 82100 μ_F = 0.796 μ_{CR} = 0.060
 Diversity = 1.1229e+00 k_t = [2, 3, 3, 2]
 Generation 830: Best = 8.367581e+03 FES = 83100 μ_F = 0.799 μ_{CR} = 0.052
 Diversity = 1.1070e+00 k_t = [1, 1, 1, 1]
 Generation 840: Best = 8.366111e+03 FES = 84100 μ_F = 0.777 μ_{CR} = 0.039
 Diversity = 1.0718e+00 k_t = [3, 2, 4, 2]
 Generation 850: Best = 8.321661e+03 FES = 85100 μ_F = 0.772 μ_{CR} = 0.036
 Diversity = 1.3273e+00 k_t = [1, 1, 1, 1]
 Generation 860: Best = 8.302124e+03 FES = 86100 μ_F = 0.783 μ_{CR} = 0.039
 Diversity = 1.2835e+00 k_t = [3, 3, 4, 3]
 Generation 870: Best = 8.270003e+03 FES = 87100 μ_F = 0.779 μ_{CR} = 0.048
 Diversity = 1.3837e+00 k_t = [1, 1, 1, 1]
 Generation 880: Best = 8.251106e+03 FES = 88100 μ_F = 0.775 μ_{CR} = 0.043
 Diversity = 1.4022e+00 k_t = [3, 3, 3, 2]
 Generation 890: Best = 8.246544e+03 FES = 89100 μ_F = 0.785 μ_{CR} = 0.043
 Diversity = 1.2998e+00 k_t = [1, 1, 1, 1]
 Generation 900: Best = 8.208903e+03 FES = 90100 μ_F = 0.787 μ_{CR} = 0.045
 Diversity = 1.0336e+00 k_t = [3, 3, 2, 3]
 Generation 910: Best = 8.182530e+03 FES = 91100 μ_F = 0.794 μ_{CR} = 0.042
 Diversity = 1.0904e+00 k_t = [1, 1, 1, 1]
 Generation 920: Best = 8.170057e+03 FES = 92100 μ_F = 0.777 μ_{CR} = 0.058
 Diversity = 9.6906e-01 k_t = [2, 2, 4, 3]
 Generation 930: Best = 8.158317e+03 FES = 93100 μ_F = 0.801 μ_{CR} = 0.054
 Diversity = 1.0867e+00 k_t = [1, 1, 1, 1]
 Generation 940: Best = 8.138997e+03 FES = 94100 μ_F = 0.796 μ_{CR} = 0.052
 Diversity = 1.0407e+00 k_t = [2, 3, 4, 3]
 Generation 950: Best = 8.097574e+03 FES = 95100 μ_F = 0.799 μ_{CR} = 0.052
 Diversity = 1.0846e+00 k_t = [1, 1, 1, 1]
 Generation 960: Best = 8.085605e+03 FES = 96100 μ_F = 0.797 μ_{CR} = 0.050
 Diversity = 1.0254e+00 k_t = [2, 4, 2, 2]
 Generation 970: Best = 8.082693e+03 FES = 97100 μ_F = 0.785 μ_{CR} = 0.056
 Diversity = 9.3131e-01 k_t = [1, 1, 1, 1]
 Generation 980: Best = 8.065835e+03 FES = 98100 μ_F = 0.786 μ_{CR} = 0.063
 Diversity = 1.0079e+00 k_t = [2, 3, 3, 2]
 Generation 990: Best = 8.051998e+03 FES = 99100 μ_F = 0.772 μ_{CR} = 0.043
 Diversity = 9.6497e-01 k_t = [1, 1, 1, 1]
 Generation 1000: Best = 8.044954e+03 FES = 100100 μ_F = 0.772 μ_{CR} = 0.049
 Diversity = 1.0093e+00 k_t = [3, 2, 3, 3]
 Generation 1010: Best = 8.031265e+03 FES = 101100 μ_F = 0.784 μ_{CR} = 0.049
 Diversity = 1.0229e+00 k_t = [1, 1, 1, 1]
 Generation 1020: Best = 8.020189e+03 FES = 102100 μ_F = 0.786 μ_{CR} = 0.046
 Diversity = 1.2048e+00 k_t = [2, 3, 3, 4]
 Generation 1030: Best = 8.004426e+03 FES = 103100 μ_F = 0.788 μ_{CR} = 0.050
 Diversity = 1.1798e+00 k_t = [1, 1, 1, 1]

Generation 1040: Best = 7.996773e+03 FES = 104100 μ_F = 0.786 μ_{CR} = 0.046
 Diversity = 1.1388e+00 k_t = [3, 3, 2, 3]
 Generation 1050: Best = 7.969733e+03 FES = 105100 μ_F = 0.799 μ_{CR} = 0.039
 Diversity = 1.1325e+00 k_t = [1, 1, 1, 1]
 Generation 1060: Best = 7.957429e+03 FES = 106100 μ_F = 0.803 μ_{CR} = 0.044
 Diversity = 1.1929e+00 k_t = [3, 4, 4, 4]
 Generation 1070: Best = 7.937050e+03 FES = 107100 μ_F = 0.806 μ_{CR} = 0.048
 Diversity = 1.2086e+00 k_t = [1, 1, 1, 1]
 Generation 1080: Best = 7.922282e+03 FES = 108100 μ_F = 0.806 μ_{CR} = 0.043
 Diversity = 1.1152e+00 k_t = [3, 3, 4, 4]
 Generation 1090: Best = 7.874163e+03 FES = 109100 μ_F = 0.820 μ_{CR} = 0.048
 Diversity = 8.5714e-01 k_t = [1, 1, 1, 1]
 Generation 1100: Best = 7.852451e+03 FES = 110100 μ_F = 0.819 μ_{CR} = 0.049
 Diversity = 8.3427e-01 k_t = [2, 4, 3, 2]
 Generation 1110: Best = 7.809556e+03 FES = 111100 μ_F = 0.814 μ_{CR} = 0.059
 Diversity = 8.8983e-01 k_t = [1, 1, 1, 1]
 Generation 1120: Best = 7.795793e+03 FES = 112100 μ_F = 0.827 μ_{CR} = 0.050
 Diversity = 9.4680e-01 k_t = [3, 4, 2, 4]
 Generation 1130: Best = 7.766464e+03 FES = 113100 μ_F = 0.813 μ_{CR} = 0.045
 Diversity = 9.5960e-01 k_t = [1, 1, 1, 1]
 Generation 1140: Best = 7.748305e+03 FES = 114100 μ_F = 0.799 μ_{CR} = 0.053
 Diversity = 8.4292e-01 k_t = [4, 4, 3, 4]
 Generation 1150: Best = 7.721652e+03 FES = 115100 μ_F = 0.810 μ_{CR} = 0.052
 Diversity = 8.8831e-01 k_t = [1, 1, 1, 1]
 Generation 1160: Best = 7.721111e+03 FES = 116100 μ_F = 0.825 μ_{CR} = 0.052
 Diversity = 8.0795e-01 k_t = [4, 3, 3, 3]
 Generation 1170: Best = 7.717408e+03 FES = 117100 μ_F = 0.814 μ_{CR} = 0.044
 Diversity = 7.5697e-01 k_t = [1, 1, 1, 1]
 Generation 1180: Best = 7.709268e+03 FES = 118100 μ_F = 0.799 μ_{CR} = 0.040
 Diversity = 6.5851e-01 k_t = [2, 3, 3, 2]
 Generation 1190: Best = 7.697180e+03 FES = 119100 μ_F = 0.780 μ_{CR} = 0.042
 Diversity = 6.8165e-01 k_t = [1, 1, 1, 1]
 Generation 1200: Best = 7.697180e+03 FES = 120100 μ_F = 0.764 μ_{CR} = 0.051
 Diversity = 6.7411e-01 k_t = [4, 3, 4, 3]
 Generation 1210: Best = 7.689380e+03 FES = 121100 μ_F = 0.752 μ_{CR} = 0.052
 Diversity = 6.1288e-01 k_t = [1, 1, 1, 1]
 Generation 1220: Best = 7.676873e+03 FES = 122100 μ_F = 0.765 μ_{CR} = 0.056
 Diversity = 6.6374e-01 k_t = [3, 4, 3, 3]
 Generation 1230: Best = 7.670041e+03 FES = 123100 μ_F = 0.786 μ_{CR} = 0.054
 Diversity = 6.7234e-01 k_t = [1, 1, 1, 1]
 Generation 1240: Best = 7.658976e+03 FES = 124100 μ_F = 0.762 μ_{CR} = 0.065
 Diversity = 5.9563e-01 k_t = [2, 2, 2, 3]
 Generation 1250: Best = 7.645059e+03 FES = 125100 μ_F = 0.773 μ_{CR} = 0.057
 Diversity = 6.4466e-01 k_t = [1, 1, 1, 1]
 Generation 1260: Best = 7.637885e+03 FES = 126100 μ_F = 0.770 μ_{CR} = 0.057
 Diversity = 6.5163e-01 k_t = [4, 2, 4, 2]
 Generation 1270: Best = 7.629715e+03 FES = 127100 μ_F = 0.801 μ_{CR} = 0.057
 Diversity = 6.3808e-01 k_t = [1, 1, 1, 1]
 Generation 1280: Best = 7.622604e+03 FES = 128100 μ_F = 0.792 μ_{CR} = 0.043
 Diversity = 6.3417e-01 k_t = [3, 4, 4, 2]
 Generation 1290: Best = 7.610753e+03 FES = 129100 μ_F = 0.806 μ_{CR} = 0.048
 Diversity = 6.1392e-01 k_t = [1, 1, 1, 1]
 Generation 1300: Best = 7.609268e+03 FES = 130100 μ_F = 0.793 μ_{CR} = 0.055
 Diversity = 5.4488e-01 k_t = [2, 2, 4, 4]

Generation 1310: Best = 7.607569e+03 FES = 131100 μ_F = 0.794 μ_{CR} = 0.049
 Diversity = 5.2231e-01 k_t = [1, 1, 1, 1]
 Generation 1320: Best = 7.599365e+03 FES = 132100 μ_F = 0.791 μ_{CR} = 0.054
 Diversity = 5.0348e-01 k_t = [2, 3, 4, 2]
 Generation 1330: Best = 7.593145e+03 FES = 133100 μ_F = 0.789 μ_{CR} = 0.052
 Diversity = 4.6321e-01 k_t = [1, 1, 1, 1]
 Generation 1340: Best = 7.592993e+03 FES = 134100 μ_F = 0.771 μ_{CR} = 0.052
 Diversity = 4.8615e-01 k_t = [3, 2, 2, 3]
 Generation 1350: Best = 7.591007e+03 FES = 135100 μ_F = 0.778 μ_{CR} = 0.048
 Diversity = 4.6223e-01 k_t = [1, 1, 1, 1]
 Generation 1360: Best = 7.586870e+03 FES = 136100 μ_F = 0.783 μ_{CR} = 0.047
 Diversity = 4.5823e-01 k_t = [4, 3, 3, 4]
 Generation 1370: Best = 7.585251e+03 FES = 137100 μ_F = 0.750 μ_{CR} = 0.038
 Diversity = 4.2319e-01 k_t = [1, 1, 1, 1]
 Generation 1380: Best = 7.579599e+03 FES = 138100 μ_F = 0.763 μ_{CR} = 0.049
 Diversity = 4.2215e-01 k_t = [4, 4, 4, 2]
 Generation 1390: Best = 7.578952e+03 FES = 139100 μ_F = 0.776 μ_{CR} = 0.061
 Diversity = 4.1193e-01 k_t = [1, 1, 1, 1]
 Generation 1400: Best = 7.575419e+03 FES = 140100 μ_F = 0.782 μ_{CR} = 0.050
 Diversity = 4.3300e-01 k_t = [3, 2, 2, 3]
 Generation 1410: Best = 7.571407e+03 FES = 141100 μ_F = 0.792 μ_{CR} = 0.051
 Diversity = 4.8848e-01 k_t = [1, 1, 1, 1]
 Generation 1420: Best = 7.569848e+03 FES = 142100 μ_F = 0.785 μ_{CR} = 0.047
 Diversity = 4.8682e-01 k_t = [2, 4, 2, 2]
 Generation 1430: Best = 7.566645e+03 FES = 143100 μ_F = 0.782 μ_{CR} = 0.048
 Diversity = 5.0144e-01 k_t = [1, 1, 1, 1]
 Generation 1440: Best = 7.564778e+03 FES = 144100 μ_F = 0.774 μ_{CR} = 0.043
 Diversity = 5.0907e-01 k_t = [2, 4, 4, 3]
 Generation 1450: Best = 7.564166e+03 FES = 145100 μ_F = 0.794 μ_{CR} = 0.049
 Diversity = 5.0956e-01 k_t = [1, 1, 1, 1]
 Generation 1460: Best = 7.561863e+03 FES = 146100 μ_F = 0.793 μ_{CR} = 0.050
 Diversity = 5.0714e-01 k_t = [2, 2, 2, 3]
 Generation 1470: Best = 7.557289e+03 FES = 147100 μ_F = 0.815 μ_{CR} = 0.039
 Diversity = 4.6915e-01 k_t = [1, 1, 1, 1]
 Generation 1480: Best = 7.557289e+03 FES = 148100 μ_F = 0.817 μ_{CR} = 0.047
 Diversity = 4.4141e-01 k_t = [2, 3, 2, 4]
 Generation 1490: Best = 7.553513e+03 FES = 149100 μ_F = 0.828 μ_{CR} = 0.049
 Diversity = 4.9710e-01 k_t = [1, 1, 1, 1]
 Generation 1500: Best = 7.553513e+03 FES = 150100 μ_F = 0.839 μ_{CR} = 0.045
 Diversity = 4.5417e-01 k_t = [3, 2, 2, 4]
 Generation 1510: Best = 7.550638e+03 FES = 151100 μ_F = 0.829 μ_{CR} = 0.041
 Diversity = 4.3435e-01 k_t = [1, 1, 1, 1]
 Generation 1520: Best = 7.550138e+03 FES = 152100 μ_F = 0.827 μ_{CR} = 0.047
 Diversity = 4.3649e-01 k_t = [3, 3, 3, 4]
 Generation 1530: Best = 7.549272e+03 FES = 153100 μ_F = 0.819 μ_{CR} = 0.044
 Diversity = 4.0710e-01 k_t = [1, 1, 1, 1]
 Generation 1540: Best = 7.538481e+03 FES = 154100 μ_F = 0.812 μ_{CR} = 0.043
 Diversity = 3.9903e-01 k_t = [3, 3, 3, 4]
 Generation 1550: Best = 7.538481e+03 FES = 155100 μ_F = 0.820 μ_{CR} = 0.047
 Diversity = 4.8483e-01 k_t = [1, 1, 1, 1]
 Generation 1560: Best = 7.532794e+03 FES = 156100 μ_F = 0.827 μ_{CR} = 0.051
 Diversity = 6.0291e-01 k_t = [4, 2, 3, 4]
 Generation 1570: Best = 7.532794e+03 FES = 157100 μ_F = 0.834 μ_{CR} = 0.050
 Diversity = 6.0951e-01 k_t = [1, 1, 1, 1]

Generation 1580: Best = 7.532783e+03 FES = 158100 μ_F = 0.852 μ_{CR} = 0.043
 Diversity = 5.8550e-01 k_t = [4, 3, 3, 2]
 Generation 1590: Best = 7.532049e+03 FES = 159100 μ_F = 0.868 μ_{CR} = 0.040
 Diversity = 5.5432e-01 k_t = [1, 1, 1, 1]
 Generation 1600: Best = 7.527581e+03 FES = 160100 μ_F = 0.853 μ_{CR} = 0.048
 Diversity = 5.1381e-01 k_t = [2, 4, 3, 3]
 Generation 1610: Best = 7.523842e+03 FES = 161100 μ_F = 0.848 μ_{CR} = 0.049
 Diversity = 5.3849e-01 k_t = [1, 1, 1, 1]
 Generation 1620: Best = 7.523671e+03 FES = 162100 μ_F = 0.844 μ_{CR} = 0.048
 Diversity = 6.1046e-01 k_t = [3, 4, 4, 3]
 Generation 1630: Best = 7.523092e+03 FES = 163100 μ_F = 0.845 μ_{CR} = 0.052
 Diversity = 5.5473e-01 k_t = [1, 1, 1, 1]
 Generation 1640: Best = 7.518979e+03 FES = 164100 μ_F = 0.846 μ_{CR} = 0.057
 Diversity = 6.0262e-01 k_t = [3, 4, 3, 4]
 Generation 1650: Best = 7.515801e+03 FES = 165100 μ_F = 0.835 μ_{CR} = 0.062
 Diversity = 5.9013e-01 k_t = [1, 1, 1, 1]
 Generation 1660: Best = 7.515801e+03 FES = 166100 μ_F = 0.841 μ_{CR} = 0.063
 Diversity = 5.4516e-01 k_t = [4, 4, 4, 4]
 Generation 1670: Best = 7.514556e+03 FES = 167100 μ_F = 0.825 μ_{CR} = 0.051
 Diversity = 5.0107e-01 k_t = [1, 1, 1, 1]
 Generation 1680: Best = 7.510870e+03 FES = 168100 μ_F = 0.824 μ_{CR} = 0.042
 Diversity = 4.9539e-01 k_t = [4, 4, 3, 4]
 Generation 1690: Best = 7.510255e+03 FES = 169100 μ_F = 0.829 μ_{CR} = 0.046
 Diversity = 4.8001e-01 k_t = [1, 1, 1, 1]
 Generation 1700: Best = 7.508604e+03 FES = 170100 μ_F = 0.808 μ_{CR} = 0.052
 Diversity = 5.2596e-01 k_t = [3, 3, 3, 3]
 Generation 1710: Best = 7.506723e+03 FES = 171100 μ_F = 0.814 μ_{CR} = 0.053
 Diversity = 6.1824e-01 k_t = [1, 1, 1, 1]
 Generation 1720: Best = 7.503713e+03 FES = 172100 μ_F = 0.796 μ_{CR} = 0.045
 Diversity = 5.9836e-01 k_t = [3, 4, 3, 3]
 Generation 1730: Best = 7.503713e+03 FES = 173100 μ_F = 0.806 μ_{CR} = 0.055
 Diversity = 6.0675e-01 k_t = [1, 1, 1, 1]
 Generation 1740: Best = 7.502357e+03 FES = 174100 μ_F = 0.798 μ_{CR} = 0.042
 Diversity = 5.9936e-01 k_t = [2, 4, 2, 2]
 Generation 1750: Best = 7.497551e+03 FES = 175100 μ_F = 0.803 μ_{CR} = 0.040
 Diversity = 5.8500e-01 k_t = [1, 1, 1, 1]
 Generation 1760: Best = 7.495041e+03 FES = 176100 μ_F = 0.806 μ_{CR} = 0.035
 Diversity = 5.2706e-01 k_t = [3, 4, 4, 4]
 Generation 1770: Best = 7.492980e+03 FES = 177100 μ_F = 0.830 μ_{CR} = 0.033
 Diversity = 6.0936e-01 k_t = [1, 1, 1, 1]
 Generation 1780: Best = 7.492280e+03 FES = 178100 μ_F = 0.809 μ_{CR} = 0.039
 Diversity = 6.9167e-01 k_t = [4, 3, 4, 4]
 Generation 1790: Best = 7.491325e+03 FES = 179100 μ_F = 0.819 μ_{CR} = 0.053
 Diversity = 7.6195e-01 k_t = [1, 1, 1, 1]
 Generation 1800: Best = 7.480844e+03 FES = 180100 μ_F = 0.811 μ_{CR} = 0.041
 Diversity = 7.5046e-01 k_t = [2, 4, 3, 2]
 Generation 1810: Best = 7.478785e+03 FES = 181100 μ_F = 0.822 μ_{CR} = 0.039
 Diversity = 7.3516e-01 k_t = [1, 1, 1, 1]
 Generation 1820: Best = 7.478390e+03 FES = 182100 μ_F = 0.821 μ_{CR} = 0.047
 Diversity = 6.8018e-01 k_t = [4, 4, 4, 4]
 Generation 1830: Best = 7.473865e+03 FES = 183100 μ_F = 0.830 μ_{CR} = 0.040
 Diversity = 5.4314e-01 k_t = [1, 1, 1, 1]
 Generation 1840: Best = 7.473720e+03 FES = 184100 μ_F = 0.826 μ_{CR} = 0.043
 Diversity = 5.2790e-01 k_t = [2, 2, 3, 4]

Generation 1850: Best = 7.471103e+03	FES = 185100	$\mu_F = 0.825$	$\mu_{CR} = 0.046$
Diversity = 5.5426e-01	k_t = [1, 1, 1, 1]		
Generation 1860: Best = 7.469950e+03	FES = 186100	$\mu_F = 0.833$	$\mu_{CR} = 0.037$
Diversity = 5.8297e-01	k_t = [3, 3, 2, 3]		
Generation 1870: Best = 7.466140e+03	FES = 187100	$\mu_F = 0.846$	$\mu_{CR} = 0.034$
Diversity = 6.9959e-01	k_t = [1, 1, 1, 1]		
Generation 1880: Best = 7.464698e+03	FES = 188100	$\mu_F = 0.824$	$\mu_{CR} = 0.041$
Diversity = 6.2196e-01	k_t = [2, 4, 2, 3]		
Generation 1890: Best = 7.448617e+03	FES = 189100	$\mu_F = 0.824$	$\mu_{CR} = 0.044$
Diversity = 7.5990e-01	k_t = [1, 1, 1, 1]		
Generation 1900: Best = 7.443825e+03	FES = 190100	$\mu_F = 0.808$	$\mu_{CR} = 0.058$
Diversity = 8.7716e-01	k_t = [4, 4, 4, 3]		
Generation 1910: Best = 7.442376e+03	FES = 191100	$\mu_F = 0.824$	$\mu_{CR} = 0.052$
Diversity = 8.9045e-01	k_t = [1, 1, 1, 1]		
Generation 1920: Best = 7.431637e+03	FES = 192100	$\mu_F = 0.817$	$\mu_{CR} = 0.048$
Diversity = 9.1063e-01	k_t = [3, 3, 2, 2]		
Generation 1930: Best = 7.428515e+03	FES = 193100	$\mu_F = 0.812$	$\mu_{CR} = 0.052$
Diversity = 9.5377e-01	k_t = [1, 1, 1, 1]		
Generation 1940: Best = 7.421929e+03	FES = 194100	$\mu_F = 0.806$	$\mu_{CR} = 0.038$
Diversity = 8.5467e-01	k_t = [3, 4, 2, 4]		
Generation 1950: Best = 7.410646e+03	FES = 195100	$\mu_F = 0.819$	$\mu_{CR} = 0.041$
Diversity = 8.9790e-01	k_t = [1, 1, 1, 1]		
Generation 1960: Best = 7.398706e+03	FES = 196100	$\mu_F = 0.807$	$\mu_{CR} = 0.037$
Diversity = 8.5102e-01	k_t = [2, 4, 3, 2]		
Generation 1970: Best = 7.398706e+03	FES = 197100	$\mu_F = 0.805$	$\mu_{CR} = 0.042$
Diversity = 8.4398e-01	k_t = [1, 1, 1, 1]		
Generation 1980: Best = 7.397643e+03	FES = 198100	$\mu_F = 0.799$	$\mu_{CR} = 0.050$
Diversity = 7.3563e-01	k_t = [2, 4, 2, 4]		
Generation 1990: Best = 7.391286e+03	FES = 199100	$\mu_F = 0.797$	$\mu_{CR} = 0.041$
Diversity = 7.2194e-01	k_t = [1, 1, 1, 1]		
Generation 2000: Best = 7.380223e+03	FES = 200100	$\mu_F = 0.797$	$\mu_{CR} = 0.045$
Diversity = 7.2807e-01	k_t = [2, 4, 3, 3]		
Generation 2010: Best = 7.377783e+03	FES = 201100	$\mu_F = 0.806$	$\mu_{CR} = 0.045$
Diversity = 8.2654e-01	k_t = [1, 1, 1, 1]		
Generation 2020: Best = 7.372677e+03	FES = 202100	$\mu_F = 0.806$	$\mu_{CR} = 0.051$
Diversity = 7.8550e-01	k_t = [2, 2, 3, 4]		
Generation 2030: Best = 7.370290e+03	FES = 203100	$\mu_F = 0.788$	$\mu_{CR} = 0.045$
Diversity = 6.9919e-01	k_t = [1, 1, 1, 1]		
Generation 2040: Best = 7.359933e+03	FES = 204100	$\mu_F = 0.806$	$\mu_{CR} = 0.046$
Diversity = 7.7978e-01	k_t = [2, 4, 3, 2]		
Generation 2050: Best = 7.357973e+03	FES = 205100	$\mu_F = 0.787$	$\mu_{CR} = 0.045$
Diversity = 7.5617e-01	k_t = [1, 1, 1, 1]		
Generation 2060: Best = 7.349507e+03	FES = 206100	$\mu_F = 0.797$	$\mu_{CR} = 0.050$
Diversity = 8.2815e-01	k_t = [3, 2, 4, 3]		
Generation 2070: Best = 7.344986e+03	FES = 207100	$\mu_F = 0.787$	$\mu_{CR} = 0.046$
Diversity = 7.7510e-01	k_t = [1, 1, 1, 1]		
Generation 2080: Best = 7.338766e+03	FES = 208100	$\mu_F = 0.773$	$\mu_{CR} = 0.051$
Diversity = 7.3346e-01	k_t = [2, 4, 4, 2]		
Generation 2090: Best = 7.338330e+03	FES = 209100	$\mu_F = 0.779$	$\mu_{CR} = 0.043$
Diversity = 7.9605e-01	k_t = [1, 1, 1, 1]		
Generation 2100: Best = 7.328473e+03	FES = 210100	$\mu_F = 0.791$	$\mu_{CR} = 0.041$
Diversity = 7.8630e-01	k_t = [4, 4, 3, 3]		
Generation 2110: Best = 7.311056e+03	FES = 211100	$\mu_F = 0.784$	$\mu_{CR} = 0.040$
Diversity = 7.4737e-01	k_t = [1, 1, 1, 1]		

Generation 2120: Best = 7.309363e+03 FES = 212100 μ_F = 0.795 μ_{CR} = 0.053
 Diversity = 8.0462e-01 k_t = [2, 4, 2, 4]
 Generation 2130: Best = 7.305113e+03 FES = 213100 μ_F = 0.775 μ_{CR} = 0.055
 Diversity = 7.9482e-01 k_t = [1, 1, 1, 1]
 Generation 2140: Best = 7.301748e+03 FES = 214100 μ_F = 0.796 μ_{CR} = 0.055
 Diversity = 7.1766e-01 k_t = [4, 4, 2, 4]
 Generation 2150: Best = 7.296122e+03 FES = 215100 μ_F = 0.759 μ_{CR} = 0.054
 Diversity = 6.6490e-01 k_t = [1, 1, 1, 1]
 Generation 2160: Best = 7.296121e+03 FES = 216100 μ_F = 0.765 μ_{CR} = 0.060
 Diversity = 6.4676e-01 k_t = [4, 3, 3, 4]
 Generation 2170: Best = 7.290697e+03 FES = 217100 μ_F = 0.764 μ_{CR} = 0.051
 Diversity = 7.2588e-01 k_t = [1, 1, 1, 1]
 Generation 2180: Best = 7.287316e+03 FES = 218100 μ_F = 0.766 μ_{CR} = 0.058
 Diversity = 6.8148e-01 k_t = [3, 2, 2, 2]
 Generation 2190: Best = 7.287316e+03 FES = 219100 μ_F = 0.772 μ_{CR} = 0.045
 Diversity = 6.2268e-01 k_t = [1, 1, 1, 1]
 Generation 2200: Best = 7.283608e+03 FES = 220100 μ_F = 0.794 μ_{CR} = 0.059
 Diversity = 5.4408e-01 k_t = [4, 3, 3, 4]
 Generation 2210: Best = 7.282111e+03 FES = 221100 μ_F = 0.800 μ_{CR} = 0.050
 Diversity = 5.0352e-01 k_t = [1, 1, 1, 1]
 Generation 2220: Best = 7.276419e+03 FES = 222100 μ_F = 0.795 μ_{CR} = 0.046
 Diversity = 4.4432e-01 k_t = [4, 3, 4, 3]
 Generation 2230: Best = 7.276167e+03 FES = 223100 μ_F = 0.799 μ_{CR} = 0.046
 Diversity = 4.1142e-01 k_t = [1, 1, 1, 1]
 Generation 2240: Best = 7.272944e+03 FES = 224100 μ_F = 0.819 μ_{CR} = 0.052
 Diversity = 3.8296e-01 k_t = [4, 2, 3, 4]
 Generation 2250: Best = 7.272205e+03 FES = 225100 μ_F = 0.800 μ_{CR} = 0.051
 Diversity = 3.8129e-01 k_t = [1, 1, 1, 1]
 Generation 2260: Best = 7.270661e+03 FES = 226100 μ_F = 0.798 μ_{CR} = 0.049
 Diversity = 4.0987e-01 k_t = [2, 3, 3, 4]
 Generation 2270: Best = 7.270661e+03 FES = 227100 μ_F = 0.804 μ_{CR} = 0.047
 Diversity = 3.8966e-01 k_t = [1, 1, 1, 1]
 Generation 2280: Best = 7.269445e+03 FES = 228100 μ_F = 0.817 μ_{CR} = 0.047
 Diversity = 3.9649e-01 k_t = [3, 2, 2, 3]
 Generation 2290: Best = 7.254580e+03 FES = 229100 μ_F = 0.821 μ_{CR} = 0.042
 Diversity = 4.4049e-01 k_t = [1, 1, 1, 1]
 Generation 2300: Best = 7.254011e+03 FES = 230100 μ_F = 0.842 μ_{CR} = 0.046
 Diversity = 5.7046e-01 k_t = [4, 4, 4, 3]
 Generation 2310: Best = 7.246124e+03 FES = 231100 μ_F = 0.841 μ_{CR} = 0.049
 Diversity = 7.7516e-01 k_t = [1, 1, 1, 1]
 Generation 2320: Best = 7.238670e+03 FES = 232100 μ_F = 0.826 μ_{CR} = 0.055
 Diversity = 8.4809e-01 k_t = [3, 2, 4, 2]
 Generation 2330: Best = 7.233876e+03 FES = 233100 μ_F = 0.813 μ_{CR} = 0.041
 Diversity = 9.6925e-01 k_t = [1, 1, 1, 1]
 Generation 2340: Best = 7.228115e+03 FES = 234100 μ_F = 0.825 μ_{CR} = 0.038
 Diversity = 8.3069e-01 k_t = [3, 3, 3, 2]
 Generation 2350: Best = 7.223445e+03 FES = 235100 μ_F = 0.821 μ_{CR} = 0.044
 Diversity = 8.4592e-01 k_t = [1, 1, 1, 1]
 Generation 2360: Best = 7.211406e+03 FES = 236100 μ_F = 0.821 μ_{CR} = 0.044
 Diversity = 1.0726e+00 k_t = [3, 3, 3, 3]
 Generation 2370: Best = 7.204540e+03 FES = 237100 μ_F = 0.827 μ_{CR} = 0.052
 Diversity = 9.3026e-01 k_t = [1, 1, 1, 1]
 Generation 2380: Best = 7.199119e+03 FES = 238100 μ_F = 0.832 μ_{CR} = 0.044
 Diversity = 1.1109e+00 k_t = [3, 2, 2, 2]

Generation 2390: Best = 7.175170e+03 FES = 239100 μ_F = 0.830 μ_{CR} = 0.044
 Diversity = 1.2818e+00 k_t = [1, 1, 1, 1]
 Generation 2400: Best = 7.170556e+03 FES = 240100 μ_F = 0.826 μ_{CR} = 0.051
 Diversity = 1.2487e+00 k_t = [2, 2, 4, 2]
 Generation 2410: Best = 7.161717e+03 FES = 241100 μ_F = 0.815 μ_{CR} = 0.049
 Diversity = 1.1547e+00 k_t = [1, 1, 1, 1]
 Generation 2420: Best = 7.140383e+03 FES = 242100 μ_F = 0.804 μ_{CR} = 0.048
 Diversity = 9.6615e-01 k_t = [4, 3, 3, 4]
 Generation 2430: Best = 7.125400e+03 FES = 243100 μ_F = 0.814 μ_{CR} = 0.045
 Diversity = 1.0792e+00 k_t = [1, 1, 1, 1]
 Generation 2440: Best = 7.119270e+03 FES = 244100 μ_F = 0.816 μ_{CR} = 0.041
 Diversity = 1.1070e+00 k_t = [4, 2, 4, 4]
 Generation 2450: Best = 7.106899e+03 FES = 245100 μ_F = 0.820 μ_{CR} = 0.051
 Diversity = 1.0200e+00 k_t = [1, 1, 1, 1]
 Generation 2460: Best = 7.095786e+03 FES = 246100 μ_F = 0.825 μ_{CR} = 0.051
 Diversity = 1.0434e+00 k_t = [3, 4, 3, 3]
 Generation 2470: Best = 7.087389e+03 FES = 247100 μ_F = 0.820 μ_{CR} = 0.037
 Diversity = 1.0068e+00 k_t = [1, 1, 1, 1]
 Generation 2480: Best = 7.082442e+03 FES = 248100 μ_F = 0.811 μ_{CR} = 0.044
 Diversity = 1.1406e+00 k_t = [4, 4, 3, 2]
 Generation 2490: Best = 7.074208e+03 FES = 249100 μ_F = 0.808 μ_{CR} = 0.049
 Diversity = 9.7922e-01 k_t = [1, 1, 1, 1]
 Generation 2500: Best = 7.066776e+03 FES = 250100 μ_F = 0.794 μ_{CR} = 0.054
 Diversity = 9.6849e-01 k_t = [2, 3, 4, 3]
 Generation 2510: Best = 7.060502e+03 FES = 251100 μ_F = 0.799 μ_{CR} = 0.048
 Diversity = 1.0979e+00 k_t = [1, 1, 1, 1]
 Generation 2520: Best = 7.054438e+03 FES = 252100 μ_F = 0.811 μ_{CR} = 0.054
 Diversity = 1.0334e+00 k_t = [3, 2, 4, 3]
 Generation 2530: Best = 7.041957e+03 FES = 253100 μ_F = 0.786 μ_{CR} = 0.045
 Diversity = 1.0891e+00 k_t = [1, 1, 1, 1]
 Generation 2540: Best = 7.024905e+03 FES = 254100 μ_F = 0.795 μ_{CR} = 0.047
 Diversity = 1.0332e+00 k_t = [4, 4, 3, 2]
 Generation 2550: Best = 7.019939e+03 FES = 255100 μ_F = 0.807 μ_{CR} = 0.040
 Diversity = 1.1458e+00 k_t = [1, 1, 1, 1]
 Generation 2560: Best = 7.011816e+03 FES = 256100 μ_F = 0.793 μ_{CR} = 0.043
 Diversity = 1.2624e+00 k_t = [4, 4, 2, 2]
 Generation 2570: Best = 7.004602e+03 FES = 257100 μ_F = 0.813 μ_{CR} = 0.044
 Diversity = 1.1716e+00 k_t = [1, 1, 1, 1]
 Generation 2580: Best = 6.989505e+03 FES = 258100 μ_F = 0.818 μ_{CR} = 0.052
 Diversity = 1.1220e+00 k_t = [2, 4, 2, 3]
 Generation 2590: Best = 6.987236e+03 FES = 259100 μ_F = 0.806 μ_{CR} = 0.051
 Diversity = 1.2072e+00 k_t = [1, 1, 1, 1]
 Generation 2600: Best = 6.976373e+03 FES = 260100 μ_F = 0.822 μ_{CR} = 0.043
 Diversity = 1.2171e+00 k_t = [3, 4, 4, 4]
 Generation 2610: Best = 6.971308e+03 FES = 261100 μ_F = 0.803 μ_{CR} = 0.053
 Diversity = 1.1888e+00 k_t = [1, 1, 1, 1]
 Generation 2620: Best = 6.968155e+03 FES = 262100 μ_F = 0.797 μ_{CR} = 0.051
 Diversity = 1.0708e+00 k_t = [4, 2, 4, 4]
 Generation 2630: Best = 6.960650e+03 FES = 263100 μ_F = 0.806 μ_{CR} = 0.053
 Diversity = 9.6238e-01 k_t = [1, 1, 1, 1]
 Generation 2640: Best = 6.958621e+03 FES = 264100 μ_F = 0.805 μ_{CR} = 0.053
 Diversity = 9.8675e-01 k_t = [4, 2, 3, 3]
 Generation 2650: Best = 6.948124e+03 FES = 265100 μ_F = 0.806 μ_{CR} = 0.050
 Diversity = 9.4259e-01 k_t = [1, 1, 1, 1]

Generation 2660: Best = 6.942413e+03 FES = 266100 μ_F = 0.792 μ_{CR} = 0.039
 Diversity = 8.7712e-01 k_t = [3, 2, 2, 2]
 Generation 2670: Best = 6.937222e+03 FES = 267100 μ_F = 0.797 μ_{CR} = 0.049
 Diversity = 8.2742e-01 k_t = [1, 1, 1, 1]
 Generation 2680: Best = 6.931122e+03 FES = 268100 μ_F = 0.791 μ_{CR} = 0.047
 Diversity = 9.0254e-01 k_t = [4, 2, 2, 2]
 Generation 2690: Best = 6.926840e+03 FES = 269100 μ_F = 0.802 μ_{CR} = 0.044
 Diversity = 8.3914e-01 k_t = [1, 1, 1, 1]
 Generation 2700: Best = 6.916033e+03 FES = 270100 μ_F = 0.810 μ_{CR} = 0.041
 Diversity = 8.3599e-01 k_t = [2, 2, 3, 4]
 Generation 2710: Best = 6.912589e+03 FES = 271100 μ_F = 0.823 μ_{CR} = 0.046
 Diversity = 8.0152e-01 k_t = [1, 1, 1, 1]
 Generation 2720: Best = 6.901756e+03 FES = 272100 μ_F = 0.819 μ_{CR} = 0.053
 Diversity = 7.8725e-01 k_t = [2, 4, 4, 2]
 Generation 2730: Best = 6.895574e+03 FES = 273100 μ_F = 0.813 μ_{CR} = 0.052
 Diversity = 8.2848e-01 k_t = [1, 1, 1, 1]
 Generation 2740: Best = 6.893196e+03 FES = 274100 μ_F = 0.793 μ_{CR} = 0.044
 Diversity = 6.9071e-01 k_t = [2, 3, 4, 3]
 Generation 2750: Best = 6.886703e+03 FES = 275100 μ_F = 0.812 μ_{CR} = 0.045
 Diversity = 8.4099e-01 k_t = [1, 1, 1, 1]
 Generation 2760: Best = 6.876226e+03 FES = 276100 μ_F = 0.797 μ_{CR} = 0.050
 Diversity = 7.7530e-01 k_t = [3, 3, 4, 2]
 Generation 2770: Best = 6.871647e+03 FES = 277100 μ_F = 0.787 μ_{CR} = 0.044
 Diversity = 8.1630e-01 k_t = [1, 1, 1, 1]
 Generation 2780: Best = 6.871647e+03 FES = 278100 μ_F = 0.791 μ_{CR} = 0.048
 Diversity = 8.8016e-01 k_t = [2, 4, 3, 2]
 Generation 2790: Best = 6.858664e+03 FES = 279100 μ_F = 0.804 μ_{CR} = 0.046
 Diversity = 9.0818e-01 k_t = [1, 1, 1, 1]
 Generation 2800: Best = 6.856713e+03 FES = 280100 μ_F = 0.804 μ_{CR} = 0.044
 Diversity = 9.7272e-01 k_t = [4, 3, 2, 2]
 Generation 2810: Best = 6.846656e+03 FES = 281100 μ_F = 0.777 μ_{CR} = 0.047
 Diversity = 9.0044e-01 k_t = [1, 1, 1, 1]
 Generation 2820: Best = 6.842028e+03 FES = 282100 μ_F = 0.782 μ_{CR} = 0.041
 Diversity = 7.1863e-01 k_t = [2, 2, 4, 3]
 Generation 2830: Best = 6.834966e+03 FES = 283100 μ_F = 0.793 μ_{CR} = 0.048
 Diversity = 6.9467e-01 k_t = [1, 1, 1, 1]
 Generation 2840: Best = 6.824349e+03 FES = 284100 μ_F = 0.785 μ_{CR} = 0.049
 Diversity = 6.8978e-01 k_t = [4, 4, 2, 3]
 Generation 2850: Best = 6.823518e+03 FES = 285100 μ_F = 0.799 μ_{CR} = 0.065
 Diversity = 7.3730e-01 k_t = [1, 1, 1, 1]
 Generation 2860: Best = 6.812658e+03 FES = 286100 μ_F = 0.795 μ_{CR} = 0.050
 Diversity = 6.8878e-01 k_t = [3, 4, 2, 2]
 Generation 2870: Best = 6.805119e+03 FES = 287100 μ_F = 0.792 μ_{CR} = 0.050
 Diversity = 6.8403e-01 k_t = [1, 1, 1, 1]
 Generation 2880: Best = 6.794411e+03 FES = 288100 μ_F = 0.781 μ_{CR} = 0.045
 Diversity = 7.7979e-01 k_t = [3, 4, 3, 2]
 Generation 2890: Best = 6.784065e+03 FES = 289100 μ_F = 0.789 μ_{CR} = 0.044
 Diversity = 8.1051e-01 k_t = [1, 1, 1, 1]
 Generation 2900: Best = 6.777768e+03 FES = 290100 μ_F = 0.785 μ_{CR} = 0.048
 Diversity = 8.4620e-01 k_t = [2, 3, 2, 2]
 Generation 2910: Best = 6.765283e+03 FES = 291100 μ_F = 0.778 μ_{CR} = 0.043
 Diversity = 8.6656e-01 k_t = [1, 1, 1, 1]
 Generation 2920: Best = 6.762300e+03 FES = 292100 μ_F = 0.780 μ_{CR} = 0.044
 Diversity = 8.7167e-01 k_t = [4, 2, 3, 2]

Generation 2930: Best = 6.760588e+03 FES = 293100 μ_F = 0.773 μ_{CR} = 0.045
 Diversity = 8.7786e-01 k_t = [1, 1, 1, 1]
 Generation 2940: Best = 6.748552e+03 FES = 294100 μ_F = 0.771 μ_{CR} = 0.041
 Diversity = 8.6704e-01 k_t = [3, 4, 3, 2]
 Generation 2950: Best = 6.743919e+03 FES = 295100 μ_F = 0.762 μ_{CR} = 0.039
 Diversity = 8.9498e-01 k_t = [1, 1, 1, 1]
 Generation 2960: Best = 6.743230e+03 FES = 296100 μ_F = 0.750 μ_{CR} = 0.036
 Diversity = 9.9358e-01 k_t = [4, 2, 3, 2]
 Generation 2970: Best = 6.734989e+03 FES = 297100 μ_F = 0.744 μ_{CR} = 0.040
 Diversity = 8.3865e-01 k_t = [1, 1, 1, 1]
 Generation 2980: Best = 6.734989e+03 FES = 298100 μ_F = 0.772 μ_{CR} = 0.045
 Diversity = 8.4639e-01 k_t = [4, 2, 3, 3]
 Generation 2990: Best = 6.729131e+03 FES = 299100 μ_F = 0.760 μ_{CR} = 0.053
 Diversity = 8.3993e-01 k_t = [1, 1, 1, 1]

Optimization complete.

Best Fitness = 6.726178e+03

Total Function Evals = 300000

Total Successful Updates = 46953

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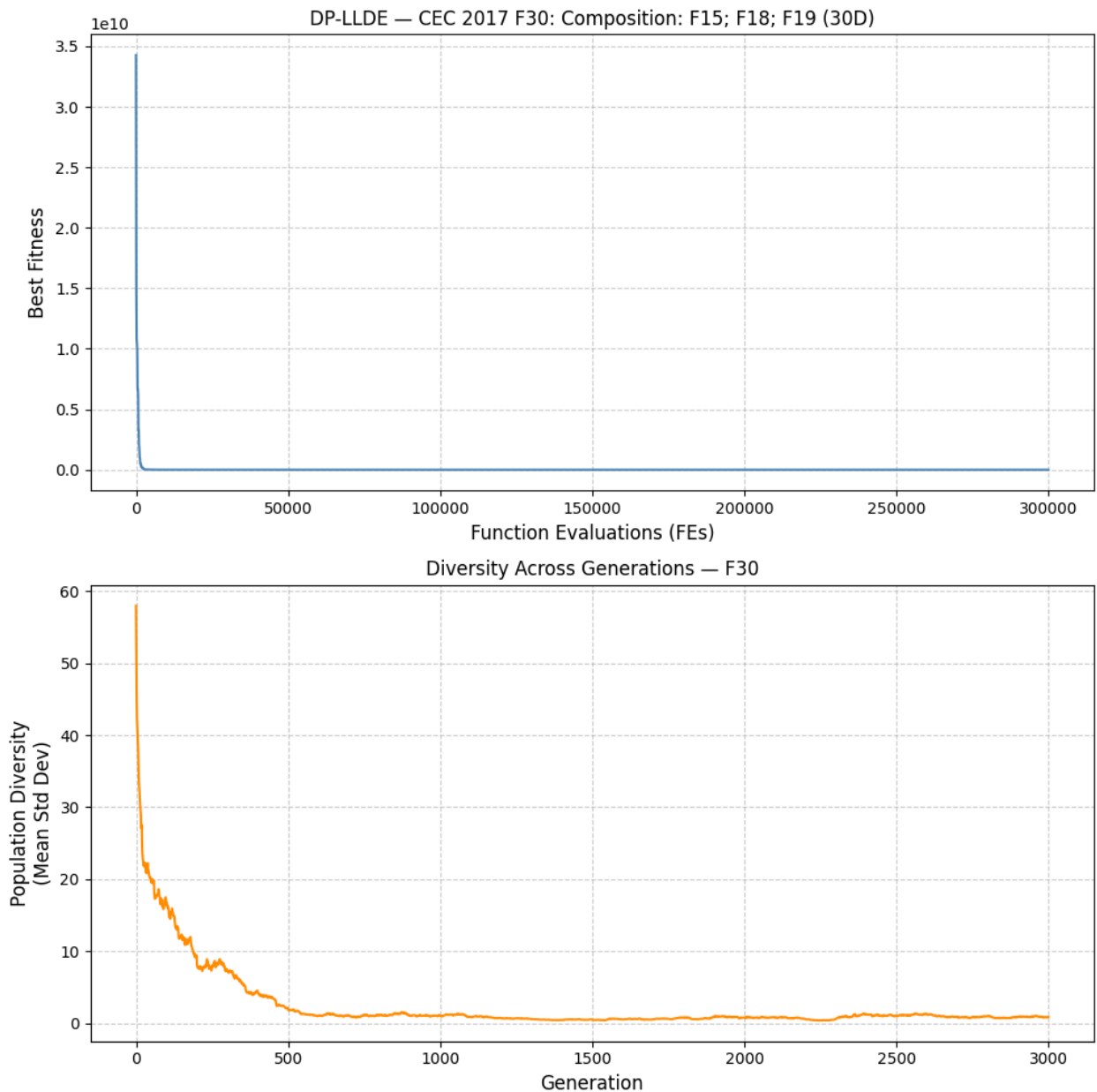
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Results:

Best Fitness : 6.726178e+03
 Global Optimum : 3.000000e+03
 Error (|best-optimum|) : 3.726178e+03
 Best Solution (5 dim) : [1.72072669 -26.17032565 86.16212817 62.47269797
 -3.69159366]
 D^(0) (init. diversity): 5.7976e+01

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Graph saved as convergence_F30.png

Section 7. Compiled Results

Print out a summary of all results in a tabular format

```
In [ ]: print('=' * 80)
print('SUMMARY — DP-LLDE on CEC 2017 (30D), Single Run, Seed=42')
print('=' * 80)
print(f'{"Function":<10} {"Best Fitness":>18} {"Global Optimum":>18} {"Error |')
print('-' * 70)
for fn in [f'F{i}' for i in range(1, 31)]:
```

```

if fn not in all_results:
    continue
r = all_results[fn]
if r['best_fit'] is None:
    print(f'{fn:<10} {"N/A (deprecated)":>18} {r["f_global"]:>18.2f} {"N/A"}')
else:
    print(f'{fn:<10} {r["best_fit"]:>18.6e} {r["f_global"]:>18.2f} {r["error"]:>18.2f}')
print('=' * 80)

```

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SUMMARY – DP-LLDE on CEC 2017 (30D), Single Run, Seed=42

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Function	Best Fitness	Global Optimum	Error best-opt
F1	1.000000e+02	100.00	2.304574e-09
F2	3.036758e+12	200.00	3.036758e+12
F3	9.542100e+02	300.00	6.542100e+02
F4	4.002958e+02	400.00	2.957904e-01
F5	5.304451e+02	500.00	3.044509e+01
F6	6.005122e+02	600.00	5.122056e-01
F7	7.652807e+02	700.00	6.528069e+01
F8	8.317910e+02	800.00	3.179096e+01
F9	9.001792e+02	900.00	1.791842e-01
F10	2.701312e+03	1000.00	1.701312e+03
F11	1.137867e+03	1100.00	3.786682e+01
F12	1.162609e+04	1200.00	1.042609e+04
F13	1.358050e+03	1300.00	5.804976e+01
F14	1.439882e+03	1400.00	3.988158e+01
F15	1.572524e+03	1500.00	7.252393e+01
F16	1.859954e+03	1600.00	2.599537e+02
F17	1.725904e+03	1700.00	2.590427e+01
F18	2.115914e+03	1800.00	3.159139e+02
F19	1.911364e+03	1900.00	1.136359e+01
F20	2.025104e+03	2000.00	2.510370e+01
F21	2.329003e+03	2100.00	2.290028e+02
F22	2.303621e+03	2200.00	1.036205e+02
F23	2.673362e+03	2300.00	3.733618e+02
F24	2.835197e+03	2400.00	4.351970e+02
F25	2.900878e+03	2500.00	4.008784e+02
F26	3.807063e+03	2600.00	1.207063e+03
F27	3.198871e+03	2700.00	4.988710e+02
F28	3.209576e+03	2800.00	4.095760e+02
F29	3.341206e+03	2900.00	4.412057e+02
F30	6.726178e+03	3000.00	3.726178e+03

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